Open Source FEA using adapy

Generated with *adapy* v0.154.0.

This report describes the main verification work of the simulation results produced by the Open Source Finite Element (FE) solvers Code Aster and Calculix.

The motivation behind producing this report is to increase confidence in the conversion of FEM models in *adapy* and also the results from the various FEA solvers.

Introduction

The results from *Code Aster* and *Calculix* and tools will be compared with proprietary FEA solvers such as;

- Abaqus
- Sestra (part of DNV's Sesam suite)

The simulations are pre- and post processed using adapy and this report is automatically generated using paradoc as part of a ci/cd pipeline within the *adapy* repositories.

For each new addition of code affecting the handling of FEM in *adapy* a new set of analyses is performed, whereas the results are gathered in this auto-generated document.

Please note! This report will not go into detail into the specifics of why the results are different between different element representations. This report will only investigate and compare the results from different FEA software.

FEA Solvers

- Calculix v2.23
- Code Aster v18.1.7
- Abaqus v2025
- Sestra v11.3.1.0

Python packages

The following python packages were instrumental in the creation of this document and the FEA results herein.

Adapy

This report was generated with *adapy* v0.154.0.

The intention behind *adapy* is to make it easier to work with finite element models and BIM models and act as a high-level scripting library for parametric and procedural modelling techniques.

Paradoc

paradoc was created to simplify the generation of reports by creating the structure of the document and text in markdown with a string substitution scheme that lets you easily pass in tables, functions and finally be able to produce production ready documents in PDF, ODT or Microsoft Word (docx).

Summary

A summary of all performed analyses will be performed in this section and a percentage difference for each FEA solver per type of analysis will be calculated and presented.

Analyses

Eigenvalue Analysis

A simple cantilevered beam is subjected to eigenvalue analysis with a total of 11 eigenmodes requested. 49 cases are compared across solvers: abaqus, calculix, code_aster, sesam.

The beam model will be analyzed by varying the following parameters:

- Mesh type (Solid, Shell, Line)
- Mesh element types (TET, HEX, TRI, QUAD)
- Mesh element order (1st and 2nd order)
- Reduced integration elements (R)
- Different FEA solvers (Calculix, Code Aster, Abaqus, Sesam)

Model Description

Property	Value
Description	Beam(“MyBeam”, [0.0, 0.5, 0.5], [3.0, 0.5, 0.5], “IPE400”, “S420”)
Length	3.00 m
Section	IPE400
Material	S420
Young’s modulus	2.100e+11 Pa
Yield stress	4.200e+08 Pa
Density	7850 kg/m³

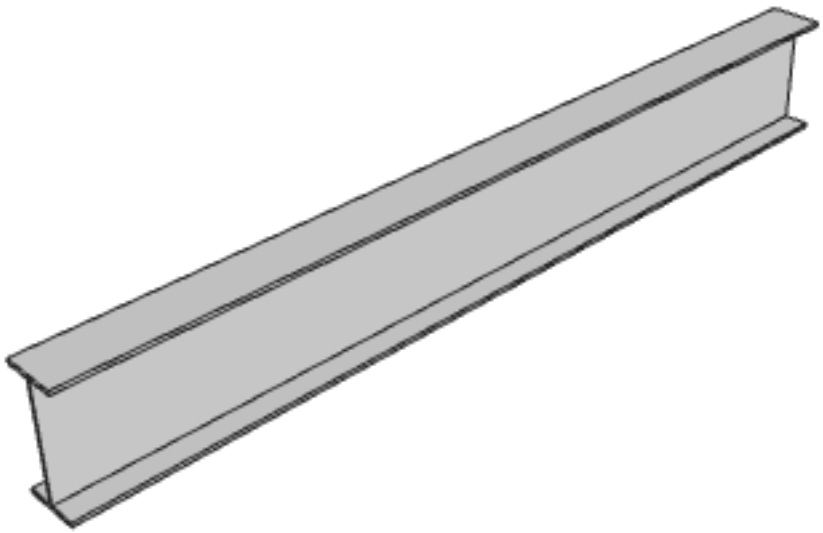

Figure 1: Cantilever beam geometry.

Summary

This section presents a comparison between the calculated results from the eigenvalue analysis of a cantilever beam using the different FEA tools.

Note! The `Sestra` and `Abaqus` analyses are not performed as part of the github actions step. The eigenmode results are kept in cached json files and imported during the compilation of the report using **paradoc**.

Eigenfrequency vs. mode number

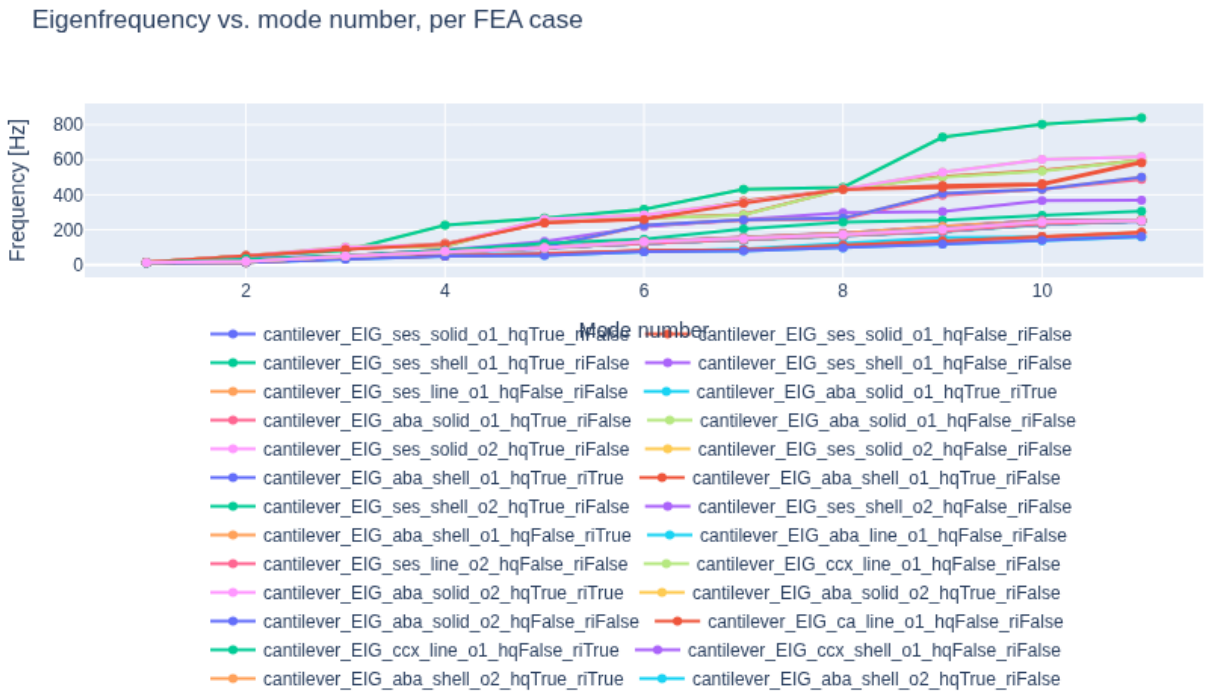

Figure 2: Eigenfrequency vs. mode number across FEA cases.

Comparison tables

Columns are sortable; the table grammar `{tbl:sortby:Mode:asc;index:no}` sets the initial sort and hides the index column.

Table 2: Eigenfrequency comparison (Hz) — solid, 1st order.

Mode	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	12.9766	15.7932	12.514	13.435	15.79	15.9285	15.9277	13.6205	13.6205	14.3018	13.5291
2	20.3015	51.5418	34.157	20.731	51.532	51.5896	51.5872	21.0685	21.0685	51.6653	34.7401
3	50.6181	97.7314	49.598	50.678	97.588	102.446	102.41	50.8252	50.8252	88.2331	50.7365
4	77.3921	105.691	55.271	82.799	104.8	116.744	116.512	84.066	84.066	116.873	82.2962
5	98.1185	260.887	64.305	100.19	260.59	261.877	261.806	102.334	102.334	240.766	123.303
6	133.016	267.131	74.866	223.14	266.26	284.697	284.465	227.25	227.25	258.286	147.139
7	153.833	287.461	90.033	254.98	285.05	357.062	356.306	255.872	255.872	351.823	205.267
8	178.565	432.347	124.58	260.1	432.31	432.36	432.35	266.519	266.519	431.955	243.985
9	214.689	503.862	155.99	396.57	500.86	529.297	528.489	407.869	407.869	439.038	254.838
10	253.463	539.506	159.95	431.8	534.26	601.943	601.586	432.207	432.207	456.762	282.888
11	256.05	598.2	162.56	486.96	596.78	617.646	616.155	500.931	500.931	582.562	305.247

Table 3: Eigenfrequency comparison (Hz) — solid, 2nd order.

Mode	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	12.9541	12.9537	12.943	12.955	12.954	12.9542	12.9542	12.9551	12.9551	12.9428	12.9427
2	20.0445	20.2945	20.029	20.15	20.294	20.2876	20.2877	20.1503	20.1503	20.0288	20.0288
3	50.5342	50.5401	50.498	50.541	50.54	50.5401	50.5402	50.5412	50.5412	50.4977	50.4977
4	76.7467	77.1278	76.687	76.89	77.128	77.2297	77.2297	76.8903	76.8903	76.6876	76.6875
5	97.4723	97.8296	97.391	97.594	97.83	97.8506	97.8505	97.5939	97.5939	97.3913	97.3912
6	130.865	132.478	130.81	131.37	132.48	133.015	133.014	131.365	131.365	130.814	130.809
7	148.496	153.631	148.47	149.96	153.63	154.804	154.802	149.958	149.958	148.478	148.474
8	172.243	178.114	172.17	173.92	178.12	181.029	181.026	173.925	173.925	172.177	172.171
9	201.904	215.815	201.84	205.29	215.82	219.501	219.497	205.292	205.292	201.846	201.839
10	246.353	253.203	246.17	251.35	253.2	253.193	253.193	251.347	251.347	246.177	246.165
11	252.742	254.466	252.57	252.92	254.47	254.685	254.684	252.917	252.917	252.573	252.573

Table 4: Eigenfrequency comparison (Hz) — shell, 1st order.

Mode	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRIR	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	12.7628	13.6414	12.373	12.78	13.6869	13.7254	13.642	12.7639	13.1463	14.2558
2	19.3943	20.1983	19.23	19.593	20.3643	36.6896	20.4181	19.6304	19.9719	53.2618
3	51.9912	52.1922	51.849	51.998	52.1874	52.2485	52.1927	51.9997	52.0595	87.9153
4	75.1918	79.5226	73.558	75.683	80.2058	83.7093	79.6596	75.4574	77.2361	121.85
5	95.9297	101.756	93.709	96.393	102.043	132.928	102.01	96.2959	98.7619	239.804
6	125.116	125.093	126.87	126.96	124.093	218.167	125.459	125.898	125.442	262.83
7	141.556	143.164	144.58	145.57	145.486	259.245	144.895	145.282	143.864	365.1
8	165.03	166.668	168.38	169.39	168.62	297.856	168.951	169.441	167.974	431.774
9	190.099	191.533	197.51	198.75	199.719	304.464	195.184	200.494	193.96	451.745
10	229.46	230.489	241.27	242.55	242.995	365.943	235.115	247.276	233.817	464.44
11	250.786	257.362	245.07	252.49	257.447	368.764	257.409	251.554	257.105	589.24

Table 5: Eigenfrequency comparison (Hz) — shell, 2nd order.

Mode	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	12.7554	12.7558	12.758	12.758	12.756	12.7562	12.7566	12.7576	12.7561	12.7559
2	19.4731	19.5048	19.486	19.486	19.567	19.5254	19.4509	19.5638	19.4736	19.5021
3	51.9702	51.9704	51.998	51.998	51.969	51.974	51.97	51.9801	51.97	51.9741
4	75.1245	75.1716	75.149	75.149	75.145	75.1848	75.1443	75.2685	75.1252	75.1399
5	95.8886	95.9337	95.92	95.92	96.001	95.9713	95.8707	96.0372	95.8893	95.9349
6	124.987	125.173	125	125	125.21	125.465	125.087	125.889	124.987	125.257
7	142.234	142.768	142.32	142.32	142.77	142.995	142.359	143.72	142.233	142.497
8	165.718	166.334	165.85	165.85	166.6	166.579	165.79	167.347	165.719	165.981
9	191.999	193.308	192.18	192.18	192.84	194.014	192.406	195.577	191.998	192.87
10	232.626	234.613	232.87	232.87	233.36	236.042	233.389	238.389	232.628	234.347
11	250.095	250.24	250.17	250.17	250.13	250.287	250.175	250.521	250.096	250.147

Table 6: Eigenfrequency comparison (Hz) — line, 1st order.

Mode	ses	aba	ccx	ca	ccxR
1	12.2454	10.785	12.9734	12.2469	12.9734
2	12.9431	12.928	52.5867	12.9678	52.5867
3	36.7525	32.342	81.1511	36.757	81.1511
4	50.5542	50.543	226.646	52.5834	226.646
5	61.3087	53.855	267.32	61.3161	267.32
6	79.793	75.296	316.915	81.0645	316.915
7	85.9466	79.59	431.04	85.9571	431.04
8	110.699	96.637	442.61	110.713	442.61
9	135.6	117.85	728.596	135.616	728.596
10	160.681	138.9	802.316	160.7	802.316
11	185.976	159.77	838.75	185.999	838.75

Table 7: Eigenfrequency comparison (Hz) — line, 2nd order.

Mode	ses	aba
1	12.2447	10.786
2	12.9377	12.932
3	36.7341	32.358
4	50.2627	50.557
5	61.2236	53.93
6	79.5712	75.502
7	85.7132	79.684
8	110.203	97.074
9	134.694	118.65
10	159.187	140.22
11	183.682	161.8

Bending modes against Euler-Bernoulli

The tables above compare the solvers with each other, which shows agreement but not accuracy. The bending modes of a clamped-free beam have a closed form, `f_n = (β_n L)² / (2π L²) √(E I / (ρ A))`. Modes are matched by the direction they move in rather than by number, because torsion modes fall between the bending modes at different places in each representation. Y

modes bend about the weak axis and Z modes about the strong axis. Only the first two in each direction are listed: past those, the shell and solid models have local modes of the web that also move mainly in Y.

Euler-Bernoulli ignores shear deformation and rotary inertia, and this beam is short: its length is 7.5 times its depth. A correct model that includes those effects lands below the closed form, by more in the higher modes. So the numbers to read are the trend and how closely the representations agree, and a large negative deviation here is the theory's limit rather than a solver's error.

Table 8: Line, 2nd order: bending frequencies against Euler-Bernoulli. Each solver column is the deviation in %. Y modes bend about the weak axis, Z about the strong axis.

Mode	closed form [Hz]	aba	ses
Y1	12.979	-0.36	-0.32
Y2	81.34	-2.04	-2.17
Z1	52.956	-4.53	-5.09

Table 9: Shell, 2nd order: bending frequencies against Euler-Bernoulli. Each solver column is the deviation in %. Y modes bend about the weak axis, Z about the strong axis.

Mode	closed form [Hz]	aba_QUAD	aba_QUADR	aba_TRI	ca_QUAD	ca_TRI	ccx_QUAD	ccx_QUADR	ccx_TRI	ses_QUAD	ses_TR
Y1	12.979	-1.71	-1.71	-1.72	-1.72	-1.72	-1.71	-1.72	-1.72	-1.73	-1.72
Y2	81.34	-7.61	-7.61	-7.62	-7.64	-7.62	-7.46	-7.62	-7.57	-7.64	-7.58
Z1	52.956	-1.81	-1.81	-1.86	-1.86	-1.86	-1.84	-1.85	-1.85	-1.86	-1.86

Table 10: Solid, 2nd order: bending frequencies against Euler-Bernoulli. Each solver column is the deviation in %. Y modes bend about the weak axis, Z about the strong axis.

Mode	closed form [Hz]	aba_HEX	aba_HEXR	aba_TET	ca_HEX	ca_HEXR	ca_TET	ccx_HEX	ccx_HEXR	ccx_TET	ses_HEX	ses_
Y1	12.979	-0.19	-0.28	-0.2	-0.19	-0.28	-0.19	-0.19	-0.28	-0.19	-0.19	-0.2
Y2	81.34	-5.47	-5.72	-5.18	-5.47	-5.72	-5.05	-5.47	-5.72	-5.05	-5.65	-5.1
Z1	52.956	-4.56	-4.64	-4.56	-4.56	-4.64	-4.56	-4.56	-4.64	-4.56	-4.57	-4.5
Z2	331.869	-23.79	-23.89	-23.7	-23.79	-23.89	-23.71	-23.79	-23.89	-23.71	-23.84	-23.

Effective modal mass

Effective modal mass [kg] per case, summed over the captured modes in the global X / Y / Z directions. Every solver reports it; Code Aster and Sesam give translational mass only. Summing over enough modes approaches the structure's total mass in each direction.

Table 11: Effective modal mass [kg] per case, summed over the captured modes in the global X/Y/Z directions. Code_Aster and Sesam report translational effective mass only.

Case	Modes	ΣMeff X	ΣMeff Y	ΣMeff Z
aba_line_o1	11	0	152.8	118.2
aba_line_o2	11	0	152.8	118.2
aba_shell_o1_QUAD	11	0	168	122
aba_shell_o1_QUADR	11	0	168.4	122
aba_shell_o2_QUAD	11	0	167.8	122
aba_shell_o2_QUADR	11	0	167.8	122
aba_shell_o2_TRI	11	0	167.9	122
aba_solid_o1_HEX	11	153.5	170.5	158.2
aba_solid_o1_HEXR	11	0	154.3	119
aba_solid_o1_TET	11	153.4	172	170.9
aba_solid_o2_HEX	11	0	163.3	158.6
aba_solid_o2_HEXR	11	0	163.2	158.7
aba_solid_o2_TET	11	0	162	158.6
ca_line_o1	11	0	152.1	114.8
ca_shell_o1_QUAD	11	0	167.6	163.7
ca_shell_o1_TRI	11	0	167.3	163.7

Case	Modes	ΣMeff X	ΣMeff Y	ΣMeff Z
ca_shell_o2_QUAD	11	0	167.8	122
ca_shell_o2_TRI	11	0	167.8	122
ca_solid_o1_HEX	11	153.5	170.8	158.3
ca_solid_o1_HEXR	11	0	164.7	158.5
ca_solid_o1_TET	11	153.4	172.6	171
ca_solid_o2_HEX	11	0	163.3	158.6
ca_solid_o2_HEXR	11	0	163.2	158.7
ca_solid_o2_TET	11	0	162.2	158.6
ccx_line_o1	11	153.9	174.4	161.2
ccx_line_o1R	11	153.9	174.4	161.2
ccx_shell_o1_QUAD	11	0	168.1	122
ccx_shell_o1_QUADR	11	158	176.8	176.5
ccx_shell_o1_TRI	11	0	174.8	163.6
ccx_shell_o2_QUAD	11	0	167.8	122
ccx_shell_o2_QUADR	11	0	167.8	122
ccx_shell_o2_TRI	11	0	167.8	122
ccx_solid_o1_HEX	11	153.4	170.7	158.2
ccx_solid_o1_HEXR	11	153.5	171.8	171.3
ccx_solid_o1_TET	11	153.4	172.6	171
ccx_solid_o2_HEX	11	0	163.3	158.6
ccx_solid_o2_HEXR	11	0	163.2	158.7
ccx_solid_o2_TET	11	0	162.2	158.6
ses_line_o1	11	0	153	120.2
ses_line_o2	11	0	152.9	118.7
ses_shell_o1_QUAD	11	0	167.6	122
ses_shell_o1_TRI	11	0	167.1	163.6
ses_shell_o2_QUAD	11	0	167.8	122
ses_shell_o2_TRI	11	0	167.9	122
ses_solid_o1_HEX	11	0	161.7	158.6
ses_solid_o1_TET	11	153.4	172.1	170.9
ses_solid_o2_HEX	11	0	163.1	158.7
ses_solid_o2_TET	11	0	162	158.6

The per-mode effective masses and participation factors, compared across solvers for every mesh configuration, are in the appendix.

Short description:

Fem formats

- ccx: Calculix
- ca: Code Aster
- ses: Sesam
- aba: Abaqus

Element Types

- TET: Tetrahedrons
- HEX: Hexahedrons
- TRI: Triangle
- QUAD: Quadrilateral

Reduced Integration

All element types ending with capital R (QUAD**R**, HEX**R** etc.) are reduced integration elements.

Nonlinear Analysis

A simple cantilever beam with a series of cutouts is subjugated to a large vertical acceleration field. The level of plastic strains are monitored for comparison across different FE tools.

Model Description

Summary

Plate Analysis

The analyses above are one beam, in three representations. This section is the plate case: an `ada.Plate` object, meshed into shell elements by adapy's own meshing module, written out as

an input deck per solver, and checked against a closed form as well as against the other solvers.

Nothing here drives a solver through its API. Each solver reads a file adapy wrote — an `.inp`, a `.fem`, a `.comm/.med` — which is the same route a user's own model takes, so what this section verifies is the thing adapy actually ships.

9 cases ran on the machine that built this report, using: calculix, code_aster. The remaining columns come from cached results measured elsewhere, the same arrangement the eigenvalue section uses for Abaqus and Sestra.

Model Description

A rectangular strip, simply supported on its two short edges and held in cylindrical bending on the two long ones, under a uniform surface pressure.

Property	Value
Span	4.00 m
Width	0.50 m
Thickness	0.010 m
Pressure	1000 Pa
Material	S355 (E = 210 GPa, ν = 0.3, ρ = 7850 kg/m^3)

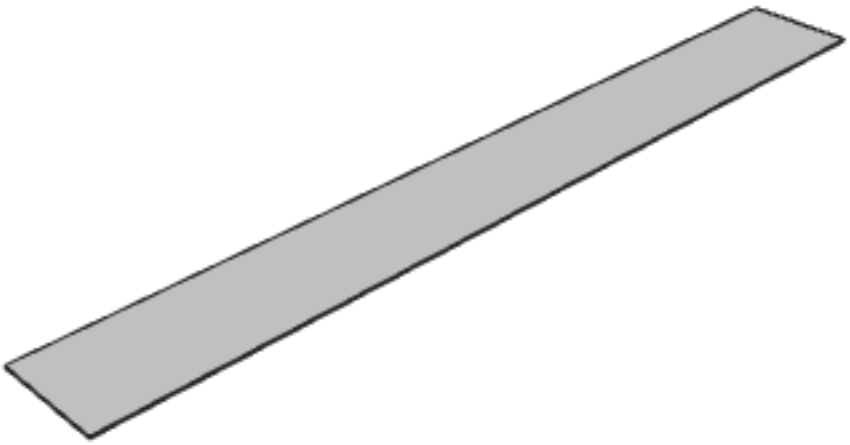

Figure 3: Plate strip geometry.

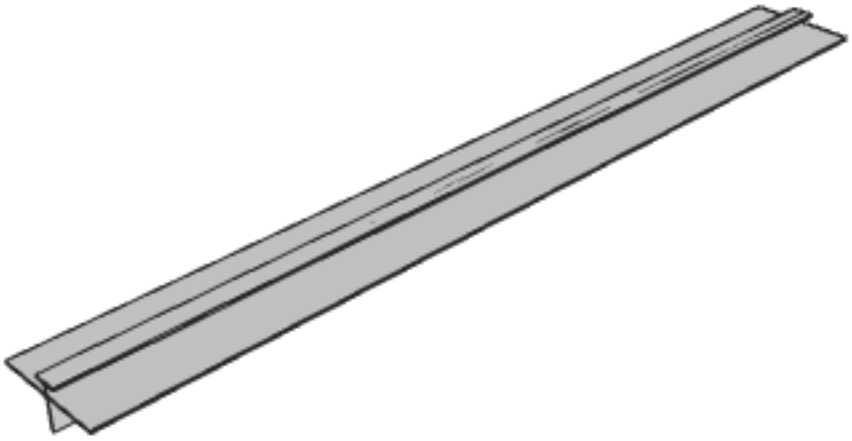

Figure 4: Plate strip with a T-profile stiffener on its centreline.

The span-to-thickness ratio is 400, so transverse shear sits several orders below the discretisation error and the thin-plate closed forms are the right reference rather than an approximation of one.

The supports are written as four node sets rather than one per edge. Simple support and cylindrical bending overlap at the four corners, and the obvious spelling constrains `u2` twice there: Abaqus and Sestra accept that, while Code_Aster refuses it outright (`<ASSEMBLA_26> … est bloqué plusieurs fois`). A model that only some solvers will read is not a cross-solver case, so the corner nodes carry only the rotation the long edges add.

Static analysis under pressure

The static case is the only one that reaches the distributed-load records — Abaqus' `*Dsload`, Calculix' `*DLOAD`, Code_Aster's `FORCE_COQUE` and Sesam's `BEUSLO`. The eigenvalue analyses reach the mass and stiffness matrices and nothing else, so a pressure written wrongly is invisible to them.

Mid-span deflection against the closed form $5\,q\,L^4 / (384\,D)$, with $D = E\,t^3 / (12\,(1-\nu^2))$ — which evaluates to 0.173333 m for this strip. Halving the seed should quarter the remaining error; the three rows are a factor of two apart so the order is readable off the table rather than asserted.

Table 13: Plate strip mid-span deflection [m] under 1 kPa, by solver and mesh seed. The closed form is 5 q L^4 / (384 D) for a strip in cylindrical bending.

Seed [m]	aba	ca	ccx	ses	closed form
0.125	0.173065	0.173198	0.173065	0.173198	0.173333
0.0625	0.173269	0.173299	0.173268	0.173299	0.173333
0.03125	0.173319	0.173325	0.173319	0.173325	0.173333

The same strip with a T-profile stiffener along its centreline, meshed as line elements sharing the shells' nodes. There is no closed form for it — a parallel-spring estimate is not one — so this is an agreement case between solvers, and it is what carries a T section through the deck writers.

Table 14: The same strip with a T-profile stiffener on its centreline [m]. No closed form: this is a cross-solver agreement case.

Seed [m]	aba_st	ca_st	ses_st
0.125	0.000584531	0.000710295	0.000720708
0.0625	0.000585478	0.000710711	0.000720976
0.03125	0.000585996	0.000710815	0.000720795

Sign convention

Every writer above places a positive pressure so that it pushes **into** the face it names. That is Abaqus' convention, and it has to be shared, because a load direction that depends on the output format makes a cross-solver comparison meaningless.

It is worth stating which face that is, because it is not a global direction: gmsh winds every element of this strip with its normal along −z, so the positive face looks *down* and a positive pressure on it moves the strip *up*. The deflections above are therefore positive. `tests/fem/test_pressure_load_cross_format.py` pins this per format.

Eigenvalue analysis

The same plate, as an eigenvalue analysis, against $f_n = n^2\,\pi / (2\,L^2)\,\sqrt{D / (\rho\,t)}$ — the simply supported beam frequencies with the beam's $EI / (\rho\,A)$ replaced by the strip's $D / (\rho\,t)$. Valid for the cylindrical modes, which are the ones the long-edge constraint admits.

Table 15: Plate strip eigenfrequencies [Hz], by solver, at a 0.0625 m seed. The closed form is f_n = n^2 pi / (2 L^2) sqrt(D / (rho t)), valid for the cylindrical modes the long-edge constraint admits.

Mode	aba	ca	ccx	ses	closed form
1	1.5371	1.53661	1.53675	1.53661	1.53661
2	6.1536	6.14644	6.14861	6.14644	6.14644
3	13.866	13.8295	13.8405	13.8295	13.8295
4	24.7	24.5858	24.6205	24.5858	24.5857
5	38.695	38.4153	38.5002	38.4153	38.4152
6	55.9	55.3181	55.4943	55.3182	55.3179

Short description of the solver column names:

- ccx: Calculix
- ca: Code Aster
- ses: Sesam
- aba: Abaqus

A `_st` suffix marks the stiffened variant.

NAFEMS Benchmarks

The cantilever and the plate are checked against formulas this report derives. These two cases are checked against values published independently, in The Standard NAFEMS Benchmarks (TNSB Rev. 3).

12 cases ran on the machine that built this report, using: calculix, code_aster. Other columns come from cached results measured elsewhere, the same arrangement the eigenvalue section uses for Abaqus and Sestra.

Model Description

Both benchmarks are a quarter of the region between two confocal ellipses, with point D on the inner ellipse at x = 2 m, y = 0.

Property	Value
Inner ellipse	$(x/2)^2 + y^2 = 1$
Outer ellipse	$(x/3.25)^2 + (y/2.75)^2 = 1$
Material	E = 210 GPa, $\nu = 0.3$
Elements	20-node hexahedra

The mesh is written directly rather than generated. Confocal ellipses map exactly onto a rectangle in elliptic coordinates, so every node sits on the true geometry and the elements are orthogonal. The error in the tables is then the element's and the solver's, not a mesher's. Each row doubles the divisions of the one before it.

The stress at D is each solver's own value at that node, averaged over the elements meeting there, the way each solver hands it out. Calculix writes nodal stresses. Code_Aster's and Abaqus' are each element's extrapolation to its nodes (Abaqus' `*Element Output, position=NODES`). Sestra stores the stresses at each element's 2×2×2 Gauss points, which adapy extrapolates to the nodes with the trilinear field through them, as a postprocessor does.

The restraints are collected per node and written once for each distinct set of degrees of freedom. One set per face would hold a degree of freedom twice along the edges where faces meet, which Code_Aster refuses.

LE1: elliptic membrane

Plane stress, under a uniform outward pressure of 10 MPa on the outer edge, with symmetry on x = 0 and y = 0. The target is the tangential stress at D, $\sigma_{yy} = 92.7$ MPa.

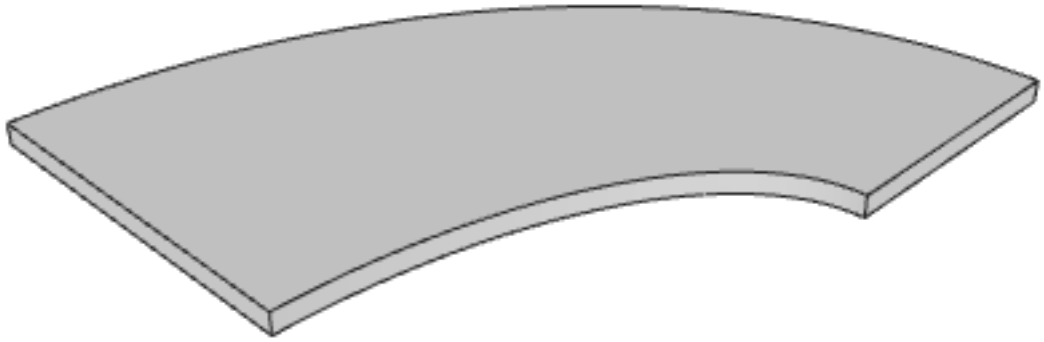

Figure 5: NAFEMS LE1 geometry.

The benchmark is defined in plane stress, and adapy's solid elements are used for it here: one layer over half the 0.1 m thickness, with a symmetry plane at mid-thickness and free faces. That leaves no through-thickness stress, so the solution is the plane-stress one.

Table 17: NAFEMS LE1, elliptic membrane: sigma_yy at point D [MPa] and its deviation from the target [%].

Divisions	Elements	aba [MPa]	aba Δ [%]	ca [MPa]	ca Δ [%]	ccx [MPa]	ccx Δ [%]	ses [MPa]	ses Δ [%]	target [MPa]
2	8	87.252	-5.88	88.883	-4.12	87.827	-5.26	89.718	-3.22	92.7
4	32	91.632	-1.15	92.205	-0.53	91.851	-0.92	92.165	-0.58	92.7
8	128	92.738	0.04	92.684	-0.02	92.764	0.07	92.643	-0.06	92.7

σyy over the plan at the finest mesh, the quantity the benchmark is judged on. All solvers share one colour scale, and D is marked with each solver's own value.

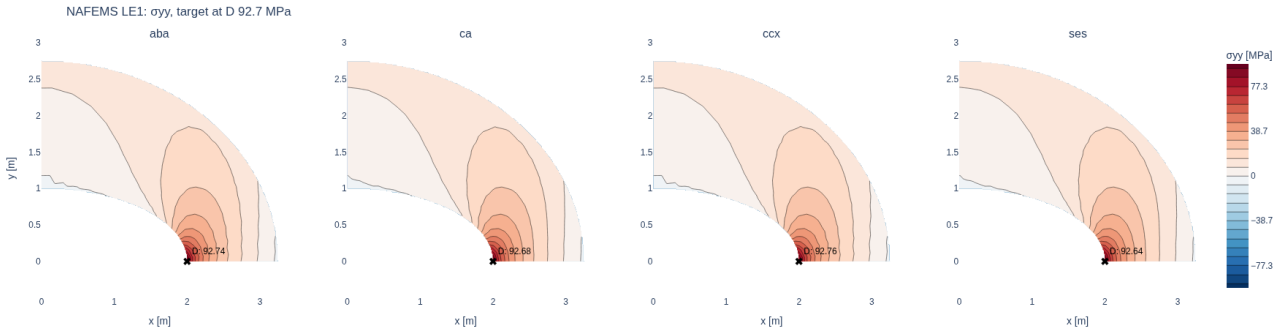

Figure 6: NAFEMS LE1: σyy at the finest mesh, by solver.

Displacement at the finest mesh, deformed, from each solver. The interactive view shows the same result in 3D.

Results at 8 radial divisions

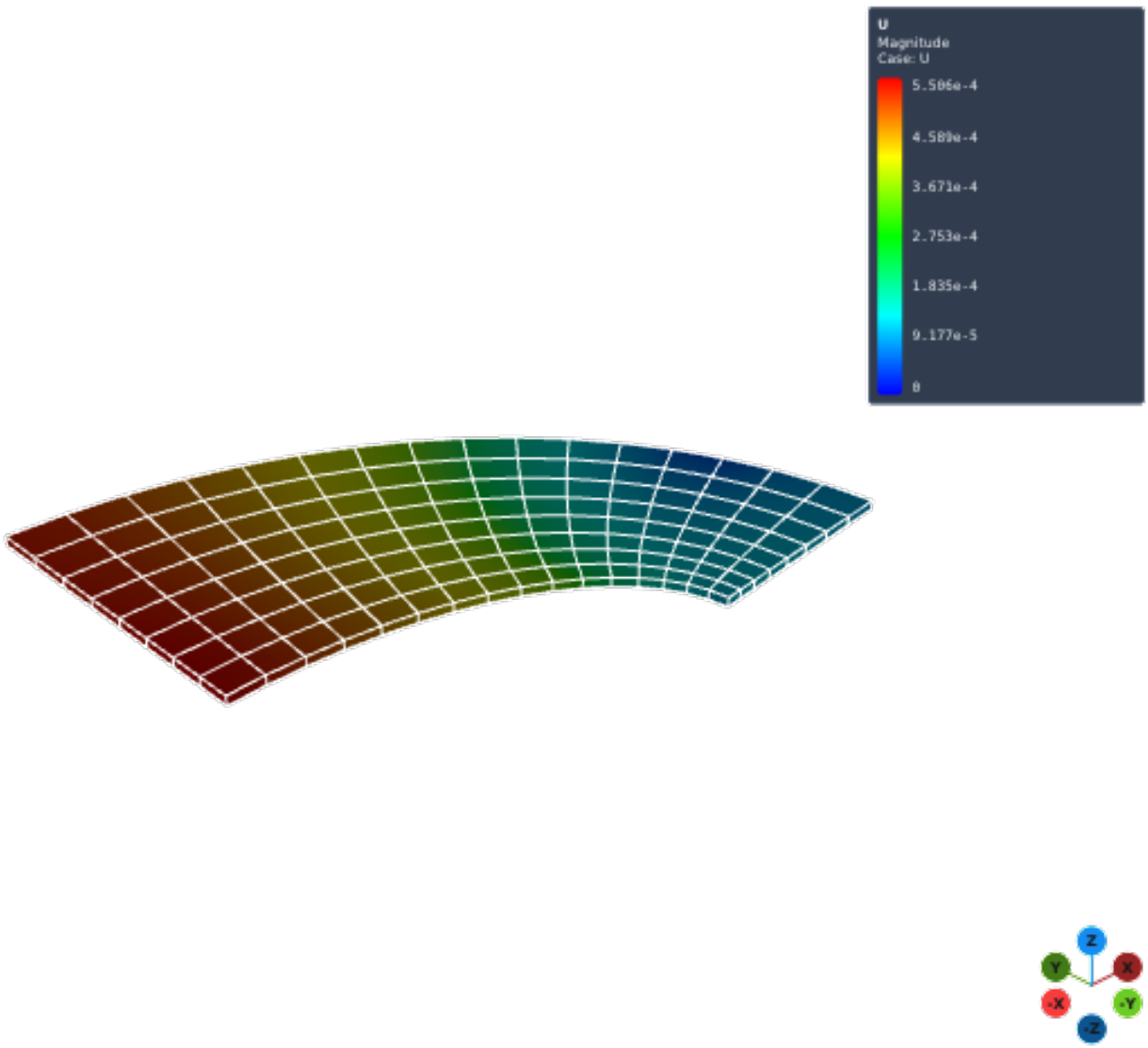

Figure 7: Abaqus — displacement

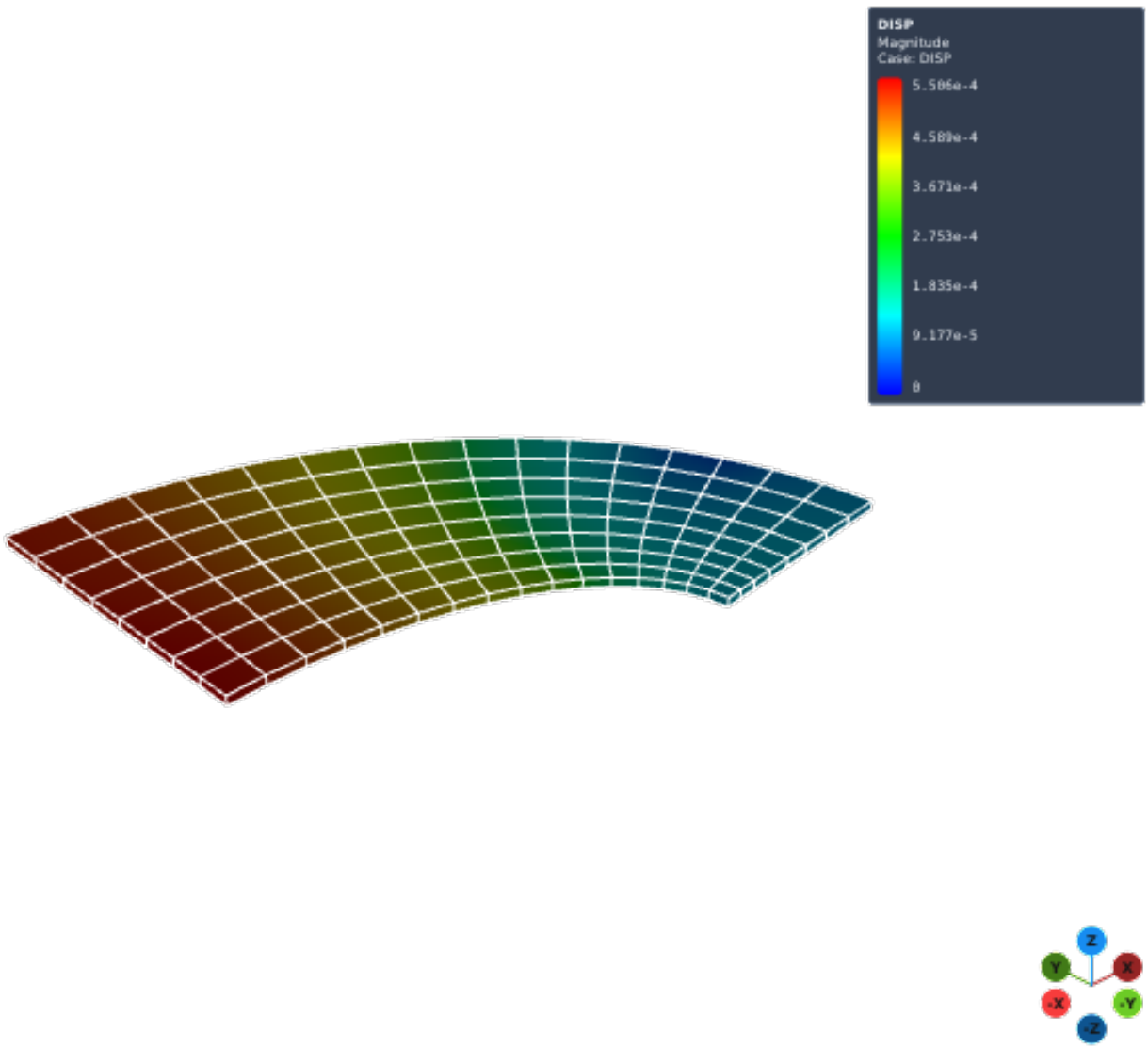

Figure 8: Calculix — displacement

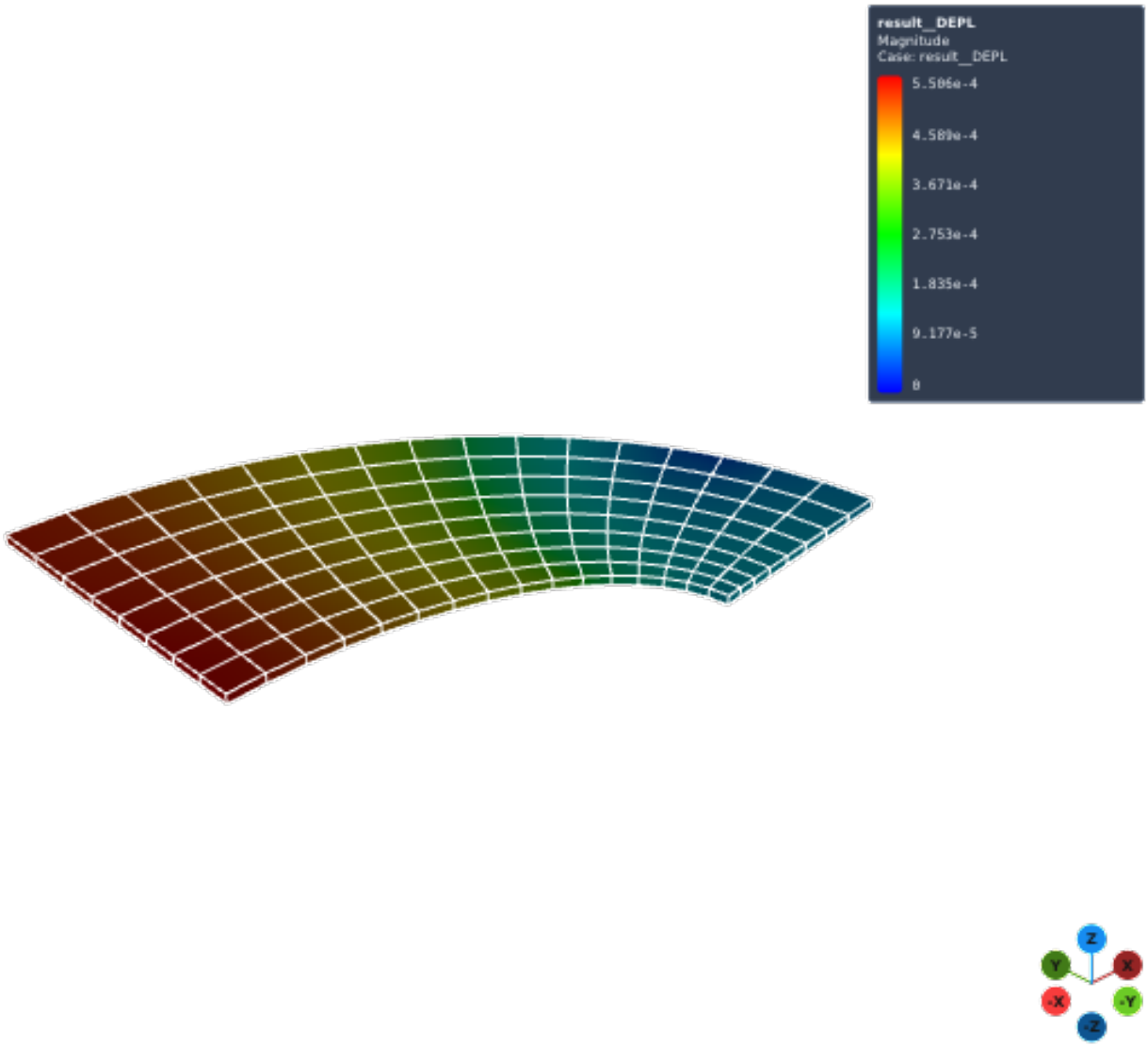

Figure 9: Code_Aster — displacement

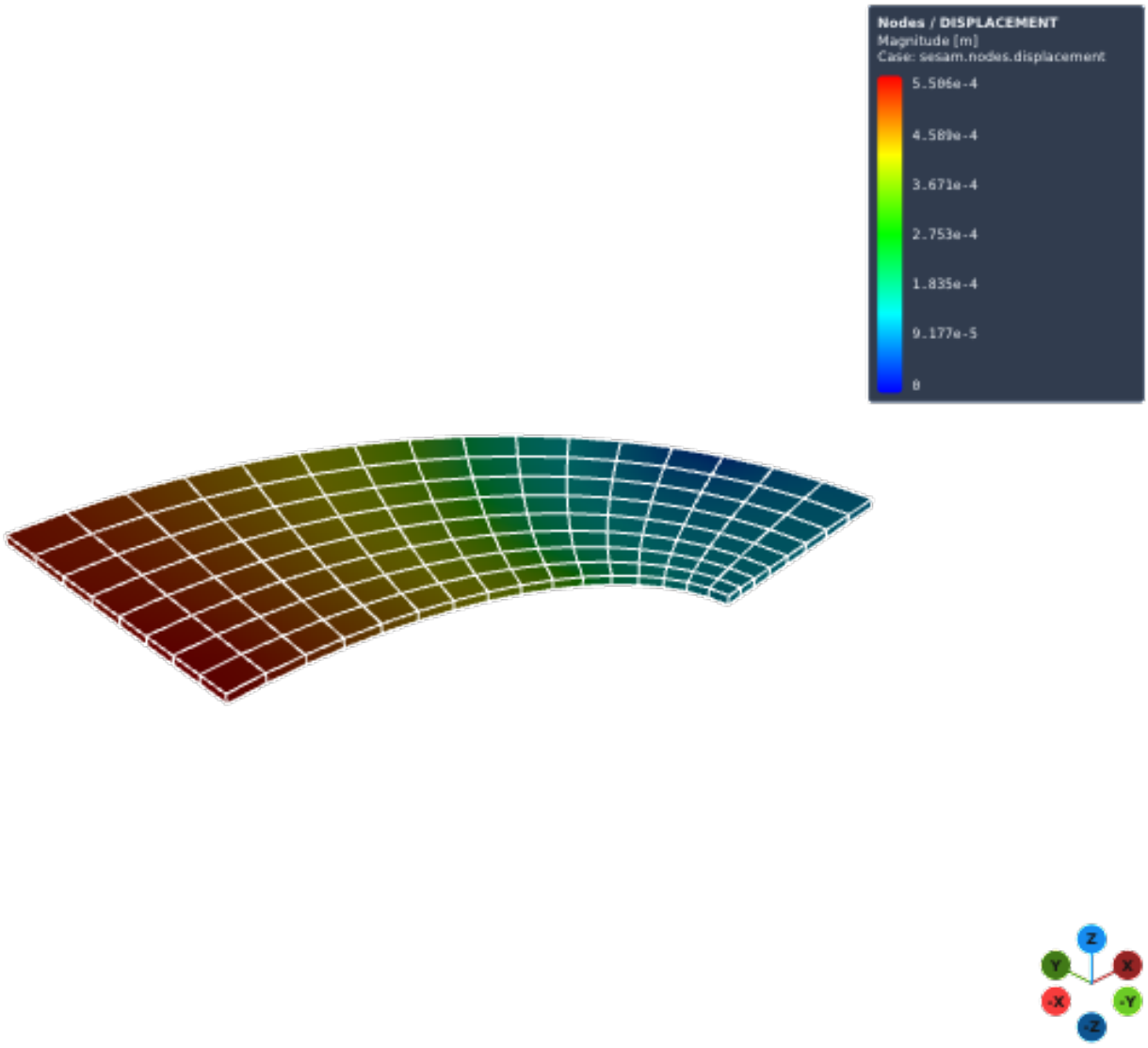

Figure 10: Sesam — displacement

Solver	Peak u_z [mm]	Peak u_x [mm]	Peak u_y [mm]
Abaqus	0.0066	0.1747	0.5506
Calculix	0.0066	0.1747	0.5506
Code_Aster	0.0066	0.1747	0.5506
Sesam	0.0066	0.1747	0.5506

LE10: thick plate under pressure

A plate 0.6 m thick, under 1 MPa on its upper face. Symmetry on x = 0 and y = 0, the outer face held in x and y, and the mid-plane of the outer face also held in z. The target is σyy on the upper face at D, -5.38 MPa.

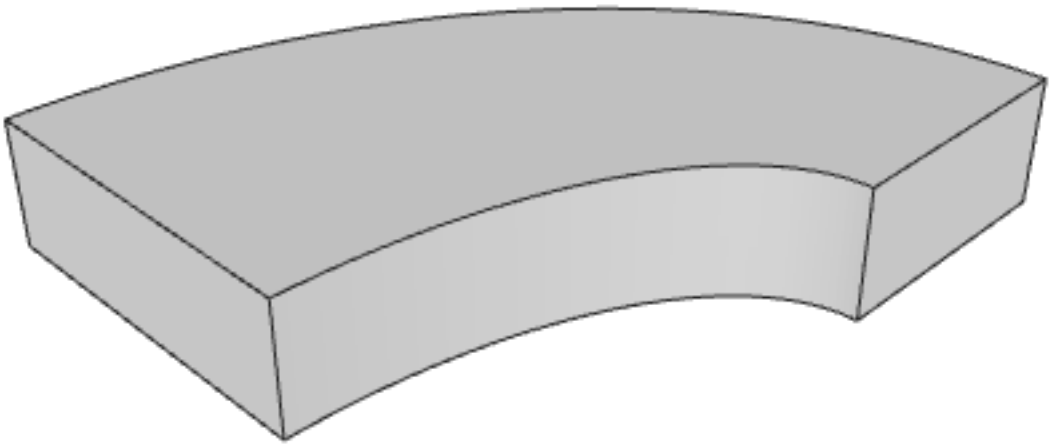

Figure 11: NAFEMS LE10 geometry.

Table 19: NAFEMS LE10, thick plate under pressure: sigma_yy at point D [MPa] and its deviation from the target [%].

Divisions	Elements	aba [MPa]	aba Δ [%]	ca [MPa]	ca Δ [%]	ccx [MPa]	ccx Δ [%]	ses [MPa]	ses Δ [%]	target [MPa]
2	16	-5.419	0.72	-5.529	2.77	-5.471	1.69	-5.606	4.2	-5.38
4	128	-5.438	1.07	-5.472	1.71	-5.454	1.38	-5.462	1.52	-5.38
8	1024	-5.418	0.71	-5.409	0.54	-5.42	0.75	-5.402	0.41	-5.38

σyy on the upper face at the finest mesh, the quantity the benchmark is judged on. All solvers share one colour scale, and D is marked with each solver's own value.

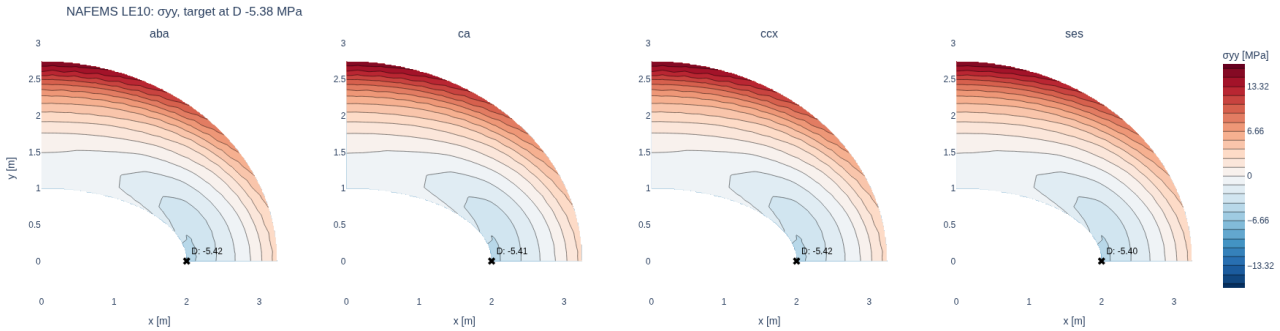

Figure 12: NAFEMS LE10: σyy at the finest mesh, by solver.

Displacement at the finest mesh, deformed, from each solver. The interactive view shows the same result in 3D.

Results at 8 radial divisions

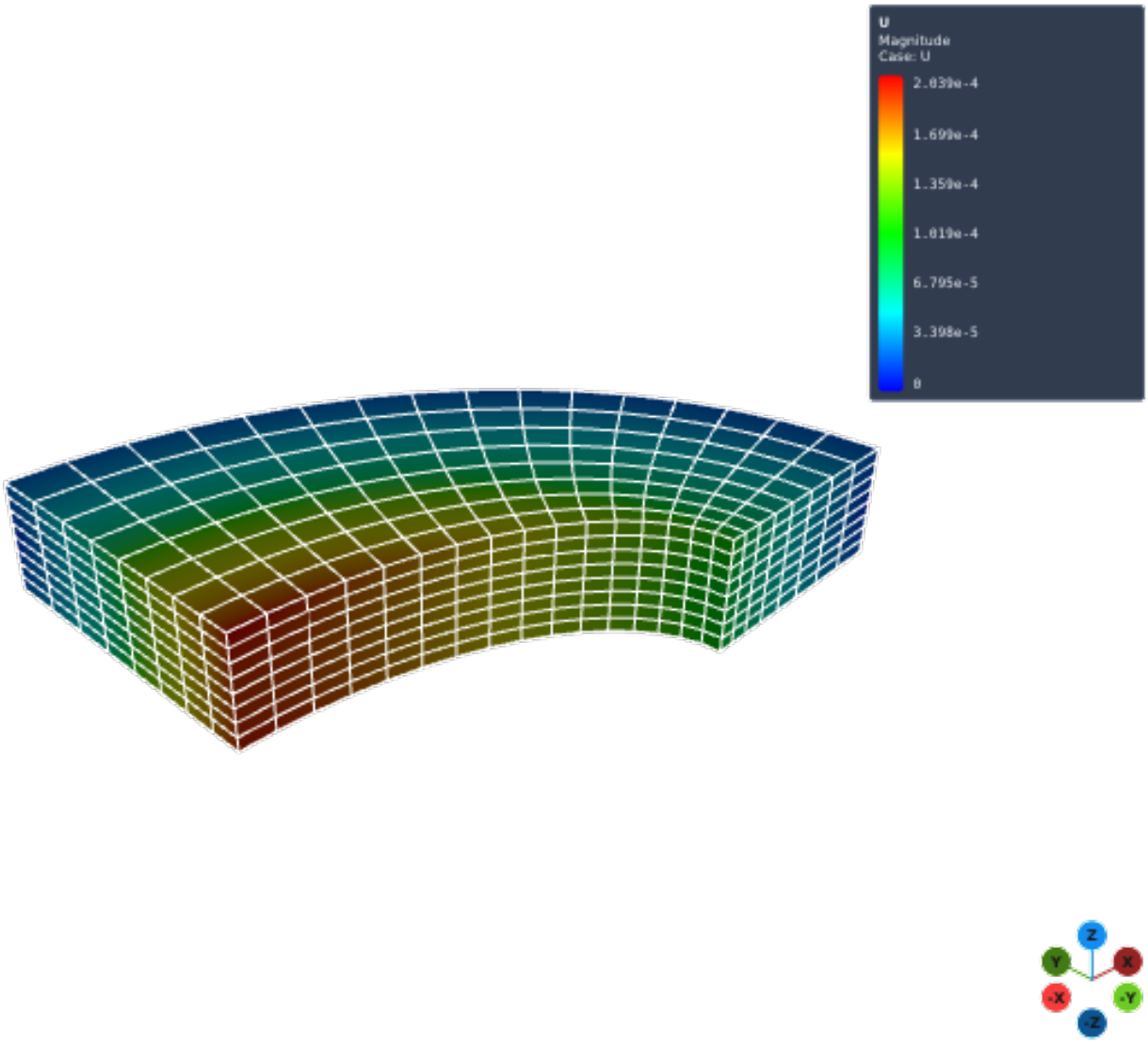

Figure 13: Abaqus — displacement

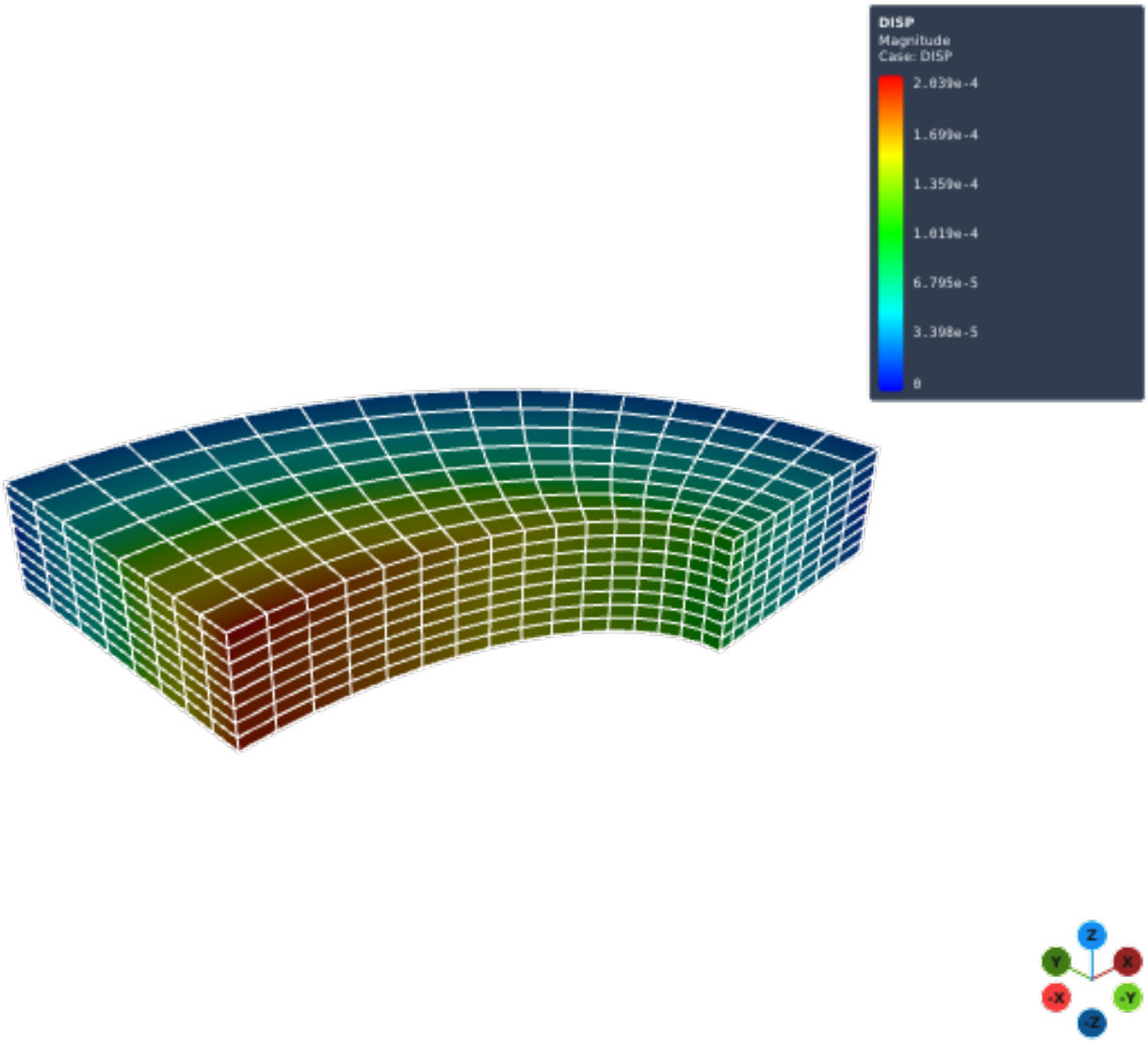

Figure 14: Calculix — displacement

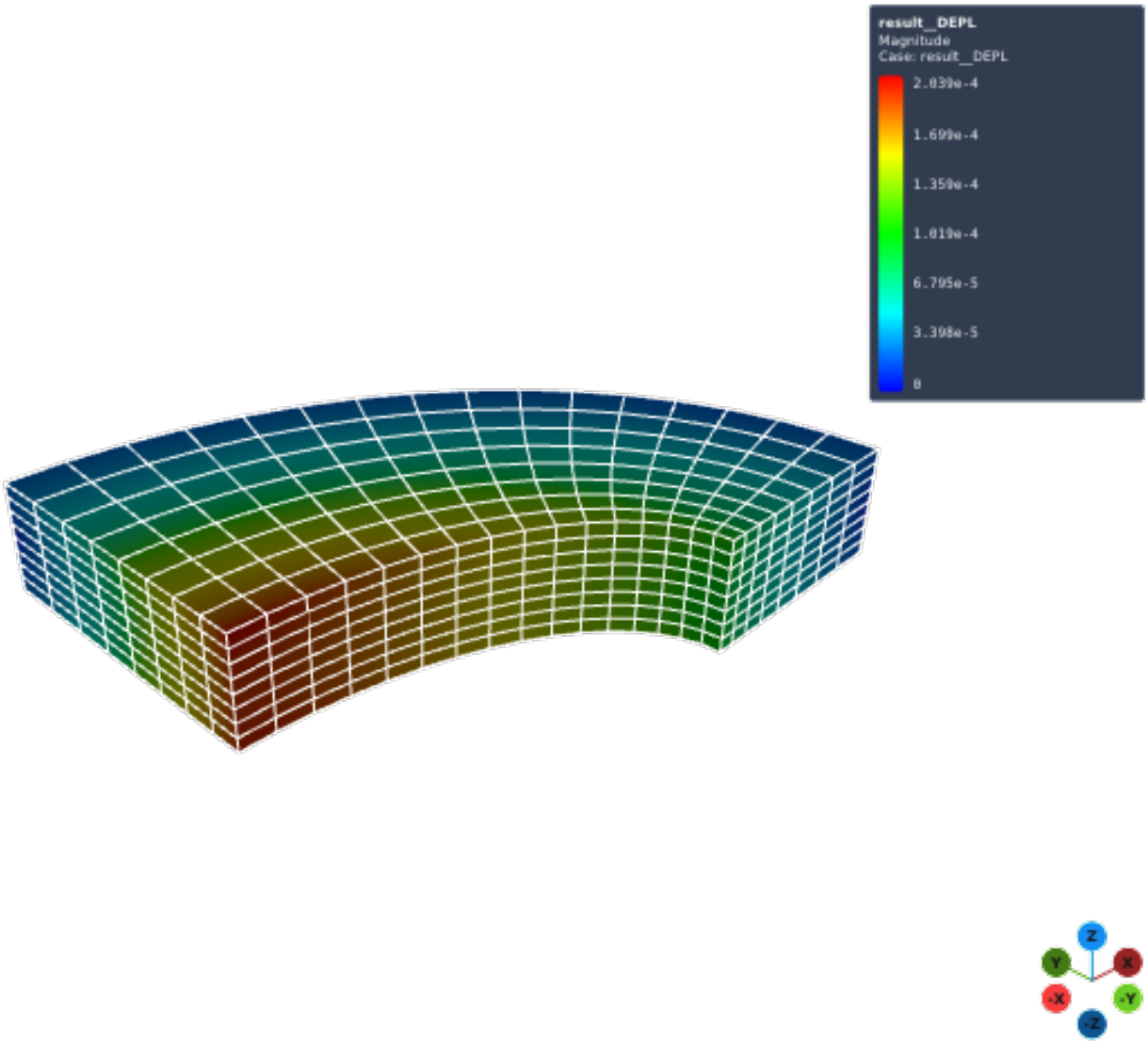

Figure 15: Code_Aster — displacement

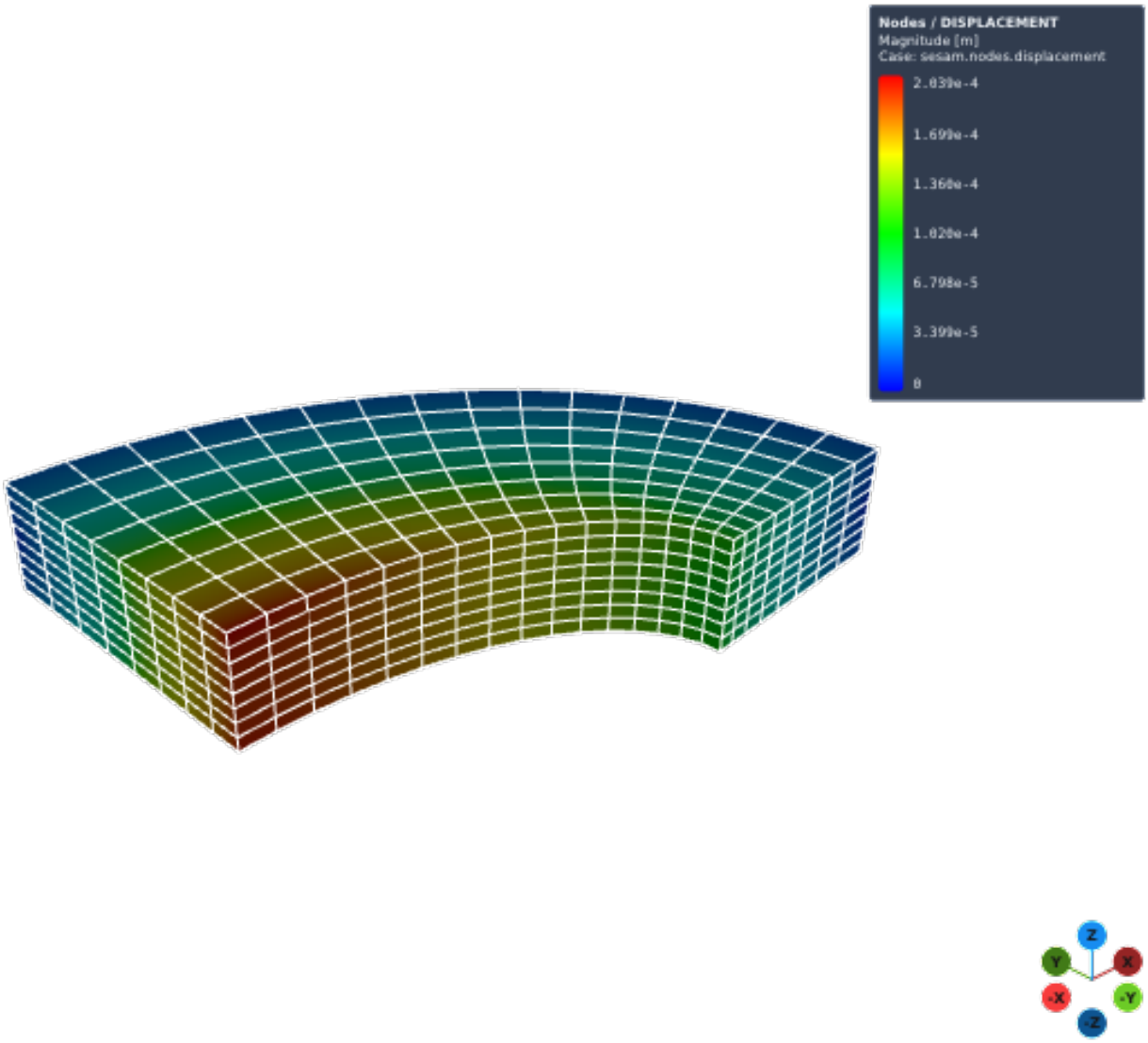

Figure 16: Sesam — displacement

Solver	Peak u_z [mm]	Peak u_x [mm]	Peak u_y [mm]
Abaqus	0.1997	0.0286	0.0418
Calculix	0.1997	0.0286	0.0418
Code_Aster	0.1997	0.0286	0.0418
Sesam	0.1997	0.0286	0.0418

Detailed results

Mode shapes and deflected shapes from every format that ran on the machine that built this report, laid out so the formats can be read against each other: one section per mesh configuration, one heading per mode, and the formats side by side under it. A swapped mode pair or a differently deflected shape shows up as two pictures that do not match.

Abaqus and Sesam cannot run where this report is built. Their mode shapes are the raw results a licensed machine committed beside its cached frequencies, and the figures are rendered from them here; a case with no committed results has no figures. Solver versions: Abaqus v2025, Calculix v2.23, Code Aster v18.1.7, Sesam v11.3.1.0.

Cantilever modal participation

Per-mode effective modal mass and participation factors in the global X / Y / Z directions, for every mesh configuration, with the formats side by side. Code Aster and Sesam report the translational directions only.

A participation factor Γ is relative to how the solver normalised its eigenvectors: Abaqus scales the largest displacement to 1, the others scale to unit generalised mass. Raw factors from different solvers therefore do not compare. The tables give the factor scaled to unit generalised mass, sign(Γ)·√Meff, which is what the others report as-is. Its sign is the solver's choice of eigenvector sign and may differ from mode to mode. The effective mass Meff = $\Gamma^2 \cdot$m_gen does not depend on the normalisation.

Solid, 1st order

Table 21: Effective modal mass [kg] per mode, global X/Y/Z — solid, order 1.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	0	0	0	0	0	0	0	0	0	0	0
1	Y	116.4	117.5	116.9	116.4	117.5	118.2	118.2	116.4	116.4	116.5	116.4
1	Z	0	0	0	0	0	0	0	0	0	0	0
2	X	0	0	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0	0	0
2	Z	0	118.3	0	0	118.2	118.4	118.4	0	0	118.5	0
3	X	0	0	0	0	0	0	0	0	0	0	0
3	Y	0	34.9	0	0	34.9	35.5	35.5	0	0	36.2	0
3	Z	118.3	0	119	118.2	0	0	0	118.2	118.2	0	118.3
4	X	0	0	0	0	0	0	0	0	0	0	0
4	Y	35.9	0	2.1	36	0	0	0	36	36.1	0	36.1
4	Z	0	0	0	0	0	0	0	0	0	0	0
5	X	0	0	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	0	12.6	0
5	Z	0	39.8	0	0	39.7	39.9	39.9	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0	0	0
6	Y	0	13.3	31.4	12.4	13.2	12.8	12.8	12.5	12.5	0	0
6	Z	0	0	0	0	0.1	0	0	0	0	40.2	0
7	X	0	0	0	0	0	0	0	0	0	0	0
7	Y	3.2	0	2.8	0	0	0	0	0	0	0	8.9
7	Z	0	0	0	40	0	0	0	40	40.1	0	0
8	X	0	153.4	0	0	153.4	153.4	153.4	0	0	153.5	0
8	Y	5.7	0	0	0	0	0	0	0	0	0	3.1
8	Z	0	0	0	0	0	0	0	0	0	0	0.3
9	X	0	0	0	0	0	0	0	0	0	0	0
9	Y	0.4	6.3	1.1	5.7	6.3	6.1	6.1	5.9	5.9	0.4	0
9	Z	0	0	0	0	0	0	0	0	0	0	39.9
10	X	0	0	0	153.5	0	0	0	153.4	153.5	0	0
10	Y	0	0	0	0	0	0	0	0	0	6.1	0
10	Z	40.3	0	0	0	0	12.8	12.8	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	0	0	0	0	0	0	0
11	Z	0	12.8	0	0	12.8	0	0	0	0	12.7	0

Table 22: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — solid, order 1.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	-0	0	0	-0	0	-0	0	-0	0	0	0
1	Y	10.79	-10.841	10.813	10.788	10.838	-10.871	10.871	10.787	10.787	10.794	10.791
1	Z	0	0.017	-0	-0	-0.017	0.019	-0.019	-0	-0	0	-0.001
2	X	-0	-0	0	0	0	0	0	-0	-0	-0	0.001
2	Y	-0	-0.028	0	0	0.028	0.03	0.03	0	0	0	0.022
2	Z	0	-10.876	-0	-0	10.874	10.88	10.88	-0	0	-10.884	-0
3	X	-0	0	-0	-0	-0	0	-0	-0	-0	0	0
3	Y	-0	5.911	0	0	-5.909	5.961	-5.961	0	0	-6.013	0
3	Z	10.876	-0.019	10.907	10.872	0.019	-0.02	0.02	10.871	10.872	-0	10.876
4	X	-0	-0.003	-0	-0	0.003	0.003	0.003	0	0	0	-0
4	Y	5.995	0.069	-1.456	-6.002	-0.075	-0.108	-0.11	6.004	-6.004	-0	-6.01
4	Z	-0	-0.003	0	0	0.003	0.007	0.008	0	-0	-0	-0.004
5	X	0	-0.001	-0	0	-0.001	0	-0	-0	-0	-0	-0.002

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
5	Y	-0	0.122	-0	-0	0.132	-0.028	0.029	0	-0	-3.549	-0.084
5	Z	0	-6.305	0	0	-6.305	6.313	-6.315	-0	-0	0	0.006
6	X	-0	0	0	-0	0	0	-0.001	0	-0	0	-0.001
6	Y	-0.114	3.642	-5.603	3.522	3.638	-3.576	3.577	3.53	3.532	-0	0.192
6	Z	0	0.207	-0	0	0.226	-0.056	0.057	0	-0	-6.34	-0.015
7	X	-0	0	-0	-0	0	-0.021	-0.021	0	0	0	0
7	Y	-1.788	0.027	-1.671	0	0.03	0.04	0.04	-0	-0	-0	2.99
7	Z	0	-0.033	0	-6.326	-0.035	-0.02	-0.02	6.327	-6.329	-0	0.071
8	X	0	12.384	-0	-0	12.385	12.384	12.386	0	-0	12.389	0.001
8	Y	2.39	-0.001	-0	0	-0.001	0	0	0	0	0	1.76
8	Z	0	0	-0	0	0	-0	-0	-0	-0	0	-0.503
9	X	-0	-0.005	-0	0	-0.006	0.001	0.001	-0	-0	0	0.001
9	Y	0.662	-2.518	-1.056	-2.385	-2.515	-2.462	-2.464	2.426	-2.43	-0.619	-0.115
9	Z	-0	0.007	-0	0	0.007	0.064	0.064	0	0	-0	-6.318
10	X	-0	-0.015	0.019	12.391	-0.016	0.001	-0.001	-12.385	12.388	-0	0.006
10	Y	0	-0.015	-0	0	-0.017	-0.045	0.045	-0	-0	2.478	0.086
10	Z	-6.346	0.003	0	-0	0.003	-3.572	3.576	0	0	-0	-0.006
11	X	0	0.001	-0	0	-0.001	-0.012	0.012	0	0	-0	0.005
11	Y	-0	0	0	0	-0	0.016	-0.016	-0	0	-0	0.009
11	Z	0	-3.578	-0	0	3.578	0.029	-0.031	0	-0	3.558	-0.001

Solid, 2nd order

Table 23: Effective modal mass [kg] per mode, global X/Y/Z — solid, order 2.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	0	0	0	0	0	0	0	0	0	0	0
1	Y	116.5	116.5	116.6	116.5	116.5	116.5	116.5	116.5	116.5	116.6	116.6
1	Z	0	0	0	0	0	0	0	0	0	0	0
2	X	0	0	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0	0	0
2	Z	0	0	0	0	0	0	0	0	0	0	0
3	X	0	0	0	0	0	0	0	0	0	0	0
3	Y	0	0	0	0	0	0	0	0	0	0	0
3	Z	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4
4	X	0	0	0	0	0	0	0	0	0	0	0
4	Y	35.9	36	35.9	35.9	36	36	36	35.9	35.9	35.9	35.9
4	Z	0	0	0	0	0	0	0	0	0	0	0
5	X	0	0	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	0	0	0
5	Z	0	0	0	0	0	0	0	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0	0	0
6	Y	0	0	0	0	0	0	0	0	0	0	0
6	Z	0	0	0	0	0	0	0	0	0	0	0
7	X	0	0	0	0	0	0	0	0	0	0	0
7	Y	3.4	3.1	3.4	3.3	3.1	3.2	3.2	3.3	3.3	3.4	3.4
7	Z	0	0	0	0	0	0	0	0	0	0	0
8	X	0	0	0	0	0	0	0	0	0	0	0
8	Y	5.3	5.8	5.3	5.4	5.8	5.9	5.9	5.4	5.4	5.3	5.3
8	Z	0	0	0	0	0	0	0	0	0	0	0
9	X	0	0	0	0	0	0	0	0	0	0	0
9	Y	0.6	0.6	0.6	0.5	0.6	0.5	0.5	0.5	0.5	0.6	0.6
9	Z	0	0	0	0	0	0	0	0	0	0	0
10	X	0	0	0	0	0	0	0	0	0	0	0
10	Y	1.5	0	1.5	1.6	0	0	0	1.6	1.6	1.5	1.5
10	Z	0	40.2	0	0	40.2	40.2	40.2	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	0	0	0	0	0	0	0
11	Z	40.3	0	40.3	40.3	0	0	0	40.3	40.3	40.3	40.3

Table 24: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — solid, order 2.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	0	-0	-0	0	0	-0	0	-0	-0	0	-0
1	Y	10.794	-10.794	10.796	10.794	10.794	-10.794	10.794	10.794	10.794	10.796	10.796
1	Z	-0	-0	0	0	0	-0	0	-0	0	0	-0
2	X	0	0	-0	-0	0	-0	0	-0	-0	-0	-0

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
2	Y	-0	-0	0	0	-0	-0	0	0	0	-0	0
2	Z	0	0	-0	0	0	-0	0	0	-0	0	0
3	X	0	0	-0	-0	0	-0	-0	-0	-0	-0	-0
3	Y	0	-0	-0	-0	-0	-0	-0	0	-0	-0	0
3	Z	10.88	10.879	10.882	10.879	10.879	10.879	10.879	10.879	10.879	10.882	10.882
4	X	-0	0	-0	-0	0	-0	-0	0	0	0	-0
4	Y	-5.991	-5.997	-5.992	-5.993	-5.997	-6.004	-6.004	-5.993	-5.993	-5.992	-5.992
4	Z	0	-0	-0	-0	-0	0	0	0	-0	-0	-0
5	X	-0	0	0	0	-0	-0	0	0	-0	-0	-0
5	Y	0	0.001	-0	-0	-0.001	0.001	-0.001	0	-0	-0	-0
5	Z	-0	-0	-0	-0	0	-0	0	-0	0	0	0
6	X	0	-0	-0	0	0	0	0	0	-0	-0	0
6	Y	-0.096	-0.119	-0.097	-0.103	0.119	-0.093	-0.093	0.103	-0.103	0.096	-0.097
6	Z	0	0	0	-0	-0	0	0	0	-0	0	-0
7	X	0	0	-0	0	-0	0	0	-0	0	0	-0
7	Y	-1.839	-1.768	1.84	1.818	1.768	1.799	1.799	1.818	1.818	1.84	1.84
7	Z	0	0	-0	0	-0	0	0	0	0	0	0
8	X	-0	0	0	0	-0	0	0	-0	-0	0	-0
8	Y	-2.305	-2.406	2.304	2.328	2.405	2.436	2.435	2.328	2.328	2.304	2.304
8	Z	-0	0	0	-0	-0	-0	-0	-0	-0	0	-0
9	X	-0	0	-0	0	-0	0	0	-0	0	-0	-0
9	Y	-0.746	-0.774	-0.747	-0.719	0.774	0.69	0.69	0.72	-0.719	0.747	-0.747
9	Z	0	-0	0	0	0	-0	-0	0	-0	0	-0
10	X	-0	-0	0	0	-0	-0	0	-0	-0	-0	0
10	Y	1.211	0	1.212	1.267	0	-0	0	-1.267	1.267	-1.213	1.212
10	Z	0	-6.344	-0	-0	-6.344	6.344	-6.344	0	-0	0	-0
11	X	-0	-0	0	0	-0	-0	0	-0	-0	0	-0
11	Y	0	-0.001	0	0	-0.001	-0.008	0.008	0	-0	-0	-0
11	Z	6.346	-0.006	-6.345	-6.344	-0.006	0.004	-0.004	6.345	-6.344	6.345	-6.345

Shell, 1st order

Table 25: Effective modal mass [kg] per mode, global X/Y/Z — shell, order 1.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	0	0	0	0	0	0	0	0	0
1	Y	119.9	119.9	120	119.9	119.7	119.9	119.9	119.9	119.9
1	Z	0	0	0	0	0	0	0	0	0
2	X	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0
2	Z	0	0	0	0	0	0	0	0	122.1
3	X	0	0	0	0	0	0	0	0	0
3	Y	0	0	0	0	0	0	0	0	37.2
3	Z	122	122	122	122	122.1	122	122	122	0
4	X	0	0	0	0	0	0	0	0	0
4	Y	36.8	36.6	37	36.9	37.3	36.6	36.9	36.8	0
4	Z	0	0	0	0	0	0	0	0	0
5	X	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	13
5	Z	0	0	0	0	0	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0
6	Y	0	0.1	0	0	12.7	0.1	0	0	0
6	Z	0	0	0	0	0	0	0	0	41.5
7	X	0	0	0	0	0	0	0	0	0
7	Y	3.5	2.7	3.8	3.3	0	2.7	3.3	3.2	0
7	Z	0	0	0	0	41.6	0	0	0	0
8	X	0	0	0	0	0	0	0	0	158
8	Y	5.4	5.9	5.4	5.7	0	6	5.8	5.7	0
8	Z	0	0	0	0	0	0	0	0	0
9	X	0	0	0	0	0	0	0	0	0
9	Y	0.6	0.6	0.6	0.6	0	0.6	0.6	0.6	5.7
9	Z	0	0	0	0	0	0	0	0	0
10	X	0	0	0	0	0	0	0	0	0
10	Y	1.3	1.3	1.6	1.5	1.5	1.3	1.7	1.3	1
10	Z	0	0	0	0	0	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	3.6	0	0	0	0
11	Z	0	41.6	0	0	0	41.6	0	41.6	12.9

Table 26: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — shell, order 1.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	-0	-0	-0	-0	0	-0	0	0	-0
1	Y	10.95	-10.951	10.952	10.951	10.941	10.951	10.95	10.951	10.95
1	Z	-0	-0	-0	-0	0	0	0	-0	0
2	X	-0	0	-0	-0	-0	0	0	-0	-0
2	Y	-0	-0	0	0	0.002	-0	0	-0	0
2	Z	-0	0	0	0	-0	0	0	0	-11.051
3	X	0	-0	-0	-0	-0	0	-0	-0	-0
3	Y	-0	0	0	0	0	-0	0	0	6.101
3	Z	-11.046	-11.047	11.046	11.046	-11.048	11.047	-11.045	11.046	0
4	X	-0	0	-0	-0	-0	0	-0	-0	0
4	Y	-6.067	-6.048	-6.081	-6.076	-6.104	-6.053	-6.072	6.065	0
4	Z	-0	-0	-0	-0	0	-0	0	0	0
5	X	0	-0	0	-0	0	0	-0	-0	-0
5	Y	0	0	-0	-0	-0.005	-0	-0	0	-3.601
5	Z	-0	-0	0	-0	-0	-0	0	-0	0
6	X	-0	-0	0	0	-0	-0	0	-0	0
6	Y	0.134	-0.252	-0.078	-0.132	-3.561	0.263	-0.174	-0.194	-0
6	Z	-0	0	-0	-0	-0	-0	-0	-0	-6.443
7	X	0	-0	-0	-0	-0	-0	-0	0	-0
7	Y	-1.86	-1.651	1.947	1.829	-0	-1.652	1.805	1.776	-0
7	Z	0	0	0	0	-6.446	0	0	0	0
8	X	0	-0	0	-0	0	0	-0	-0	12.568
8	Y	2.334	2.438	2.332	2.383	0.003	-2.455	-2.406	2.398	0
8	Z	0	0	-0	-0	0	-0	0	-0	0
9	X	-0	-0	-0	0	-0	0	-0	-0	-0
9	Y	-0.798	-0.781	-0.786	-0.771	-0.156	0.788	-0.76	-0.795	-2.388
9	Z	0	-0	0	0	0	-0	0	0	-0
10	X	0	0	-0	0	-0	0	-0	-0	-0
10	Y	-1.15	-1.128	1.269	1.232	1.218	-1.148	1.31	1.152	1.006
10	Z	0	-0	0	0	-0	0	0	-0	-0
11	X	0	-0	-0	0	-0	0	0	0	-0
11	Y	-0	-0	0	0	-1.9	-0	0	0	0
11	Z	0	-6.451	0	0	0	6.453	0	6.453	3.589

Shell, 2nd order

Table 27: Effective modal mass [kg] per mode, global X/Y/Z — shell, order 2.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	0	0	0	0	0	0	0	0	0	0
1	Y	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
1	Z	0	0	0	0	0	0	0	0	0	0
2	X	0	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0	0
2	Z	0	0	0	0	0	0	0	0	0	0
3	X	0	0	0	0	0	0	0	0	0	0
3	Y	0	0	0	0	0	0	0	0	0	0
3	Z	122	122	122	122	122	122	122	122	122	122
4	X	0	0	0	0	0	0	0	0	0	0
4	Y	36.9	36.9	36.9	36.9	36.9	36.9	36.9	36.9	36.9	36.9
4	Z	0	0	0	0	0	0	0	0	0	0
5	X	0	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	0	0
5	Z	0	0	0	0	0	0	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0	0
6	Y	0	0	0	0	0	0	0	0	0	0
6	Z	0	0	0	0	0	0	0	0	0	0
7	X	0	0	0	0	0	0	0	0	0	0
7	Y	3.5	3.4	3.5	3.5	3.5	3.4	3.4	3.4	3.5	3.4
7	Z	0	0	0	0	0	0	0	0	0	0
8	X	0	0	0	0	0	0	0	0	0	0
8	Y	5.5	5.6	5.5	5.5	5.5	5.6	5.5	5.6	5.5	5.5
8	Z	0	0	0	0	0	0	0	0	0	0
9	X	0	0	0	0	0	0	0	0	0	0
9	Y	0.7	0.6	0.7	0.7	0.7	0.6	0.6	0.6	0.7	0.6
9	Z	0	0	0	0	0	0	0	0	0	0
10	X	0	0	0	0	0	0	0	0	0	0
10	Y	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
10	Z	0	0	0	0	0	0	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	0	0	0	0	0	0
11	Z	0	0	0	0	0	0	0	0	0	0

Table 28: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — shell, order 2.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	-0	-0	0	0	0	-0	0	0	-0	-0
1	Y	-10.952	10.952	10.951	10.951	10.952	10.952	10.952	10.951	10.952	10.951
1	Z	-0	-0	0	0	0	0	-0	0	0	0
2	X	-0	0	0	0	0	-0	-0	0	0	0
2	Y	0	0	0	0	0	0	-0	0	-0	0
2	Z	0	0	0	0	-0	-0	0	-0	-0	-0
3	X	-0	-0	0	0	0	-0	0	0	0	0
3	Y	-0	-0	0	0	0	0	0	0	0	0
3	Z	11.047	-11.047	11.044	11.044	11.047	-11.046	11.047	-11.046	11.047	-11.046
4	X	-0	-0	-0	-0	0	-0	-0	-0	0	0
4	Y	-6.071	6.072	-6.071	-6.071	-6.073	6.072	6.071	6.073	-6.071	6.071
4	Z	0	-0	0	0	0	-0	-0	-0	0	-0
5	X	0	-0	-0	-0	-0	0	-0	-0	0	-0
5	Y	0	0	-0	-0	-0	0	-0	-0	-0	-0
5	Z	-0	0	-0	-0	0	-0	-0	-0	-0	0
6	X	-0	0	-0	-0	-0	-0	0	-0	0	0
6	Y	-0.137	-0.144	-0.137	-0.137	-0.138	-0.144	0.142	0.141	-0.137	0.138
6	Z	0	0	0	0	-0	-0	-0	0	0	0
7	X	0	-0	-0	-0	-0	-0	-0	-0	-0	-0
7	Y	1.859	-1.85	1.858	1.858	1.88	1.848	-1.849	1.839	-1.859	1.852
7	Z	-0	0	-0	-0	0	0	0	0	0	0
8	X	-0	0	-0	-0	-0	0	-0	-0	-0	0
8	Y	-2.35	-2.358	2.351	2.351	2.349	2.356	2.353	-2.362	2.35	-2.35
8	Z	-0	-0	0	0	0	-0	0	0	0	0
9	X	0	0	-0	-0	0	-0	-0	0	-0	0
9	Y	-0.809	-0.801	-0.808	-0.808	-0.818	0.793	0.801	0.781	-0.809	0.799
9	Z	0	0	0	0	-0	-0	-0	-0	-0	-0
10	X	-0	-0	-0	-0	-0	-0	0	-0	-0	-0
10	Y	-1.171	-1.185	1.17	1.17	1.175	1.193	1.178	-1.201	1.17	1.178
10	Z	-0	-0	-0	-0	0.001	0	0	-0	0	0
11	X	-0	-0	0	0	0	-0	-0	-0	-0	-0
11	Y	-0	0.001	0	0	0	0.001	-0.001	0	0	0
11	Z	0	-0	-0	-0	0	-0	-0	0	-0	-0

Line, 1st order

Table 29: Effective modal mass [kg] per mode, global X/Y/Z — line, order 1.

Mode	Direction	ses	aba	ccx	ca	ccxR
1	X	0	0	0	0	0
1	Y	0	0	116.4	0	116.4
1	Z	0	0	0	0	0
2	X	0	0	0	0	0
2	Y	116.7	116.6	0	116.4	0
2	Z	0	0	114.8	0	114.8
3	X	0	0	0	0	0
3	Y	0	0	35.7	0	35.7
3	Z	0	0	0	0	0
4	X	0	0	0	0	0
4	Y	0	0	12.3	0	12.3
4	Z	120.2	118.2	0	114.8	0
5	X	0	0	0	0	0
5	Y	0	0	0	0	0
5	Z	0	0	0	0	0
6	X	0	0	0	0	0
6	Y	36.3	0	0	35.7	0
6	Z	0	0	35	0	35
7	X	0	0	153.9	0	153.9
7	Y	0	36.2	0	0	0

Mode	Direction	ses	aba	ccx	ca	ccxR
7	Z	0	0	0	0	0
8	X	0	0	0	0	0
8	Y	0	0	6.2	0	6.2
8	Z	0	0	0	0	0
9	X	0	0	0	0	0
9	Y	0	0	3.8	0	3.8
9	Z	0	0	0	0	0
10	X	0	0	0	0	0
10	Y	0	0	0	0	0
10	Z	0	0	0	0	0
11	X	0	0	0	0	0
11	Y	0	0	0	0	0
11	Z	0	0	11.4	0	11.4

Table 30: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — line, order 1.

Mode	Direction	ses	aba	ccx	ca	ccxR
1	X	0	-0	0	-0	0
1	Y	-0	-0	-10.789	-0	-10.789
1	Z	-0	0	0	0	0
2	X	0	-0	0	-0	0
2	Y	10.803	10.797	-0	10.788	-0
2	Z	-0	-0	-10.716	0	-10.716
3	X	0	0	-0	0	-0
3	Y	-0	-0	5.979	-0	5.979
3	Z	-0	0	-0	-0	-0
4	X	0	0	-0	0	-0
4	Y	-0	-0	3.5	0	3.5
4	Z	-10.963	10.87	-0	10.714	-0
5	X	-0	-0	0	0	0
5	Y	-0	0	-0	-0	-0
5	Z	0	0	0	0	0
6	X	0	-0	-0	0	-0
6	Y	-6.024	-0	-0	-5.977	-0
6	Z	0	0	-5.913	-0	-5.913
7	X	-0	0	12.407	0	12.407
7	Y	-0	-6.016	-0	-0	-0
7	Z	-0	-0	-0	-0	-0
8	X	0	-0	0	0	0
8	Y	-0	-0	2.498	0	2.498
8	Z	-0	0	0	-0	0
9	X	-0	0	0	0	0
9	Y	0	0	-1.937	-0	-1.937
9	Z	0	0	-0	0	-0
10	X	-0	0	-0	-0	-0
10	Y	-0	0	0	-0	0
10	Z	-0	-0	0	0	0
11	X	-0	0	0	-0	0
11	Y	-0	-0	0	-0	0
11	Z	-0	0	-3.374	0	-3.374

Line, 2nd order

Table 31: Effective modal mass [kg] per mode, global X/Y/Z — line, order 2.

Mode	Direction	ses	aba
1	X	0	0
1	Y	0	0
1	Z	0	0
2	X	0	0
2	Y	116.6	116.6
2	Z	0	0
3	X	0	0
3	Y	0	0
3	Z	0	0
4	X	0	0
4	Y	0	0

Mode	Direction	ses	aba
4	Z	118.7	118.2
5	X	0	0
5	Y	0	0
5	Z	0	0
6	X	0	0
6	Y	36.2	0
6	Z	0	0
7	X	0	0
7	Y	0	36.2
7	Z	0	0
8	X	0	0
8	Y	0	0
8	Z	0	0
9	X	0	0
9	Y	0	0
9	Z	0	0
10	X	0	0
10	Y	0	0
10	Z	0	0
11	X	0	0
11	Y	0	0
11	Z	0	0

Table 32: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — line, order 2.

Mode	Direction	ses	aba
1	X	0	0
1	Y	0	0
1	Z	-0	-0
2	X	0	-0
2	Y	-10.799	10.798
2	Z	0	-0
3	X	-0	0
3	Y	-0	-0
3	Z	0	-0
4	X	-0	-0
4	Y	-0	-0
4	Z	-10.895	10.871
5	X	-0	-0
5	Y	-0	0
5	Z	-0	0
6	X	0	-0
6	Y	-6.02	0
6	Z	0	-0
7	X	-0	0
7	Y	0	-6.015
7	Z	-0	-0
8	X	0	0
8	Y	0	0
8	Z	-0	-0
9	X	0	0
9	Y	0	0
9	Z	0	-0
10	X	0	-0
10	Y	-0	0
10	Z	-0	0
11	X	0	-0
11	Y	-0	-0
11	Z	-0	0

Cantilever eigenvalue analysis

Line, 1st order

Mode 1

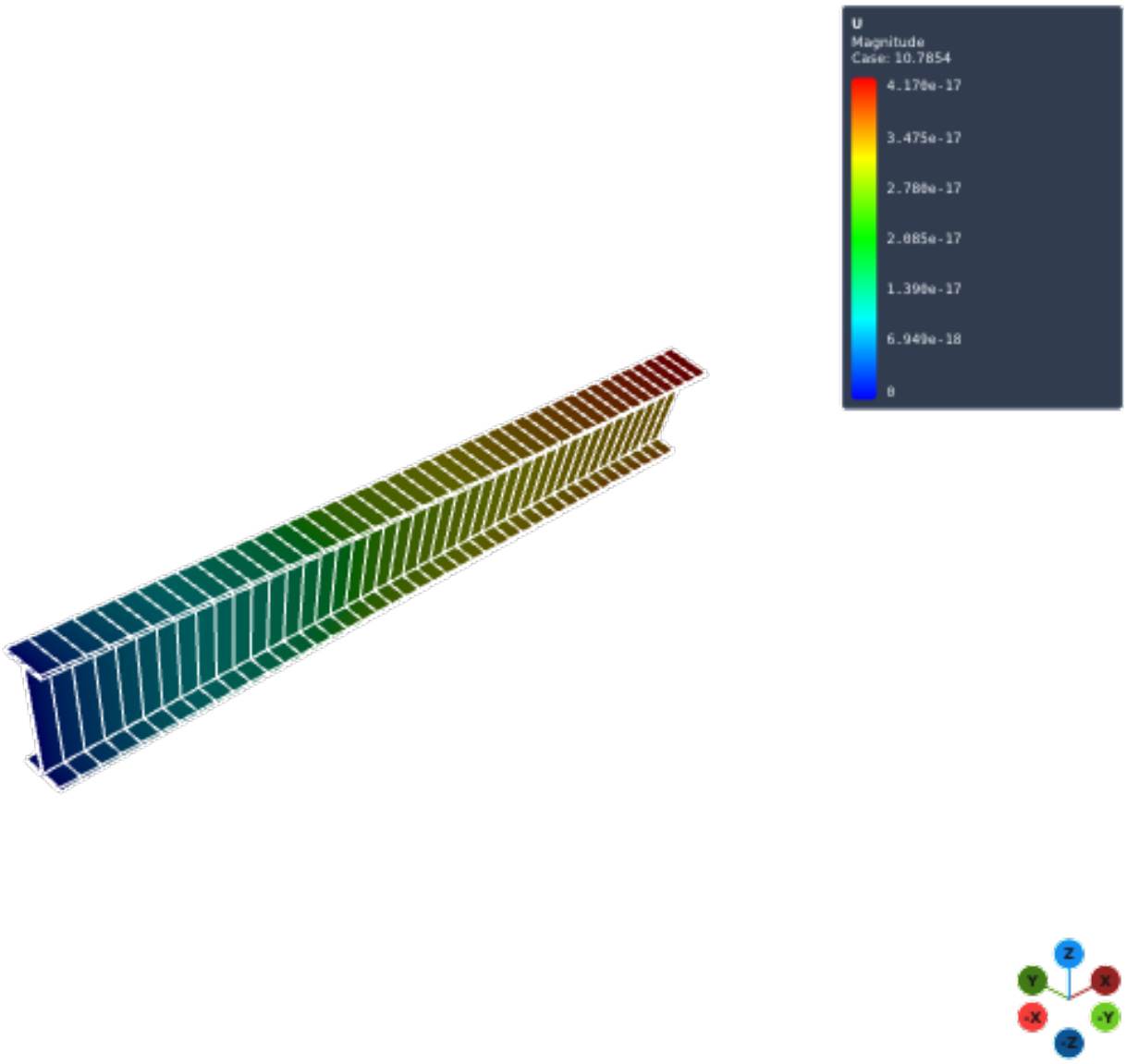

Figure 17: Abaqus — mode 1, 10.785 Hz

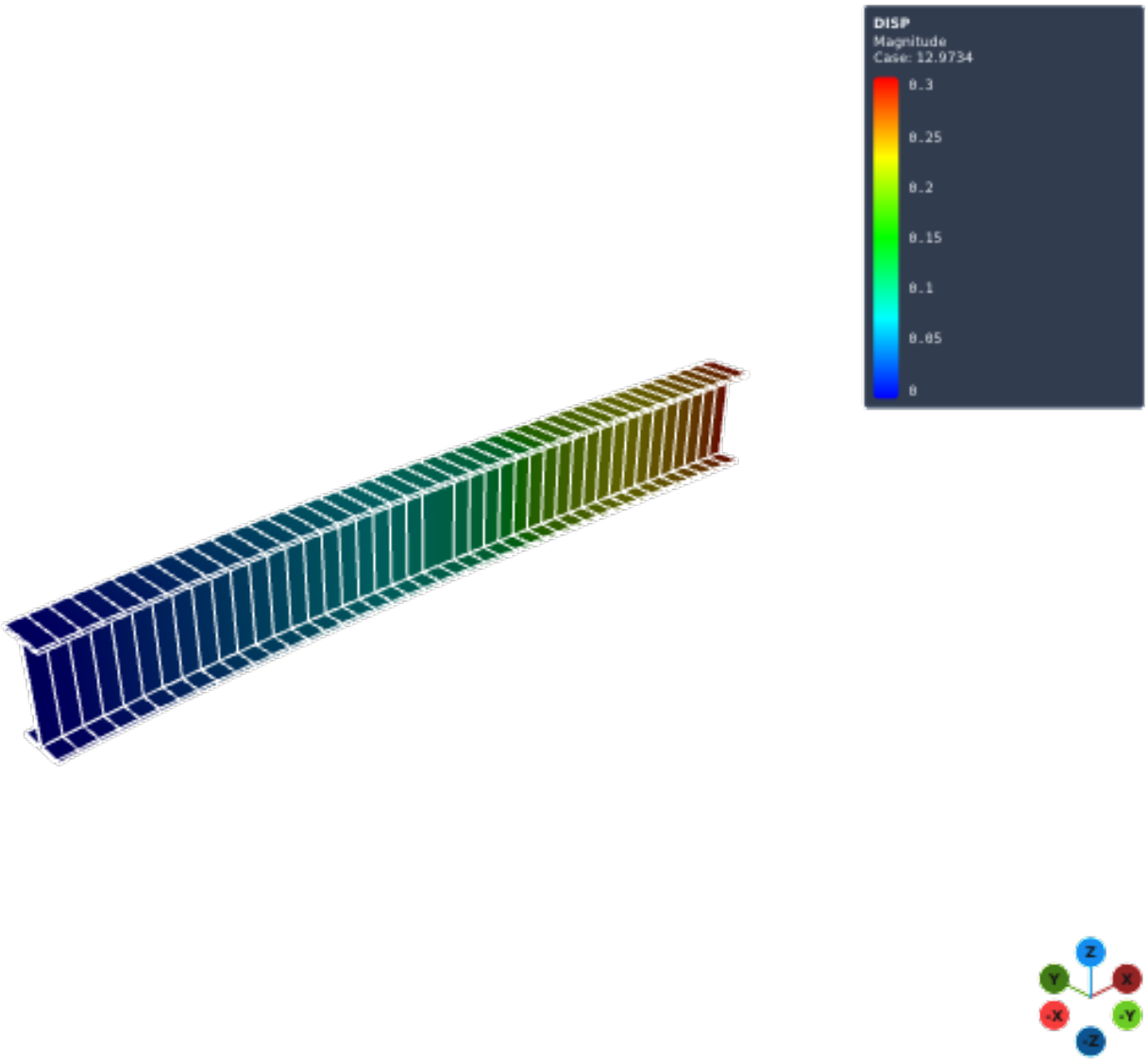

Figure 18: Calculix — mode 1, 12.973 Hz

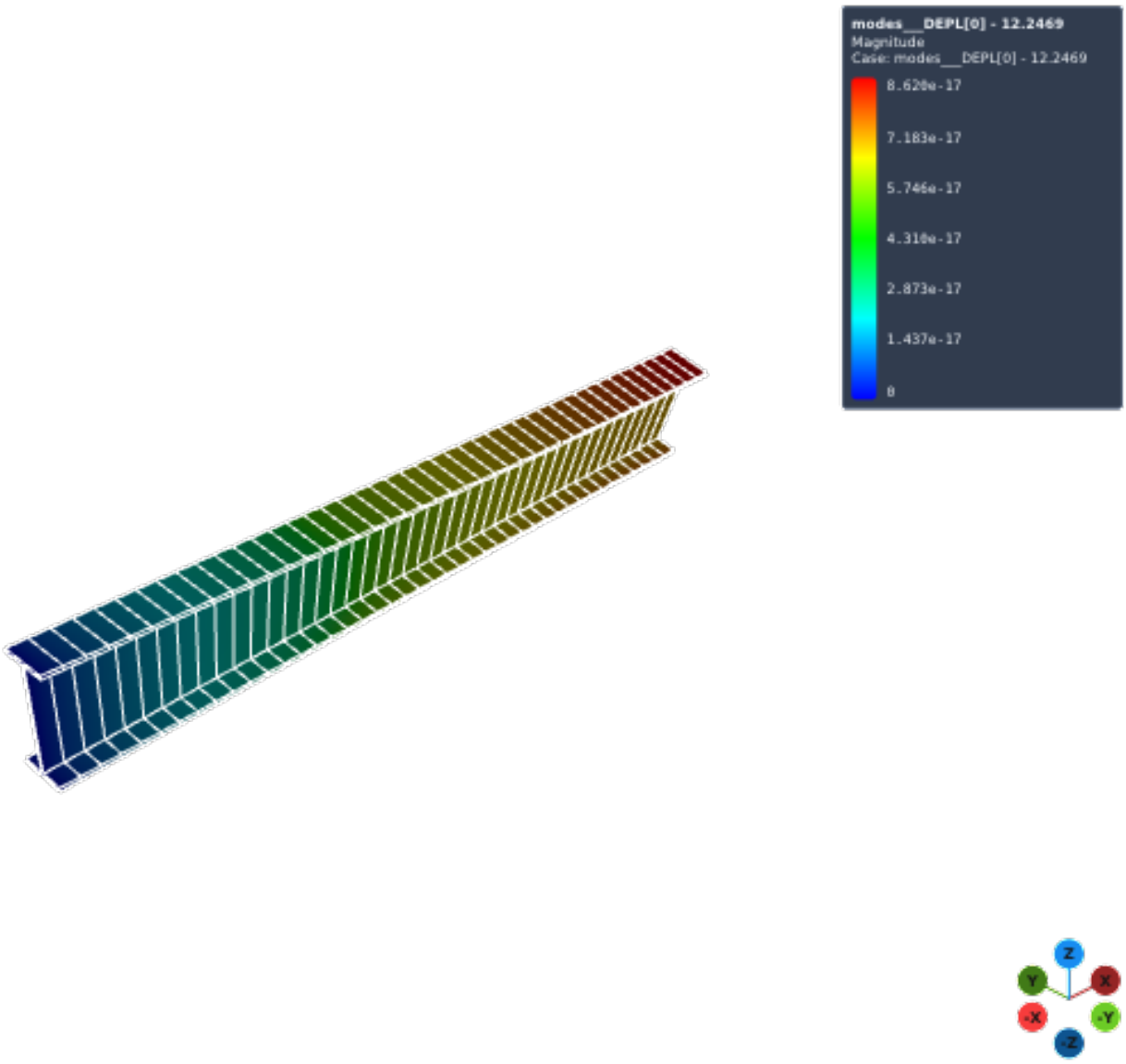

Figure 19: Code_Aster — mode 1, 12.247 Hz

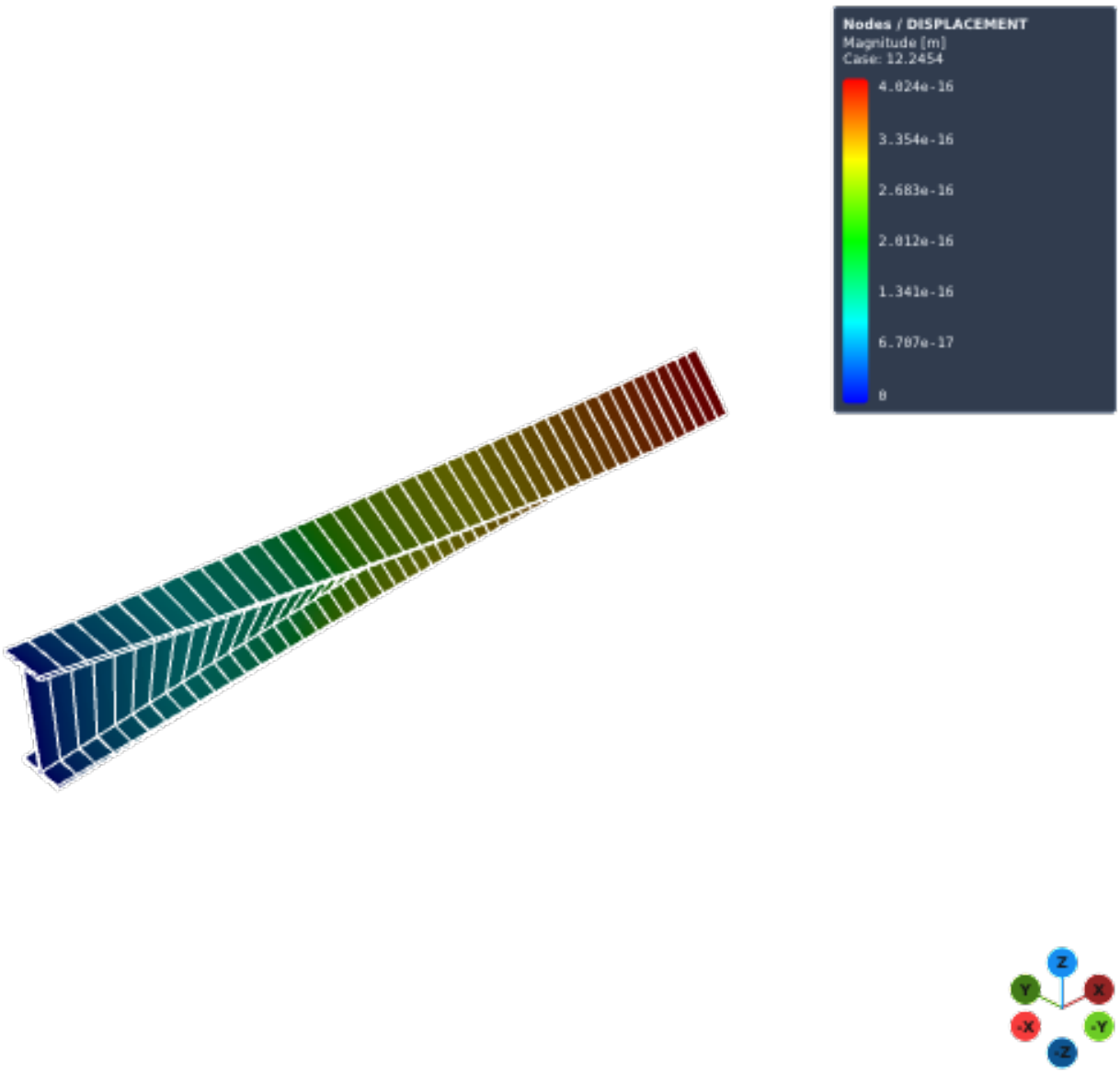

Figure 20: Sesam — mode 1, 12.245 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	10.785	-10.59	4592.3	4.83e-34	1.00e-29	1.05e-31	-2.20e-17	-3.17e-15	3.24e-16
Calculix	12.973	+7.55	6644.56	1.74e-37	116.40	9.37e-33	4.17e-19	-10.789	9.68e-17
Code_Aster	12.247	+1.53	5921.21	9.09e-44	9.42e-29	5.71e-35	-3.01e-22	-9.71e-15	7.55e-18
Sesam	12.245	+1.51	5919.77	1.37e-35	1.81e-28	4.01e-31	3.70e-18	-1.35e-14	-6.33e-16

Mode 2

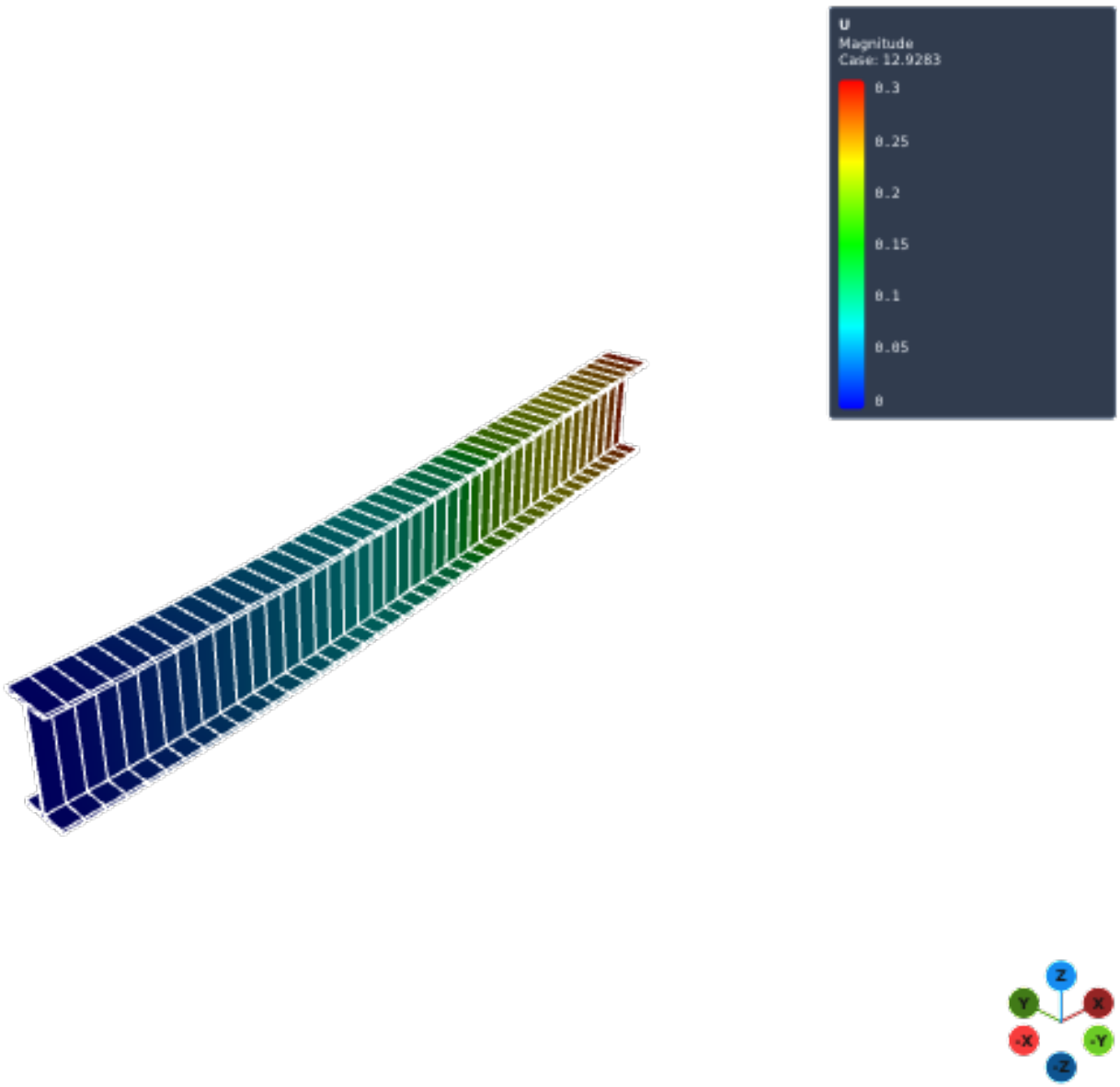

Figure 21: Abaqus — mode 2, 12.928 Hz

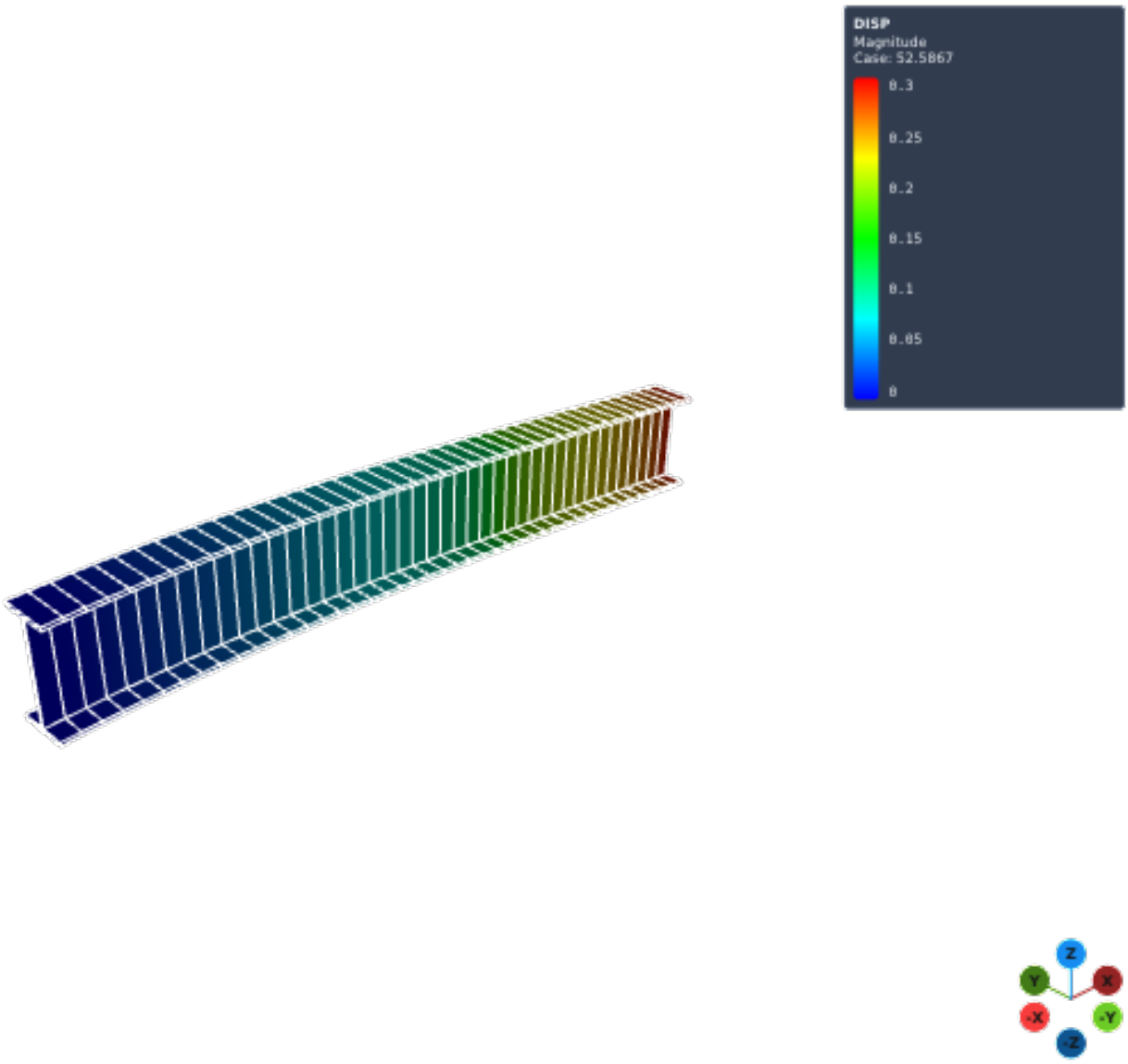

Figure 22: Calculix — mode 2, 52.587 Hz

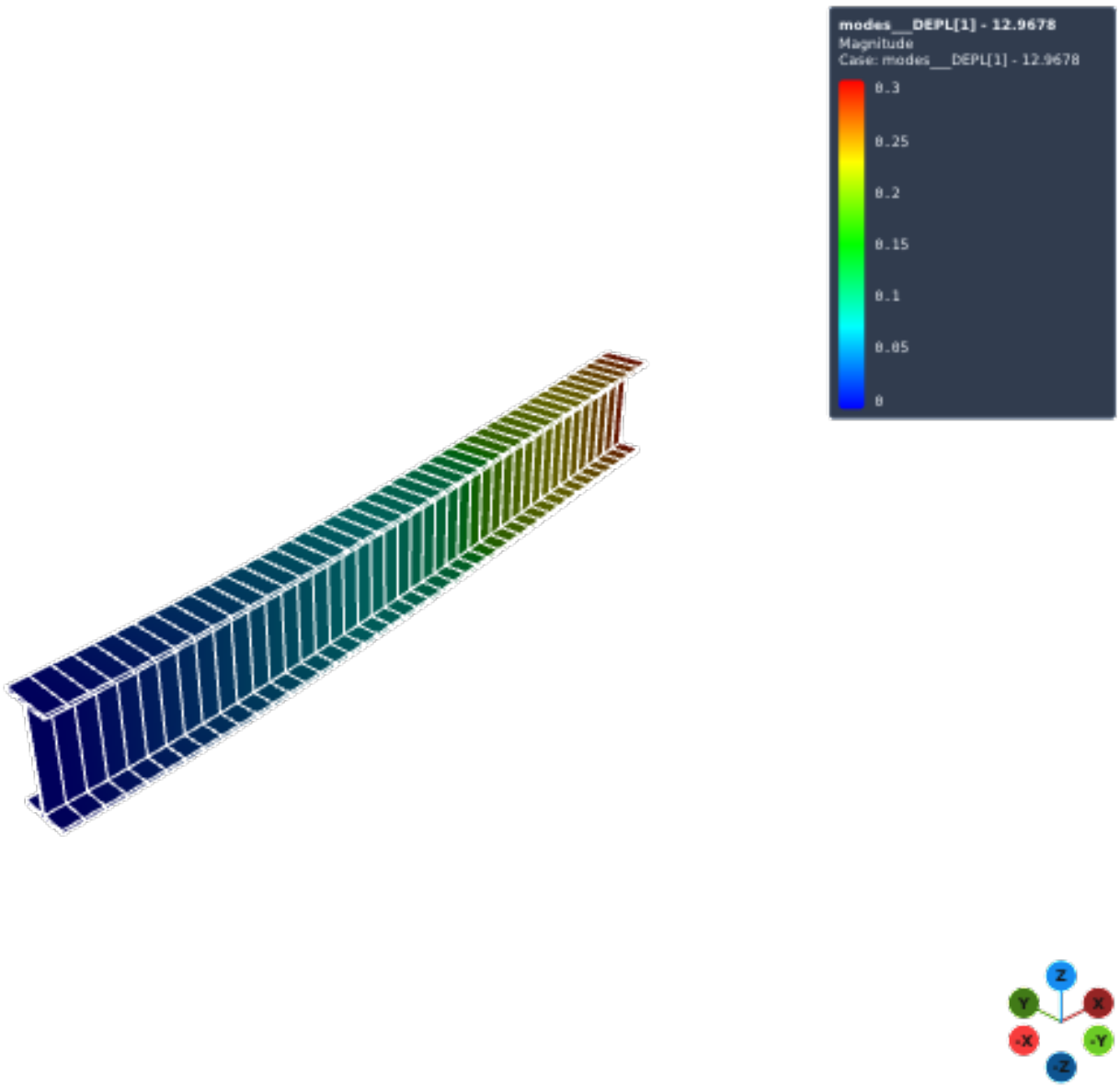

Figure 23: Code_Aster — mode 2, 12.968 Hz

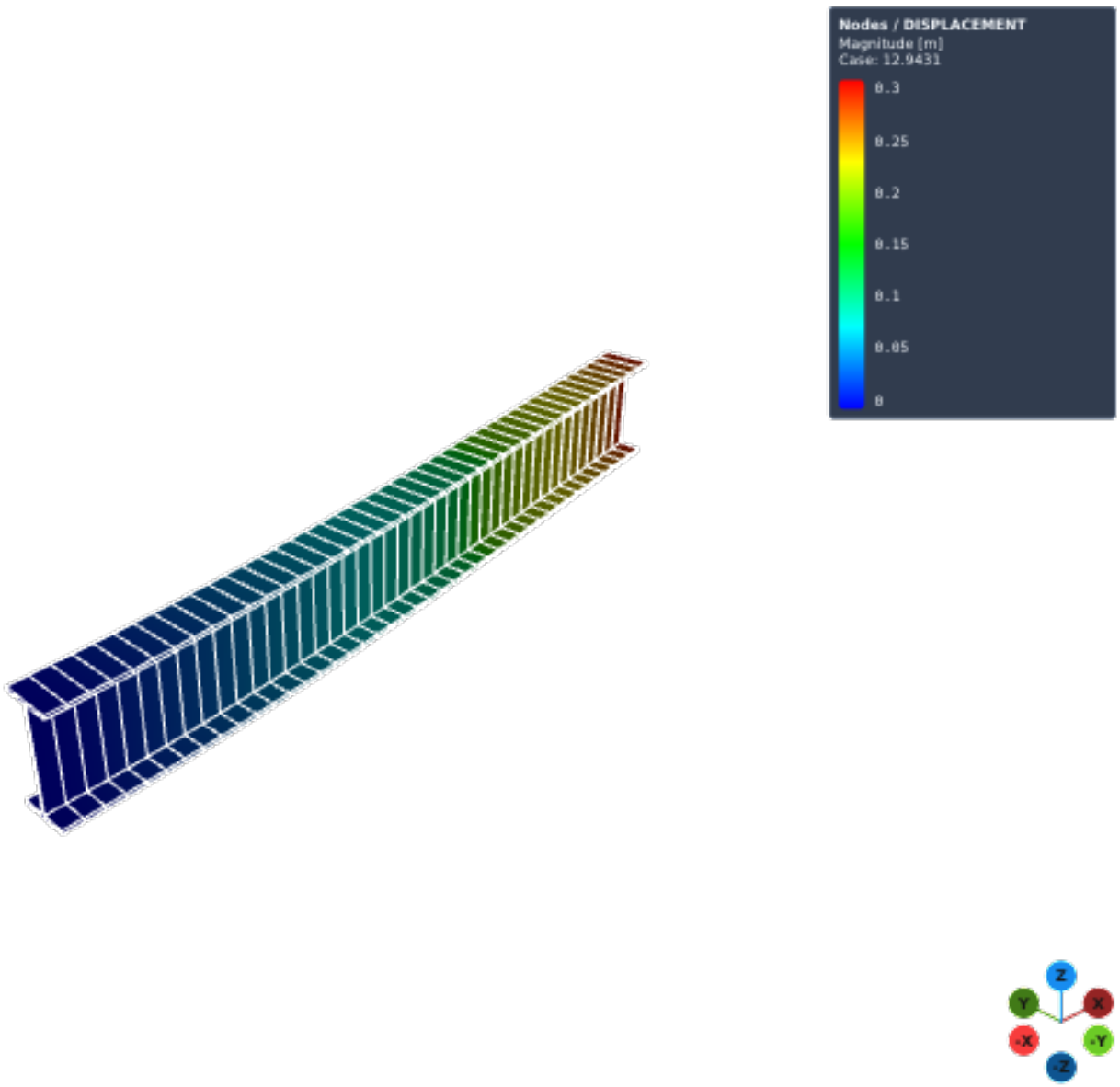

Figure 24: Sesam — mode 2, 12.943 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.928	-43.44	6598.4	6.75e-31	116.57	3.27e-31	-8.21e-16	10.797	-5.72e-16
Calculix	52.587	+130.07	109172	2.95e-34	6.28e-31	114.82	1.72e-17	-7.92e-16	-10.716
Code_Aster	12.968	-43.26	6638.82	1.87e-42	116.38	5.02e-32	-1.37e-21	10.788	2.24e-16
Sesam	12.943	-43.37	6613.55	4.26e-36	116.71	1.68e-31	2.06e-18	10.803	-4.10e-16

Mode 3

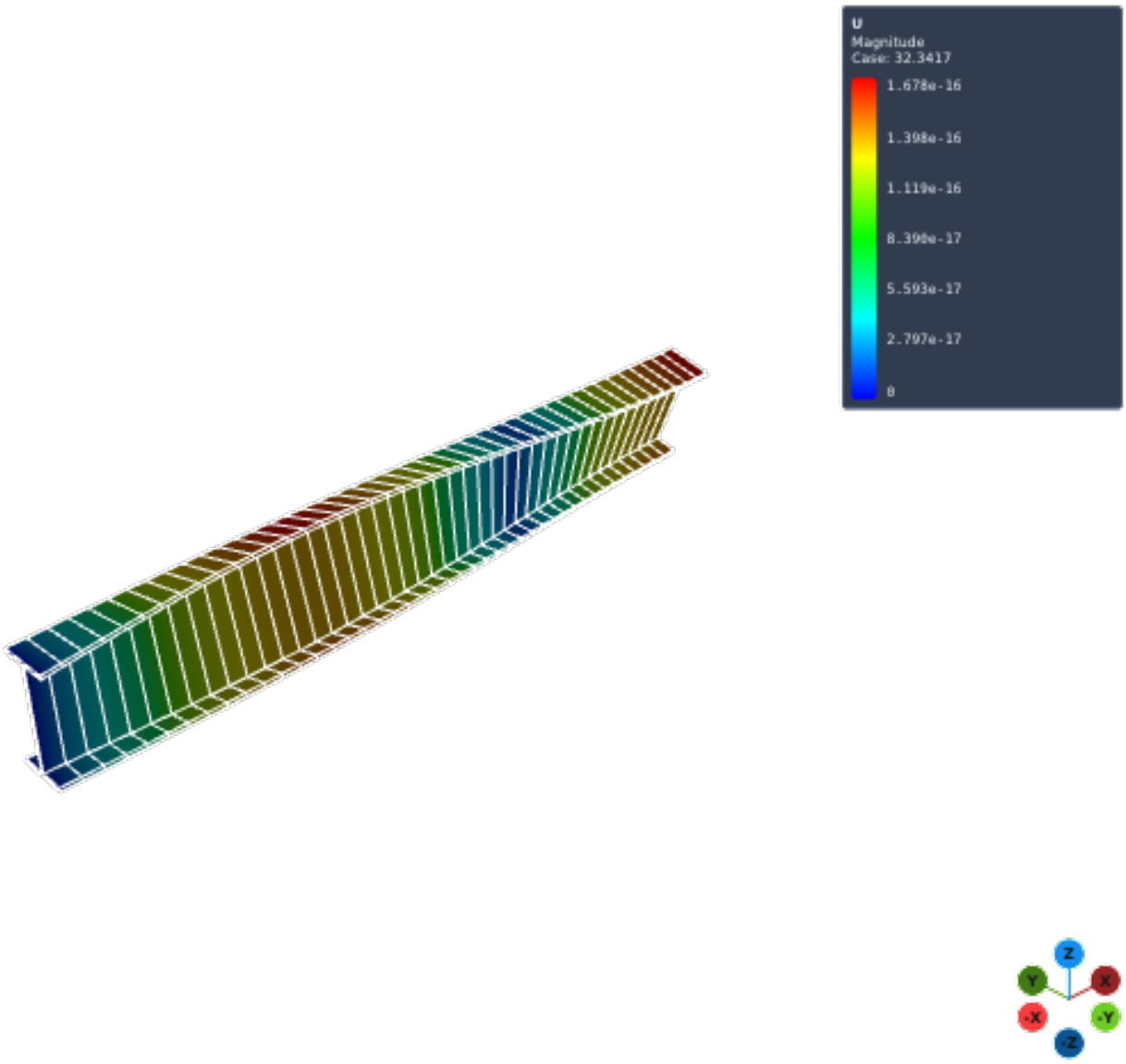

Figure 25: Abaqus — mode 3, 32.342 Hz

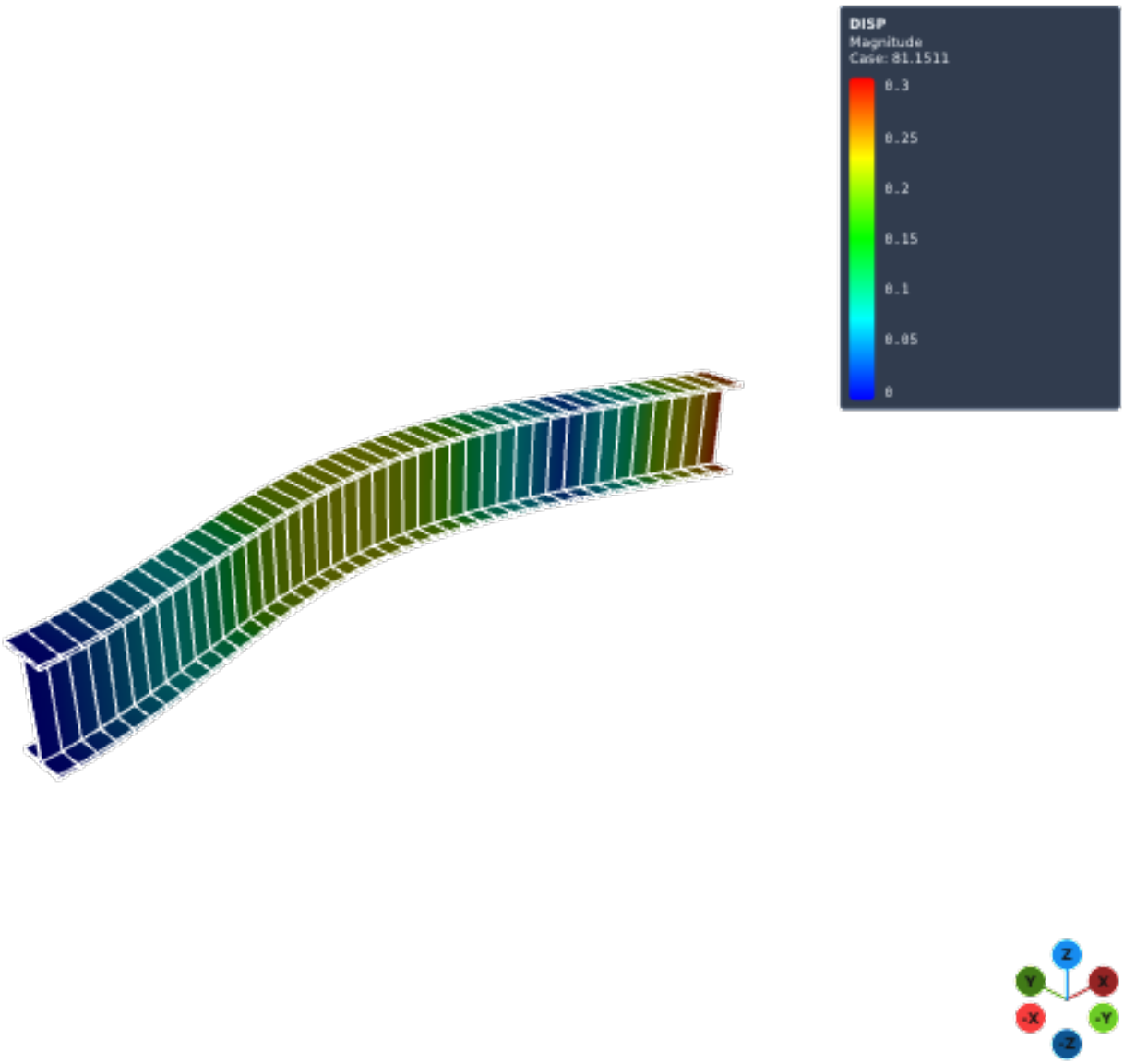

Figure 26: Calculix — mode 3, 81.151 Hz

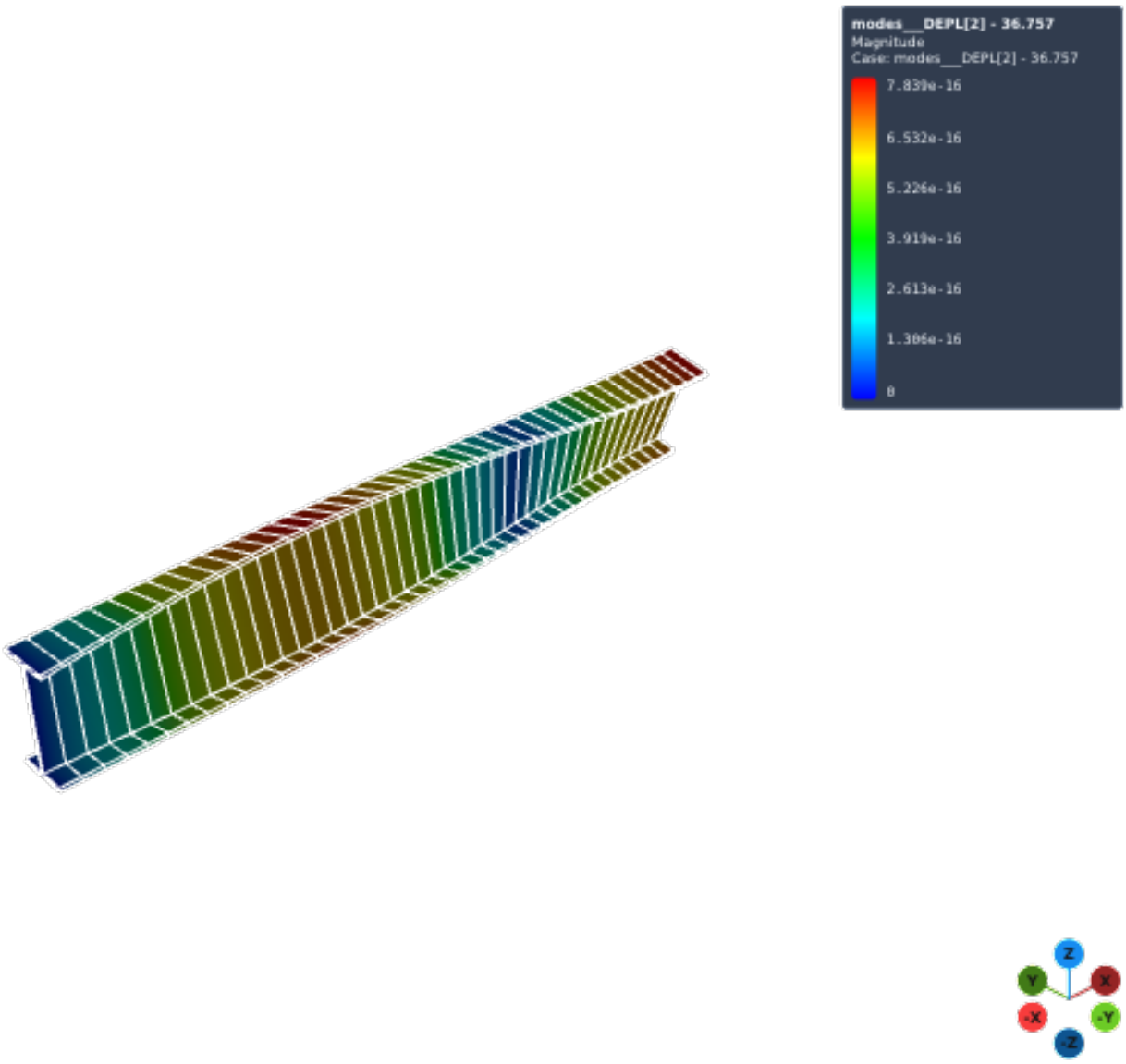

Figure 27: Code_Aster — mode 3, 36.757 Hz

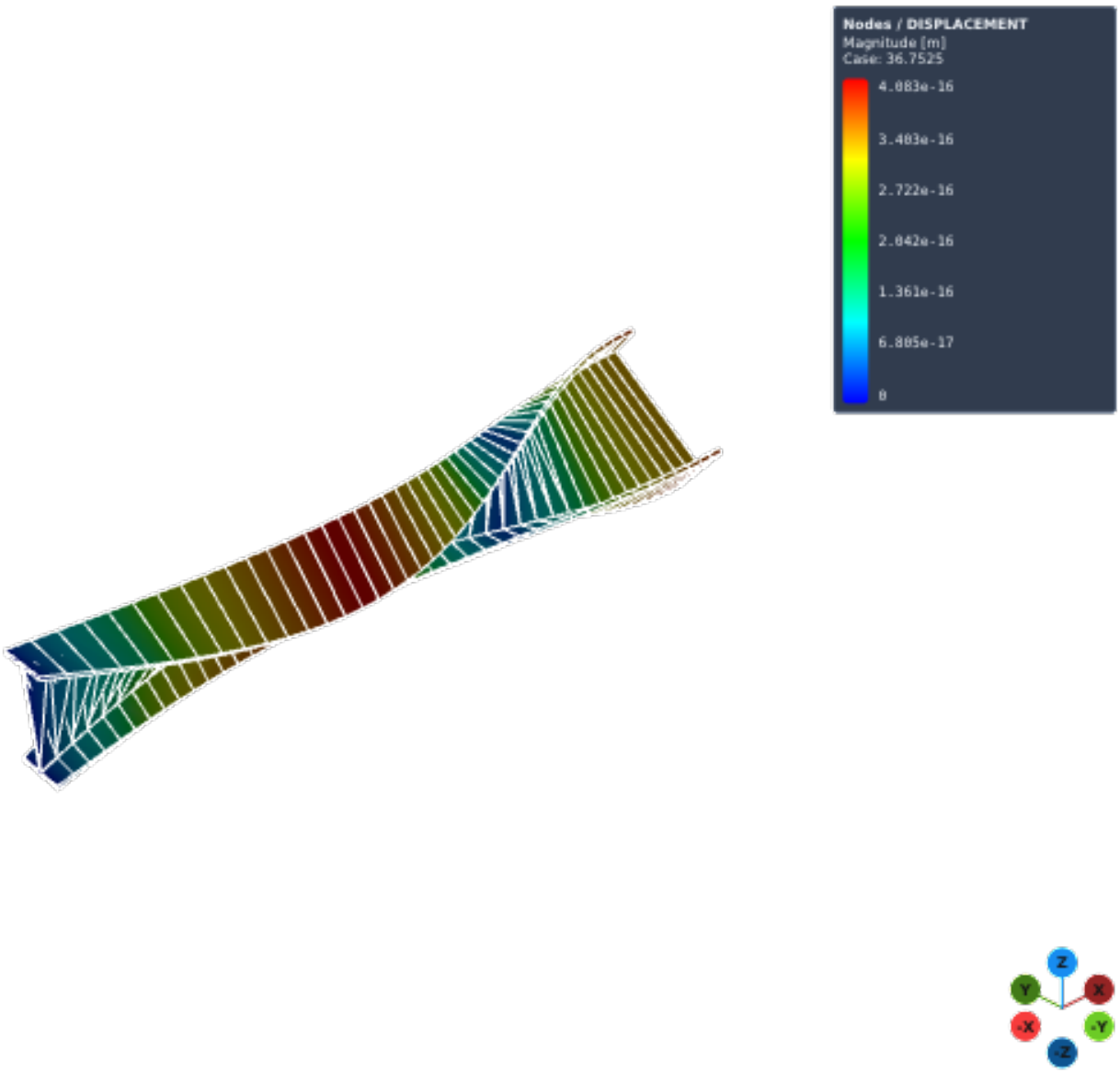

Figure 28: Sesam — mode 3, 36.752 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	32.342	-30.82	41294	7.85e-31	8.23e-32	1.14e-28	8.86e-16	-2.87e-16	1.07e-14
Calculix	81.151	+73.58	259985	4.06e-33	35.74	4.38e-30	-6.37e-17	5.979	-2.09e-15
Code_Aster	36.757	-21.38	53338.3	2.78e-40	4.16e-27	6.14e-32	1.67e-20	-6.45e-14	-2.48e-16
Sesam	36.752	-21.39	53325.3	9.01e-34	4.75e-30	1.52e-28	3.00e-17	-2.18e-15	-1.23e-14

Mode 4

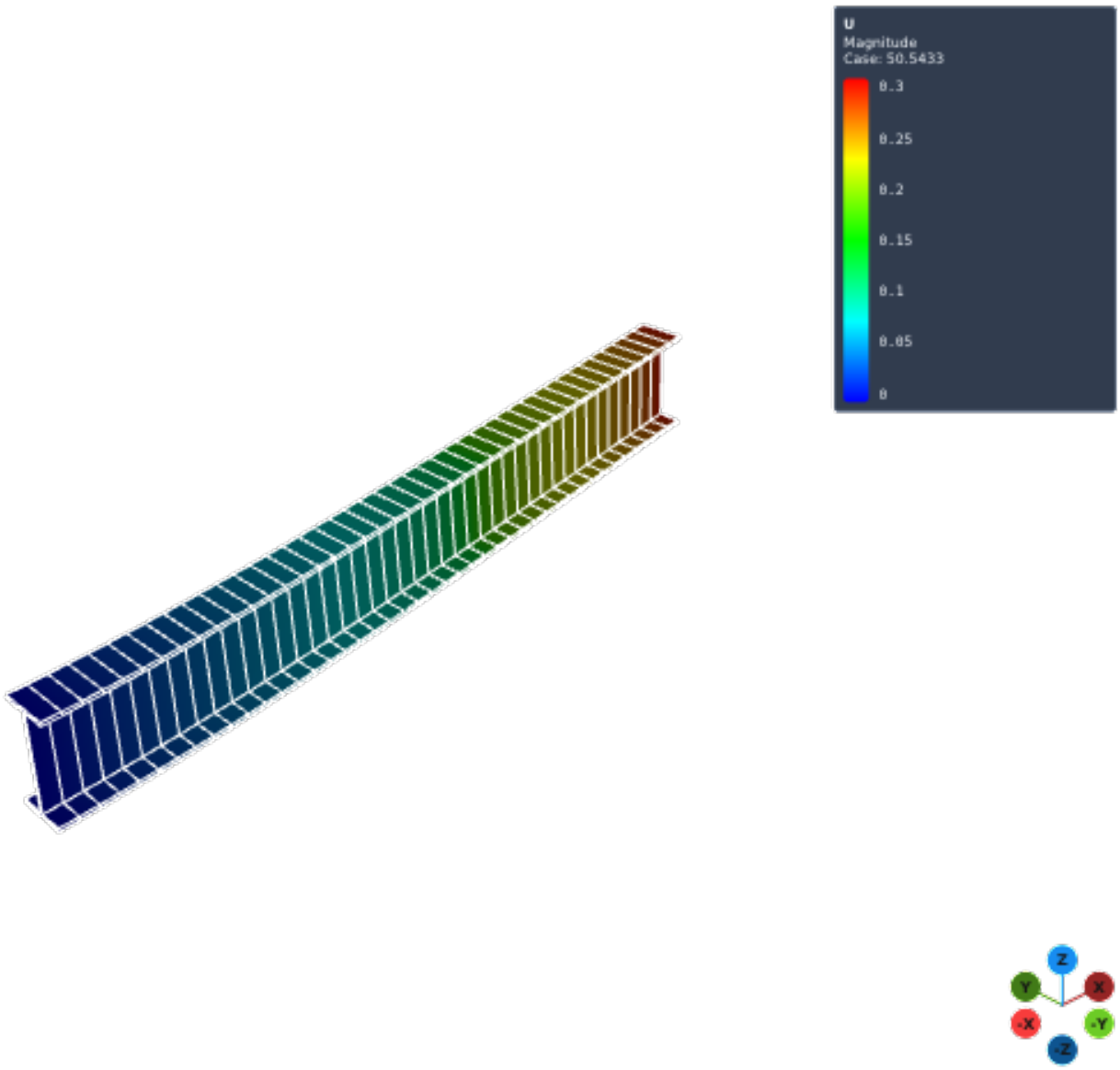

Figure 29: Abaqus — mode 4, 50.543 Hz

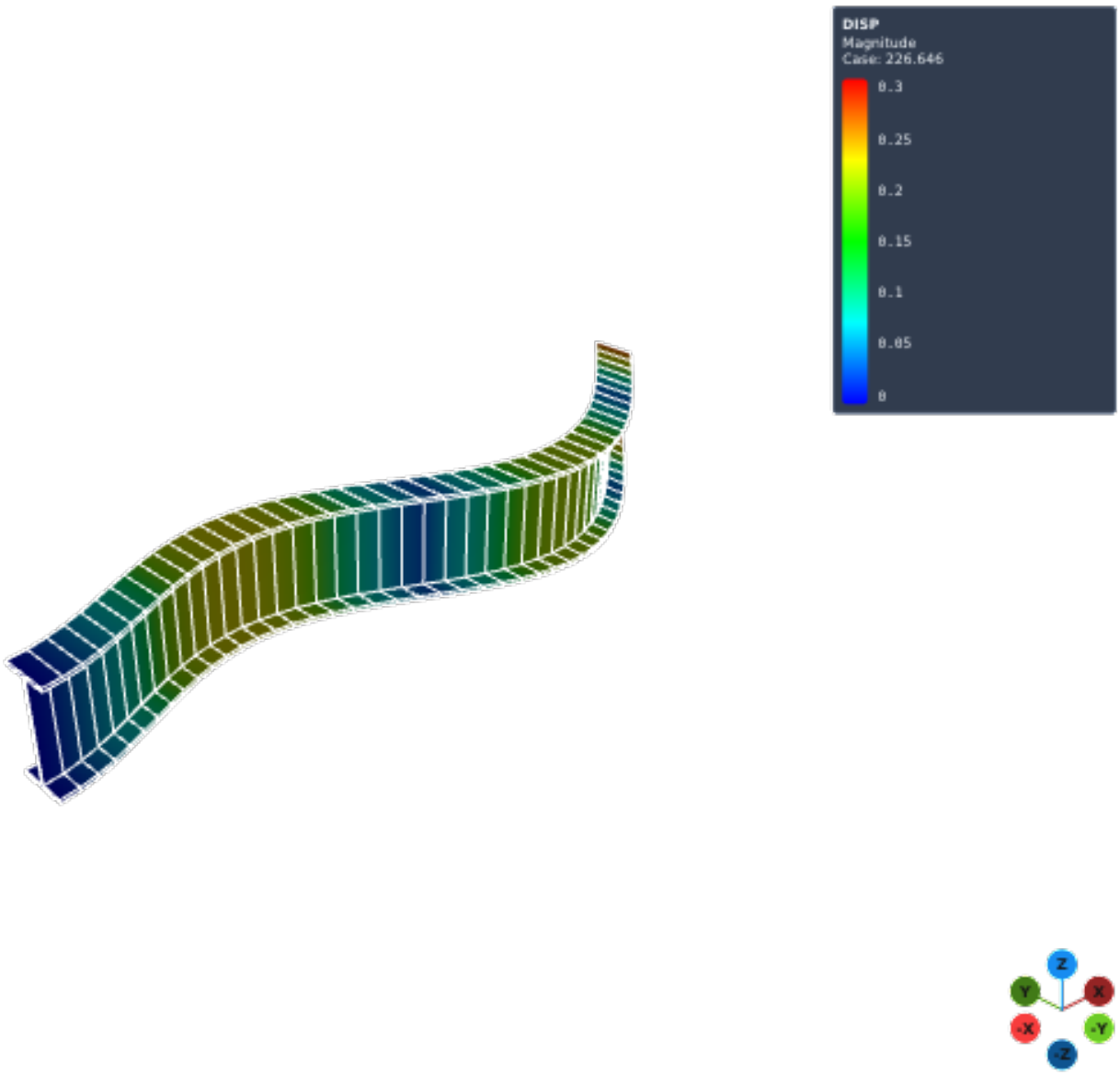

Figure 30: Calculix — mode 4, 226.646 Hz

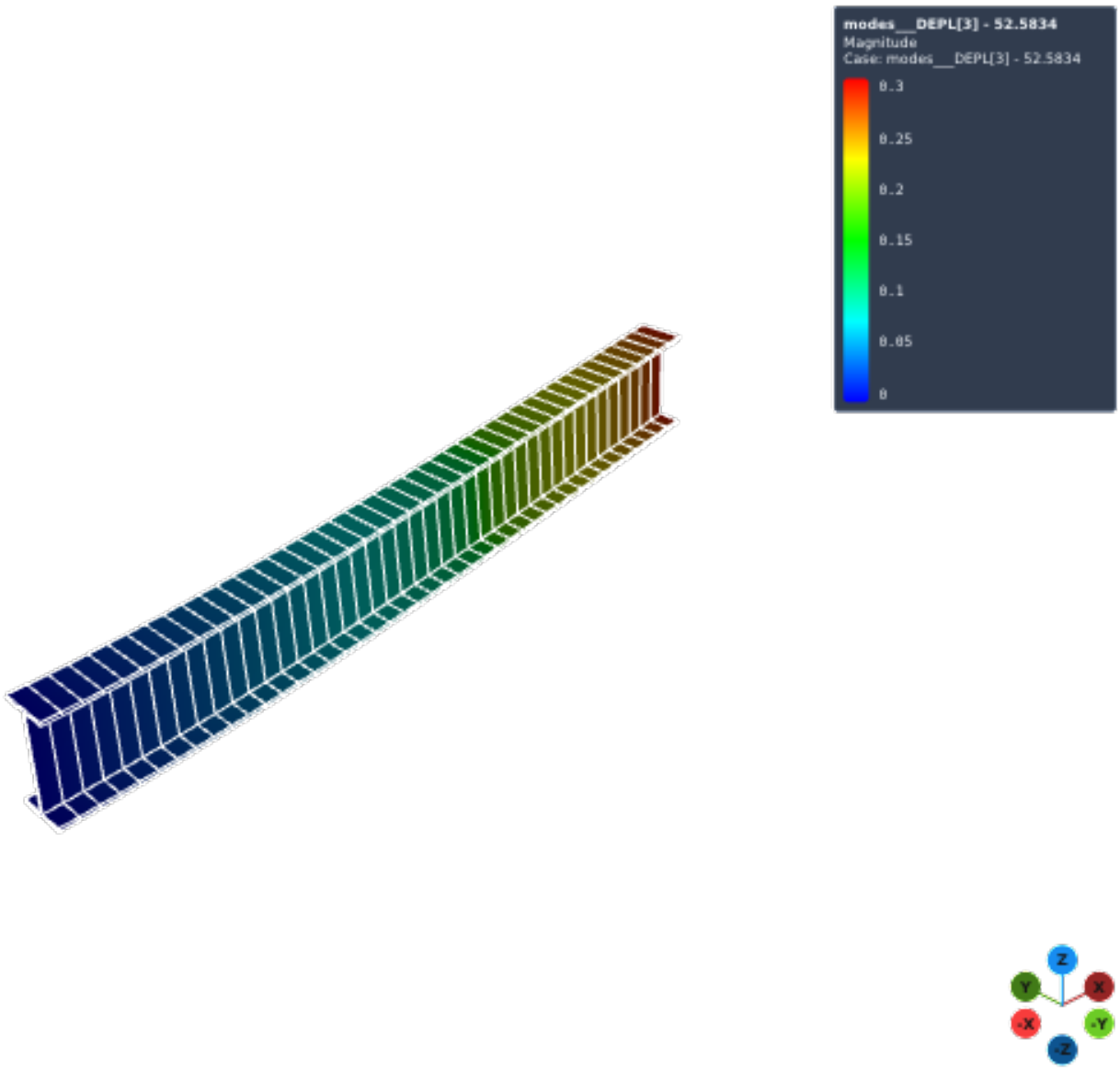

Figure 31: Code_Aster — mode 4, 52.583 Hz

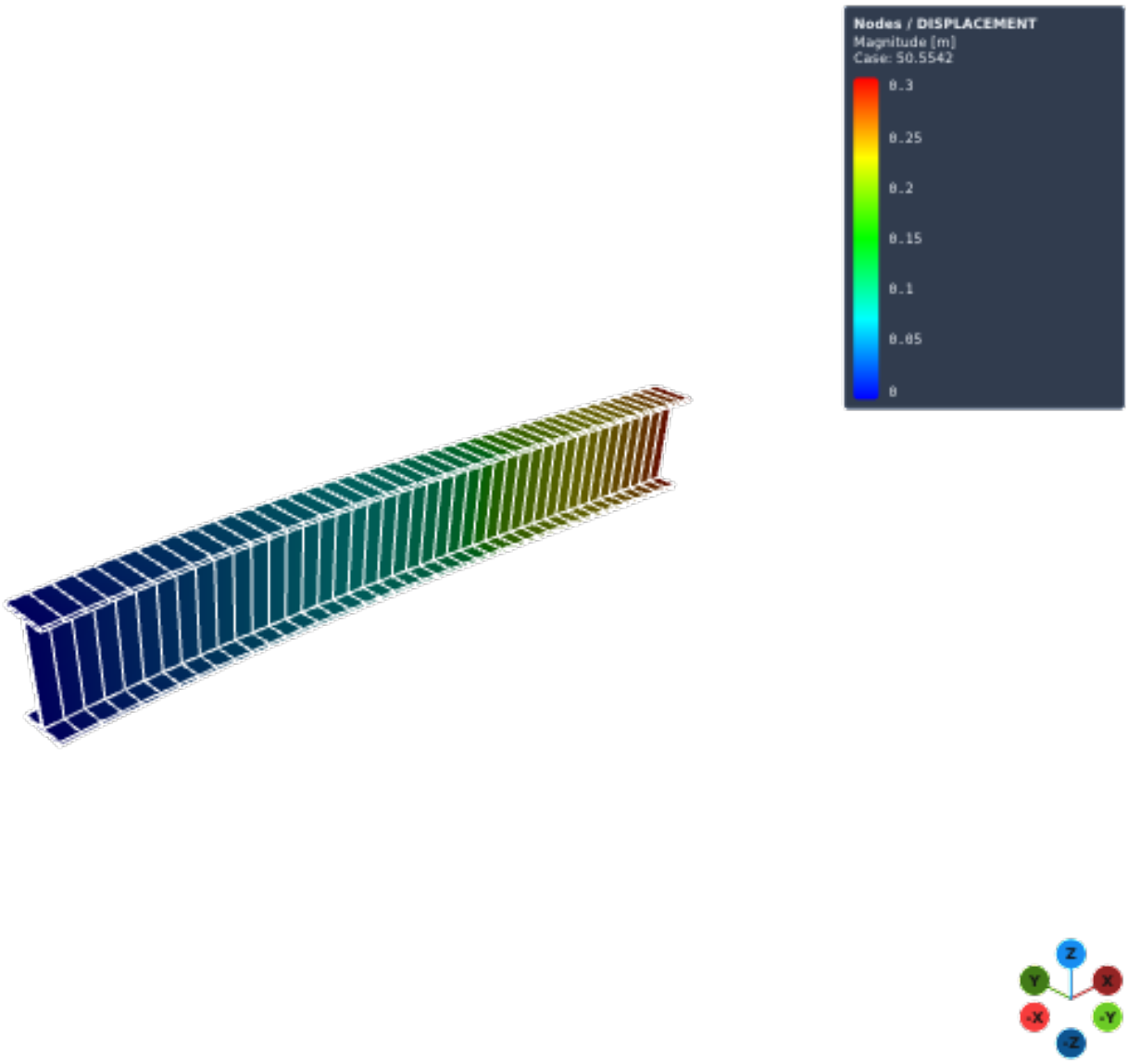

Figure 32: Sesam — mode 4, 50.554 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.543	-46.84	100852	2.51e-30	1.42e-29	118.16	1.58e-15	-3.76e-15	10.870
Calculix	226.646	+138.37	2.02794e+06	1.30e-30	12.25	2.11e-29	-1.14e-15	3.500	-4.59e-15
Code_Aster	52.583	-44.70	109158	5.72e-39	1.70e-30	114.80	7.56e-20	1.30e-15	10.714
Sesam	50.554	-46.83	100896	1.69e-33	1.87e-30	120.18	4.11e-17	-1.37e-15	-10.963

Mode 5

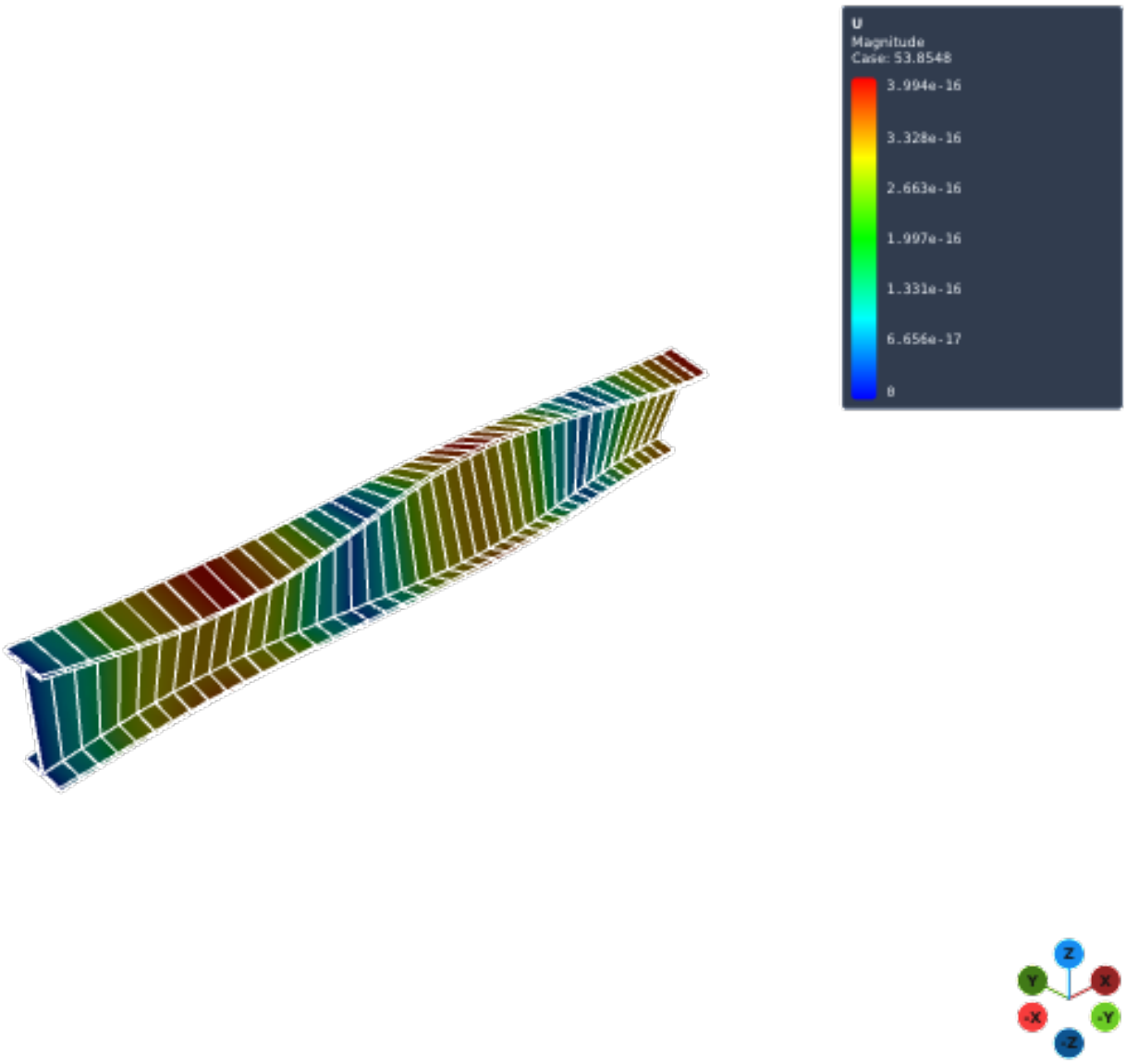

Figure 33: Abaqus — mode 5, 53.855 Hz

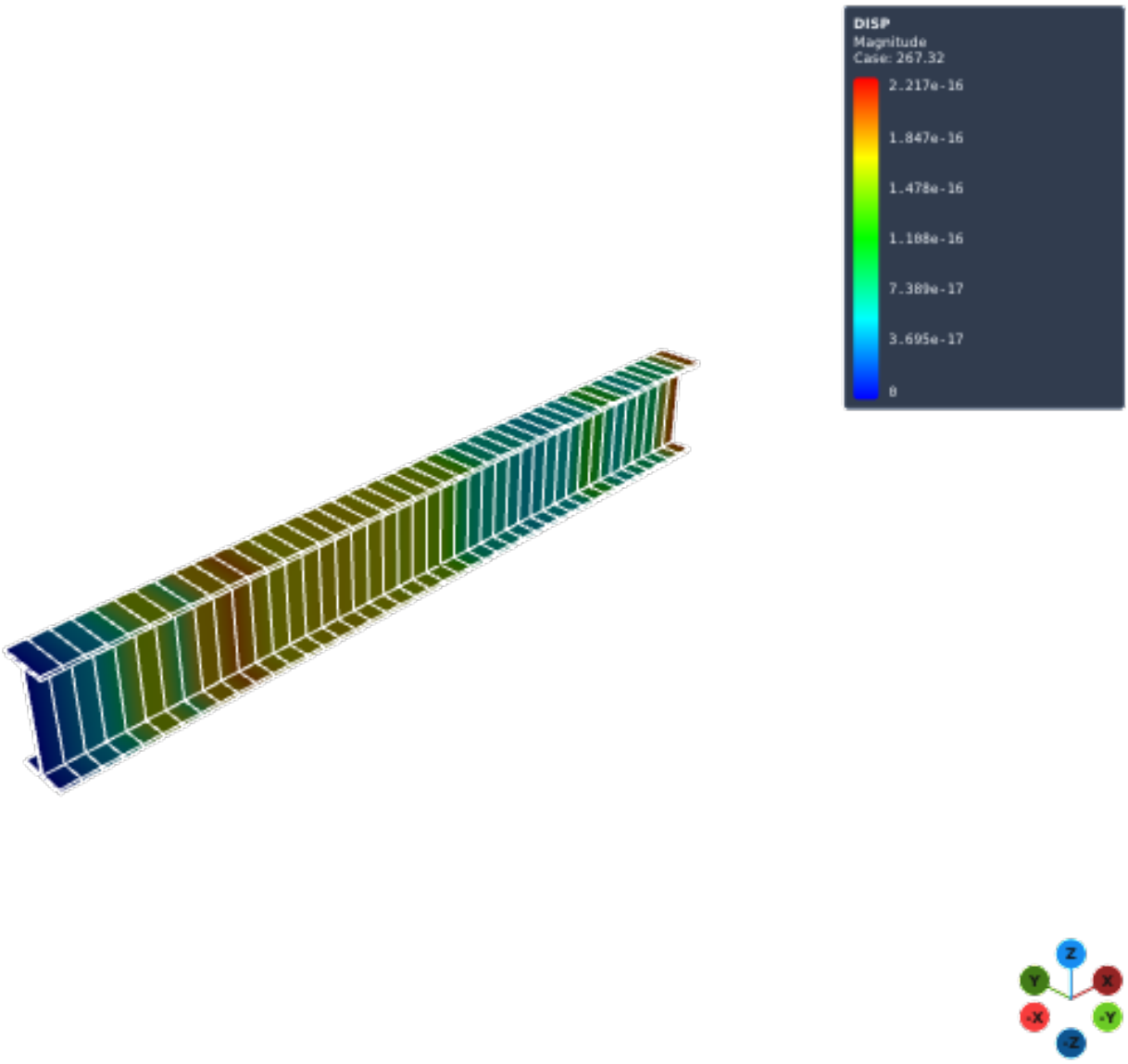

Figure 34: Calculix — mode 5, 267.320 Hz

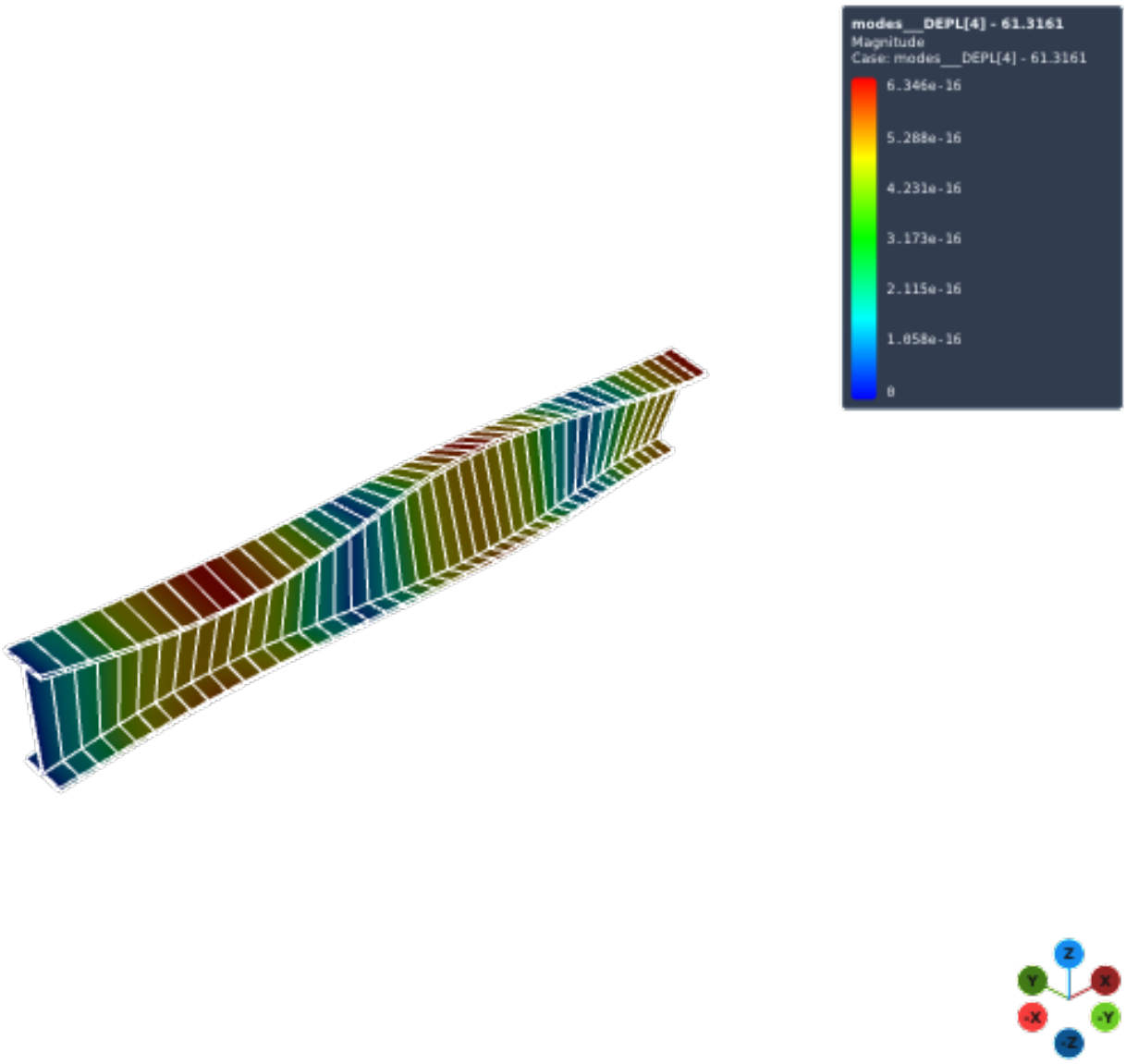

Figure 35: Code_Aster — mode 5, 61.316 Hz

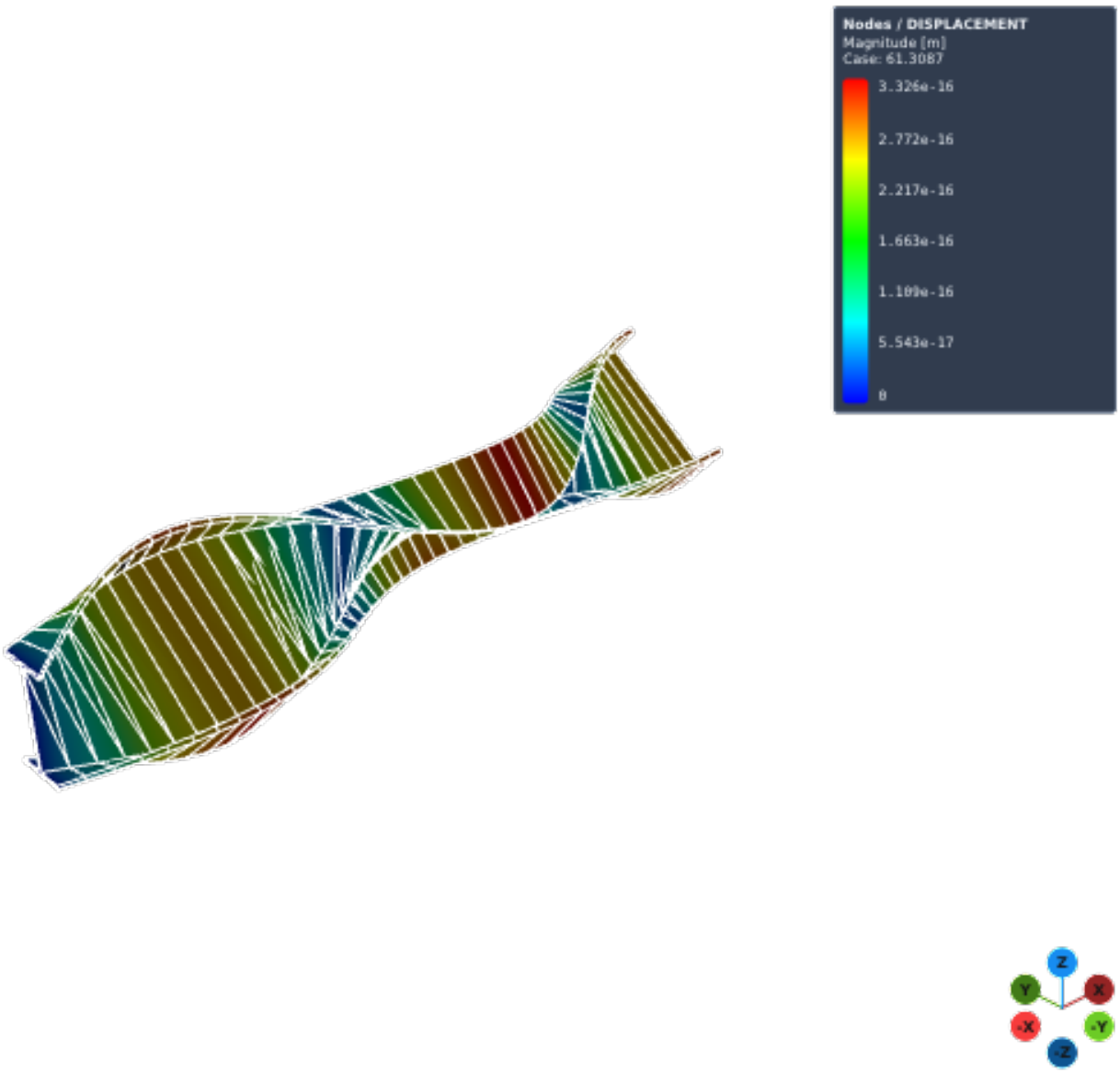

Figure 36: Sesam — mode 5, 61.309 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	53.855	-51.46	114501	1.75e-29	8.90e-30	2.61e-27	-4.18e-15	2.98e-15	5.11e-14
Calculix	267.320	+140.94	2.82112e+06	3.28e-30	4.98e-30	2.38e-29	1.81e-15	-2.23e-15	4.88e-15
Code_Aster	61.316	-44.74	148426	6.01e-37	2.98e-27	3.86e-27	7.75e-19	-5.46e-14	6.21e-14
Sesam	61.309	-44.74	148390	5.33e-36	2.55e-29	4.34e-29	-2.31e-18	-5.05e-15	6.59e-15

Mode 6

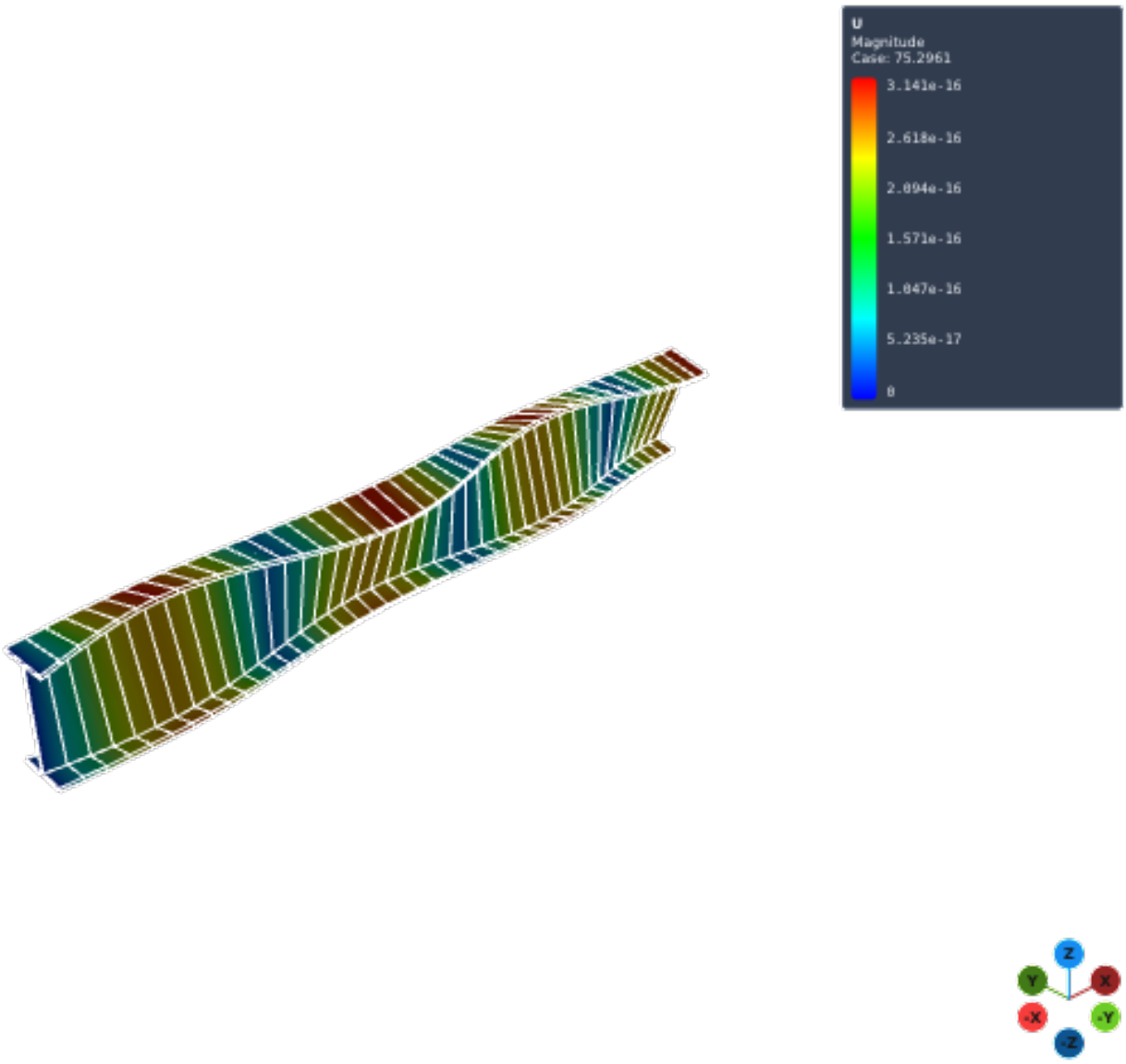

Figure 37: Abaqus — mode 6, 75.296 Hz

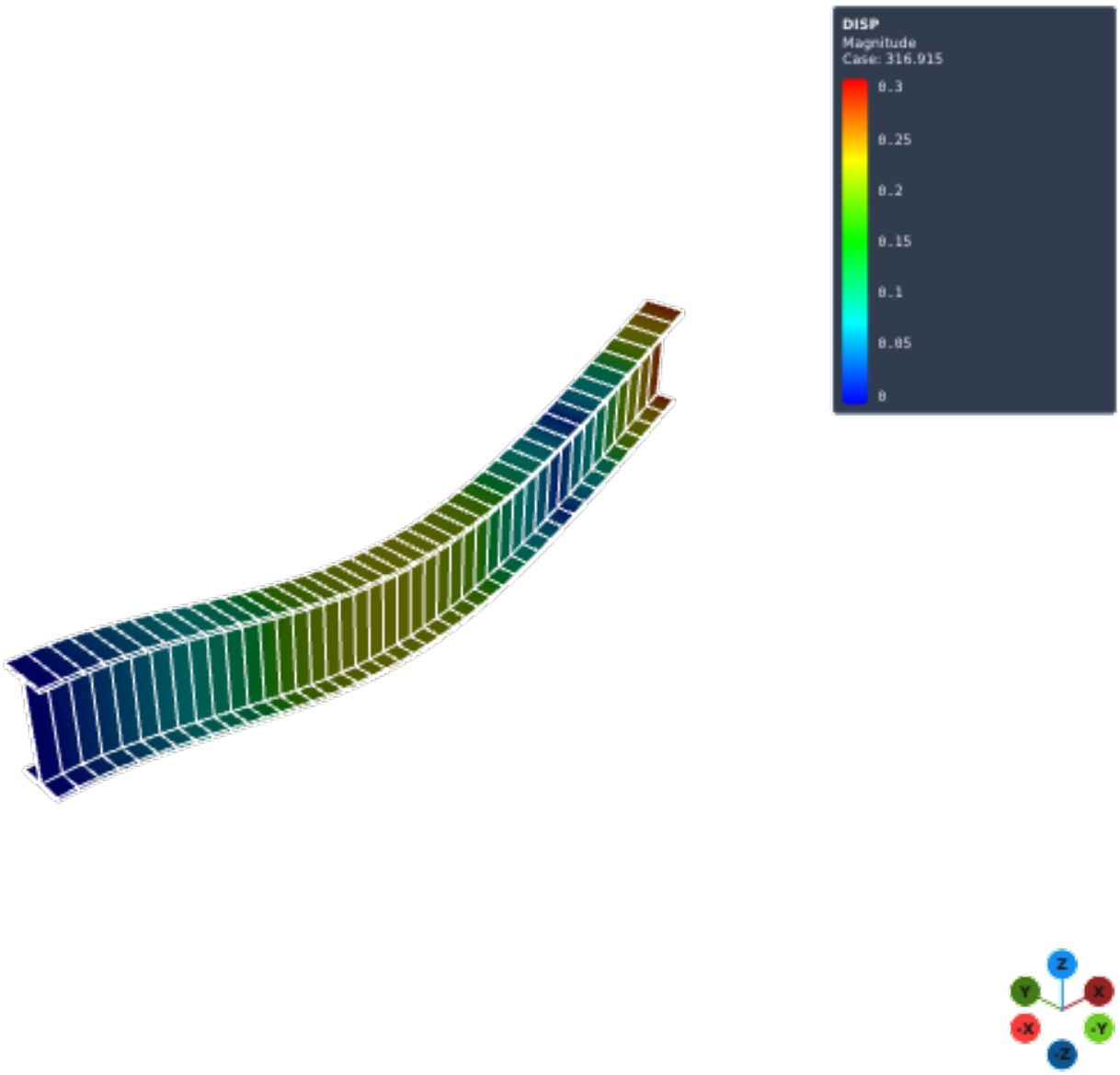

Figure 38: Calculix — mode 6, 316.915 Hz

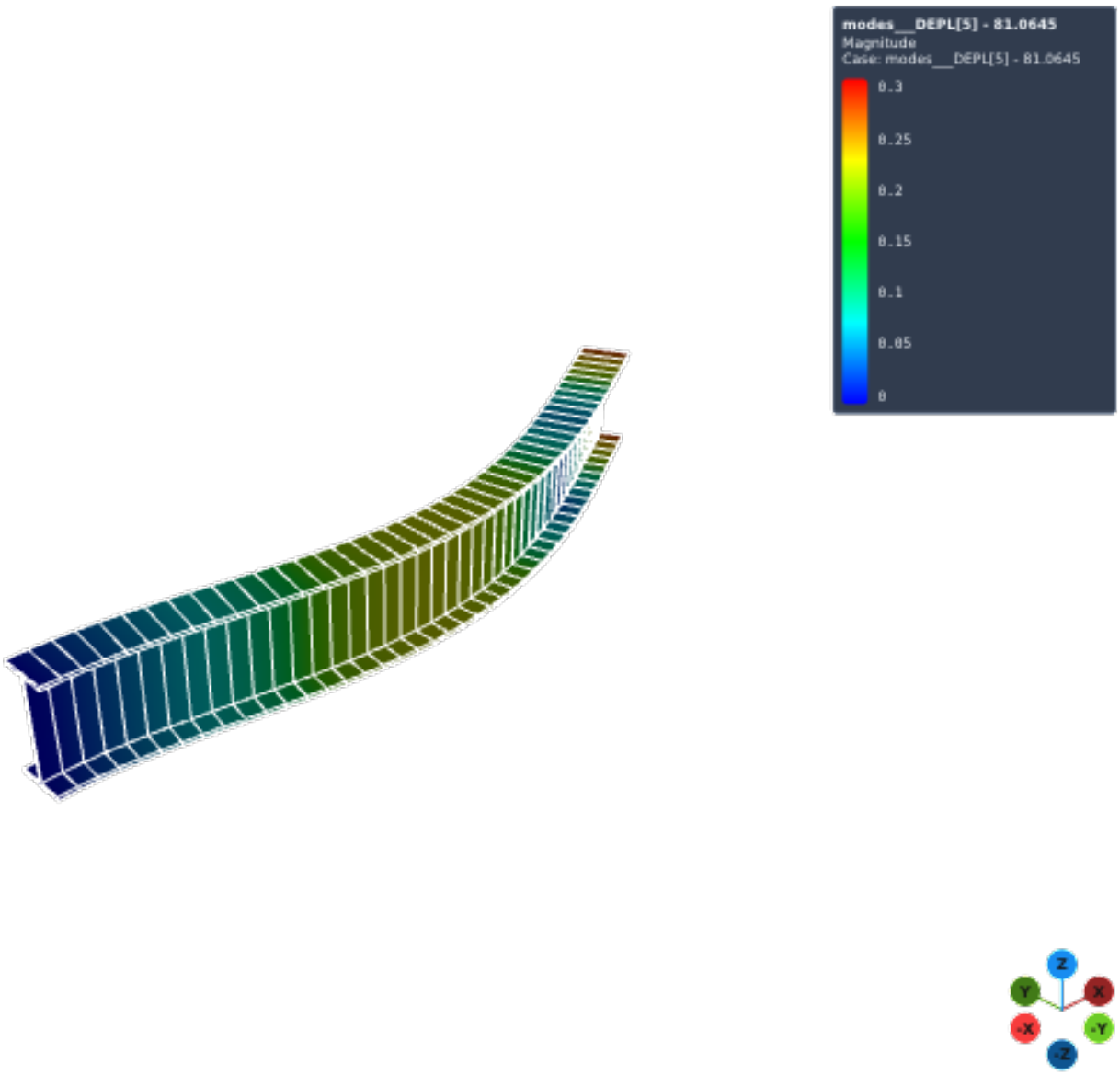

Figure 39: Code_Aster — mode 6, 81.065 Hz

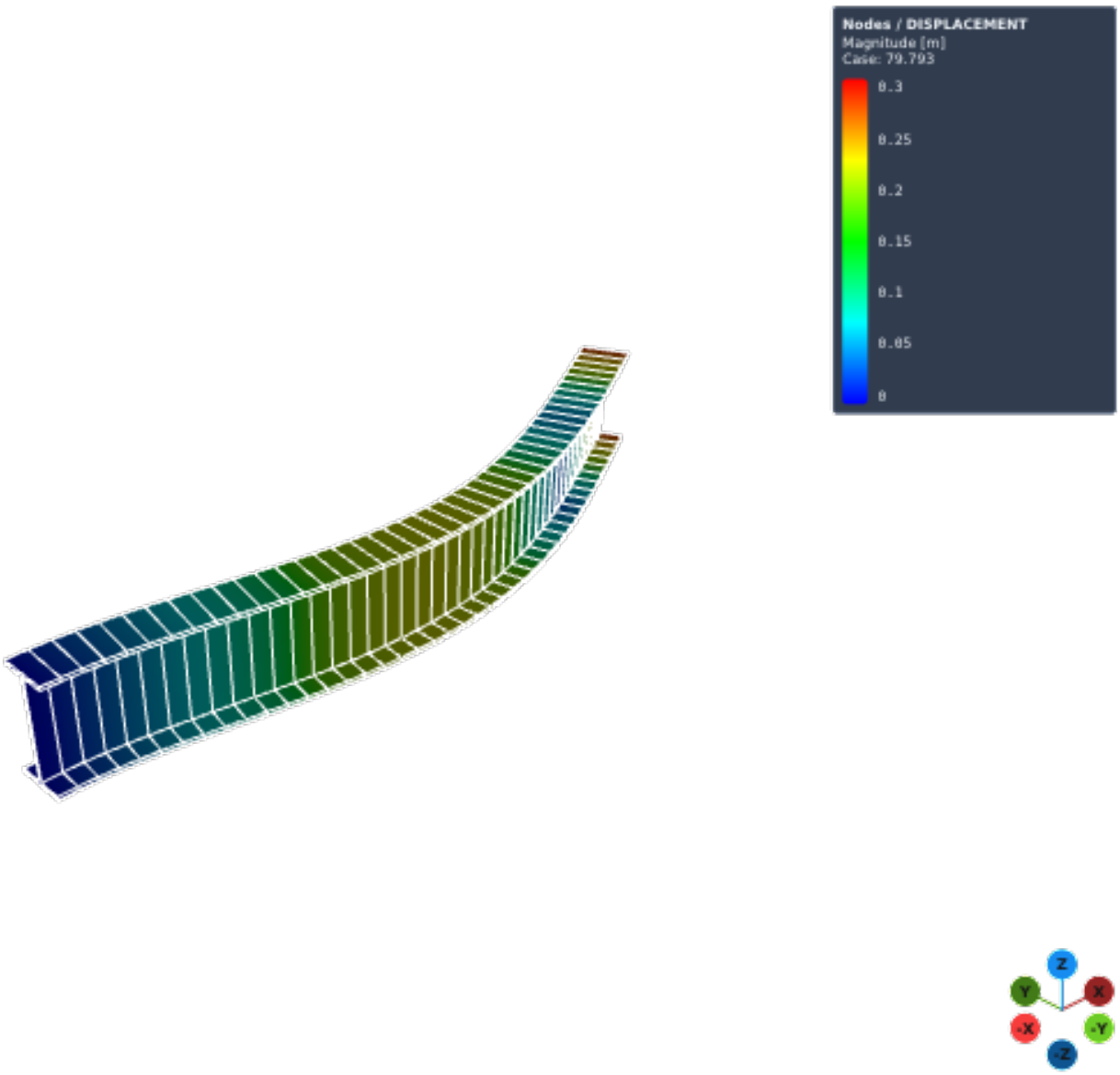

Figure 40: Sesam — mode 6, 79.793 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.296	-45.54	223823	3.82e-29	2.17e-28	3.67e-28	-6.18e-15	-1.47e-14	1.92e-14
Calculix	316.915	+129.20	3.96503e+06	1.31e-33	7.09e-30	34.97	-3.62e-17	-2.66e-15	-5.913
Code_Aster	81.065	-41.37	259431	1.92e-36	35.72	4.12e-29	1.38e-18	-5.977	-6.42e-15
Sesam	79.793	-42.29	251356	6.55e-33	36.29	2.49e-29	8.09e-17	-6.024	4.99e-15

Mode 7

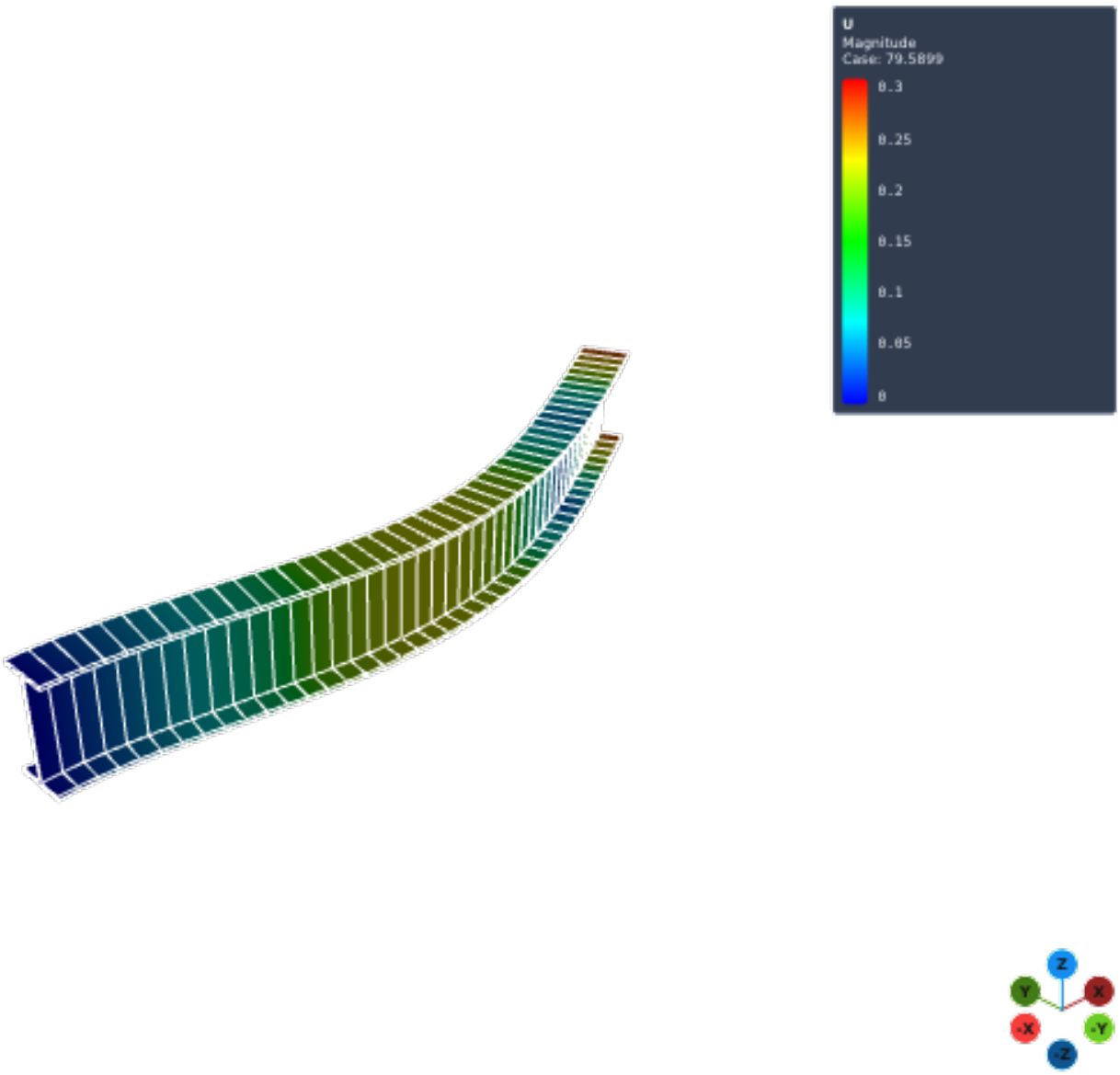

Figure 41: Abaqus — mode 7, 79.590 Hz

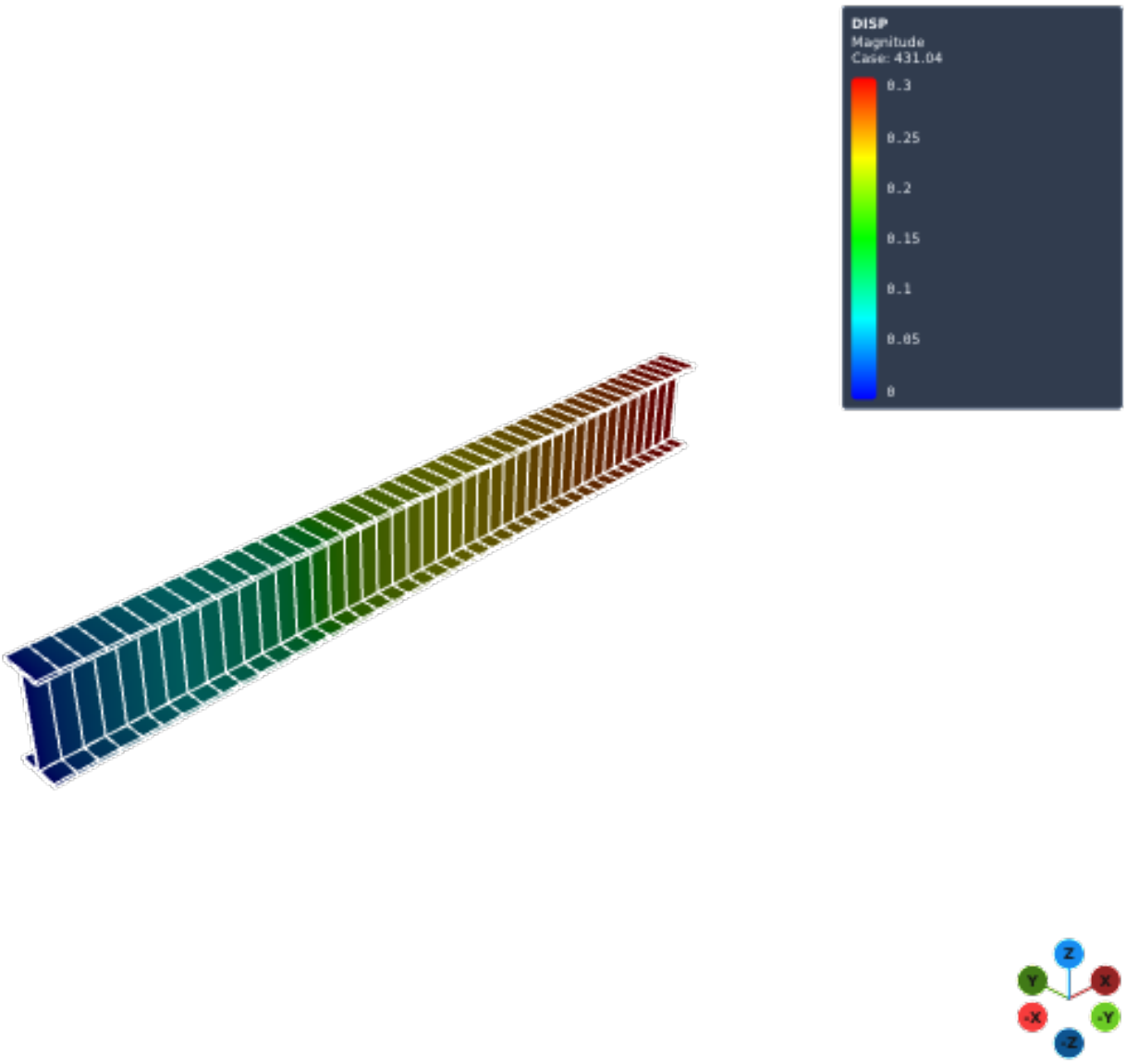

Figure 42: Calculix — mode 7, 431.040 Hz

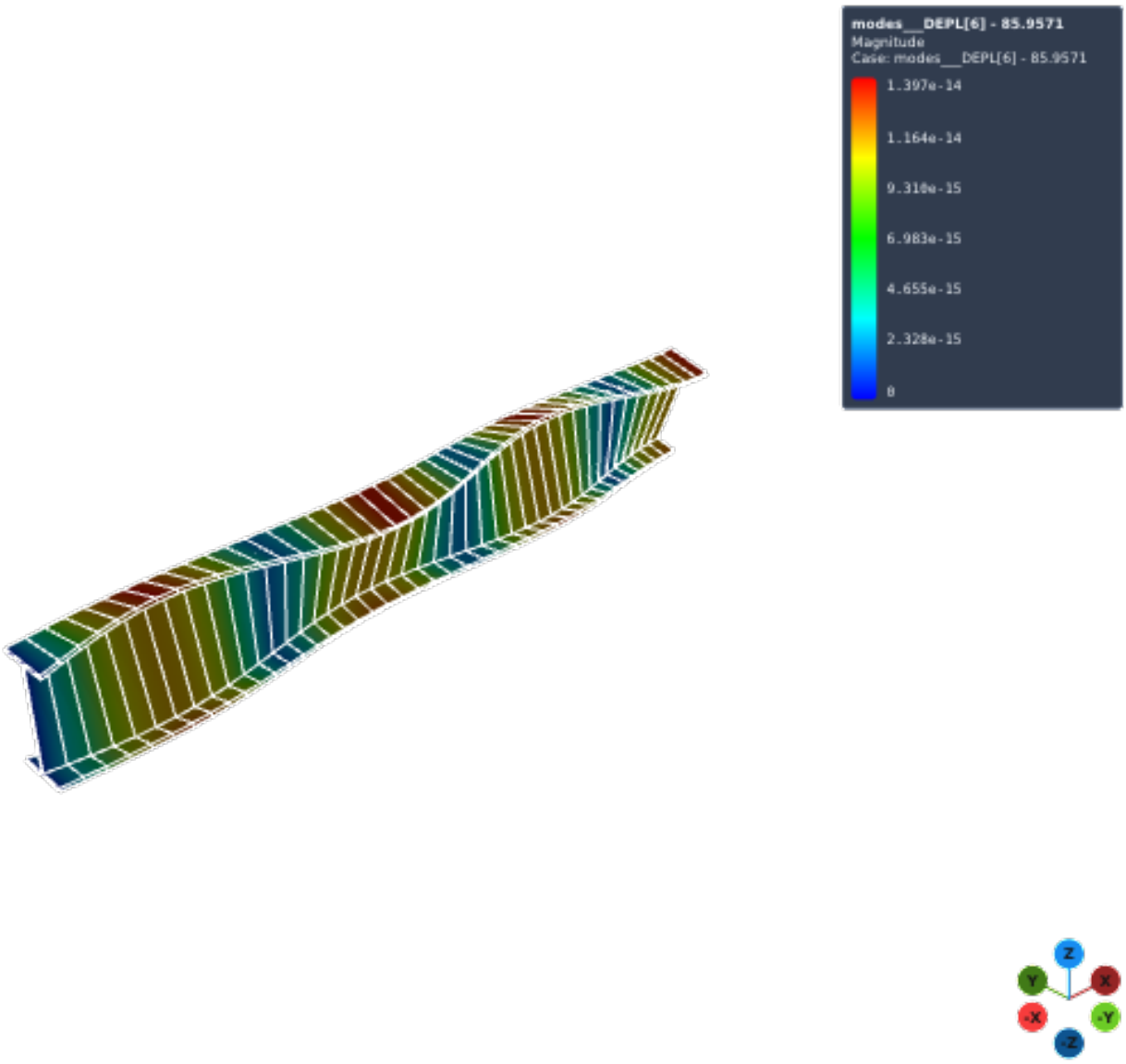

Figure 43: Code_Aster — mode 7, 85.957 Hz

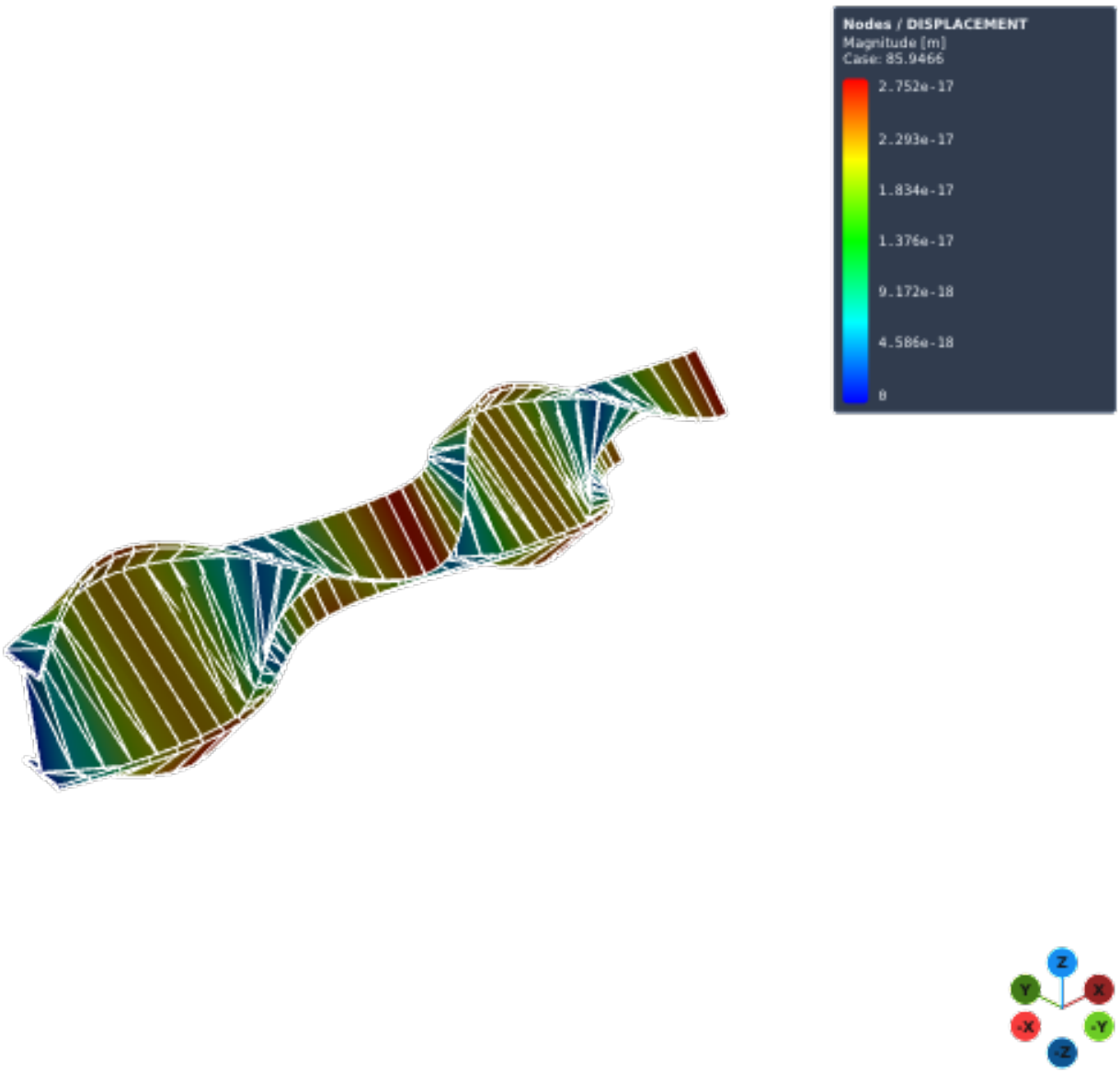

Figure 44: Sesam — mode 7, 85.947 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	79.590	-53.36	250078	1.01e-30	36.20	1.38e-29	1.01e-15	-6.016	-3.72e-15
Calculix	431.040	+152.61	7.33492e+06	153.94	1.29e-27	4.83e-31	12.407	-3.60e-14	-6.95e-16
Code_Aster	85.957	-49.62	291691	1.51e-37	1.30e-24	1.56e-27	3.89e-19	-1.14e-12	-3.95e-14
Sesam	85.947	-49.63	291620	1.01e-32	4.87e-31	2.32e-31	-1.01e-16	-6.98e-16	-4.81e-16

Mode 8

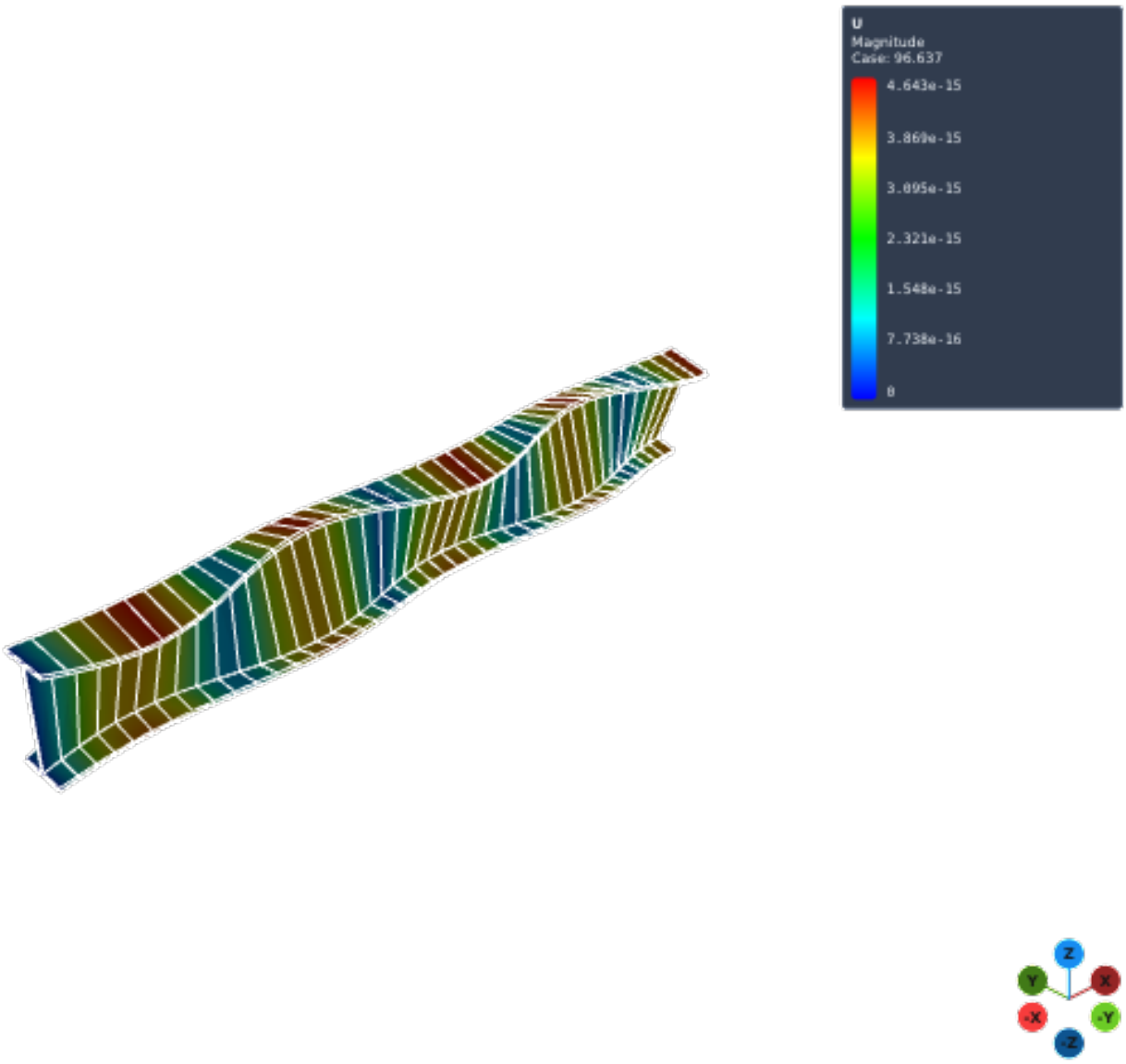

Figure 45: Abaqus — mode 8, 96.637 Hz

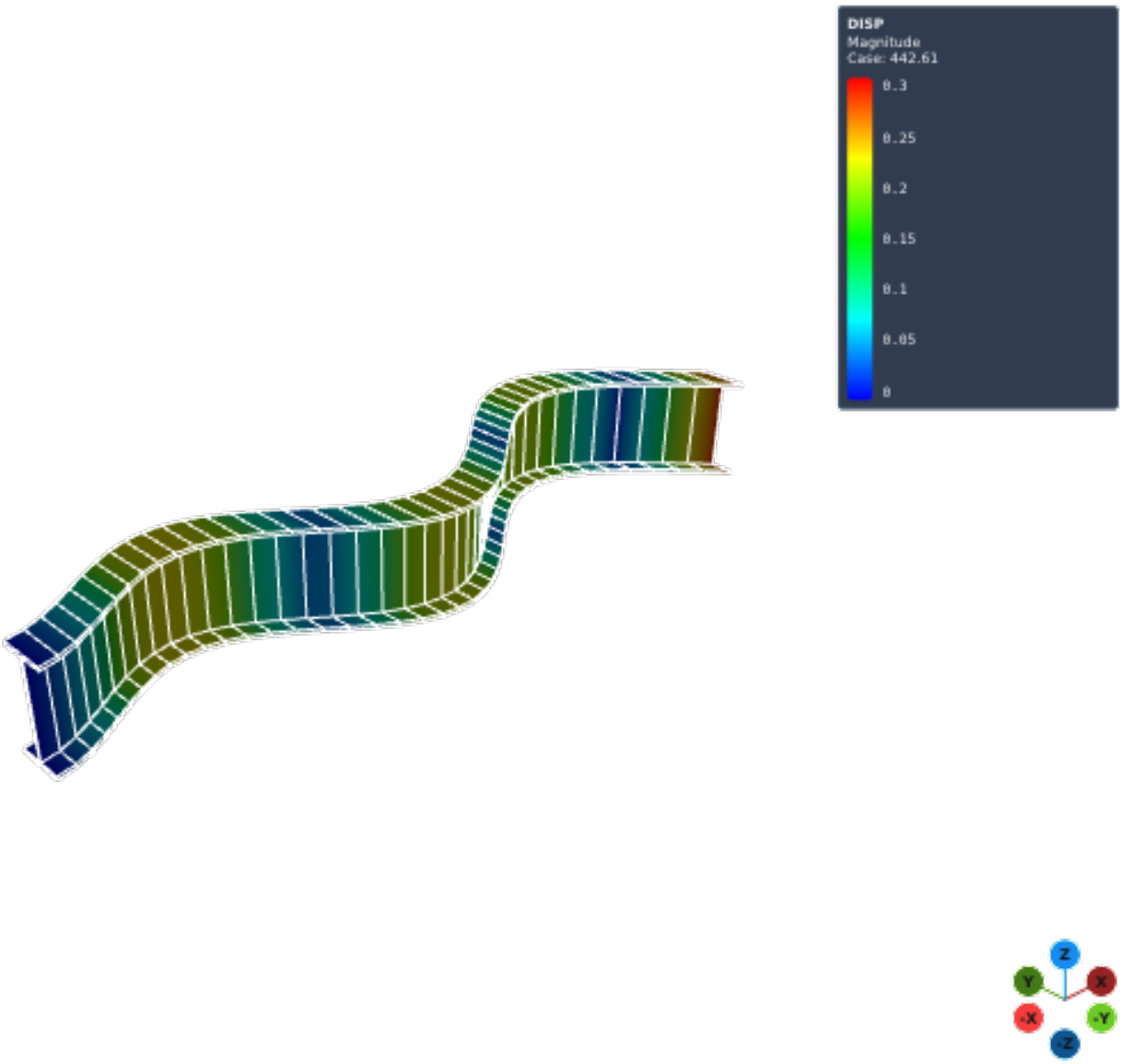

Figure 46: Calculix — mode 8, 442.610 Hz

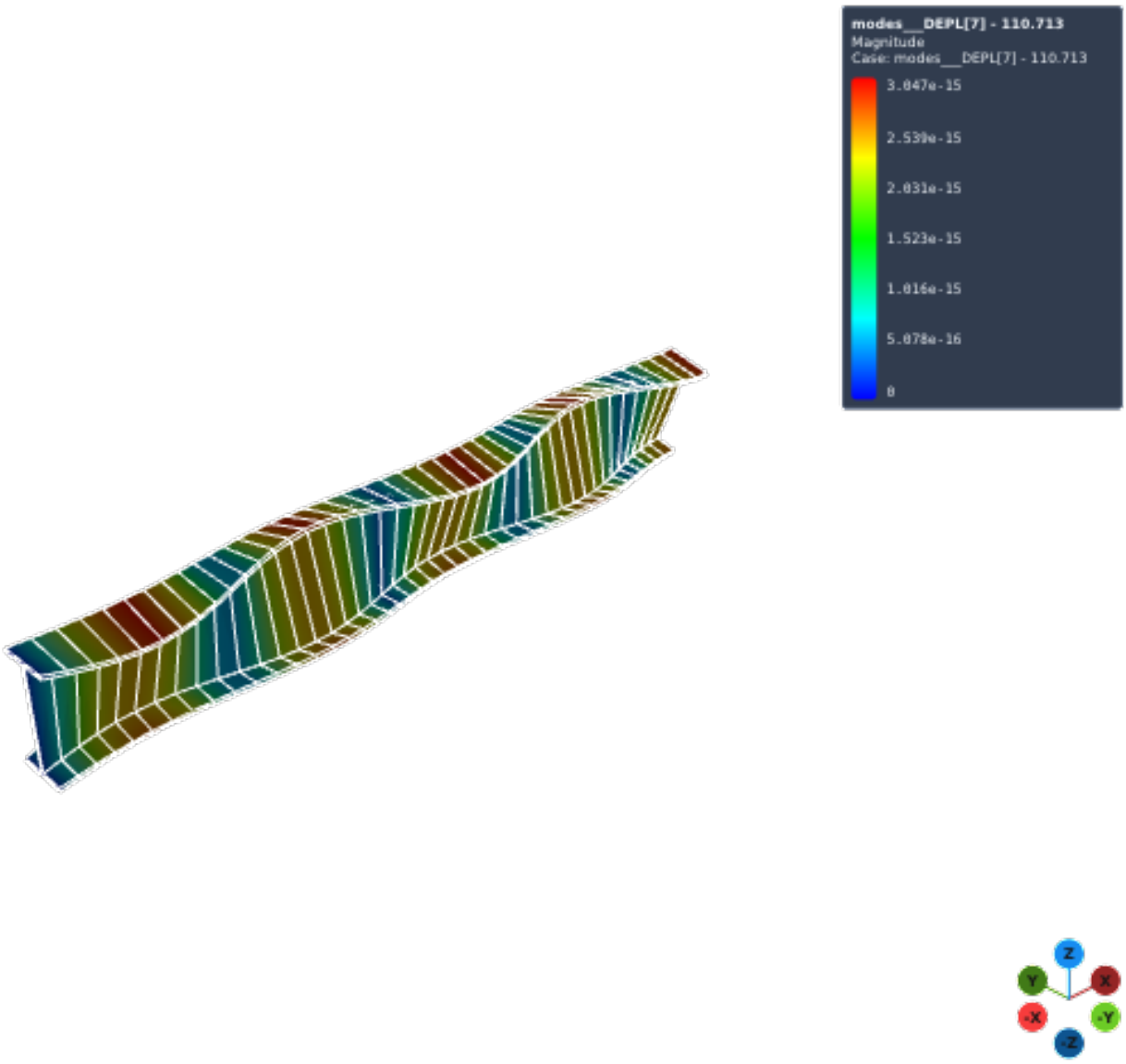

Figure 47: Code_Aster — mode 8, 110.713 Hz

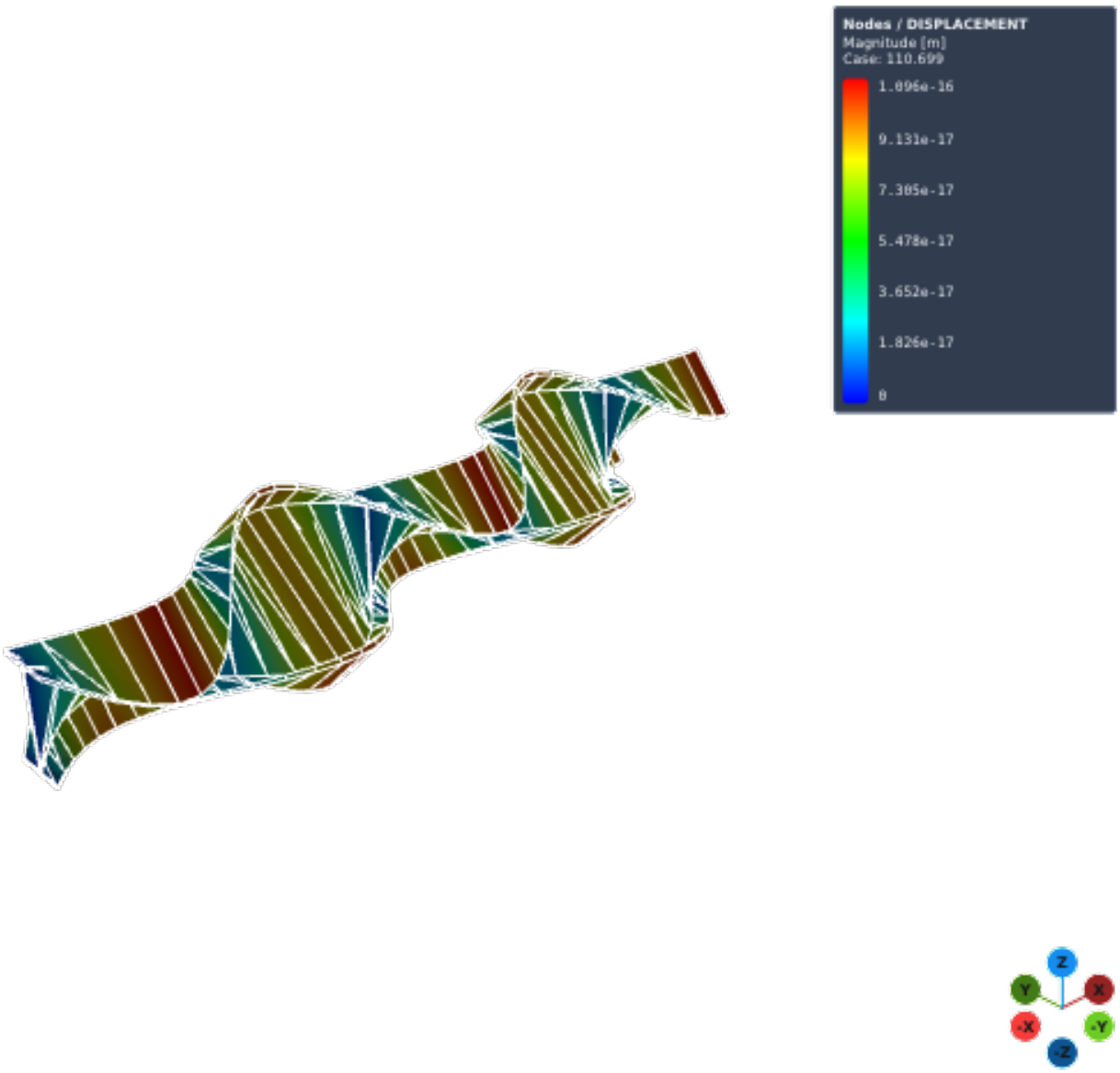

Figure 48: Sesam — mode 8, 110.699 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	96.637	-49.18	368677	4.04e-26	1.03e-27	4.21e-27	-2.01e-13	-3.20e-14	6.49e-14
Calculix	442.610	+132.75	7.73397e+06	3.15e-26	6.24	6.00e-30	1.77e-13	2.498	2.45e-15
Code_Aster	110.713	-41.78	483900	4.19e-35	1.70e-26	1.21e-28	6.48e-18	1.30e-13	-1.10e-14
Sesam	110.699	-41.79	483782	5.75e-32	4.96e-30	6.35e-30	2.40e-16	-2.23e-15	-2.52e-15

Mode 9

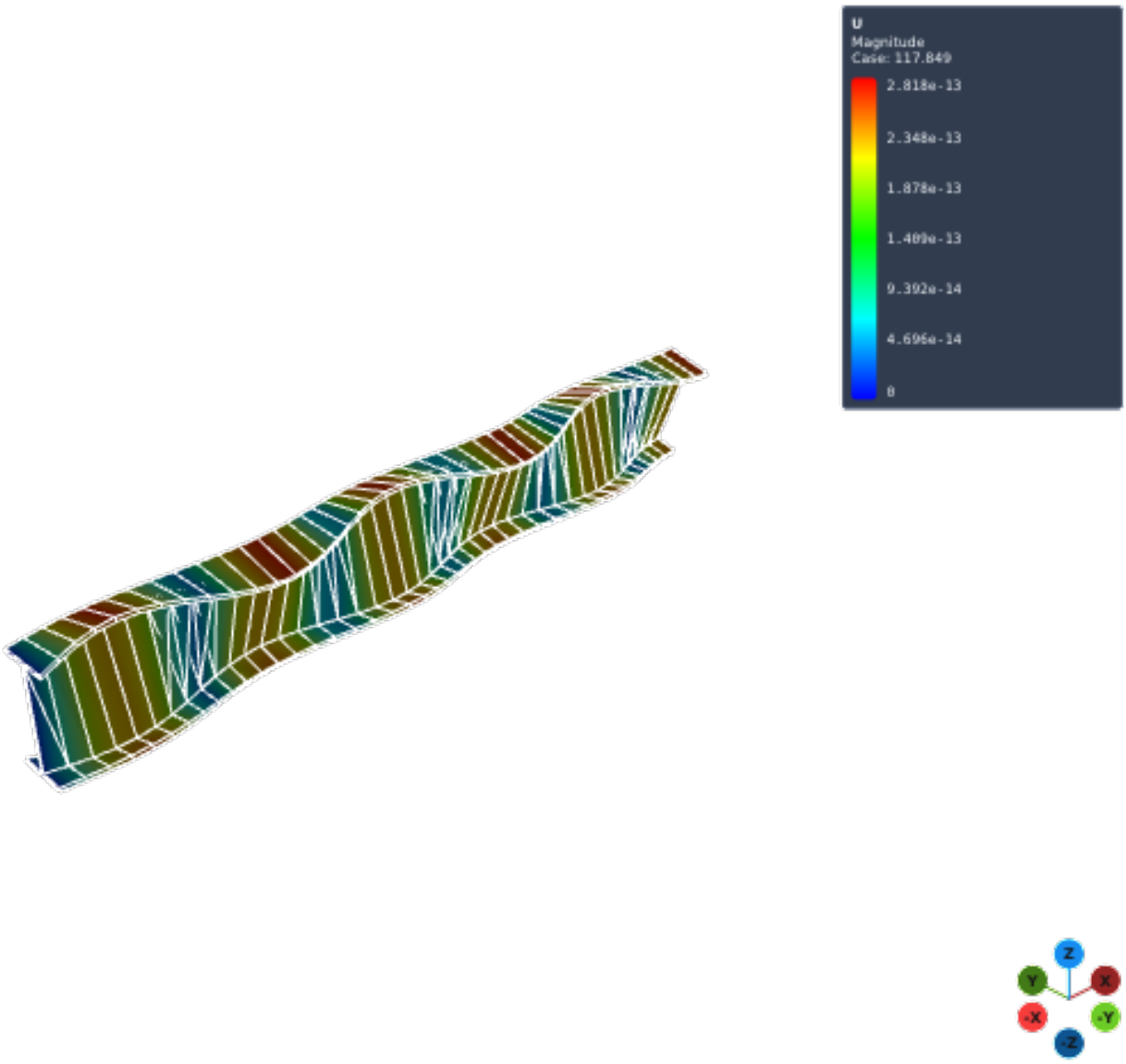

Figure 49: Abaqus — mode 9, 117.849 Hz

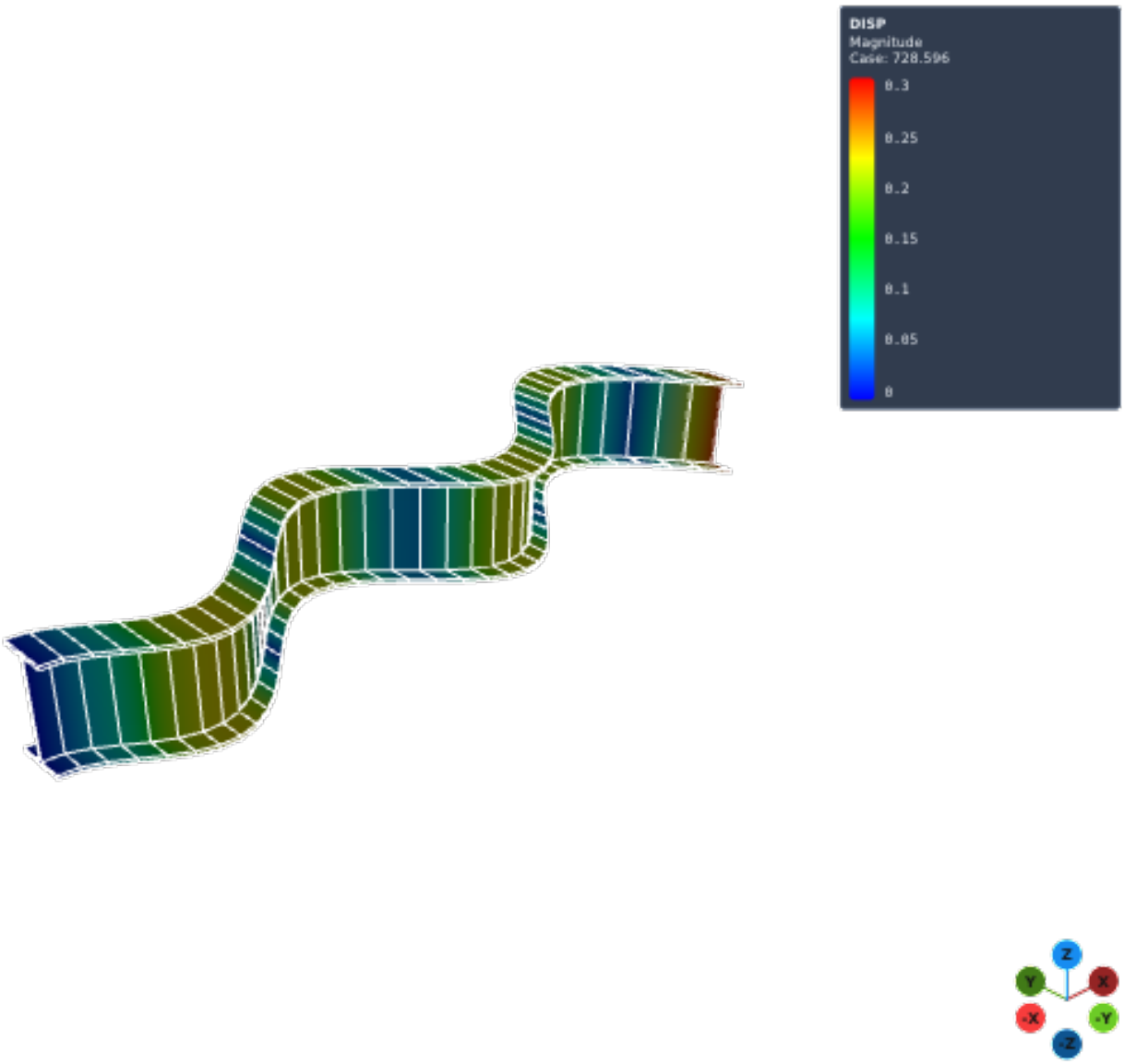

Figure 50: Calculix — mode 9, 728.596 Hz

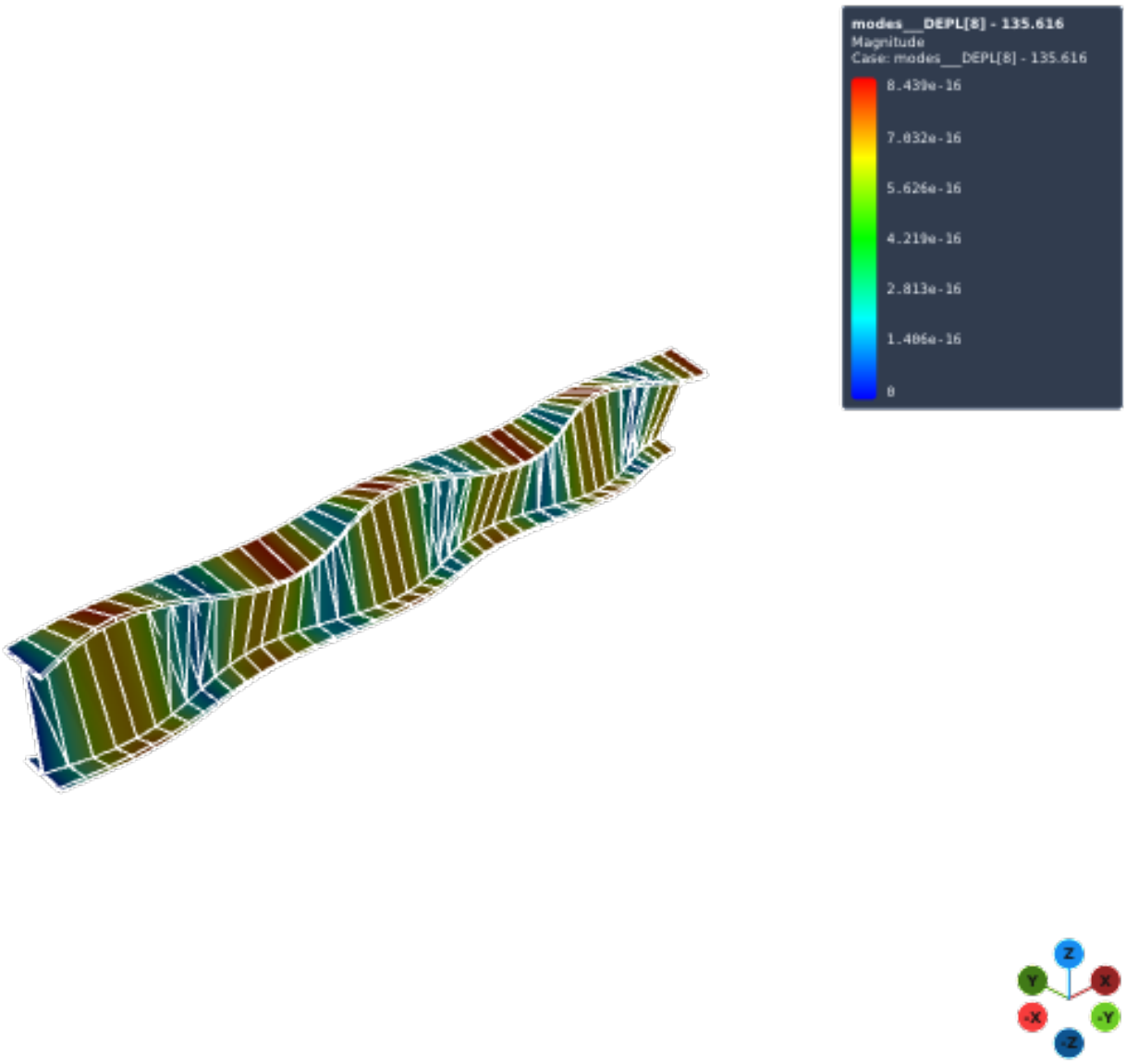

Figure 51: Code_Aster — mode 9, 135.616 Hz

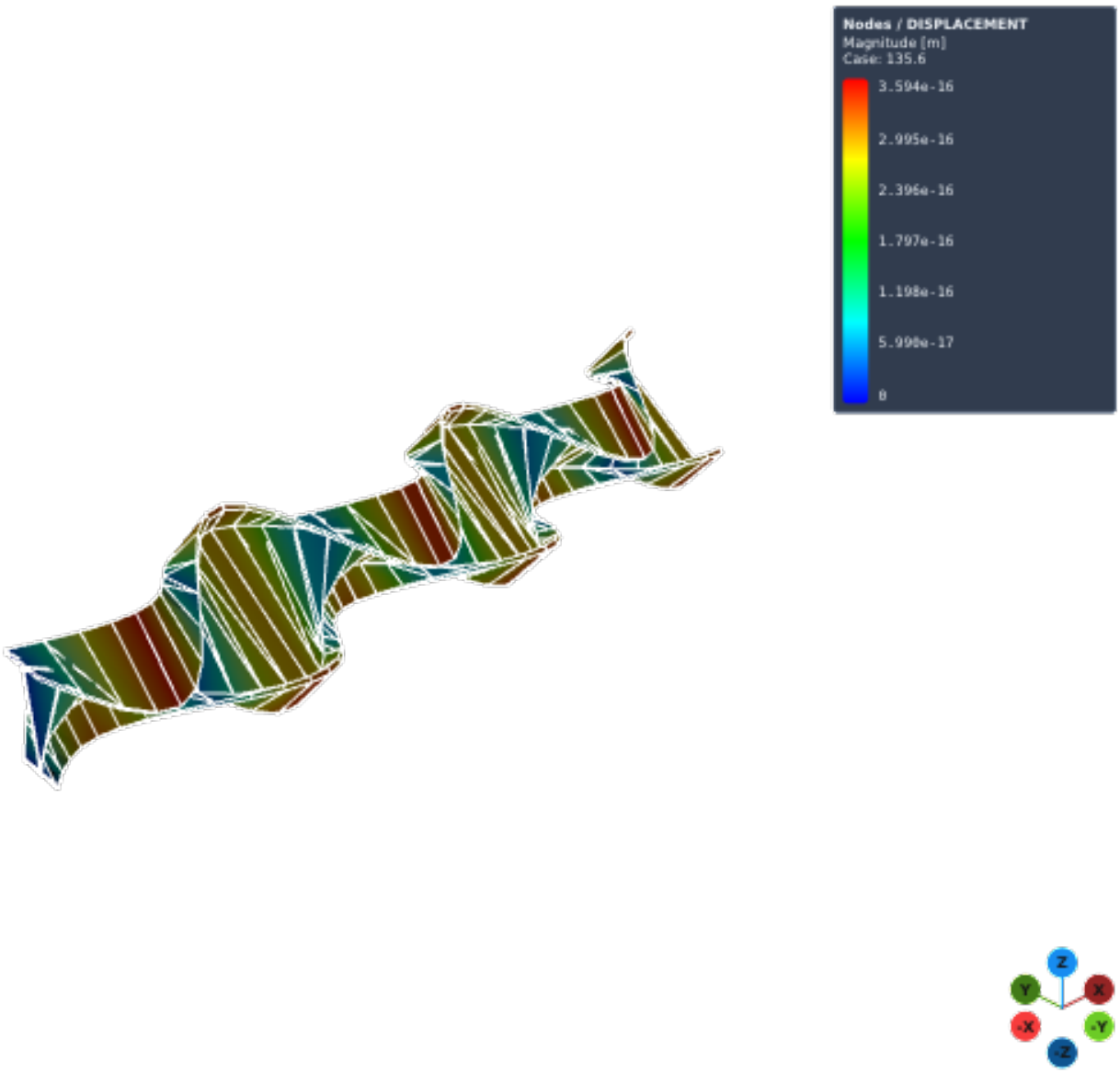

Figure 52: Sesam — mode 9, 135.600 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	117.850	-57.82	548290	9.24e-23	1.72e-25	7.37e-24	9.61e-12	4.15e-13	2.71e-12
Calculix	728.596	+160.76	2.09572e+07	1.49e-29	3.75	1.24e-28	3.85e-15	-1.937	-1.11e-14
Code_Aster	135.616	-51.46	726077	1.04e-33	3.56e-27	1.01e-29	3.23e-17	-5.96e-14	3.18e-15
Sesam	135.600	-51.47	725900	8.08e-31	4.13e-29	6.76e-29	-8.99e-16	6.43e-15	8.22e-15

Mode 10

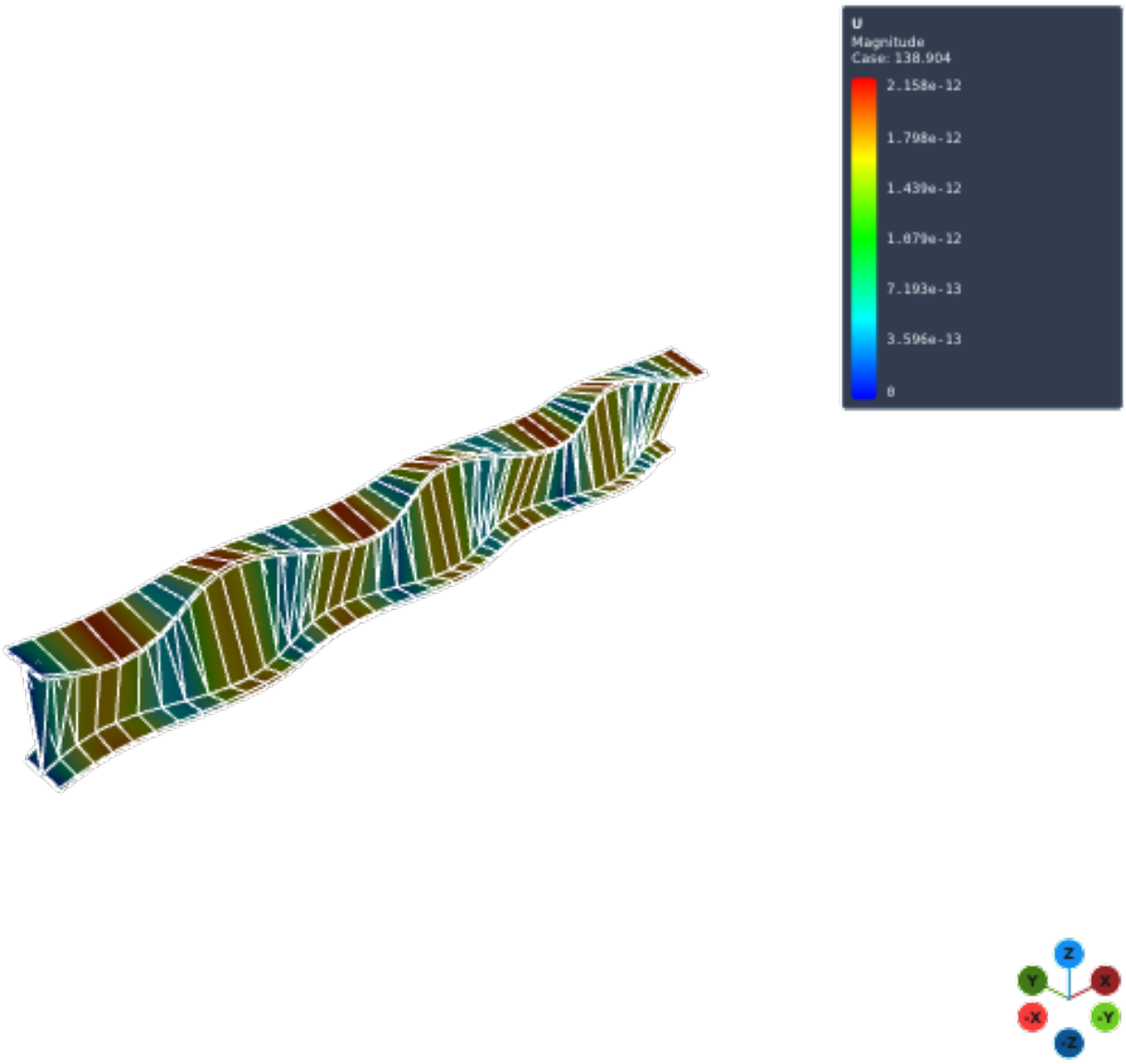

Figure 53: Abaqus — mode 10, 138.904 Hz

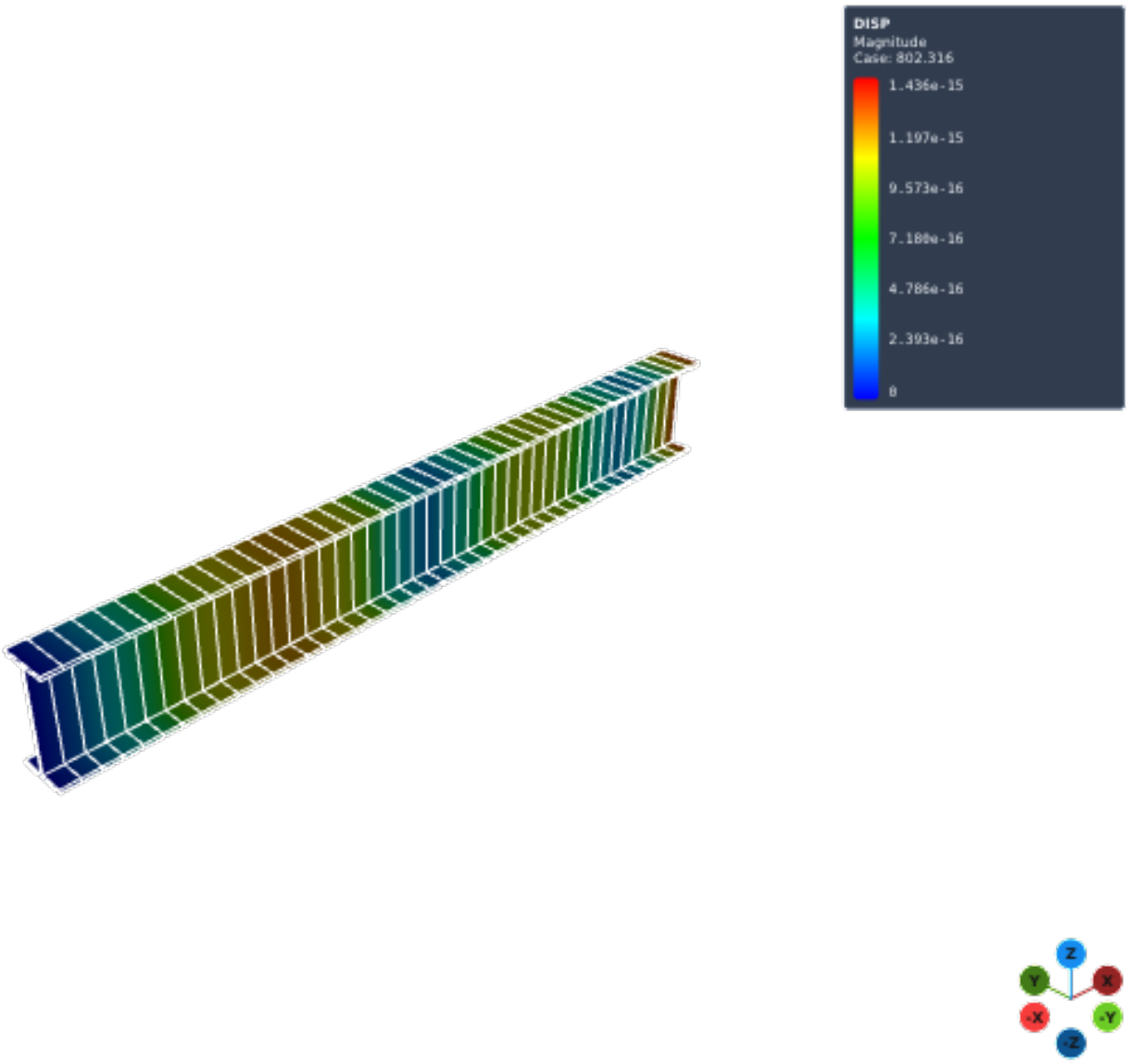

Figure 54: Calculix — mode 10, 802.316 Hz

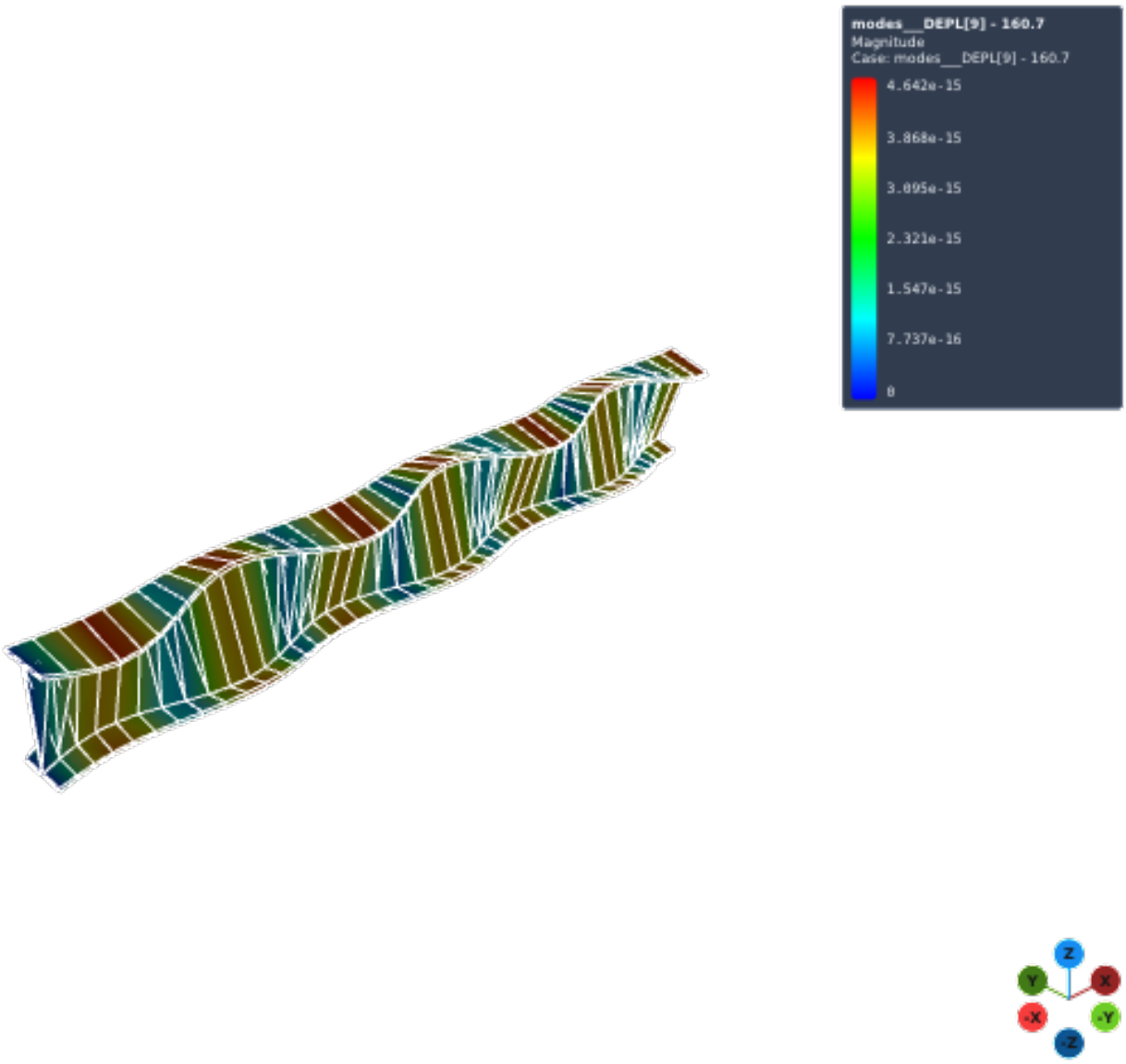

Figure 55: Code_Aster — mode 10, 160.700 Hz

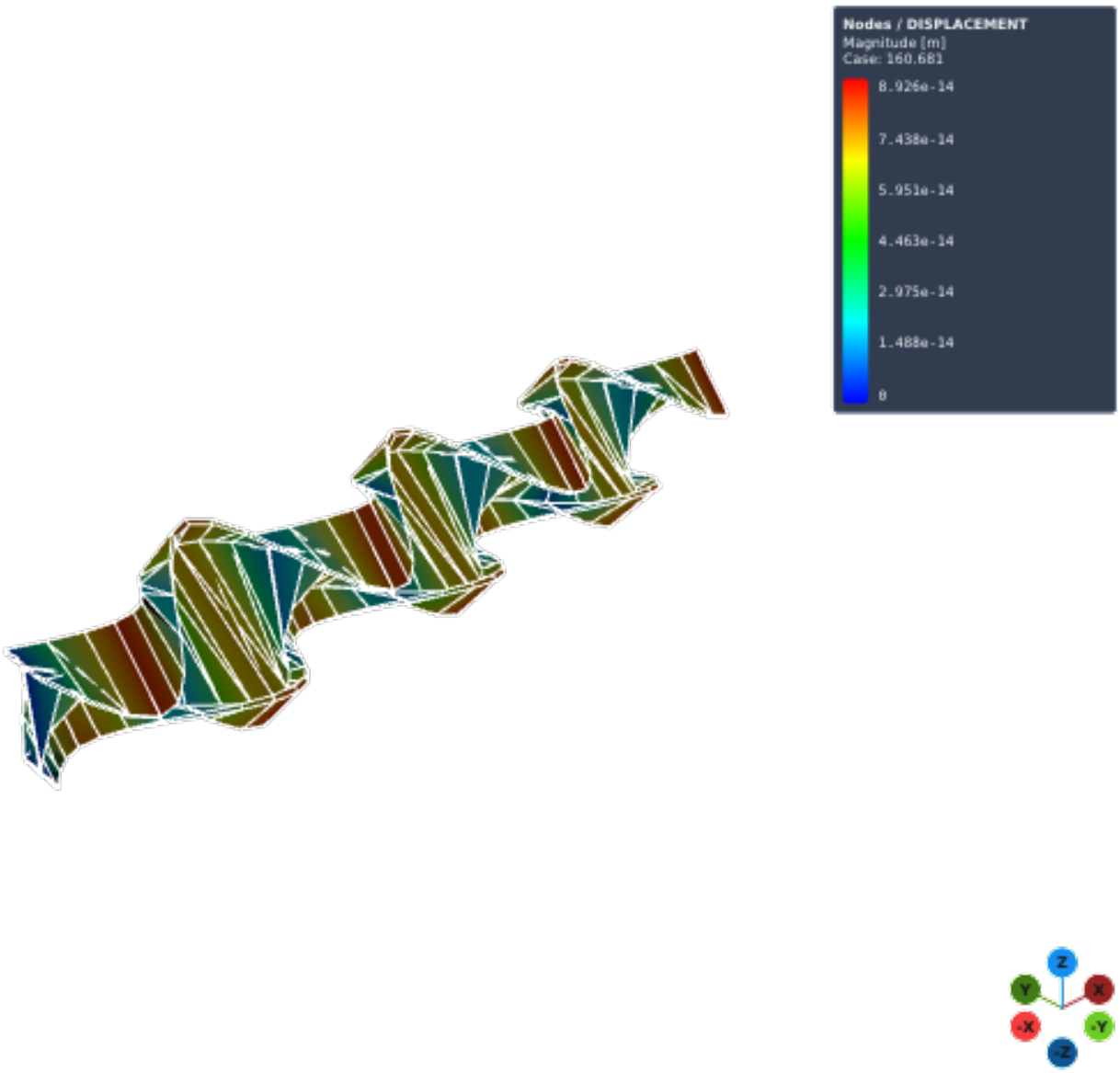

Figure 56: Sesam — mode 10, 160.681 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	138.900	-56.00	761704	1.00e-20	2.09e-22	2.61e-21	1.00e-10	1.45e-11	-5.11e-11
Calculix	802.316	+154.18	2.54127e+07	3.17e-30	9.74e-30	3.78e-28	-1.78e-15	3.12e-15	1.94e-14
Code_Aster	160.700	-49.09	1.01952e+06	6.07e-34	7.26e-27	6.87e-28	-2.46e-17	-8.52e-14	2.62e-14
Sesam	160.681	-49.10	1.01927e+06	1.48e-24	9.00e-28	2.96e-25	-1.22e-12	-3.00e-14	-5.44e-13

Mode 11

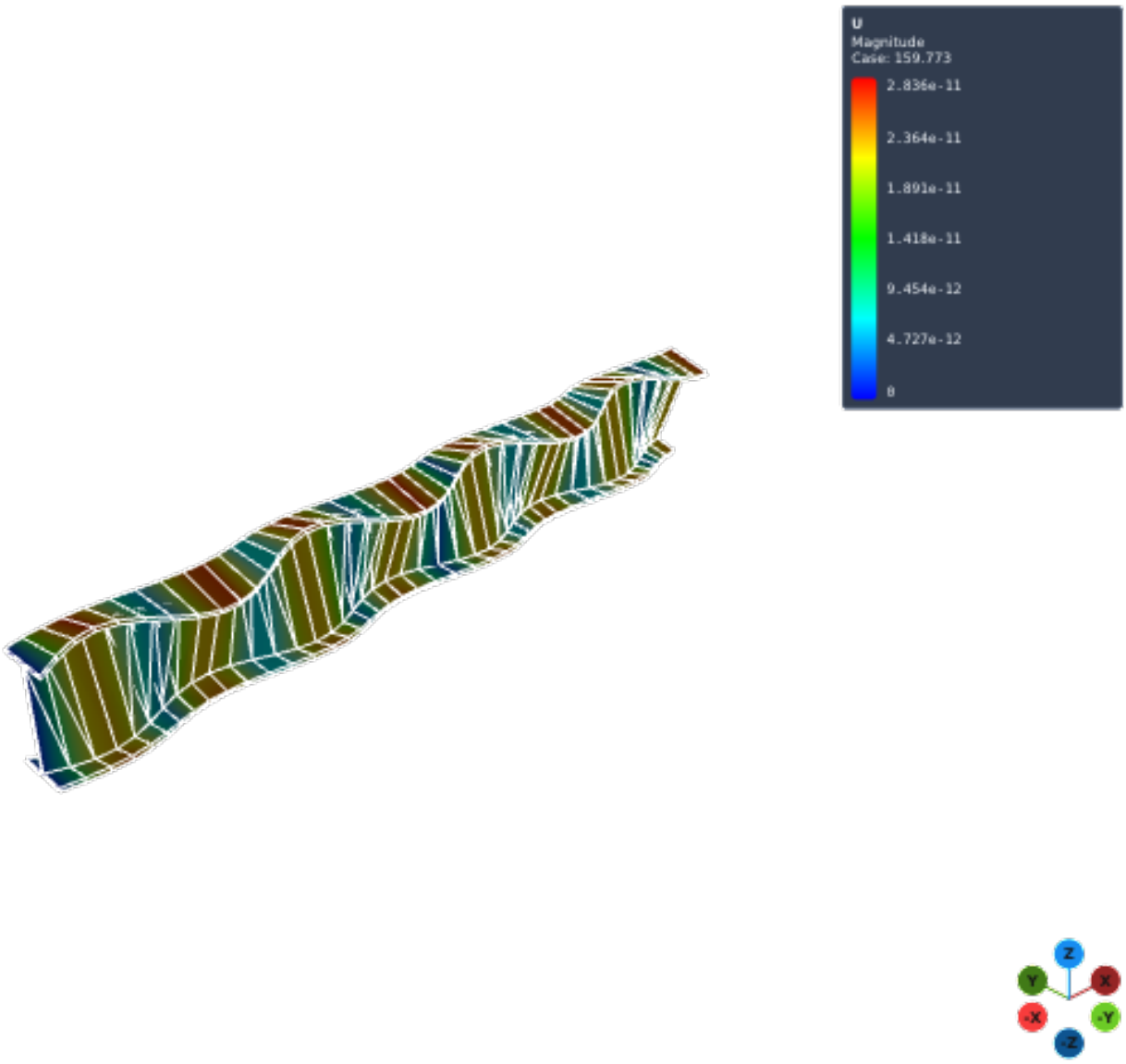

Figure 57: Abaqus — mode 11, 159.773 Hz

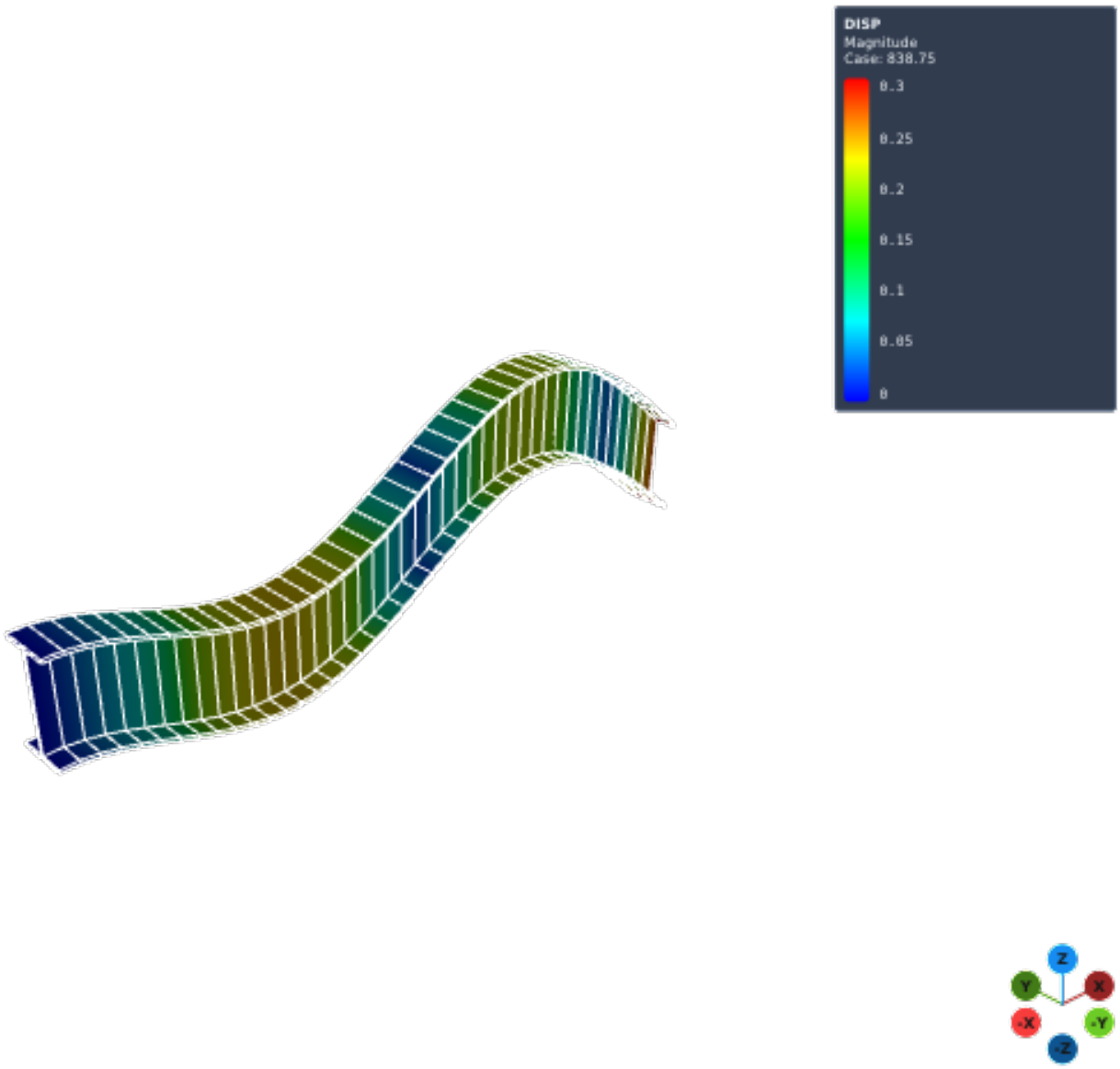

Figure 58: Calculix — mode 11, 838.750 Hz

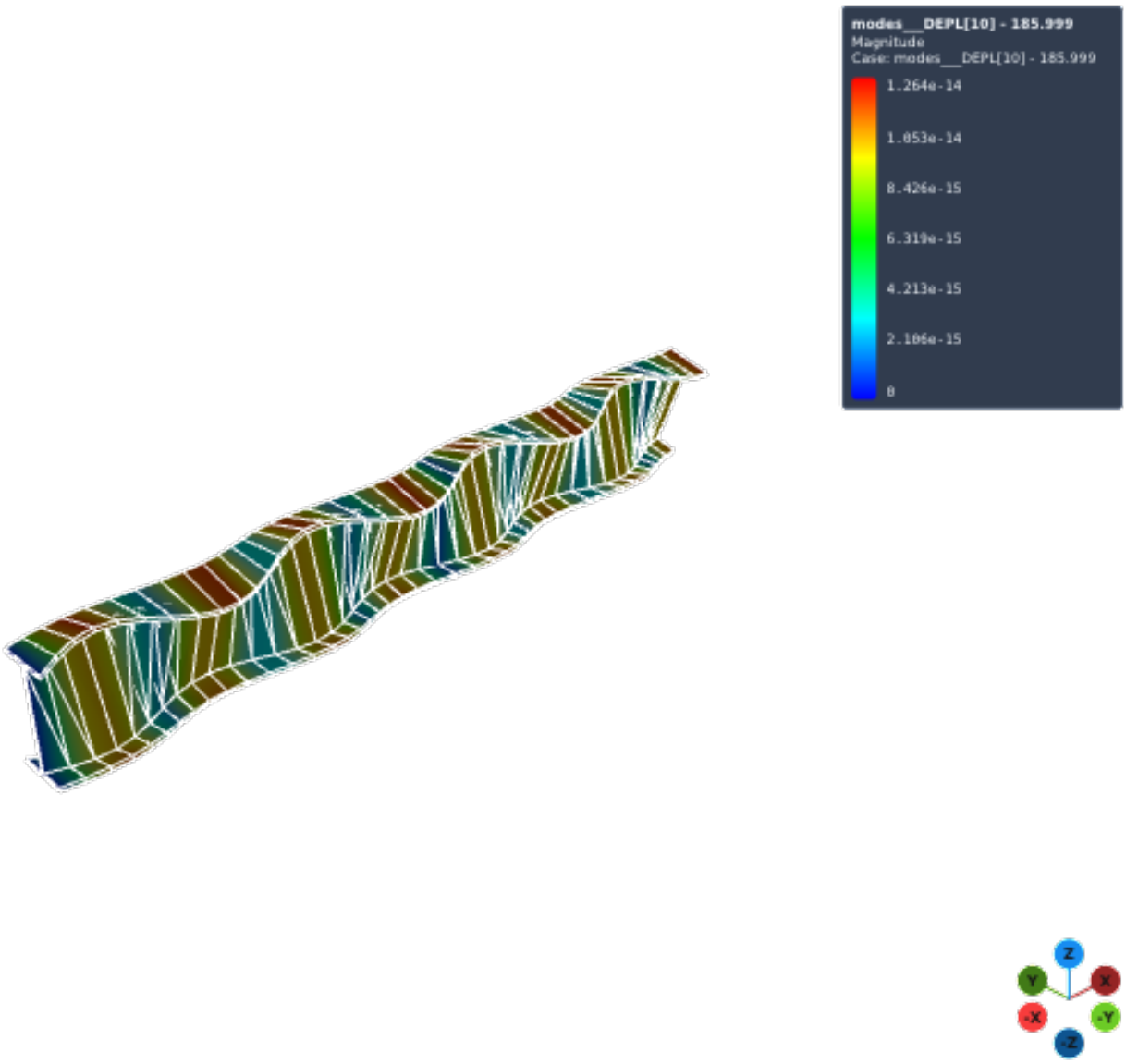

Figure 59: Code_Aster — mode 11, 185.999 Hz

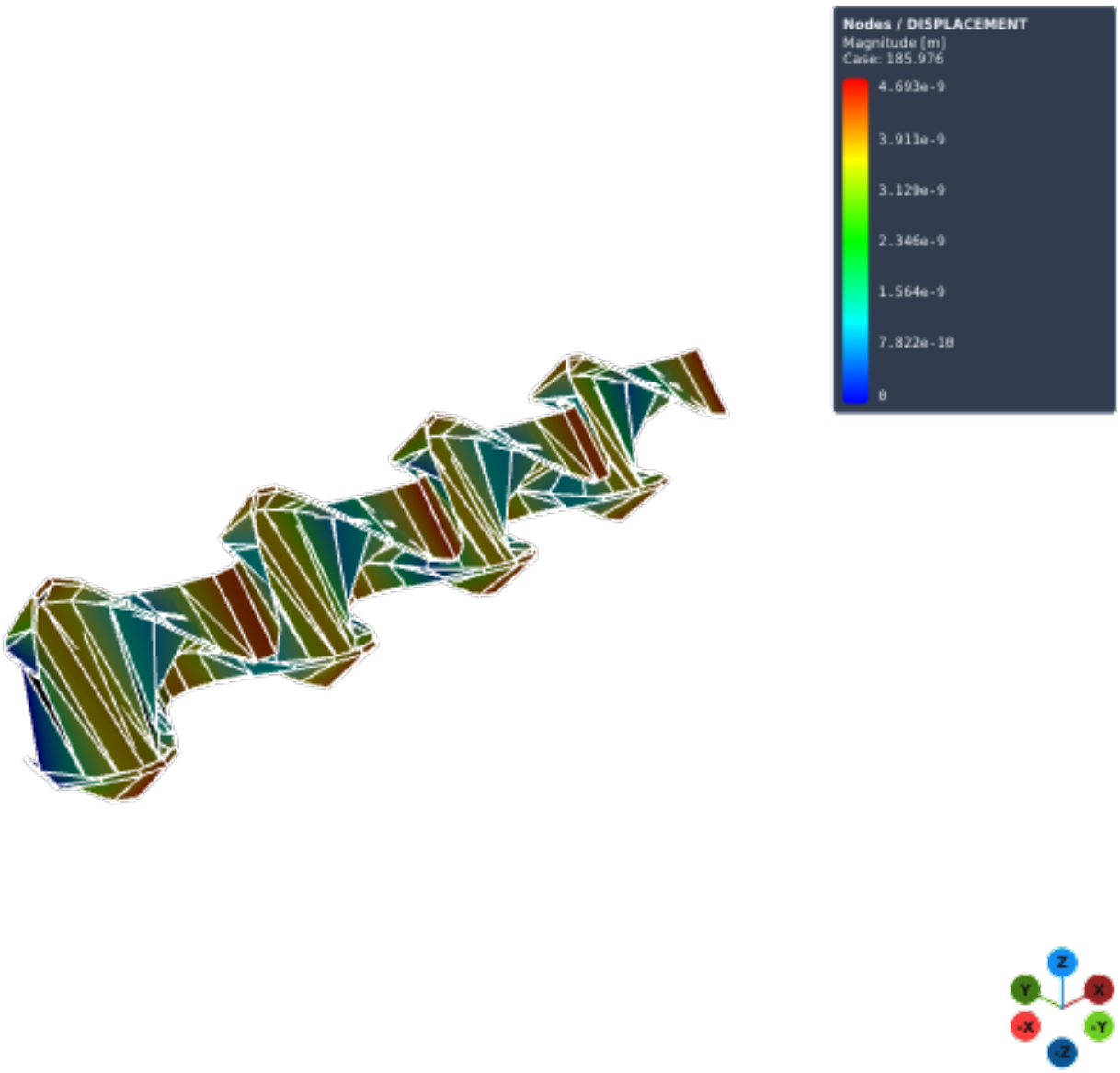

Figure 60: Sesam — mode 11, 185.976 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	159.770	-53.37	1.00778e+06	1.05e-18	1.96e-22	2.87e-21	1.02e-09	-1.40e-11	5.35e-11
Calculix	838.750	+144.80	2.77731e+07	1.58e-31	5.81e-29	11.38	3.98e-16	7.62e-15	-3.374
Code_Aster	185.999	-45.71	1.36578e+06	1.79e-34	1.67e-26	1.05e-26	-1.34e-17	-1.29e-13	1.02e-13
Sesam	185.976	-45.72	1.36544e+06	4.16e-15	1.83e-18	8.10e-16	-6.45e-08	-1.35e-09	-2.85e-08

Line, 1st order, reduced integration

Mode 1

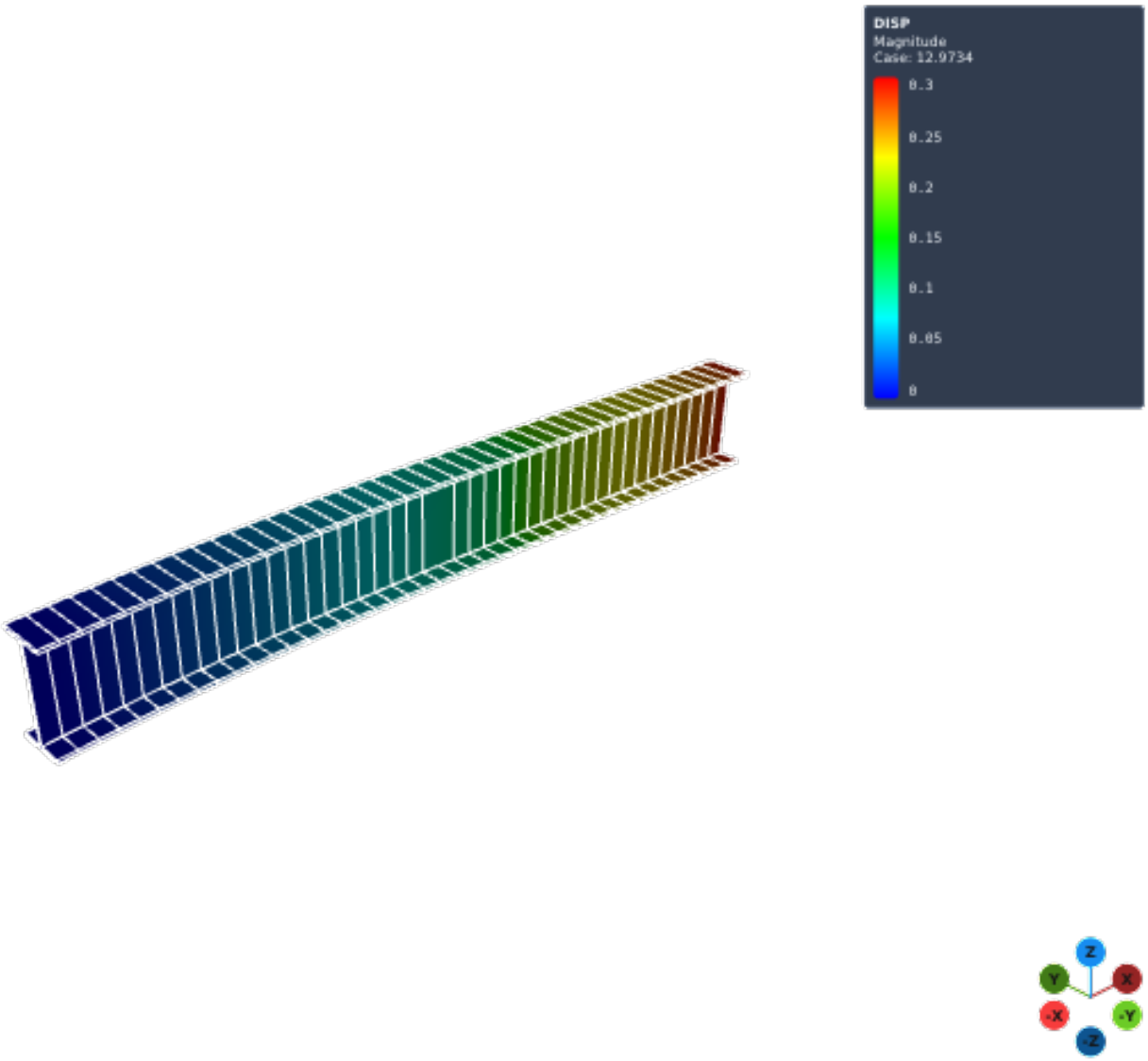

Figure 61: Calculix — mode 1, 12.973 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	12.973	6644.56	1.74e-37	116.40	9.37e-33	4.17e-19	-10.789	9.68e-17

Mode 2

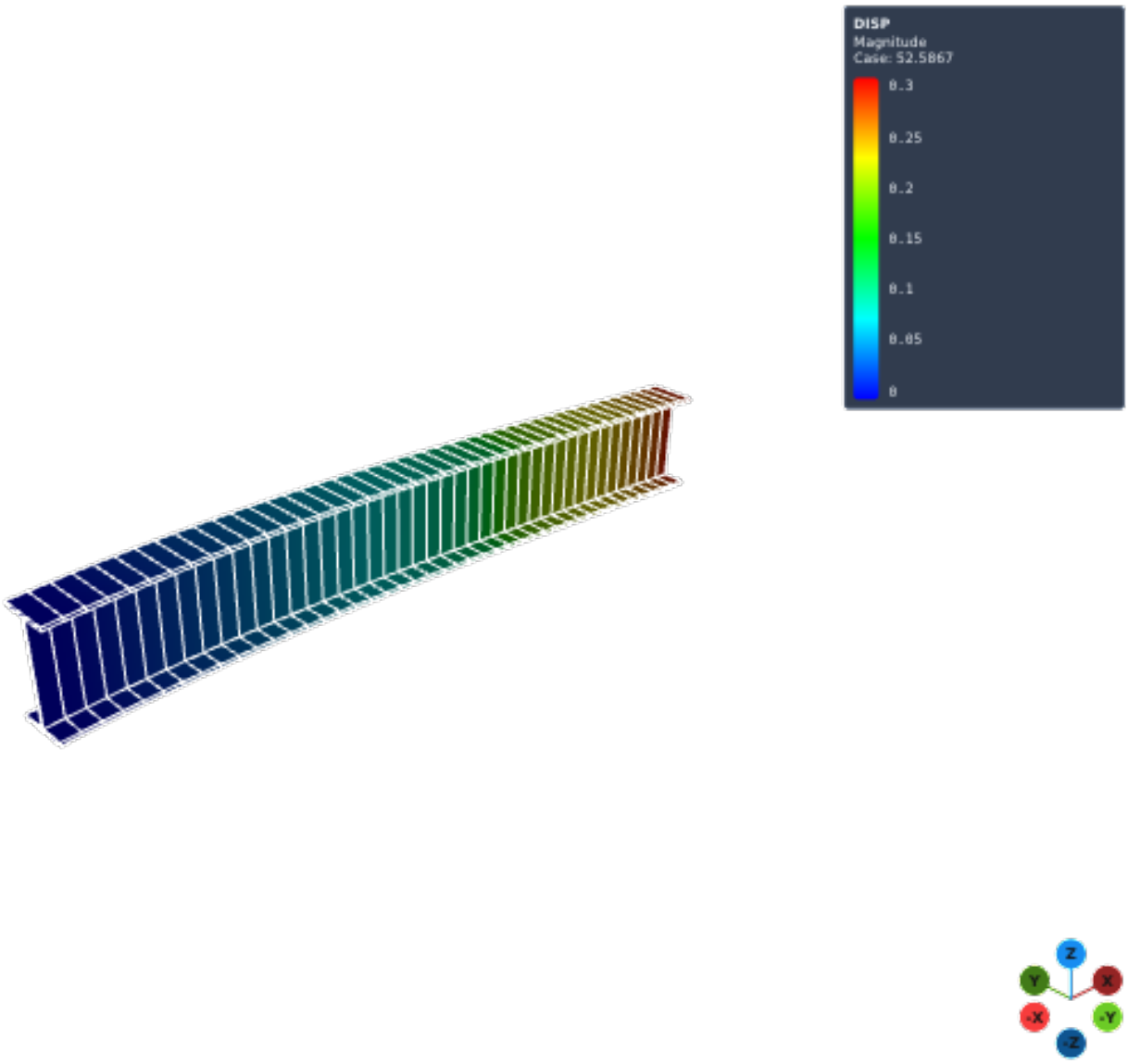

Figure 62: Calculix — mode 2, 52.587 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	52.587	109172	2.95e-34	6.28e-31	114.82	1.72e-17	-7.92e-16	-10.716

Mode 3

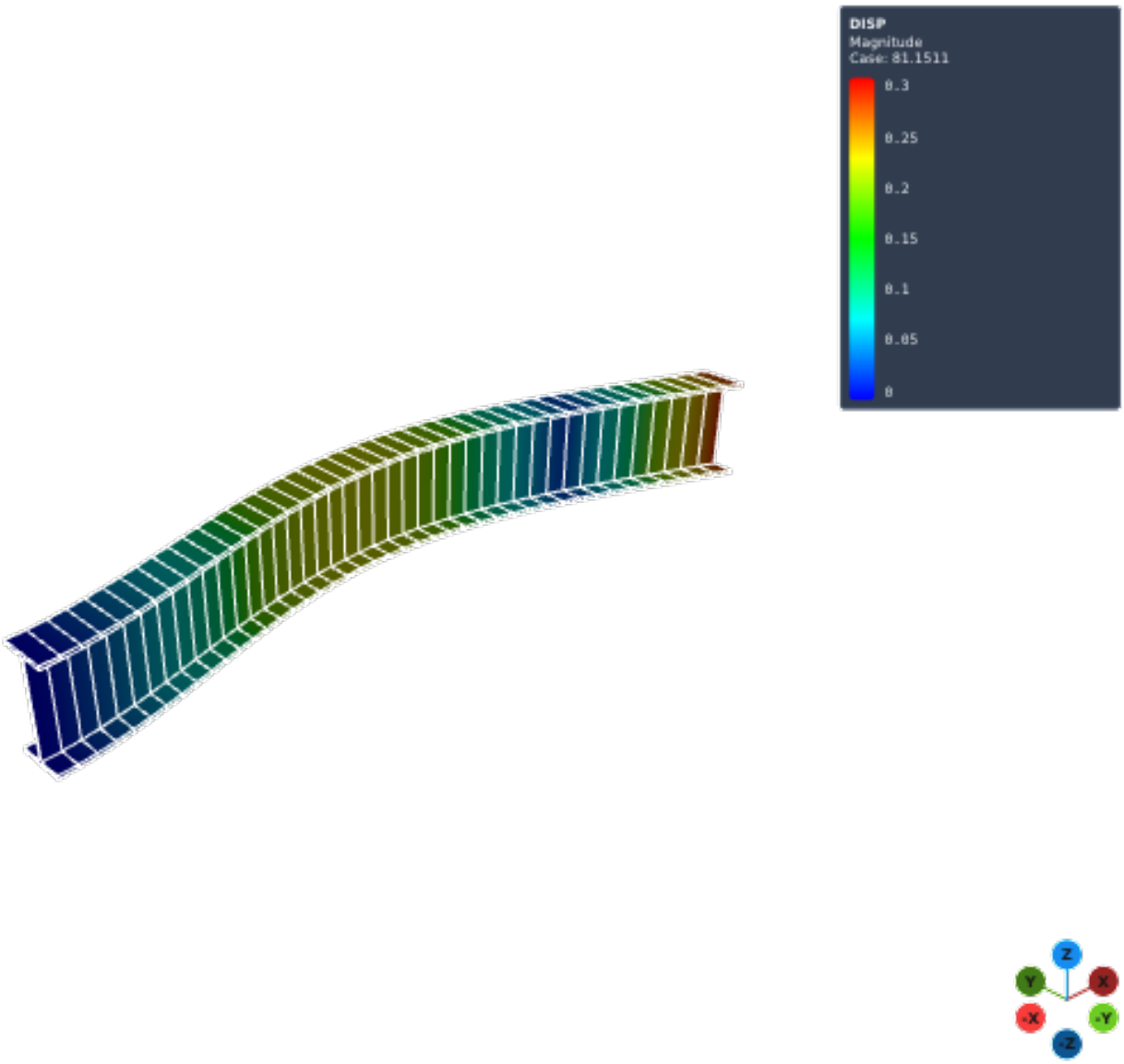

Figure 63: Calculix — mode 3, 81.151 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	81.151	259985	4.06e-33	35.74	4.38e-30	-6.37e-17	5.979	-2.09e-15

Mode 4

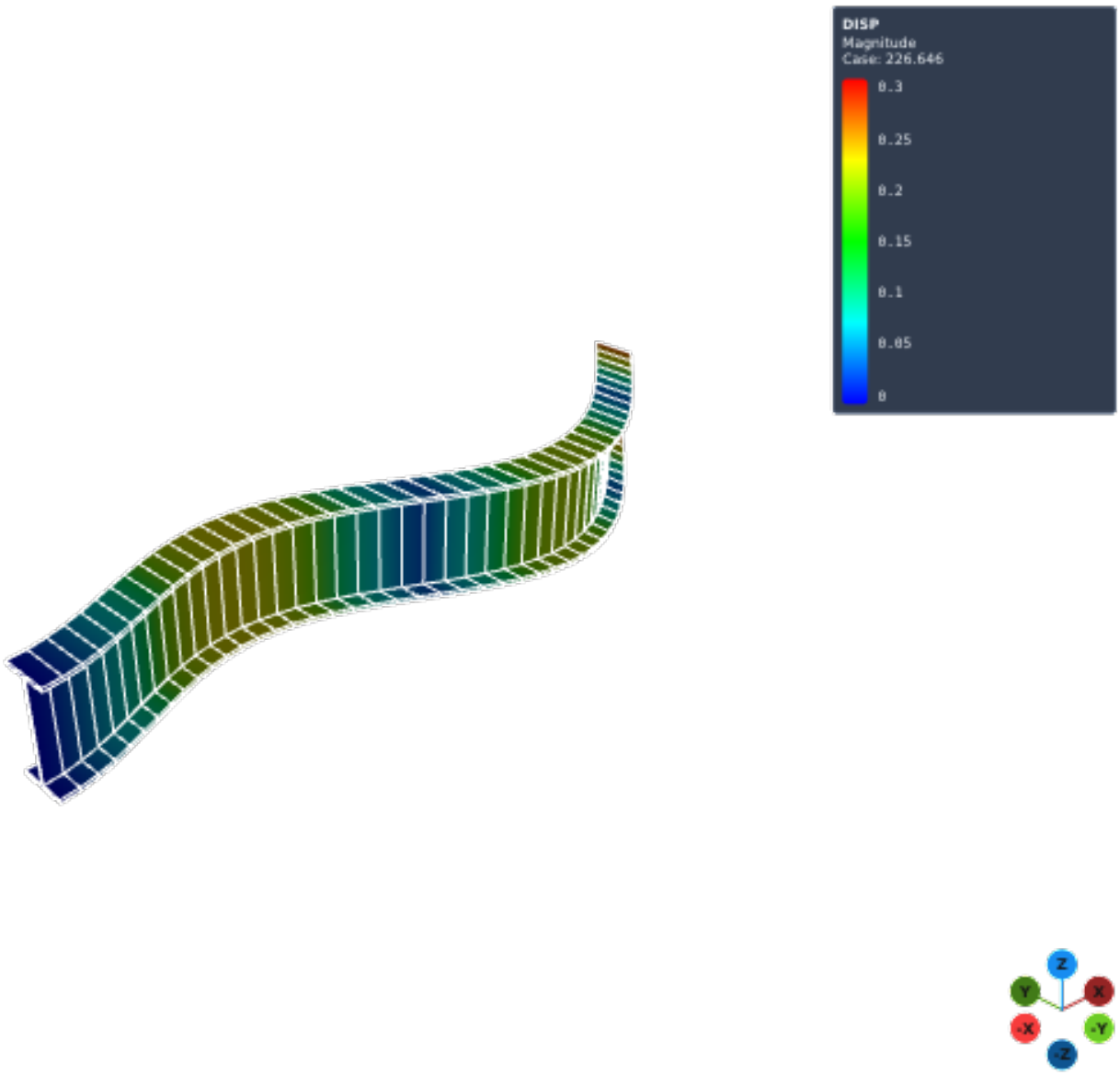

Figure 64: Calculix — mode 4, 226.646 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	226.646	2.02794e+06	1.30e-30	12.25	2.11e-29	-1.14e-15	3.500	-4.59e-15

Mode 5

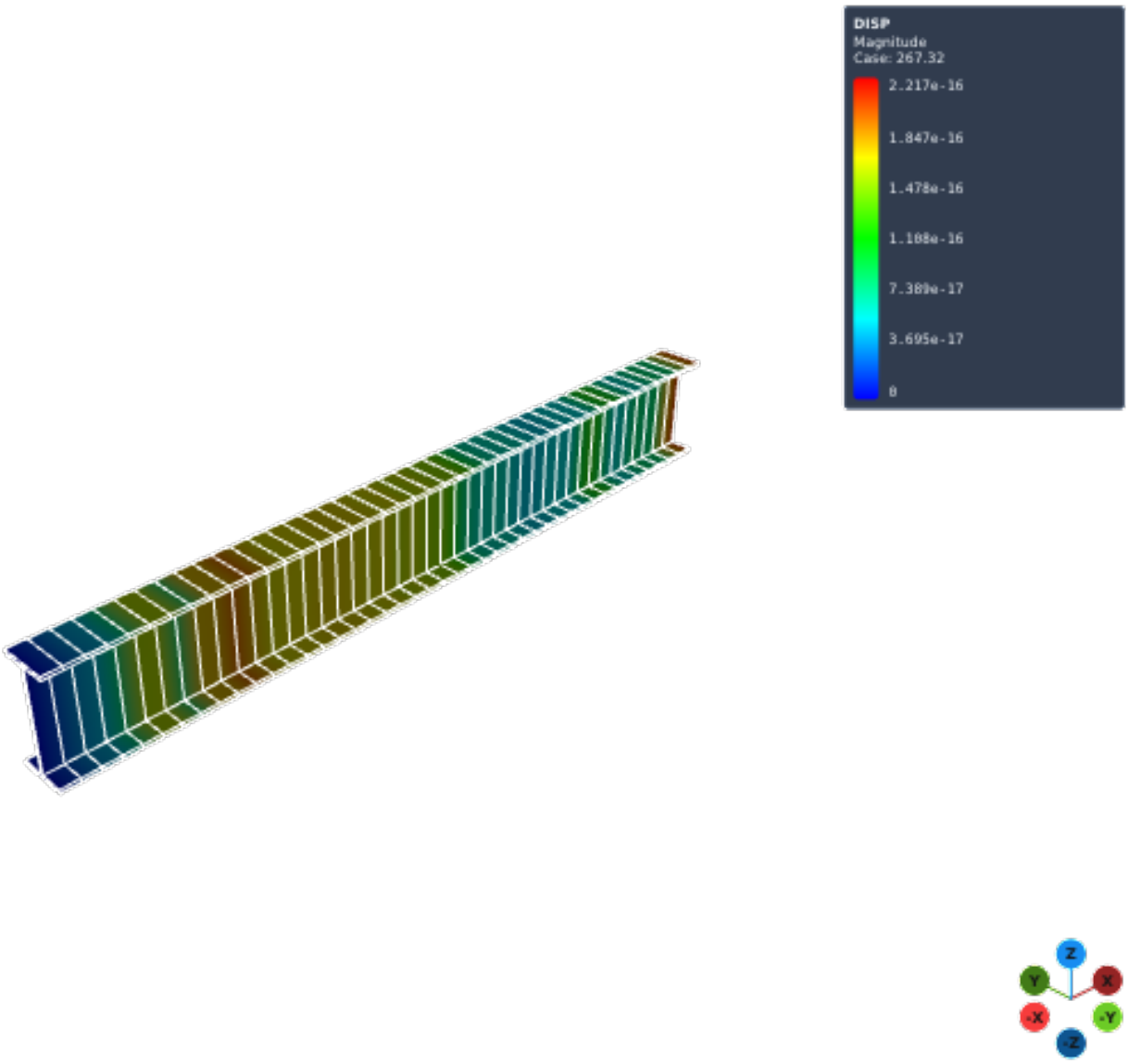

Figure 65: Calculix — mode 5, 267.320 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	267.320	2.82112e+06	3.28e-30	4.98e-30	2.38e-29	1.81e-15	-2.23e-15	4.88e-15

Mode 6

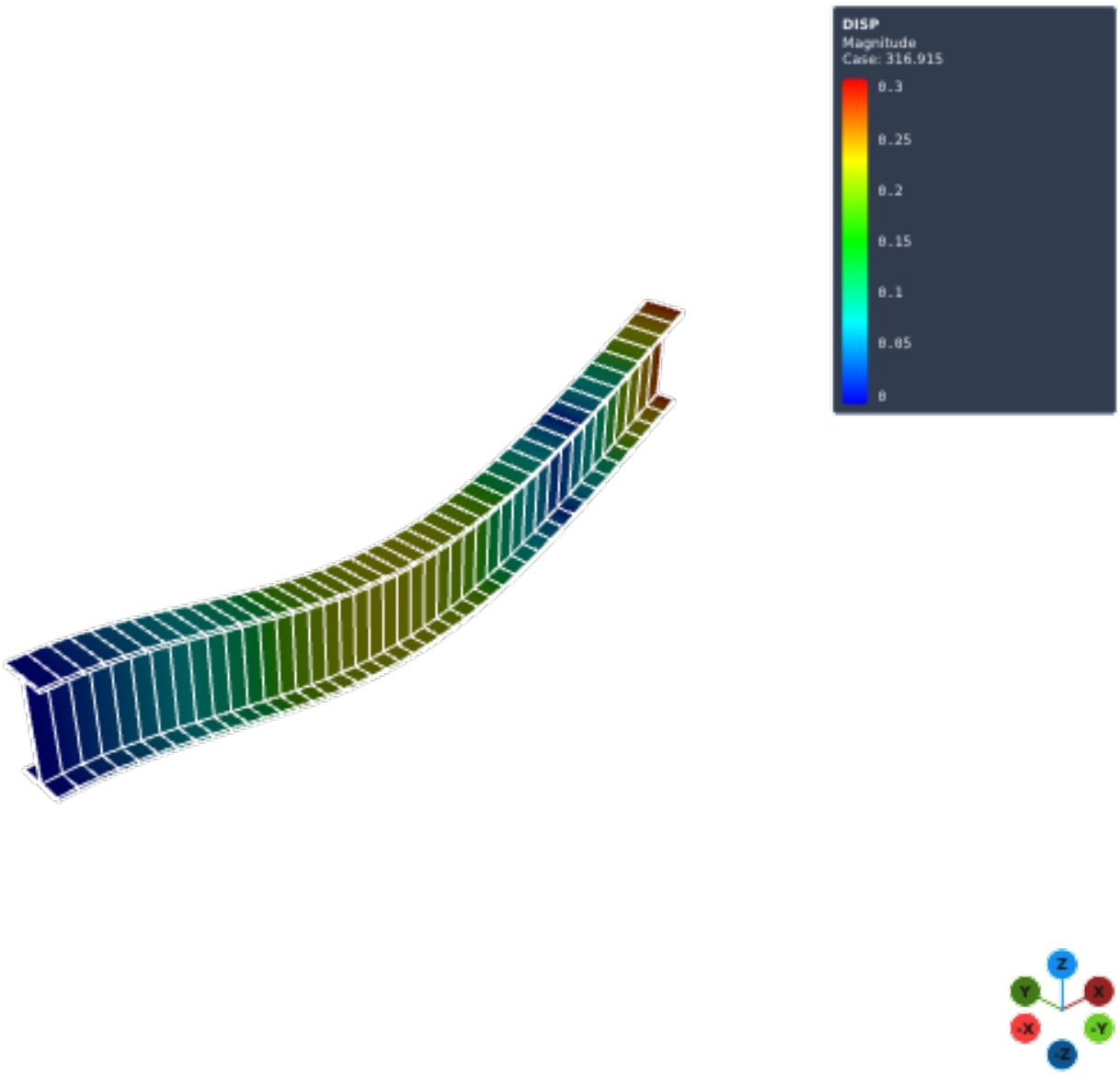

Figure 66: Calculix — mode 6, 316.915 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	316.915	3.96503e+06	1.31e-33	7.09e-30	34.97	-3.62e-17	-2.66e-15	-5.913

Mode 7

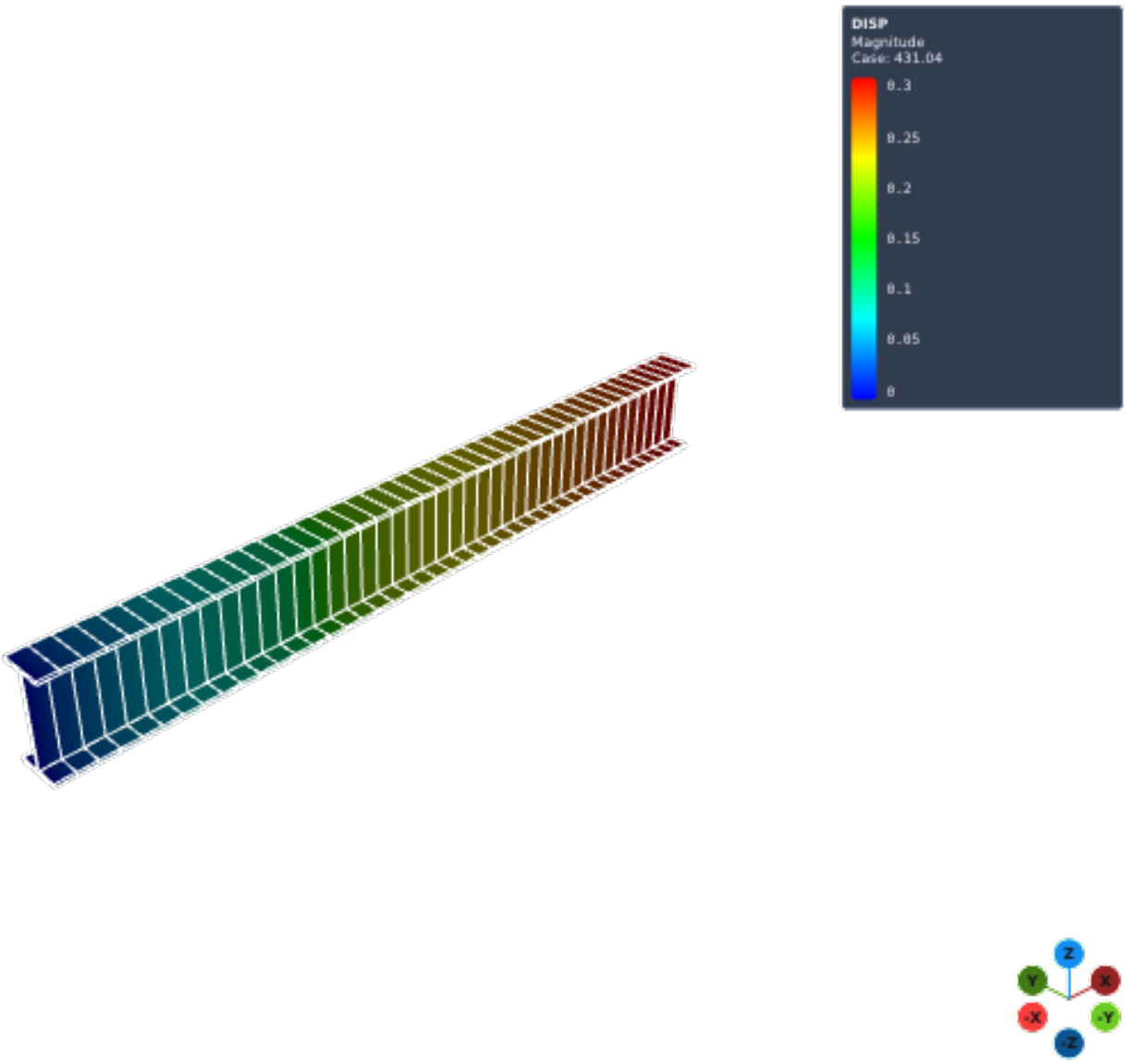

Figure 67: Calculix — mode 7, 431.040 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	431.040	7.33492e+06	153.94	1.29e-27	4.83e-31	12.407	-3.60e-14	-6.95e-16

Mode 8

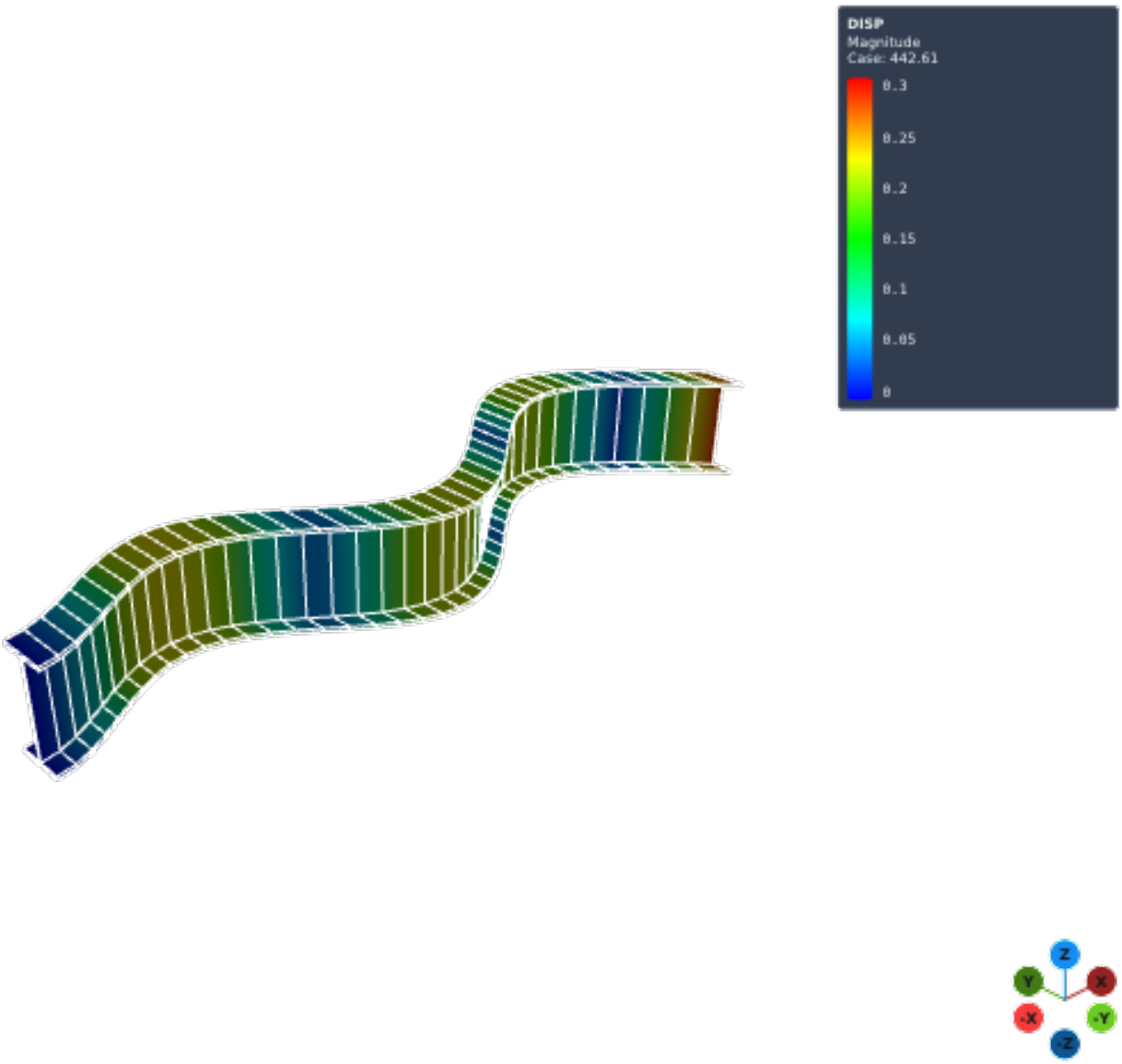

Figure 68: Calculix — mode 8, 442.610 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	442.610	7.73397e+06	3.15e-26	6.24	6.00e-30	1.77e-13	2.498	2.45e-15

Mode 9

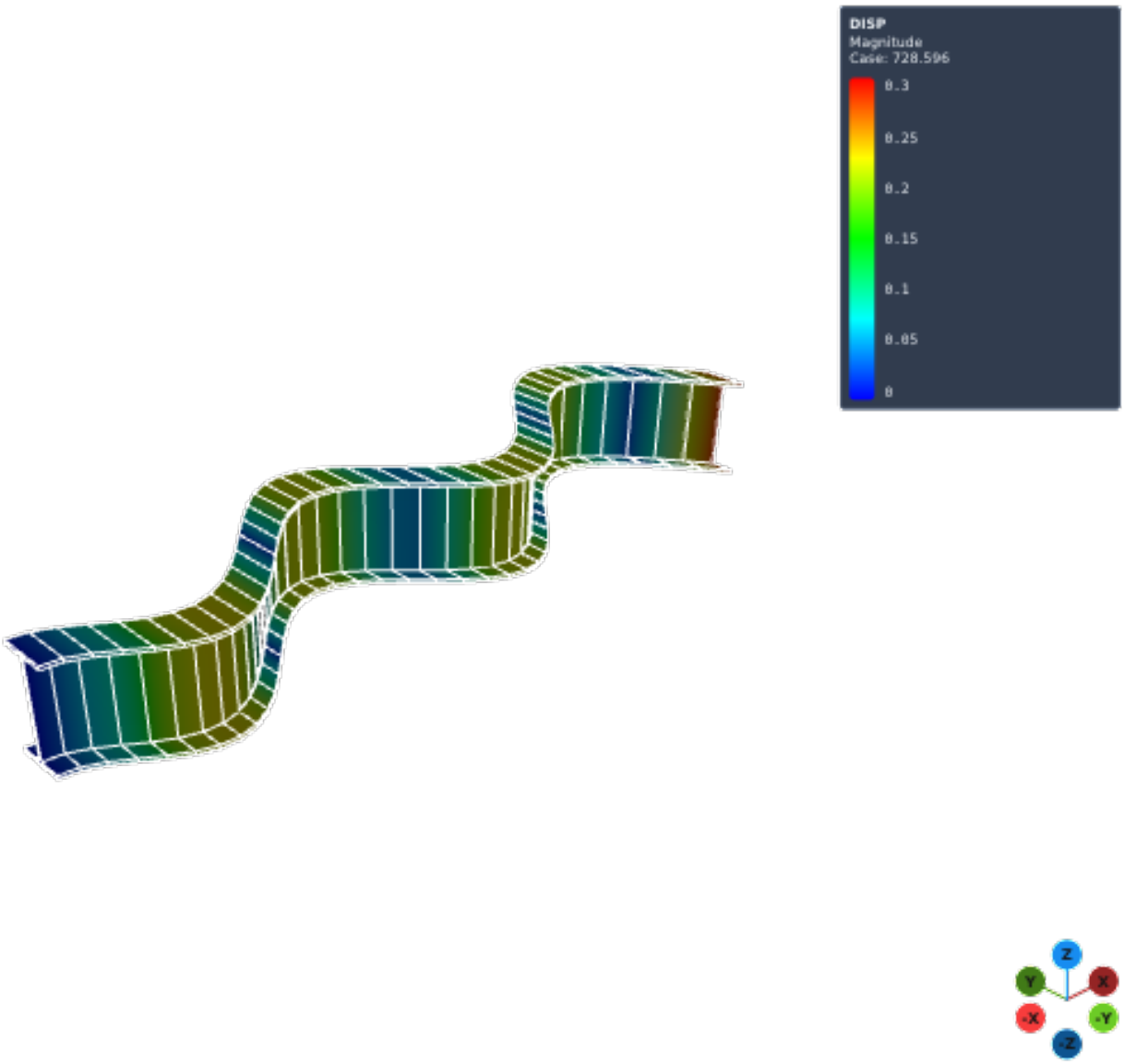

Figure 69: Calculix — mode 9, 728.596 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	728.596	2.09572e+07	1.49e-29	3.75	1.24e-28	3.85e-15	-1.937	-1.11e-14

Mode 10

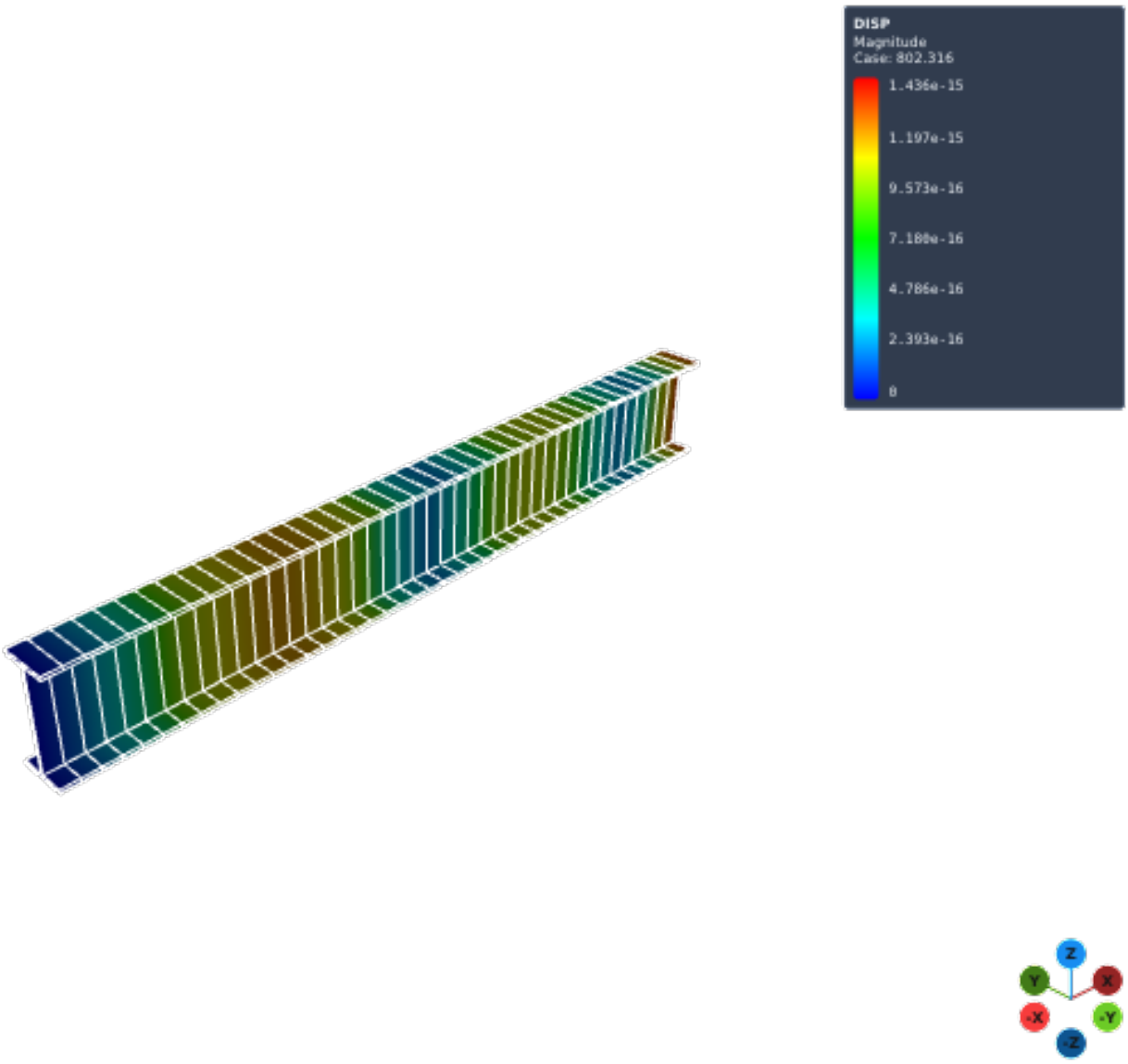

Figure 70: Calculix — mode 10, 802.316 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	802.316	2.54127e+07	3.17e-30	9.74e-30	3.78e-28	-1.78e-15	3.12e-15	1.94e-14

Mode 11

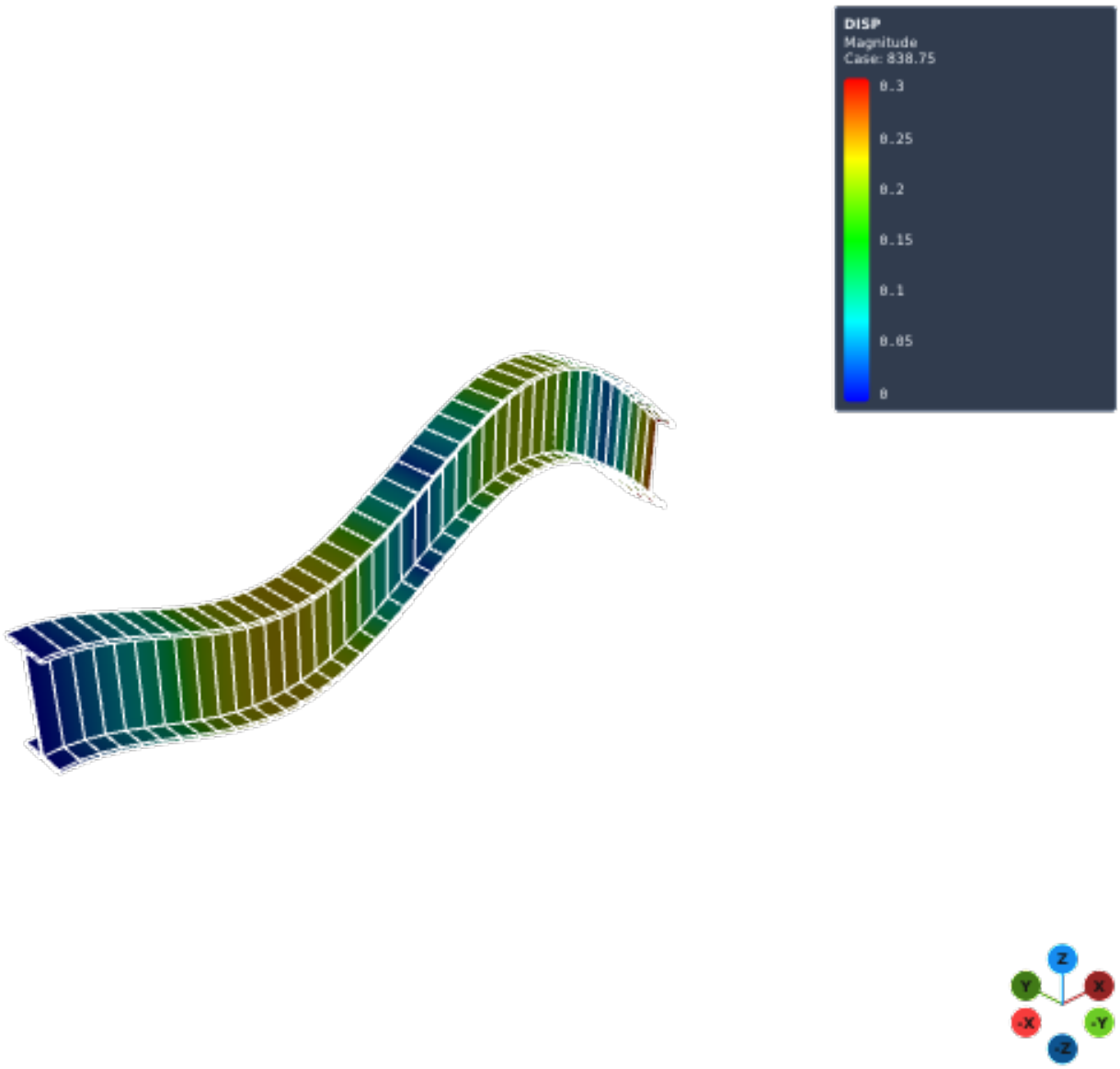

Figure 71: Calculix — mode 11, 838.750 Hz

Solver	f [Hz]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	838.750	2.77731e+07	1.58e-31	5.81e-29	11.38	3.98e-16	7.62e-15	-3.374

Line, 2nd order

Mode 1

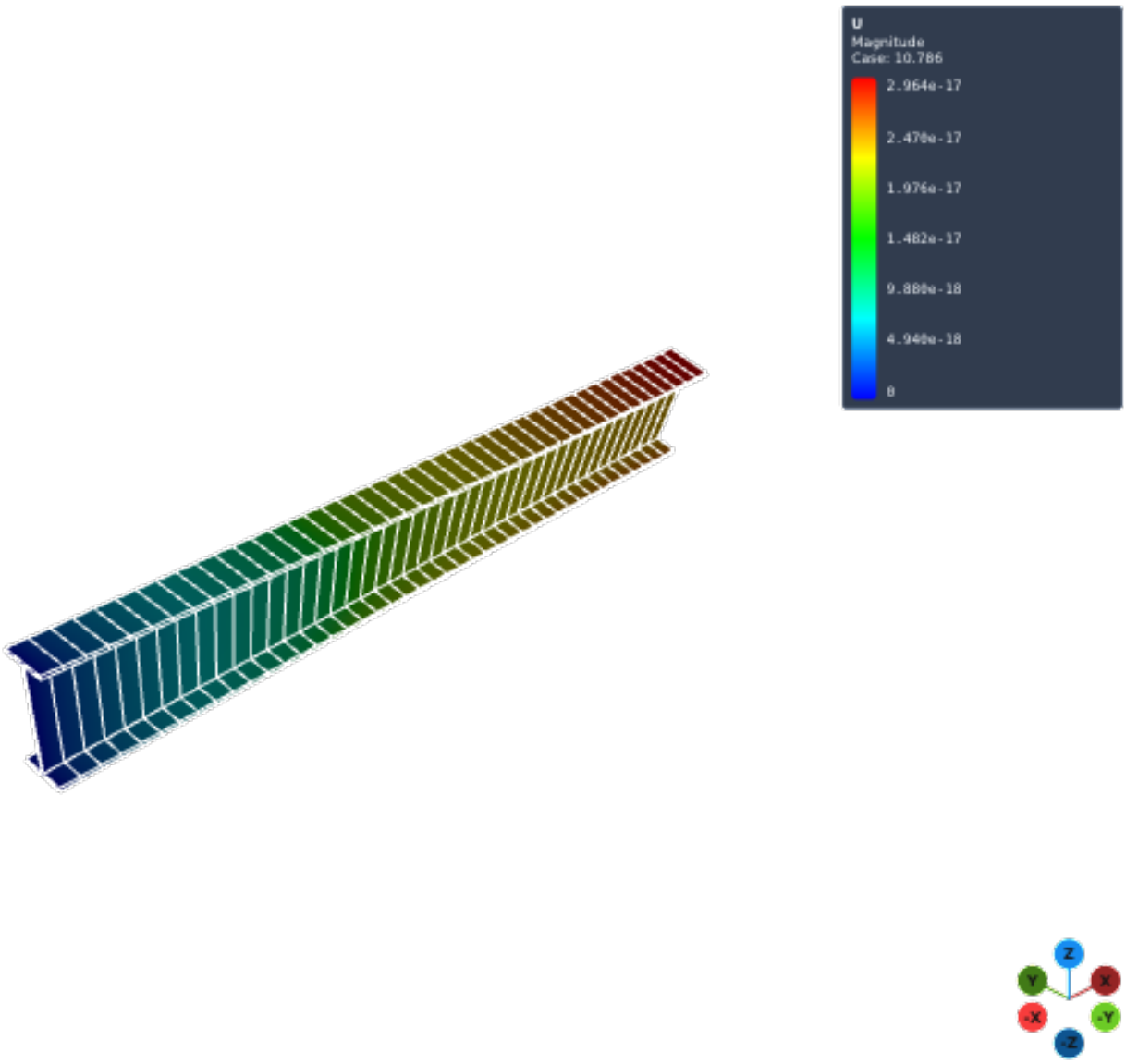

Figure 72: Abaqus — mode 1, 10.786 Hz

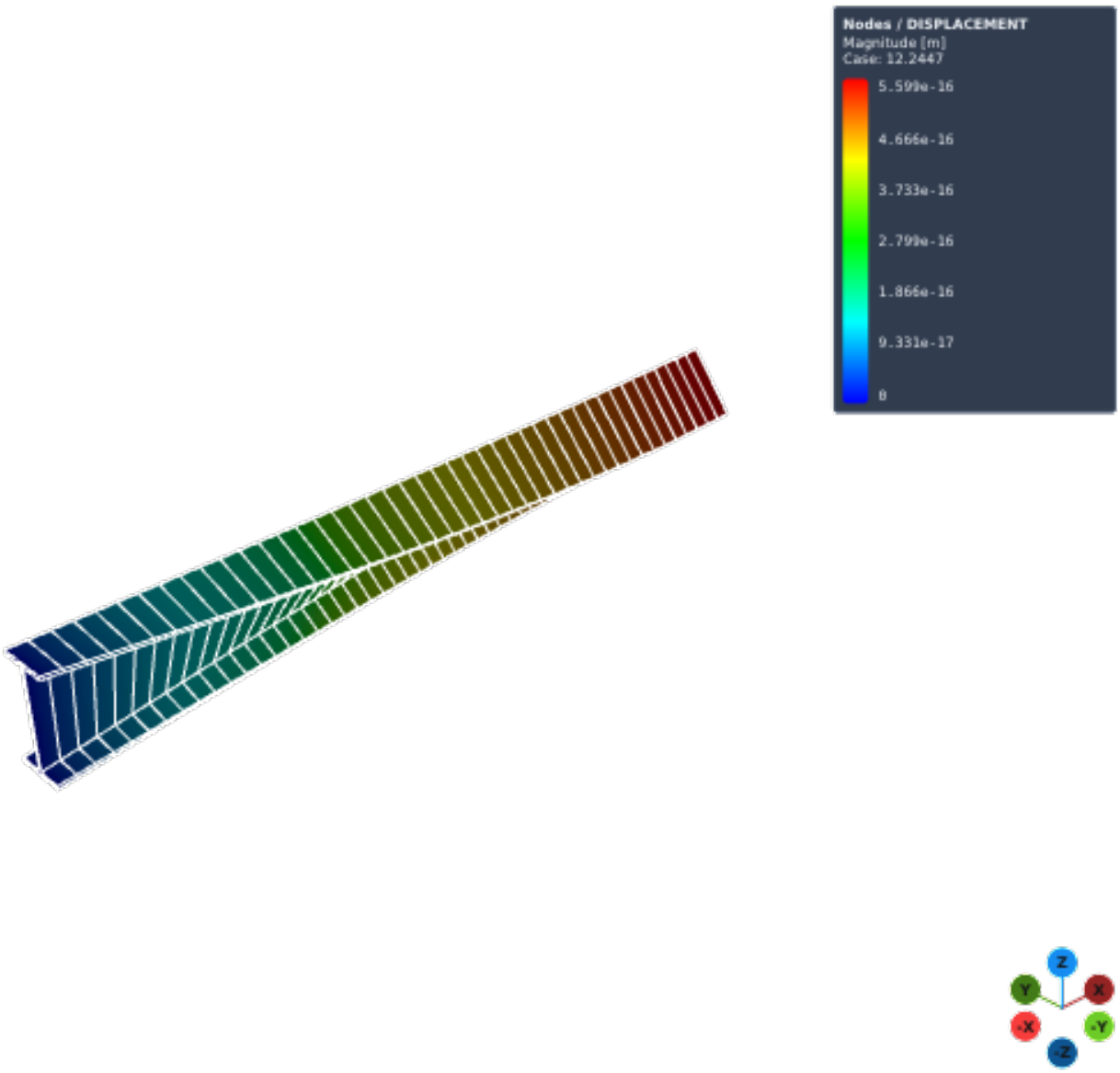

Figure 73: Sesam — mode 1, 12.245 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	10.786	-6.33	4592.8	3.02e-31	1.31e-30	3.09e-30	5.50e-16	1.14e-15	-1.76e-15
Sesam	12.245	+6.33	5919.11	5.46e-33	3.21e-28	1.94e-29	7.39e-17	1.79e-14	-4.40e-15

Mode 2

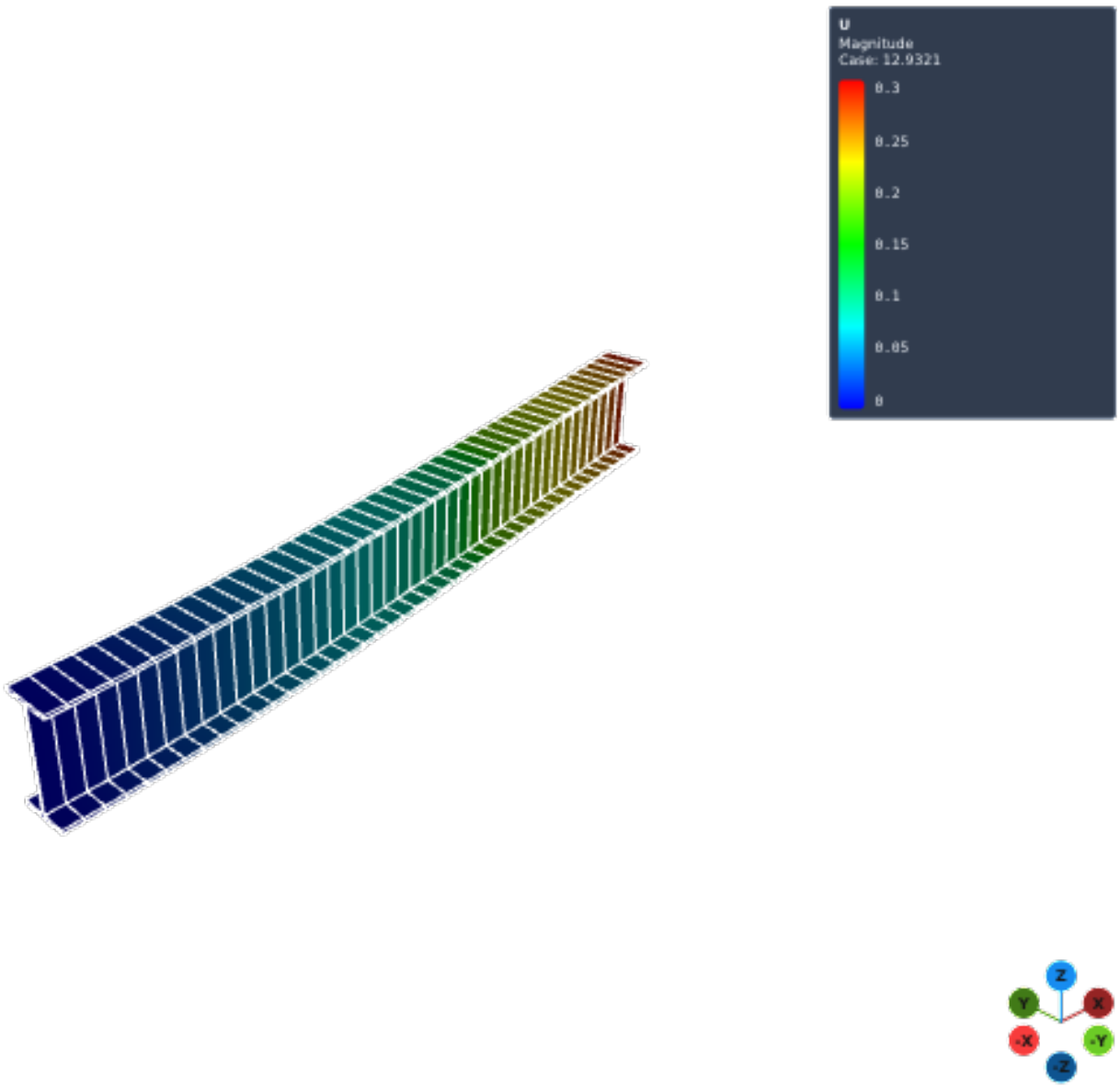

Figure 74: Abaqus — mode 2, 12.932 Hz

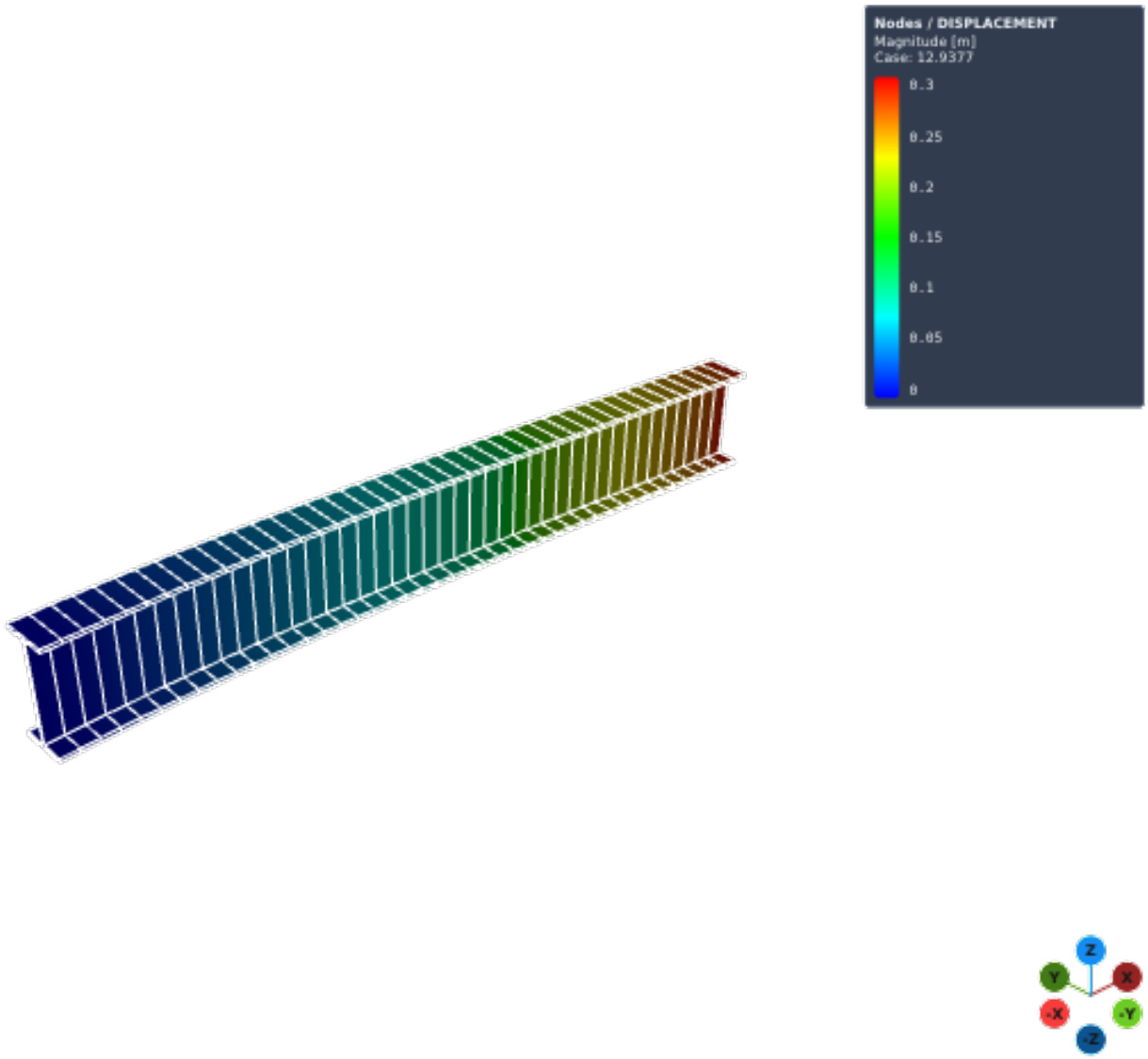

Figure 75: Sesam — mode 2, 12.938 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.932	-0.02	6602.4	1.50e-32	116.59	2.60e-32	-1.22e-16	10.798	-1.61e-16
Sesam	12.938	+0.02	6608.05	3.10e-34	116.61	1.28e-31	1.76e-17	-10.799	3.57e-16

Mode 3

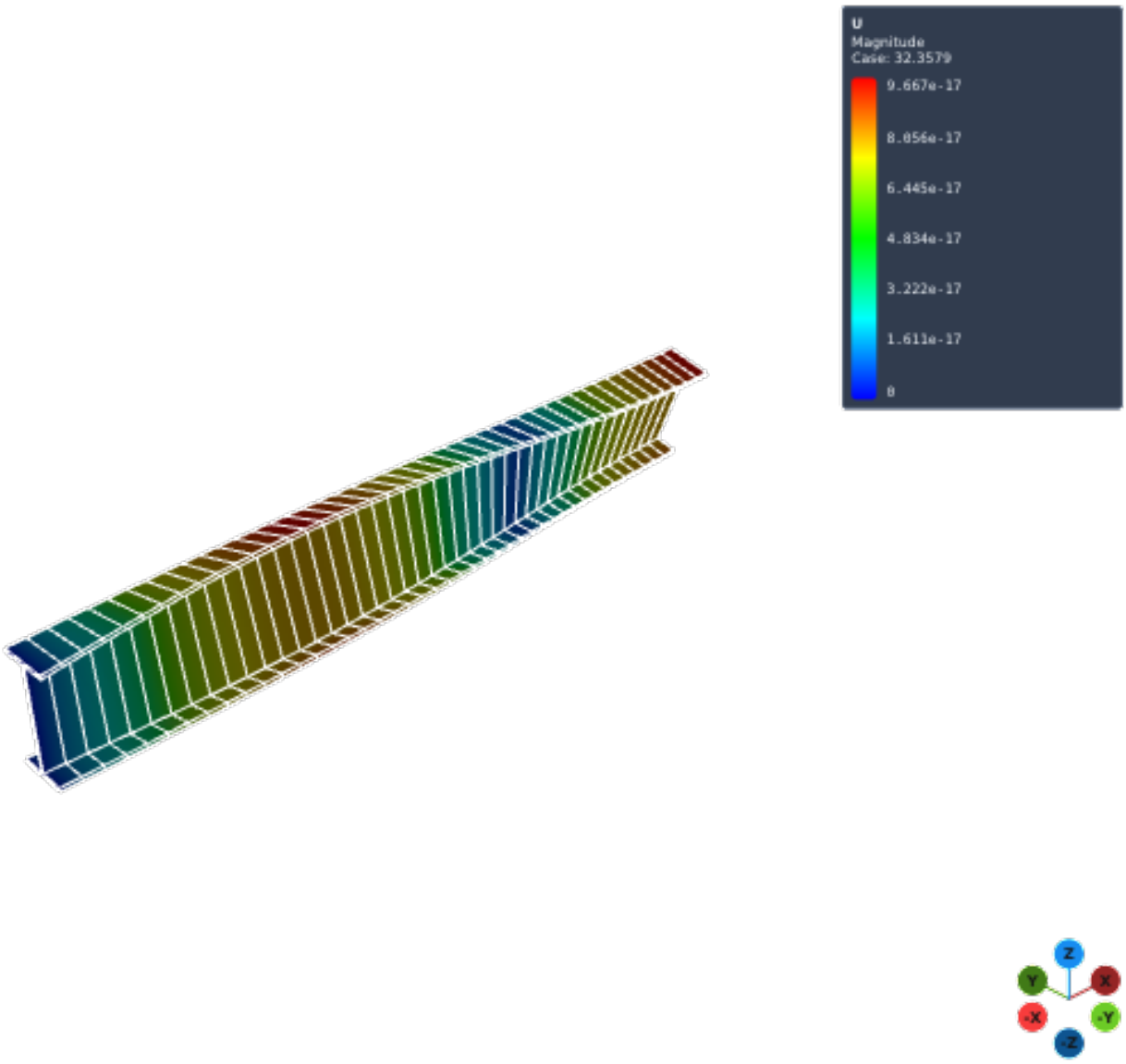

Figure 76: Abaqus — mode 3, 32.358 Hz

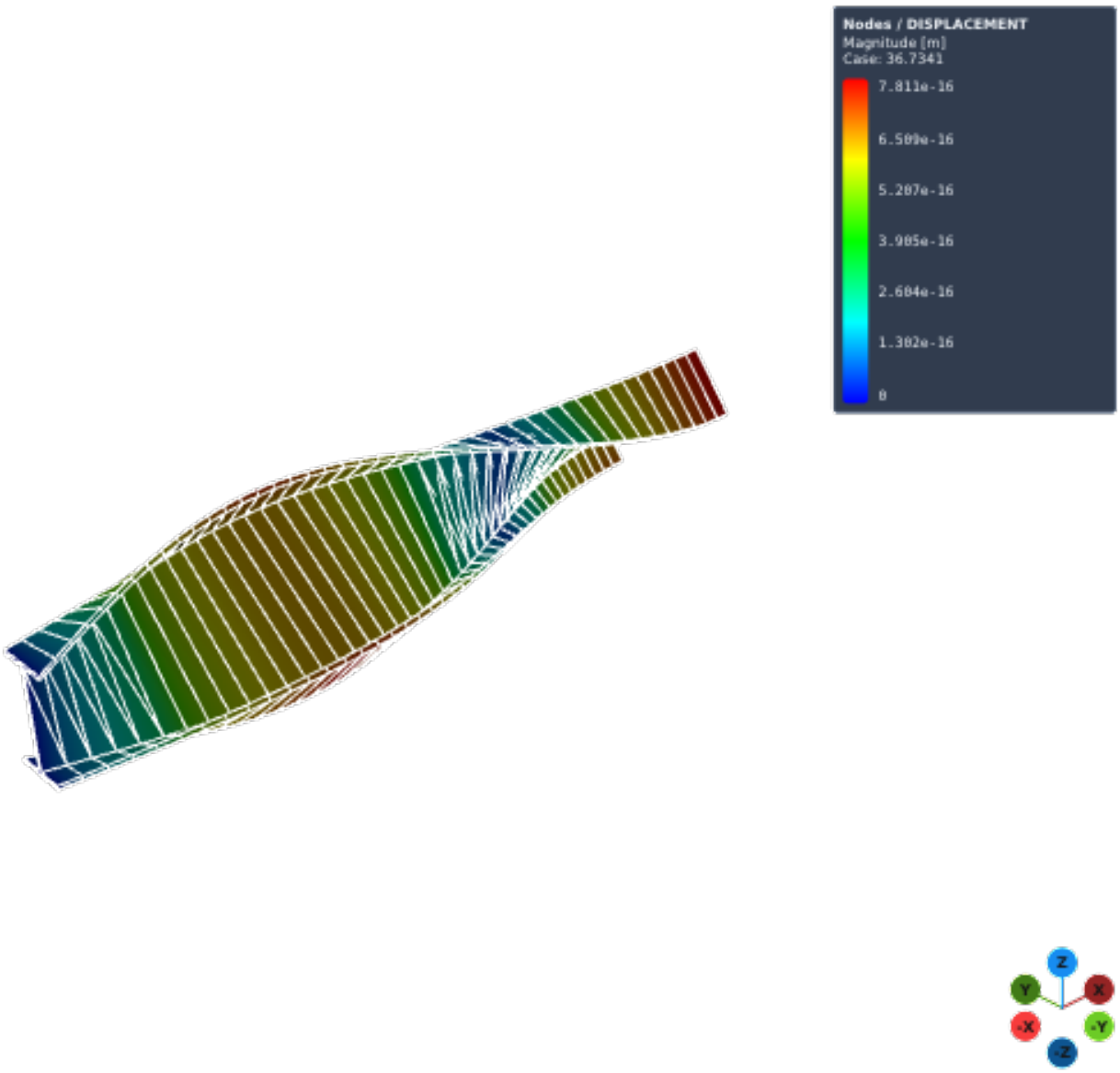

Figure 77: Sesam — mode 3, 36.734 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	32.358	-6.33	41335	2.64e-31	5.69e-29	8.91e-29	5.14e-16	-7.54e-15	-9.44e-15
Sesam	36.734	+6.33	53272	3.14e-32	1.27e-30	1.00e-27	-1.77e-16	-1.12e-15	3.17e-14

Mode 4

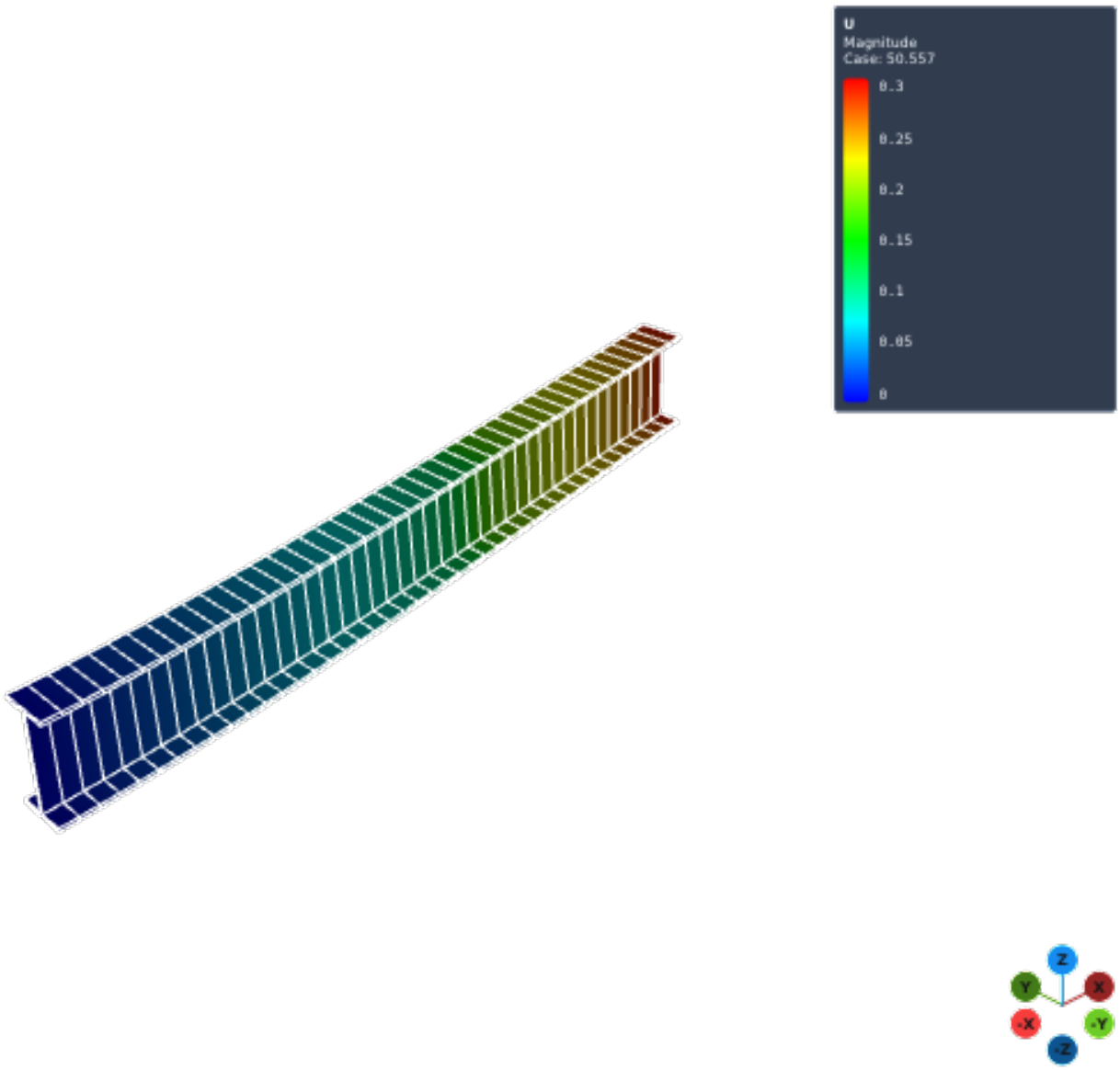

Figure 78: Abaqus — mode 4, 50.557 Hz

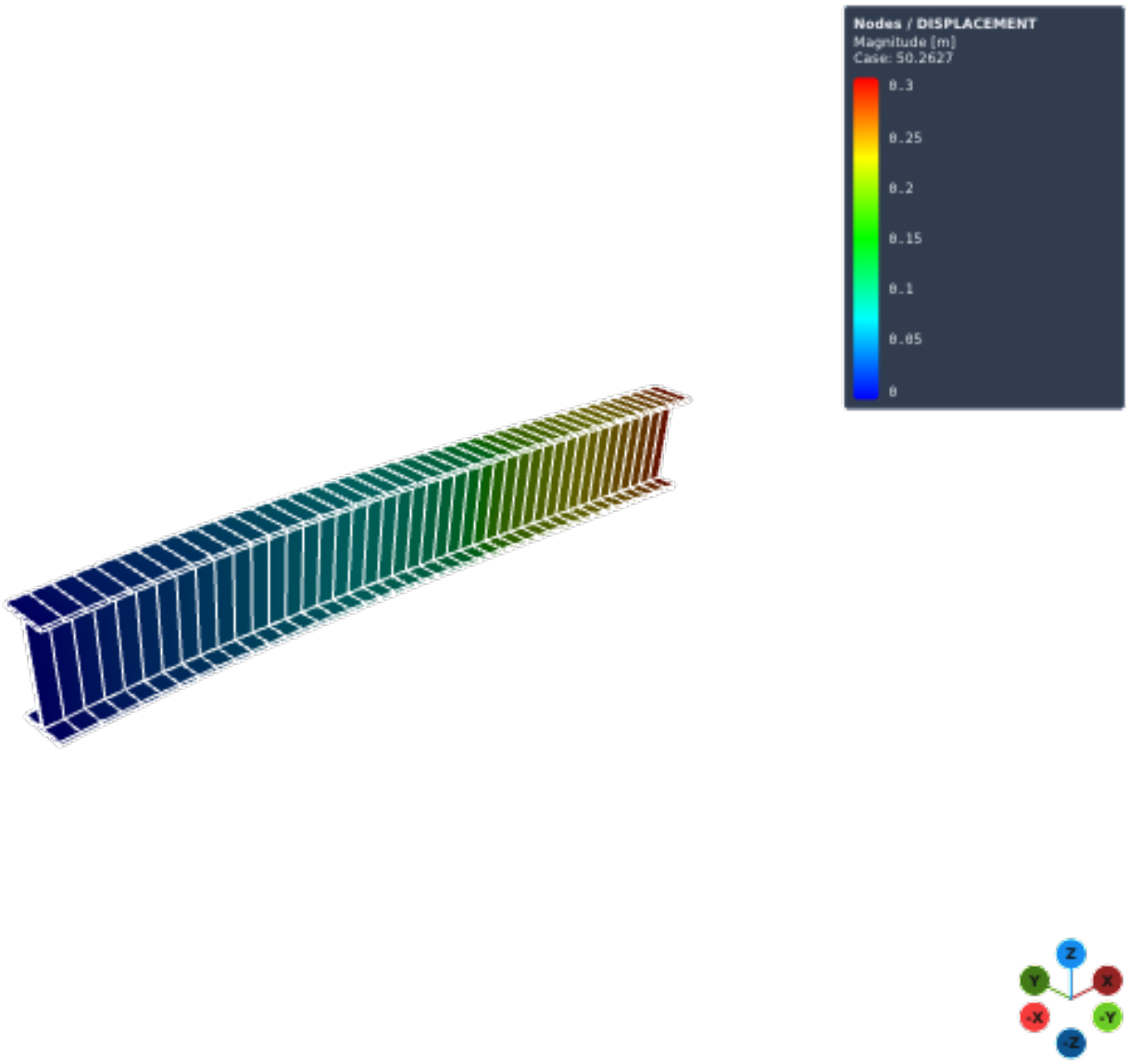

Figure 79: Sesam — mode 4, 50.263 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.557	+0.29	100907	6.29e-36	9.63e-28	118.18	-2.51e-18	-3.10e-14	10.871
Sesam	50.263	-0.29	99735.8	3.32e-32	8.75e-30	118.70	-1.82e-16	-2.96e-15	-10.895

Mode 5

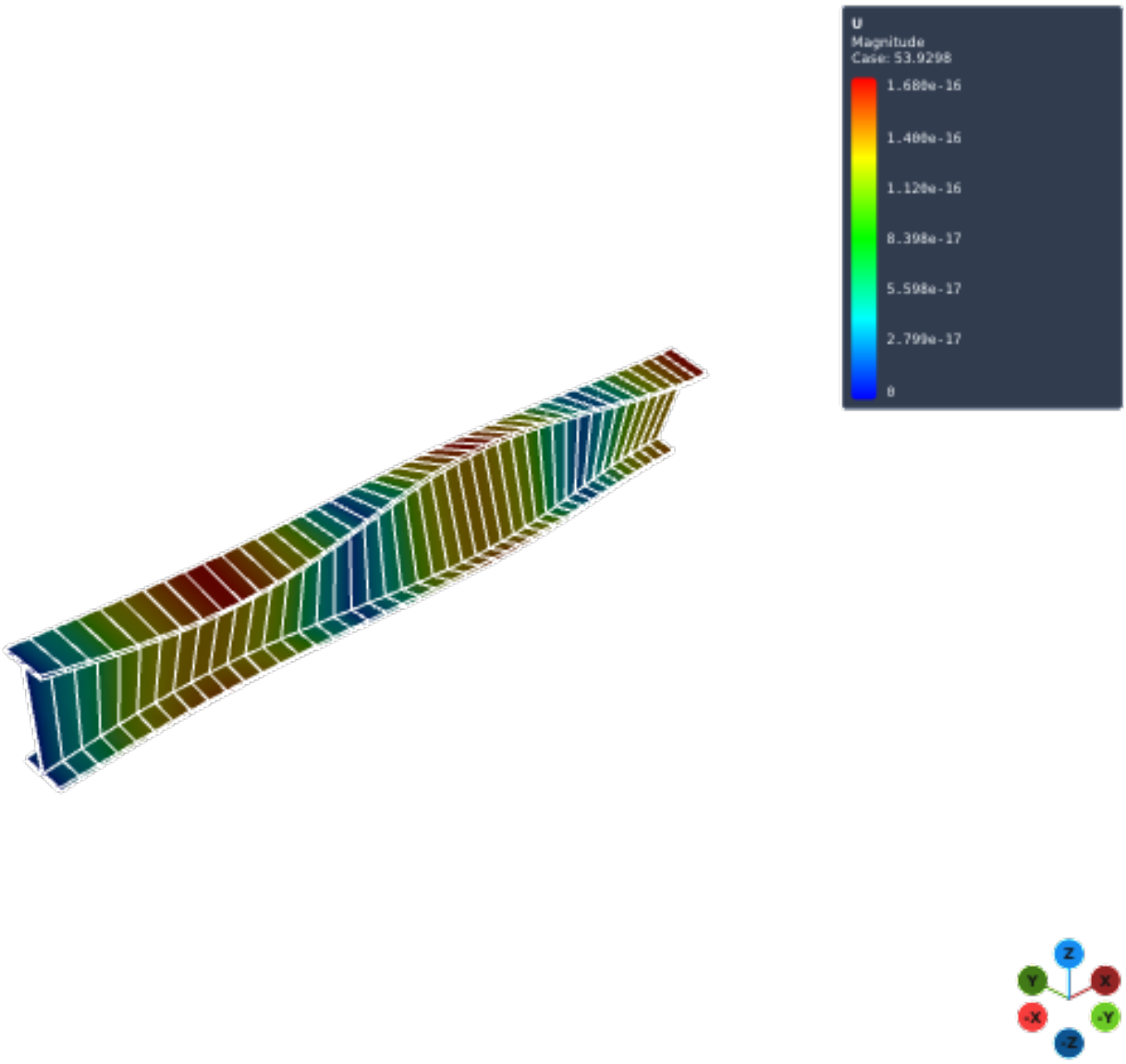

Figure 80: Abaqus — mode 5, 53.930 Hz

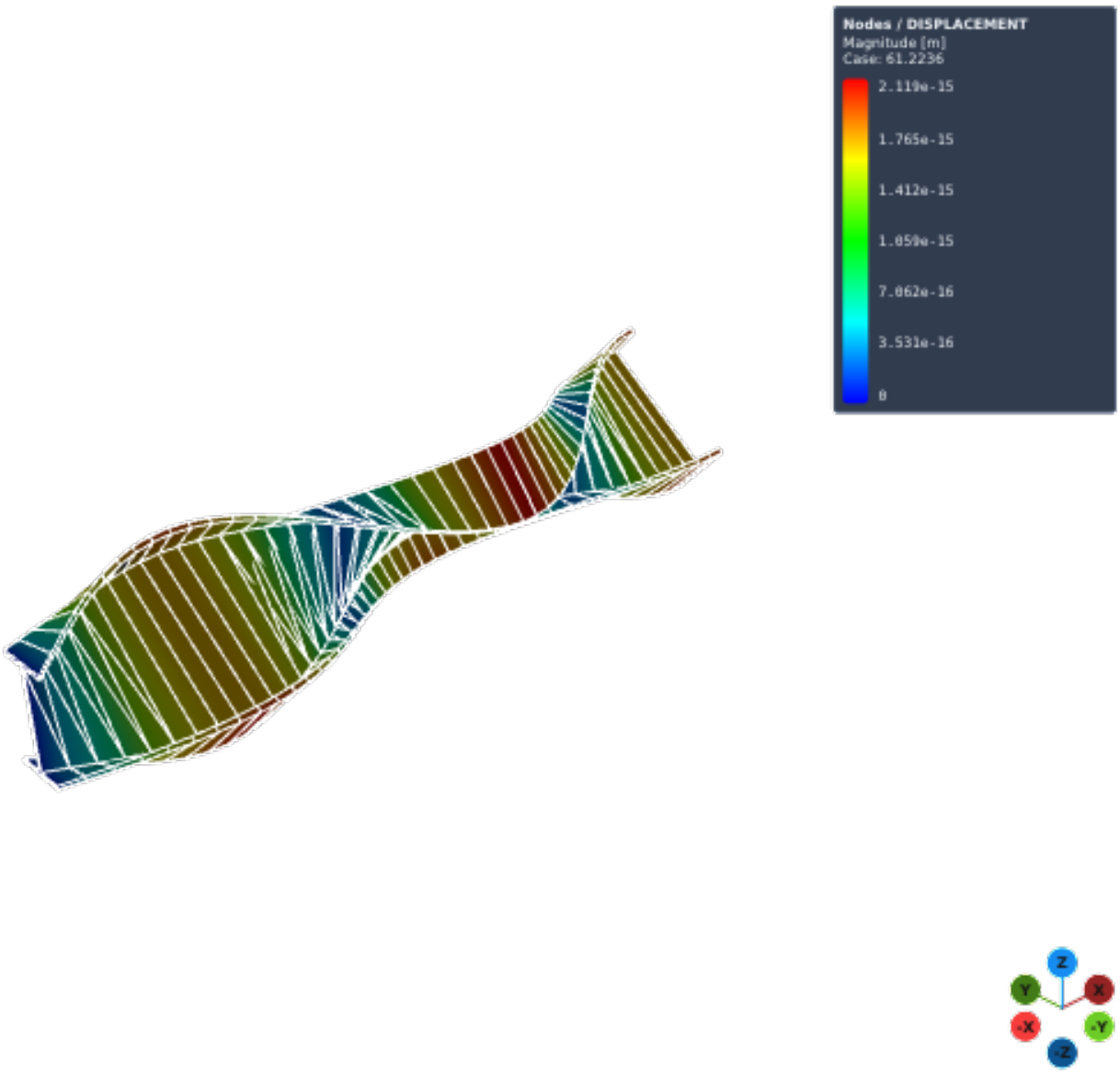

Figure 81: Sesam — mode 5, 61.224 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	53.930	-6.33	114820	1.13e-29	6.97e-31	2.94e-28	-3.36e-15	8.35e-16	1.72e-14
Sesam	61.224	+6.33	147978	1.75e-34	5.90e-30	4.72e-27	-1.32e-17	-2.43e-15	-6.87e-14

Mode 6

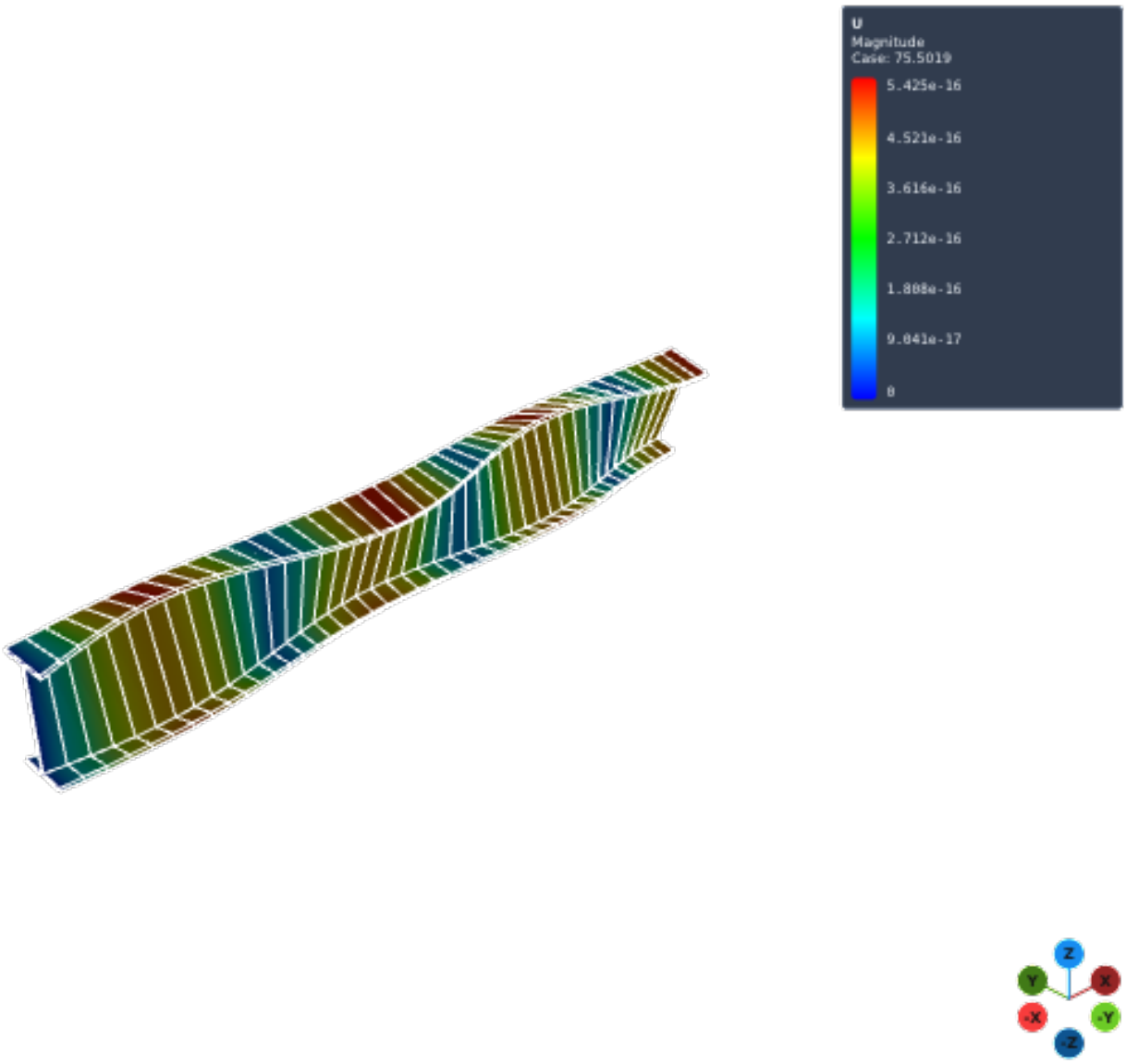

Figure 82: Abaqus — mode 6, 75.502 Hz

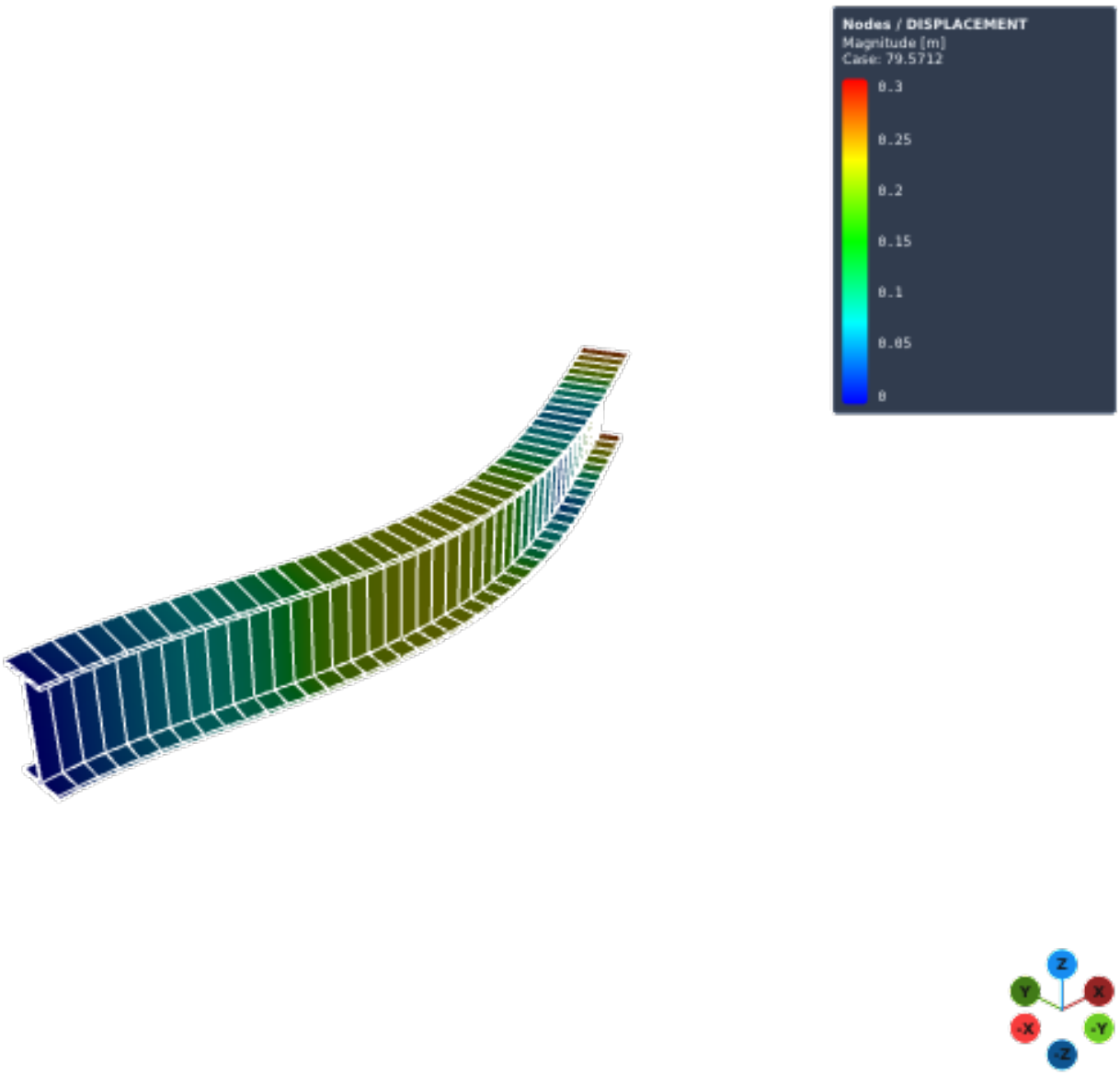

Figure 83: Sesam — mode 6, 79.571 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.502	-2.62	225048	6.44e-34	3.07e-27	1.85e-29	-2.54e-17	5.54e-14	-4.30e-15
Sesam	79.571	+2.62	249960	2.75e-31	36.24	2.23e-30	5.25e-16	-6.020	1.49e-15

Mode 7

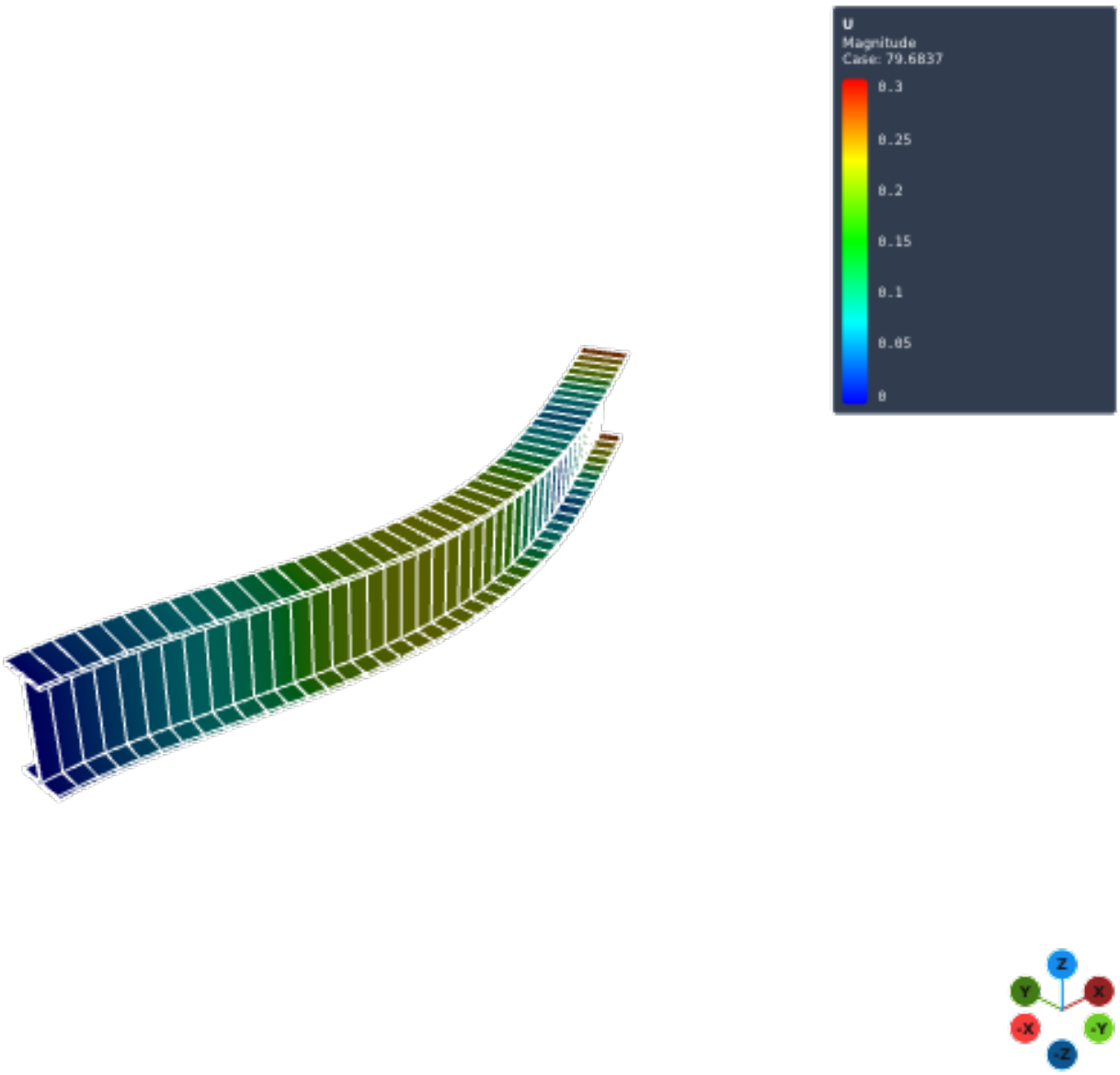

Figure 84: Abaqus — mode 7, 79.684 Hz

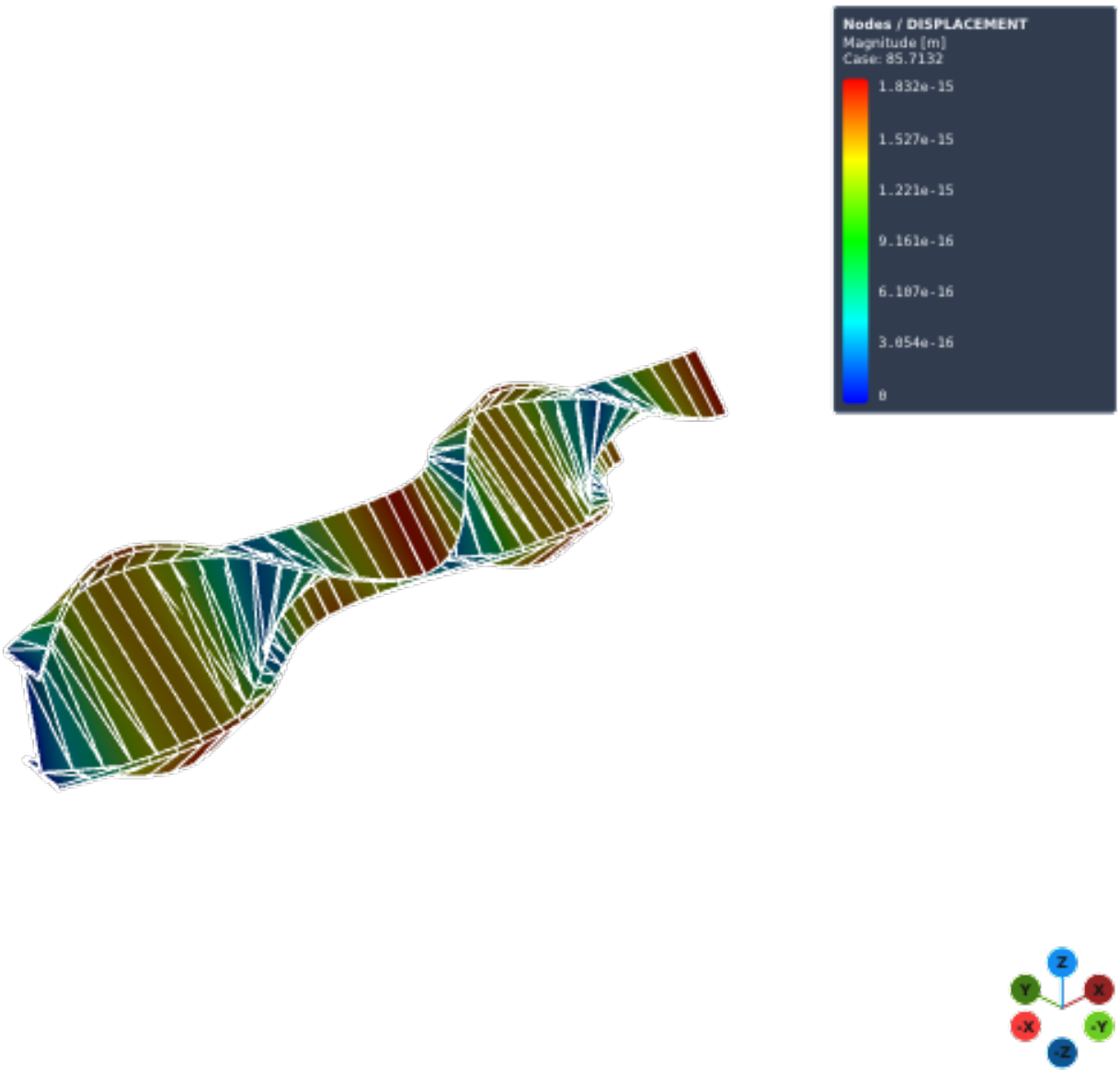

Figure 85: Sesam — mode 7, 85.713 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	79.684	-3.65	250668	9.57e-31	36.19	1.97e-27	9.78e-16	-6.015	-4.44e-14
Sesam	85.713	+3.65	290038	5.31e-31	1.33e-27	7.16e-29	-7.29e-16	3.65e-14	-8.46e-15

Mode 8

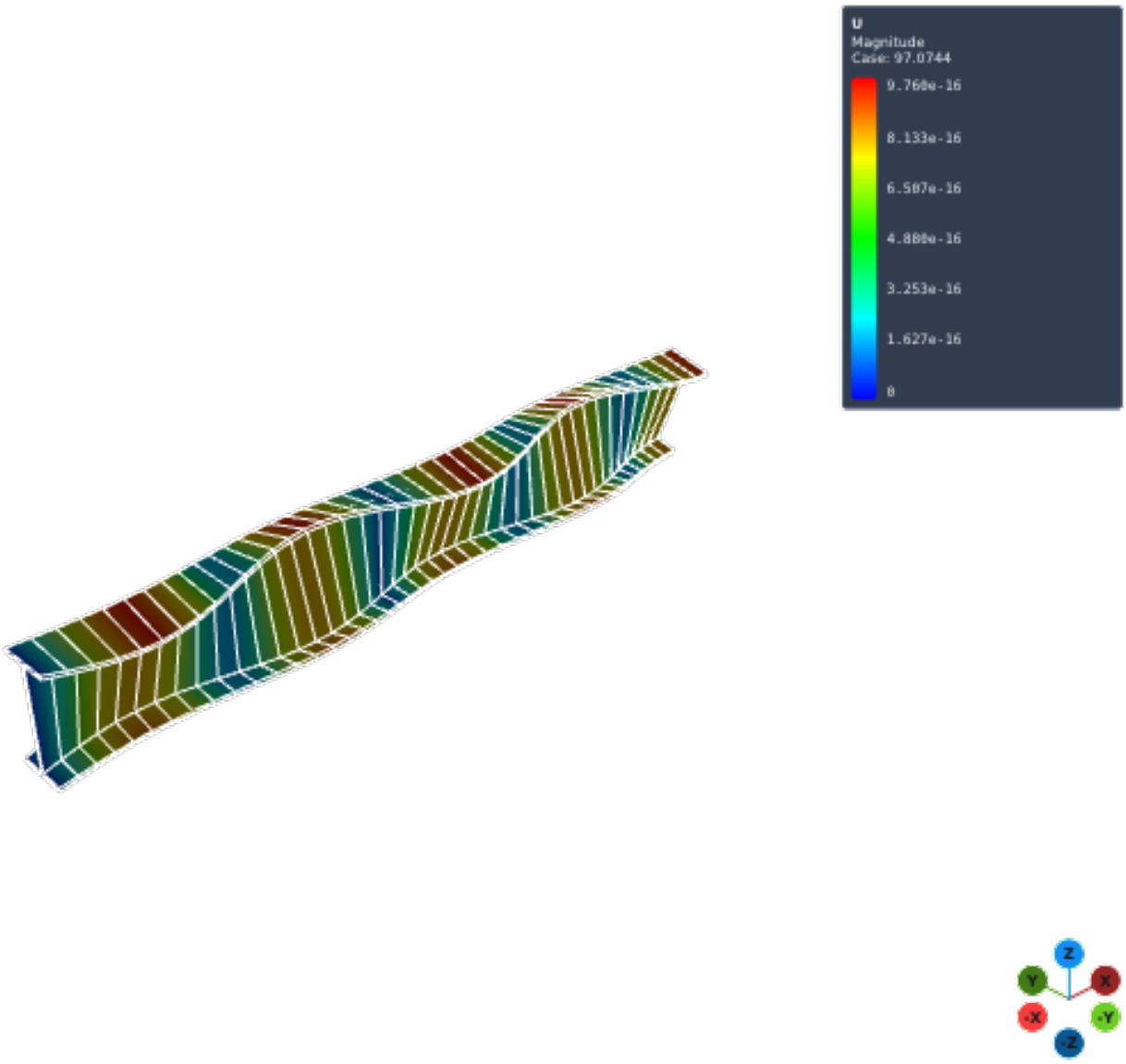

Figure 86: Abaqus — mode 8, 97.074 Hz

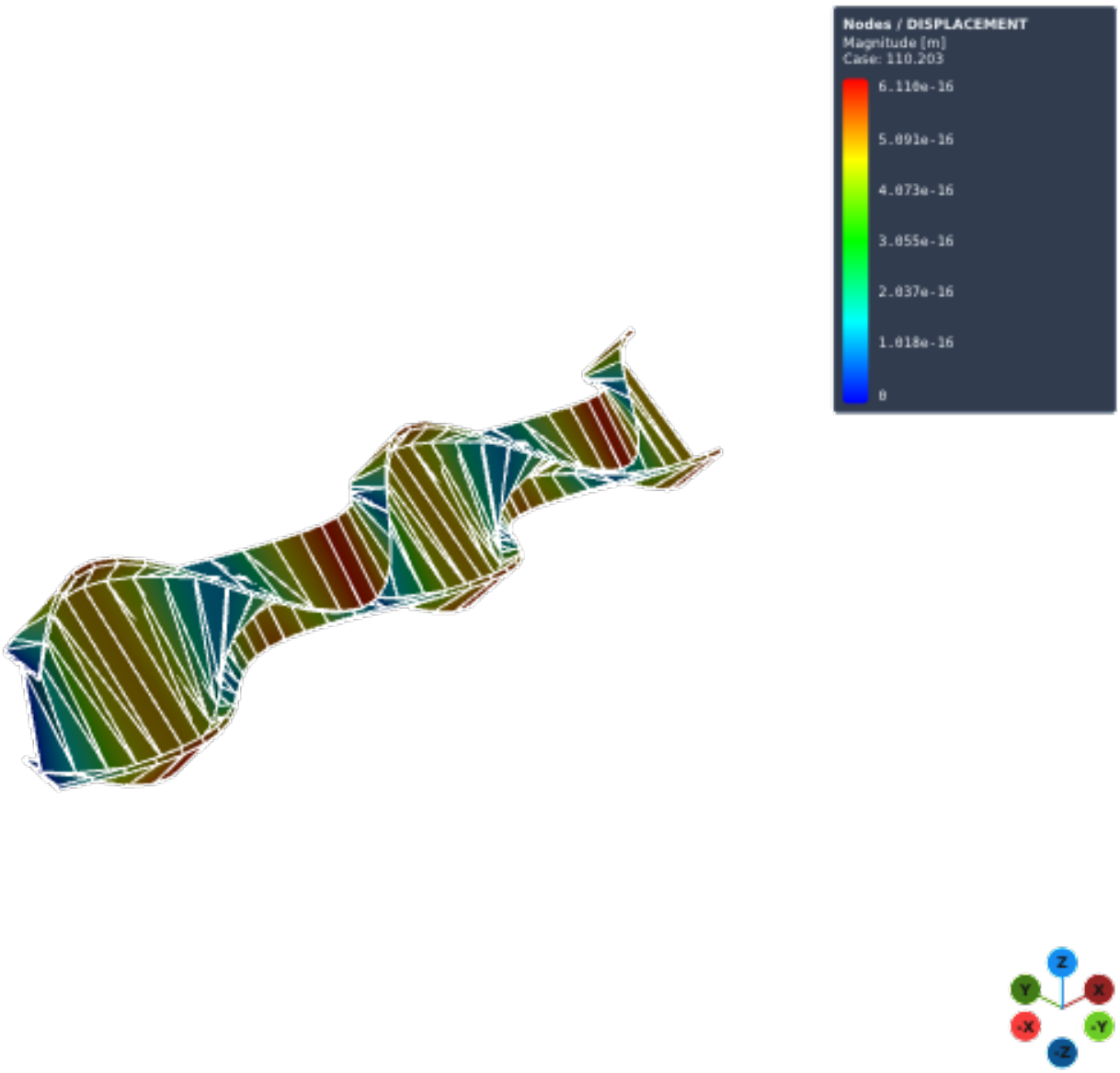

Figure 87: Sesam — mode 8, 110.203 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	97.074	-6.33	372022	4.46e-28	1.86e-30	1.09e-27	2.11e-14	1.36e-15	-3.31e-14
Sesam	110.203	+6.33	479455	1.22e-31	3.86e-29	2.48e-30	3.49e-16	6.21e-15	-1.57e-15

Mode 9

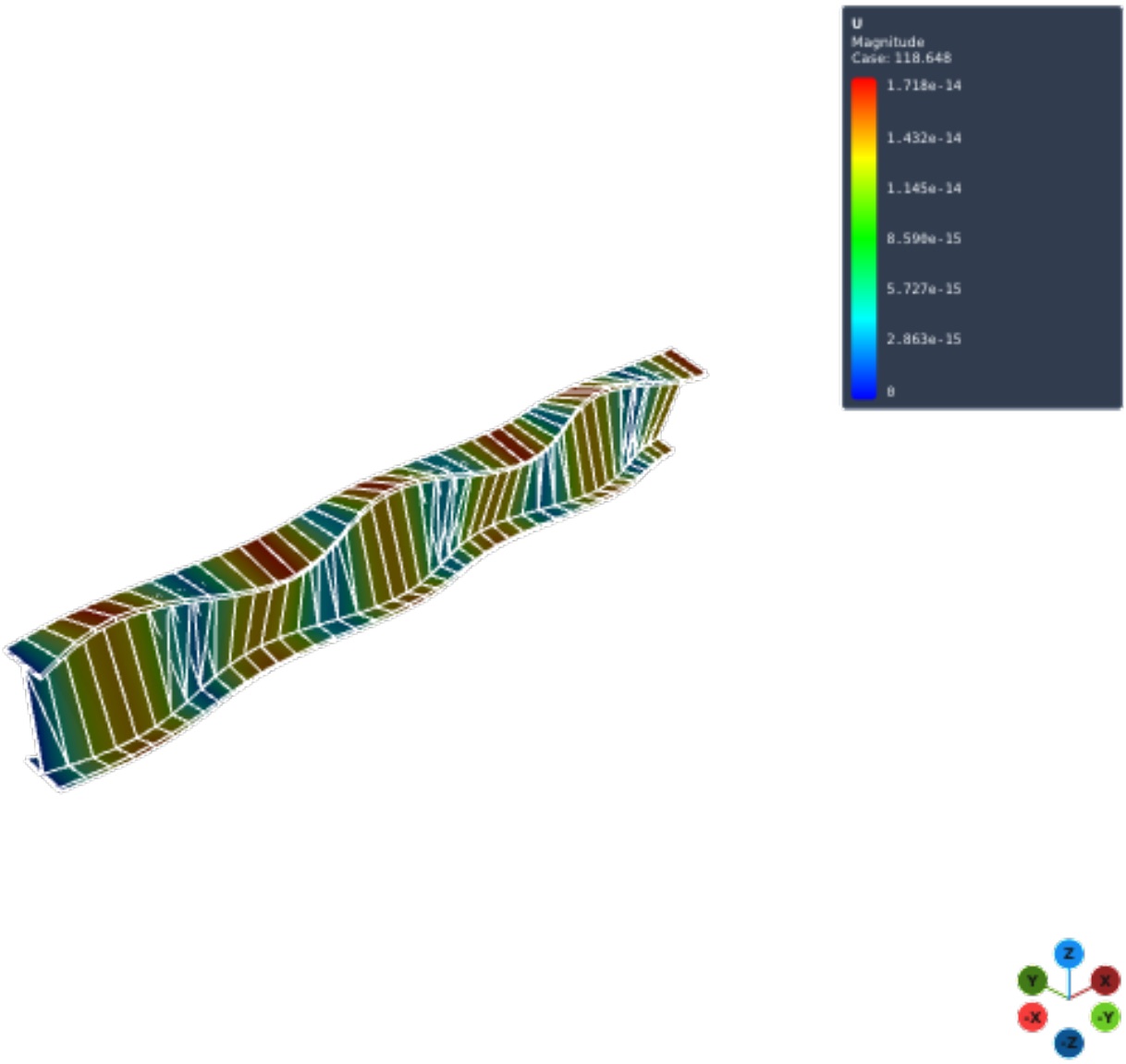

Figure 88: Abaqus — mode 9, 118.648 Hz

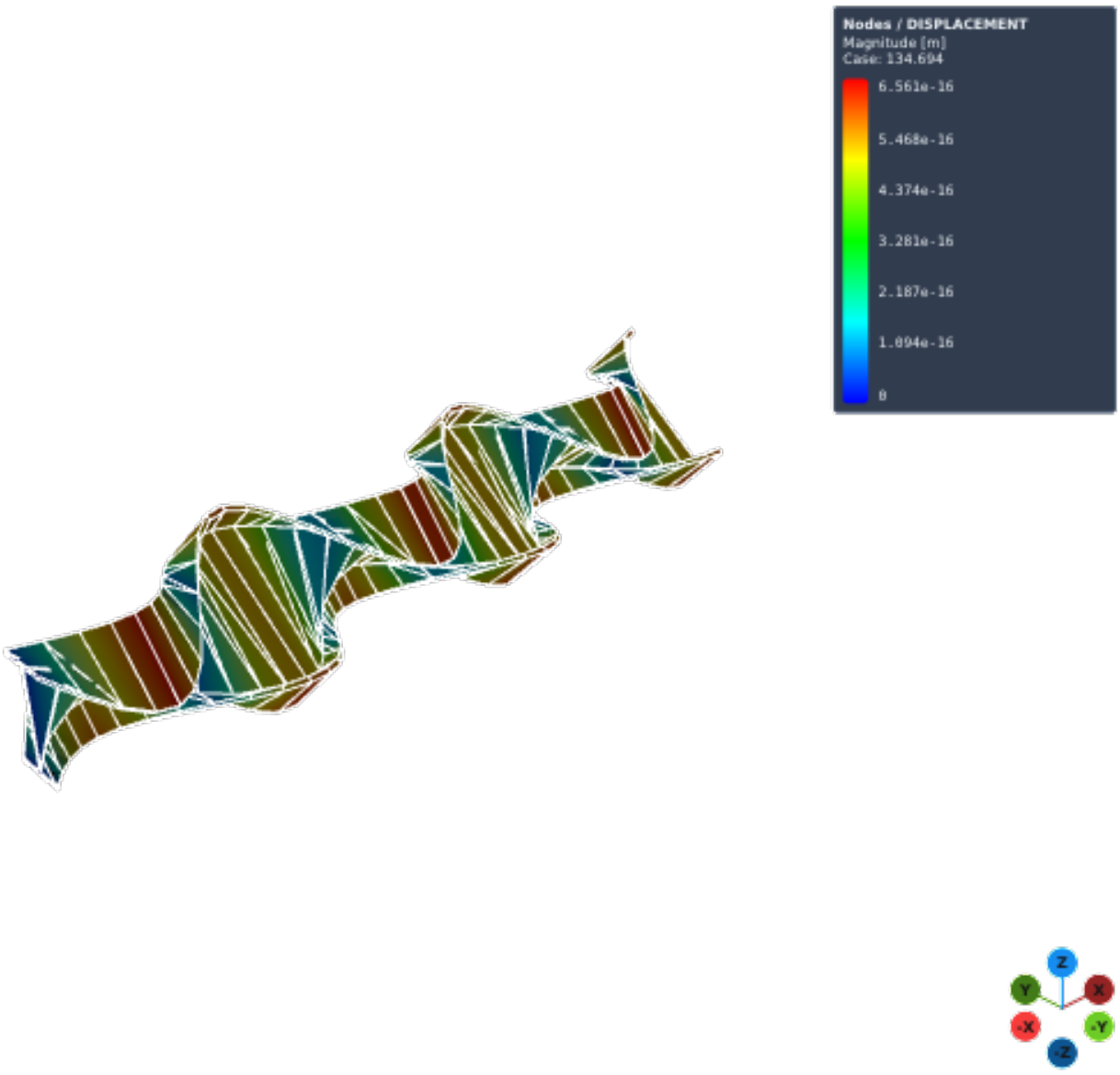

Figure 89: Sesam — mode 9, 134.694 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	118.650	-6.33	555748	3.94e-25	6.15e-26	2.26e-25	6.28e-13	2.48e-13	-4.75e-13
Sesam	134.694	+6.33	716238	2.09e-30	1.91e-30	3.39e-29	1.45e-15	1.38e-15	5.82e-15

Mode 10

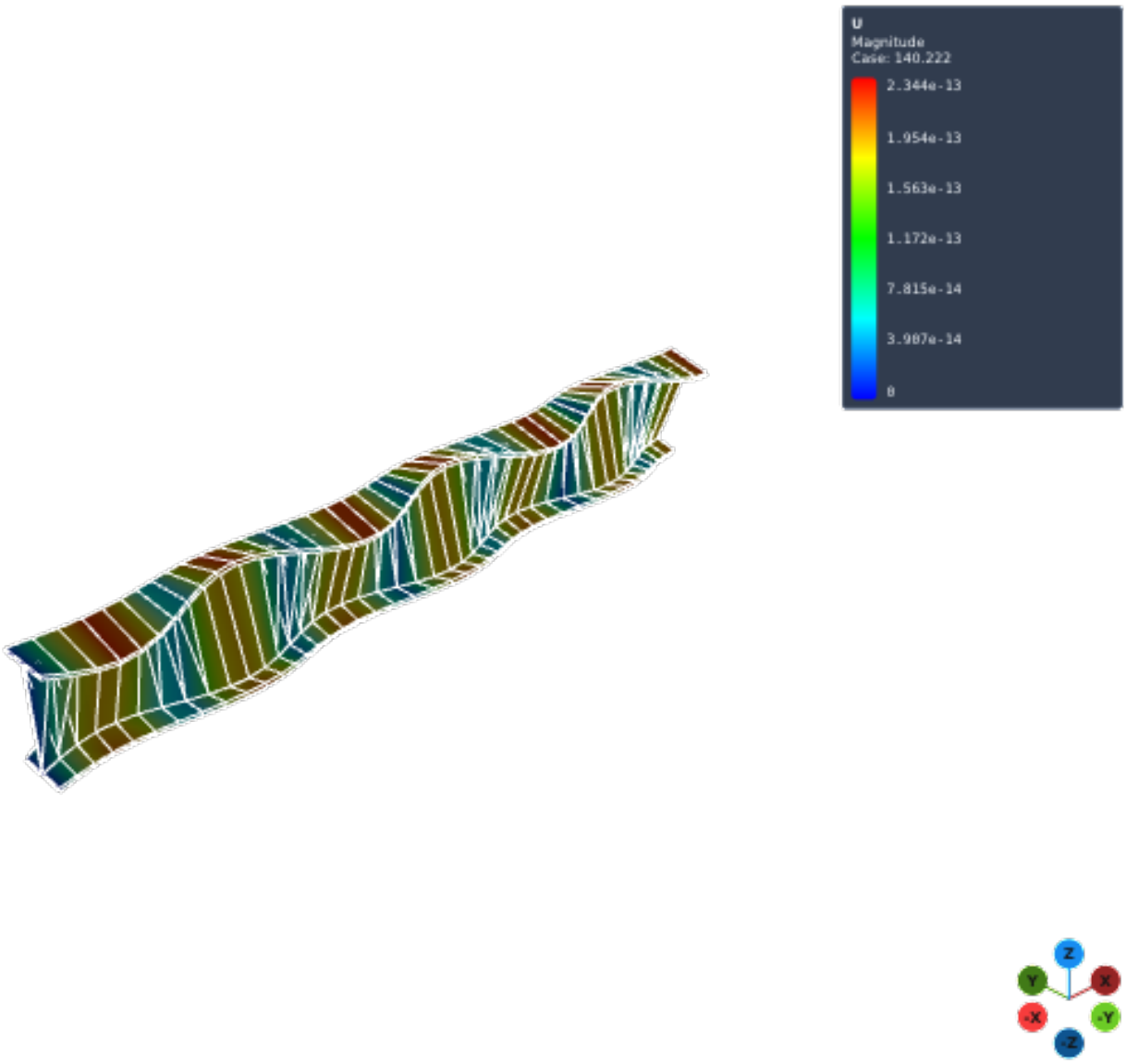

Figure 90: Abaqus — mode 10, 140.222 Hz

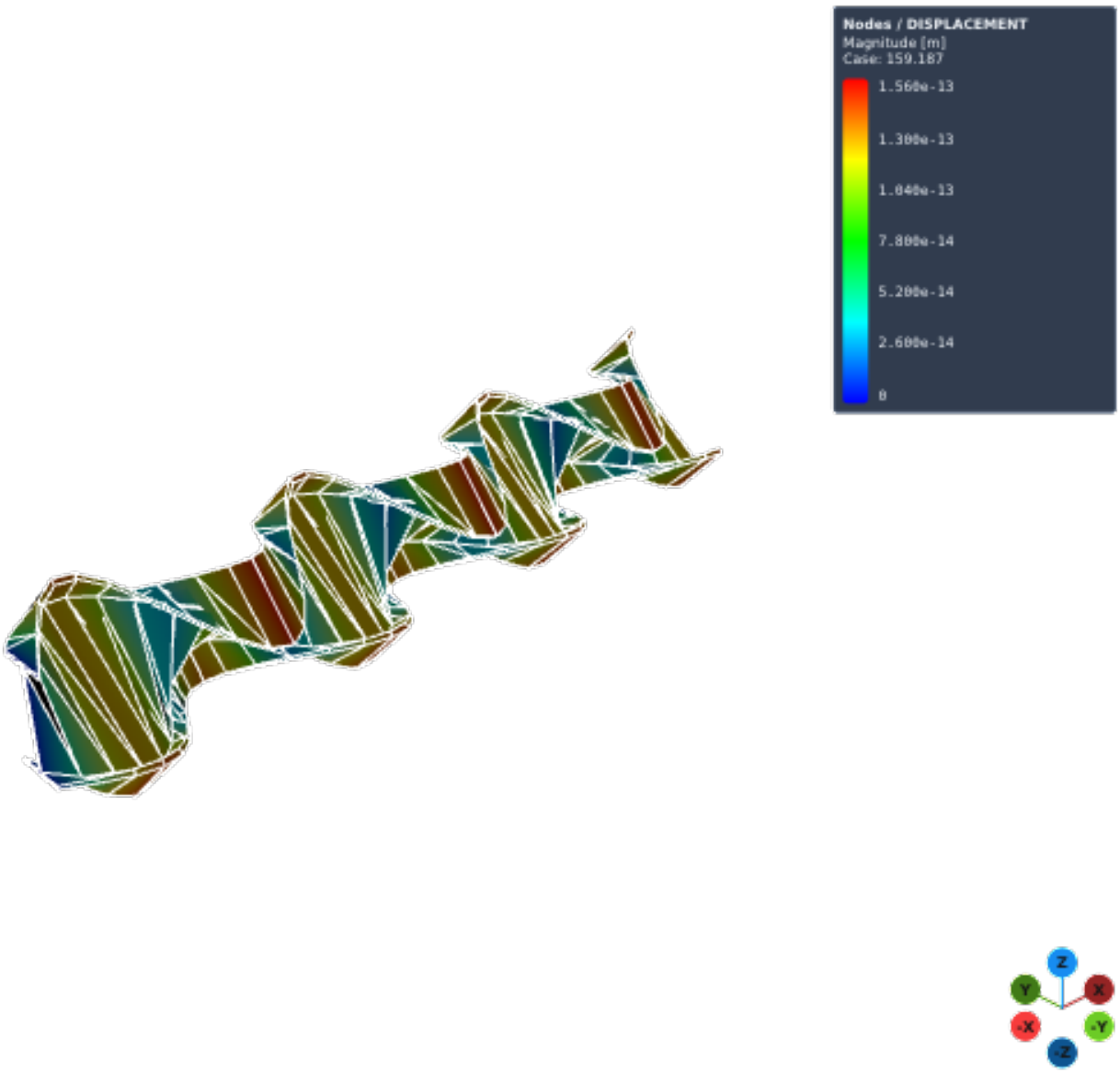

Figure 91: Sesam — mode 10, 159.187 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	140.220	-6.33	776236	2.58e-24	4.27e-23	1.99e-23	-1.61e-12	6.54e-12	4.46e-12
Sesam	159.187	+6.33	1.0004e+06	2.05e-24	2.36e-24	1.97e-27	1.43e-12	-1.54e-12	-4.44e-14

Mode 11

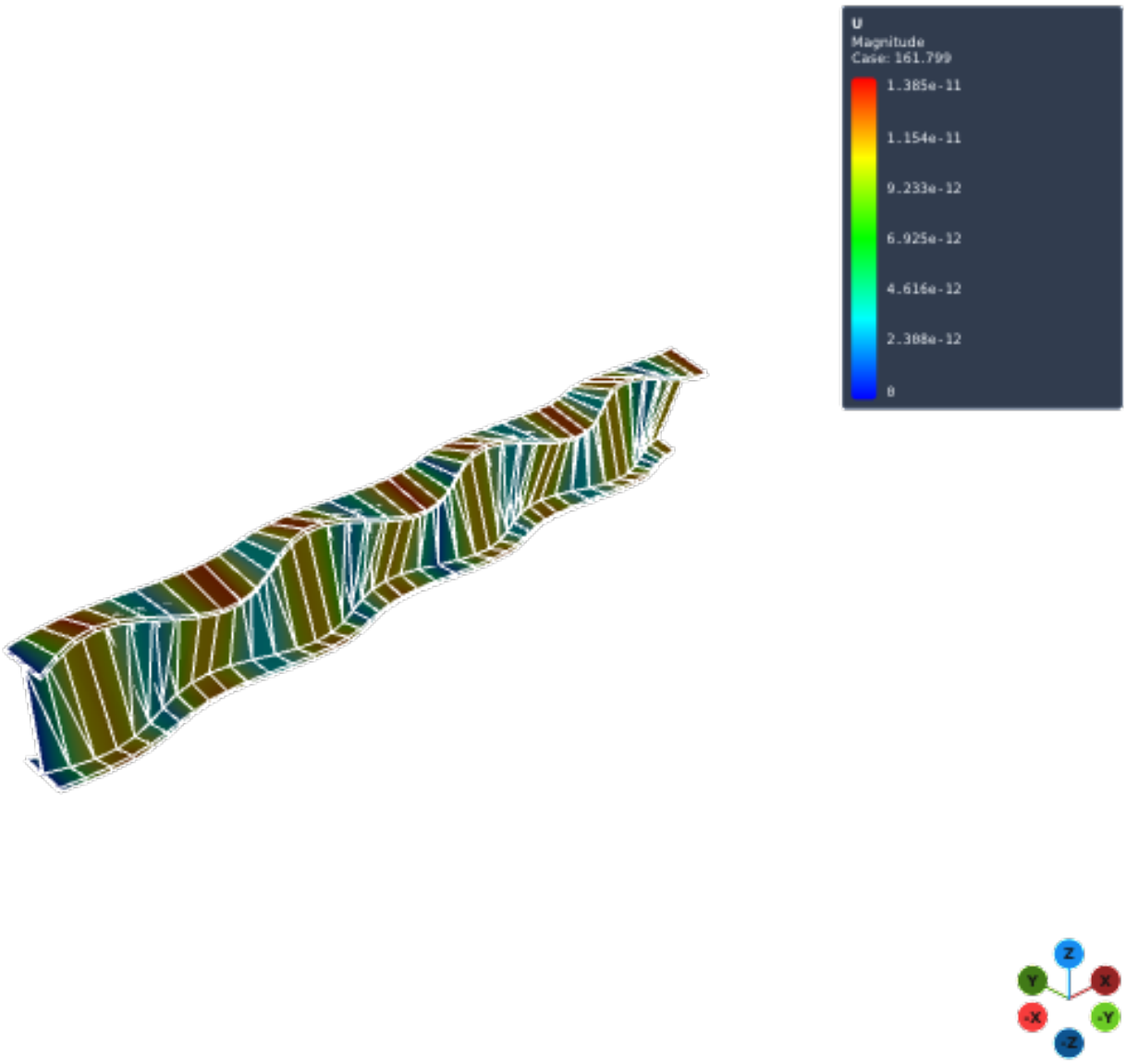

Figure 92: Abaqus — mode 11, 161.799 Hz

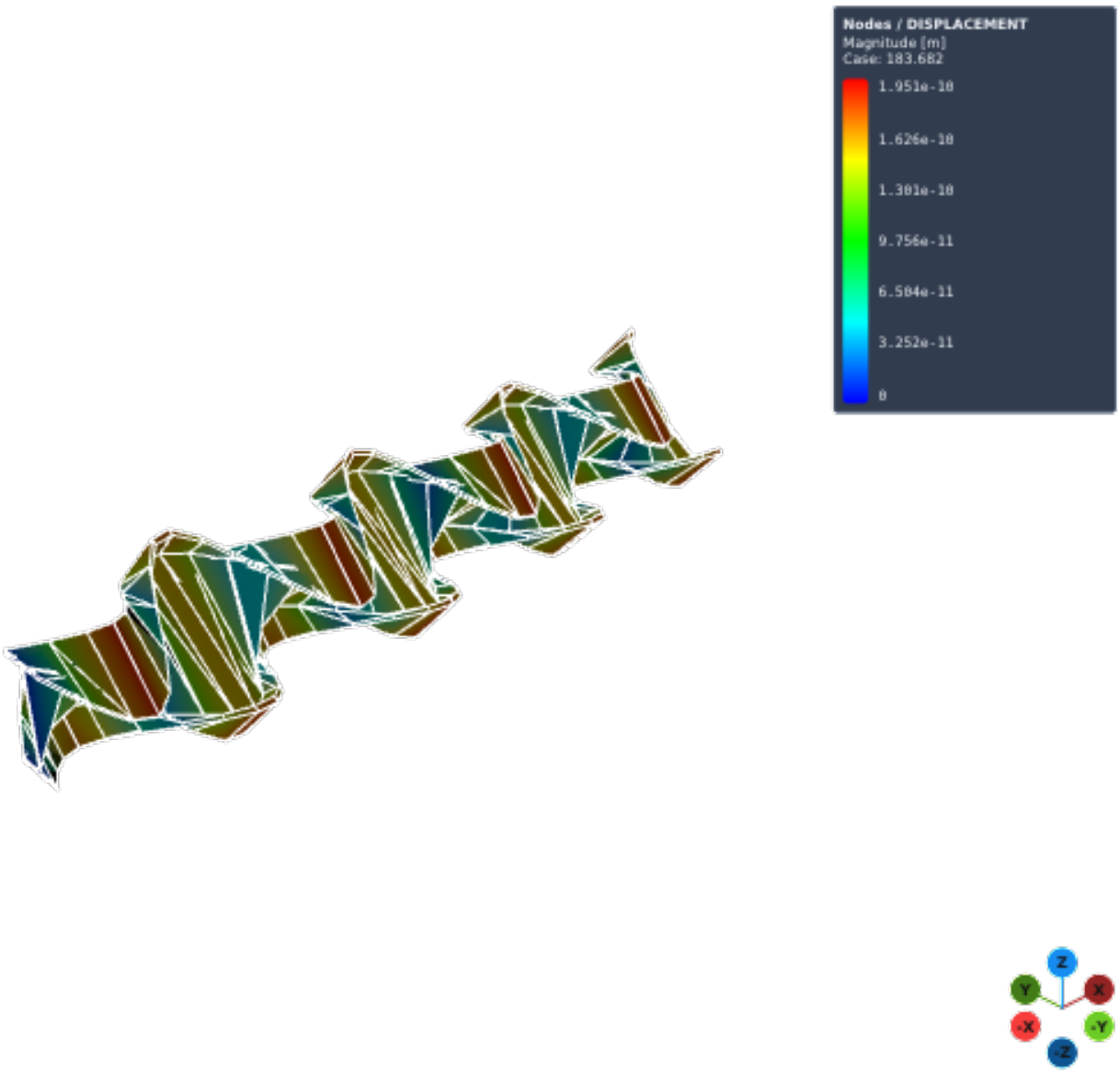

Figure 93: Sesam — mode 11, 183.682 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	161.800	-6.33	1.03351e+06	2.30e-19	1.18e-19	3.67e-20	-4.80e-10	-3.43e-10	1.92e-10
Sesam	183.682	+6.33	1.33196e+06	3.28e-18	3.71e-18	2.15e-21	1.81e-09	-1.93e-09	-4.64e-11

Shell, 1st order, TRI

Mode 1

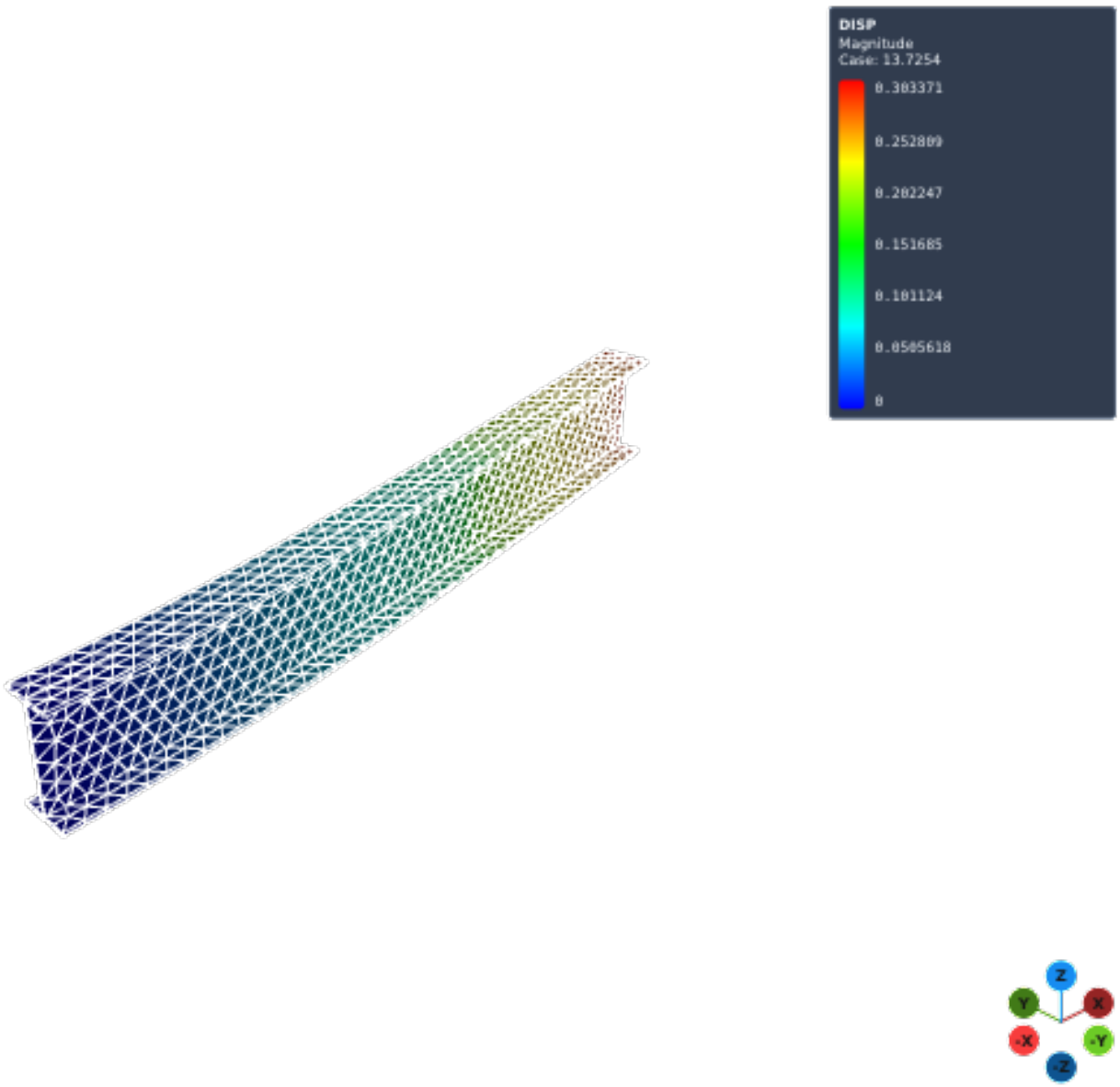

Figure 94: Calculix — mode 1, 13.725 Hz

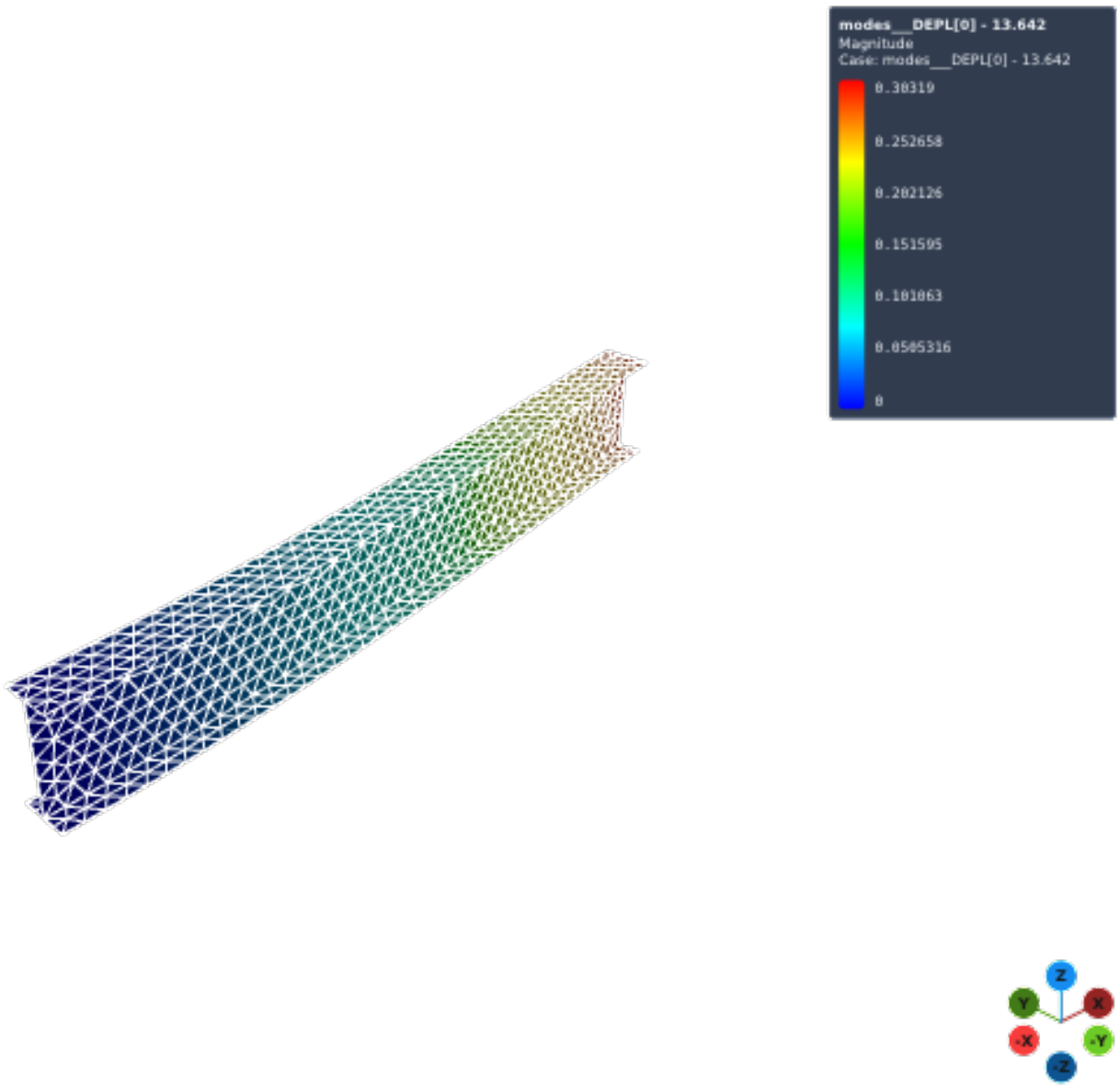

Figure 95: Code_Aster — mode 1, 13.642 Hz

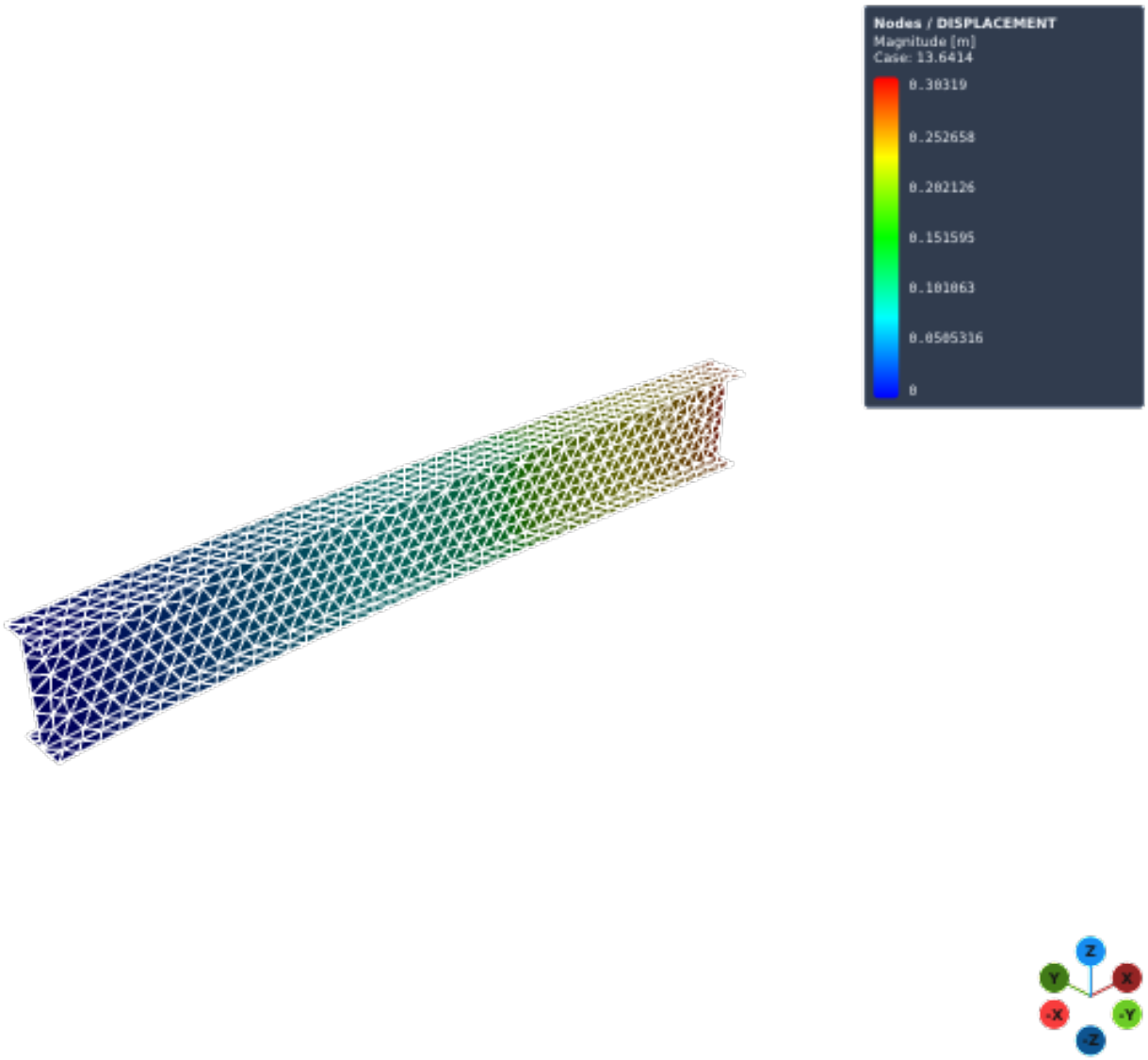

Figure 96: Sesam — mode 1, 13.641 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	13.725	+0.41	7437.24	9.89e-19	119.69	1.11e-11	9.95e-10	10.941	3.34e-06
Code_Aster	13.642	-0.20	7347.12	1.48e-27	119.93	1.17e-25	-3.85e-14	10.951	3.42e-13
Sesam	13.641	-0.21	7346.4	2.31e-27	119.93	9.89e-25	-4.81e-14	-10.951	-9.94e-13

Mode 2

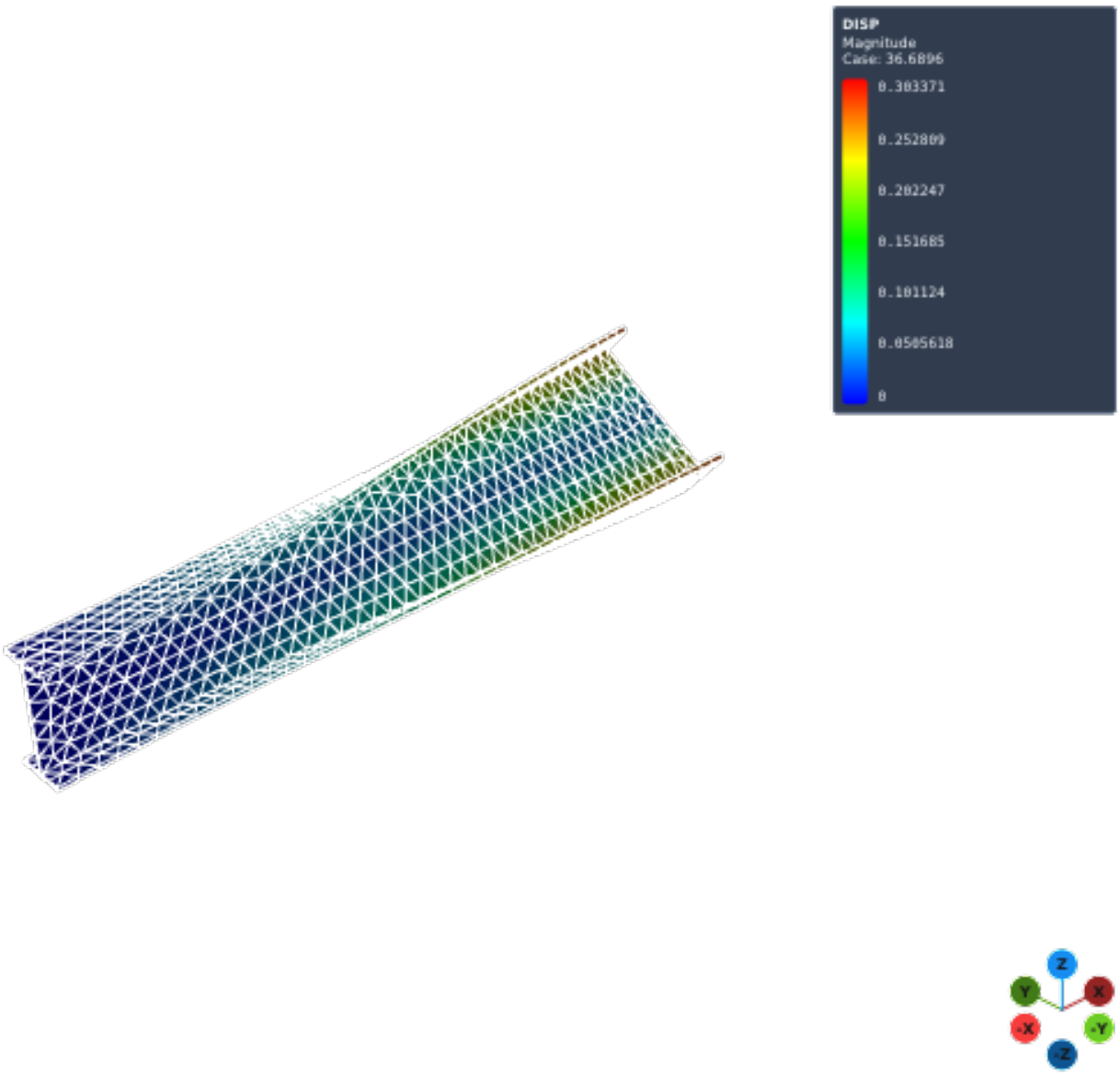

Figure 97: Calculix — mode 2, 36.690 Hz

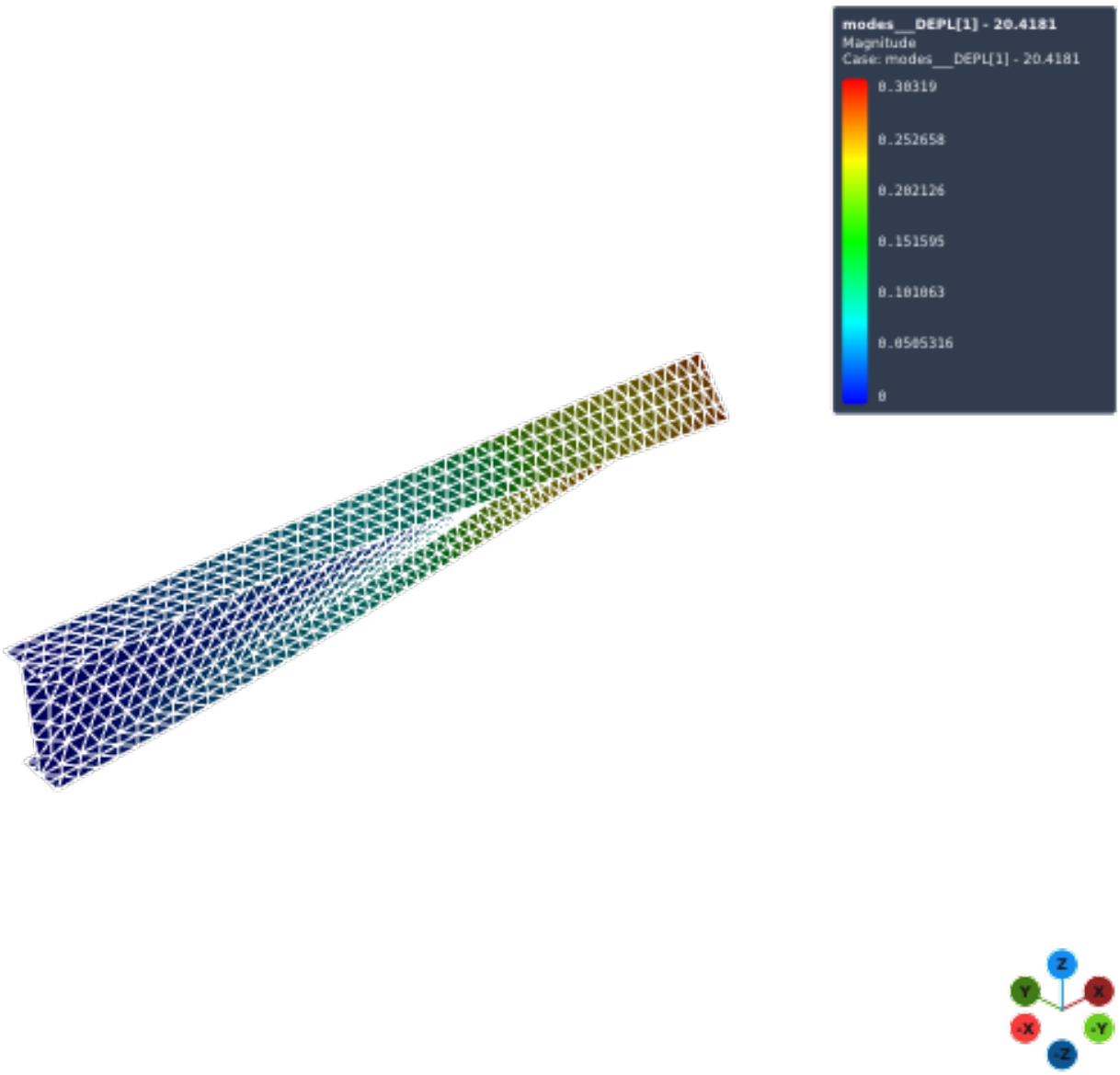

Figure 98: Code_Aster — mode 2, 20.418 Hz

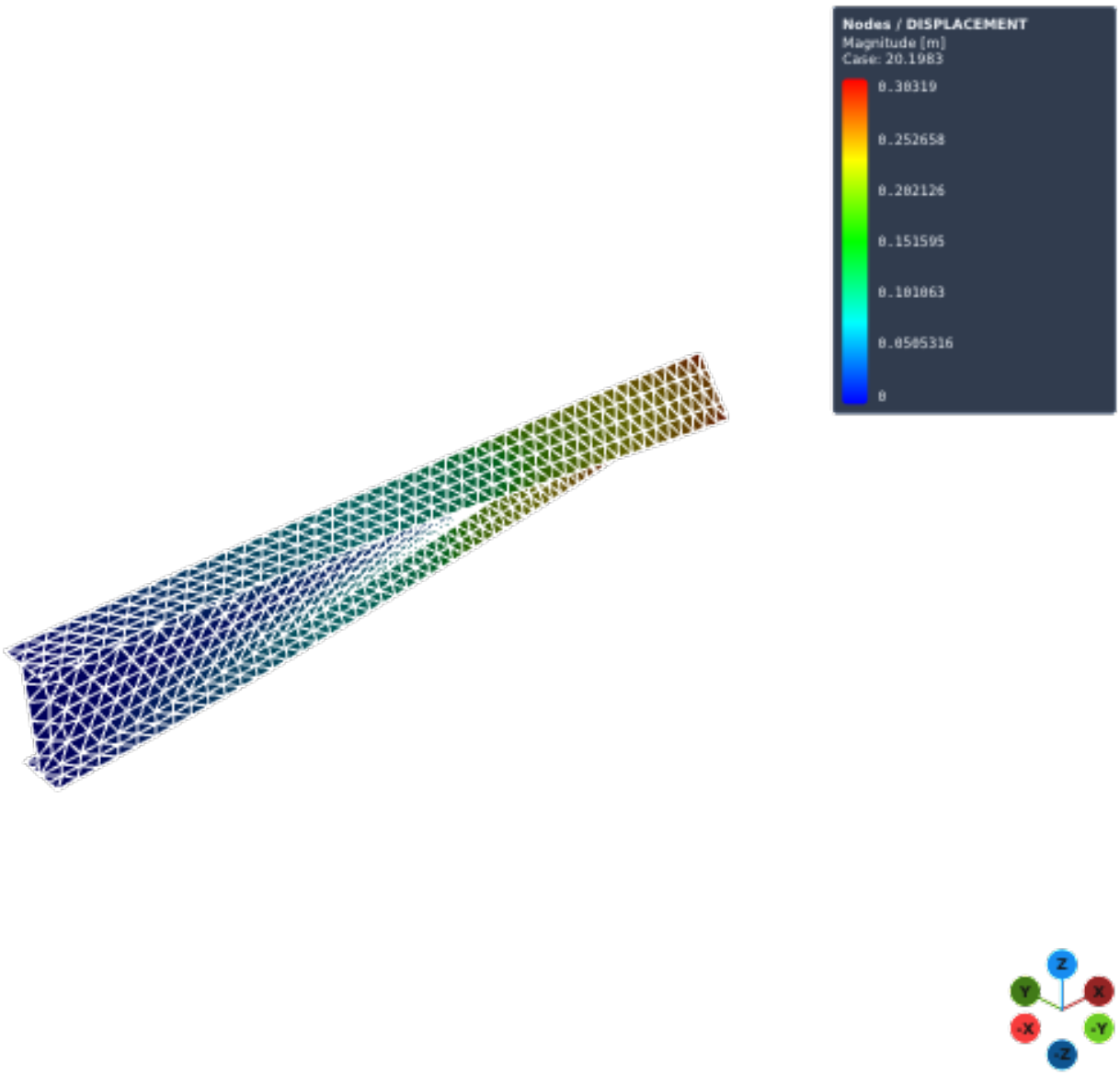

Figure 99: Sesam — mode 2, 20.198 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	36.690	+42.38	53142.9	2.89e-08	4.24e-06	5.81e-14	-1.70e-04	0.002	-2.41e-07
Code_Aster	20.418	-20.76	16458.5	4.91e-27	1.63e-08	7.48e-26	7.01e-14	-1.28e-04	2.73e-13
Sesam	20.198	-21.62	16106.1	2.48e-26	3.02e-11	1.06e-25	1.57e-13	-5.49e-06	3.25e-13

Mode 3

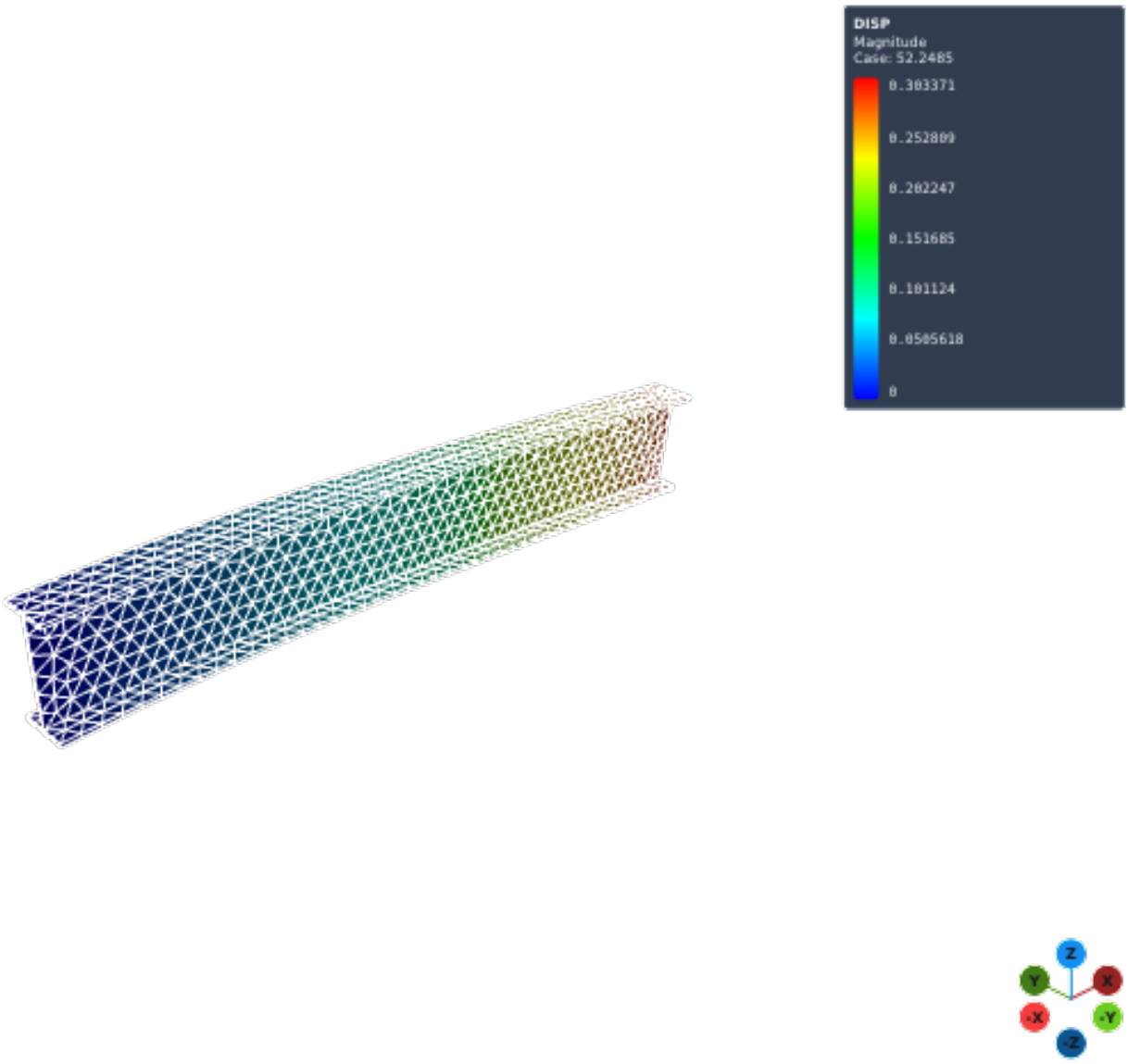

Figure 100: Calculix — mode 3, 52.249 Hz

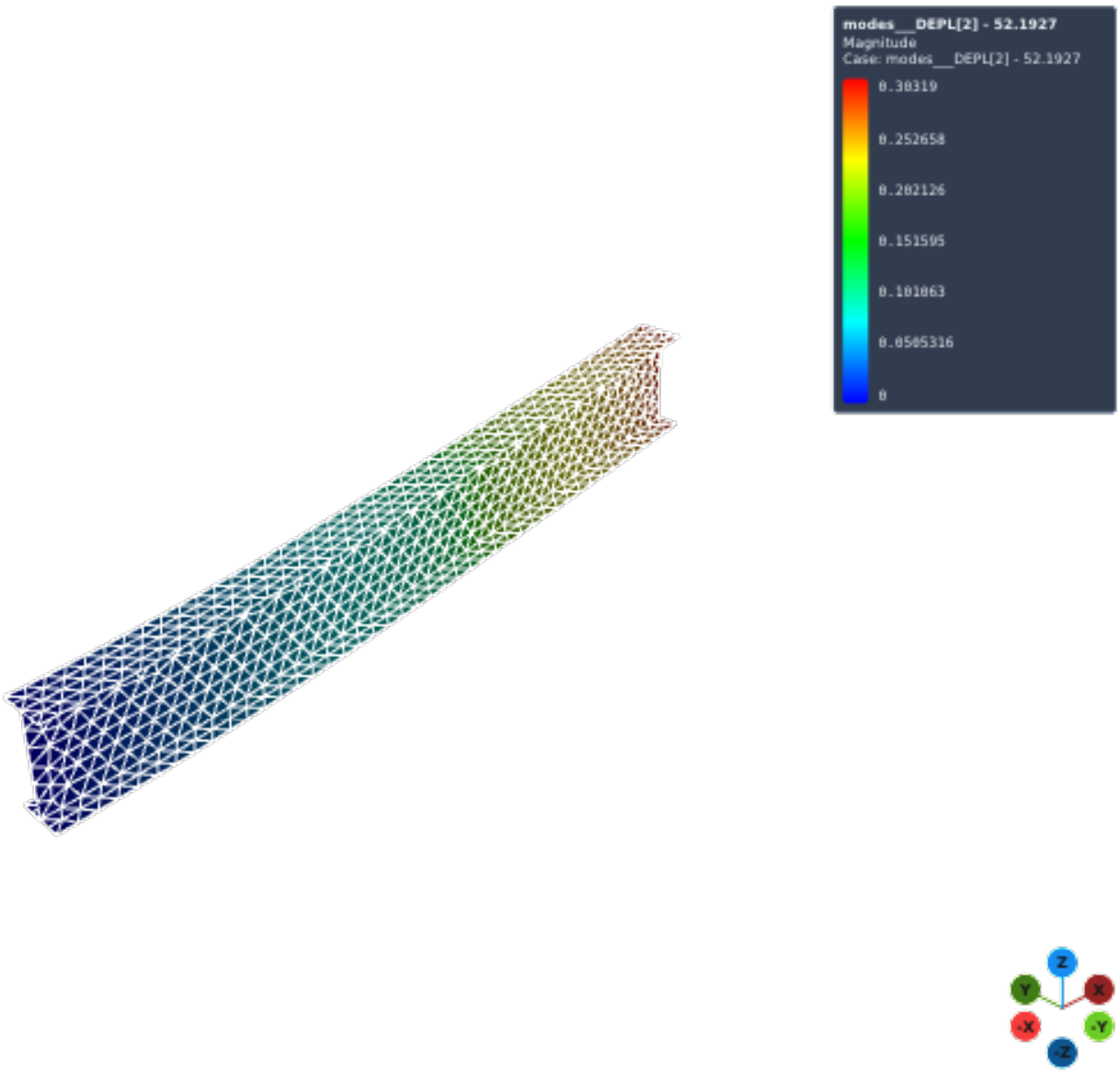

Figure 101: Code_Aster — mode 3, 52.193 Hz

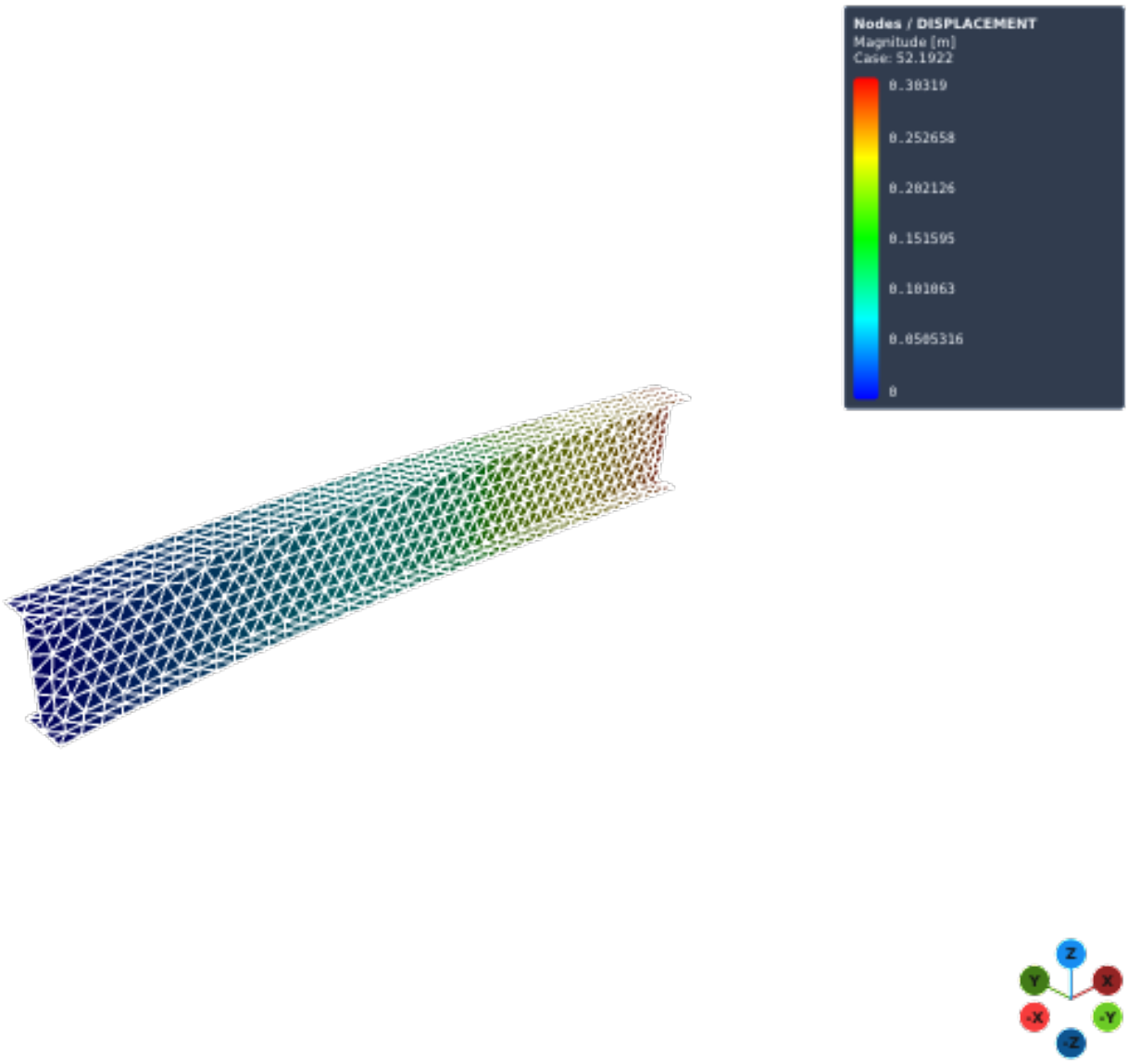

Figure 102: Sesam — mode 3, 52.192 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	52.249	+0.07	107772	1.80e-10	4.08e-10	122.06	-1.34e-05	2.02e-05	-11.048
Code_Aster	52.193	-0.04	107542	1.54e-10	3.07e-25	122.03	1.24e-05	-5.54e-13	11.047
Sesam	52.192	-0.04	107540	4.79e-10	1.53e-24	122.03	-2.19e-05	1.24e-12	-11.047

Mode 4

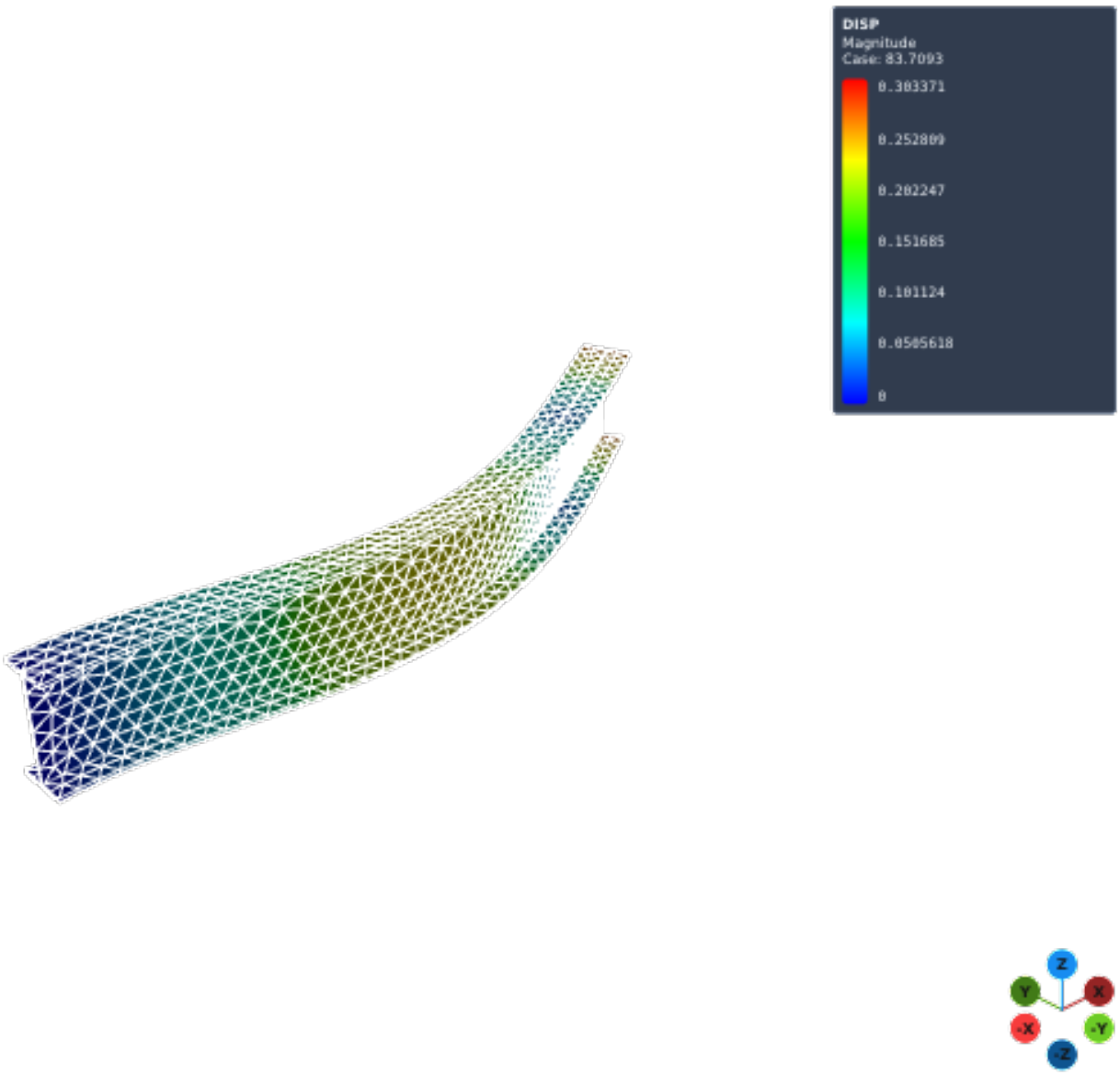

Figure 103: Calculix — mode 4, 83.709 Hz

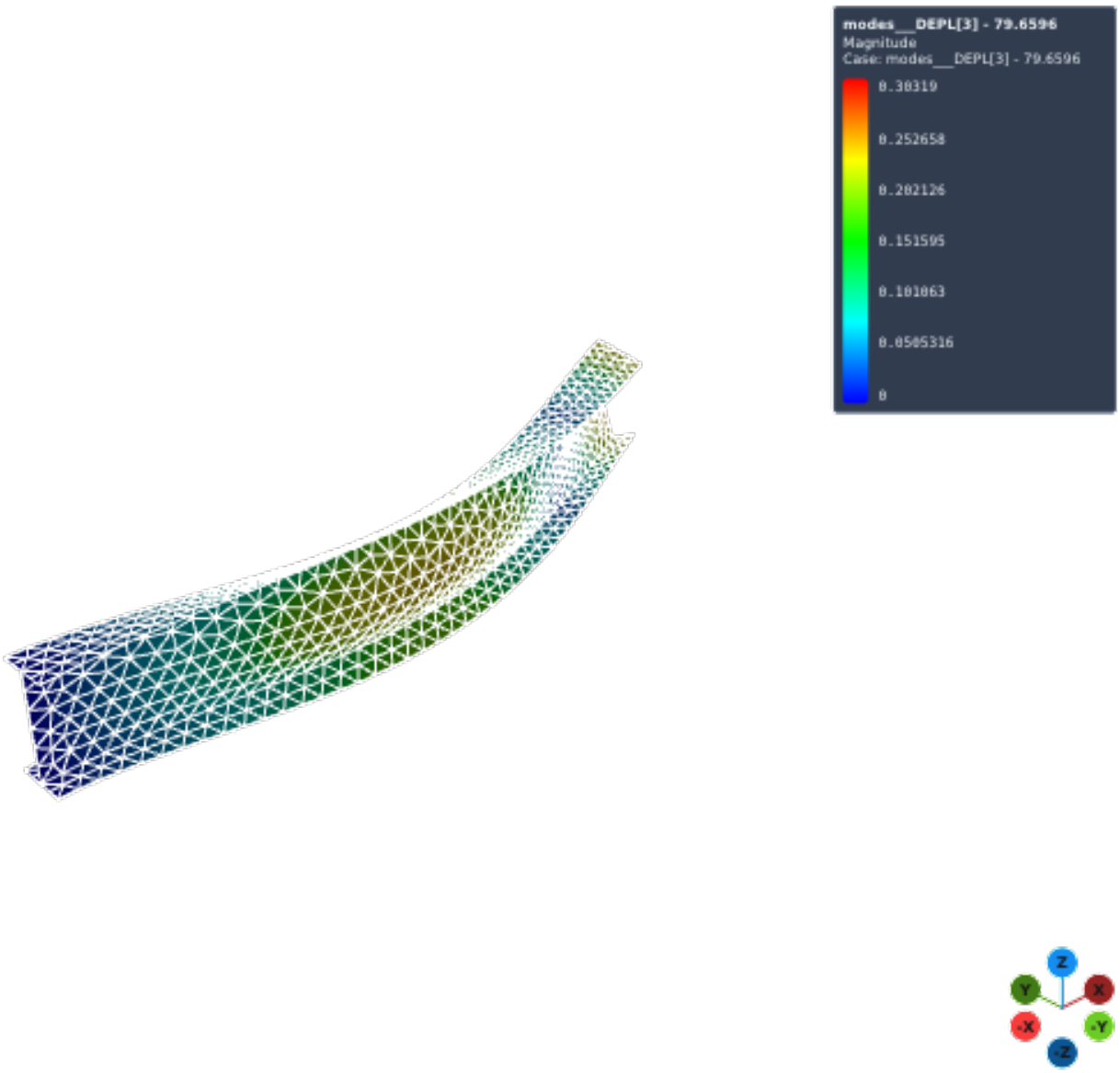

Figure 104: Code_Aster — mode 4, 79.660 Hz

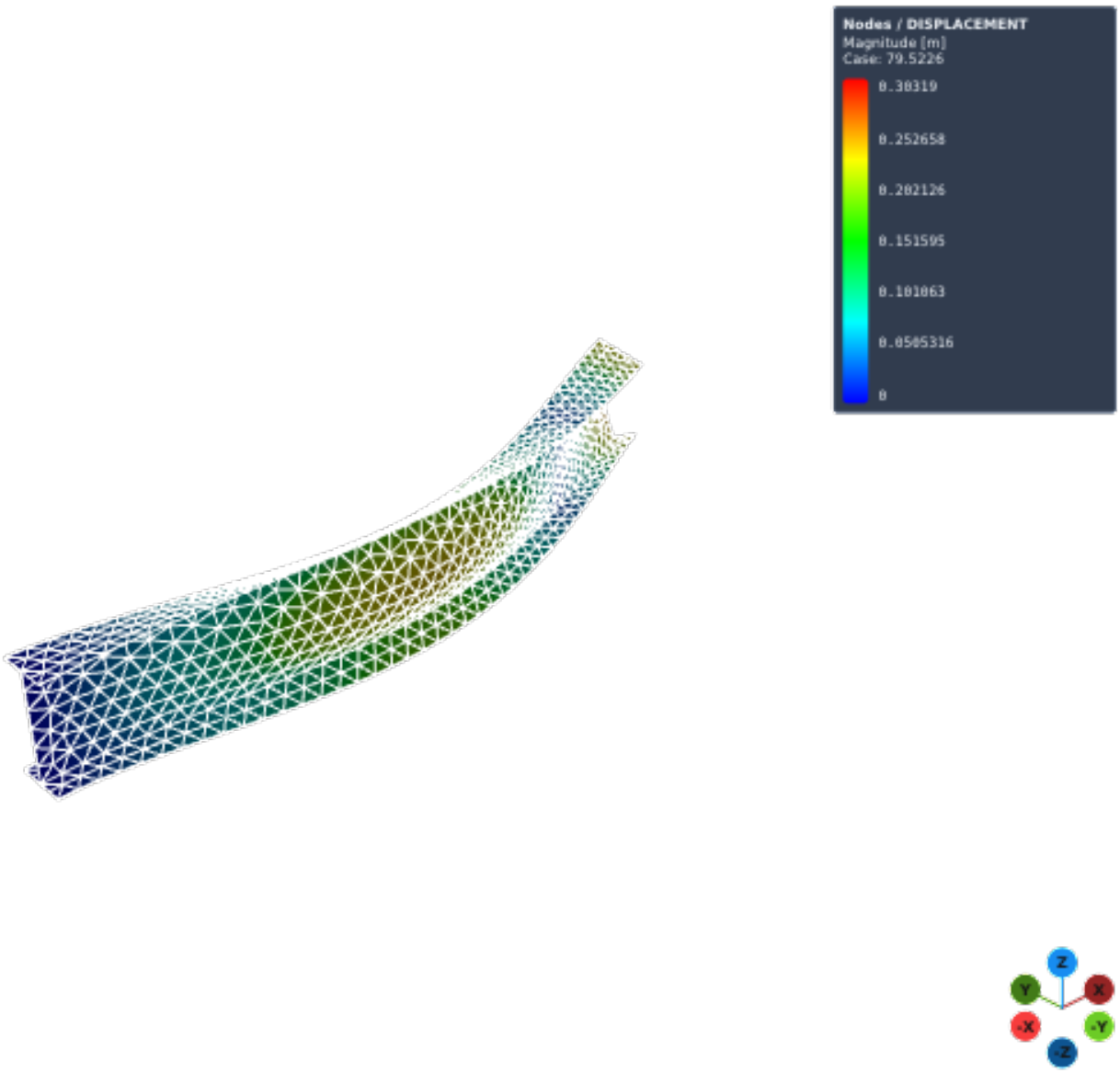

Figure 105: Sesam — mode 4, 79.523 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	83.709	+3.39	276635	1.88e-12	37.26	1.12e-11	-1.37e-06	-6.104	3.35e-06
Code_Aster	79.660	-1.61	250516	1.53e-26	36.64	1.82e-25	1.24e-13	-6.053	-4.26e-13
Sesam	79.523	-1.78	249655	1.60e-28	36.58	1.25e-25	1.26e-14	-6.048	-3.54e-13

Mode 5

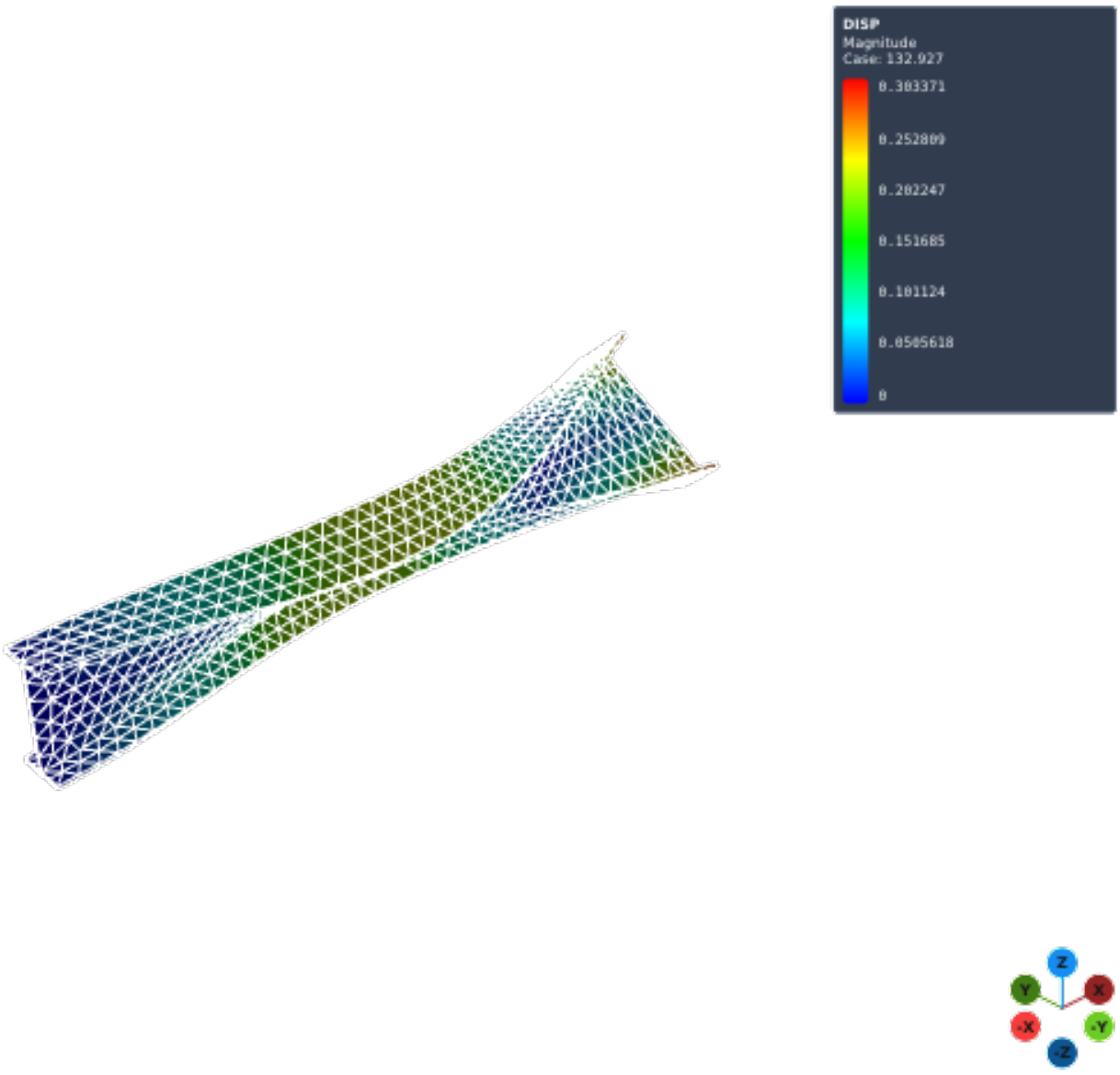

Figure 106: Calculix — mode 5, 132.927 Hz

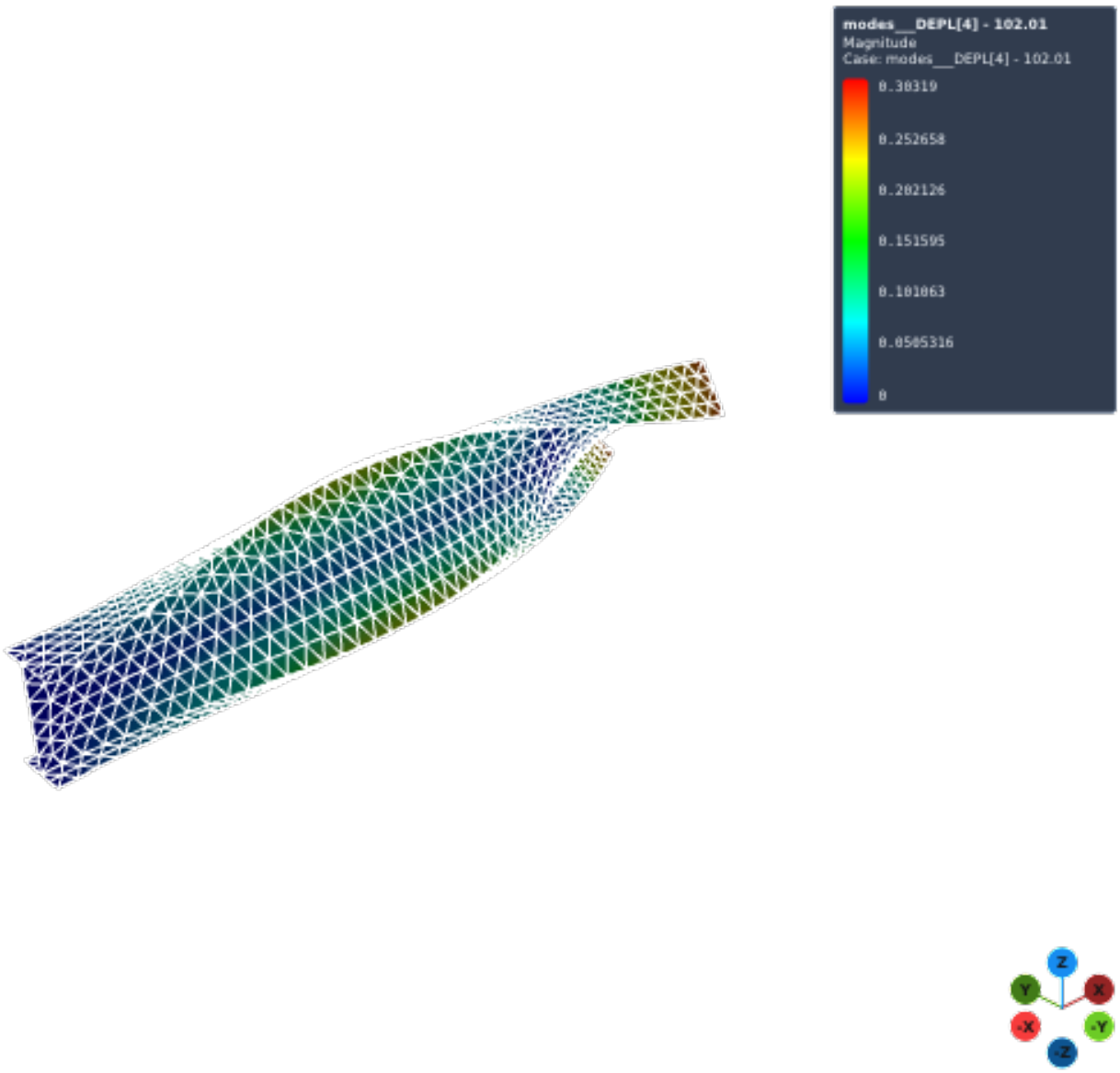

Figure 107: Code_Aster — mode 5, 102.010 Hz

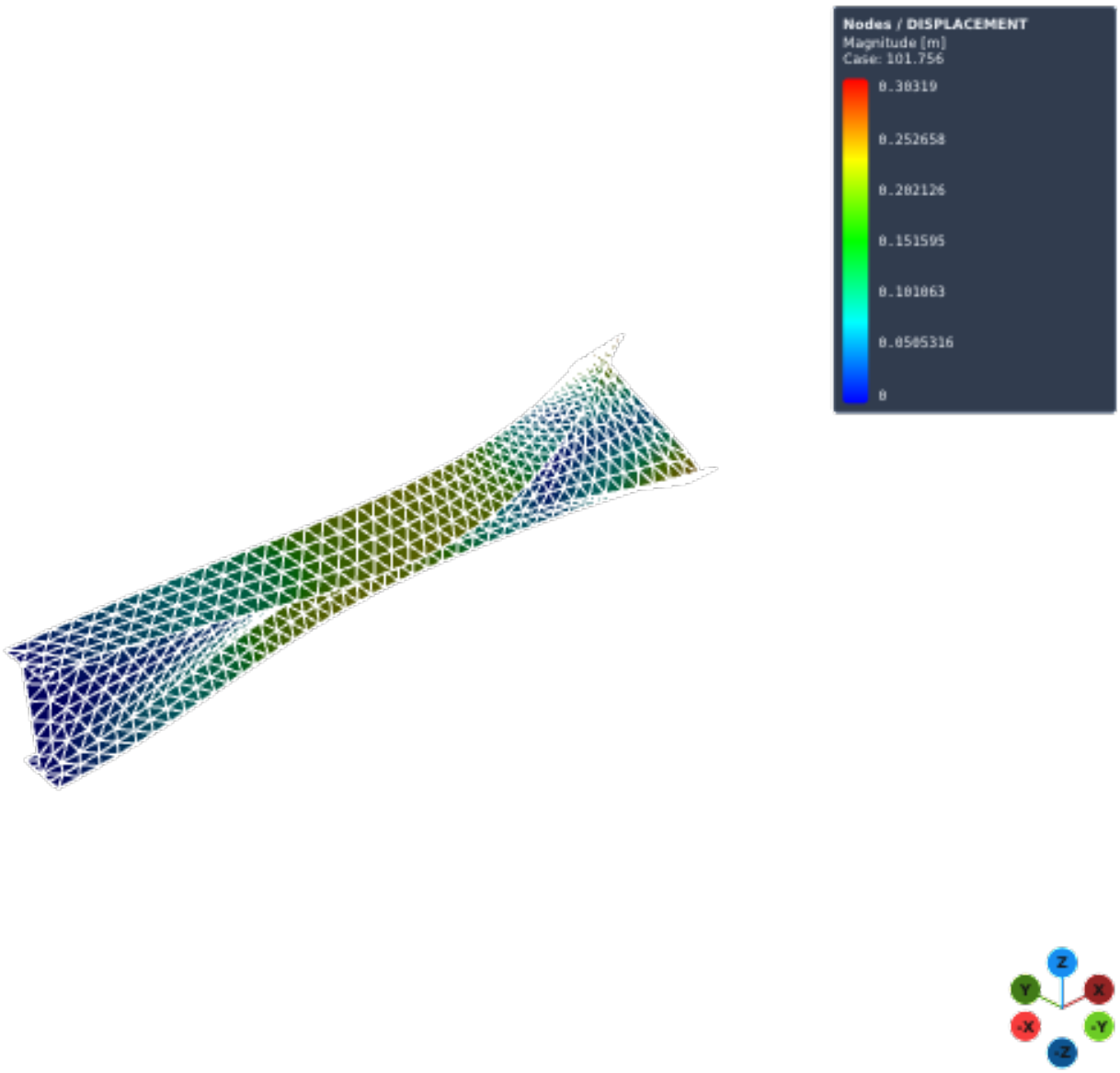

Figure 108: Sesam — mode 5, 101.756 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	132.928	+18.44	697572	4.77e-09	2.89e-05	2.99e-14	6.91e-05	-0.005	-1.73e-07
Code_Aster	102.010	-9.11	410817	1.16e-26	3.22e-09	2.68e-25	1.08e-13	-5.68e-05	-5.18e-13
Sesam	101.756	-9.33	408770	4.49e-27	4.27e-09	6.32e-27	-6.70e-14	6.54e-05	-7.95e-14

Mode 6

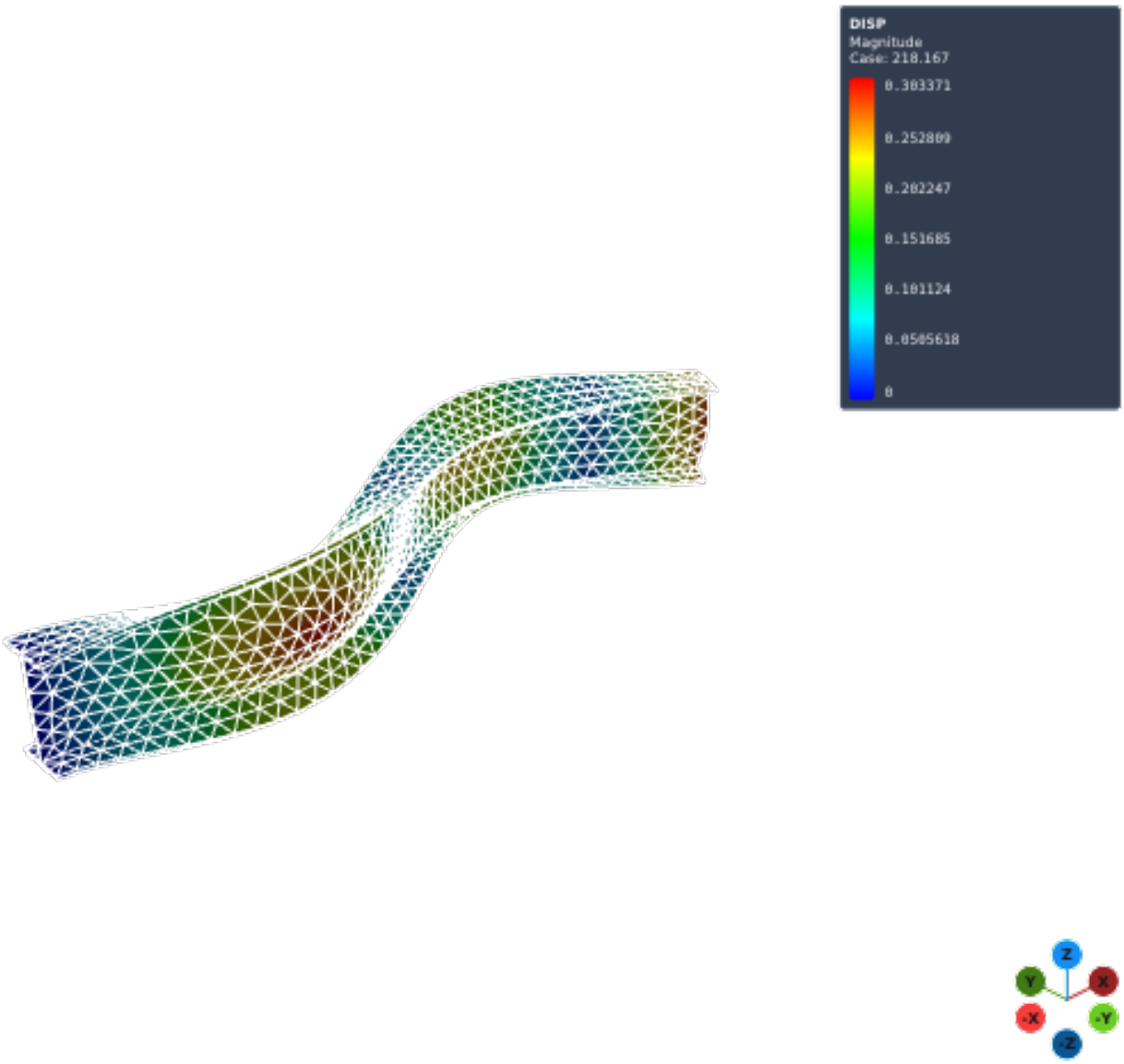

Figure 109: Calculix — mode 6, 218.167 Hz

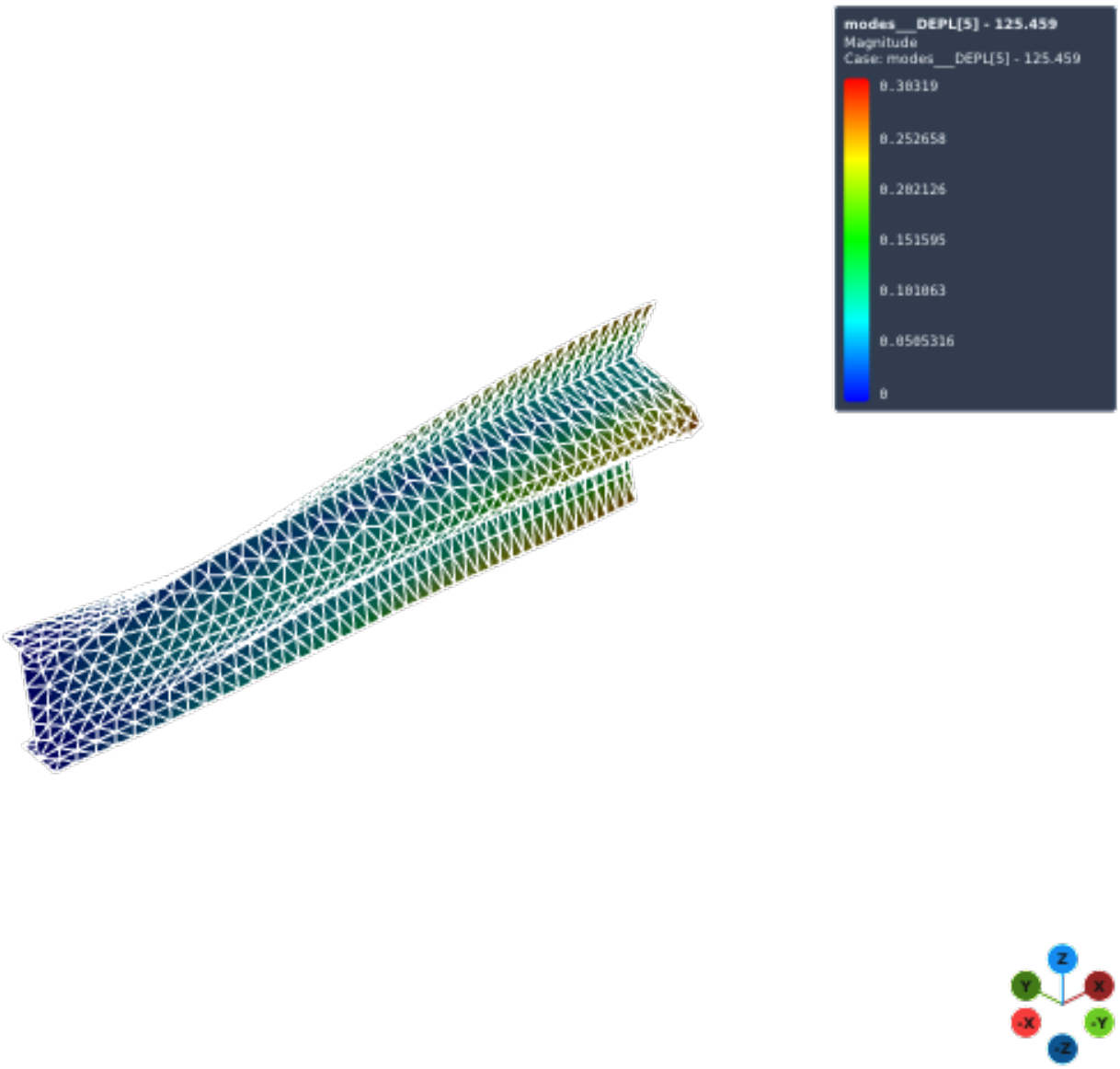

Figure 110: Code_Aster — mode 6, 125.459 Hz

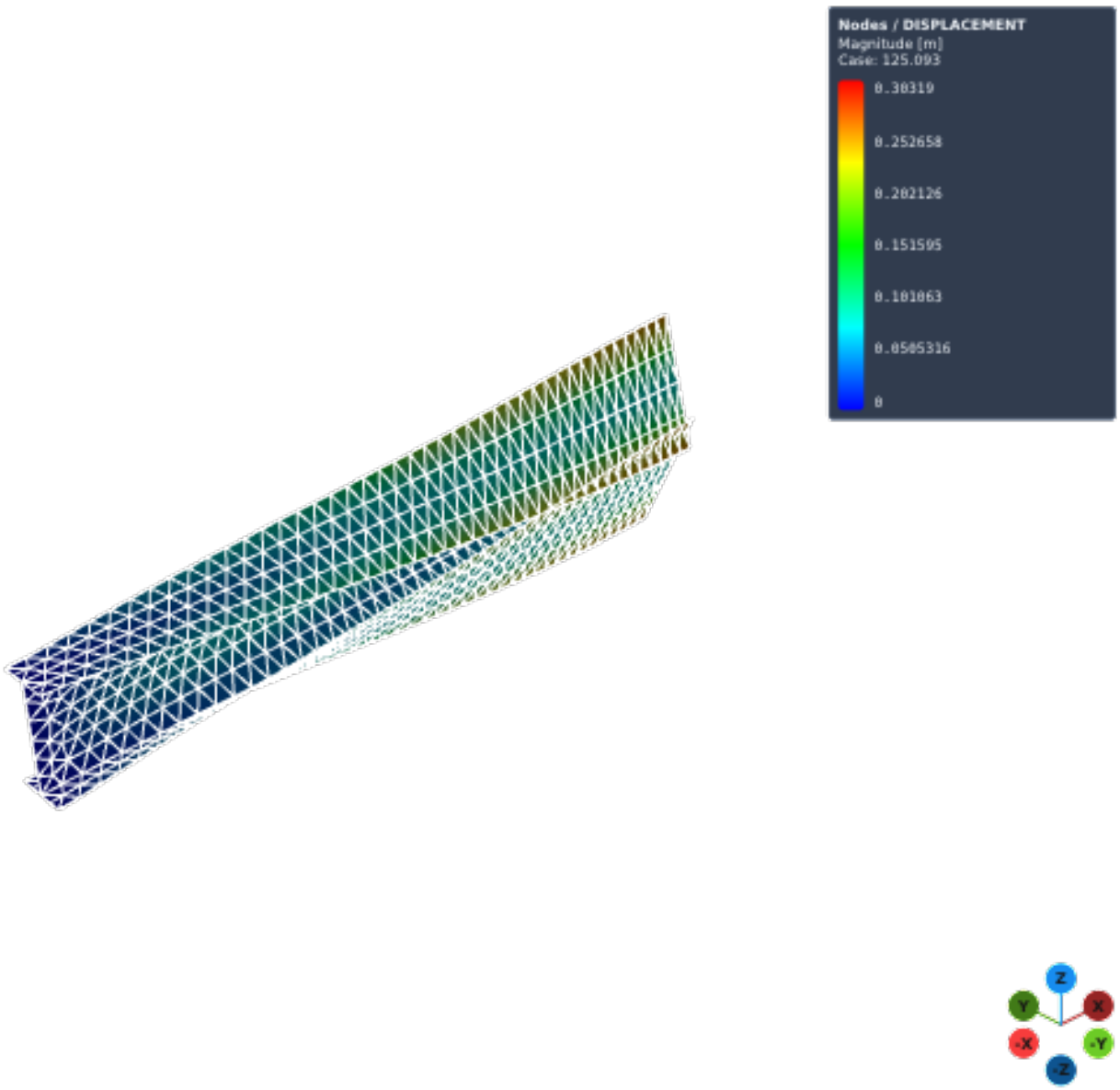

Figure 111: Sesam — mode 6, 125.093 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	218.167	+39.64	1.87905e+06	1.01e-11	12.68	1.01e-11	-3.18e-06	-3.561	-3.17e-06
Code_Aster	125.459	-19.70	621390	5.78e-28	0.07	7.34e-25	-2.40e-14	0.263	-8.57e-13
Sesam	125.093	-19.94	617772	7.12e-28	0.06	1.10e-26	-2.67e-14	-0.252	1.05e-13

Mode 7

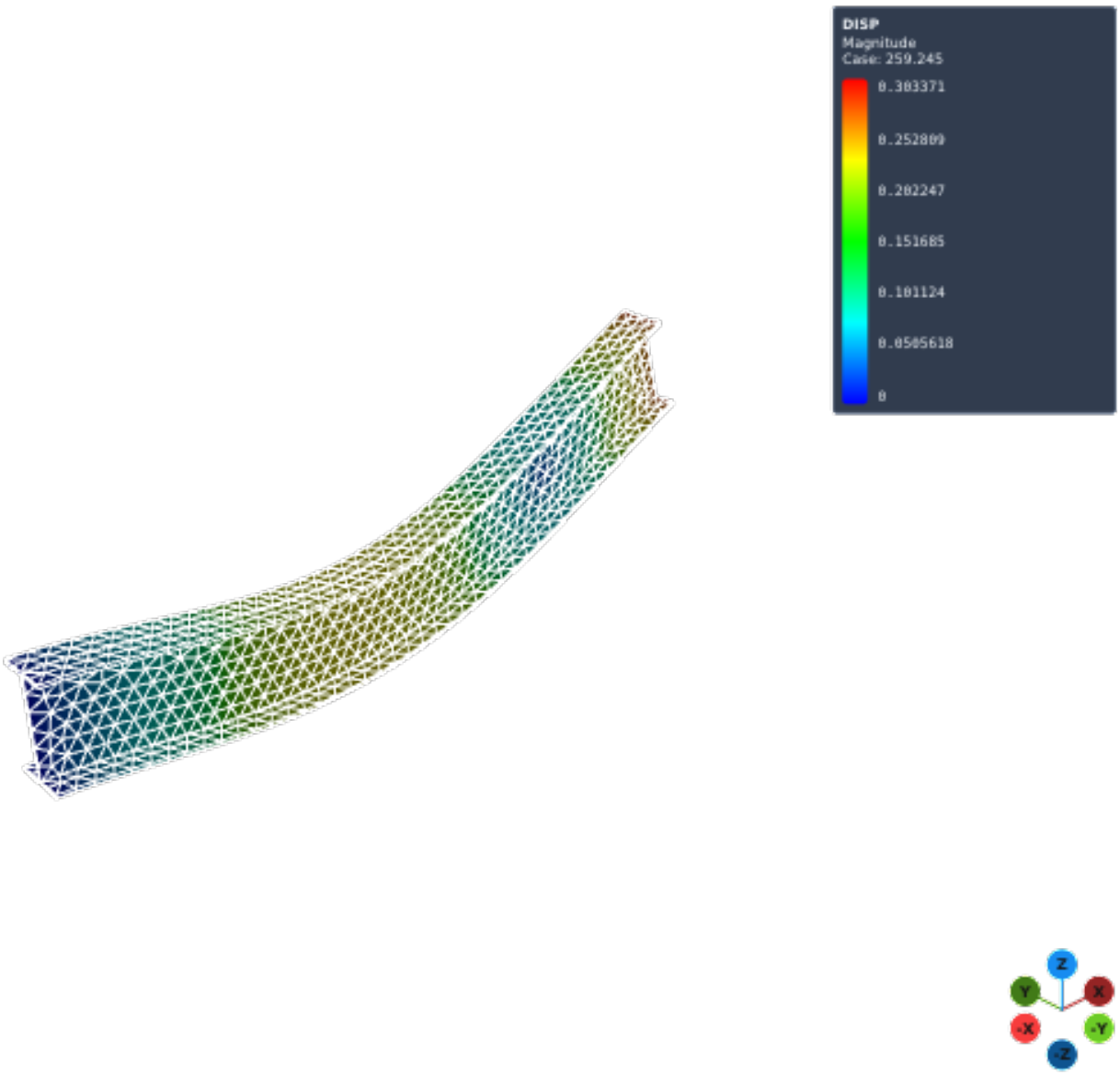

Figure 112: Calculix — mode 7, 259.245 Hz

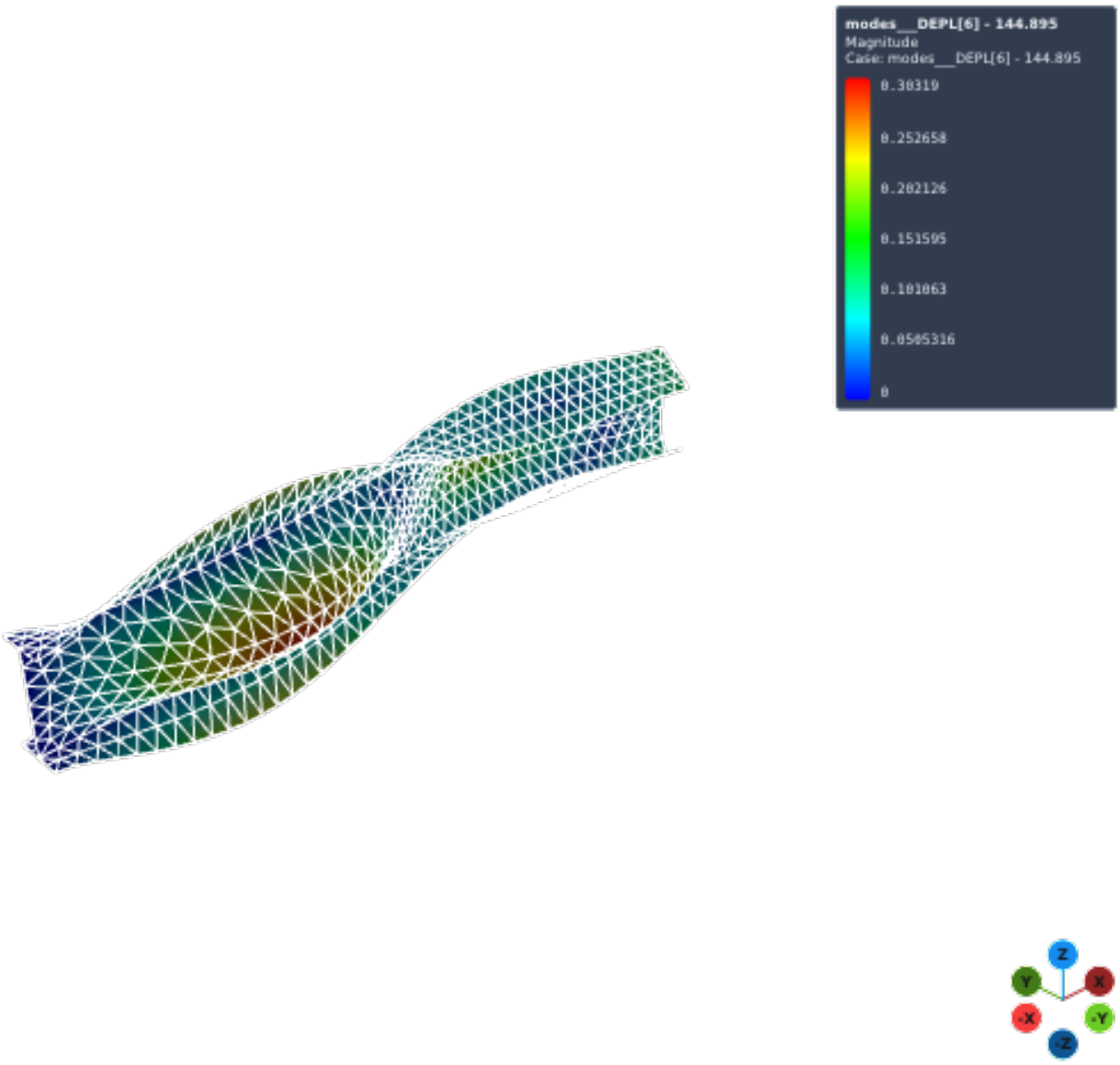

Figure 113: Code_Aster — mode 7, 144.895 Hz

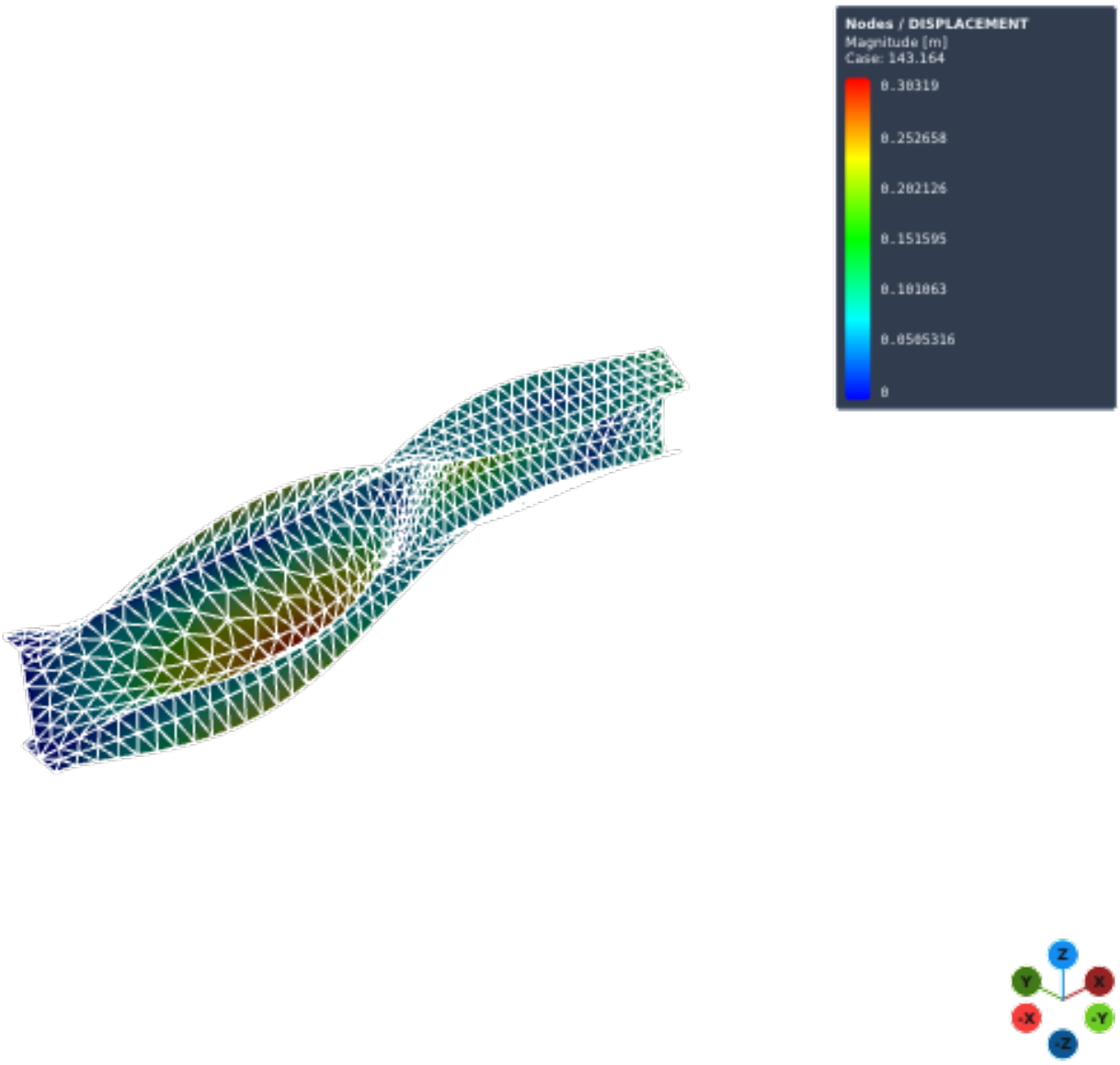

Figure 114: Sesam — mode 7, 143.164 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	259.245	+42.10	2.65327e+06	6.03e-11	3.23e-10	41.55	-7.77e-06	-1.80e-05	-6.446
Code_Aster	144.895	-20.58	828828	1.71e-27	2.73	9.34e-26	-4.13e-14	-1.652	3.06e-13
Sesam	143.164	-21.53	809150	7.95e-30	2.72	2.49e-27	-2.82e-15	-1.651	4.99e-14

Mode 8

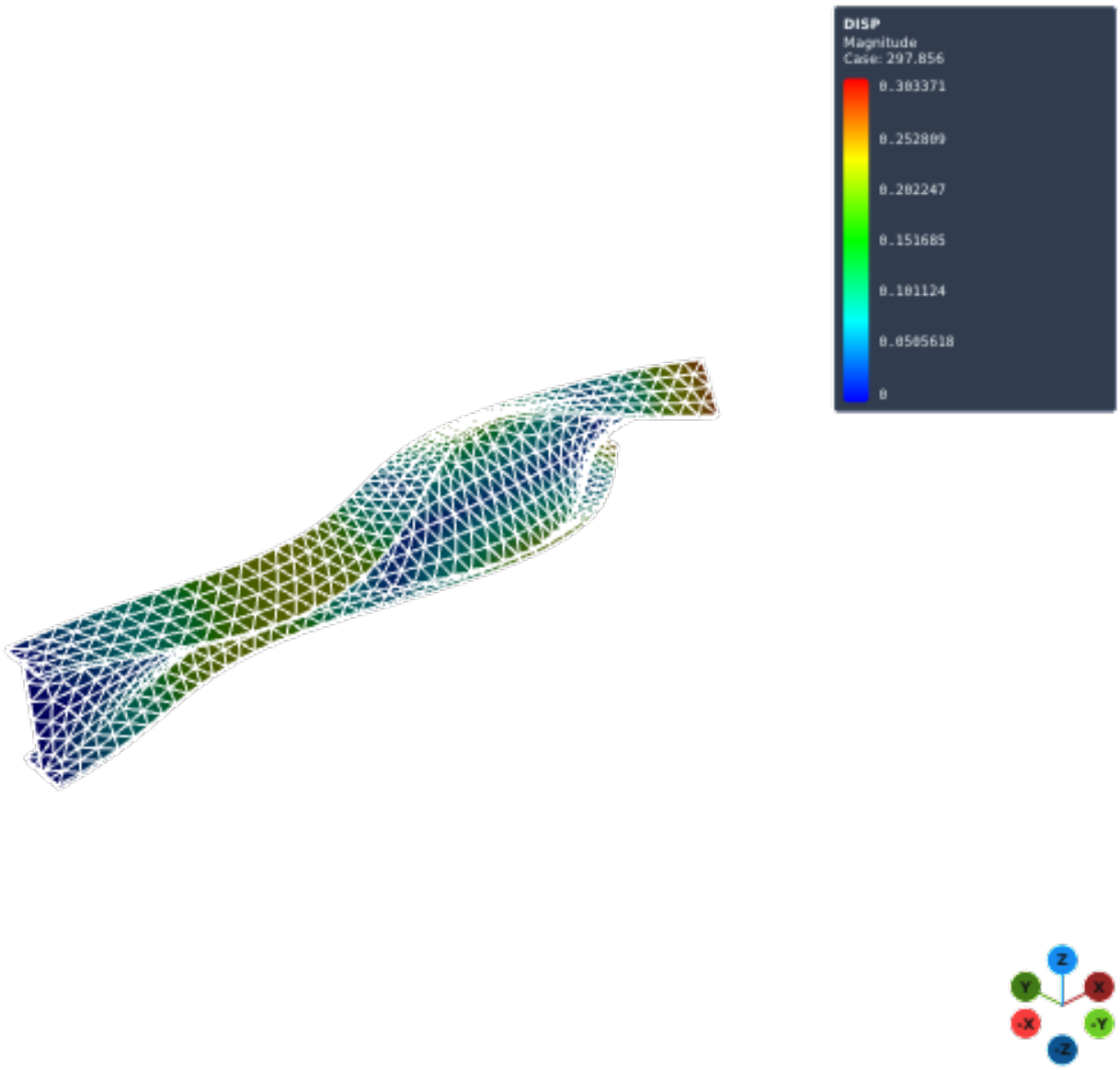

Figure 115: Calculix — mode 8, 297.856 Hz

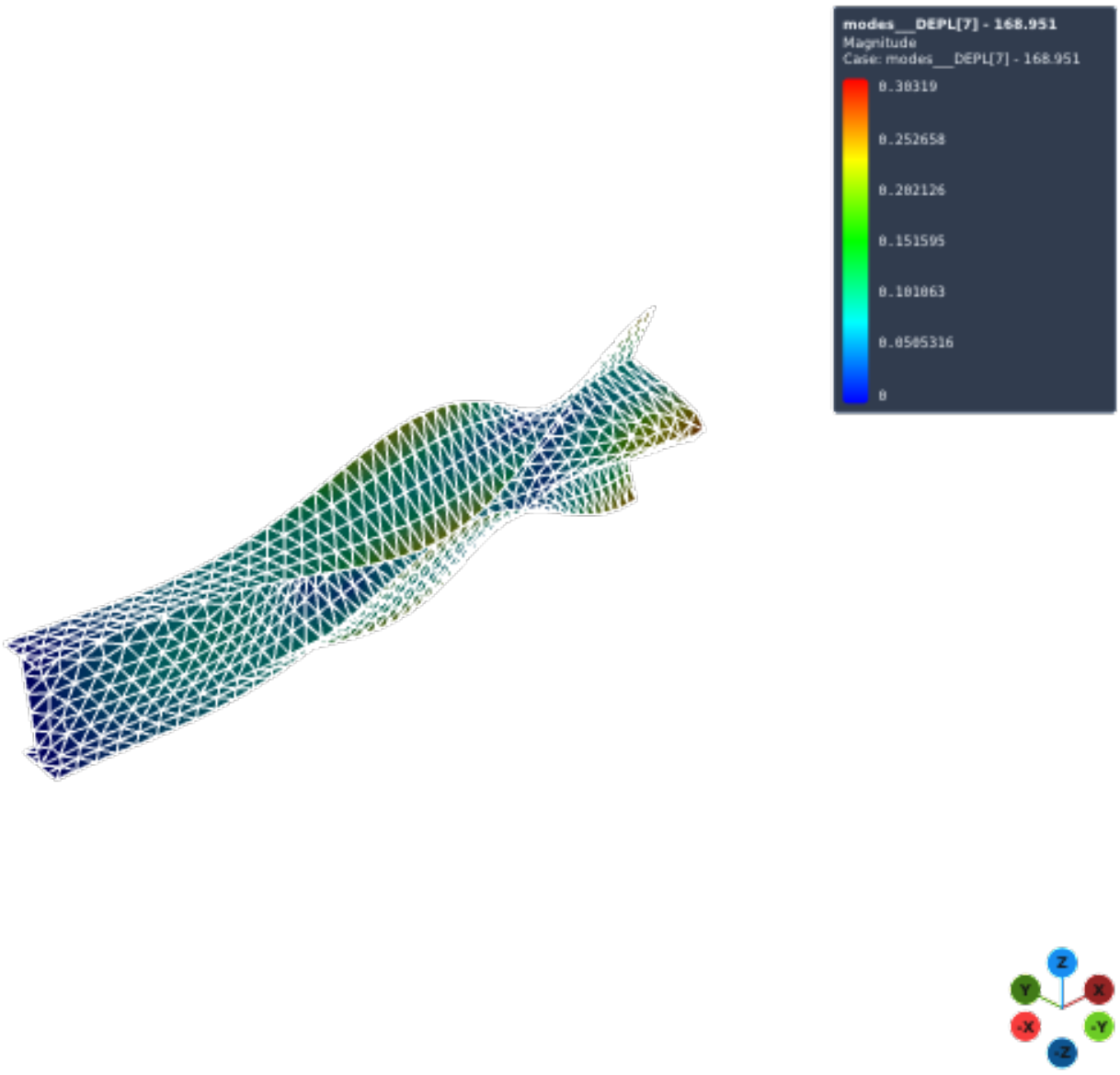

Figure 116: Code_Aster — mode 8, 168.951 Hz

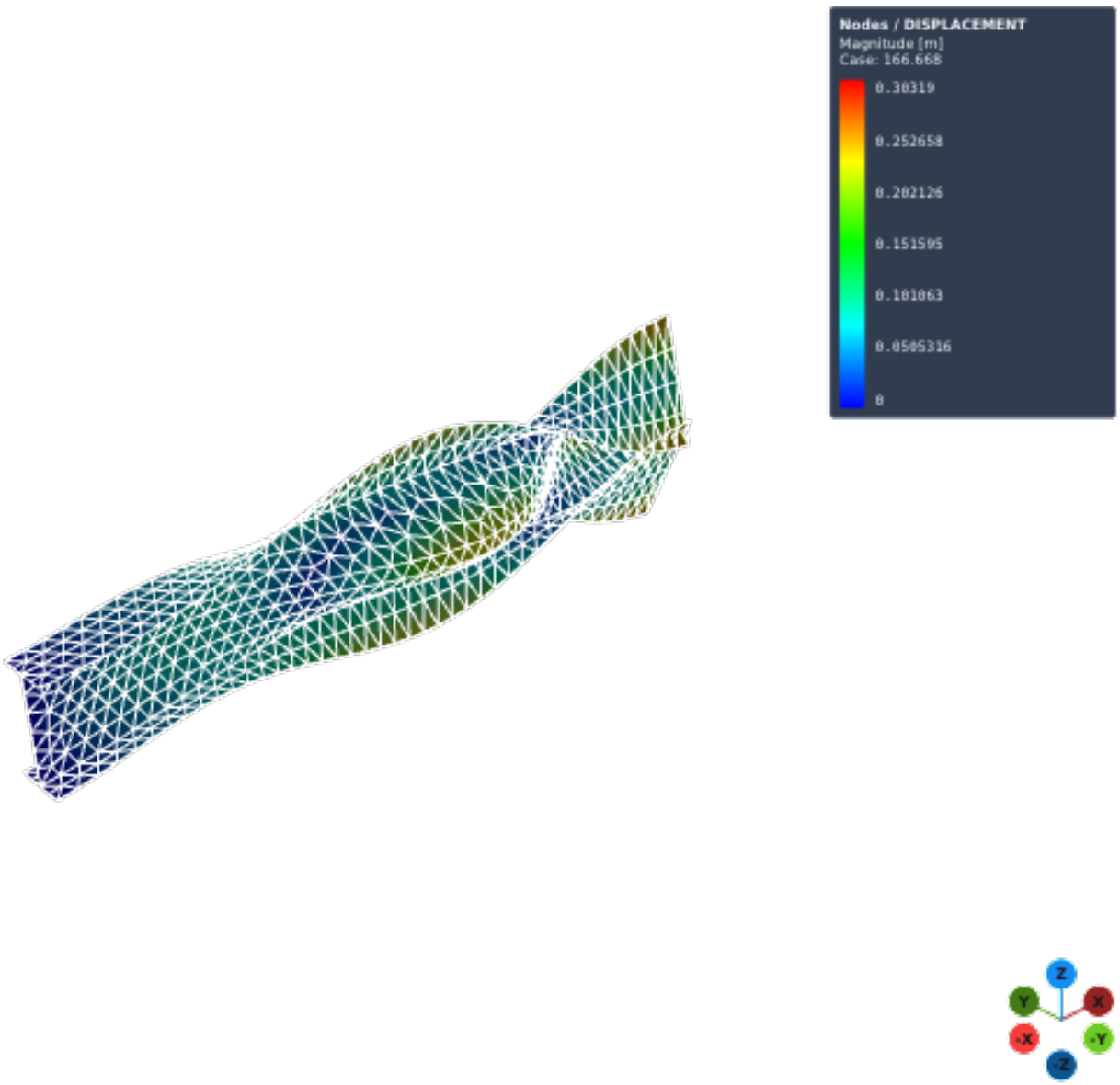

Figure 117: Sesam — mode 8, 166.668 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	297.856	+41.06	3.50246e+06	2.44e-09	7.20e-06	2.53e-14	4.94e-05	0.003	1.59e-07
Code_Aster	168.951	-19.99	1.12689e+06	5.89e-27	6.03	3.09e-26	7.67e-14	-2.455	-1.76e-13
Sesam	166.668	-21.07	1.09664e+06	4.67e-28	5.94	7.43e-31	-2.16e-14	2.438	8.62e-16

Mode 9

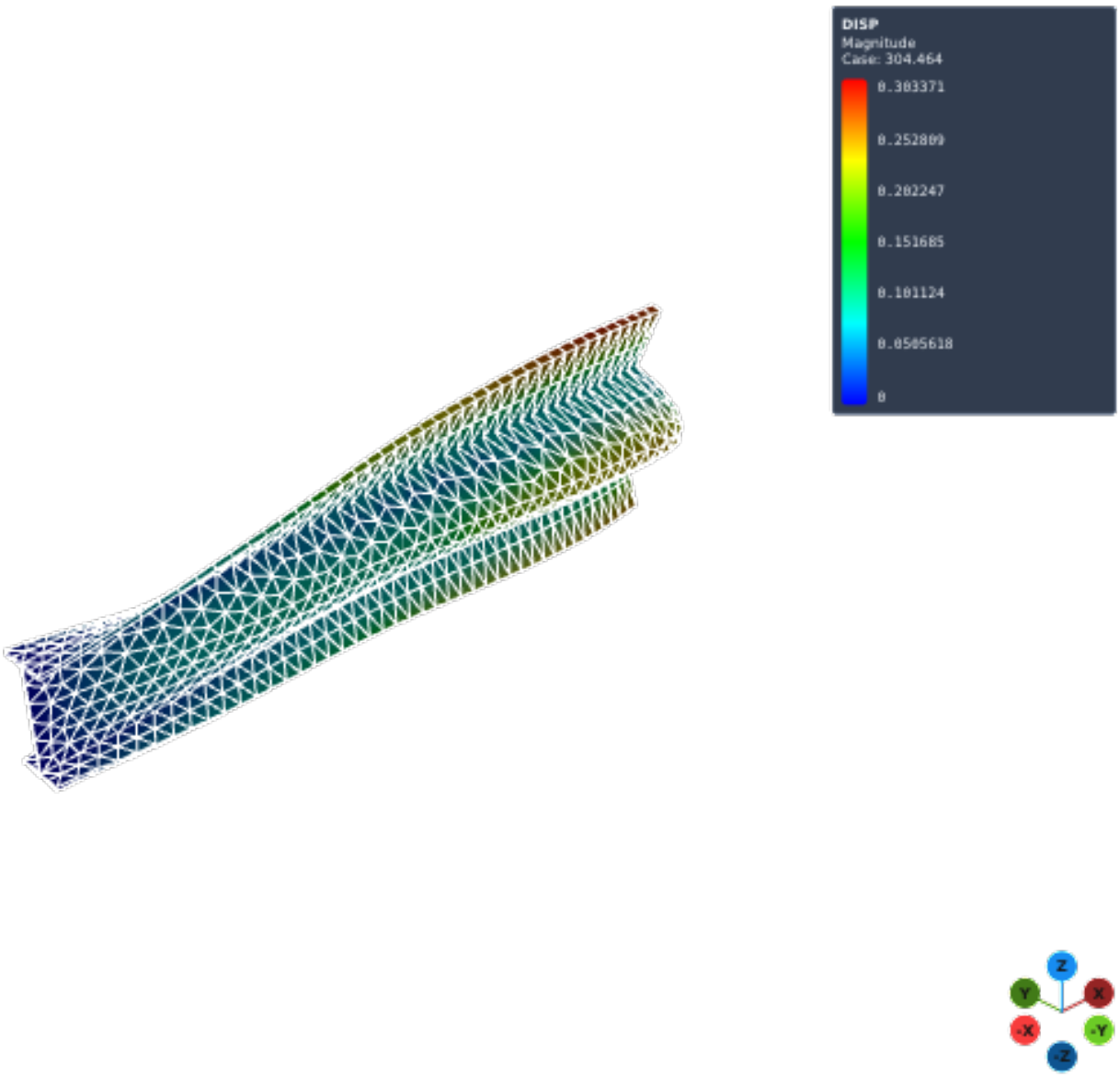

Figure 118: Calculix — mode 9, 304.464 Hz

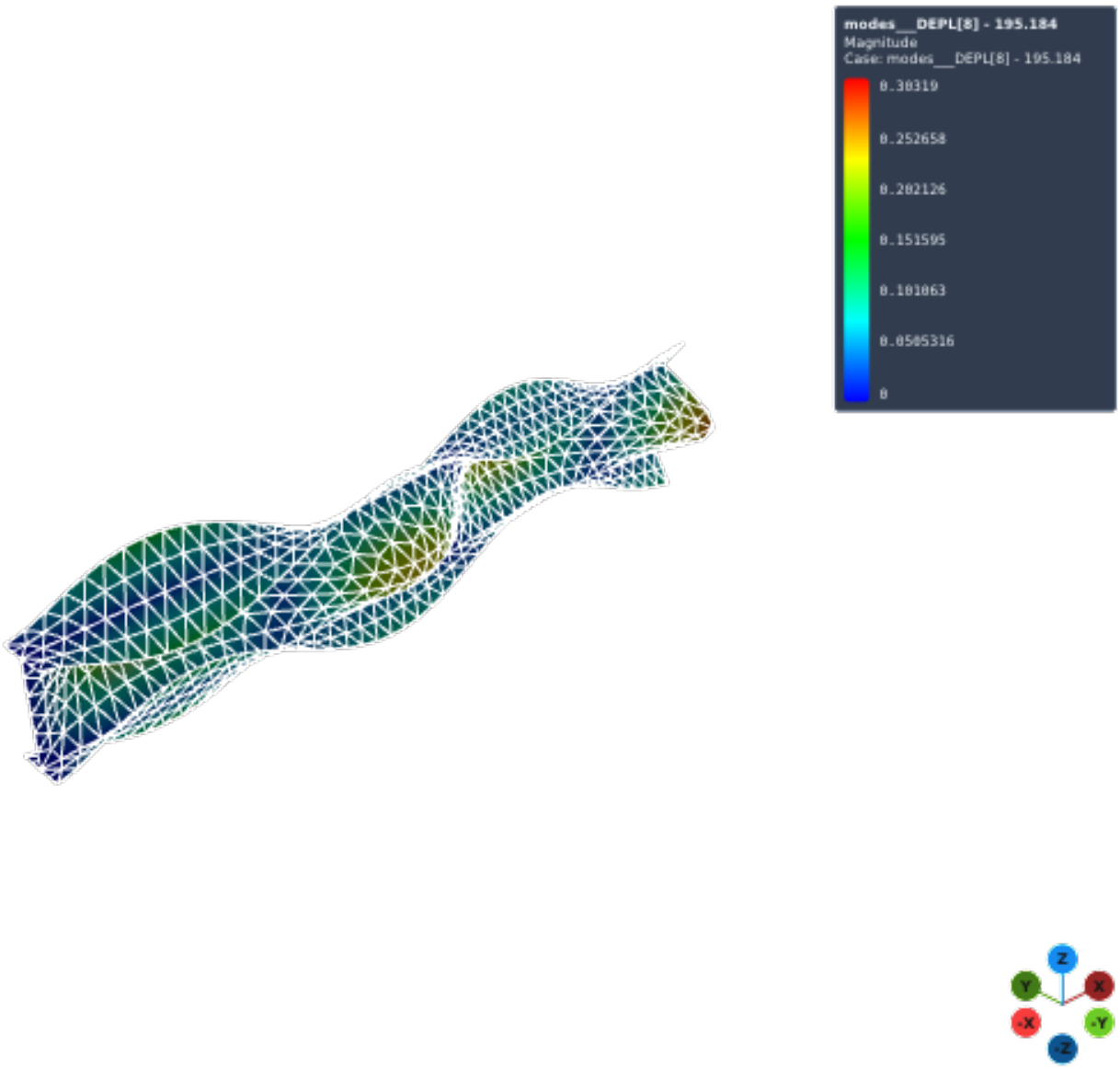

Figure 119: Code_Aster — mode 9, 195.184 Hz

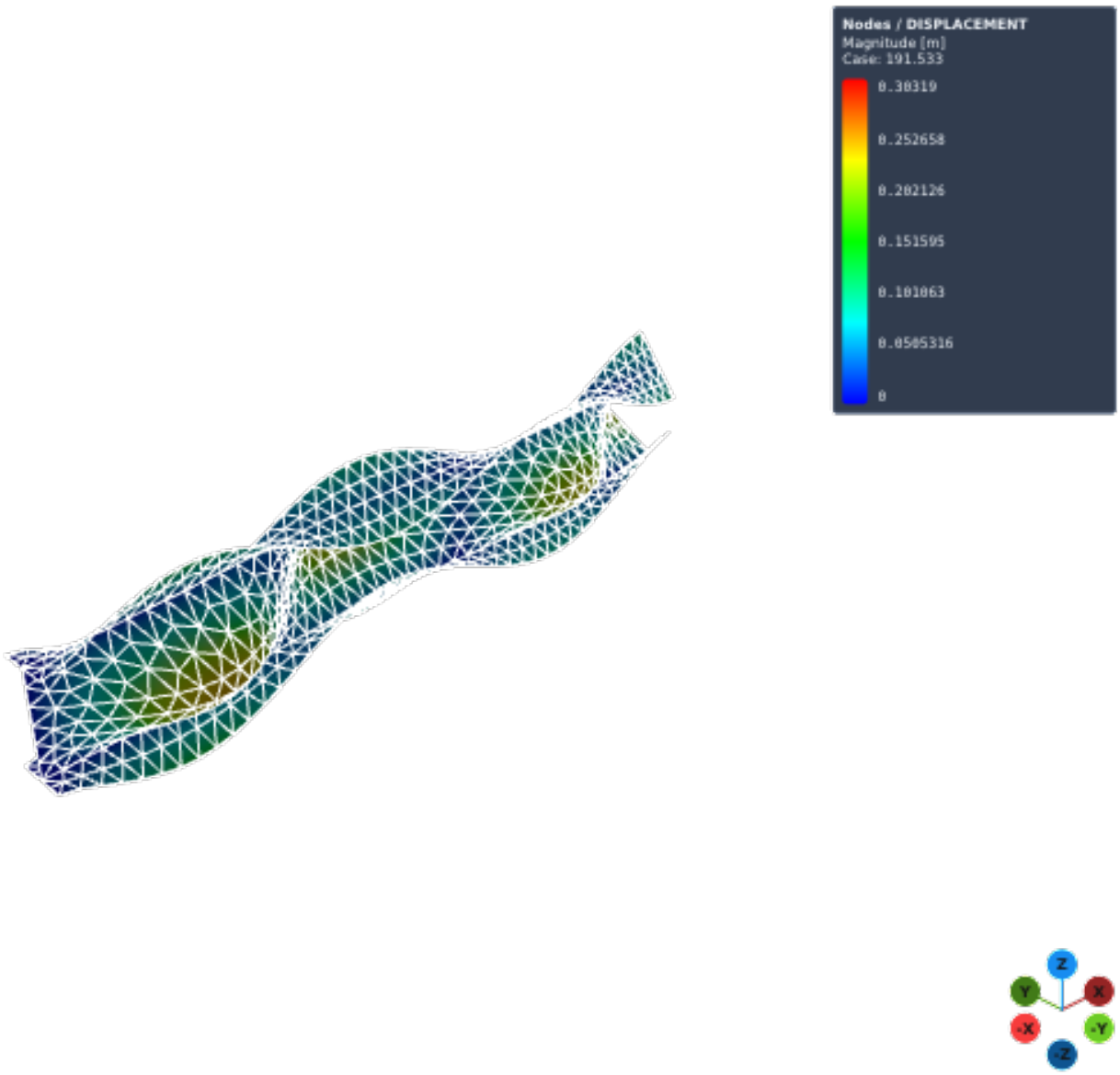

Figure 120: Sesam — mode 9, 191.533 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	304.464	+32.15	3.65958e+06	2.70e-09	0.02	8.66e-13	-5.19e-05	-0.156	9.30e-07
Code_Aster	195.184	-15.28	1.504e+06	5.26e-28	0.62	7.49e-27	2.29e-14	0.788	-8.66e-14
Sesam	191.533	-16.87	1.44826e+06	2.49e-28	0.61	5.28e-28	-1.58e-14	-0.781	-2.30e-14

Mode 10

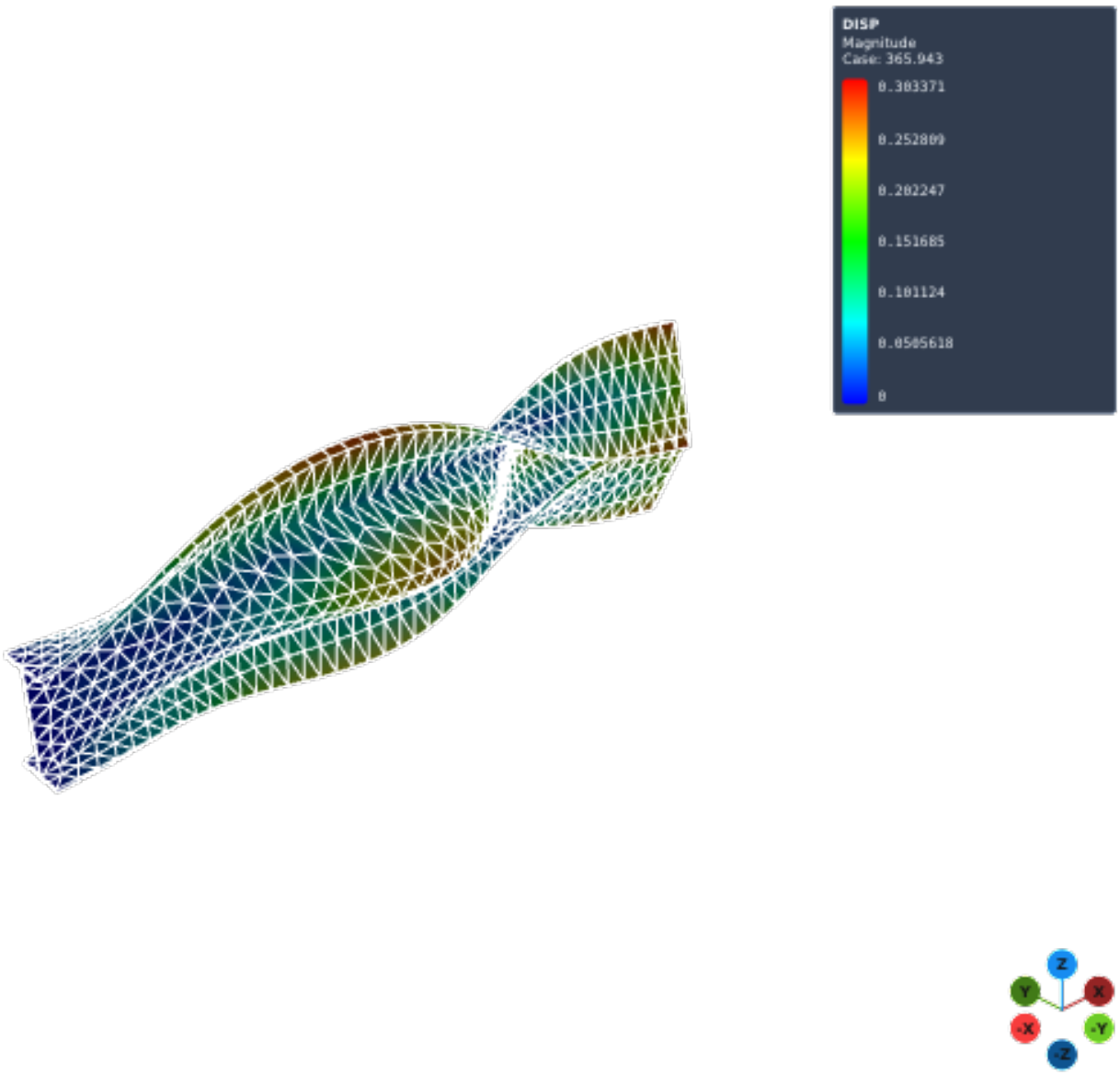

Figure 121: Calculix — mode 10, 365.943 Hz

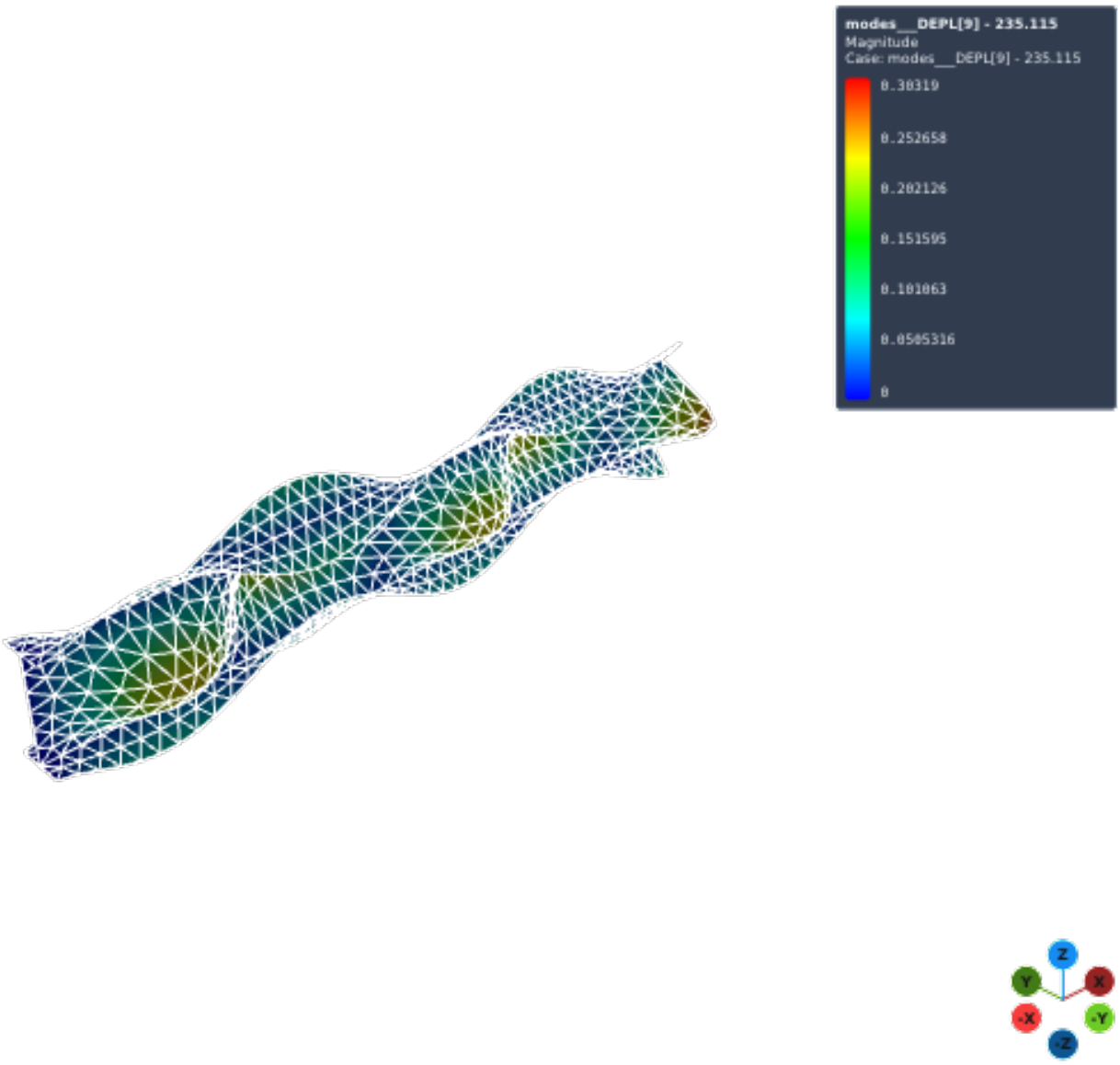

Figure 122: Code_Aster — mode 10, 235.115 Hz

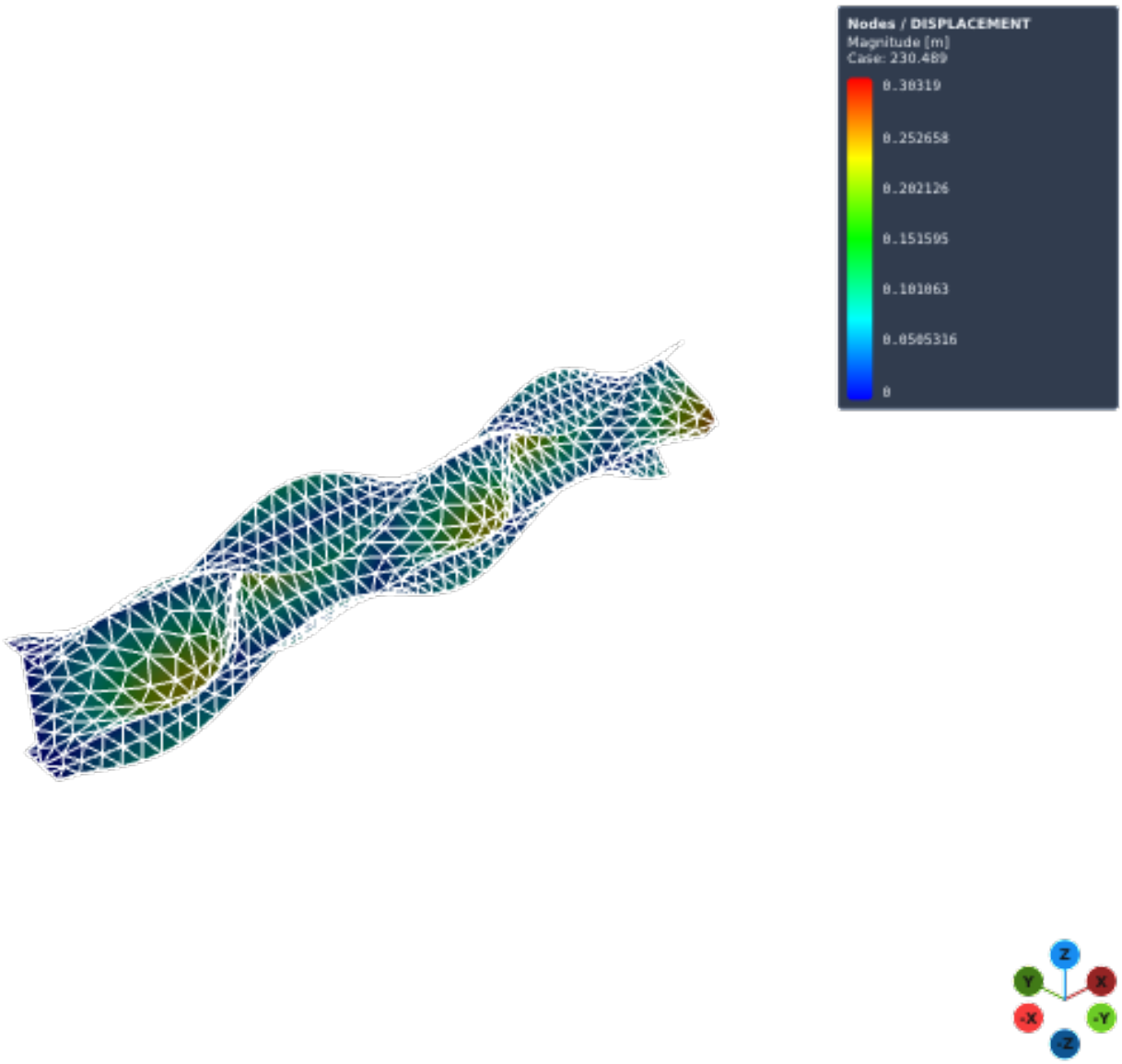

Figure 123: Sesam — mode 10, 230.489 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	365.943	+32.02	5.28673e+06	1.72e-09	1.48	3.23e-12	-4.15e-05	1.218	-1.80e-06
Code_Aster	235.115	-15.18	2.18233e+06	2.11e-27	1.32	4.83e-27	4.59e-14	-1.148	6.95e-14
Sesam	230.489	-16.85	2.09731e+06	1.88e-28	1.27	4.80e-28	1.37e-14	-1.128	-2.19e-14

Mode 11

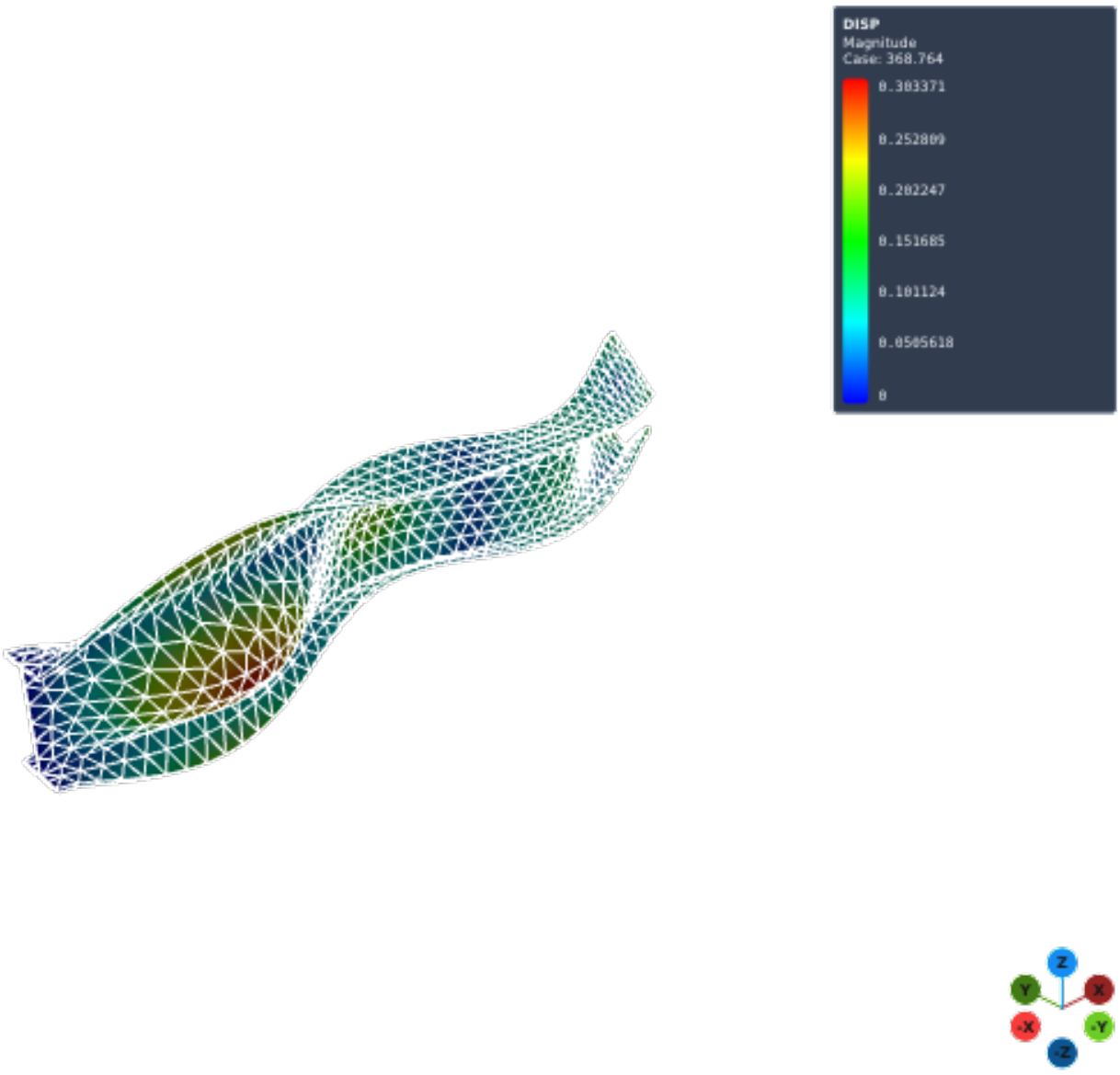

Figure 124: Calculix — mode 11, 368.764 Hz

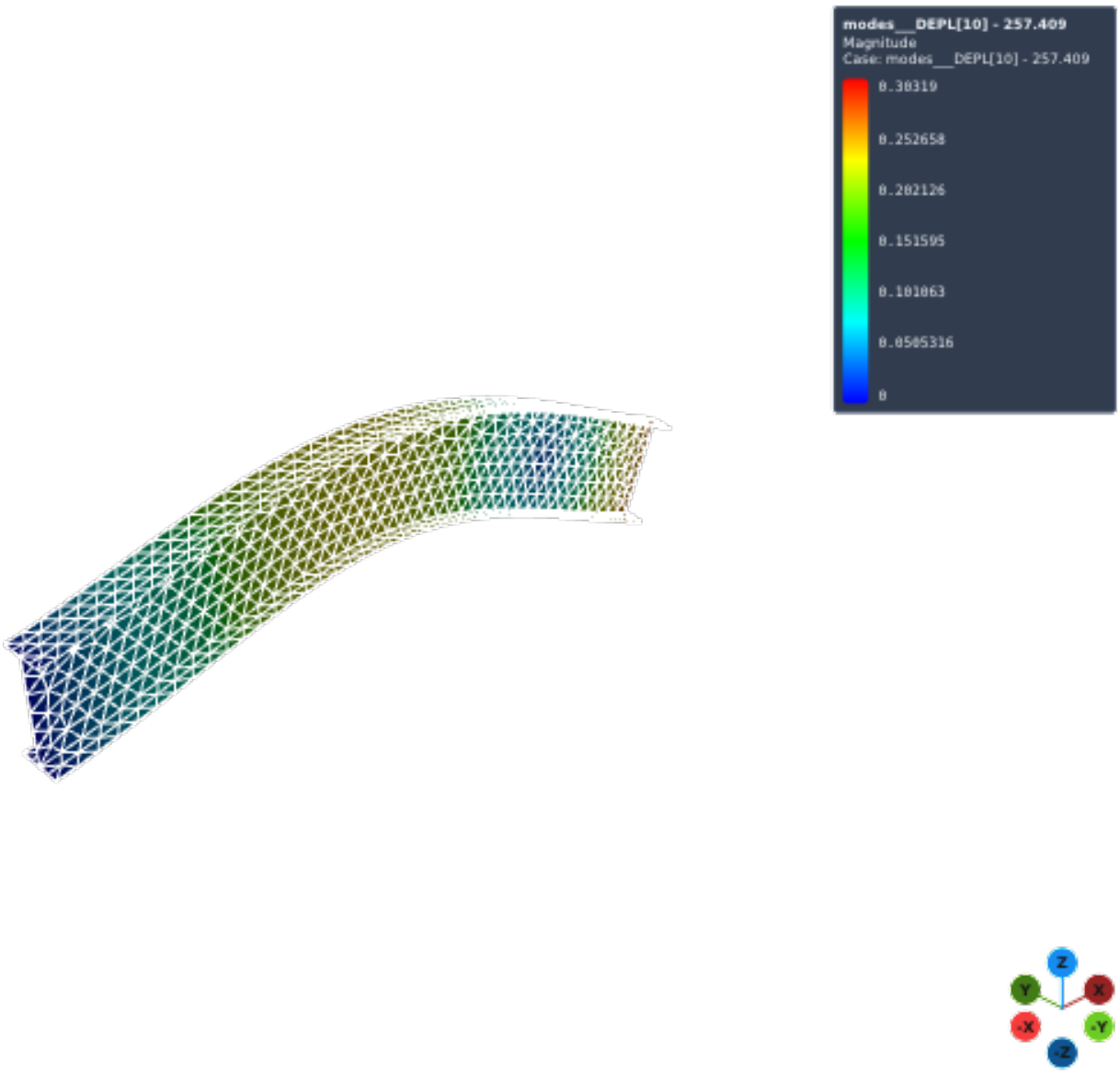

Figure 125: Code_Aster — mode 11, 257.409 Hz

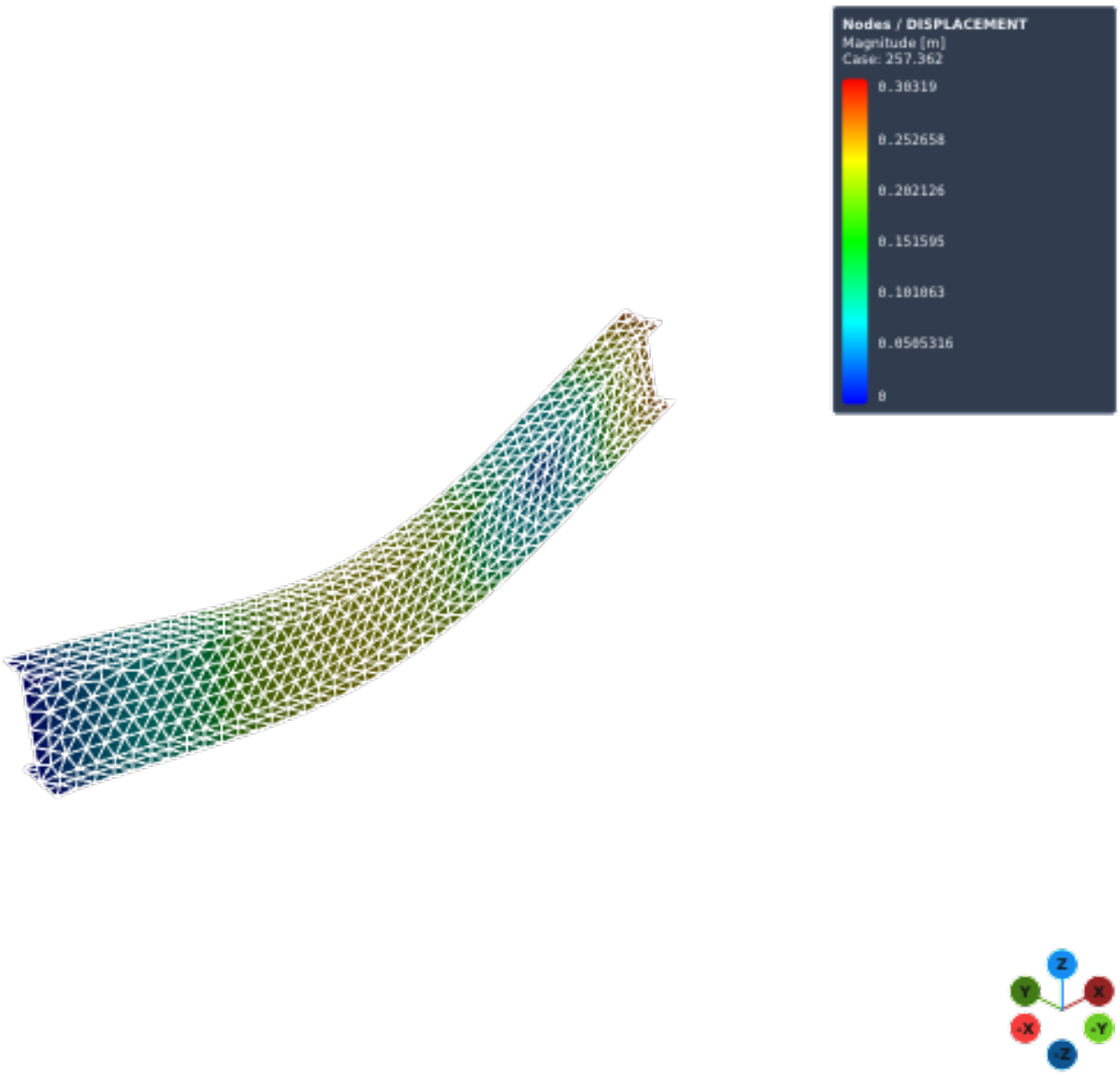

Figure 126: Sesam — mode 11, 257.362 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Calculix	368.764	+25.21	5.36853e+06	1.81e-10	3.61	4.88e-12	-1.34e-05	-1.900	2.21e-06
Code_Aster	257.409	-12.60	2.61582e+06	1.20e-09	1.38e-26	41.64	3.47e-05	-1.17e-13	6.453
Sesam	257.362	-12.61	2.61487e+06	6.51e-09	3.57e-22	41.62	-8.07e-05	-1.89e-11	-6.451

Shell, 1st order, QUAD

Mode 1

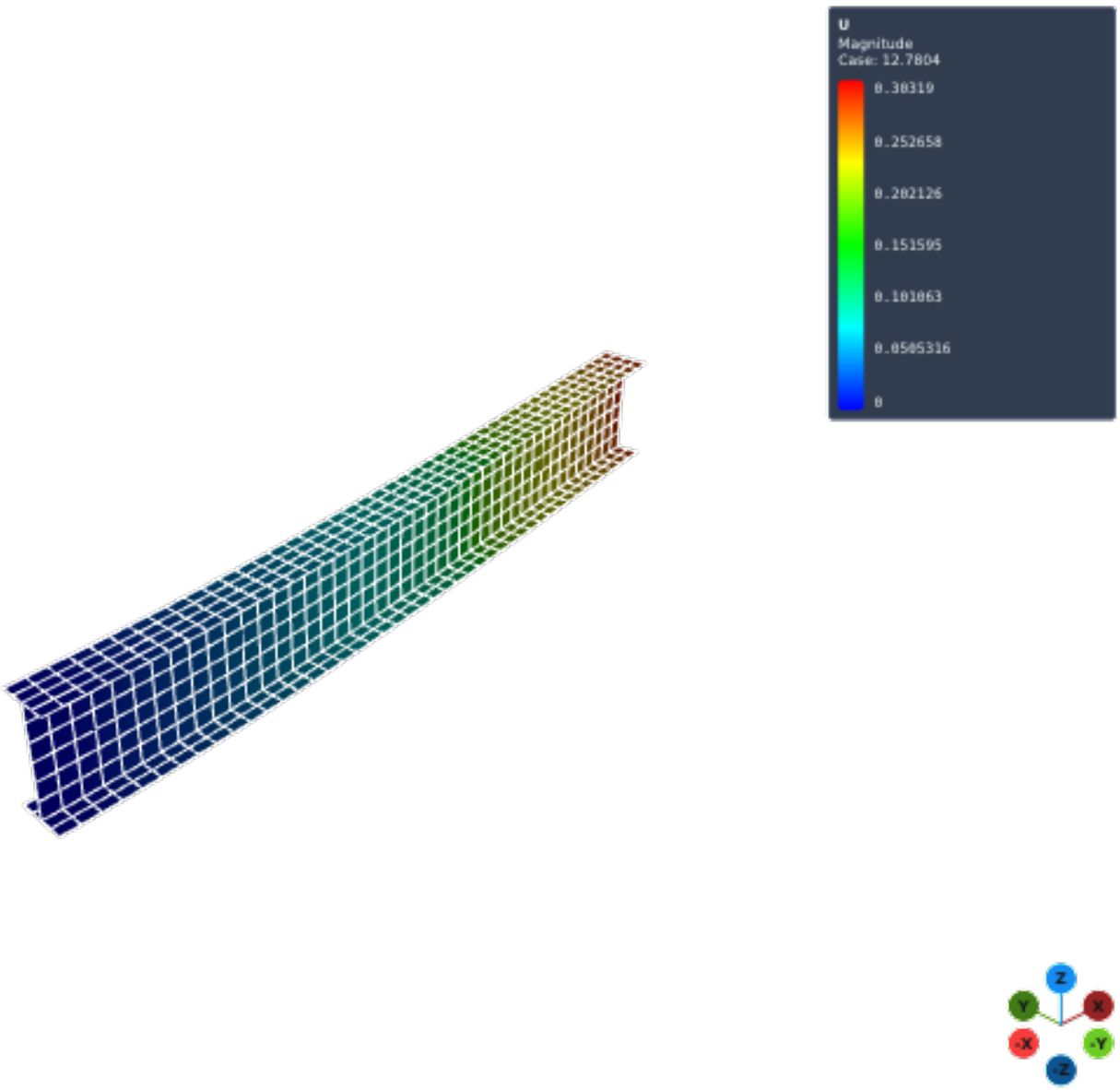

Figure 127: Abaqus — mode 1, 12.780 Hz

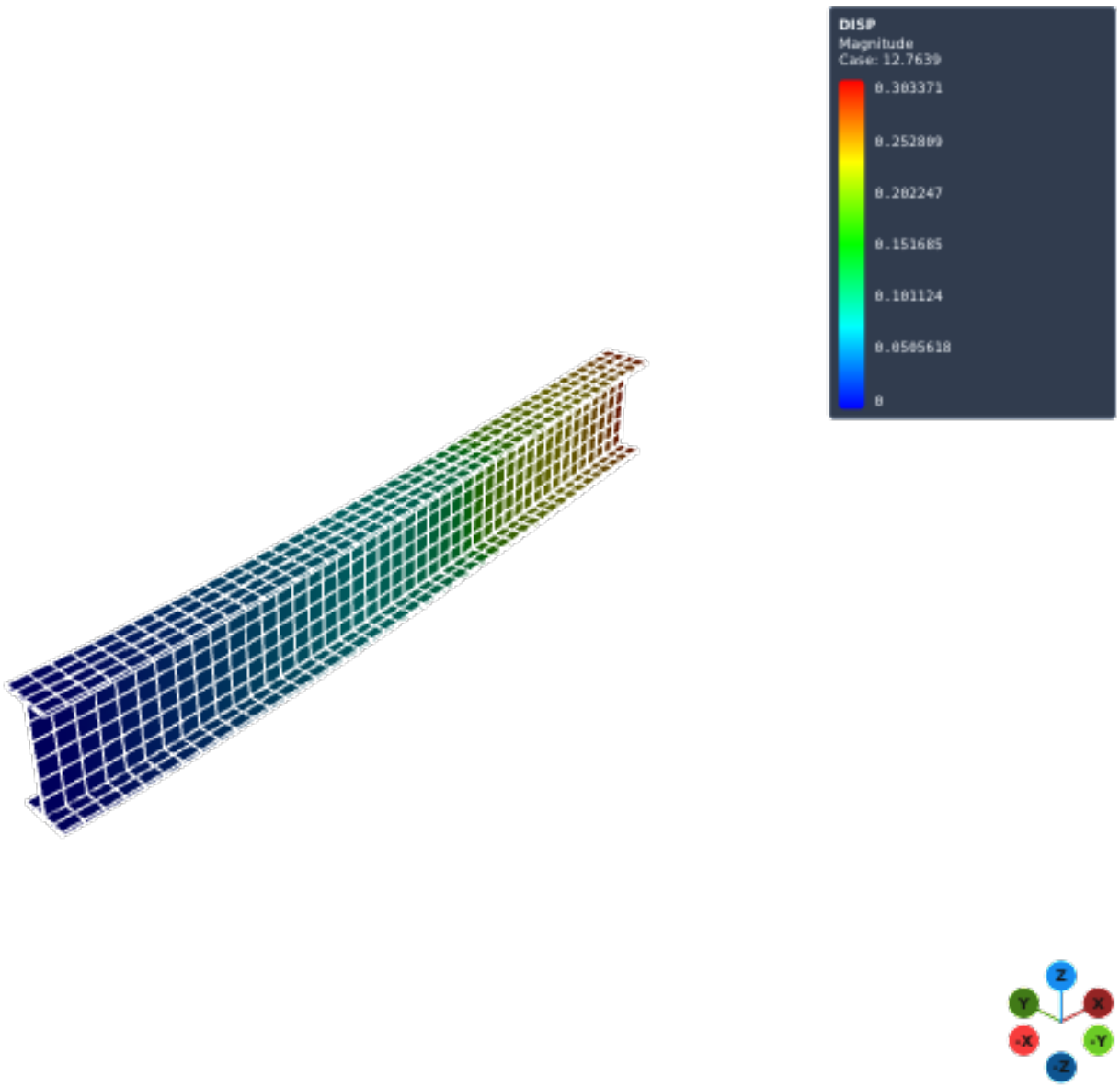

Figure 128: Calculix — mode 1, 12.764 Hz

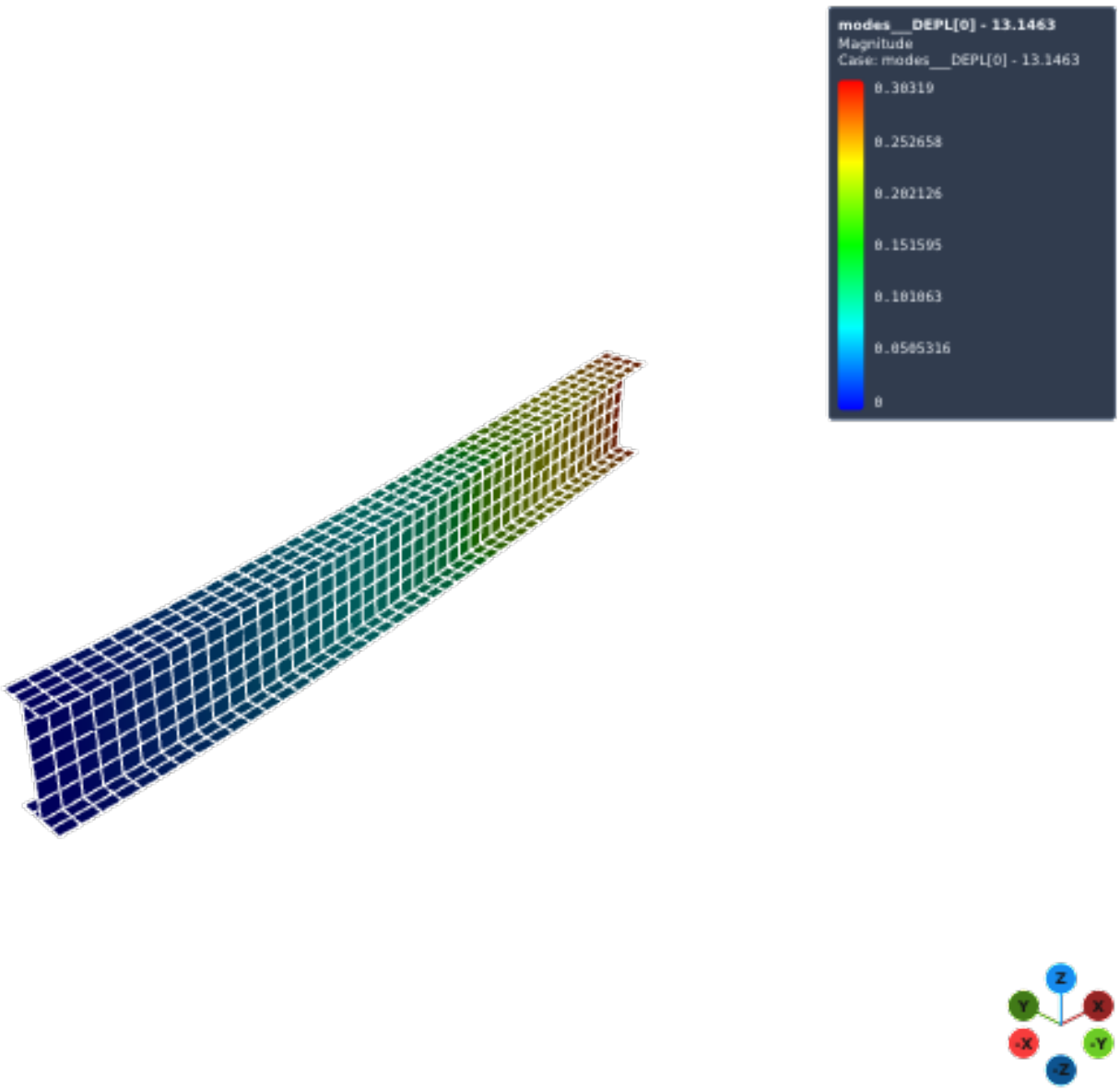

Figure 129: Code_Aster — mode 1, 13.146 Hz

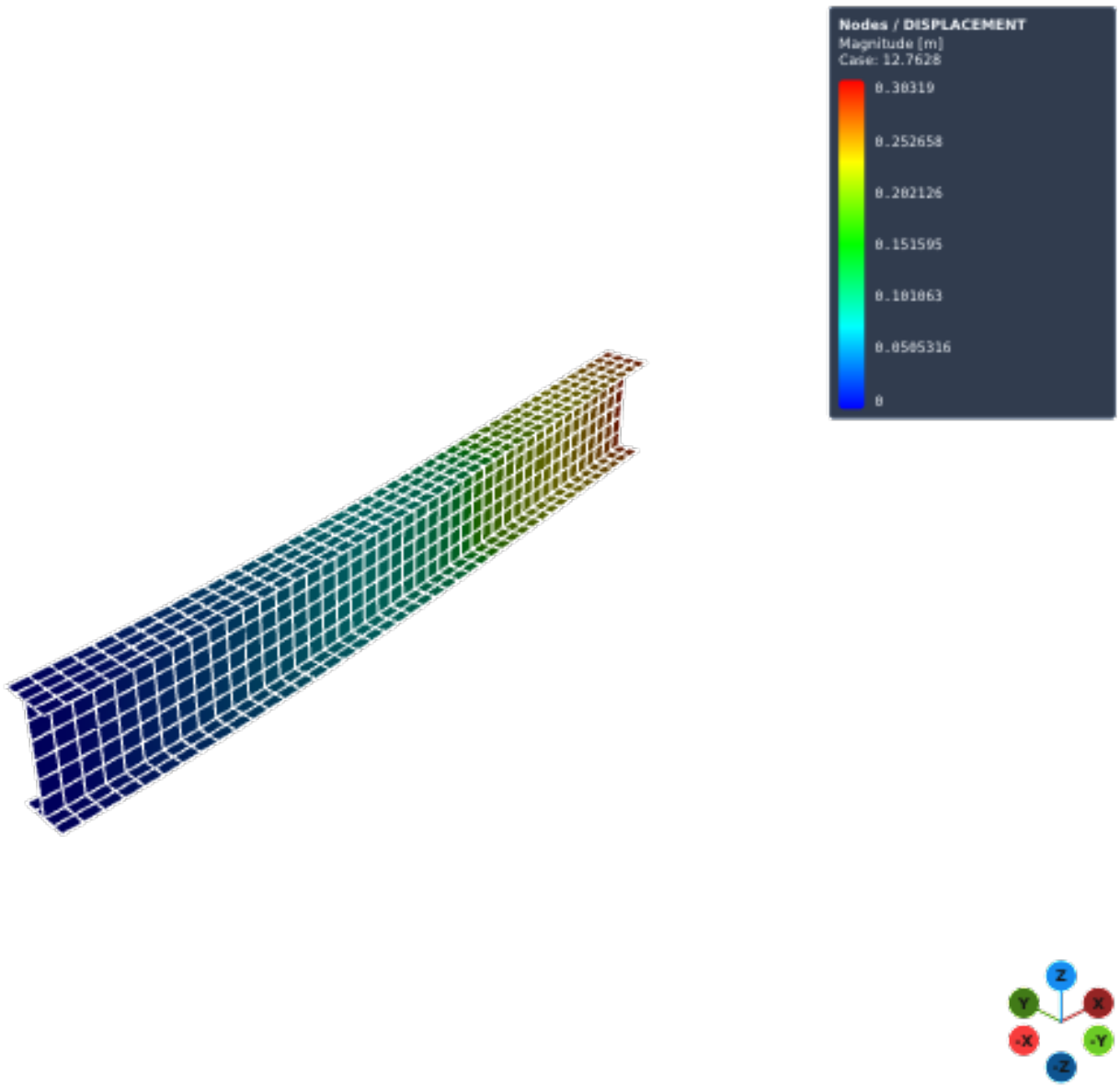

Figure 130: Sesam — mode 1, 12.763 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.780	-0.65	6448.4	7.41e-27	119.92	1.38e-25	-8.61e-14	10.951	-3.71e-13
Calculix	12.764	-0.77	6431.71	1.18e-24	119.90	1.74e-12	1.09e-12	10.950	1.32e-06
Code_Aster	13.146	+2.20	6822.87	5.73e-28	119.92	3.53e-25	2.39e-14	10.951	-5.94e-13
Sesam	12.763	-0.78	6430.64	2.41e-28	119.90	3.46e-25	-1.55e-14	10.950	-5.88e-13

Mode 2

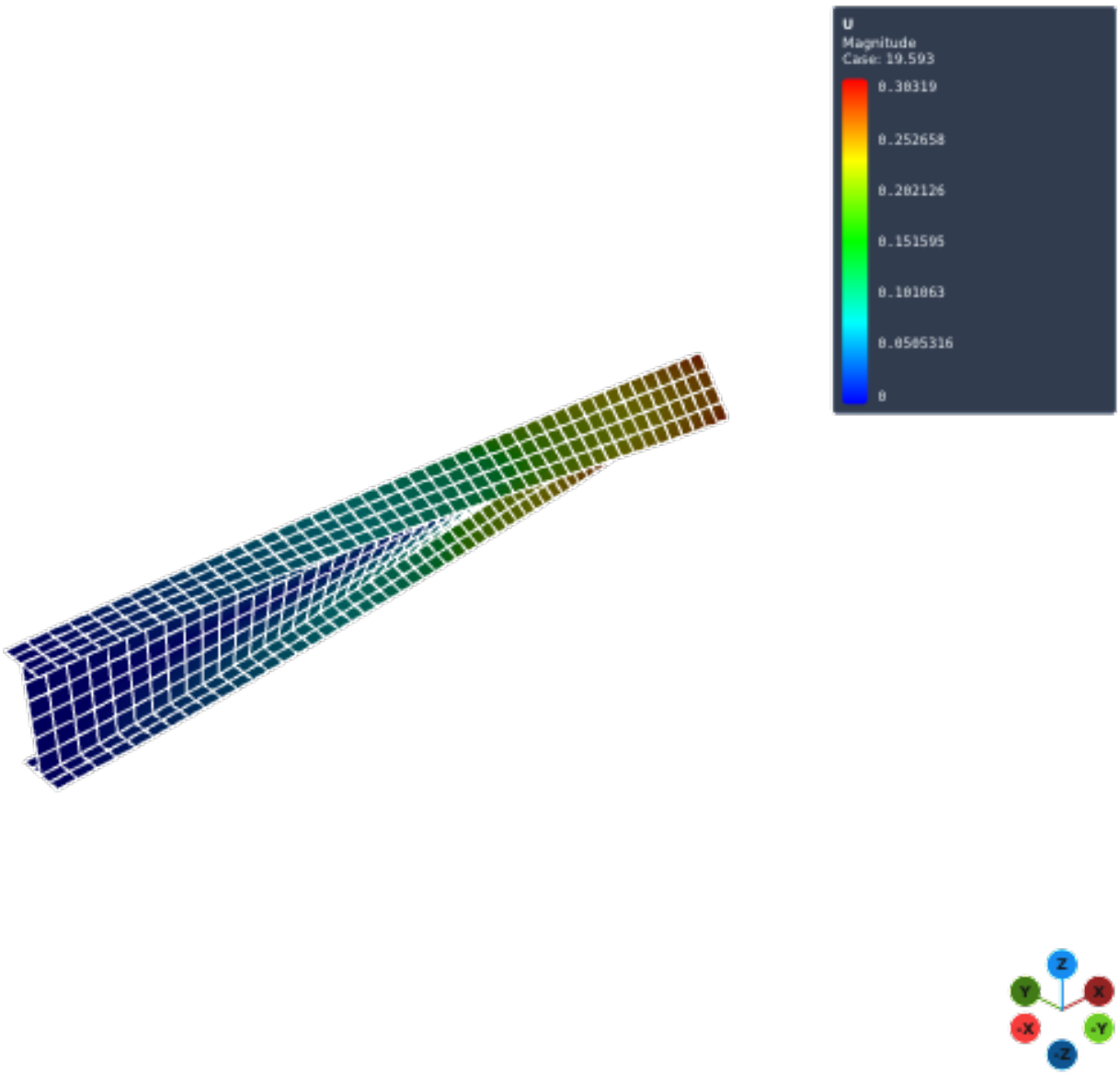

Figure 131: Abaqus — mode 2, 19.593 Hz

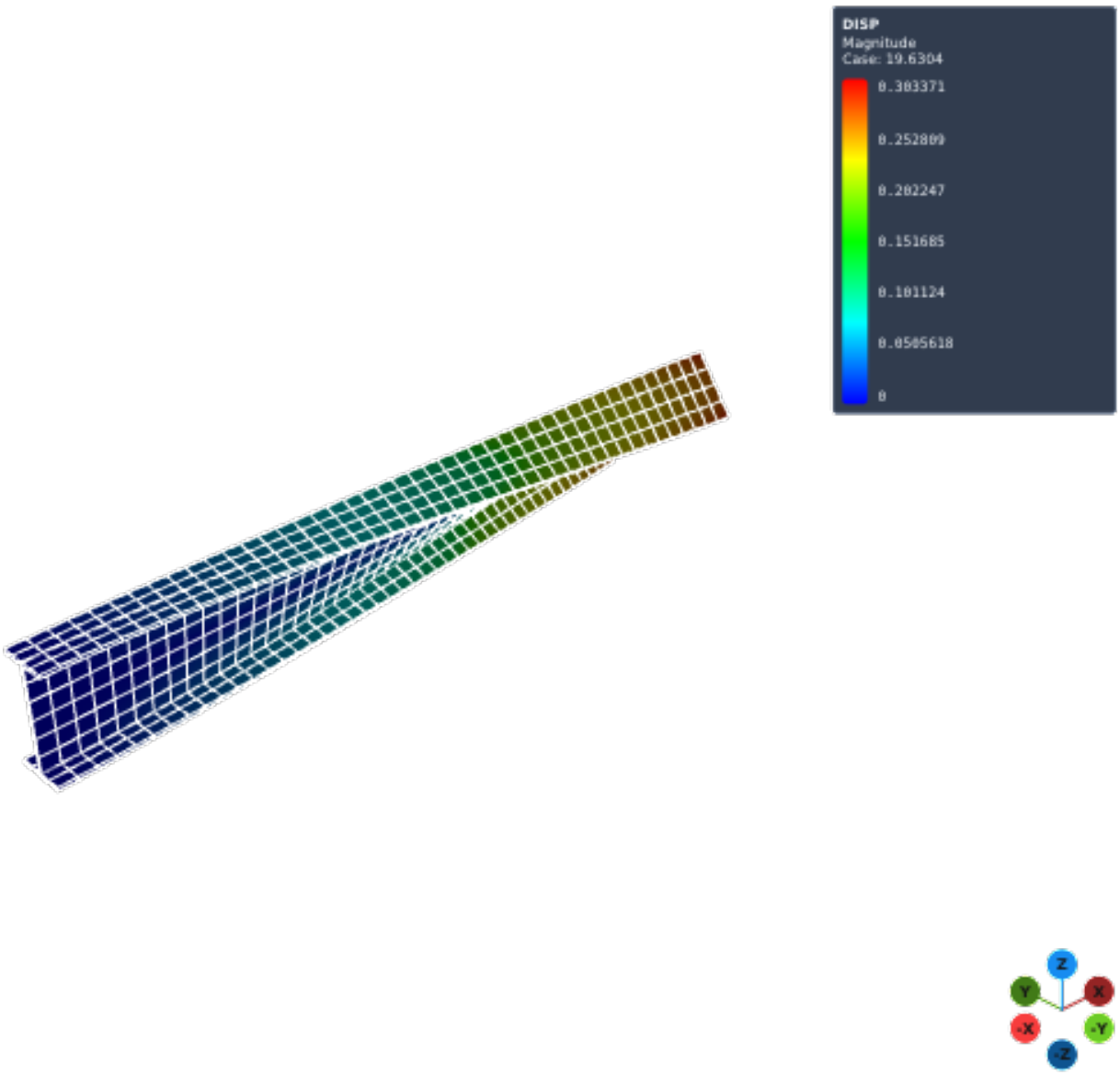

Figure 132: Calculix — mode 2, 19.630 Hz

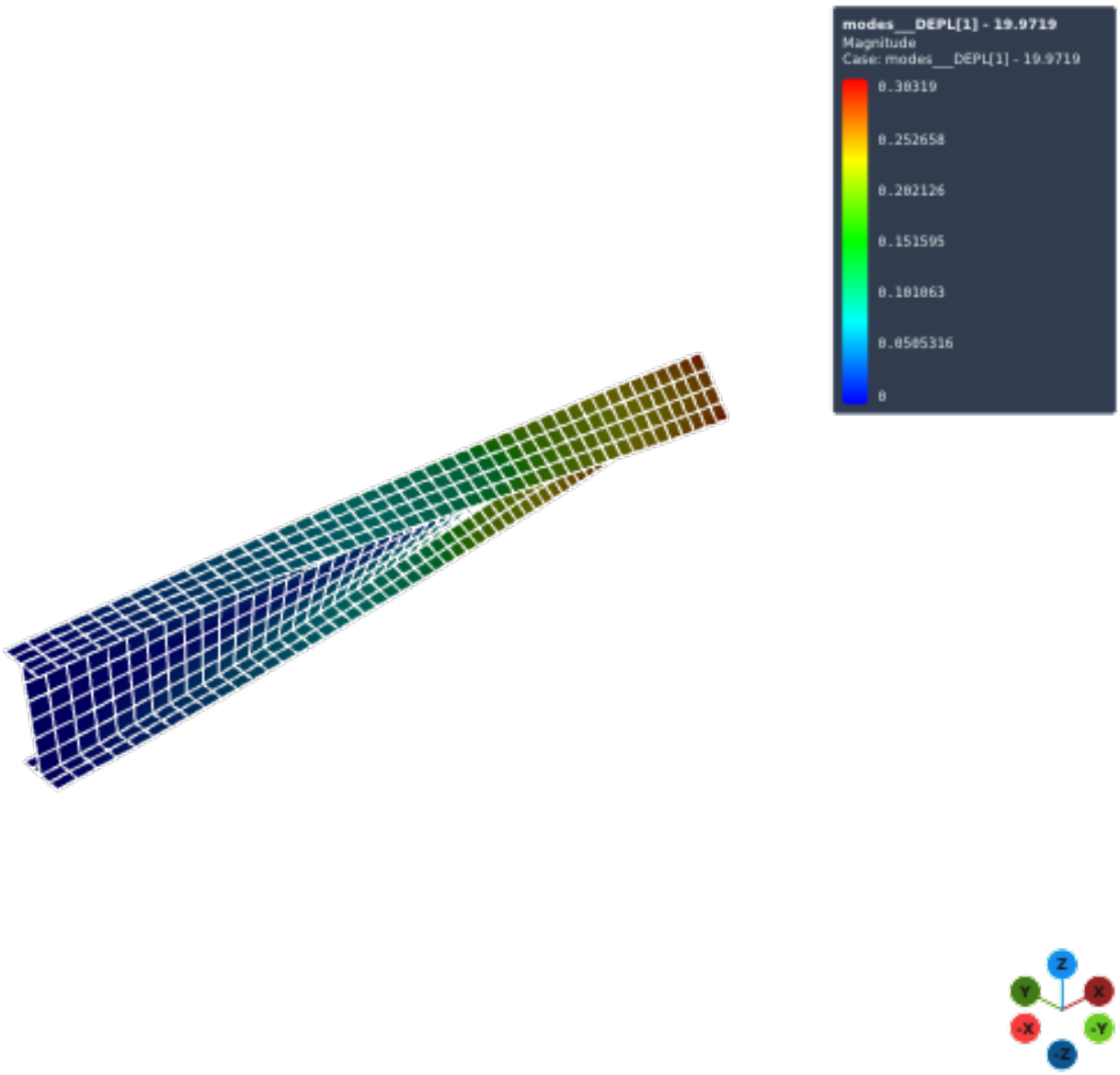

Figure 133: Code_Aster — mode 2, 19.972 Hz

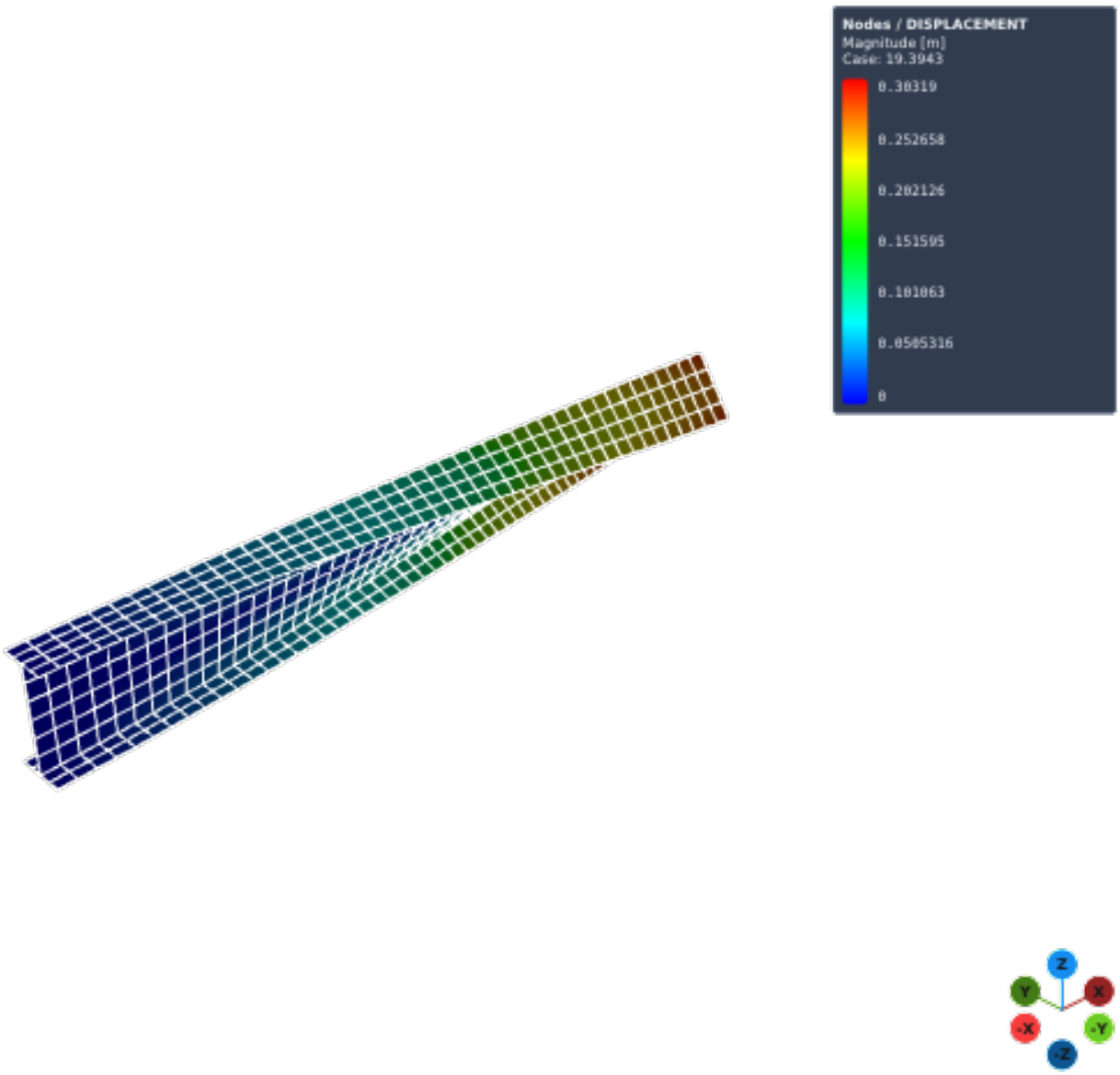

Figure 134: Sesam — mode 2, 19.394 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	19.593	-0.28	15155	1.46e-27	1.08e-19	7.67e-28	-3.83e-14	3.28e-10	2.77e-14
Calculix	19.630	-0.09	15213	3.11e-09	1.86e-17	2.79e-20	5.58e-05	4.31e-09	1.67e-10
Code_Aster	19.972	+1.65	15747.1	8.93e-27	4.15e-19	1.50e-24	-9.45e-14	-6.44e-10	1.23e-12
Sesam	19.394	-1.29	14849.3	5.82e-27	6.29e-19	1.98e-24	-7.63e-14	-7.93e-10	-1.41e-12

Mode 3

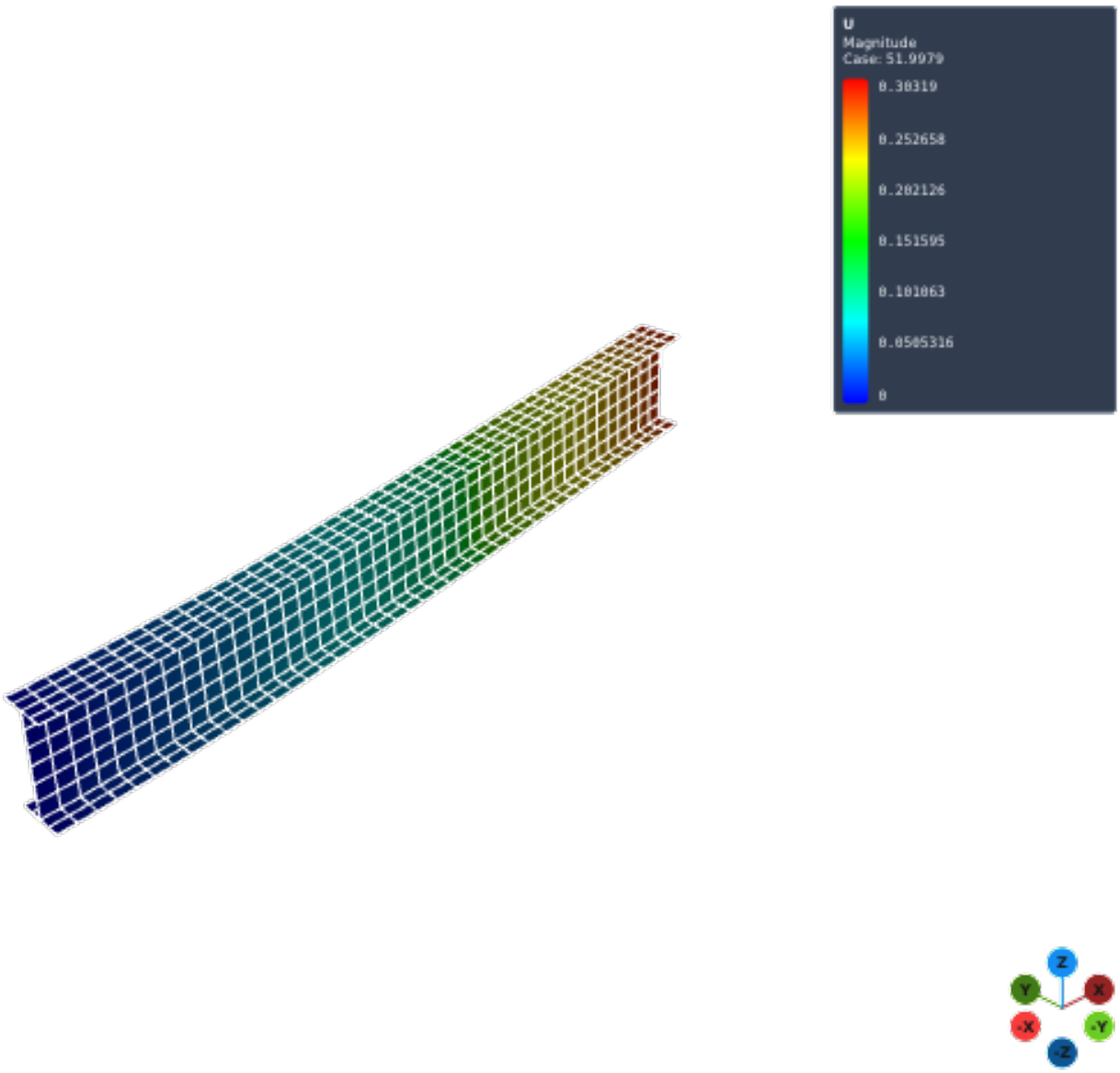

Figure 135: Abaqus — mode 3, 51.998 Hz

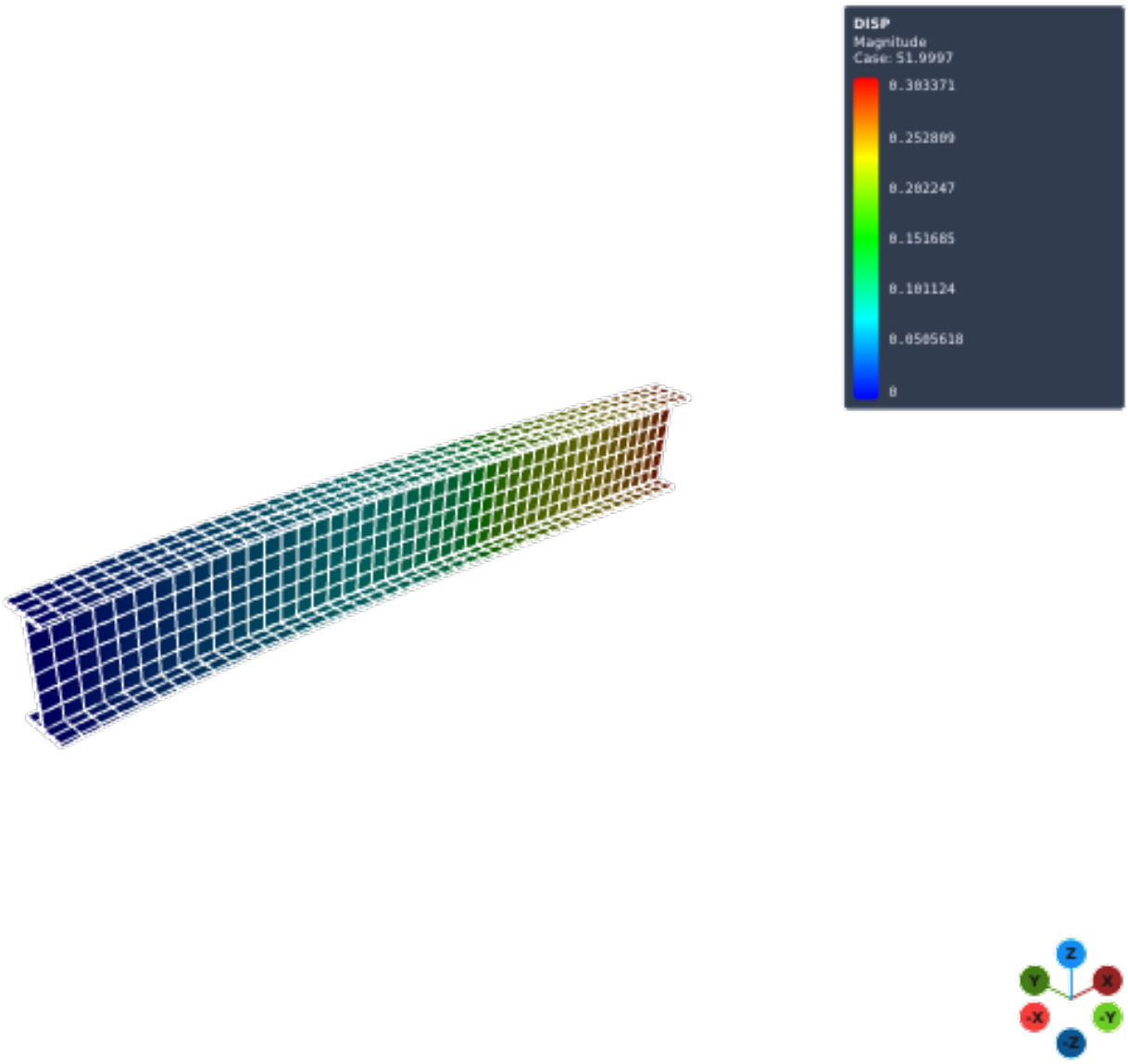

Figure 136: Calculix — mode 3, 52.000 Hz

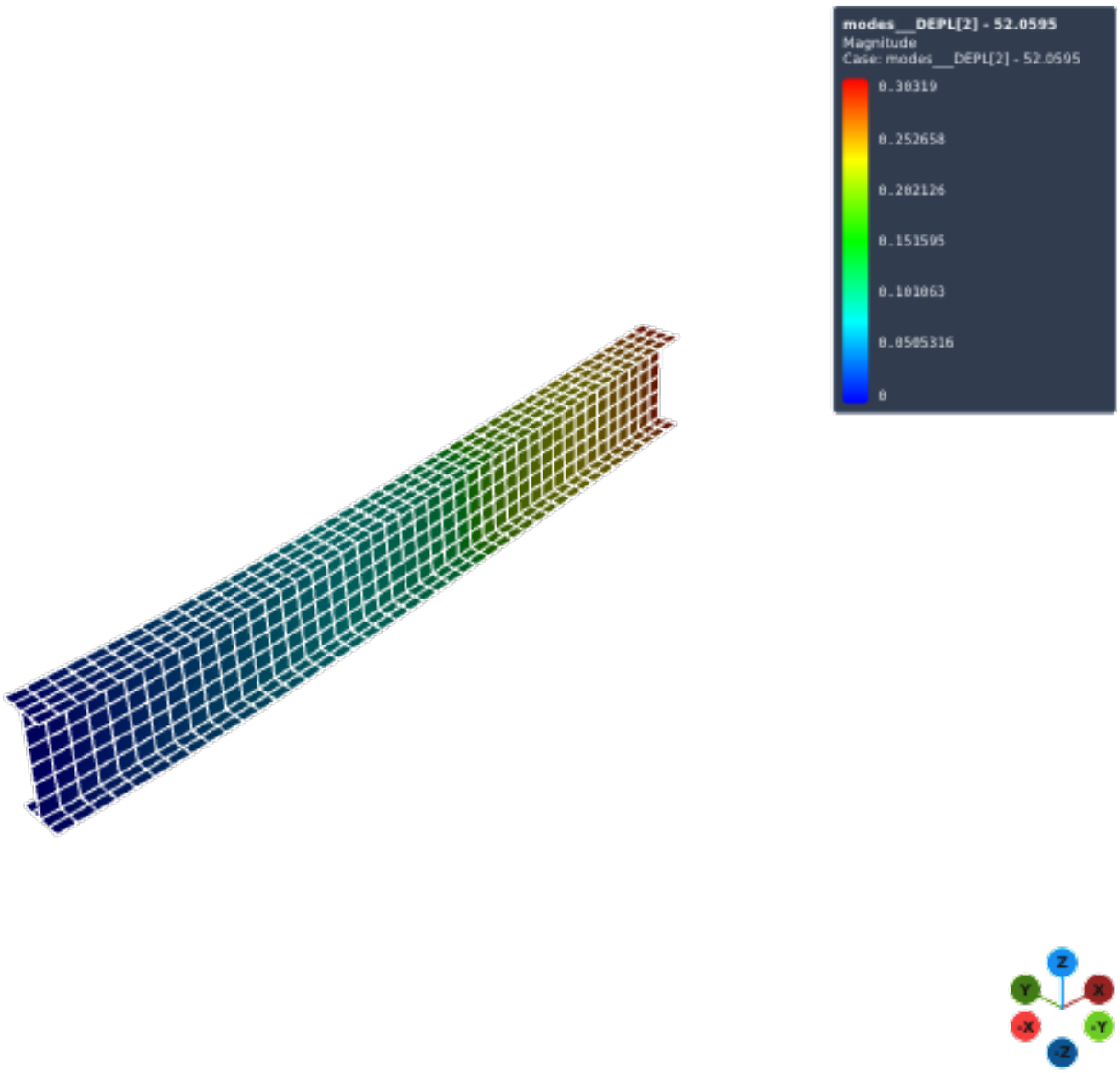

Figure 137: Code_Aster — mode 3, 52.060 Hz

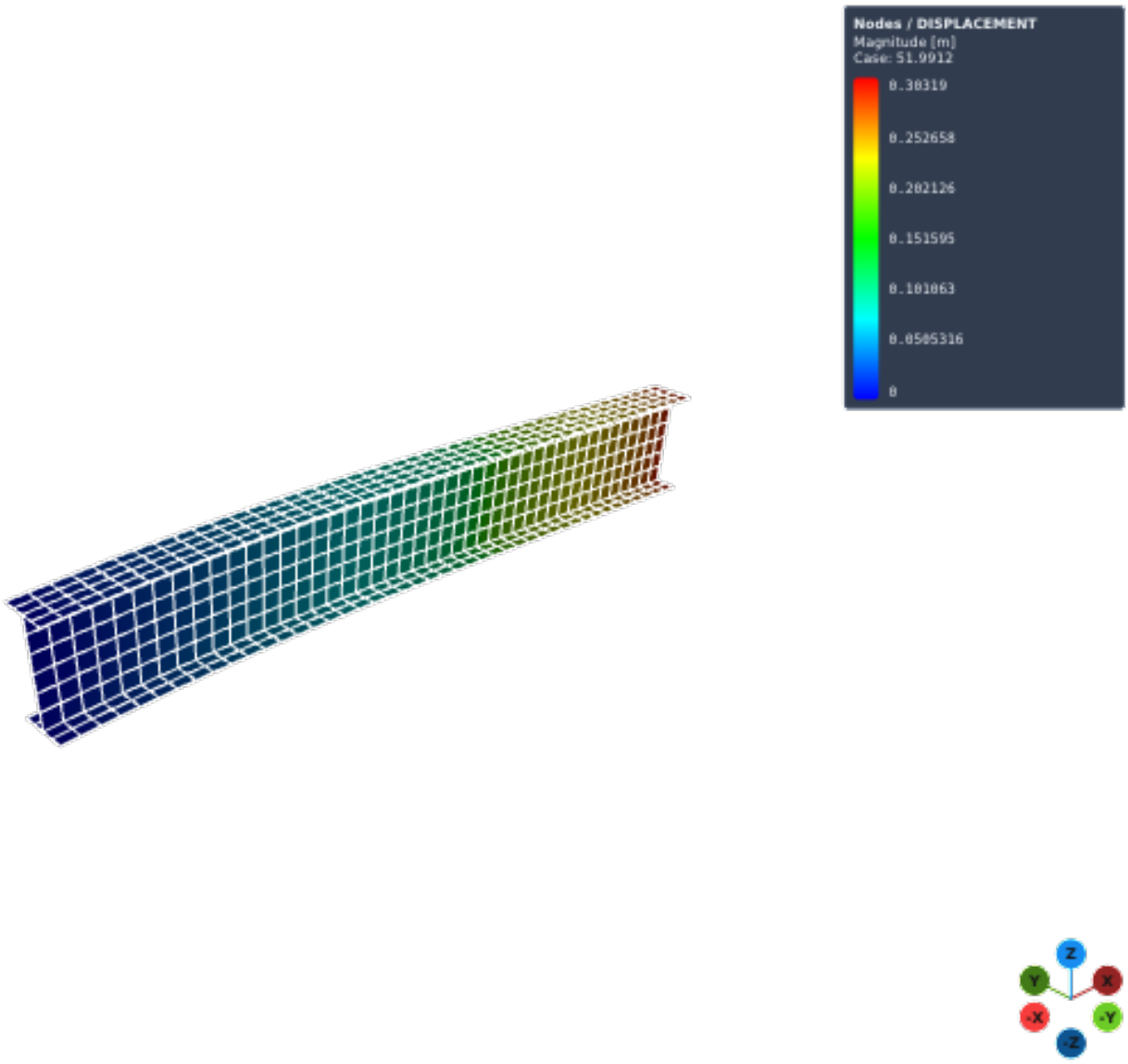

Figure 138: Sesam — mode 3, 51.991 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.998	-0.03	106741	1.35e-26	1.05e-26	122.01	-1.16e-13	1.03e-13	11.046
Calculix	52.000	-0.02	106749	9.22e-26	4.50e-11	121.98	-3.04e-13	6.71e-06	-11.045
Code_Aster	52.060	+0.09	106994	9.26e-28	2.72e-25	122.02	-3.04e-14	5.21e-13	11.046
Sesam	51.991	-0.04	106714	3.83e-27	1.19e-25	122.00	6.19e-14	-3.45e-13	-11.046

Mode 4

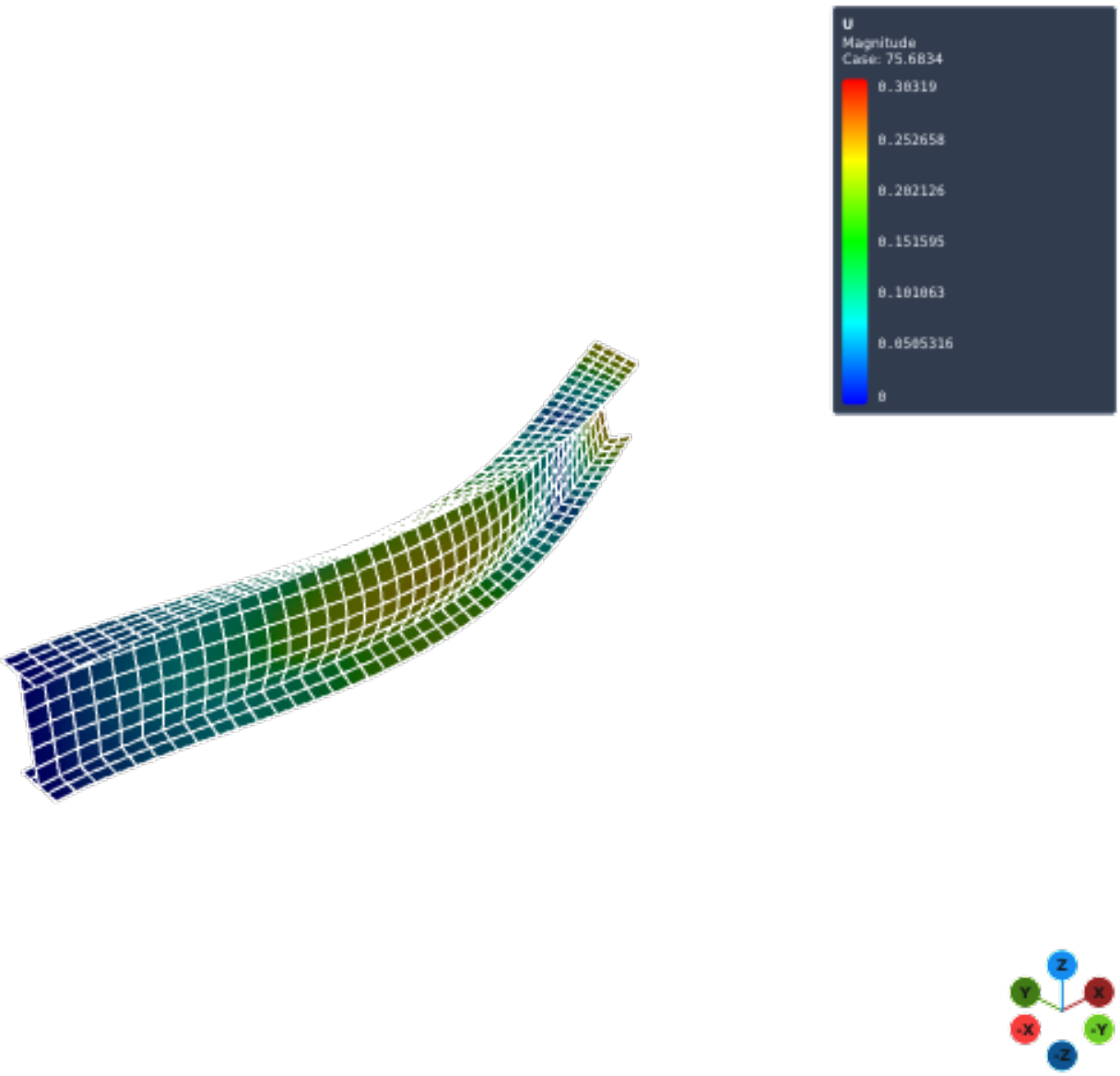

Figure 139: Abaqus — mode 4, 75.683 Hz

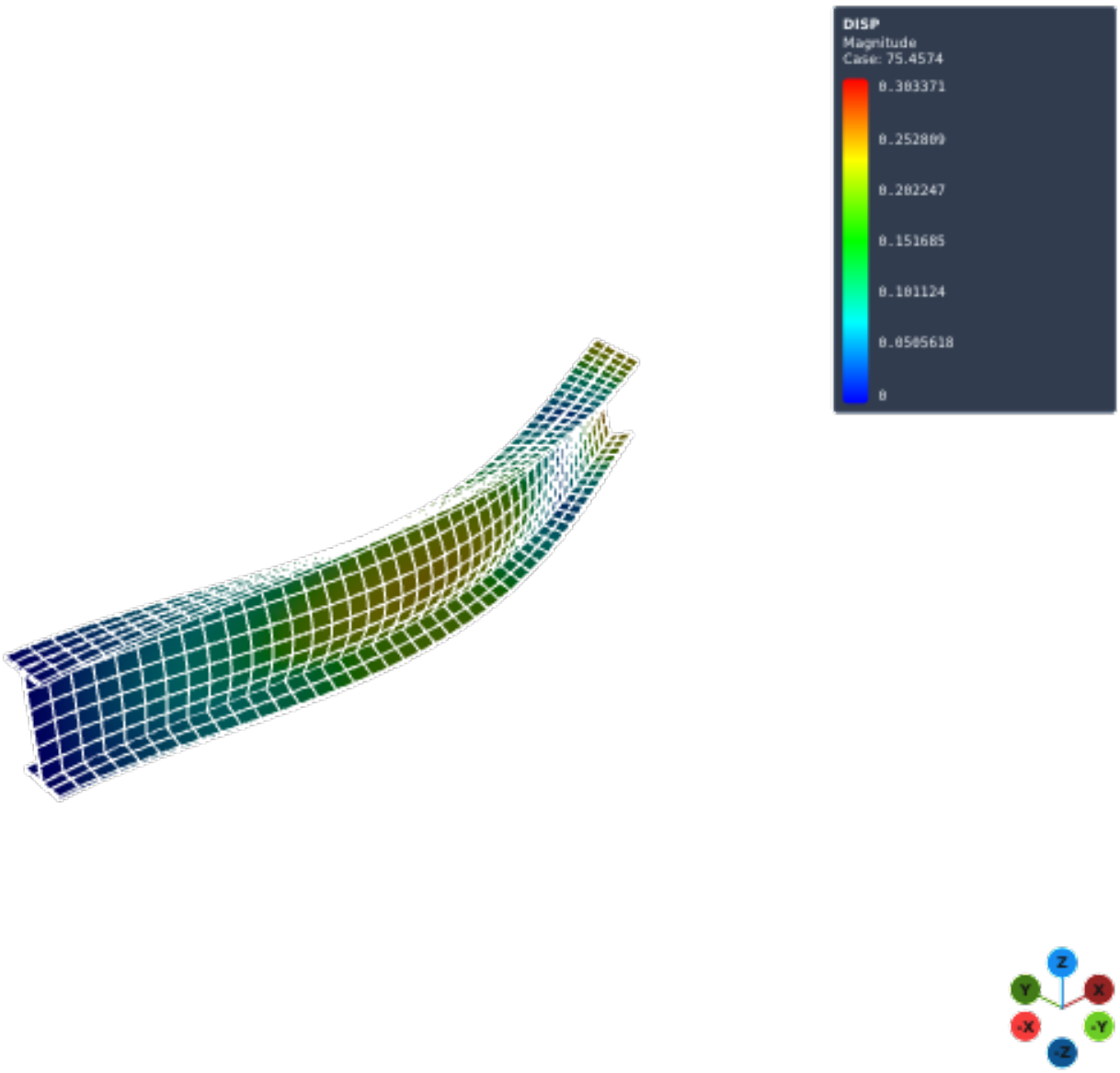

Figure 140: Calculix — mode 4, 75.457 Hz

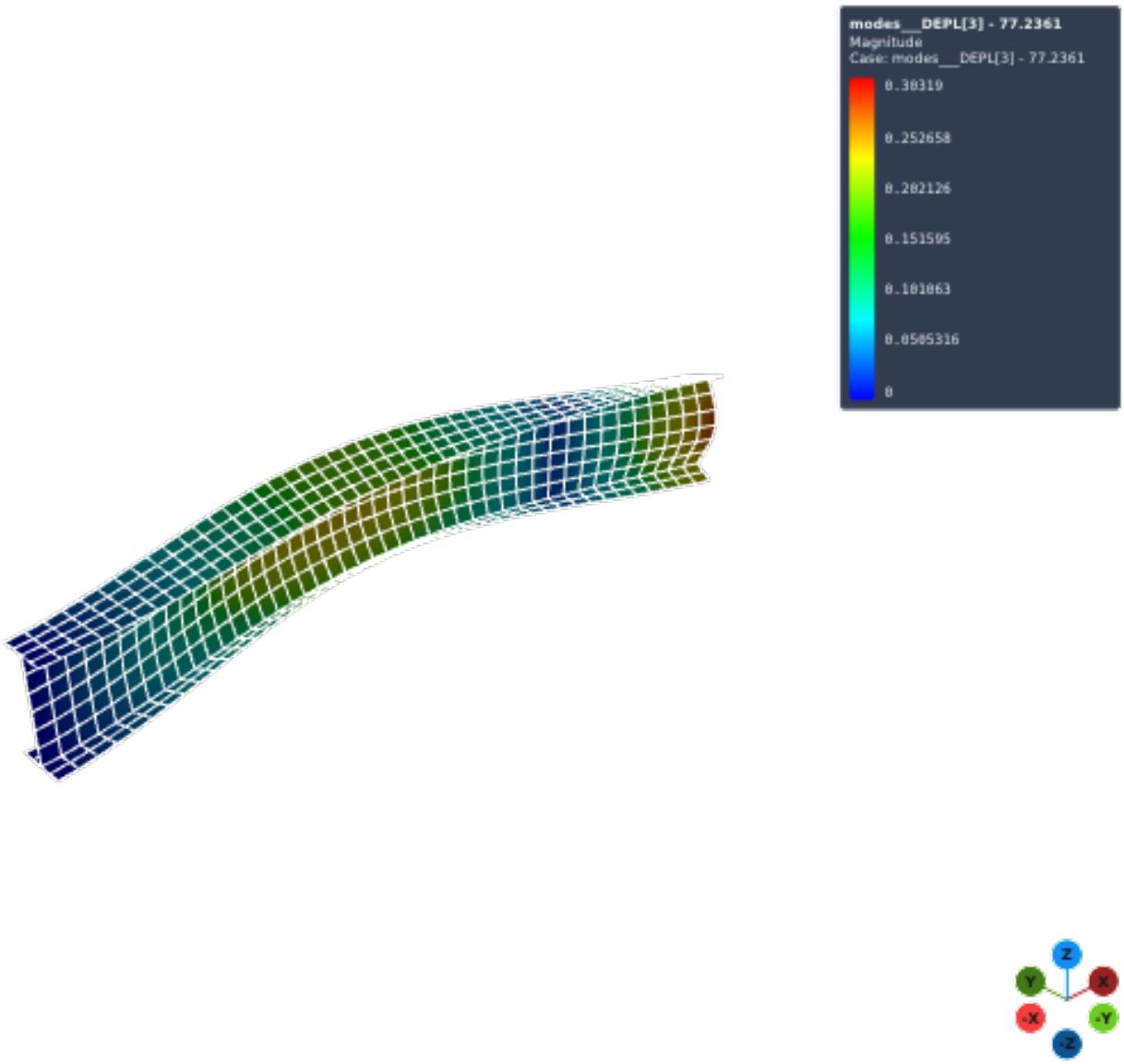

Figure 141: Code_Aster — mode 4, 77.236 Hz

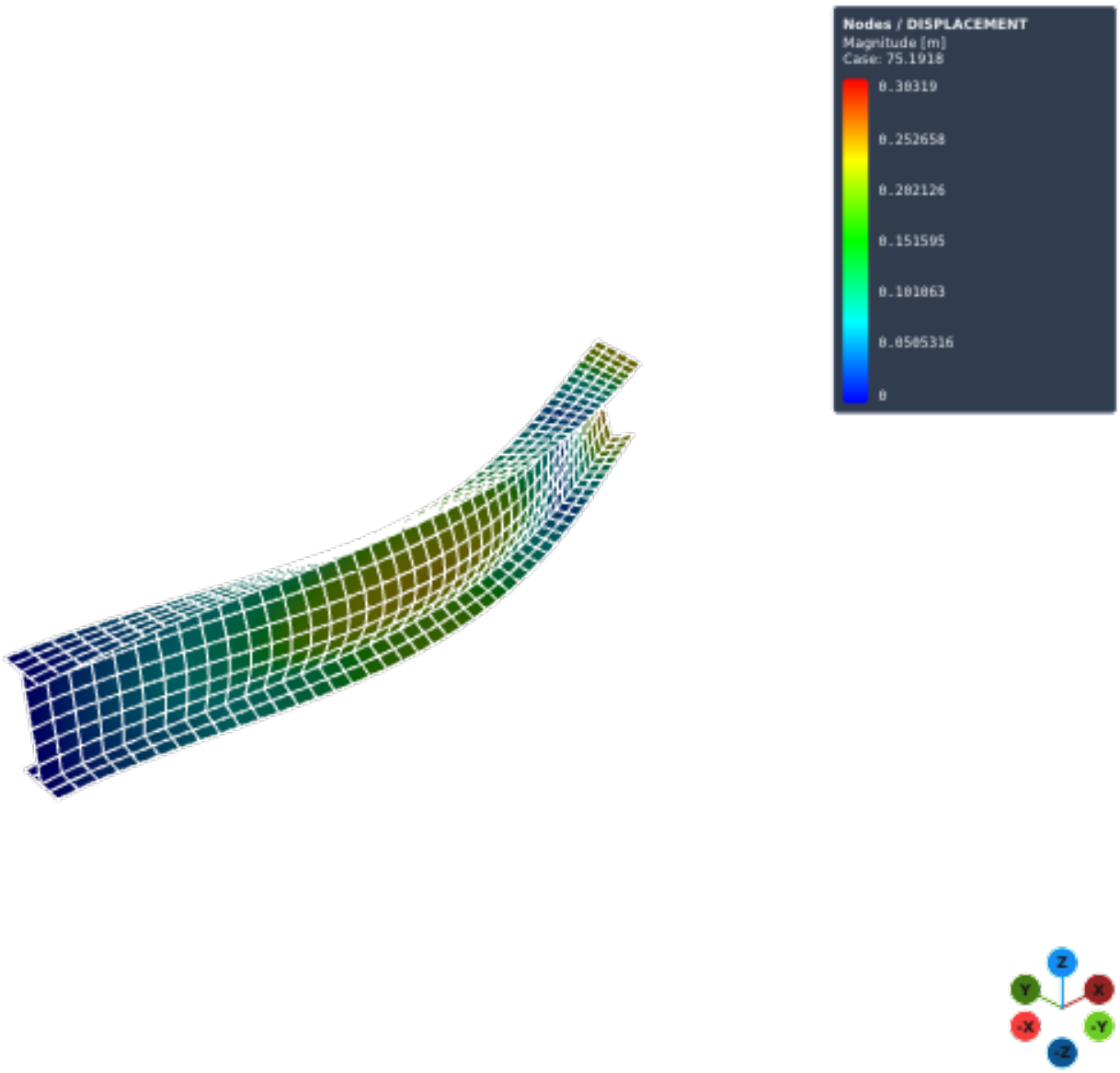

Figure 142: Sesam — mode 4, 75.192 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.683	-0.28	226132	1.88e-28	36.92	2.68e-25	-1.37e-14	-6.076	-5.18e-13
Calculix	75.457	-0.57	224783	1.01e-25	36.87	1.53e-12	-3.18e-13	-6.072	1.24e-06
Code_Aster	77.236	+1.77	235505	3.33e-27	36.78	3.14e-26	-5.77e-14	6.065	1.77e-13
Sesam	75.192	-0.92	223203	1.83e-27	36.80	1.95e-25	-4.28e-14	-6.067	-4.41e-13

Mode 5

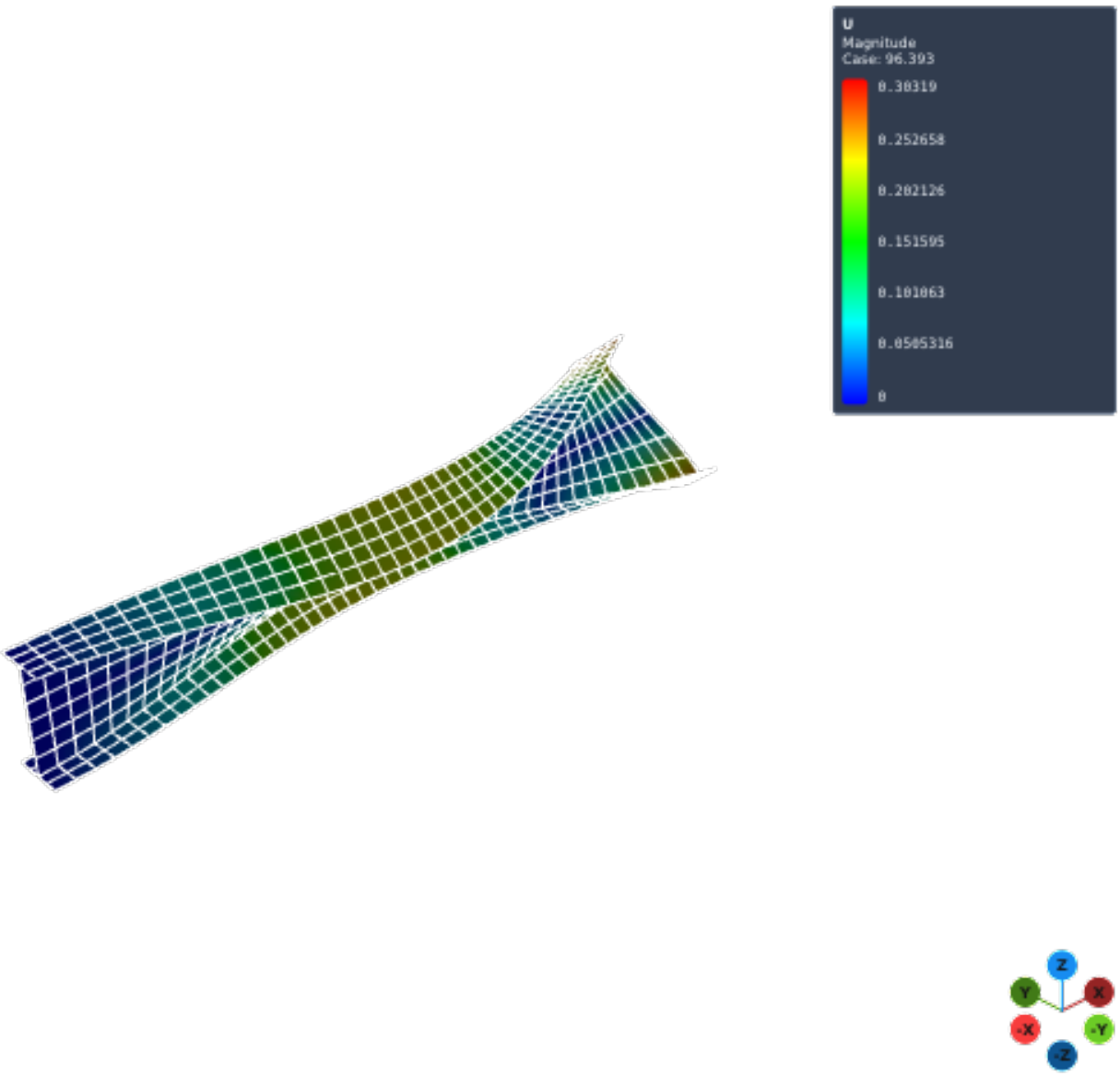

Figure 143: Abaqus — mode 5, 96.393 Hz

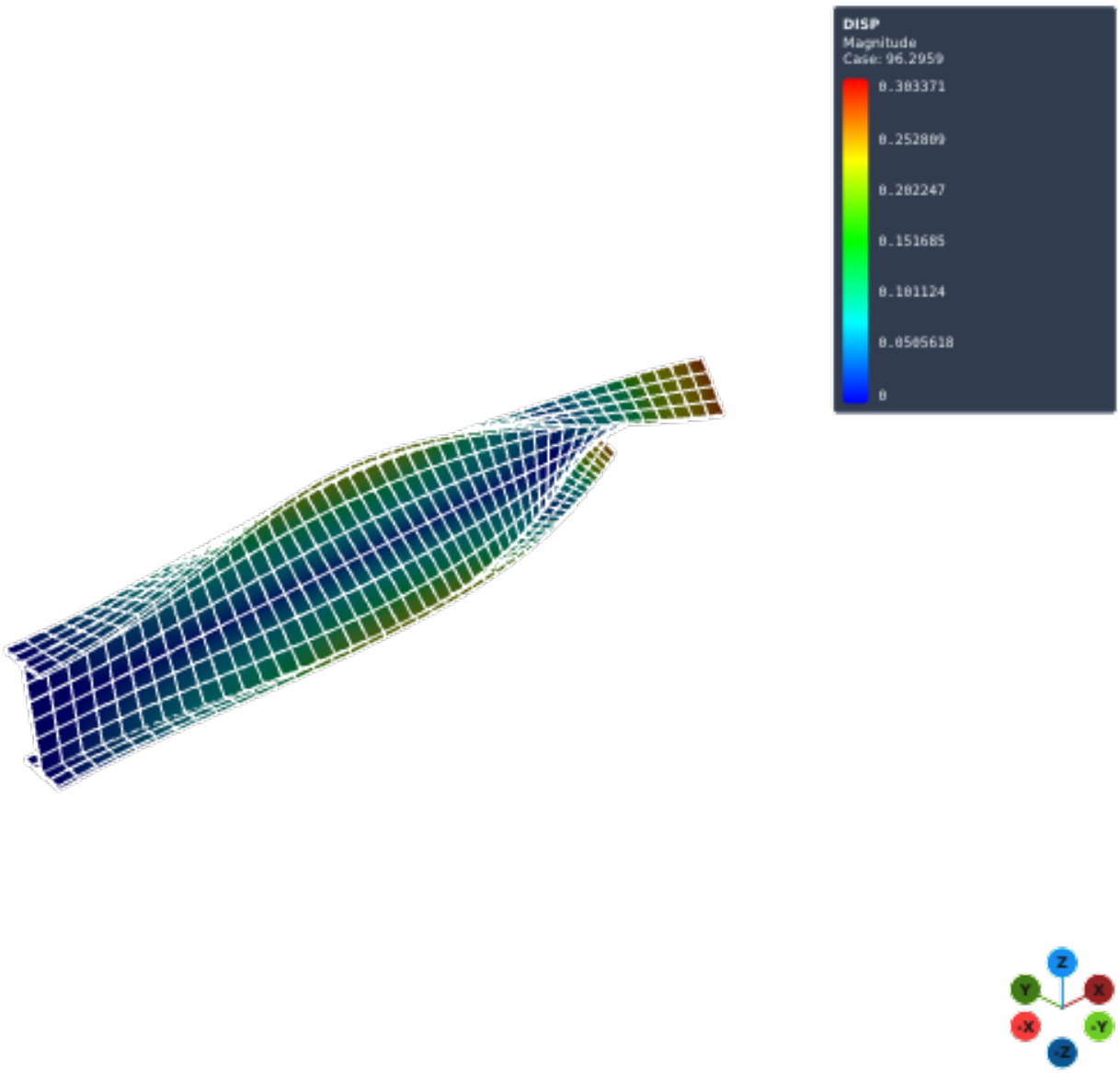

Figure 144: Calculix — mode 5, 96.296 Hz

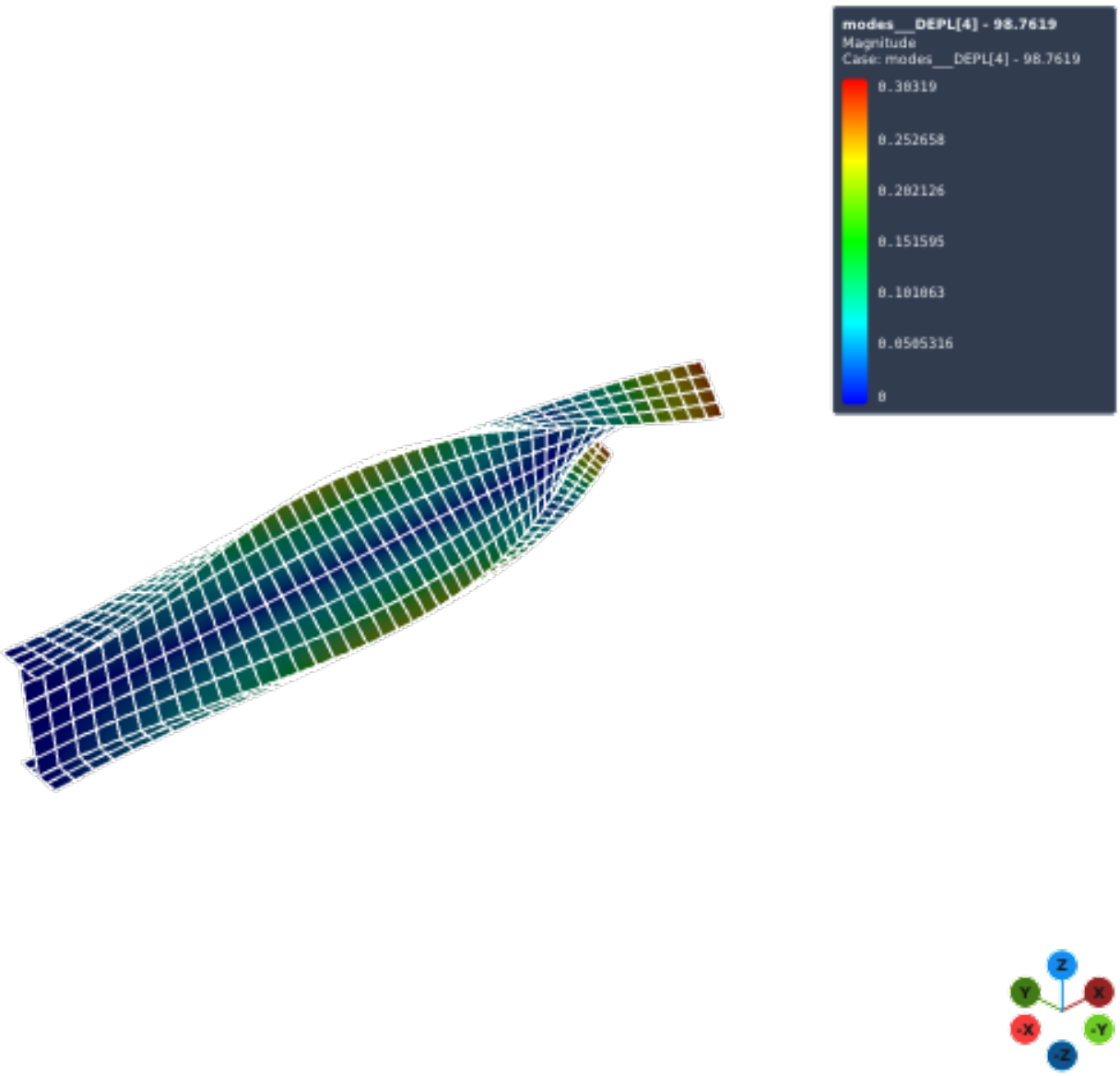

Figure 145: Code_Aster — mode 5, 98.762 Hz

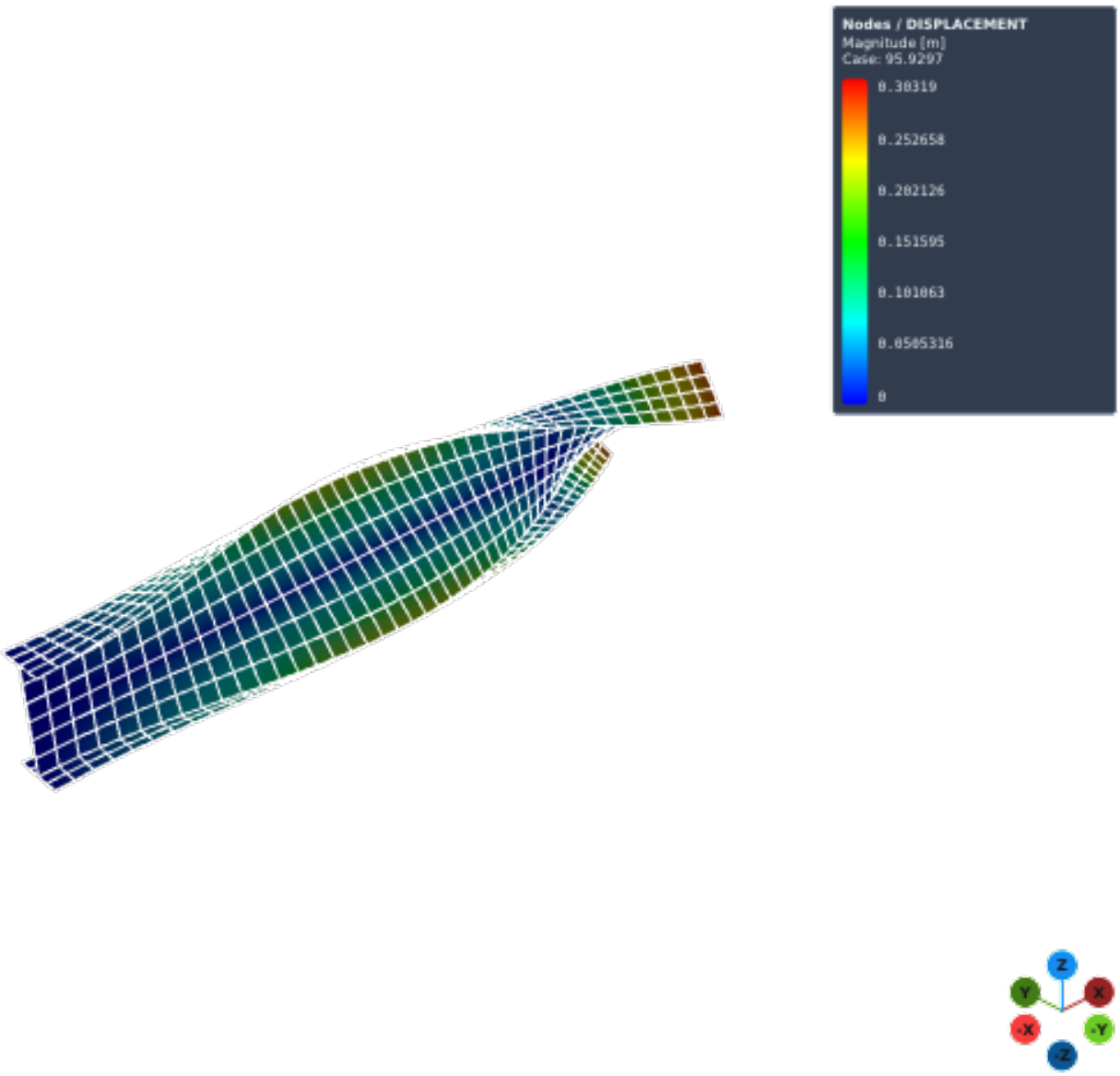

Figure 146: Sesam — mode 5, 95.930 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	96.393	-0.47	366818	7.25e-28	1.11e-22	7.70e-26	-2.69e-14	-1.05e-11	-2.77e-13
Calculix	96.296	-0.57	366079	8.46e-10	5.61e-21	1.48e-22	-2.91e-05	-7.49e-11	1.22e-11
Code_Aster	98.762	+1.98	385069	6.28e-29	9.74e-23	7.49e-26	-7.92e-15	9.87e-12	-2.74e-13
Sesam	95.930	-0.95	363300	3.32e-27	2.02e-23	9.04e-27	5.76e-14	4.49e-12	-9.51e-14

Mode 6

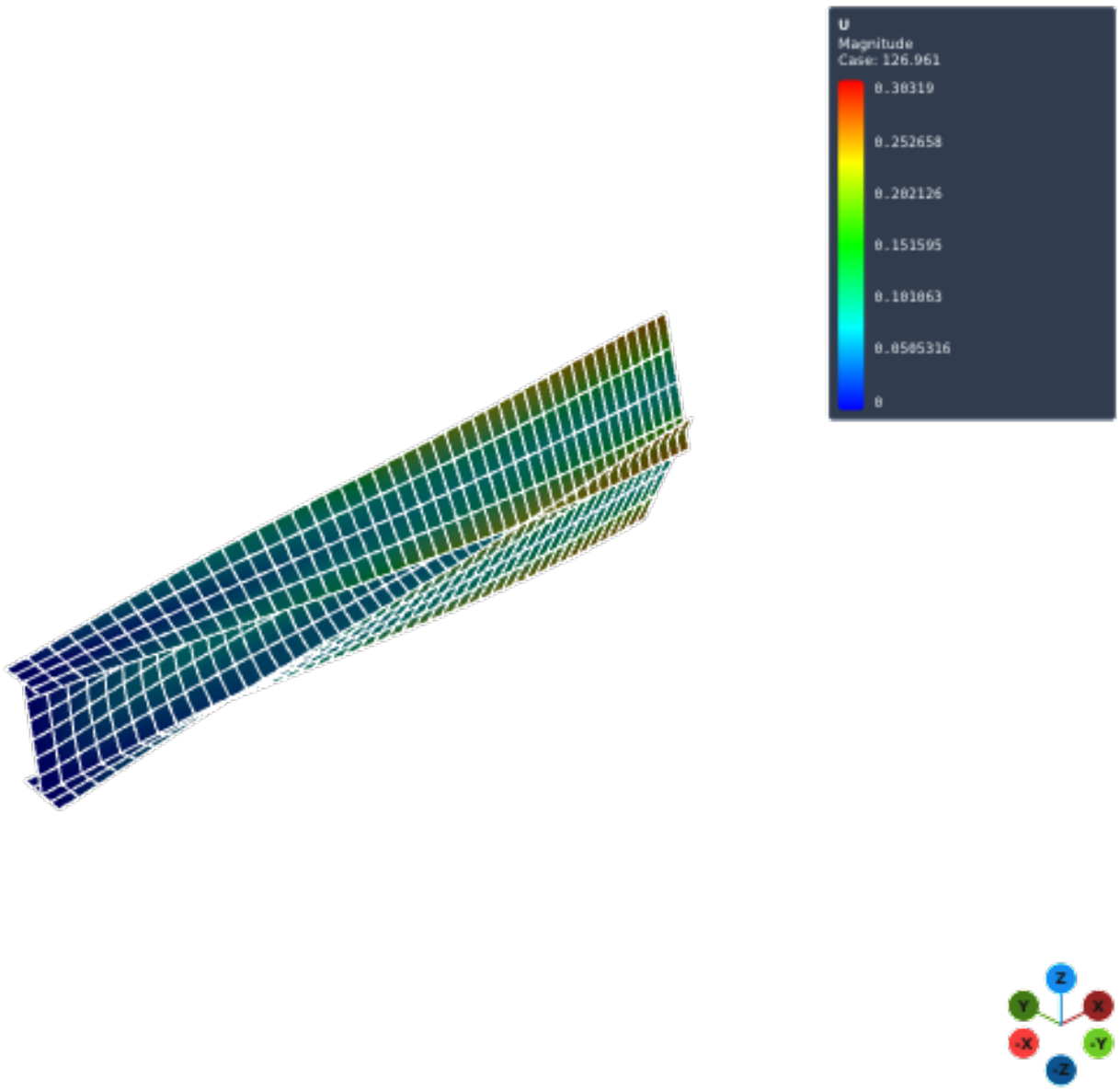

Figure 147: Abaqus — mode 6, 126.961 Hz

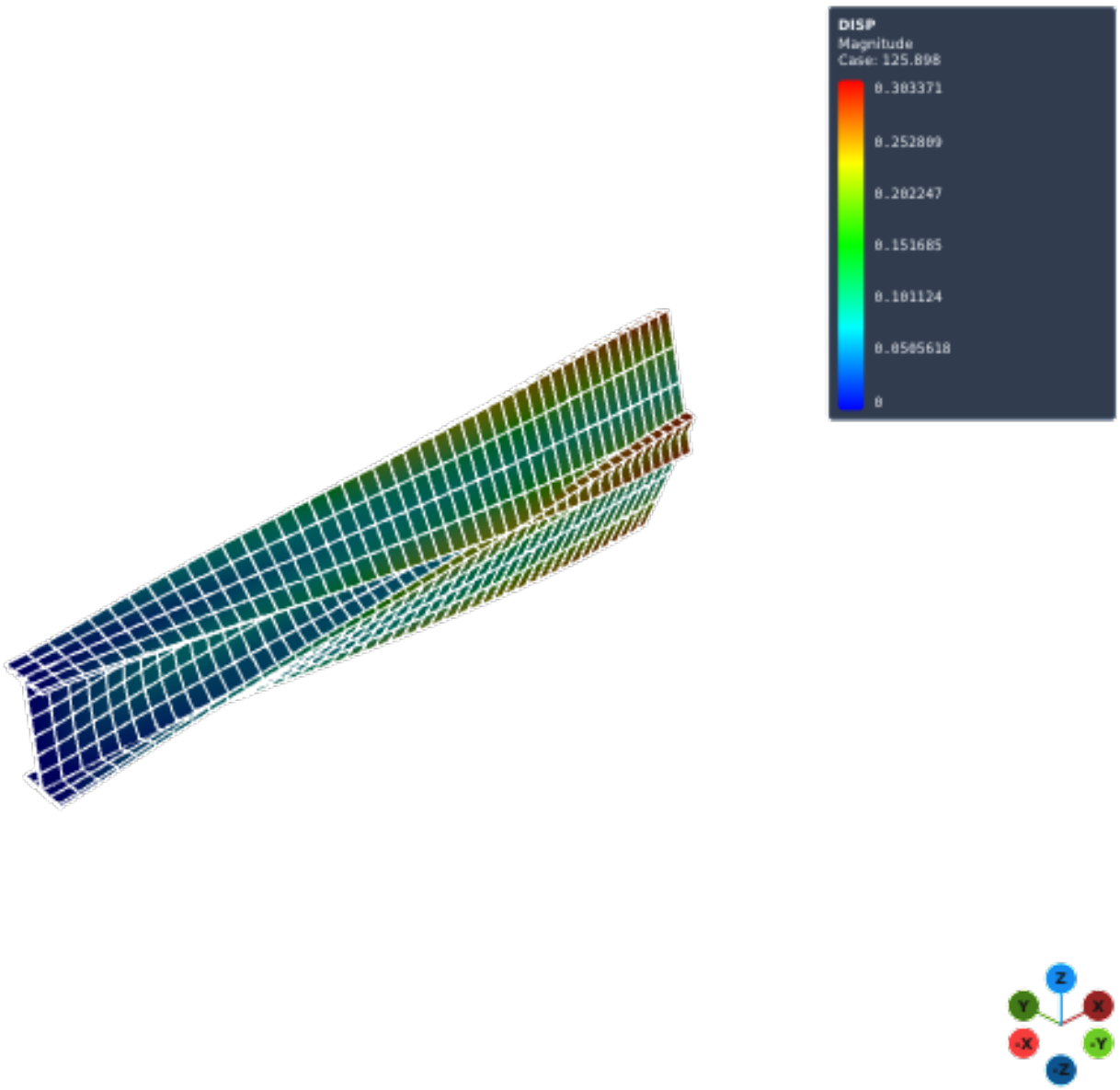

Figure 148: Calculix — mode 6, 125.898 Hz

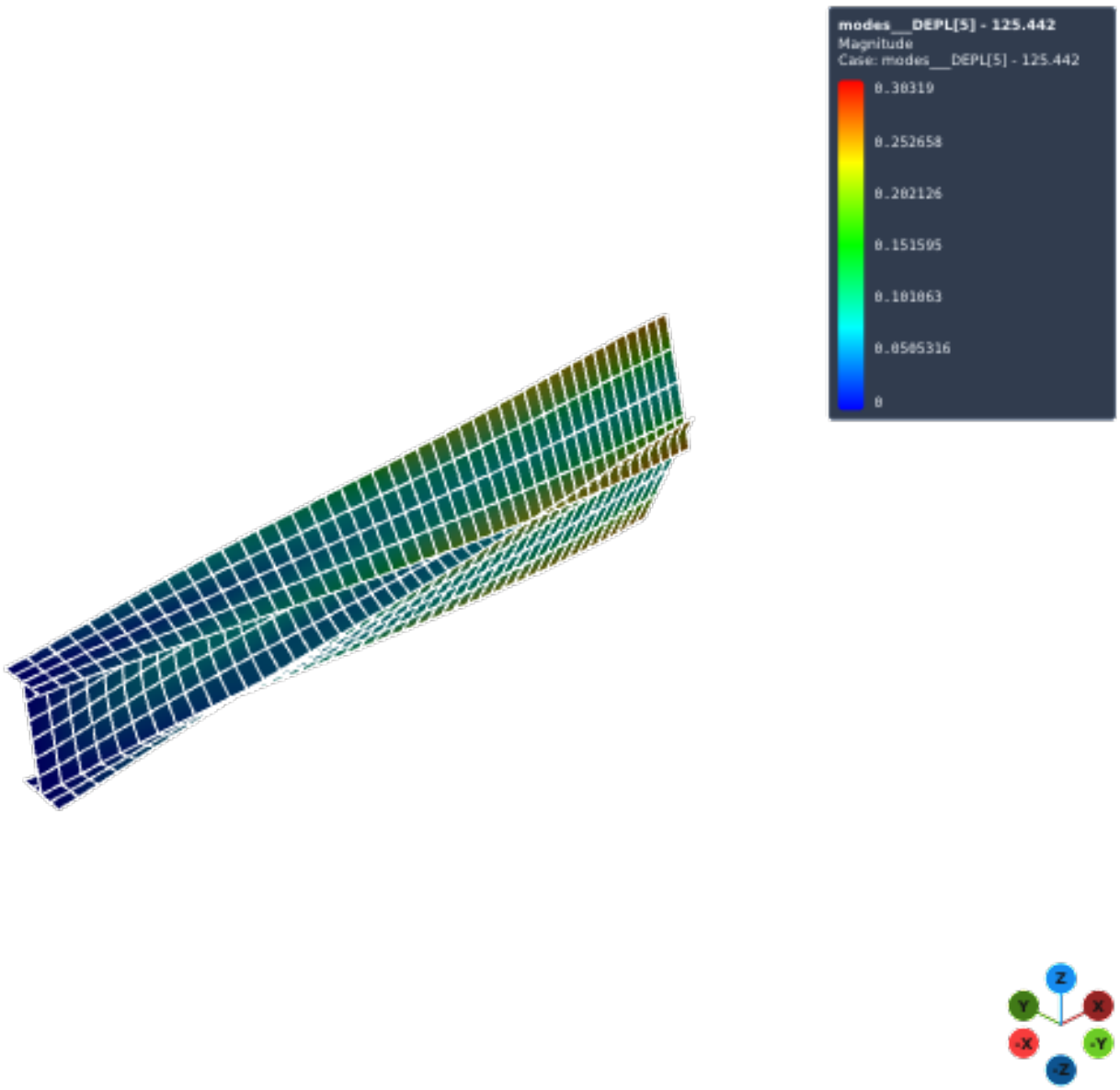

Figure 149: Code_Aster — mode 6, 125.442 Hz

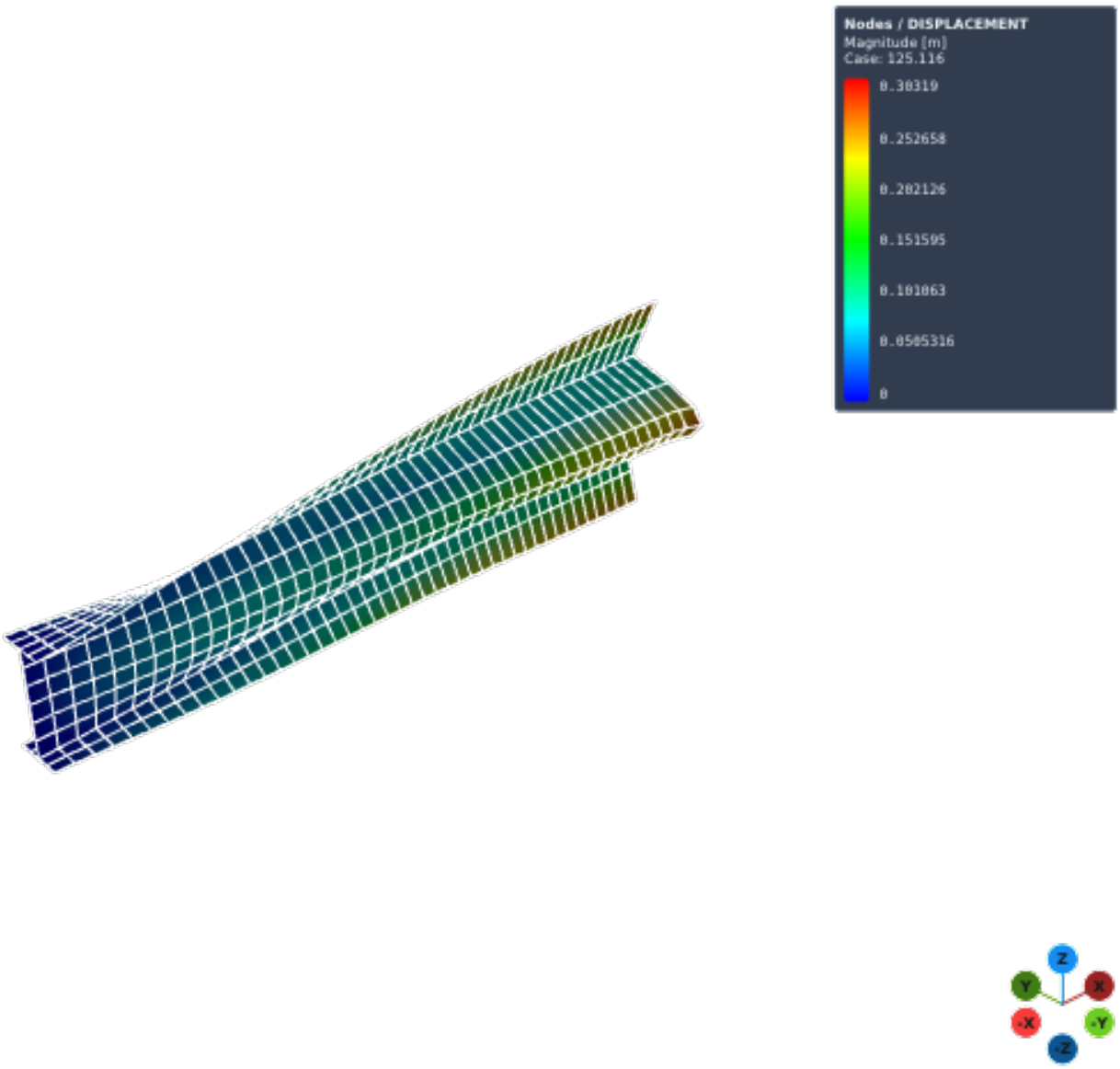

Figure 150: Sesam — mode 6, 125.116 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	126.960	+0.88	636351	1.52e-27	0.02	2.42e-26	3.90e-14	-0.132	-1.56e-13
Calculix	125.898	+0.03	625743	4.03e-28	0.03	4.13e-15	2.01e-14	-0.174	-6.43e-08
Code_Aster	125.442	-0.33	621220	2.55e-28	0.04	5.76e-25	-1.60e-14	-0.194	-7.59e-13
Sesam	125.116	-0.59	617993	9.40e-30	0.02	2.19e-27	-3.07e-15	0.134	-4.68e-14

Mode 7

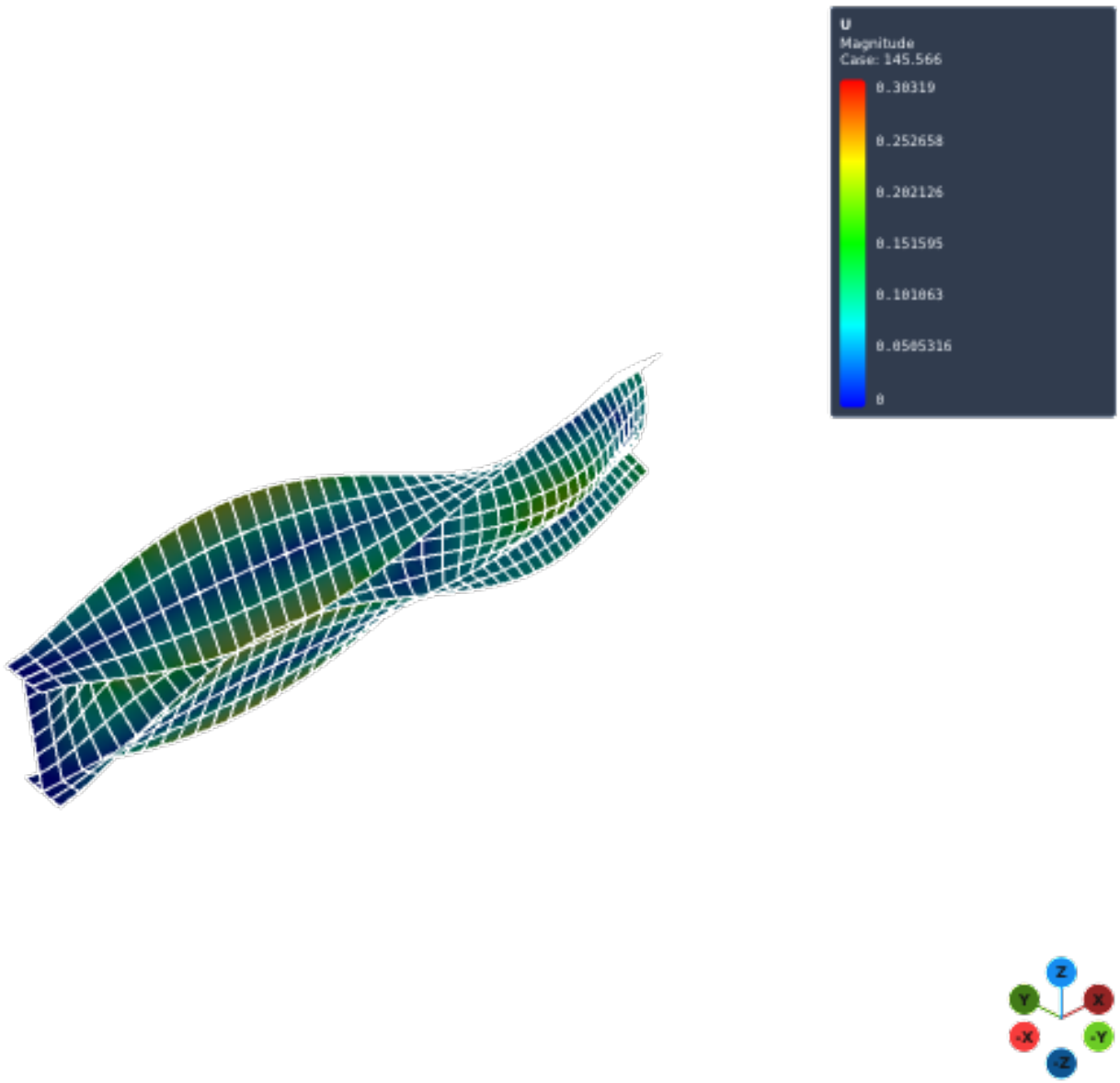

Figure 151: Abaqus — mode 7, 145.566 Hz

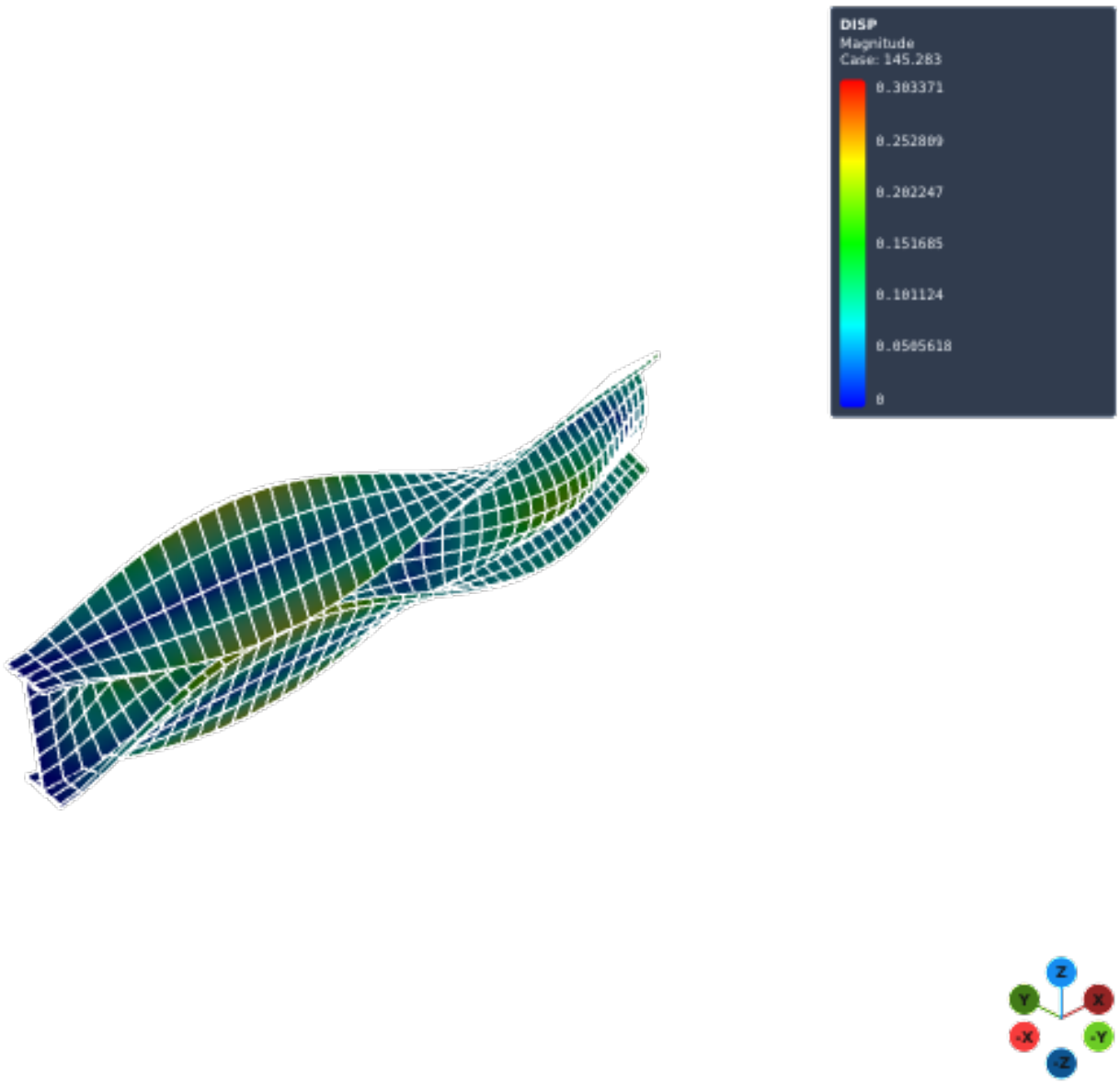

Figure 152: Calculix — mode 7, 145.283 Hz

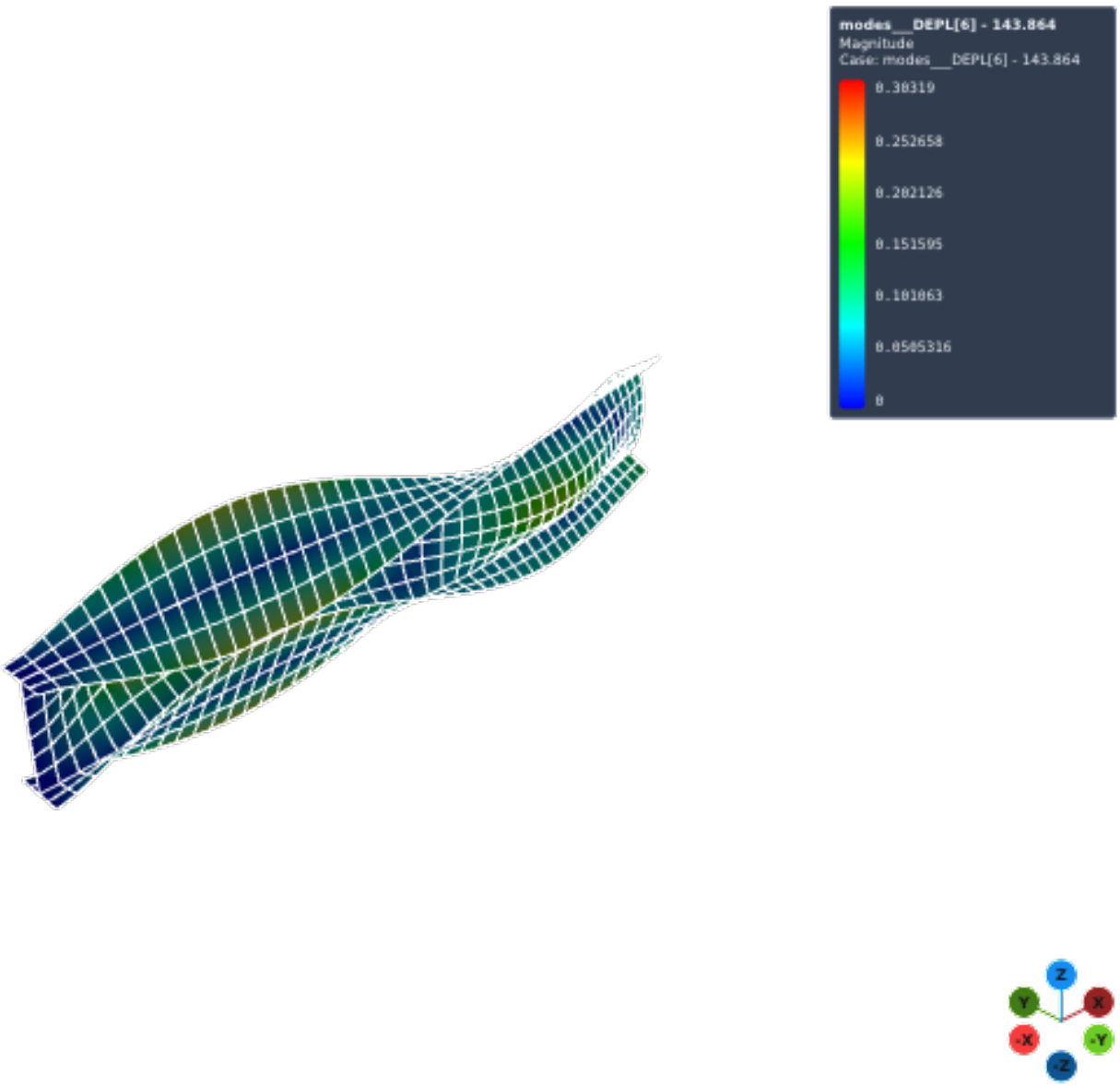

Figure 153: Code_Aster — mode 7, 143.864 Hz

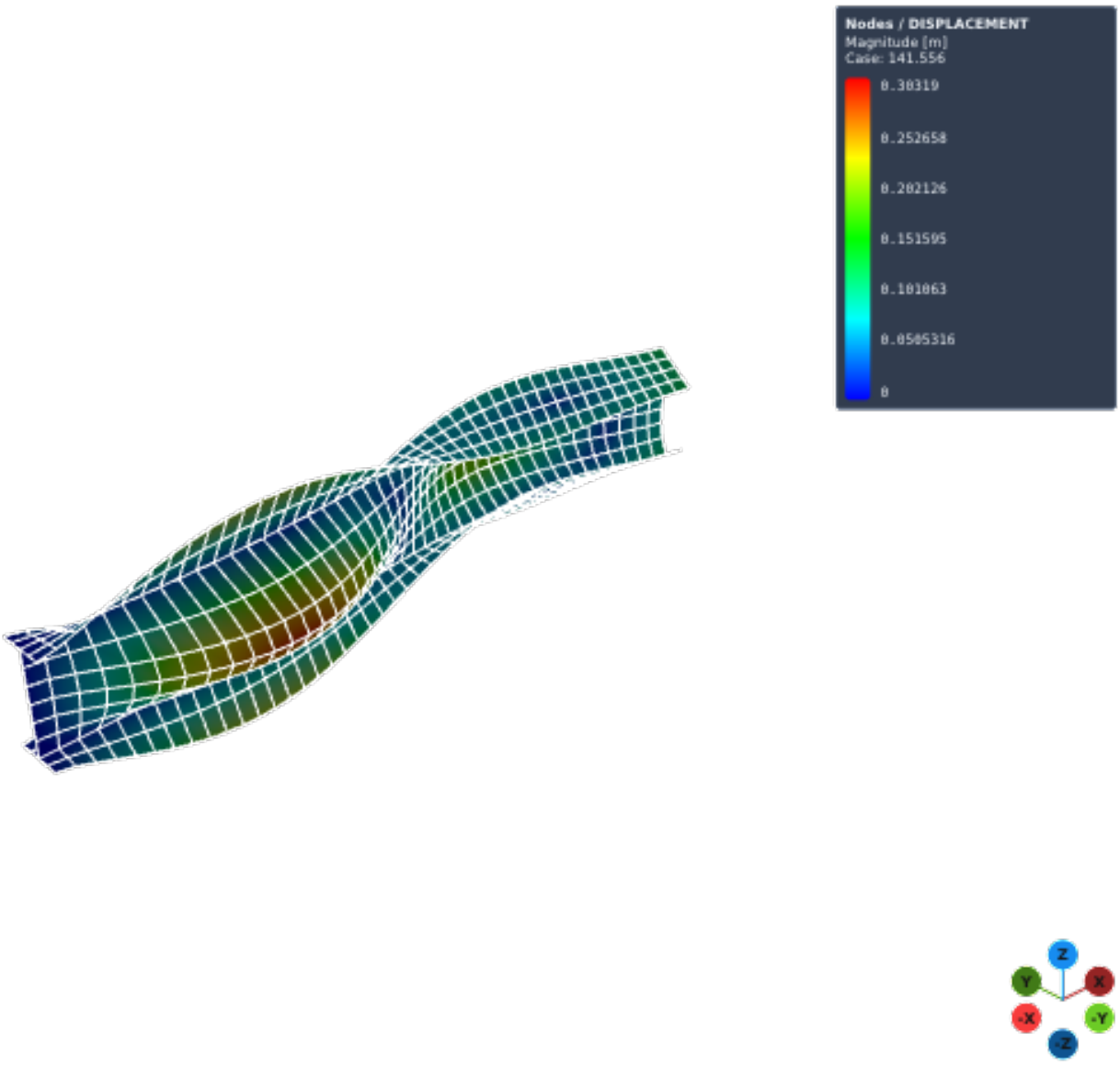

Figure 154: Sesam — mode 7, 141.556 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	145.570	+1.04	836523	2.58e-27	3.34	1.12e-26	-5.08e-14	1.829	1.06e-13
Calculix	145.282	+0.84	833272	1.59e-26	3.26	6.68e-13	-1.26e-13	1.805	8.17e-07
Code_Aster	143.864	-0.14	817074	5.14e-28	3.15	8.15e-26	2.27e-14	1.776	2.86e-13
Sesam	141.556	-1.74	791068	4.10e-28	3.46	8.46e-30	2.02e-14	-1.860	2.91e-15

Mode 8

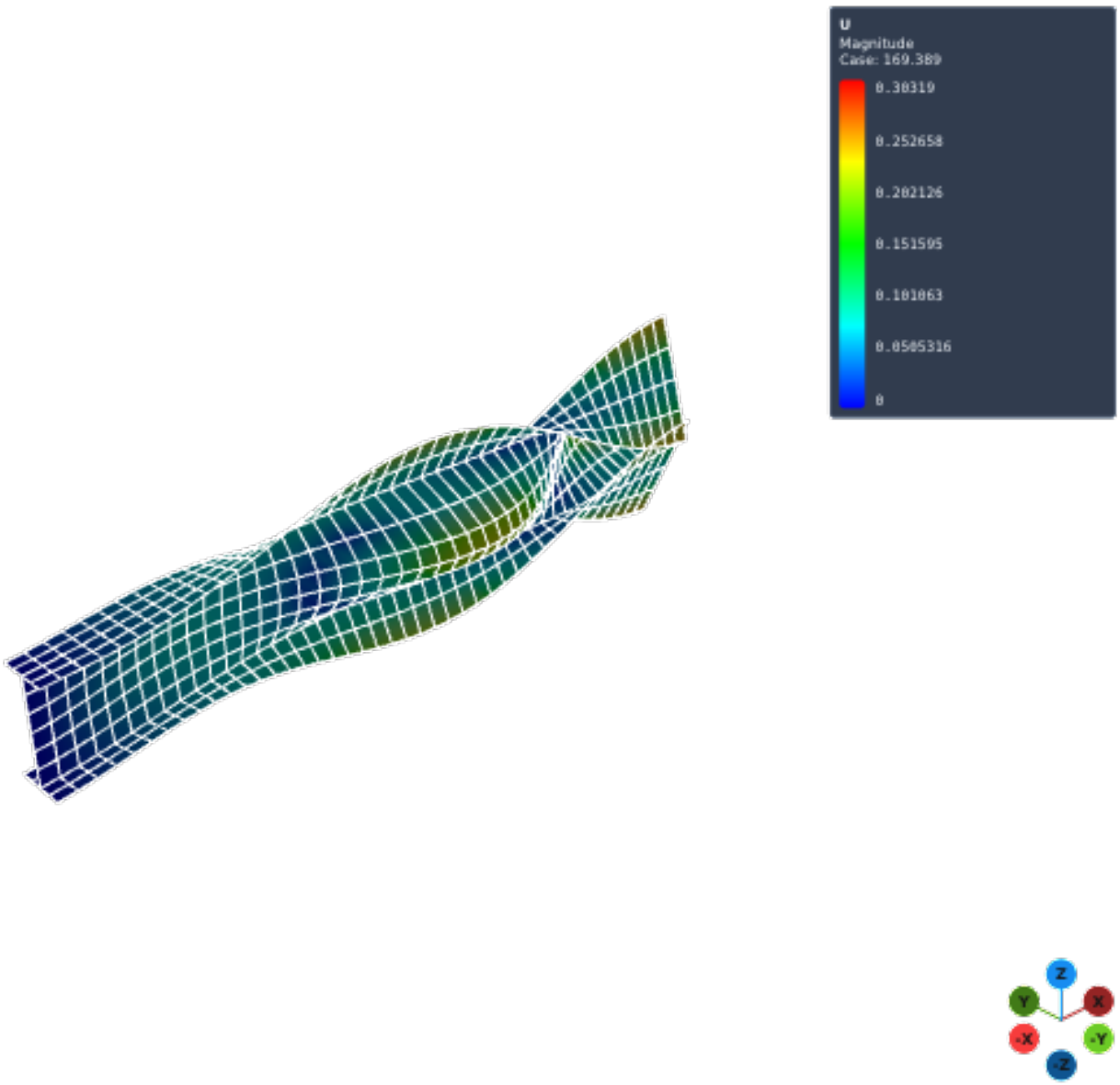

Figure 155: Abaqus — mode 8, 169.389 Hz

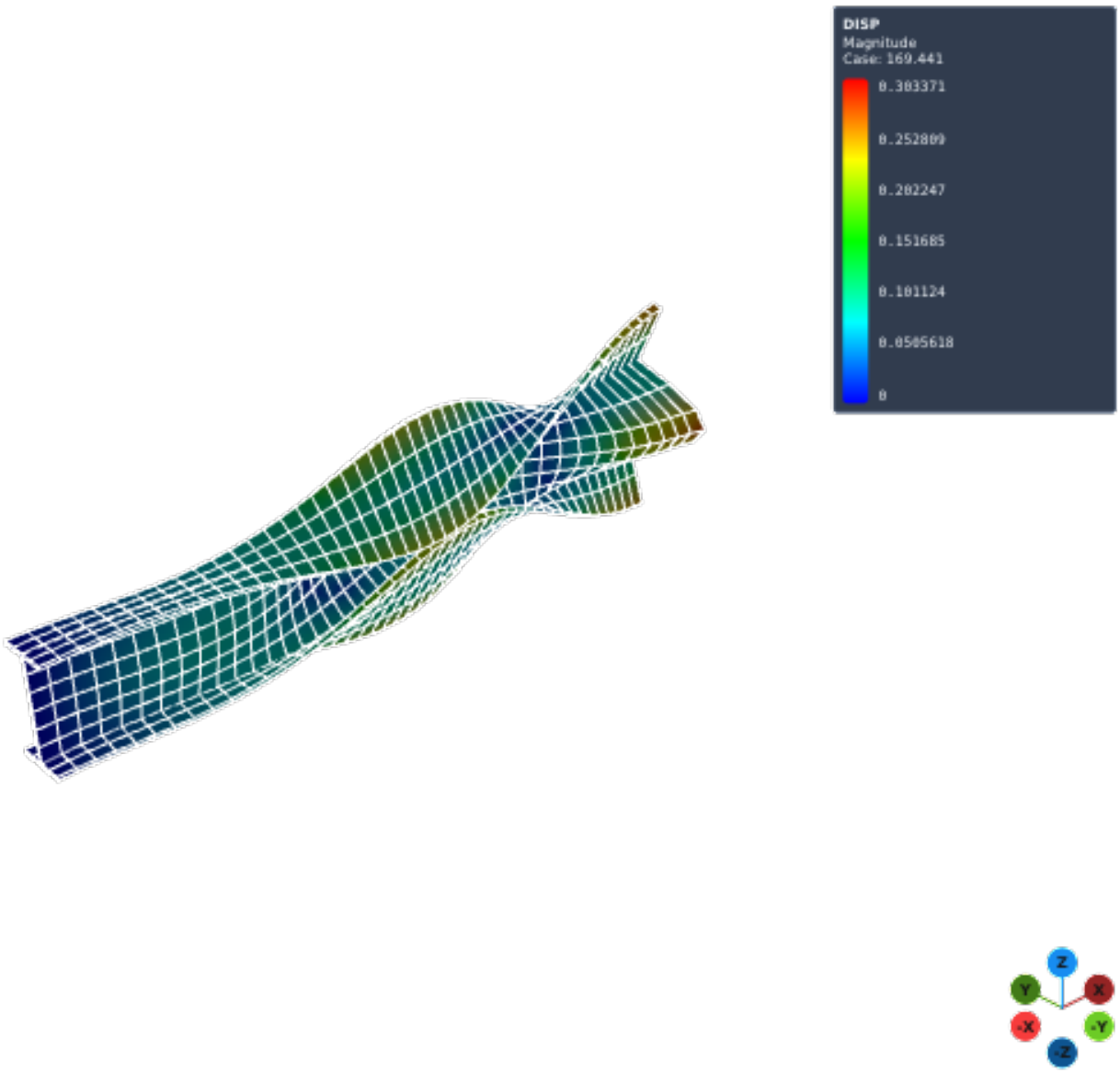

Figure 156: Calculix — mode 8, 169.441 Hz

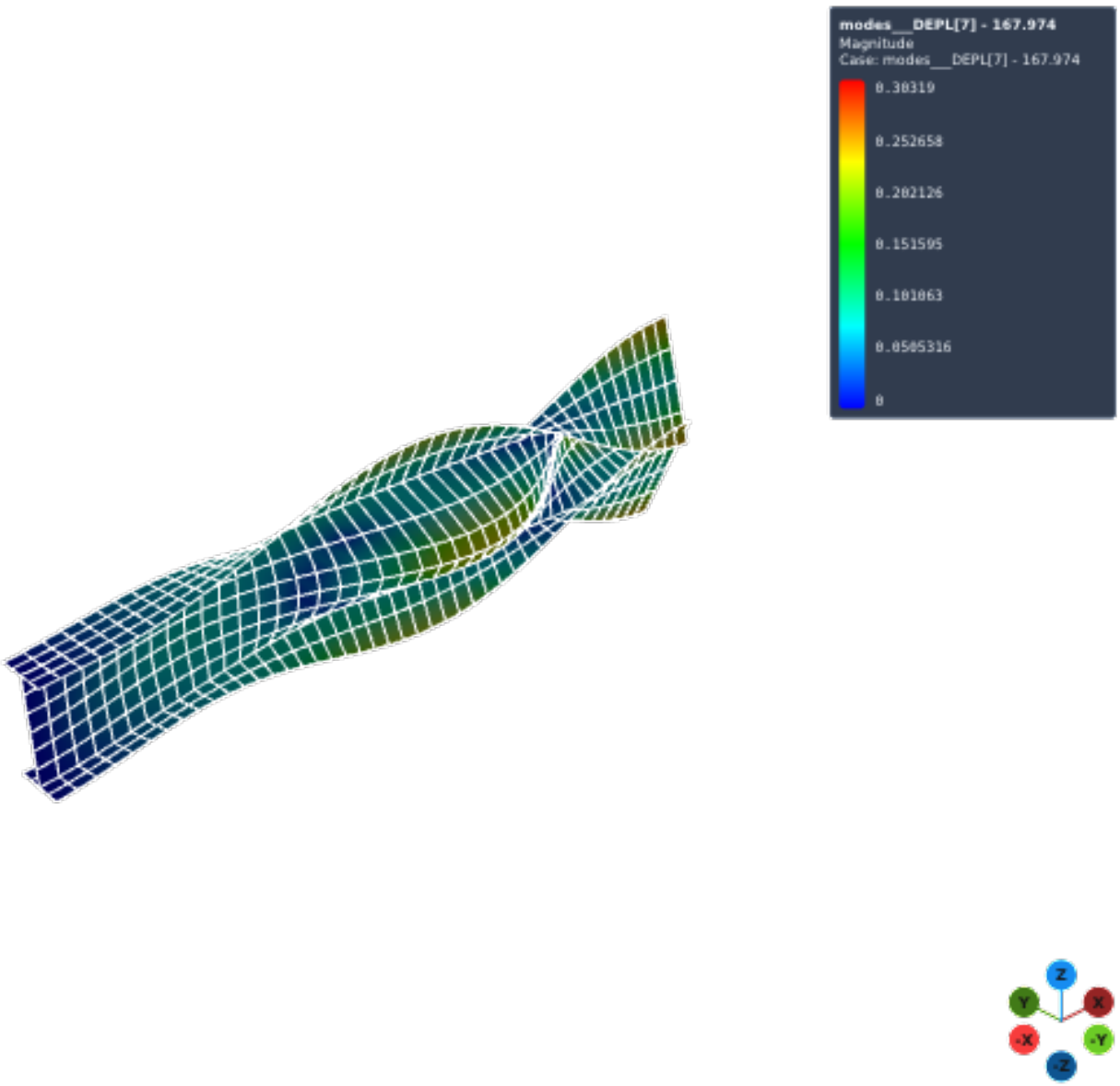

Figure 157: Code_Aster — mode 8, 167.974 Hz

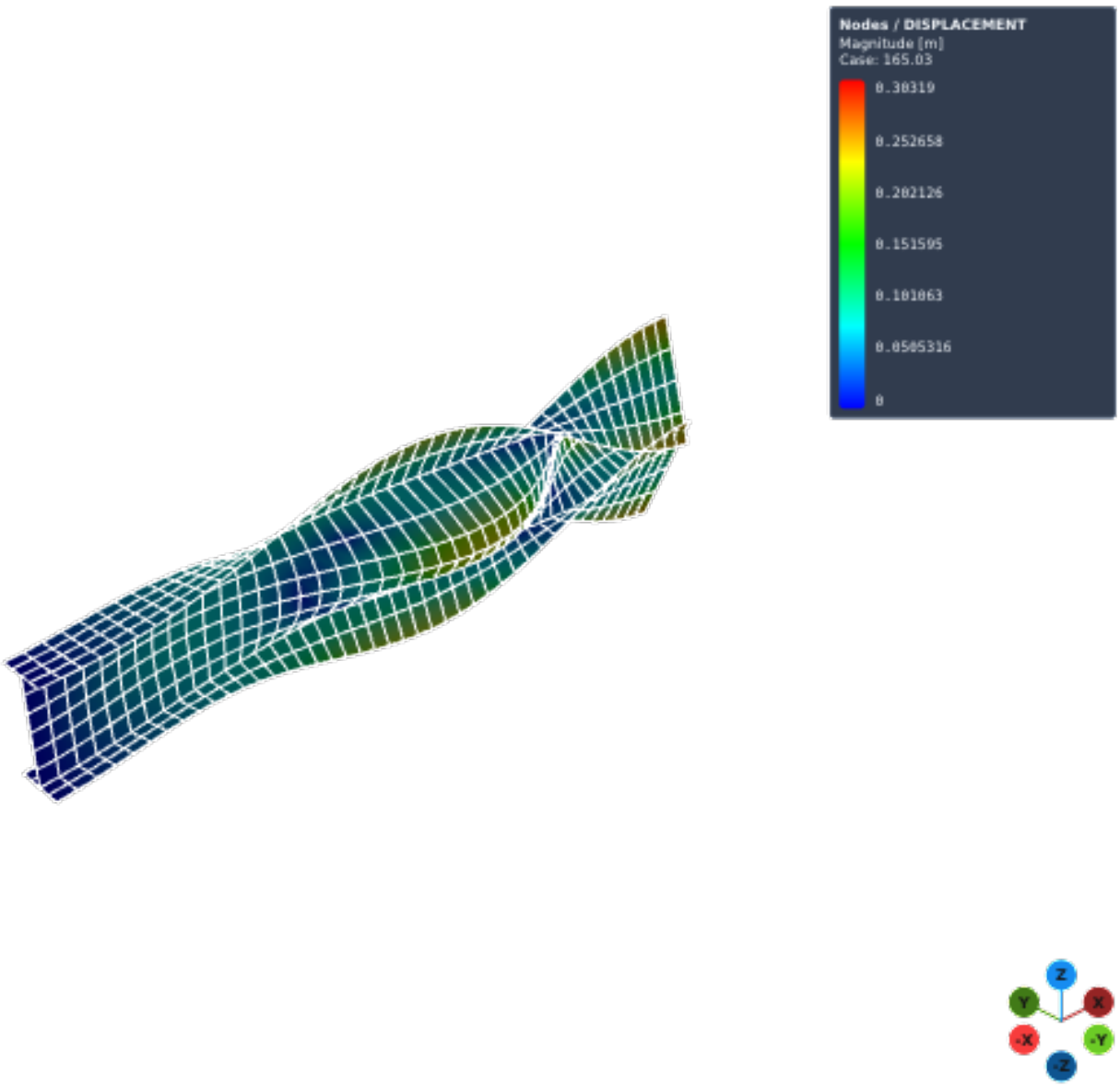

Figure 158: Sesam — mode 8, 165.030 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	169.390	+0.85	1.13275e+06	2.66e-25	5.68	6.71e-24	-5.16e-13	2.383	-2.59e-12
Calculix	169.441	+0.88	1.13343e+06	1.08e-25	5.79	1.37e-14	-3.29e-13	-2.406	1.17e-07
Code_Aster	167.974	+0.01	1.11389e+06	3.09e-27	5.75	3.62e-25	-5.56e-14	2.398	-6.01e-13
Sesam	165.030	-1.74	1.07519e+06	8.05e-28	5.45	1.35e-26	2.84e-14	2.334	1.16e-13

Mode 9

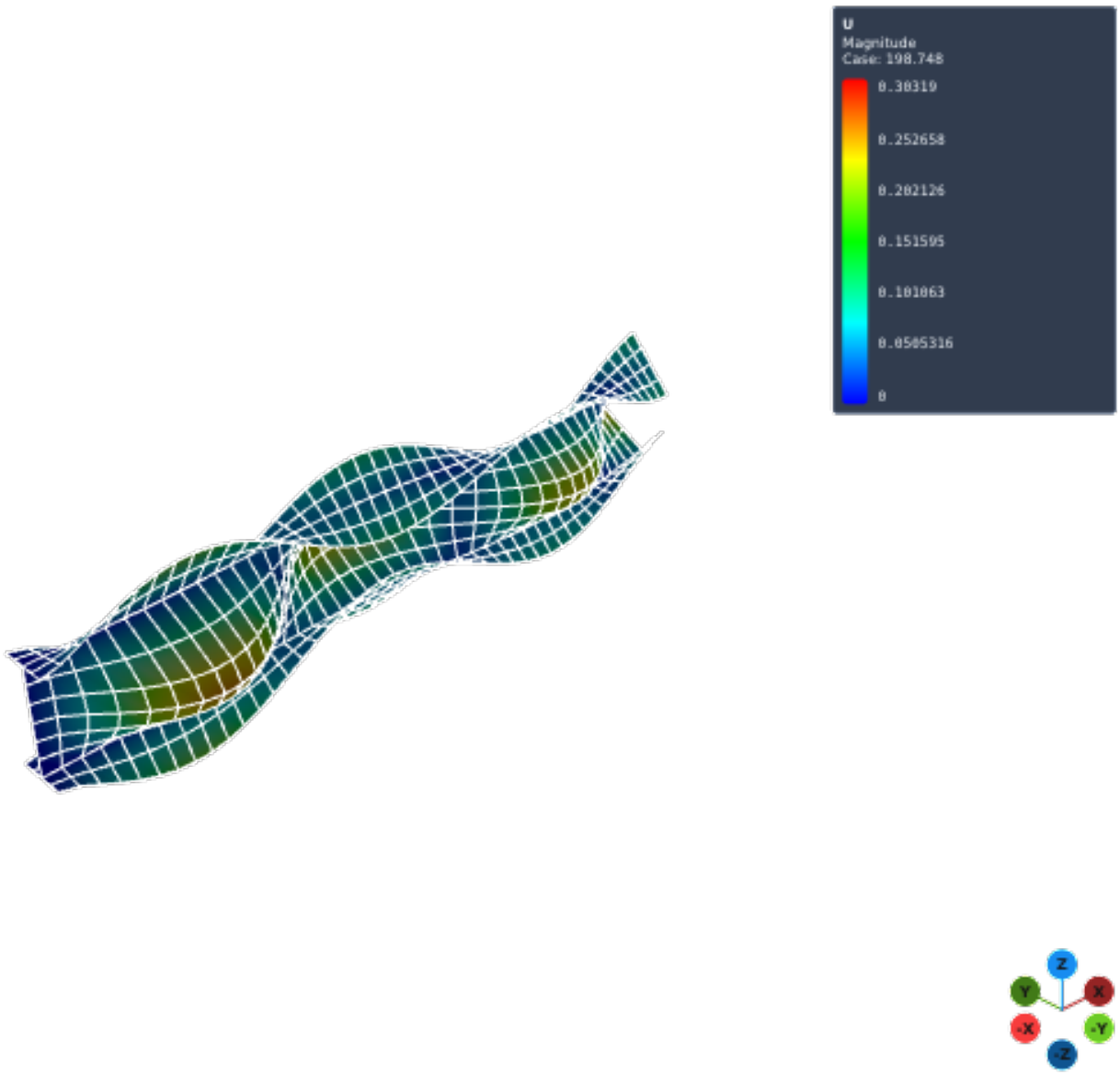

Figure 159: Abaqus — mode 9, 198.748 Hz

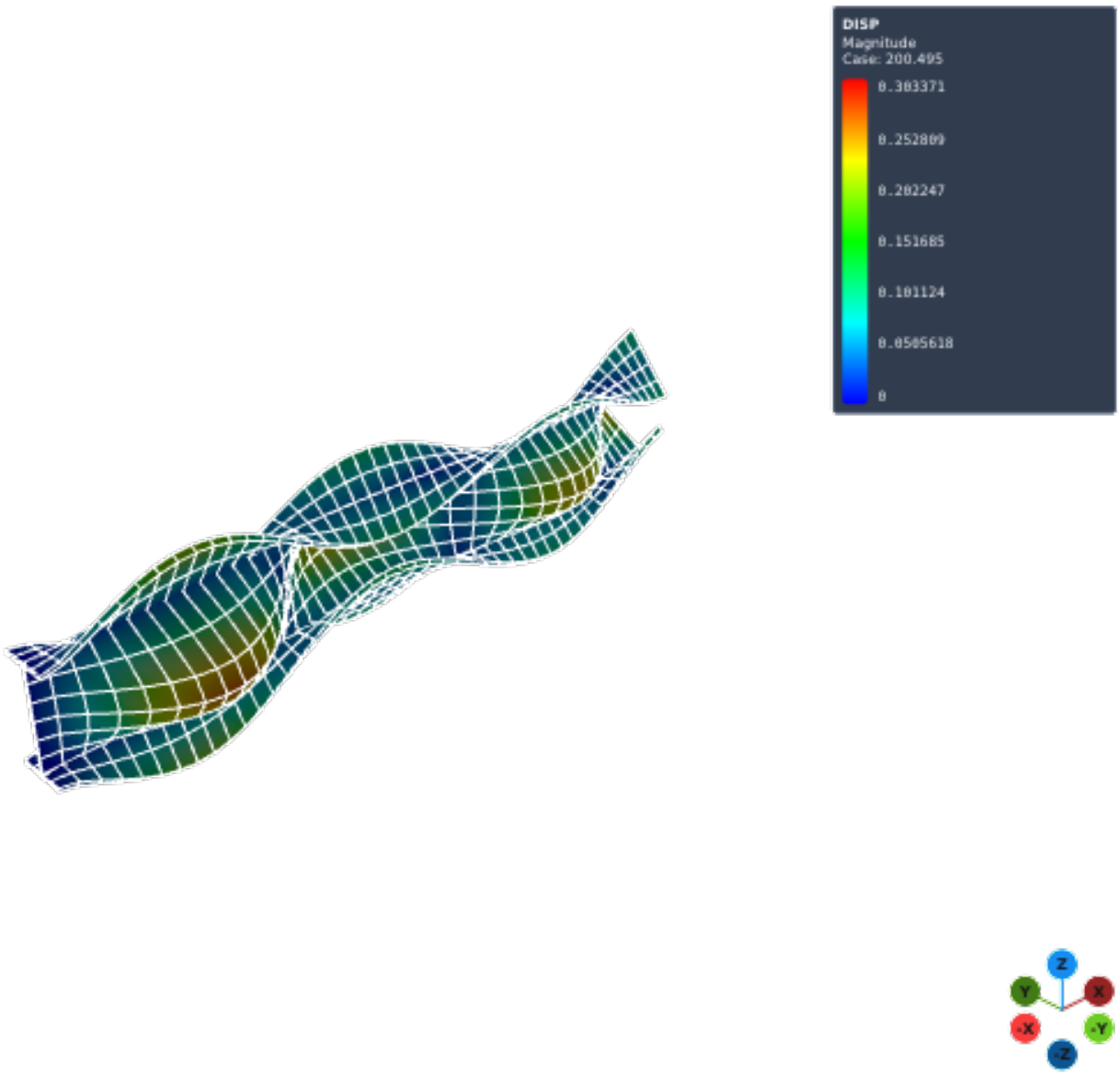

Figure 160: Calculix — mode 9, 200.495 Hz

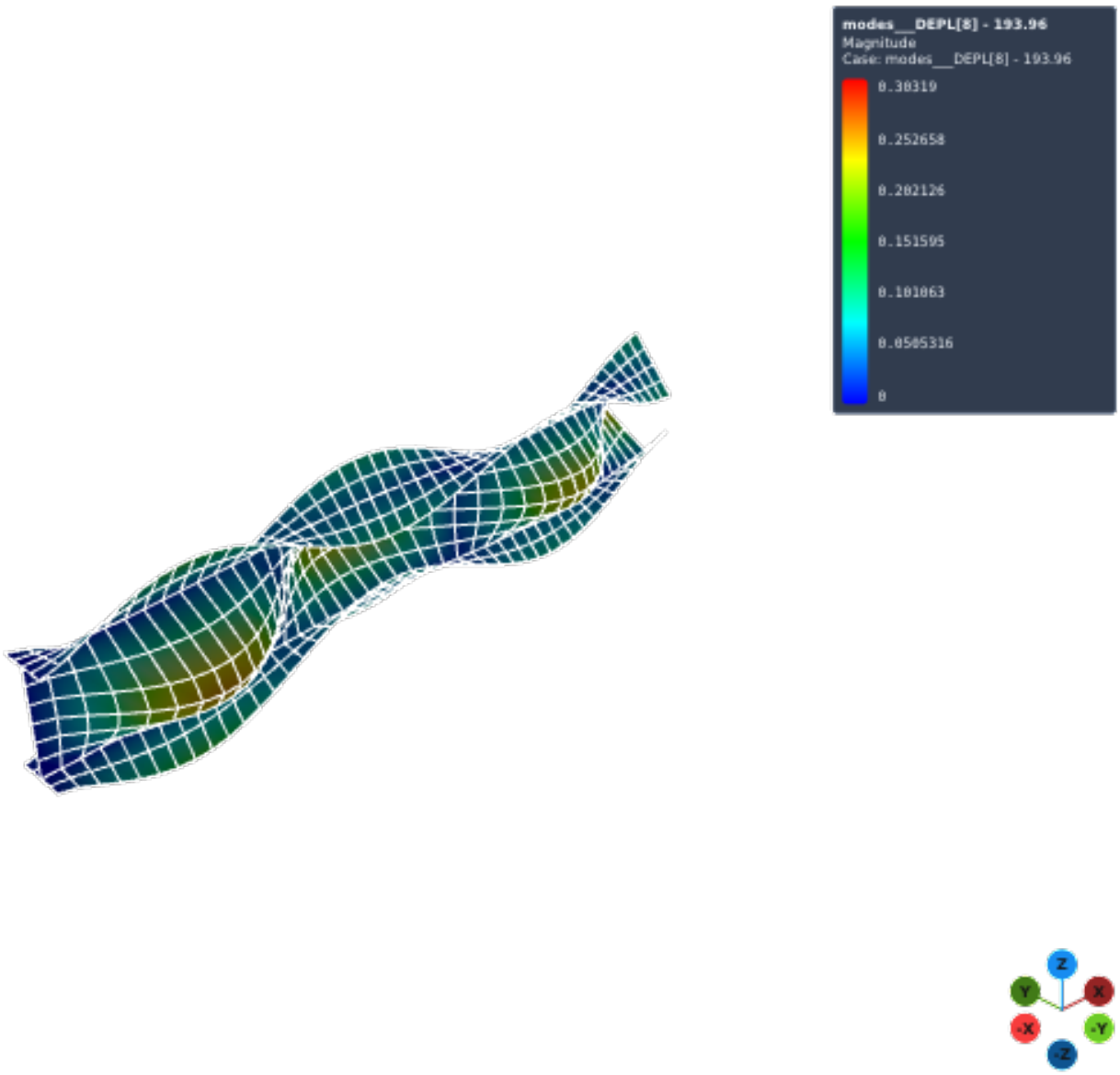

Figure 161: Code_Aster — mode 9, 193.960 Hz

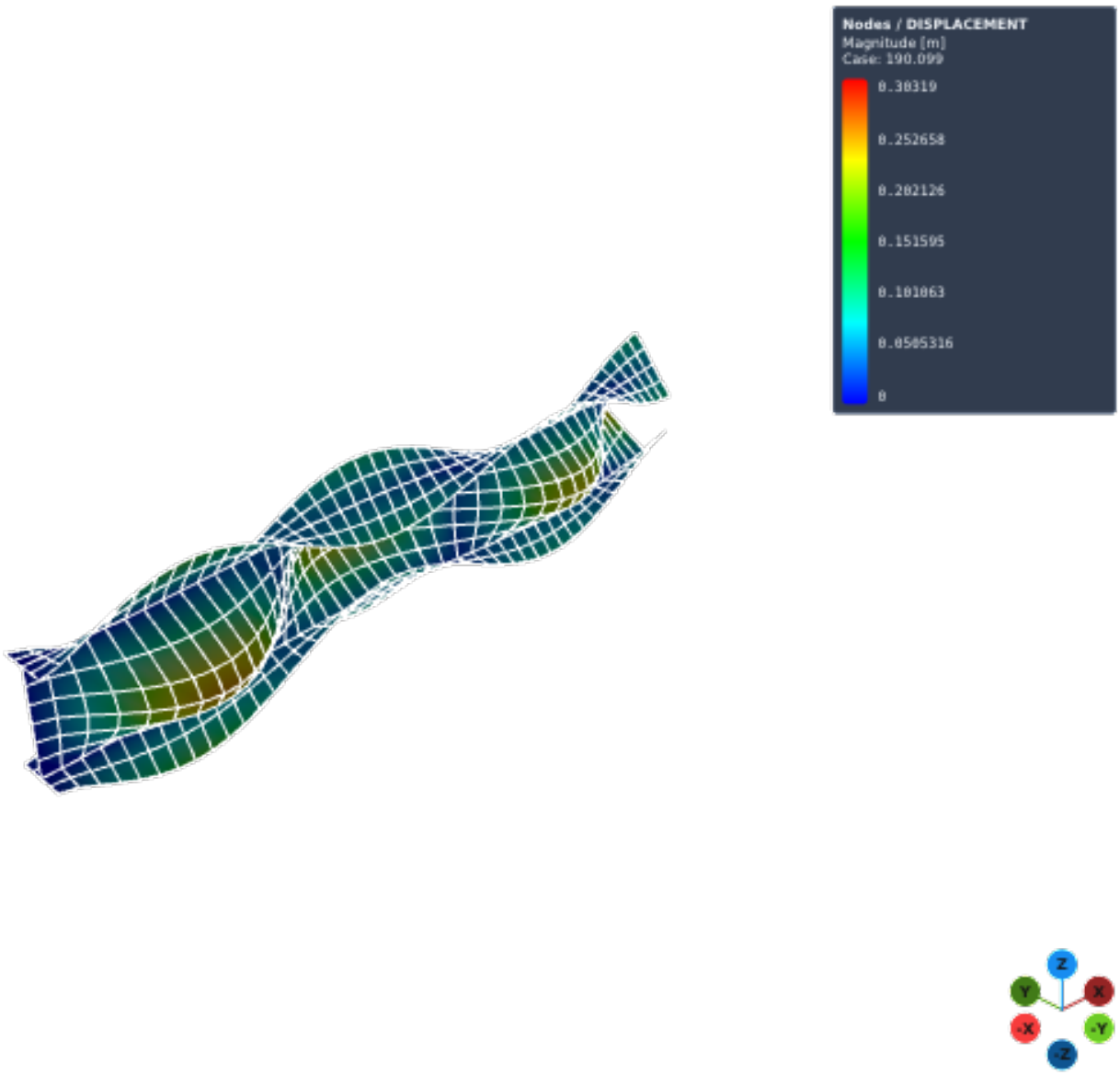

Figure 162: Sesam — mode 9, 190.099 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	198.750	+1.49	1.55943e+06	6.12e-22	0.59	1.15e-20	2.47e-11	-0.771	1.07e-10
Calculix	200.494	+2.38	1.58696e+06	1.04e-25	0.58	3.83e-13	-3.23e-13	-0.760	6.19e-07
Code_Aster	193.960	-0.95	1.48519e+06	2.45e-27	0.63	5.78e-29	-4.95e-14	-0.795	7.61e-15
Sesam	190.099	-2.92	1.42665e+06	1.48e-29	0.64	2.81e-27	-3.85e-15	-0.798	5.30e-14

Mode 10

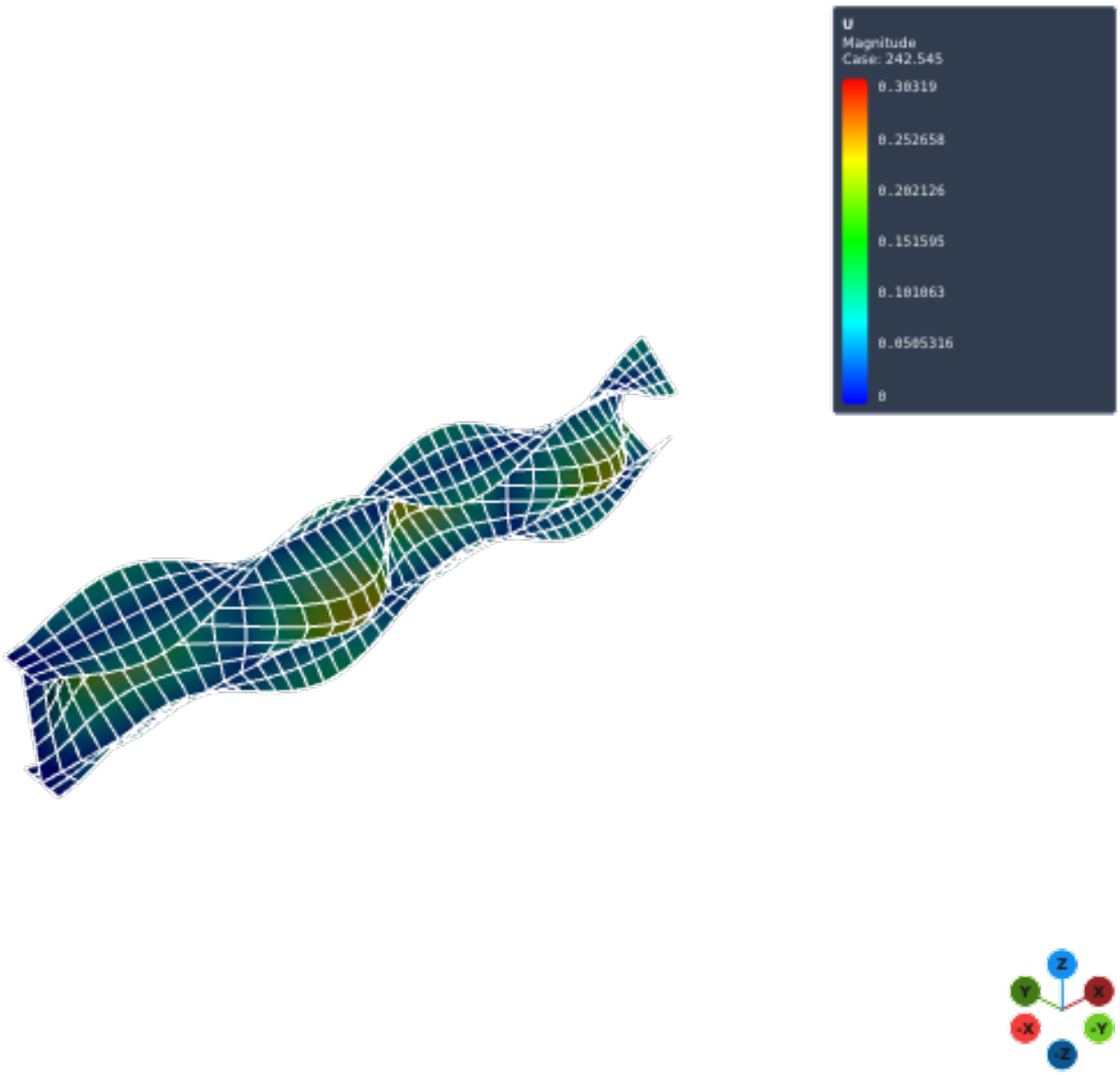

Figure 163: Abaqus — mode 10, 242.545 Hz

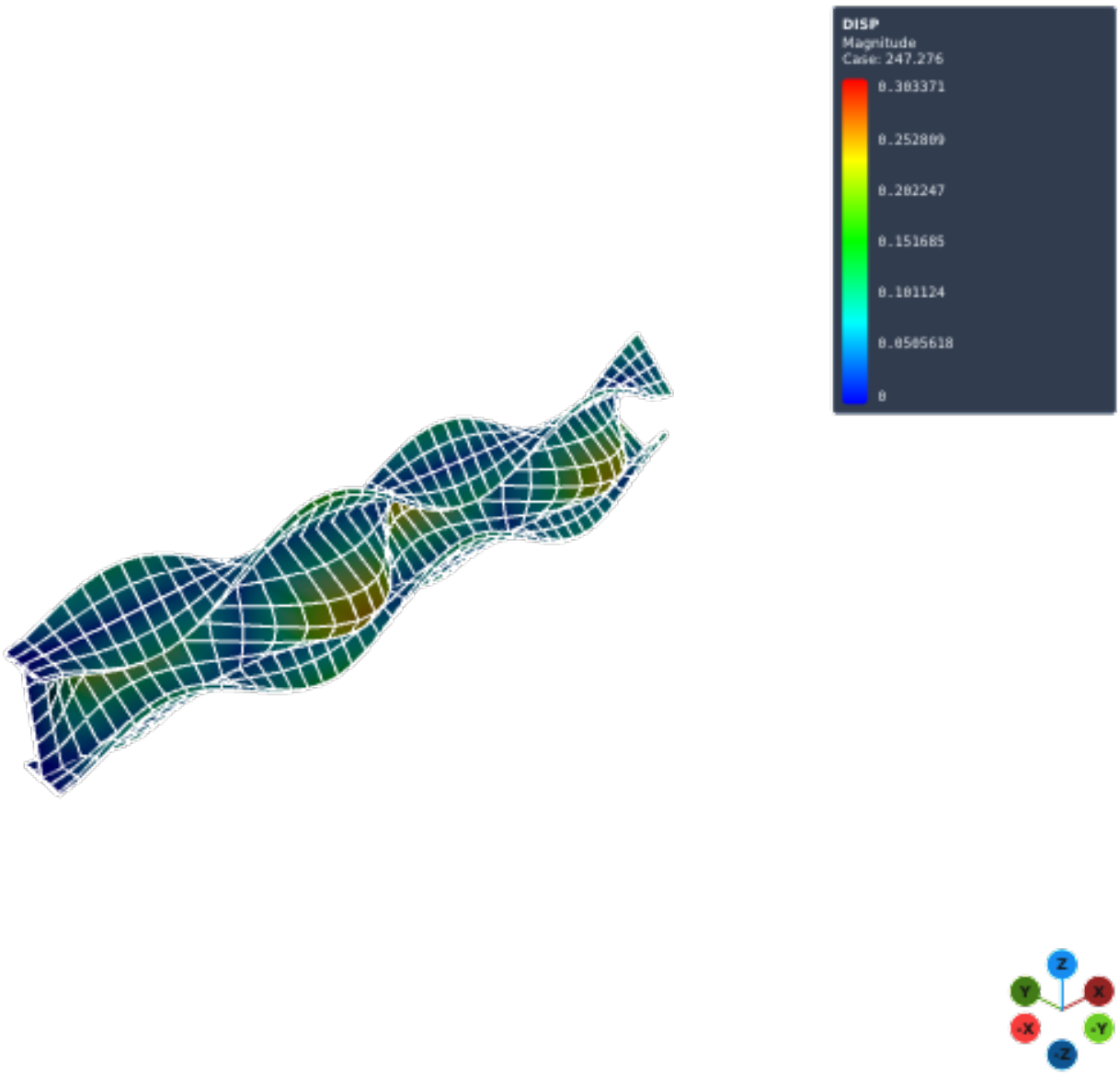

Figure 164: Calculix — mode 10, 247.276 Hz

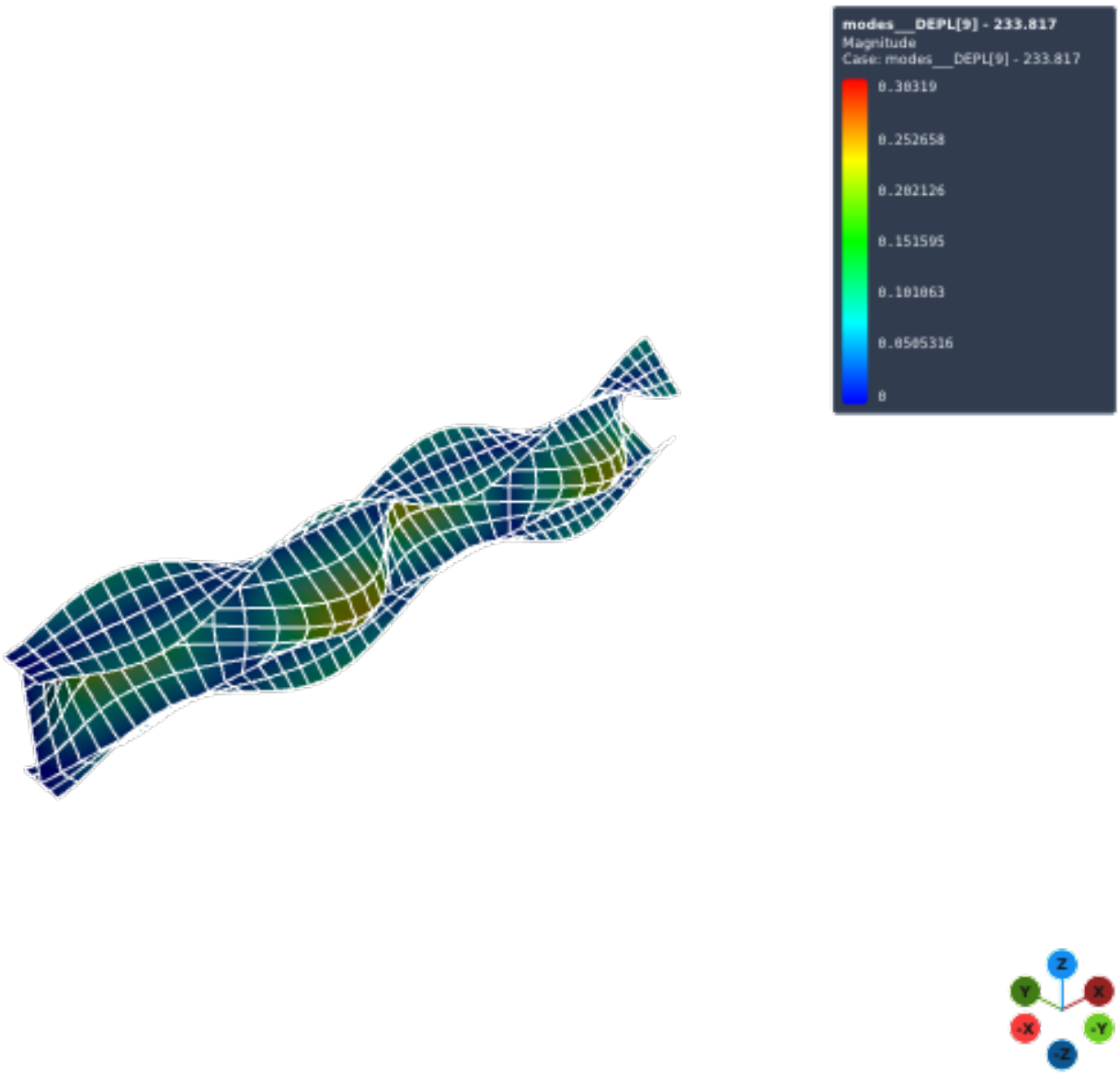

Figure 165: Code_Aster — mode 10, 233.817 Hz

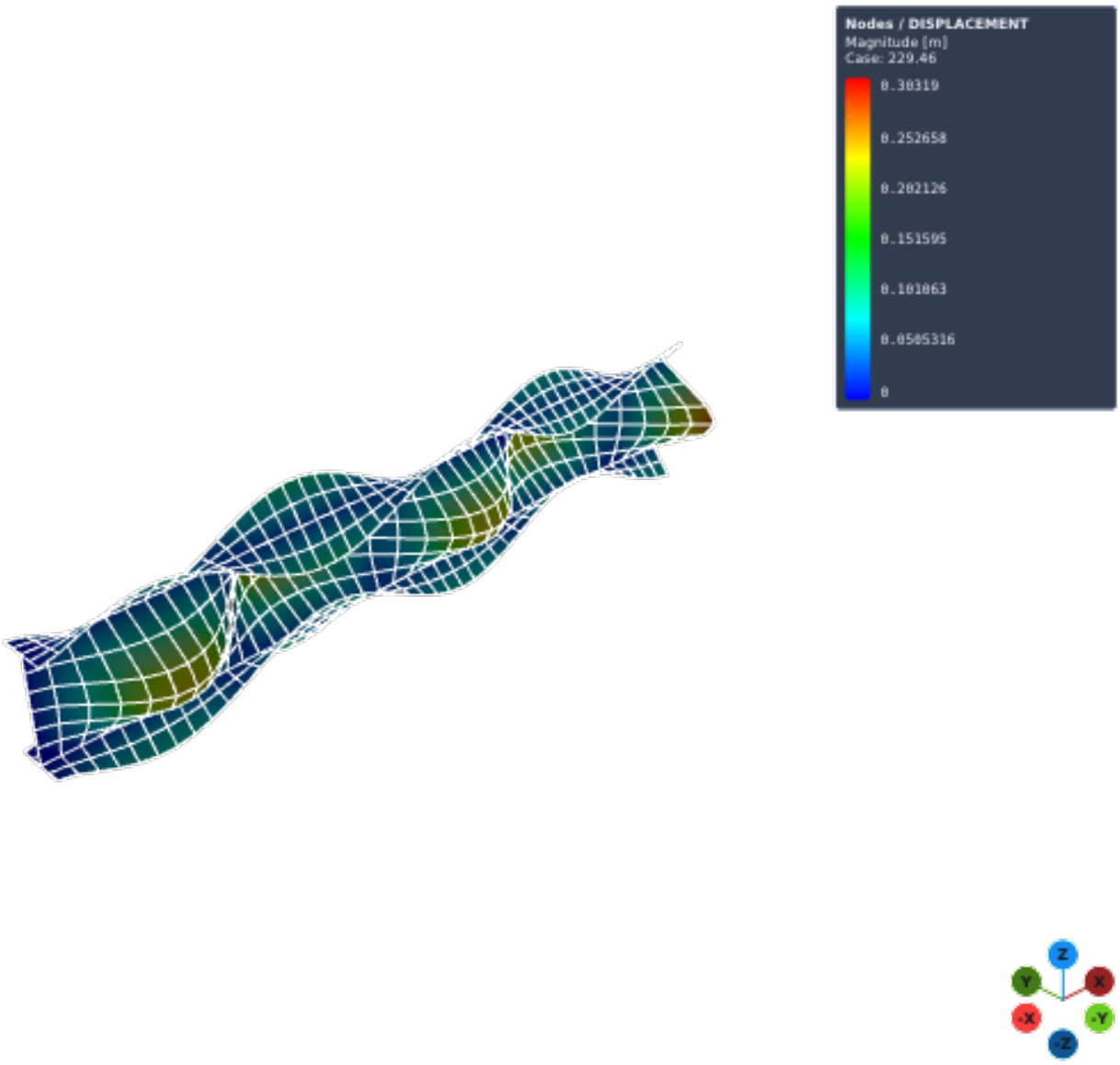

Figure 166: Sesam — mode 10, 229.460 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	242.550	+1.79	2.32244e+06	3.62e-17	1.52	5.68e-16	6.02e-09	1.232	2.38e-08
Calculix	247.276	+3.78	2.41392e+06	6.50e-25	1.71	2.90e-13	-8.06e-13	1.310	5.39e-07
Code_Aster	233.817	-1.87	2.1583e+06	3.50e-27	1.33	2.79e-26	-5.92e-14	1.152	-1.67e-13
Sesam	229.460	-3.70	2.07861e+06	5.01e-29	1.32	1.10e-28	7.08e-15	-1.150	1.05e-14

Mode 11

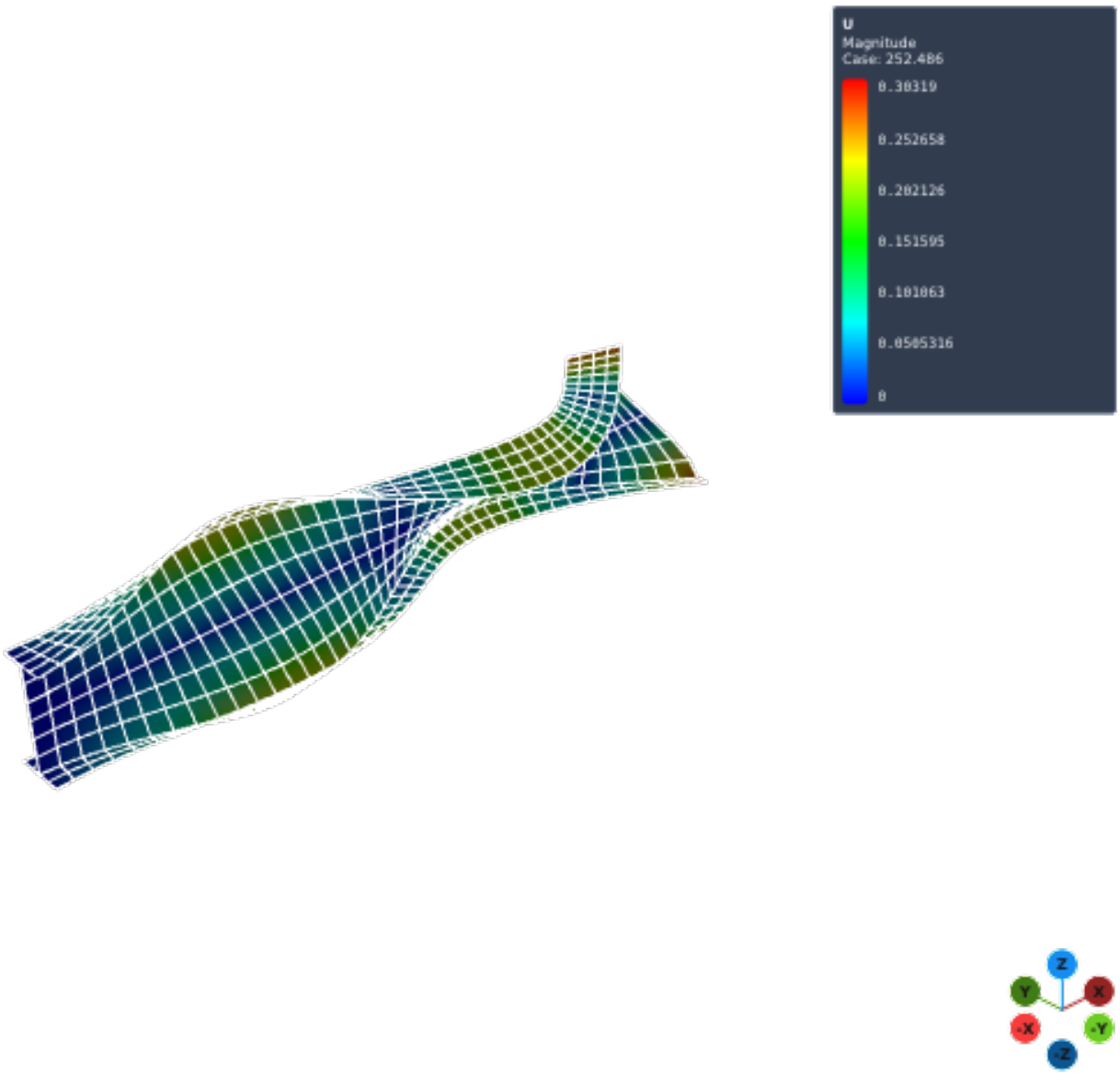

Figure 167: Abaqus — mode 11, 252.486 Hz

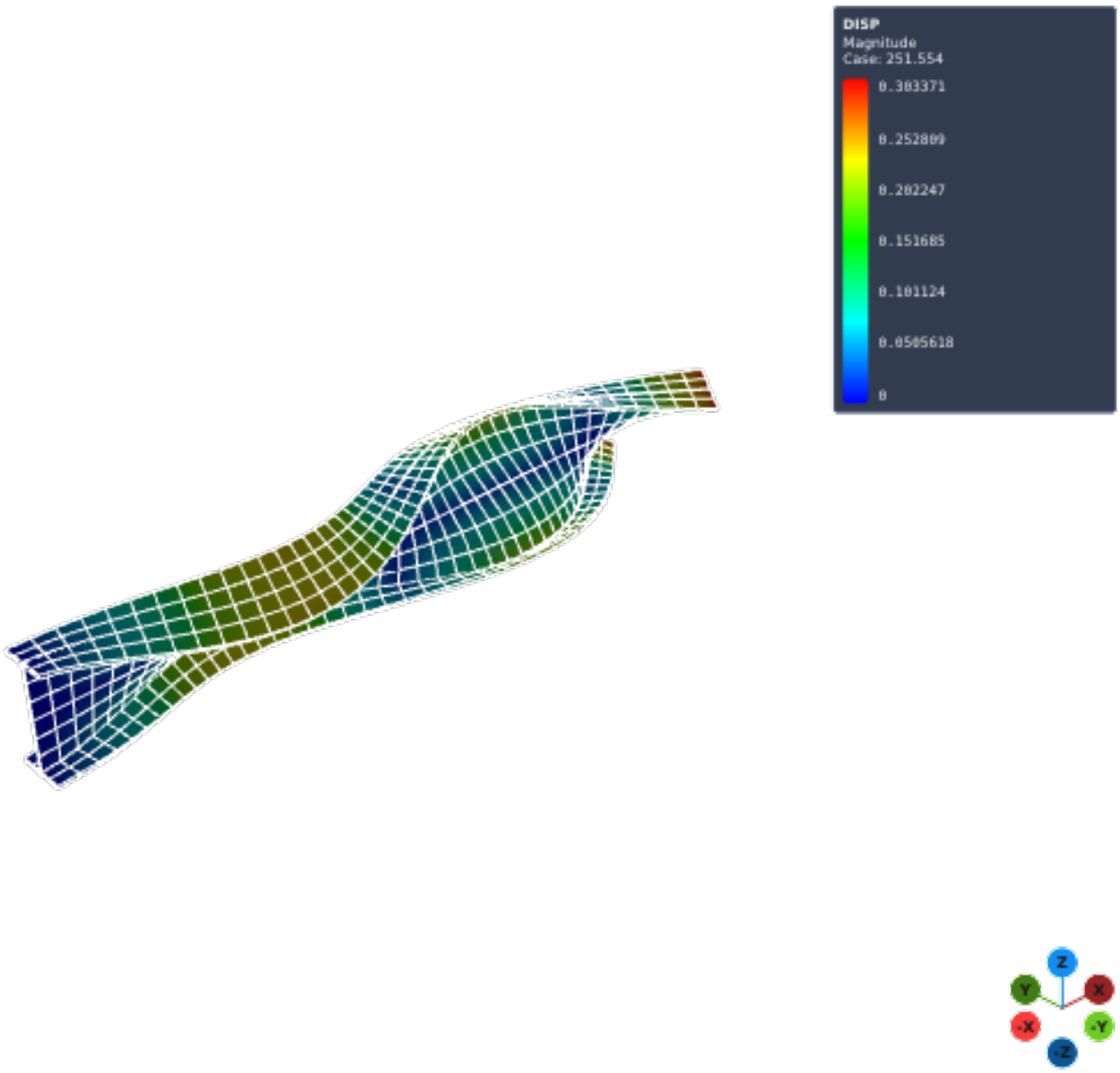

Figure 168: Calculix — mode 11, 251.554 Hz

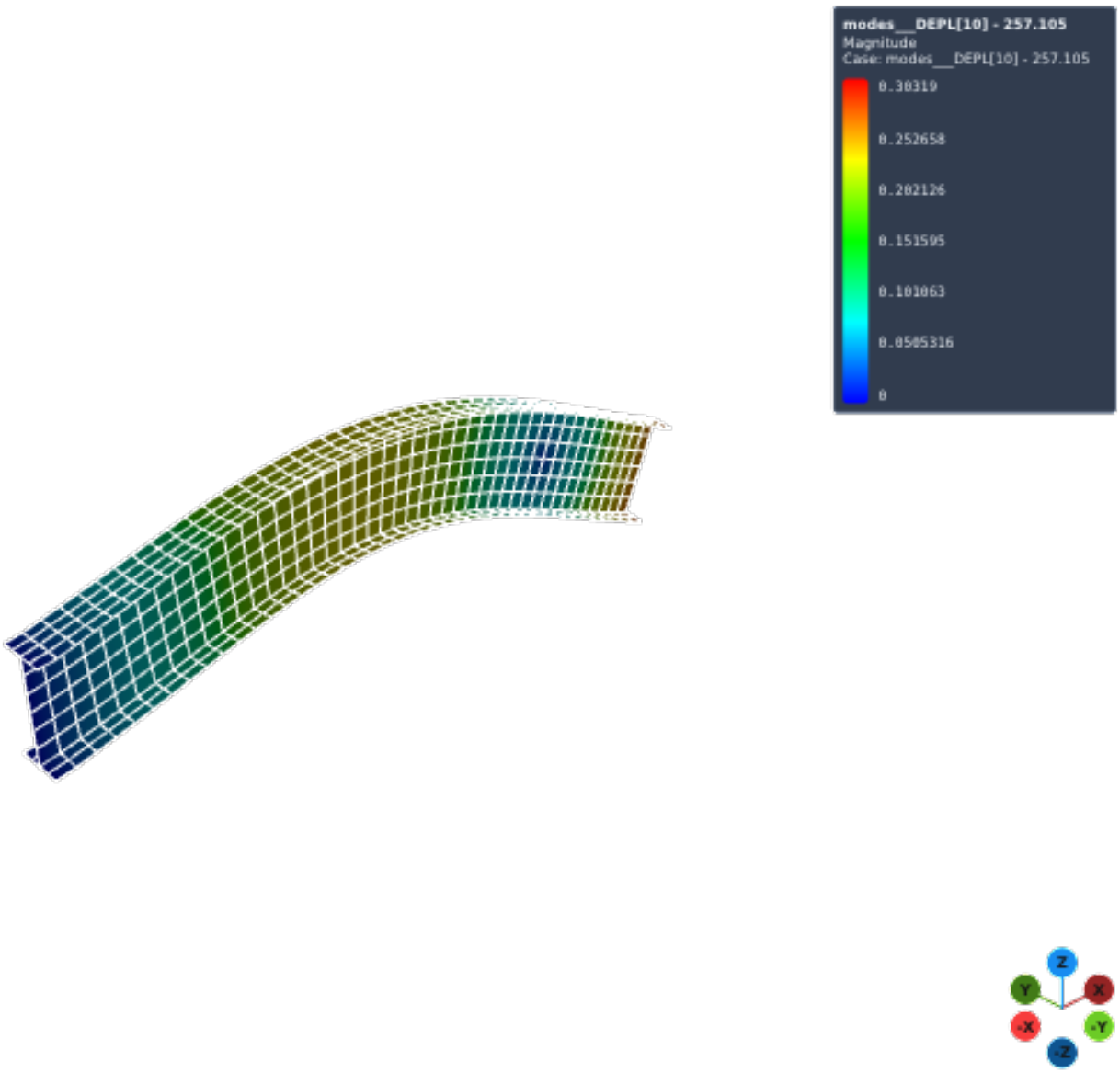

Figure 169: Code_Aster — mode 11, 257.105 Hz

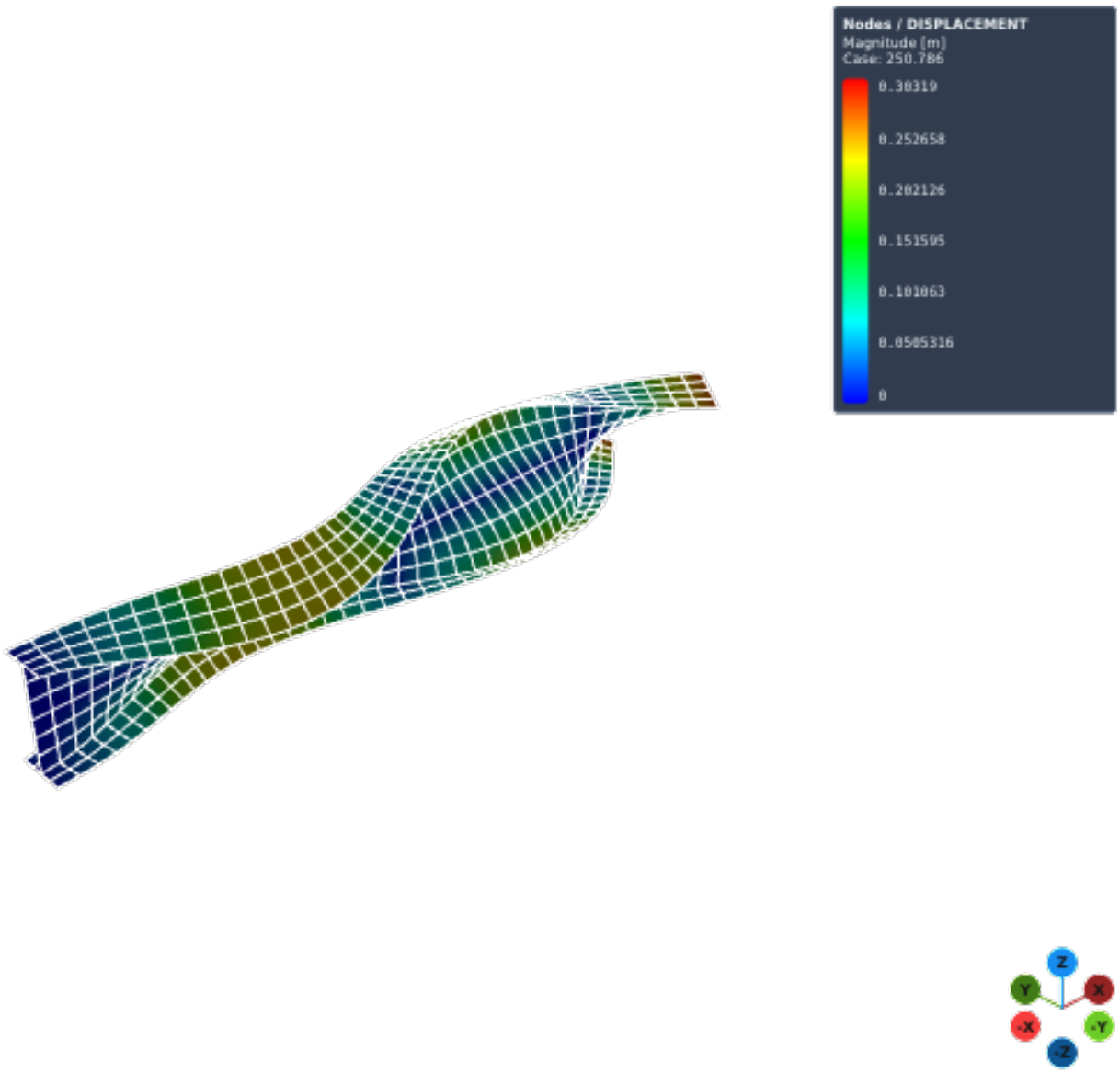

Figure 170: Sesam — mode 11, 250.786 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	252.490	-0.20	2.51672e+06	4.88e-17	7.27e-17	7.10e-16	6.99e-09	8.53e-09	2.67e-08
Calculix	251.554	-0.57	2.49818e+06	2.66e-10	6.32e-23	1.13e-21	1.63e-05	7.95e-12	3.37e-11
Code_Aster	257.105	+1.63	2.60965e+06	1.80e-26	2.95e-31	41.64	1.34e-13	5.43e-16	6.453
Sesam	250.786	-0.87	2.48294e+06	6.32e-21	7.50e-20	8.72e-20	7.95e-11	-2.74e-10	2.95e-10

Shell, 1st order, QUAD, reduced integration

Mode 1

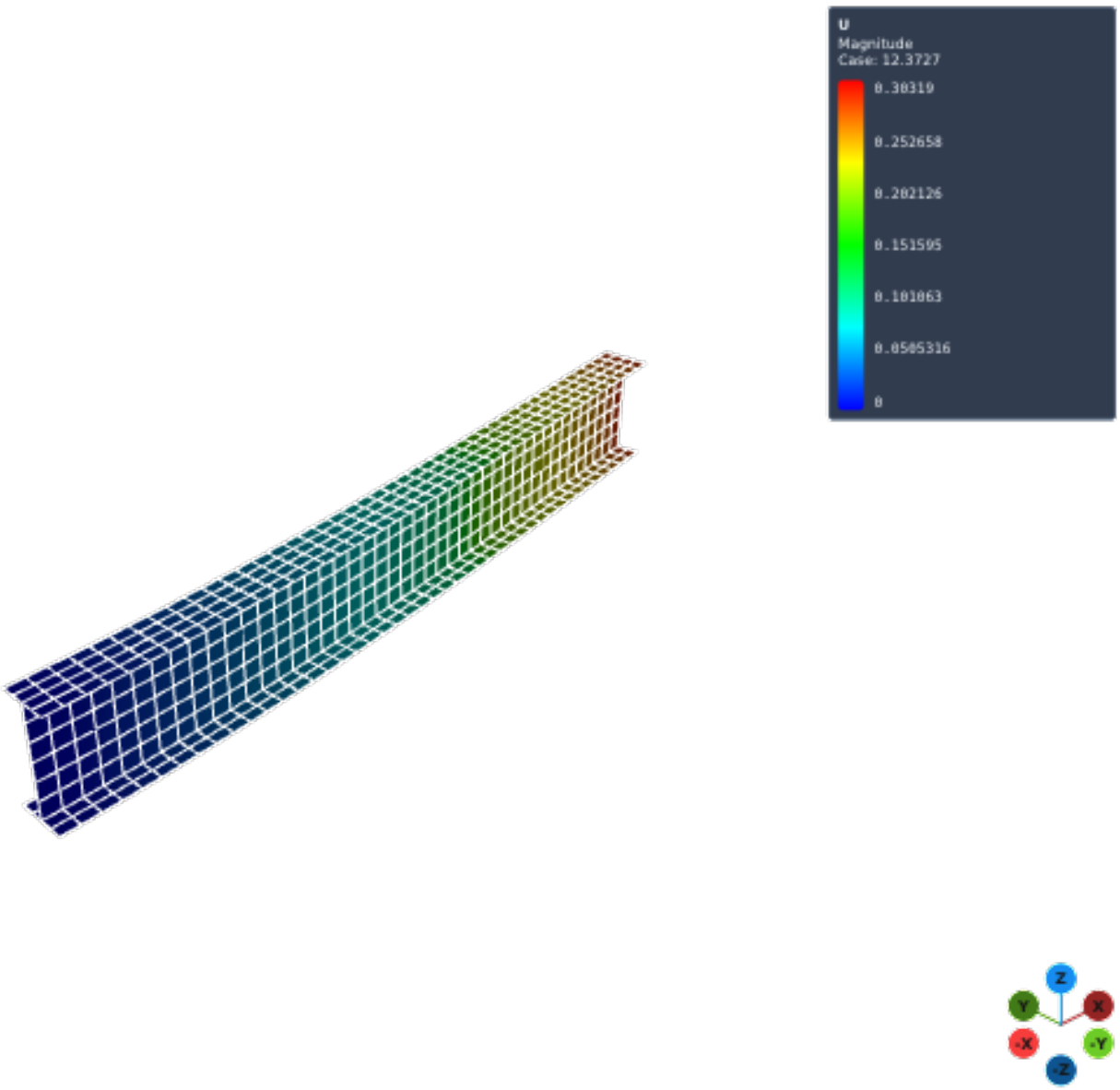

Figure 171: Abaqus — mode 1, 12.373 Hz

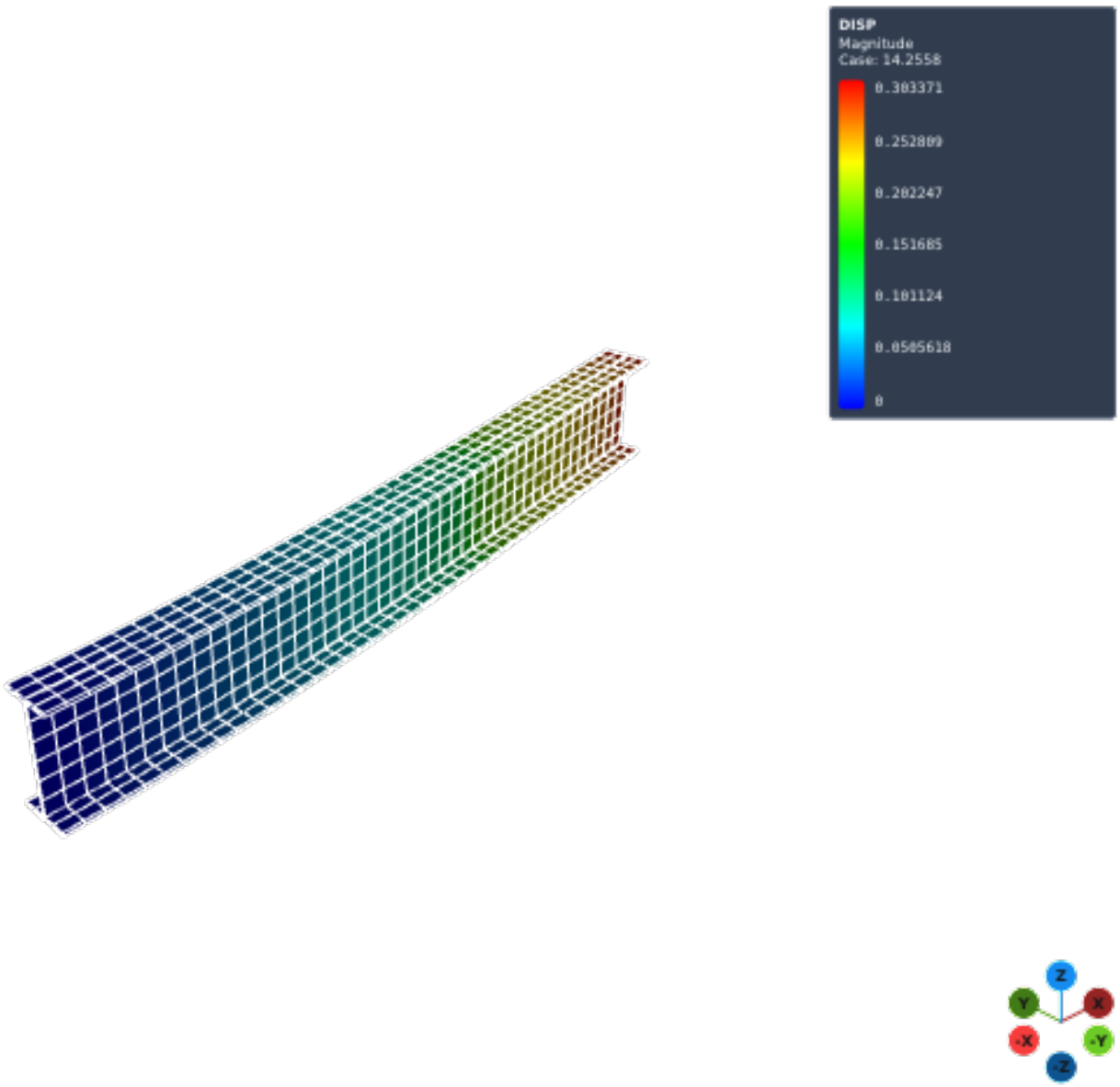

Figure 172: Calculix — mode 1, 14.256 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.373	-7.07	6043.5	1.78e-28	119.95	5.35e-25	-1.33e-14	10.952	-7.31e-13
Calculix	14.256	+7.07	8023.13	1.87e-26	119.90	6.12e-22	-1.37e-13	10.950	2.47e-11

Mode 2

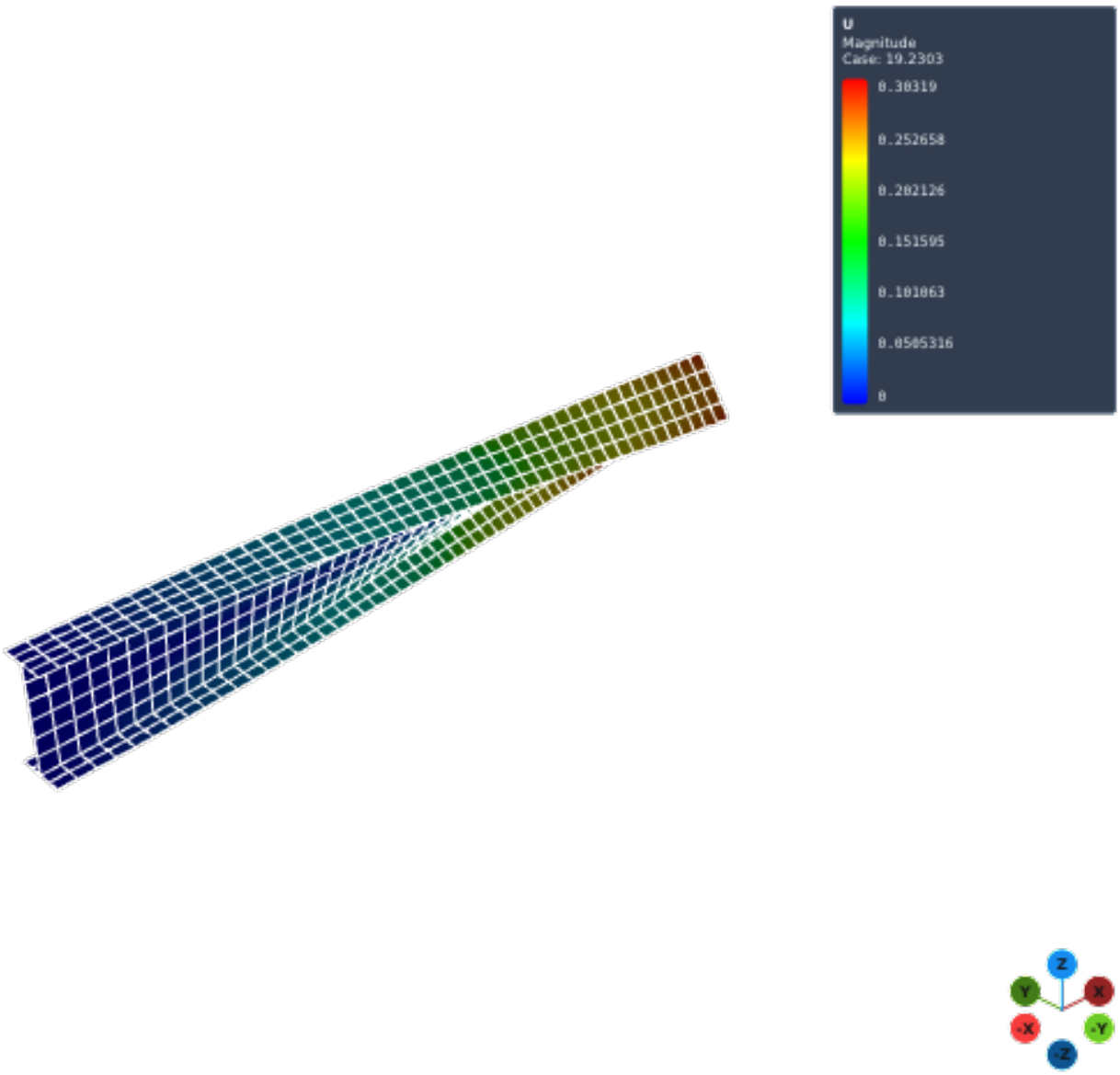

Figure 173: Abaqus — mode 2, 19.230 Hz

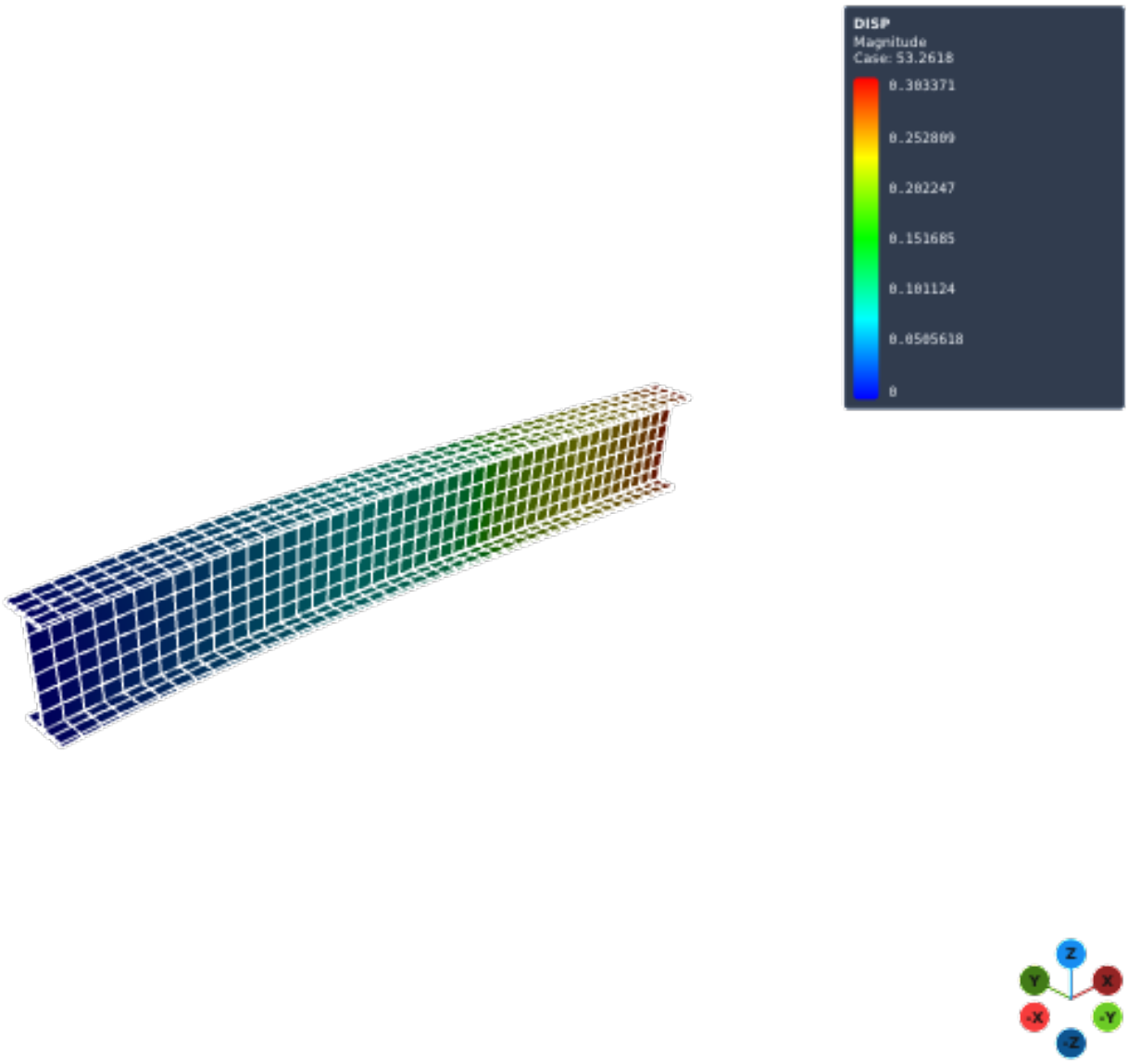

Figure 174: Calculix — mode 2, 53.262 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	19.230	-46.95	14599	1.25e-26	4.60e-21	3.10e-26	-1.12e-13	6.78e-11	1.76e-13
Calculix	53.262	+46.95	111993	4.93e-27	8.03e-22	122.13	-7.02e-14	2.83e-11	-11.051

Mode 3

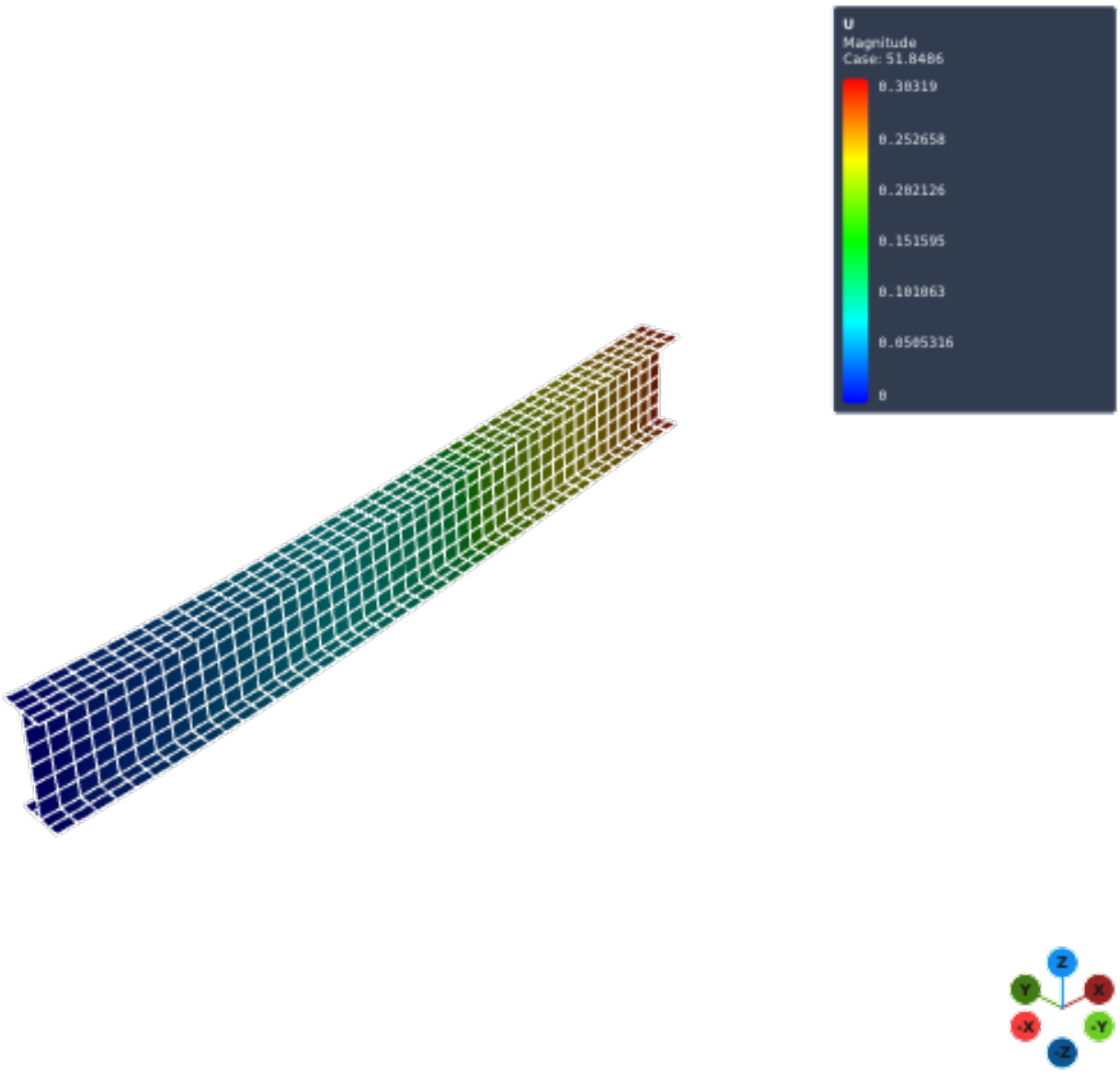

Figure 175: Abaqus — mode 3, 51.849 Hz

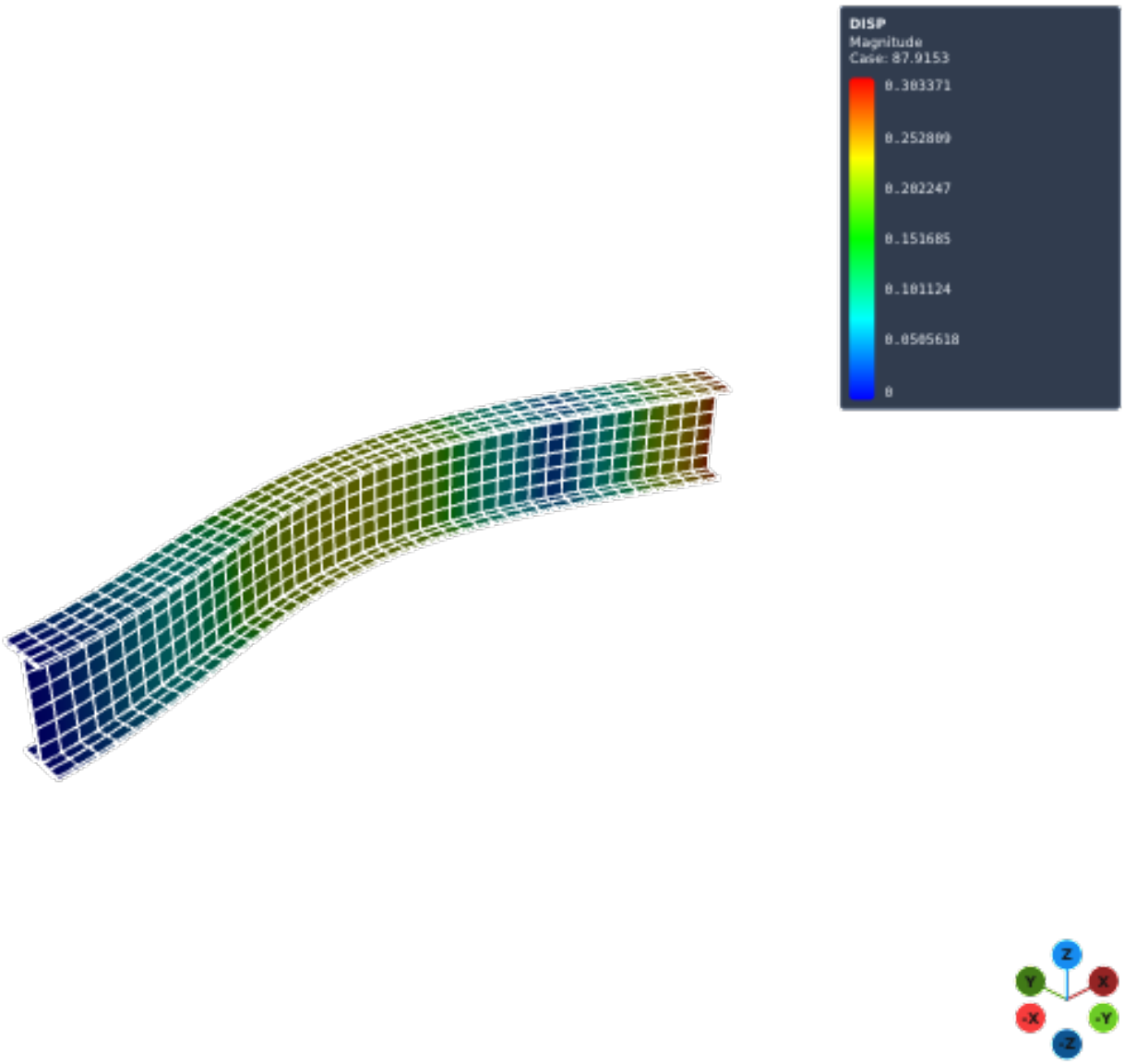

Figure 176: Calculix — mode 3, 87.915 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.849	-25.81	106129	2.48e-26	1.19e-25	122.02	-1.58e-13	3.45e-13	11.046
Calculix	87.915	+25.81	305133	8.49e-26	37.23	4.65e-23	-2.91e-13	6.101	6.82e-12

Mode 4

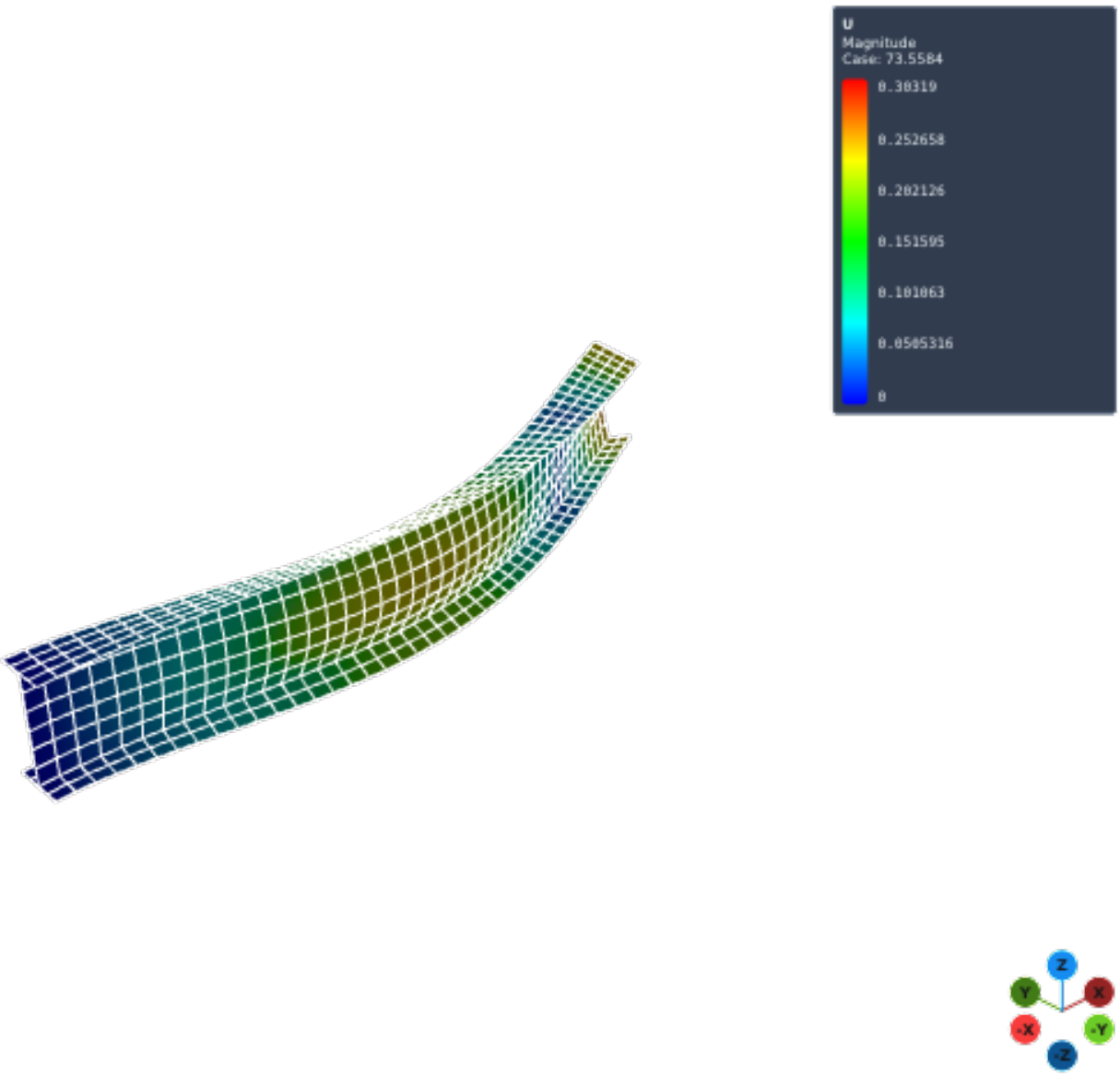

Figure 177: Abaqus — mode 4, 73.558 Hz

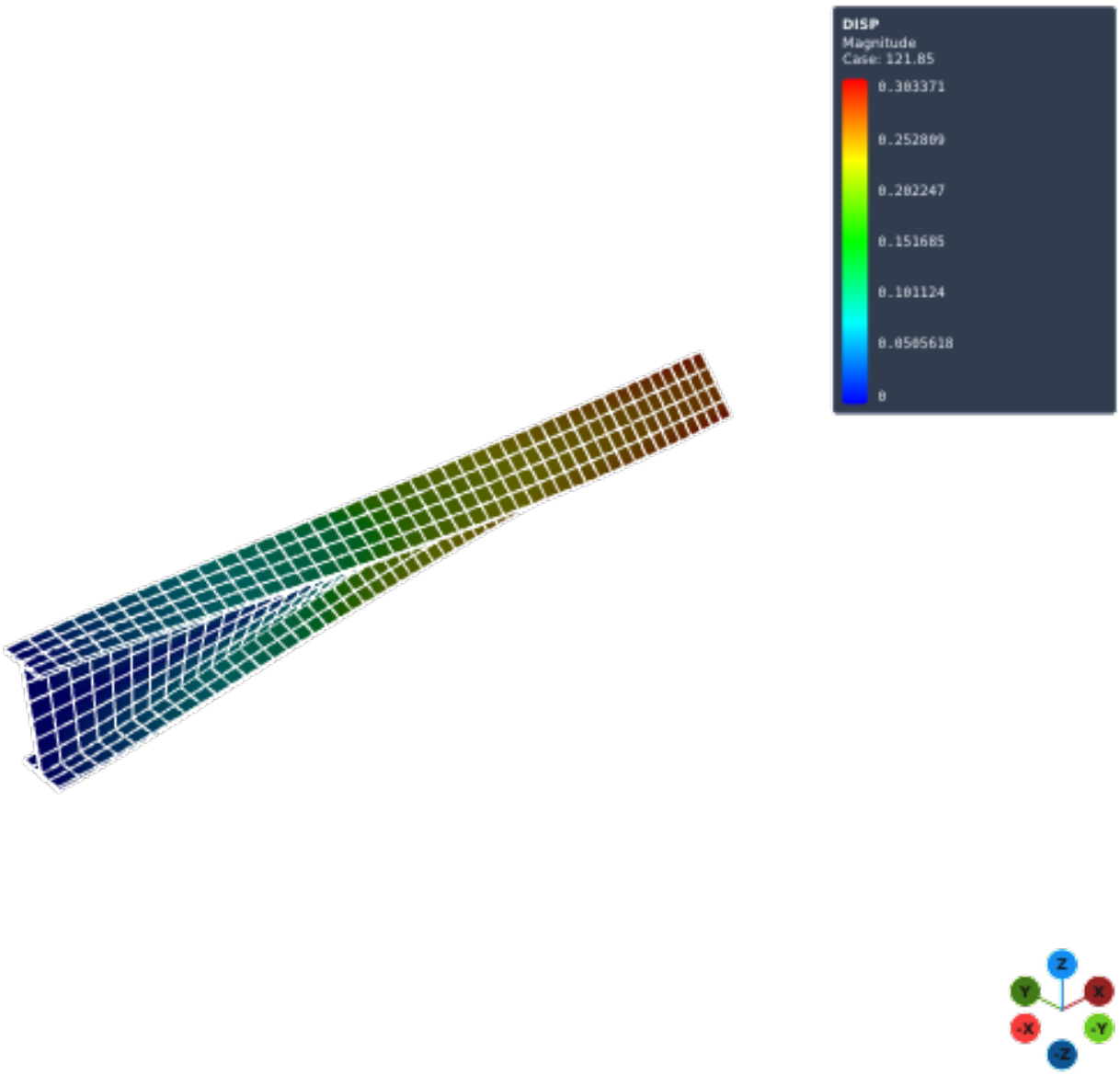

Figure 178: Calculix — mode 4, 121.850 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	73.558	-24.71	213611	2.90e-28	36.98	6.94e-25	-1.70e-14	-6.081	-8.33e-13
Calculix	121.850	+24.71	586156	6.61e-28	4.01e-22	9.84e-24	2.57e-14	2.00e-11	3.14e-12

Mode 5

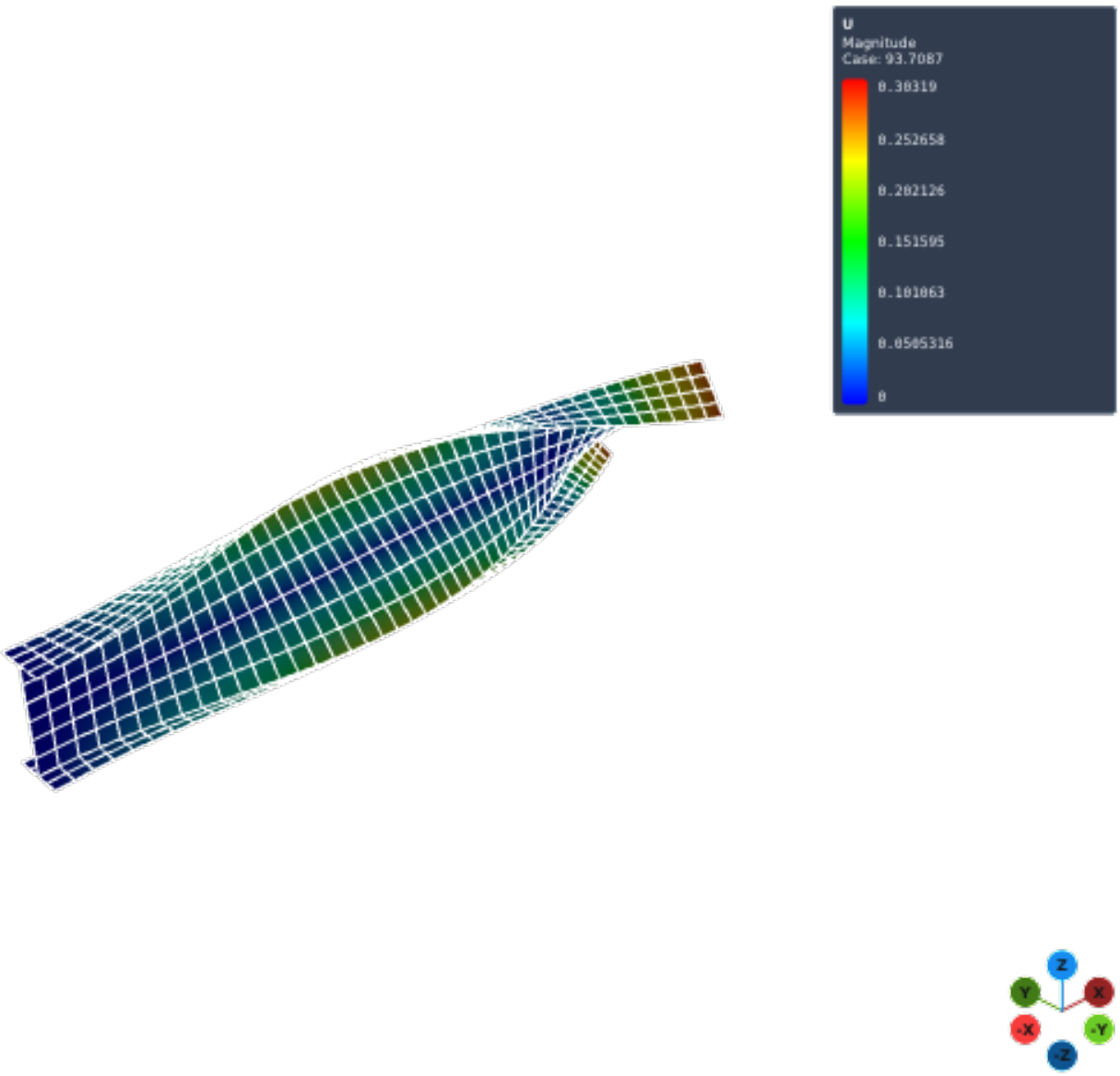

Figure 179: Abaqus — mode 5, 93.709 Hz

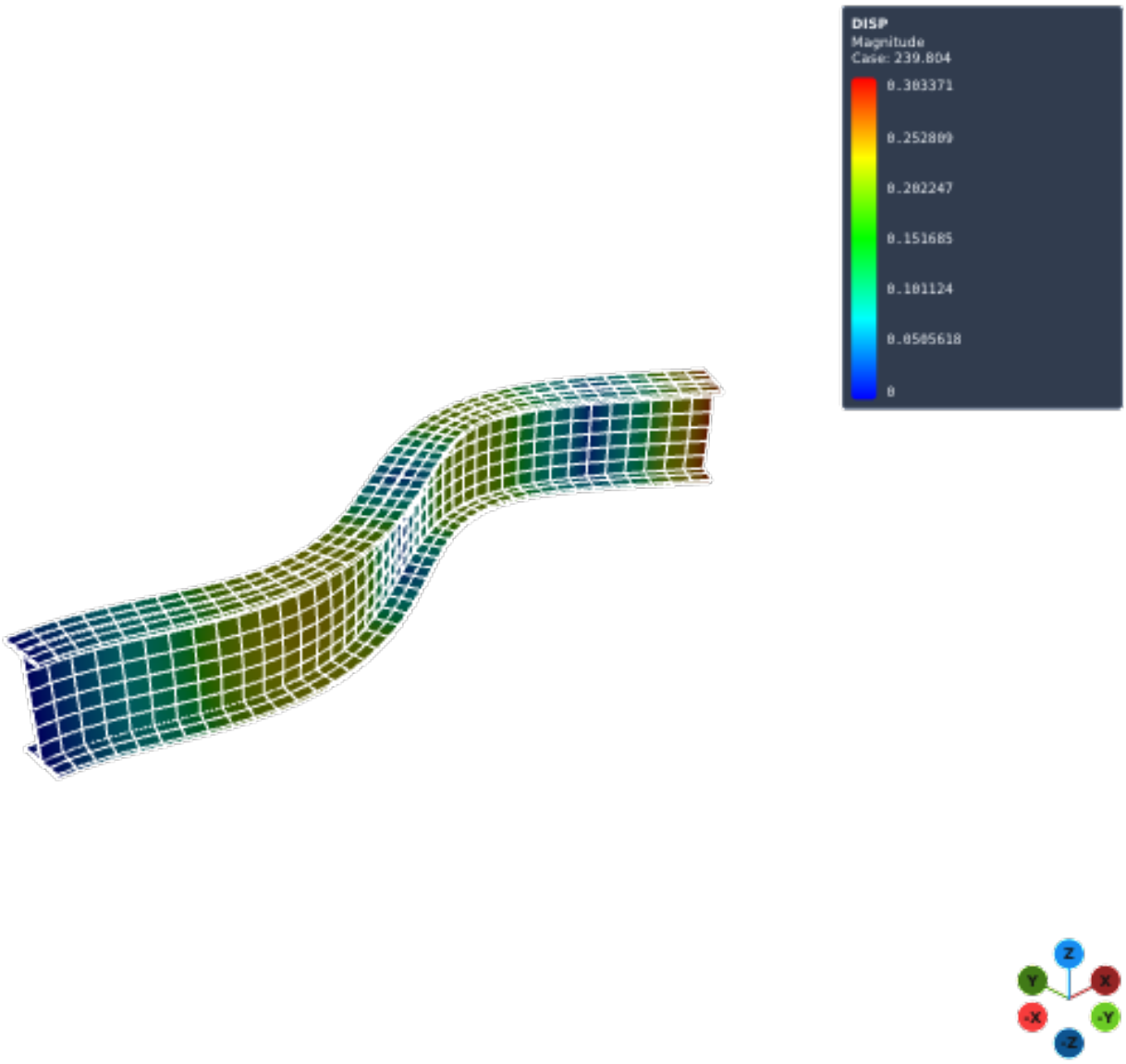

Figure 180: Calculix — mode 5, 239.804 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	93.709	-43.80	346673	9.12e-28	8.02e-23	5.37e-26	3.02e-14	-8.96e-12	2.32e-13
Calculix	239.804	+43.80	2.27025e+06	1.53e-26	12.96	3.22e-24	-1.24e-13	-3.601	1.80e-12

Mode 6

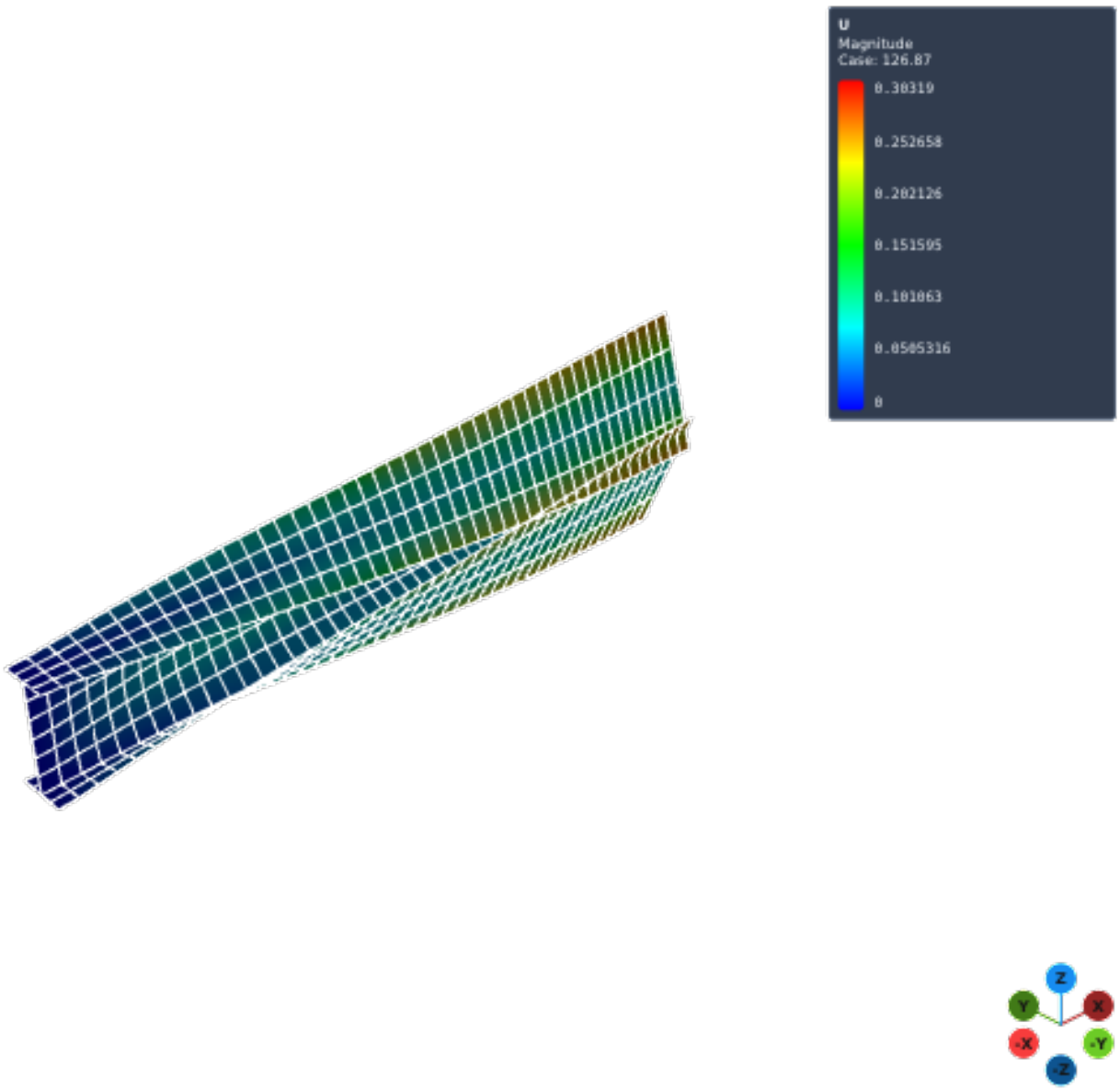

Figure 181: Abaqus — mode 6, 126.870 Hz

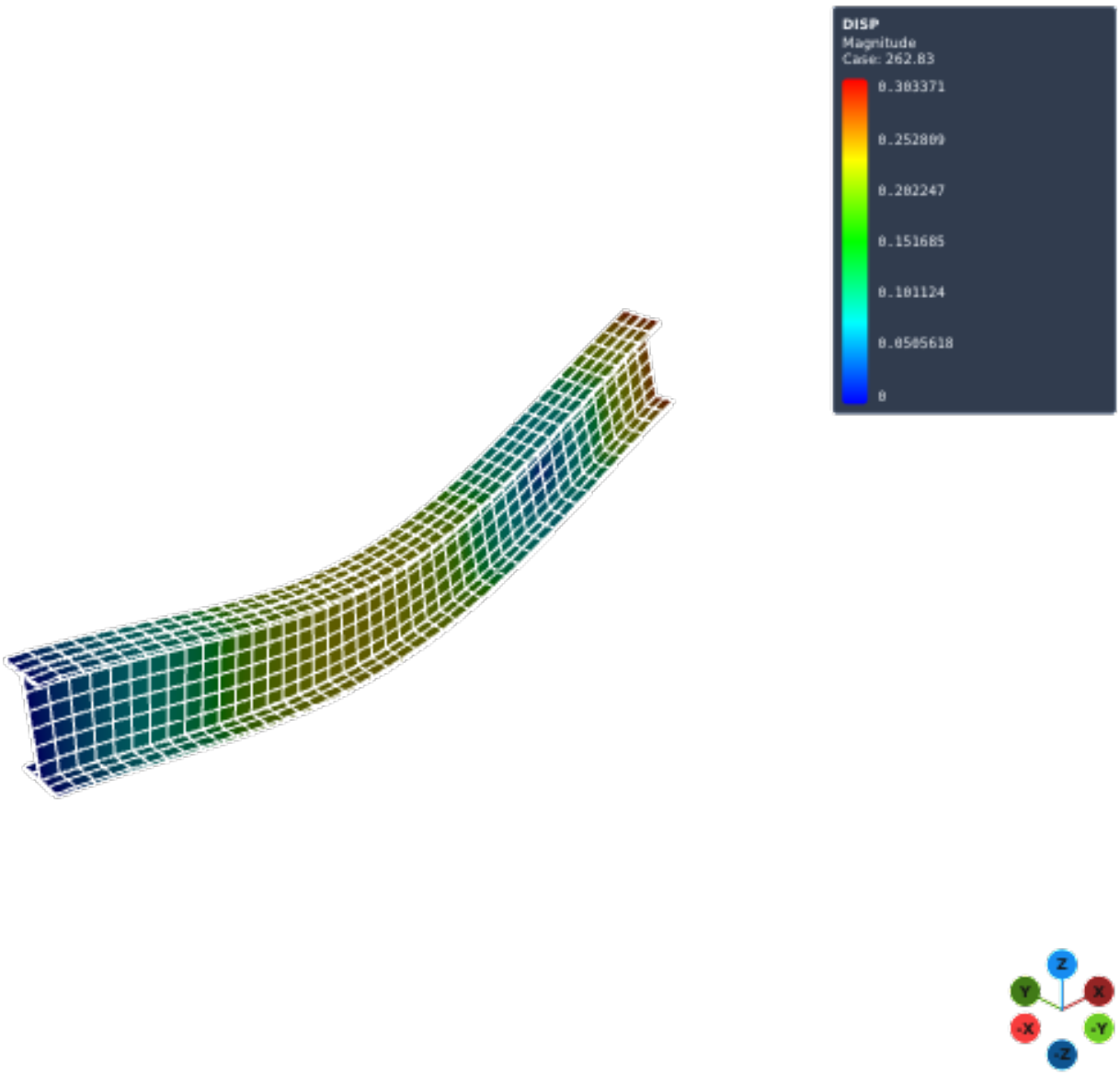

Figure 182: Calculix — mode 6, 262.830 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	126.870	-34.89	635447	1.80e-30	0.01	1.84e-25	1.34e-15	-0.078	-4.29e-13
Calculix	262.830	+34.89	2.72716e+06	2.80e-26	1.10e-24	41.52	1.67e-13	-1.05e-12	-6.443

Mode 7

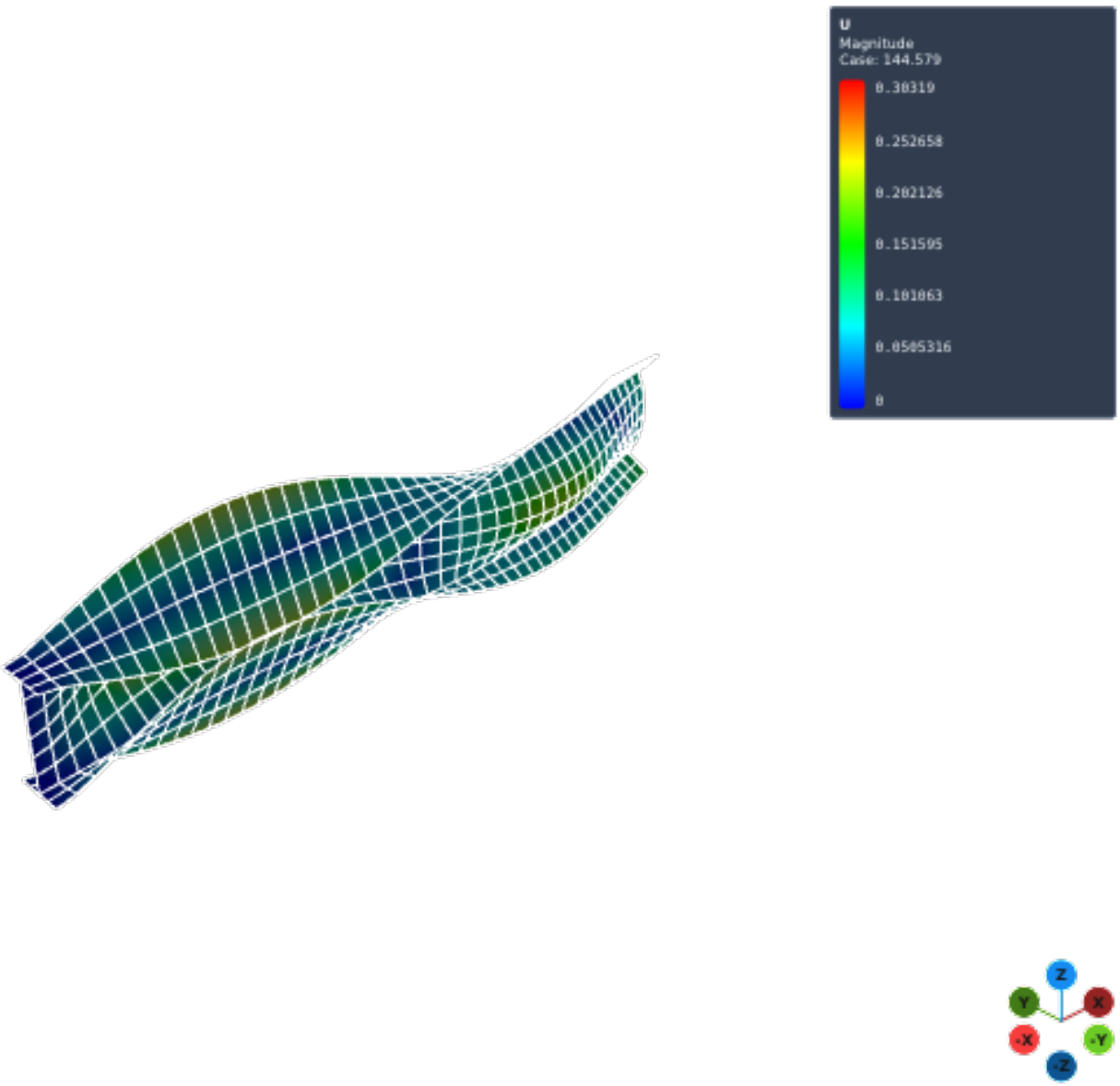

Figure 183: Abaqus — mode 7, 144.579 Hz

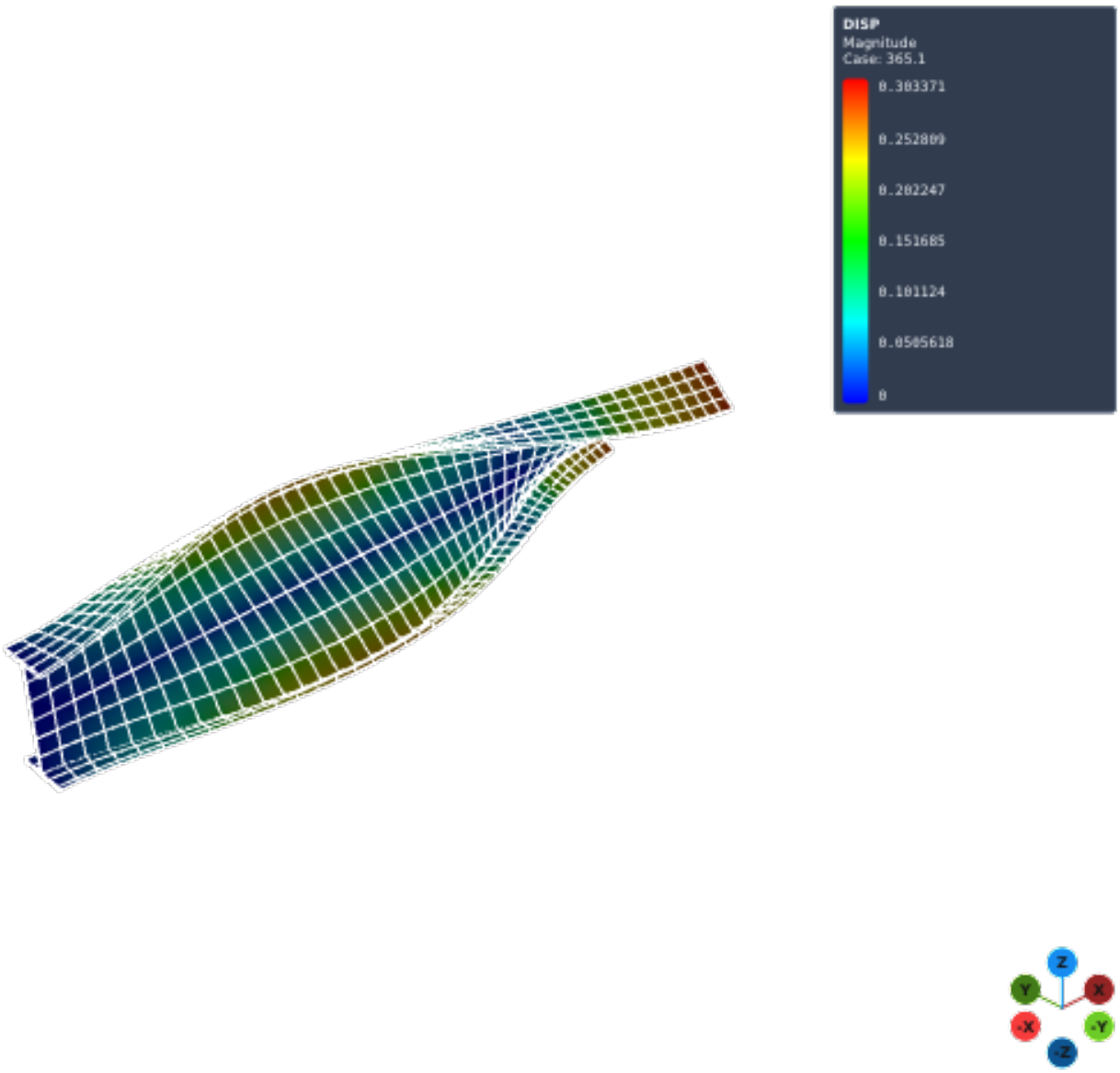

Figure 184: Calculix — mode 7, 365.100 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	144.580	-43.27	825226	1.99e-29	3.79	2.06e-26	-4.46e-15	1.947	1.43e-13
Calculix	365.100	+43.27	5.26239e+06	1.49e-25	6.79e-24	3.09e-25	-3.86e-13	-2.61e-12	5.56e-13

Mode 8

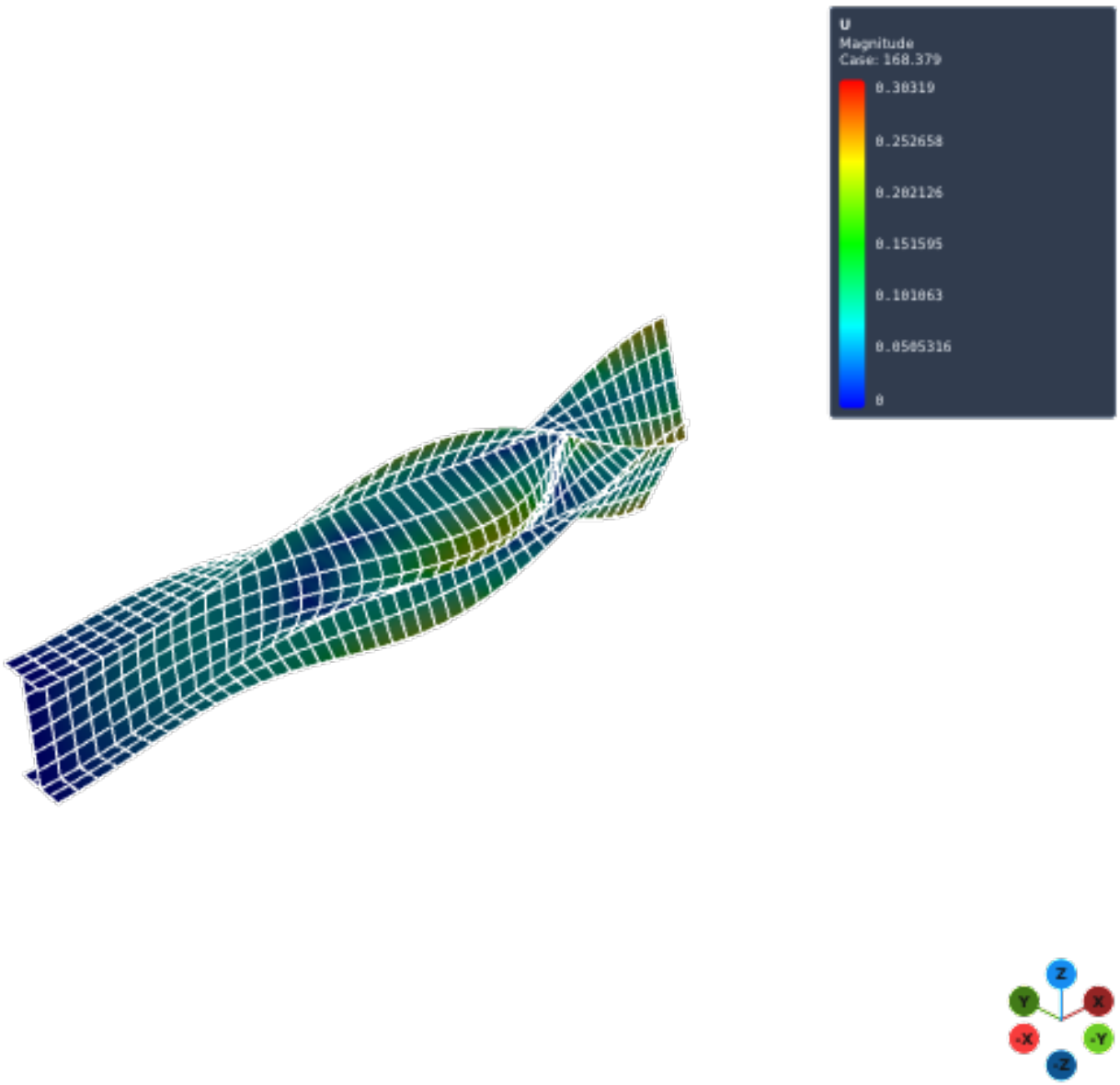

Figure 185: Abaqus — mode 8, 168.379 Hz

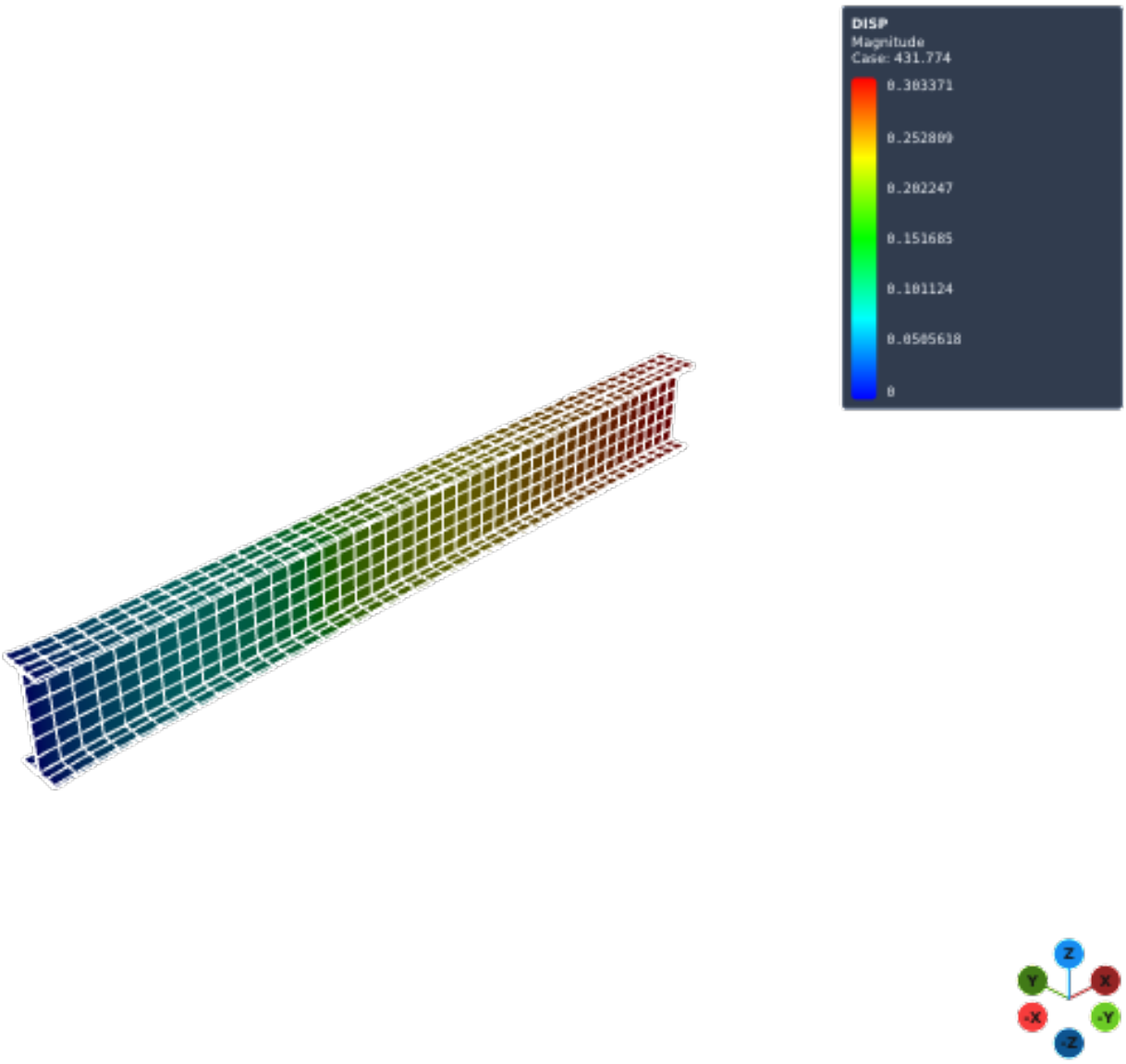

Figure 186: Calculix — mode 8, 431.774 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	168.380	-43.89	1.11927e+06	2.51e-28	5.44	1.97e-25	1.58e-14	2.332	-4.43e-13
Calculix	431.774	+43.89	7.35992e+06	157.96	2.40e-26	6.49e-27	12.568	1.55e-13	8.05e-14

Mode 9

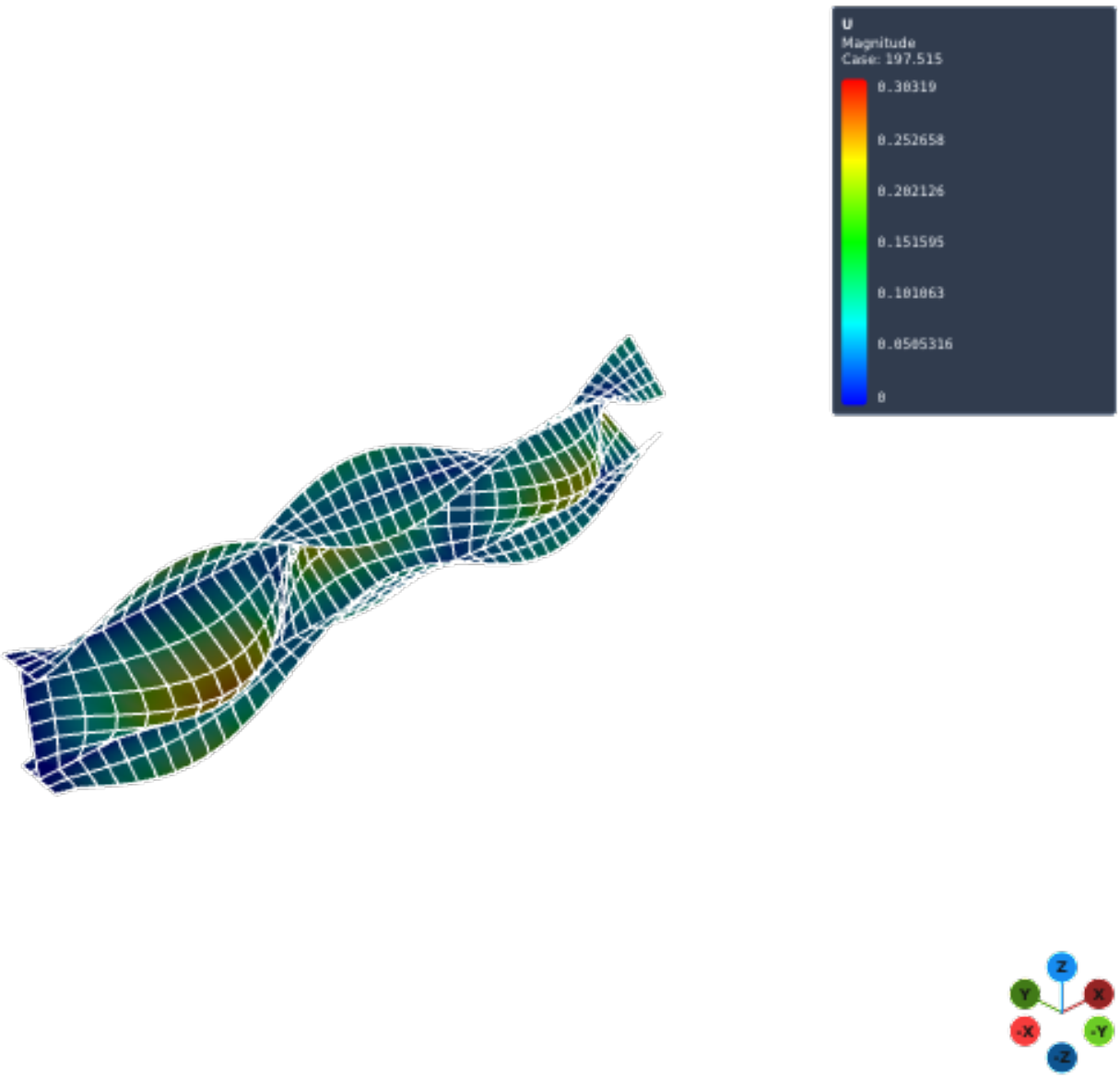

Figure 187: Abaqus — mode 9, 197.515 Hz

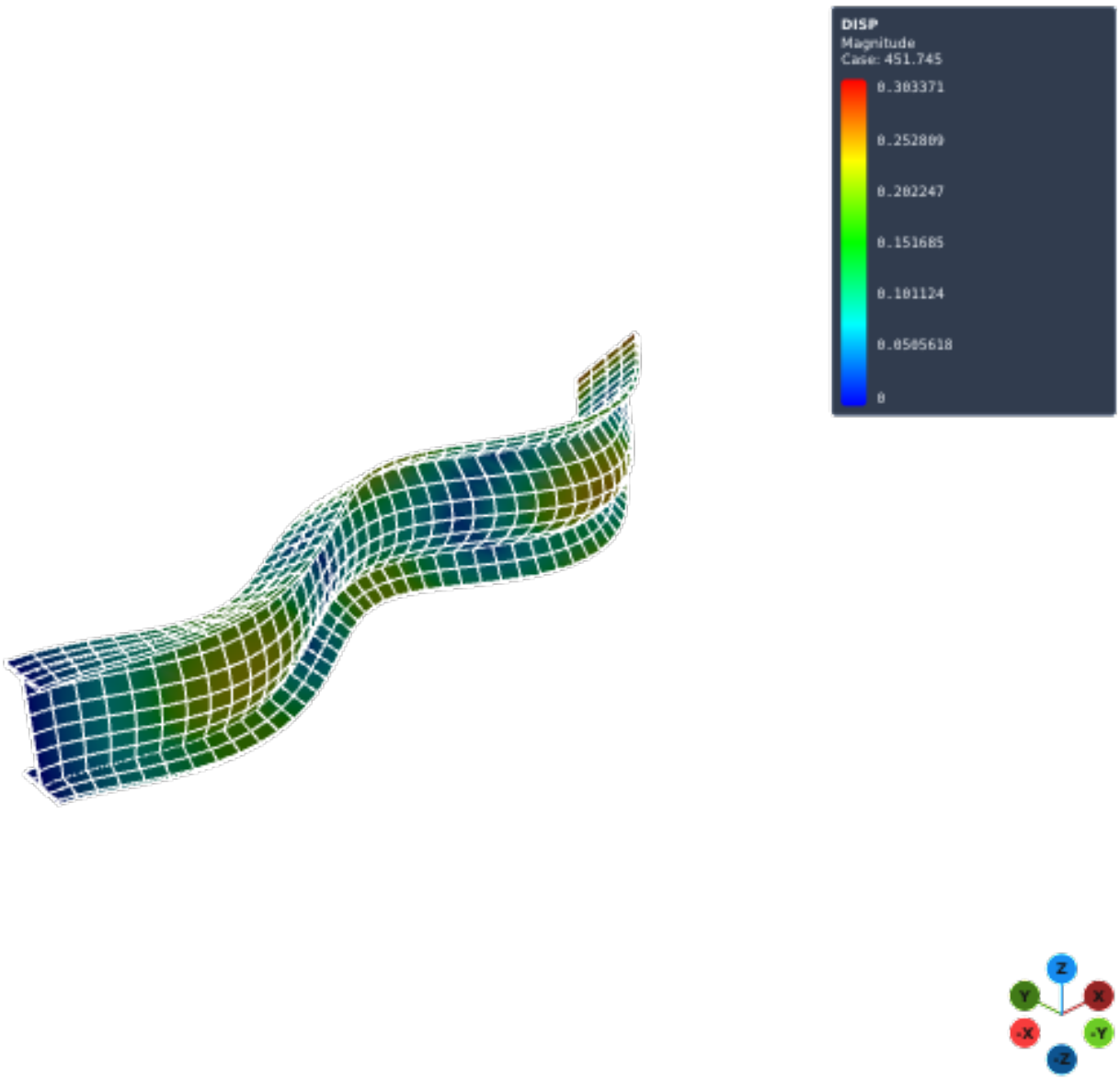

Figure 188: Calculix — mode 9, 451.745 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	197.510	-39.16	1.54014e+06	1.19e-26	0.62	1.84e-25	-1.09e-13	-0.786	4.29e-13
Calculix	451.745	+39.16	8.0565e+06	7.61e-25	5.70	3.22e-25	-8.72e-13	-2.388	-5.67e-13

Mode 10

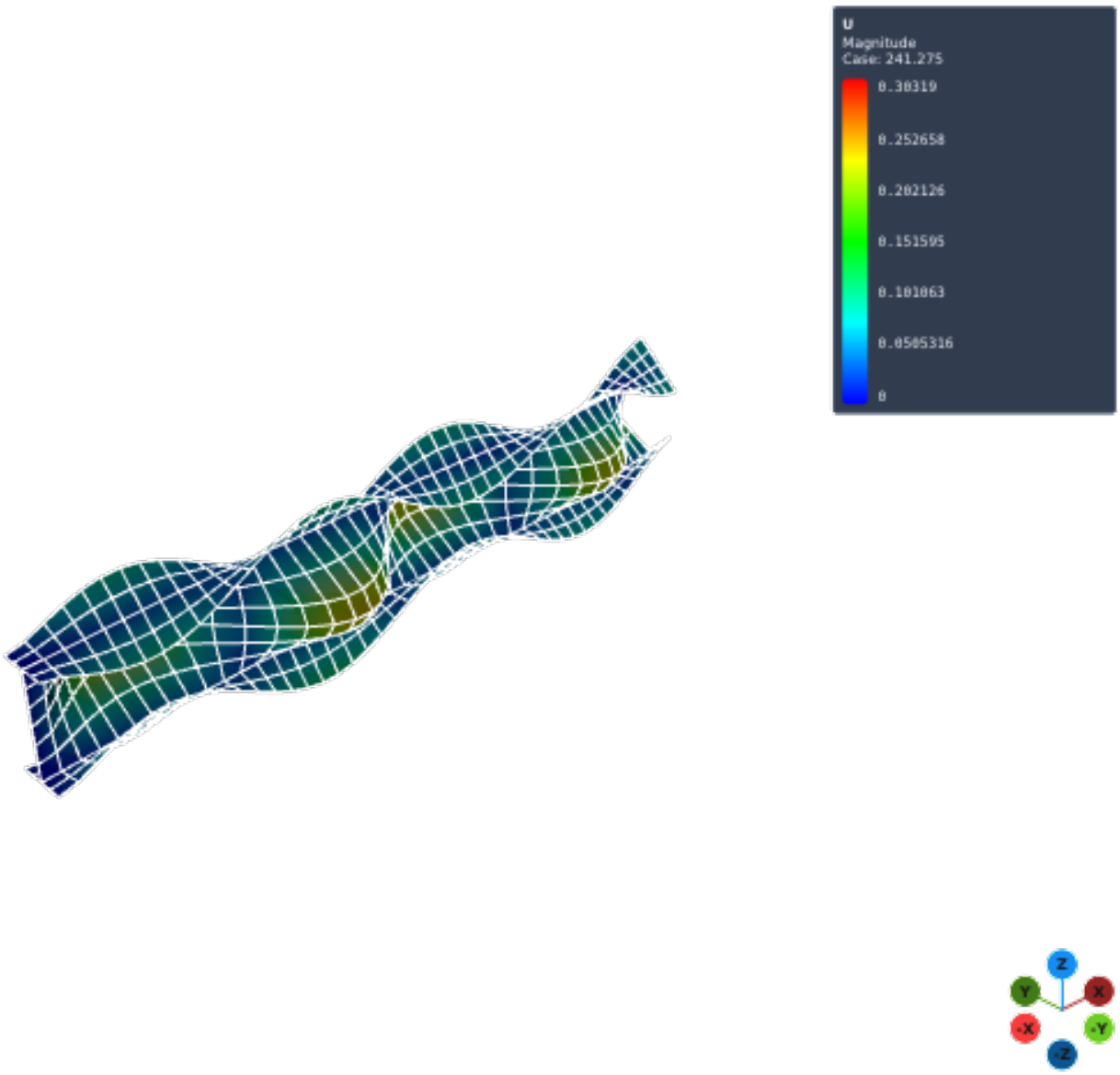

Figure 189: Abaqus — mode 10, 241.275 Hz

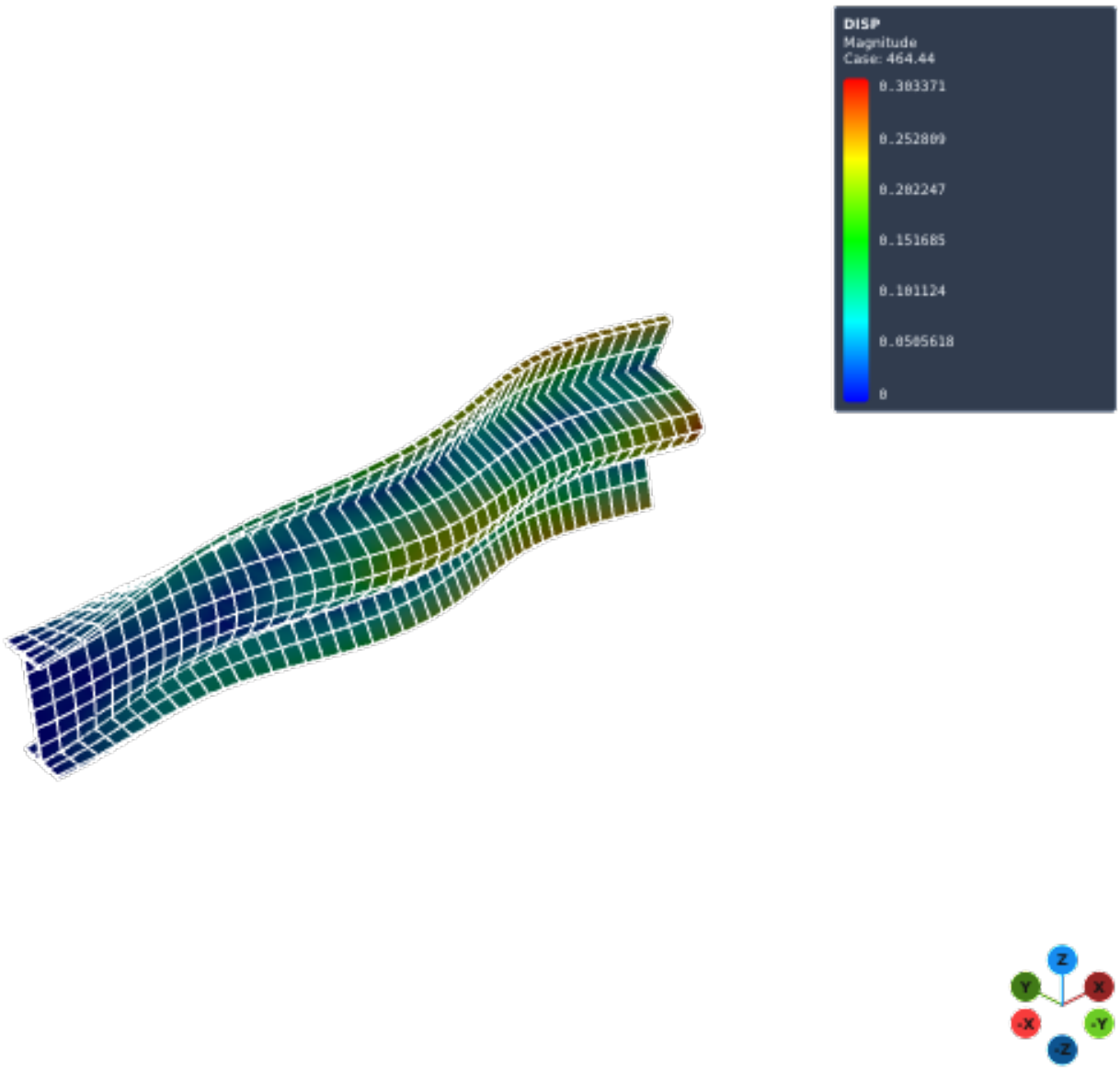

Figure 190: Calculix — mode 10, 464.440 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	241.270	-31.62	2.29818e+06	2.67e-22	1.61	1.69e-20	-1.63e-11	1.269	1.30e-10
Calculix	464.440	+31.62	8.51568e+06	1.15e-24	1.01	2.87e-24	-1.07e-12	1.006	-1.69e-12

Mode 11

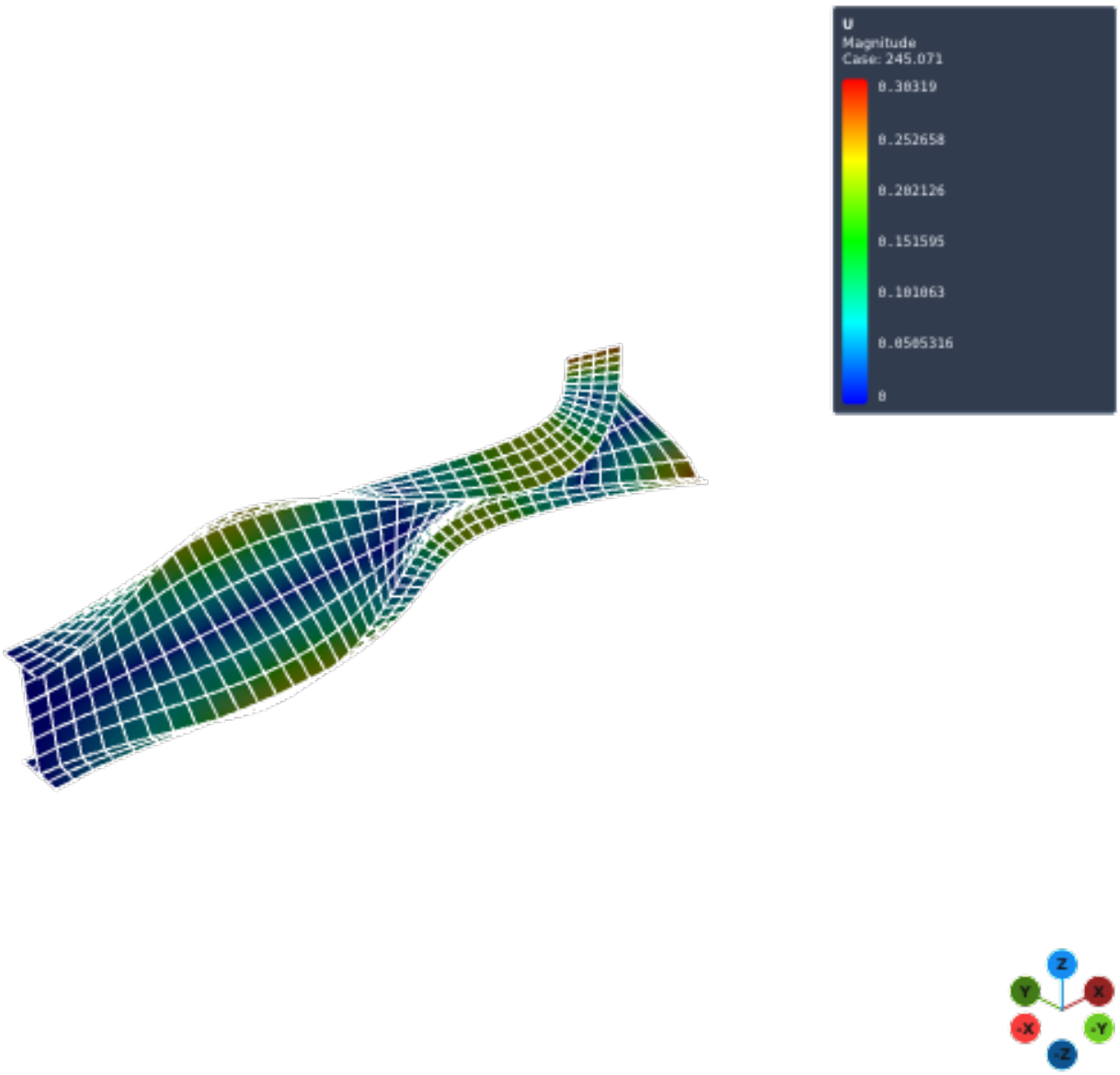

Figure 191: Abaqus — mode 11, 245.071 Hz

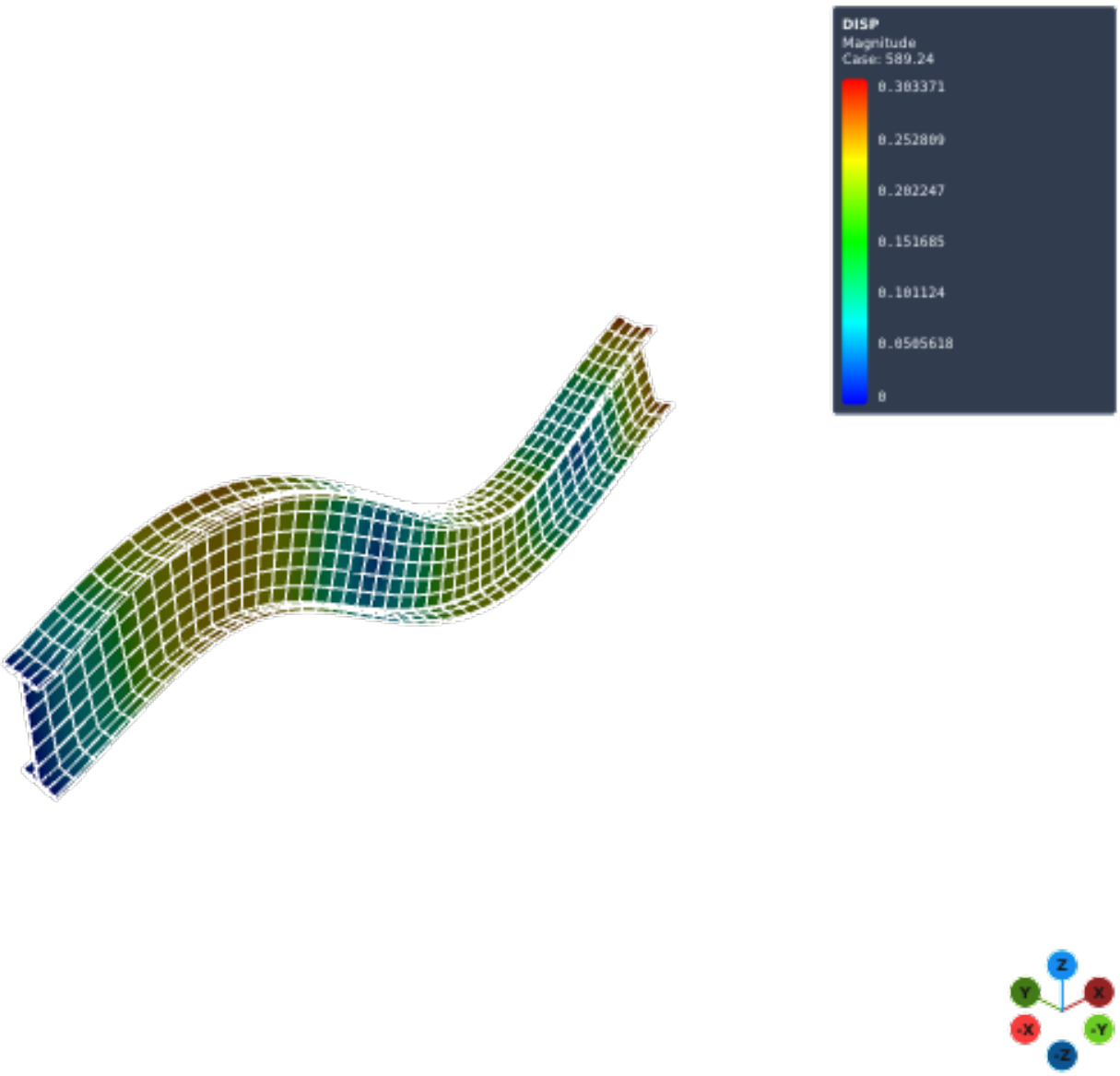

Figure 192: Calculix — mode 11, 589.240 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	245.070	-41.25	2.37106e+06	3.55e-23	8.35e-21	8.47e-21	-5.95e-12	9.14e-11	9.20e-11
Calculix	589.240	+41.25	1.3707e+07	3.93e-26	1.86e-26	12.88	-1.98e-13	1.36e-13	3.589

Shell, 2nd order, TRI

Mode 1

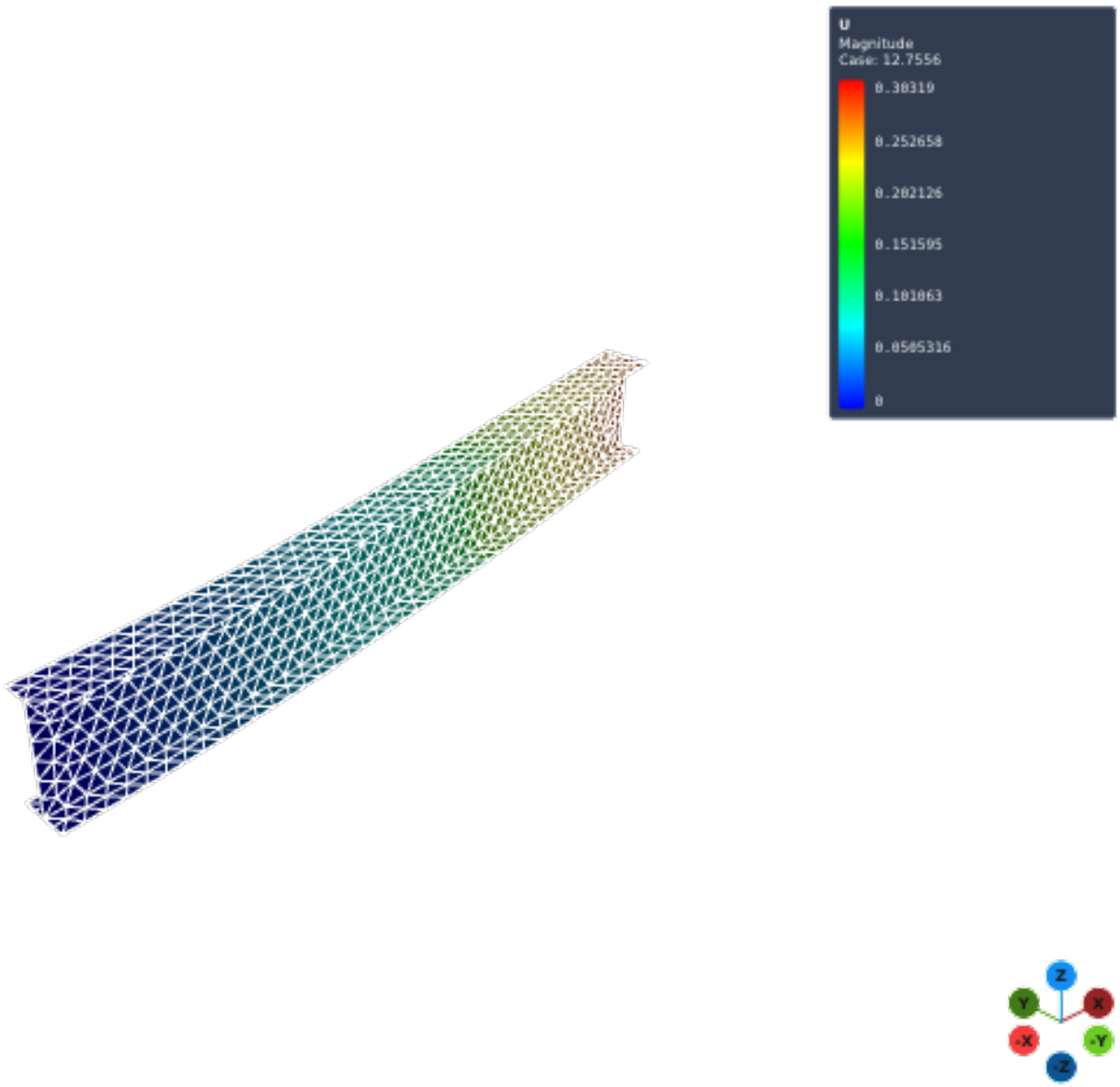

Figure 193: Abaqus — mode 1, 12.756 Hz

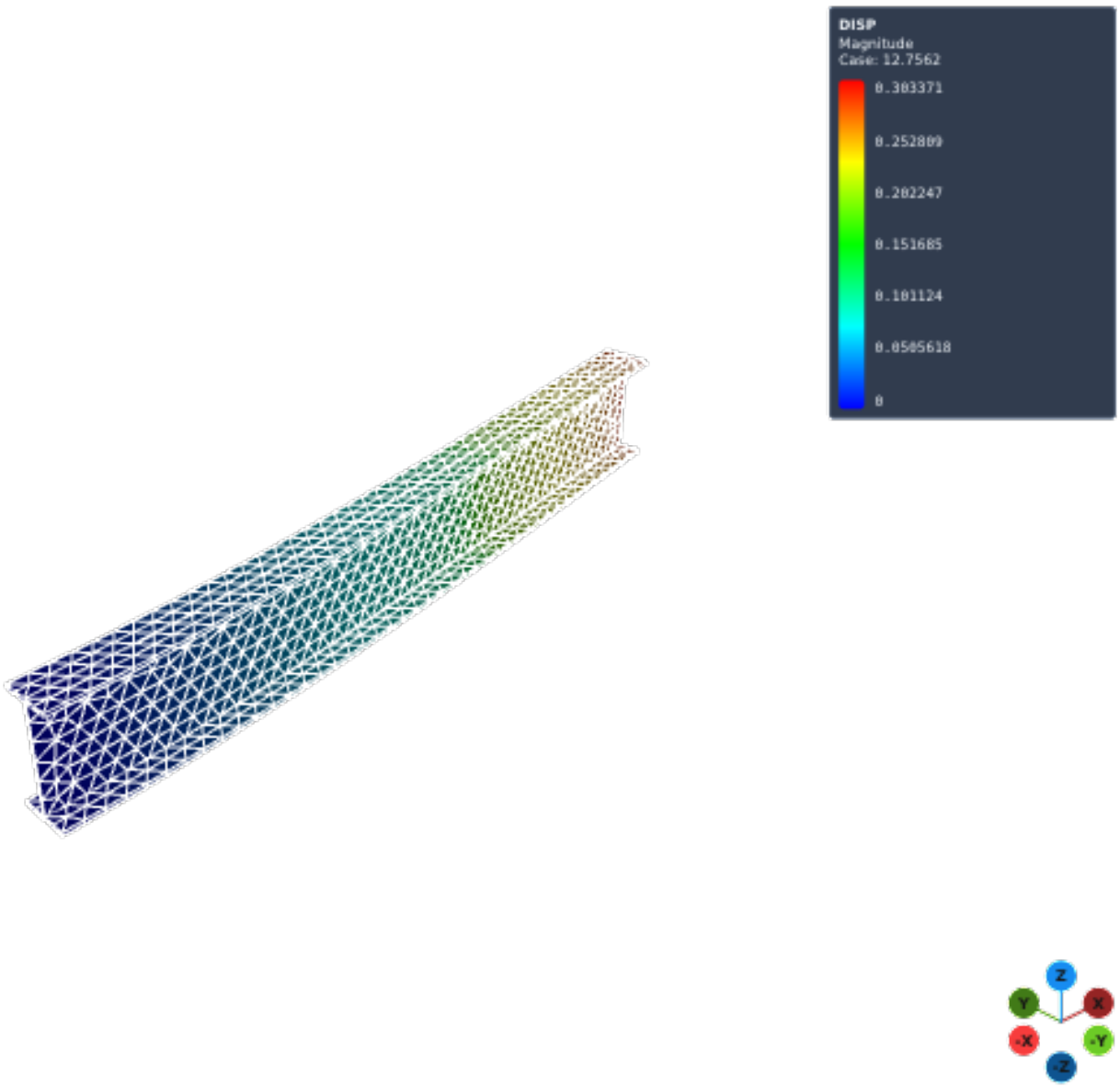

Figure 194: Calculix — mode 1, 12.756 Hz

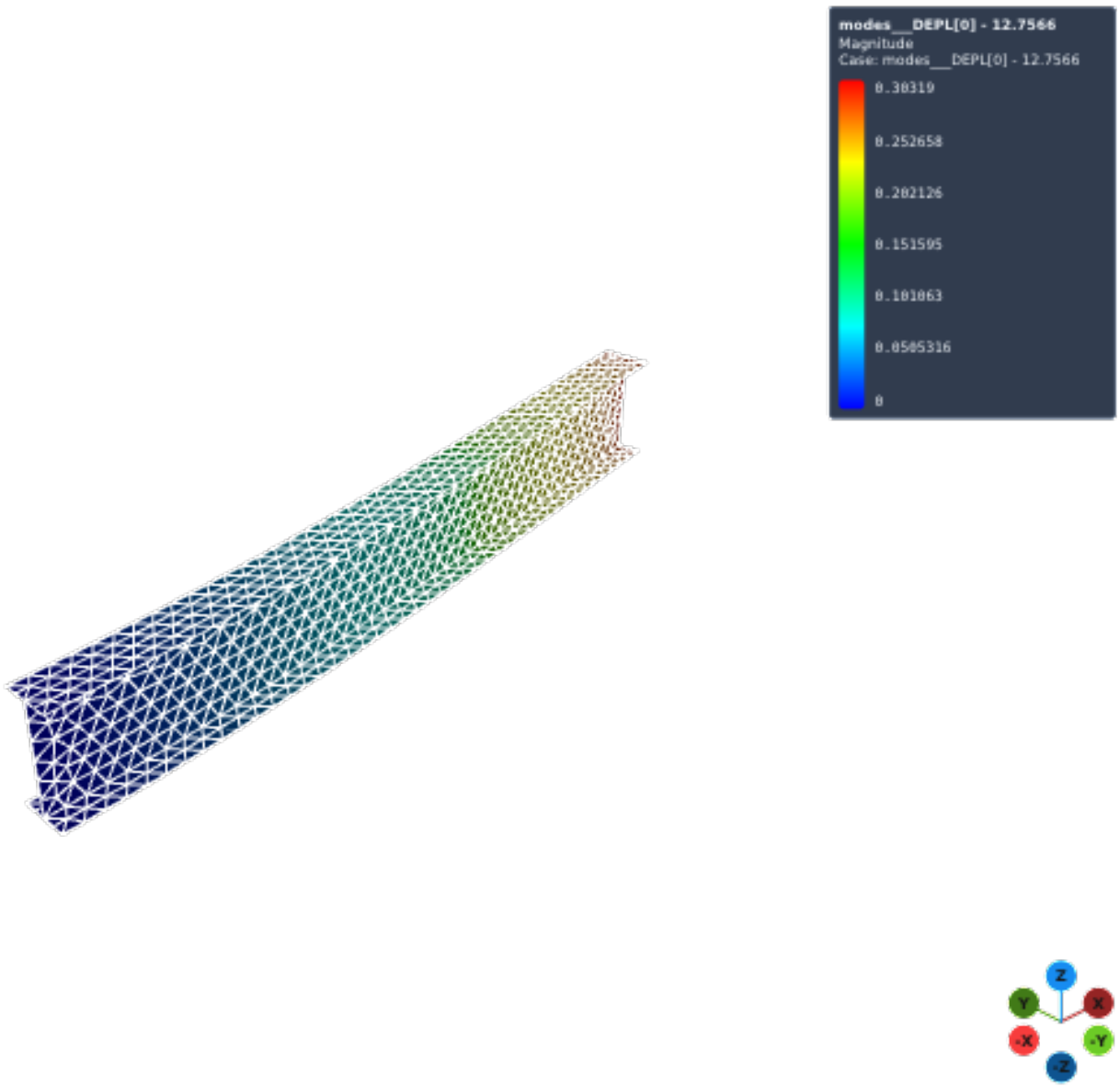

Figure 195: Code_Aster — mode 1, 12.757 Hz

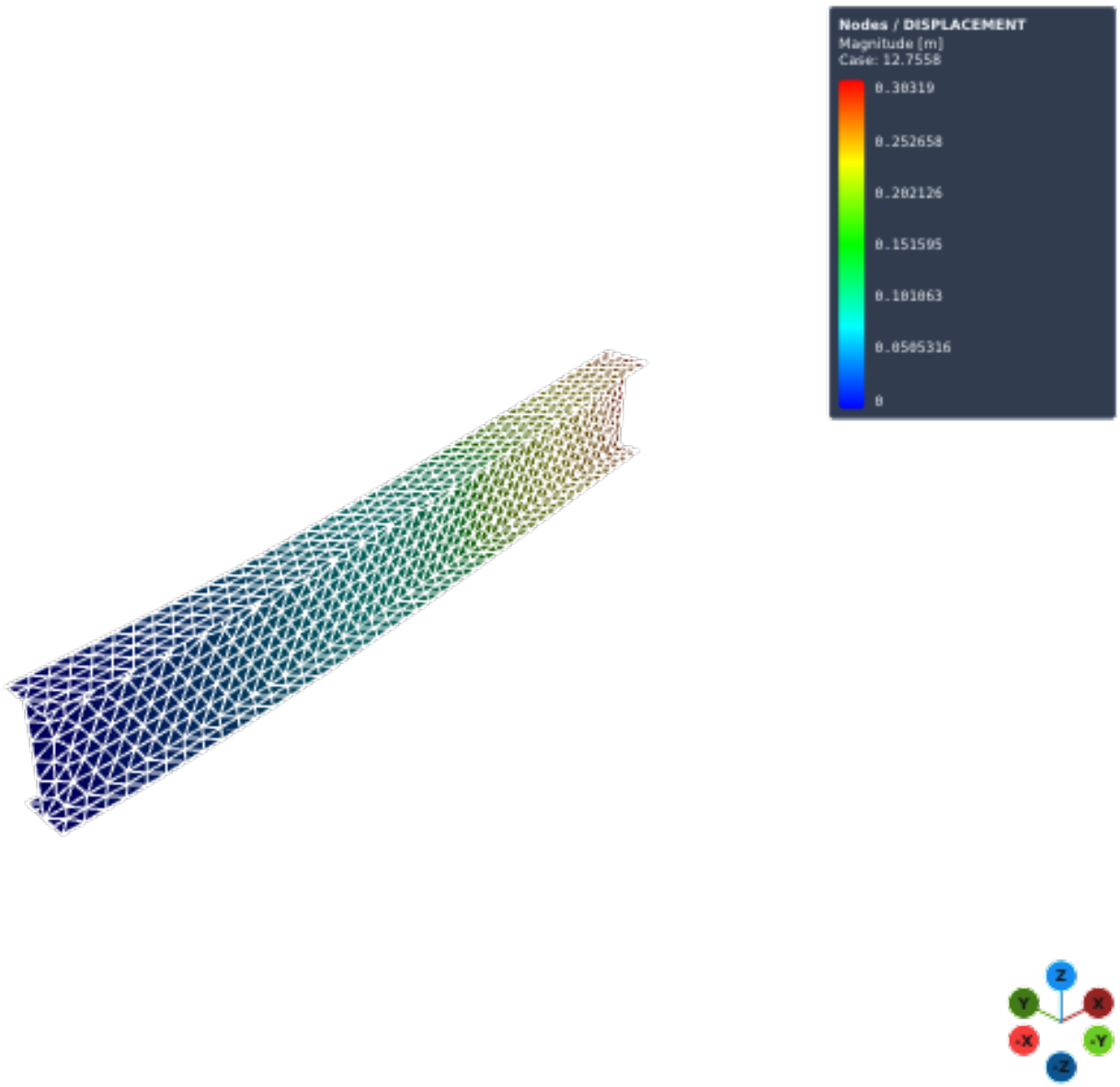

Figure 196: Sesam — mode 1, 12.756 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.756	-1.04e-03	6423.3	2.07e-18	119.94	3.02e-12	1.44e-09	10.952	1.74e-06
Calculix	12.756	4.50e-04	6423.95	1.62e-16	119.94	1.22e-12	-1.27e-08	10.952	1.11e-06
Code_Aster	12.757	3.43e-03	6424.33	2.51e-26	119.94	4.99e-26	1.59e-13	10.952	-2.23e-13
Sesam	12.756	-2.84e-03	6423.52	2.49e-28	119.94	4.86e-24	-1.58e-14	10.952	-2.20e-12

Mode 2

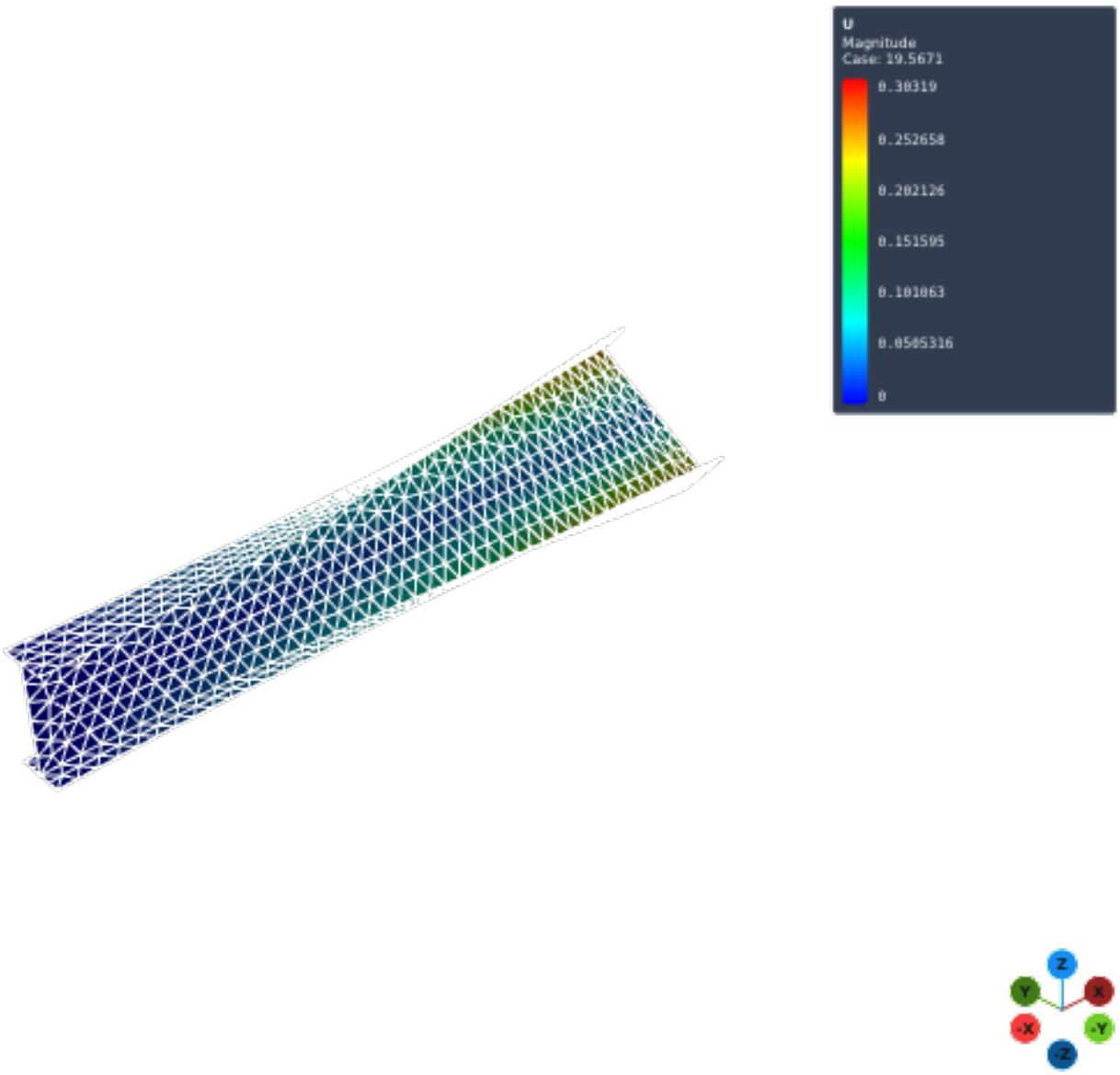

Figure 197: Abaqus — mode 2, 19.567 Hz

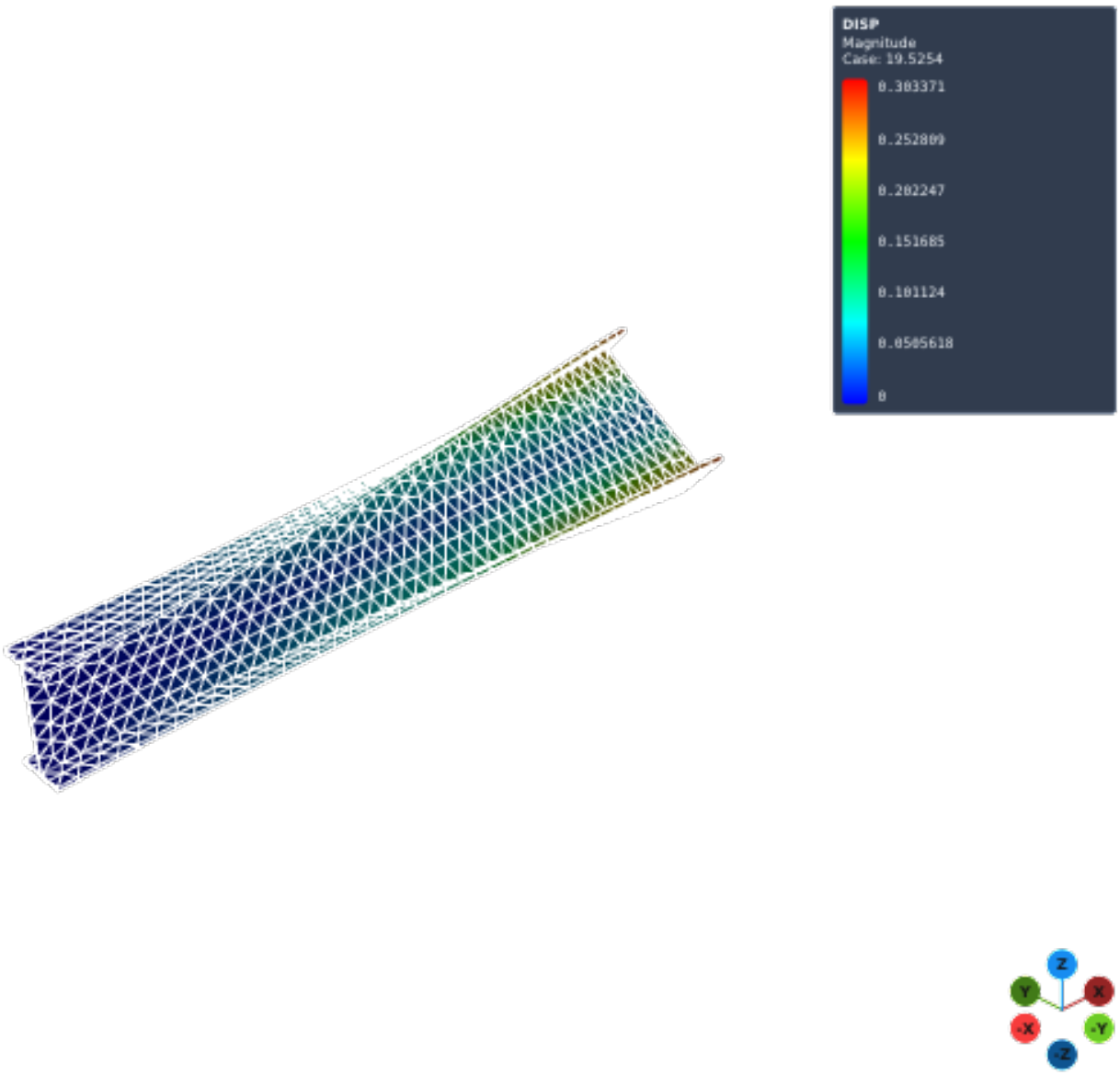

Figure 198: Calculix — mode 2, 19.525 Hz

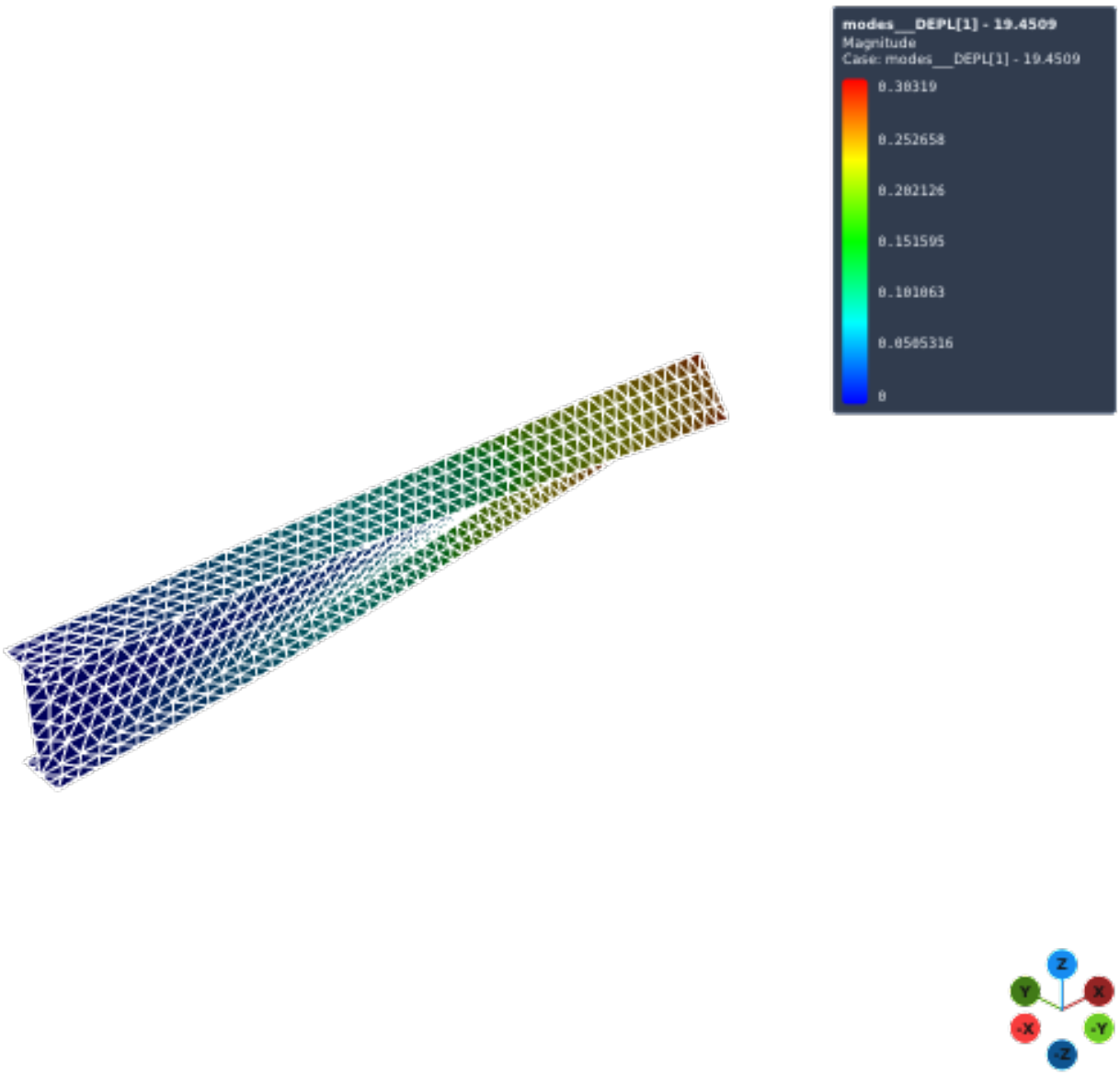

Figure 199: Code_Aster — mode 2, 19.451 Hz

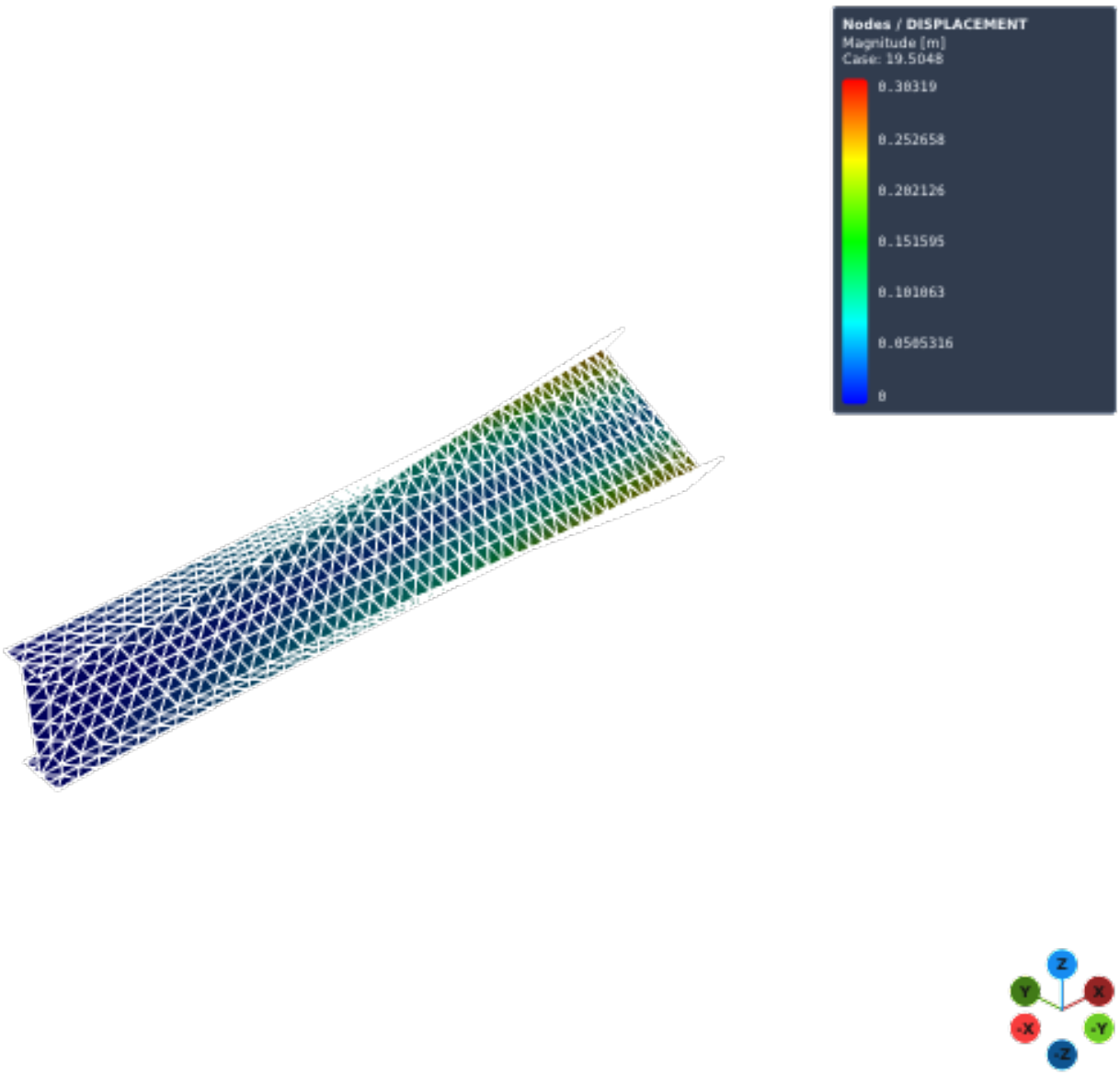

Figure 200: Sesam — mode 2, 19.505 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	19.567	+0.28	15115	3.71e-13	3.08e-10	4.41e-12	6.09e-07	1.75e-05	-2.10e-06
Calculix	19.525	+0.07	15050.8	2.96e-09	2.59e-09	1.08e-14	-5.44e-05	5.09e-05	-1.04e-07
Code_Aster	19.451	-0.31	14936.2	1.12e-26	4.57e-09	2.69e-23	-1.06e-13	-6.76e-05	5.19e-12
Sesam	19.505	-0.04	15019.1	1.90e-25	1.78e-08	2.63e-25	4.36e-13	1.33e-04	5.13e-13

Mode 3

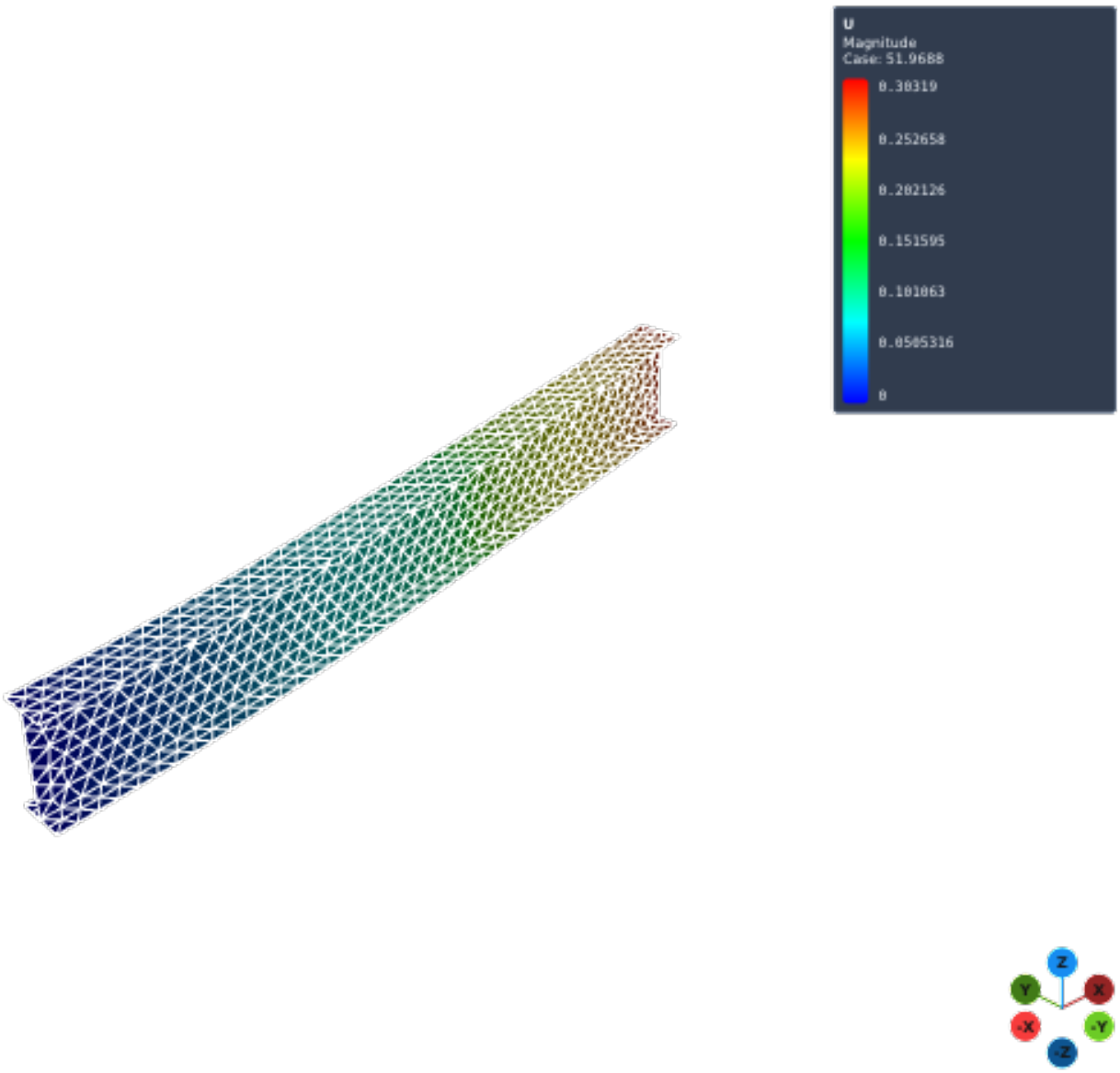

Figure 201: Abaqus — mode 3, 51.969 Hz

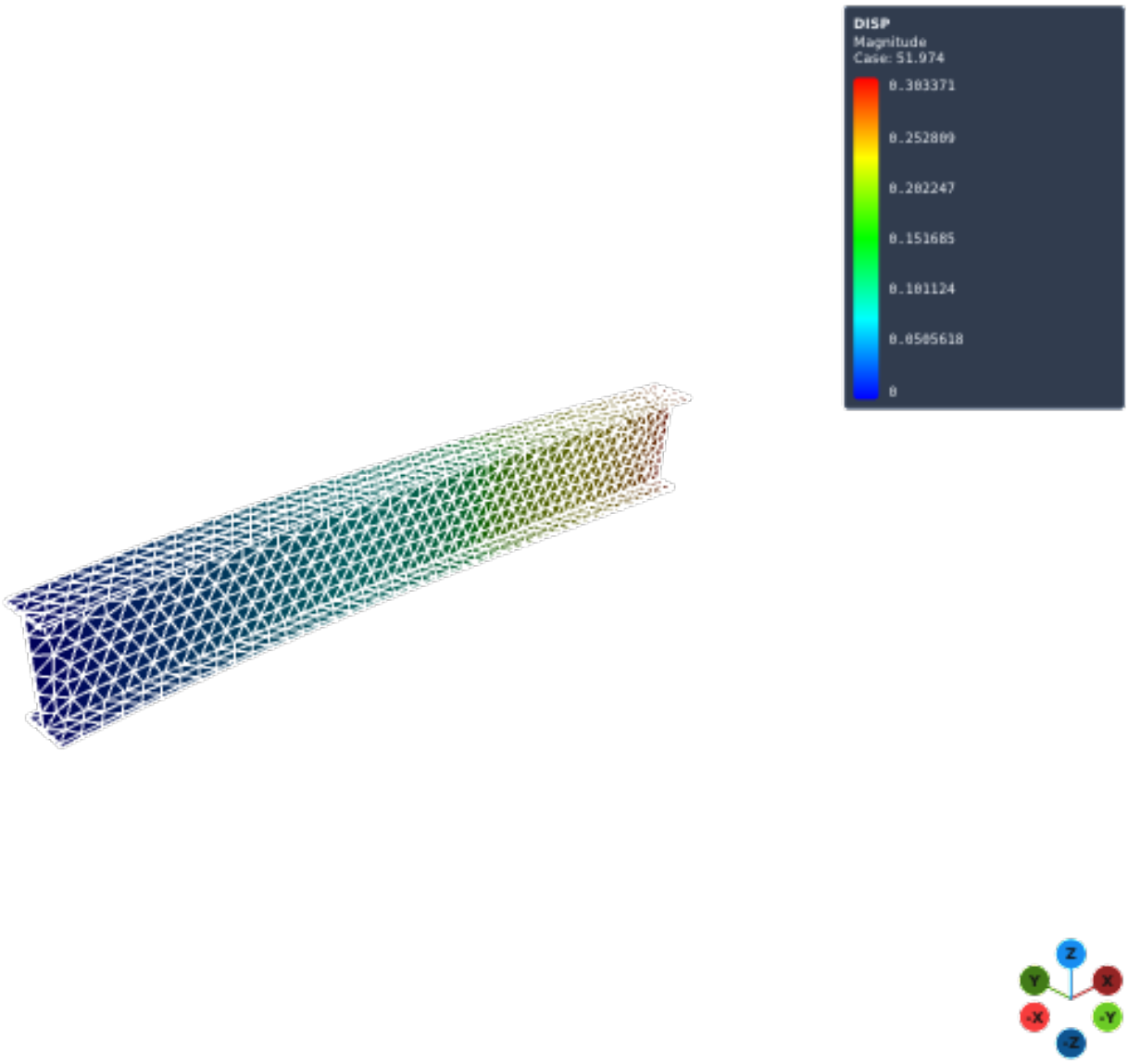

Figure 202: Calculix — mode 3, 51.974 Hz

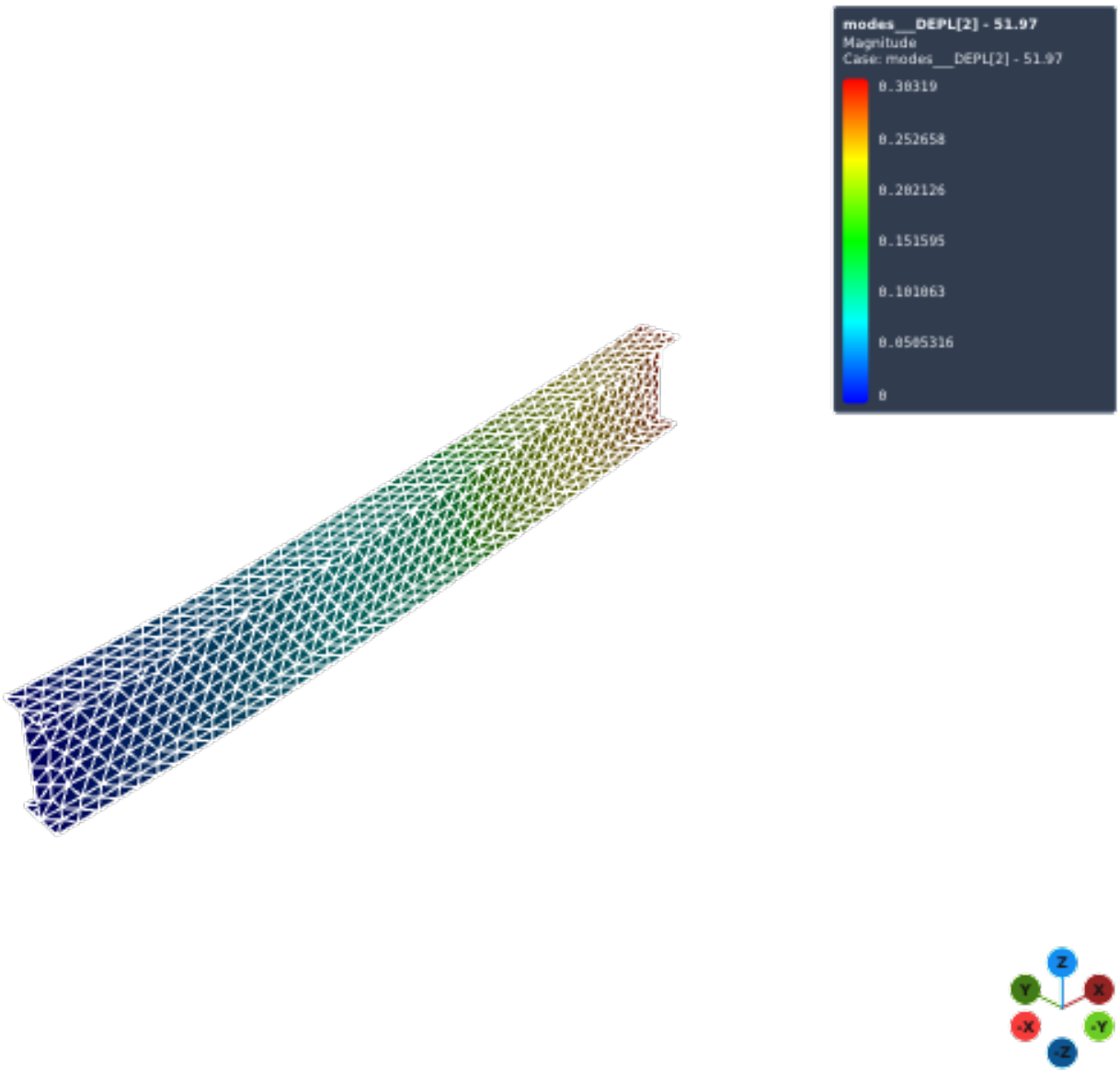

Figure 203: Code_Aster — mode 3, 51.970 Hz

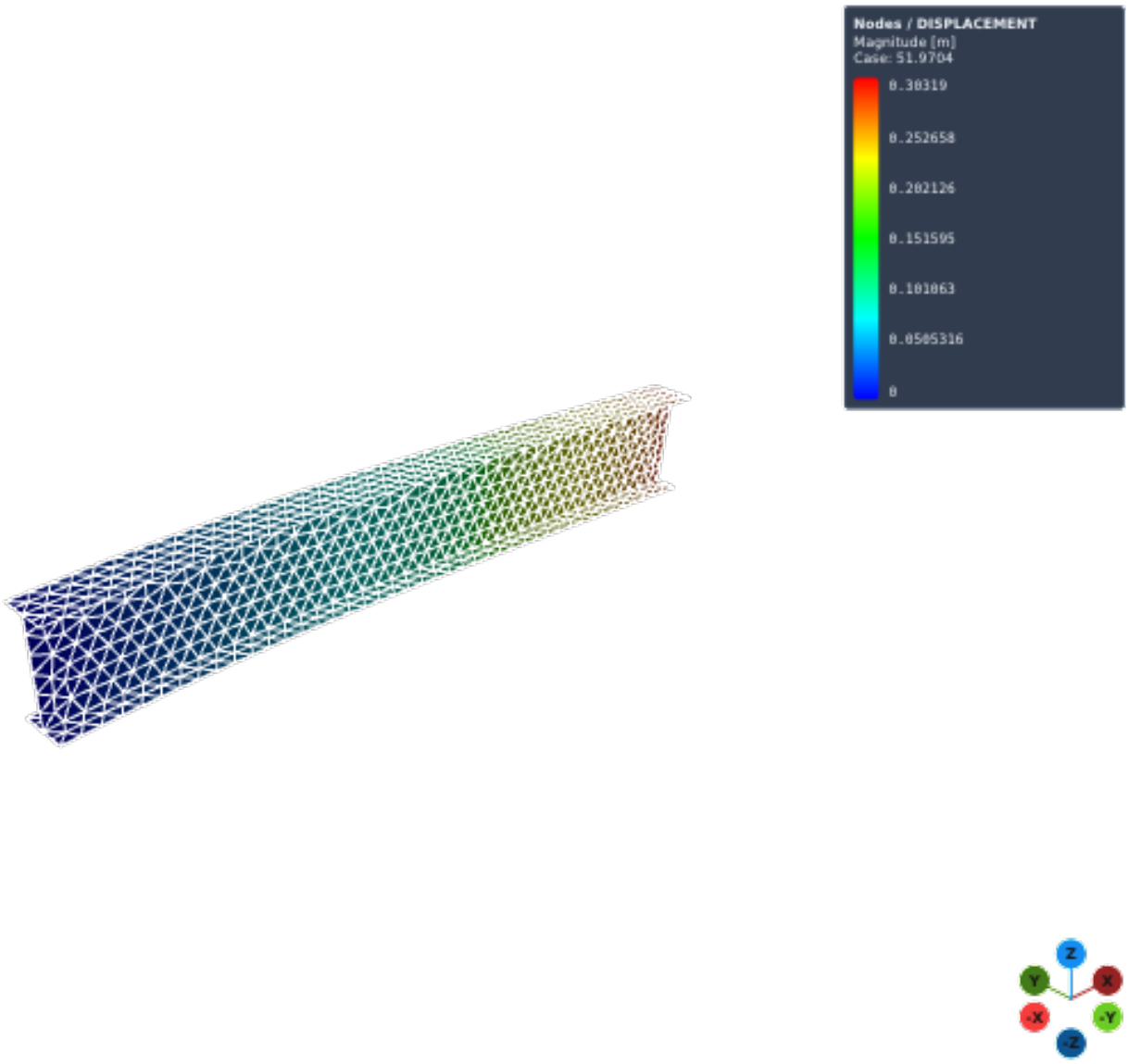

Figure 204: Sesam — mode 3, 51.970 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.969	-3.58e-03	106622	4.50e-14	9.00e-10	122.03	2.12e-07	3.00e-05	11.047
Calculix	51.974	+0.01	106643	5.80e-14	4.56e-11	122.02	-2.41e-07	6.75e-06	-11.046
Code_Aster	51.970	-1.63e-03	106627	3.55e-14	1.48e-24	122.03	1.89e-07	1.21e-12	11.047
Sesam	51.970	-9.06e-04	106628	6.59e-14	1.01e-23	122.03	-2.57e-07	-3.18e-12	-11.047

Mode 4

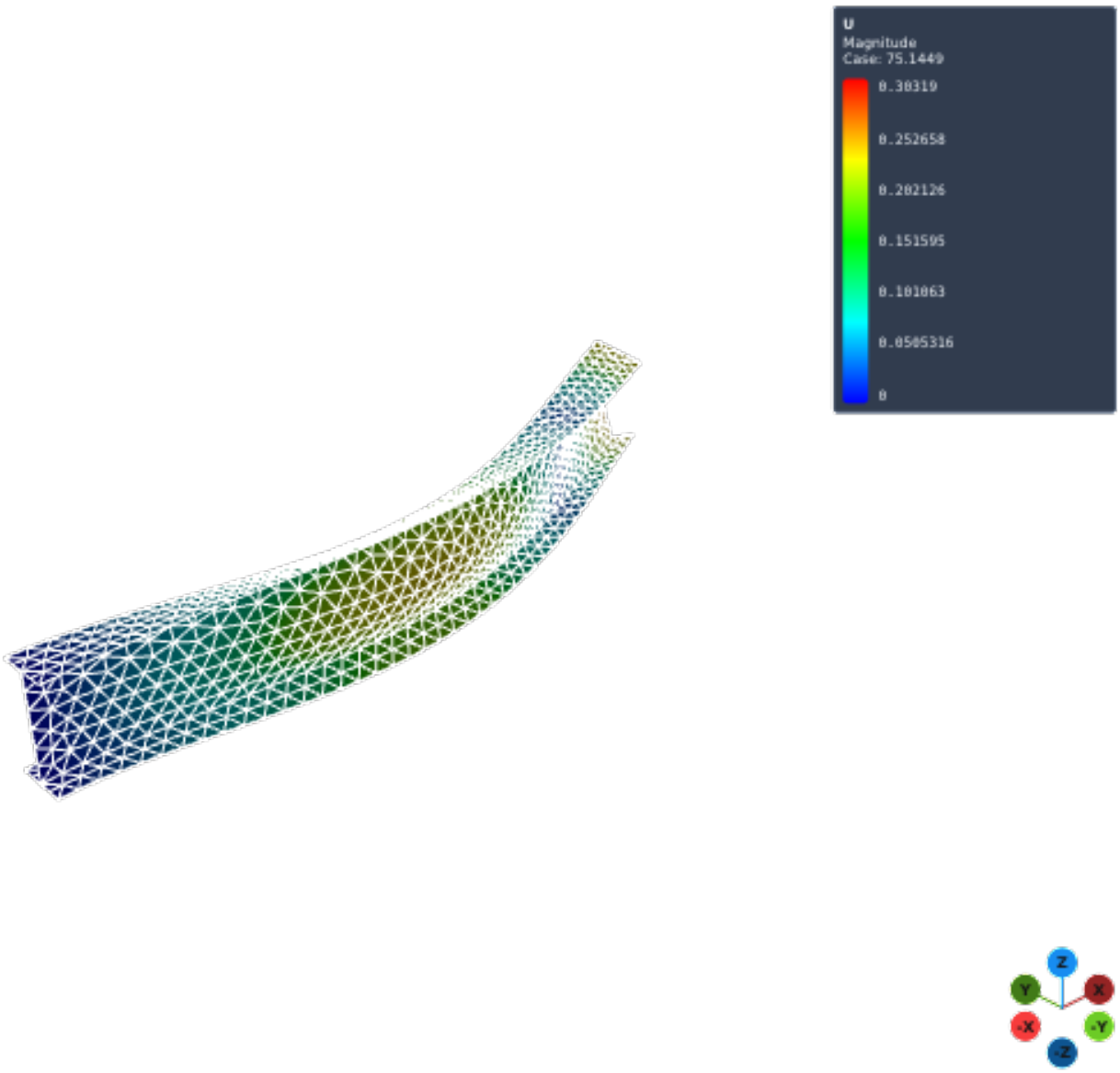

Figure 205: Abaqus — mode 4, 75.145 Hz

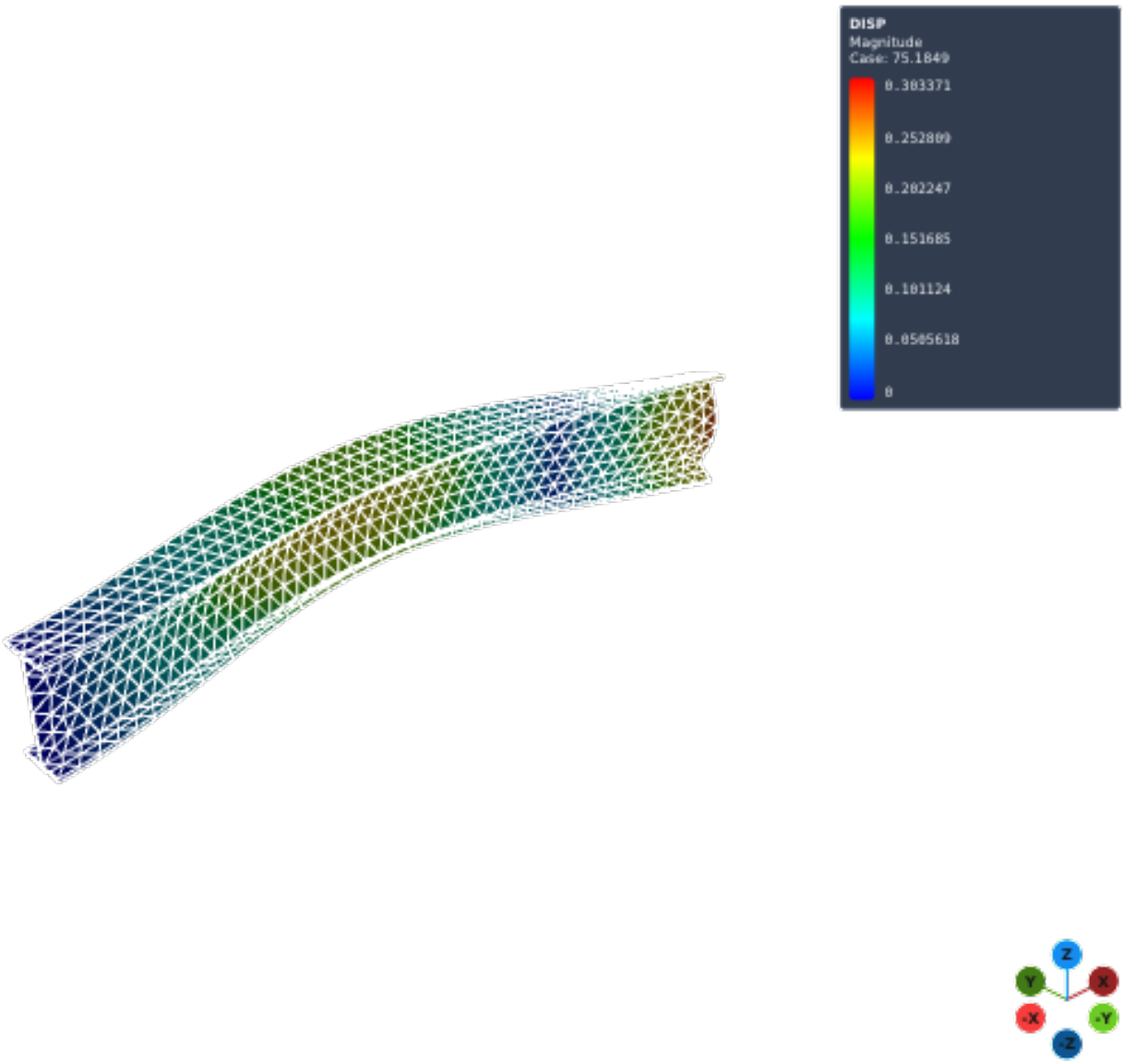

Figure 206: Calculix — mode 4, 75.185 Hz

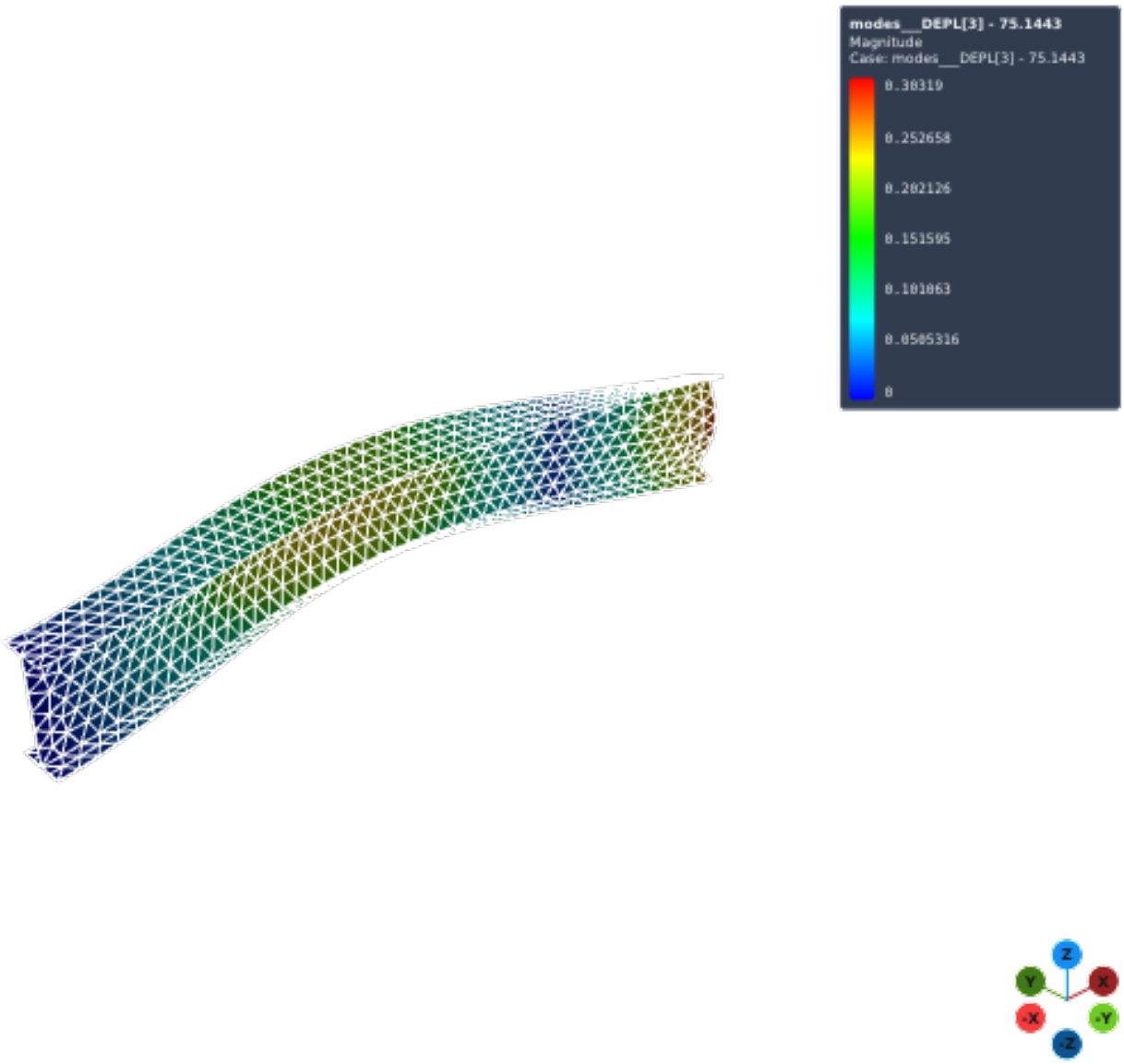

Figure 207: Code_Aster — mode 4, 75.144 Hz

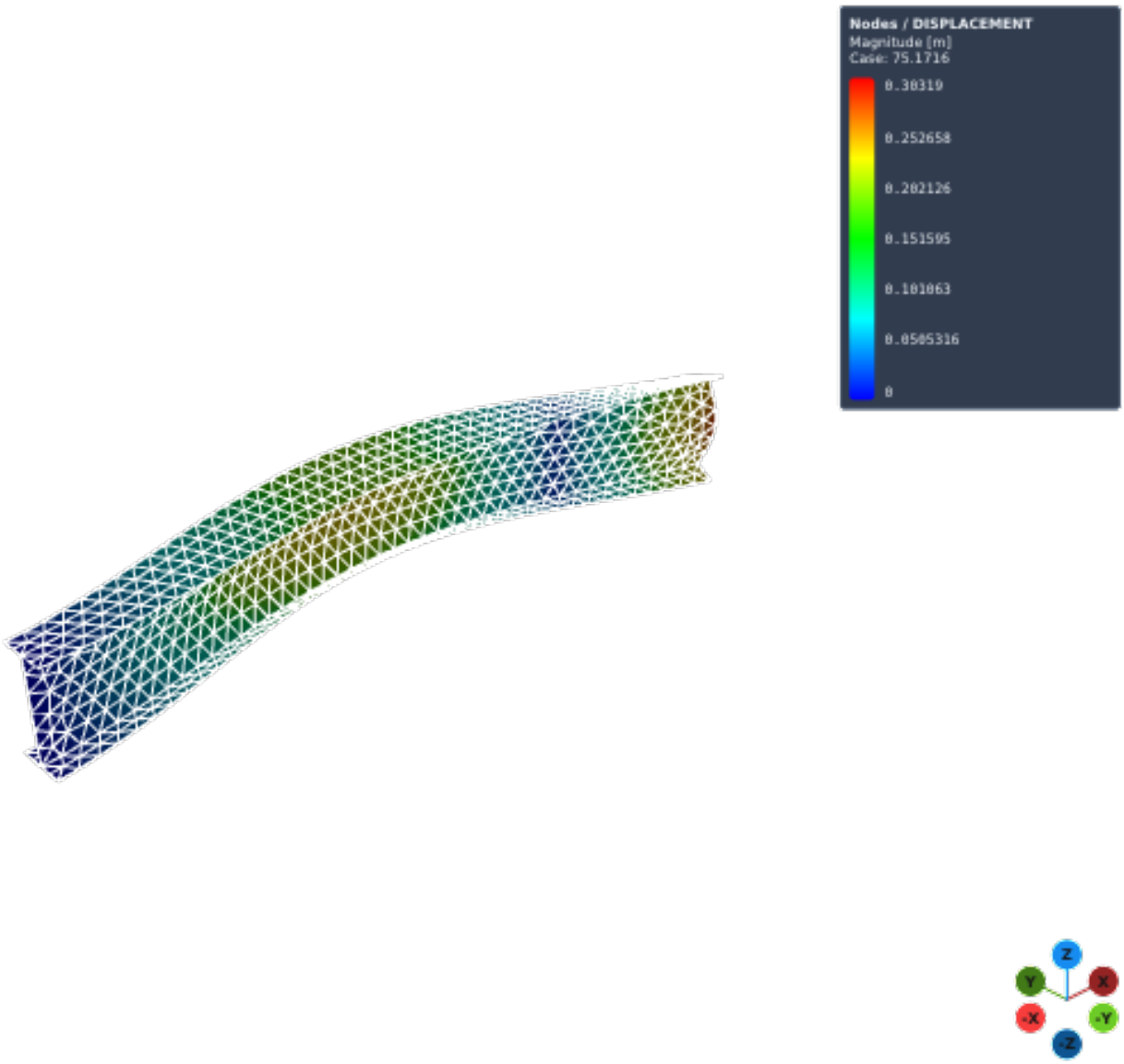

Figure 208: Sesam — mode 4, 75.172 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.145	-0.02	222925	1.24e-16	36.88	1.80e-09	1.11e-08	-6.073	4.24e-05
Calculix	75.185	+0.03	223162	9.99e-14	36.87	1.02e-12	-3.16e-07	6.072	-1.01e-06
Code_Aster	75.144	-0.02	222921	1.20e-26	36.86	3.28e-24	-1.10e-13	6.071	-1.81e-12
Sesam	75.172	+0.01	223083	1.70e-25	36.87	2.56e-24	-4.12e-13	6.072	-1.60e-12

Mode 5

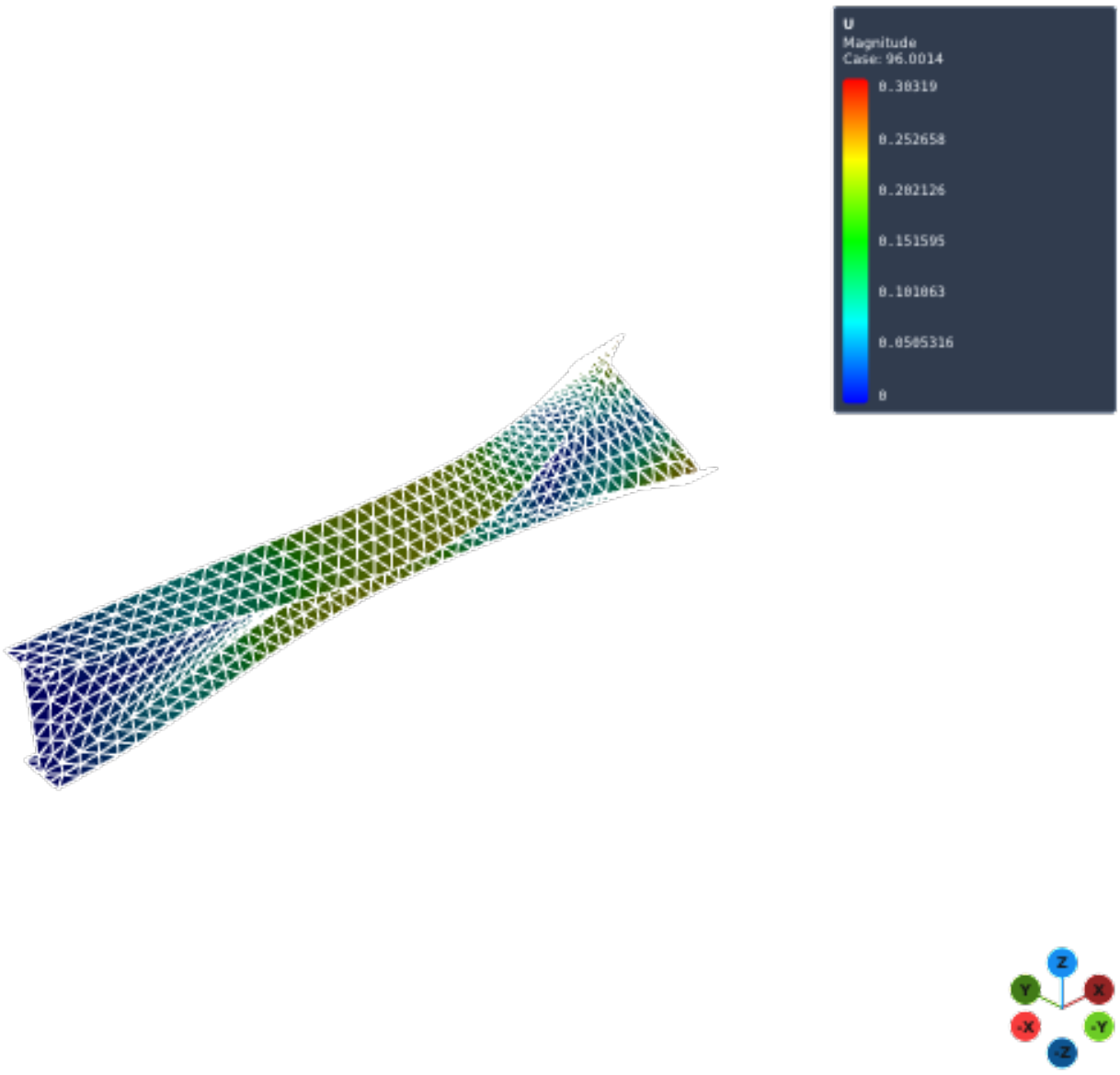

Figure 209: Abaqus — mode 5, 96.001 Hz

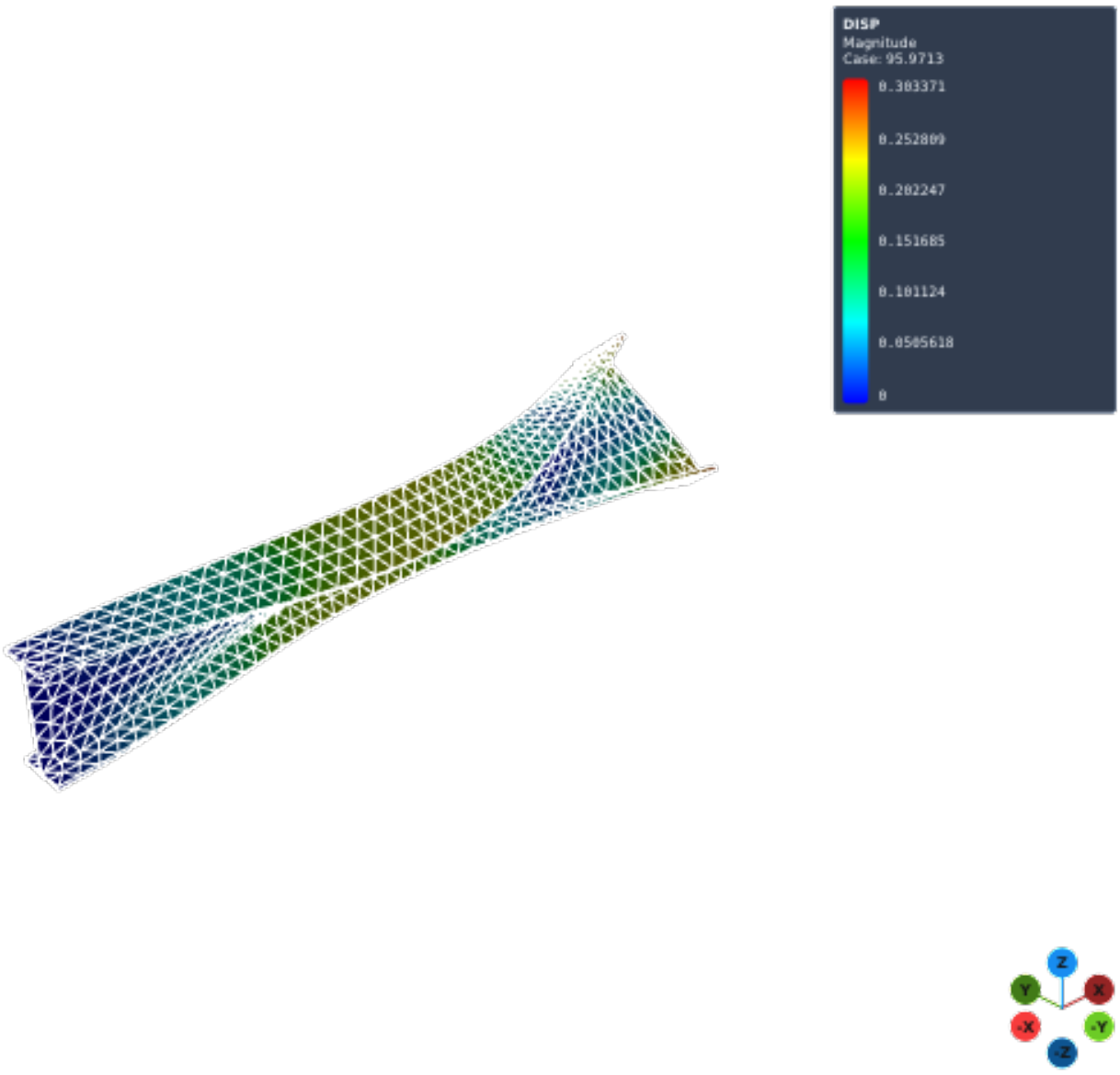

Figure 210: Calculix — mode 5, 95.971 Hz

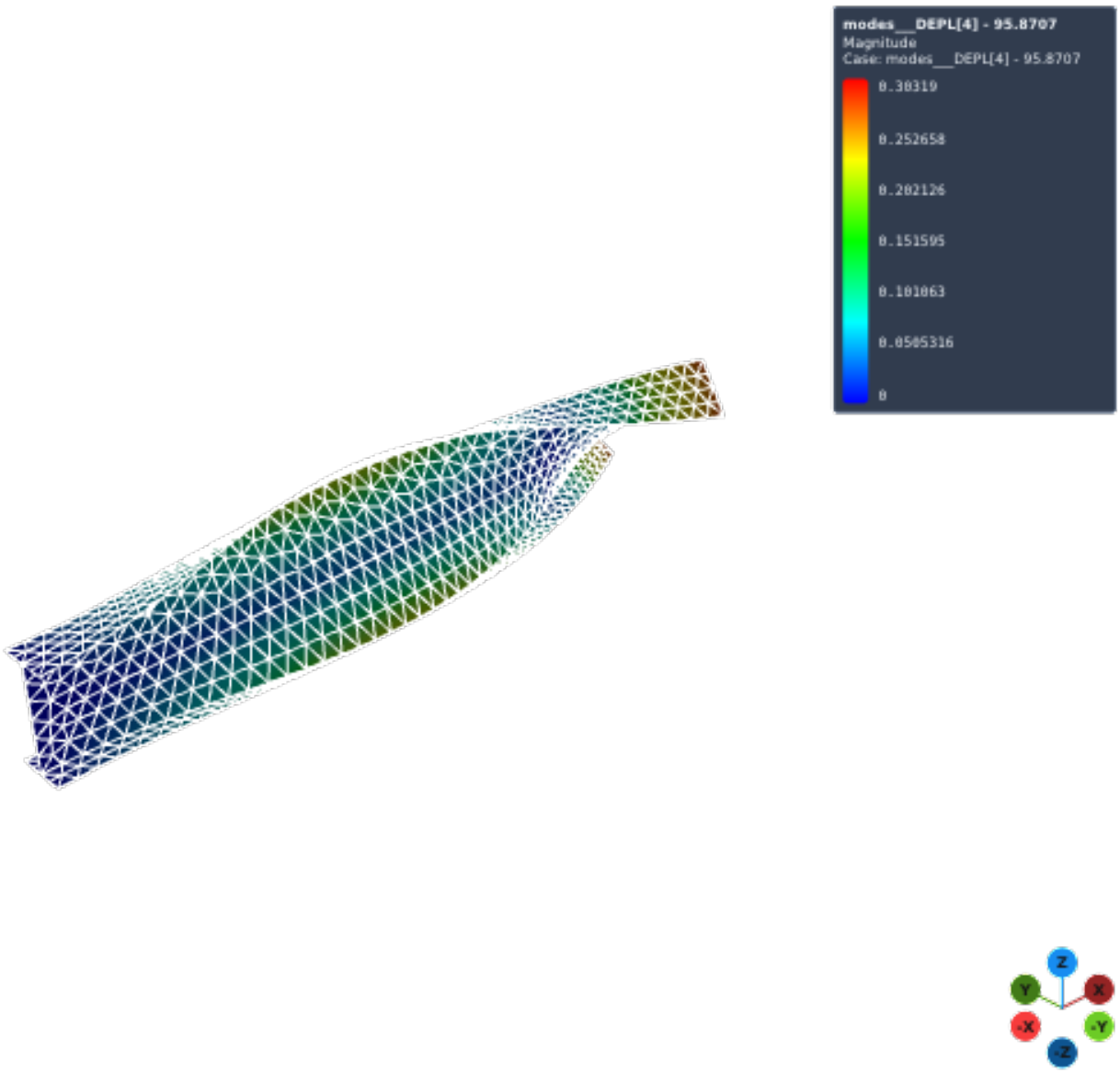

Figure 211: Code_Aster — mode 5, 95.871 Hz

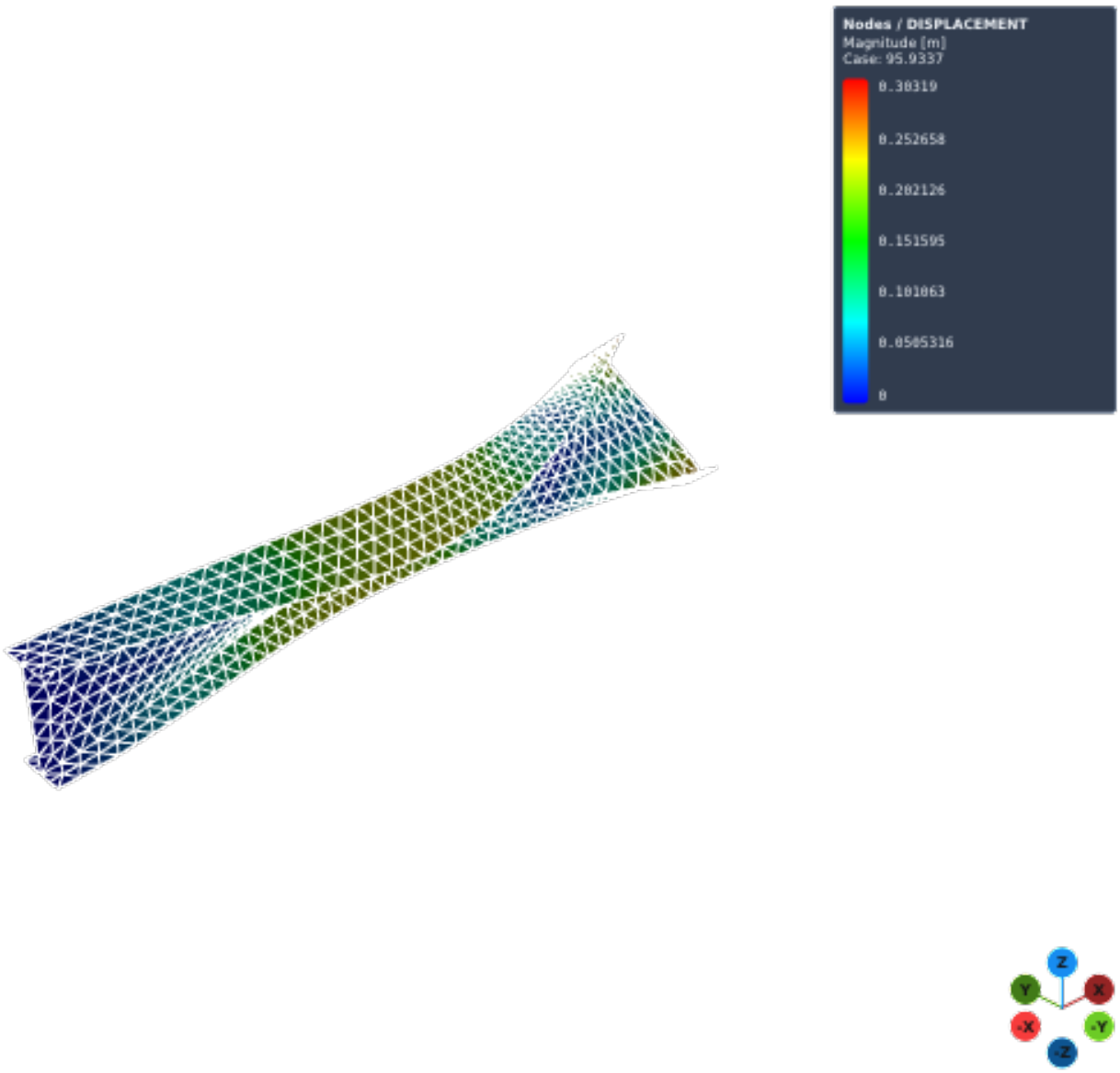

Figure 212: Sesam — mode 5, 95.934 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	96.001	+0.06	363843	8.08e-12	6.17e-09	5.02e-12	-2.84e-06	-7.85e-05	2.24e-06
Calculix	95.971	+0.03	363616	8.10e-10	1.26e-10	3.55e-15	2.85e-05	1.12e-05	-5.96e-08
Code_Aster	95.871	-0.08	362854	5.98e-26	1.61e-08	1.06e-24	-2.45e-13	-1.27e-04	-1.03e-12
Sesam	95.934	-0.01	363330	5.45e-26	1.06e-09	5.71e-24	-2.34e-13	3.26e-05	2.39e-12

Mode 6

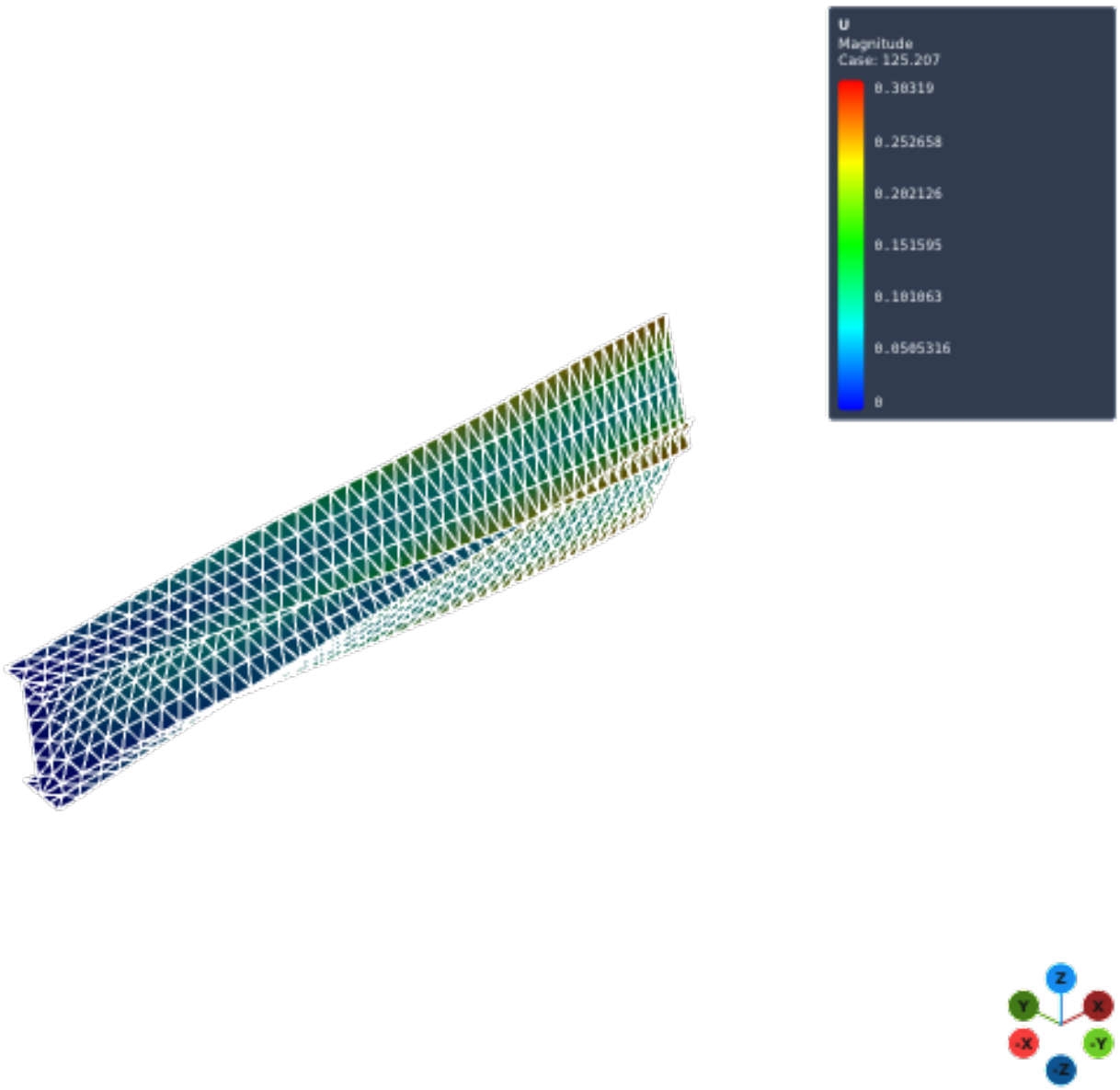

Figure 213: Abaqus — mode 6, 125.207 Hz

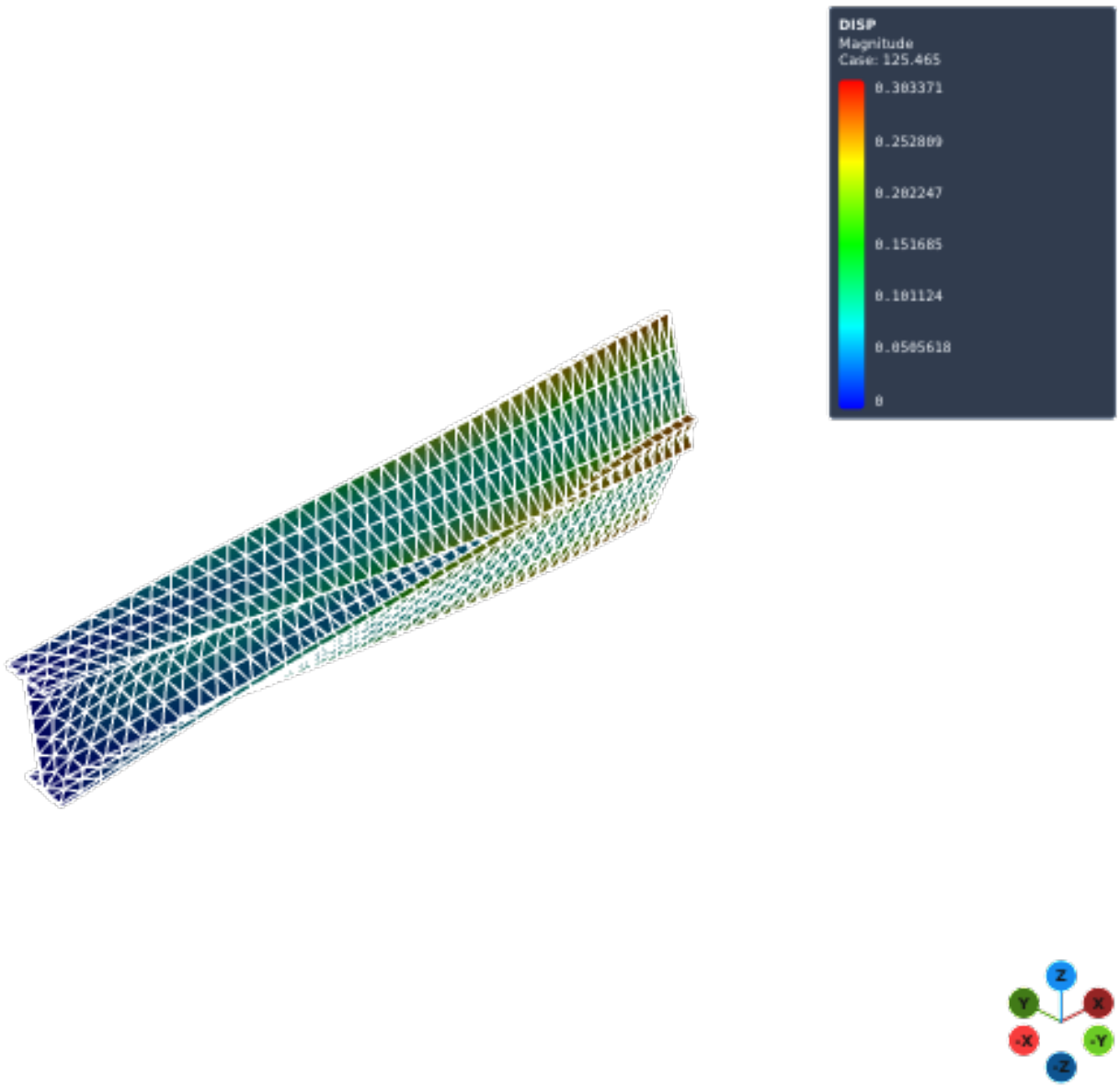

Figure 214: Calculix — mode 6, 125.465 Hz

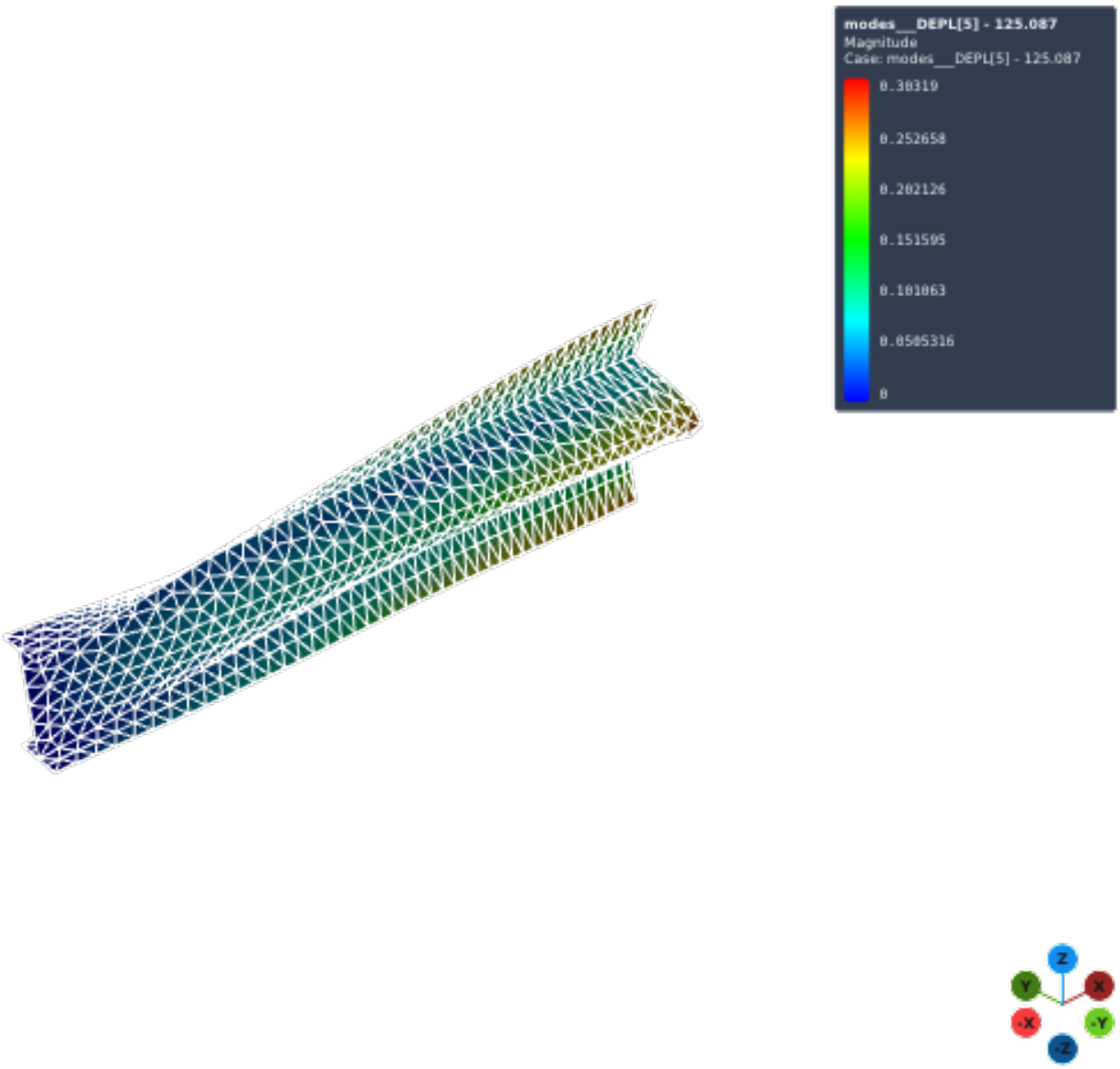

Figure 215: Code_Aster — mode 6, 125.087 Hz

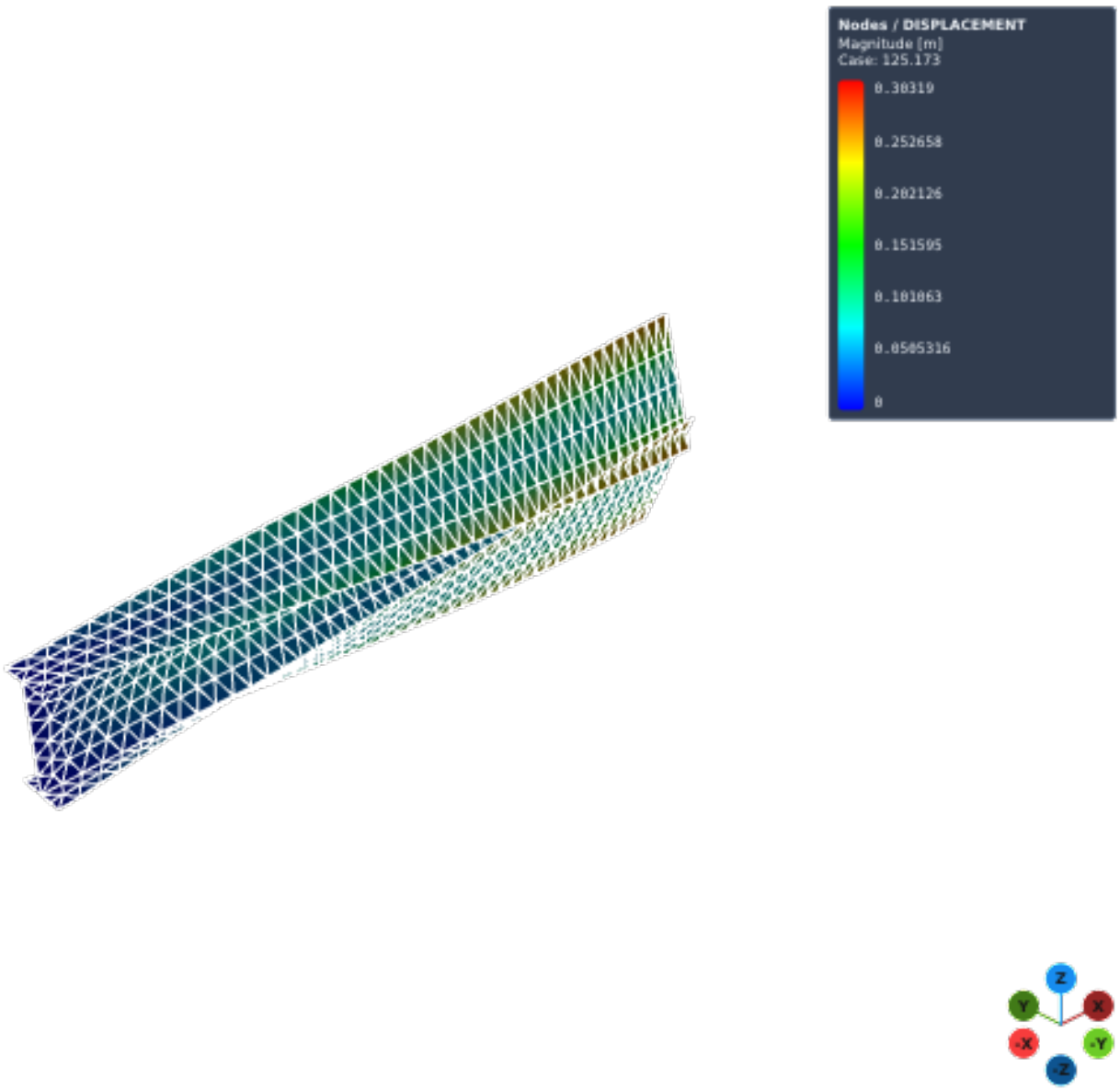

Figure 216: Sesam — mode 6, 125.173 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	125.210	-0.02	618895	4.62e-16	0.02	1.76e-10	-2.15e-08	-0.138	-1.33e-05
Calculix	125.465	+0.18	621448	3.80e-12	0.02	1.15e-13	-1.95e-06	-0.144	-3.39e-07
Code_Aster	125.087	-0.12	617709	1.54e-26	0.02	4.07e-24	1.24e-13	0.142	-2.02e-12
Sesam	125.173	-0.05	618555	1.15e-26	0.02	8.07e-23	1.07e-13	-0.144	8.98e-12

Mode 7

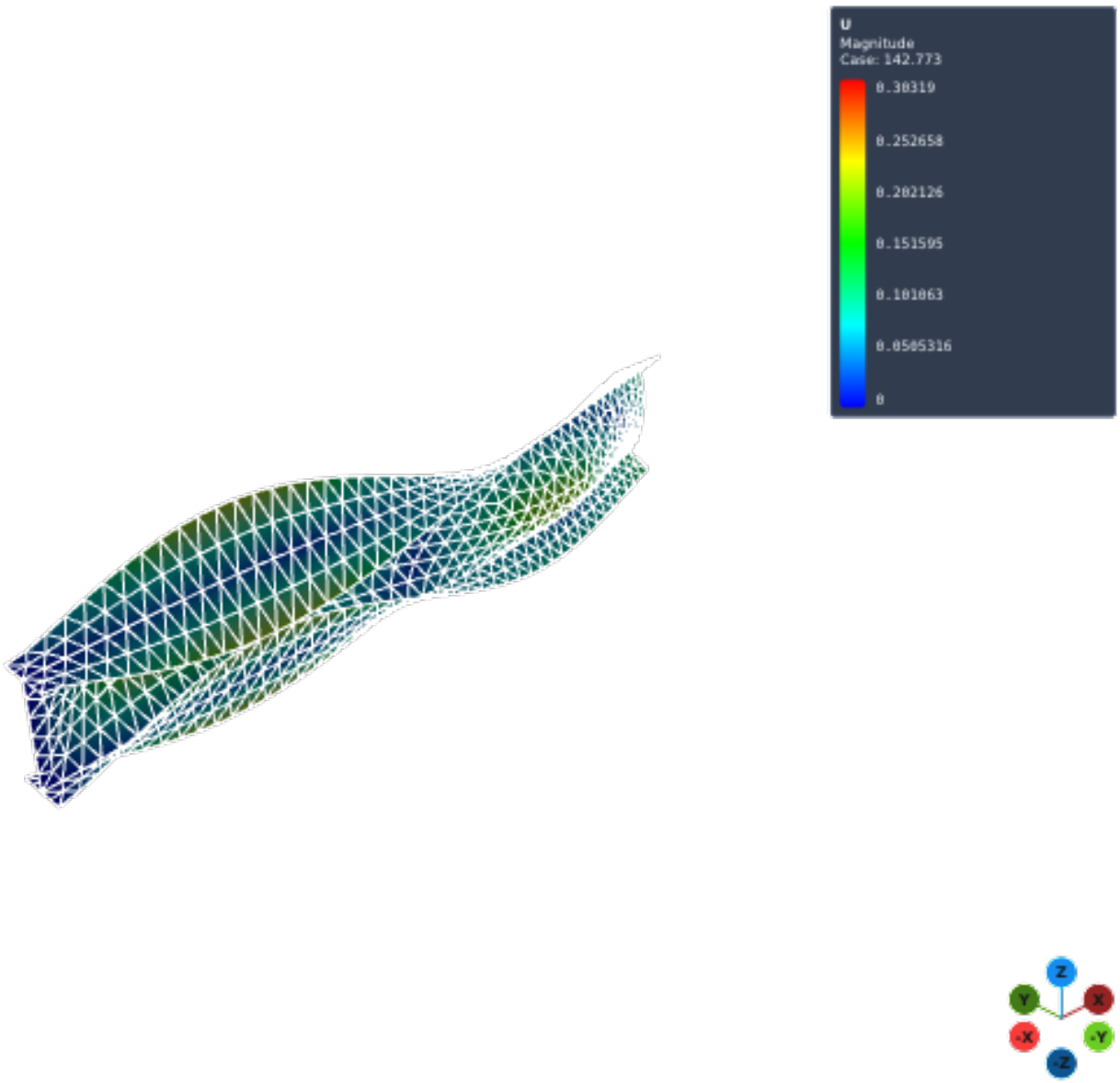

Figure 217: Abaqus — mode 7, 142.773 Hz

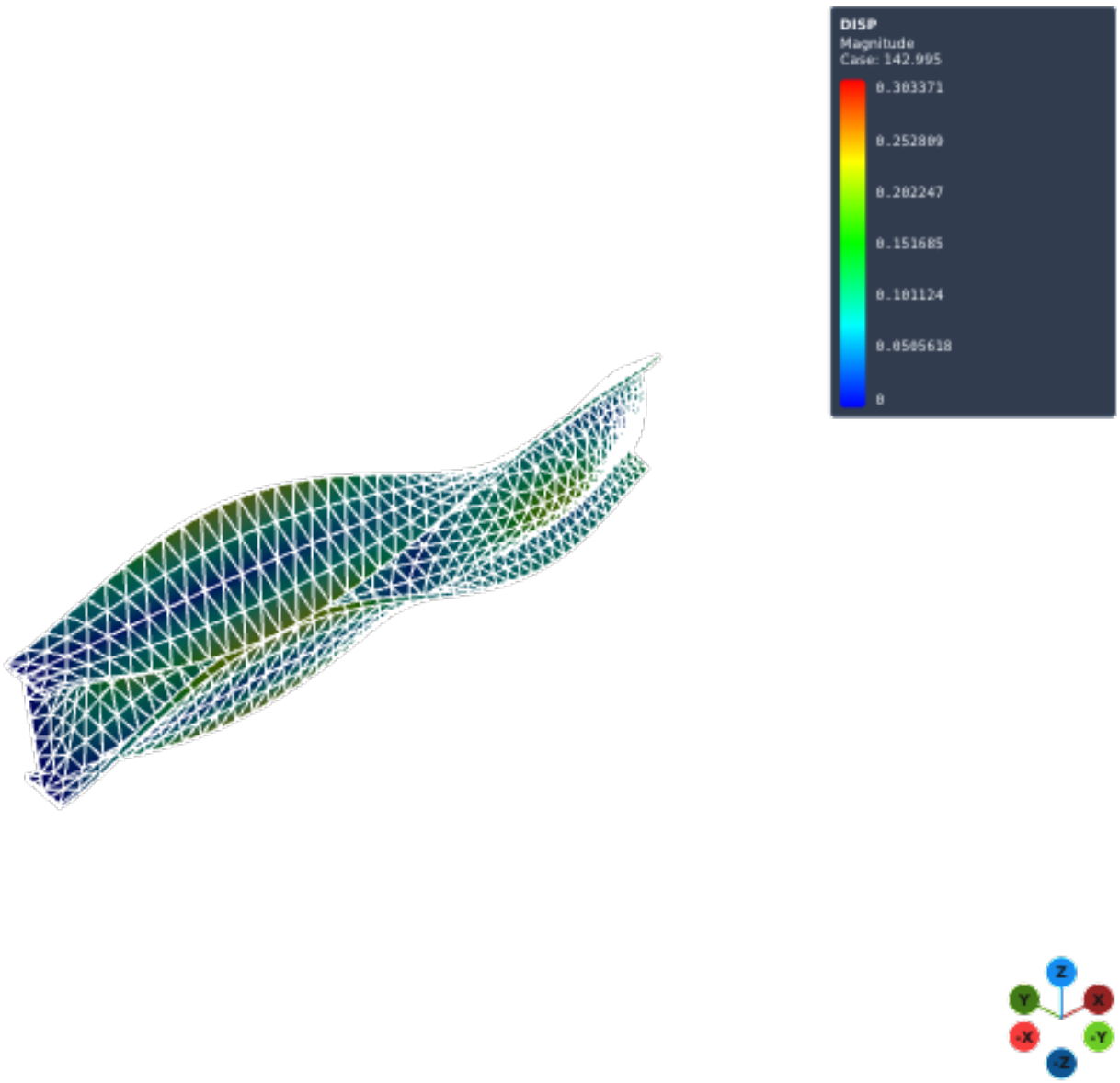

Figure 218: Calculix — mode 7, 142.995 Hz

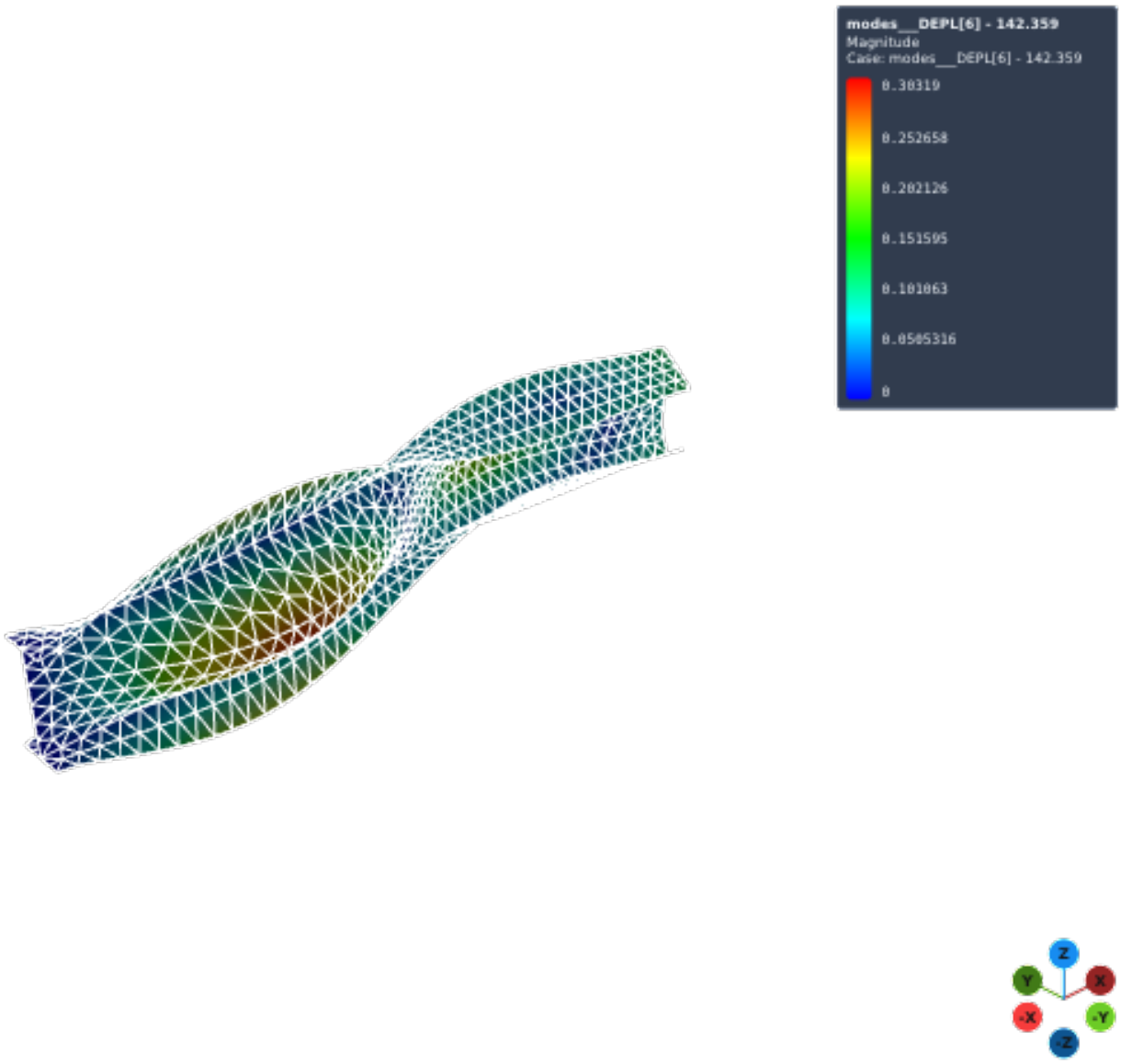

Figure 219: Code_Aster — mode 7, 142.359 Hz

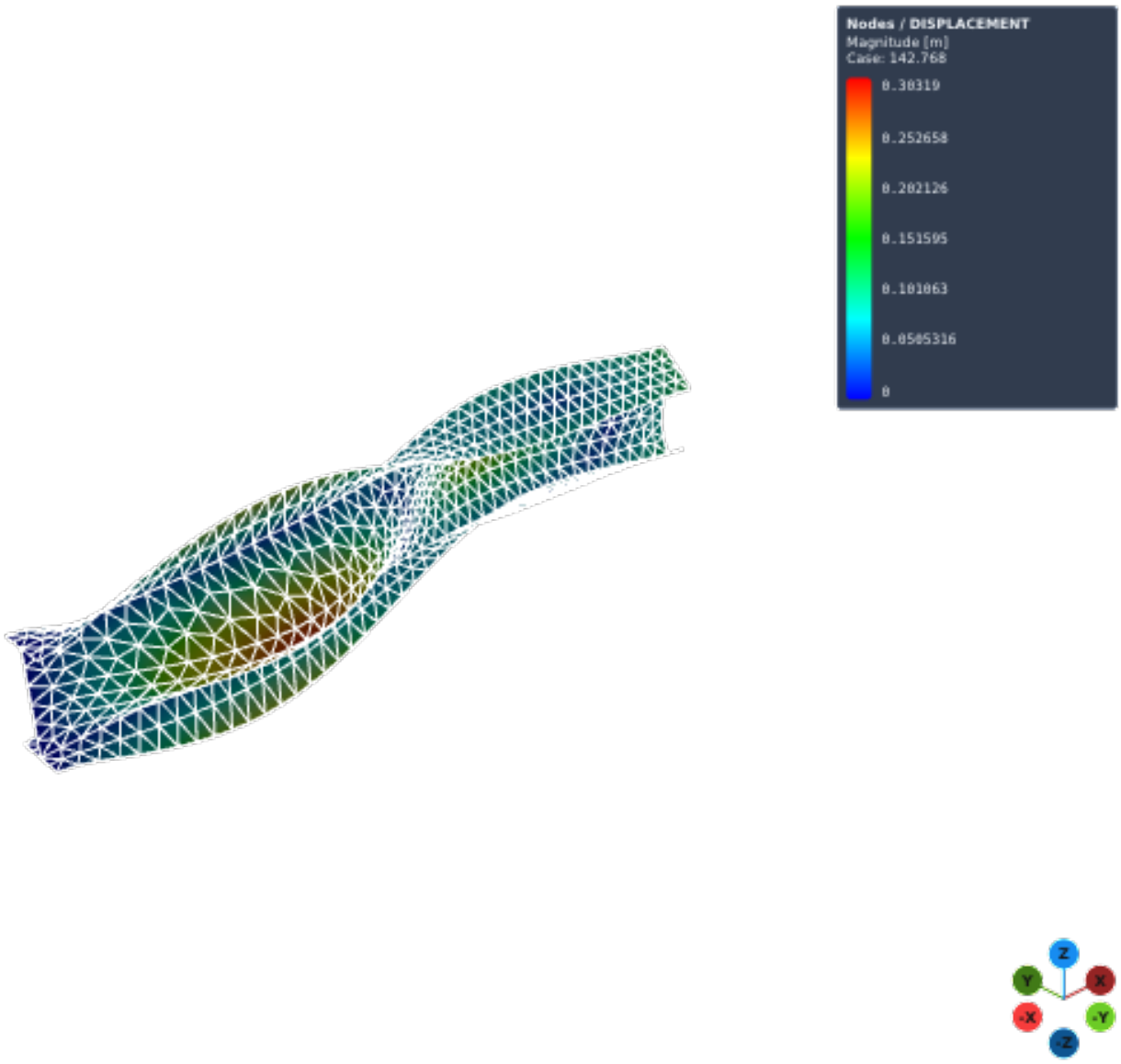

Figure 220: Sesam — mode 7, 142.768 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	142.770	+0.03	804737	9.41e-15	3.54	1.08e-09	-9.70e-08	1.880	3.28e-05
Calculix	142.995	+0.19	807237	7.68e-13	3.41	5.68e-13	-8.76e-07	1.848	7.54e-07
Code_Aster	142.359	-0.26	800074	2.54e-26	3.42	3.14e-25	-1.59e-13	-1.849	5.60e-13
Sesam	142.768	+0.03	804681	4.61e-26	3.42	1.06e-23	-2.15e-13	-1.850	3.26e-12

Mode 8

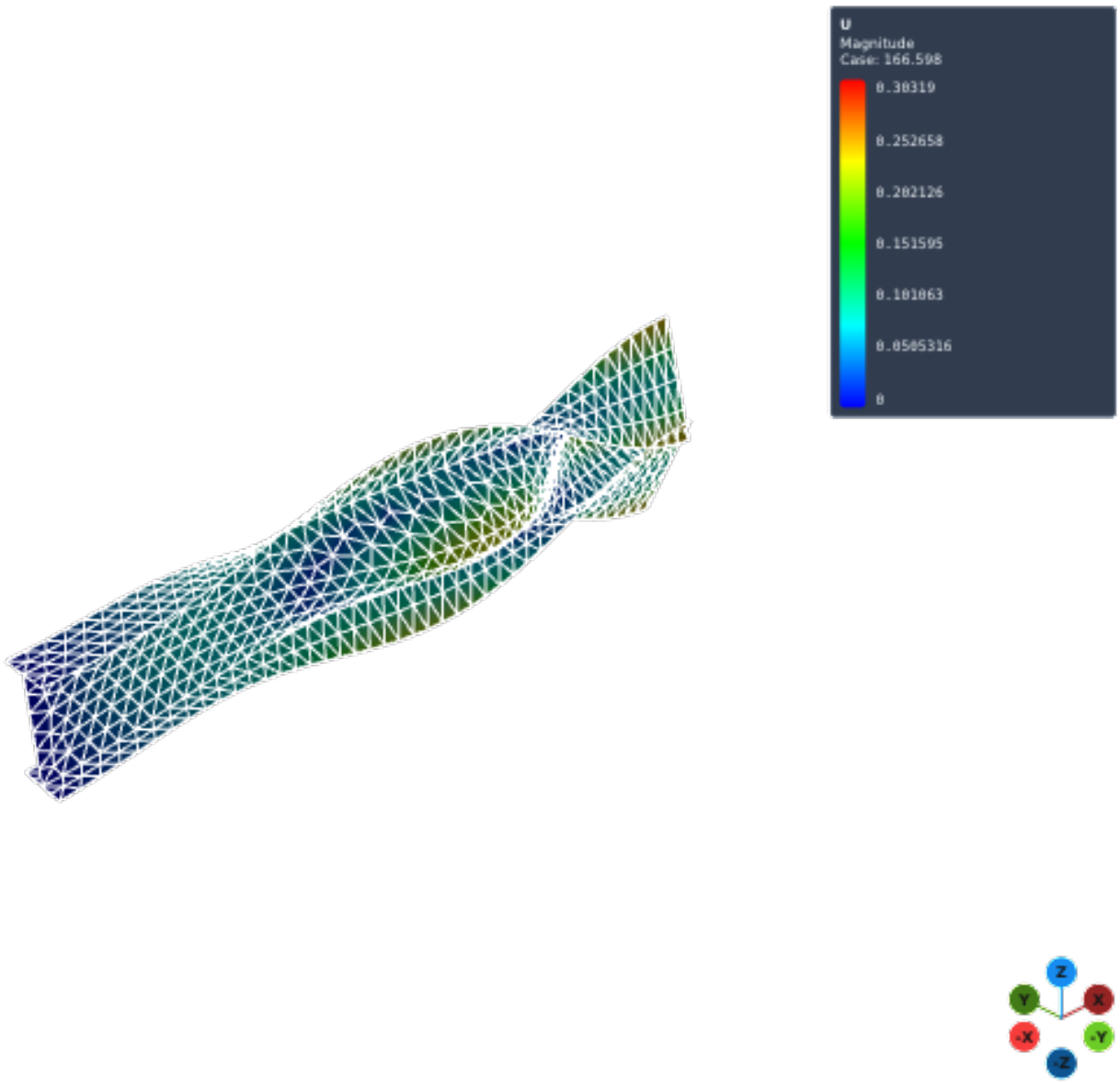

Figure 221: Abaqus — mode 8, 166.598 Hz

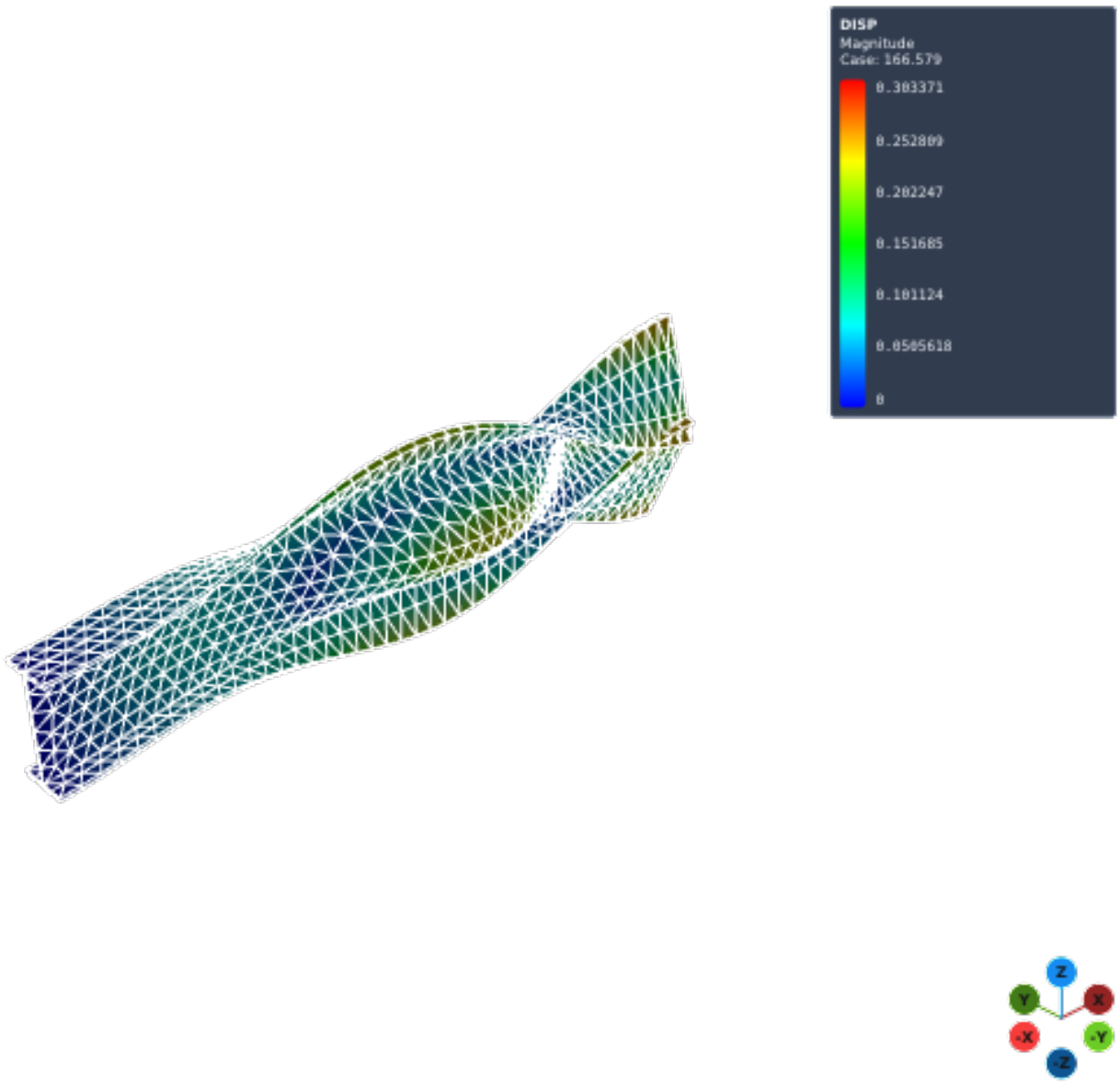

Figure 222: Calculix — mode 8, 166.579 Hz

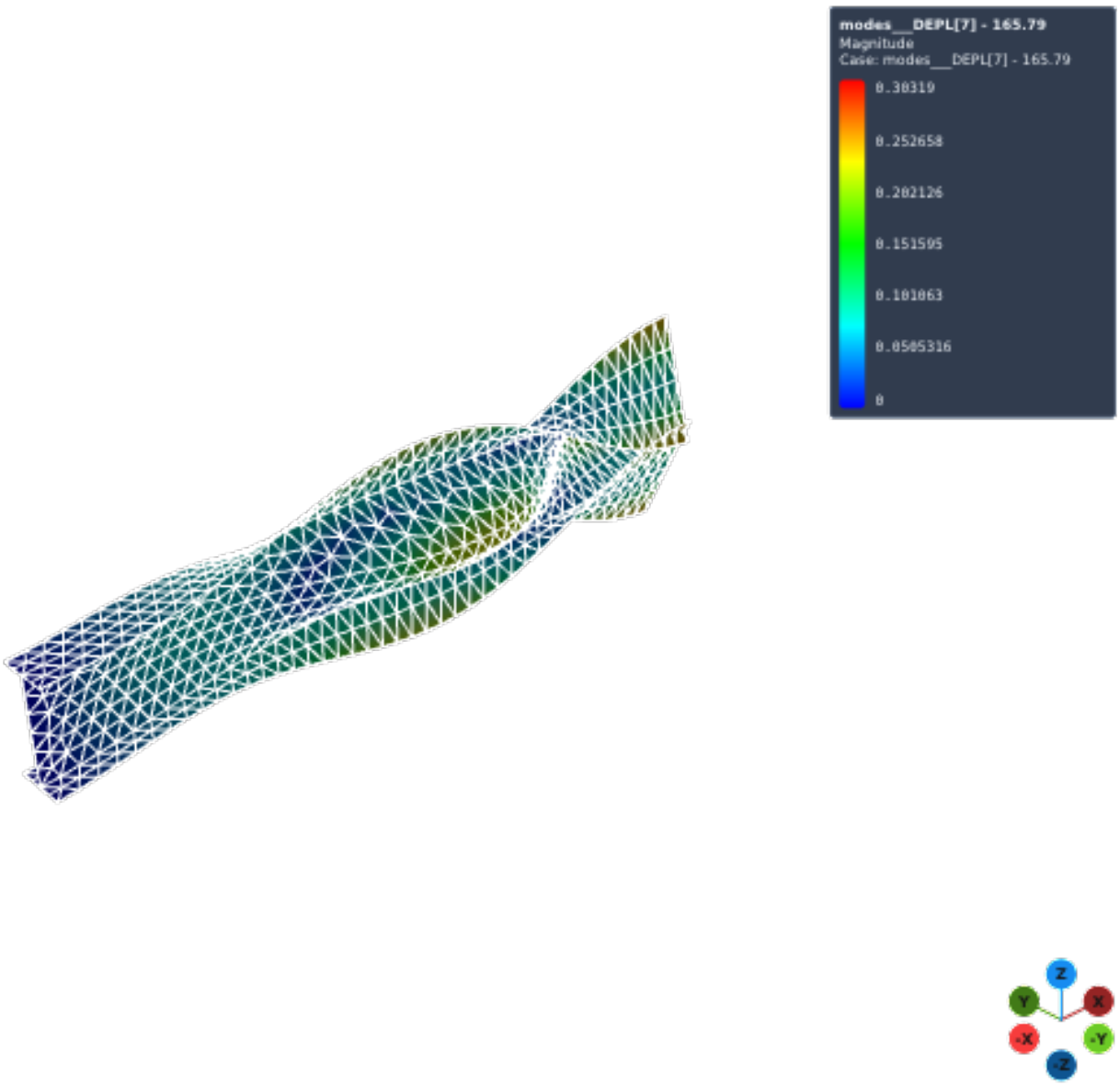

Figure 223: Code_Aster — mode 8, 165.790 Hz

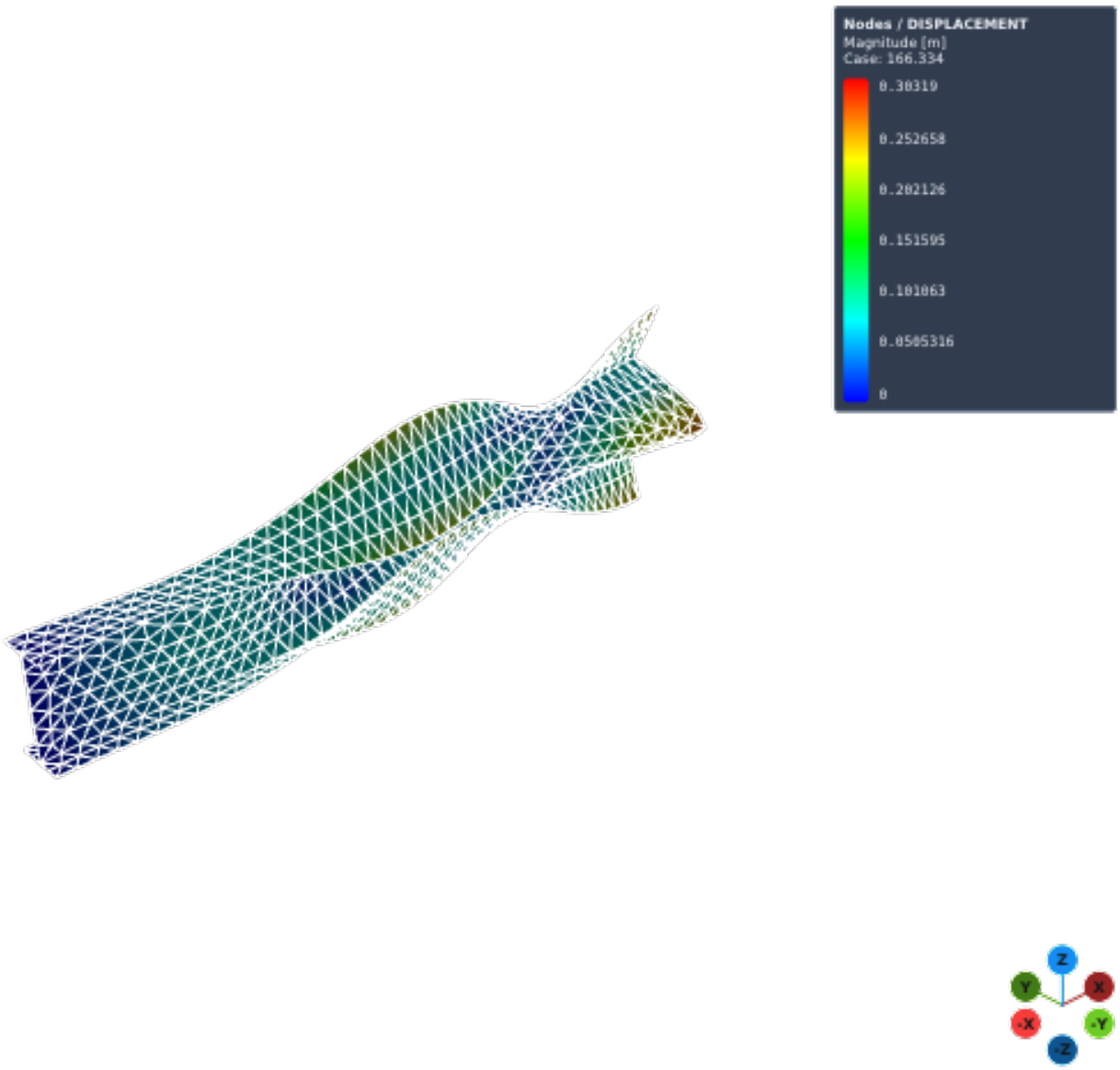

Figure 224: Sesam — mode 8, 166.334 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	166.600	+0.17	1.09572e+06	1.06e-14	5.52	1.13e-08	-1.03e-07	2.349	1.06e-04
Calculix	166.579	+0.15	1.09547e+06	5.11e-13	5.55	2.25e-13	7.15e-07	2.356	-4.75e-07
Code_Aster	165.790	-0.32	1.08511e+06	1.18e-26	5.54	3.14e-25	-1.09e-13	2.353	5.60e-13
Sesam	166.334	4.85e-03	1.09224e+06	7.25e-26	5.56	9.43e-24	2.69e-13	-2.358	-3.07e-12

Mode 9

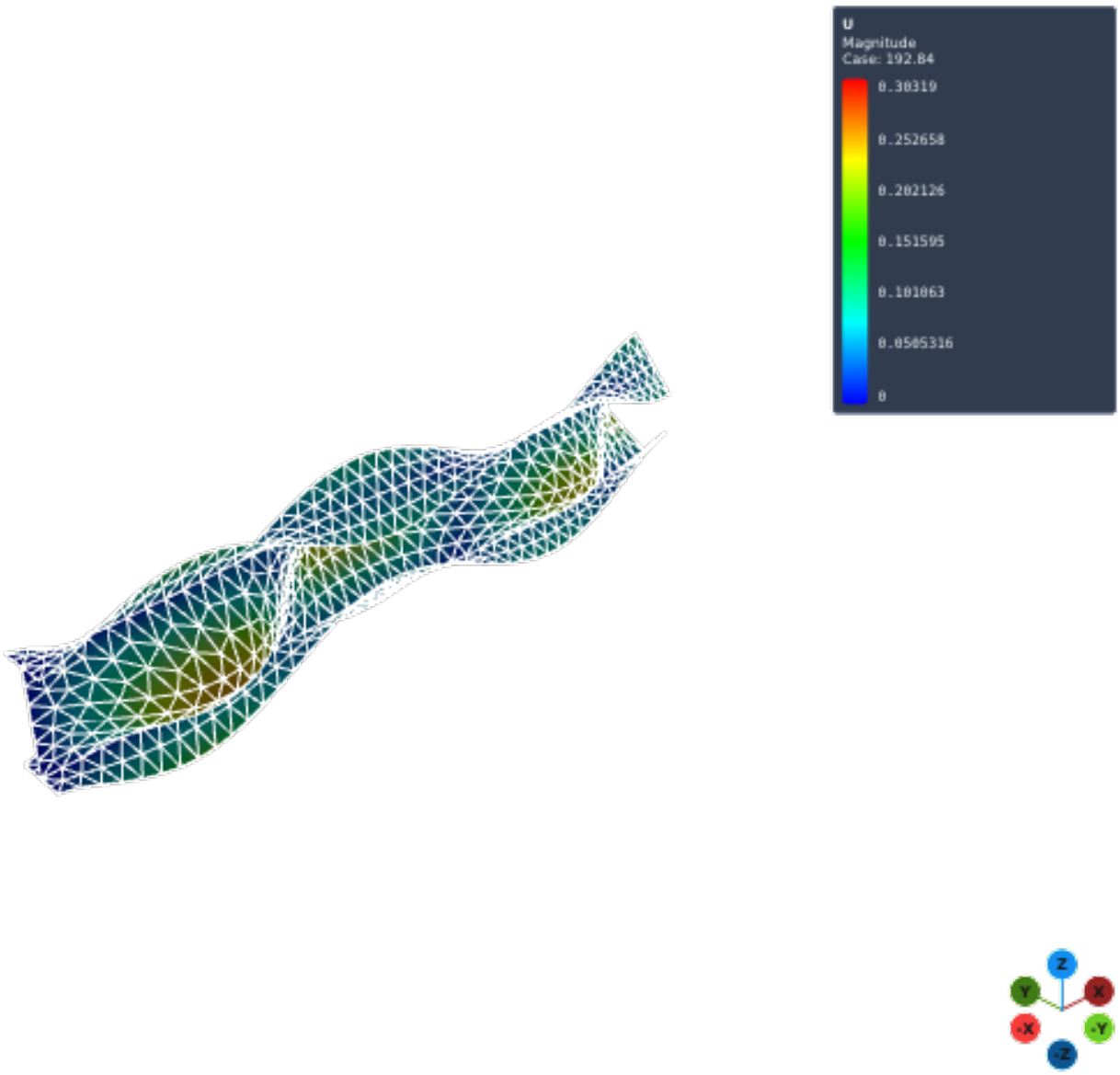

Figure 225: Abaqus — mode 9, 192.840 Hz

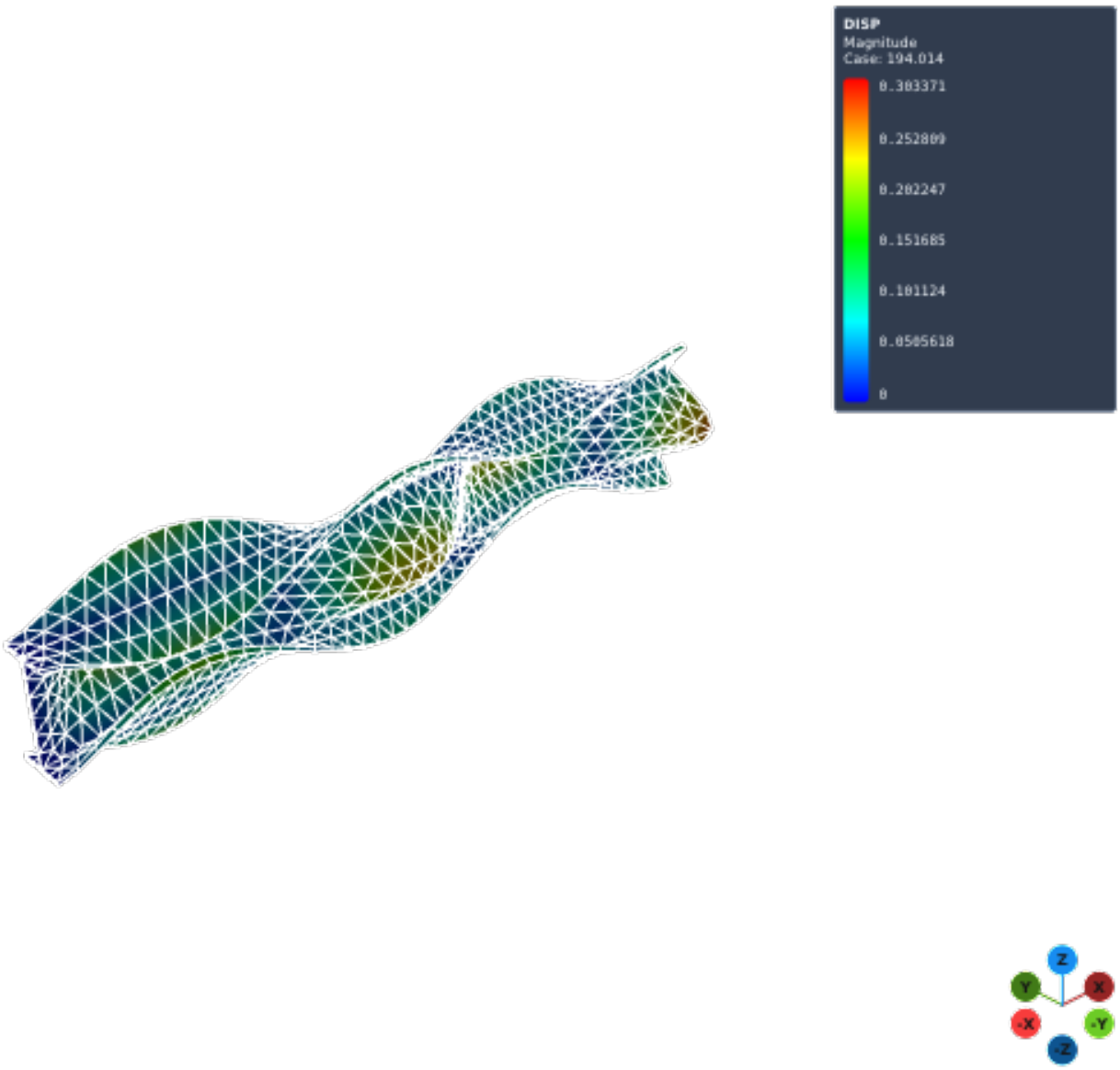

Figure 226: Calculix — mode 9, 194.014 Hz

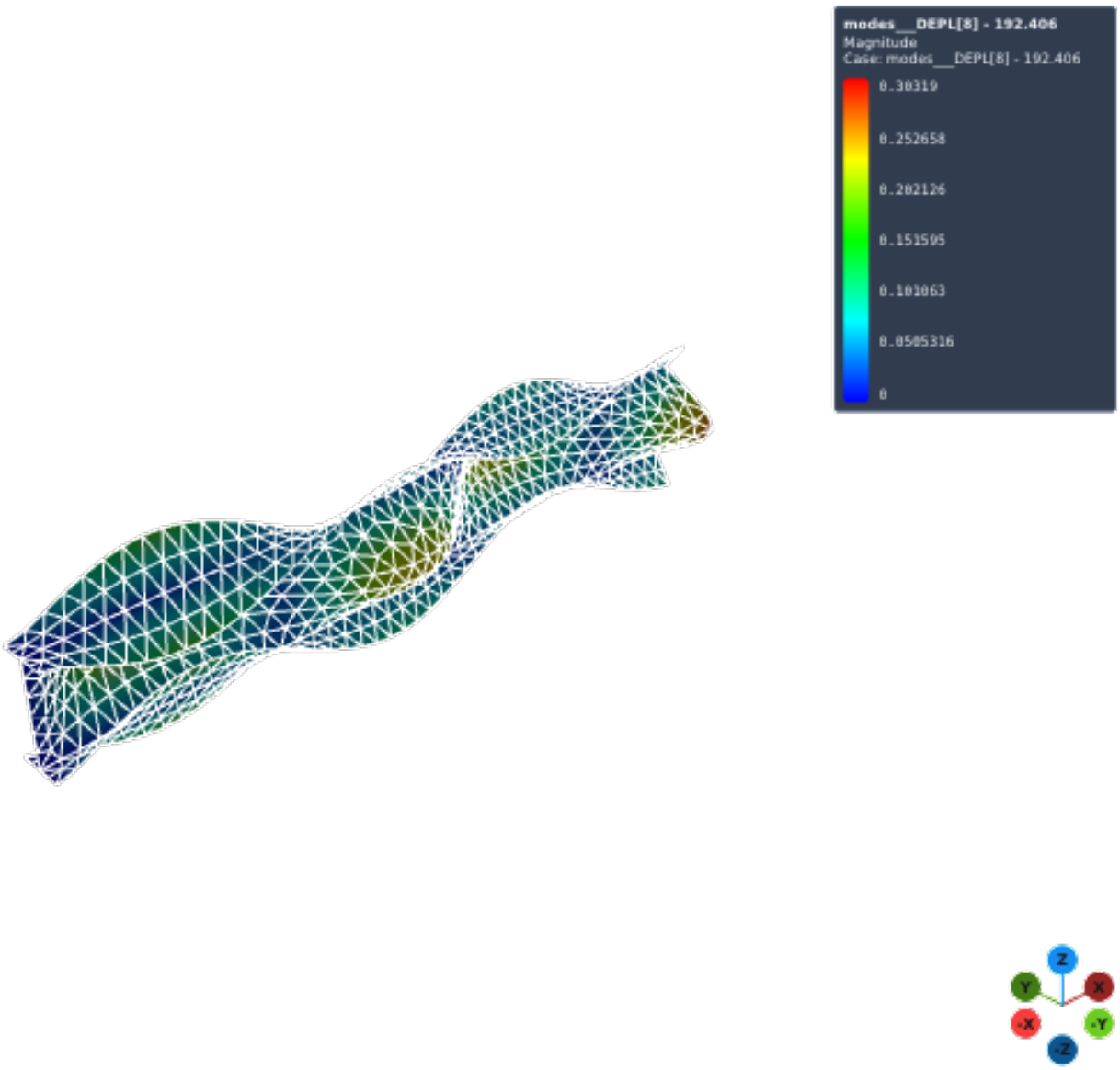

Figure 227: Code_Aster — mode 9, 192.406 Hz

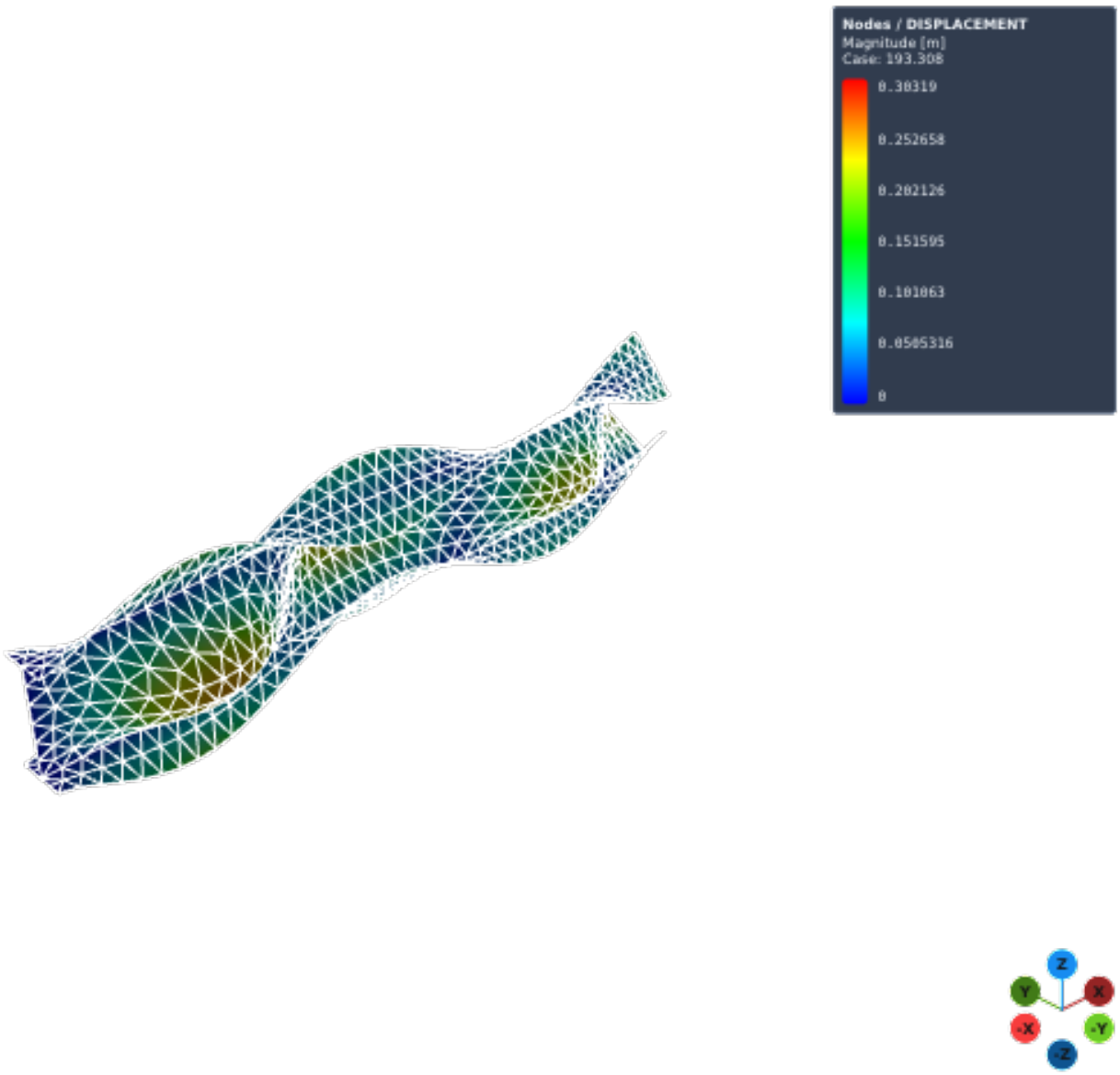

Figure 228: Sesam — mode 9, 193.308 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	192.840	-0.16	1.4681e+06	5.91e-14	0.67	5.79e-09	2.43e-07	-0.818	-7.61e-05
Calculix	194.014	+0.45	1.48603e+06	4.33e-13	0.63	7.51e-14	-6.58e-07	0.793	-2.74e-07
Code_Aster	192.406	-0.38	1.46149e+06	2.47e-27	0.64	8.63e-26	-4.97e-14	0.801	-2.94e-13
Sesam	193.308	+0.09	1.47524e+06	1.19e-27	0.64	3.81e-25	3.44e-14	-0.801	6.18e-13

Mode 10

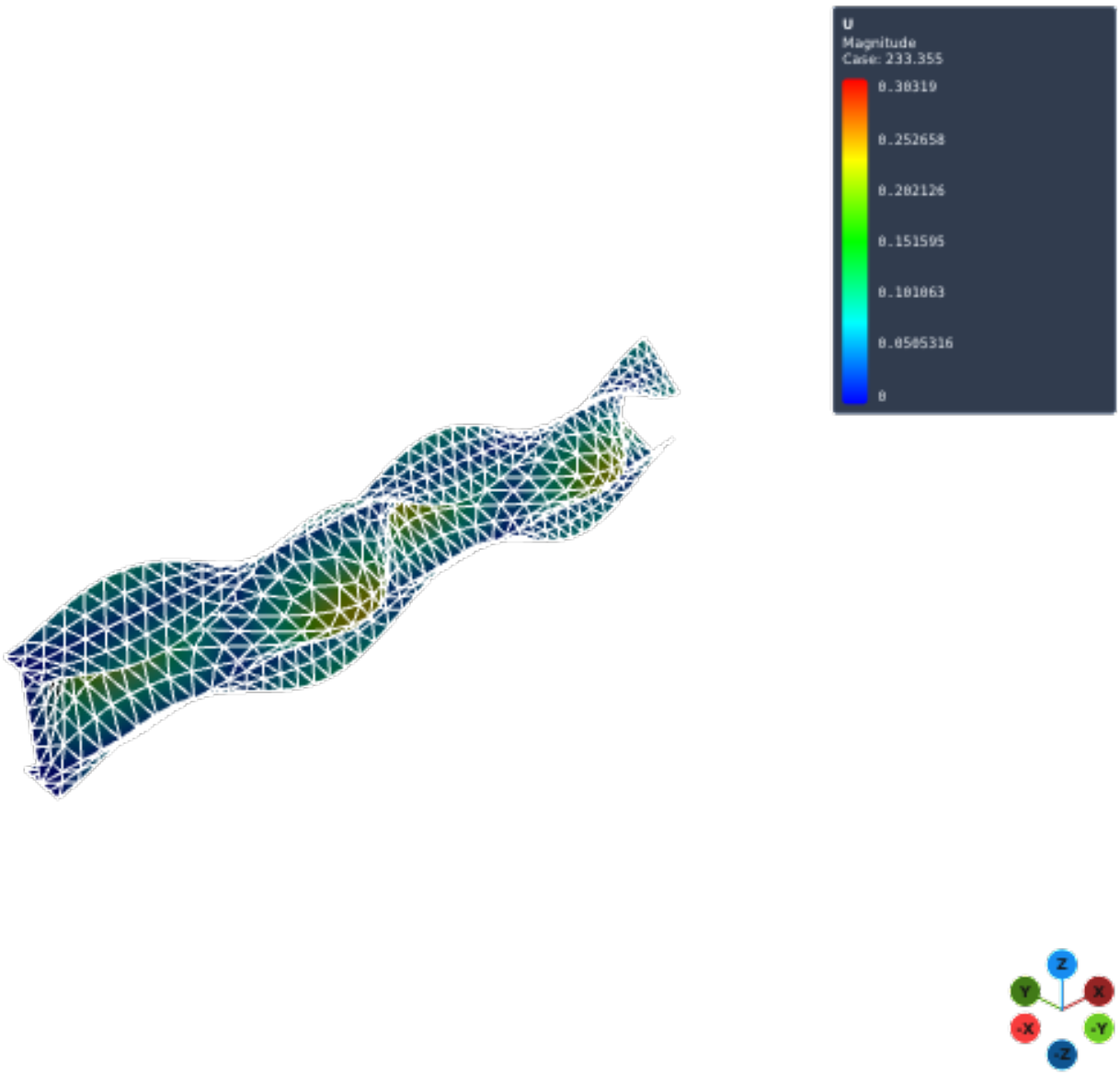

Figure 229: Abaqus — mode 10, 233.355 Hz

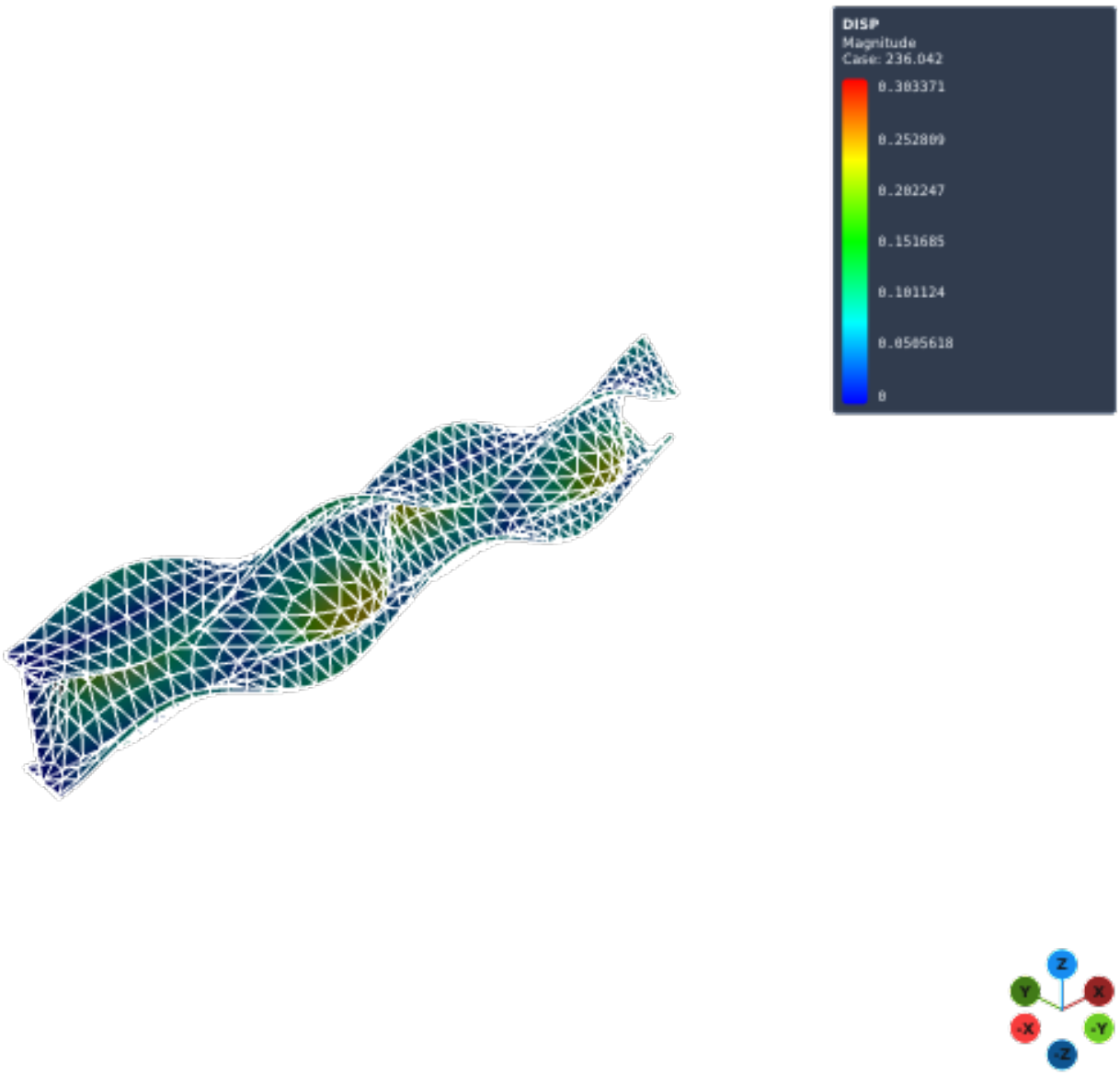

Figure 230: Calculix — mode 10, 236.042 Hz

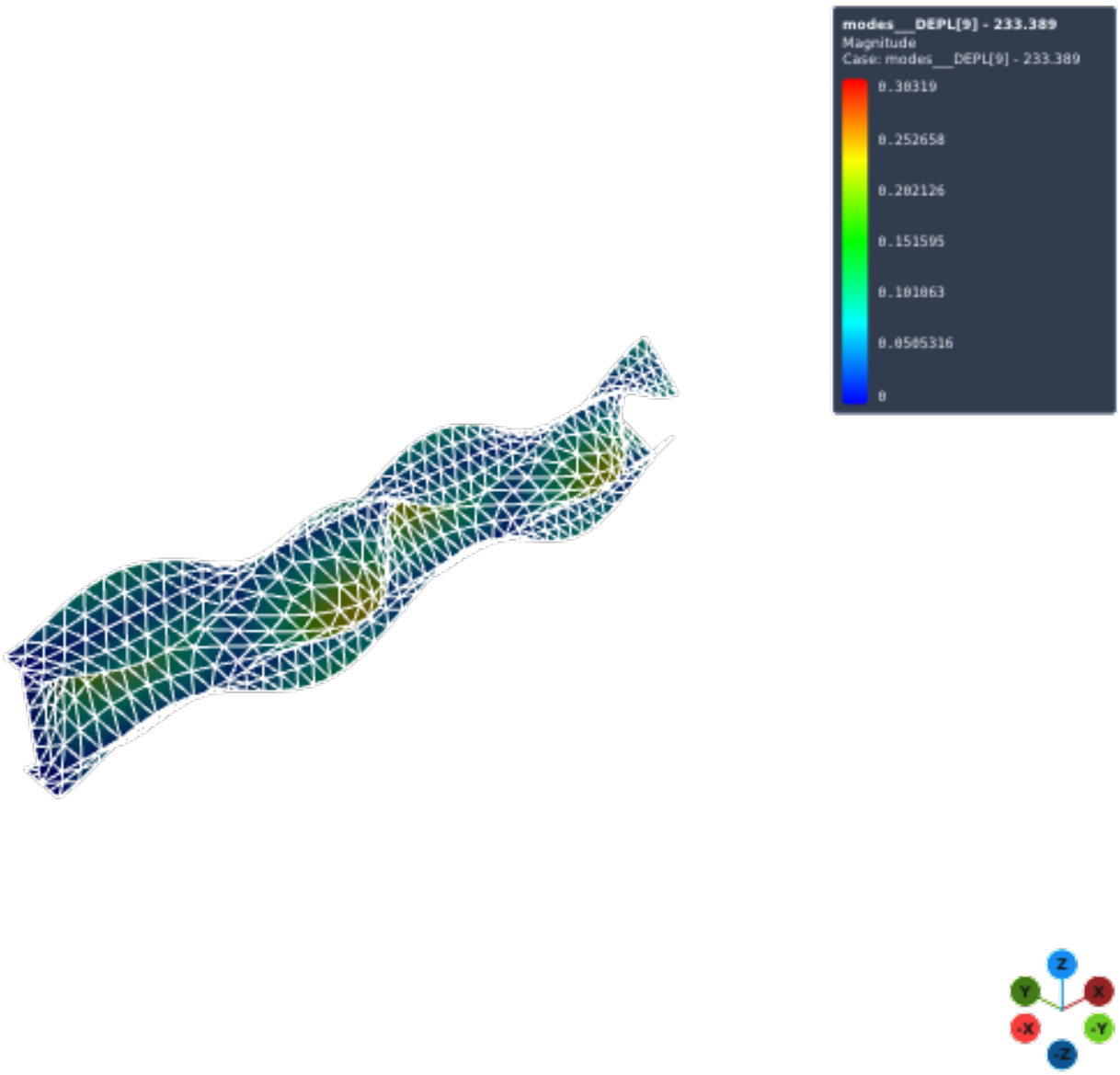

Figure 231: Code_Aster — mode 10, 233.389 Hz

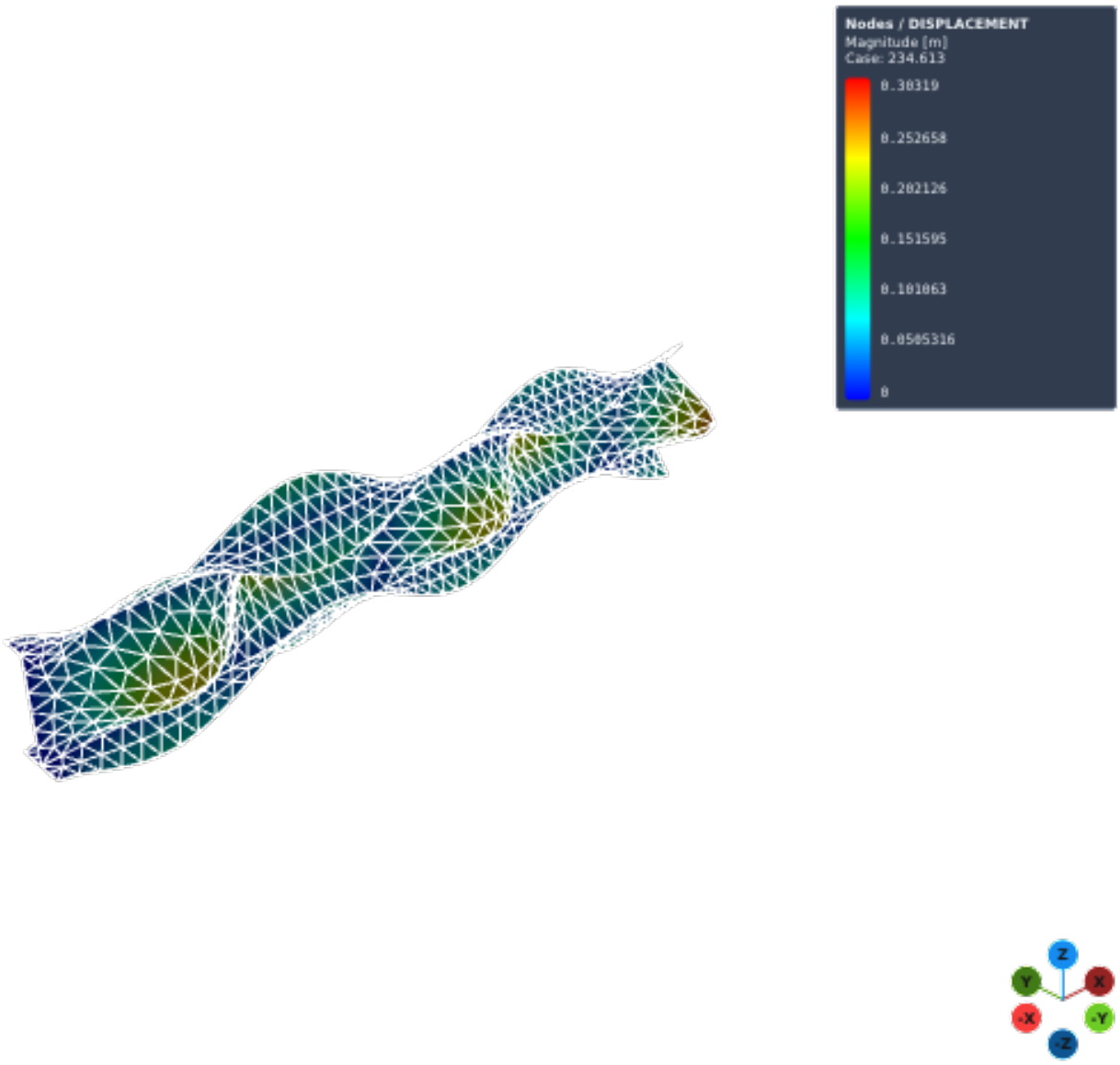

Figure 232: Sesam — mode 10, 234.613 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	233.360	-0.42	2.14979e+06	3.28e-15	1.38	3.26e-07	-5.73e-08	1.175	0.001
Calculix	236.042	+0.72	2.19956e+06	1.09e-14	1.42	2.89e-14	-1.04e-07	1.193	1.70e-07
Code_Aster	233.389	-0.41	2.15041e+06	2.31e-29	1.39	6.58e-25	4.80e-15	1.178	8.11e-13
Sesam	234.613	+0.11	2.17302e+06	1.23e-18	1.40	2.14e-20	-1.11e-09	-1.185	-1.46e-10

Mode 11

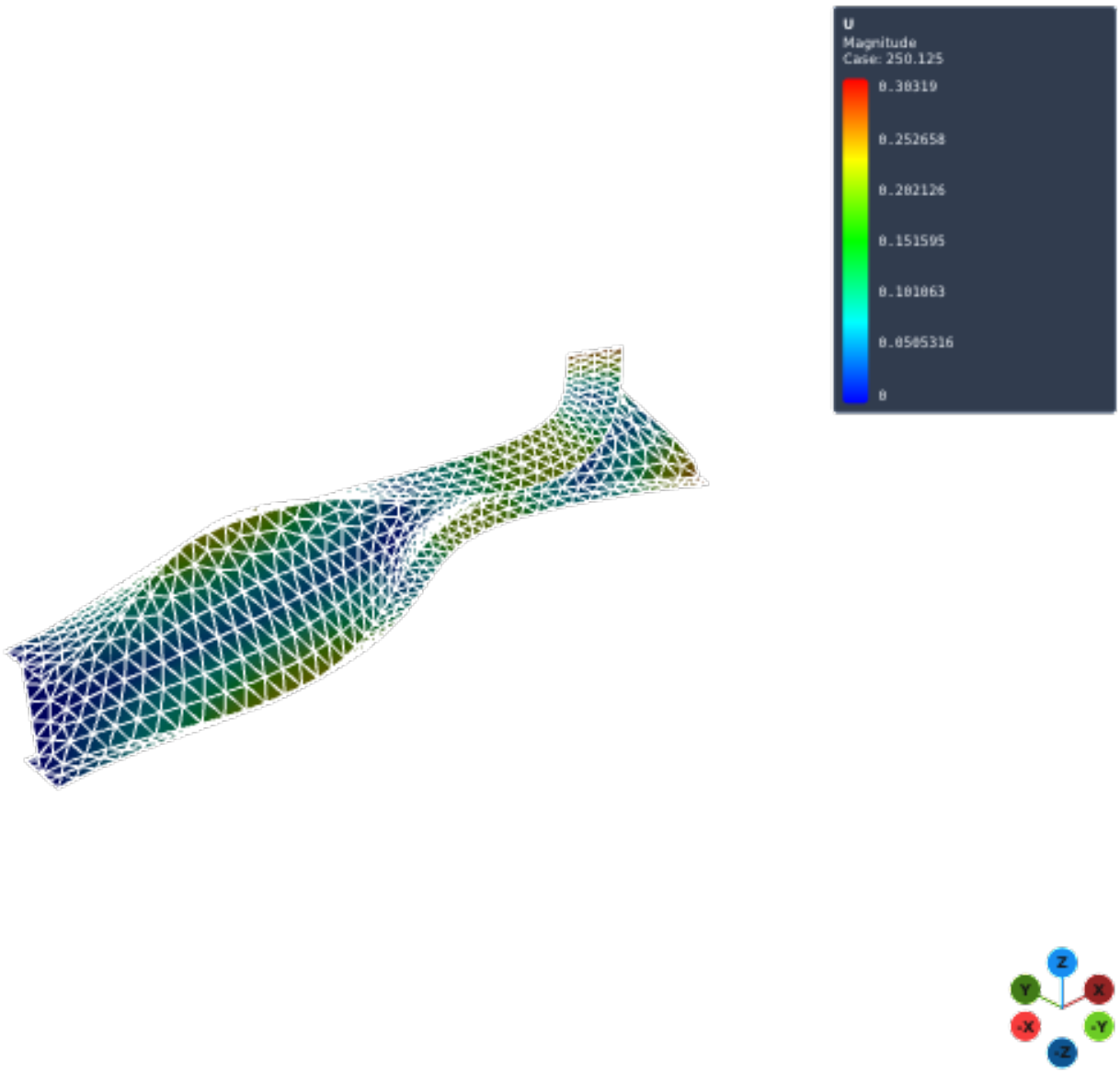

Figure 233: Abaqus — mode 11, 250.125 Hz

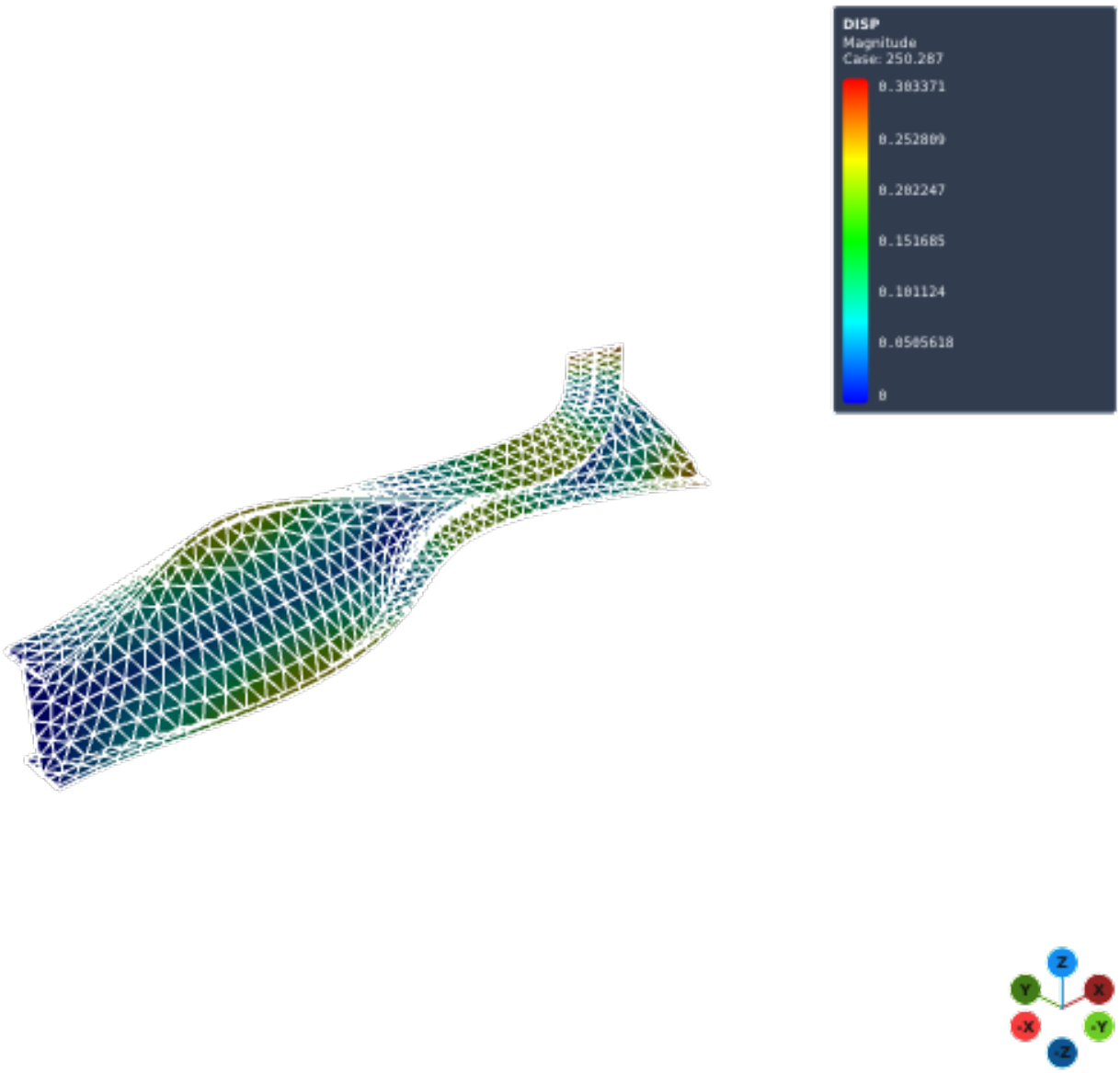

Figure 234: Calculix — mode 11, 250.287 Hz

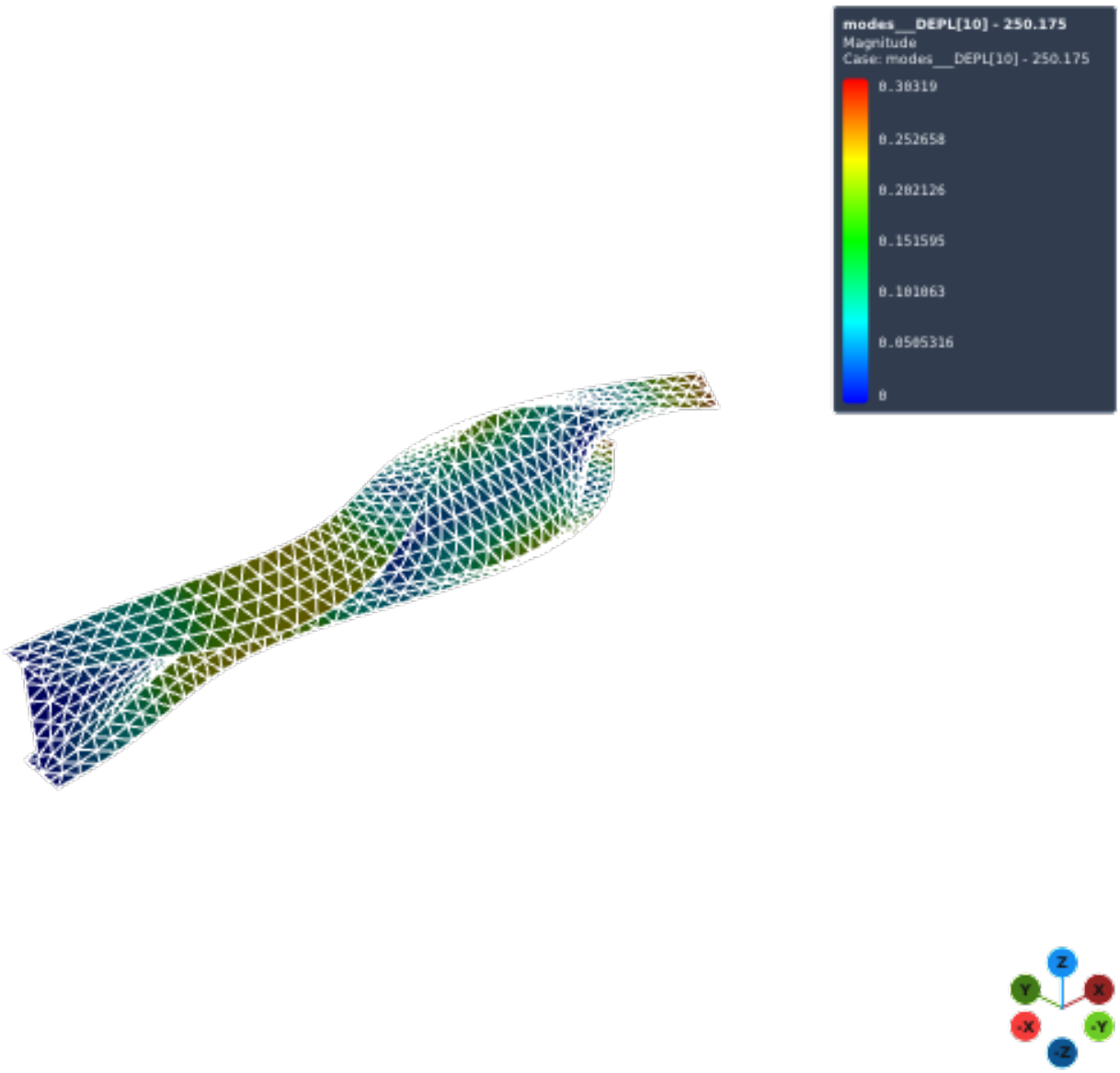

Figure 235: Code_Aster — mode 11, 250.175 Hz

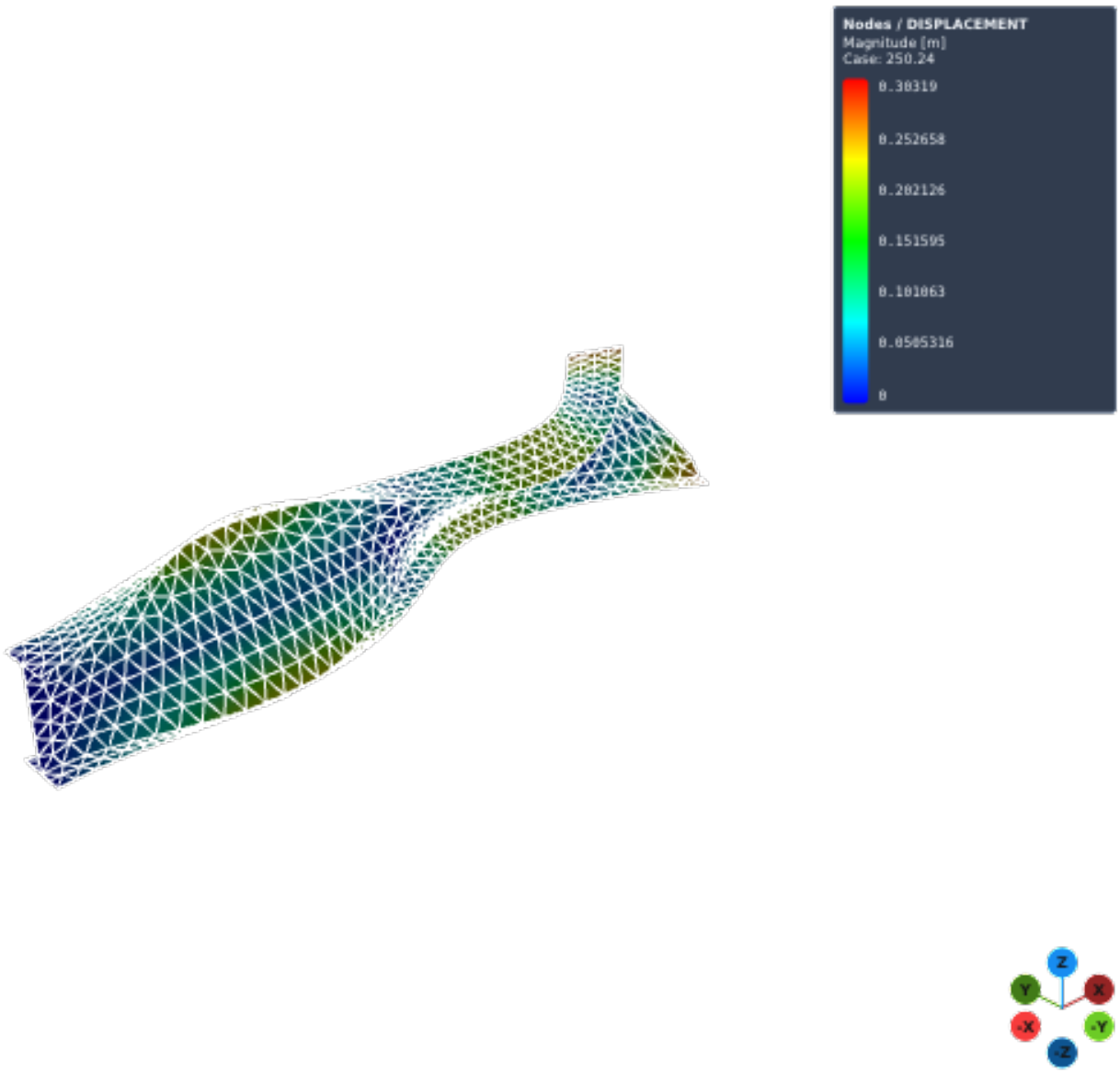

Figure 236: Sesam — mode 11, 250.240 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	250.130	-0.03	2.46987e+06	9.03e-11	3.18e-08	4.03e-09	9.50e-06	1.78e-04	6.35e-05
Calculix	250.287	+0.03	2.47306e+06	2.16e-10	5.05e-07	3.53e-15	-1.47e-05	0.001	-5.94e-08
Code_Aster	250.175	-0.01	2.47085e+06	1.31e-27	5.41e-07	2.47e-24	-3.62e-14	-0.001	-1.57e-12
Sesam	250.240	+0.01	2.47215e+06	1.60e-13	1.65e-06	2.77e-15	-4.01e-07	0.001	-5.26e-08

Shell, 2nd order, QUAD

Mode 1

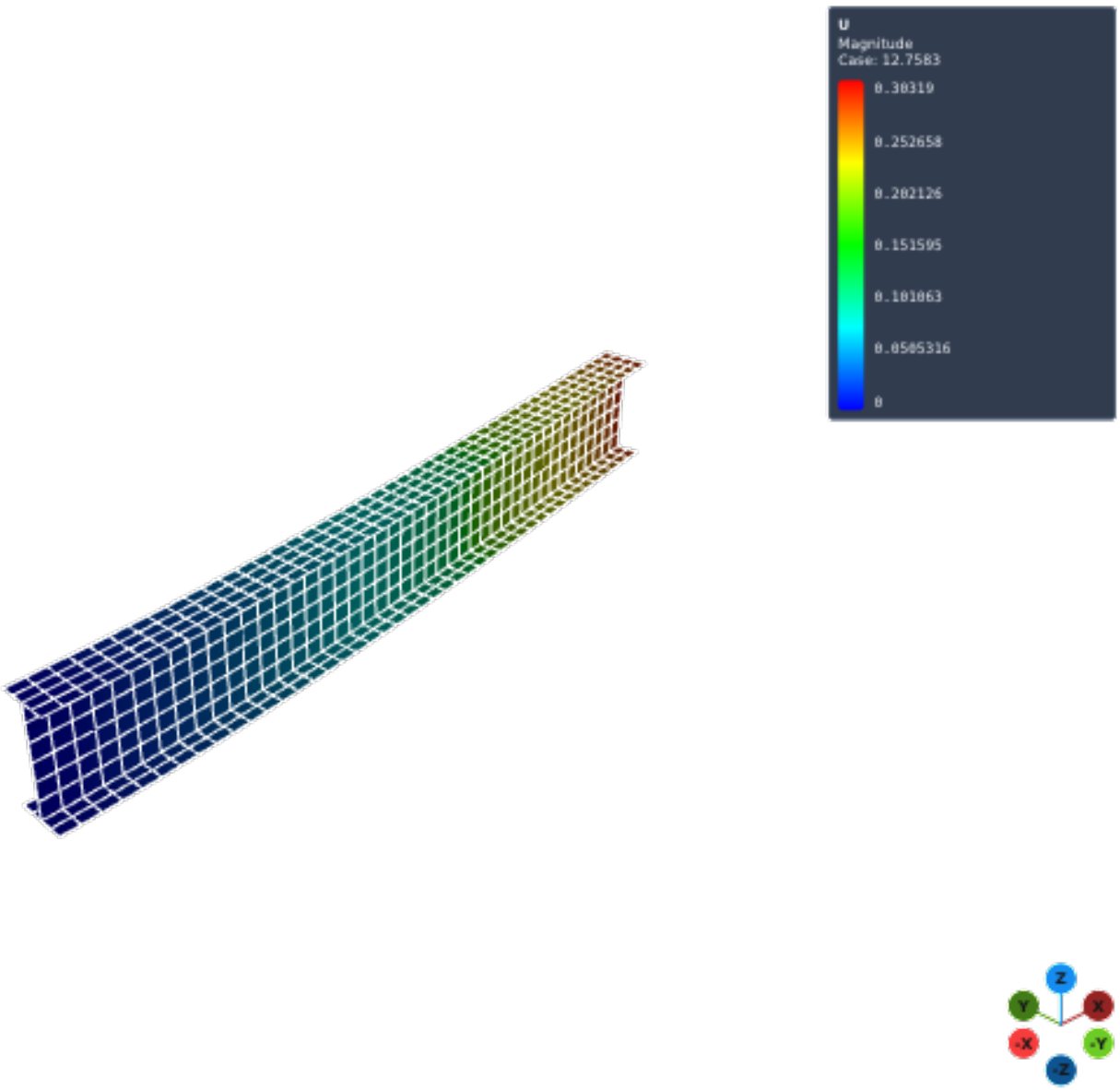

Figure 237: Abaqus — mode 1, 12.758 Hz

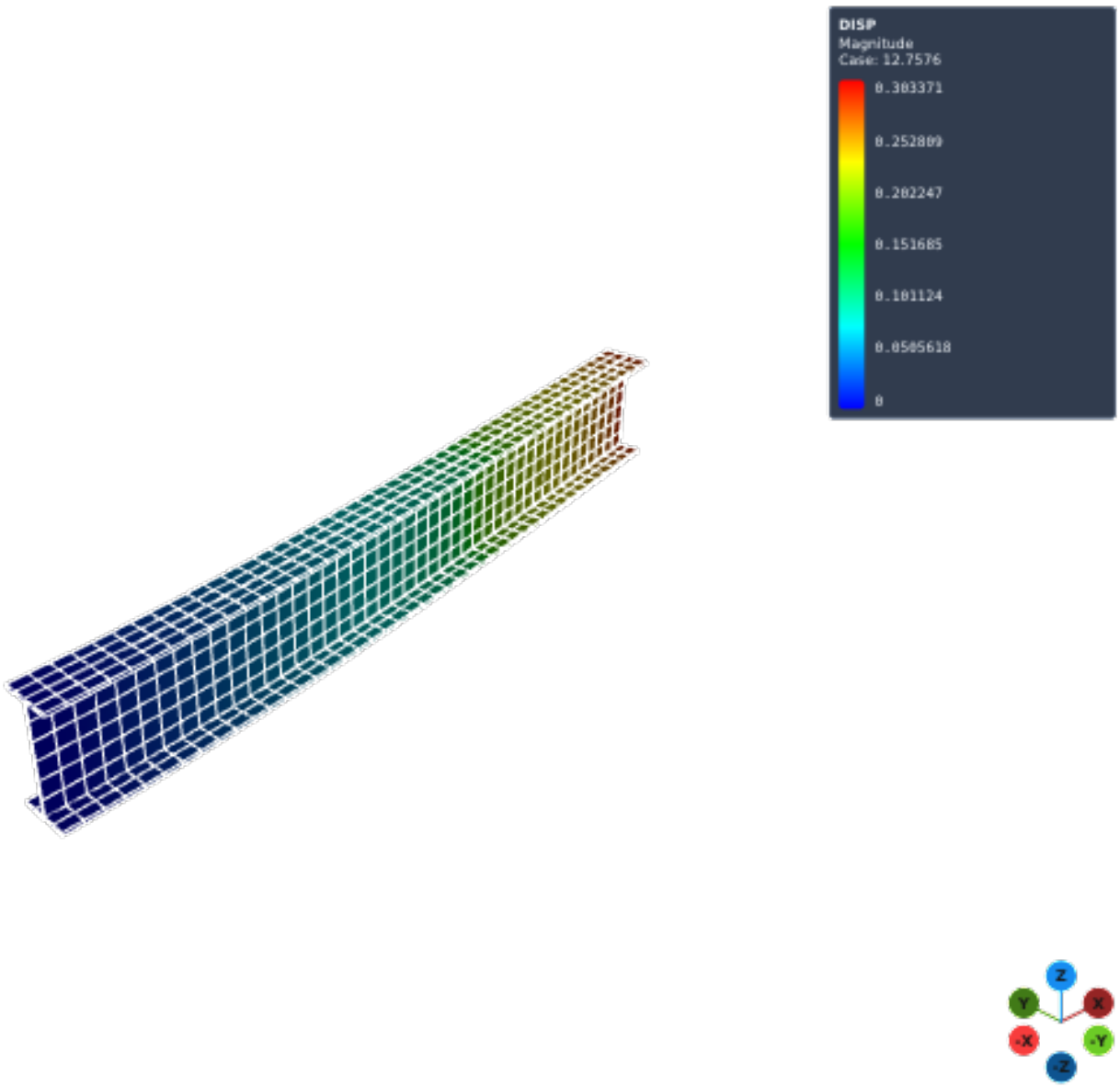

Figure 238: Calculix — mode 1, 12.758 Hz

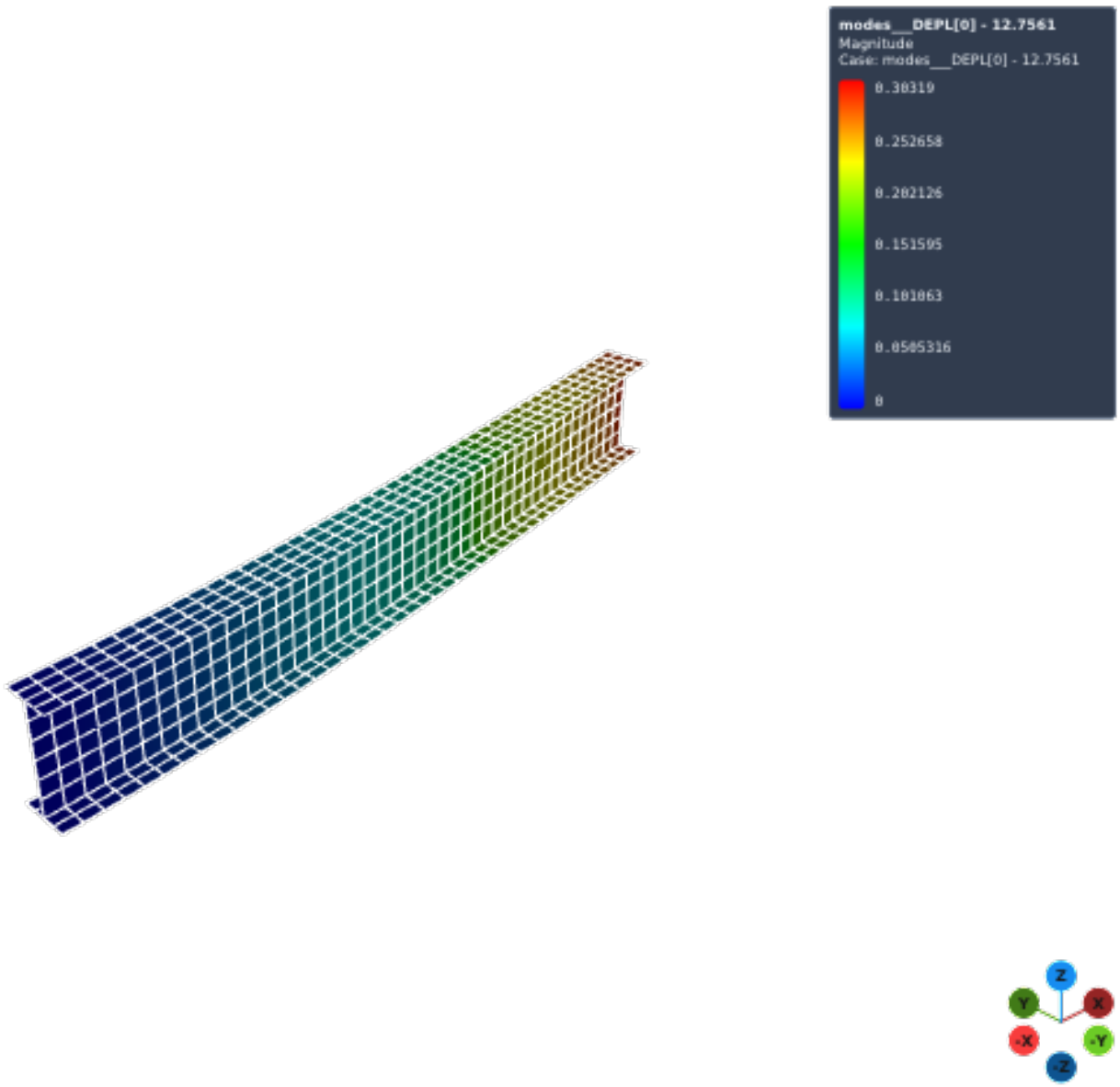

Figure 239: Code_Aster — mode 1, 12.756 Hz

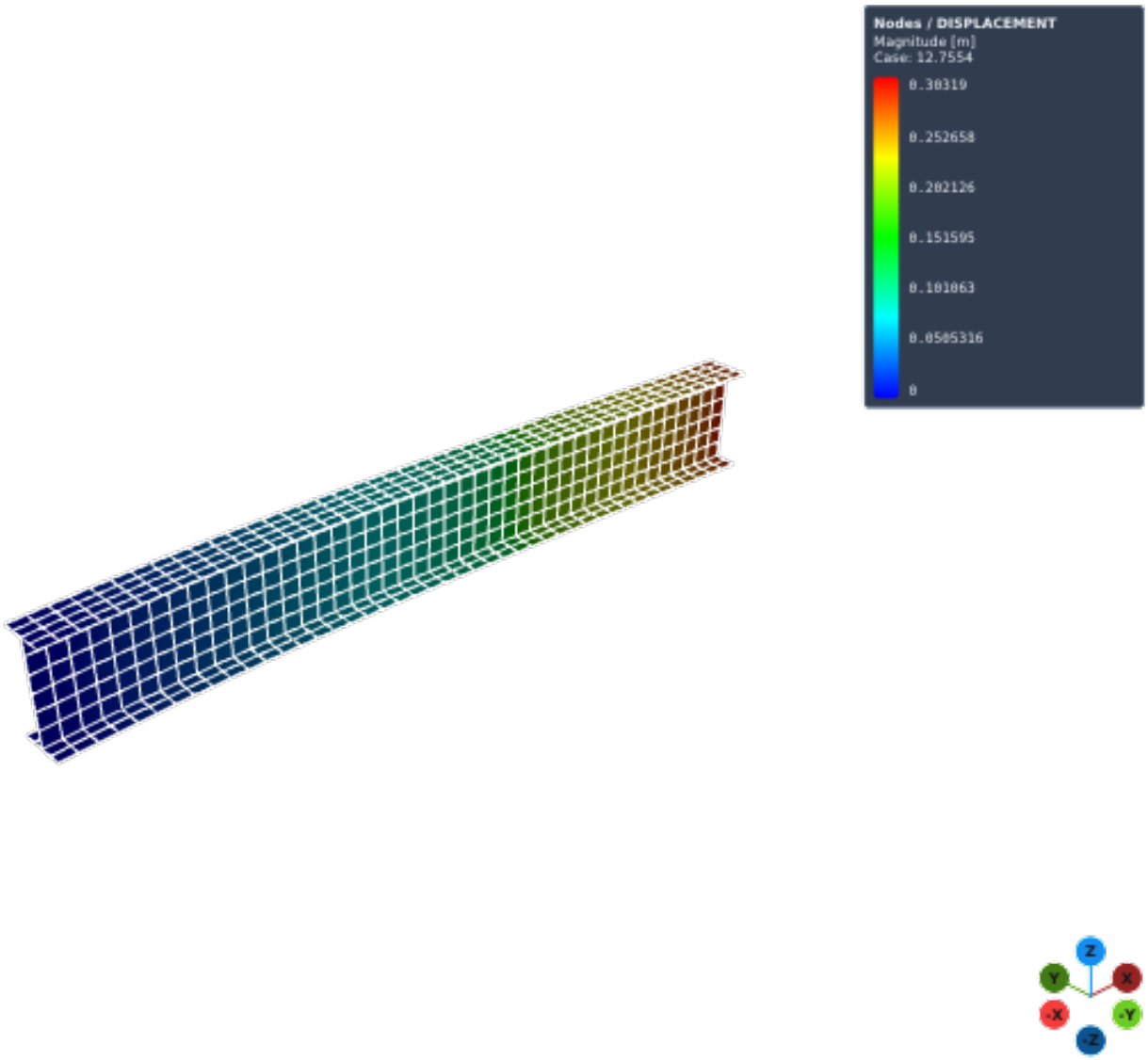

Figure 240: Sesam — mode 1, 12.755 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.758	+0.01	6426.1	1.23e-25	119.92	9.18e-25	3.51e-13	10.951	9.58e-13
Calculix	12.758	+0.01	6425.39	1.29e-23	119.93	1.74e-12	3.59e-12	10.951	1.32e-06
Code_Aster	12.756	-0.01	6423.86	3.33e-27	119.94	1.40e-25	-5.77e-14	10.952	3.74e-13
Sesam	12.755	-0.01	6423.15	3.17e-26	119.94	2.23e-24	-1.78e-13	-10.952	-1.49e-12

Mode 2

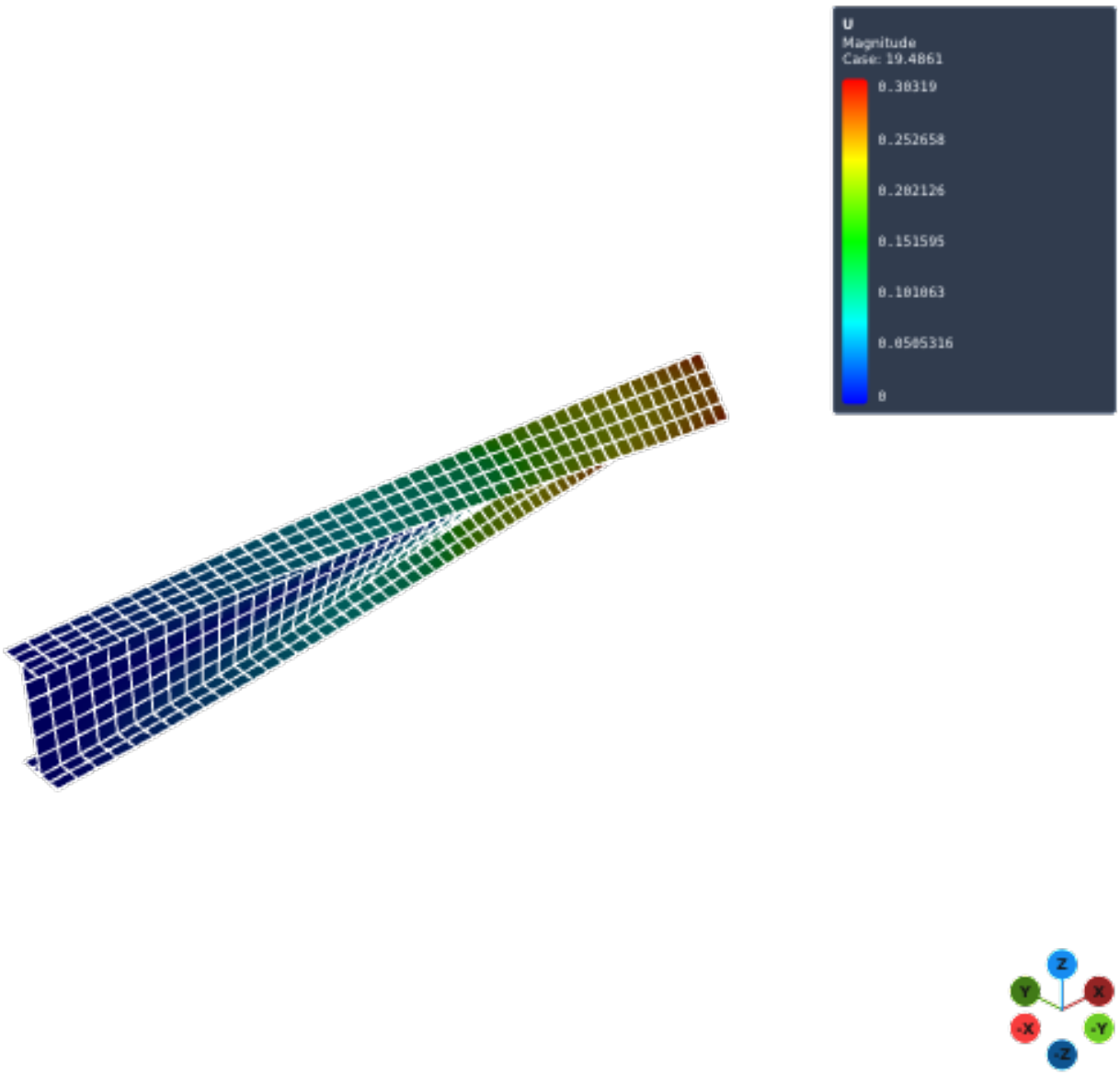

Figure 241: Abaqus — mode 2, 19.486 Hz

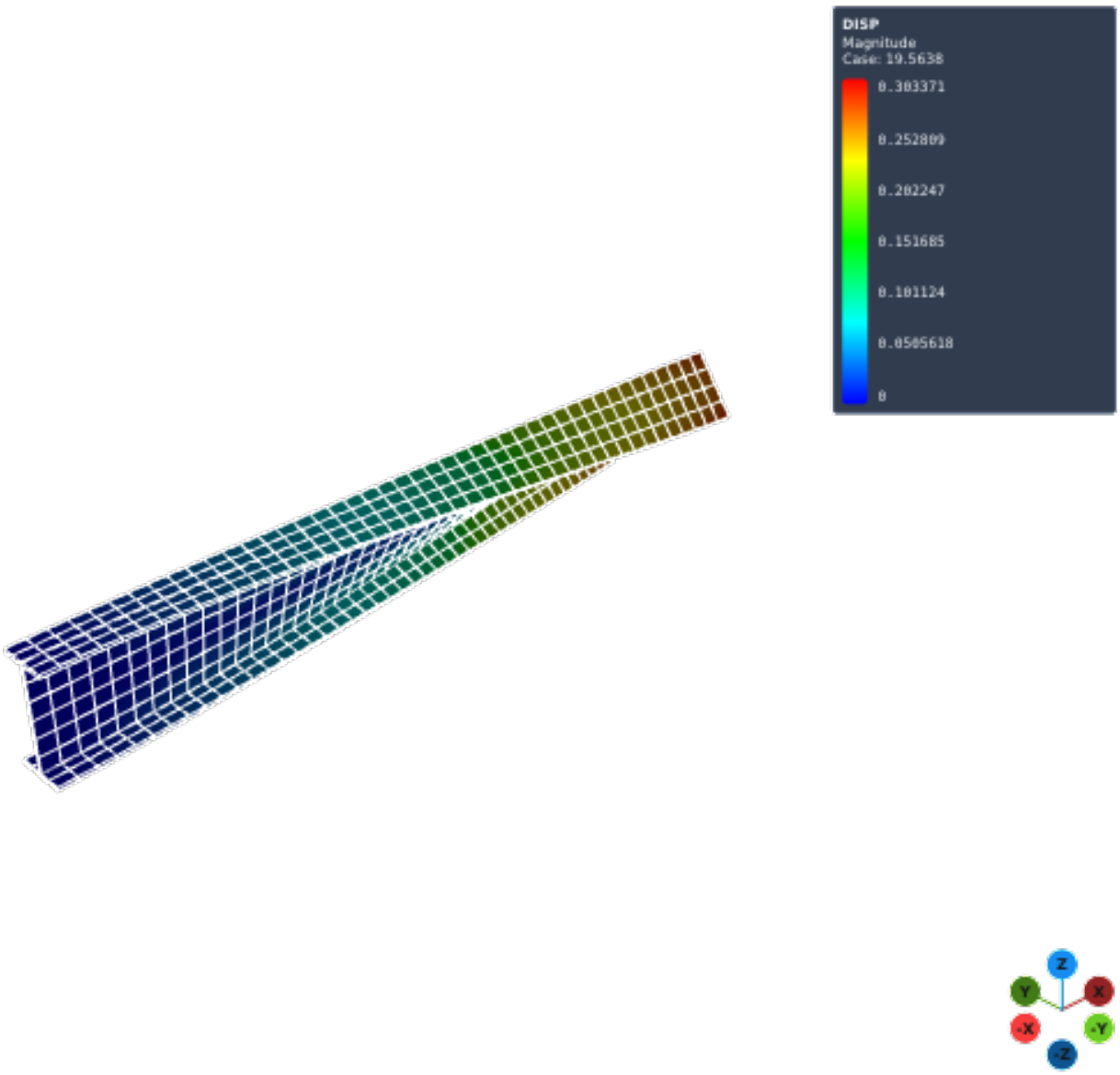

Figure 242: Calculix — mode 2, 19.564 Hz

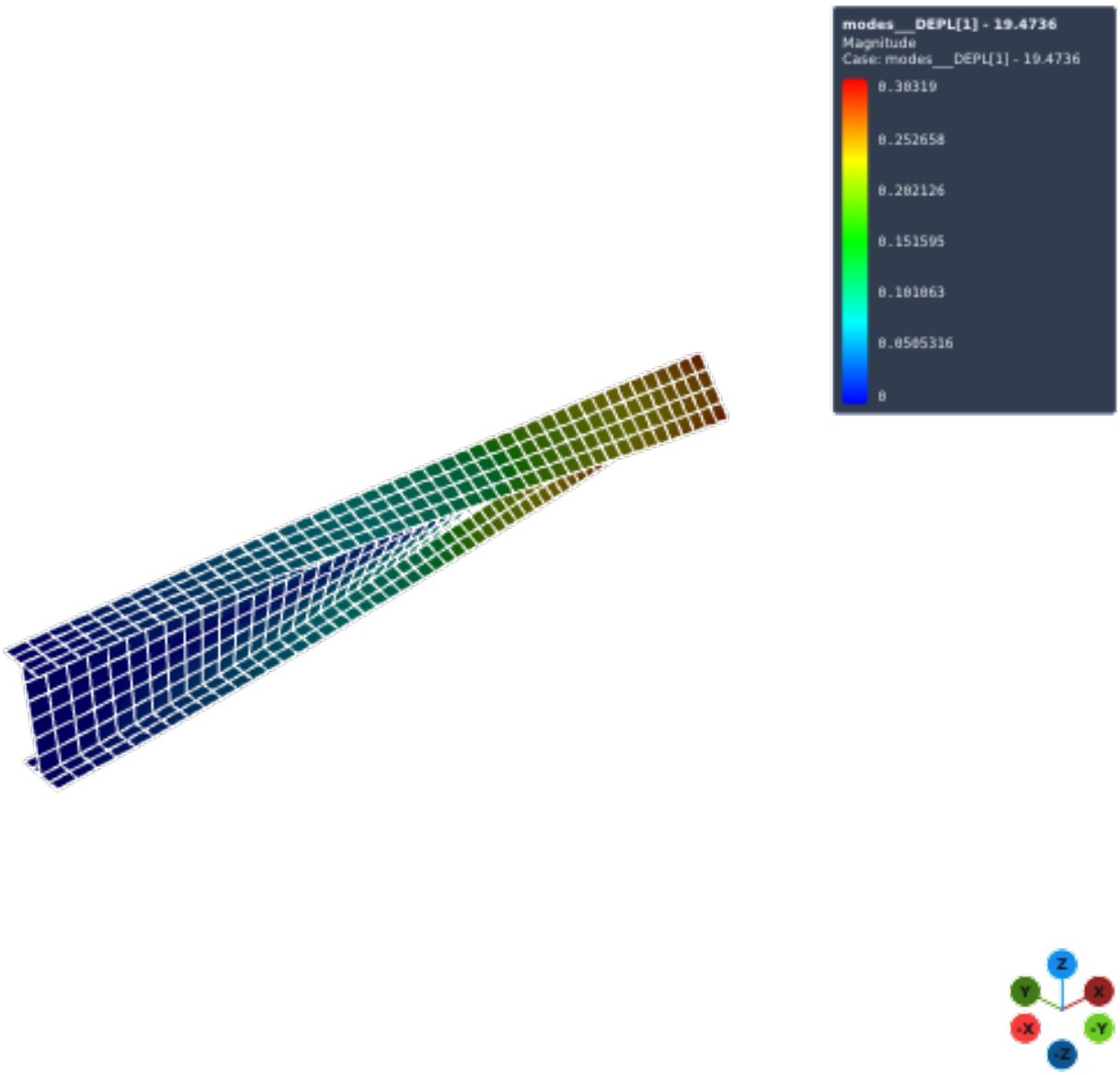

Figure 243: Code_Aster — mode 2, 19.474 Hz

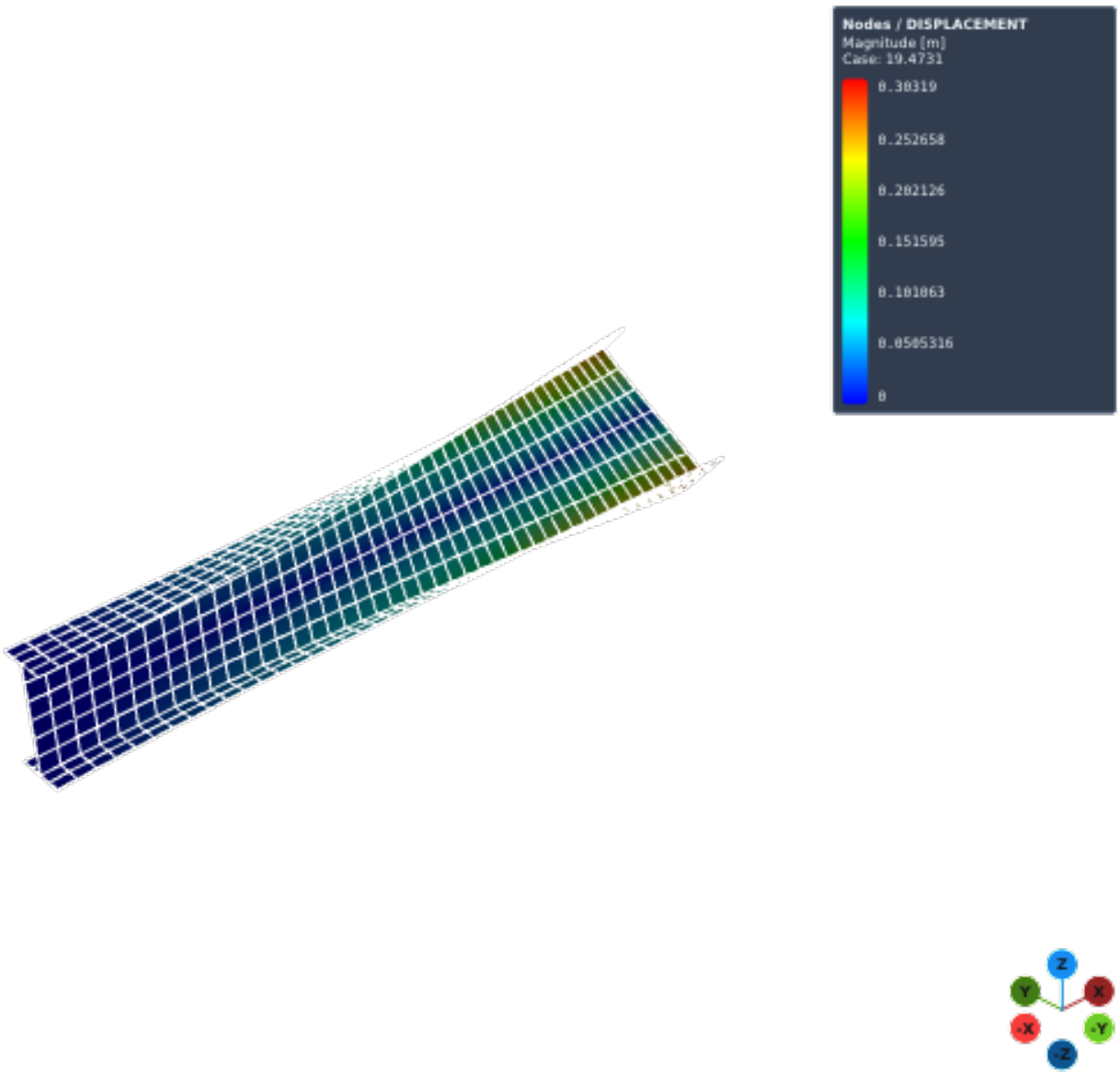

Figure 244: Sesam — mode 2, 19.473 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	19.486	-0.07	14990	1.04e-26	4.65e-20	2.40e-23	1.02e-13	2.16e-10	4.90e-12
Calculix	19.564	+0.33	15110.1	3.11e-09	1.08e-15	6.33e-19	5.58e-05	3.28e-08	-7.95e-10
Code_Aster	19.474	-0.13	14971	1.19e-26	2.41e-18	1.48e-22	1.09e-13	-1.55e-09	-1.22e-11
Sesam	19.473	-0.13	14970.3	4.18e-26	1.44e-18	3.35e-23	-2.04e-13	1.20e-09	5.79e-12

Mode 3

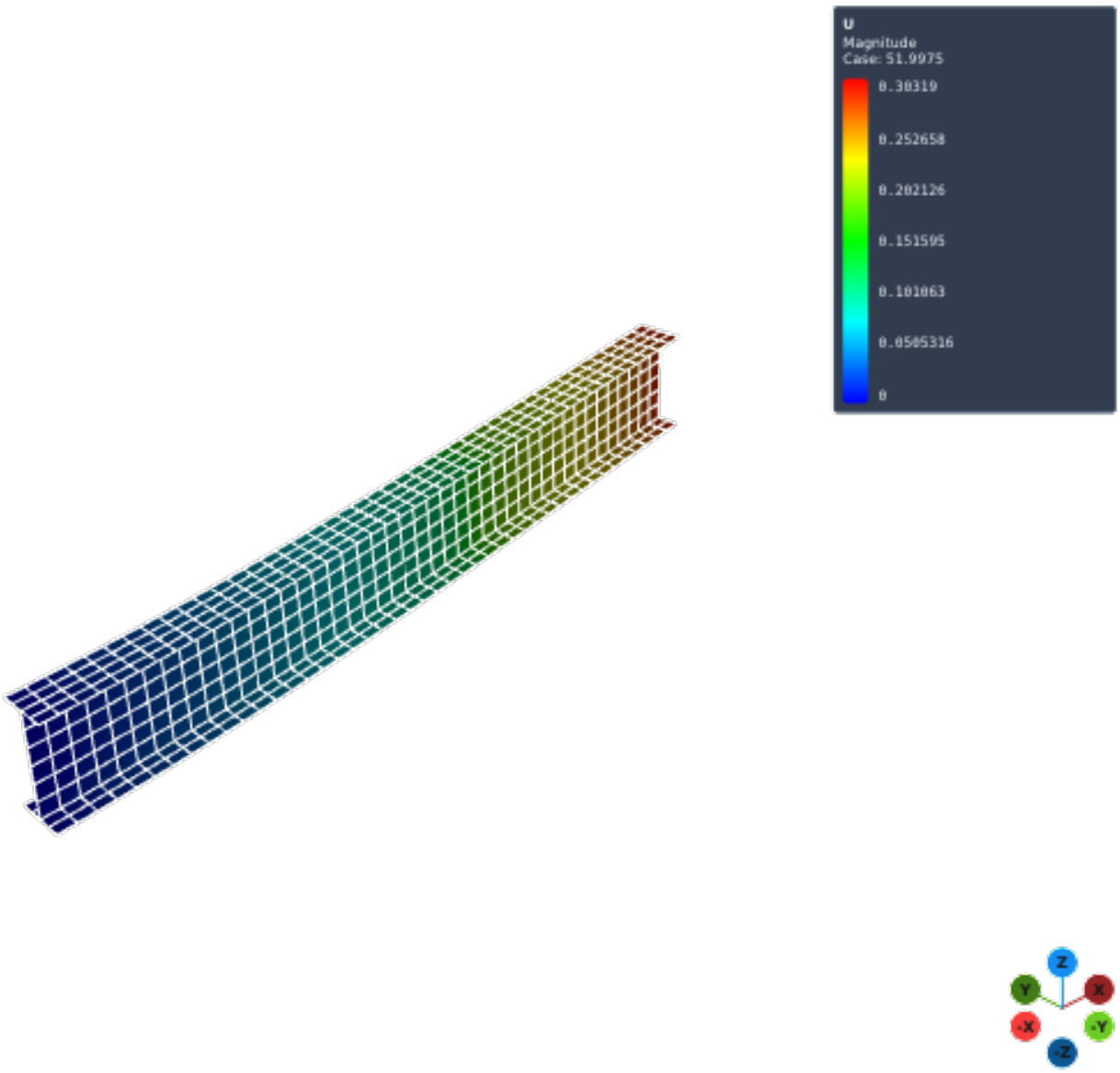

Figure 245: Abaqus — mode 3, 51.998 Hz

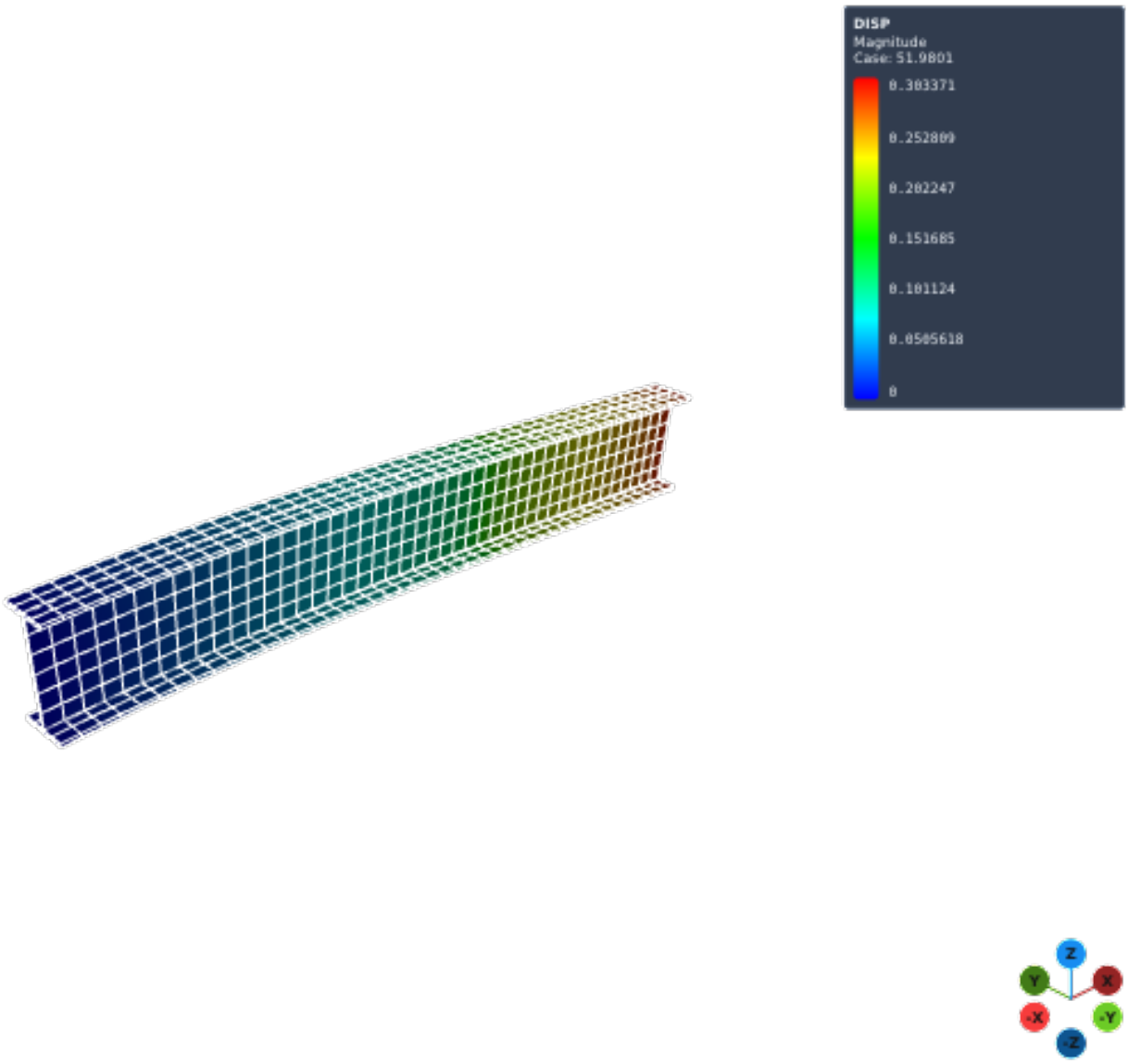

Figure 246: Calculix — mode 3, 51.980 Hz

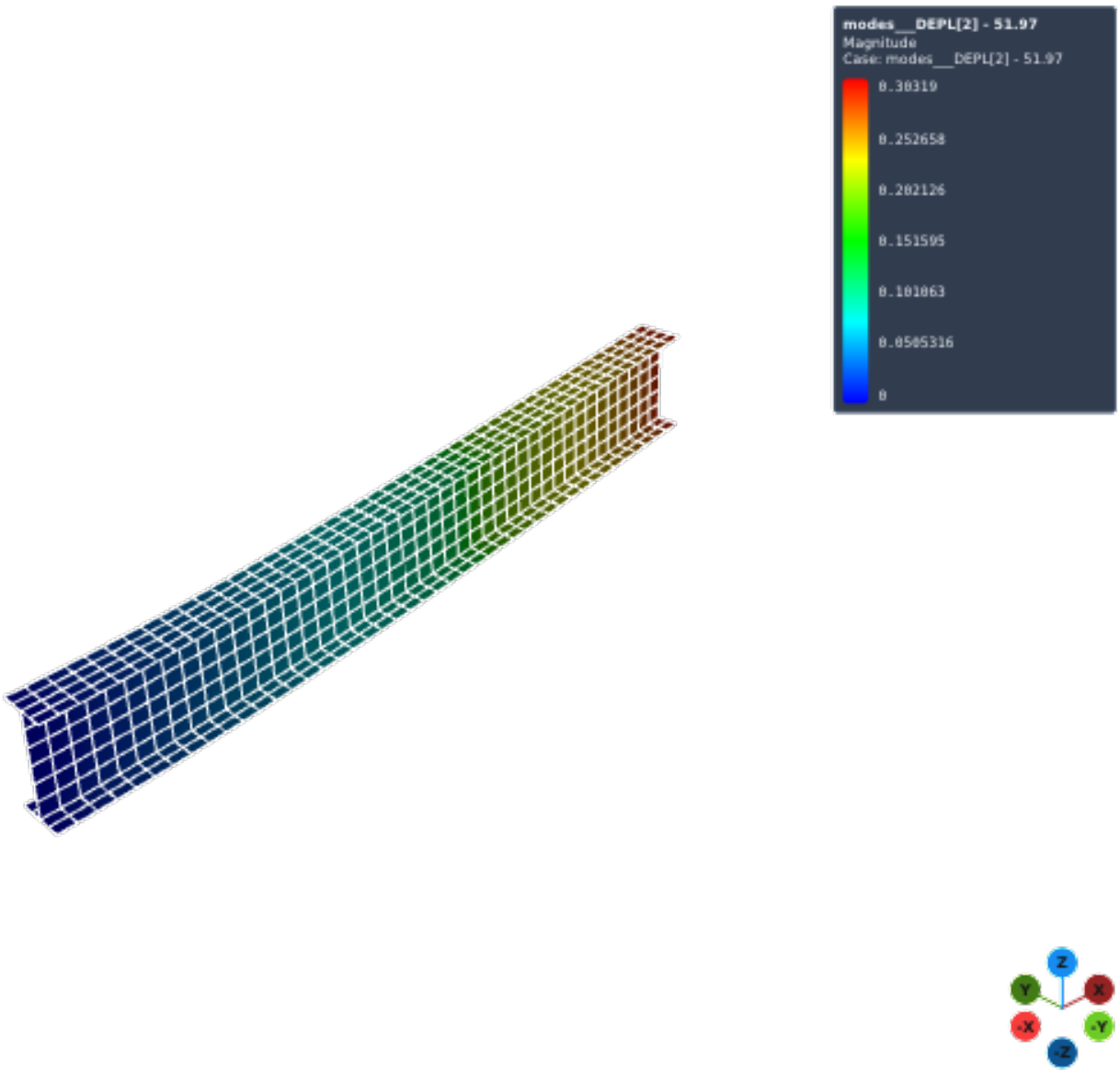

Figure 247: Code_Aster — mode 3, 51.970 Hz

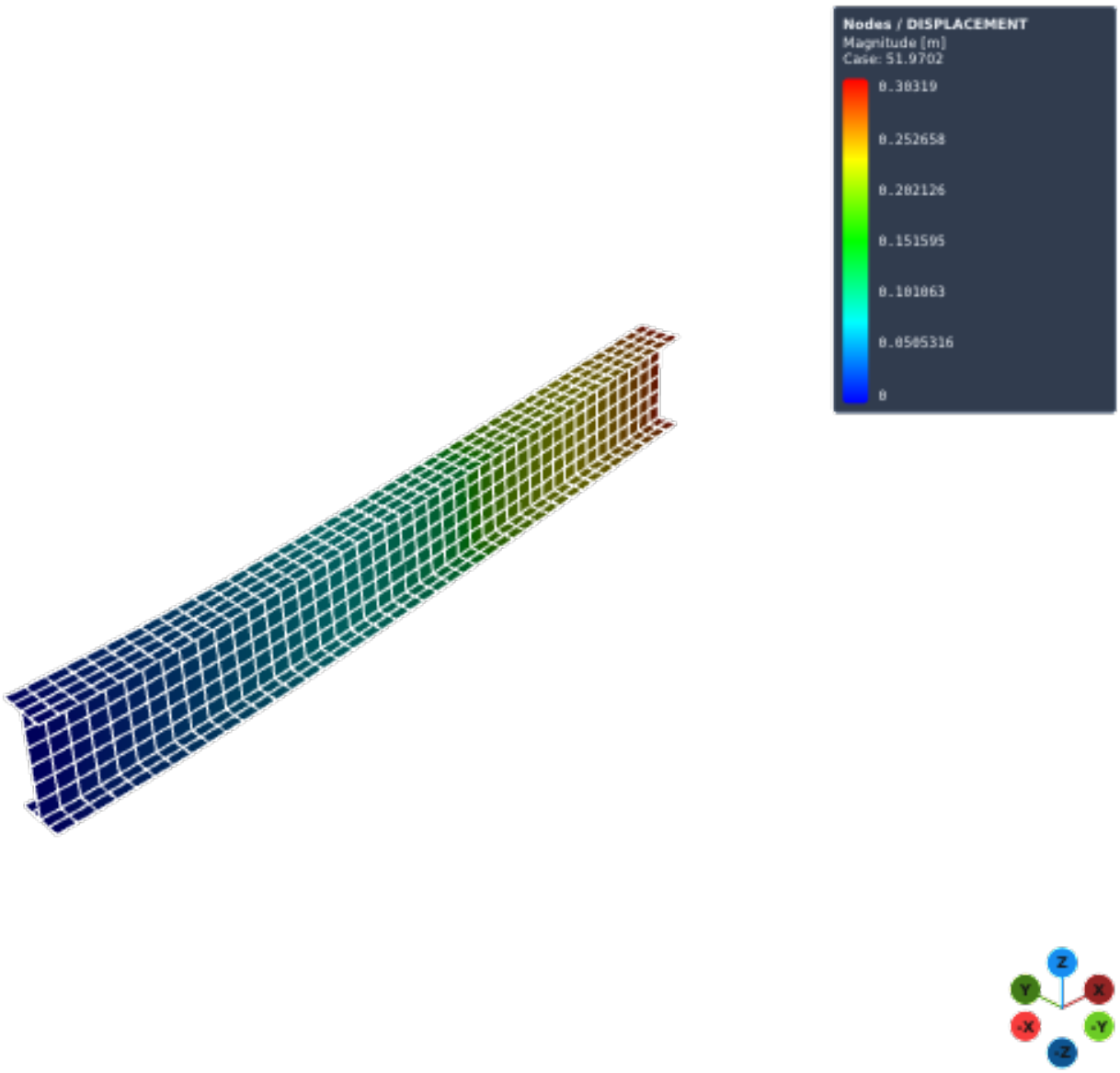

Figure 248: Sesam — mode 3, 51.970 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.998	+0.04	106740	5.15e-27	9.66e-26	121.97	7.18e-14	3.11e-13	11.044
Calculix	51.980	1.01e-03	106668	1.38e-23	4.52e-11	122.01	3.72e-12	6.73e-06	-11.046
Code_Aster	51.970	-0.02	106626	1.82e-26	1.94e-26	122.03	1.35e-13	1.39e-13	11.047
Sesam	51.970	-0.02	106628	4.55e-26	4.93e-25	122.03	-2.13e-13	-7.02e-13	11.047

Mode 4

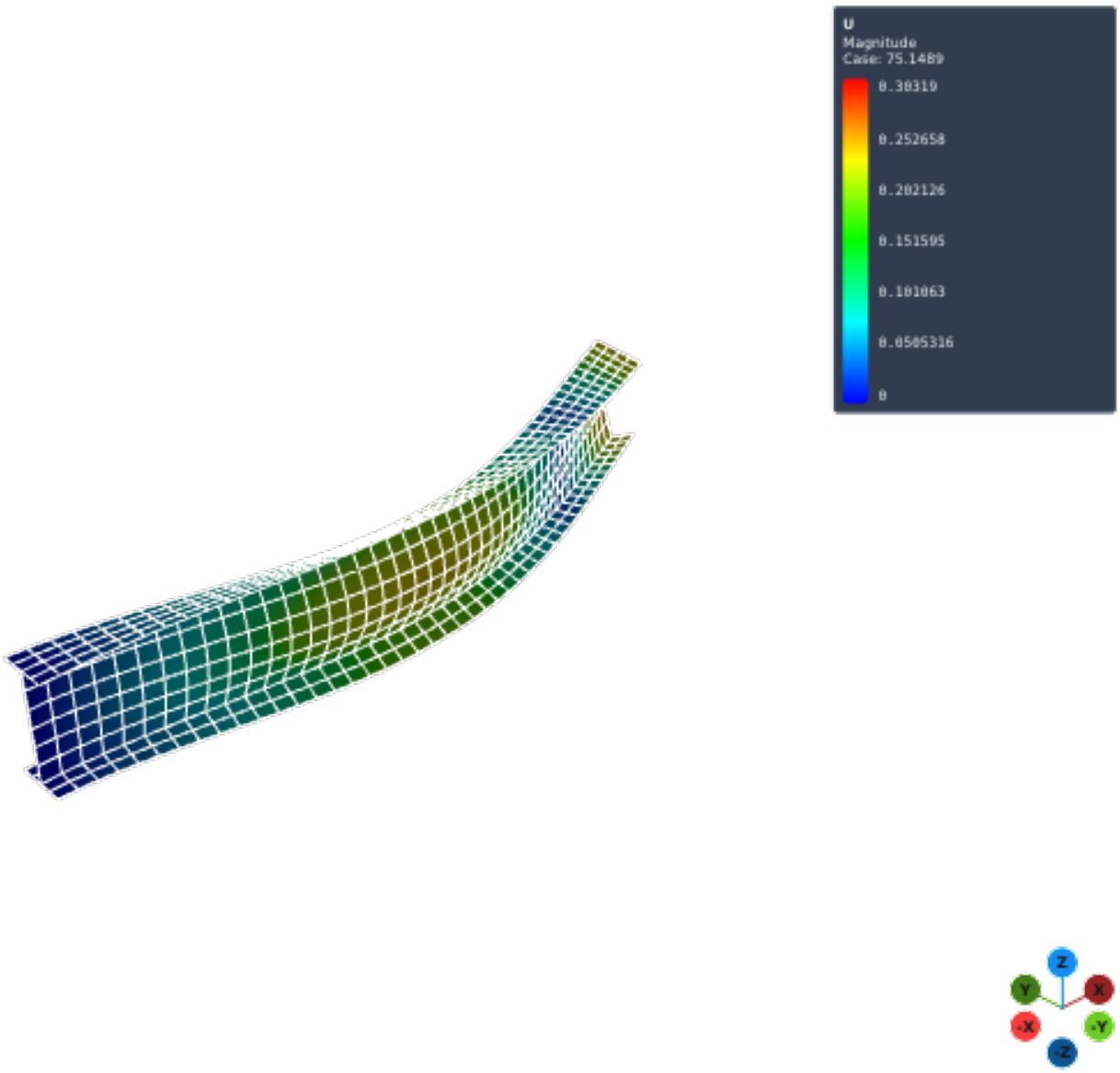

Figure 249: Abaqus — mode 4, 75.149 Hz

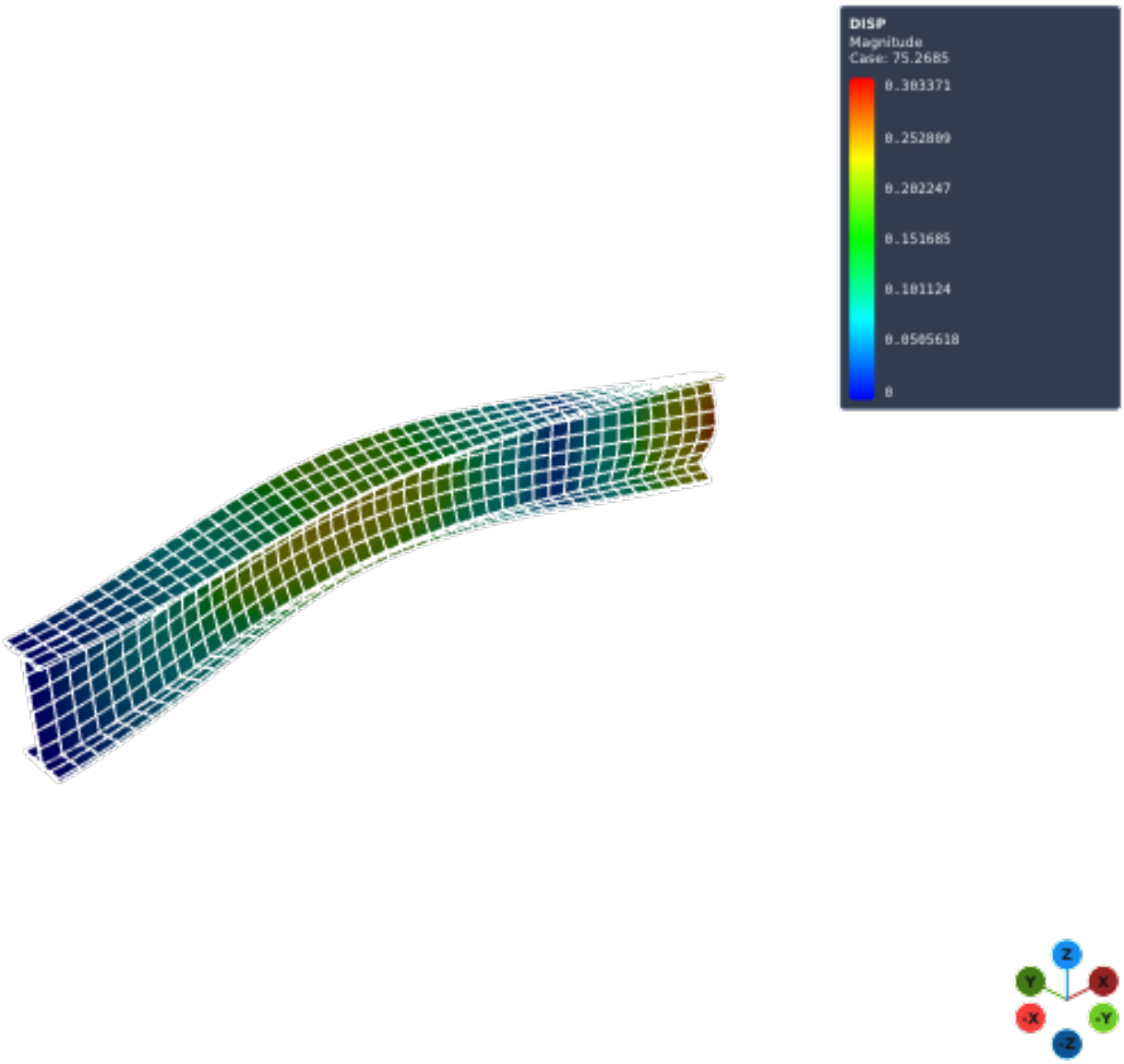

Figure 250: Calculix — mode 4, 75.269 Hz

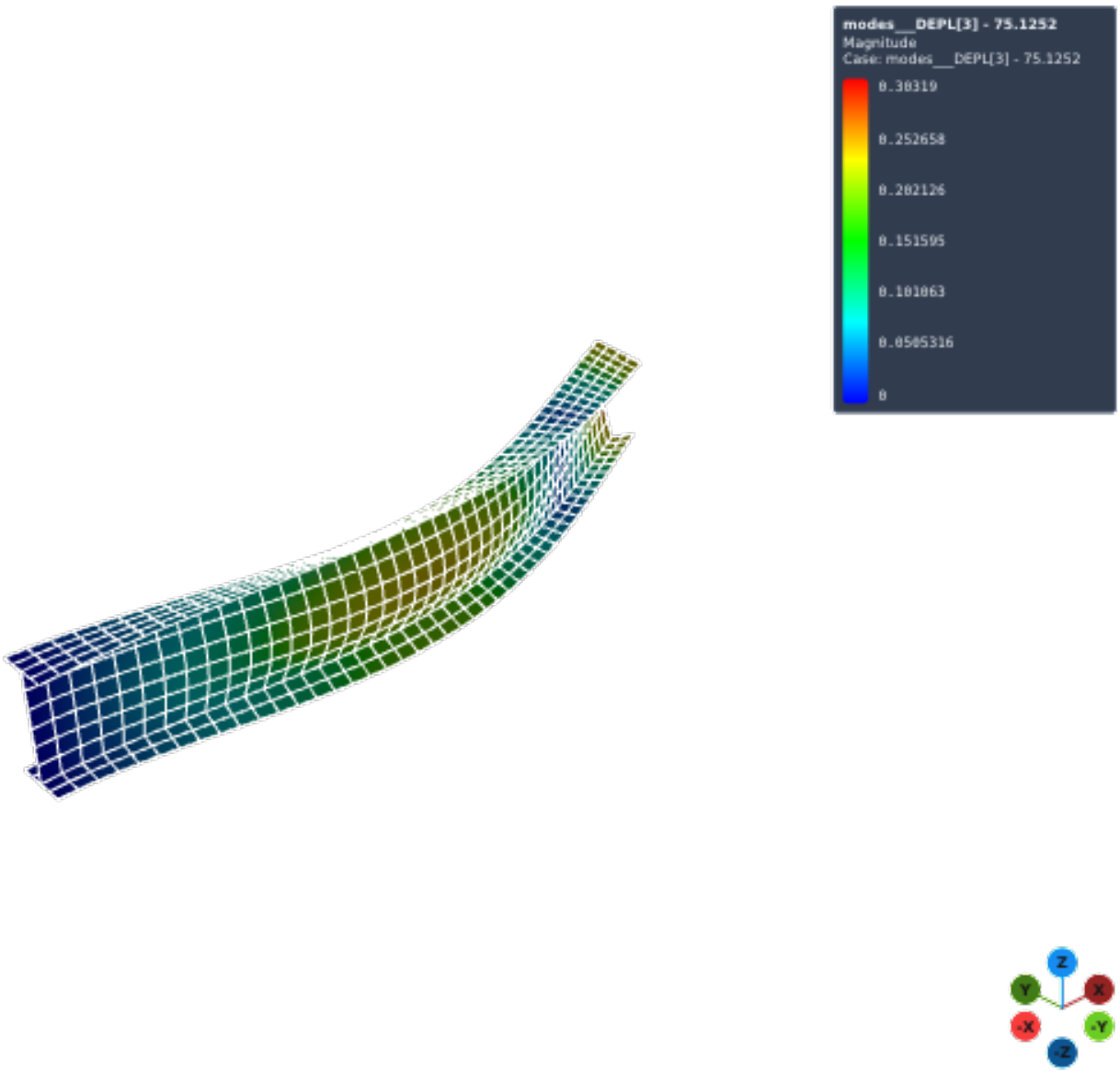

Figure 251: Code_Aster — mode 4, 75.125 Hz

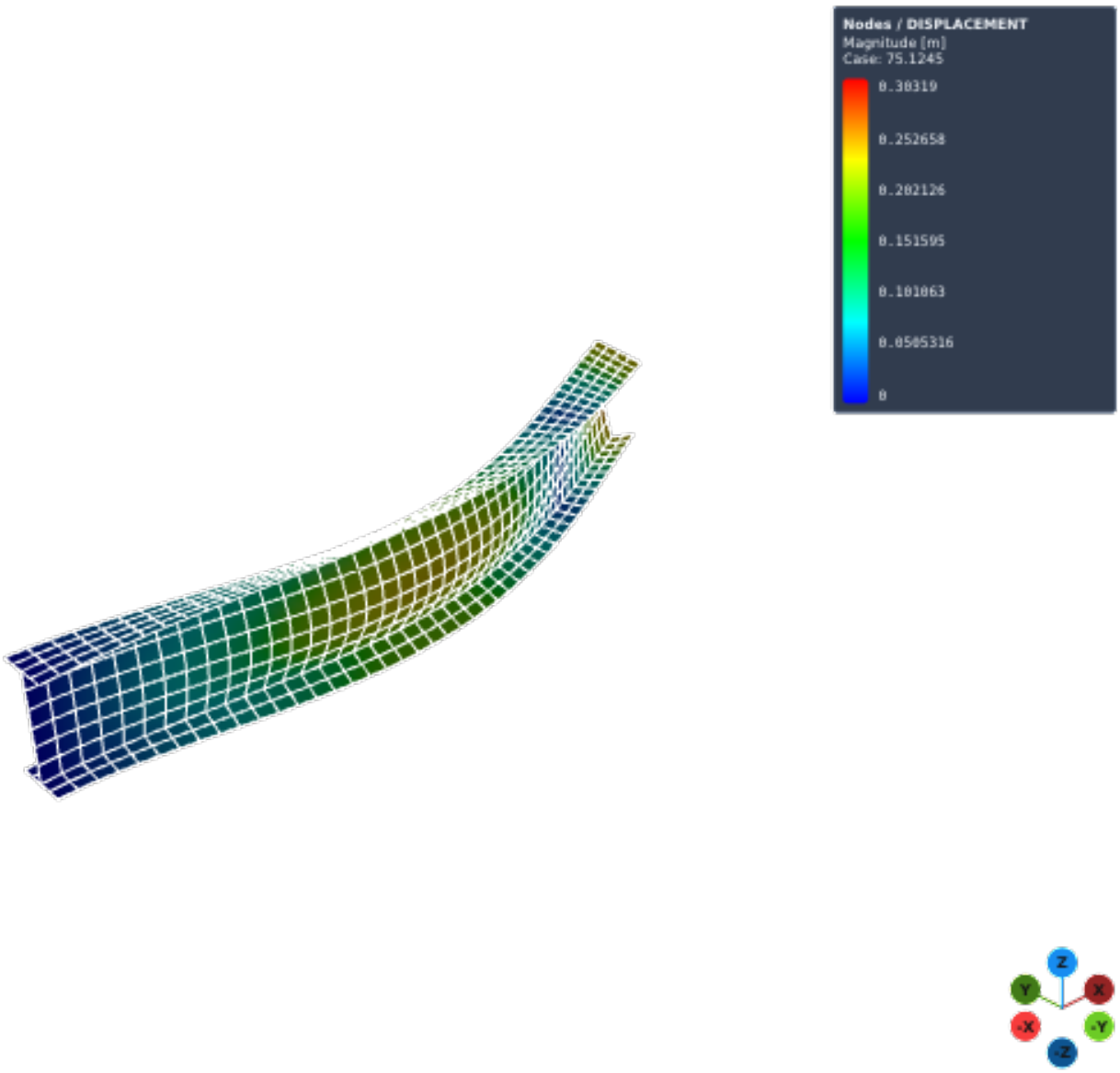

Figure 252: Sesam — mode 4, 75.125 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.149	-0.02	222949	1.40e-27	36.85	6.29e-24	-3.75e-14	-6.071	2.51e-12
Calculix	75.269	+0.14	223659	1.71e-24	36.88	1.39e-12	-1.31e-12	6.073	-1.18e-06
Code_Aster	75.125	-0.06	222808	2.82e-26	36.86	8.50e-25	1.68e-13	-6.071	9.22e-13
Sesam	75.125	-0.06	222804	3.41e-28	36.86	2.78e-24	-1.85e-14	-6.071	1.67e-12

Mode 5

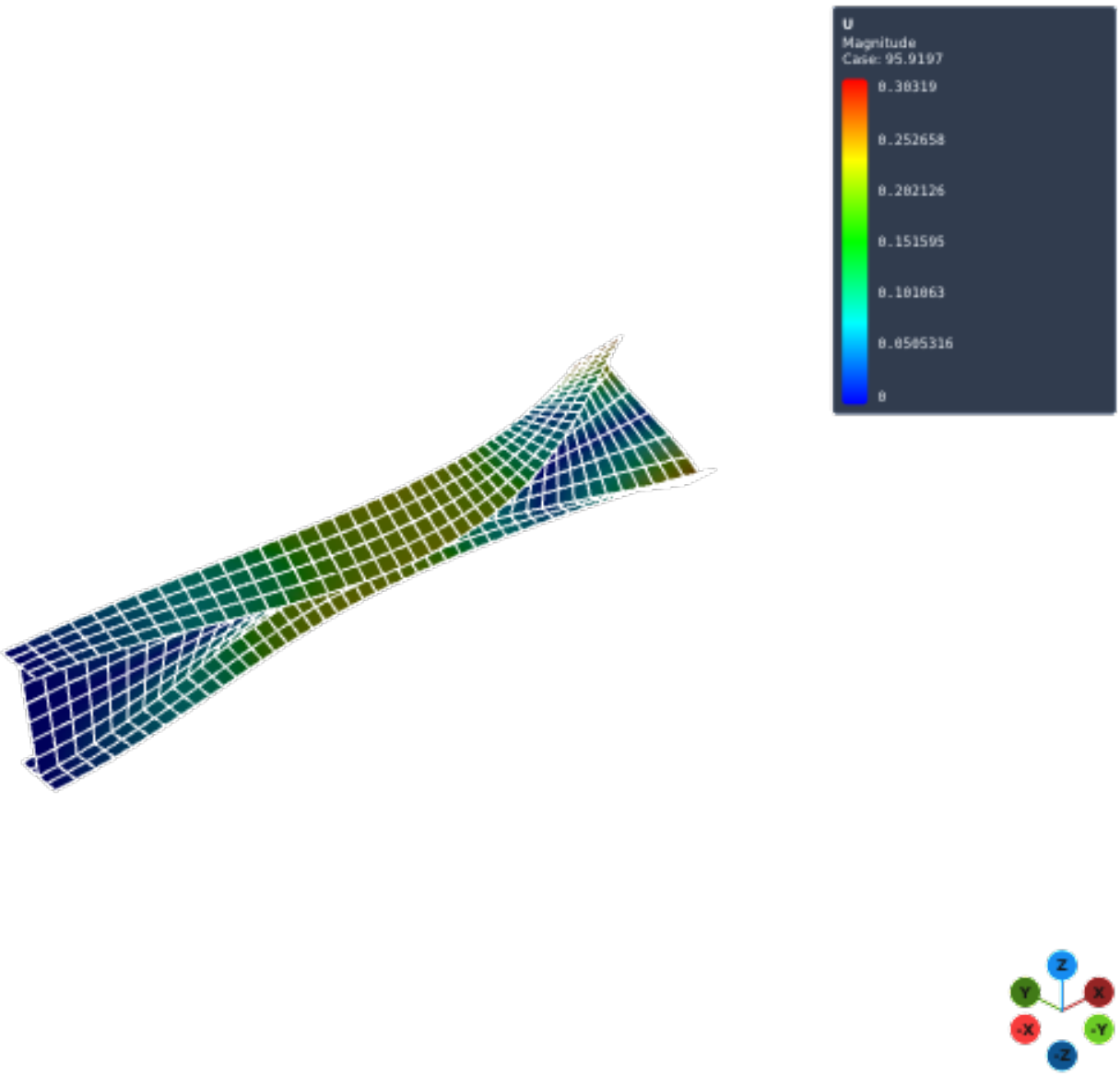

Figure 253: Abaqus — mode 5, 95.920 Hz

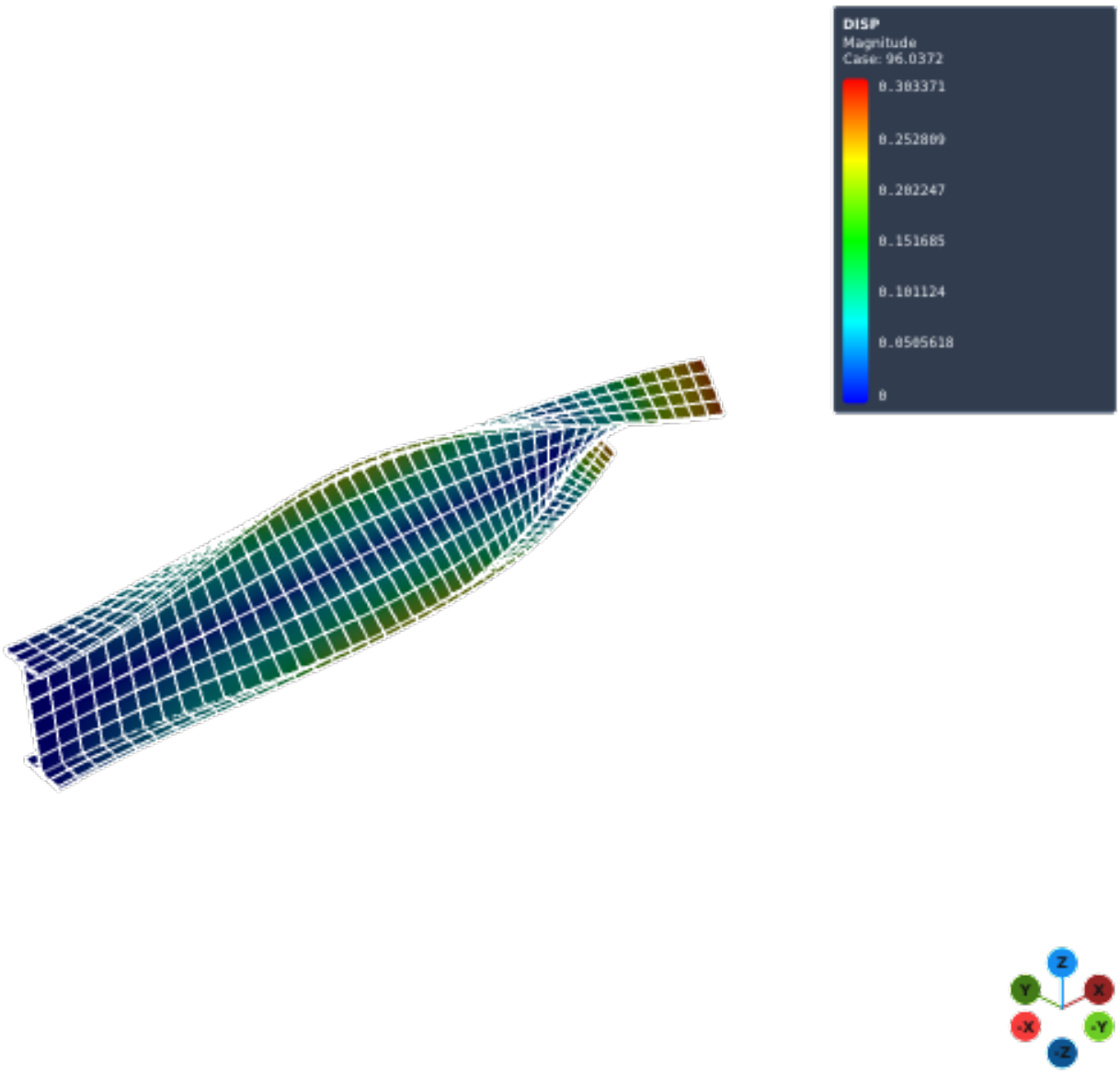

Figure 254: Calculix — mode 5, 96.037 Hz

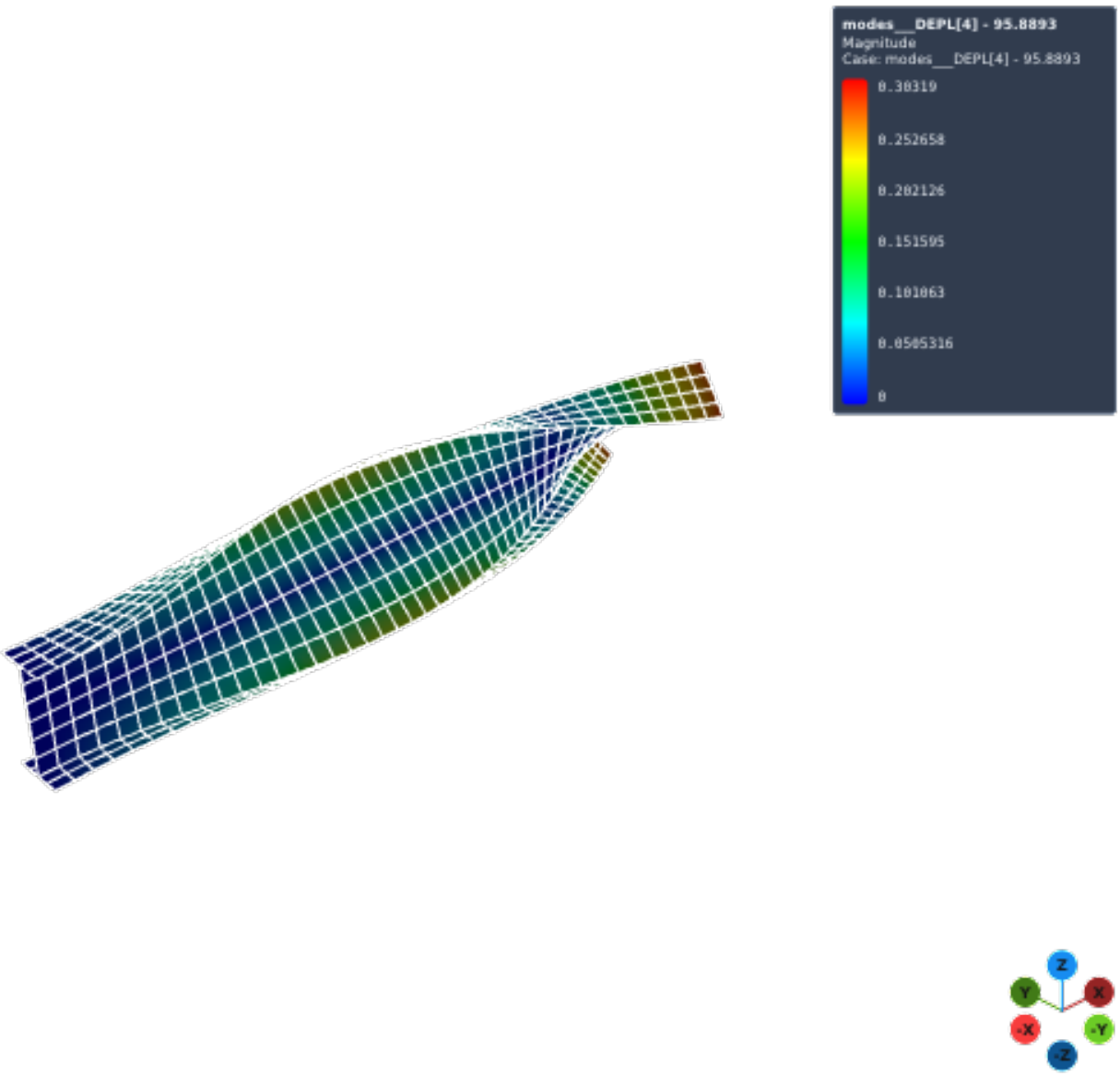

Figure 255: Code_Aster — mode 5, 95.889 Hz

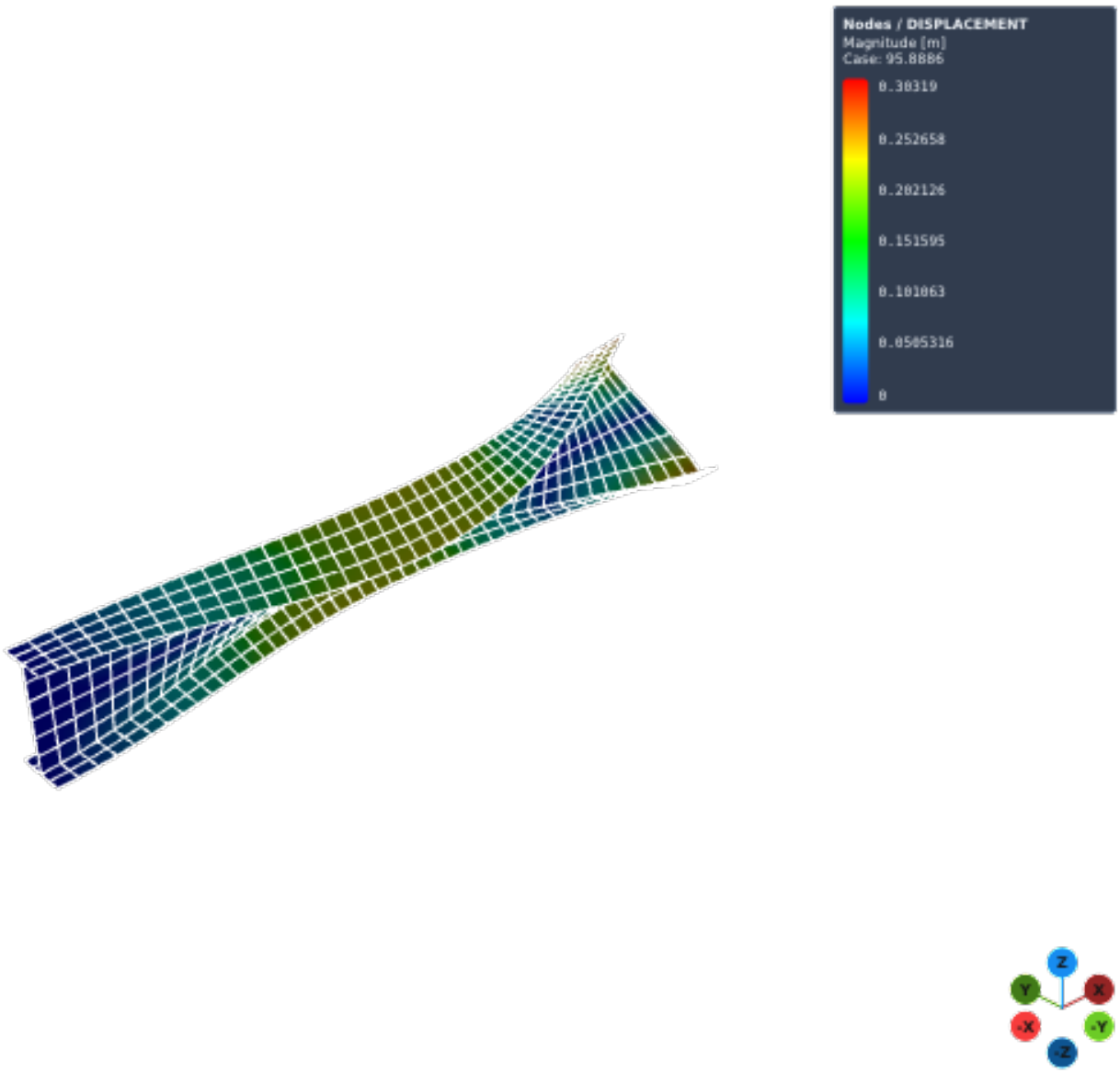

Figure 256: Sesam — mode 5, 95.889 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	95.920	-0.01	363224	7.50e-27	2.54e-21	8.04e-25	-8.66e-14	-5.04e-11	-8.97e-13
Calculix	96.037	+0.11	364115	8.58e-10	7.52e-19	4.09e-21	-2.93e-05	-8.67e-10	-6.40e-11
Code_Aster	95.889	-0.05	362994	7.82e-27	1.77e-21	5.02e-24	8.84e-14	-4.21e-11	-2.24e-12
Sesam	95.889	-0.05	362989	3.04e-26	1.19e-21	4.30e-24	1.74e-13	3.45e-11	-2.07e-12

Mode 6

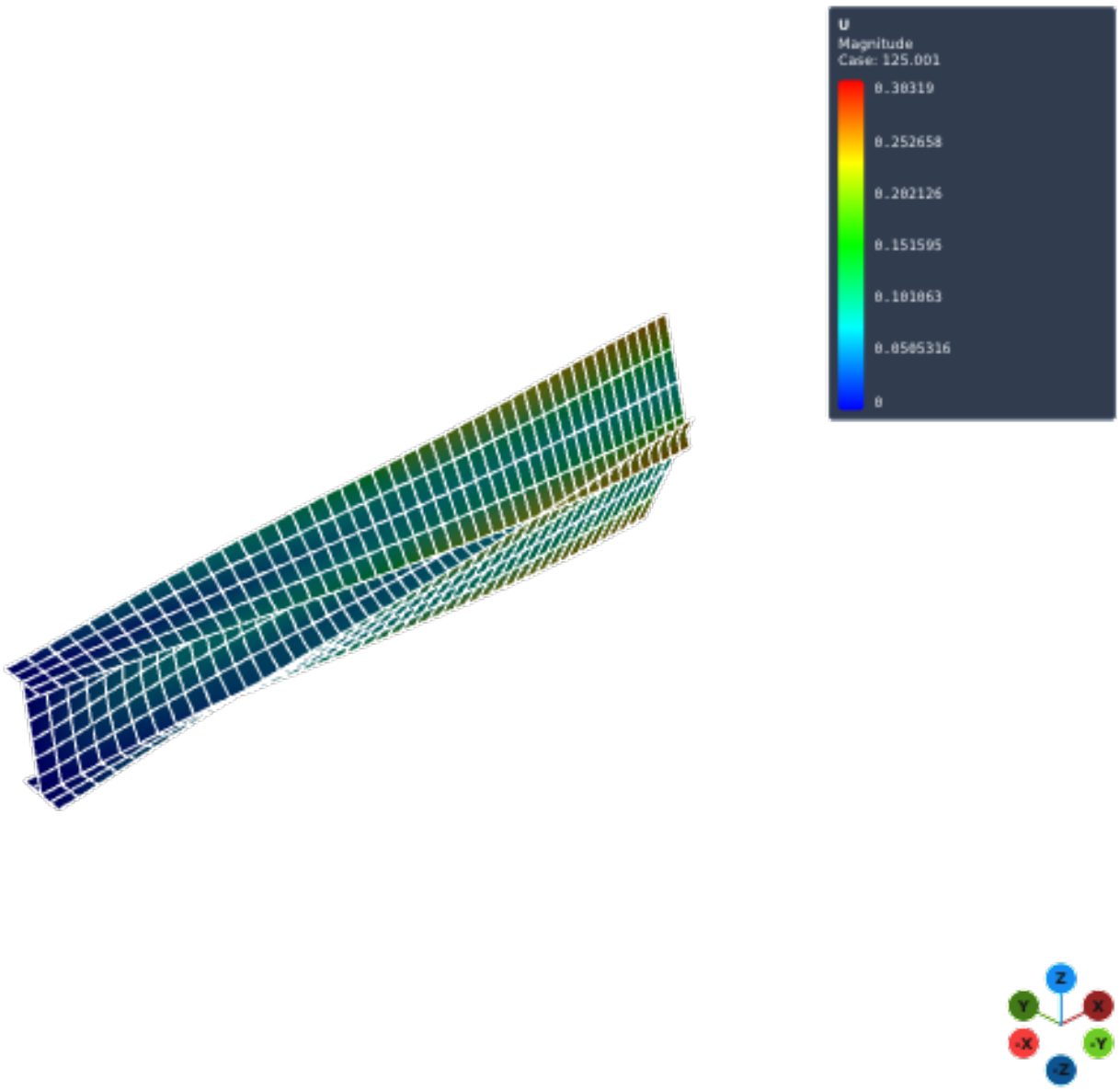

Figure 257: Abaqus — mode 6, 125.001 Hz

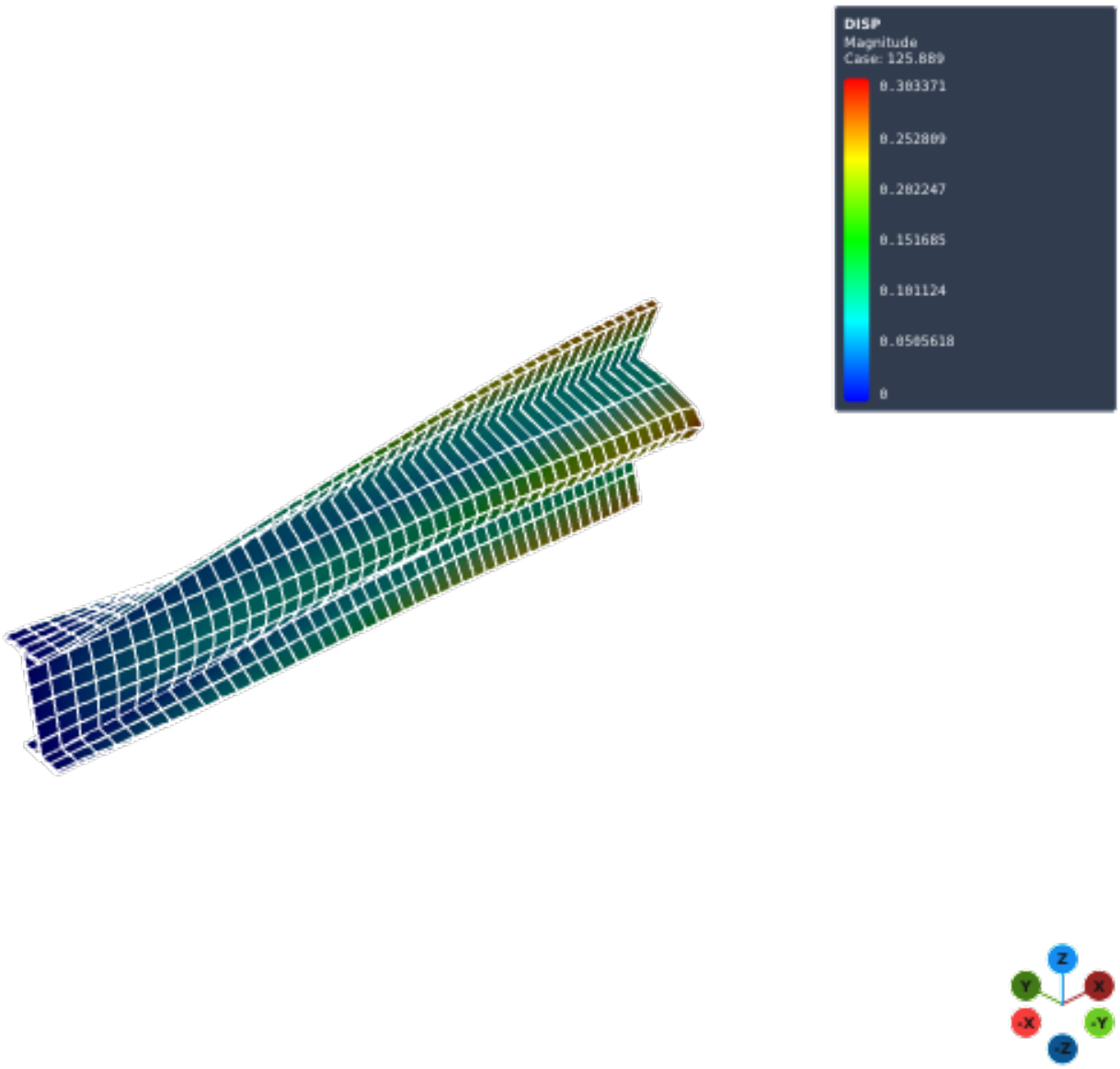

Figure 258: Calculix — mode 6, 125.889 Hz

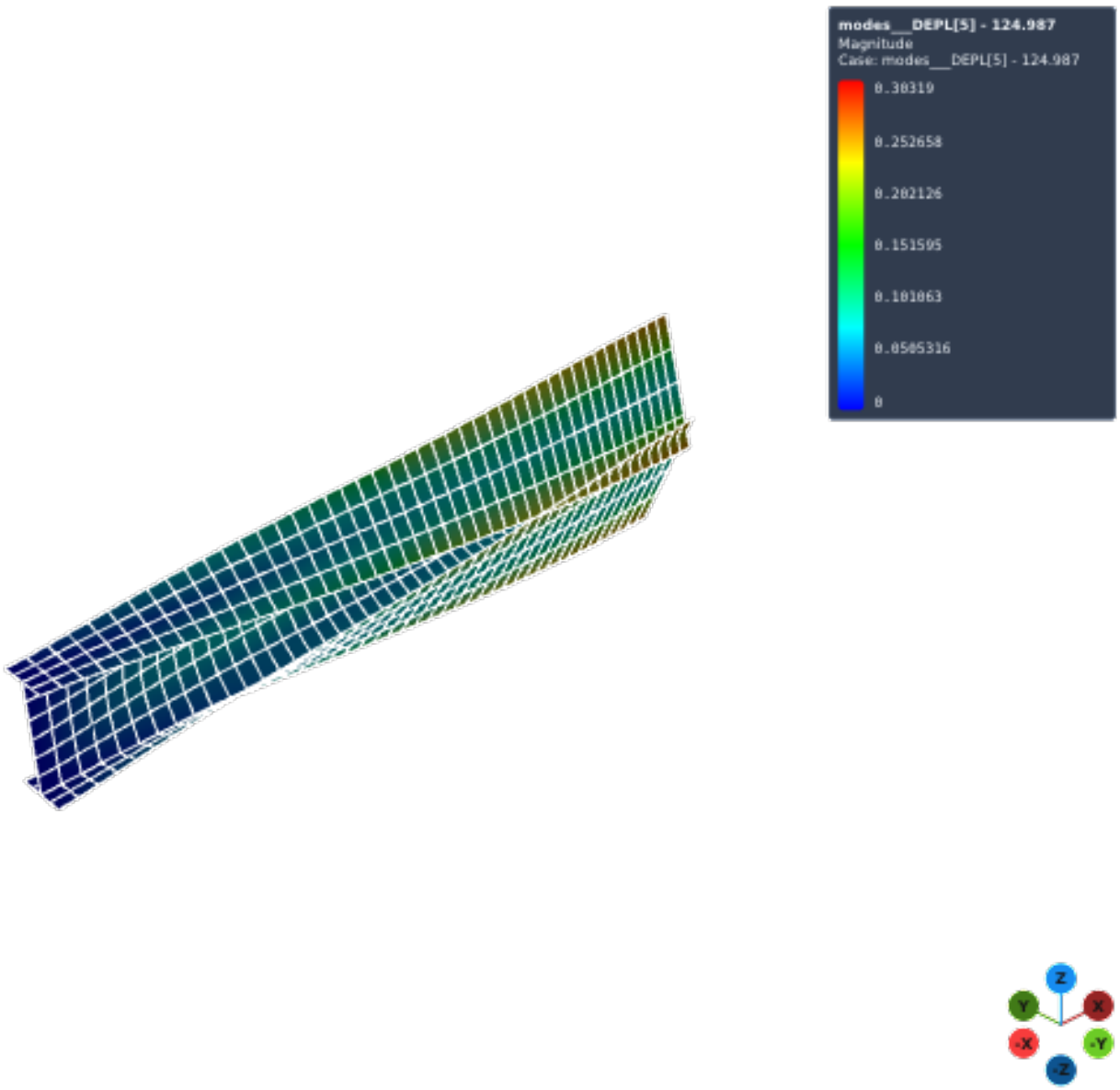

Figure 259: Code_Aster — mode 6, 124.987 Hz

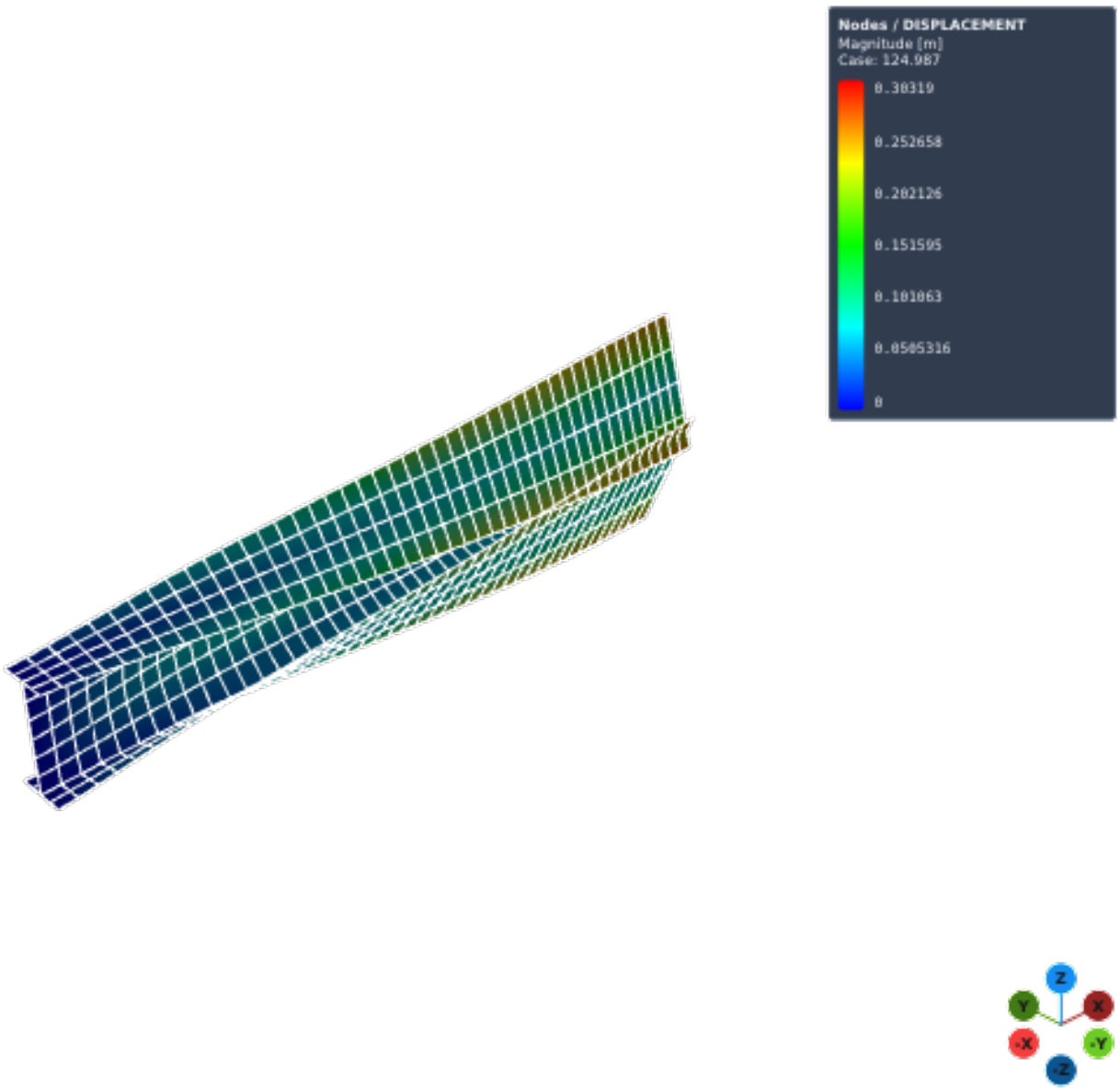

Figure 260: Sesam — mode 6, 124.987 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	125.000	-0.17	616856	5.44e-26	0.02	4.78e-23	-2.33e-13	-0.137	6.92e-12
Calculix	125.889	+0.54	625652	1.05e-24	0.02	1.64e-13	-1.03e-12	0.141	4.05e-07
Code_Aster	124.987	-0.18	616718	4.03e-27	0.02	3.10e-23	6.35e-14	-0.137	5.57e-12
Sesam	124.987	-0.18	616722	1.24e-26	0.02	1.35e-24	-1.11e-13	-0.137	1.16e-12

Mode 7

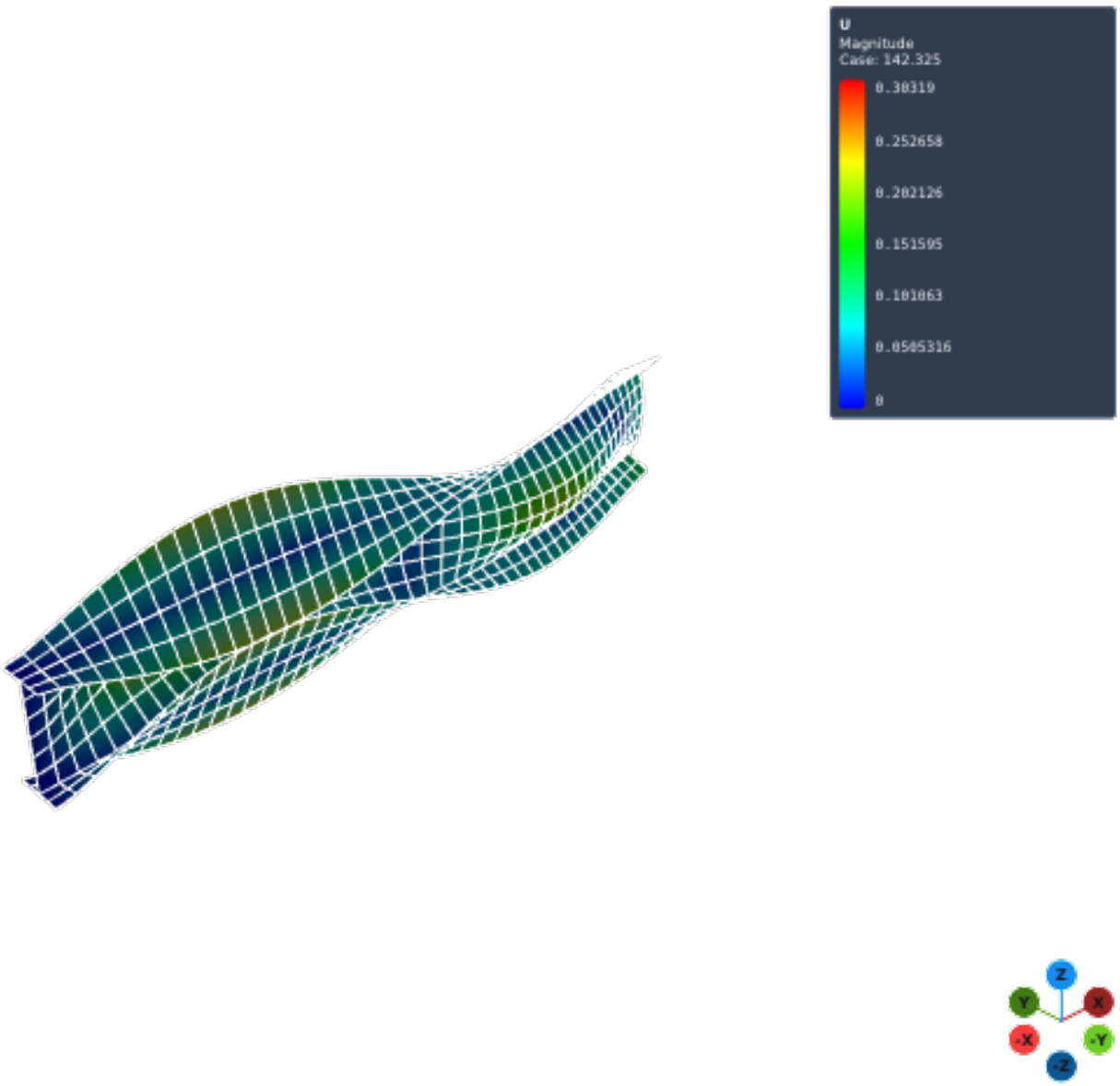

Figure 261: Abaqus — mode 7, 142.325 Hz

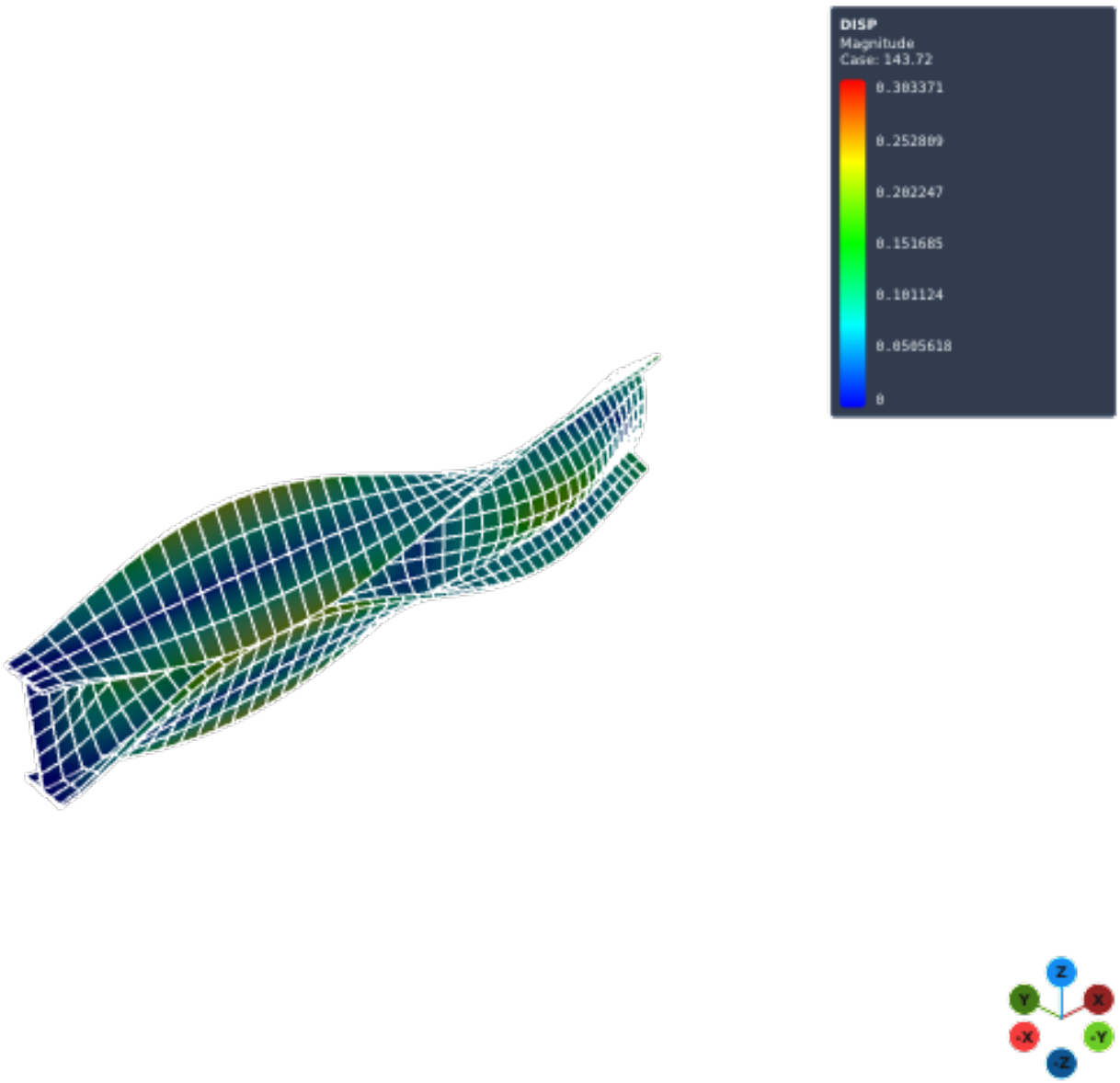

Figure 262: Calculix — mode 7, 143.720 Hz

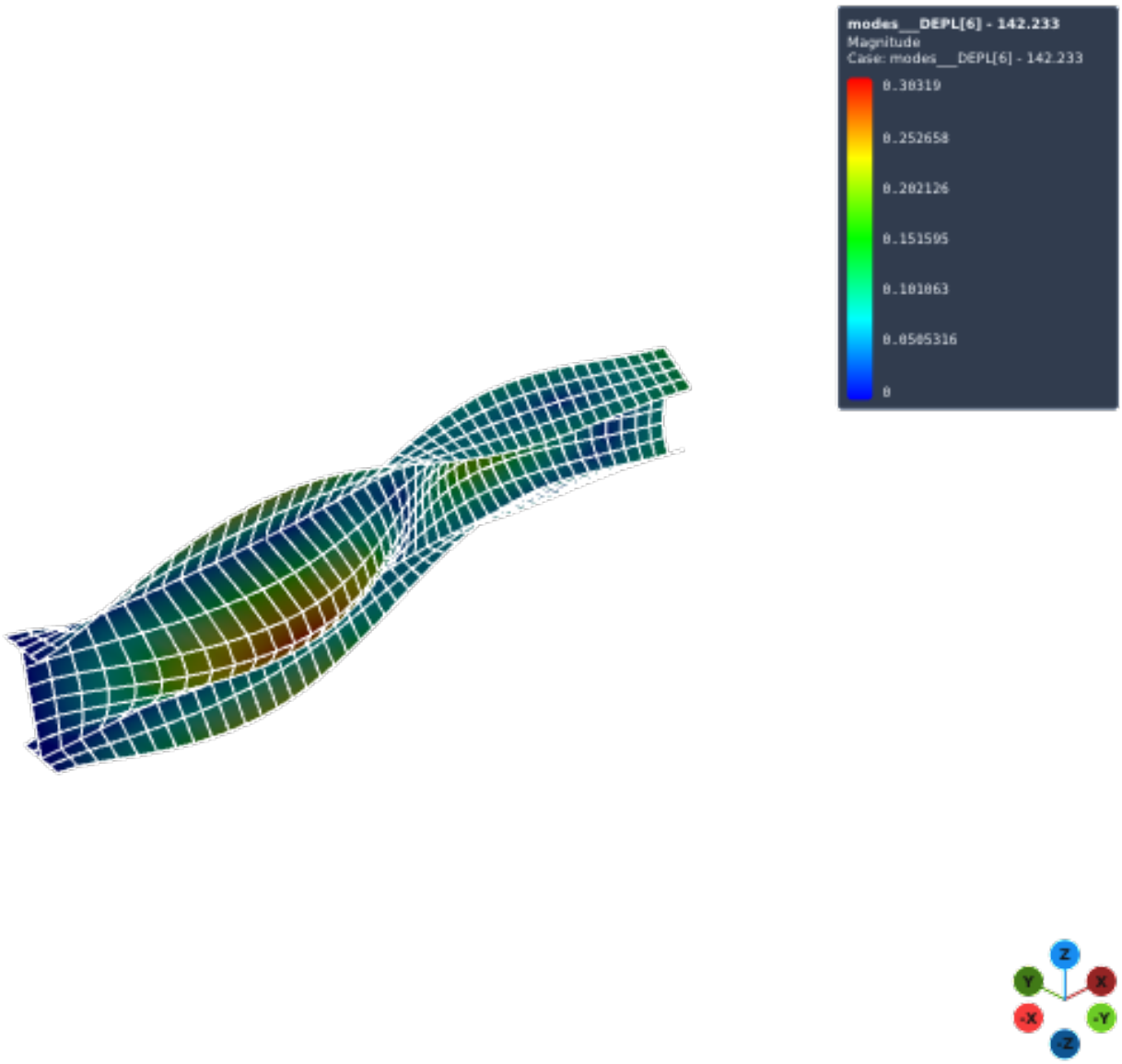

Figure 263: Code_Aster — mode 7, 142.233 Hz

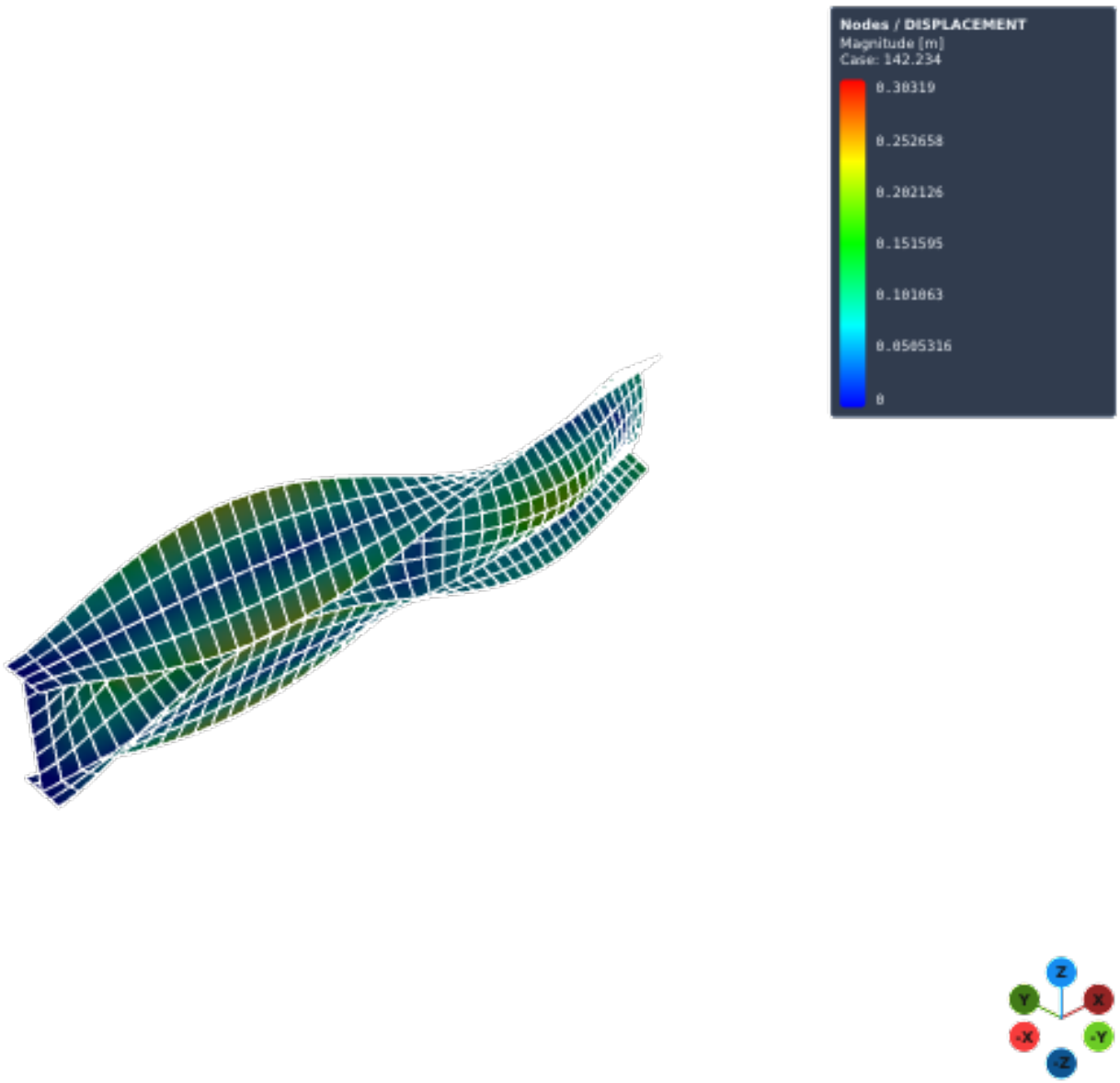

Figure 264: Sesam — mode 7, 142.234 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	142.320	-0.21	799690	2.49e-27	3.45	4.42e-24	-4.99e-14	1.858	-2.10e-12
Calculix	143.720	+0.77	815440	6.94e-26	3.38	8.05e-13	-2.63e-13	1.839	8.97e-07
Code_Aster	142.233	-0.28	798656	2.75e-27	3.45	2.74e-24	-5.24e-14	-1.859	1.66e-12
Sesam	142.234	-0.28	798665	3.85e-28	3.45	6.92e-26	1.96e-14	1.859	-2.63e-13

Mode 8

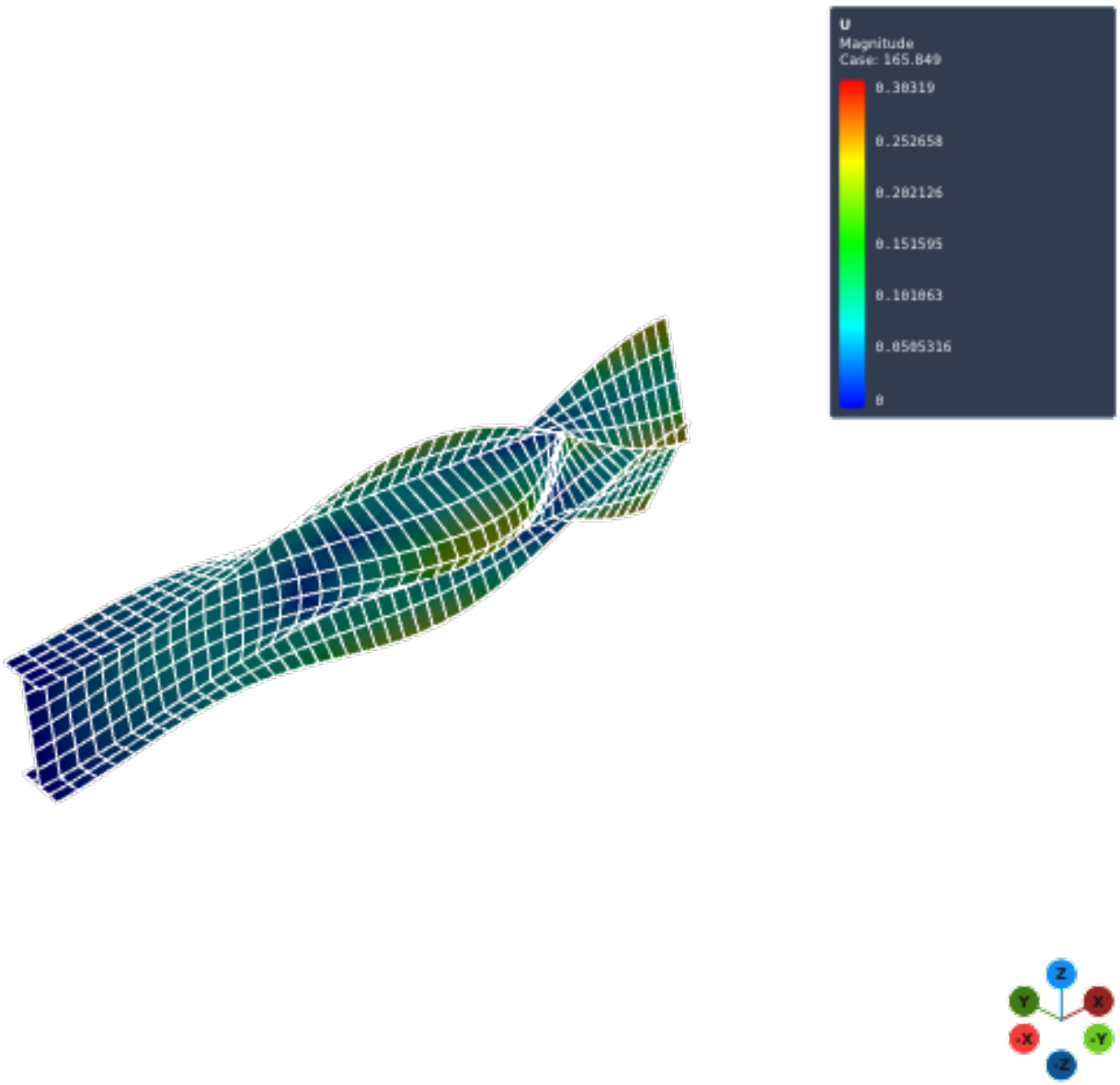

Figure 265: Abaqus — mode 8, 165.849 Hz

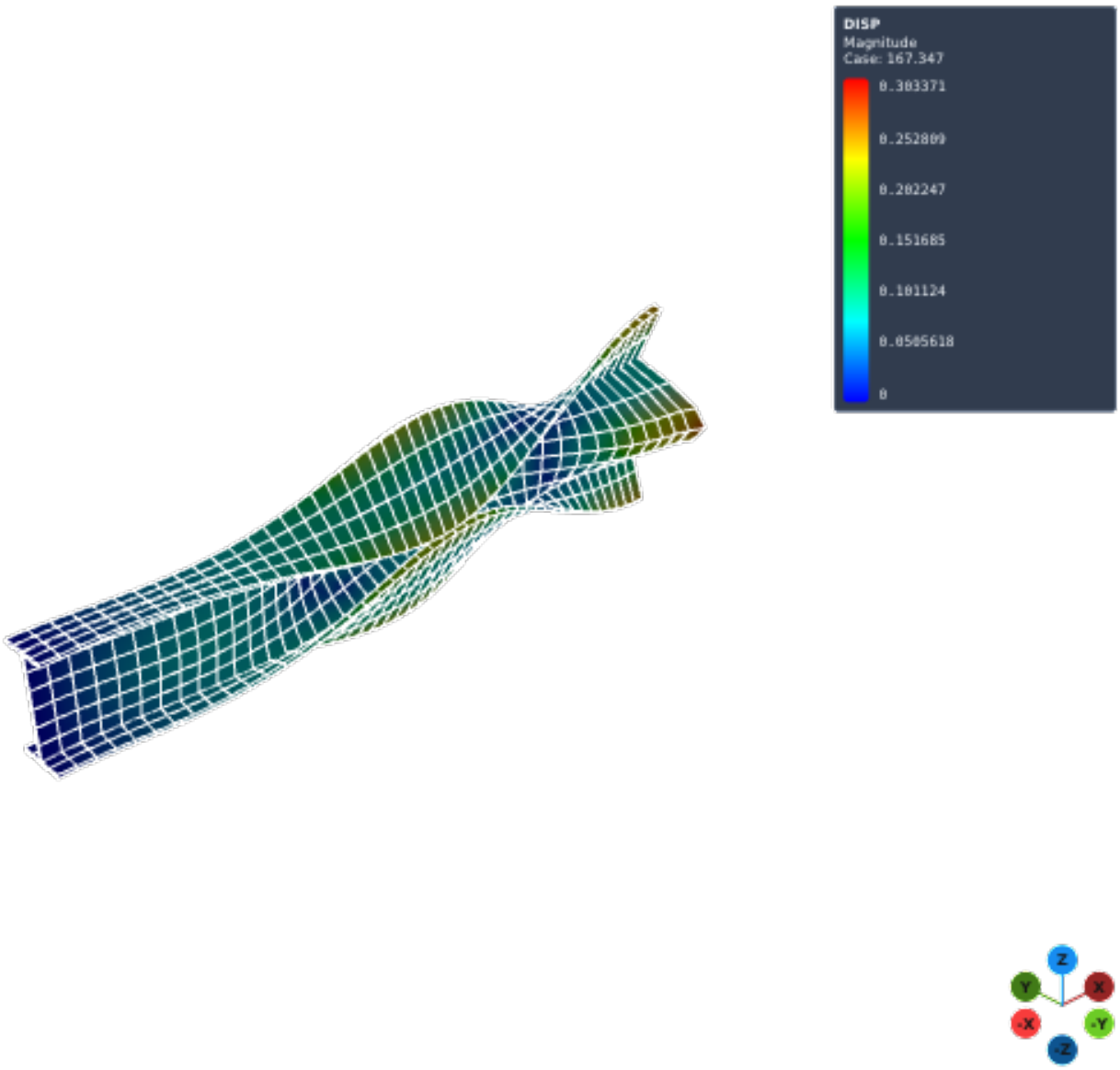

Figure 266: Calculix — mode 8, 167.347 Hz

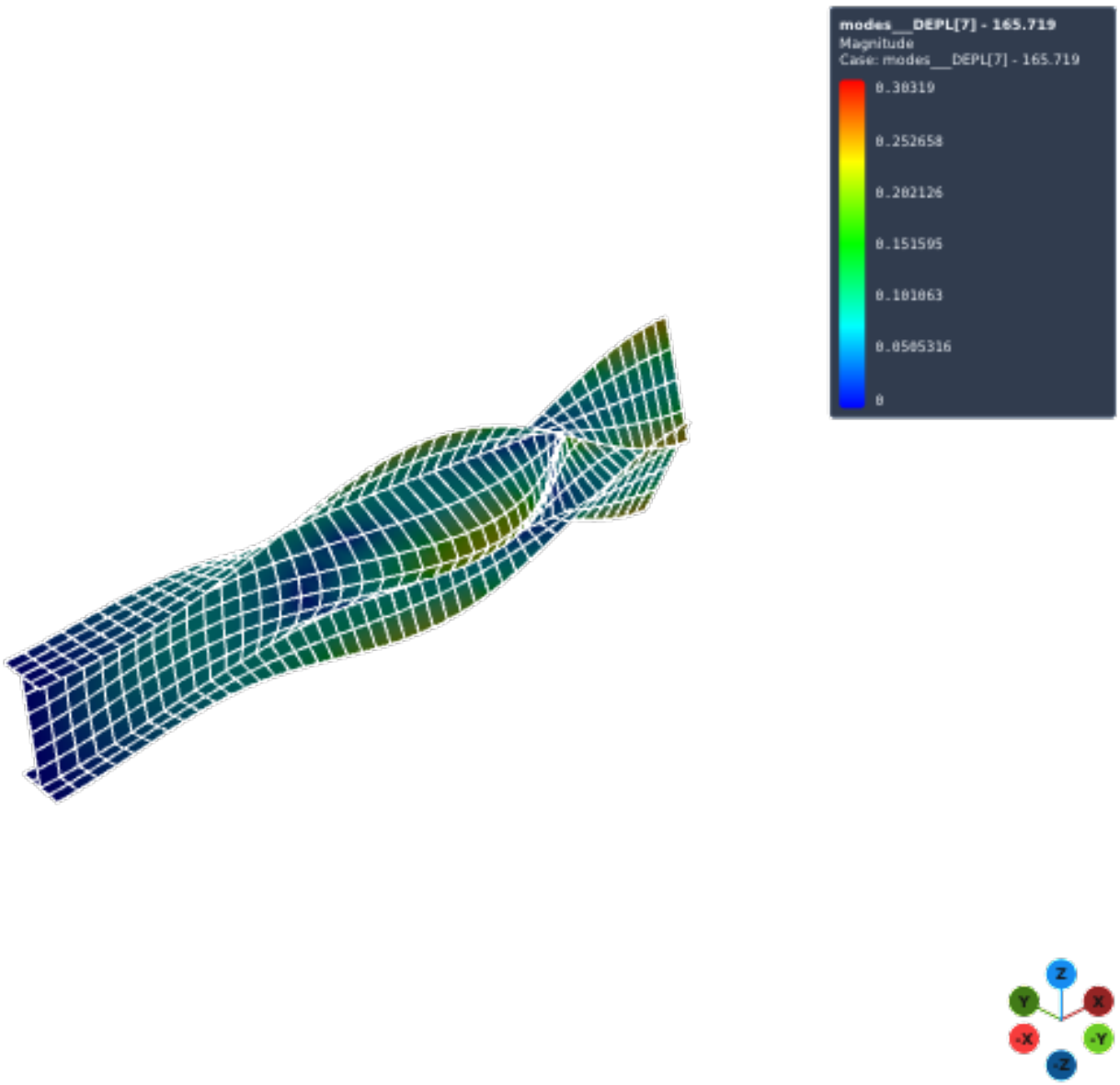

Figure 267: Code_Aster — mode 8, 165.719 Hz

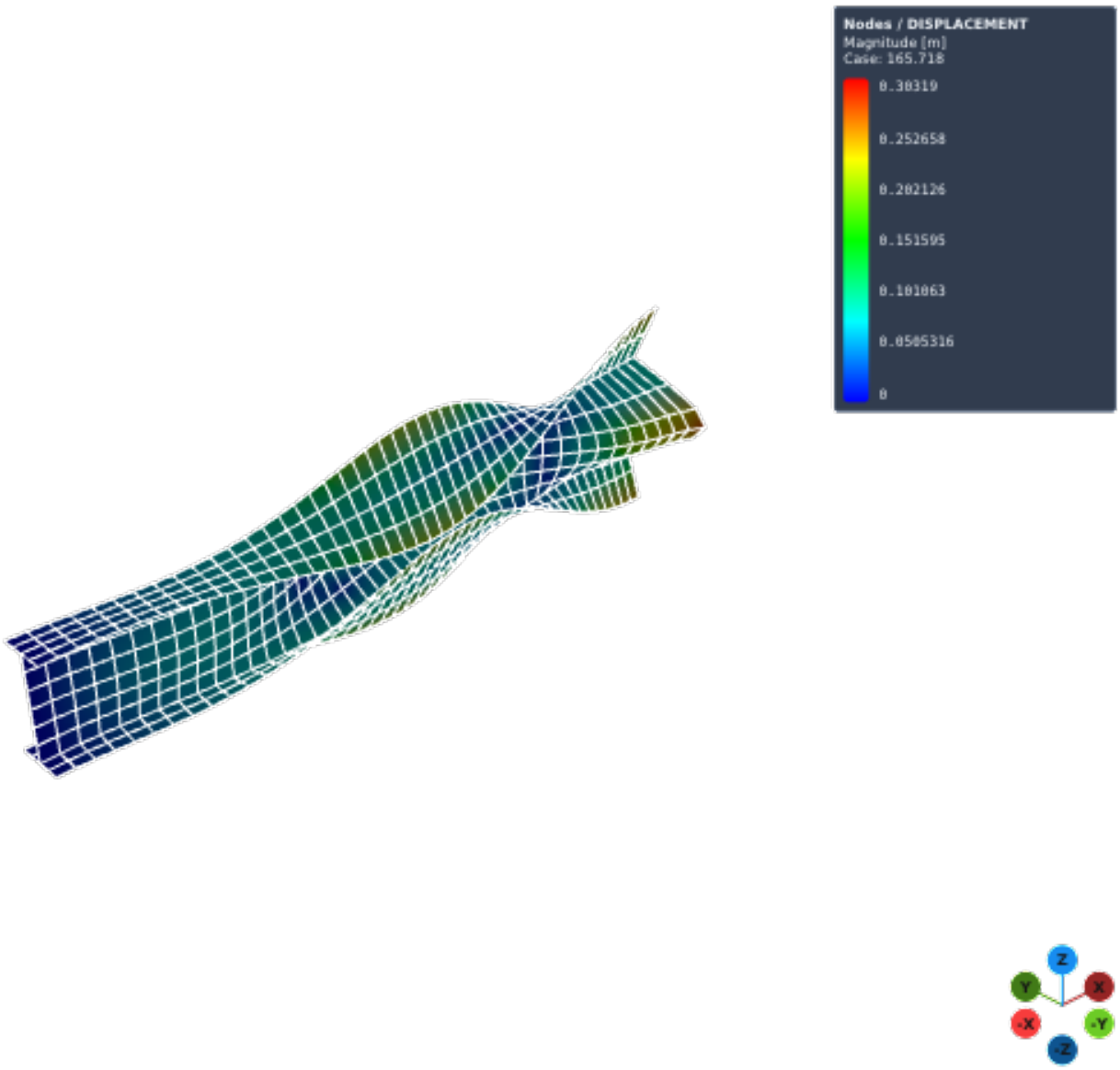

Figure 268: Sesam — mode 8, 165.718 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	165.850	-0.19	1.08589e+06	2.43e-27	5.53	1.06e-23	-4.93e-14	2.351	3.25e-12
Calculix	167.347	+0.72	1.10559e+06	2.91e-24	5.58	2.92e-13	-1.71e-12	-2.362	5.40e-07
Code_Aster	165.719	-0.26	1.08419e+06	2.11e-26	5.52	3.83e-24	-1.45e-13	2.350	1.96e-12
Sesam	165.718	-0.26	1.08418e+06	3.12e-28	5.52	6.16e-25	-1.77e-14	-2.350	-7.85e-13

Mode 9

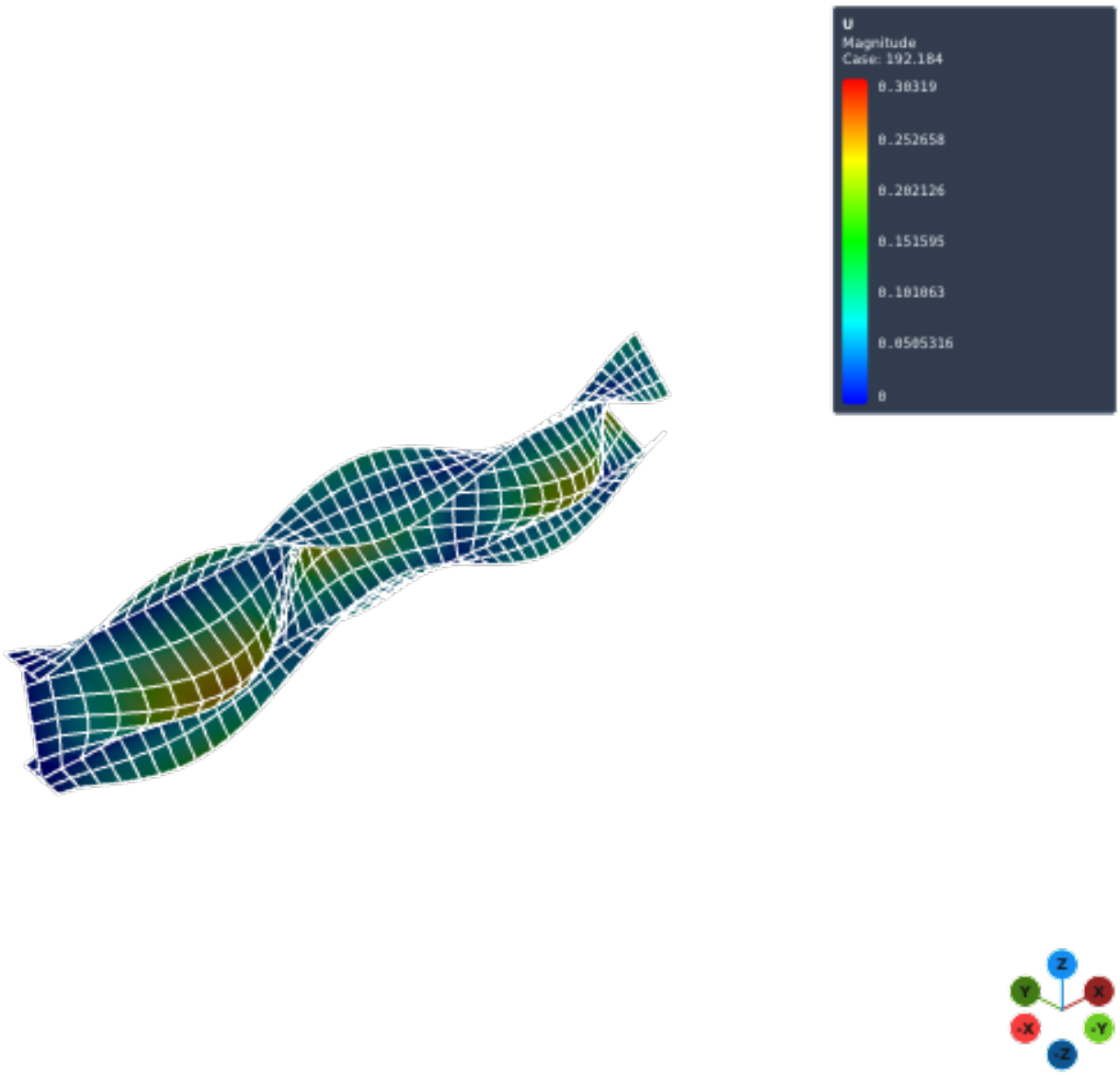

Figure 269: Abaqus — mode 9, 192.184 Hz

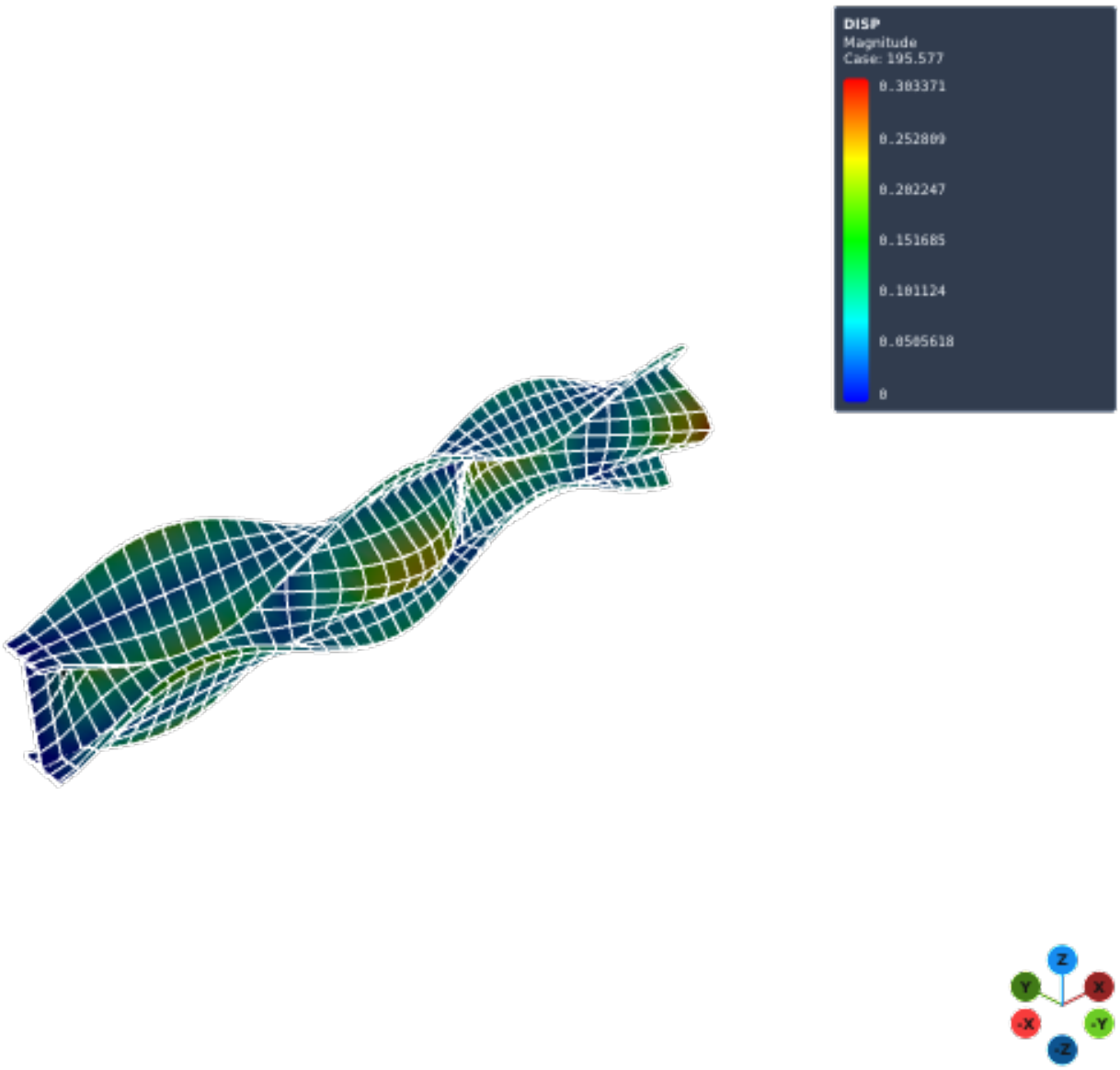

Figure 270: Calculix — mode 9, 195.577 Hz

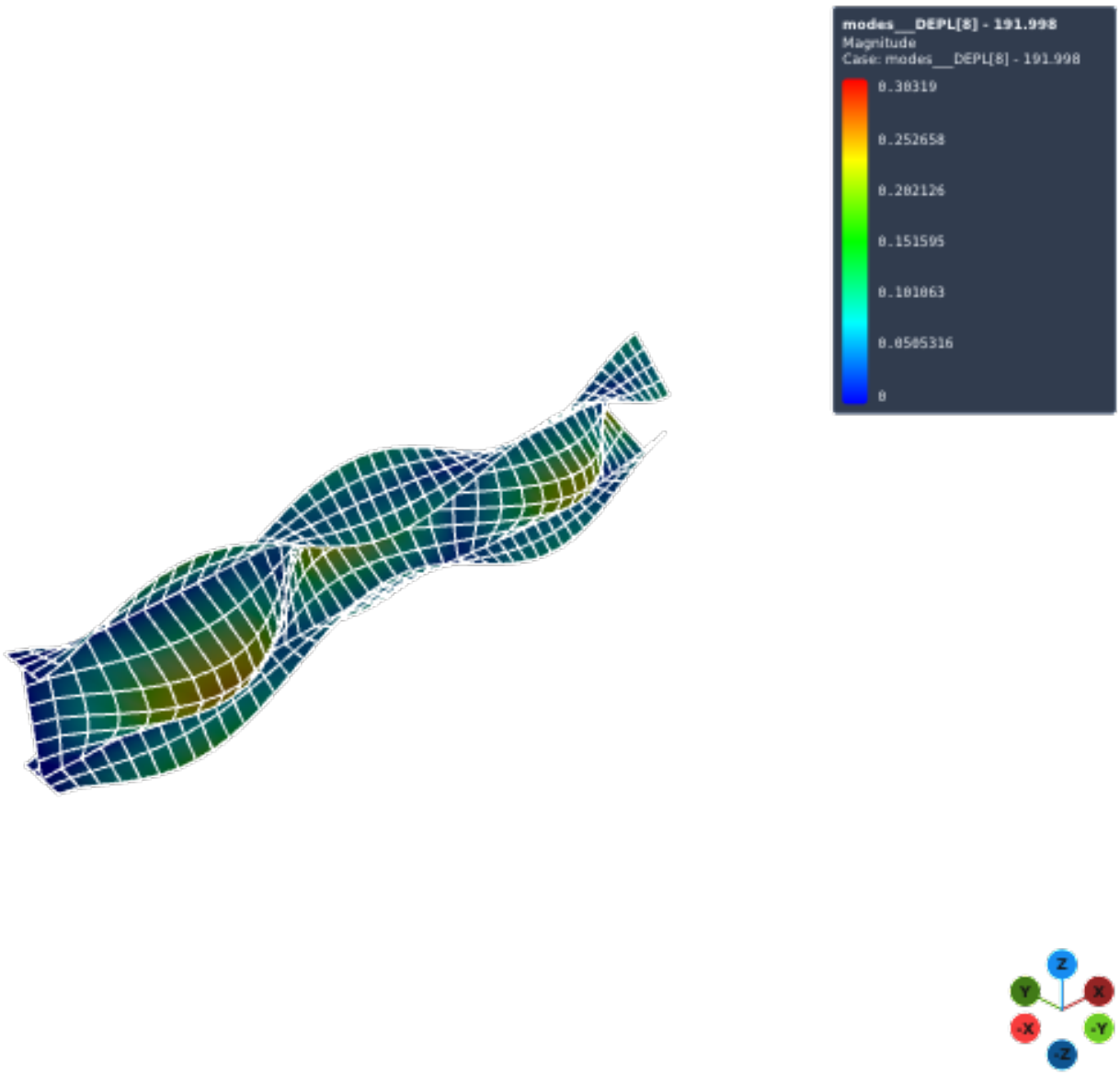

Figure 271: Code_Aster — mode 9, 191.998 Hz

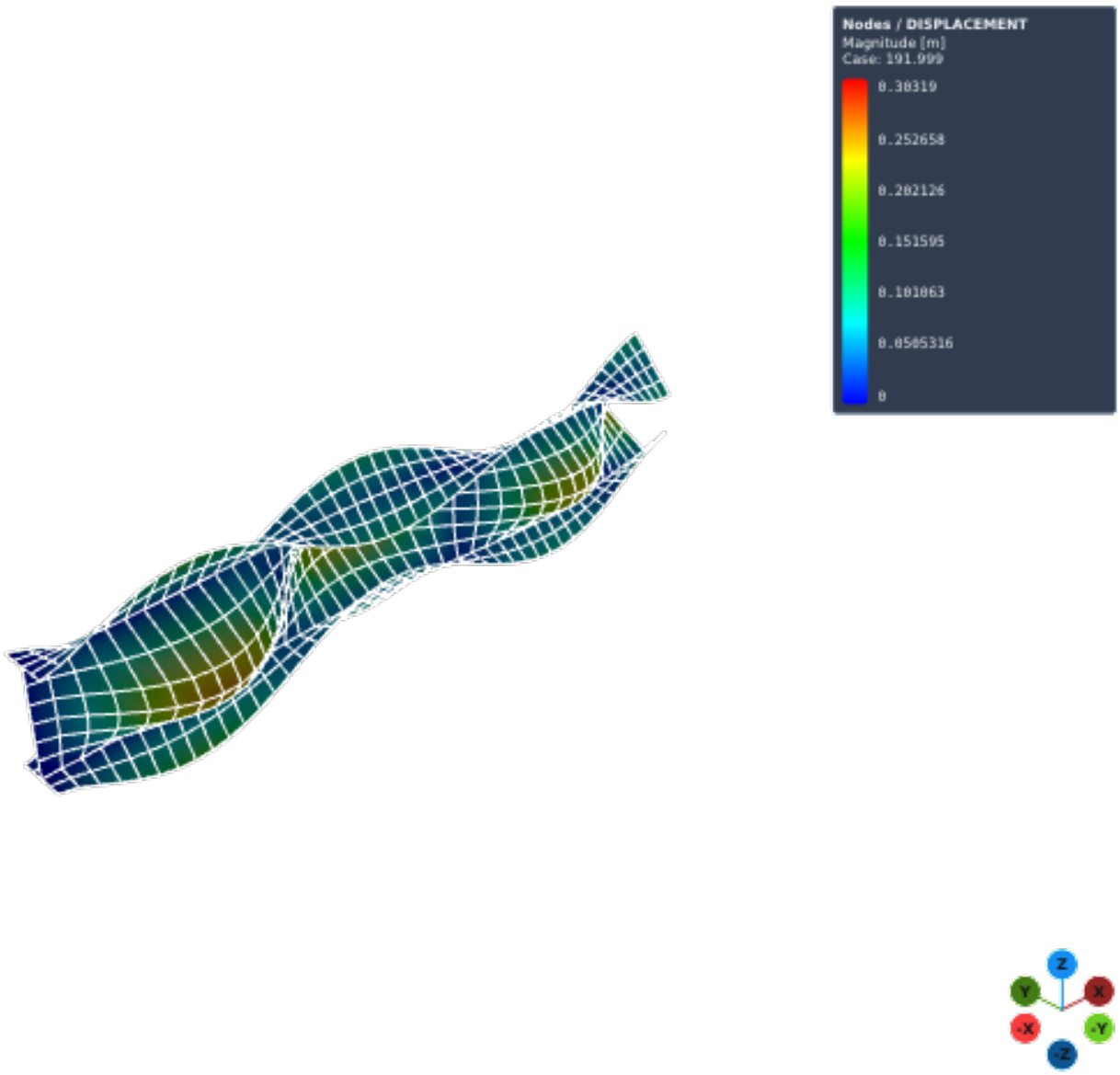

Figure 272: Sesam — mode 9, 191.999 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	192.180	-0.39	1.45812e+06	1.15e-27	0.65	1.20e-25	-3.40e-14	-0.808	3.47e-13
Calculix	195.577	+1.37	1.51006e+06	3.54e-25	0.61	9.24e-14	5.95e-13	0.781	-3.04e-07
Code_Aster	191.998	-0.49	1.45531e+06	4.91e-28	0.65	4.66e-25	-2.22e-14	-0.809	-6.82e-13
Sesam	191.999	-0.49	1.45531e+06	2.21e-28	0.65	1.82e-28	1.49e-14	-0.809	1.35e-14

Mode 10

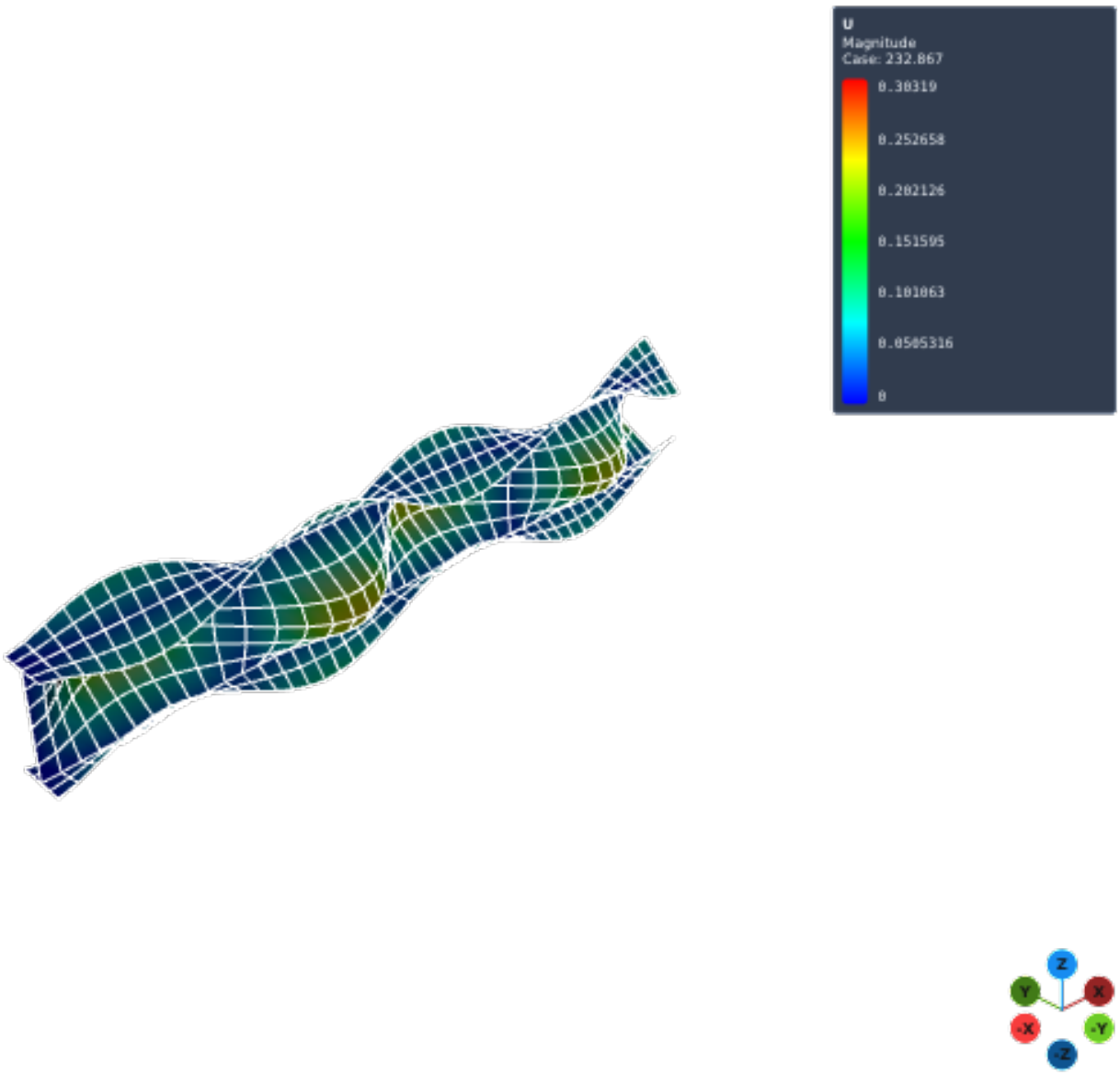

Figure 273: Abaqus — mode 10, 232.867 Hz

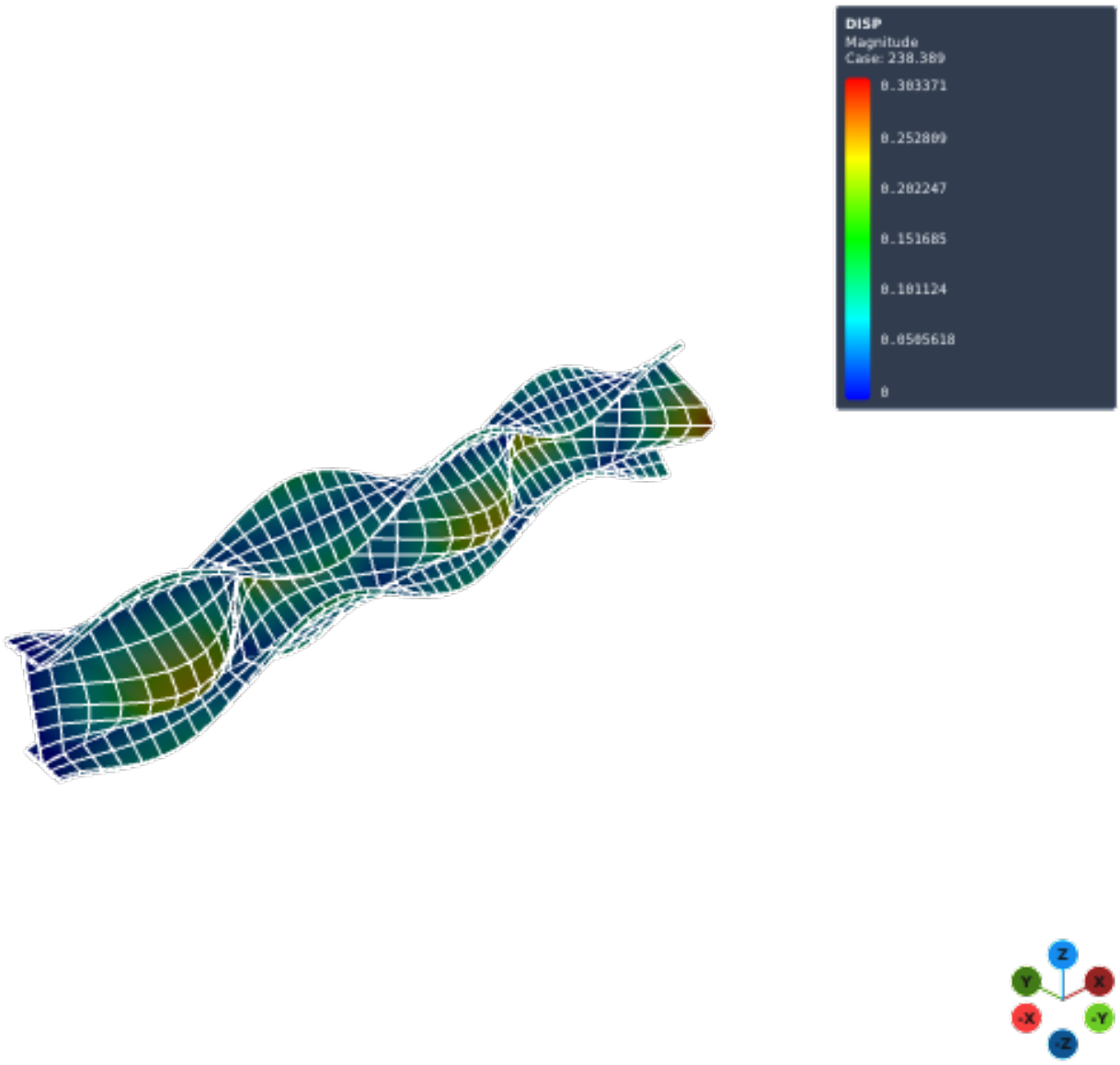

Figure 274: Calculix — mode 10, 238.389 Hz

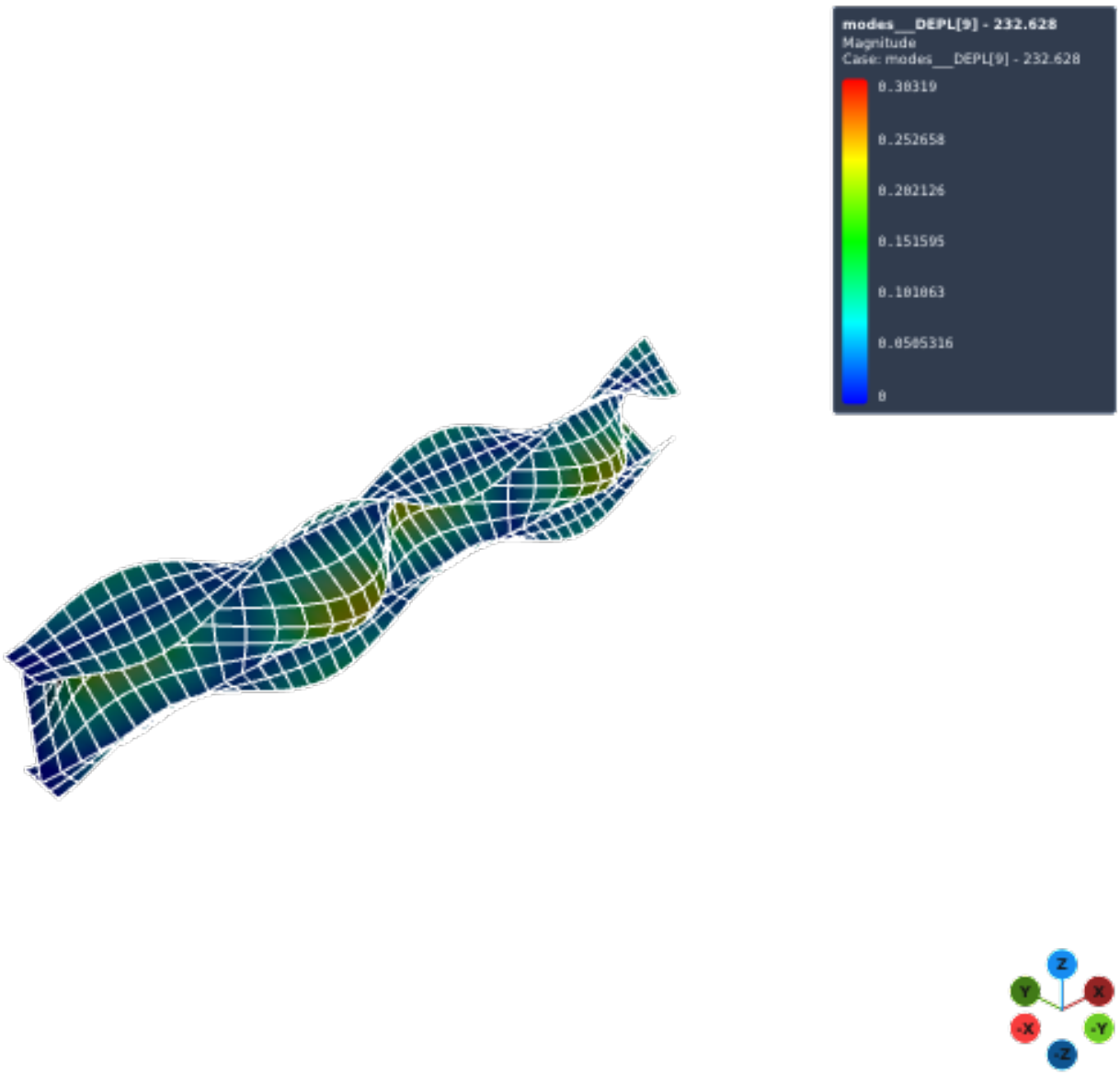

Figure 275: Code_Aster — mode 10, 232.628 Hz

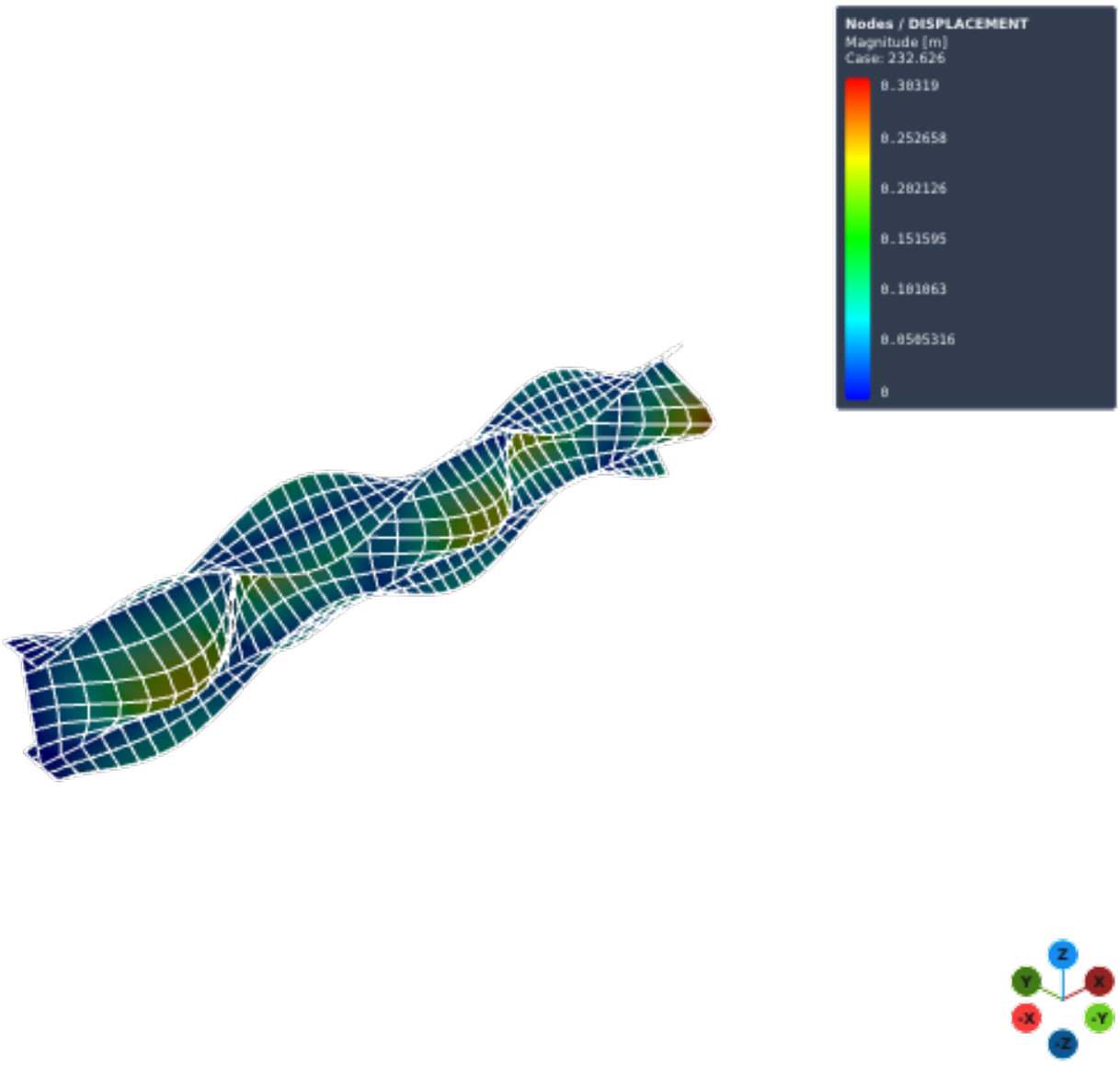

Figure 276: Sesam — mode 10, 232.626 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	232.870	-0.54	2.14079e+06	4.88e-26	1.37	1.33e-23	-2.21e-13	1.170	-3.65e-12
Calculix	238.389	+1.82	2.24354e+06	7.93e-24	1.44	5.13e-14	-2.82e-12	-1.201	-2.27e-07
Code_Aster	232.628	-0.64	2.13641e+06	3.71e-27	1.37	1.27e-24	-6.09e-14	1.170	1.13e-12
Sesam	232.626	-0.64	2.13637e+06	1.17e-26	1.37	1.70e-25	-1.08e-13	-1.171	-4.12e-13

Mode 11

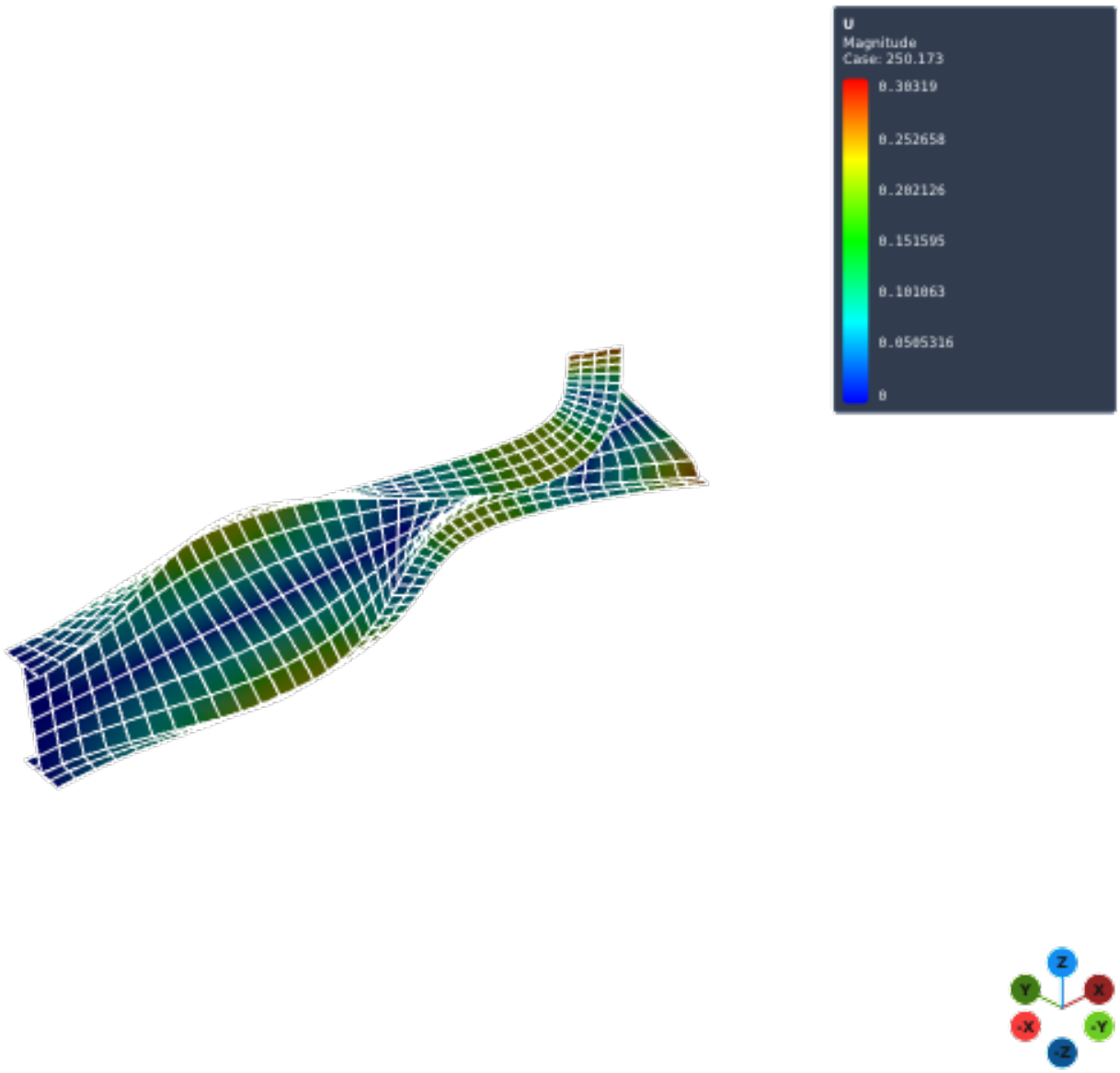

Figure 277: Abaqus — mode 11, 250.173 Hz

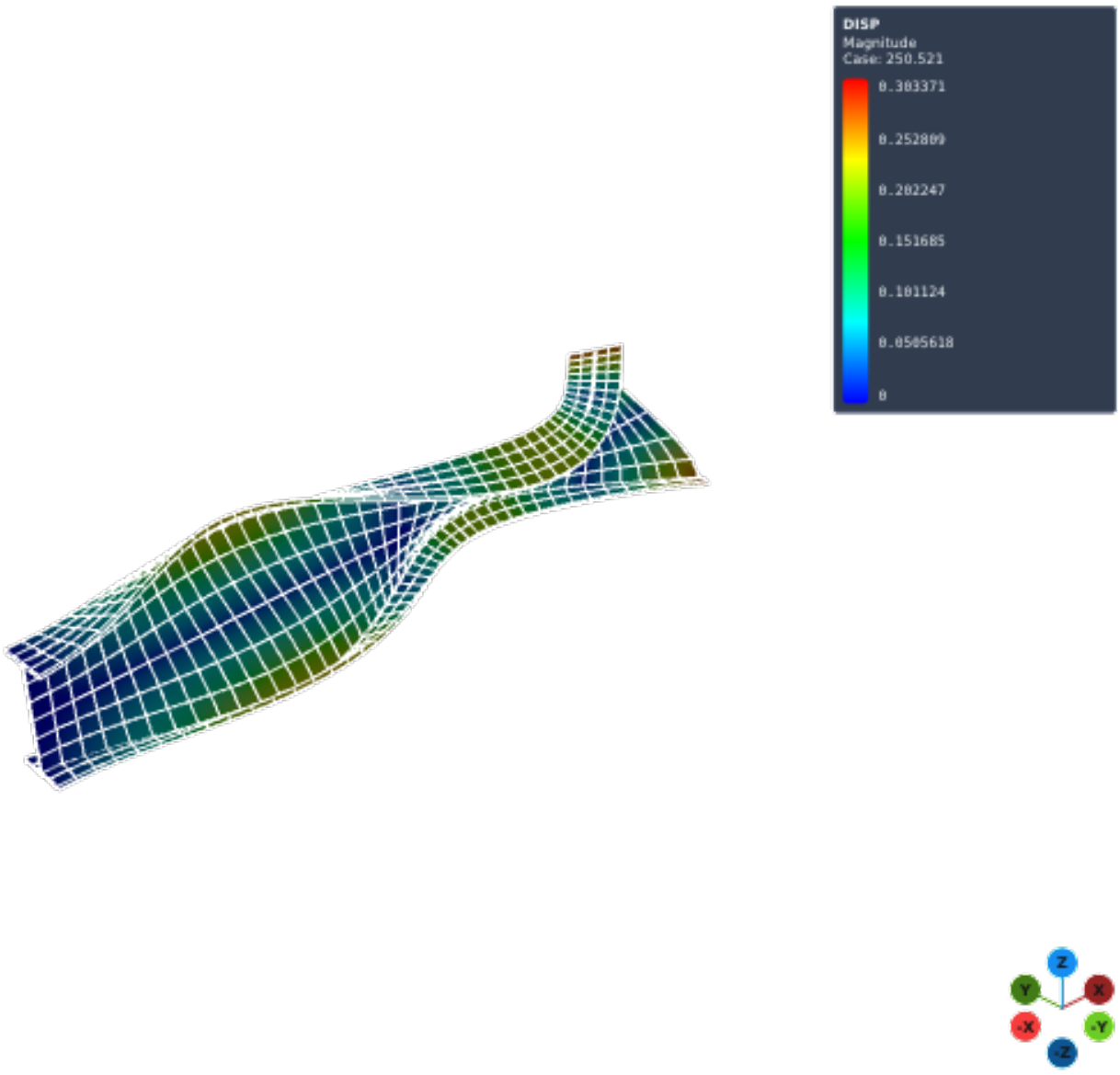

Figure 278: Calculix — mode 11, 250.521 Hz

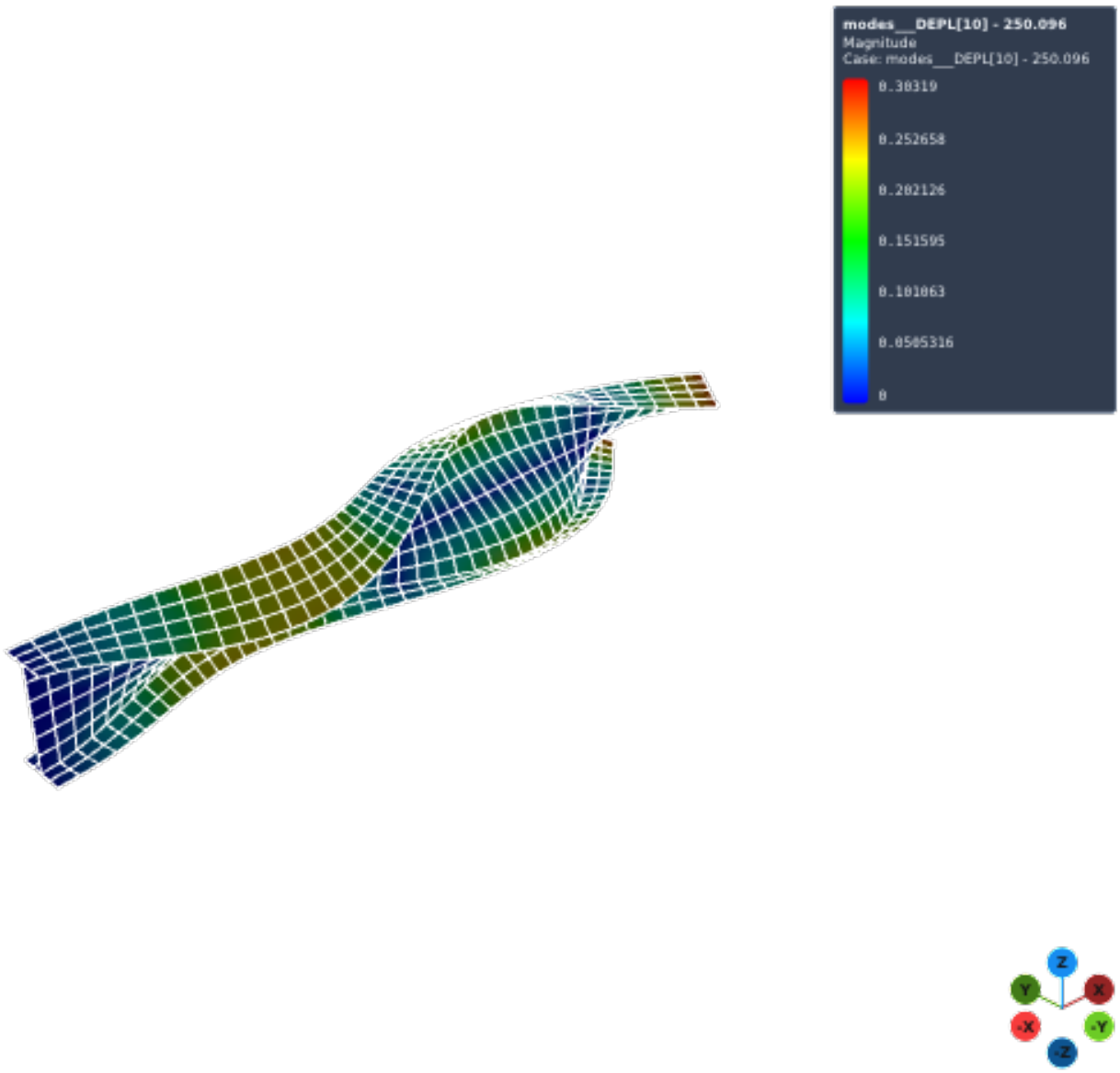

Figure 279: Code_Aster — mode 11, 250.096 Hz

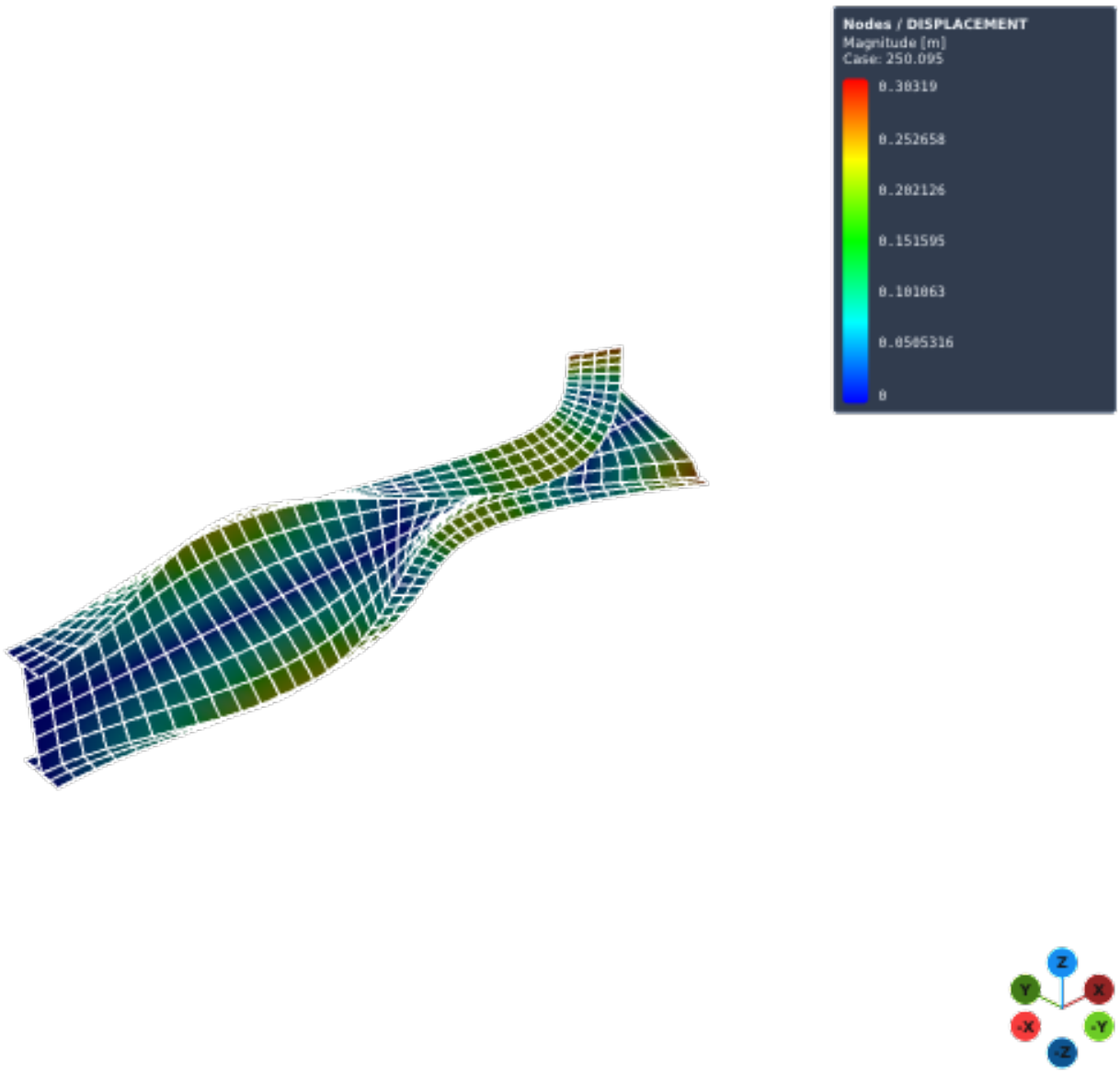

Figure 280: Sesam — mode 11, 250.095 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	250.170	-0.02	2.47082e+06	3.28e-21	3.98e-19	6.25e-19	5.72e-11	6.31e-10	-7.90e-10
Calculix	250.521	+0.12	2.47769e+06	2.31e-10	2.30e-22	4.17e-20	-1.52e-05	1.52e-11	2.04e-10
Code_Aster	250.096	-0.05	2.46929e+06	6.87e-28	1.30e-24	7.49e-25	-2.62e-14	1.14e-12	-8.65e-13
Sesam	250.095	-0.05	2.46928e+06	1.10e-19	4.31e-22	2.22e-21	-3.32e-10	-2.08e-11	4.71e-11

Shell, 2nd order, QUAD, reduced integration

Mode 1

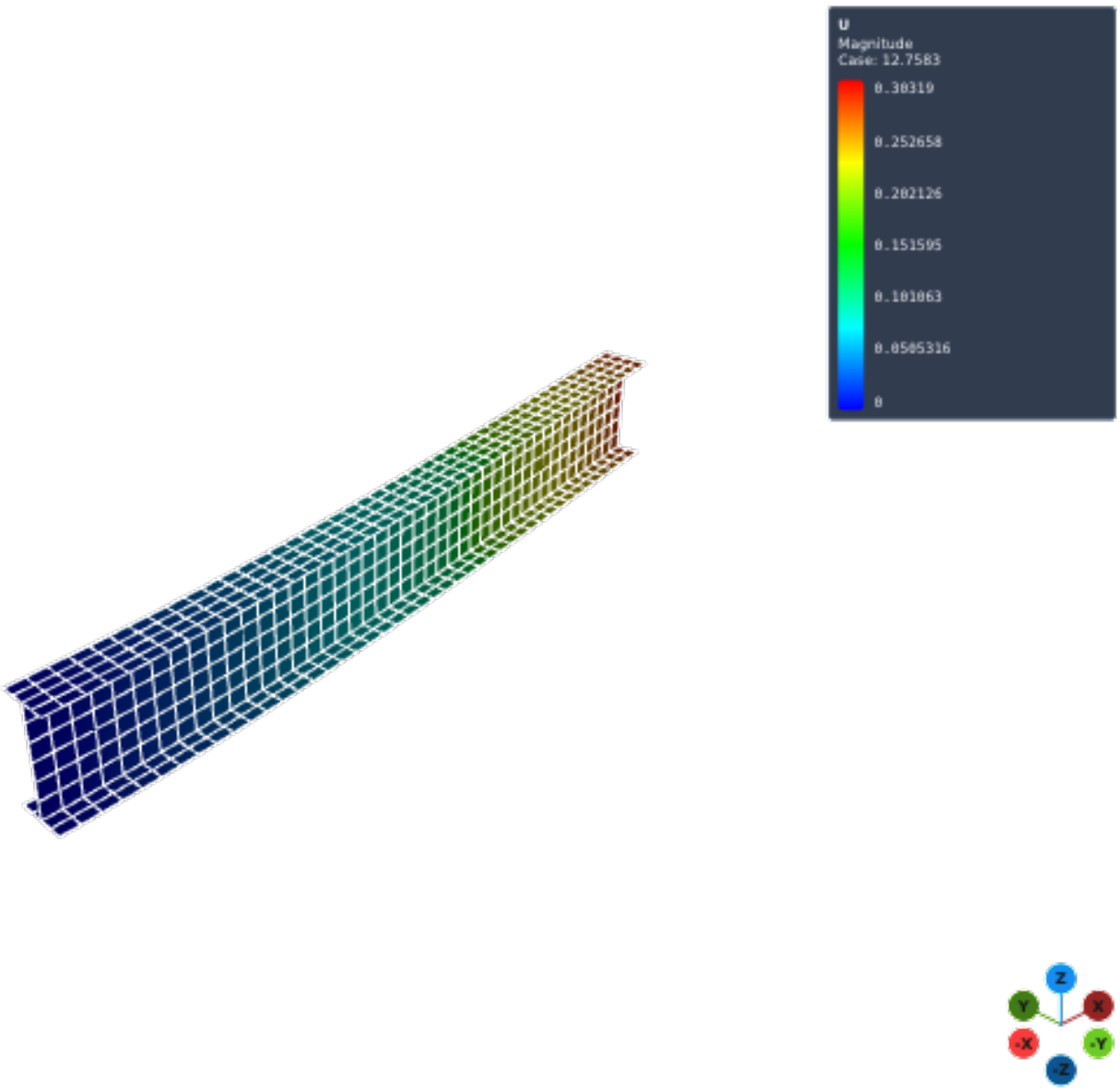

Figure 281: Abaqus — mode 1, 12.758 Hz

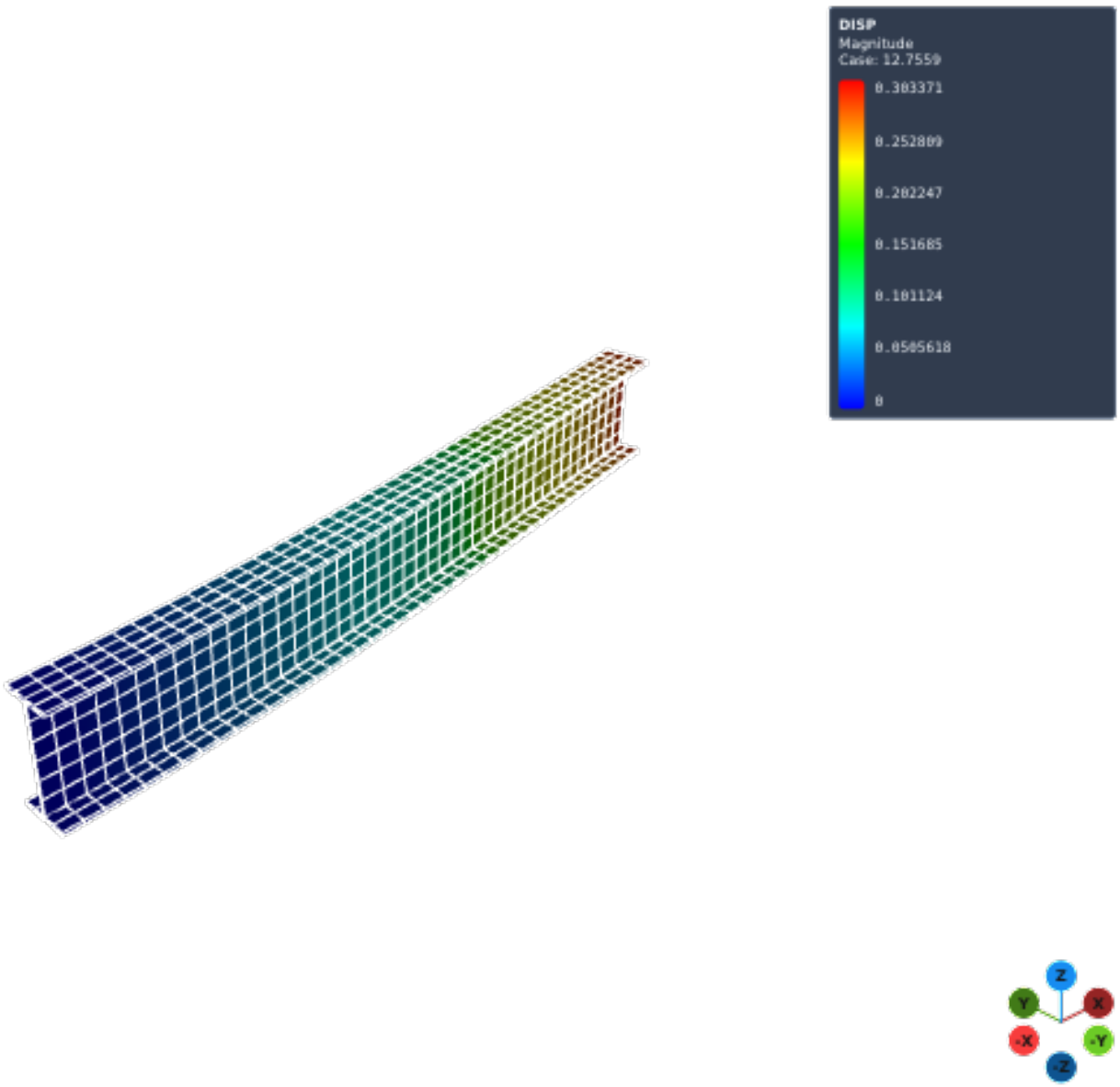

Figure 282: Calculix — mode 1, 12.756 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.758	+0.01	6426.1	1.23e-25	119.92	9.18e-25	3.51e-13	10.951	9.58e-13
Calculix	12.756	-0.01	6423.65	4.95e-25	119.93	1.74e-12	-7.04e-13	10.951	1.32e-06

Mode 2

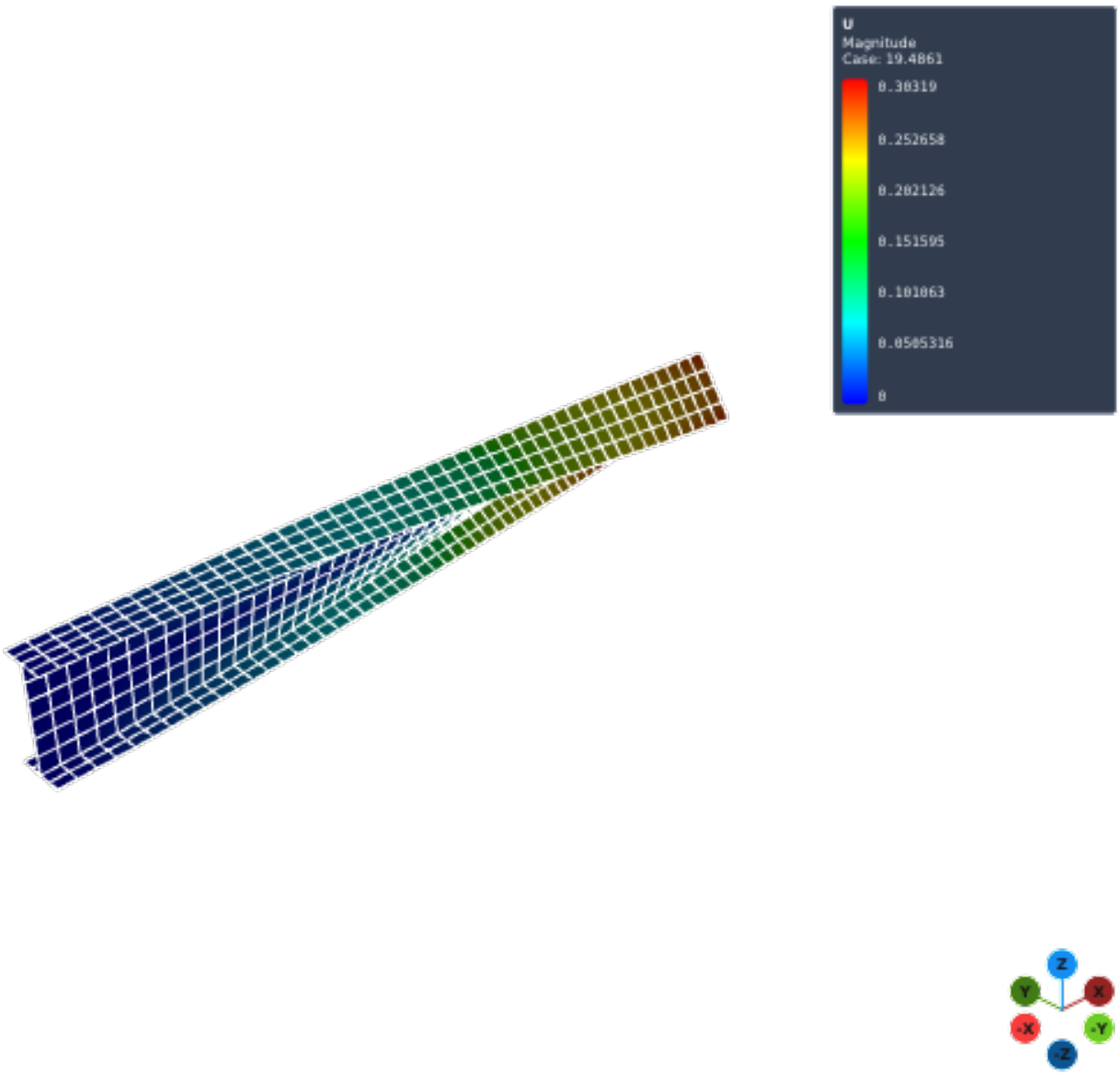

Figure 283: Abaqus — mode 2, 19.486 Hz

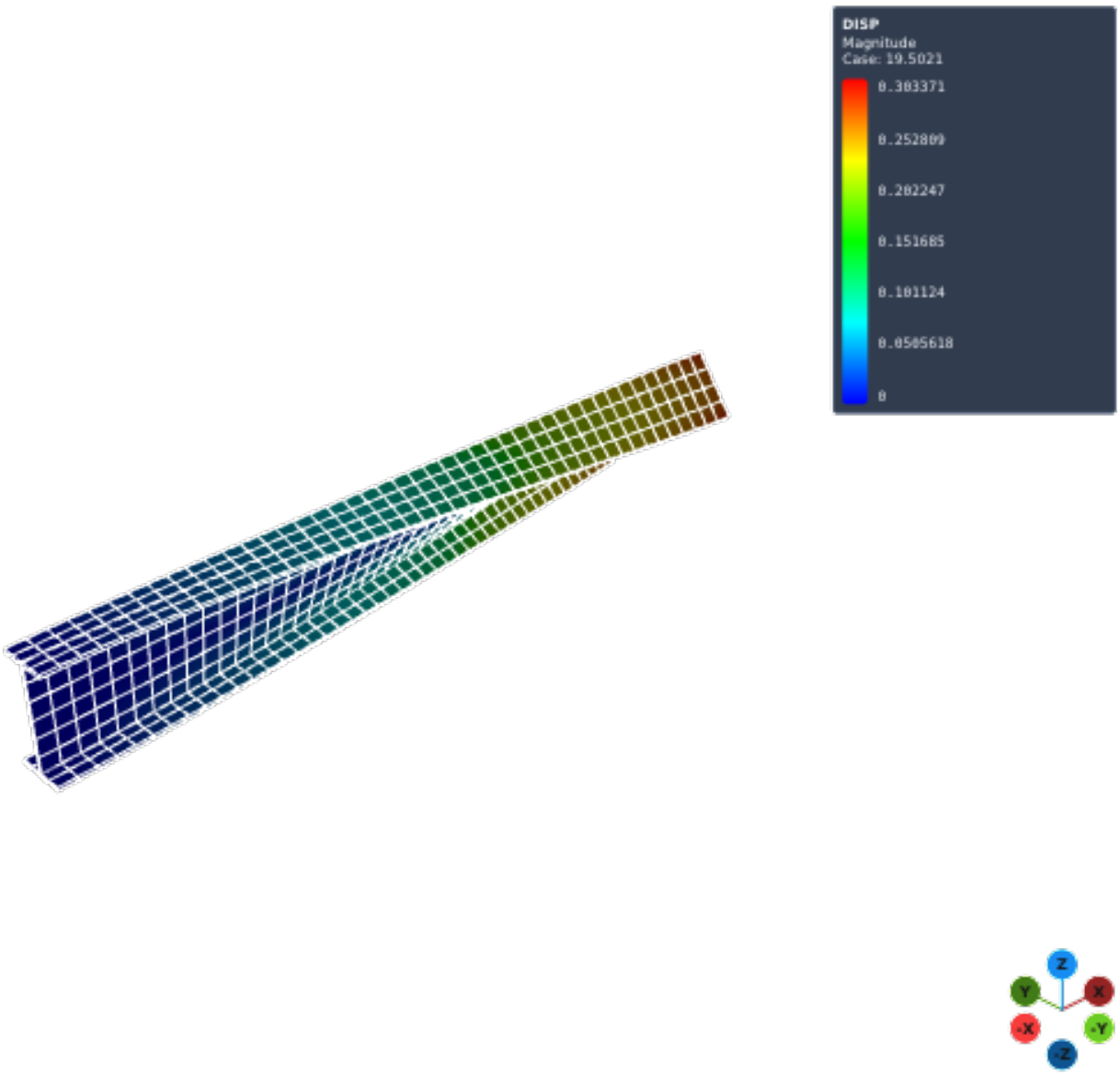

Figure 284: Calculix — mode 2, 19.502 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	19.486	-0.04	14990	1.04e-26	4.65e-20	2.40e-23	1.02e-13	2.16e-10	4.90e-12
Calculix	19.502	+0.04	15015	3.11e-09	2.97e-16	8.11e-20	5.58e-05	1.72e-08	-2.85e-10

Mode 3

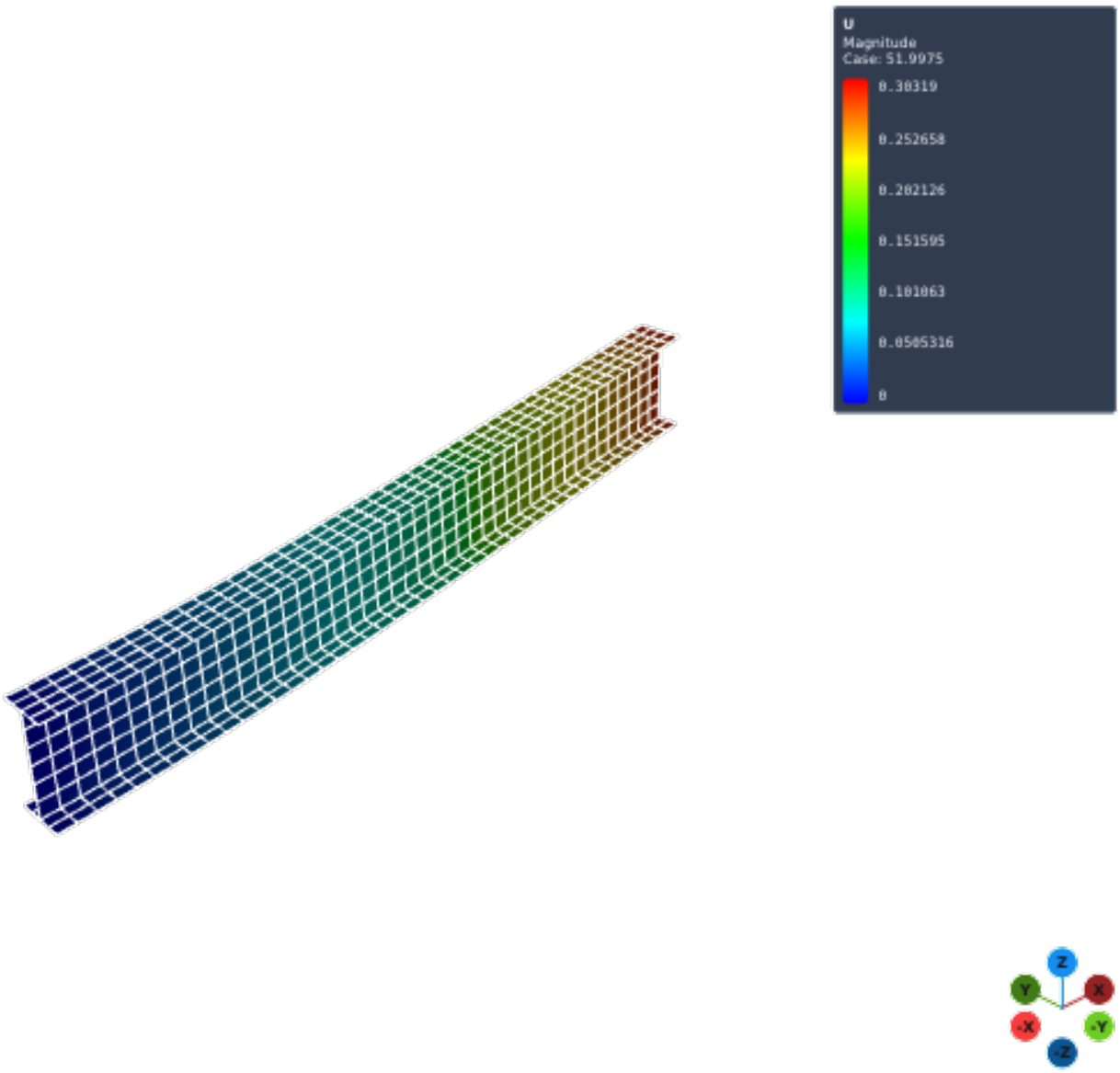

Figure 285: Abaqus — mode 3, 51.998 Hz

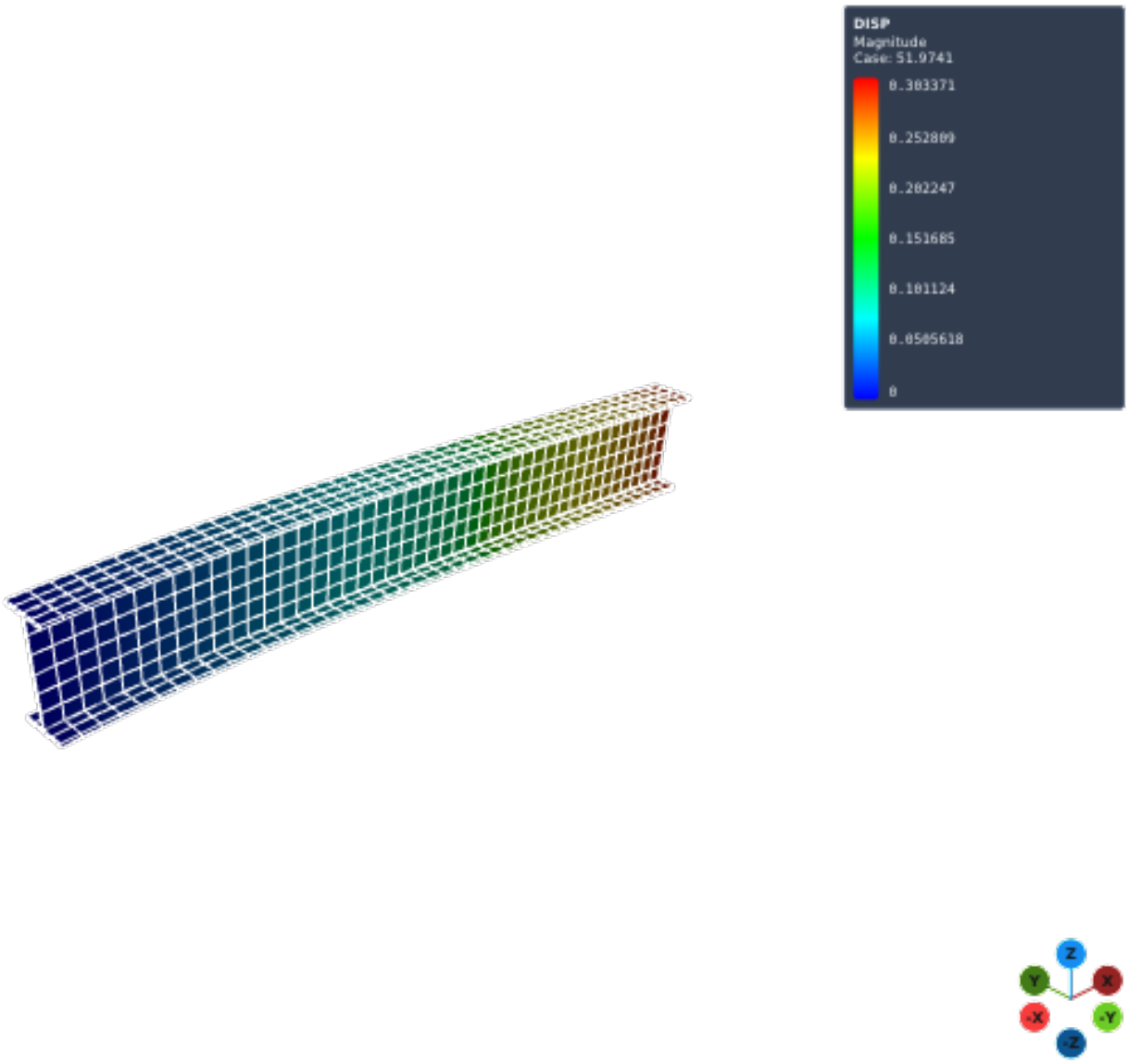

Figure 286: Calculix — mode 3, 51.974 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.998	+0.02	106740	5.15e-27	9.66e-26	121.97	7.18e-14	3.11e-13	11.044
Calculix	51.974	-0.02	106644	2.79e-24	4.52e-11	122.02	1.67e-12	6.73e-06	-11.046

Mode 4

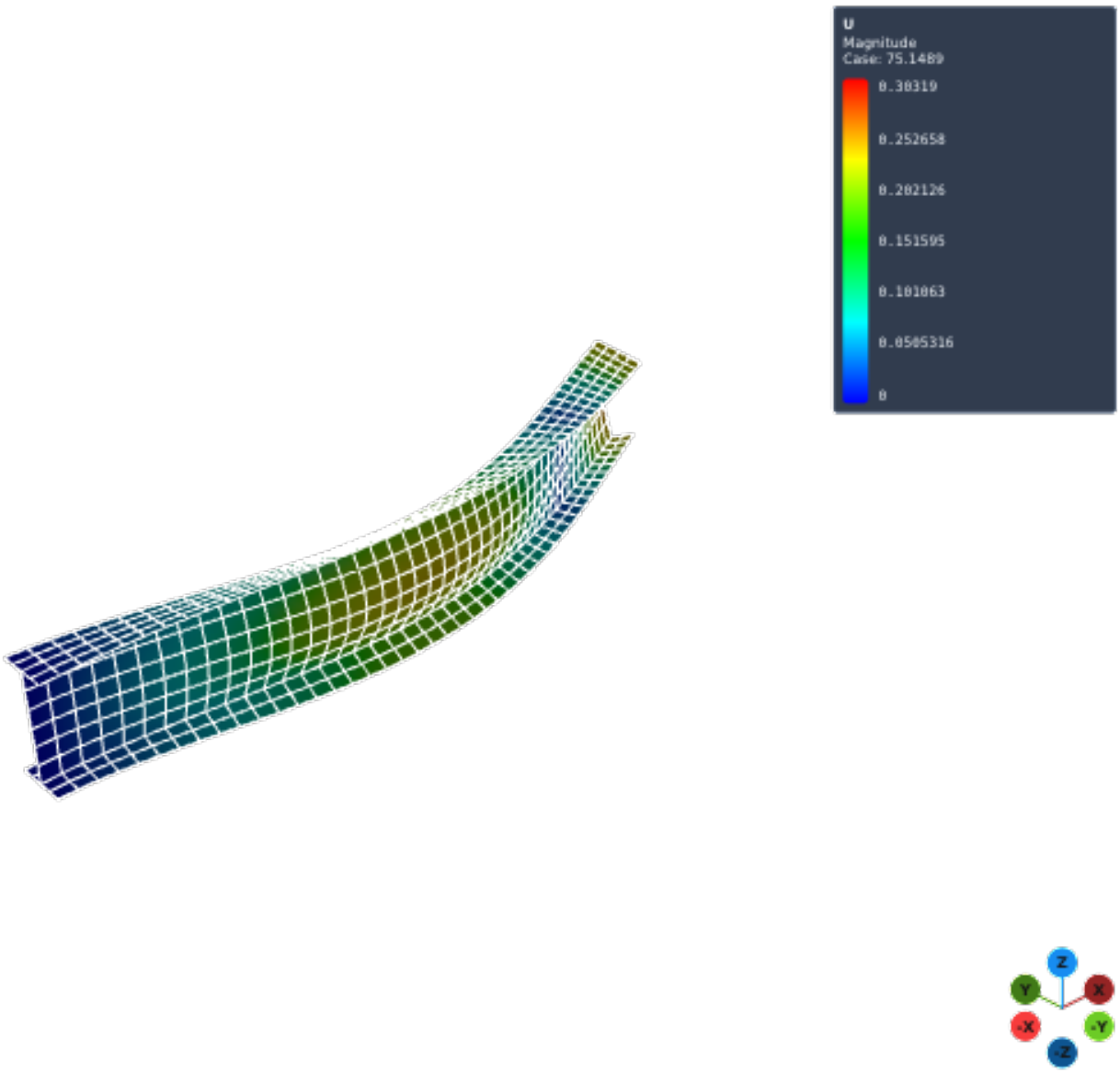

Figure 287: Abaqus — mode 4, 75.149 Hz

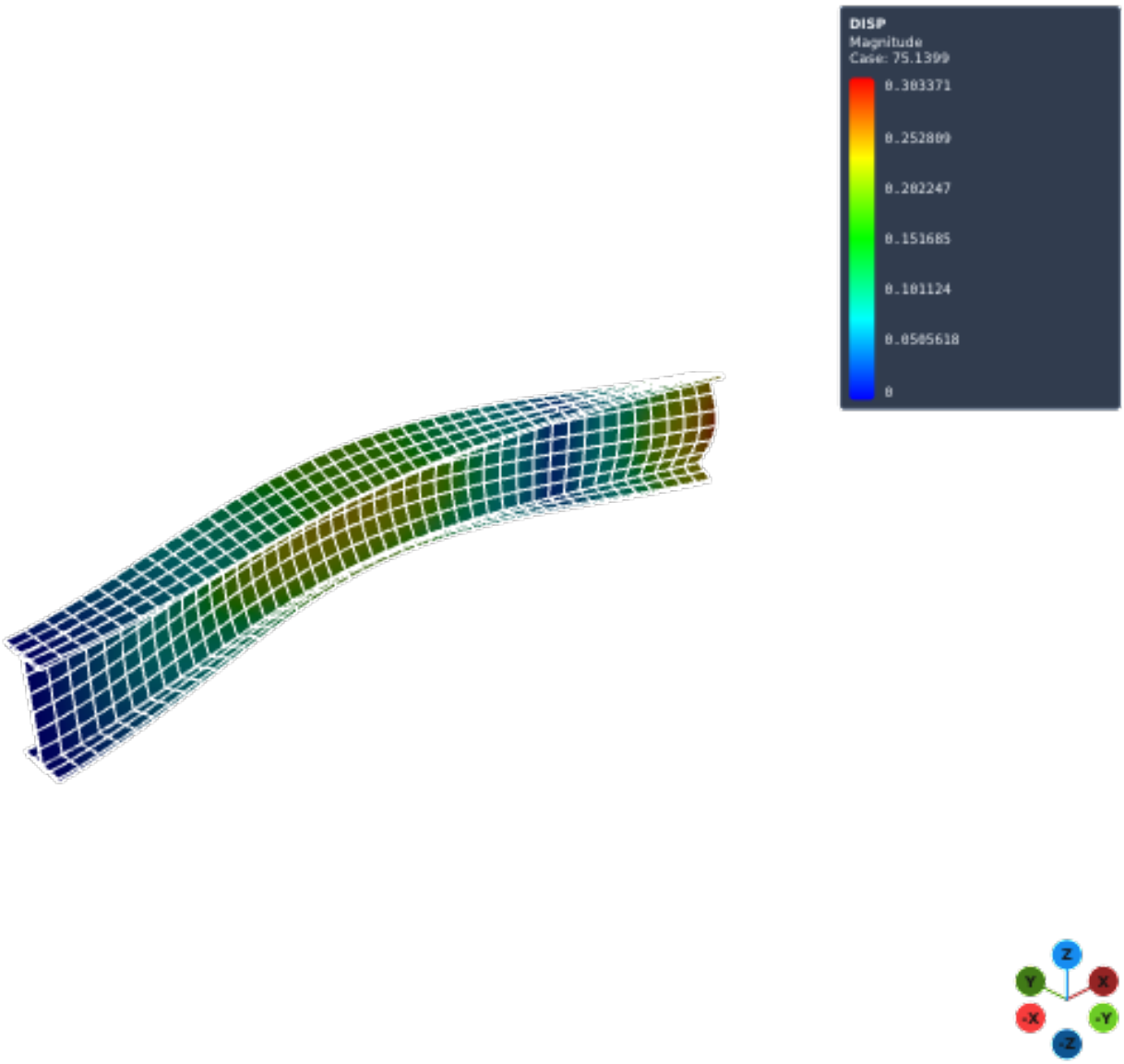

Figure 288: Calculix — mode 4, 75.140 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	75.149	+0.01	222949	1.40e-27	36.85	6.29e-24	-3.75e-14	-6.071	2.51e-12
Calculix	75.140	-0.01	222895	2.73e-24	36.85	1.38e-12	1.65e-12	6.071	-1.18e-06

Mode 5

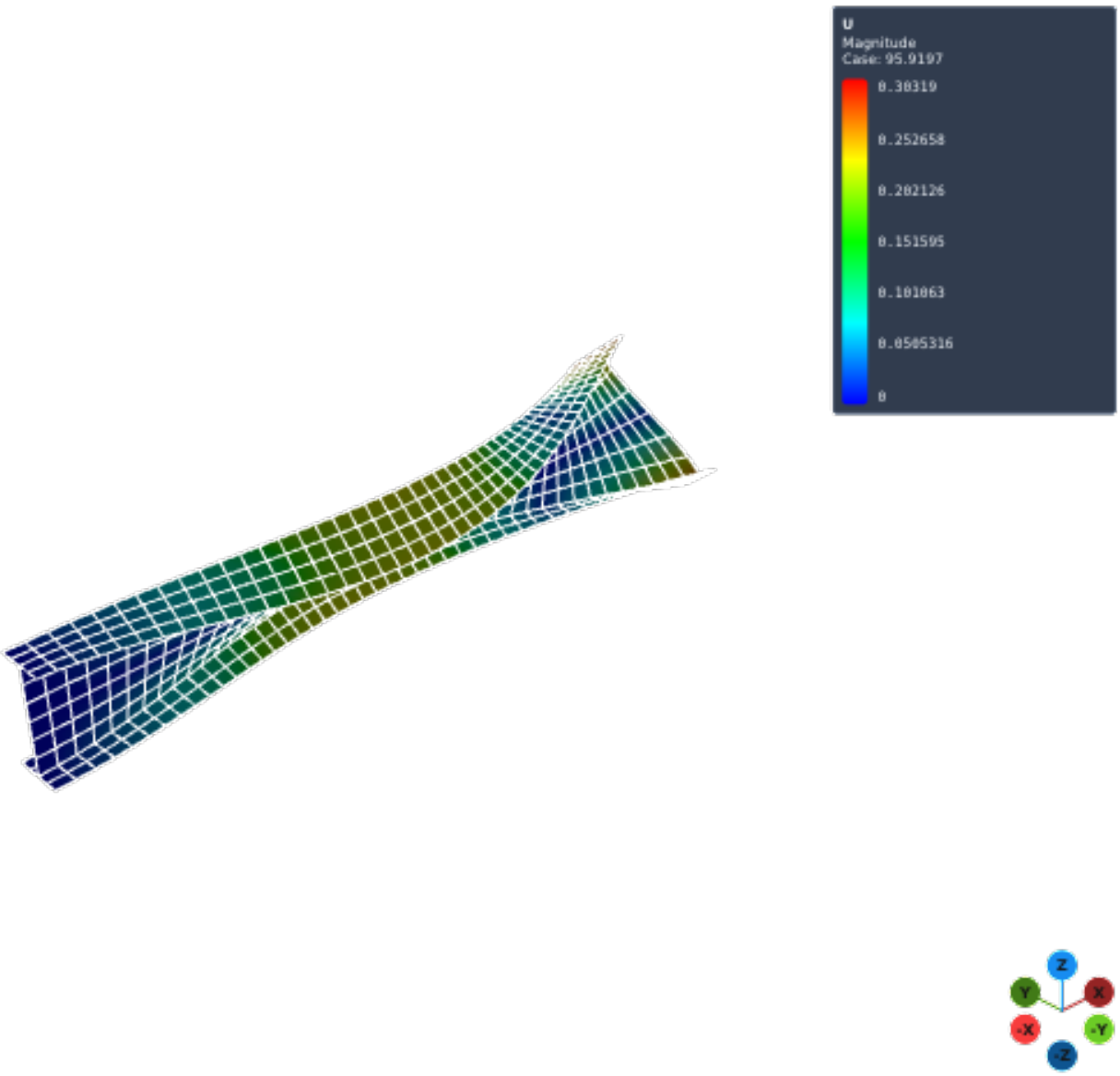

Figure 289: Abaqus — mode 5, 95.920 Hz

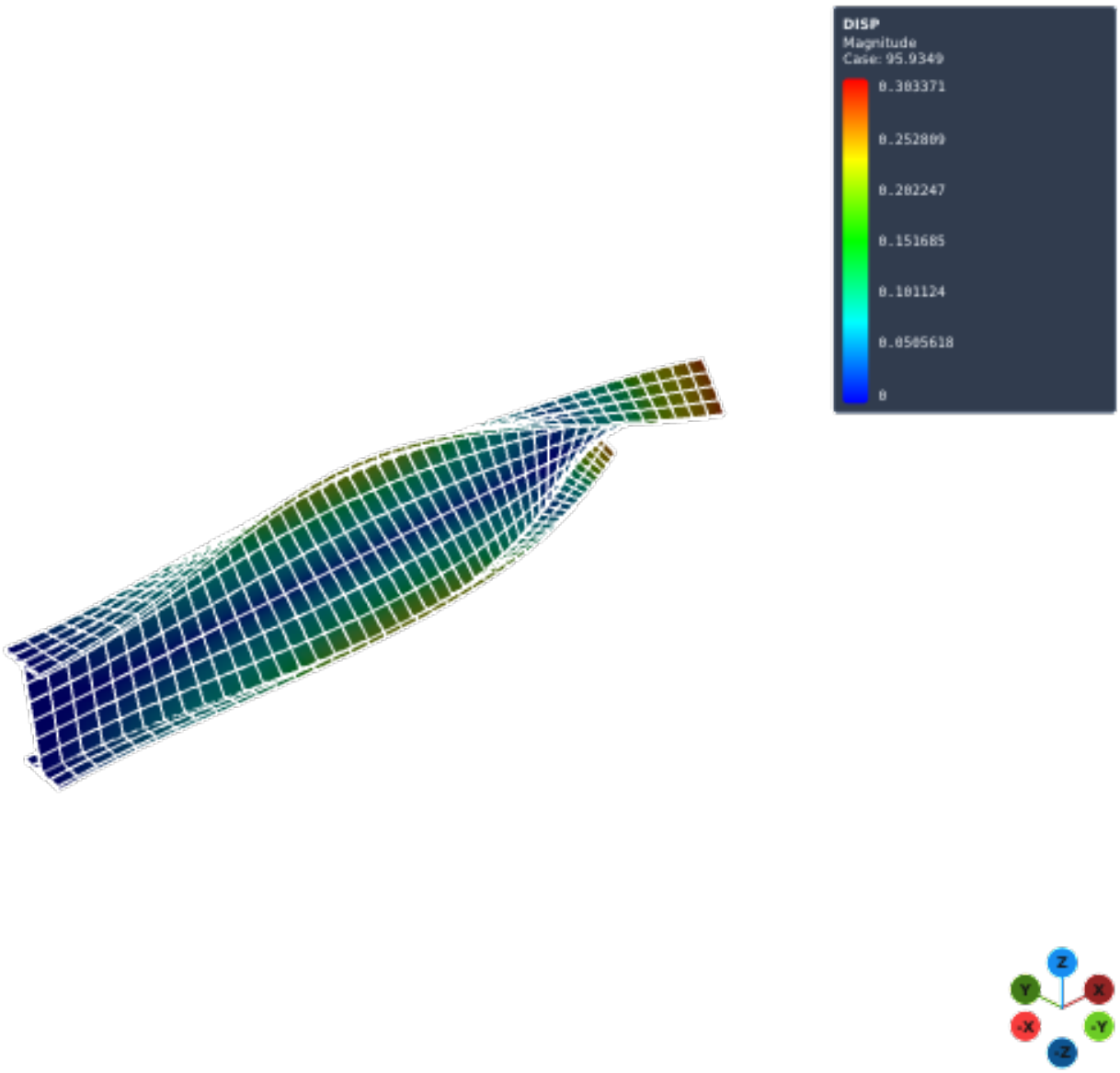

Figure 290: Calculix — mode 5, 95.935 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	95.920	-0.01	363224	7.50e-27	2.54e-21	8.04e-25	-8.66e-14	-5.04e-11	-8.97e-13
Calculix	95.935	+0.01	363340	8.60e-10	2.62e-19	7.05e-22	-2.93e-05	-5.11e-10	2.66e-11

Mode 6

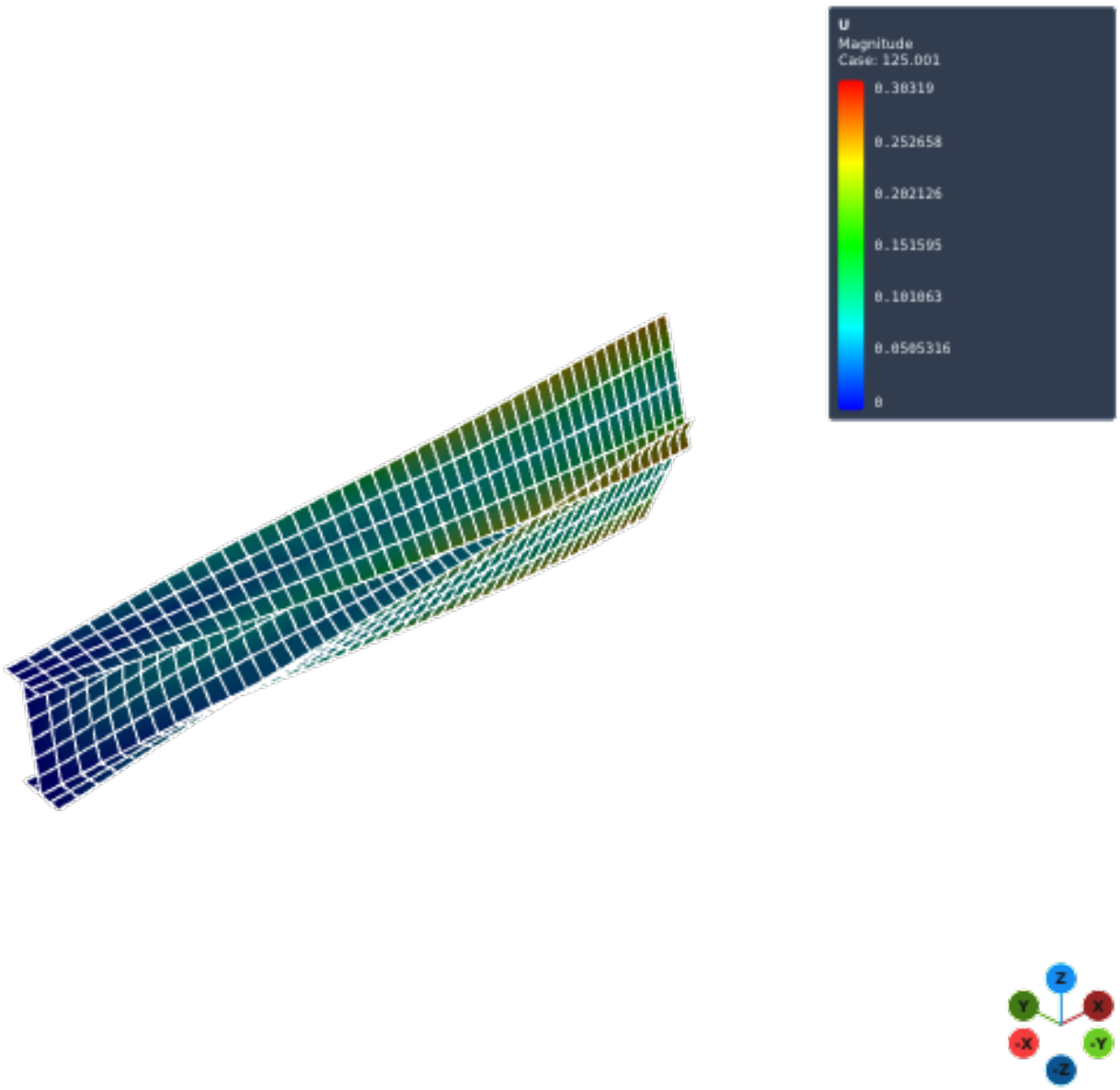

Figure 291: Abaqus — mode 6, 125.001 Hz

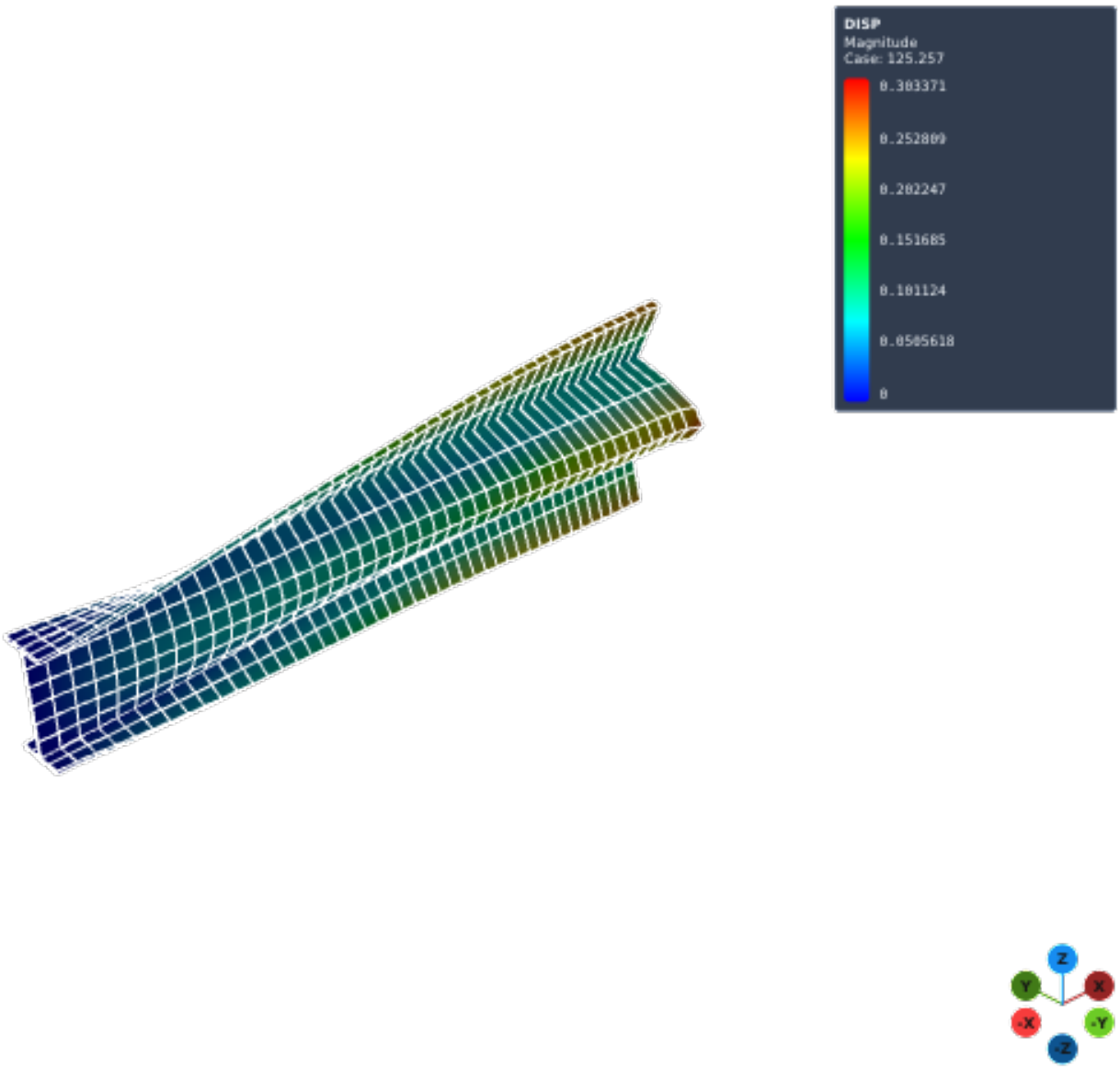

Figure 292: Calculix — mode 6, 125.257 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	125.000	-0.10	616856	5.44e-26	0.02	4.78e-23	-2.33e-13	-0.137	6.92e-12
Calculix	125.257	+0.10	619385	5.54e-25	0.02	1.65e-13	7.44e-13	0.138	4.06e-07

Mode 7

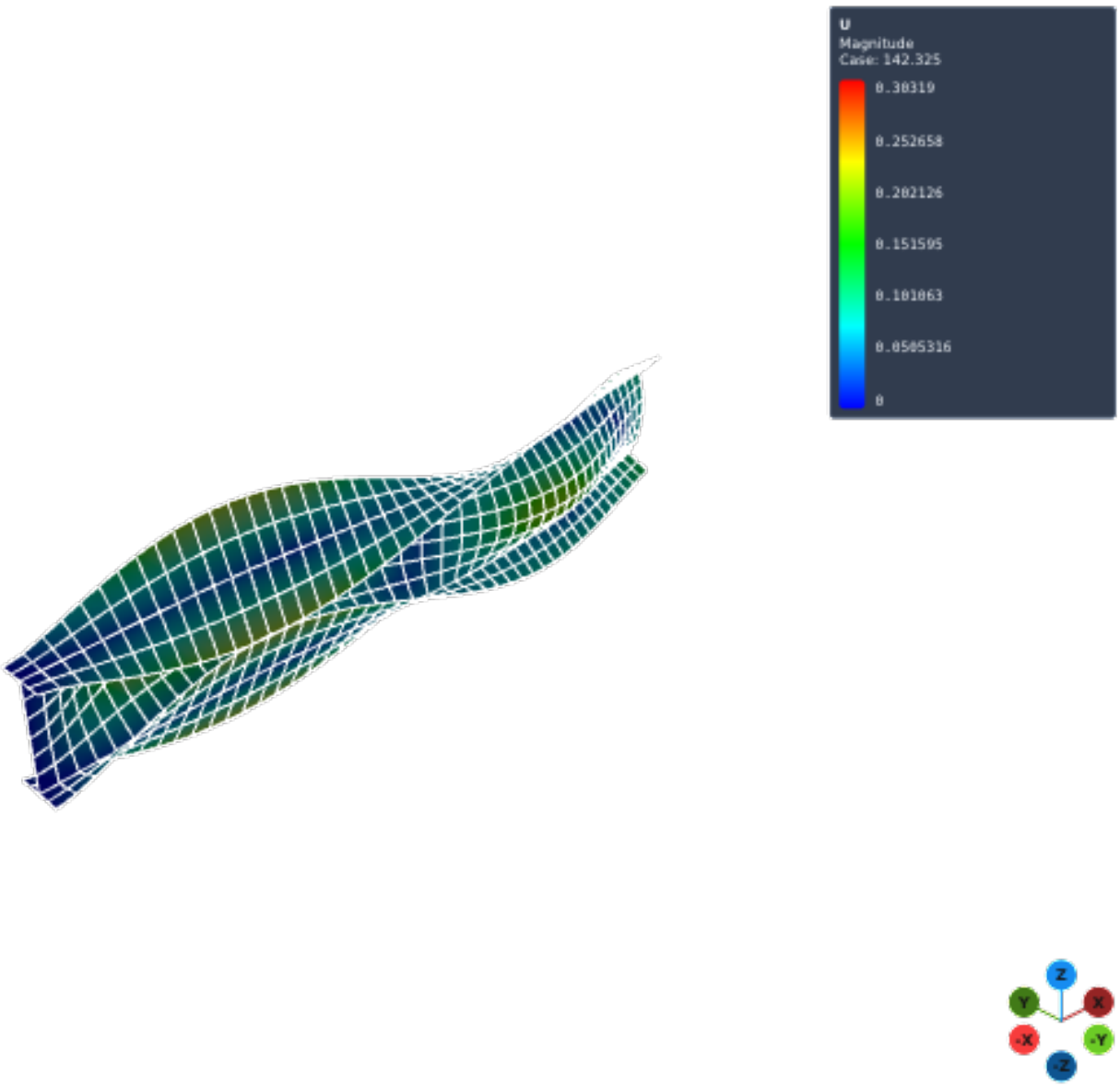

Figure 293: Abaqus — mode 7, 142.325 Hz

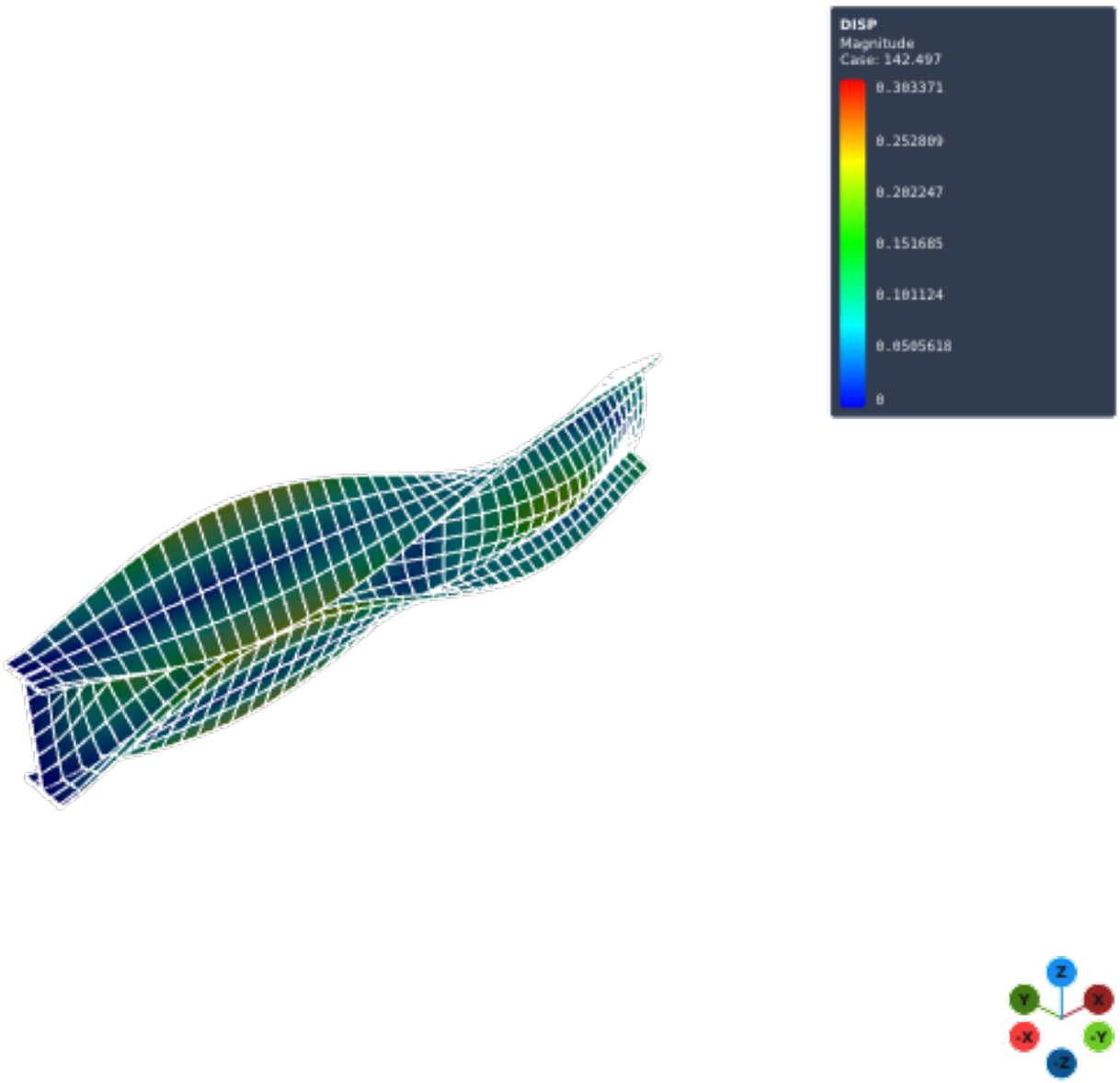

Figure 294: Calculix — mode 7, 142.497 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	142.320	-0.06	799690	2.49e-27	3.45	4.42e-24	-4.99e-14	1.858	-2.10e-12
Calculix	142.497	+0.06	801628	6.39e-25	3.43	7.82e-13	-7.99e-13	1.852	8.84e-07

Mode 8

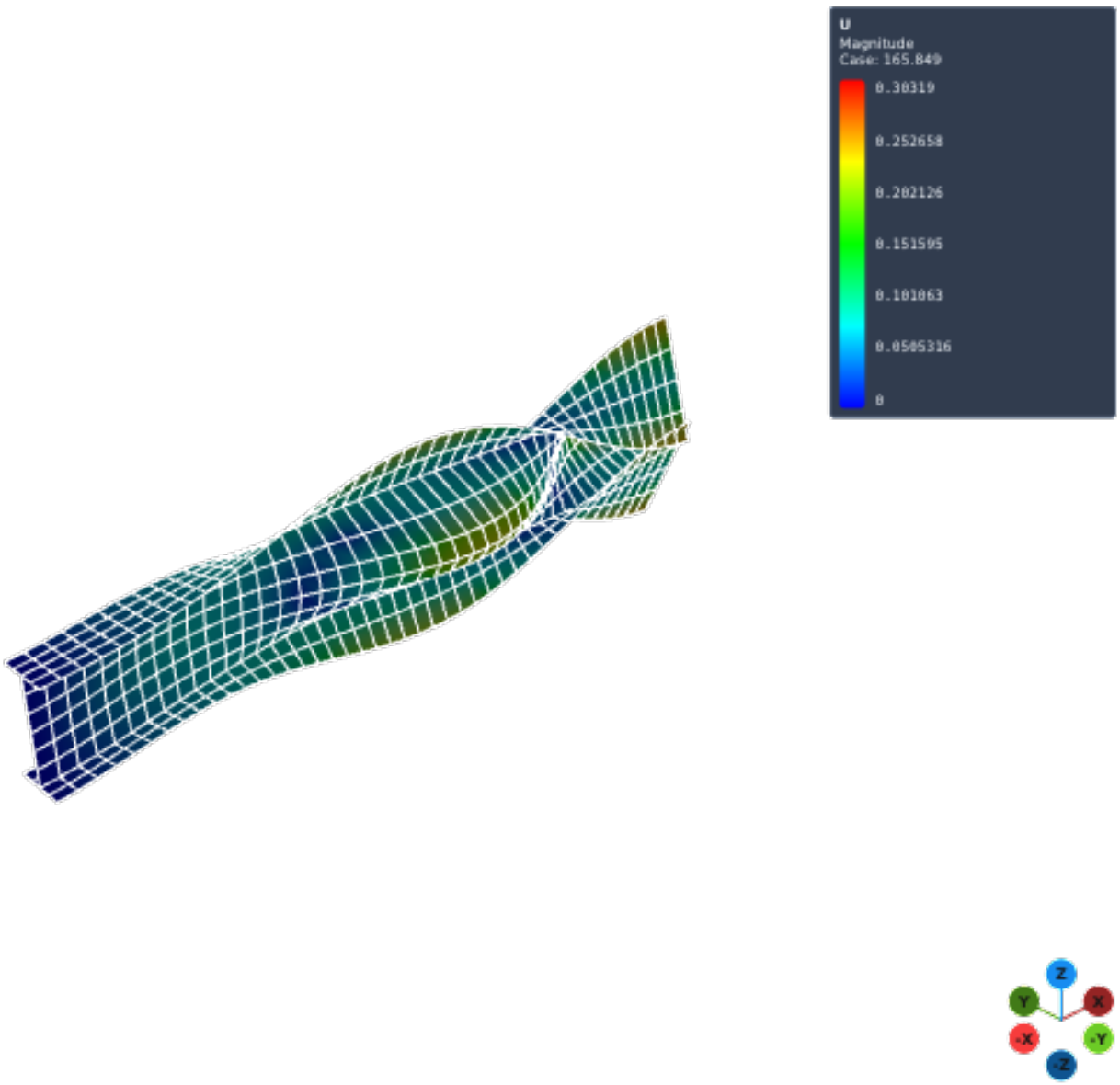

Figure 295: Abaqus — mode 8, 165.849 Hz

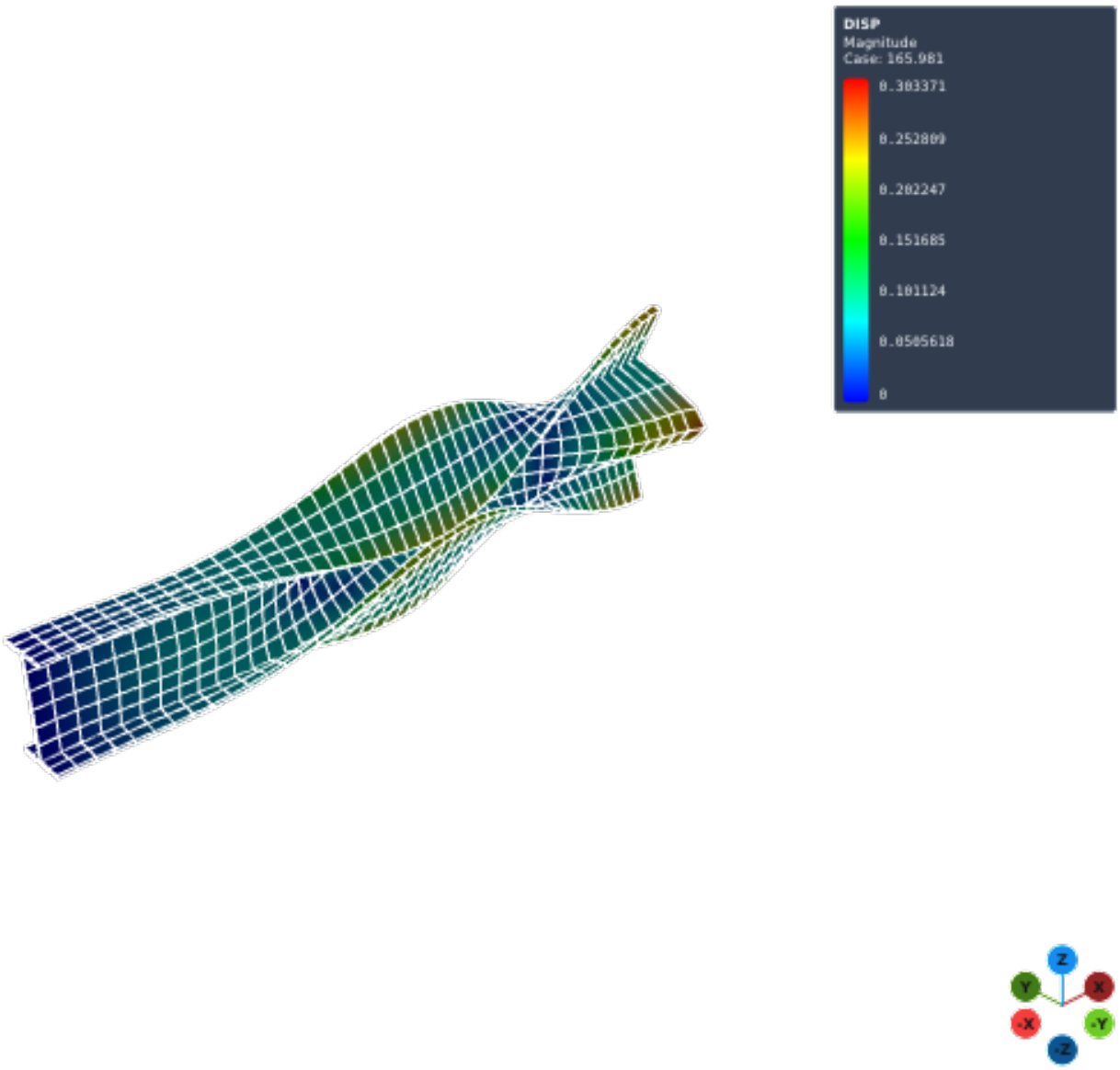

Figure 296: Calculix — mode 8, 165.981 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	165.850	-0.04	1.08589e+06	2.43e-27	5.53	1.06e-23	-4.93e-14	2.351	3.25e-12
Calculix	165.981	+0.04	1.08762e+06	9.68e-25	5.52	3.21e-13	9.84e-13	-2.350	5.66e-07

Mode 9

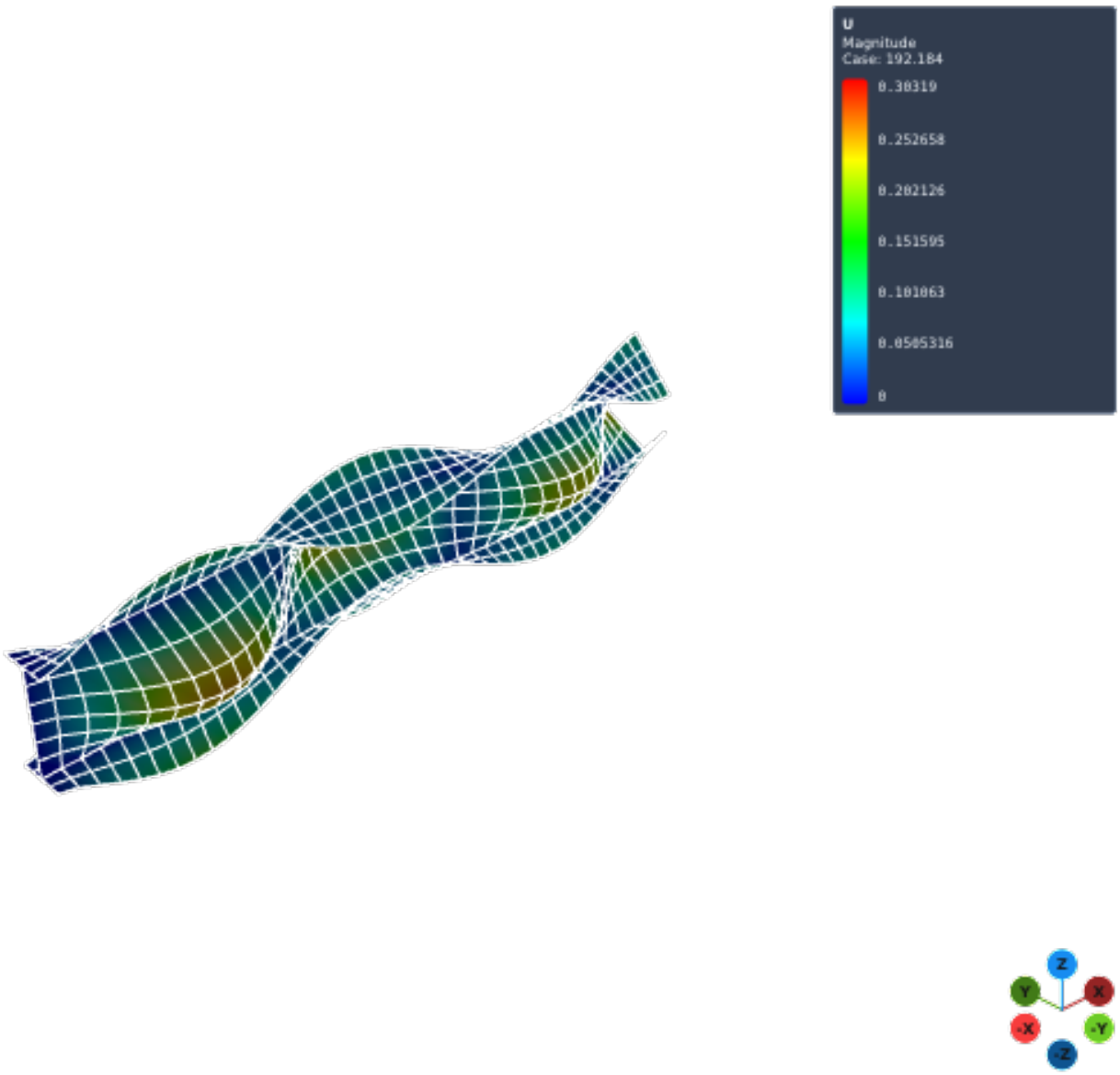

Figure 297: Abaqus — mode 9, 192.184 Hz

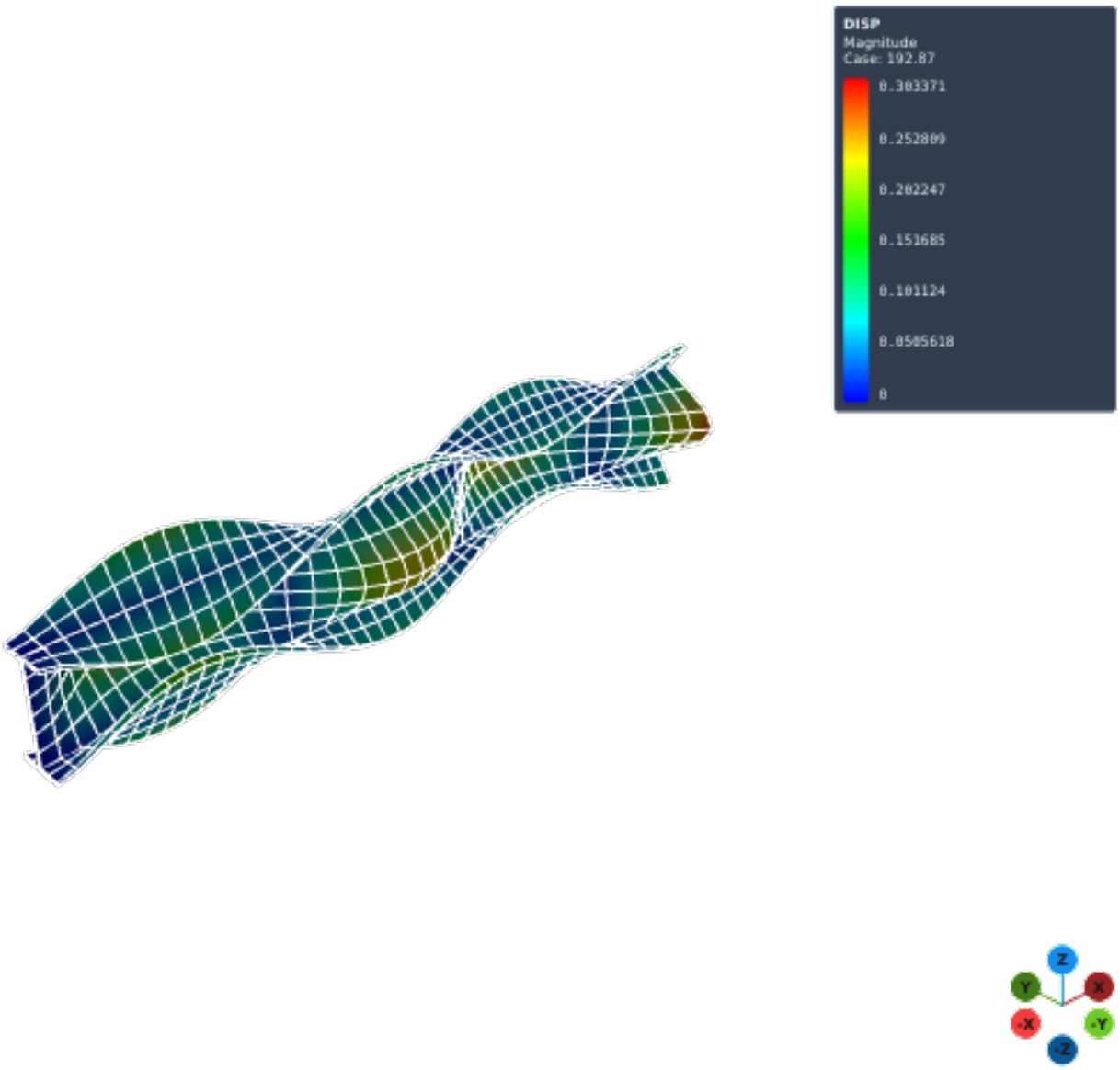

Figure 298: Calculix — mode 9, 192.870 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	192.180	-0.18	1.45812e+06	1.15e-27	0.65	1.20e-25	-3.40e-14	-0.808	3.47e-13
Calculix	192.870	+0.18	1.46855e+06	3.29e-24	0.64	7.71e-14	1.81e-12	0.799	-2.78e-07

Mode 10

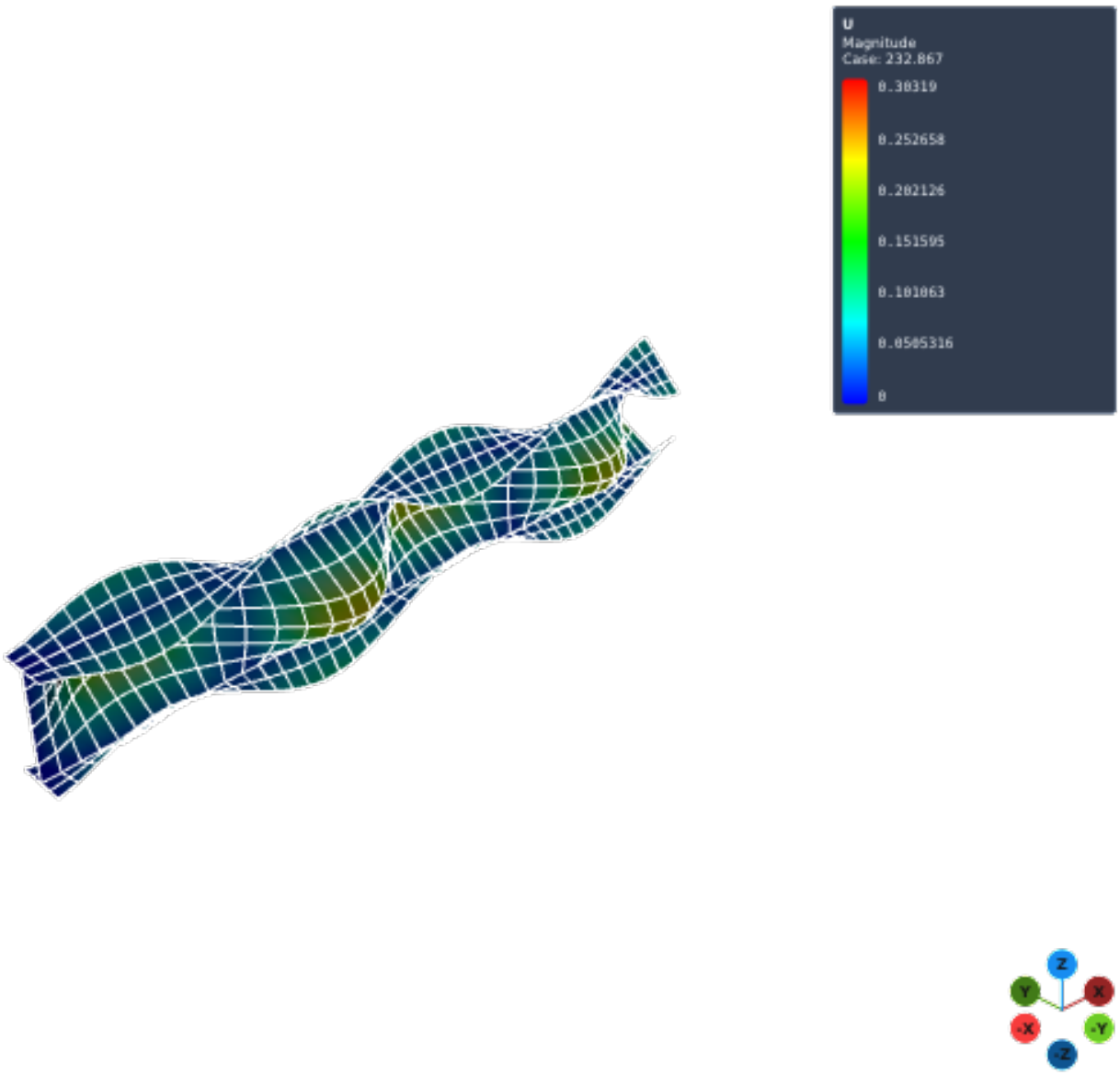

Figure 299: Abaqus — mode 10, 232.867 Hz

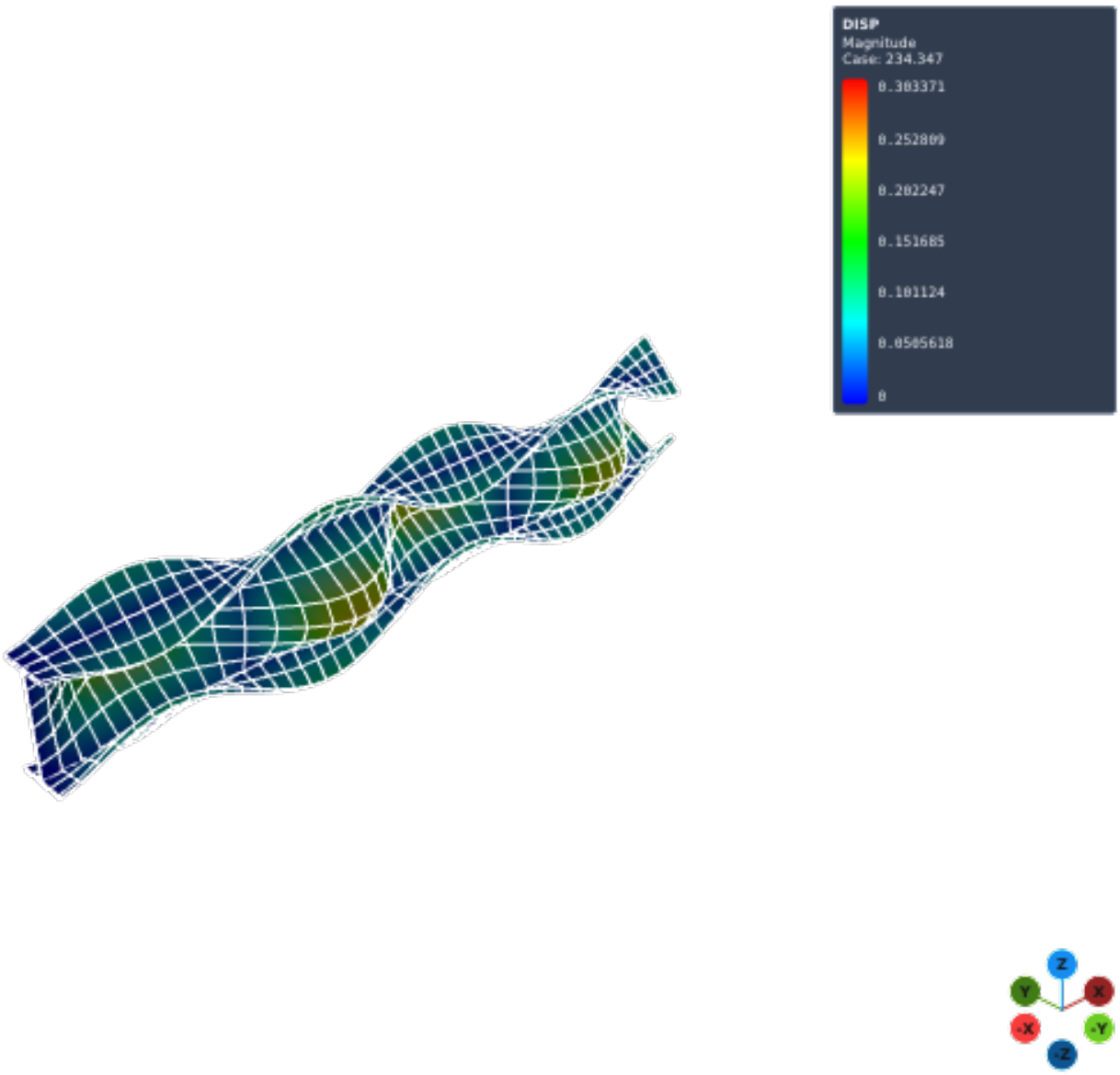

Figure 300: Calculix — mode 10, 234.347 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	232.870	-0.32	2.14079e+06	4.88e-26	1.37	1.33e-23	-2.21e-13	1.170	-3.65e-12
Calculix	234.347	+0.32	2.1681e+06	5.08e-24	1.39	3.57e-14	-2.25e-12	1.178	1.89e-07

Mode 11

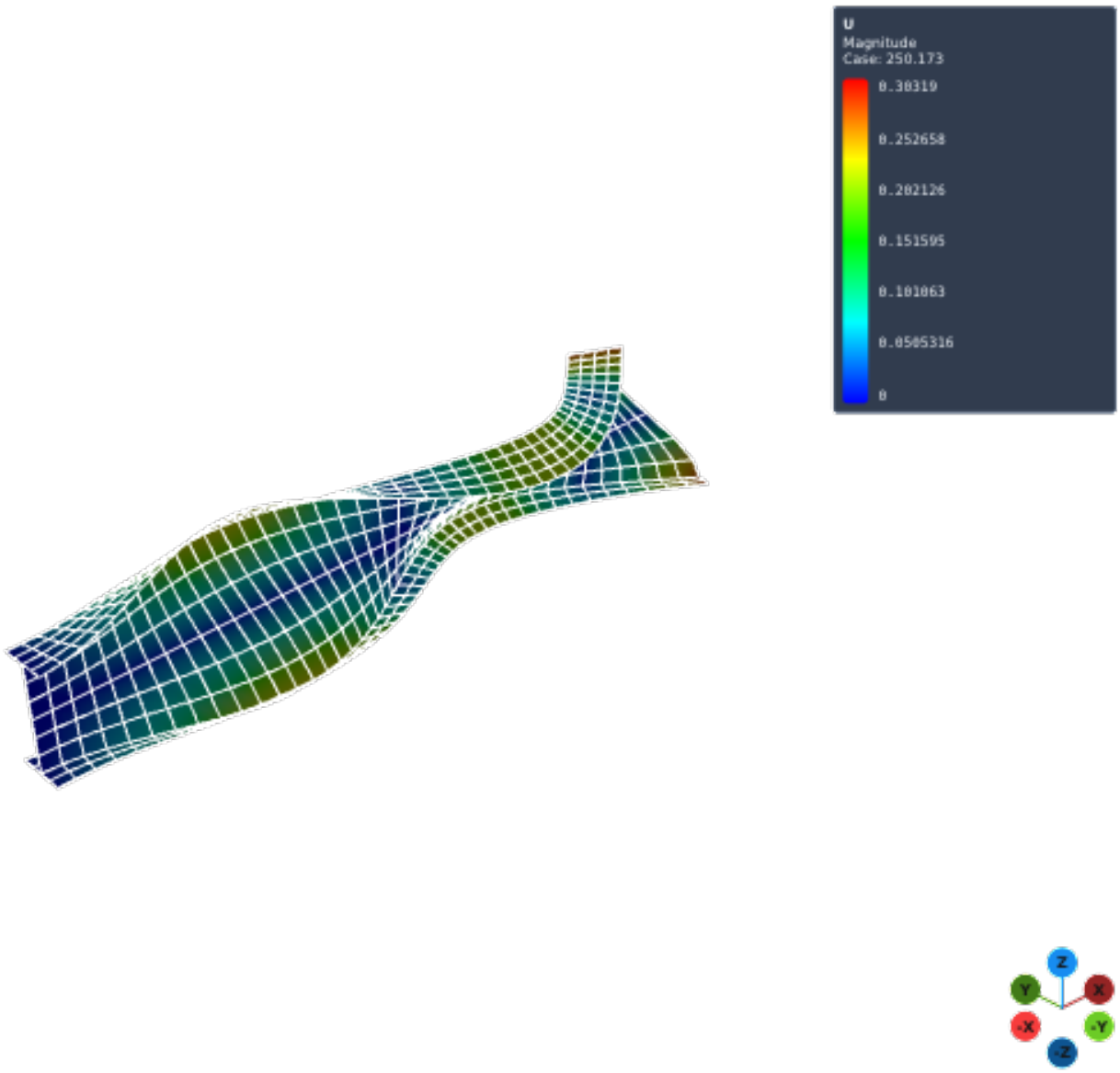

Figure 301: Abaqus — mode 11, 250.173 Hz

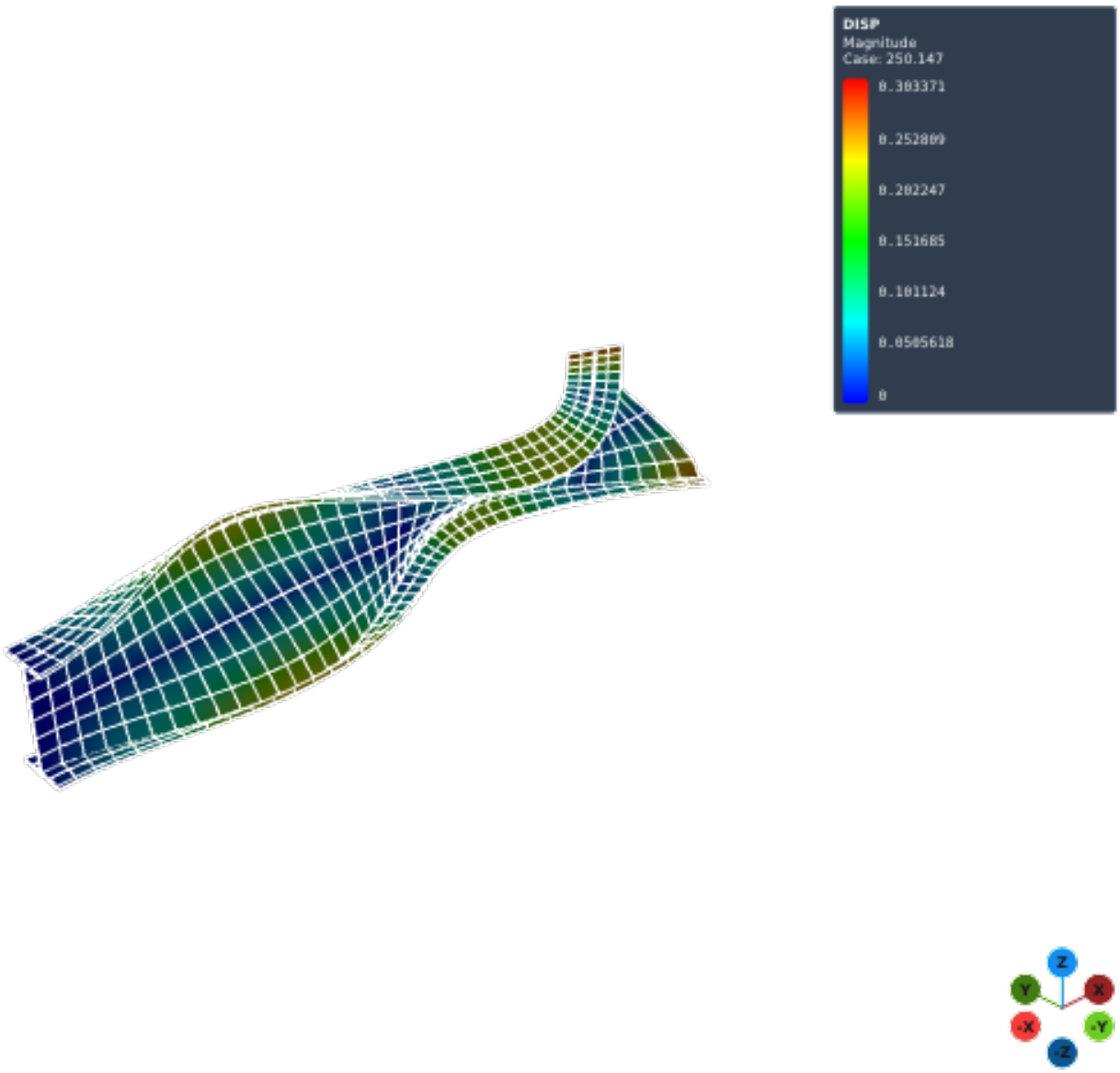

Figure 302: Calculix — mode 11, 250.147 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	250.170	4.62e-03	2.47082e+06	3.28e-21	3.98e-19	6.25e-19	5.72e-11	6.31e-10	-7.90e-10
Calculix	250.147	-4.62e-03	2.4703e+06	2.11e-10	3.23e-21	1.17e-22	-1.45e-05	5.68e-11	-1.08e-11

Solid, 1st order, TET

Mode 1

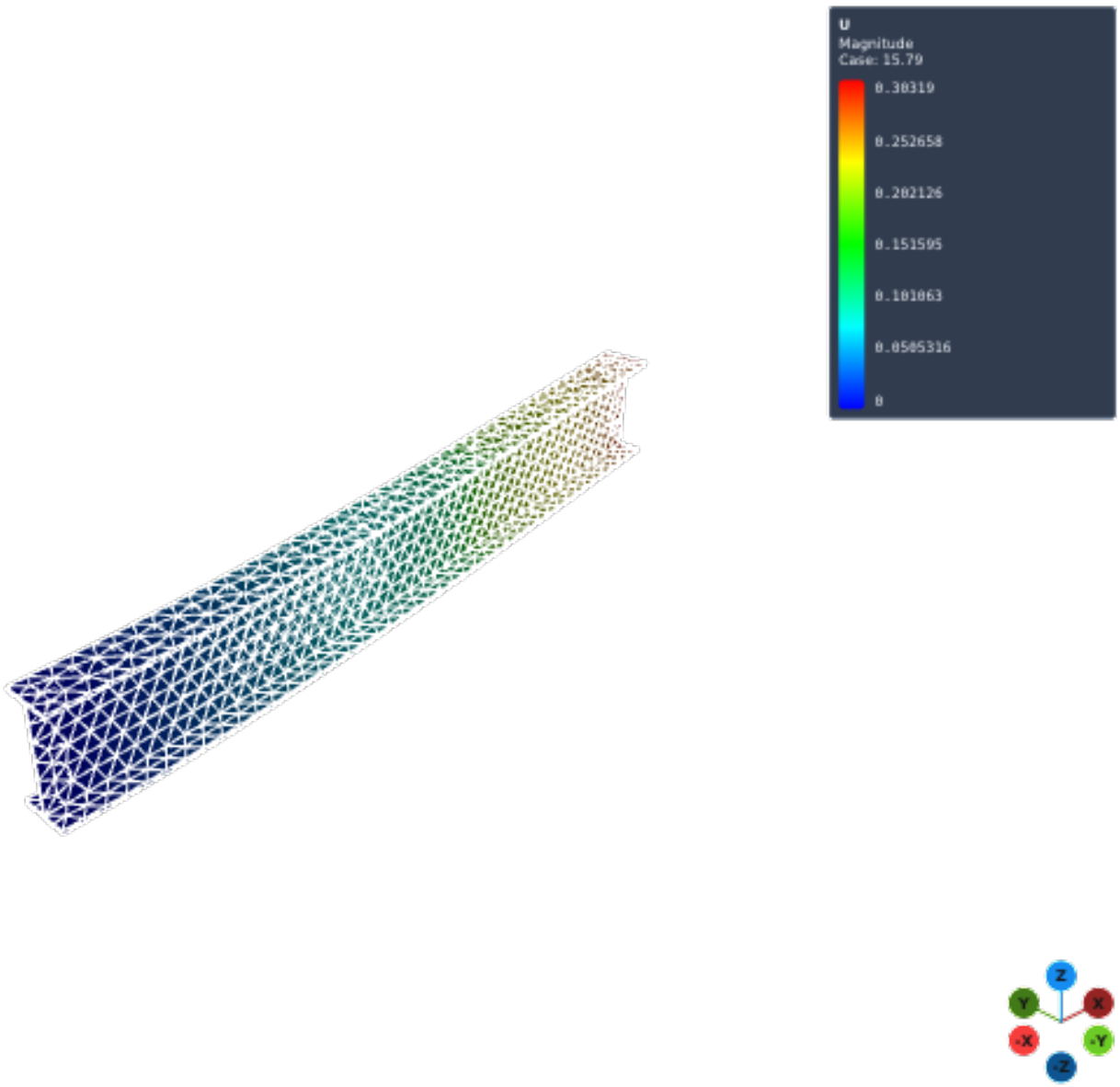

Figure 303: Abaqus — mode 1, 15.790 Hz

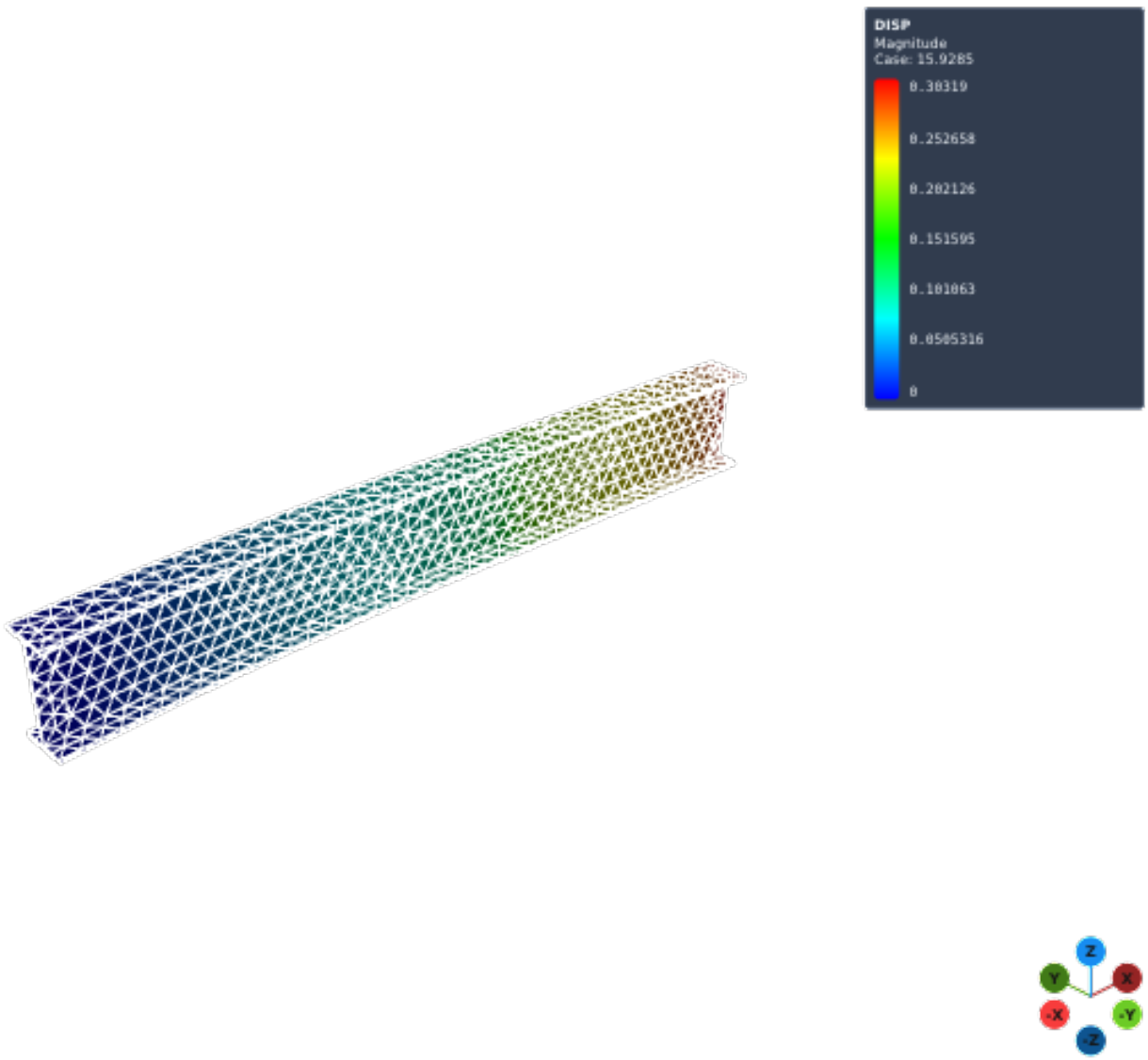

Figure 304: Calculix — mode 1, 15.928 Hz

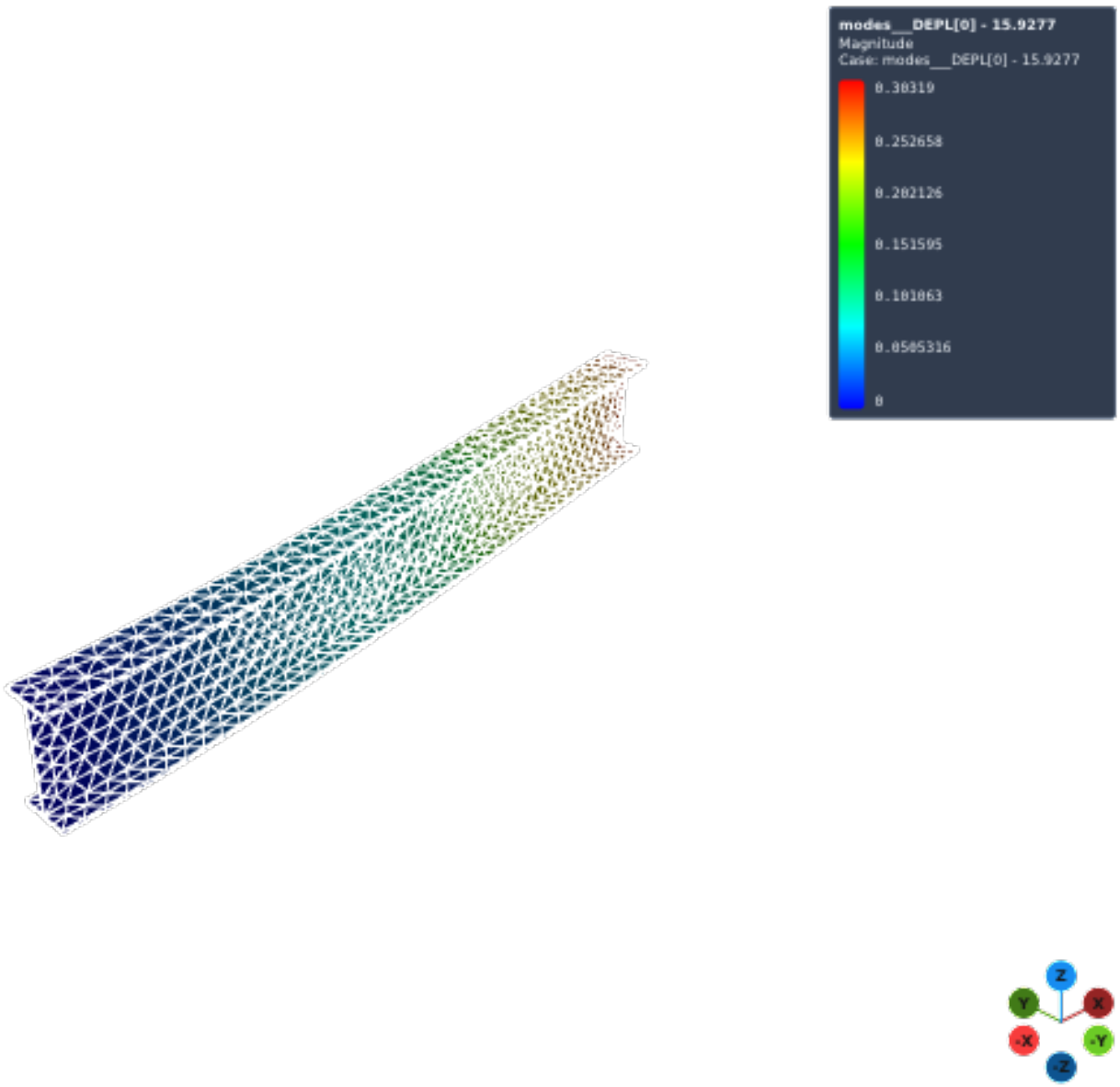

Figure 305: Code_Aster — mode 1, 15.928 Hz

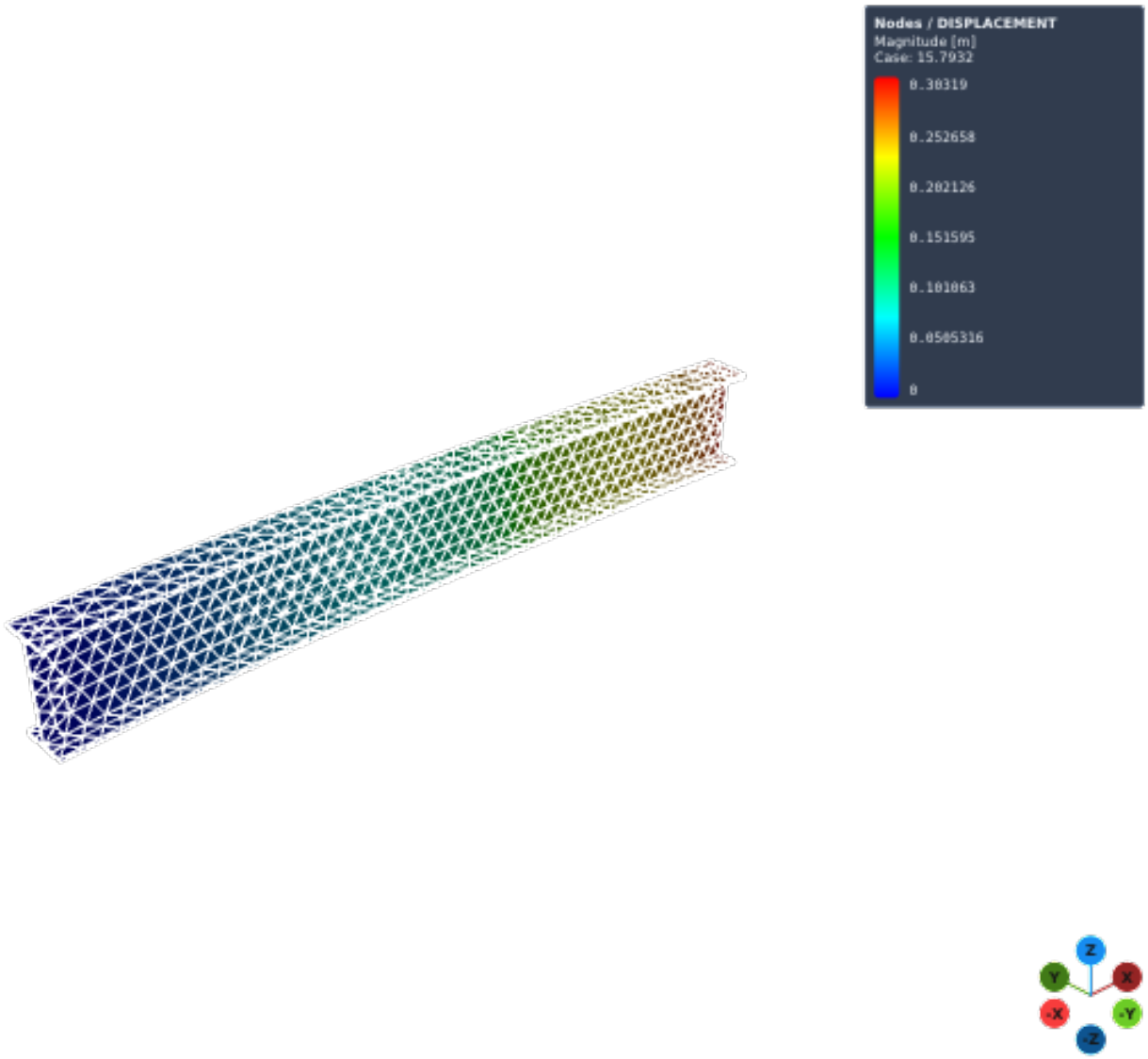

Figure 306: Sesam — mode 1, 15.793 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	15.790	-0.44	9842.9	2.50e-14	117.47	2.82e-04	1.58e-07	10.838	-0.017
Calculix	15.928	+0.43	10016.3	1.60e-09	118.18	3.43e-04	-4.00e-05	-10.871	0.019
Code_Aster	15.928	+0.43	10015.3	1.52e-09	118.17	3.43e-04	3.89e-05	10.871	-0.019
Sesam	15.793	-0.42	9846.92	5.82e-13	117.52	2.82e-04	7.63e-07	-10.841	0.017

Mode 2

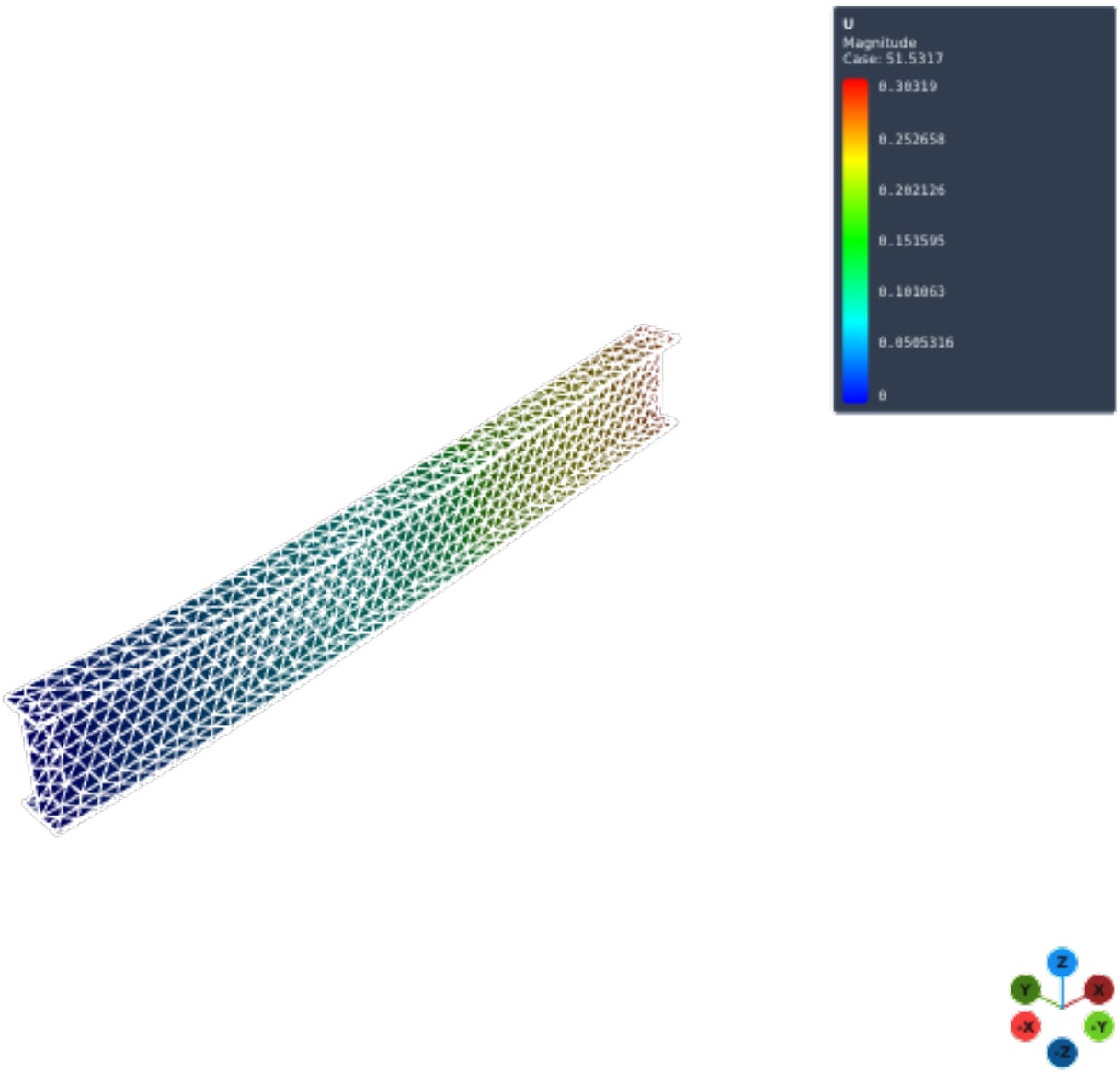

Figure 307: Abaqus — mode 2, 51.532 Hz

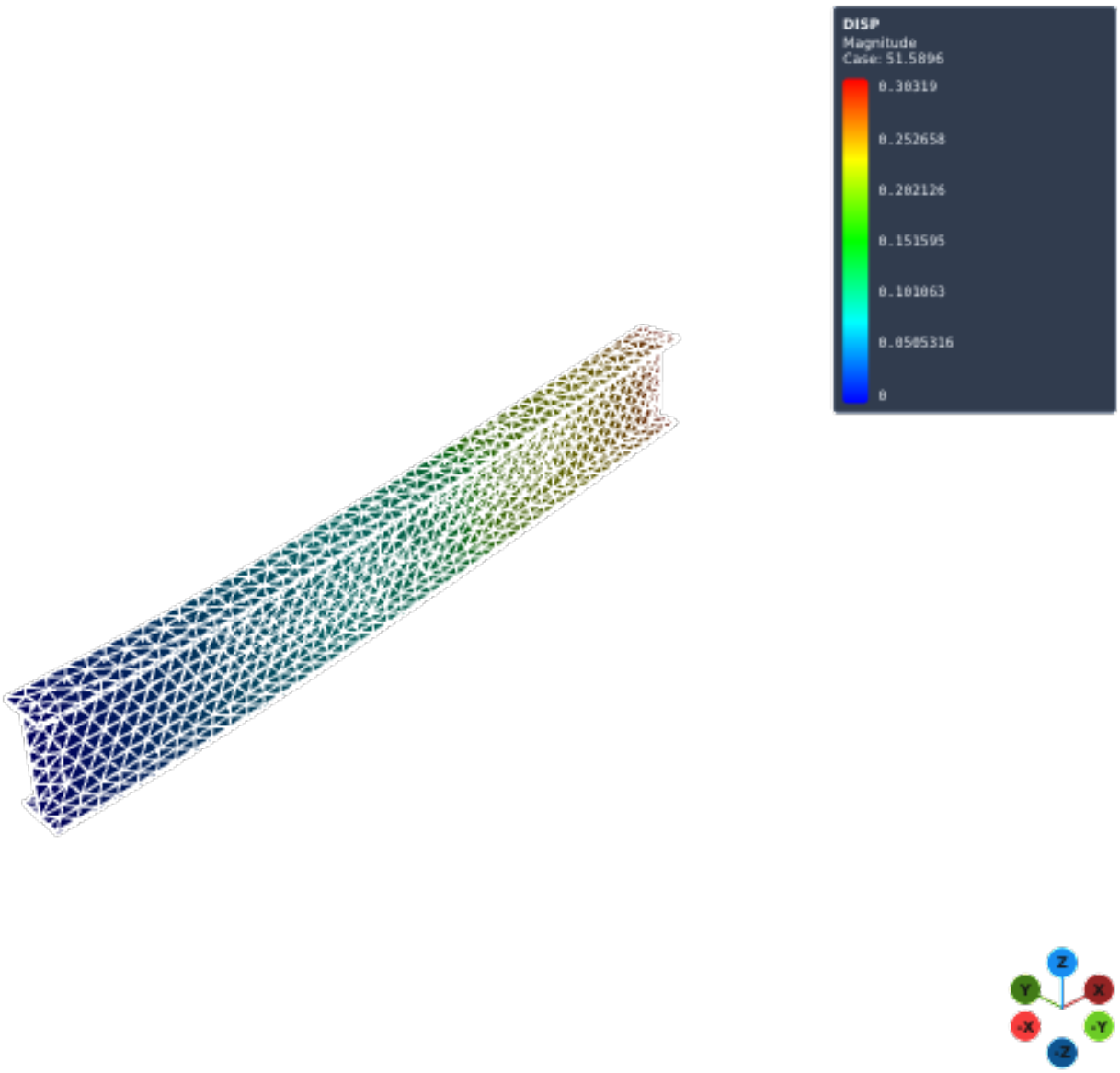

Figure 308: Calculix — mode 2, 51.590 Hz

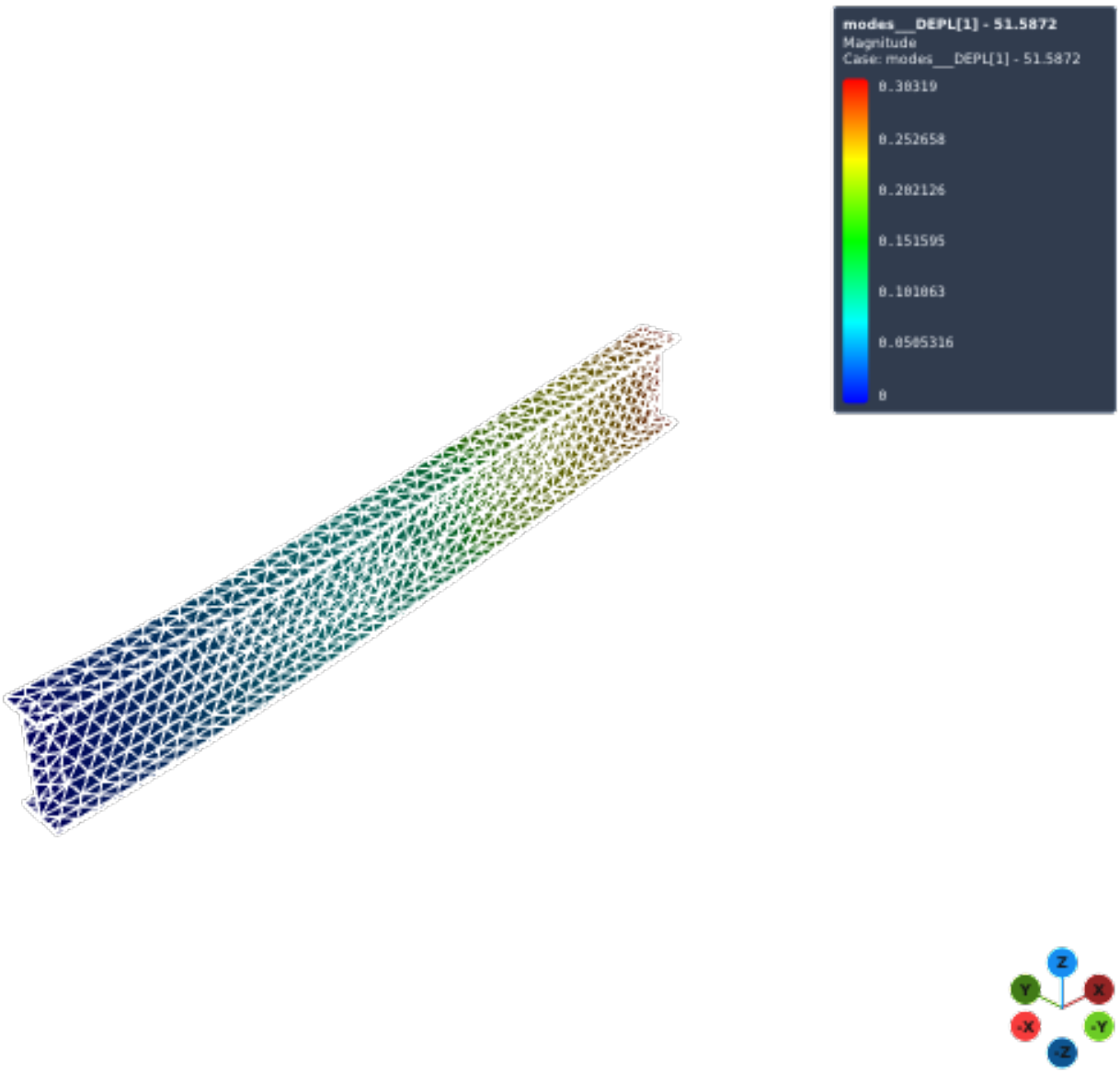

Figure 309: Code_Aster — mode 2, 51.587 Hz

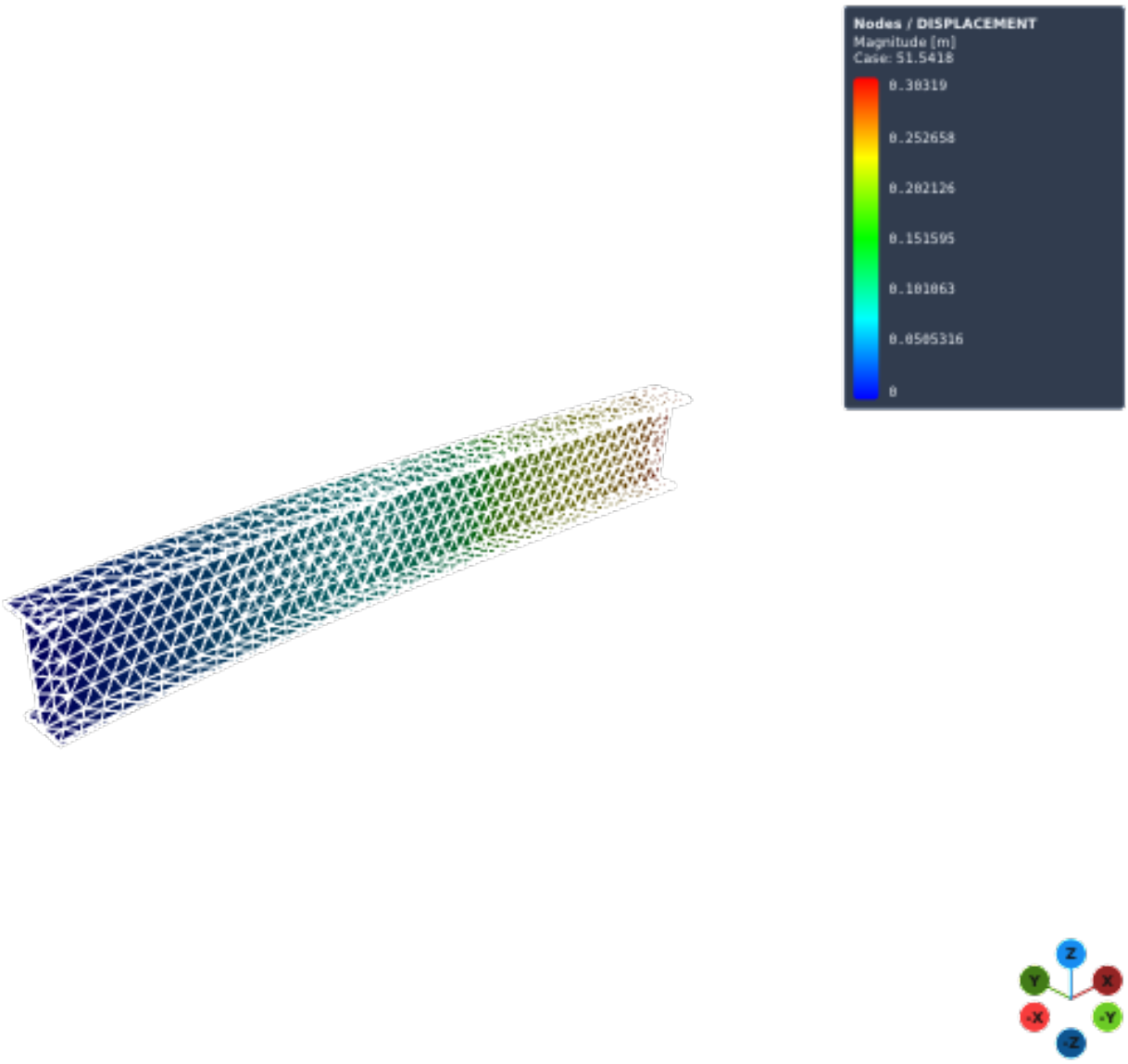

Figure 310: Sesam — mode 2, 51.542 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	51.532	-0.06	104836	1.44e-08	7.90e-04	118.25	1.20e-04	0.028	10.874
Calculix	51.590	+0.05	105071	6.78e-09	9.26e-04	118.38	8.23e-05	0.030	10.880
Code_Aster	51.587	+0.05	105061	7.19e-09	9.26e-04	118.37	8.48e-05	0.030	10.880
Sesam	51.542	-0.04	104877	1.53e-08	7.86e-04	118.29	-1.24e-04	-0.028	-10.876

Mode 3

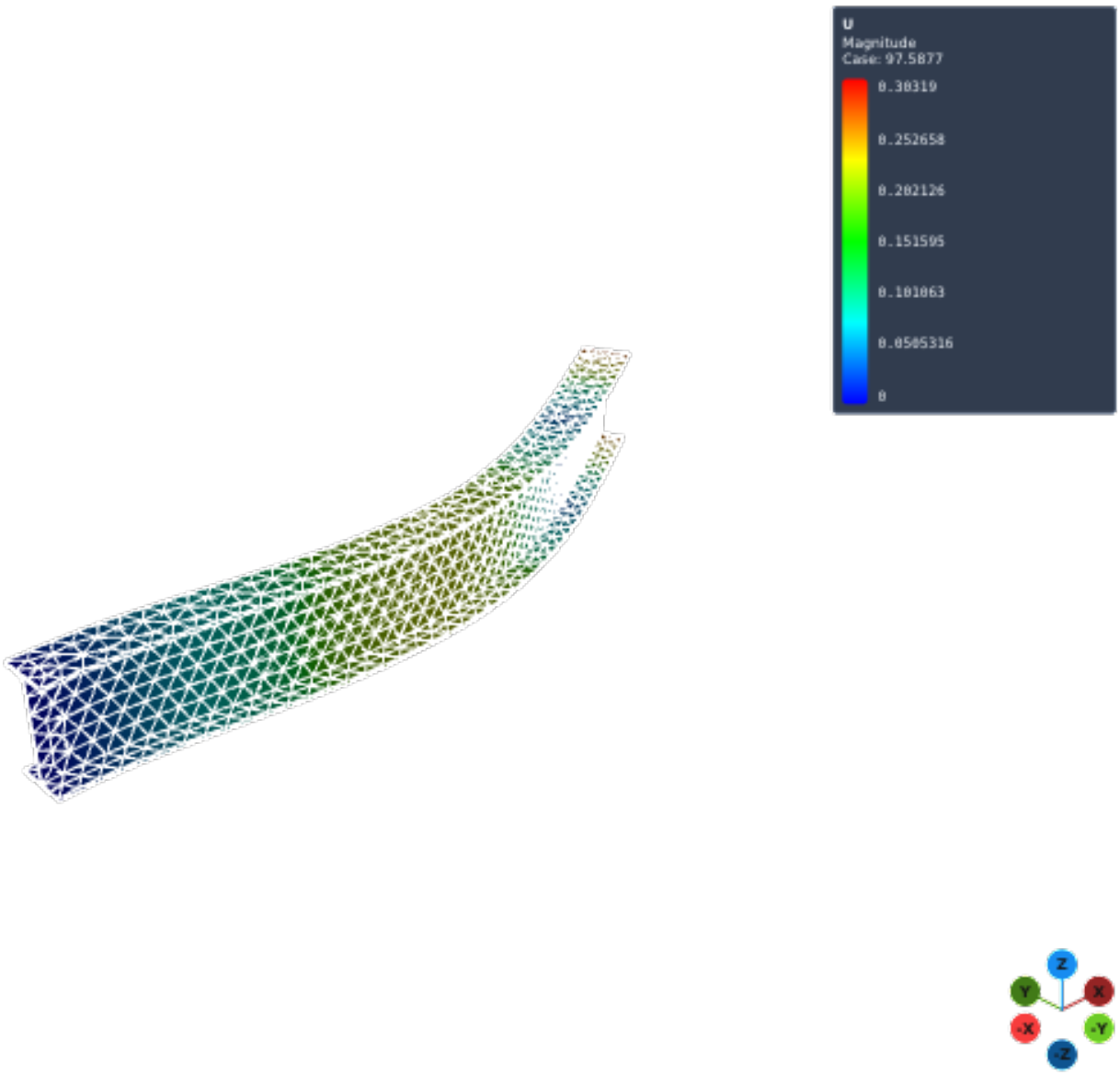

Figure 311: Abaqus — mode 3, 97.588 Hz

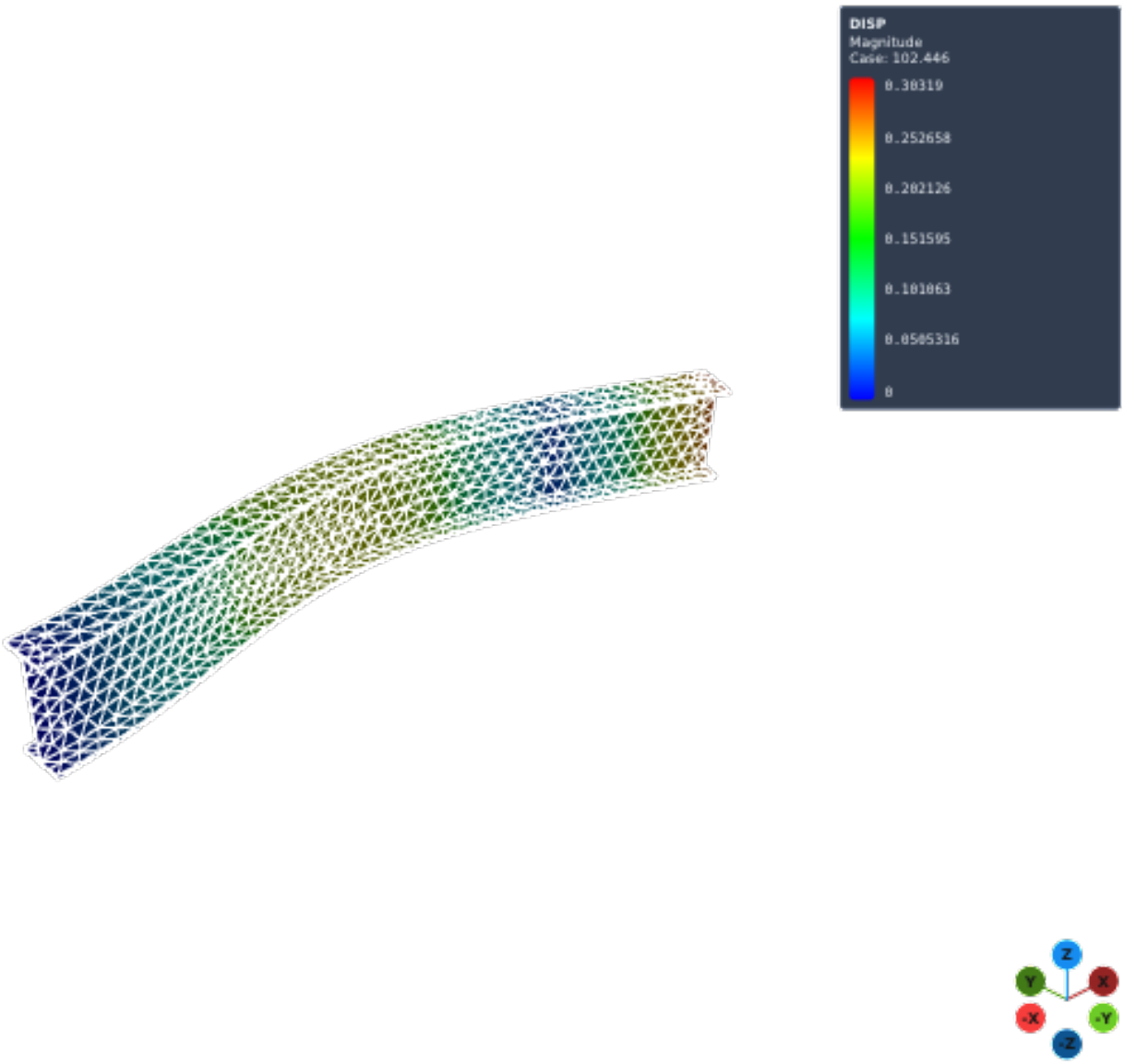

Figure 312: Calculix — mode 3, 102.446 Hz

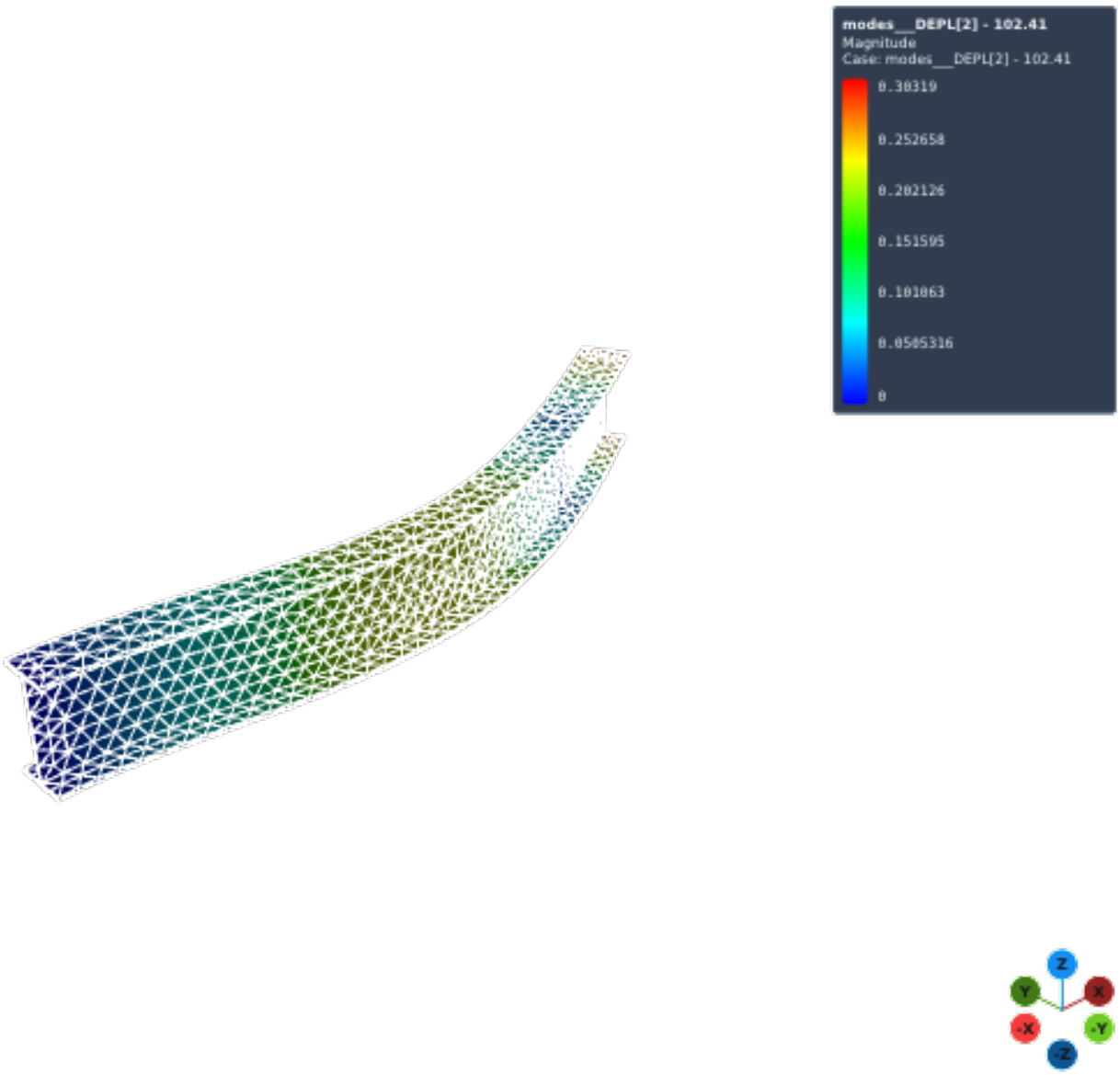

Figure 313: Code_Aster — mode 3, 102.410 Hz

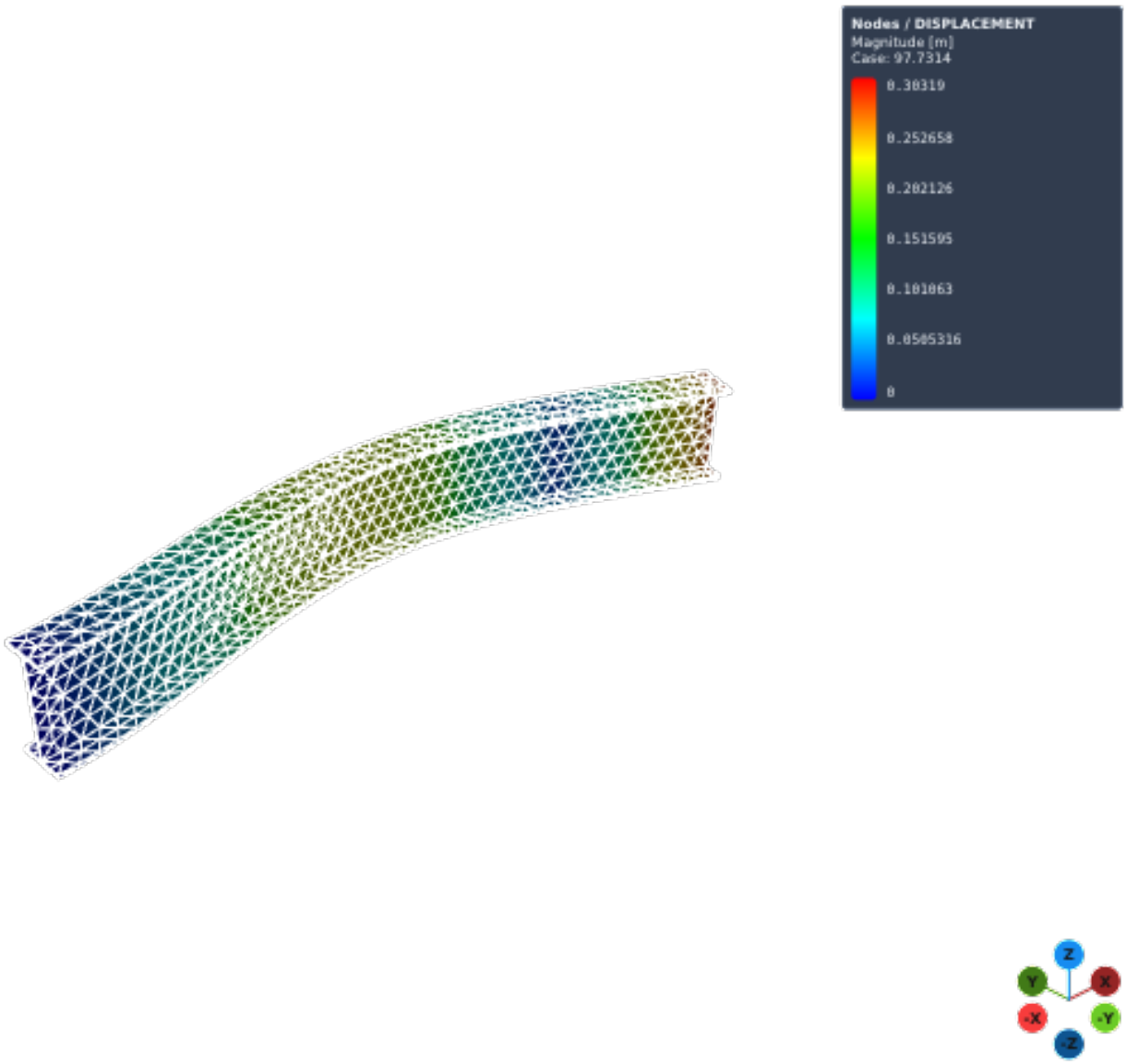

Figure 314: Sesam — mode 3, 97.731 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	97.588	-2.45	375967	4.61e-08	34.91	3.71e-04	-2.15e-04	-5.909	0.019
Calculix	102.446	+2.40	414334	1.14e-07	35.54	4.01e-04	3.38e-04	5.961	-0.020
Code_Aster	102.410	+2.37	414043	1.10e-07	35.54	3.99e-04	-3.32e-04	-5.961	0.020
Sesam	97.731	-2.31	377075	4.30e-08	34.94	3.65e-04	2.07e-04	5.911	-0.019

Mode 4

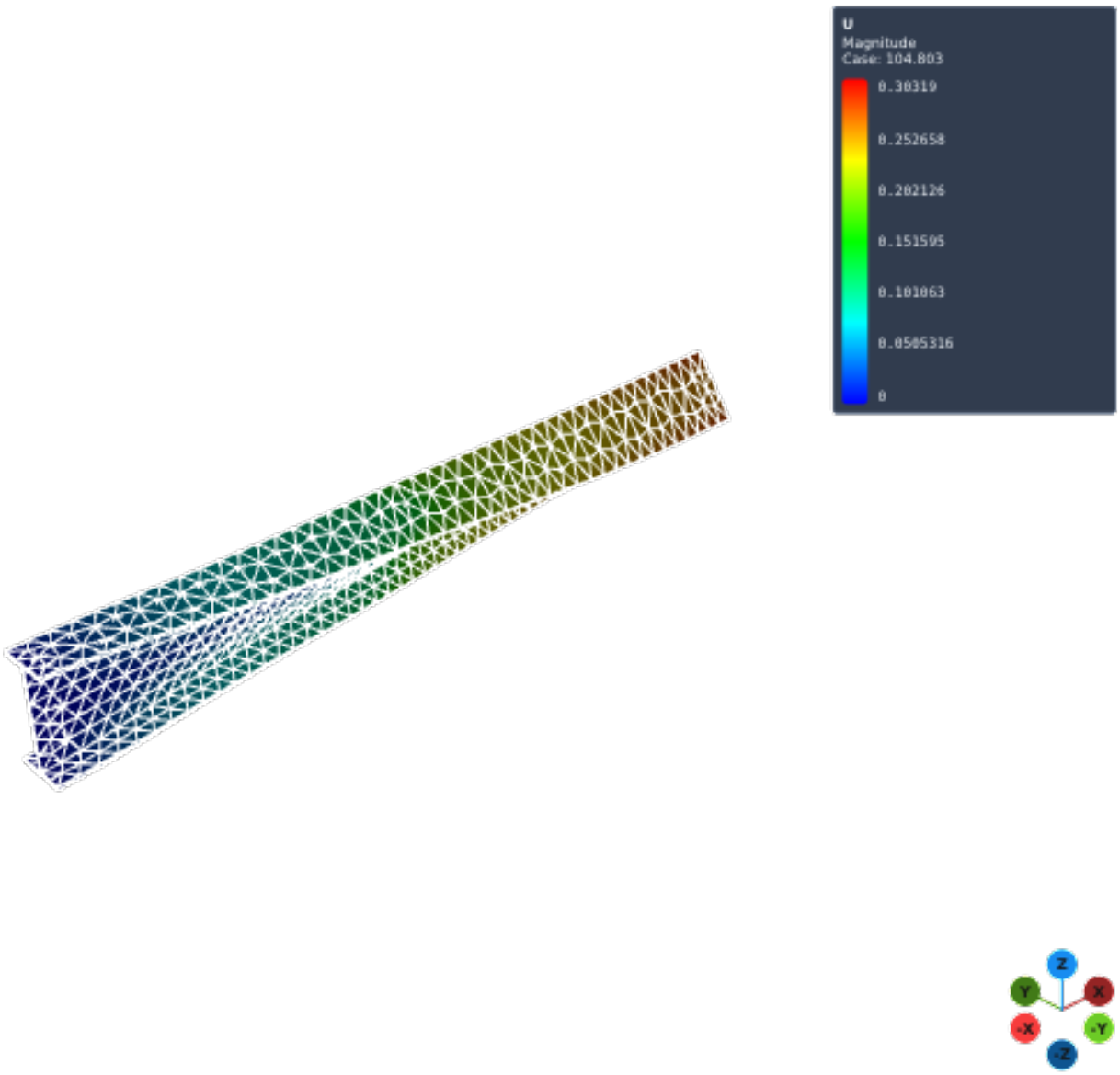

Figure 315: Abaqus — mode 4, 104.803 Hz

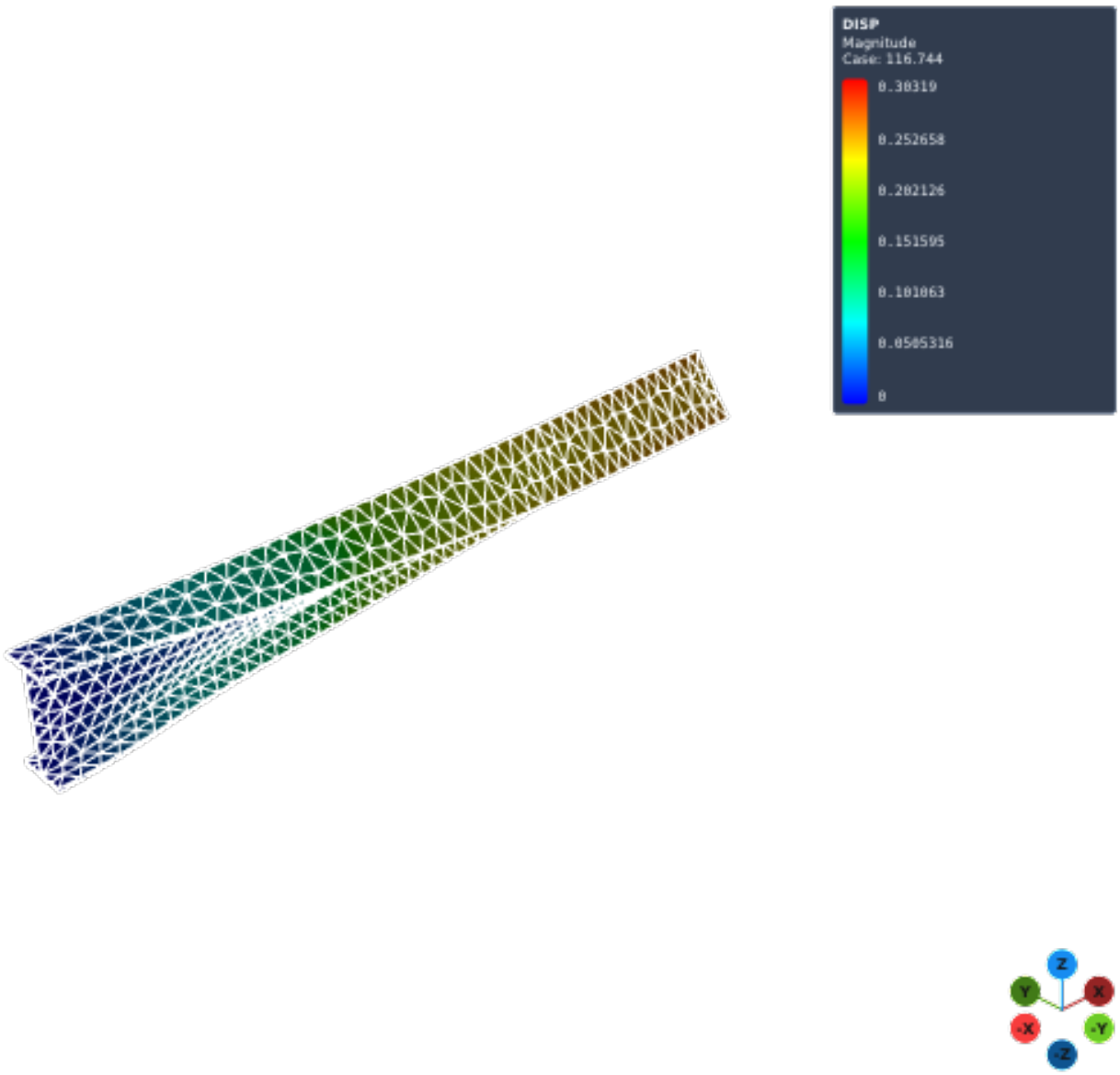

Figure 316: Calculix — mode 4, 116.744 Hz

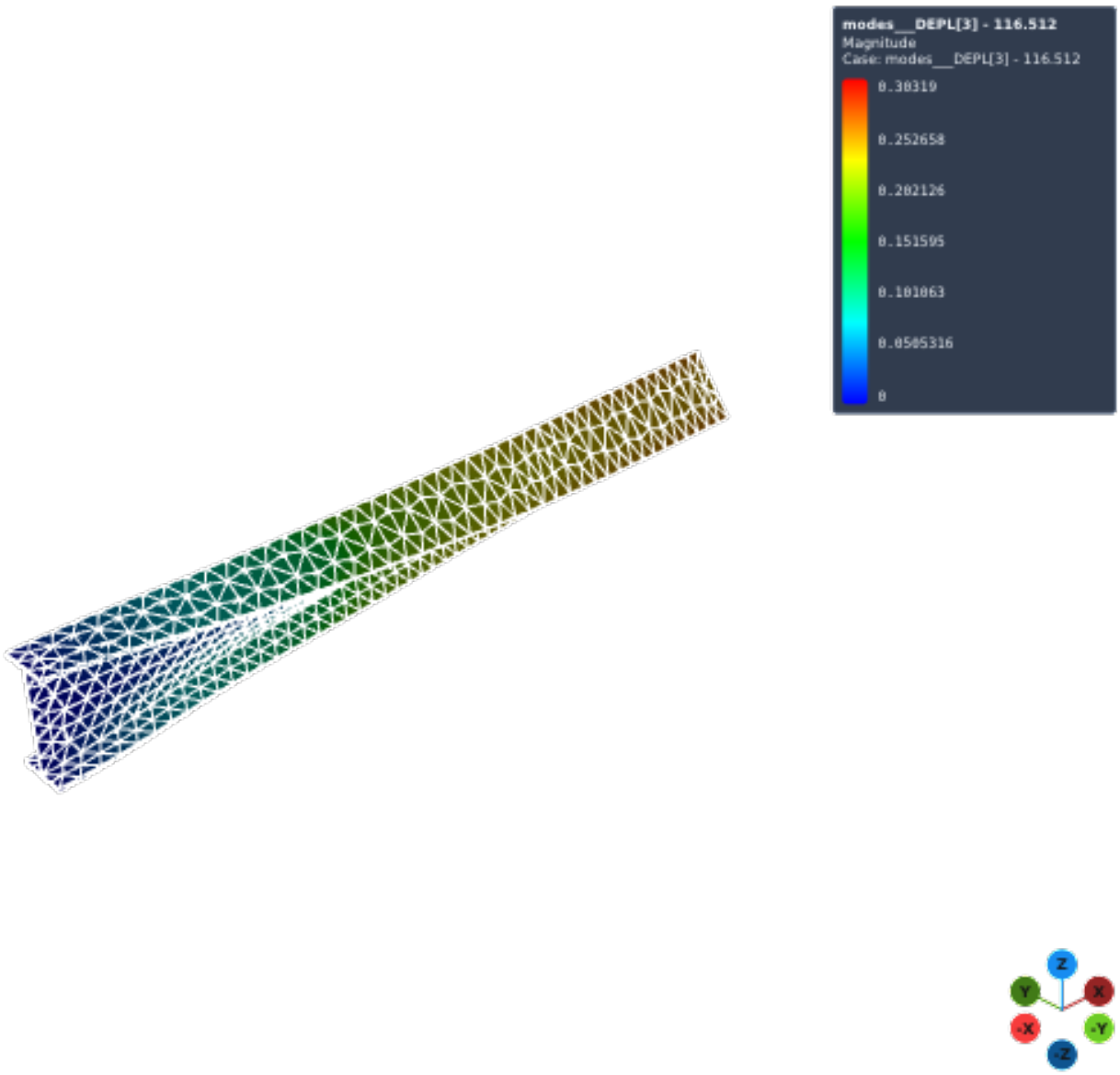

Figure 317: Code_Aster — mode 4, 116.512 Hz

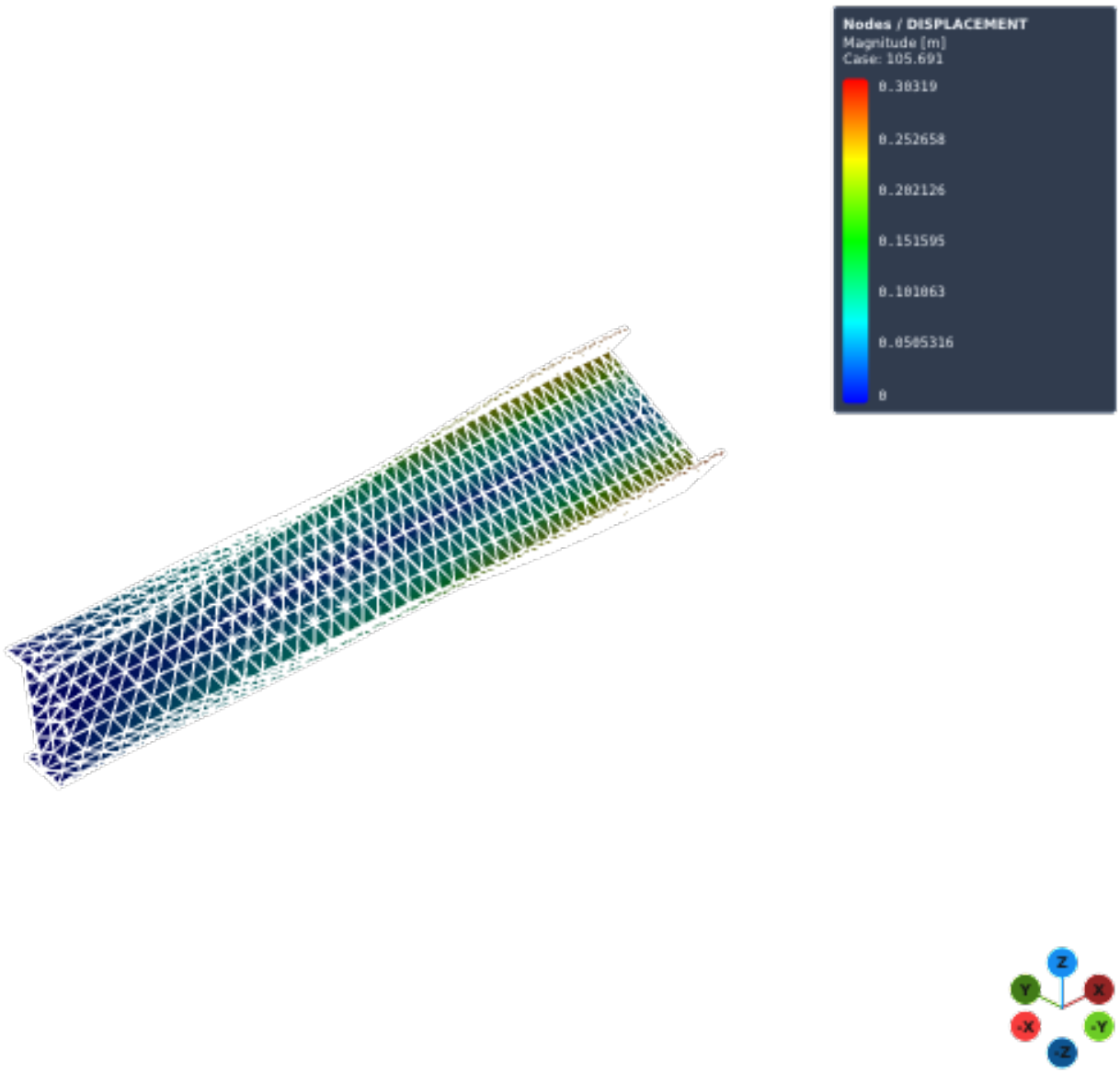

Figure 318: Sesam — mode 4, 105.691 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	104.800	-5.53	433617	6.83e-06	0.01	1.07e-05	0.003	-0.075	0.003
Calculix	116.744	+5.24	538061	9.63e-06	0.01	5.55e-05	0.003	-0.108	0.007
Code_Aster	116.512	+5.03	535924	9.53e-06	0.01	5.63e-05	0.003	-0.110	0.008
Sesam	105.691	-4.73	440993	6.99e-06	4.82e-03	1.08e-05	-0.003	0.069	-0.003

Mode 5

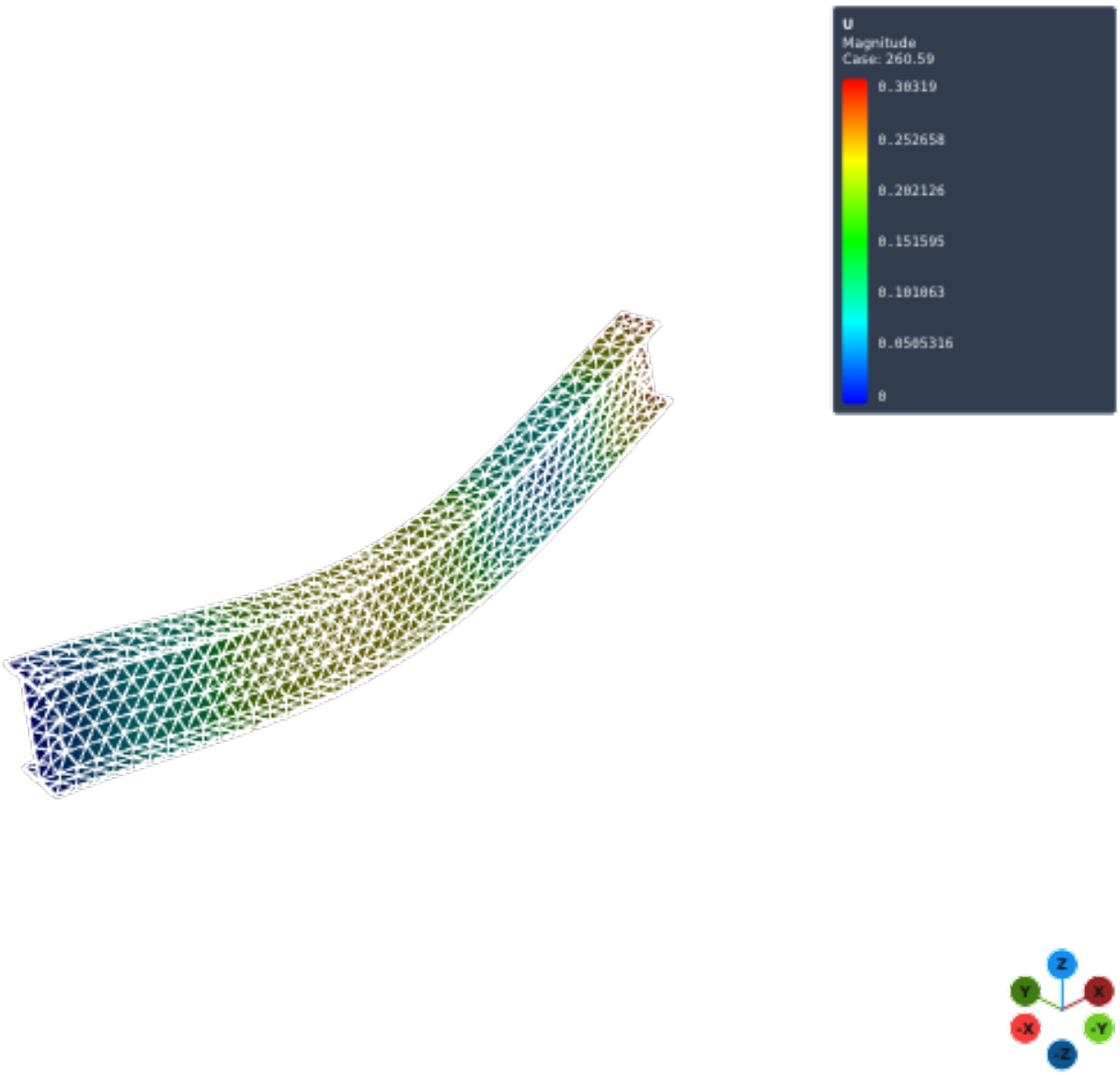

Figure 319: Abaqus — mode 5, 260.590 Hz

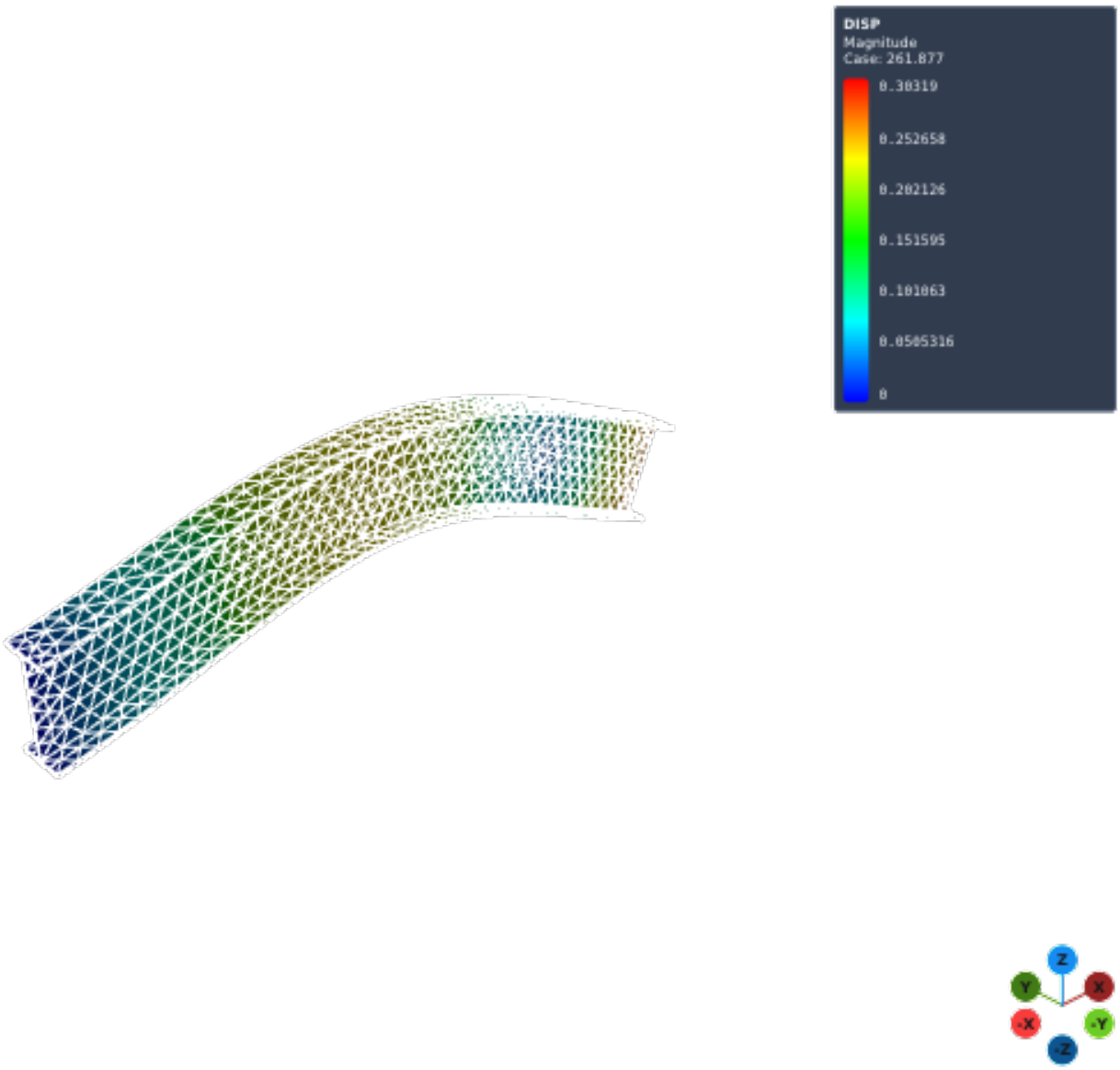

Figure 320: Calculix — mode 5, 261.877 Hz

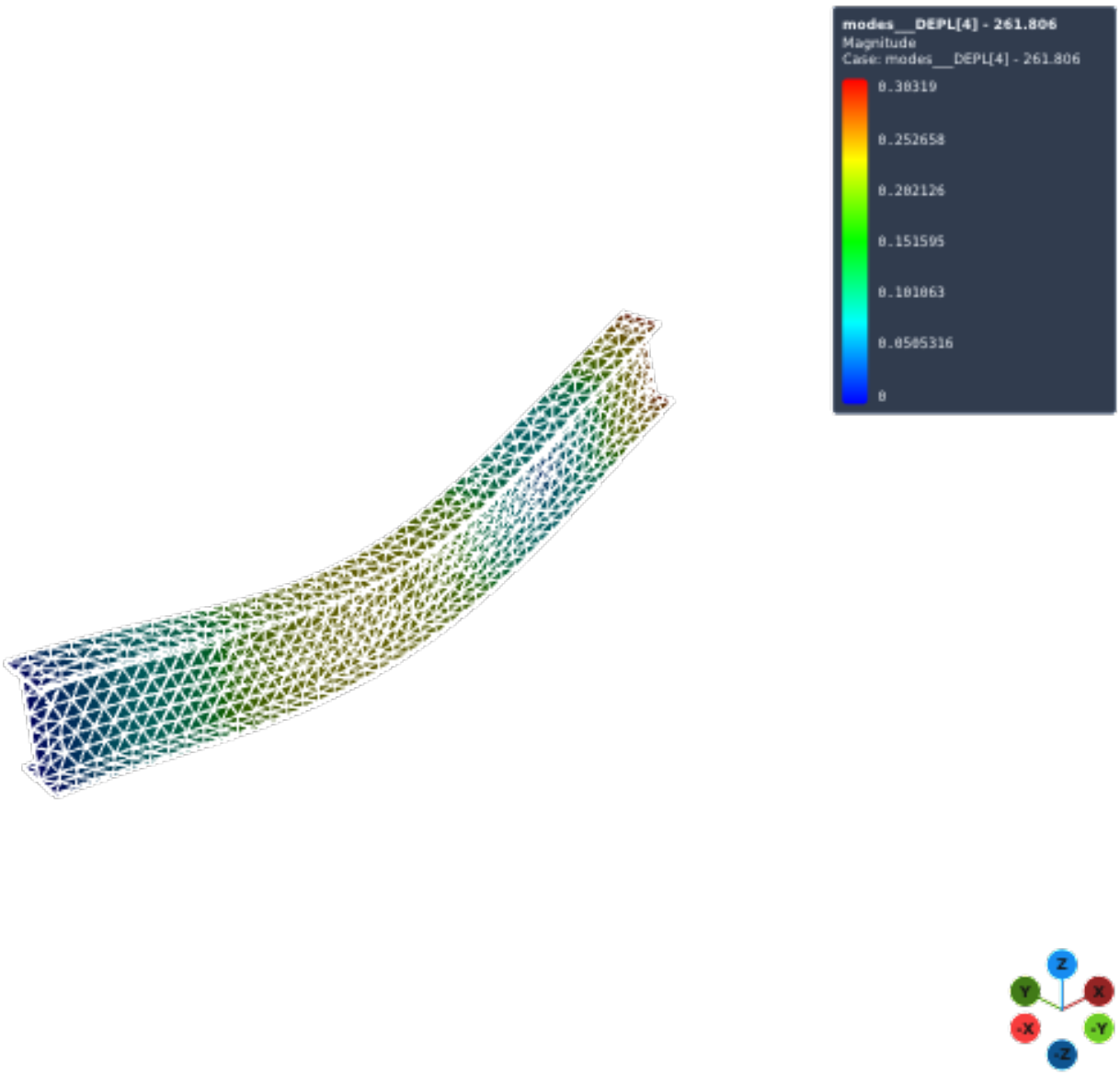

Figure 321: Code_Aster — mode 5, 261.806 Hz

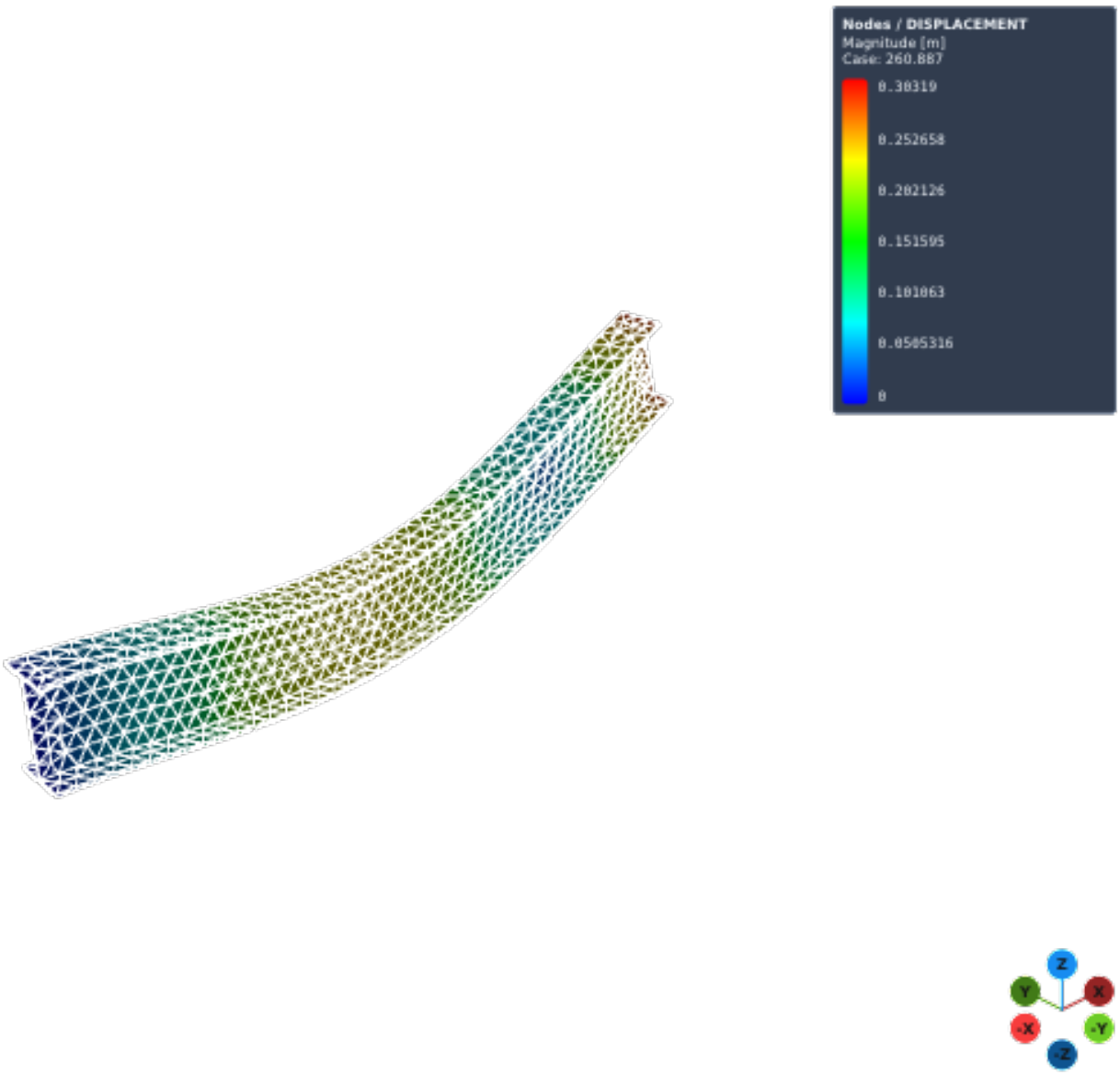

Figure 322: Sesam — mode 5, 260.887 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	260.590	-0.27	2.68086e+06	3.30e-07	0.02	39.75	-0.001	0.132	-6.305
Calculix	261.877	+0.22	2.70742e+06	9.91e-08	7.99e-04	39.86	3.15e-04	-0.028	6.313
Code_Aster	261.806	+0.20	2.70594e+06	1.05e-07	8.24e-04	39.88	-3.24e-04	0.029	-6.315
Sesam	260.887	-0.15	2.68699e+06	3.61e-07	0.01	39.76	-0.001	0.122	-6.305

Mode 6

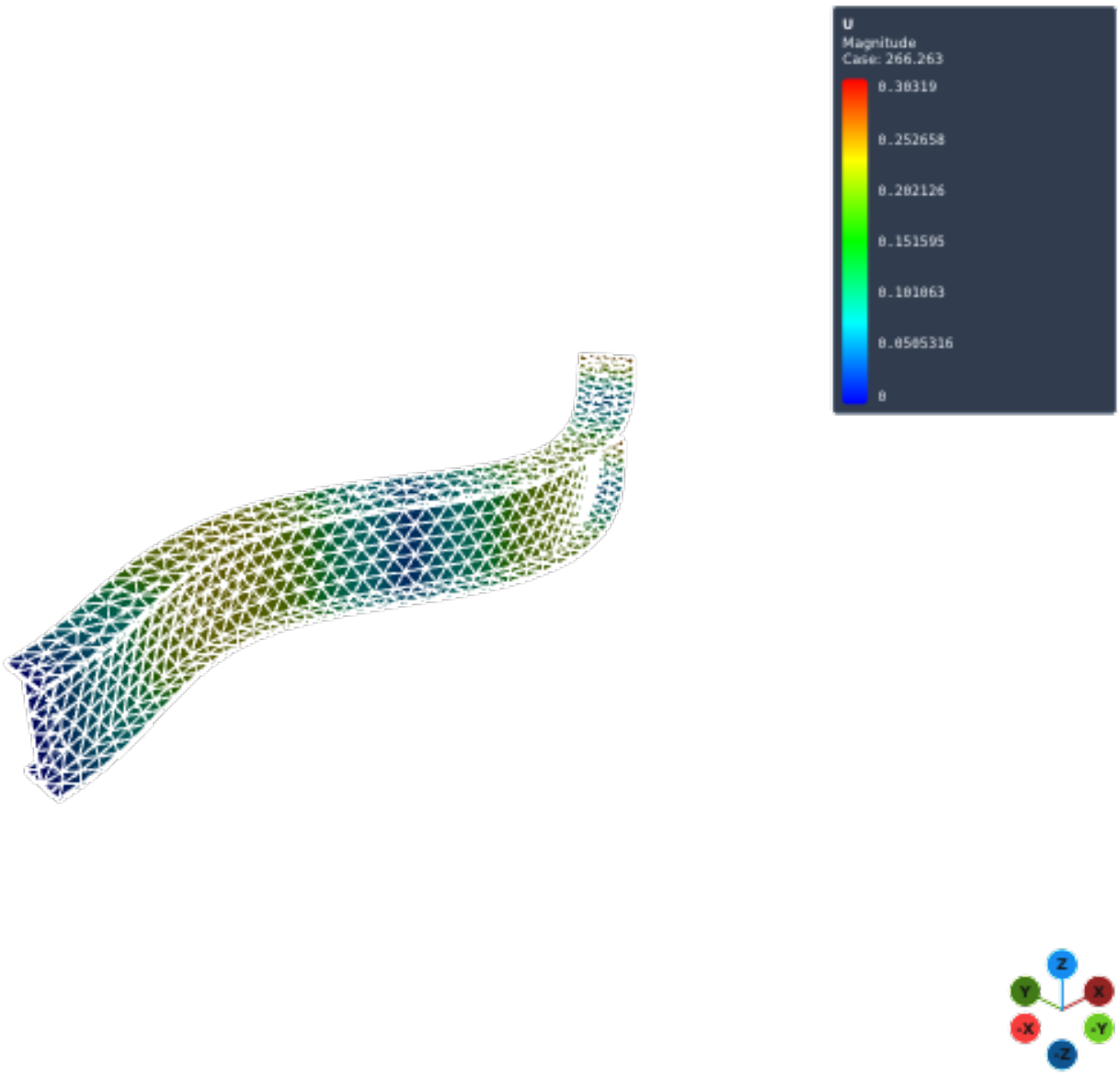

Figure 323: Abaqus — mode 6, 266.263 Hz

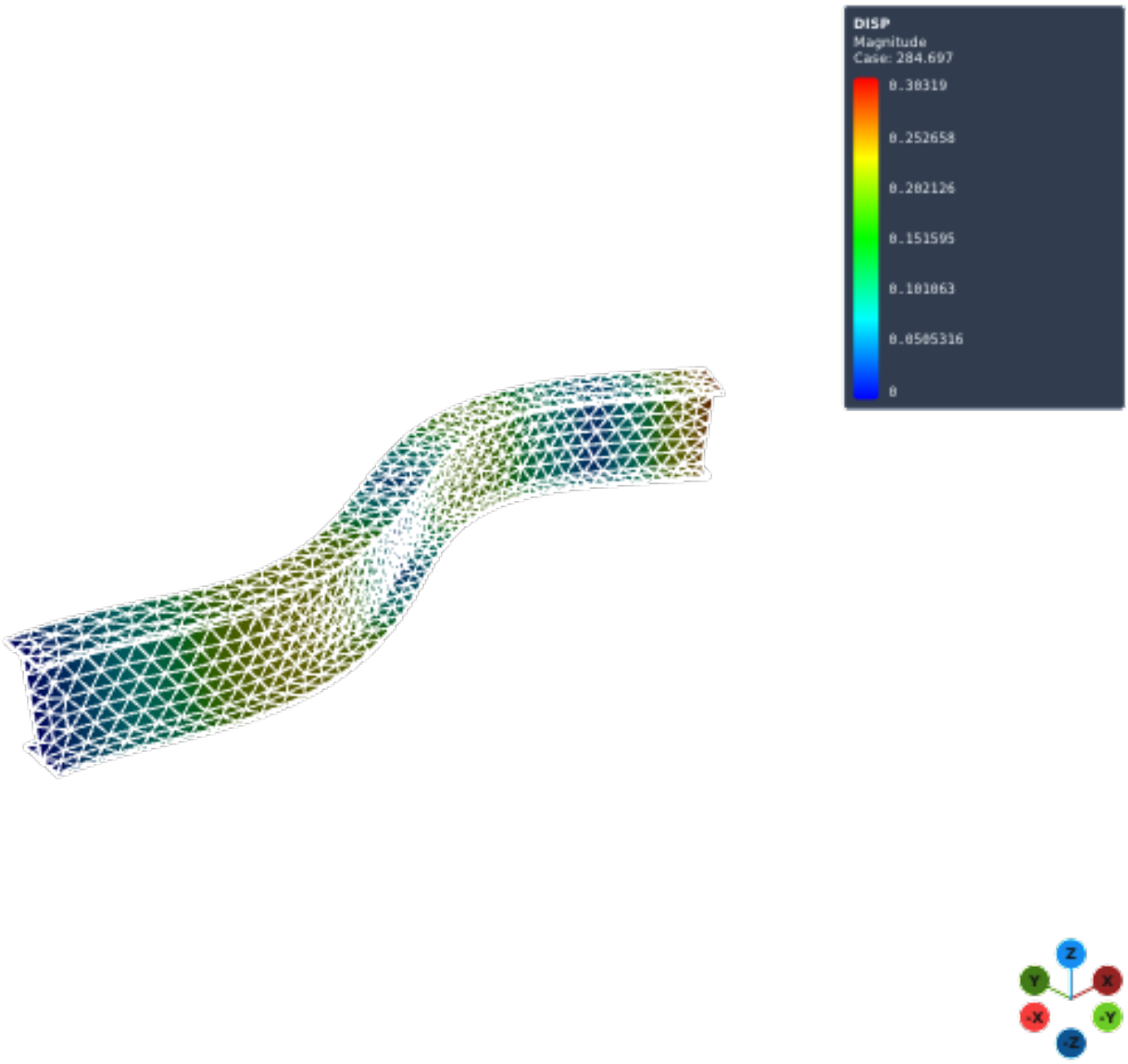

Figure 324: Calculix — mode 6, 284.697 Hz

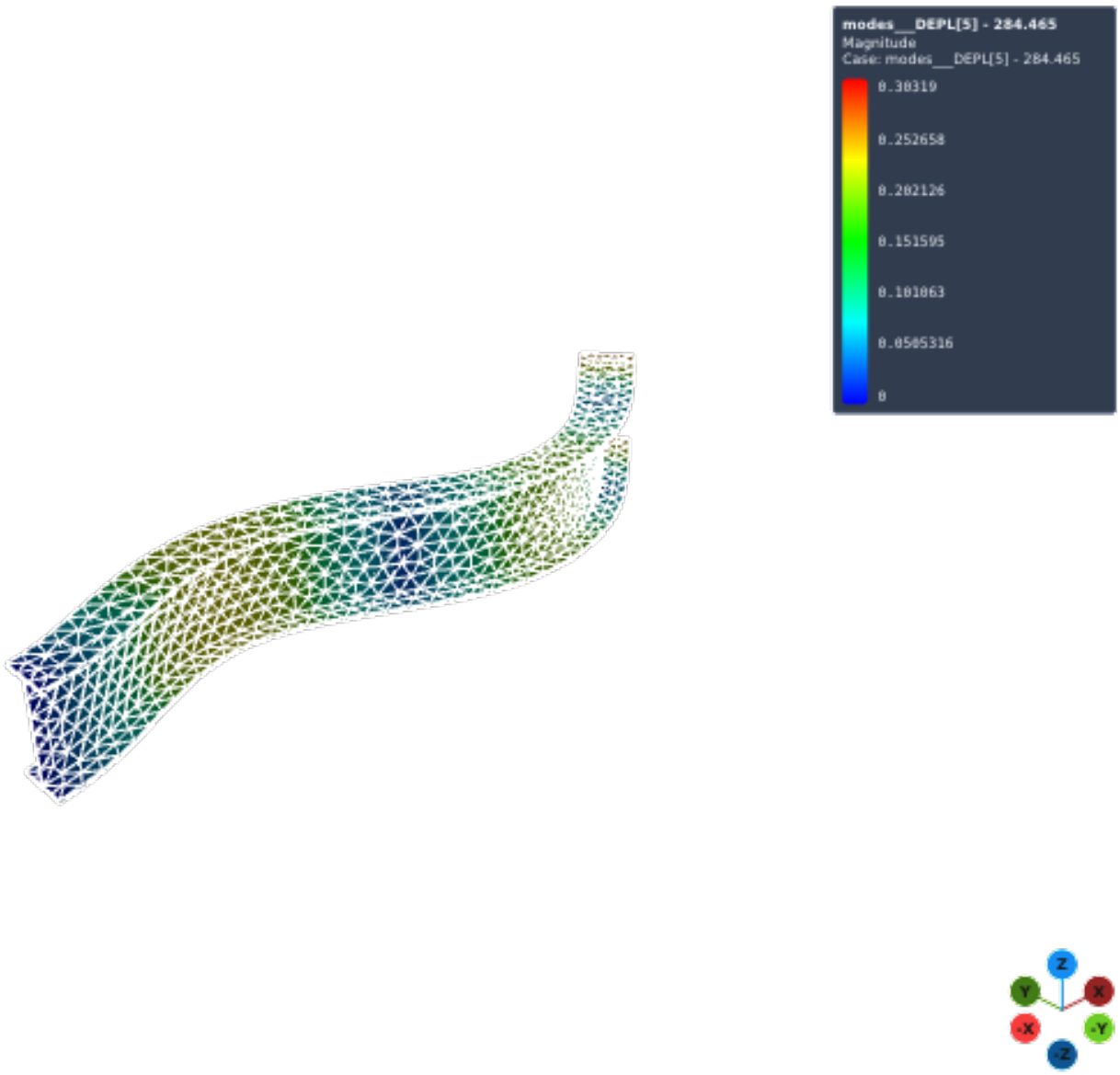

Figure 325: Code_Aster — mode 6, 284.465 Hz

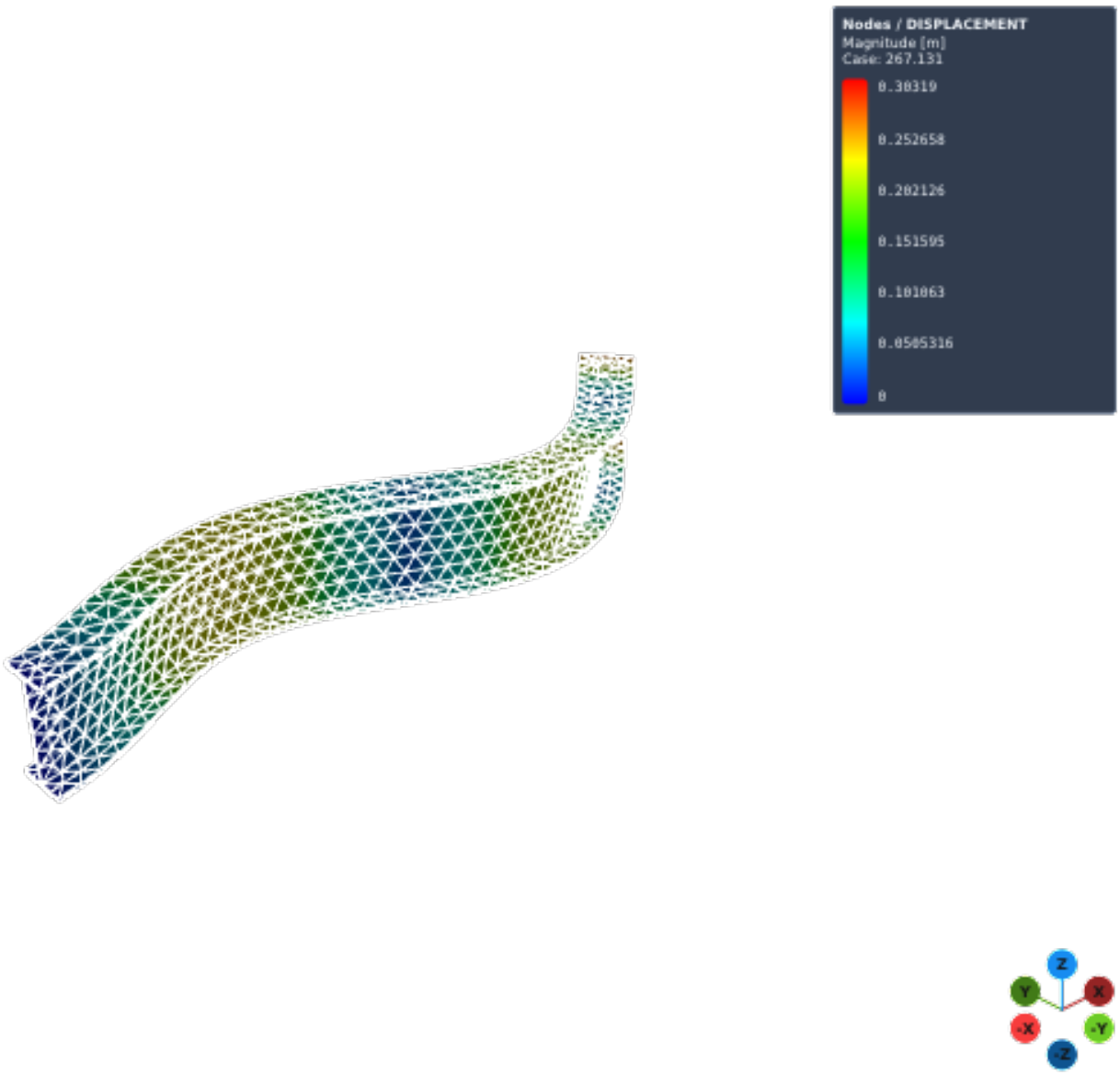

Figure 326: Sesam — mode 6, 267.131 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	266.260	-3.40	2.79886e+06	1.72e-08	13.24	0.05	1.31e-04	3.638	0.226
Calculix	284.697	+3.29	3.19983e+06	2.41e-07	12.79	3.17e-03	4.91e-04	-3.576	-0.056
Code_Aster	284.465	+3.20	3.19461e+06	2.62e-07	12.80	3.27e-03	-0.001	3.577	0.057
Sesam	267.131	-3.09	2.81714e+06	1.49e-08	13.26	0.04	1.22e-04	3.642	0.207

Mode 7

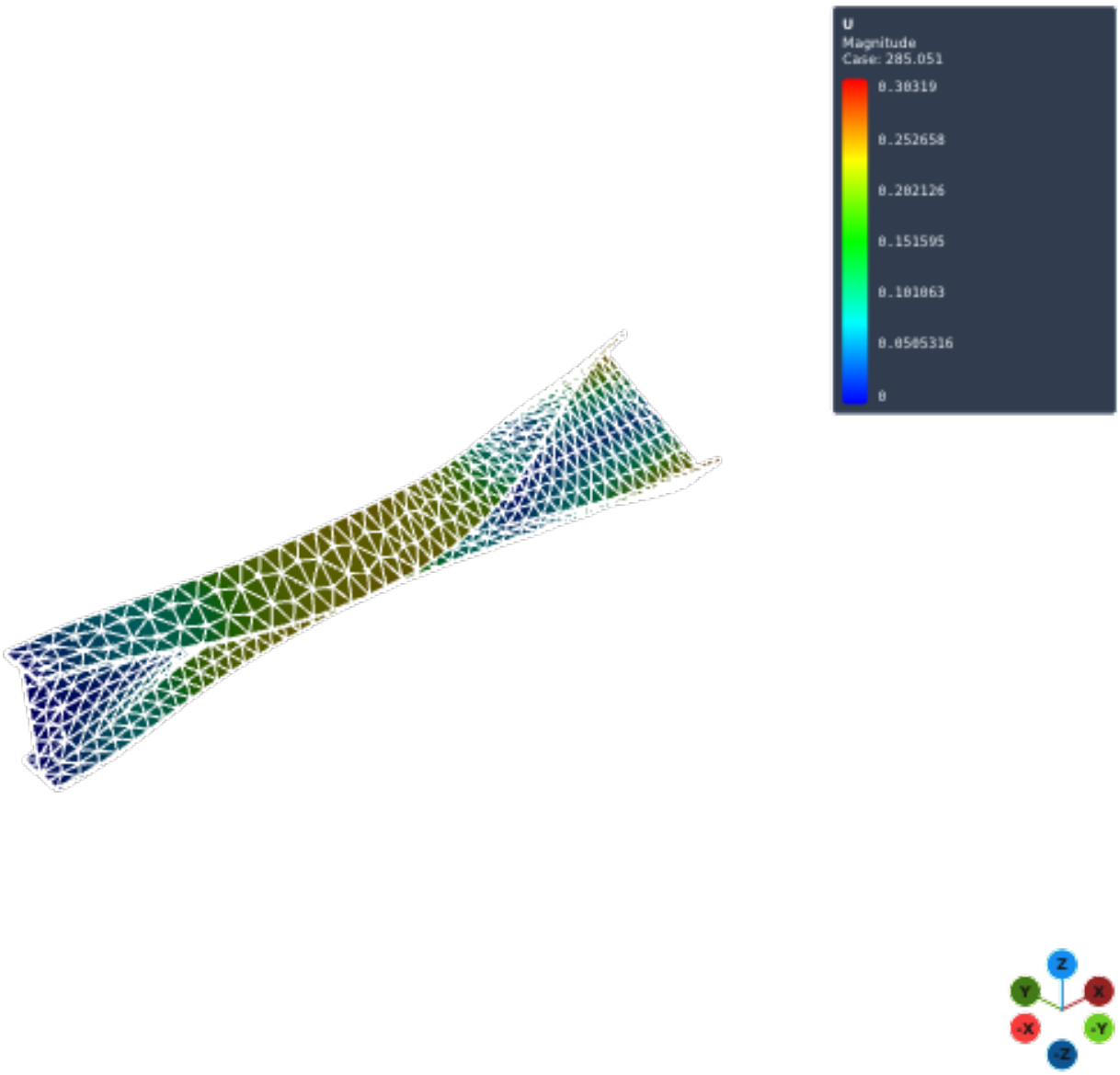

Figure 327: Abaqus — mode 7, 285.051 Hz

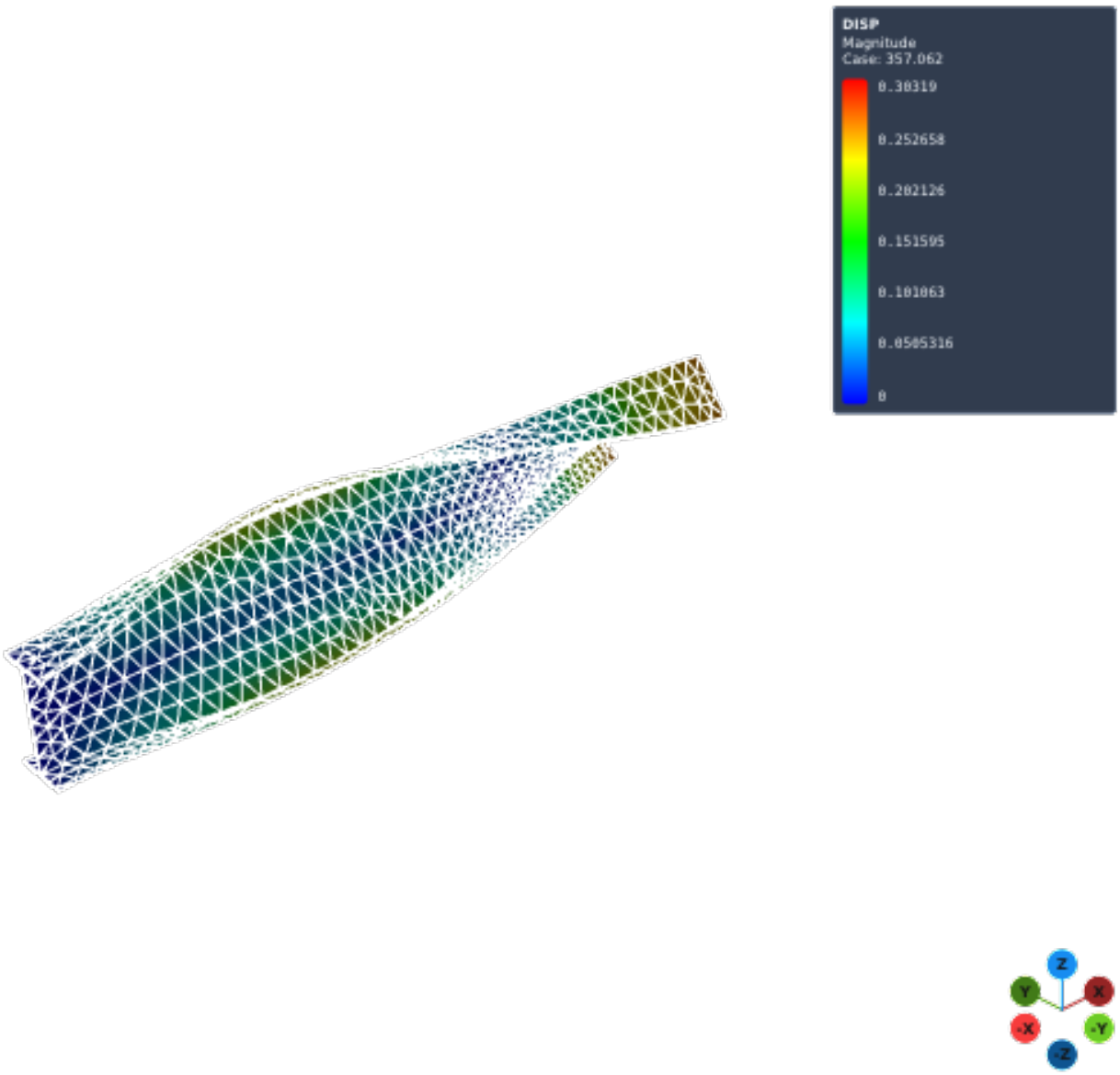

Figure 328: Calculix — mode 7, 357.062 Hz

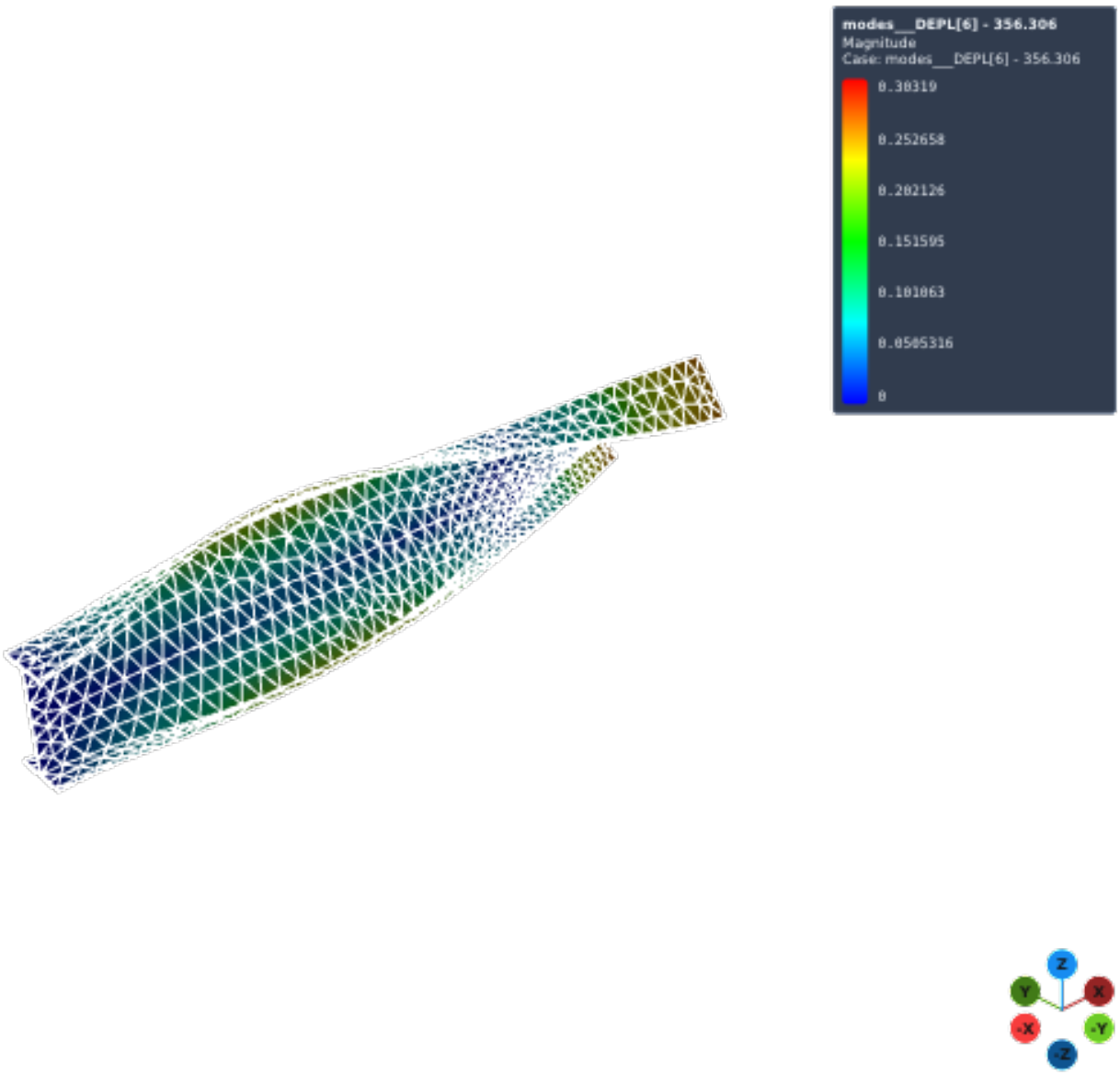

Figure 329: Code_Aster — mode 7, 356.306 Hz

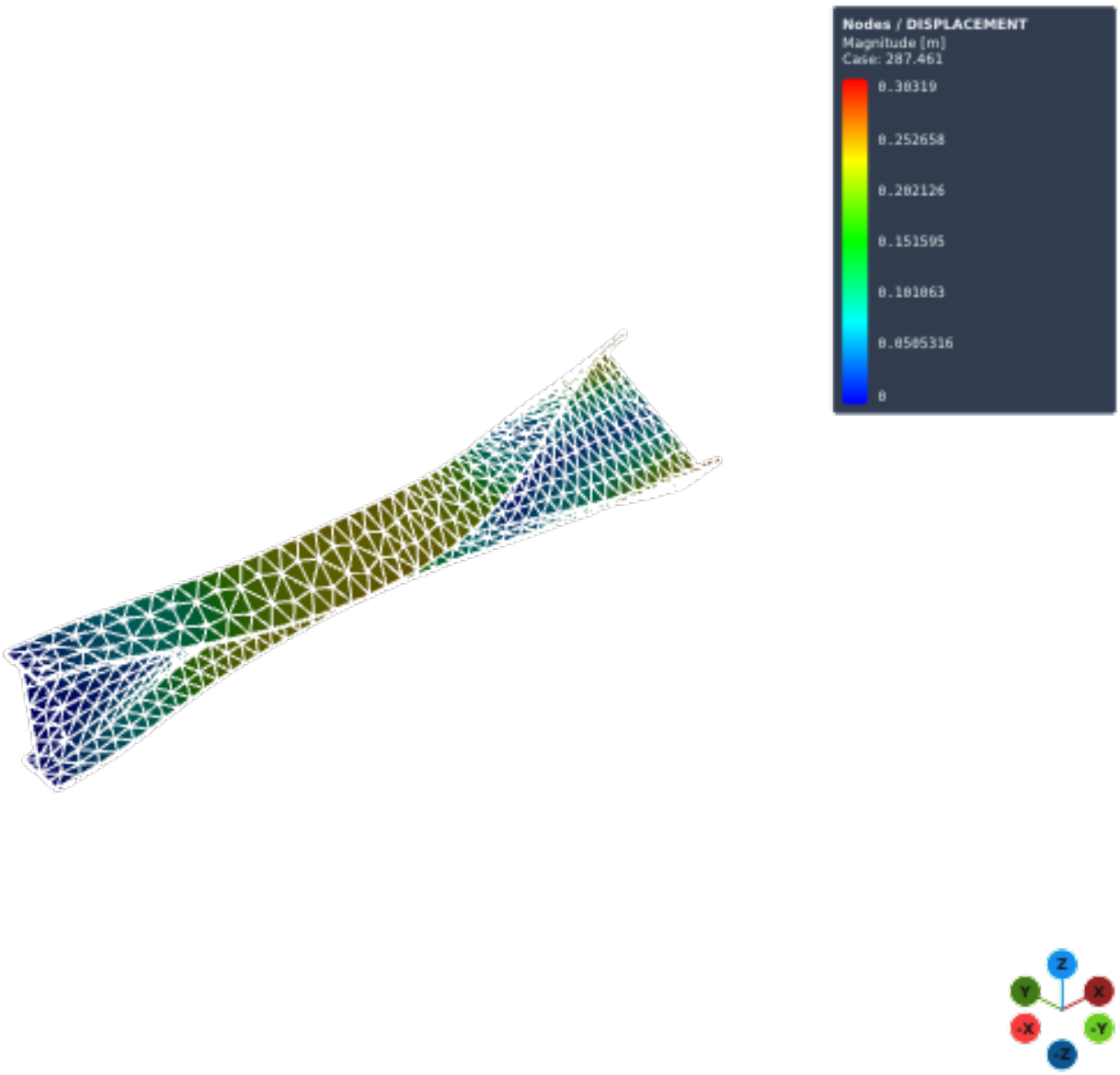

Figure 330: Sesam — mode 7, 287.461 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	285.050	-11.33	3.20779e+06	1.44e-07	9.28e-04	1.25e-03	3.79e-04	0.030	-0.035
Calculix	357.062	+11.07	5.03322e+06	4.30e-04	1.56e-03	3.86e-04	-0.021	0.040	-0.020
Code_Aster	356.306	+10.84	5.01194e+06	4.21e-04	1.60e-03	3.90e-04	-0.021	0.040	-0.020
Sesam	287.461	-10.58	3.26225e+06	1.26e-07	7.47e-04	1.07e-03	3.54e-04	0.027	-0.033

Mode 8

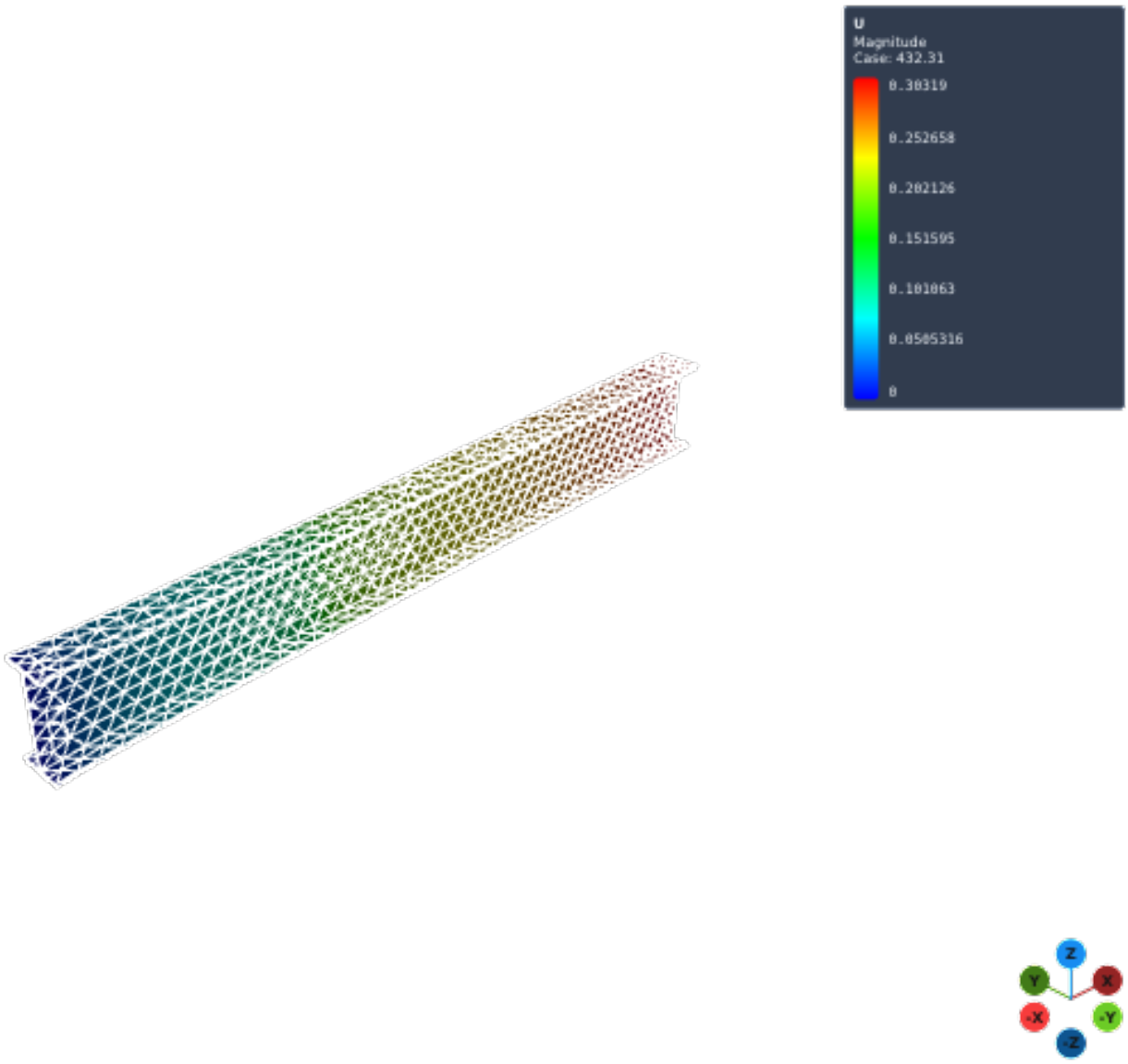

Figure 331: Abaqus — mode 8, 432.310 Hz

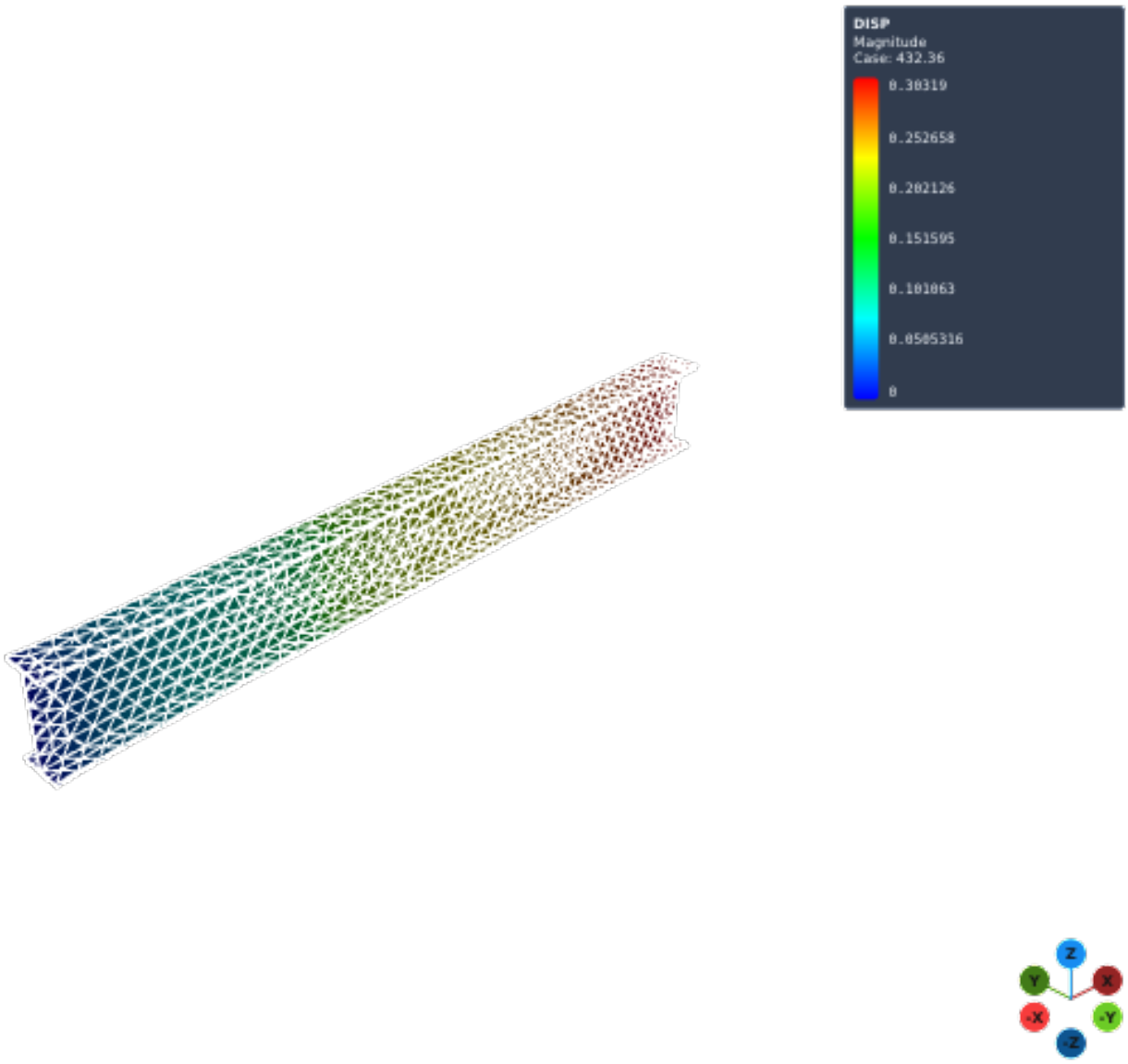

Figure 332: Calculix — mode 8, 432.360 Hz

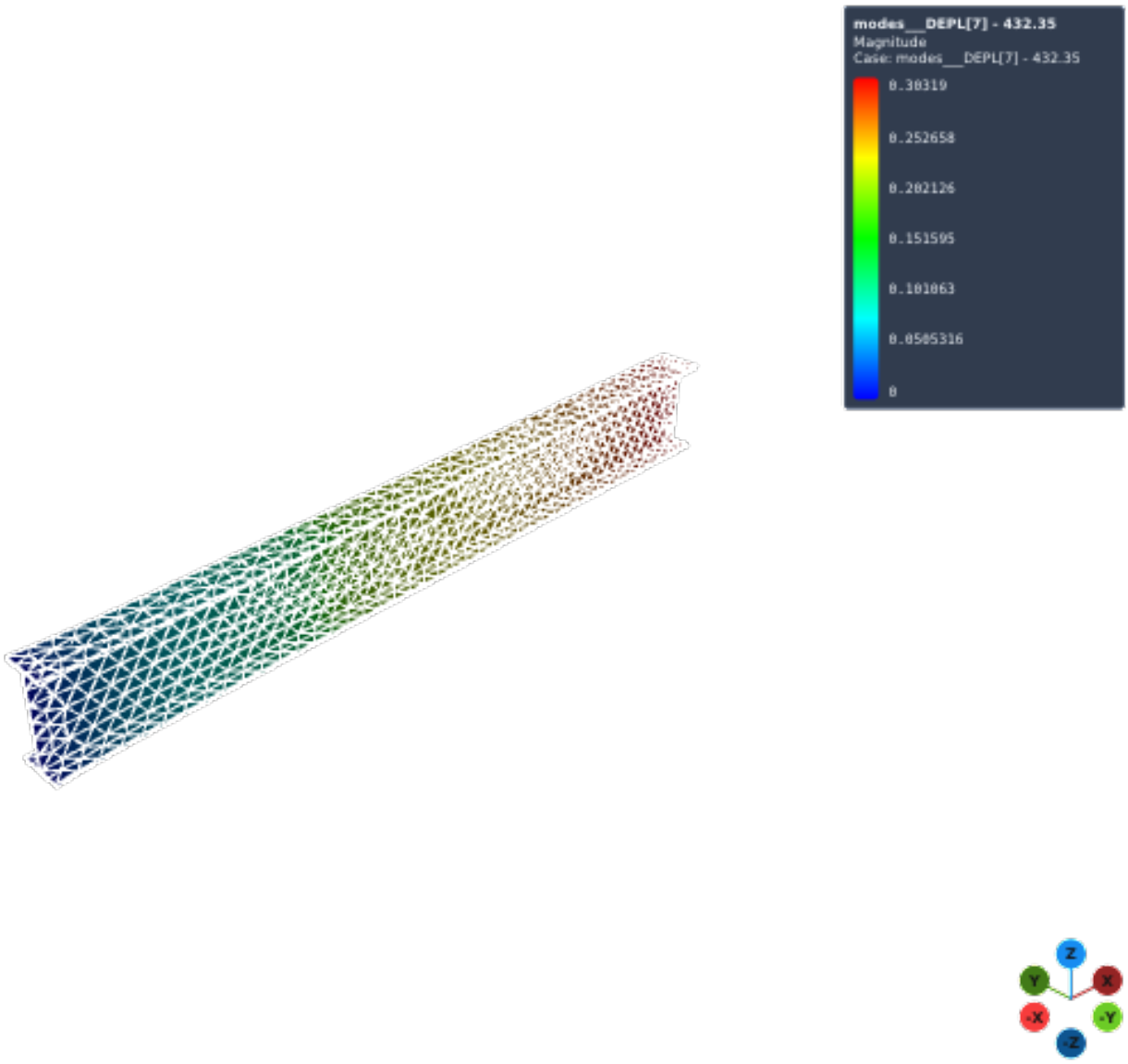

Figure 333: Code_Aster — mode 8, 432.350 Hz

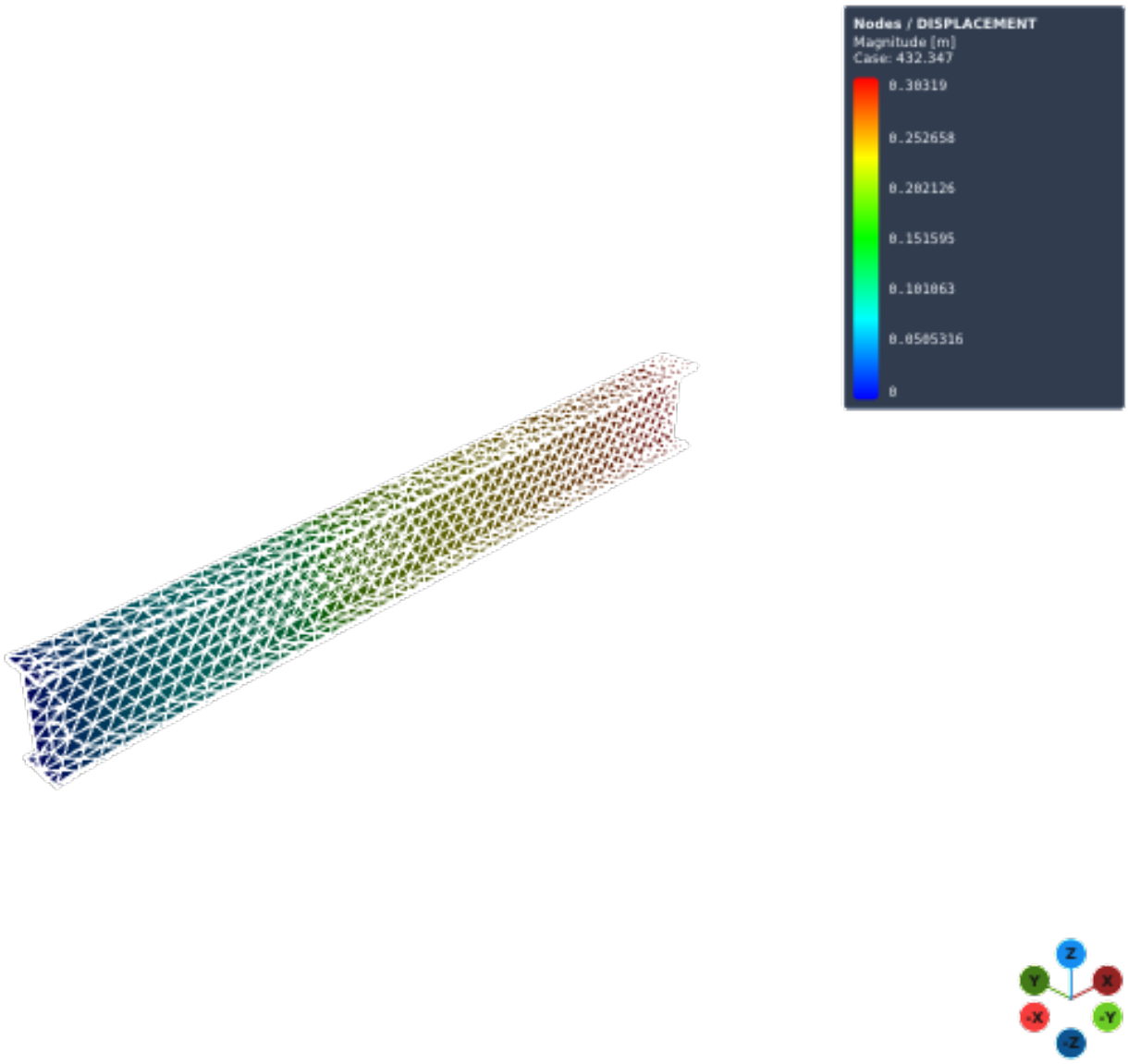

Figure 334: Sesam — mode 8, 432.347 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	432.310	-0.01	7.37819e+06	153.38	2.06e-06	3.31e-08	12.385	-0.001	1.82e-04
Calculix	432.360	4.12e-03	7.37989e+06	153.38	9.52e-08	2.76e-08	12.384	3.09e-04	-1.66e-04
Code_Aster	432.350	1.96e-03	7.37957e+06	153.41	9.68e-08	2.69e-08	12.386	3.11e-04	-1.64e-04
Sesam	432.347	1.27e-03	7.37947e+06	153.37	1.91e-06	2.88e-08	12.384	-0.001	1.70e-04

Mode 9

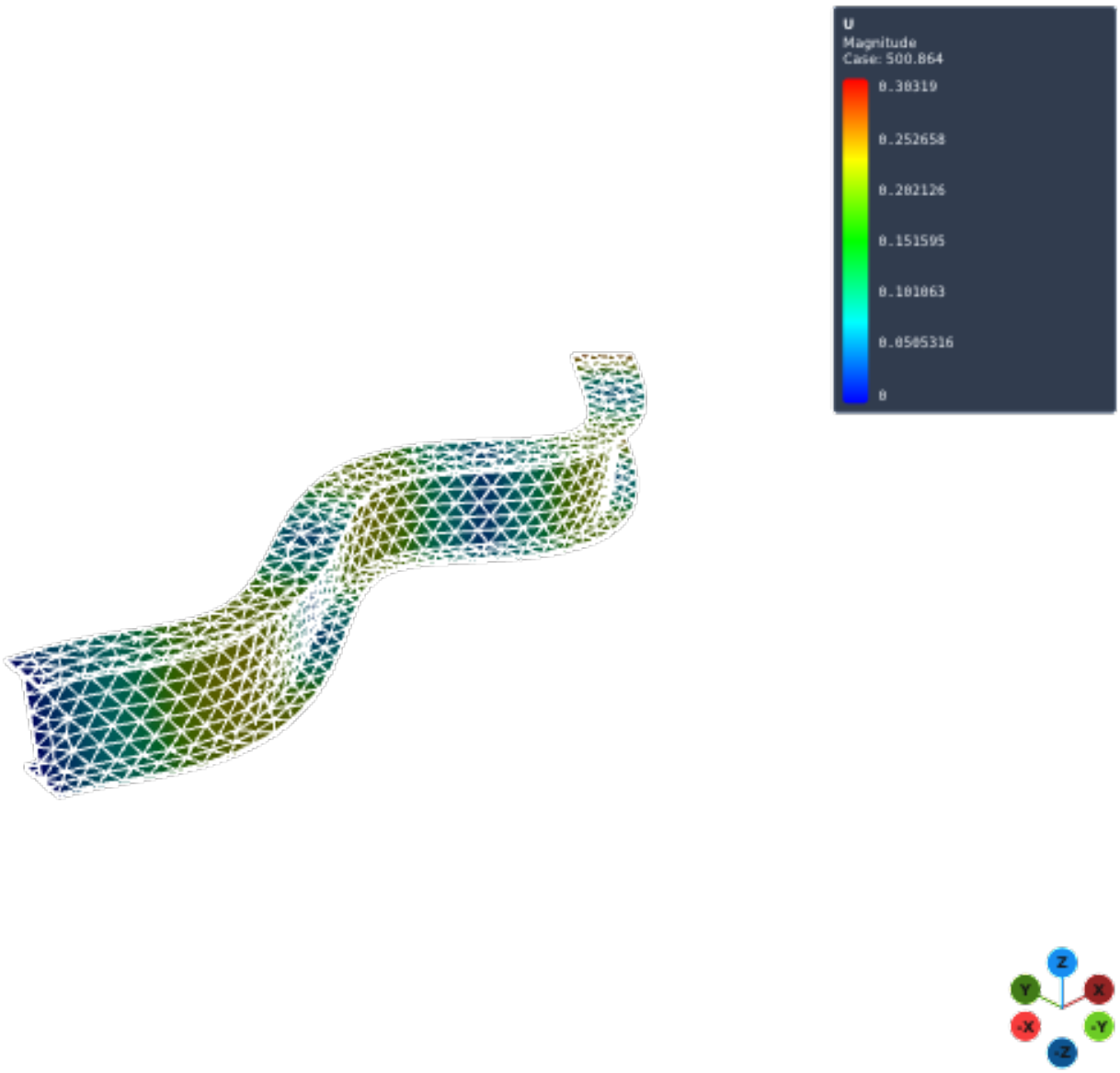

Figure 335: Abaqus — mode 9, 500.864 Hz

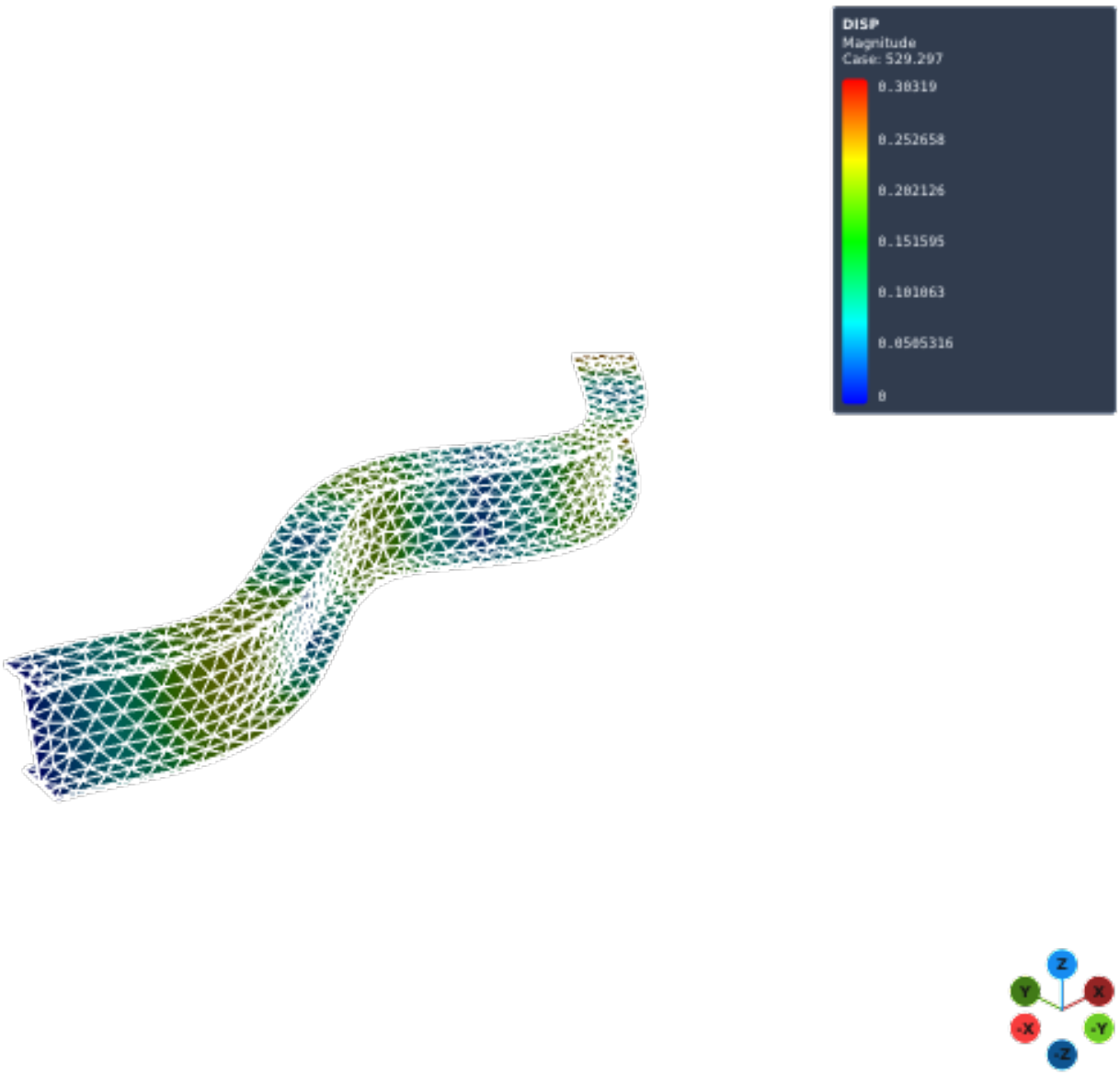

Figure 336: Calculix — mode 9, 529.297 Hz

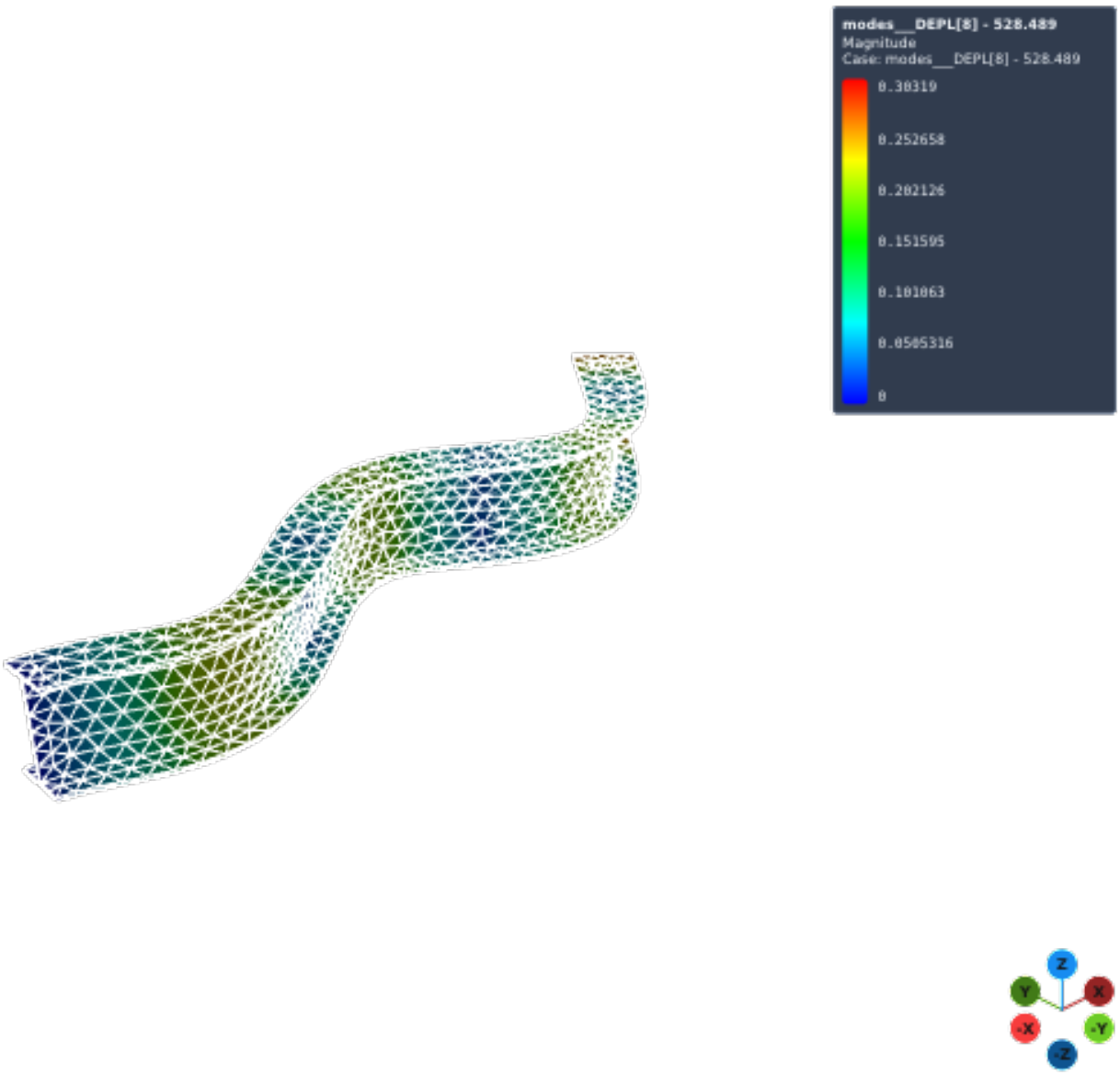

Figure 337: Code_Aster — mode 9, 528.489 Hz

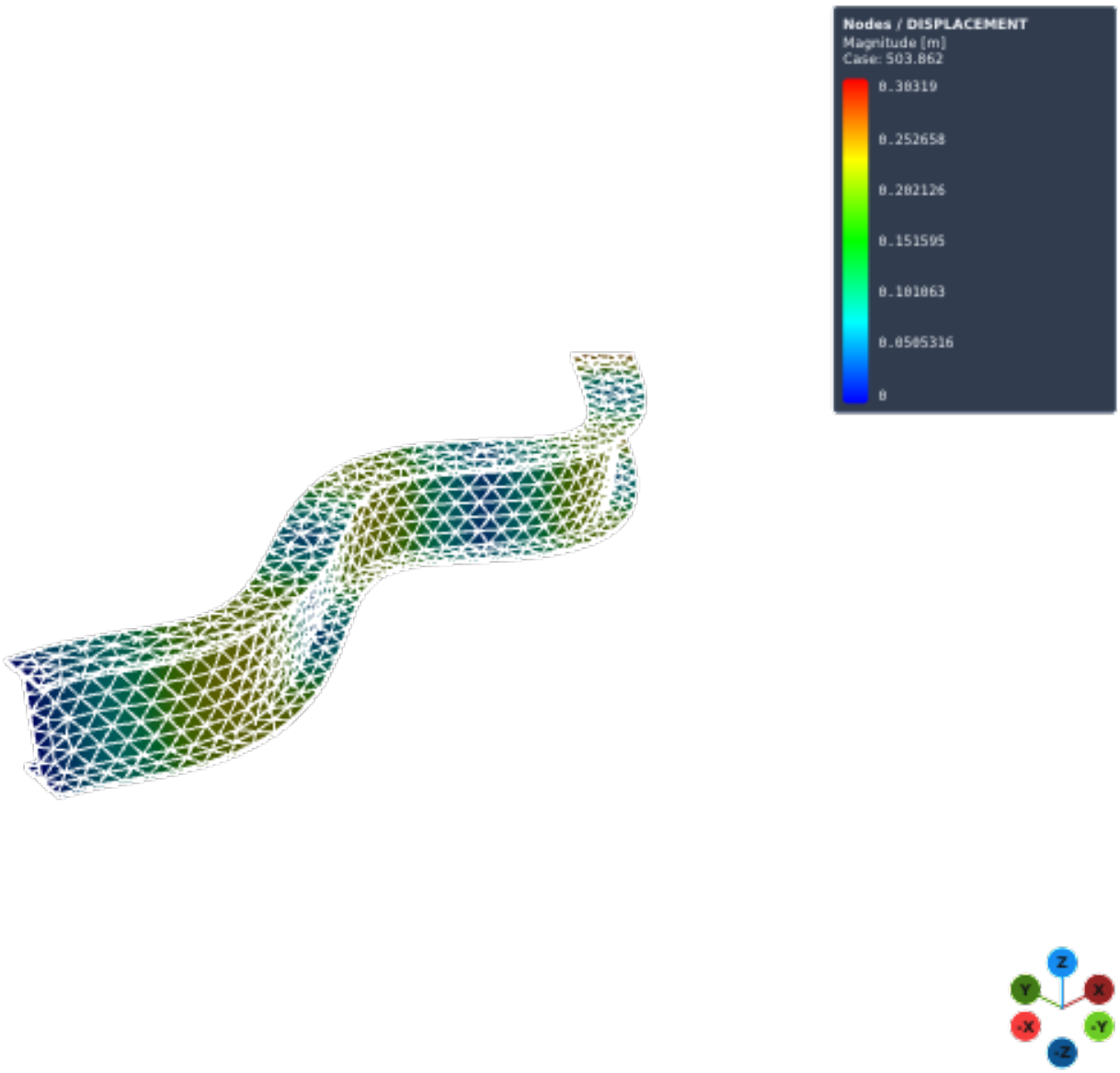

Figure 338: Sesam — mode 9, 503.862 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	500.860	-2.86	9.90373e+06	3.39e-05	6.33	4.73e-05	-0.006	-2.515	0.007
Calculix	529.297	+2.65	1.10601e+07	8.02e-07	6.06	4.08e-03	0.001	-2.462	0.064
Code_Aster	528.489	+2.49	1.10263e+07	7.98e-07	6.07	4.07e-03	0.001	-2.464	0.064
Sesam	503.862	-2.28	1.00226e+07	3.01e-05	6.34	4.92e-05	-0.005	-2.518	0.007

Mode 10

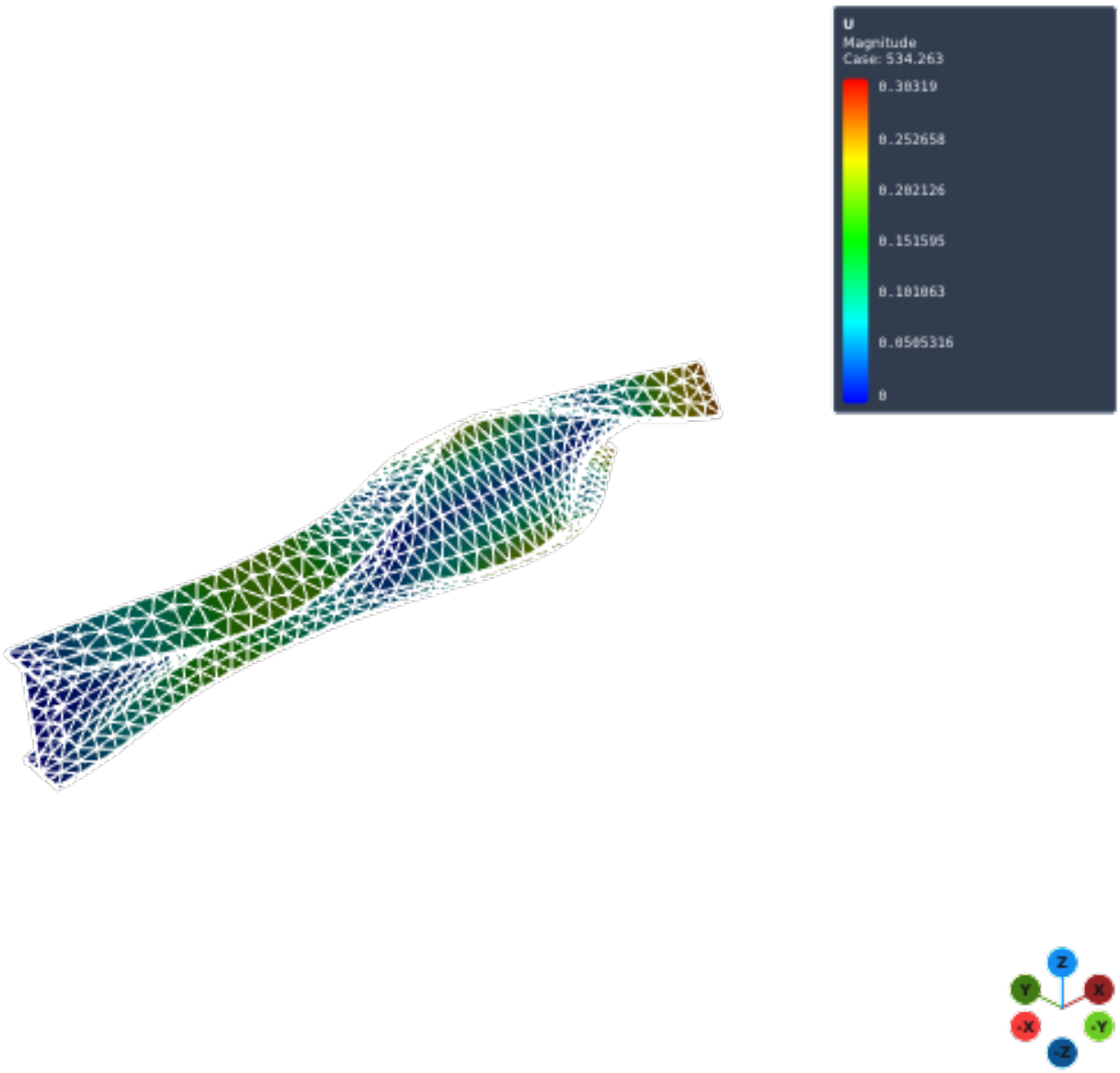

Figure 339: Abaqus — mode 10, 534.263 Hz

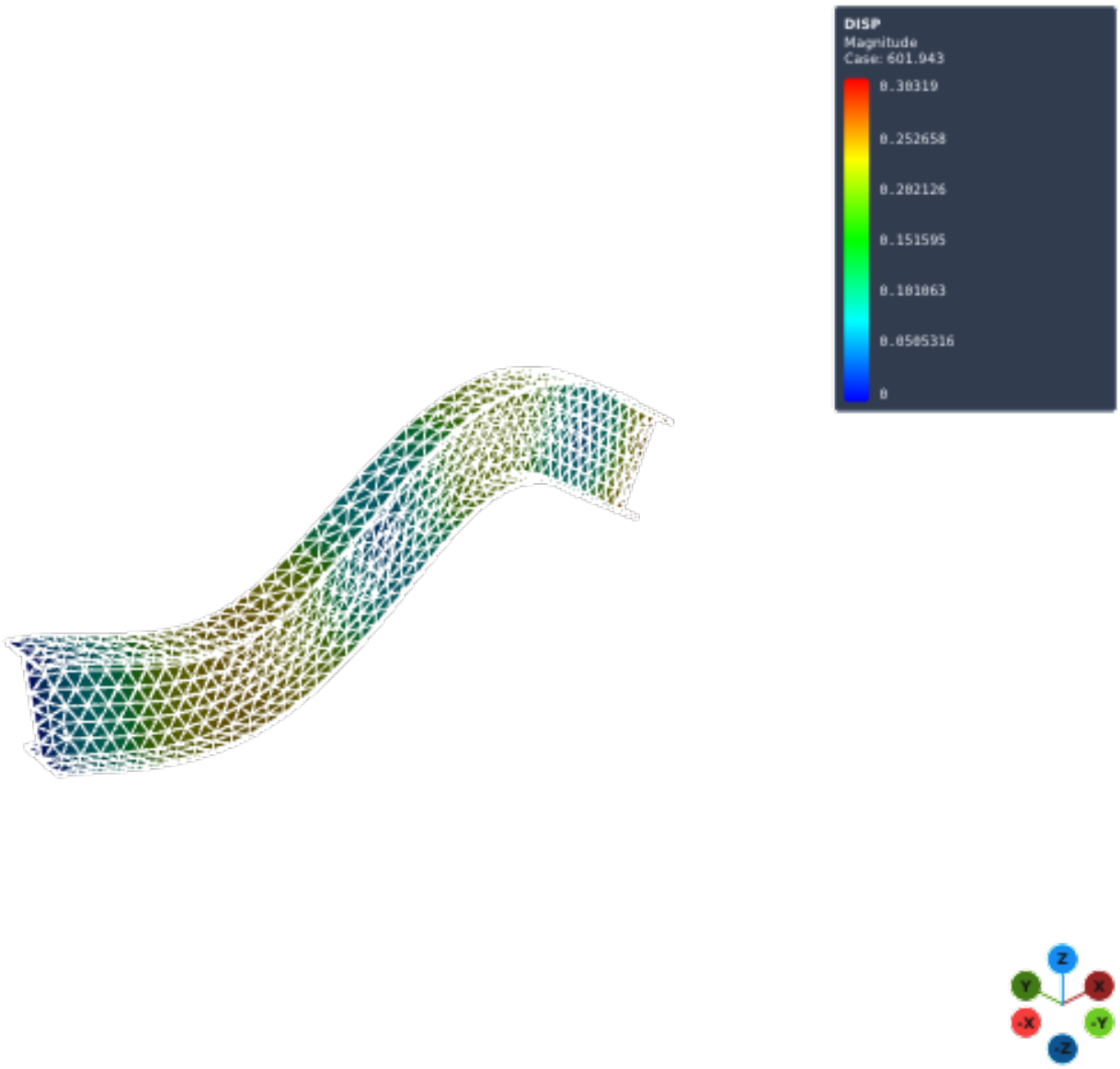

Figure 340: Calculix — mode 10, 601.943 Hz

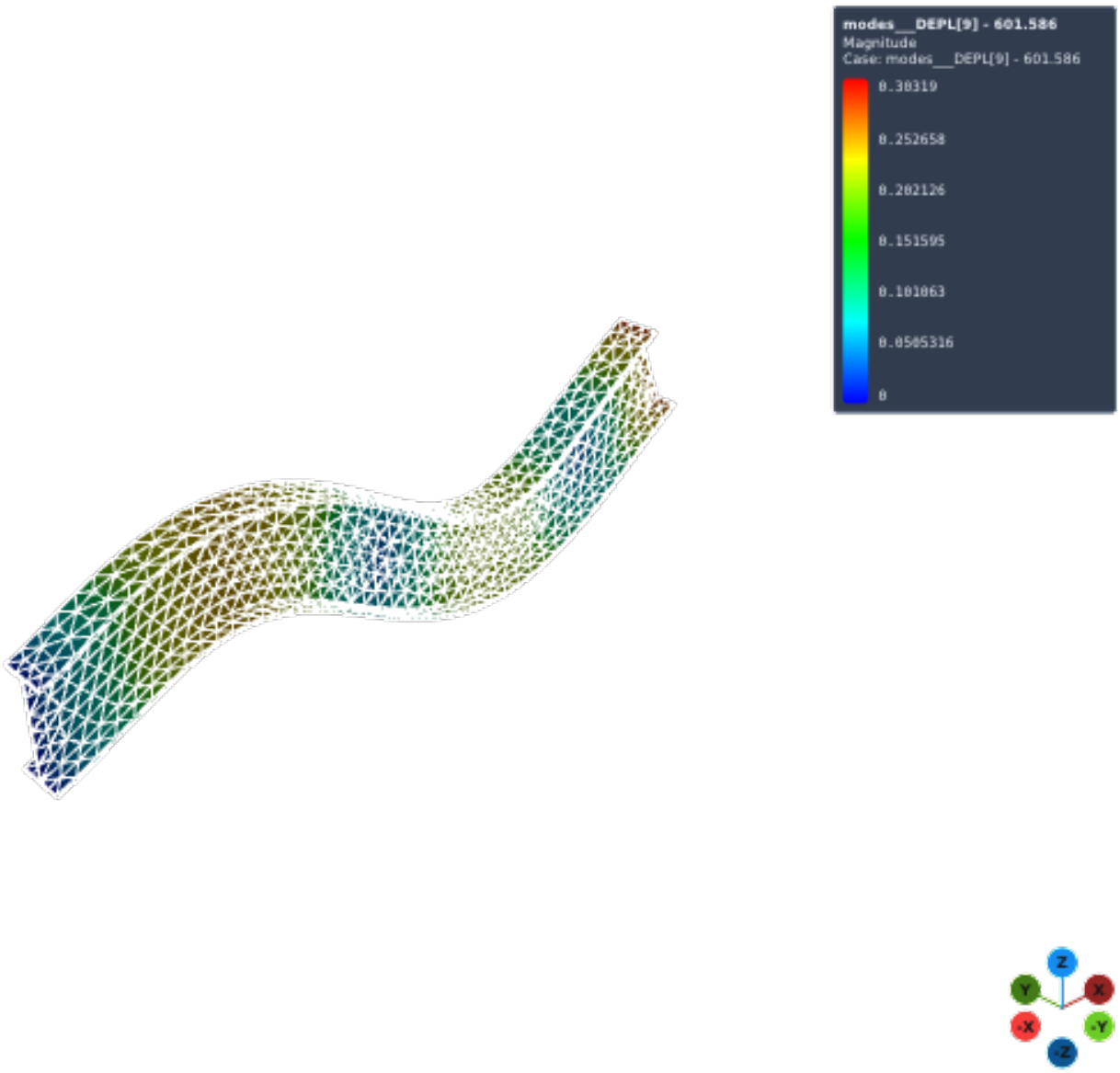

Figure 341: Code_Aster — mode 10, 601.586 Hz

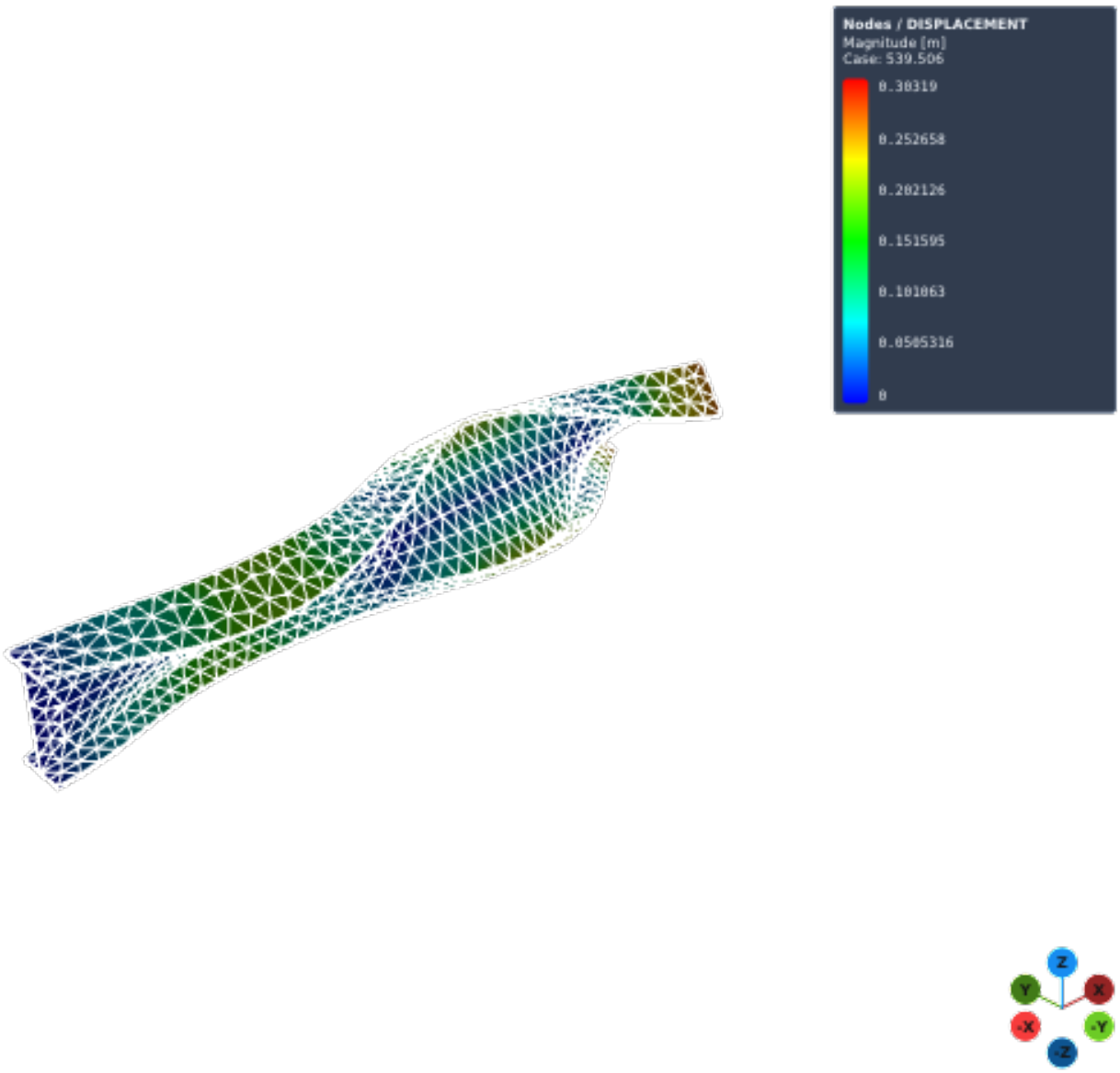

Figure 342: Sesam — mode 10, 539.506 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	534.260	-6.16	1.12686e+07	2.60e-04	2.77e-04	6.63e-06	-0.016	-0.017	0.003
Calculix	601.943	+5.73	1.43044e+07	3.09e-07	2.06e-03	12.76	0.001	-0.045	-3.572
Code_Aster	601.586	+5.67	1.42875e+07	3.19e-07	2.07e-03	12.79	-0.001	0.045	3.576
Sesam	539.506	-5.24	1.14909e+07	2.36e-04	2.31e-04	6.55e-06	-0.015	-0.015	0.003

Mode 11

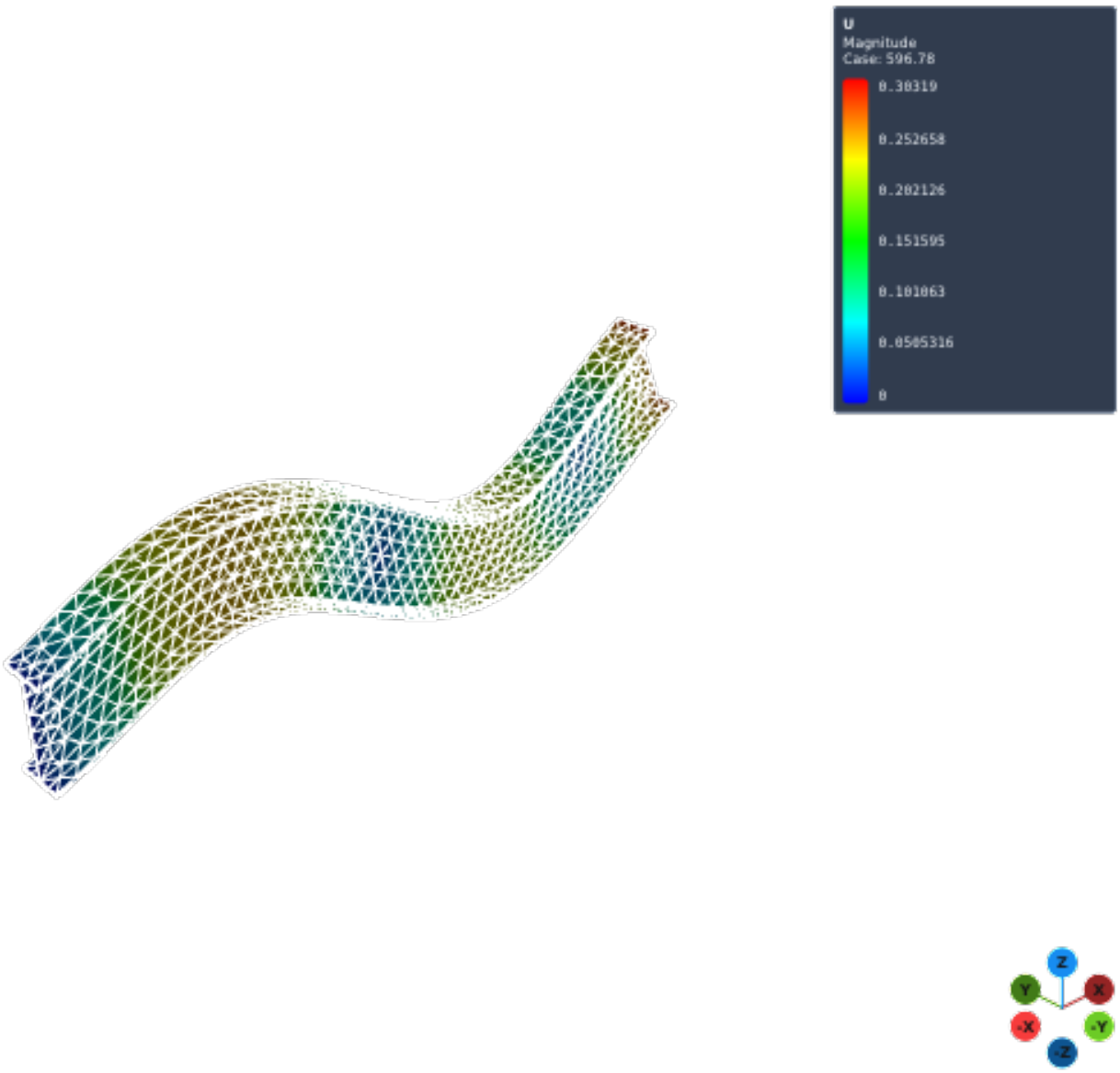

Figure 343: Abaqus — mode 11, 596.780 Hz

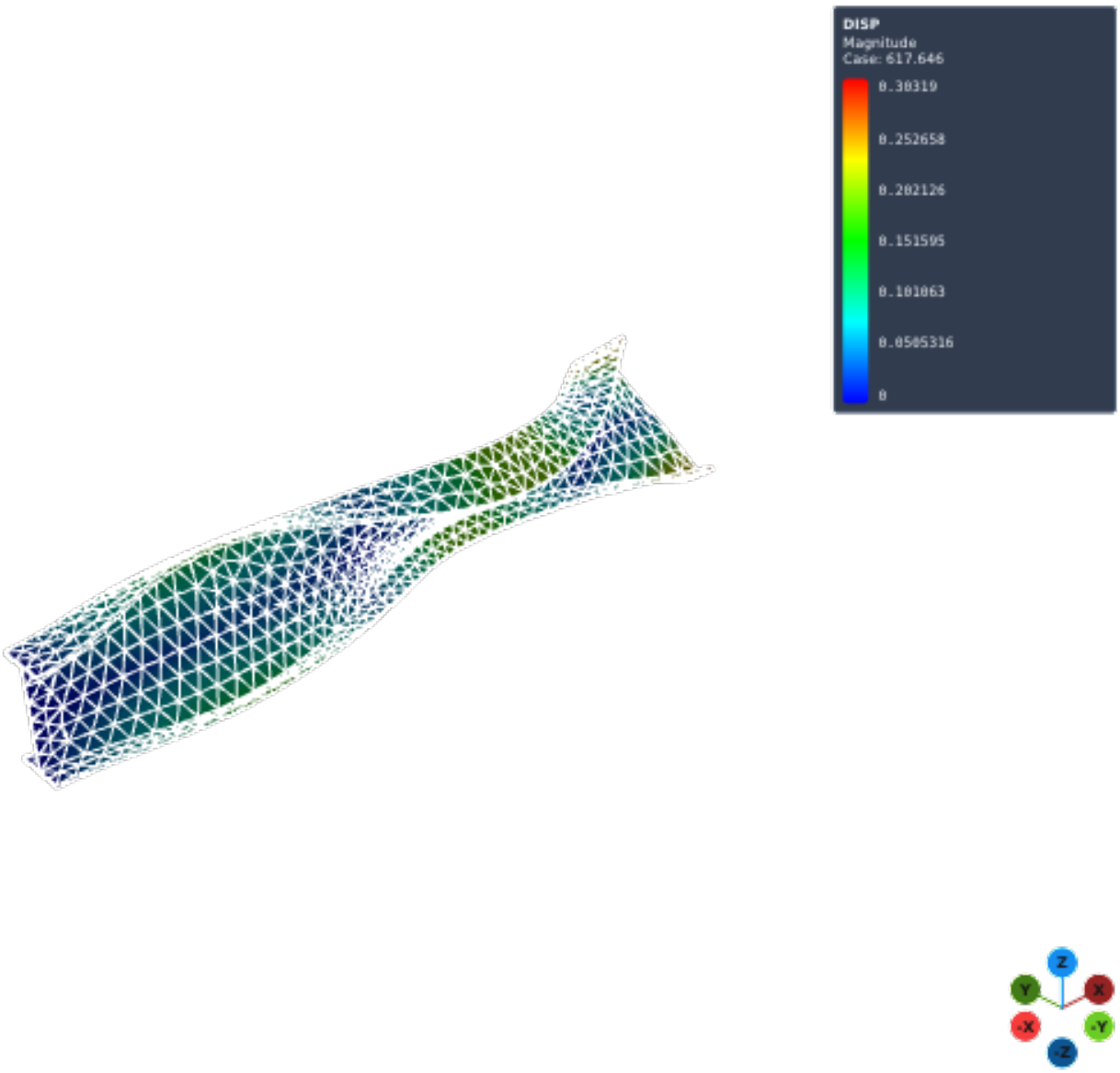

Figure 344: Calculix — mode 11, 617.646 Hz

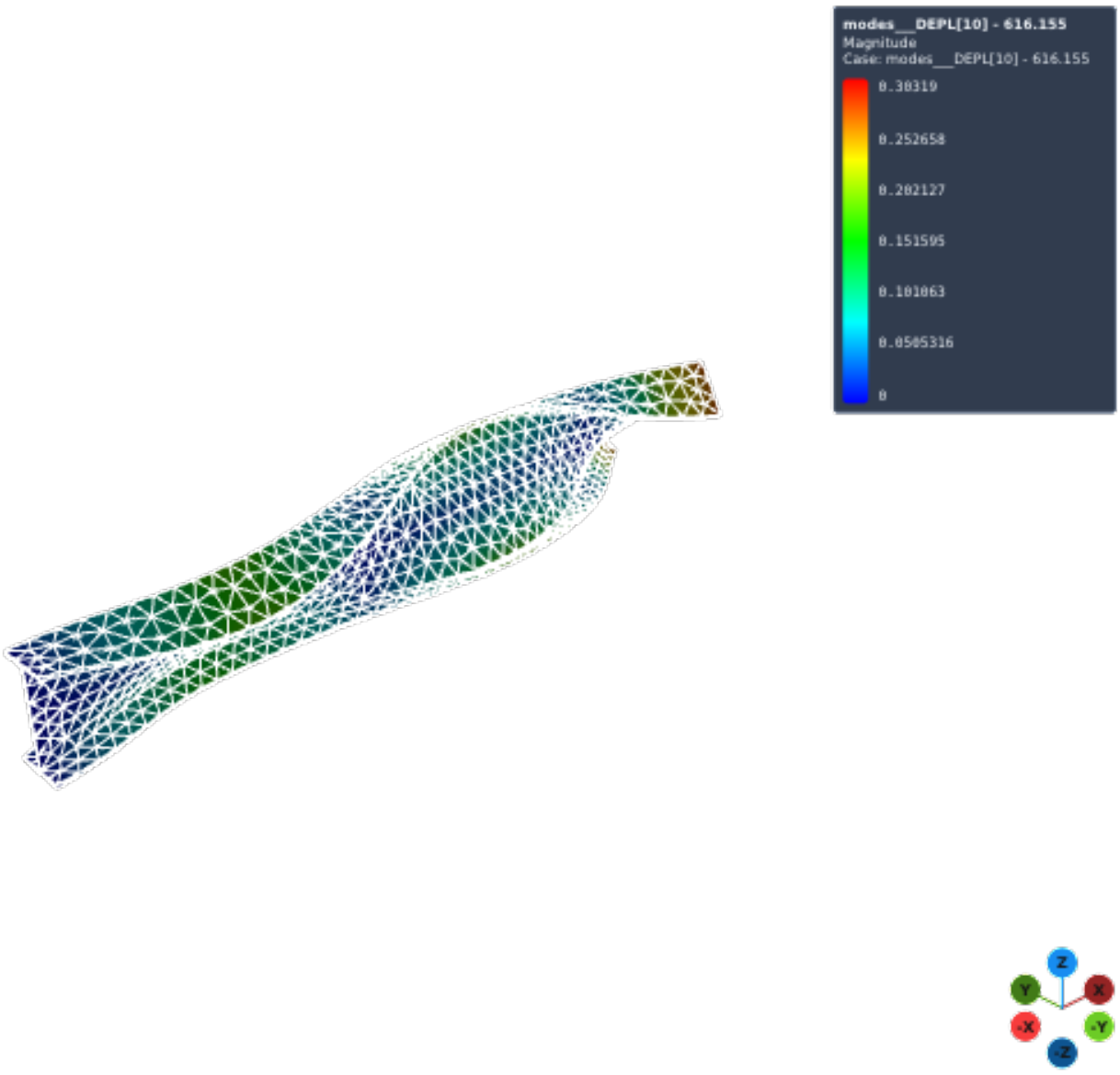

Figure 345: Code_Aster — mode 11, 616.155 Hz

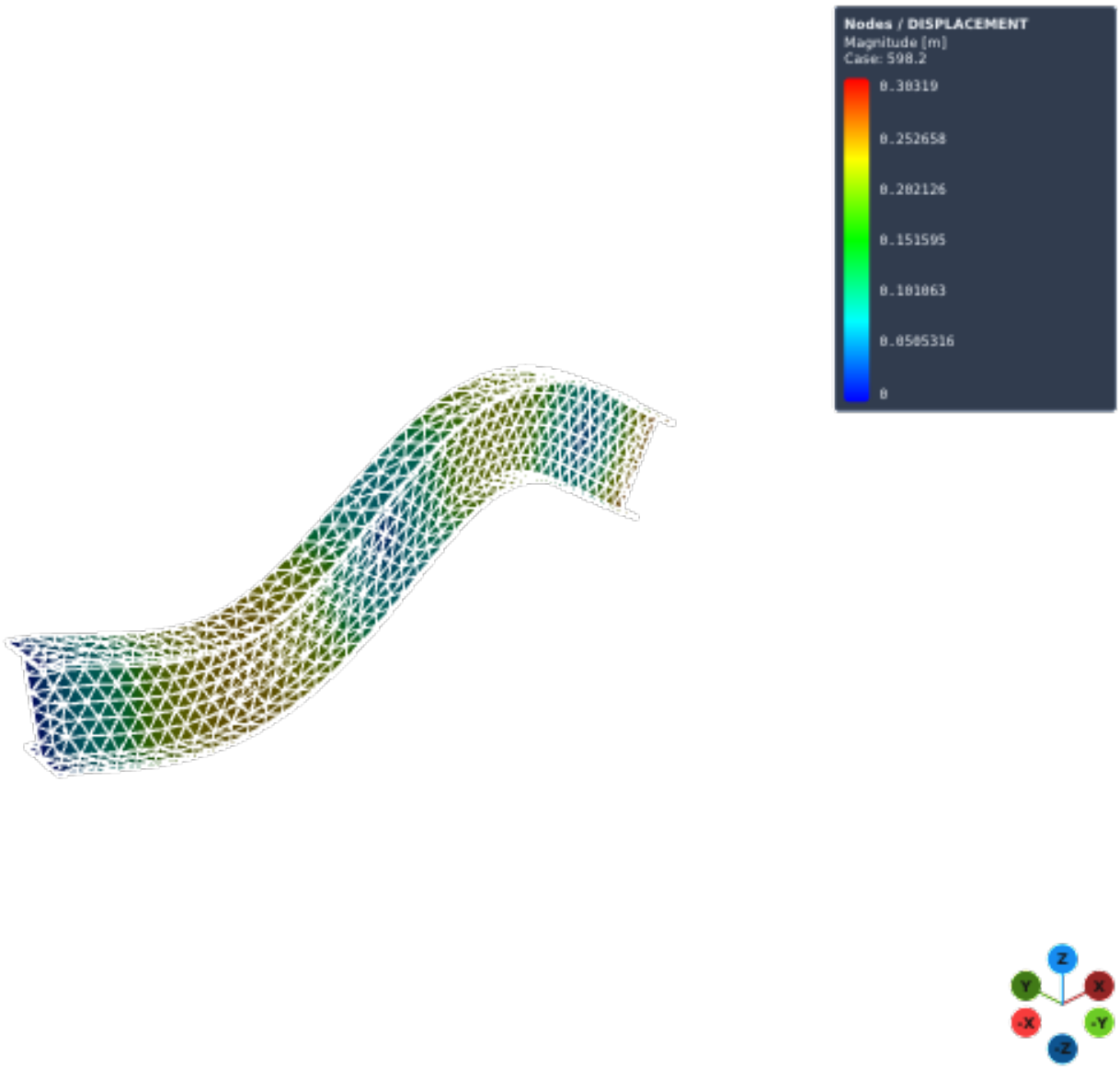

Figure 346: Sesam — mode 11, 598.200 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	596.780	-1.72	1.40601e+07	4.55e-07	1.33e-07	12.80	-0.001	-3.65e-04	3.578
Calculix	617.646	+1.72	1.50605e+07	1.44e-04	2.59e-04	8.16e-04	-0.012	0.016	0.029
Code_Aster	616.155	+1.48	1.49879e+07	1.45e-04	2.65e-04	9.64e-04	0.012	-0.016	-0.031
Sesam	598.200	-1.48	1.41271e+07	4.36e-07	3.62e-08	12.80	0.001	1.90e-04	-3.578

Solid, 1st order, HEX

Mode 1

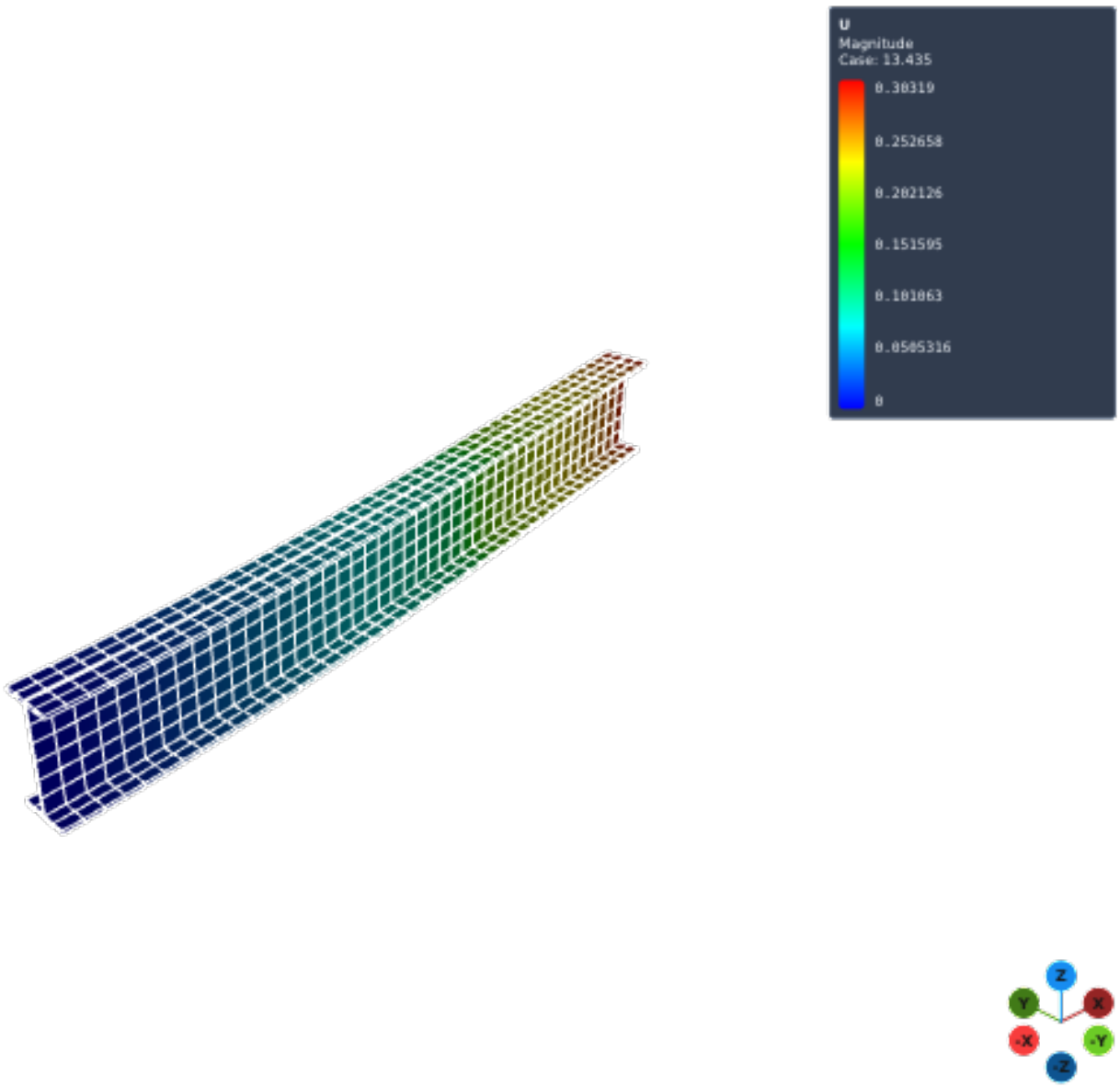

Figure 347: Abaqus — mode 1, 13.435 Hz

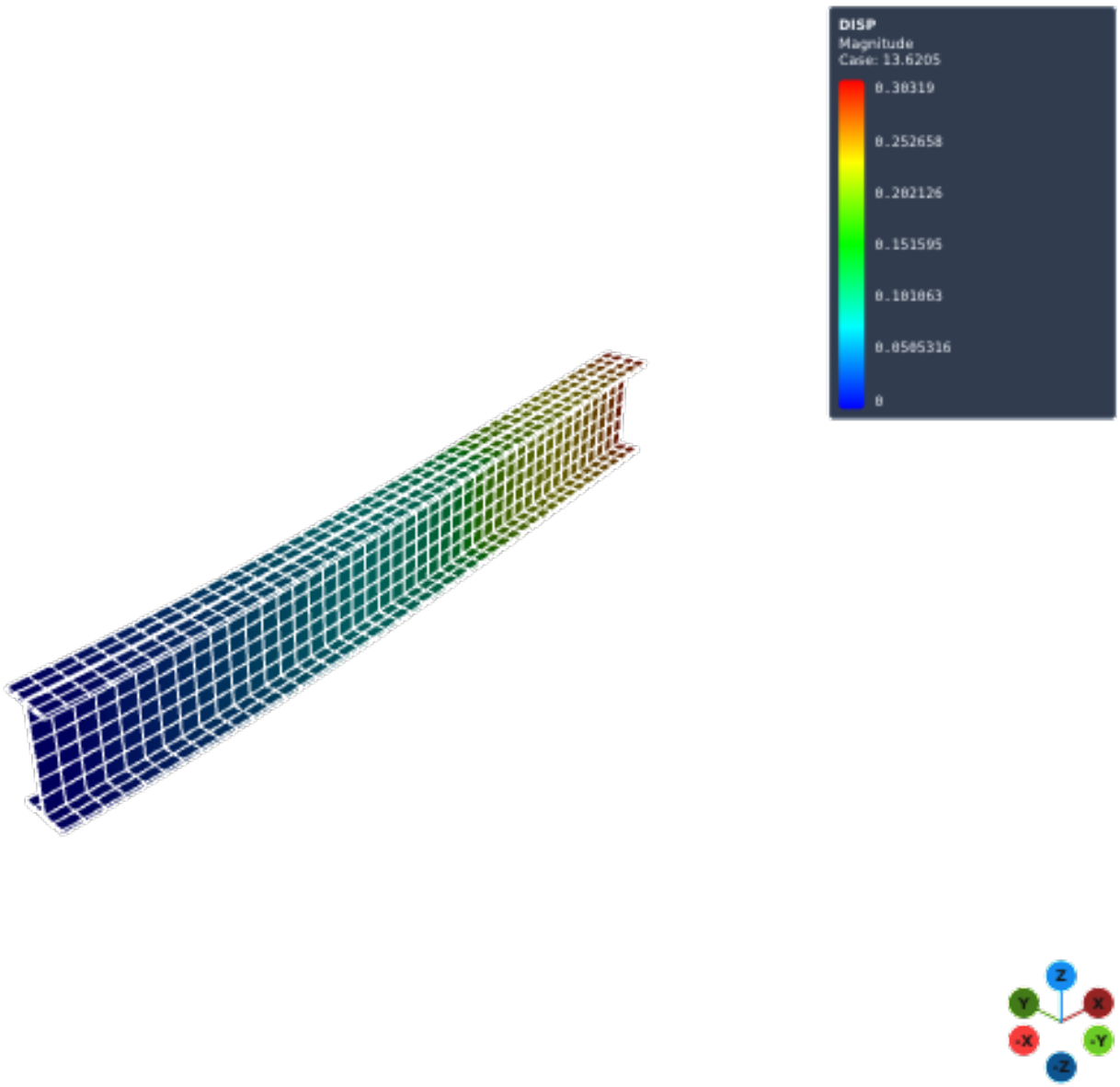

Figure 348: Calculix — mode 1, 13.620 Hz

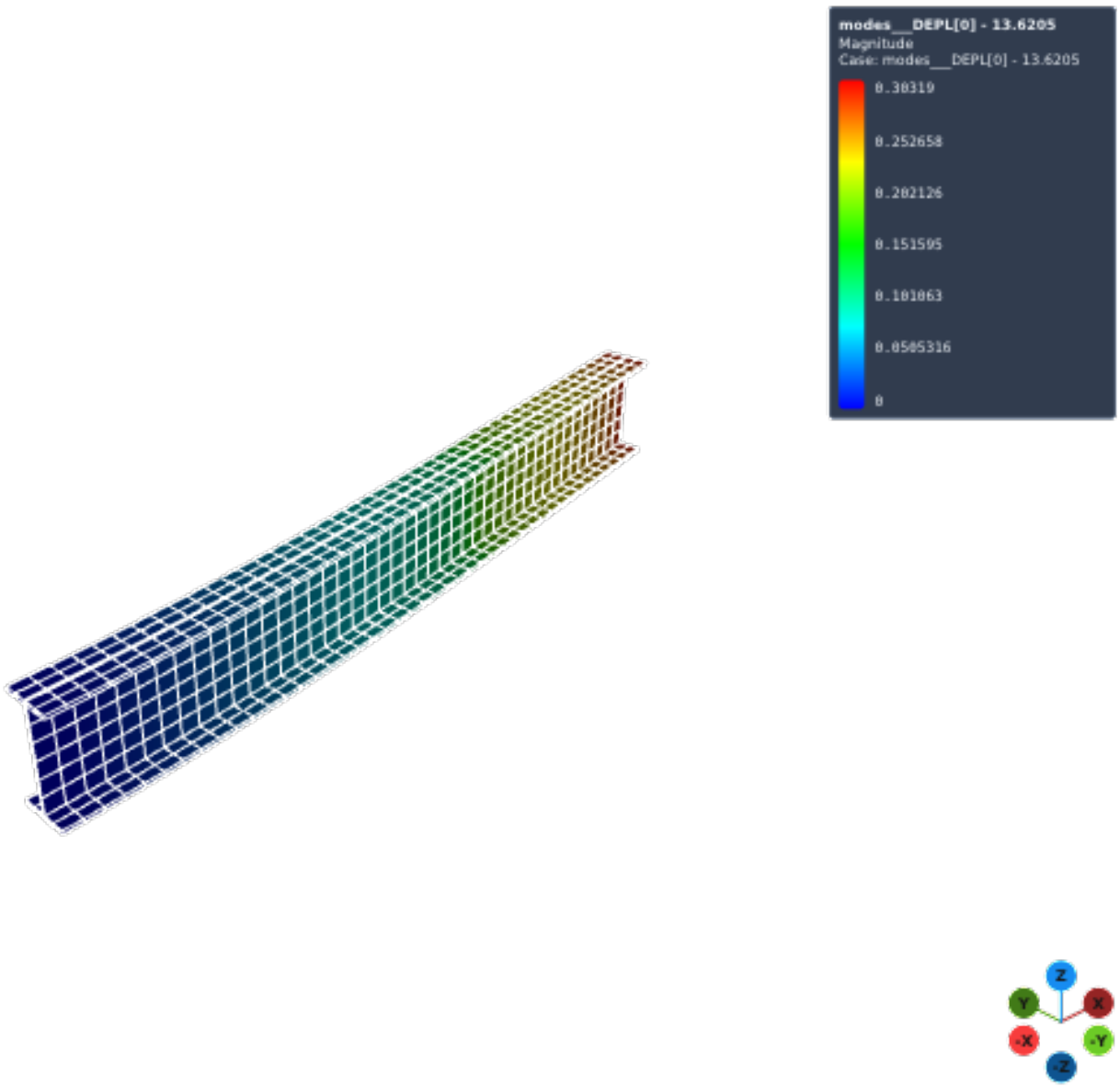

Figure 349: Code_Aster — mode 1, 13.620 Hz

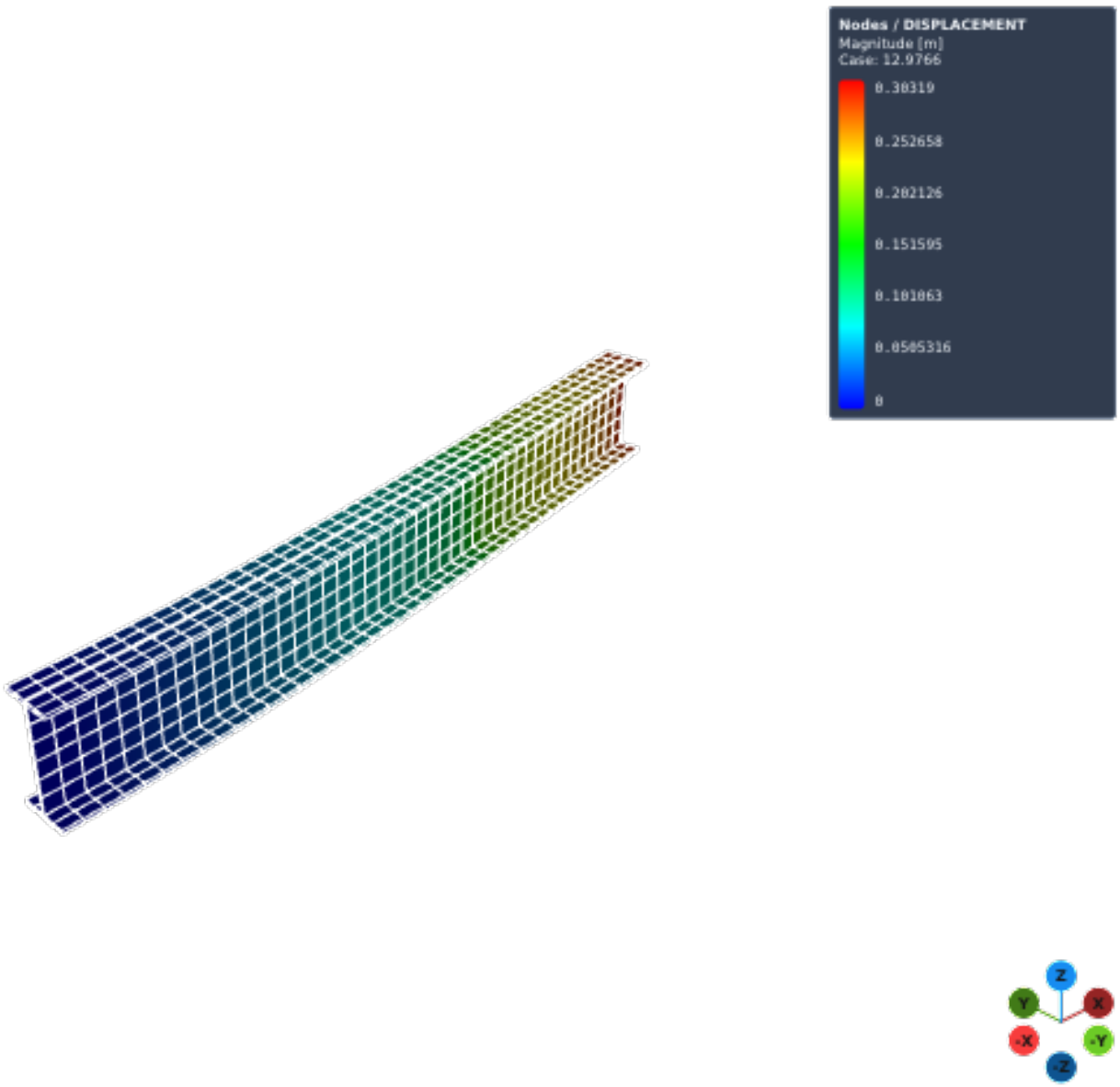

Figure 350: Sesam — mode 1, 12.977 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	13.435	+0.16	7125.8	4.92e-29	116.39	5.47e-23	-7.01e-15	10.788	-7.39e-12
Calculix	13.620	+1.55	7323.93	3.07e-25	116.36	7.93e-23	-5.54e-13	10.787	-8.91e-12
Code_Aster	13.620	+1.55	7323.93	3.34e-26	116.36	1.98e-22	1.83e-13	10.787	-1.41e-11
Sesam	12.977	-3.25	6647.9	9.71e-25	116.43	6.43e-21	-9.85e-13	10.790	8.02e-11

Mode 2

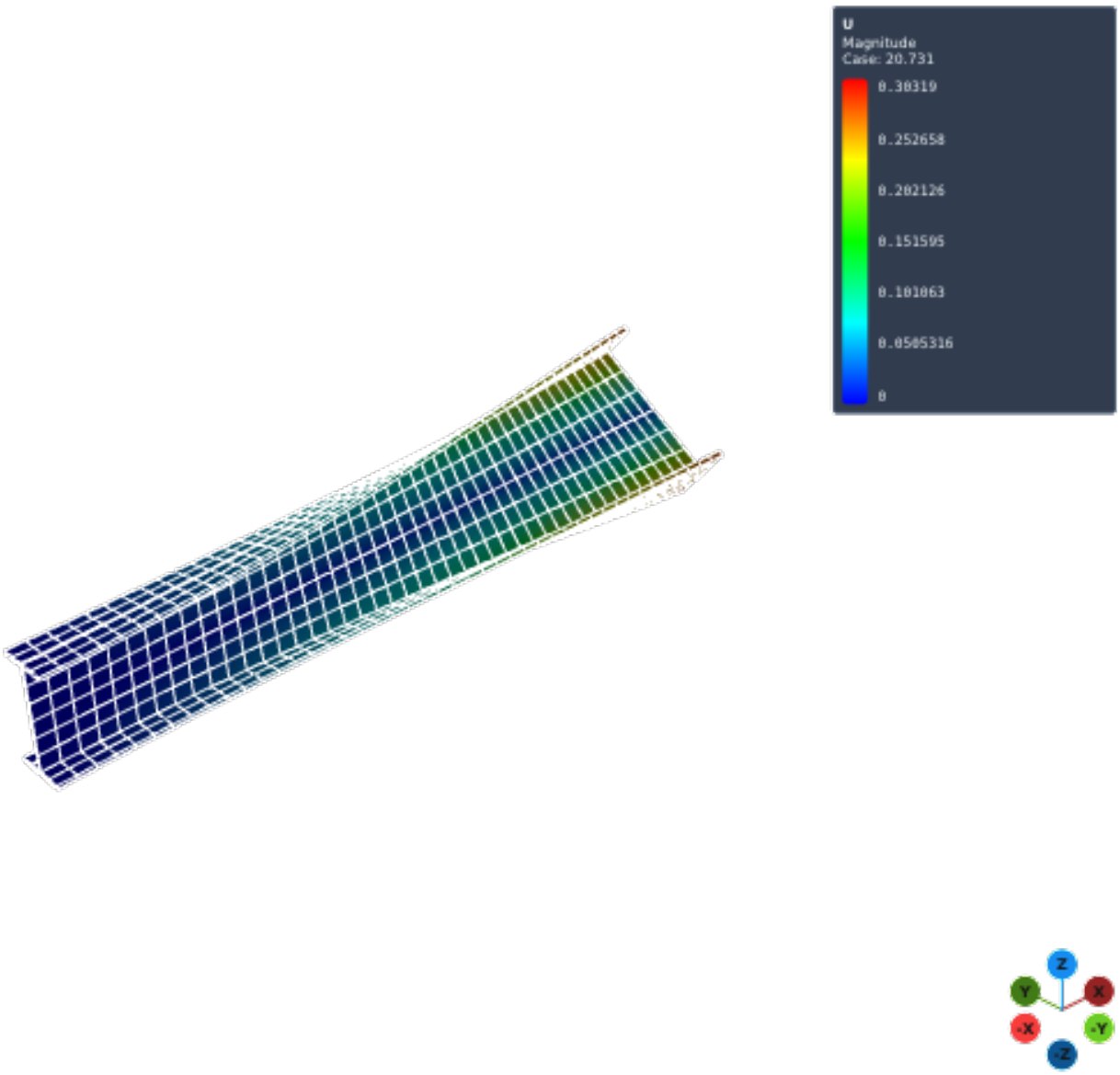

Figure 351: Abaqus — mode 2, 20.731 Hz

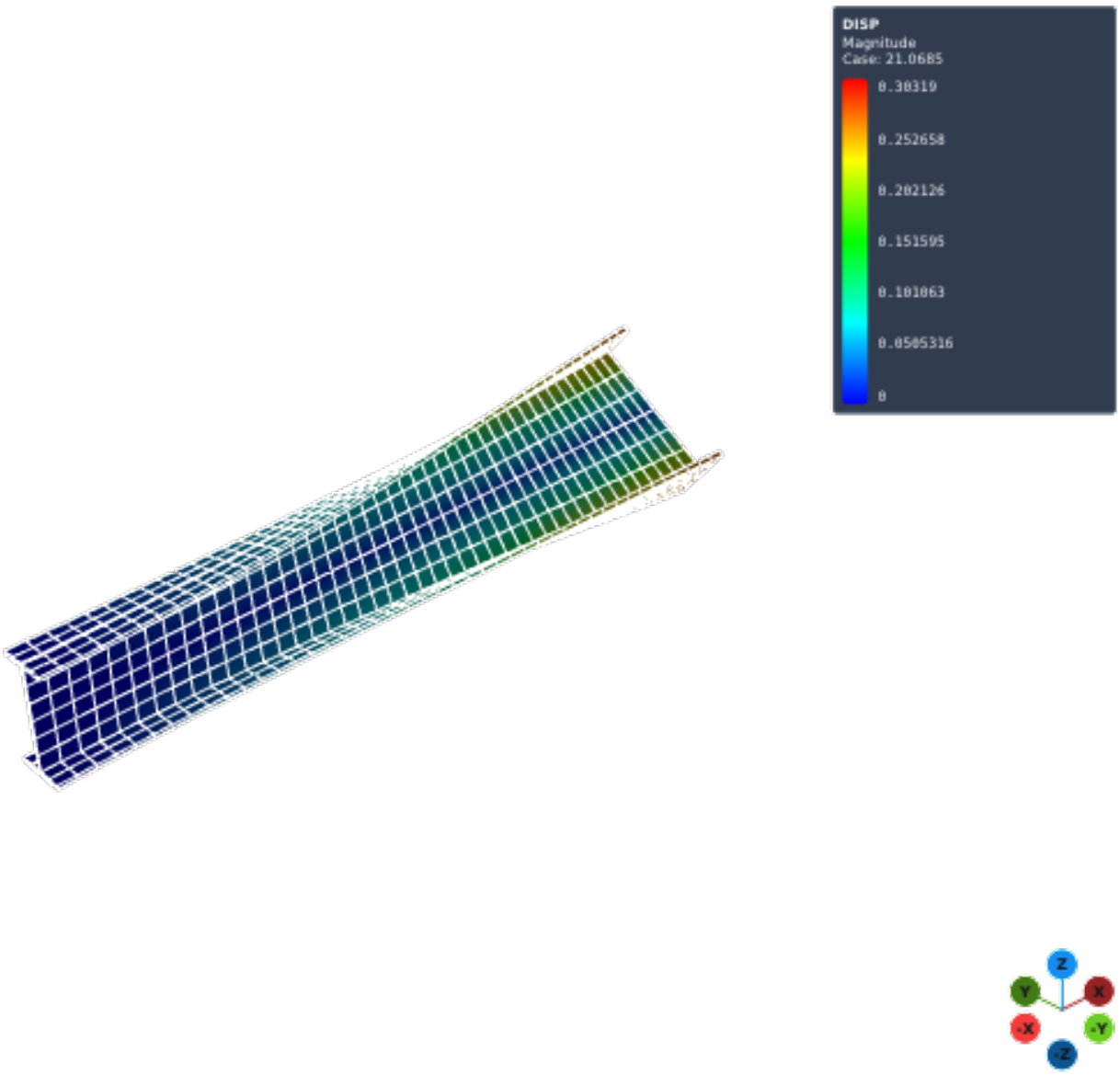

Figure 352: Calculix — mode 2, 21.069 Hz

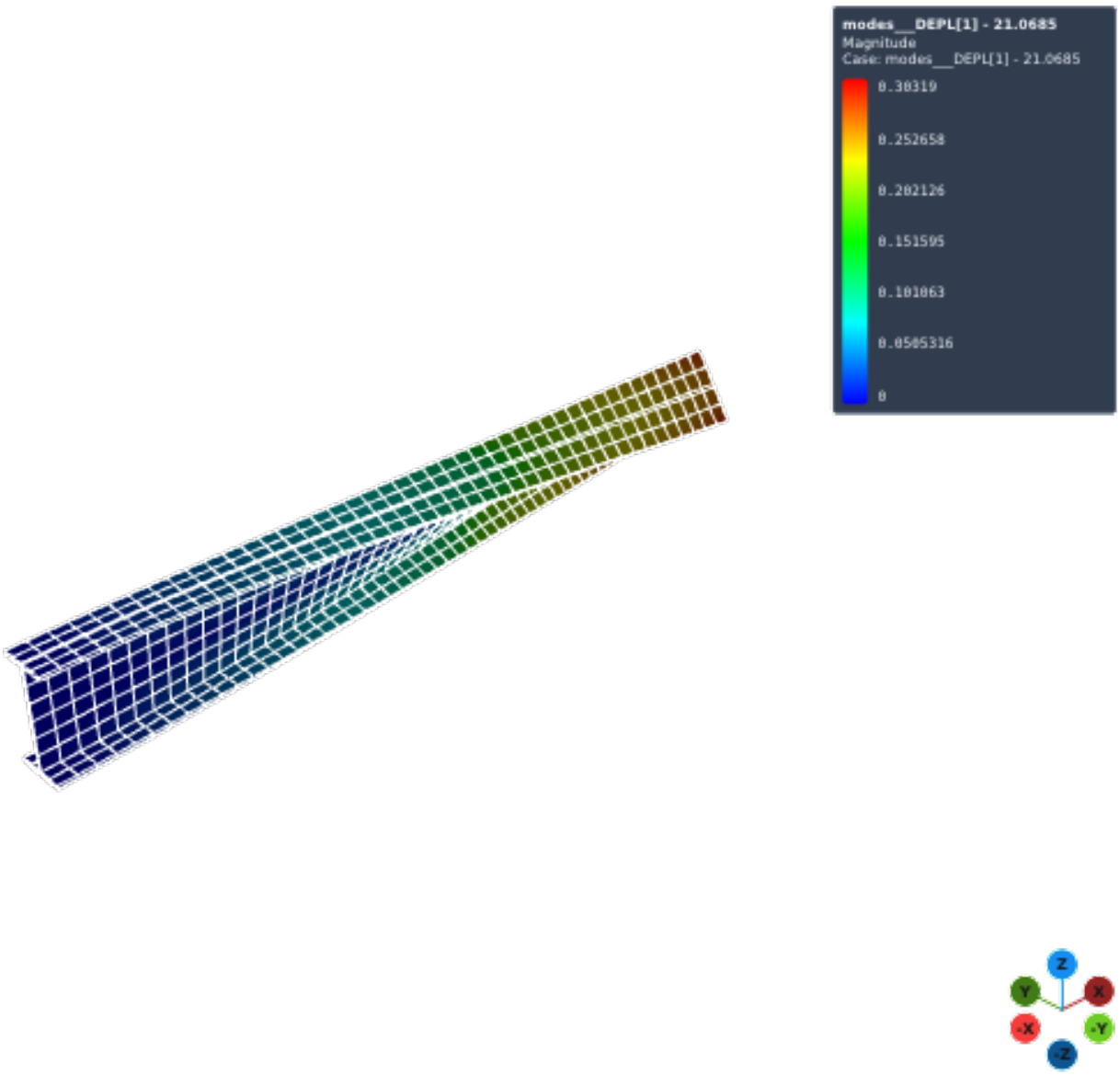

Figure 353: Code_Aster — mode 2, 21.069 Hz

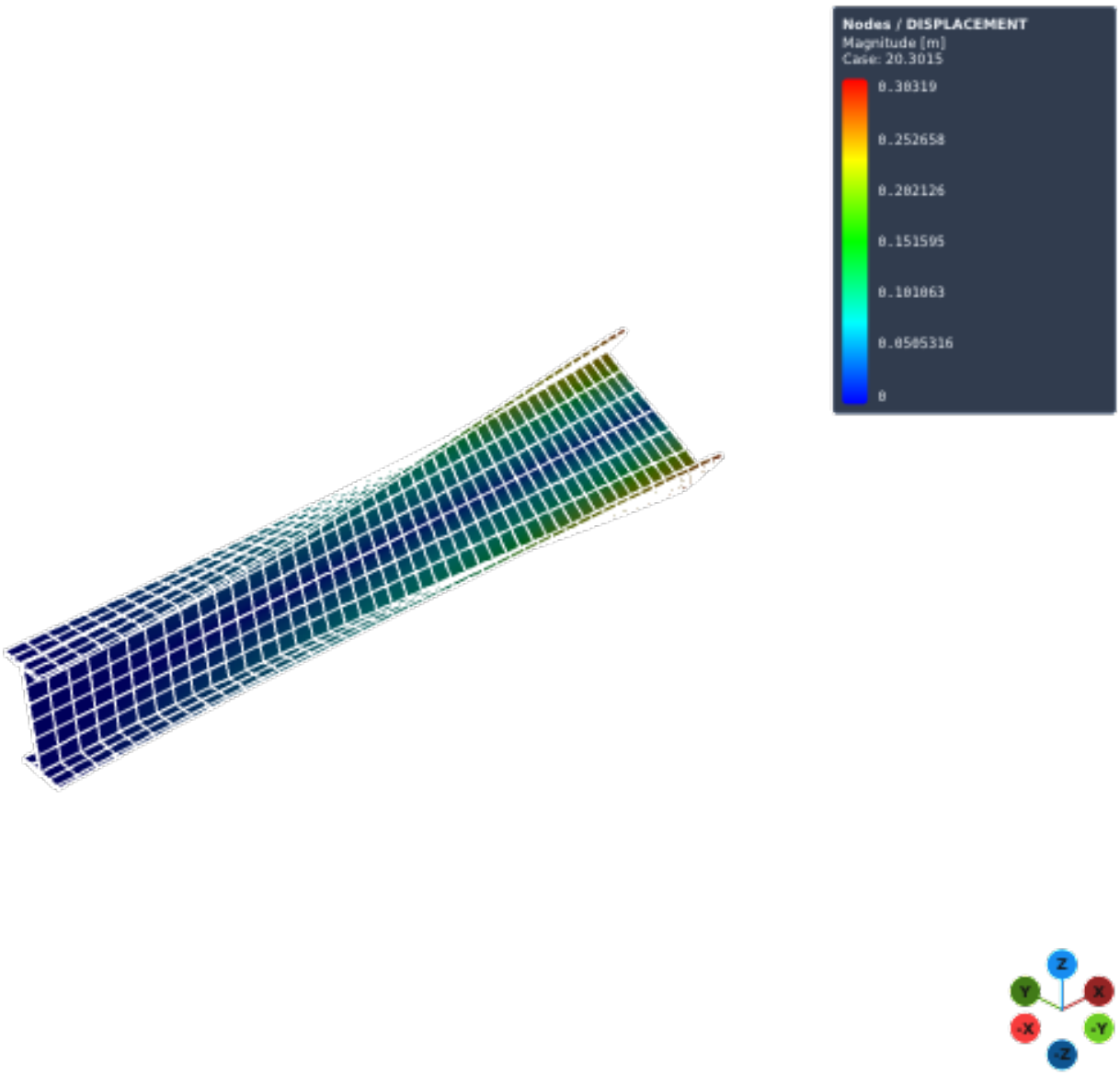

Figure 354: Sesam — mode 2, 20.302 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	20.731	-0.30	16967	7.03e-26	1.48e-18	2.87e-22	2.65e-13	1.22e-09	-1.69e-11
Calculix	21.069	+1.33	17523.8	2.84e-30	5.85e-19	9.43e-21	-1.68e-15	7.65e-10	-9.71e-11
Code_Aster	21.069	+1.33	17523.8	1.97e-28	1.10e-17	3.14e-22	-1.40e-14	3.32e-09	1.77e-11
Sesam	20.302	-2.36	16271.1	1.70e-26	3.10e-18	2.42e-20	-1.31e-13	-1.76e-09	1.56e-10

Mode 3

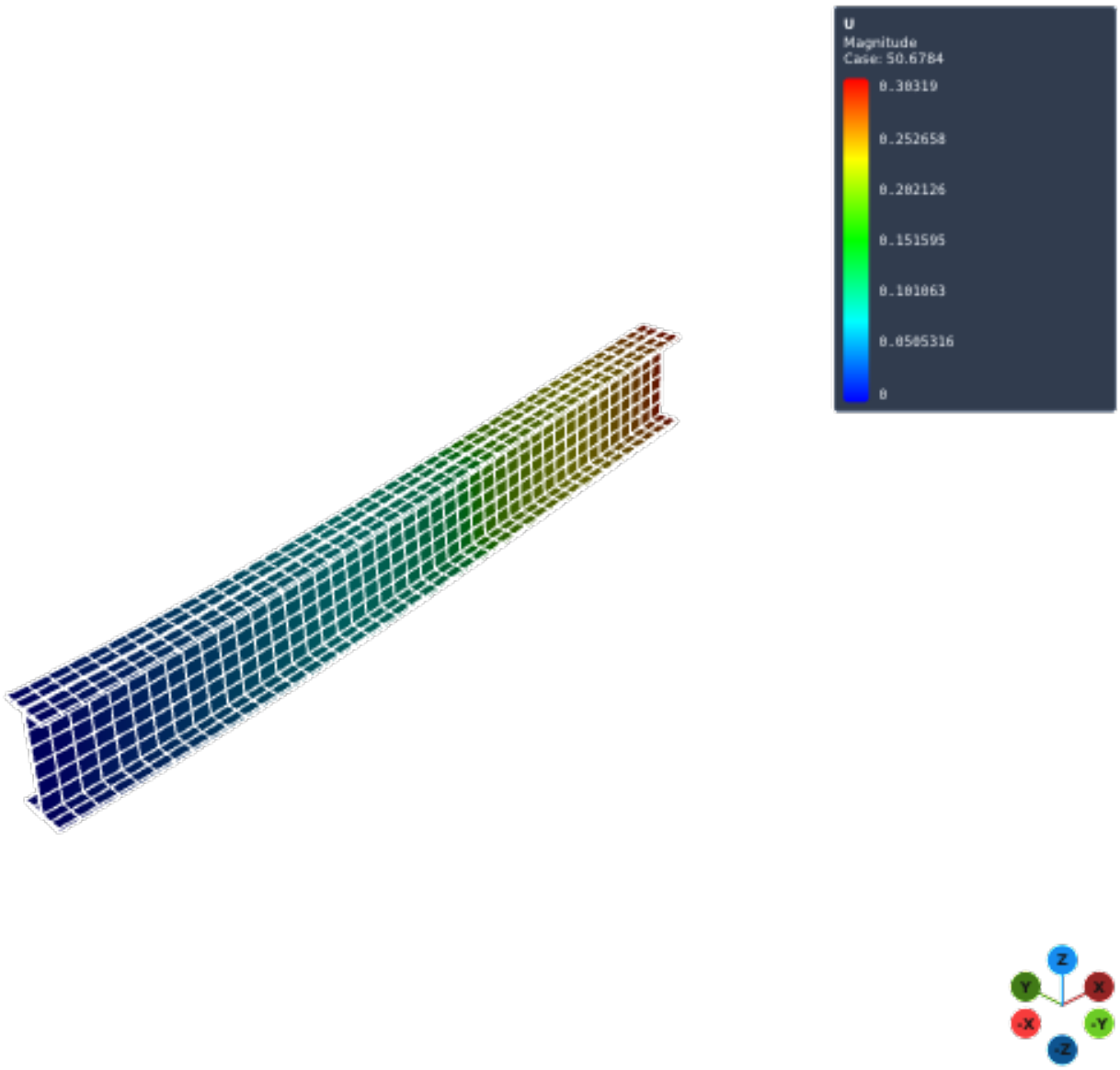

Figure 355: Abaqus — mode 3, 50.678 Hz

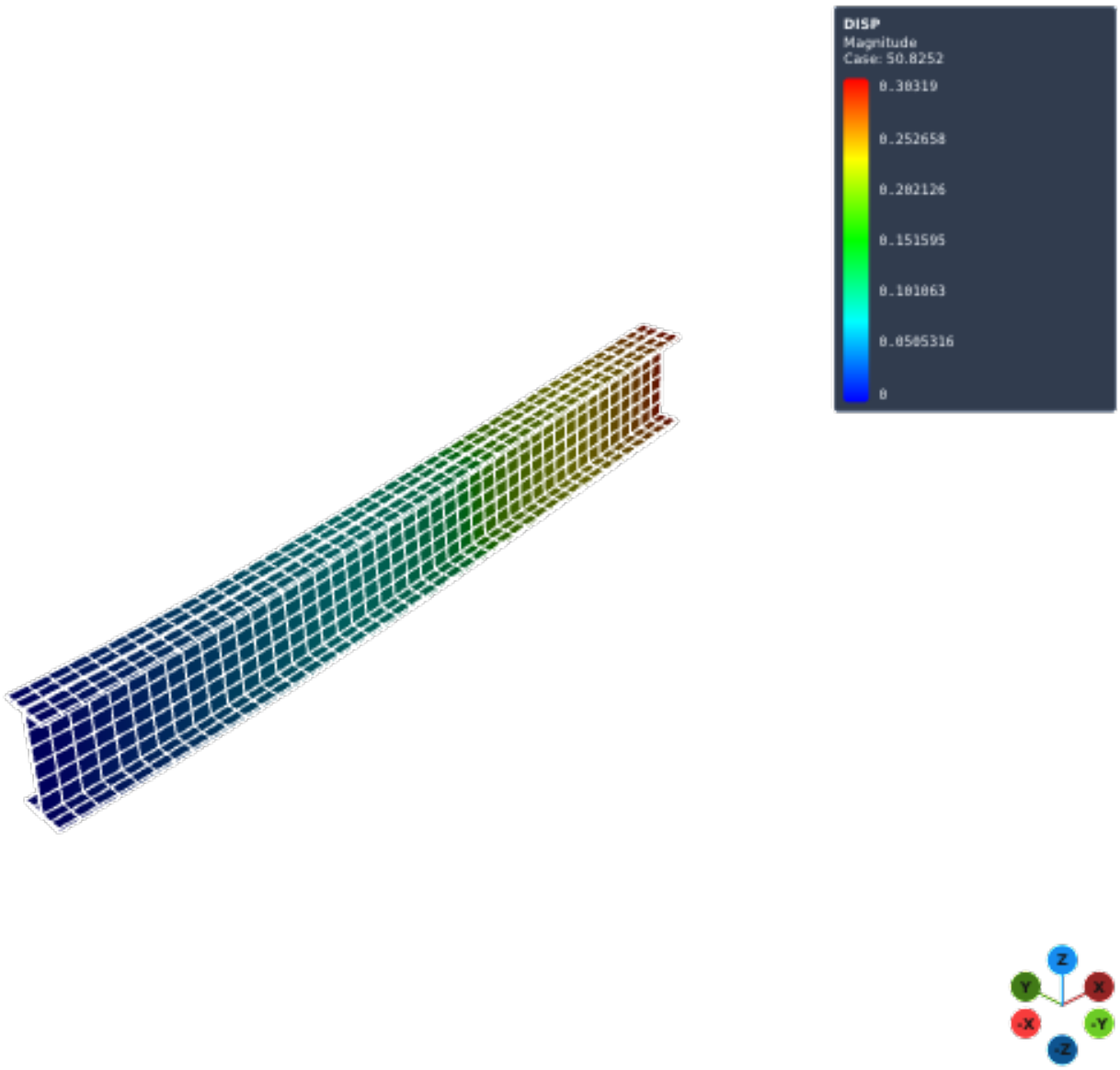

Figure 356: Calculix — mode 3, 50.825 Hz

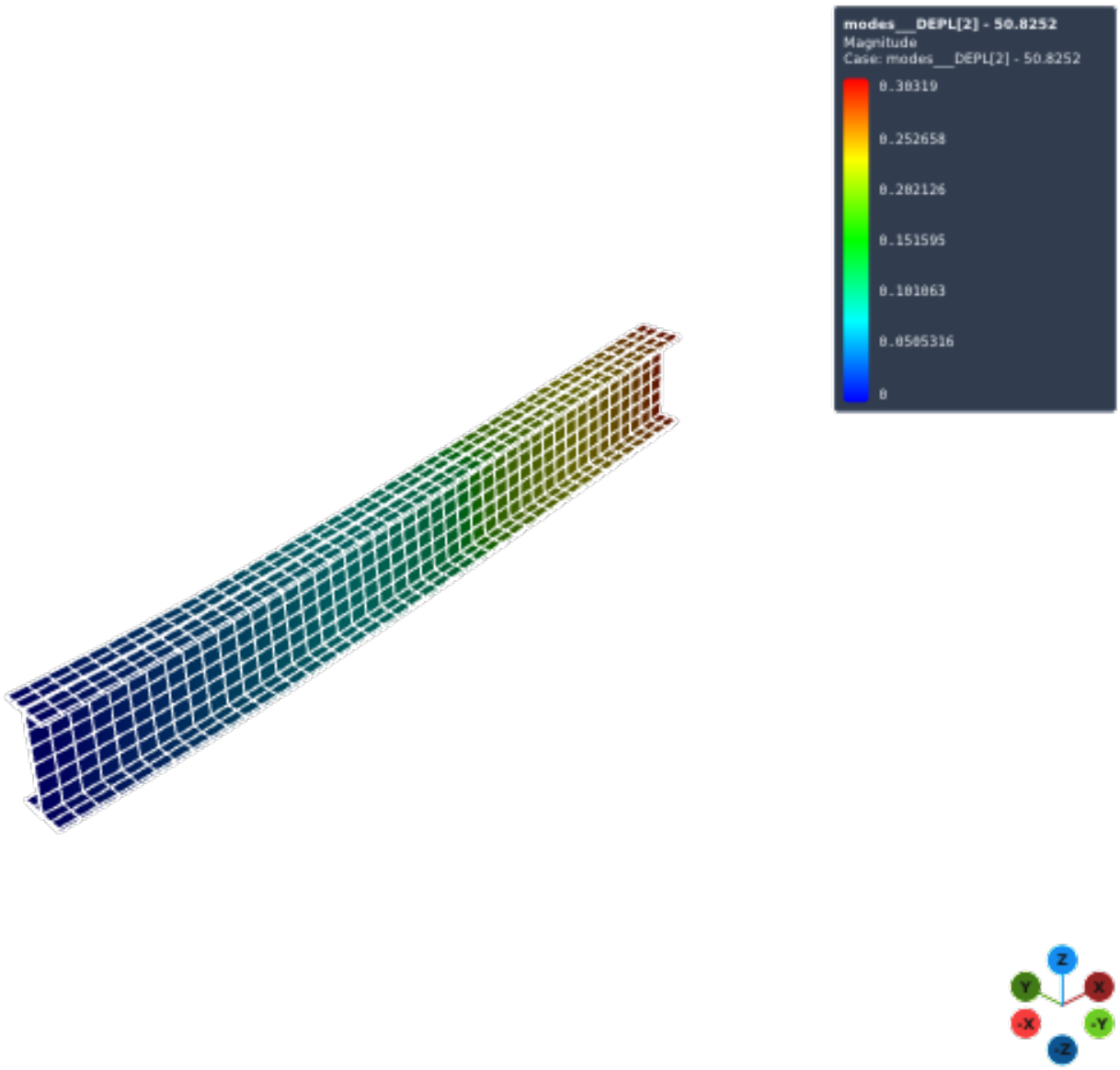

Figure 357: Code_Aster — mode 3, 50.825 Hz

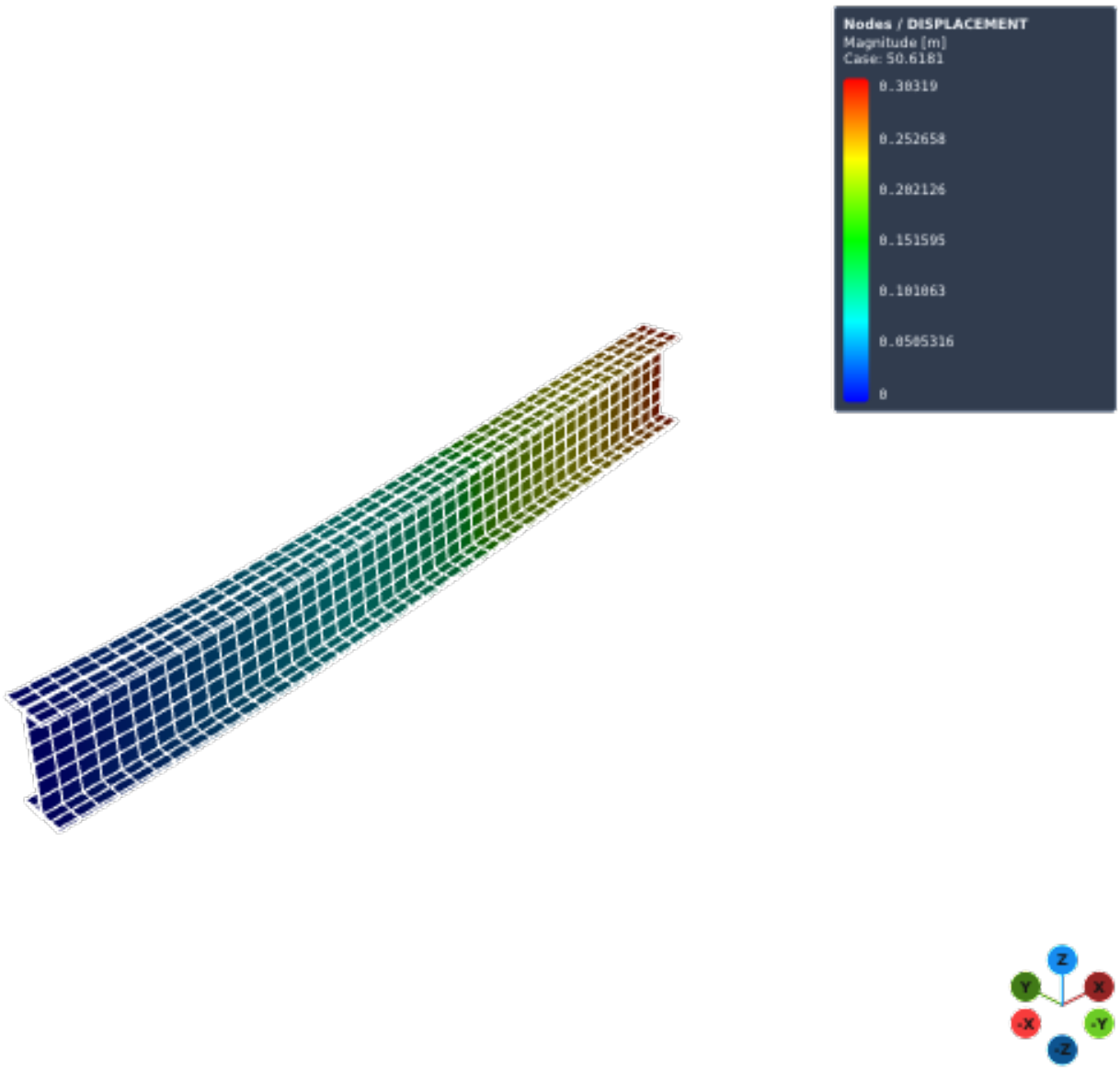

Figure 358: Sesam — mode 3, 50.618 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.678	-0.12	101392	1.09e-25	2.57e-22	118.20	-3.30e-13	1.60e-11	10.872
Calculix	50.825	+0.17	101981	3.20e-26	2.09e-23	118.19	-1.79e-13	4.57e-12	10.871
Code_Aster	50.825	+0.17	101981	3.12e-25	3.28e-23	118.19	-5.58e-13	5.73e-12	10.872
Sesam	50.618	-0.23	101151	1.20e-24	4.37e-21	118.29	-1.10e-12	-6.61e-11	10.876

Mode 4

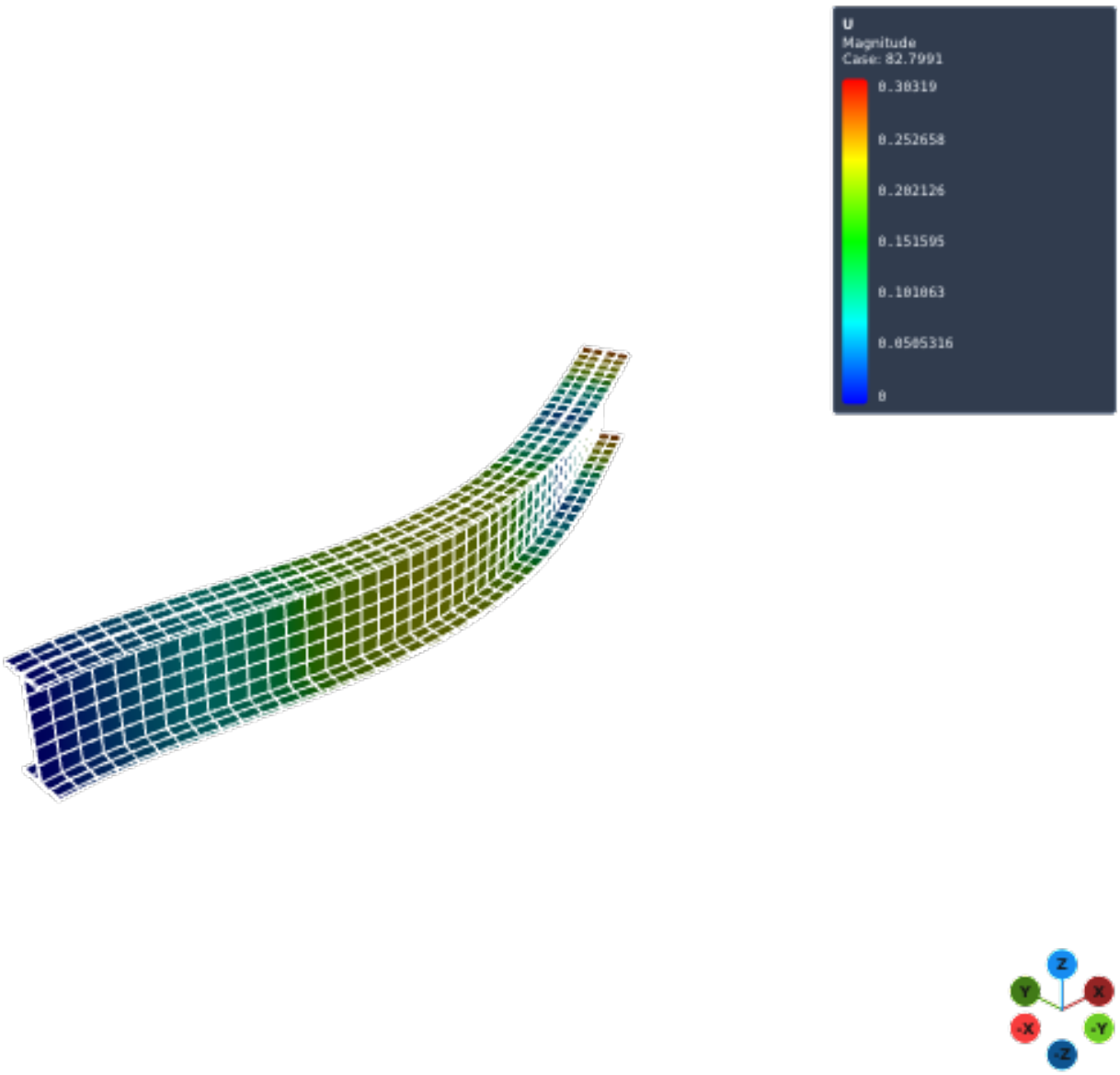

Figure 359: Abaqus — mode 4, 82.799 Hz

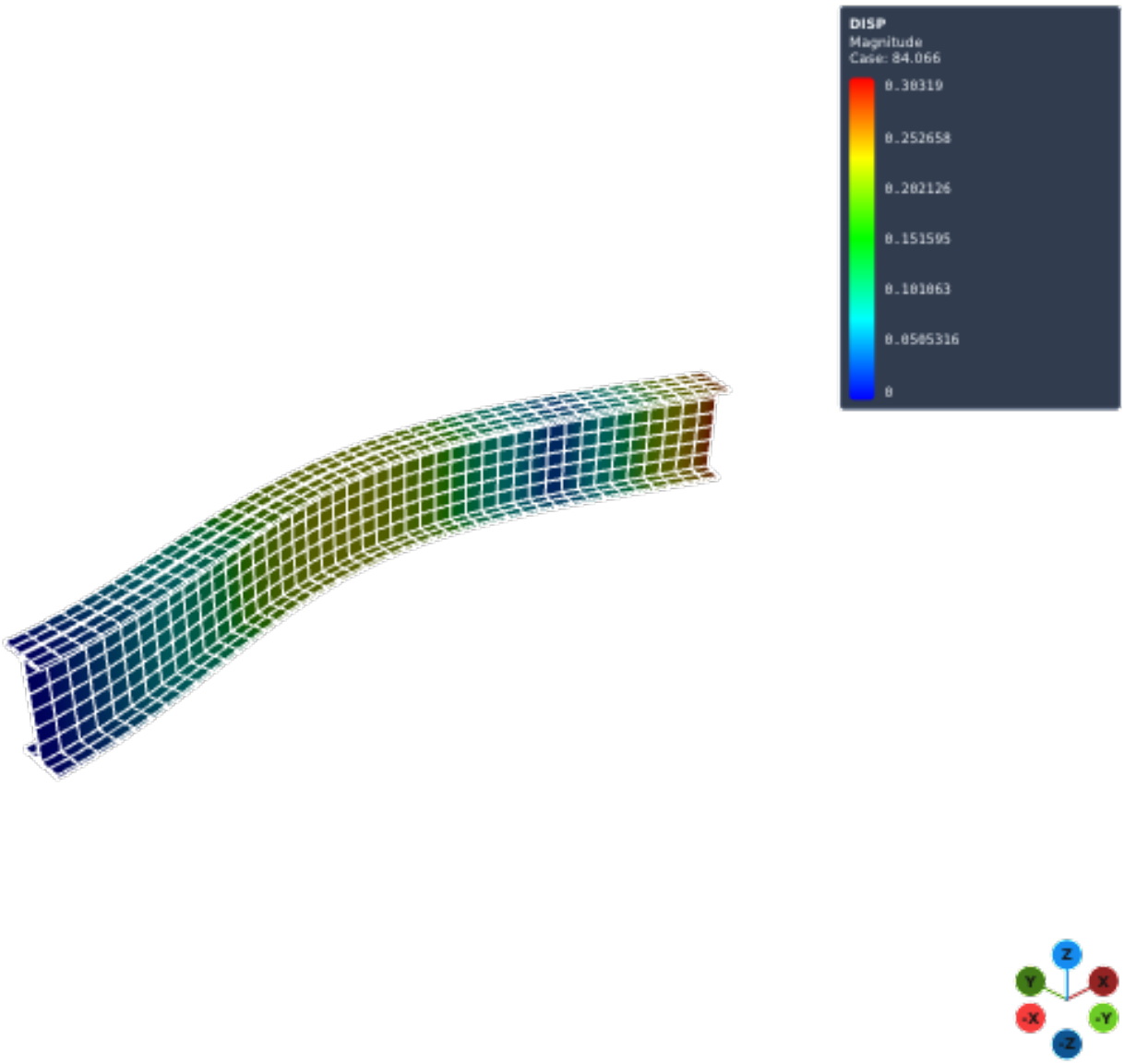

Figure 360: Calculix — mode 4, 84.066 Hz

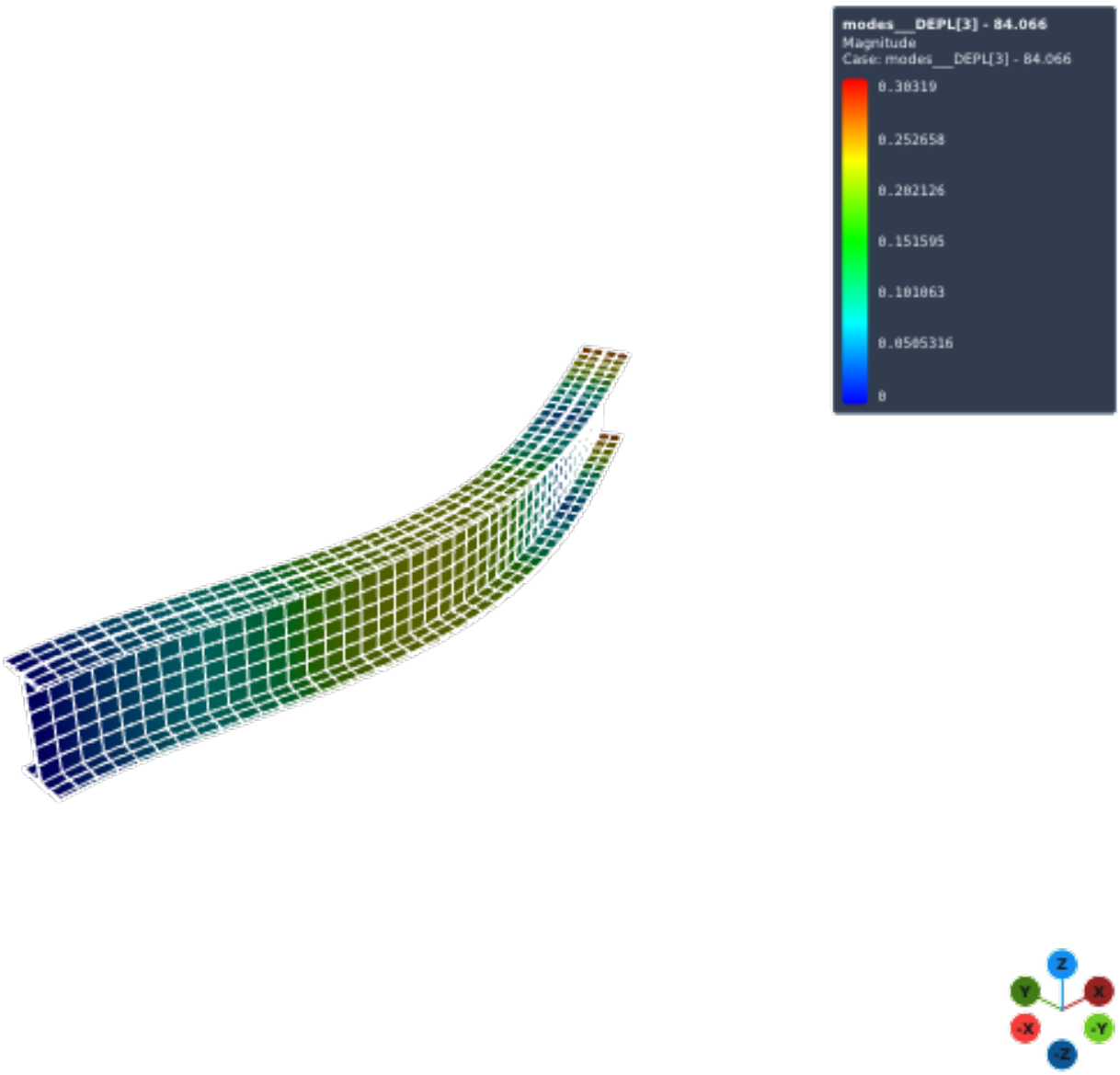

Figure 361: Code_Aster — mode 4, 84.066 Hz

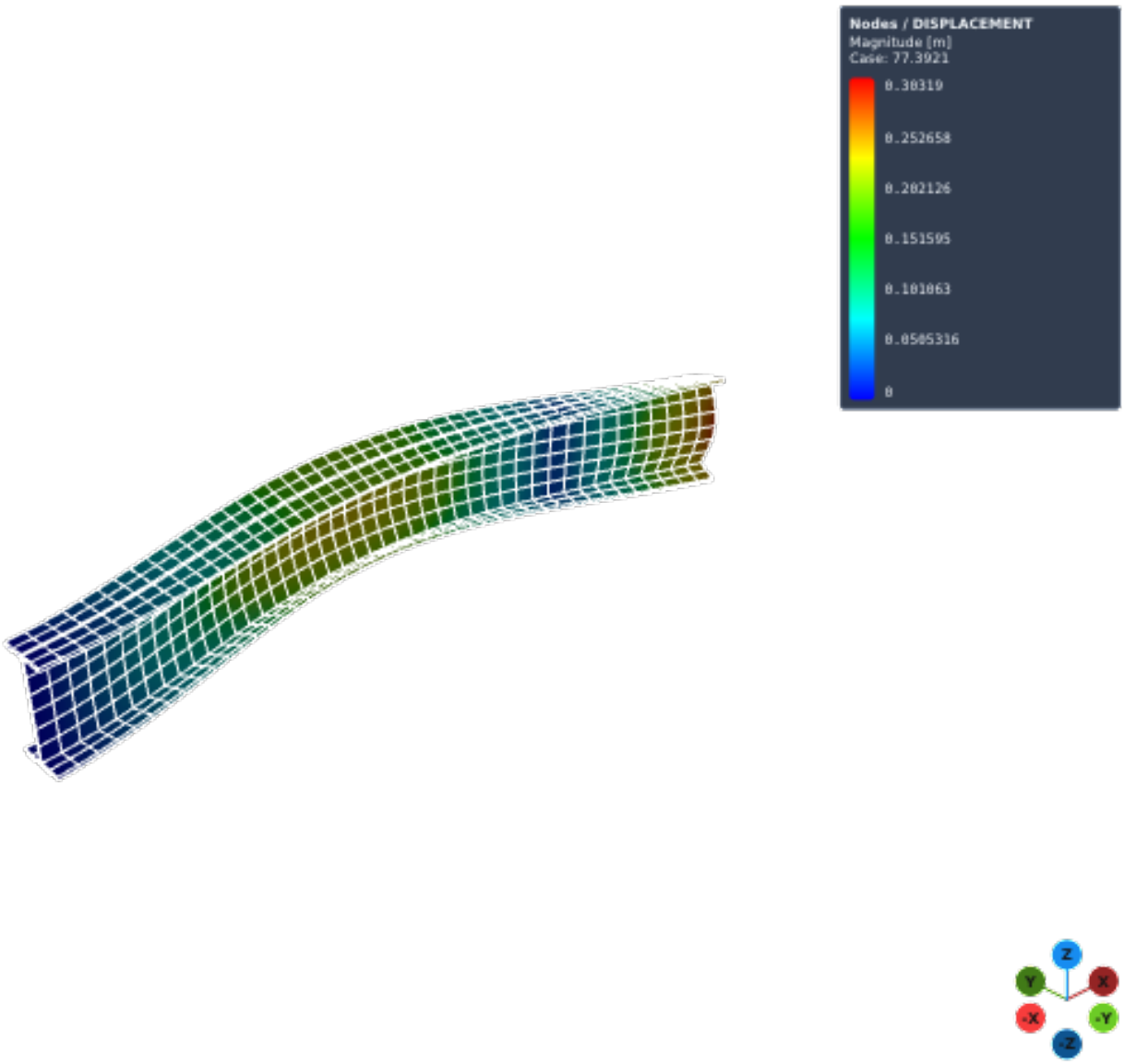

Figure 362: Sesam — mode 4, 77.392 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	82.799	+0.87	270652	7.52e-26	36.02	2.70e-22	-2.74e-13	-6.002	1.64e-11
Calculix	84.066	+2.42	278998	4.64e-26	36.05	4.98e-23	2.15e-13	6.004	7.06e-12
Code_Aster	84.066	+2.42	278998	2.15e-26	36.05	2.12e-22	1.47e-13	-6.004	-1.46e-11
Sesam	77.392	-5.71	236458	2.29e-26	35.94	6.77e-22	-1.51e-13	5.995	-2.60e-11

Mode 5

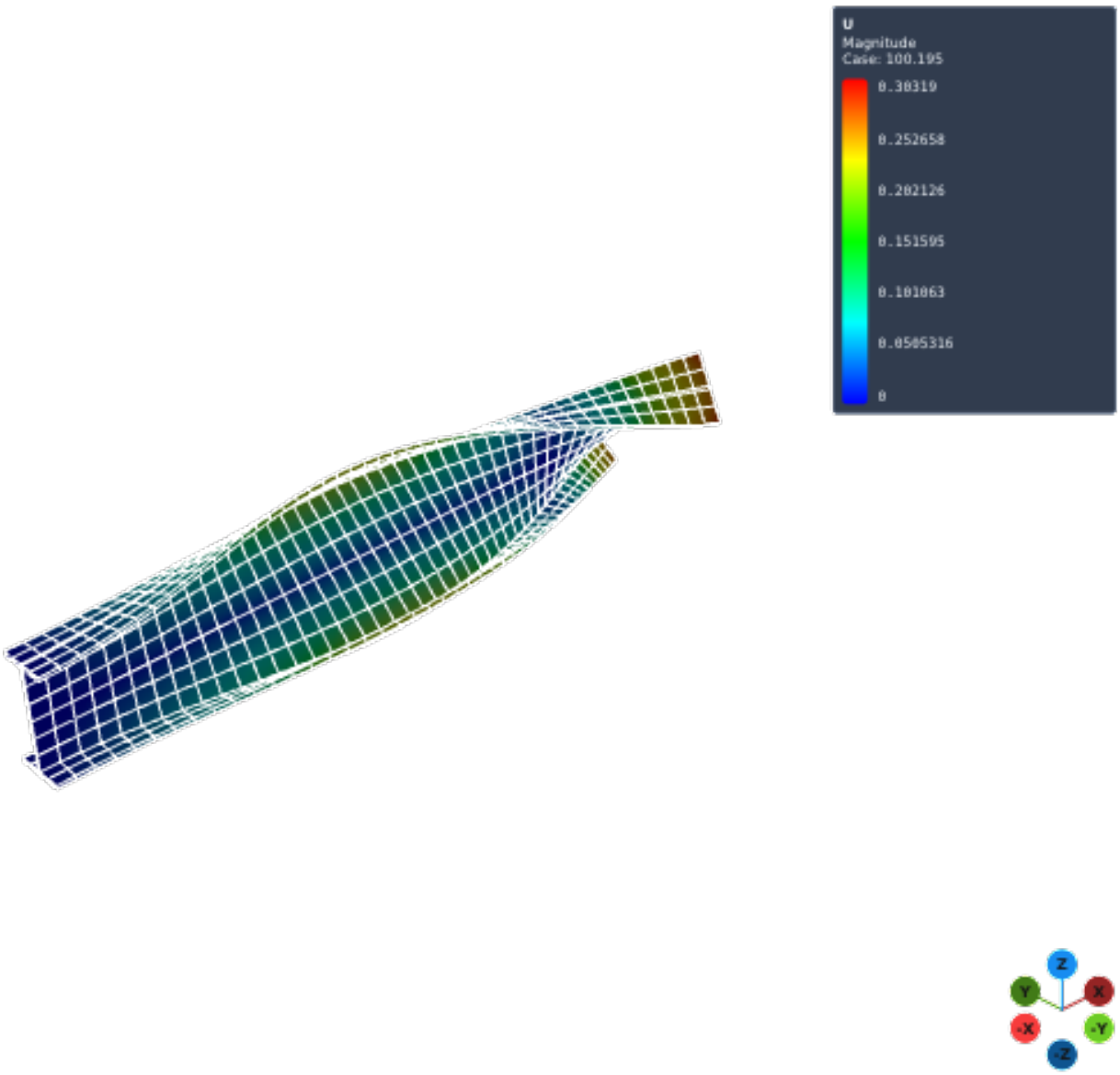

Figure 363: Abaqus — mode 5, 100.195 Hz

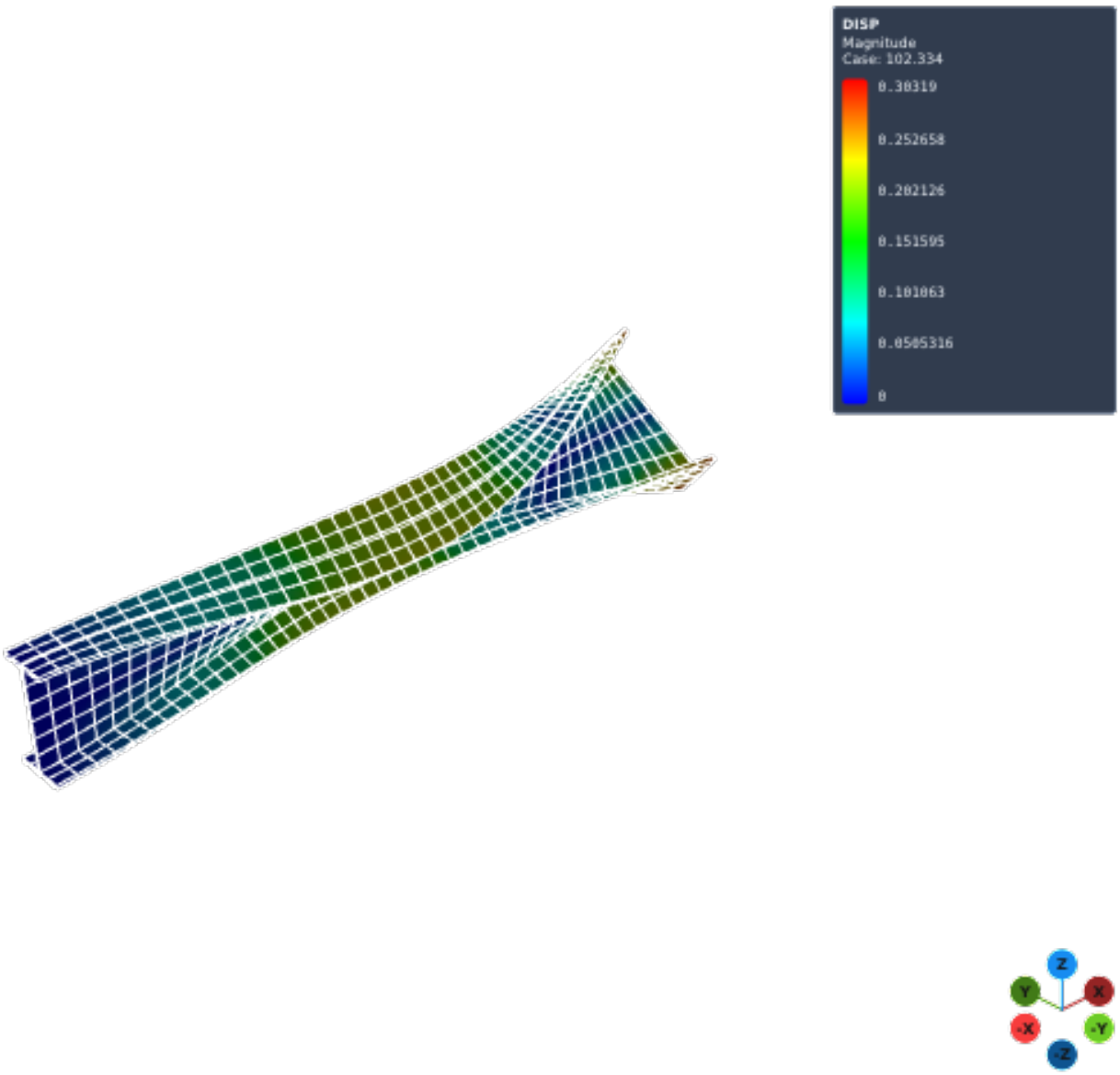

Figure 364: Calculix — mode 5, 102.334 Hz

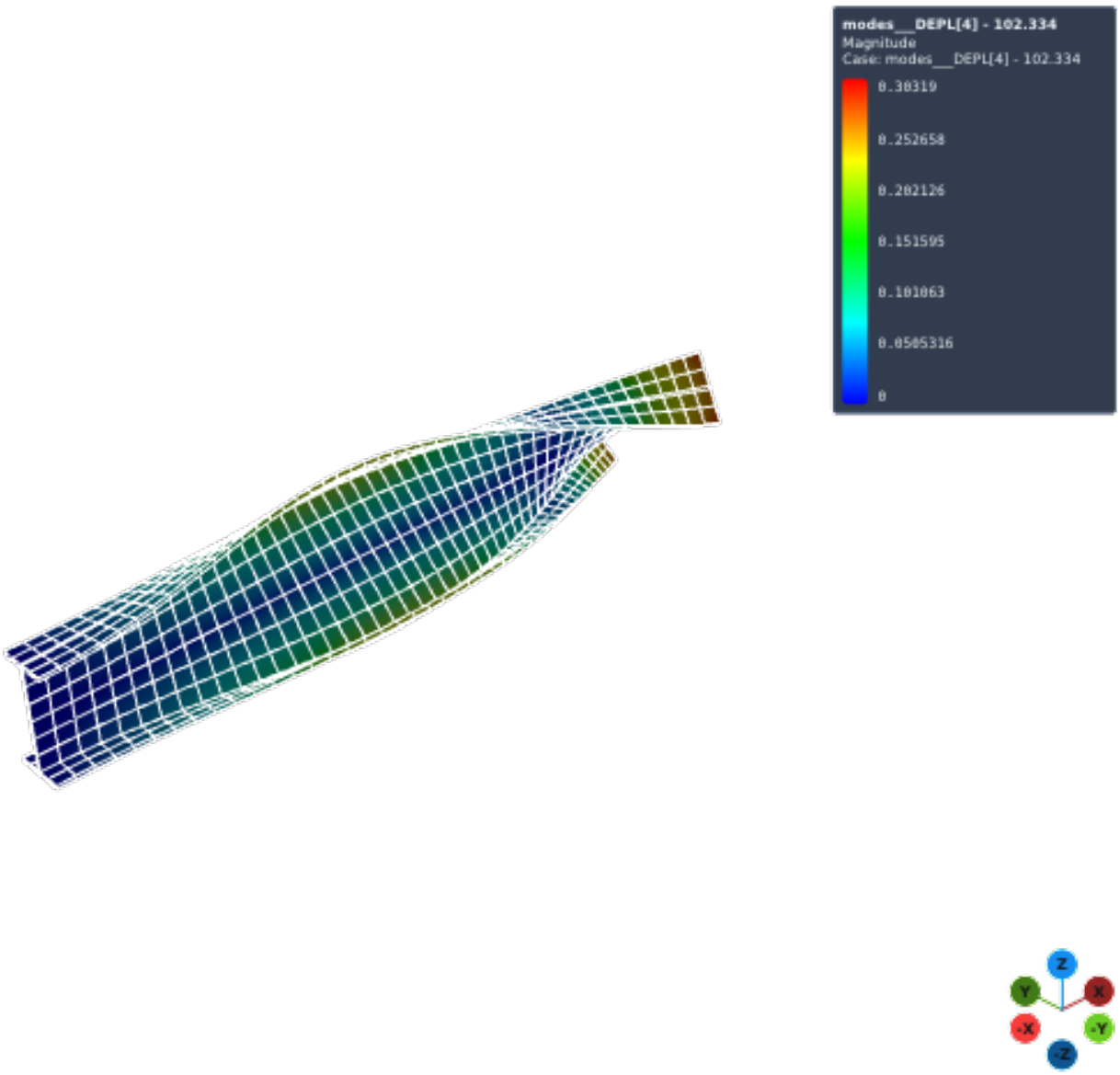

Figure 365: Code_Aster — mode 5, 102.334 Hz

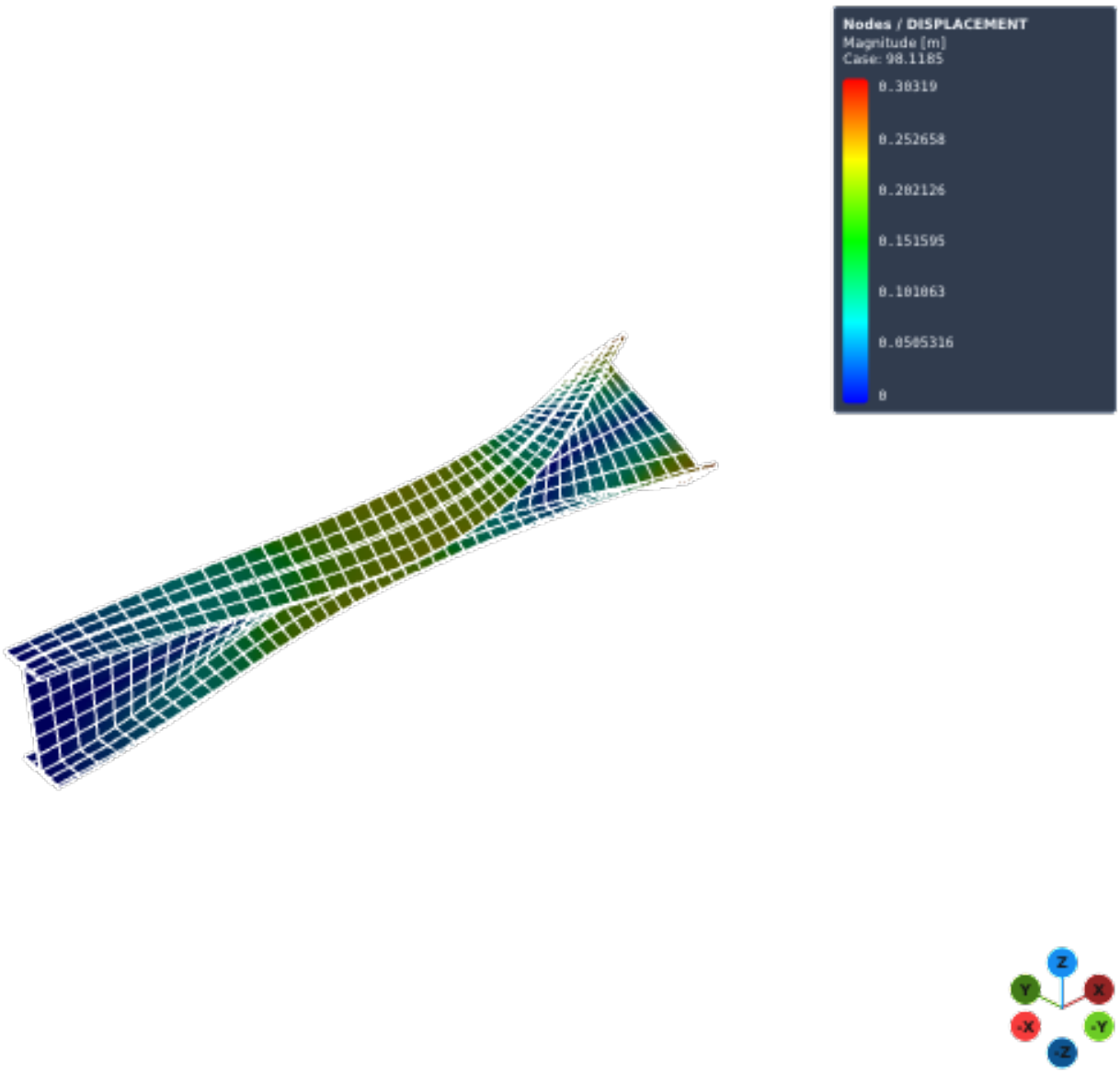

Figure 366: Sesam — mode 5, 98.119 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	100.190	-0.55	396325	4.10e-26	6.87e-21	7.52e-23	2.03e-13	-8.29e-11	8.67e-12
Calculix	102.334	+1.58	413428	1.70e-27	6.42e-21	4.55e-23	-4.12e-14	8.01e-11	-6.74e-12
Code_Aster	102.334	+1.58	413428	9.62e-27	3.93e-20	3.02e-24	-9.81e-14	-1.98e-10	-1.74e-12
Sesam	98.119	-2.61	380068	1.41e-25	3.09e-21	8.31e-23	3.76e-13	-5.56e-11	9.12e-12

Mode 6

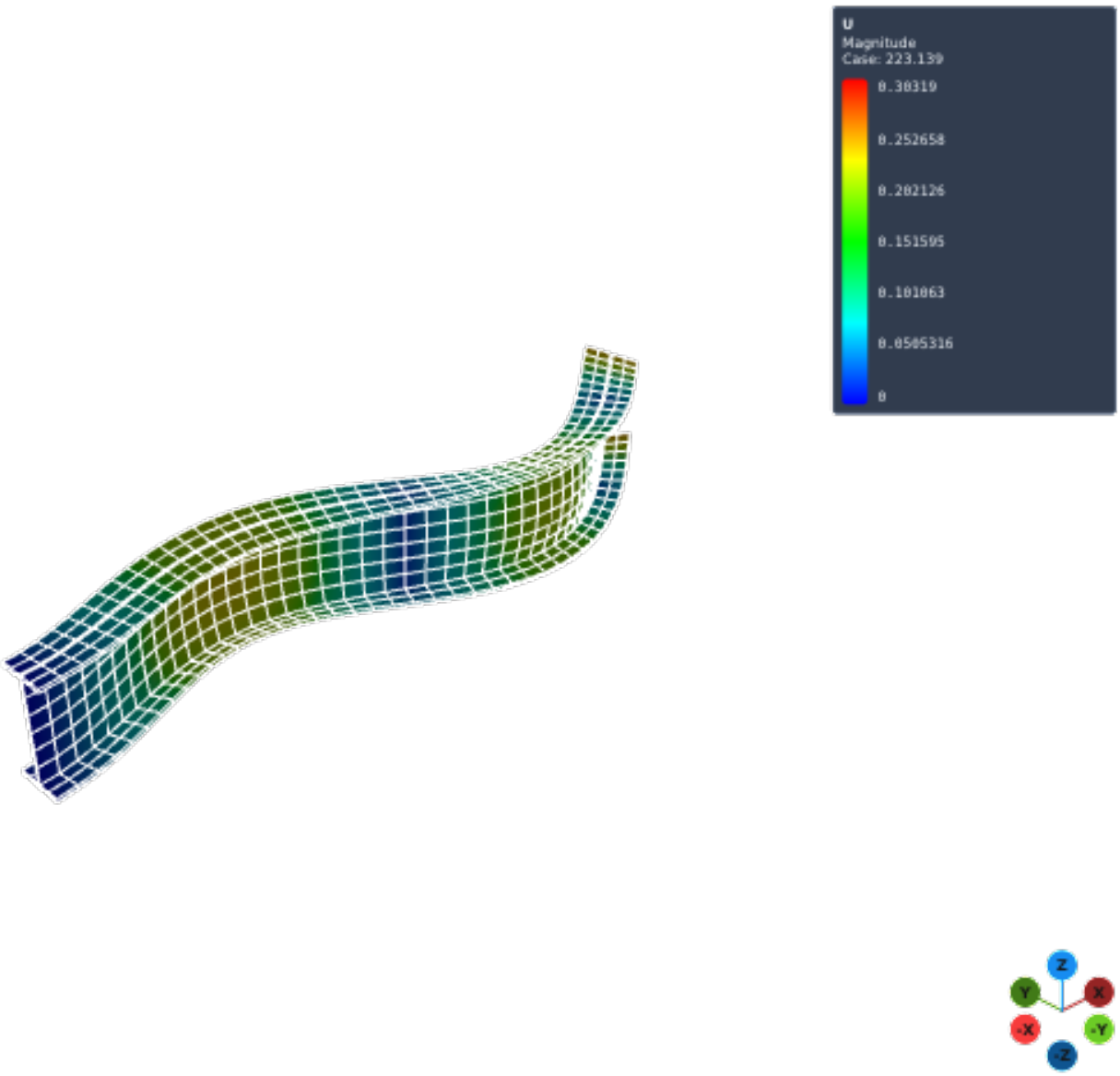

Figure 367: Abaqus — mode 6, 223.139 Hz

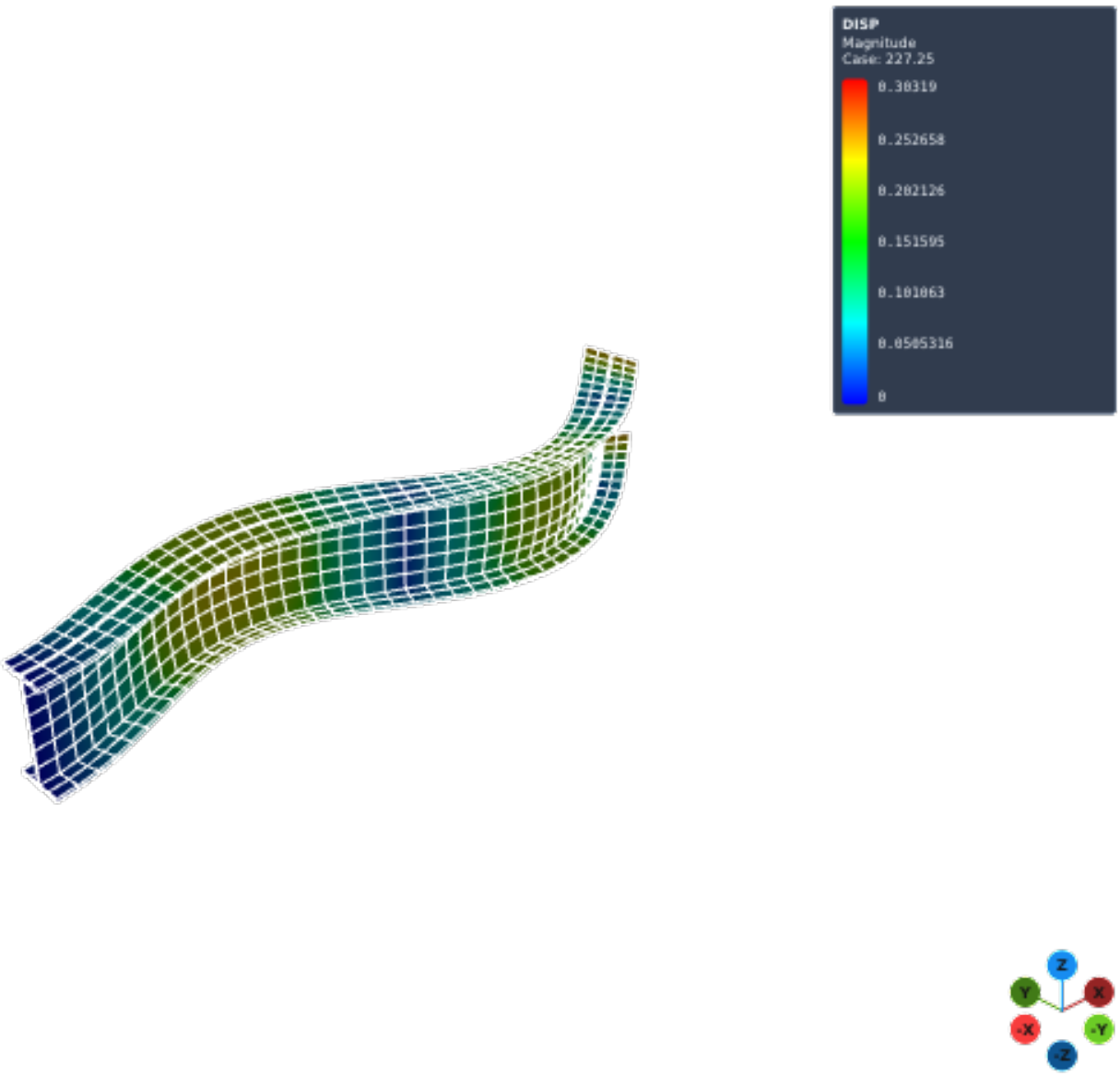

Figure 368: Calculix — mode 6, 227.250 Hz

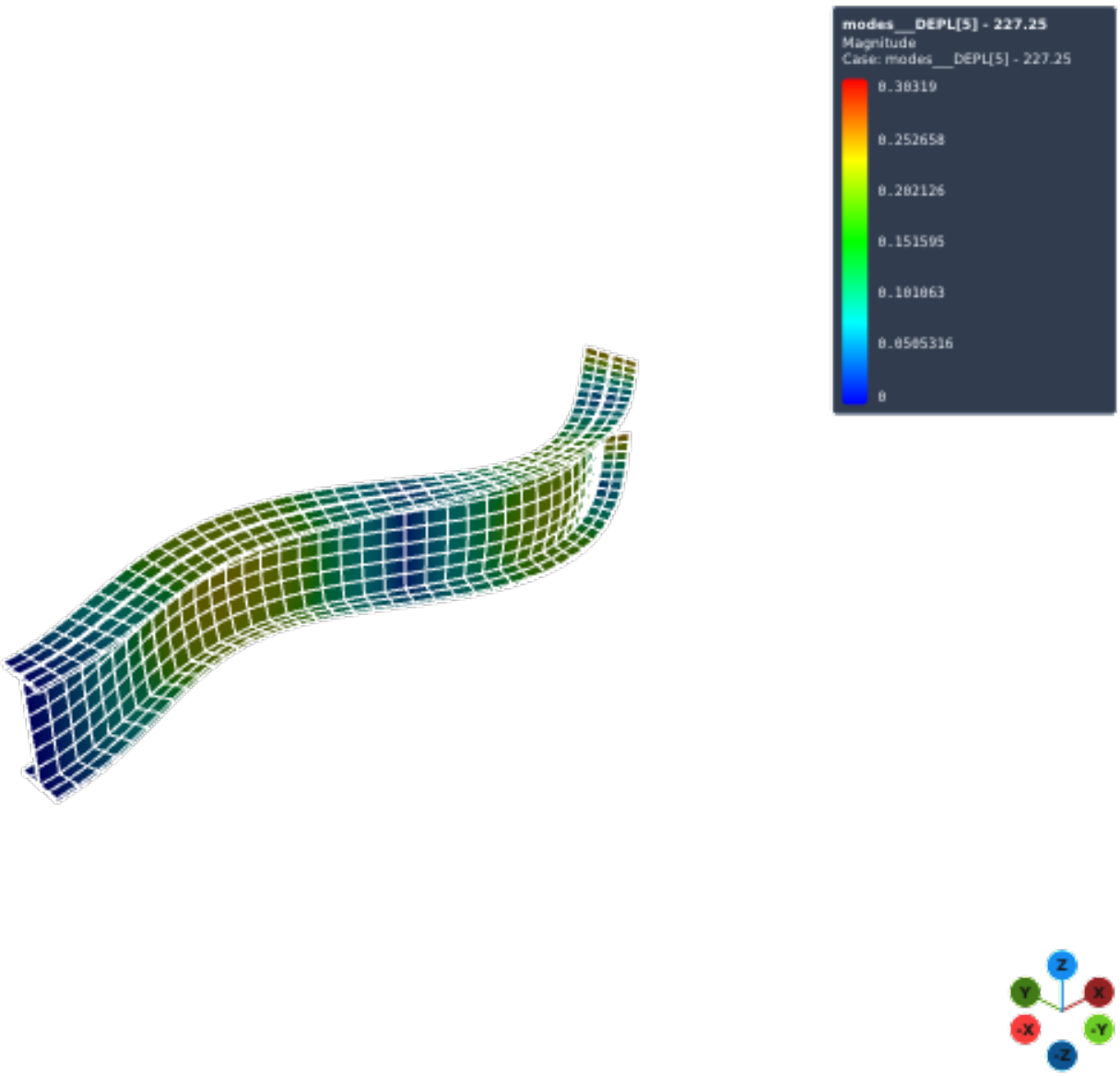

Figure 369: Code_Aster — mode 6, 227.250 Hz

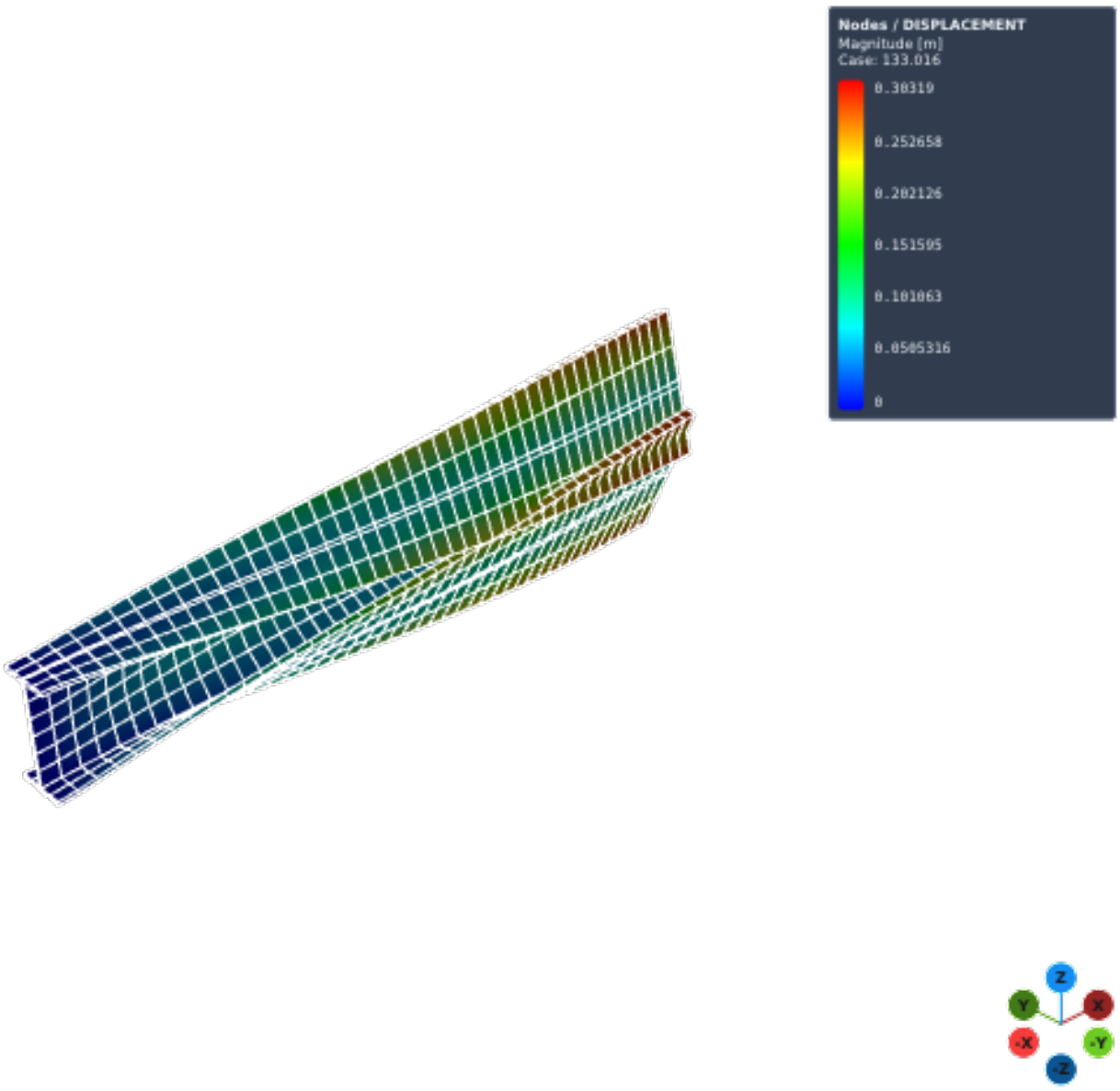

Figure 370: Sesam — mode 6, 133.016 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	223.140	+10.10	1.96567e+06	2.33e-25	12.40	1.64e-23	-4.83e-13	3.522	4.05e-12
Calculix	227.250	+12.13	2.03876e+06	1.37e-26	12.46	3.21e-24	1.17e-13	3.530	1.79e-12
Code_Aster	227.250	+12.13	2.03876e+06	4.29e-26	12.47	5.64e-25	-2.07e-13	3.532	-7.51e-13
Sesam	133.016	-34.37	698501	5.75e-26	0.01	1.11e-20	-2.40e-13	-0.114	1.05e-10

Mode 7

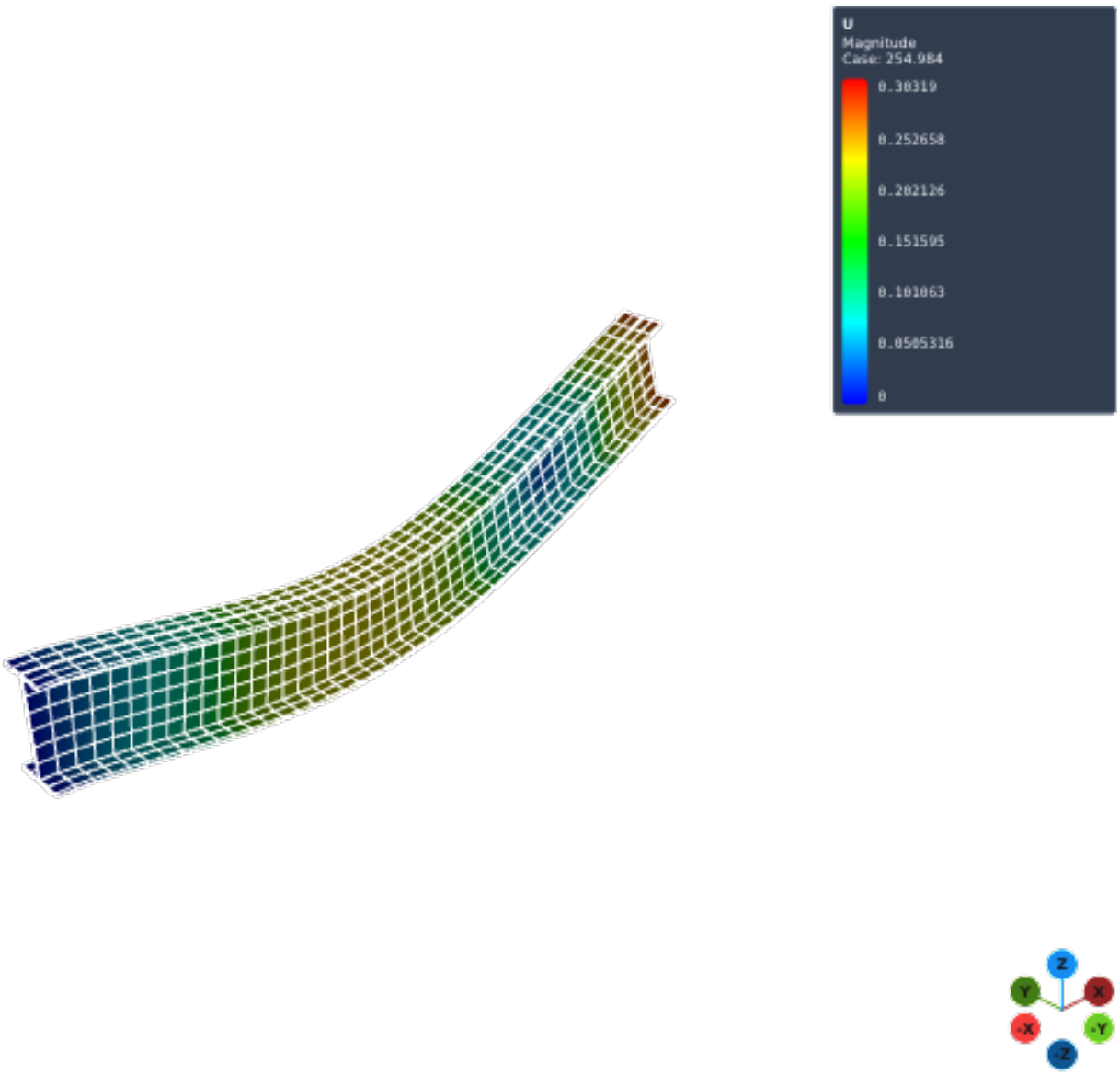

Figure 371: Abaqus — mode 7, 254.984 Hz

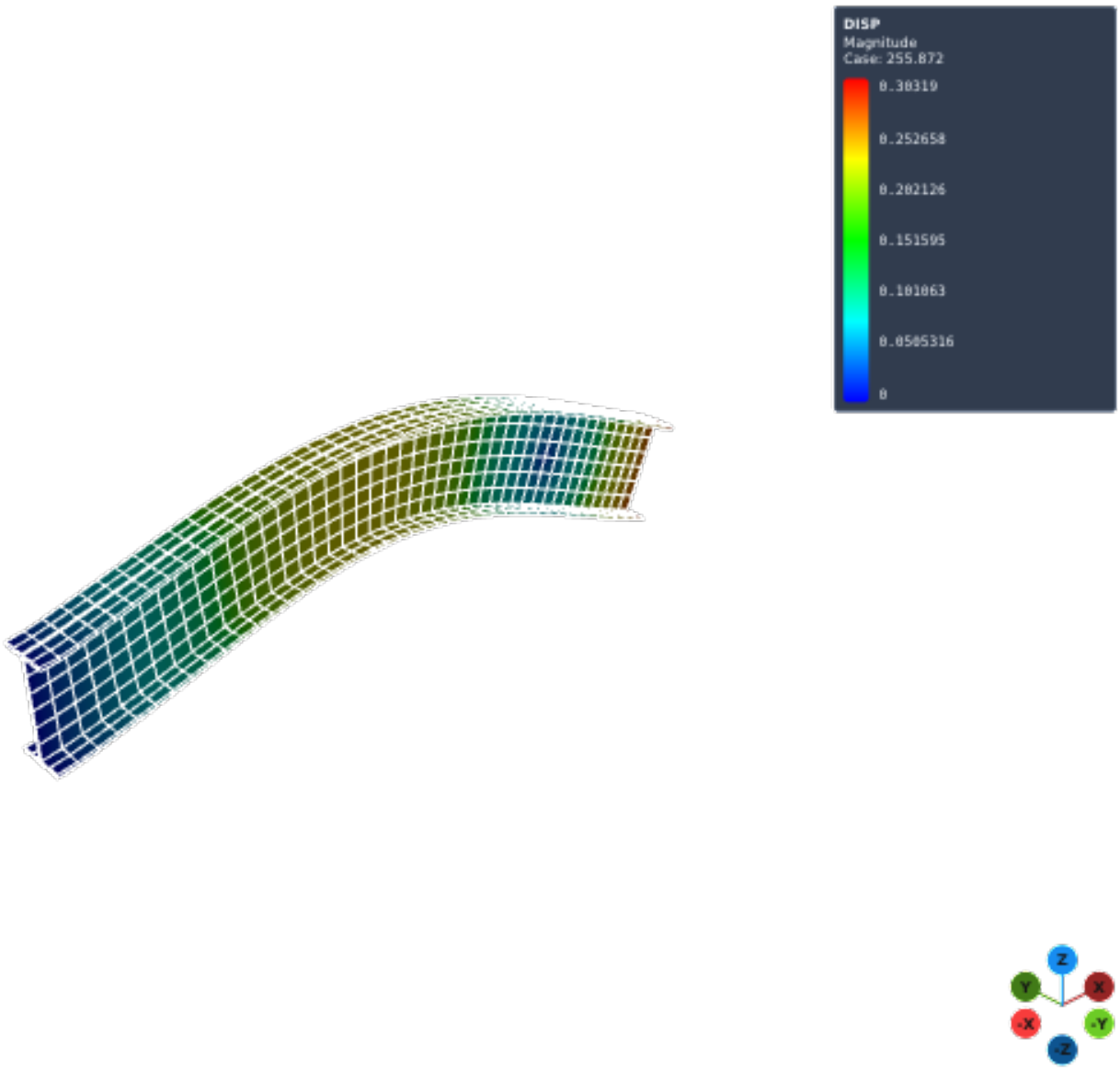

Figure 372: Calculix — mode 7, 255.872 Hz

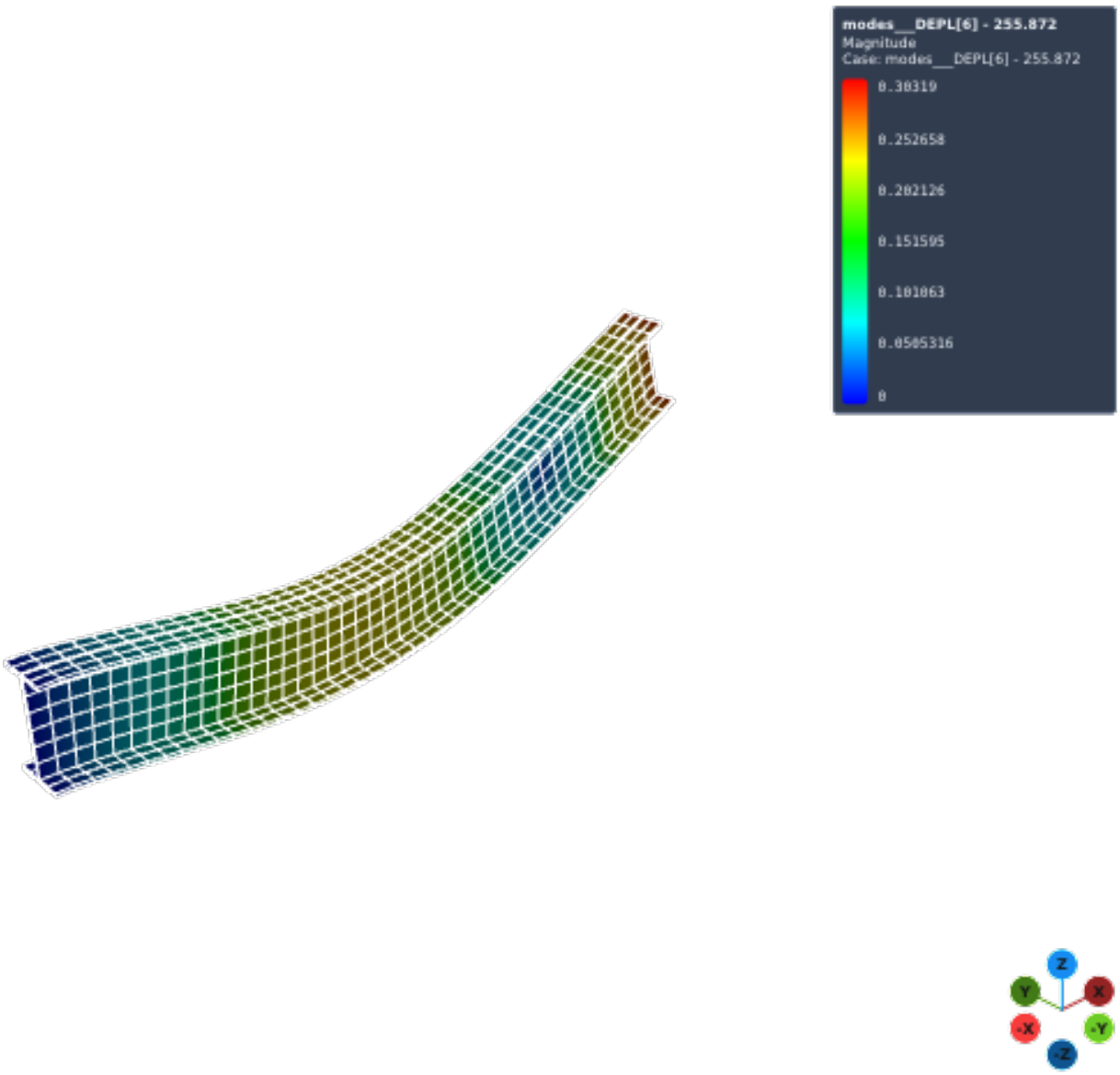

Figure 373: Code_Aster — mode 7, 255.872 Hz

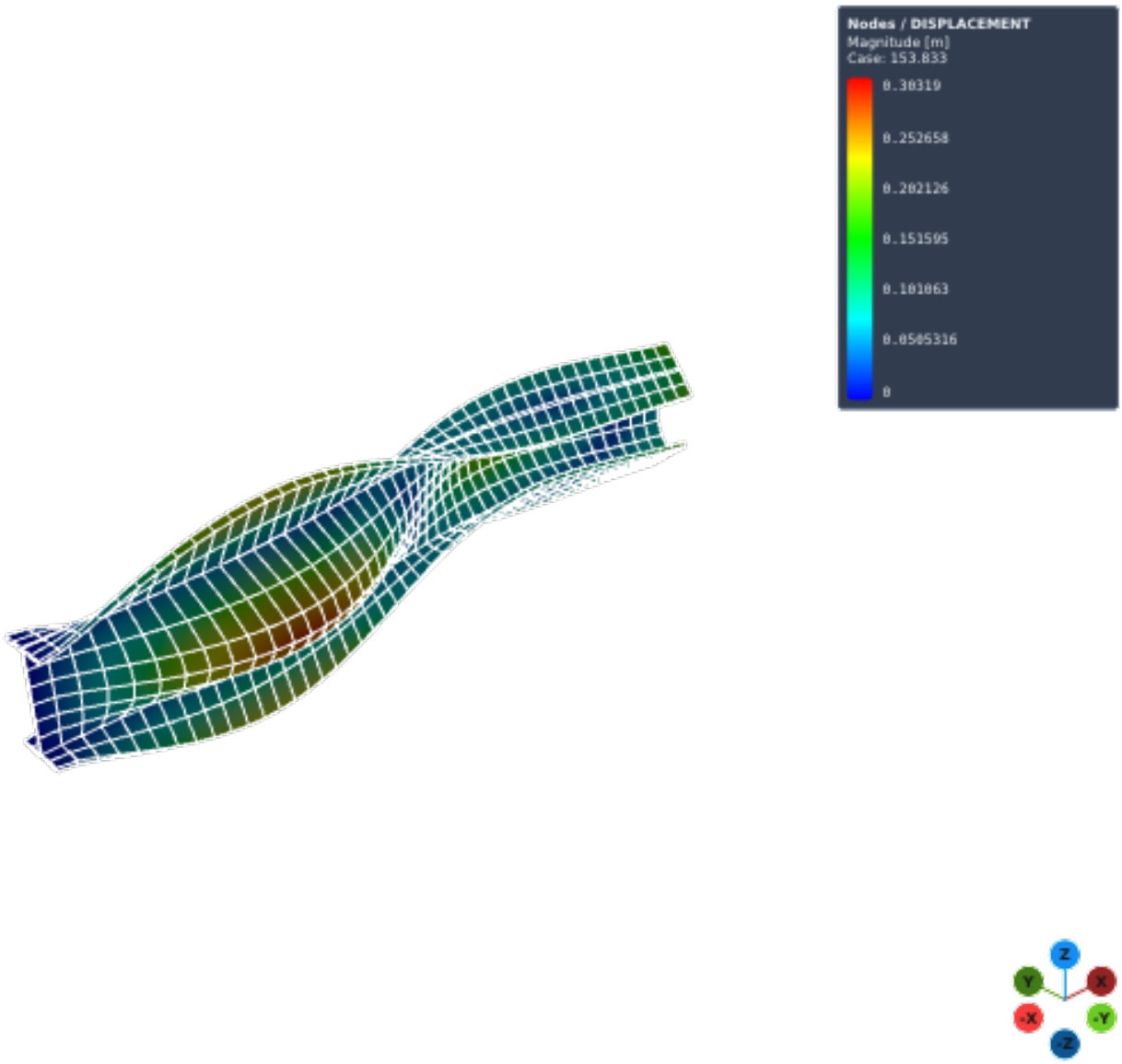

Figure 374: Sesam — mode 7, 153.833 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	254.980	+10.79	2.56676e+06	2.24e-25	2.85e-24	40.01	-4.74e-13	1.69e-12	-6.326
Calculix	255.872	+11.18	2.58466e+06	7.68e-25	2.60e-25	40.03	8.76e-13	-5.10e-13	6.327
Code_Aster	255.872	+11.18	2.58466e+06	7.10e-26	5.53e-25	40.06	2.66e-13	-7.43e-13	-6.329
Sesam	153.833	-33.16	934235	1.46e-25	3.20	1.07e-21	-3.82e-13	-1.788	3.27e-11

Mode 8

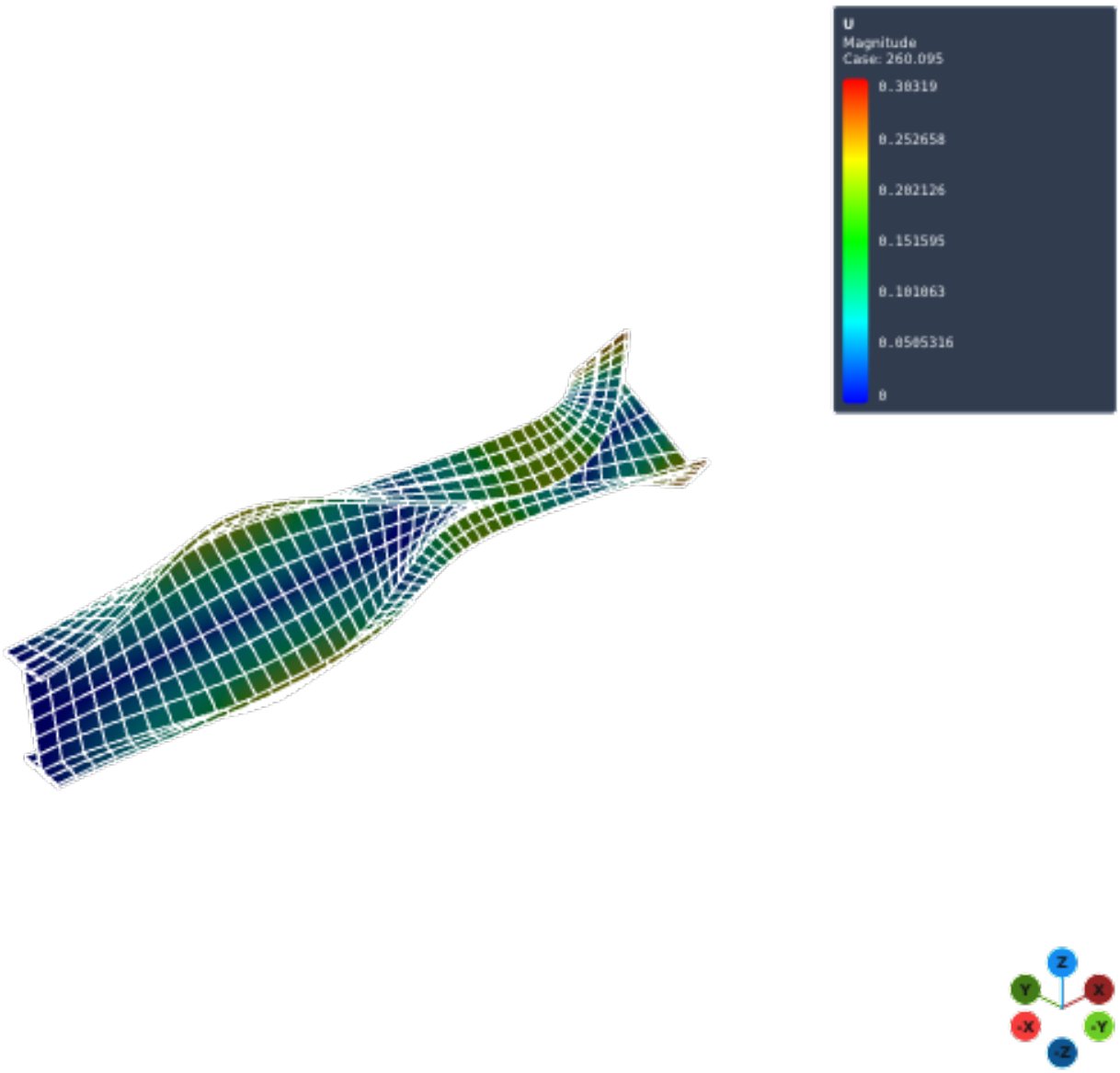

Figure 375: Abaqus — mode 8, 260.095 Hz

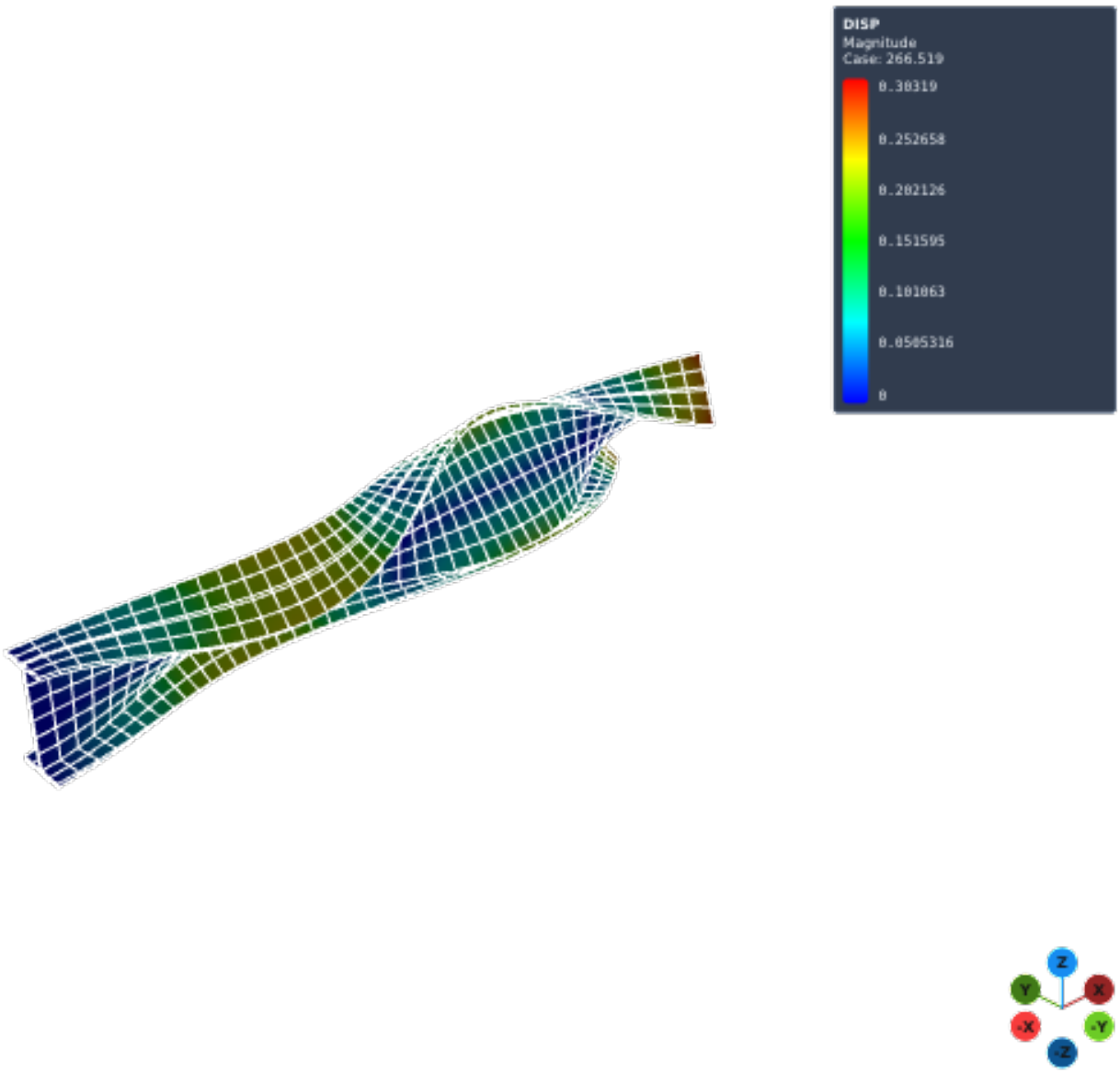

Figure 376: Calculix — mode 8, 266.519 Hz

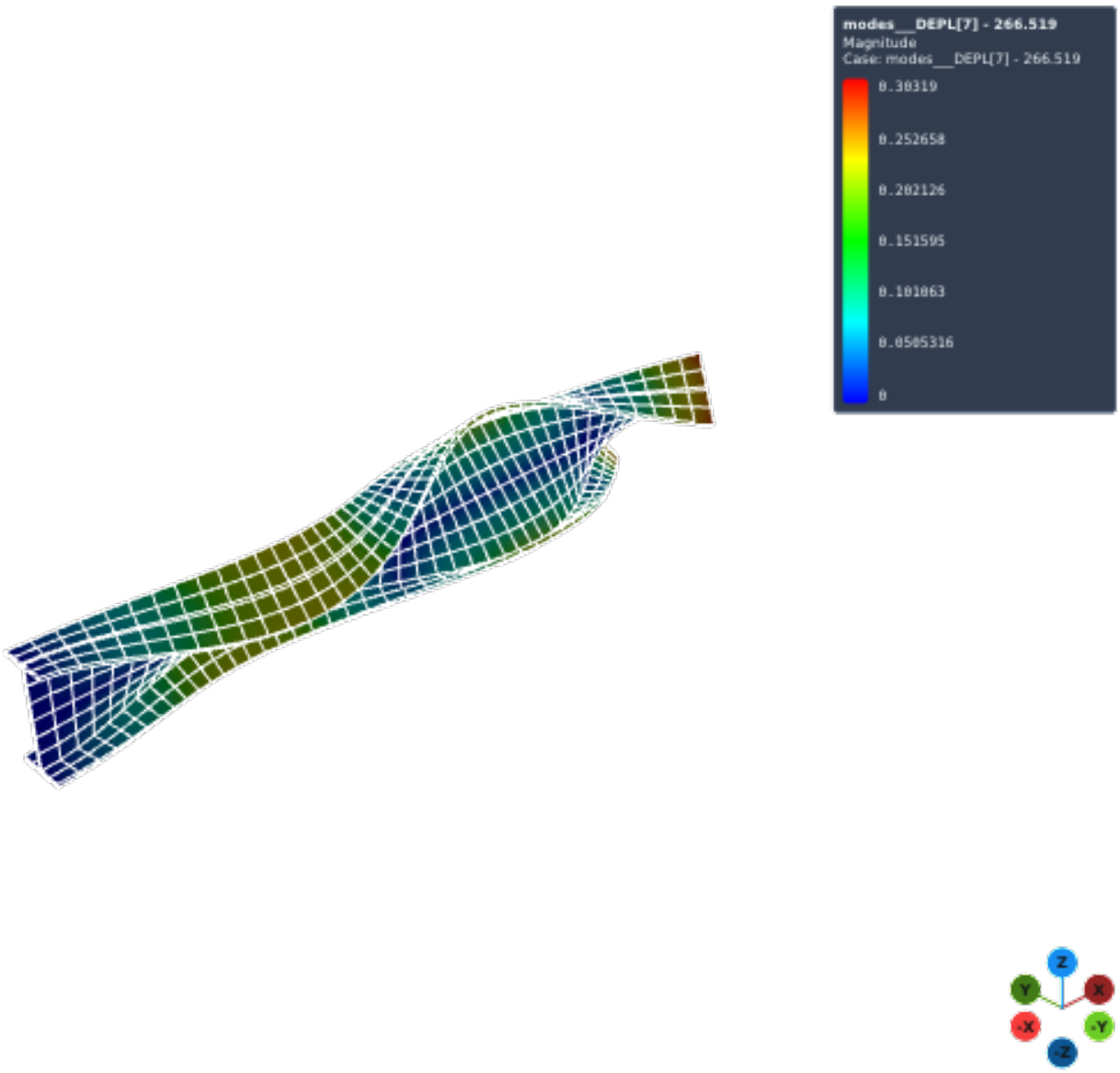

Figure 377: Code_Aster — mode 8, 266.519 Hz

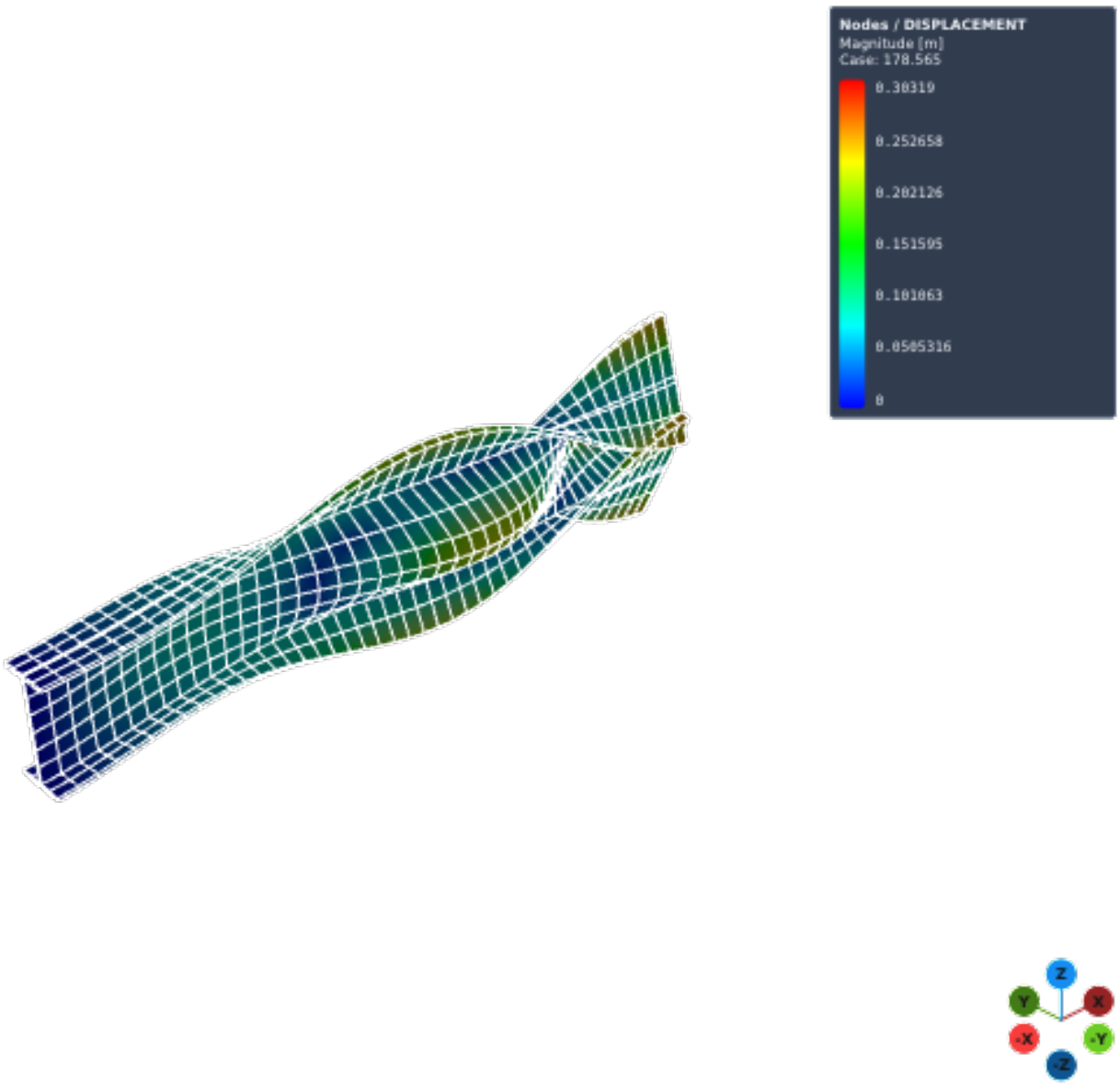

Figure 378: Sesam — mode 8, 178.565 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	260.100	+7.07	2.6707e+06	2.66e-26	6.24e-23	3.82e-21	-1.63e-13	7.90e-12	6.18e-11
Calculix	266.519	+9.71	2.80424e+06	5.05e-27	2.40e-22	2.23e-22	7.10e-14	1.55e-11	-1.49e-11
Code_Aster	266.519	+9.71	2.80424e+06	3.00e-26	3.09e-23	8.16e-23	-1.73e-13	5.56e-12	-9.03e-12
Sesam	178.565	-26.49	1.25879e+06	2.86e-28	5.71	2.39e-21	1.69e-14	2.390	4.89e-11

Mode 9

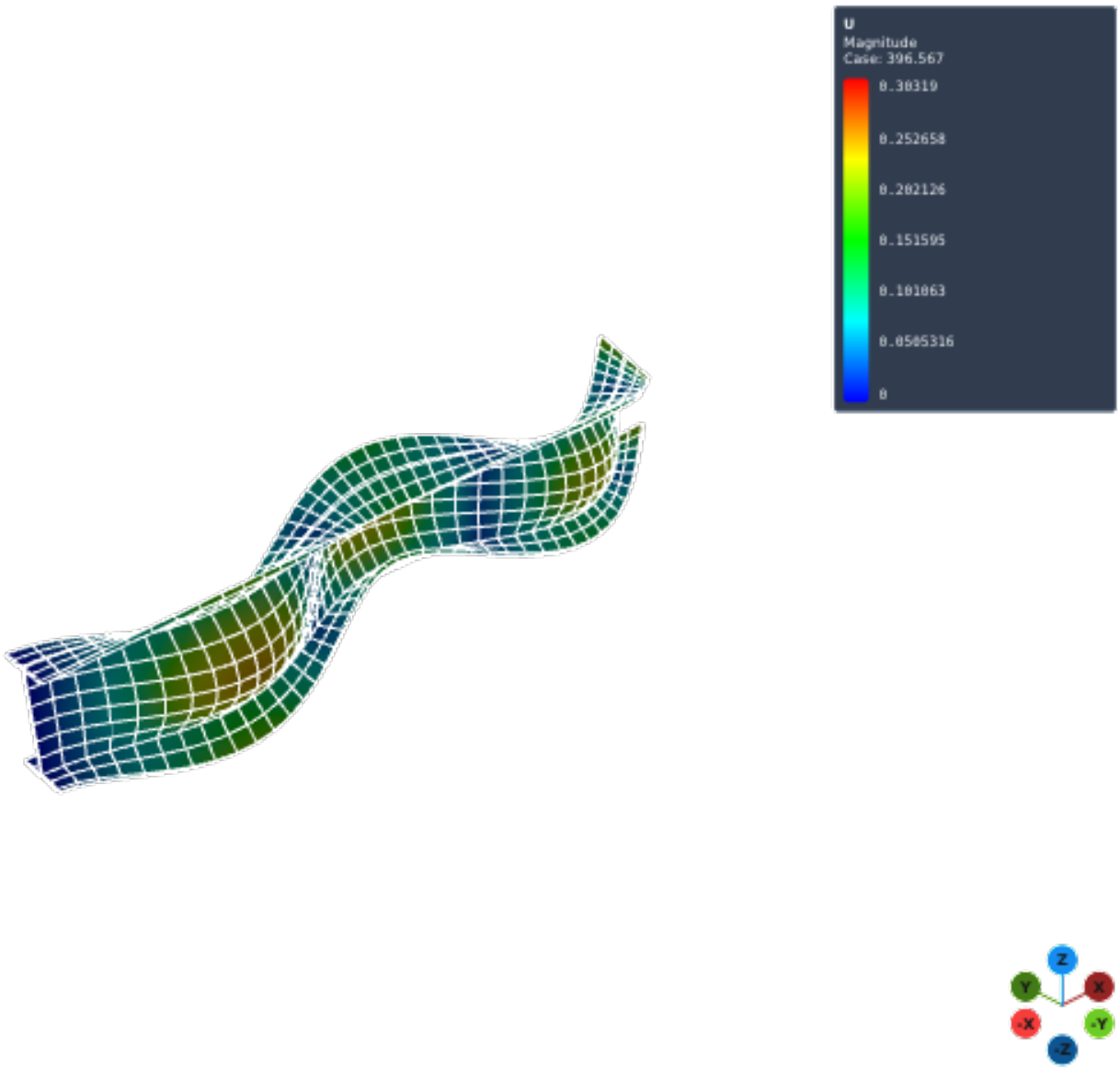

Figure 379: Abaqus — mode 9, 396.567 Hz

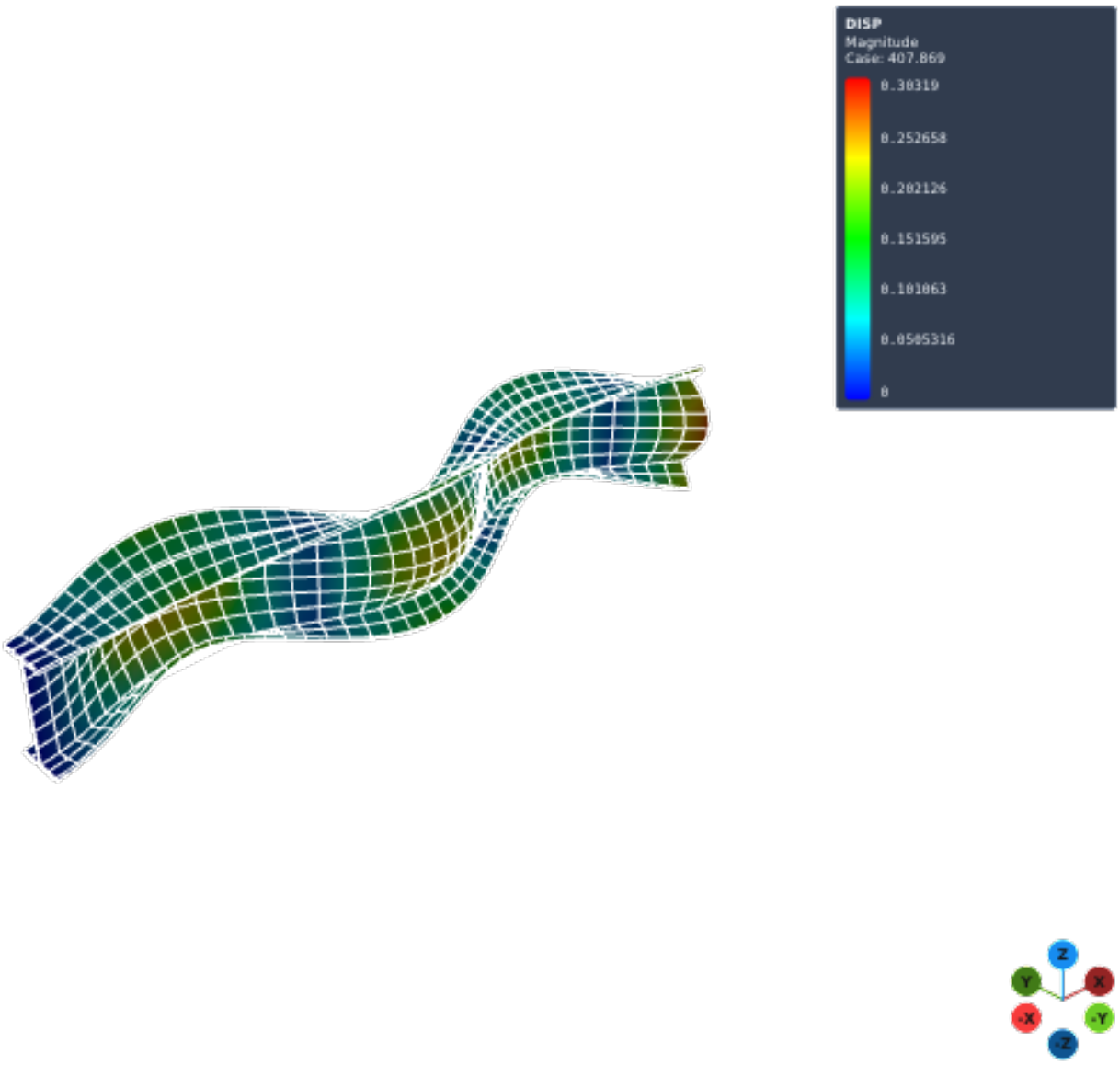

Figure 380: Calculix — mode 9, 407.869 Hz

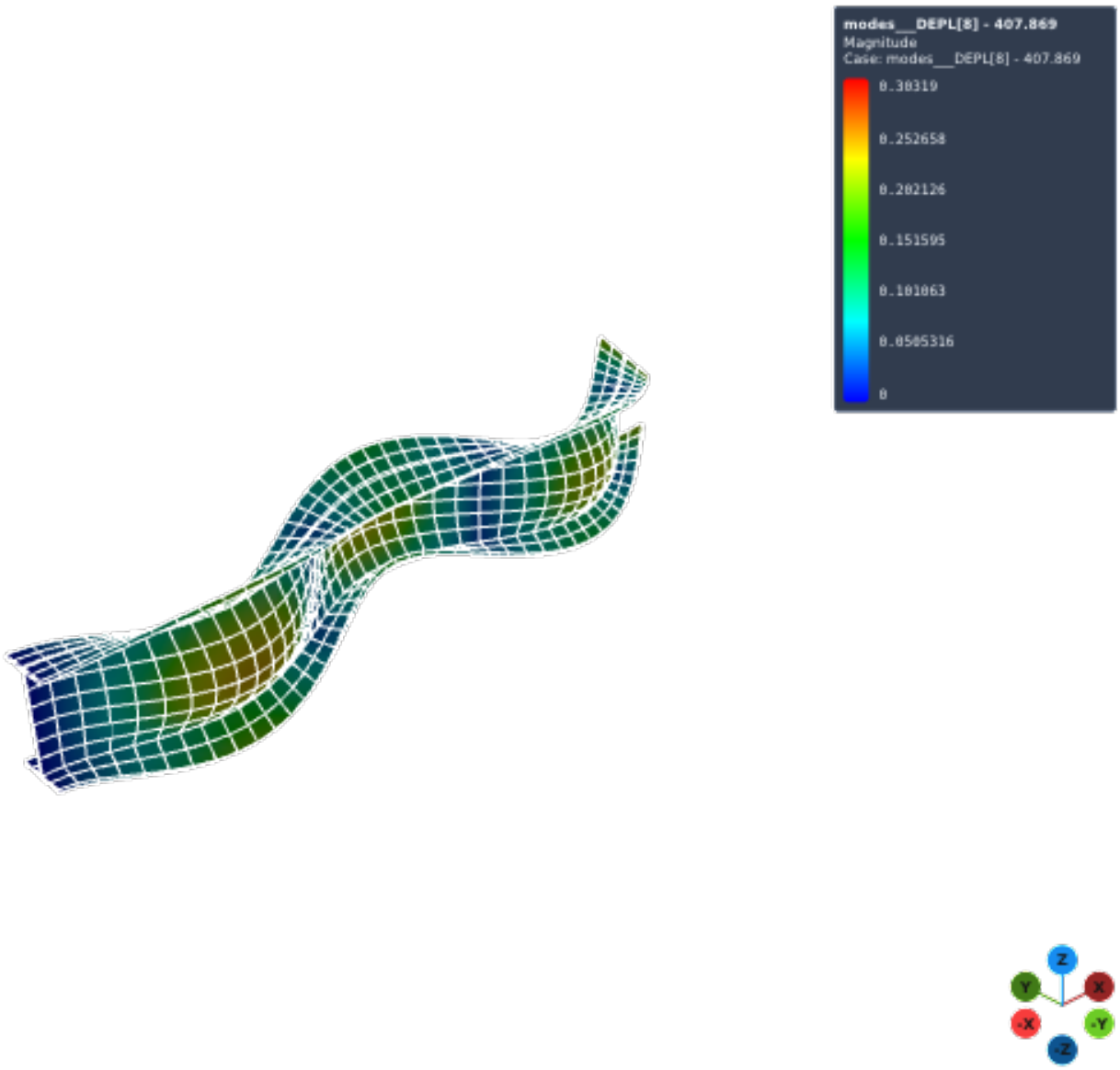

Figure 381: Code_Aster — mode 9, 407.869 Hz

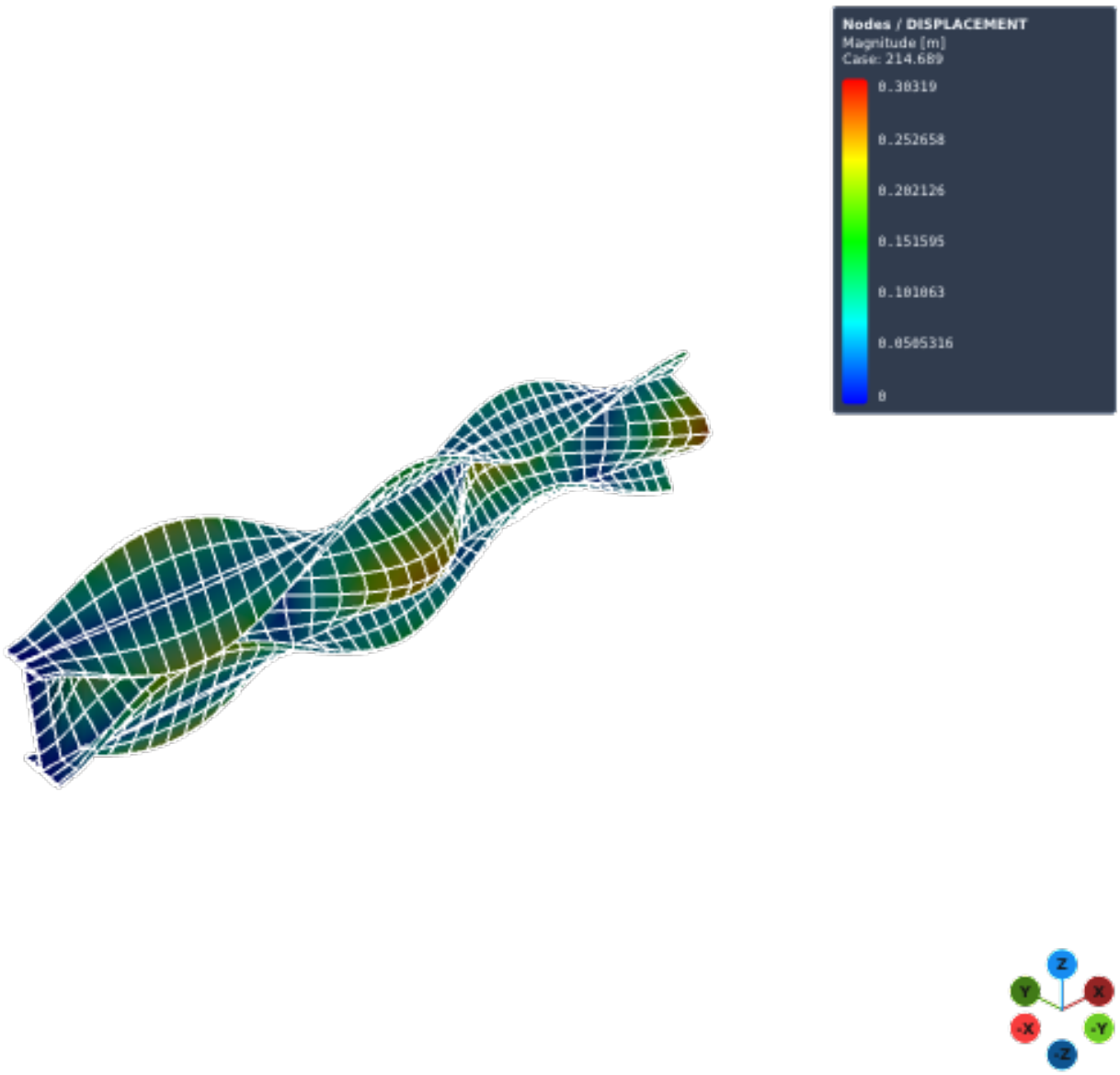

Figure 382: Sesam — mode 9, 214.689 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	396.570	+11.16	6.20858e+06	2.52e-22	5.69	2.13e-23	1.59e-11	-2.385	4.61e-12
Calculix	407.869	+14.33	6.56752e+06	2.22e-24	5.88	4.86e-25	-1.49e-12	2.426	6.97e-13
Code_Aster	407.869	+14.33	6.56753e+06	1.12e-23	5.90	3.42e-25	-3.34e-12	-2.430	5.84e-13
Sesam	214.689	-39.82	1.81962e+06	3.48e-26	0.44	2.56e-22	-1.87e-13	0.662	-1.60e-11

Mode 10

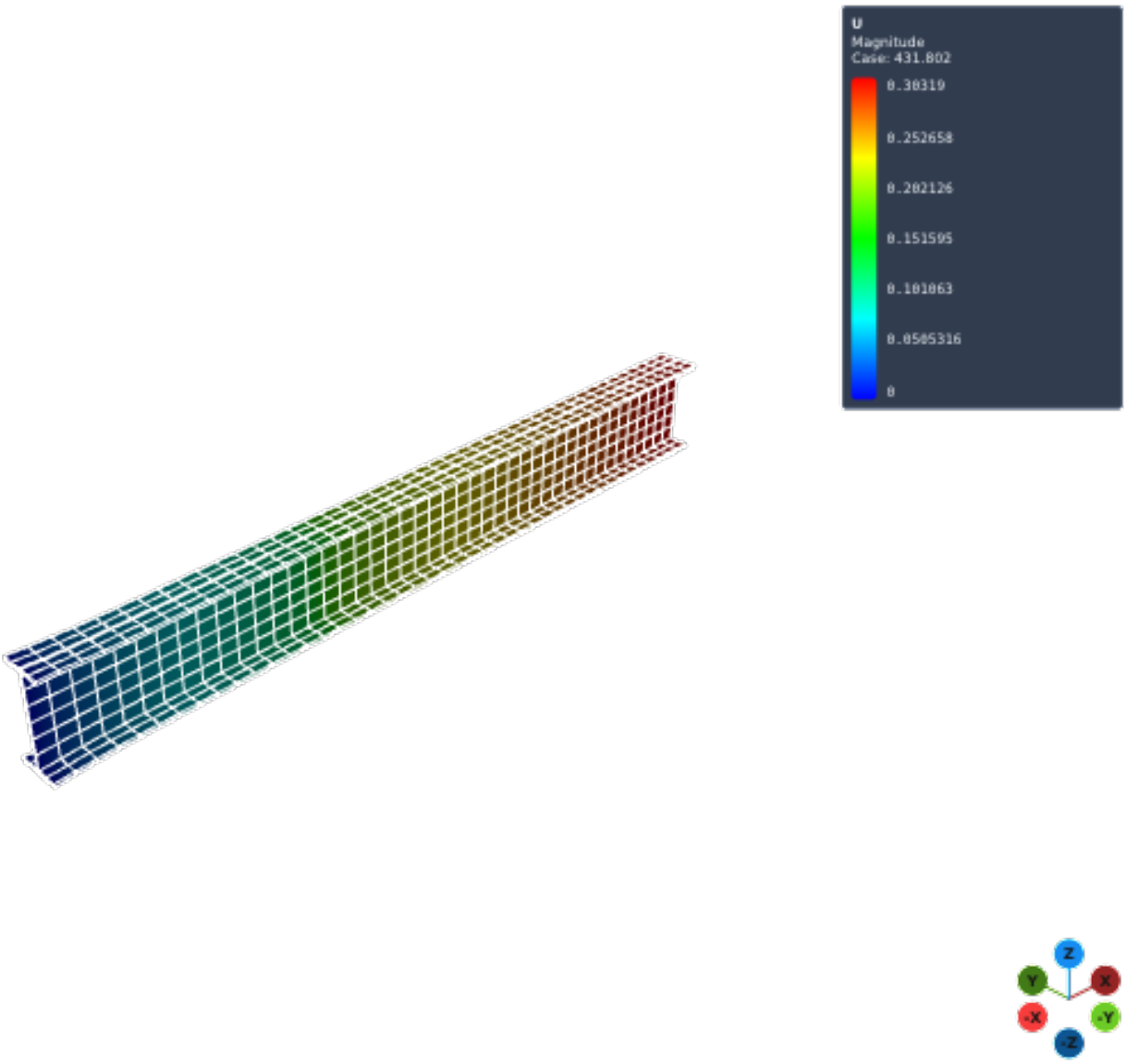

Figure 383: Abaqus — mode 10, 431.802 Hz

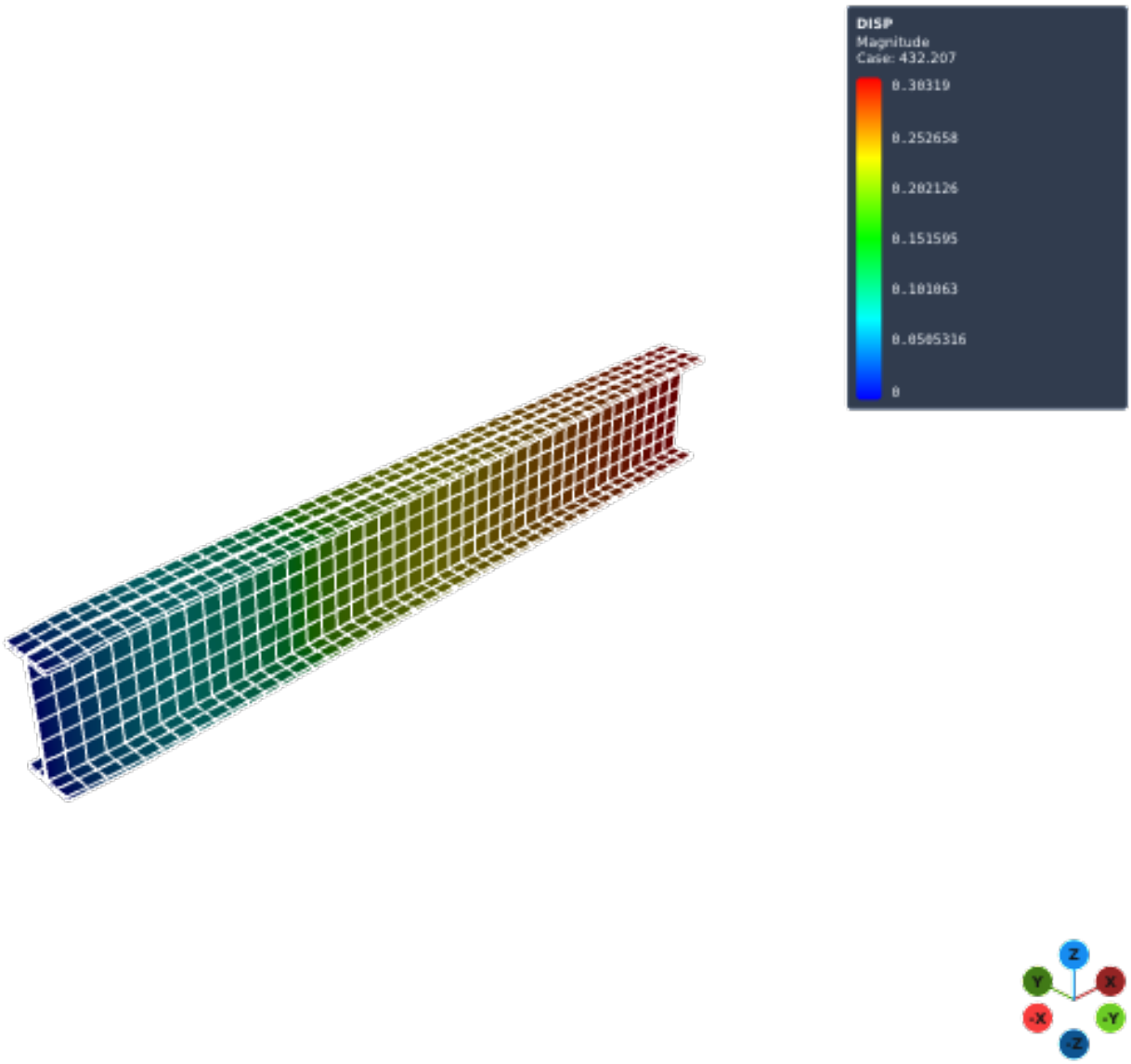

Figure 384: Calculix — mode 10, 432.207 Hz

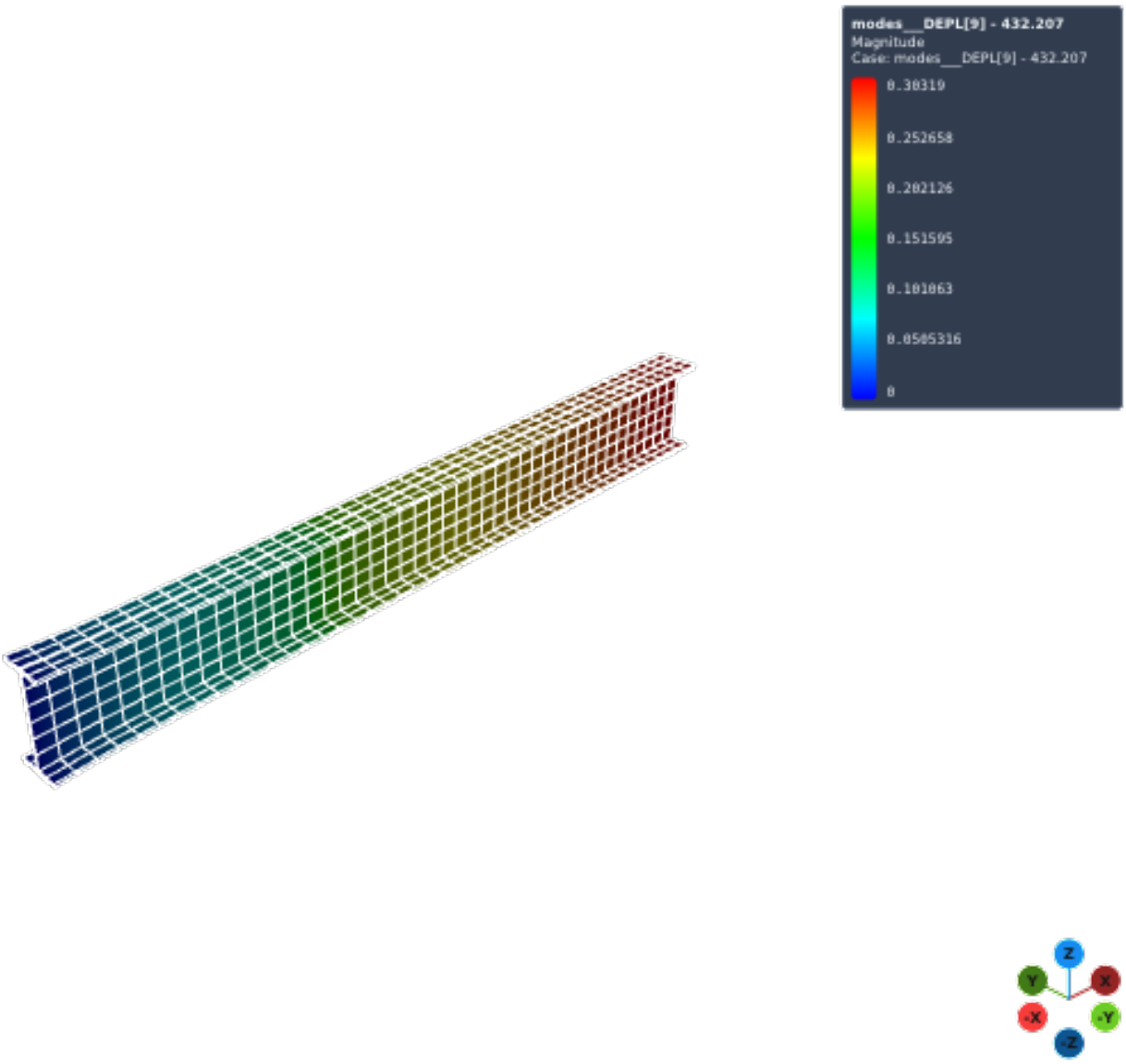

Figure 385: Code_Aster — mode 10, 432.207 Hz

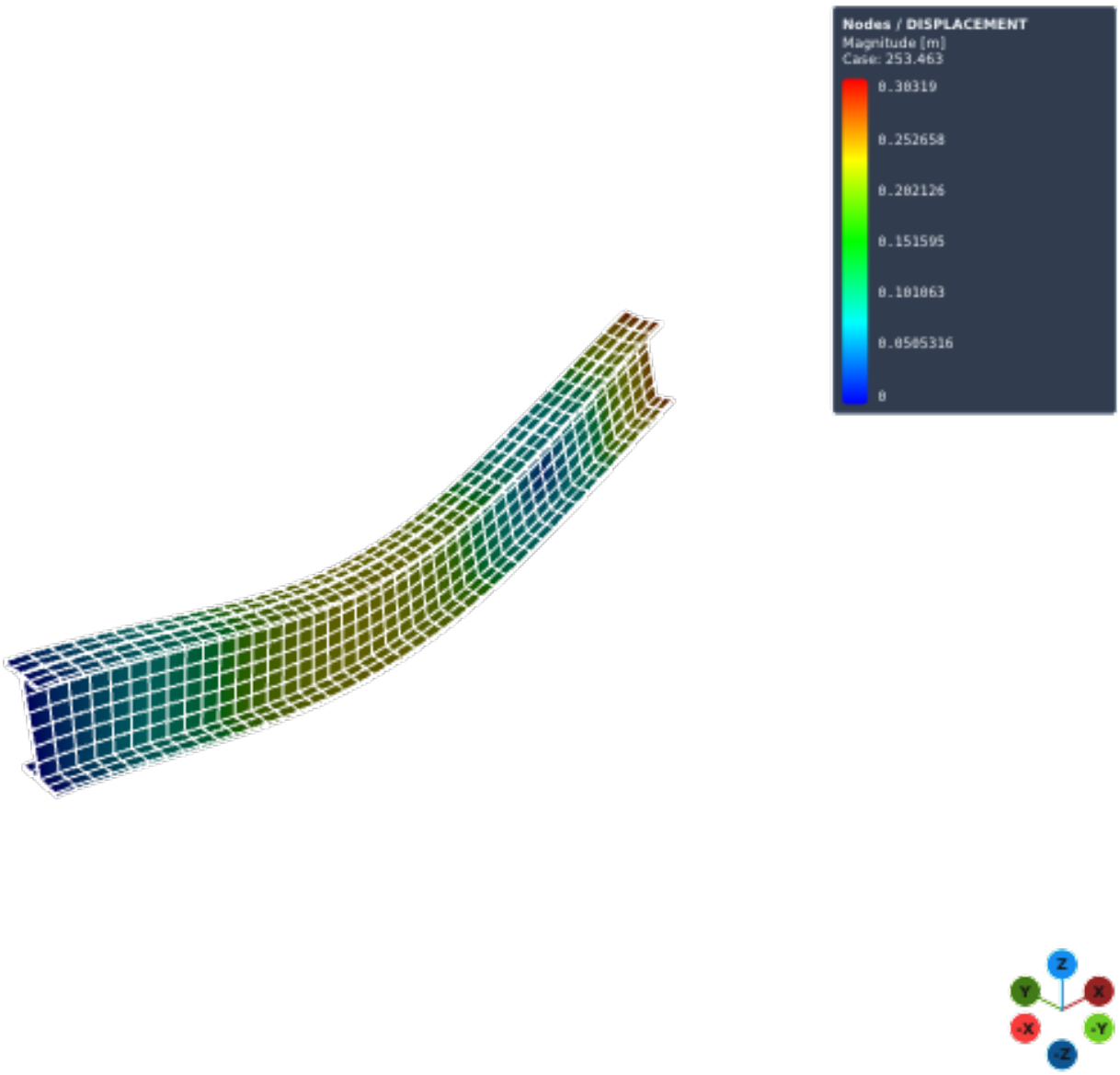

Figure 386: Sesam — mode 10, 253.463 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	431.800	+11.46	7.36085e+06	153.54	1.07e-21	9.26e-21	12.391	3.27e-11	-9.62e-11
Calculix	432.207	+11.56	7.3747e+06	153.40	3.75e-25	1.02e-25	-12.385	-6.13e-13	3.19e-13
Code_Aster	432.207	+11.56	7.3747e+06	153.46	4.59e-25	3.59e-25	12.388	-6.77e-13	5.99e-13
Sesam	253.463	-34.58	2.53623e+06	1.48e-19	5.08e-20	40.27	-3.85e-10	2.25e-10	-6.346

Mode 11

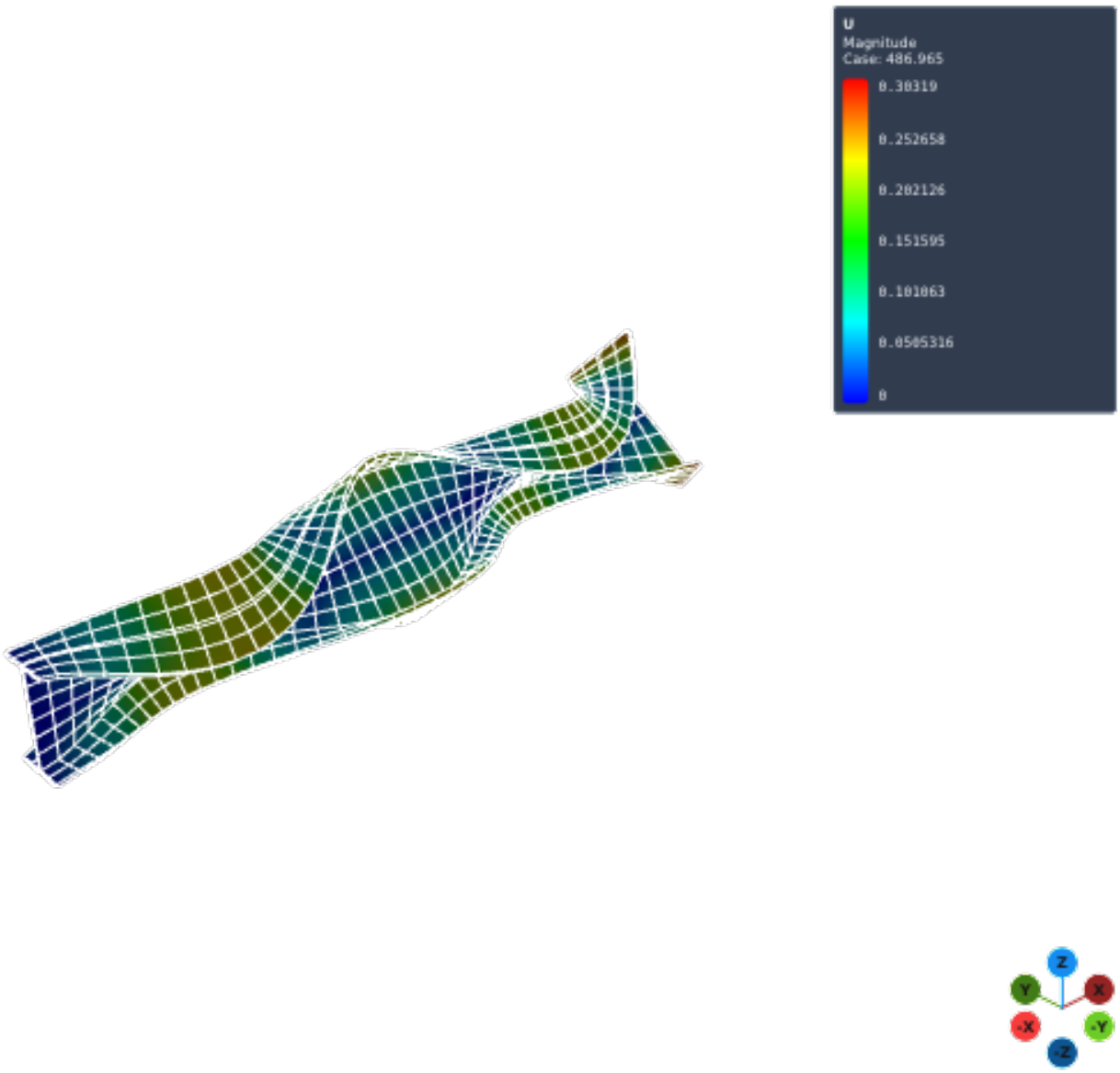

Figure 387: Abaqus — mode 11, 486.965 Hz

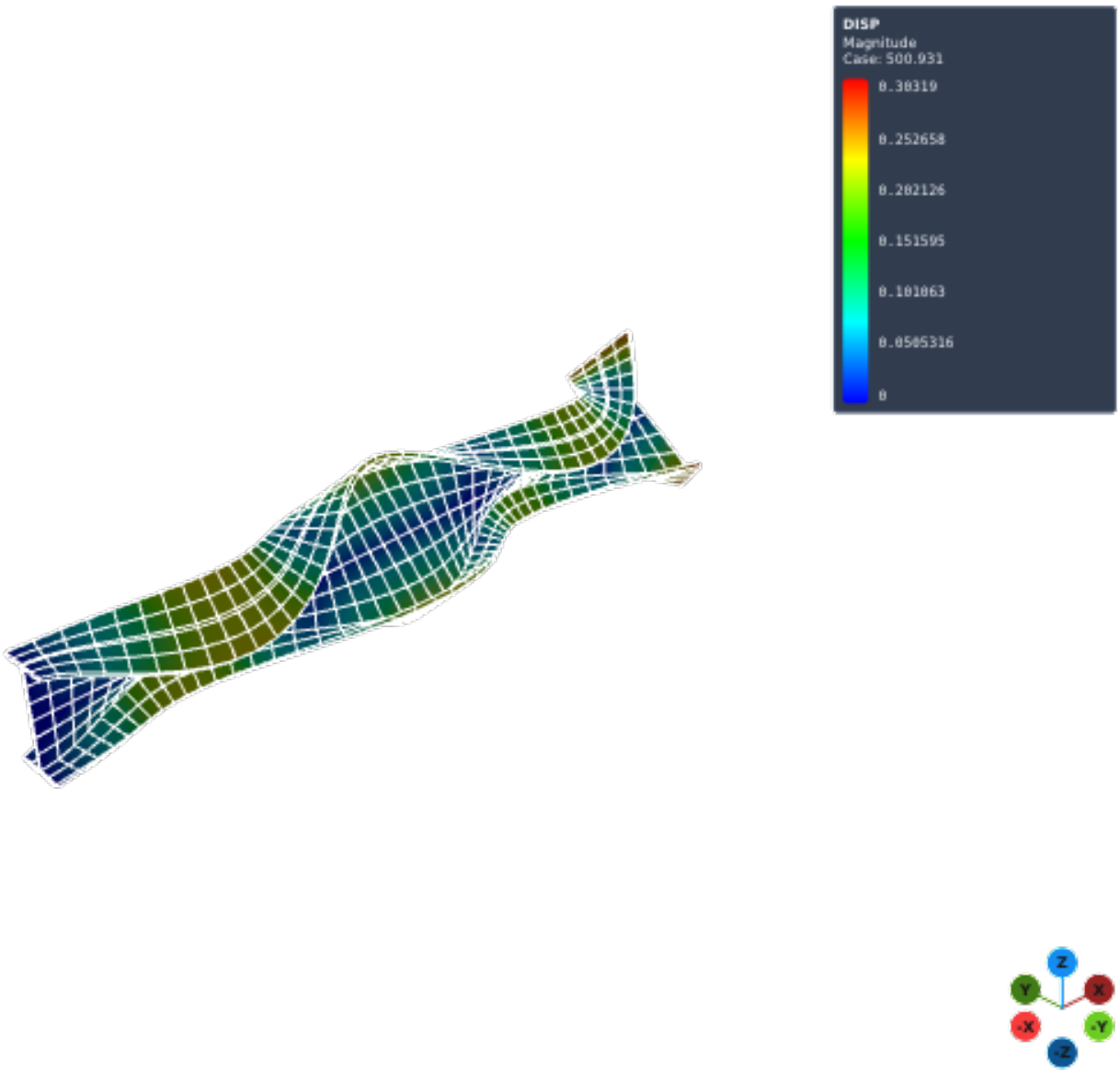

Figure 388: Calculix — mode 11, 500.931 Hz

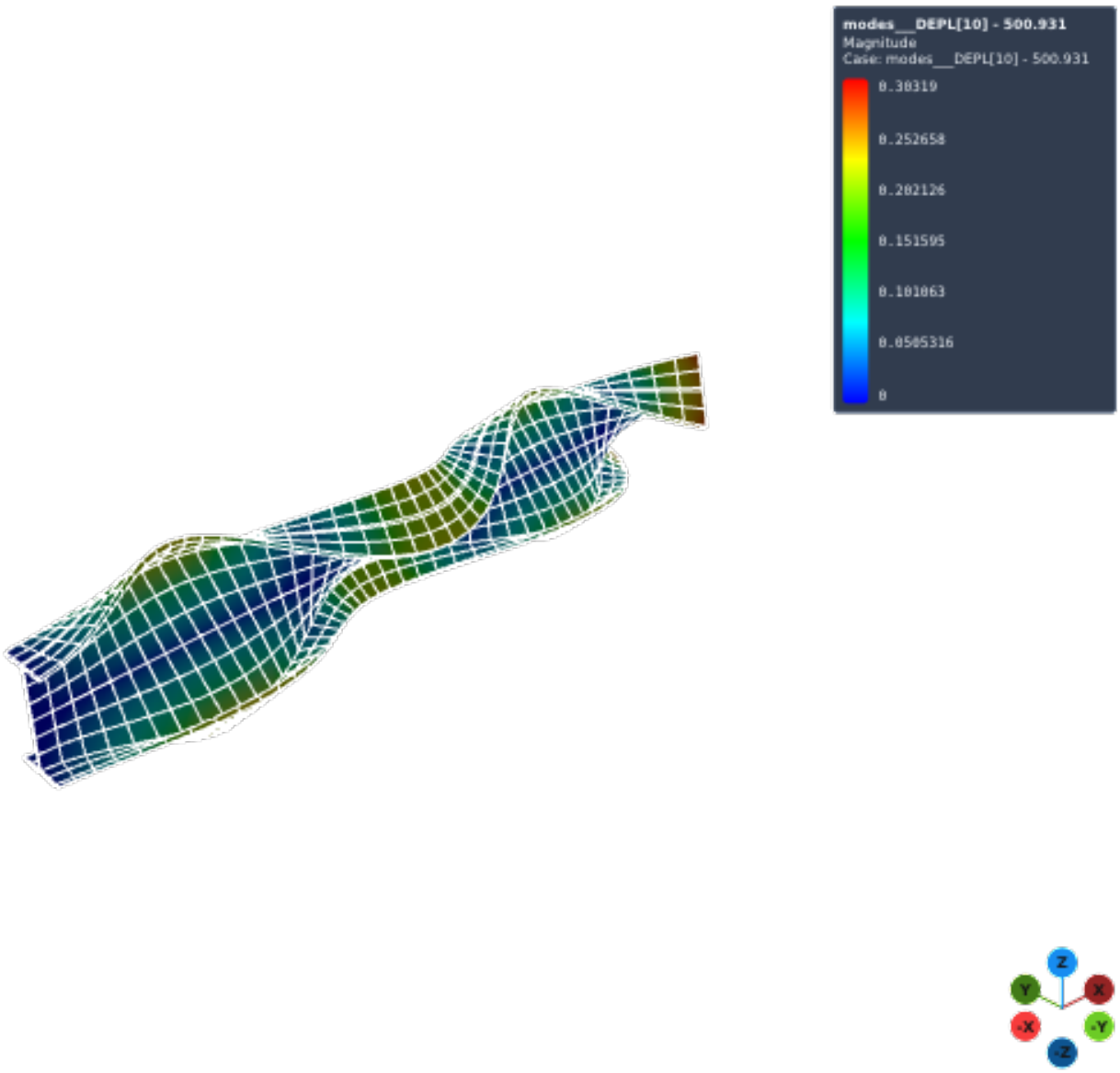

Figure 389: Code_Aster — mode 11, 500.931 Hz

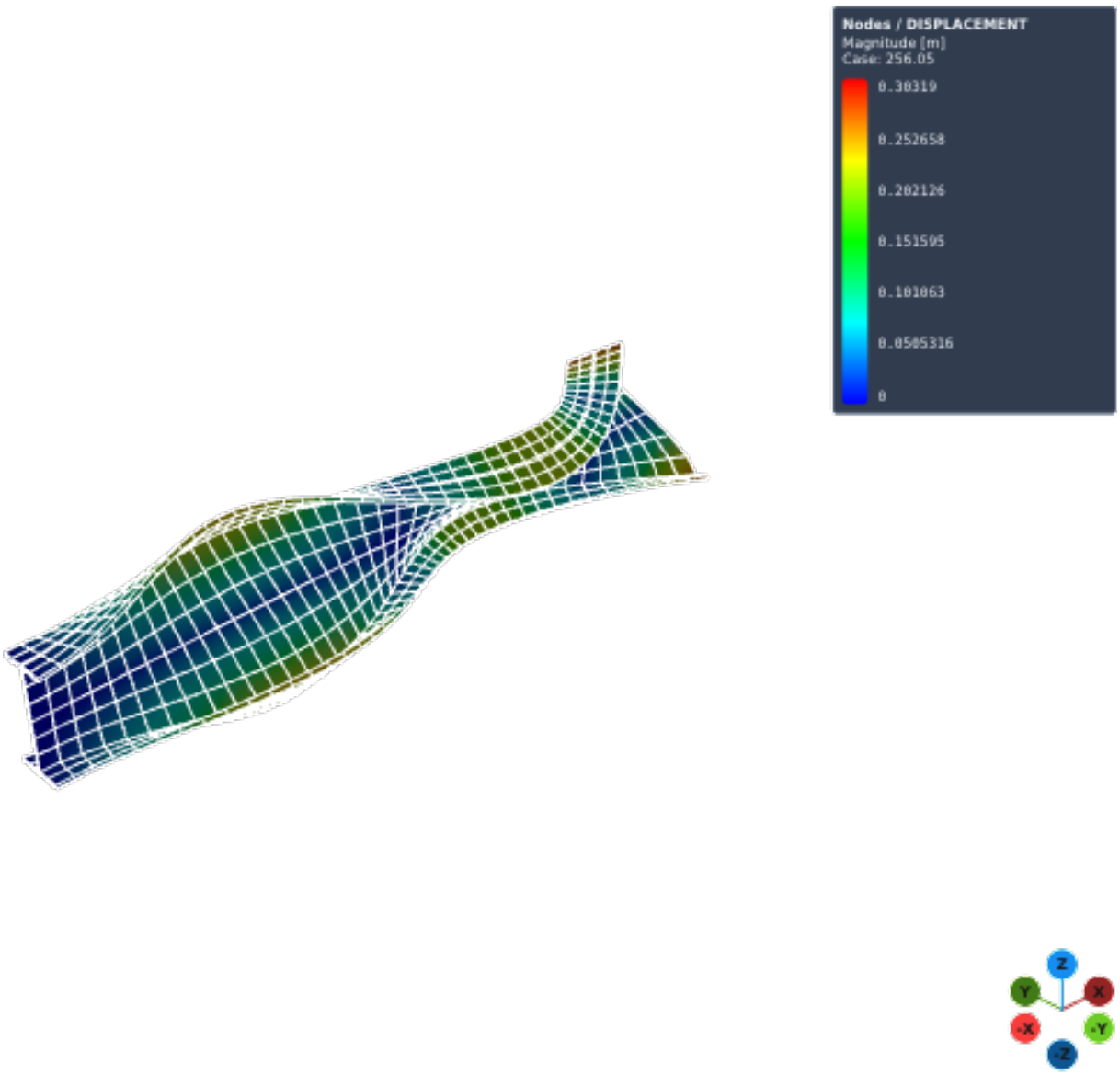

Figure 390: Sesam — mode 11, 256.050 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	486.960	+11.63	9.3617e+06	1.09e-16	2.28e-18	1.17e-18	1.04e-08	1.51e-09	1.08e-09
Calculix	500.931	+14.84	9.90641e+06	2.23e-25	1.21e-24	2.22e-24	4.72e-13	-1.10e-12	1.49e-12
Code_Aster	500.931	+14.84	9.90641e+06	2.35e-25	2.66e-25	5.46e-25	4.85e-13	5.16e-13	-7.39e-13
Sesam	256.050	-41.30	2.58827e+06	4.62e-20	1.26e-20	3.01e-23	2.15e-10	-1.12e-10	5.49e-12

Solid, 1st order, HEX, reduced integration

Mode 1

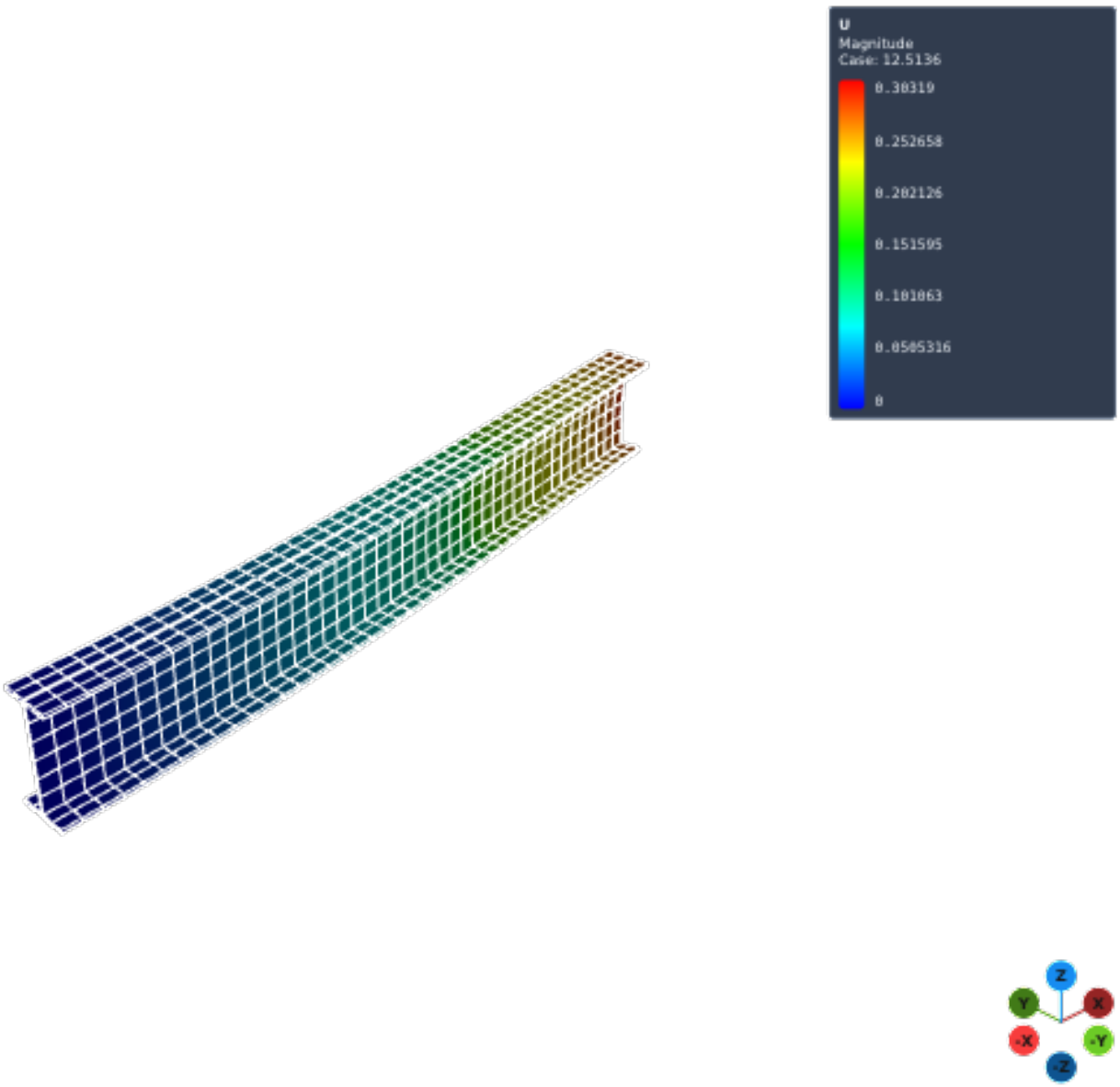

Figure 391: Abaqus — mode 1, 12.514 Hz

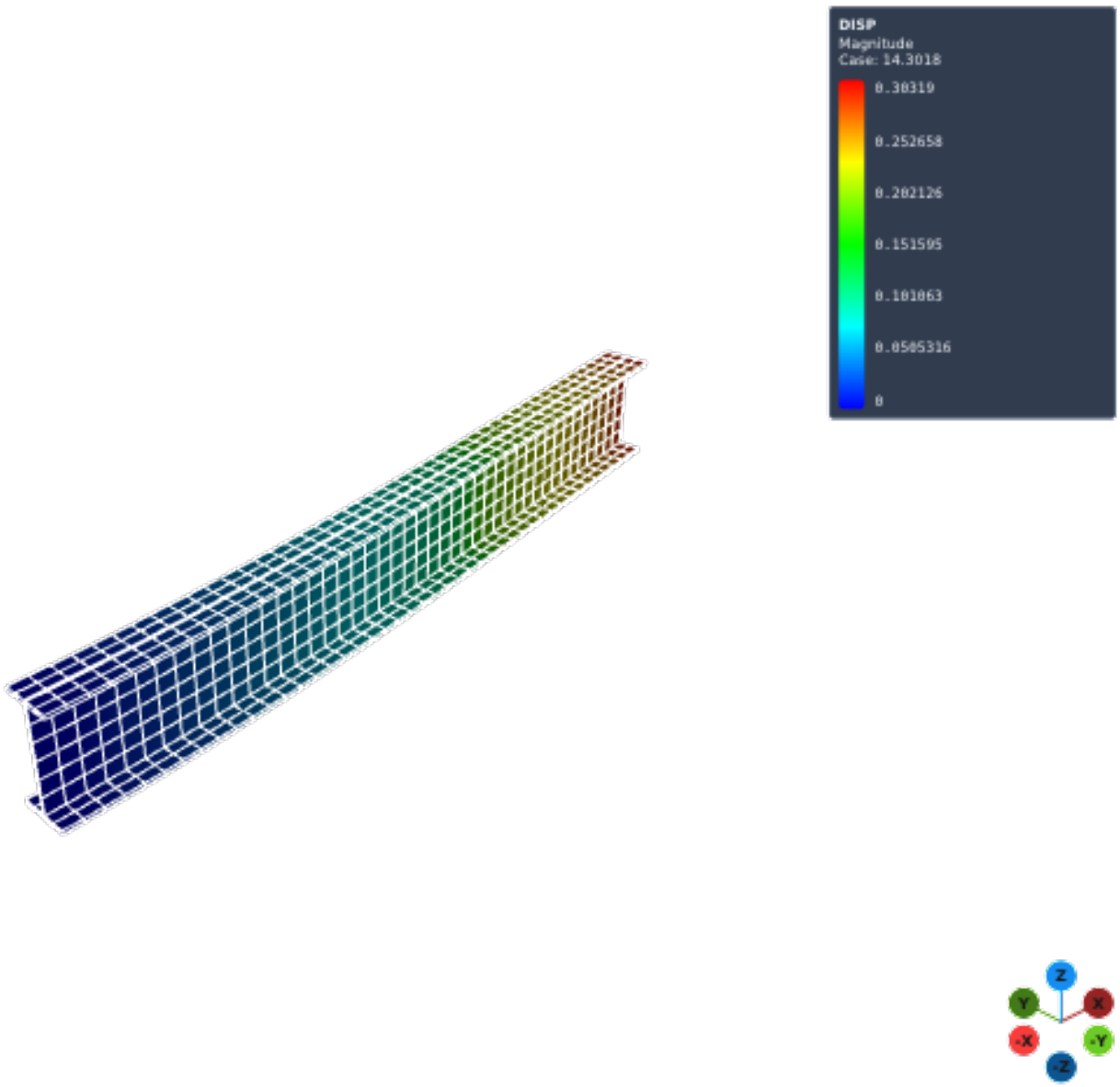

Figure 392: Calculix — mode 1, 14.302 Hz

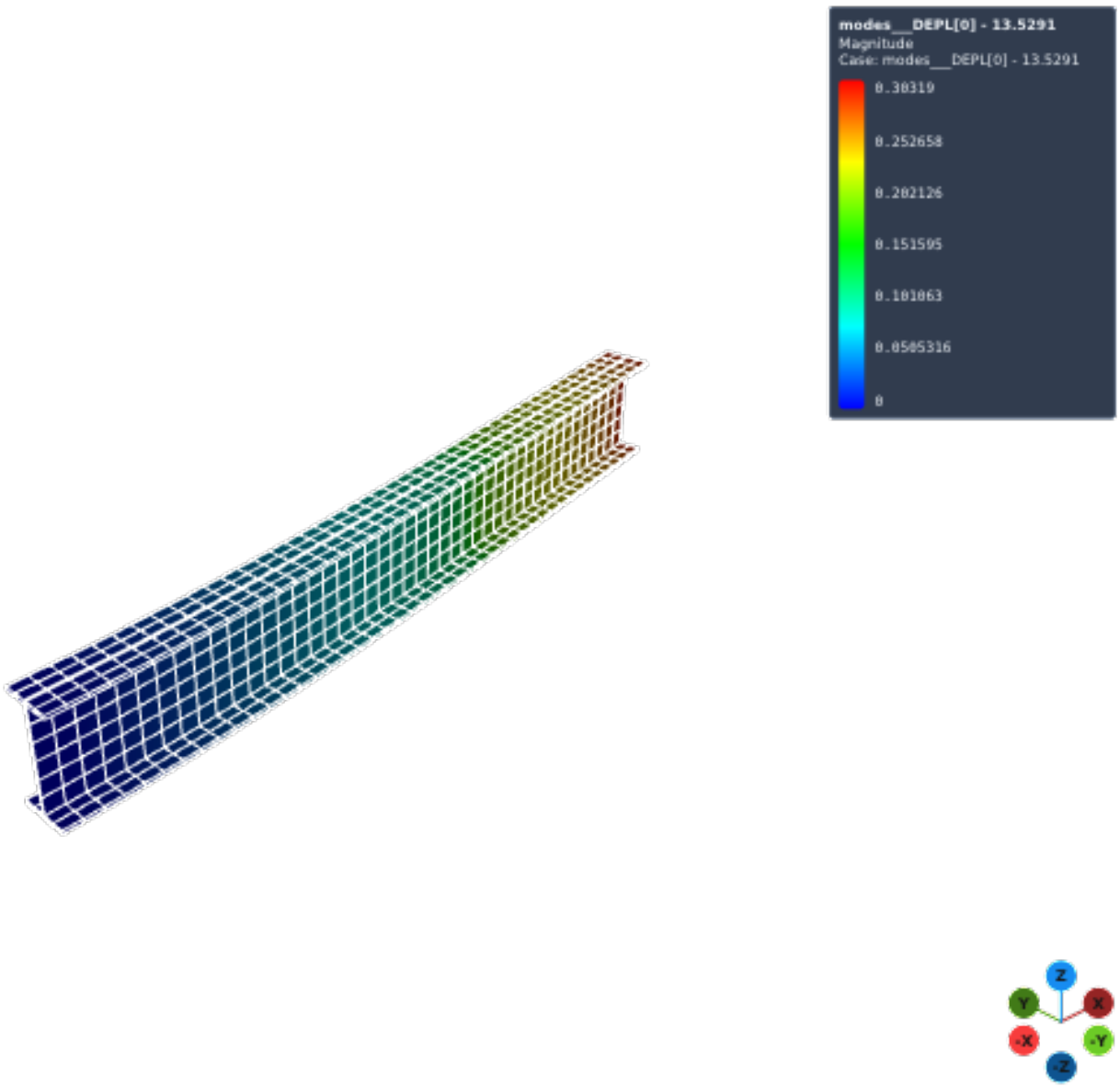

Figure 393: Code_Aster — mode 1, 13.529 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.514	-6.95	6182	1.92e-26	116.92	5.56e-23	1.39e-13	10.813	-7.45e-12
Calculix	14.302	+6.35	8074.92	5.38e-26	116.50	6.38e-22	2.32e-13	10.794	2.53e-11
Code_Aster	13.529	+0.60	7225.96	1.74e-09	116.44	1.64e-06	4.17e-05	10.791	-0.001

Mode 2

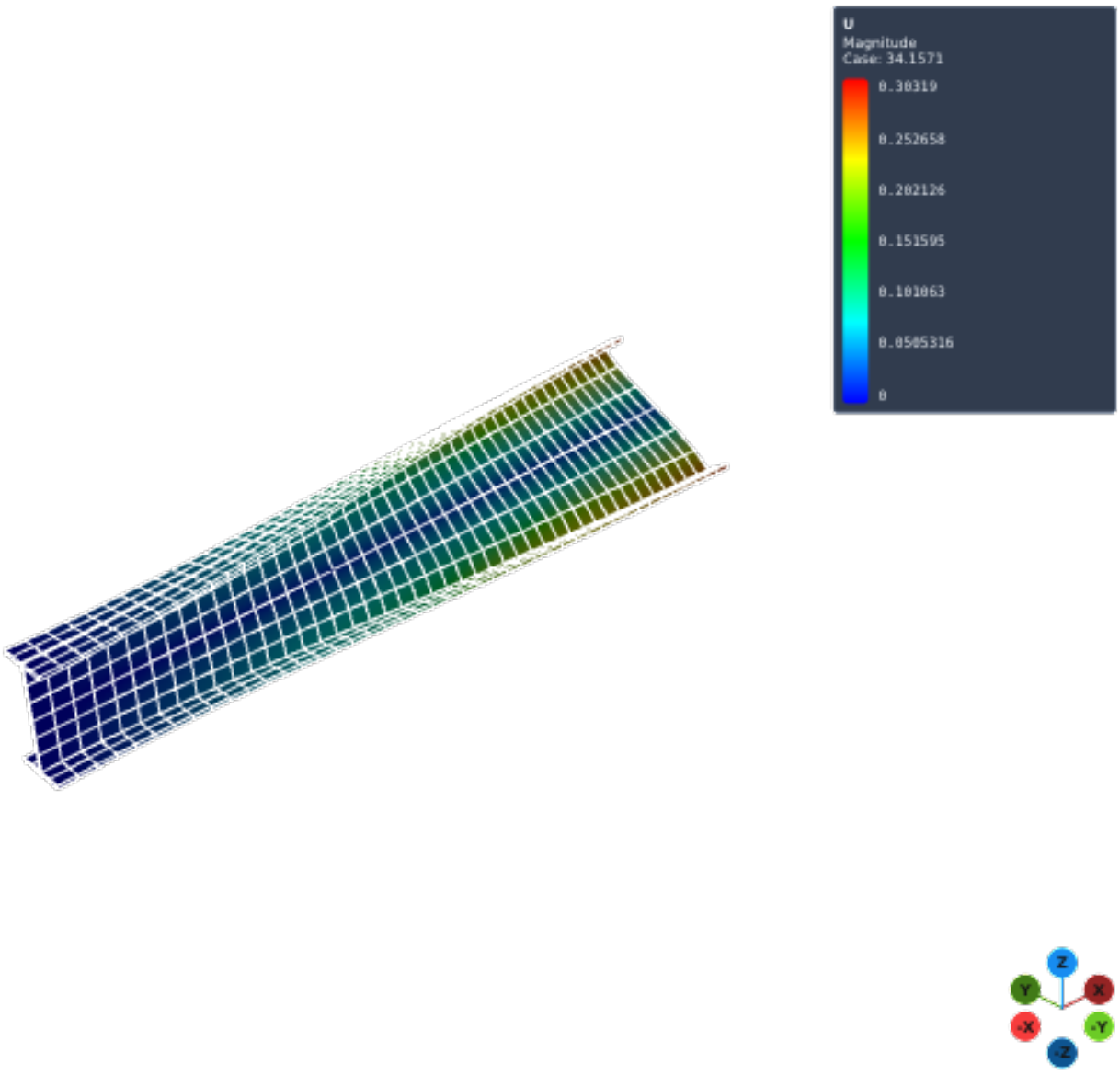

Figure 394: Abaqus — mode 2, 34.157 Hz

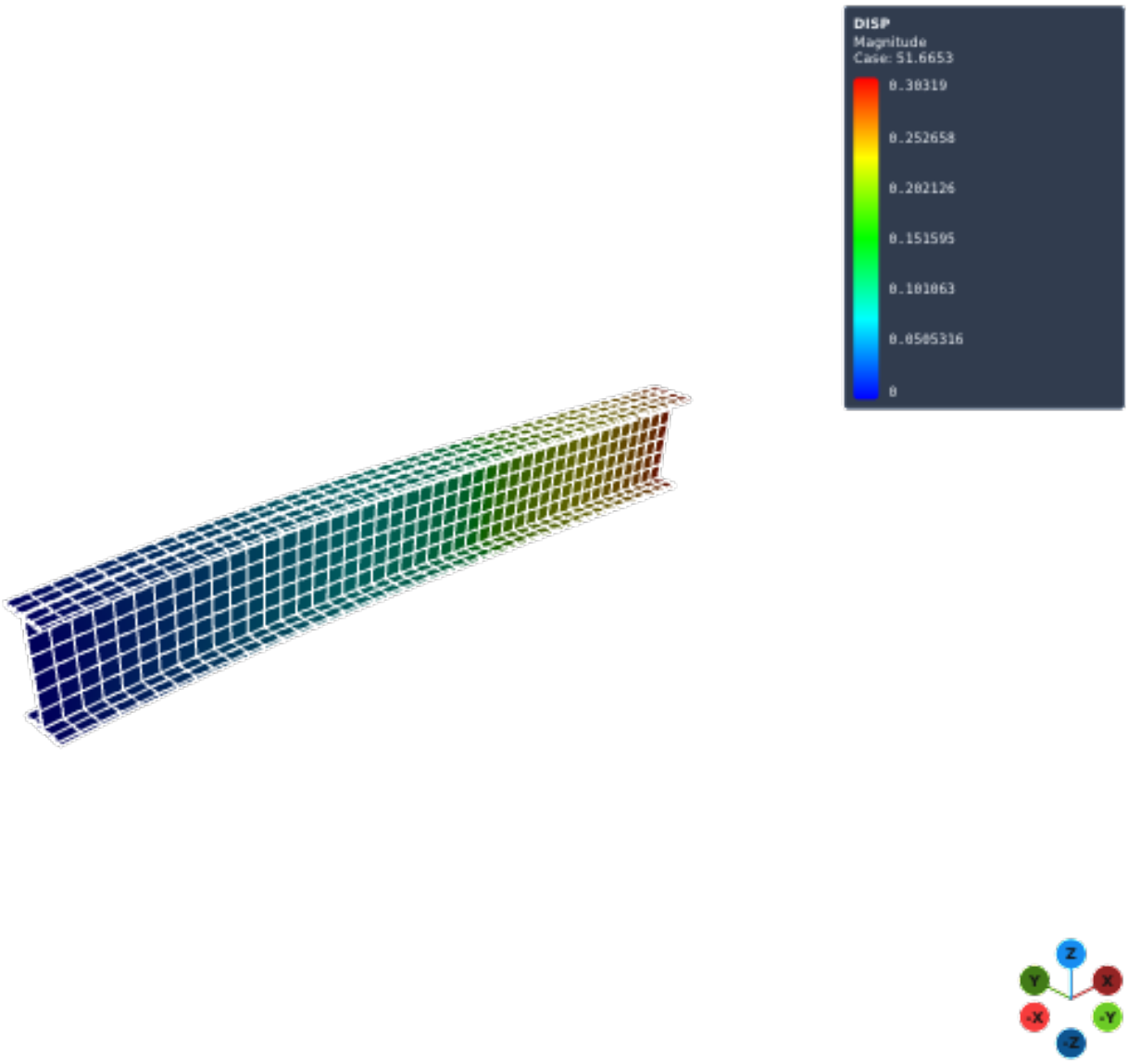

Figure 395: Calculix — mode 2, 51.665 Hz

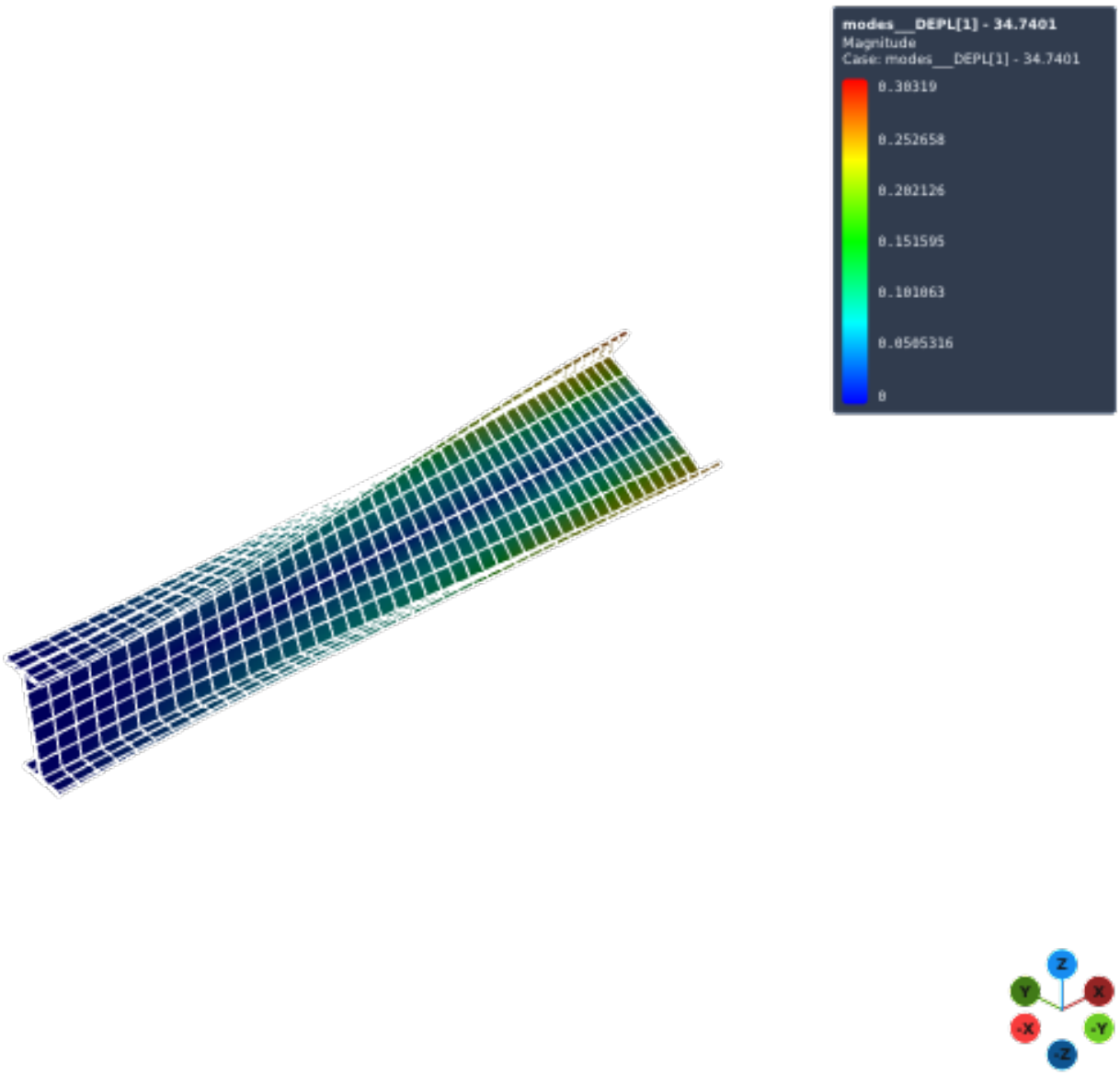

Figure 396: Code_Aster — mode 2, 34.740 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	34.157	-15.01	46060	4.54e-27	1.11e-19	8.80e-22	6.74e-14	3.33e-10	-2.97e-11
Calculix	51.665	+28.56	105380	6.55e-26	8.26e-22	118.46	-2.56e-13	2.87e-11	-10.884
Code_Aster	34.740	-13.55	47645.5	3.35e-07	5.01e-04	1.48e-07	0.001	0.022	-3.84e-04

Mode 3

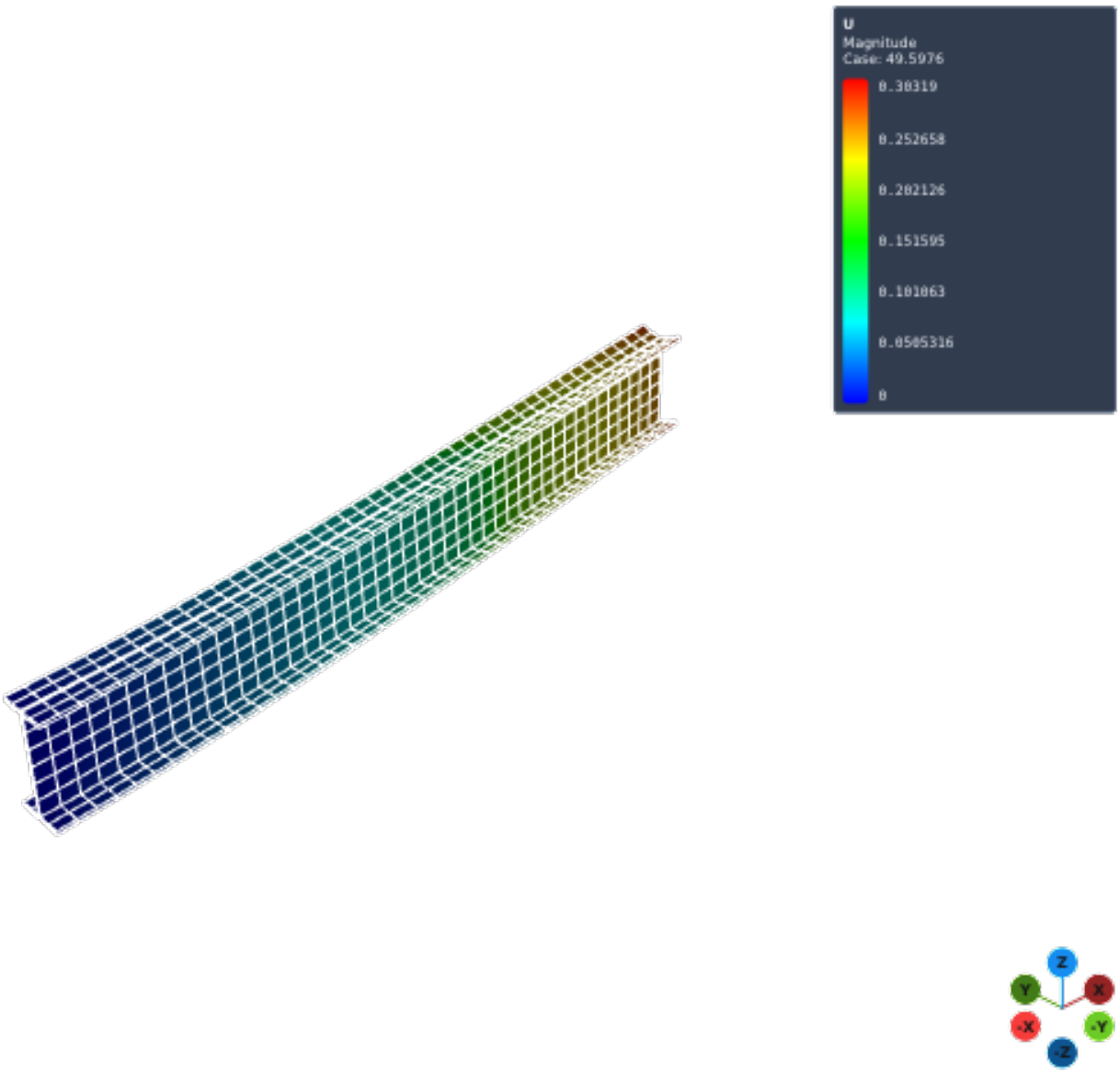

Figure 397: Abaqus — mode 3, 49.598 Hz

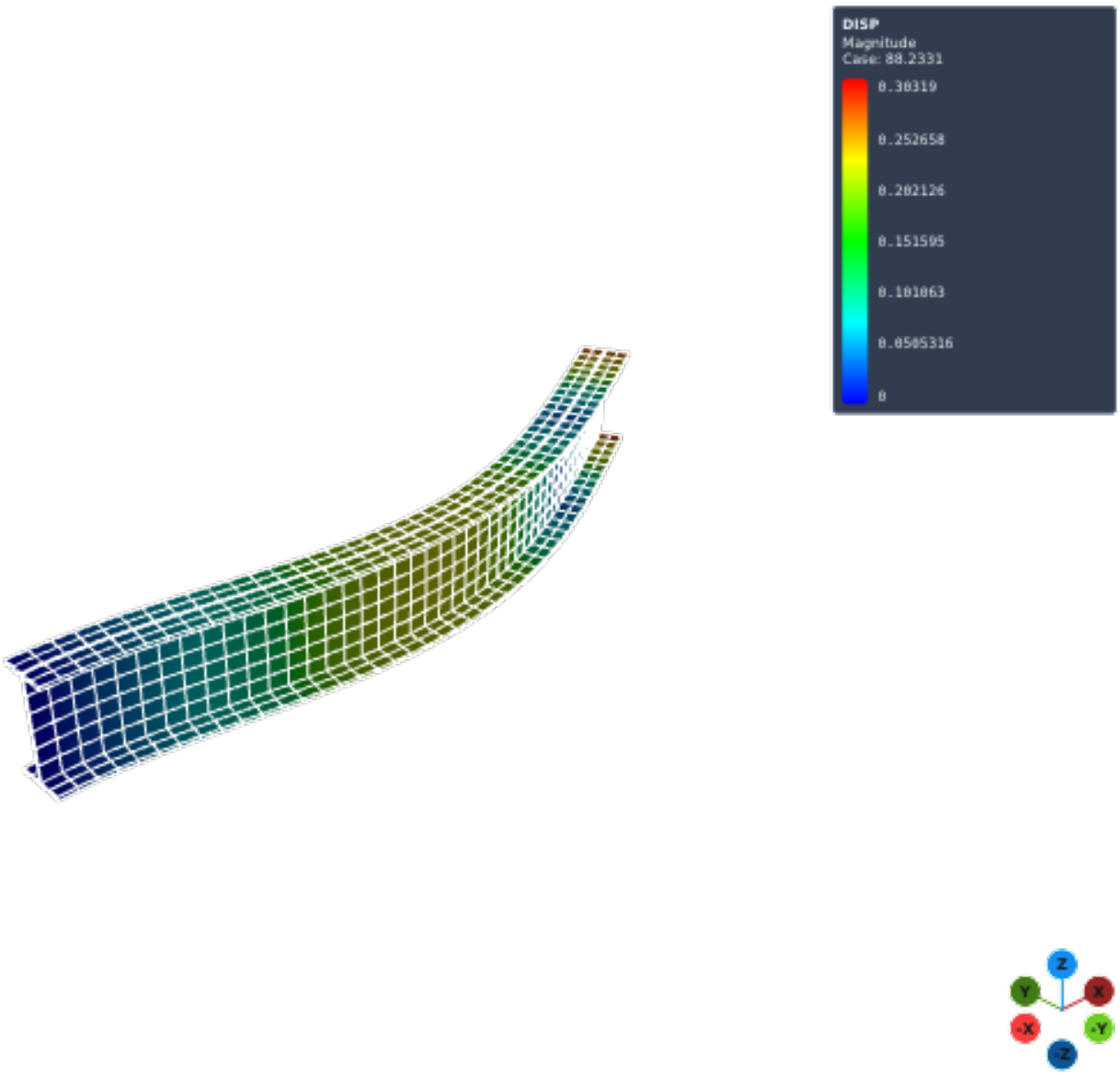

Figure 398: Calculix — mode 3, 88.233 Hz

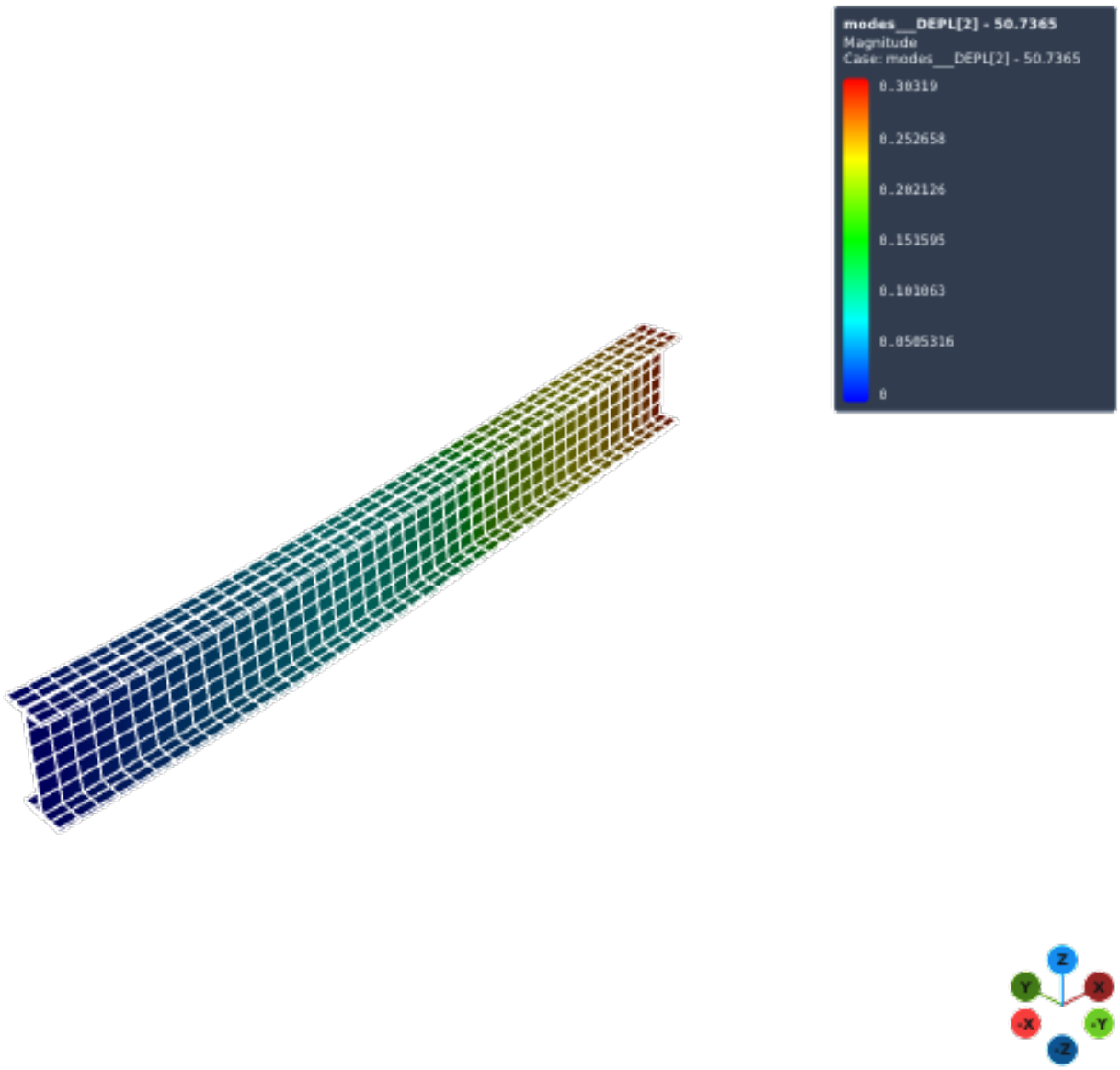

Figure 399: Code_Aster — mode 3, 50.737 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	49.598	-21.09	97114	1.57e-26	1.07e-20	118.97	-1.25e-13	1.03e-10	10.907
Calculix	88.233	+40.37	307343	1.15e-26	36.16	4.83e-23	1.07e-13	-6.013	-6.95e-12
Code_Aster	50.737	-19.28	101625	5.88e-09	8.10e-08	118.29	7.67e-05	2.85e-04	10.876

Mode 4

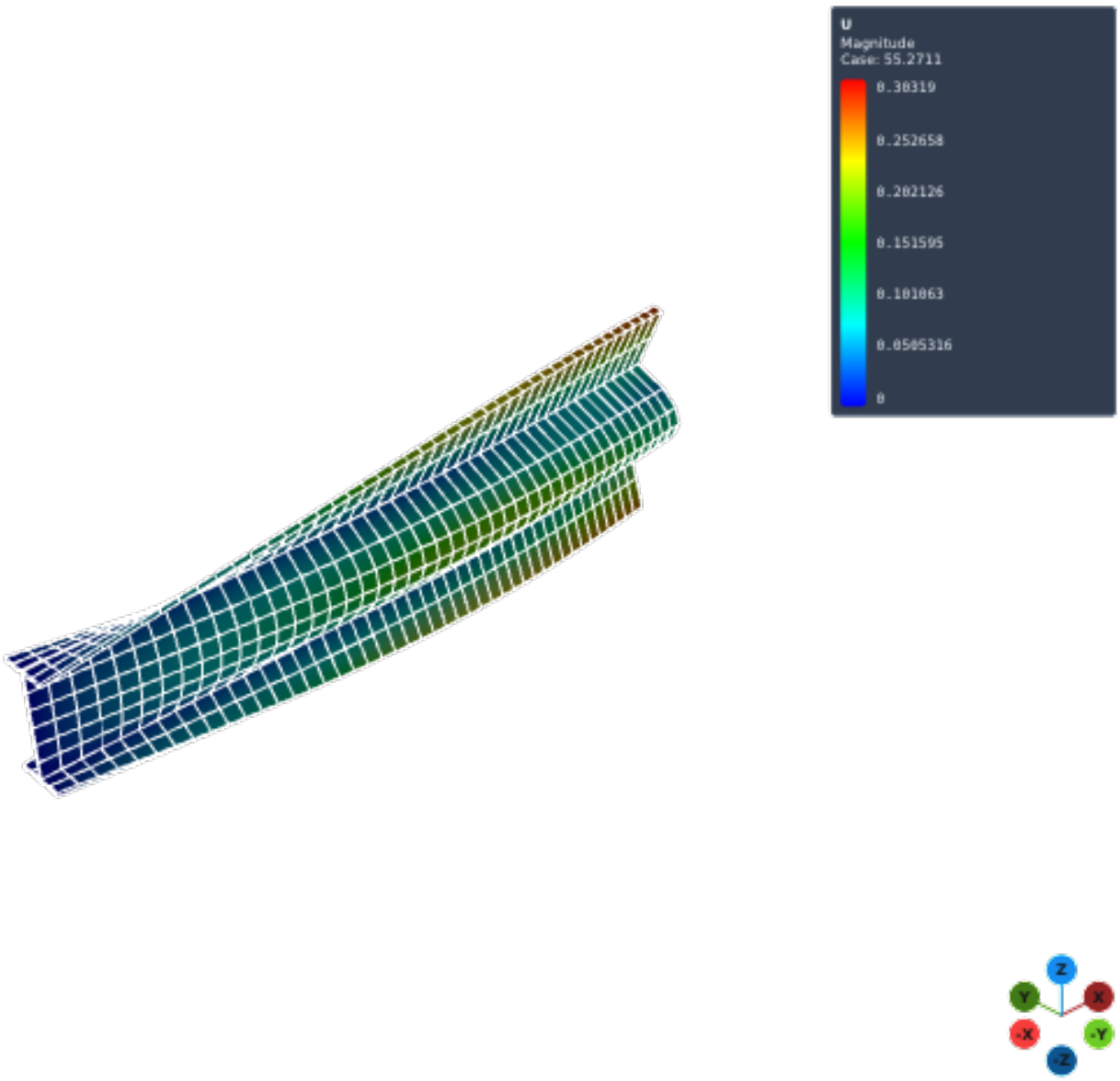

Figure 400: Abaqus — mode 4, 55.271 Hz

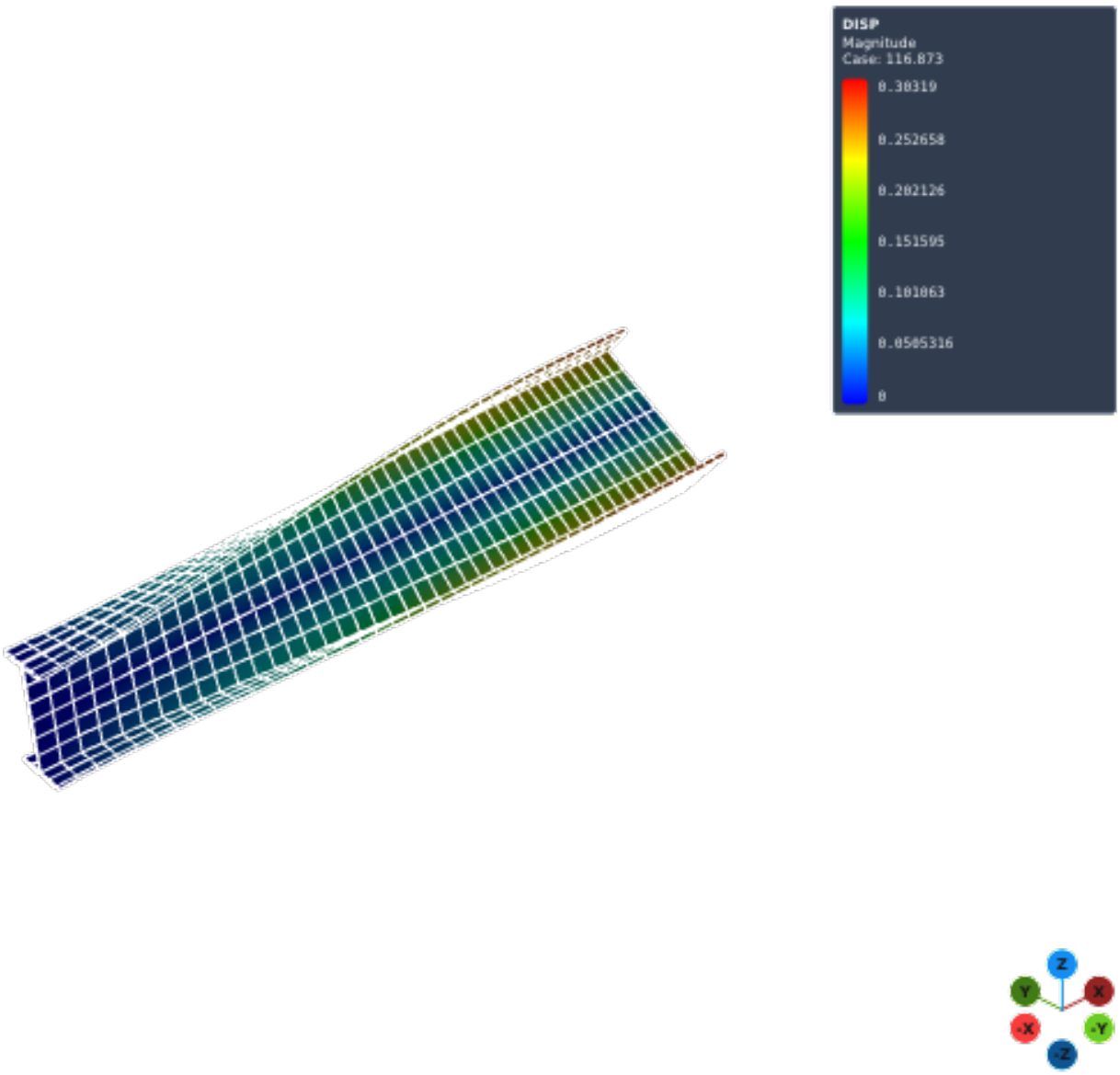

Figure 401: Calculix — mode 4, 116.873 Hz

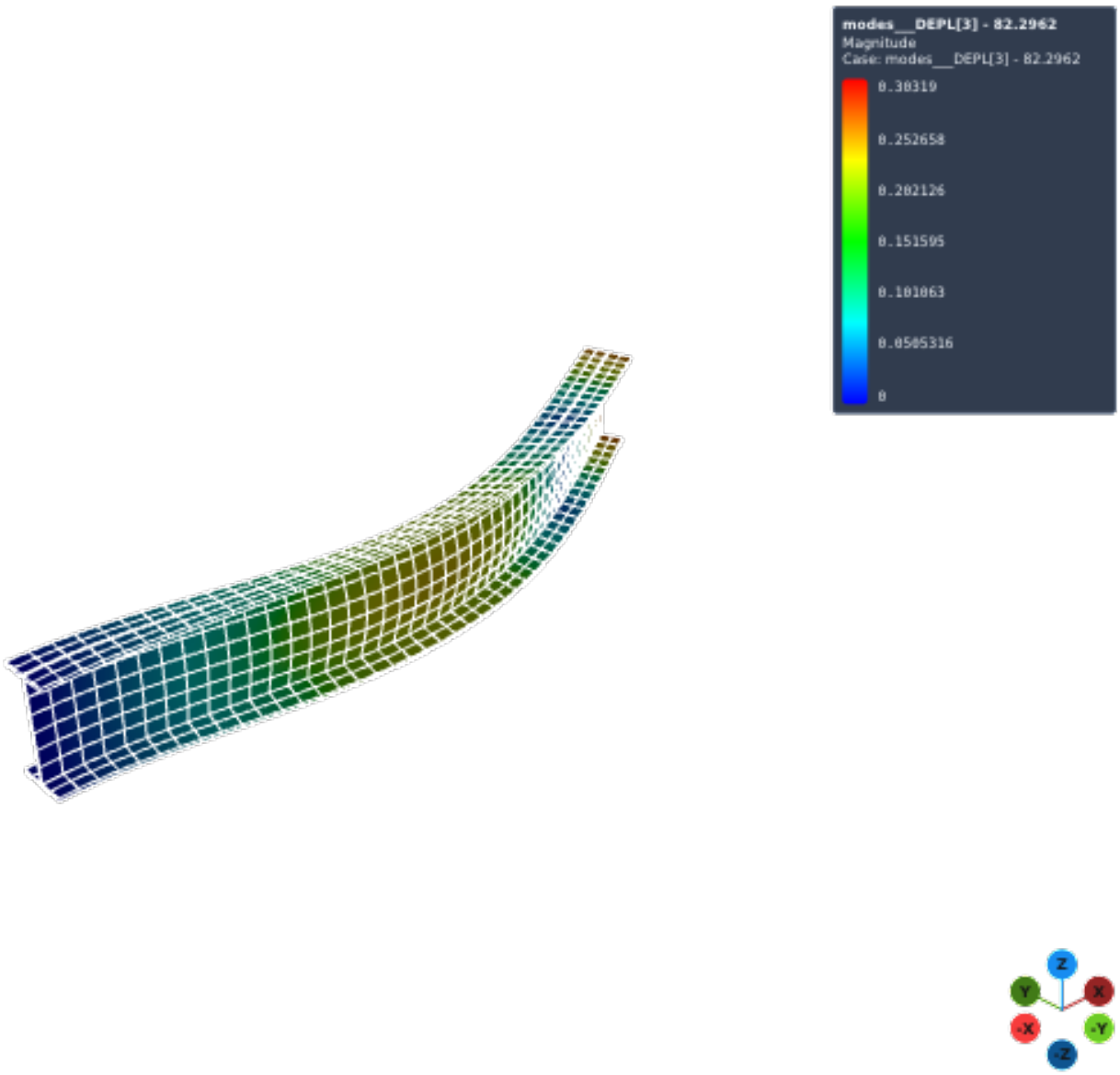

Figure 402: Code_Aster — mode 4, 82.296 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	55.271	-34.83	120602	1.14e-25	2.12	1.05e-18	-3.37e-13	-1.456	1.02e-09
Calculix	116.873	+37.80	539244	3.53e-27	9.54e-22	1.90e-26	5.94e-14	-3.09e-11	-1.38e-13
Code_Aster	82.296	-2.97	267374	6.08e-08	36.12	1.81e-05	-2.47e-04	-6.010	-0.004

Mode 5

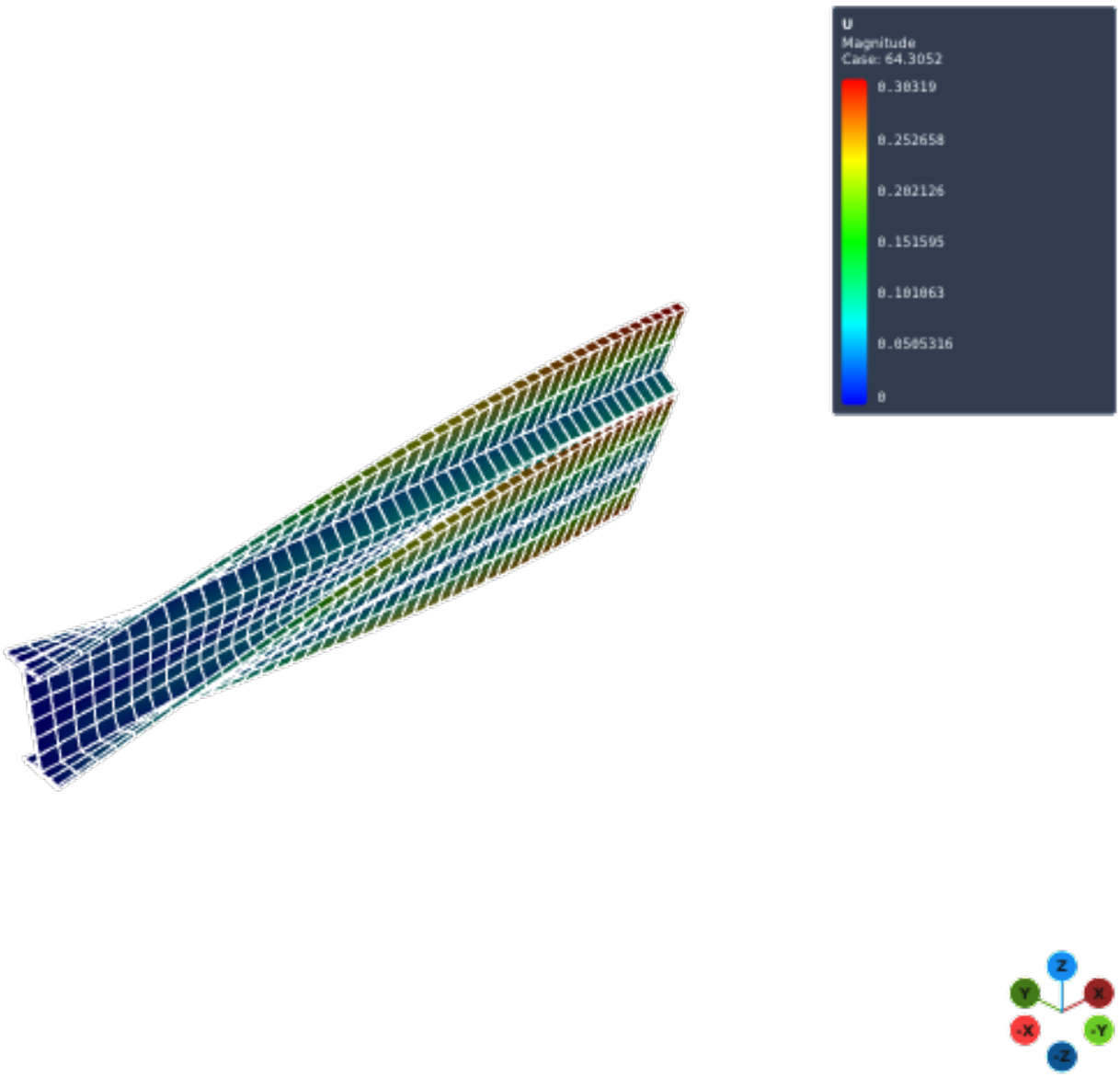

Figure 403: Abaqus — mode 5, 64.305 Hz

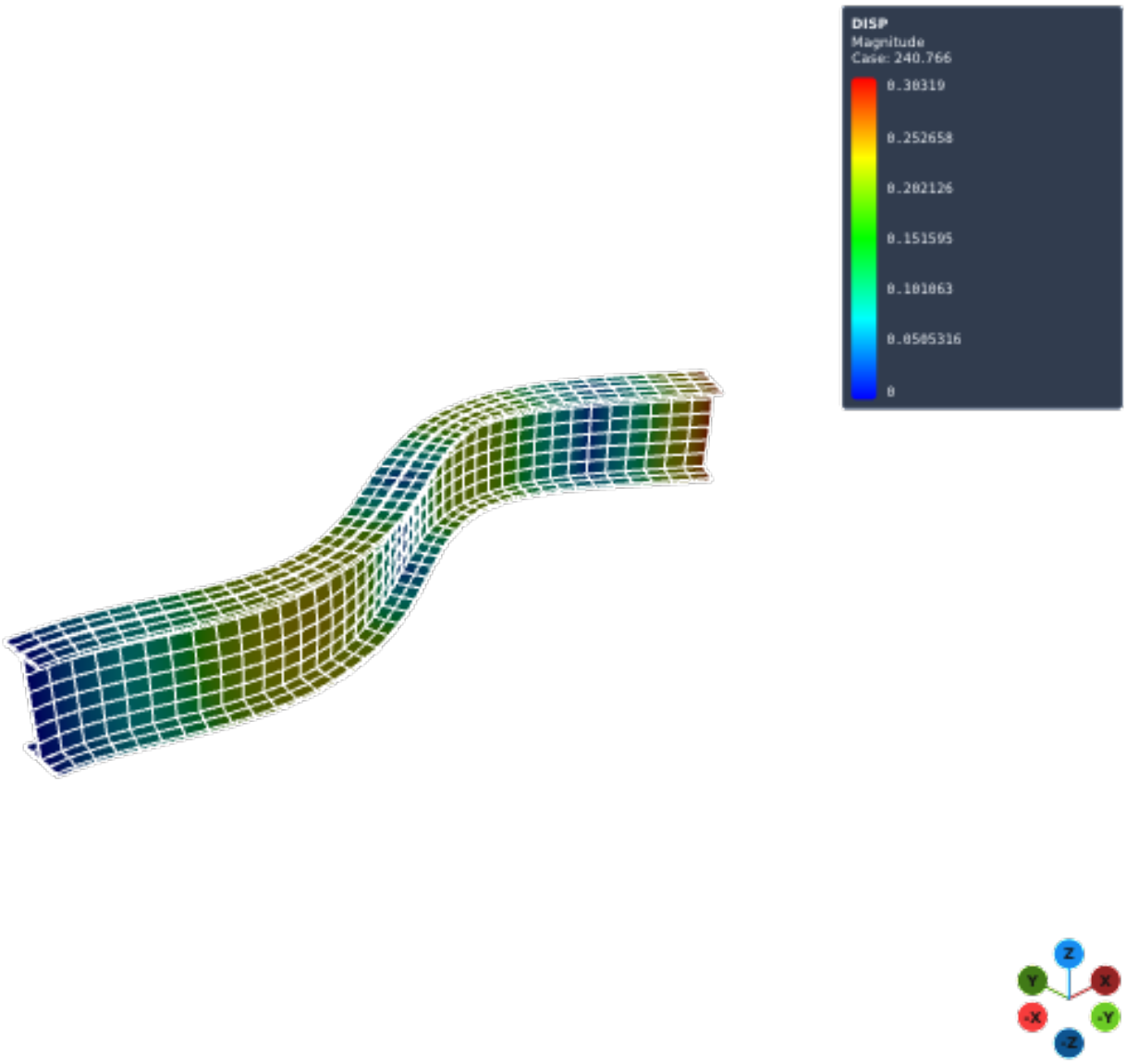

Figure 404: Calculix — mode 5, 240.766 Hz

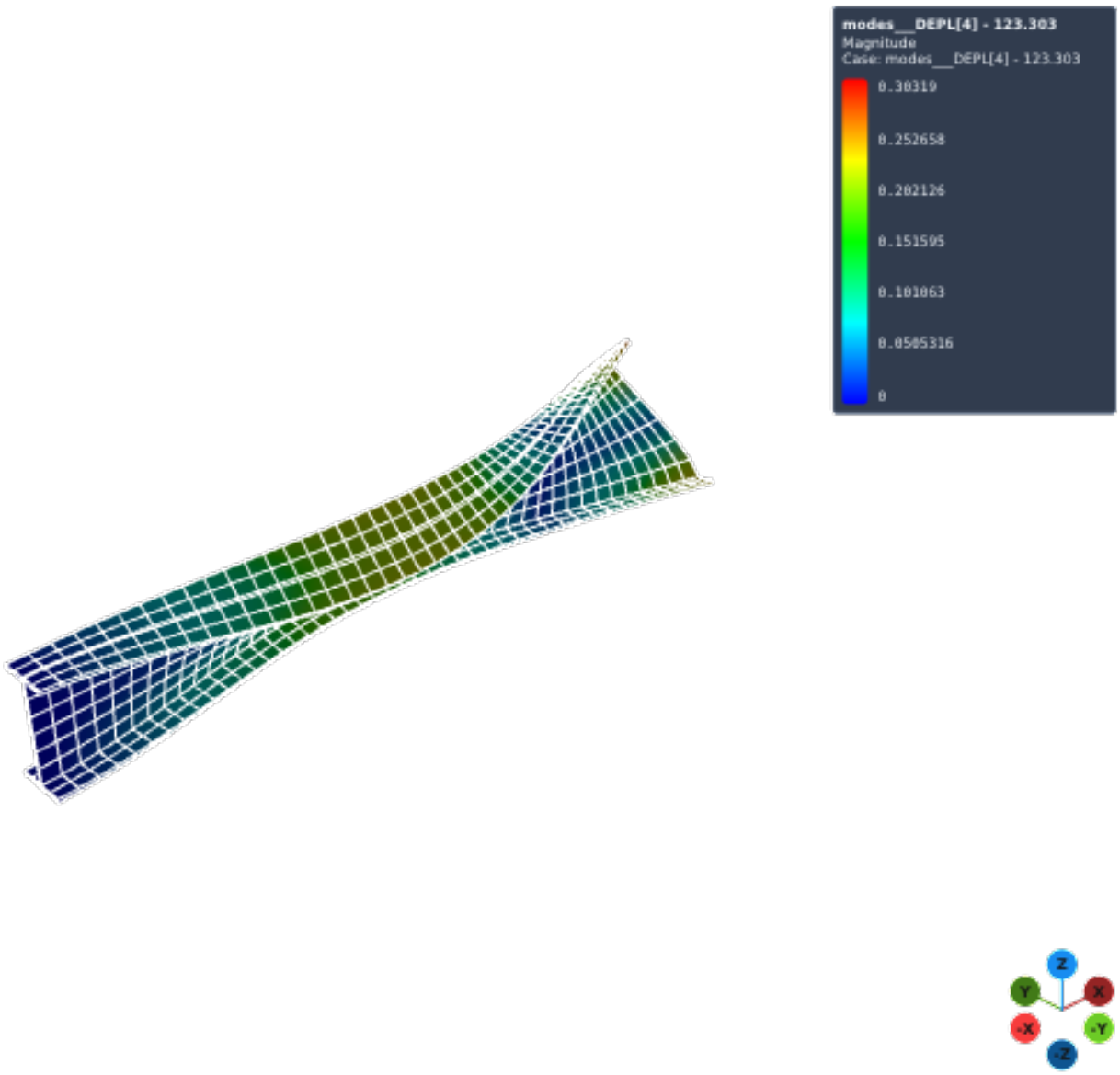

Figure 405: Code_Aster — mode 5, 123.303 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	64.305	-54.97	163249	1.13e-26	8.64e-20	1.24e-20	-1.06e-13	-2.94e-10	1.11e-10
Calculix	240.766	+68.61	2.2885e+06	3.27e-26	12.59	3.11e-25	-1.81e-13	-3.549	5.58e-13
Code_Aster	123.303	-13.65	600212	3.19e-06	0.01	4.14e-05	-0.002	-0.084	0.006

Mode 6

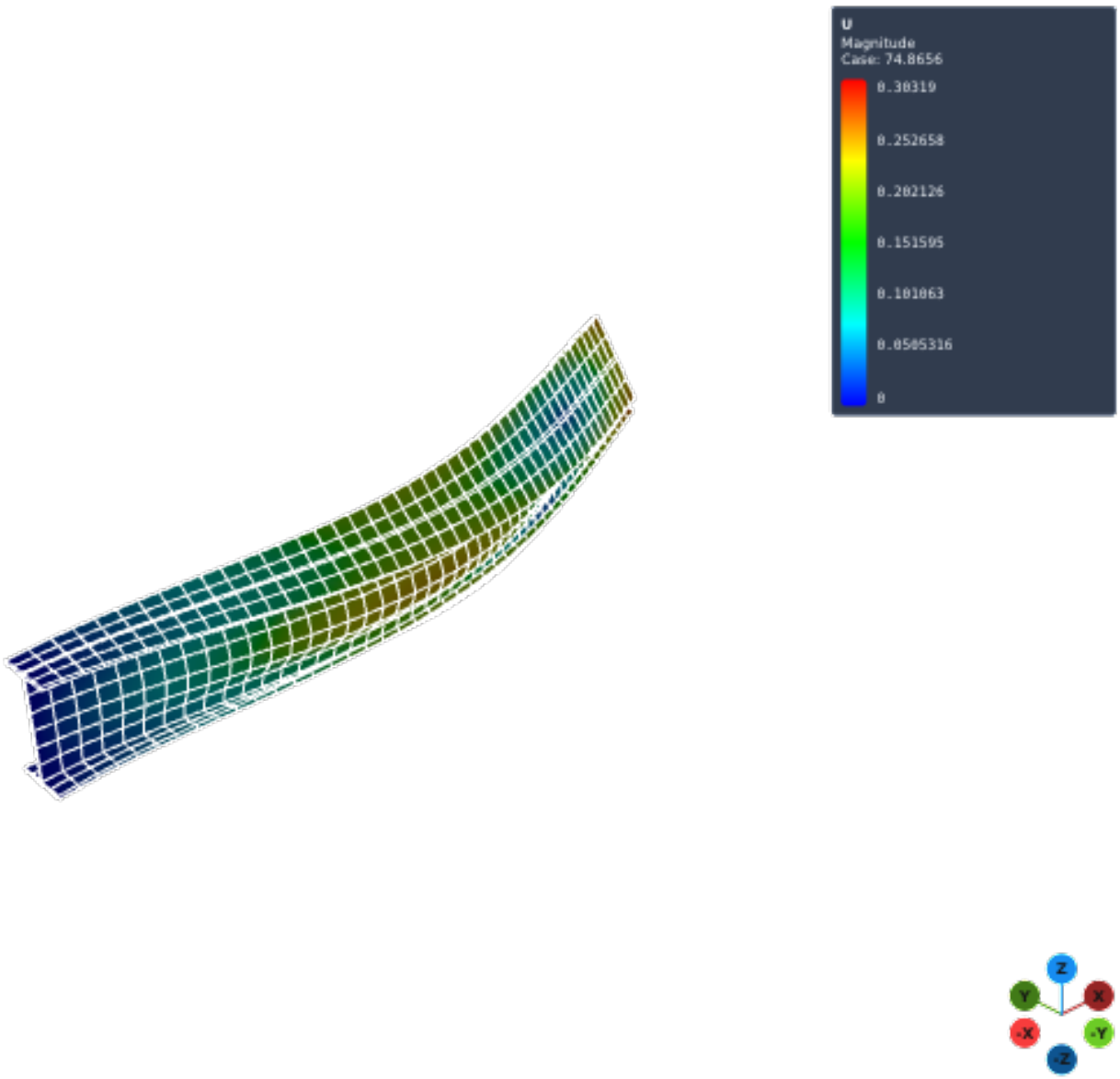

Figure 406: Abaqus — mode 6, 74.866 Hz

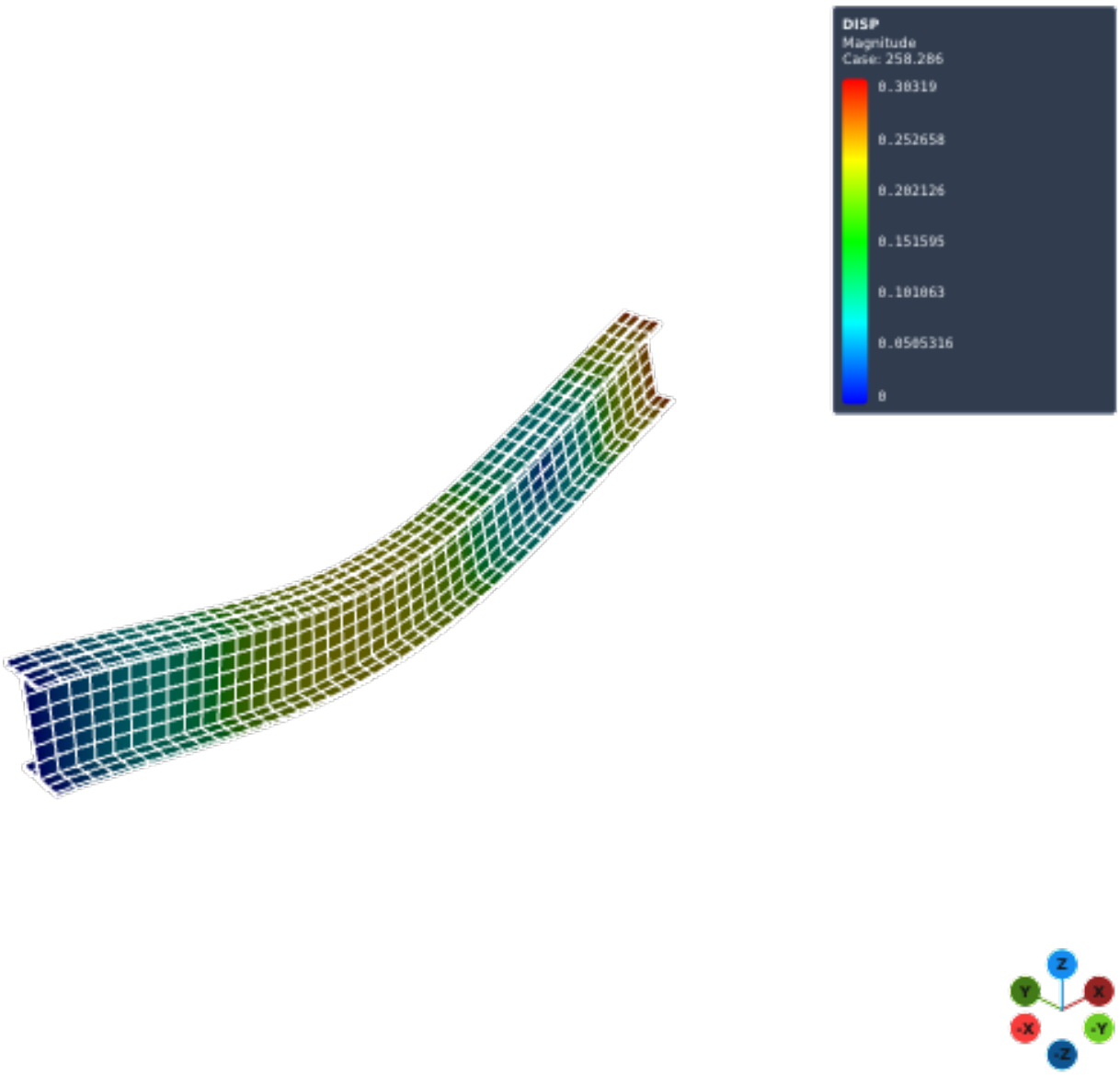

Figure 407: Calculix — mode 6, 258.286 Hz

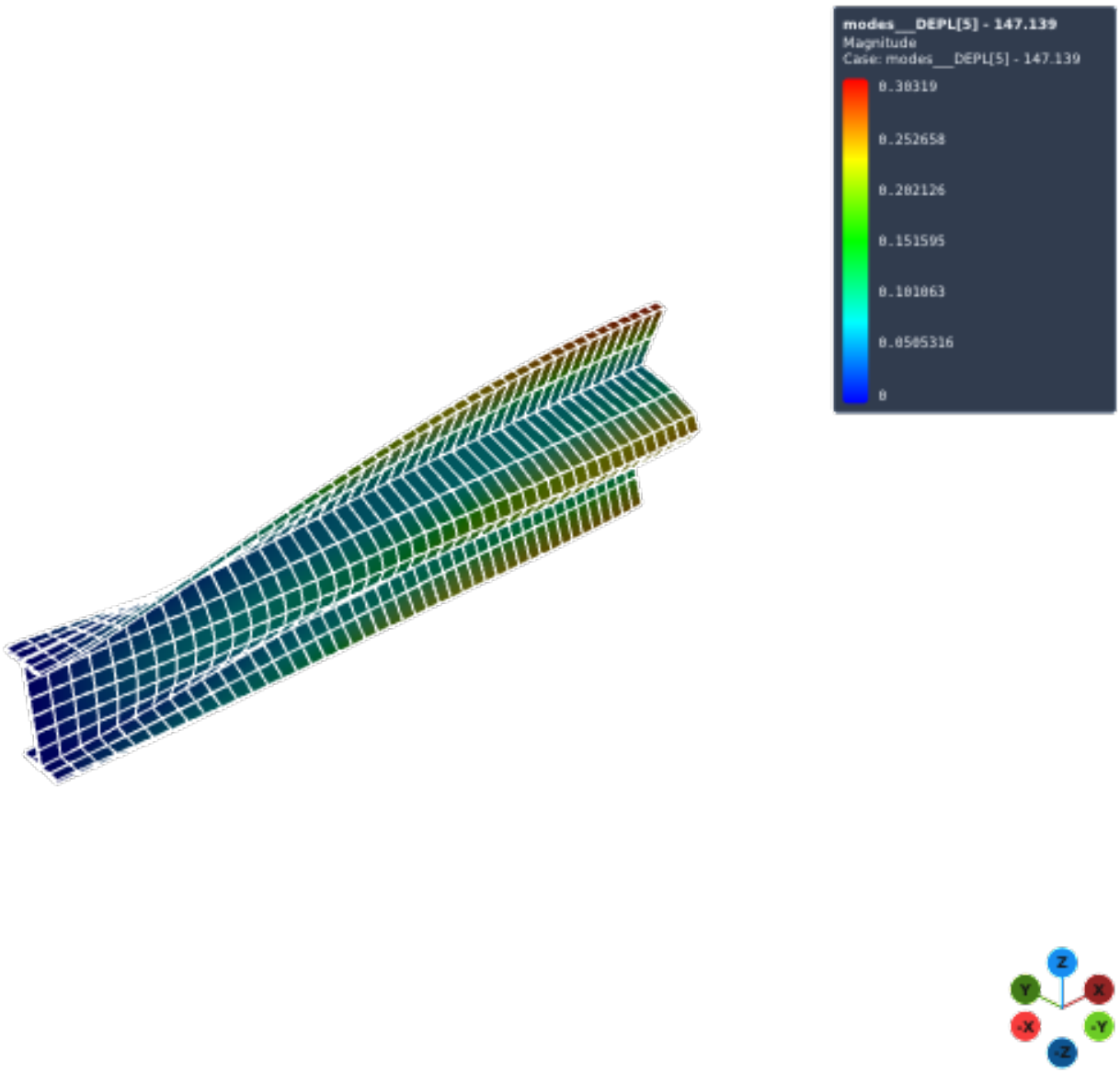

Figure 408: Code_Aster — mode 6, 147.139 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	74.866	-53.24	221271	1.12e-26	31.40	1.02e-20	1.06e-13	-5.603	-1.01e-10
Calculix	258.286	+61.33	2.63368e+06	2.30e-25	2.45e-26	40.19	4.80e-13	-1.56e-13	-6.340
Code_Aster	147.139	-8.09	854706	3.65e-07	0.04	2.32e-04	-0.001	0.192	-0.015

Mode 7

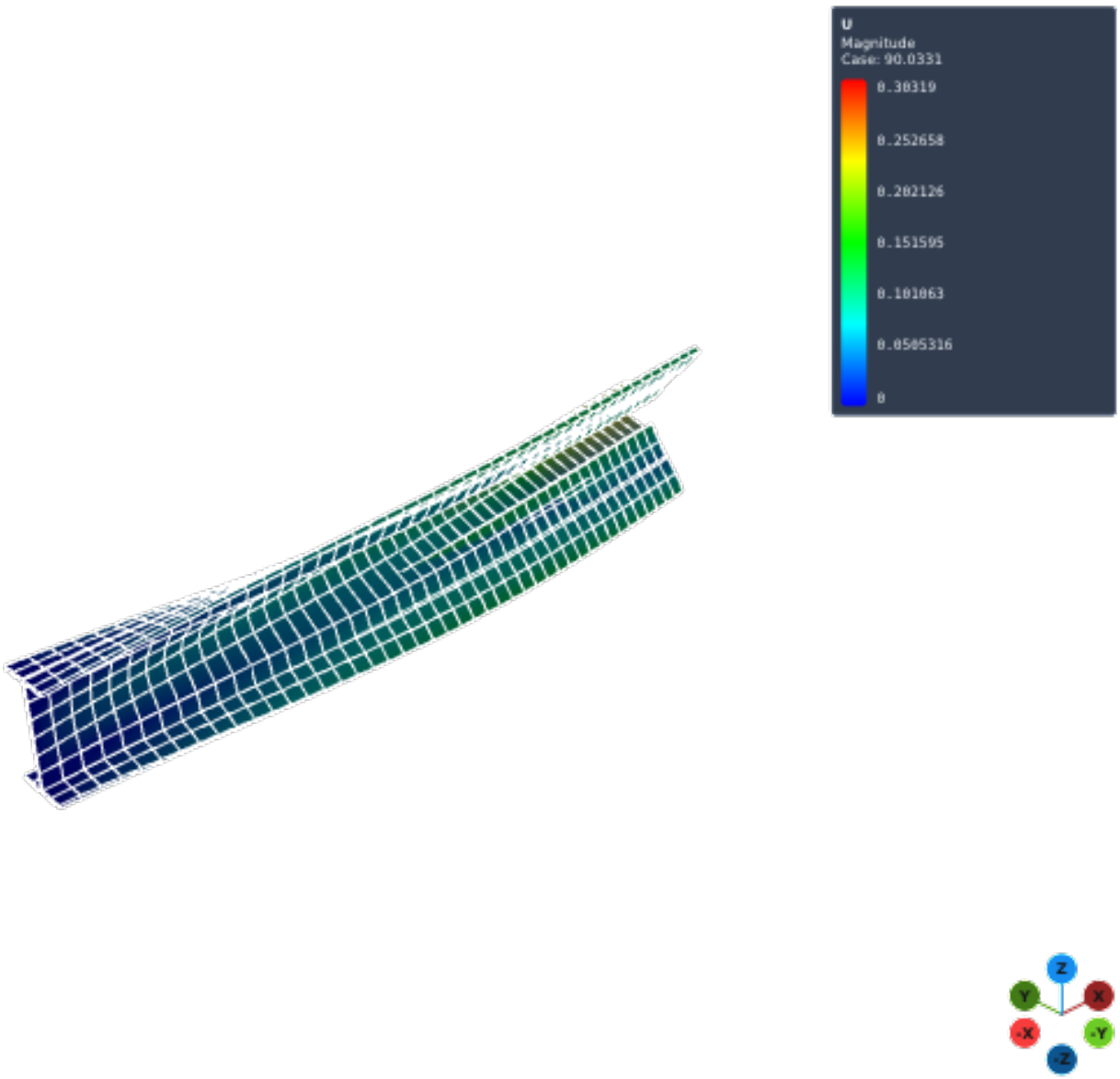

Figure 409: Abaqus — mode 7, 90.033 Hz

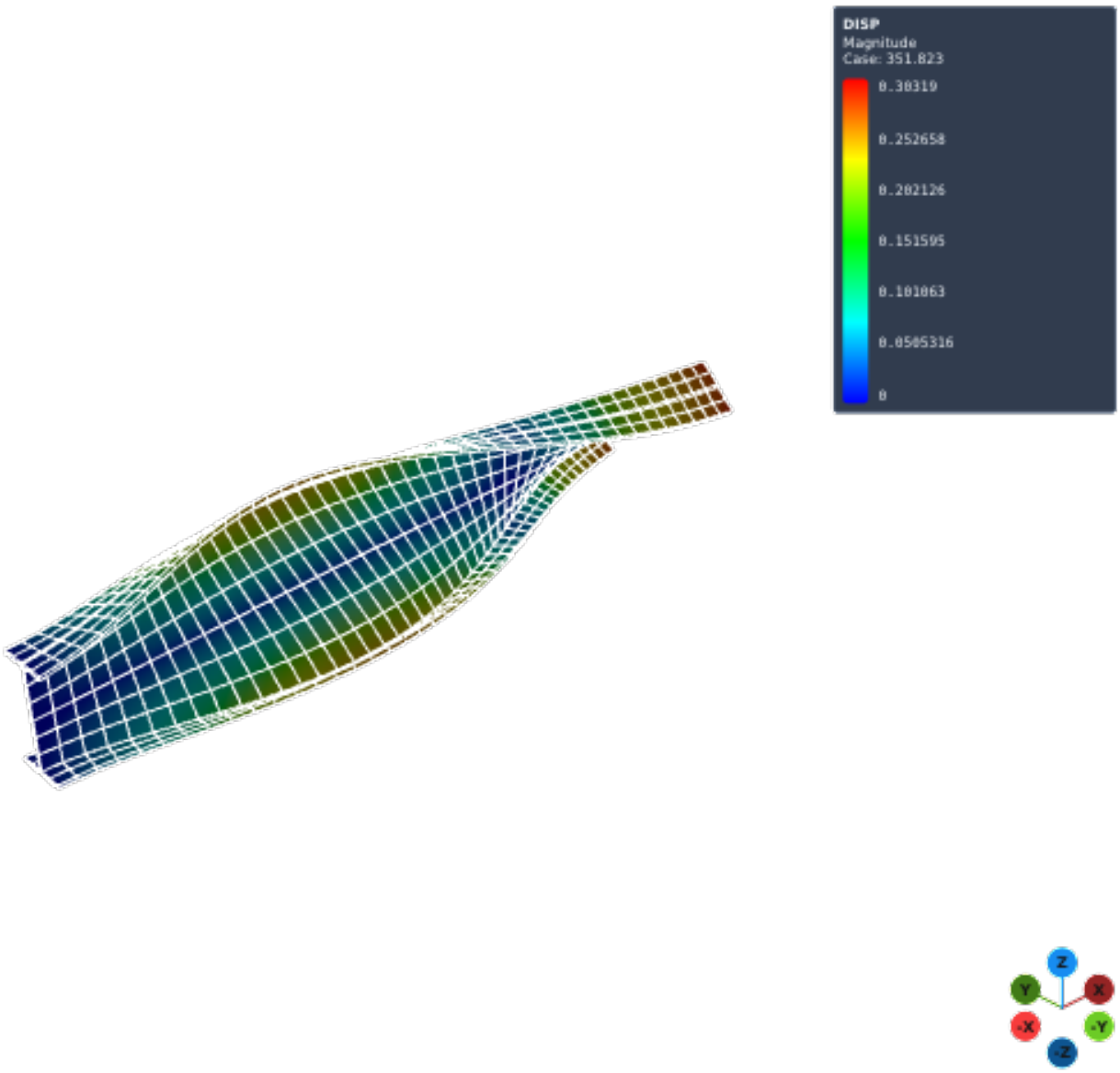

Figure 410: Calculix — mode 7, 351.823 Hz

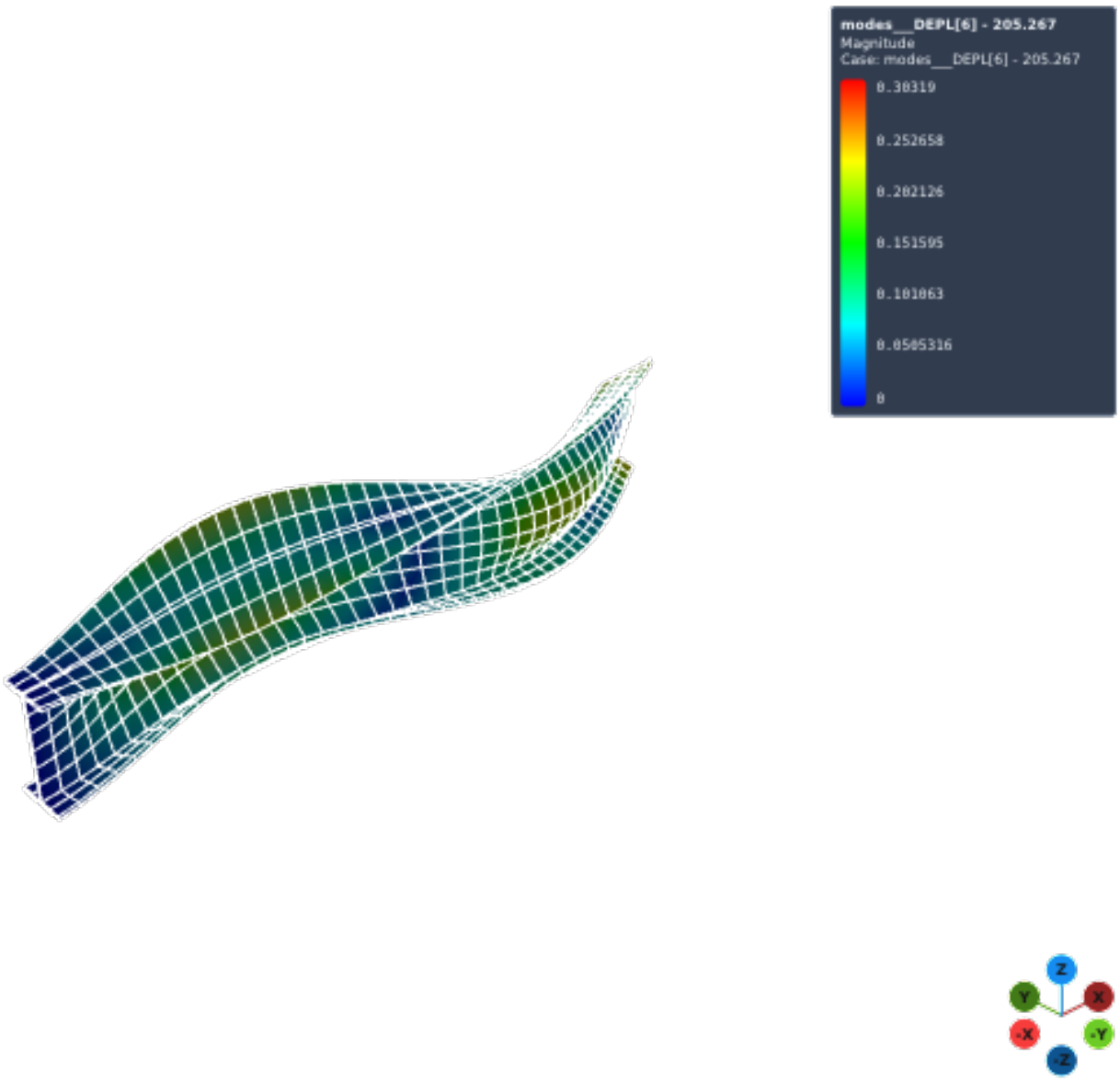

Figure 411: Code_Aster — mode 7, 205.267 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	90.033	-58.26	320010	1.17e-28	2.79	5.27e-21	-1.08e-14	-1.671	7.26e-11
Calculix	351.823	+63.10	4.88661e+06	7.91e-27	2.11e-24	1.60e-27	8.89e-14	-1.45e-12	-4.00e-14
Code_Aster	205.267	-4.84	1.66341e+06	2.28e-09	8.94	0.01	4.77e-05	2.990	0.071

Mode 8

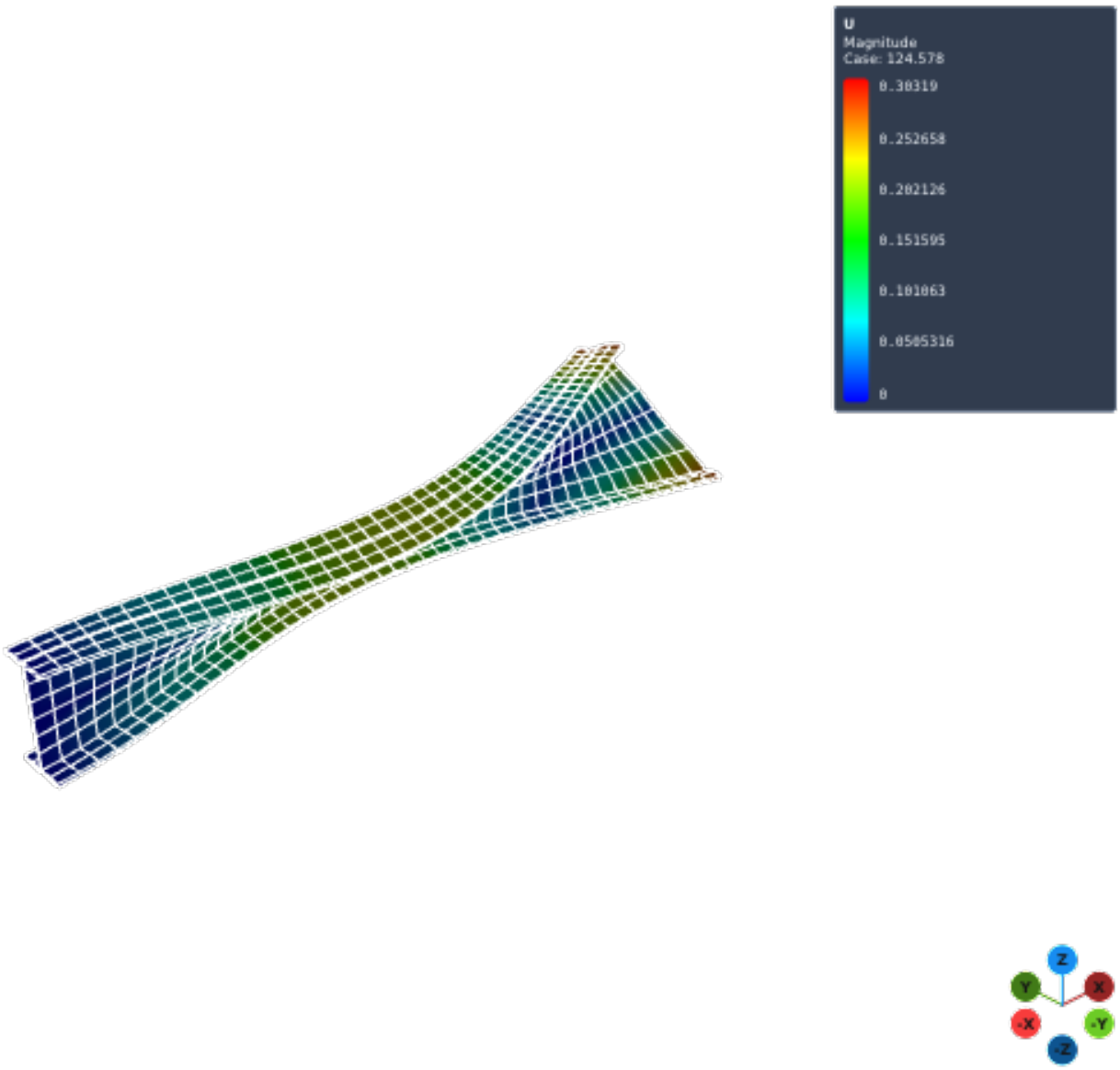

Figure 412: Abaqus — mode 8, 124.578 Hz

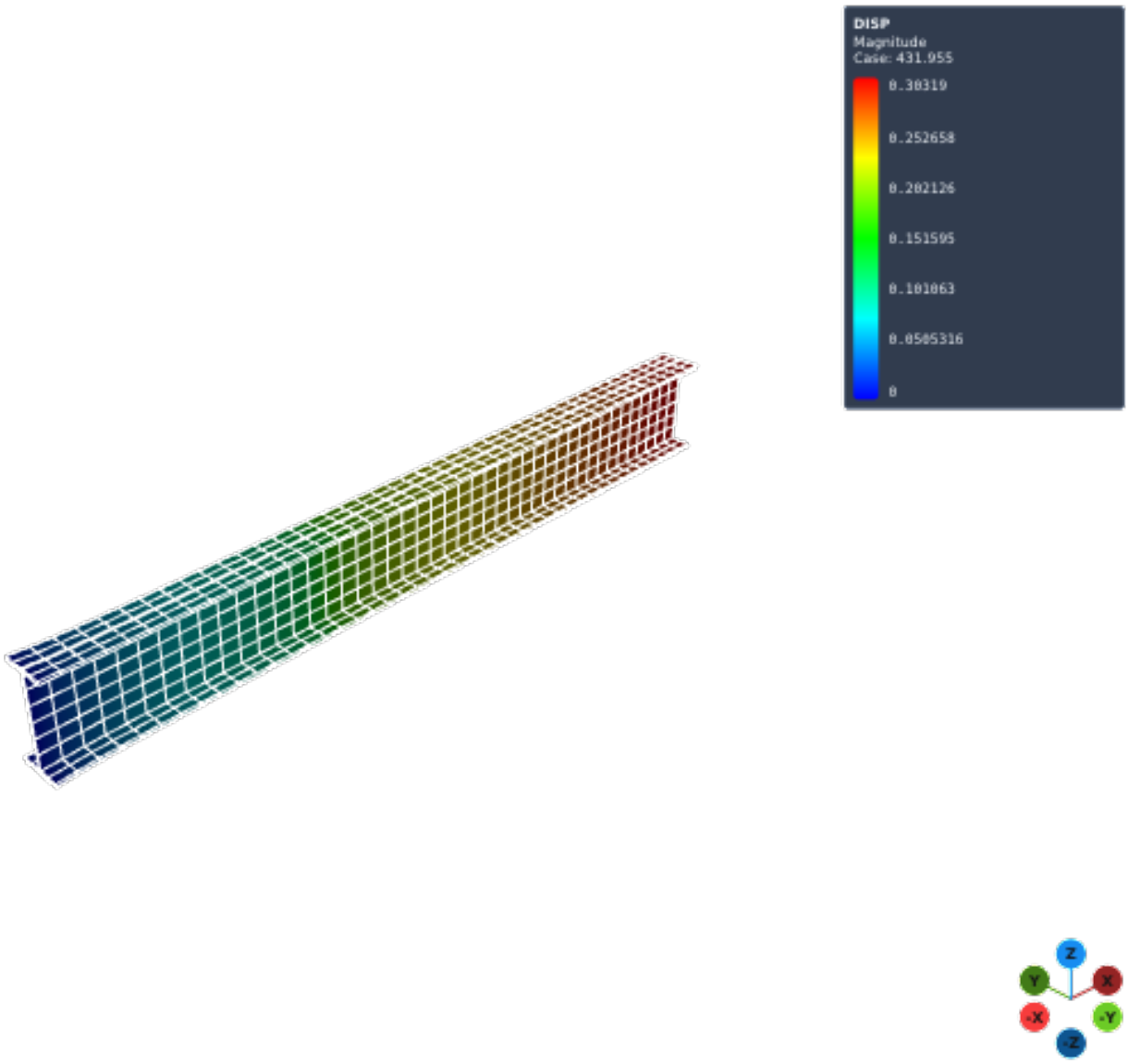

Figure 413: Calculix — mode 8, 431.955 Hz

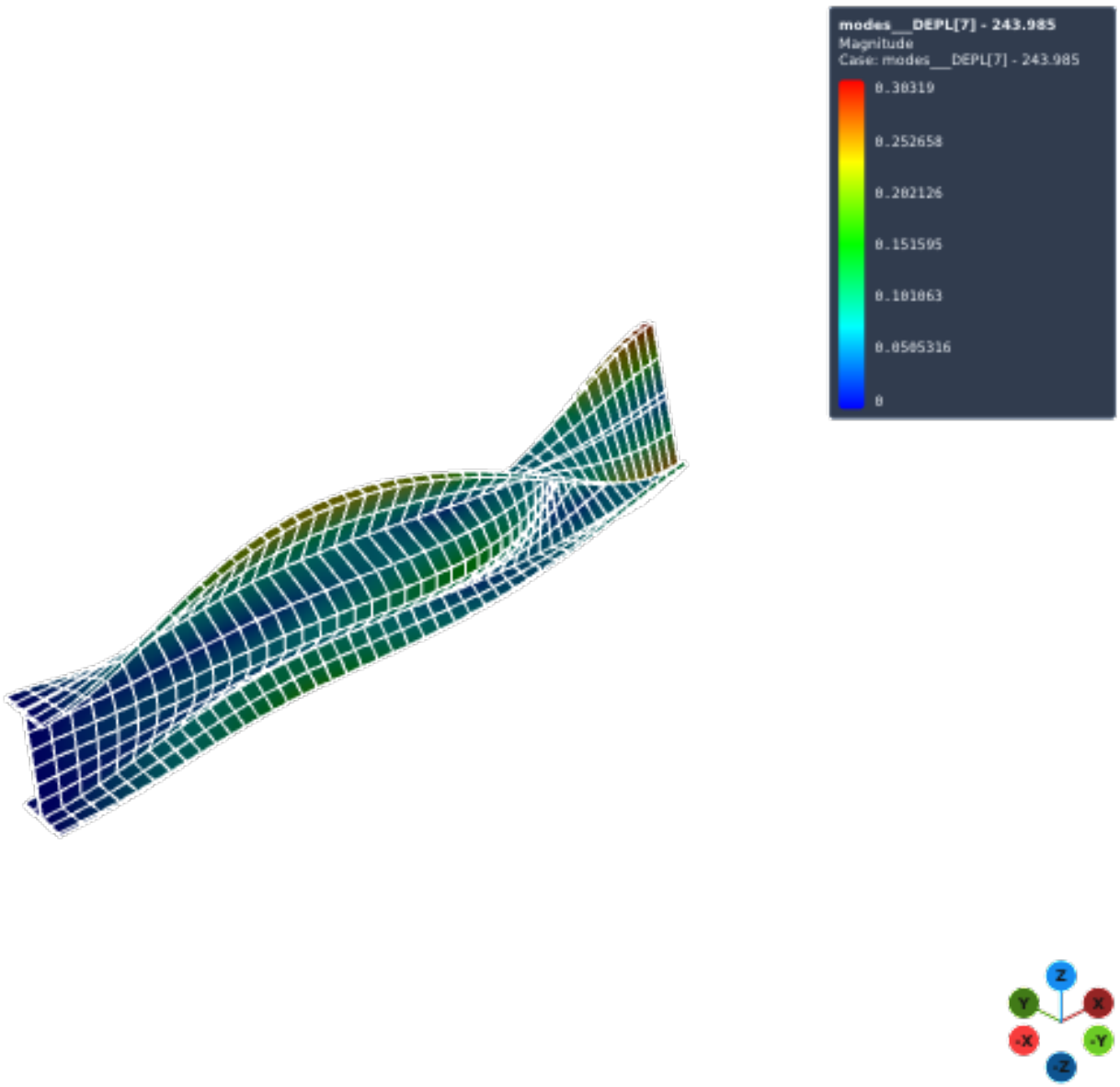

Figure 414: Code_Aster — mode 8, 243.985 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	124.580	-53.31	612693	3.87e-24	3.38e-21	2.90e-24	-1.97e-12	-5.82e-11	-1.70e-12
Calculix	431.955	+61.88	7.36609e+06	153.48	1.64e-27	2.24e-26	12.389	4.05e-14	1.50e-13
Code_Aster	243.985	-8.57	2.3501e+06	6.66e-07	3.10	0.25	0.001	1.760	-0.503

Mode 9

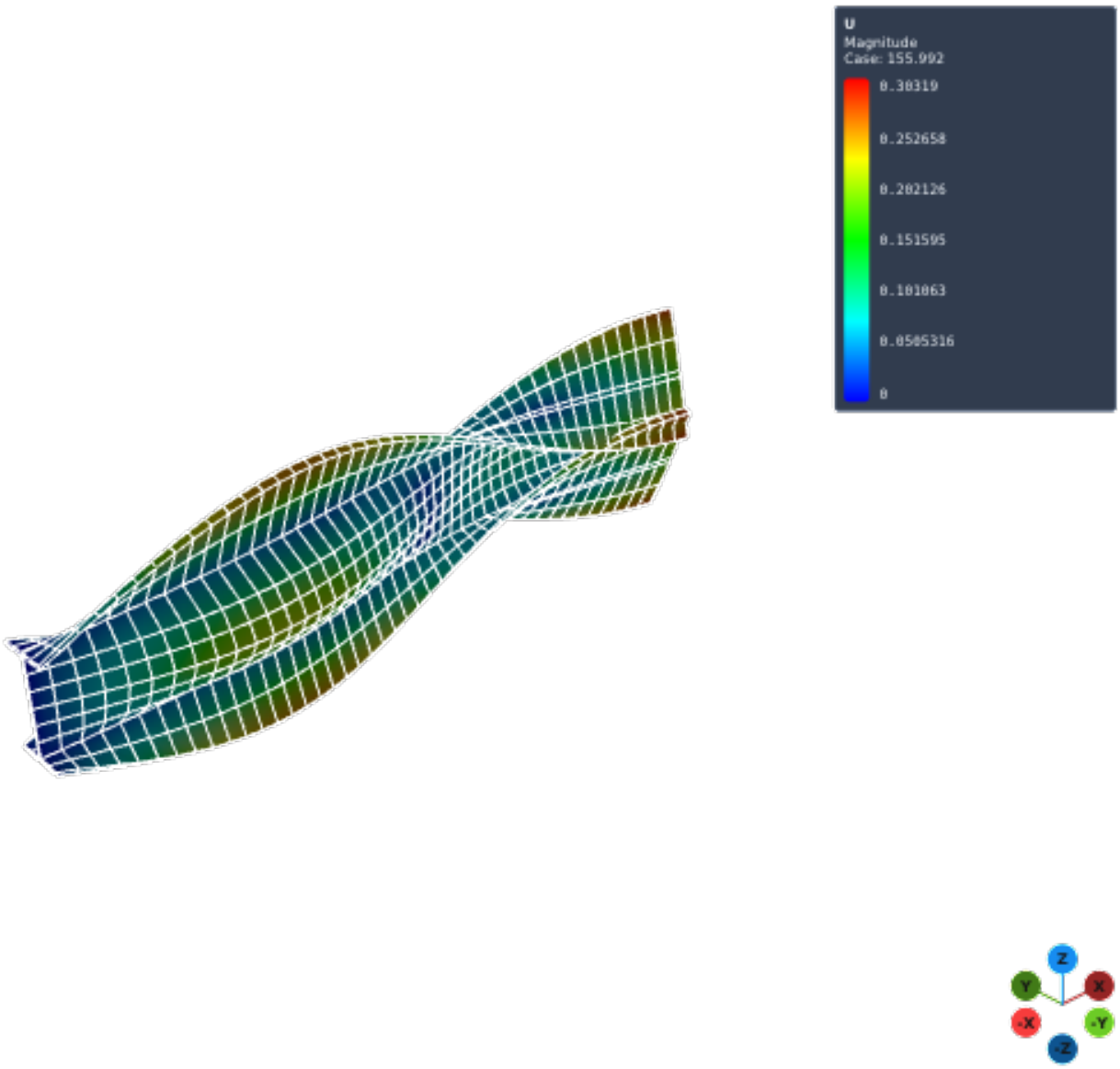

Figure 415: Abaqus — mode 9, 155.992 Hz

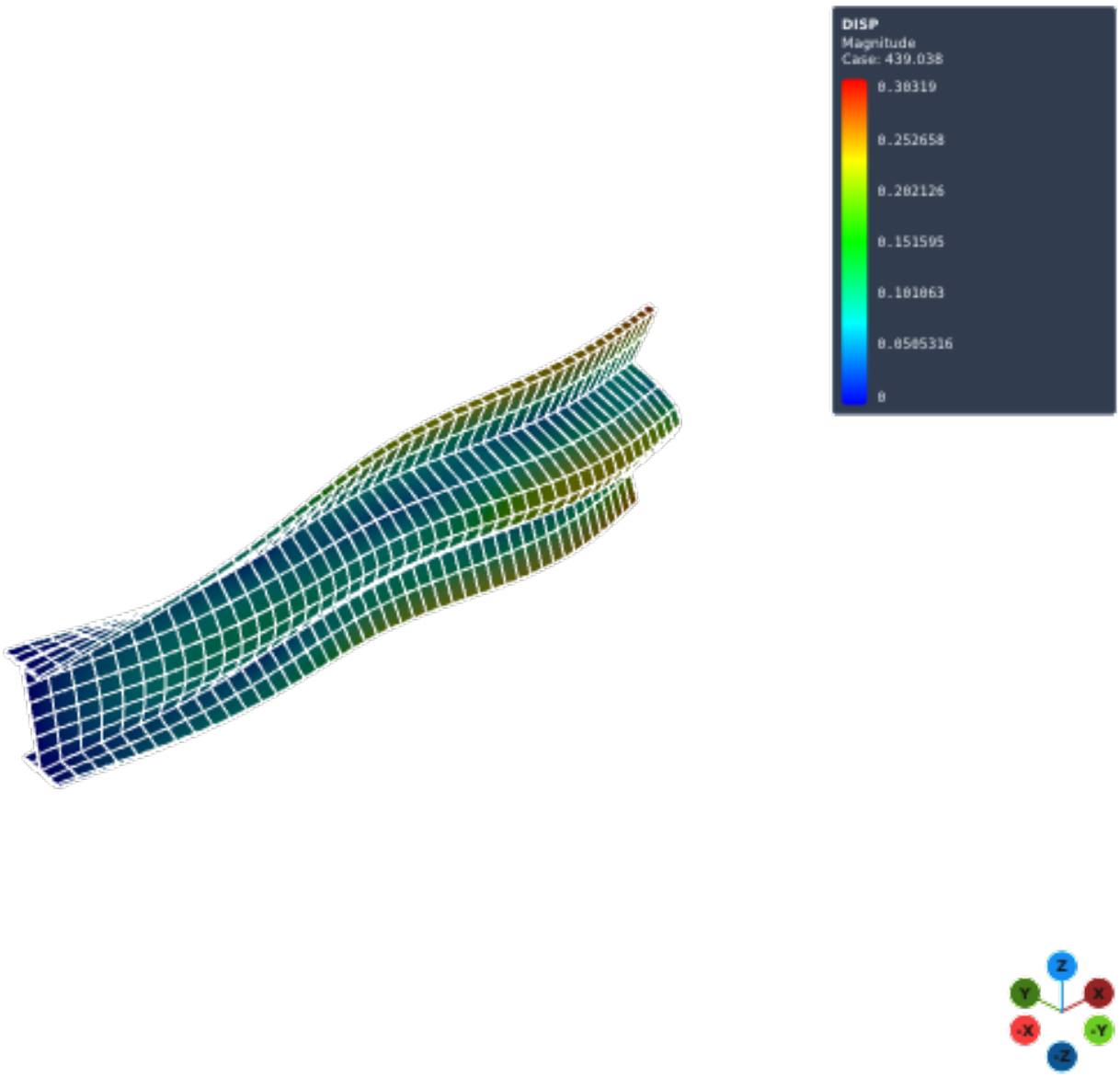

Figure 416: Calculix — mode 9, 439.038 Hz

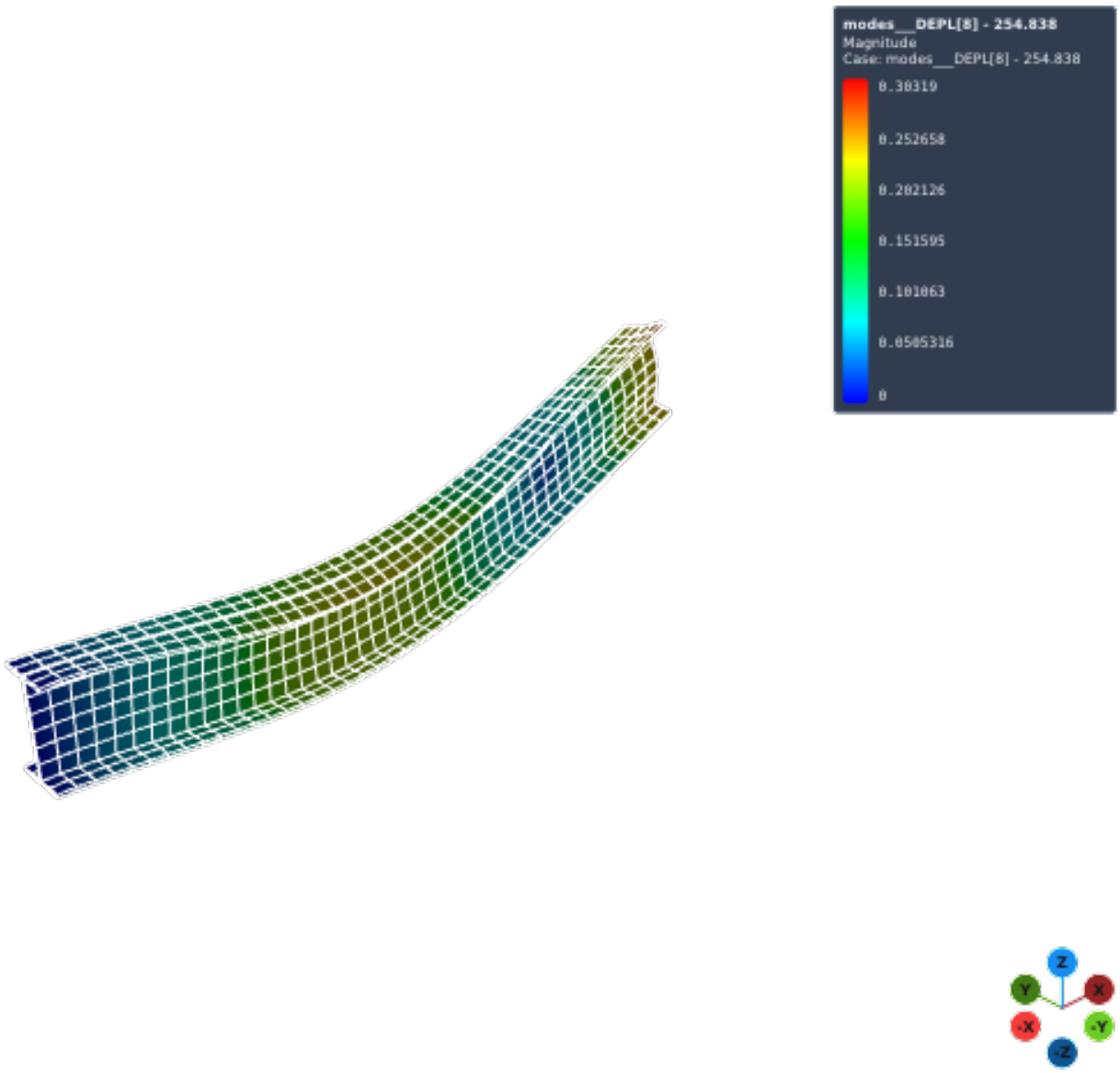

Figure 417: Code_Aster — mode 9, 254.838 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	155.990	-44.94	960645	7.43e-21	1.12	2.07e-20	-8.62e-11	-1.056	-1.44e-10
Calculix	439.038	+54.98	7.60964e+06	1.40e-23	0.38	1.42e-27	3.74e-12	-0.619	-3.77e-14
Code_Aster	254.838	-10.04	2.56382e+06	1.17e-06	0.01	39.92	0.001	-0.115	-6.318

Mode 10

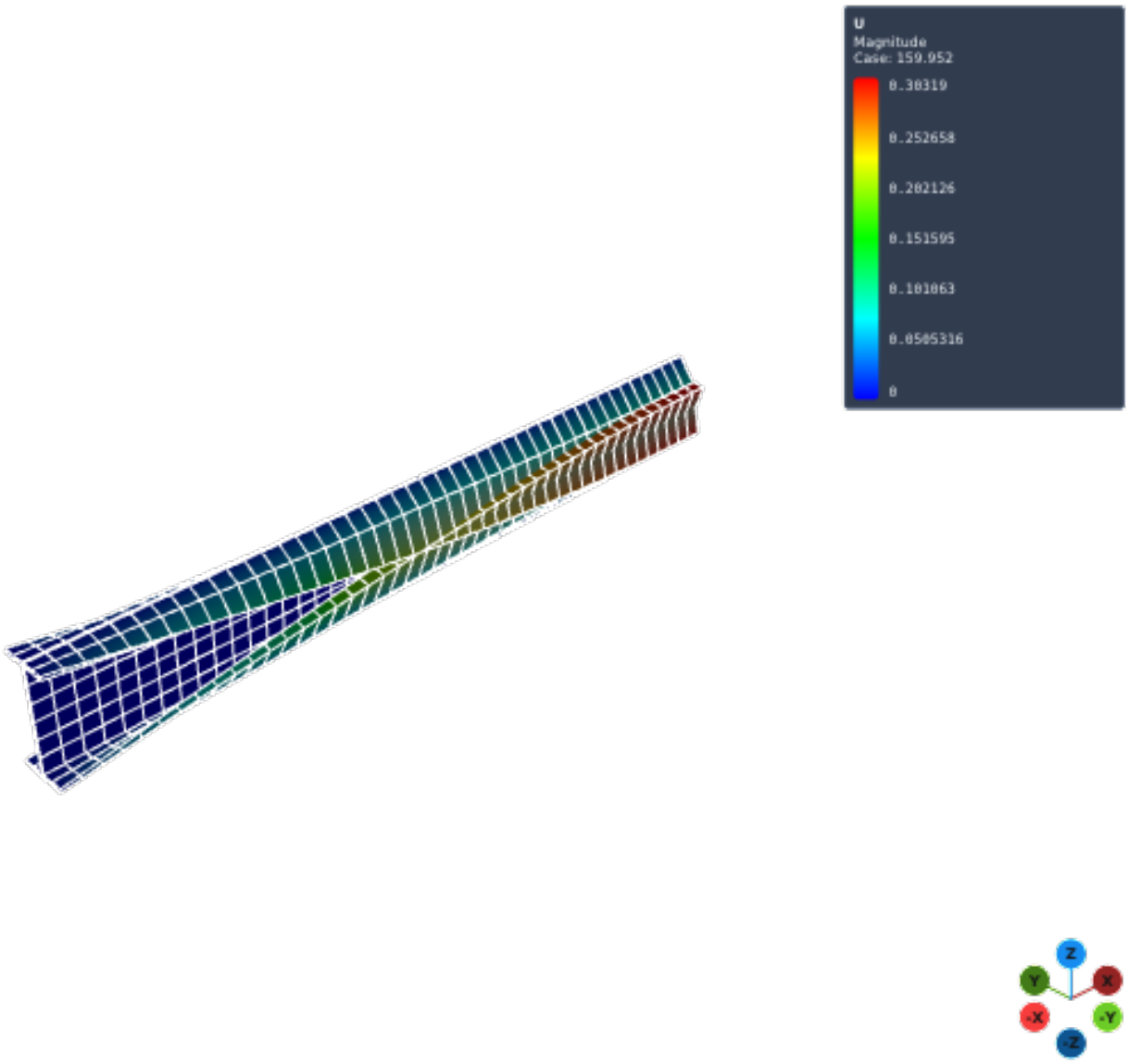

Figure 418: Abaqus — mode 10, 159.952 Hz

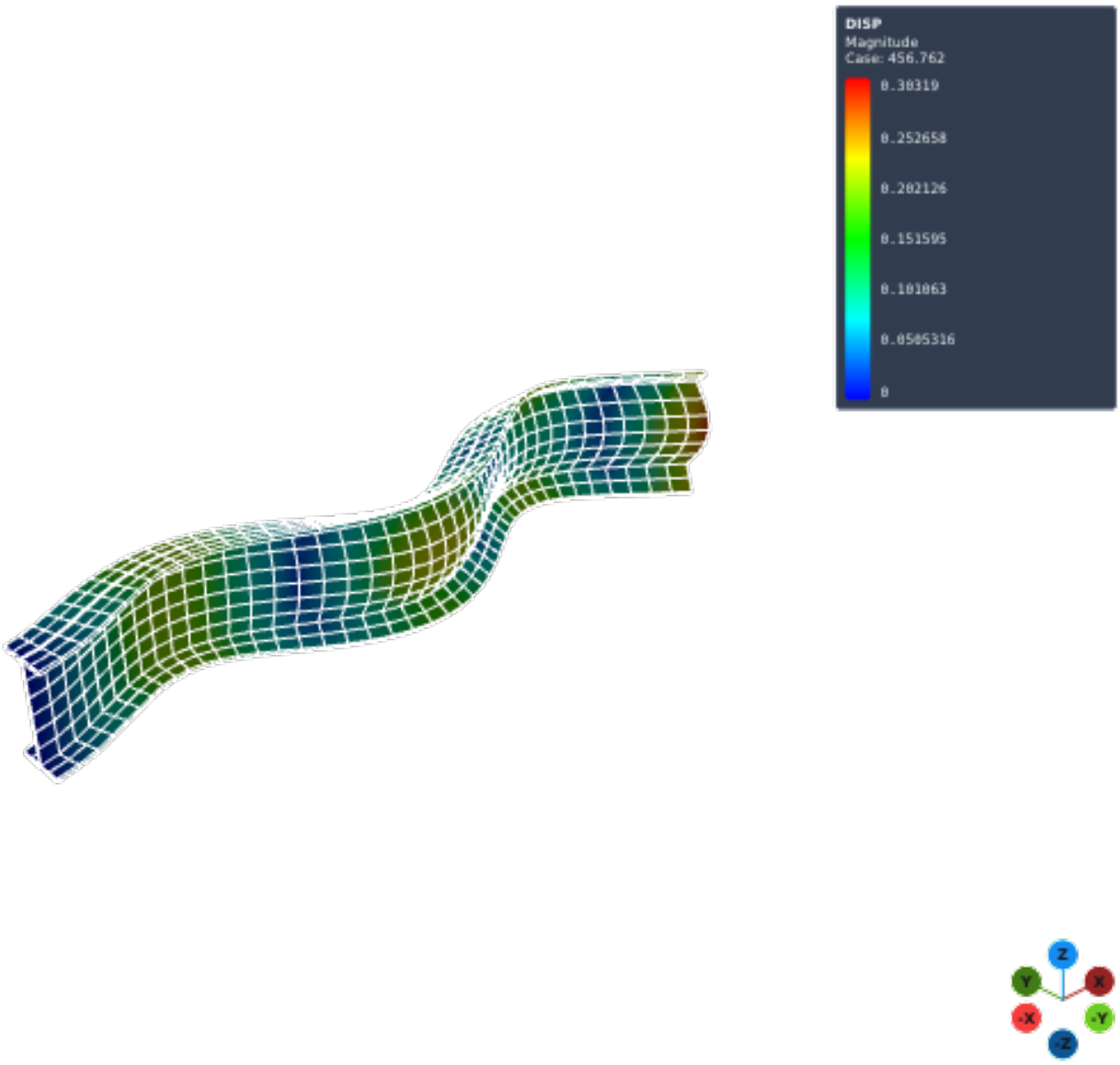

Figure 419: Calculix — mode 10, 456.762 Hz

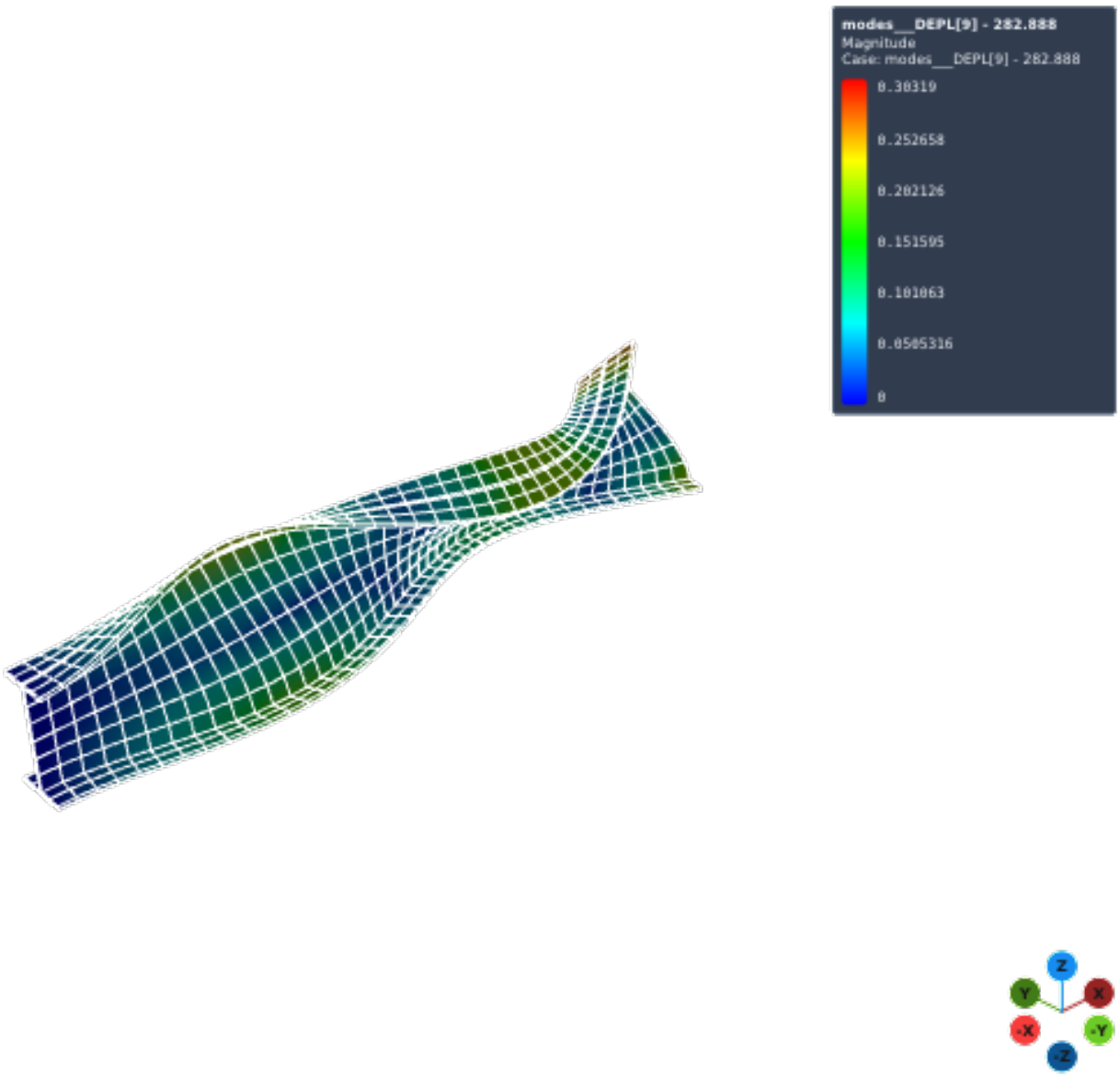

Figure 420: Code_Aster — mode 10, 282.888 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	159.950	-46.66	1.01004e+06	3.65e-04	8.80e-19	3.04e-19	0.019	-9.38e-10	5.51e-10
Calculix	456.762	+52.32	8.23646e+06	1.29e-25	6.14	7.58e-30	-3.59e-13	2.478	-2.75e-15
Code_Aster	282.888	-5.66	3.15929e+06	3.28e-05	0.01	3.08e-05	0.006	0.086	-0.006

Mode 11

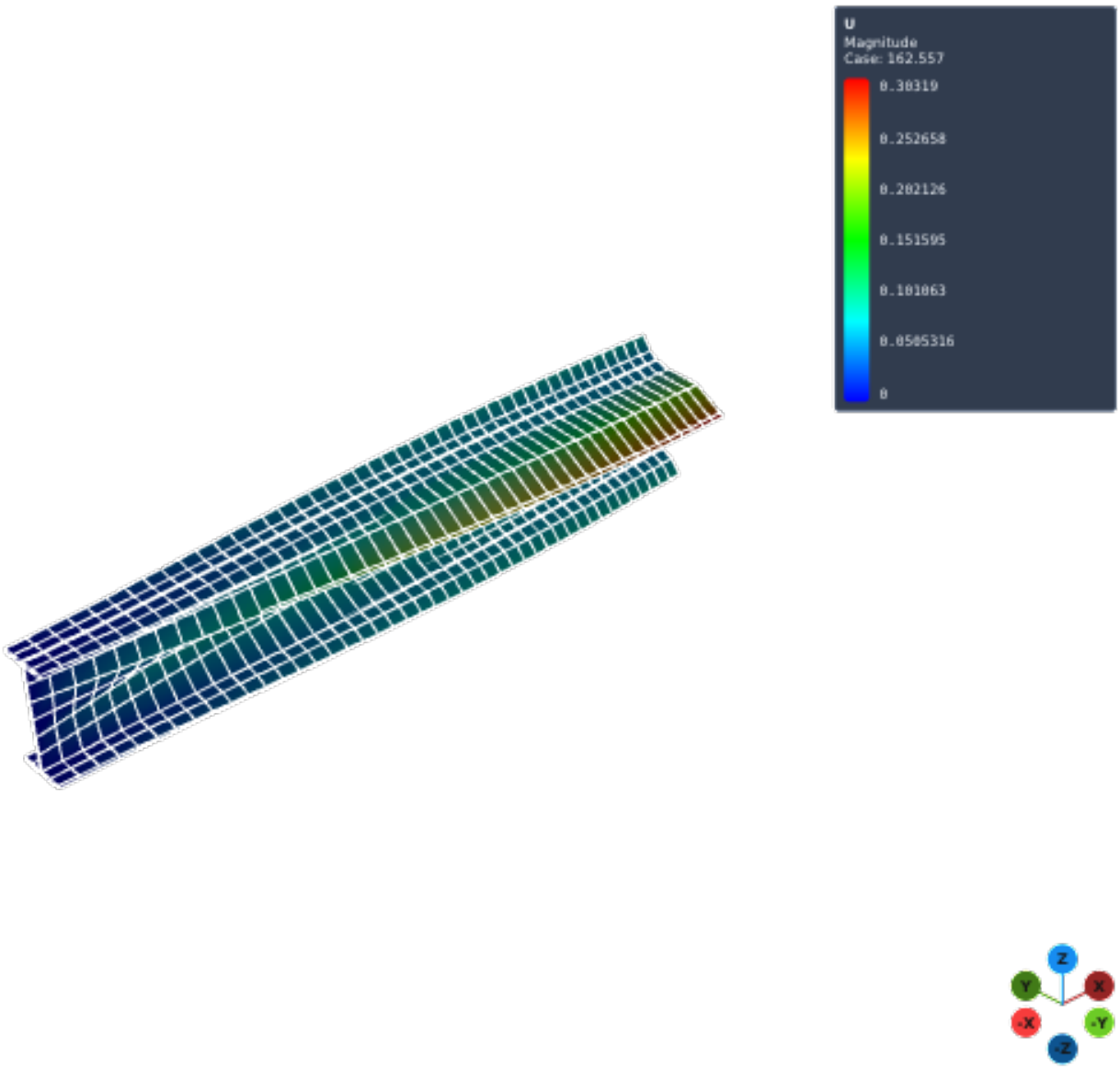

Figure 421: Abaqus — mode 11, 162.557 Hz

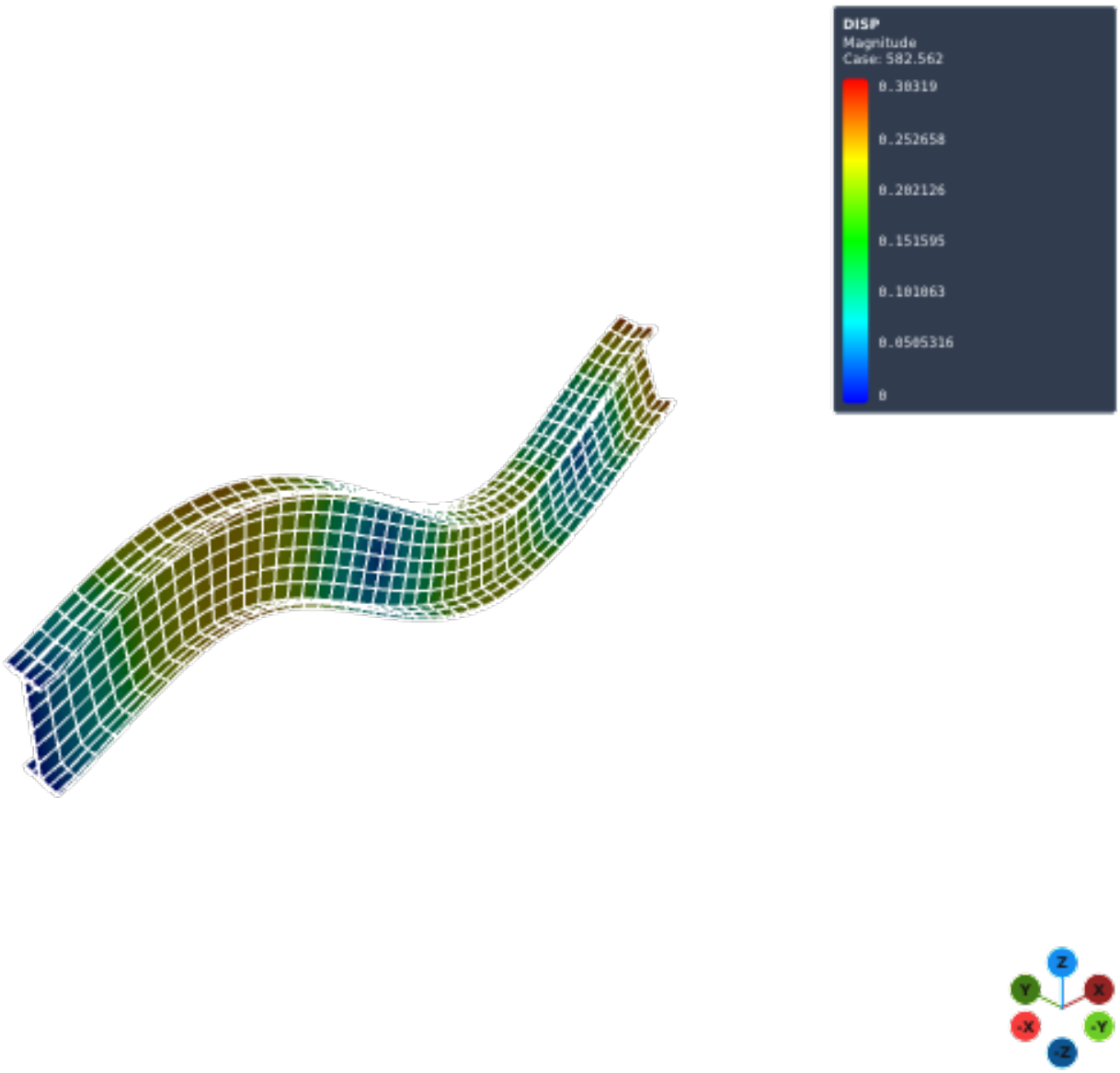

Figure 422: Calculix — mode 11, 582.562 Hz

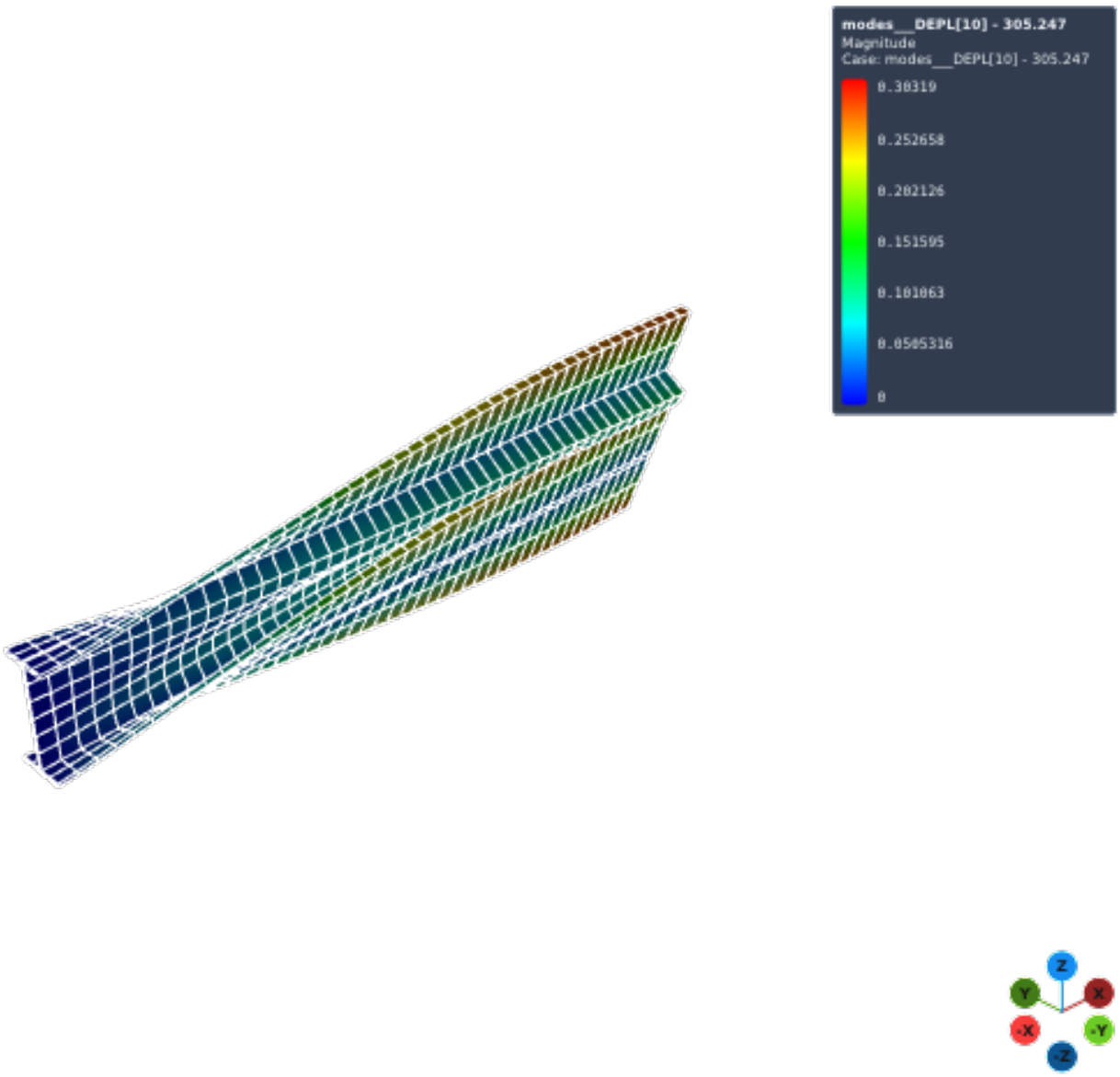

Figure 423: Code_Aster — mode 11, 305.247 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	162.560	-53.57	1.04321e+06	4.88e-19	1.57e-19	3.29e-20	-6.98e-10	3.96e-10	-1.81e-10
Calculix	582.562	+66.39	1.33981e+07	2.76e-25	1.21e-26	12.66	-5.25e-13	-1.10e-13	3.558
Code_Aster	305.247	-12.82	3.67844e+06	2.38e-05	8.68e-05	3.72e-07	0.005	0.009	-0.001

Solid, 2nd order, TET

Mode 1

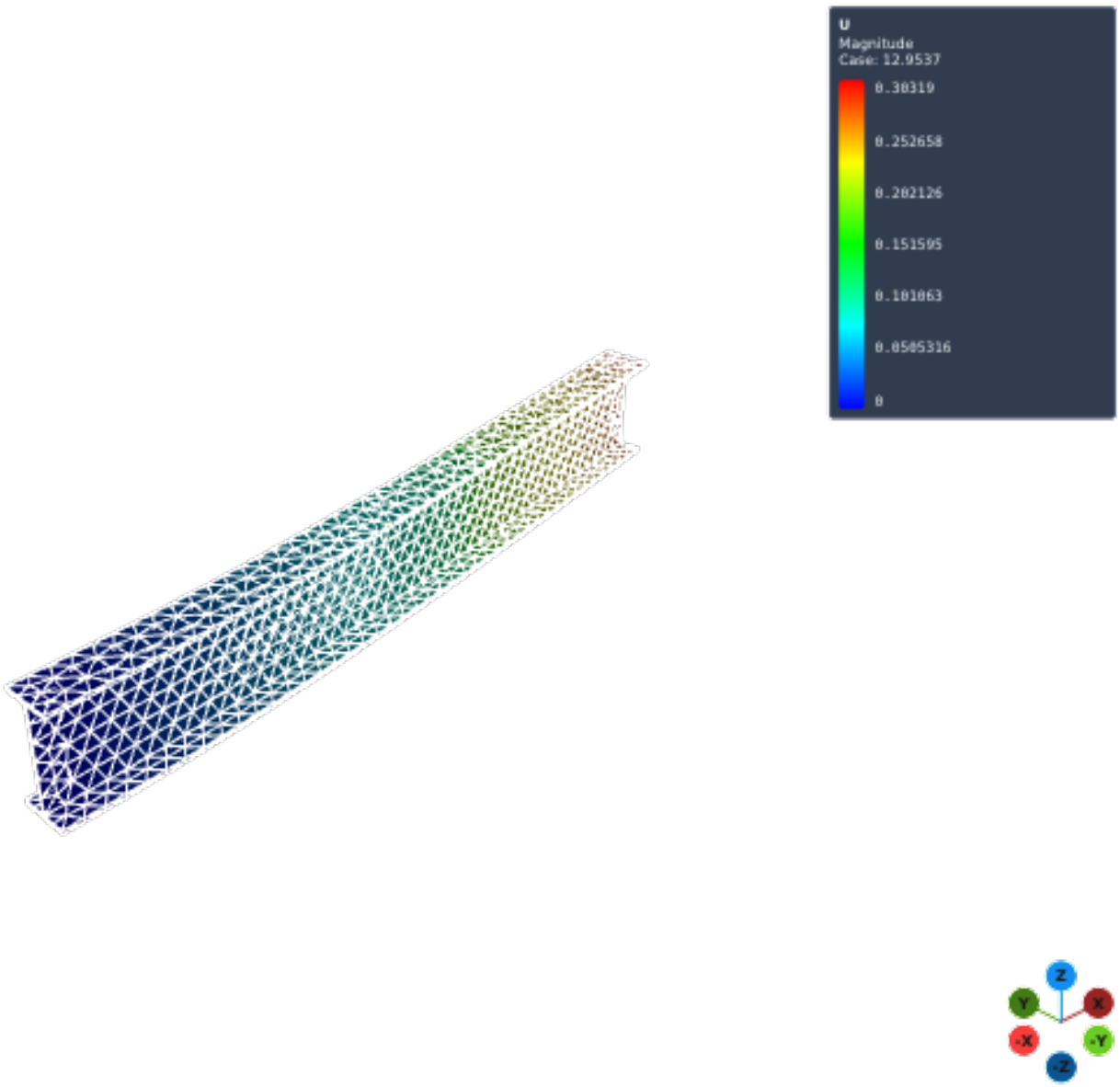

Figure 424: Abaqus — mode 1, 12.954 Hz

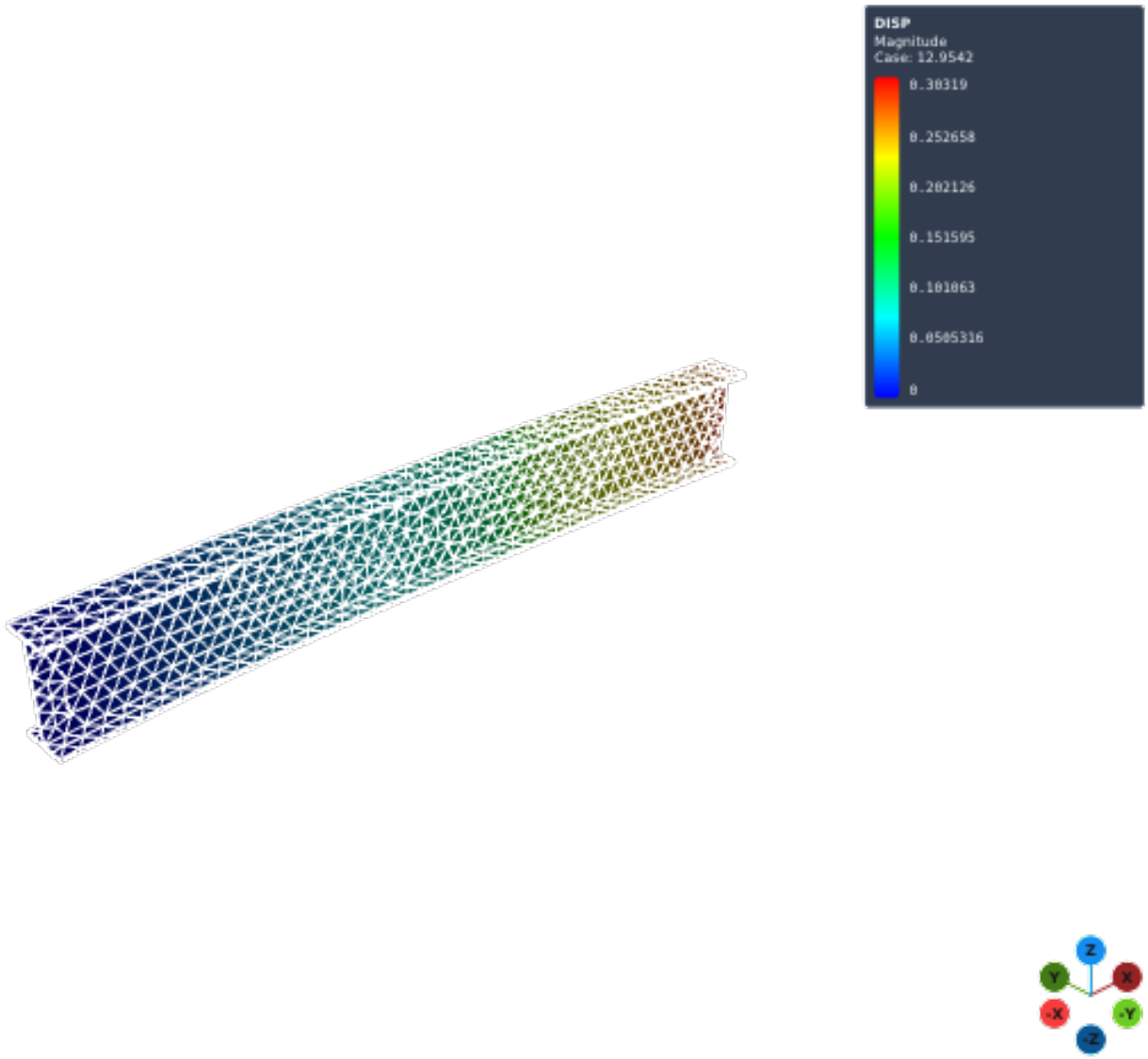

Figure 425: Calculix — mode 1, 12.954 Hz

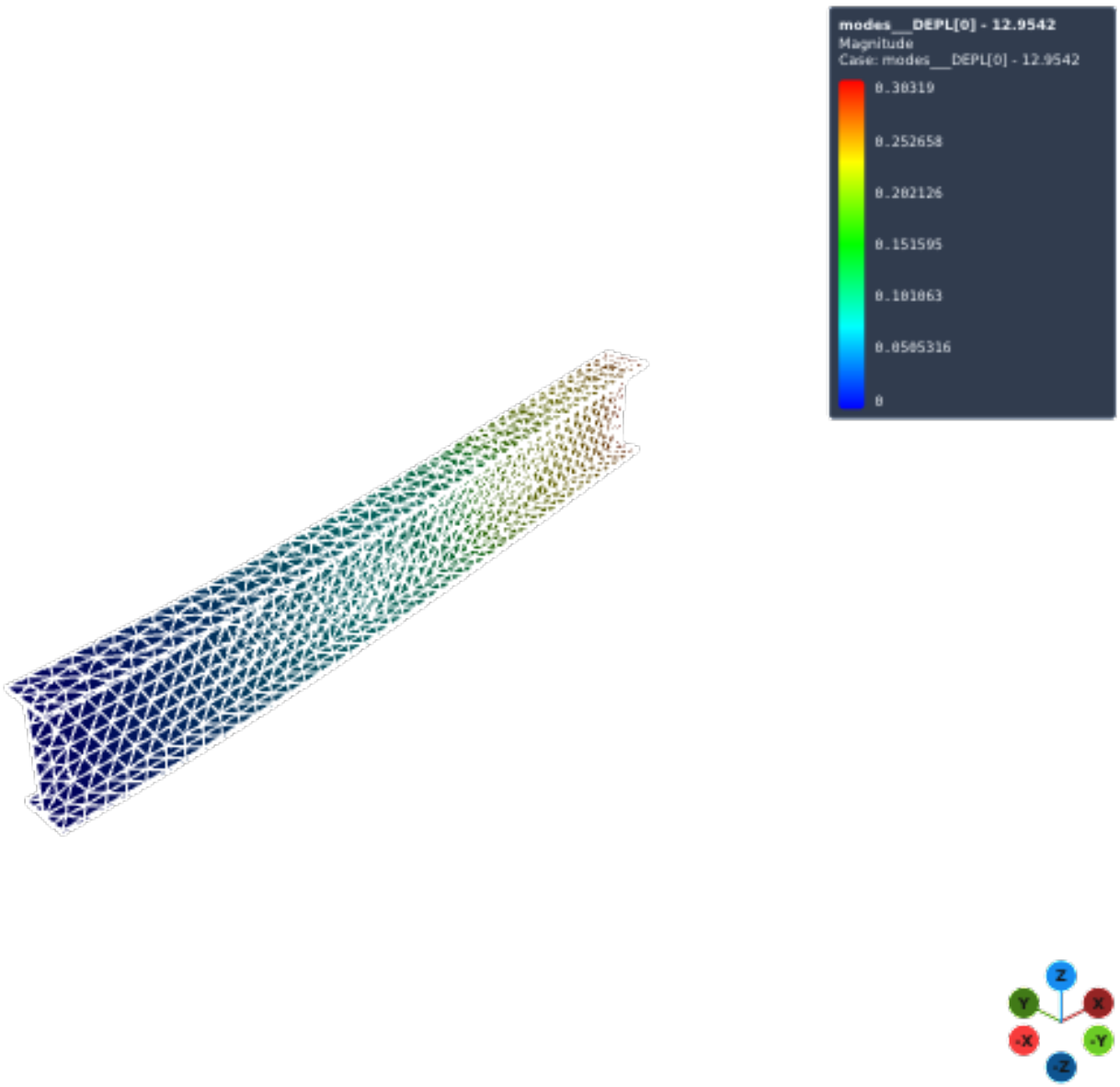

Figure 426: Code_Aster — mode 1, 12.954 Hz

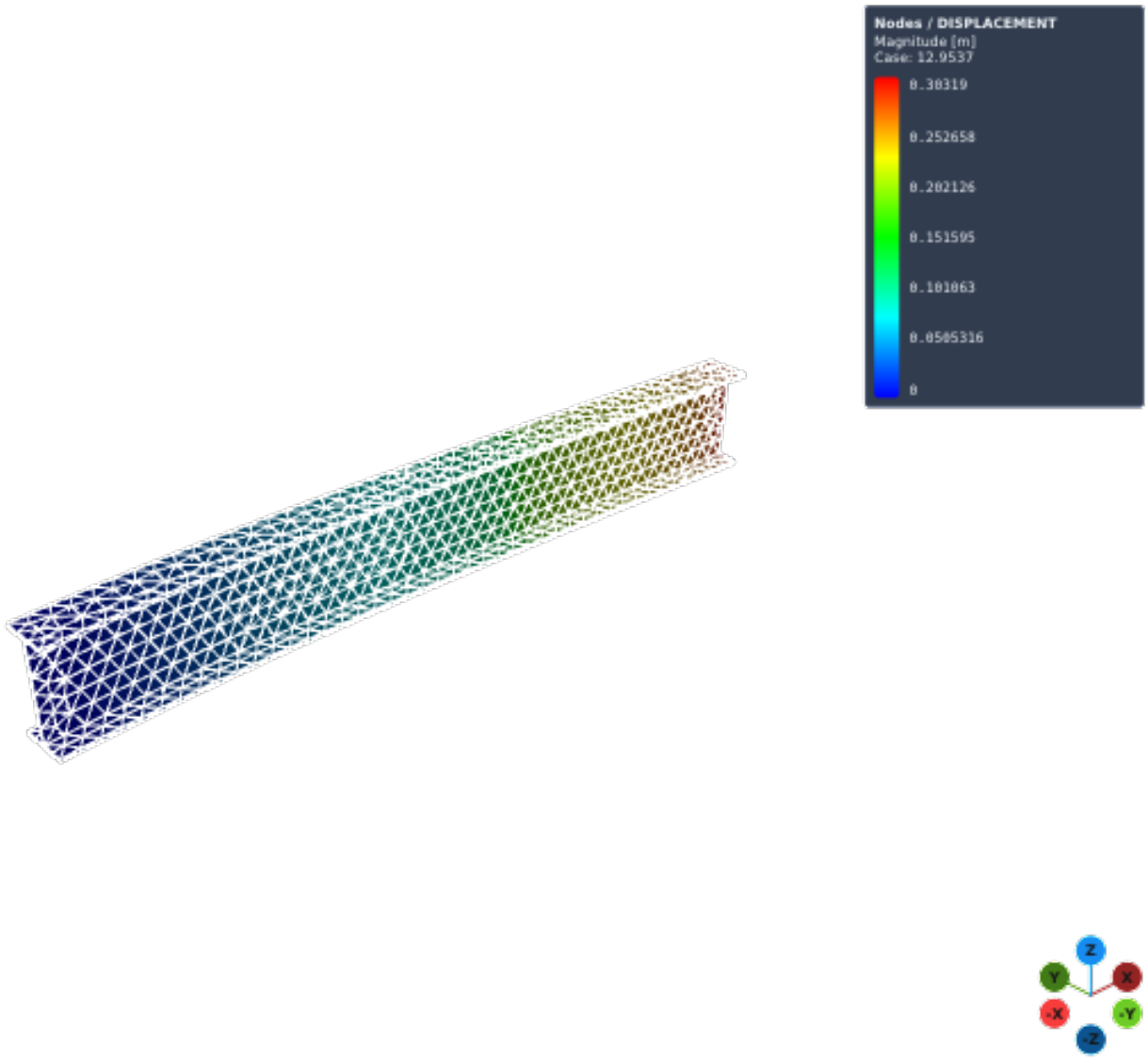

Figure 427: Sesam — mode 1, 12.954 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.954	-1.55e-04	6624.5	3.65e-13	116.51	2.81e-11	6.04e-07	10.794	5.30e-06
Calculix	12.954	1.16e-03	6624.89	4.94e-13	116.51	2.88e-10	-7.03e-07	-10.794	-1.70e-05
Code_Aster	12.954	1.16e-03	6624.89	5.04e-13	116.51	2.91e-10	7.10e-07	10.794	1.71e-05
Sesam	12.954	-2.16e-03	6624.46	3.01e-13	116.51	2.70e-11	-5.49e-07	-10.794	-5.20e-06

Mode 2

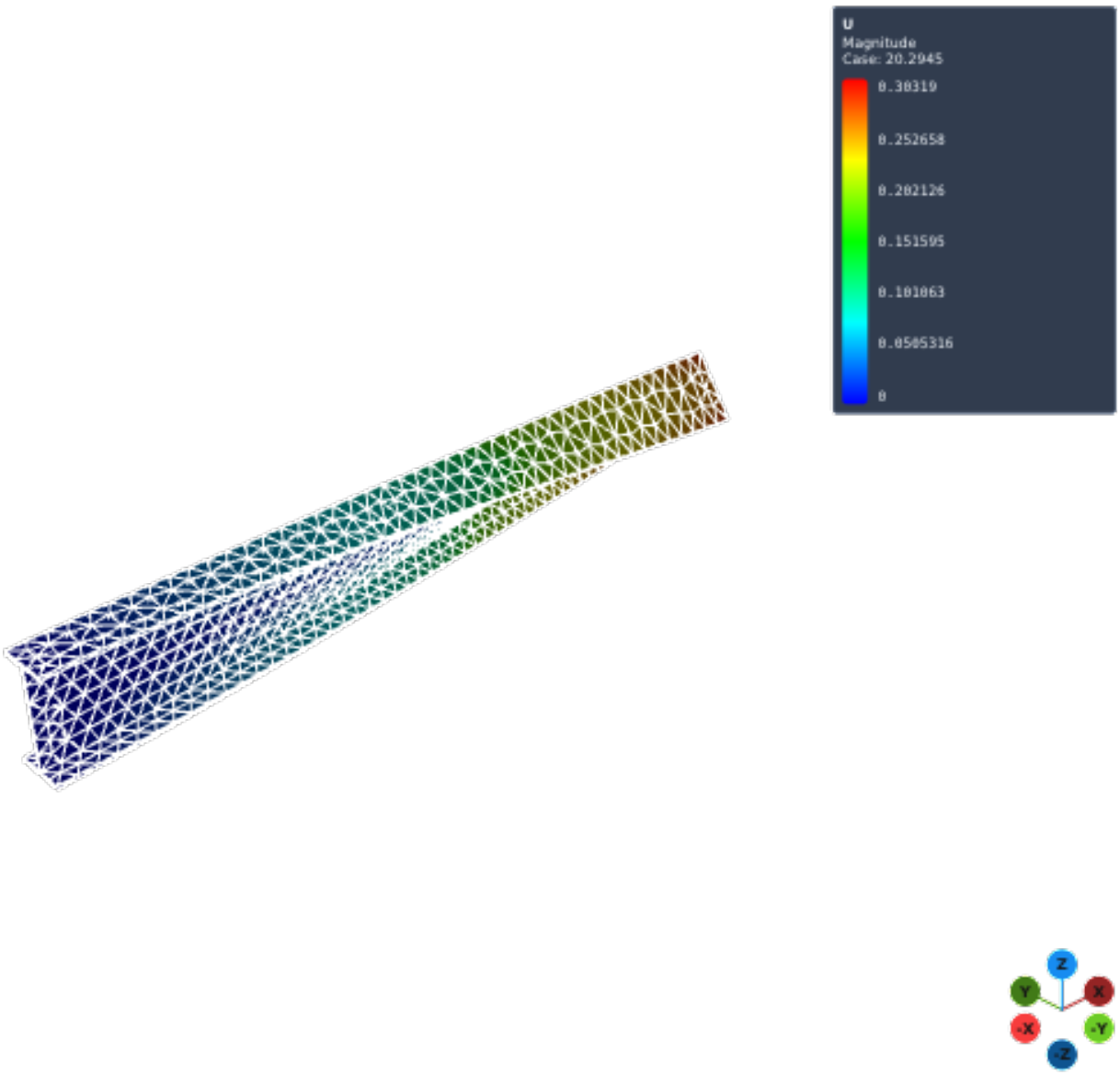

Figure 428: Abaqus — mode 2, 20.294 Hz

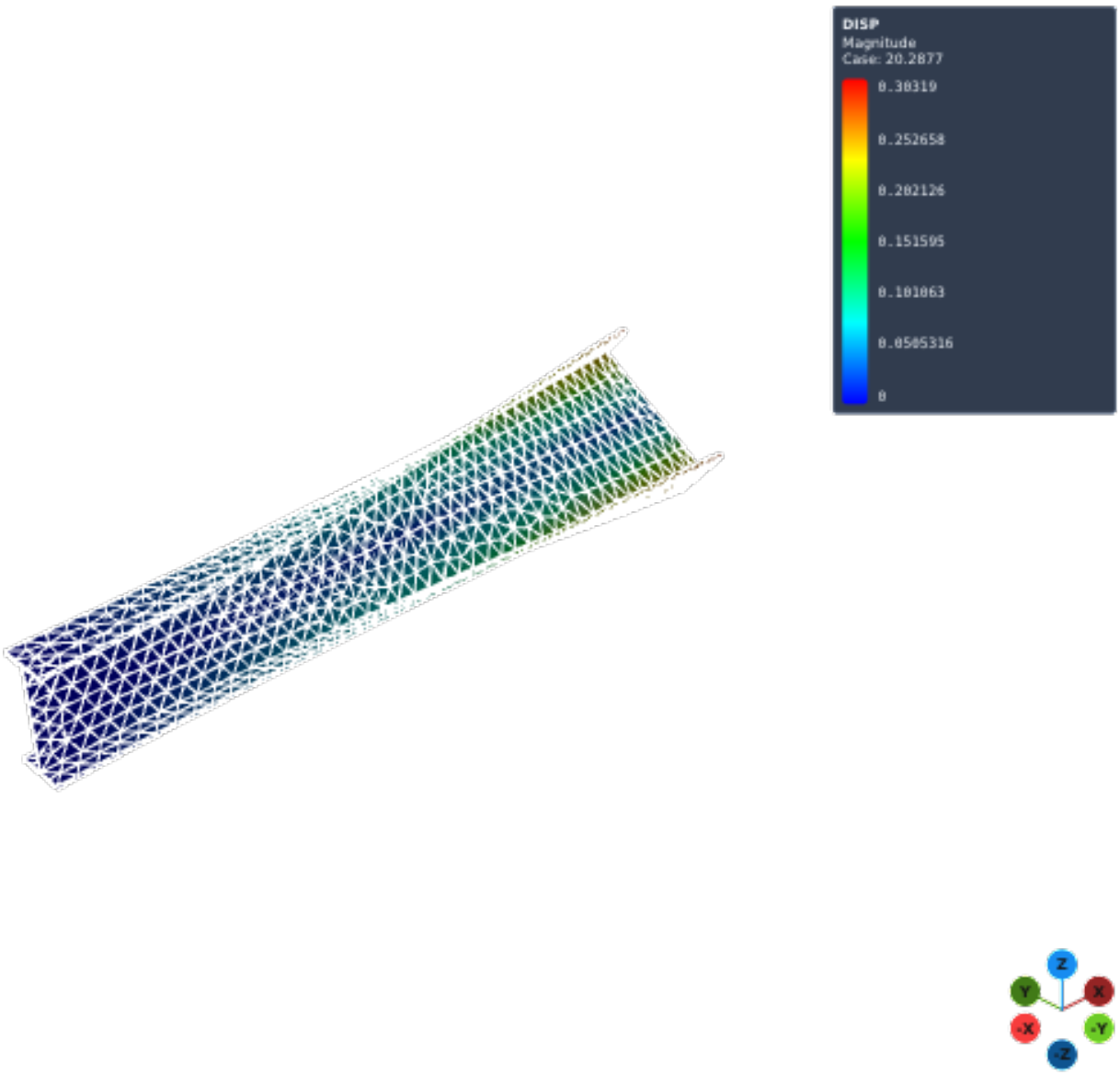

Figure 429: Calculix — mode 2, 20.288 Hz

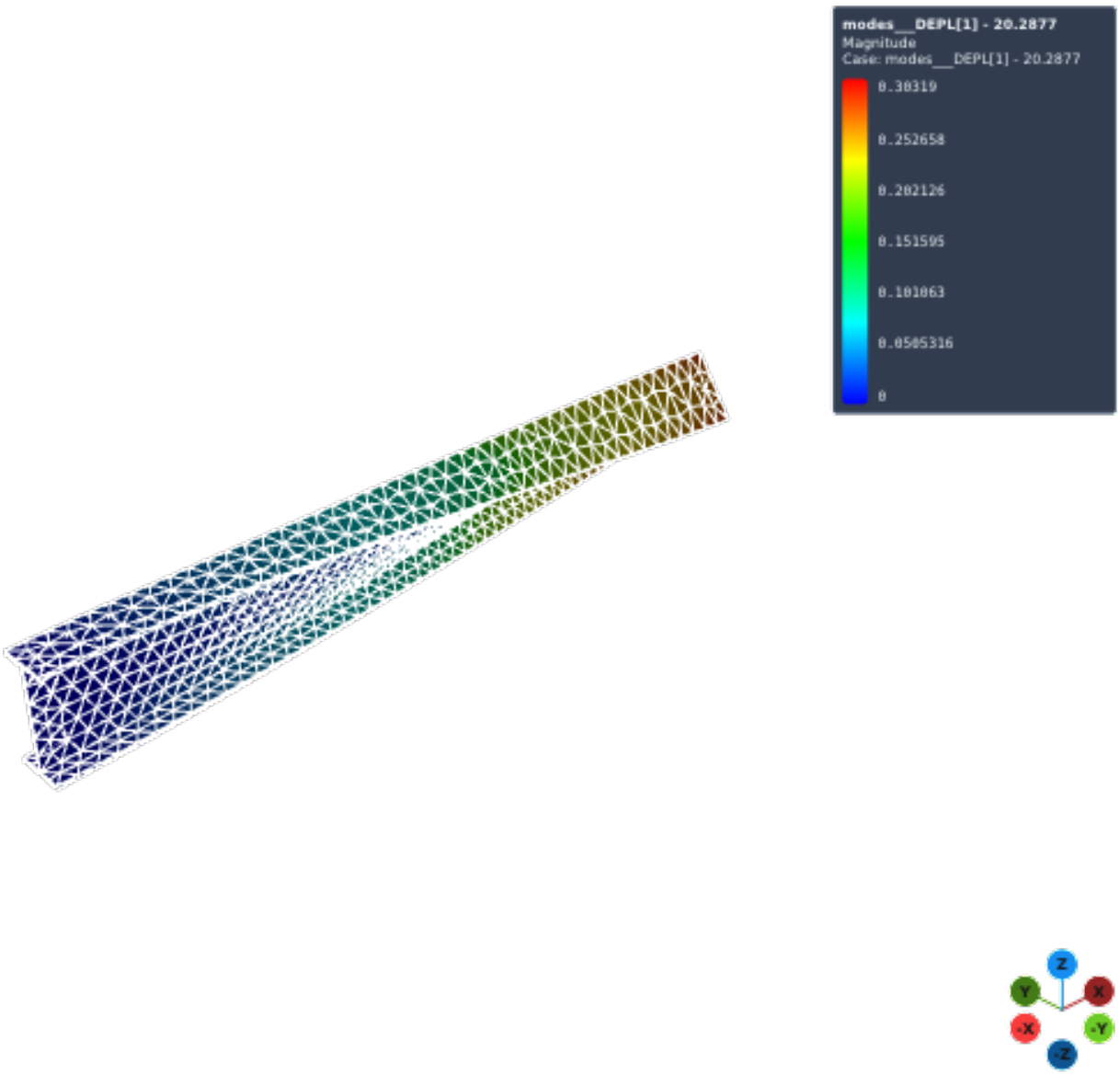

Figure 430: Code_Aster — mode 2, 20.288 Hz

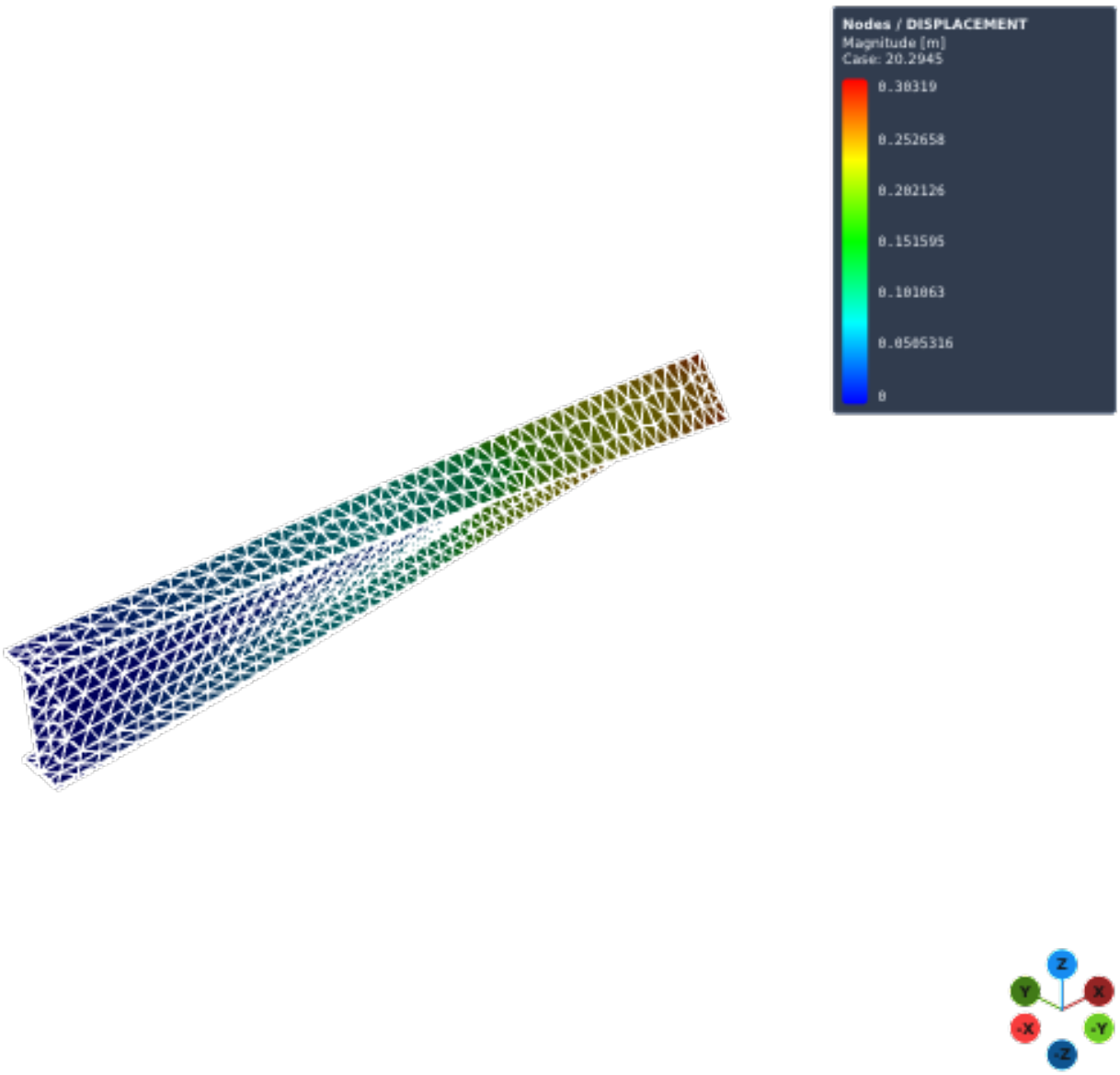

Figure 431: Sesam — mode 2, 20.294 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	20.294	+0.02	16260	8.19e-12	5.86e-08	1.99e-09	2.86e-06	-2.42e-04	4.47e-05
Calculix	20.288	-0.02	16248.9	3.07e-11	1.64e-07	3.90e-10	-5.54e-06	-4.05e-04	-1.98e-05
Code_Aster	20.288	-0.02	16248.9	3.26e-11	1.64e-07	3.90e-10	5.71e-06	4.05e-04	1.98e-05
Sesam	20.294	+0.02	16259.8	8.59e-12	5.86e-08	1.98e-09	2.93e-06	-2.42e-04	4.45e-05

Mode 3

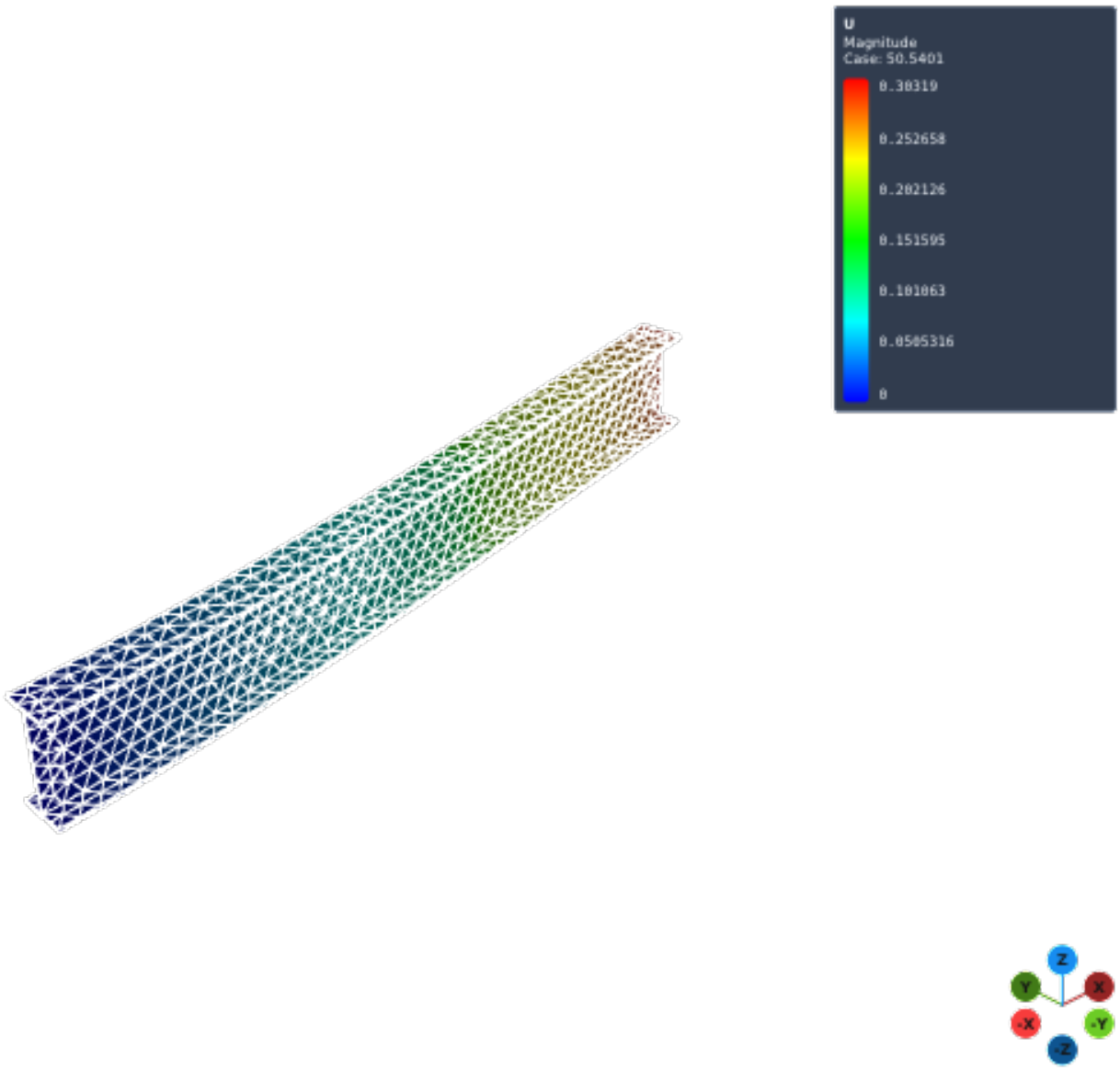

Figure 432: Abaqus — mode 3, 50.540 Hz

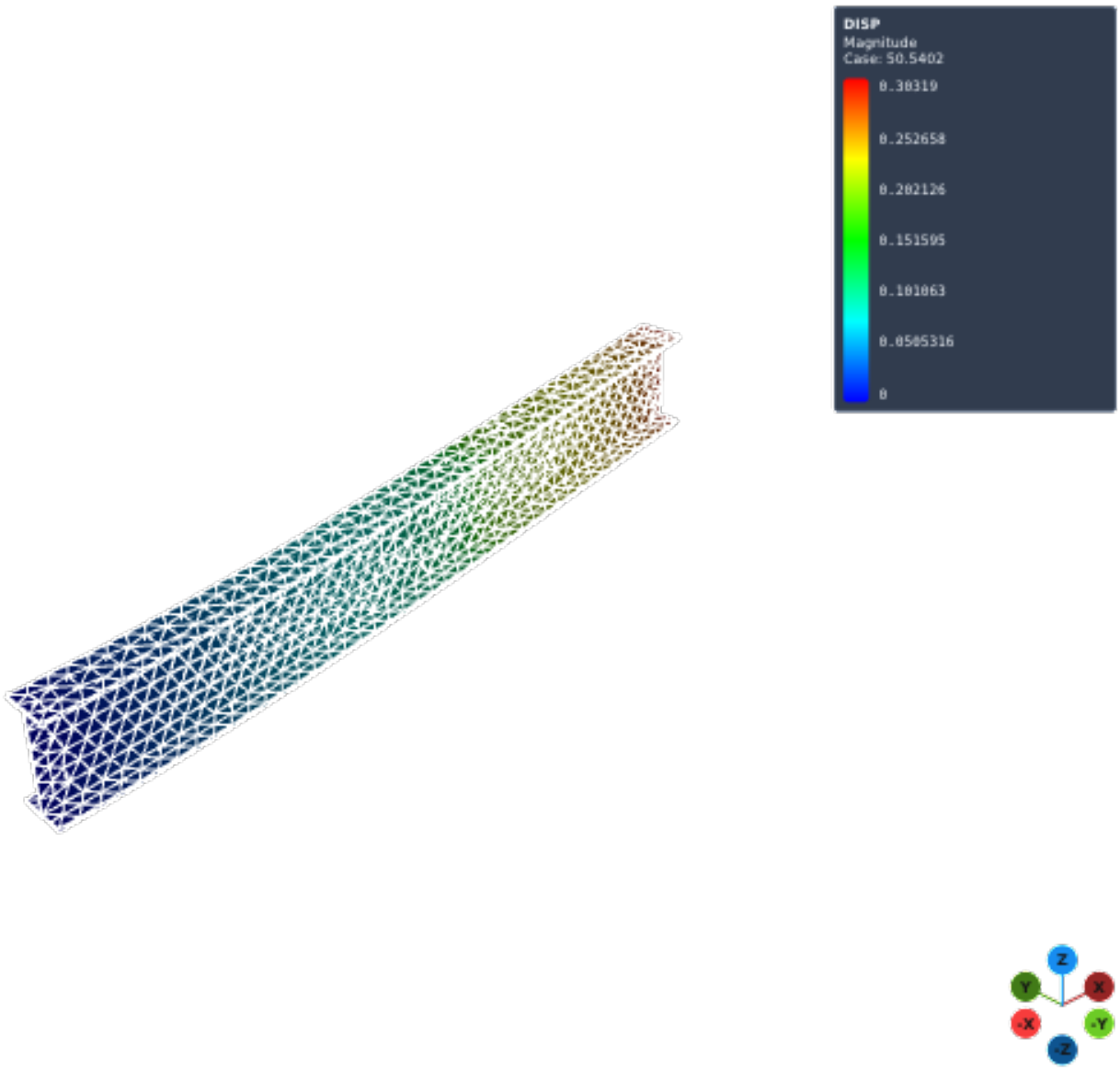

Figure 433: Calculix — mode 3, 50.540 Hz

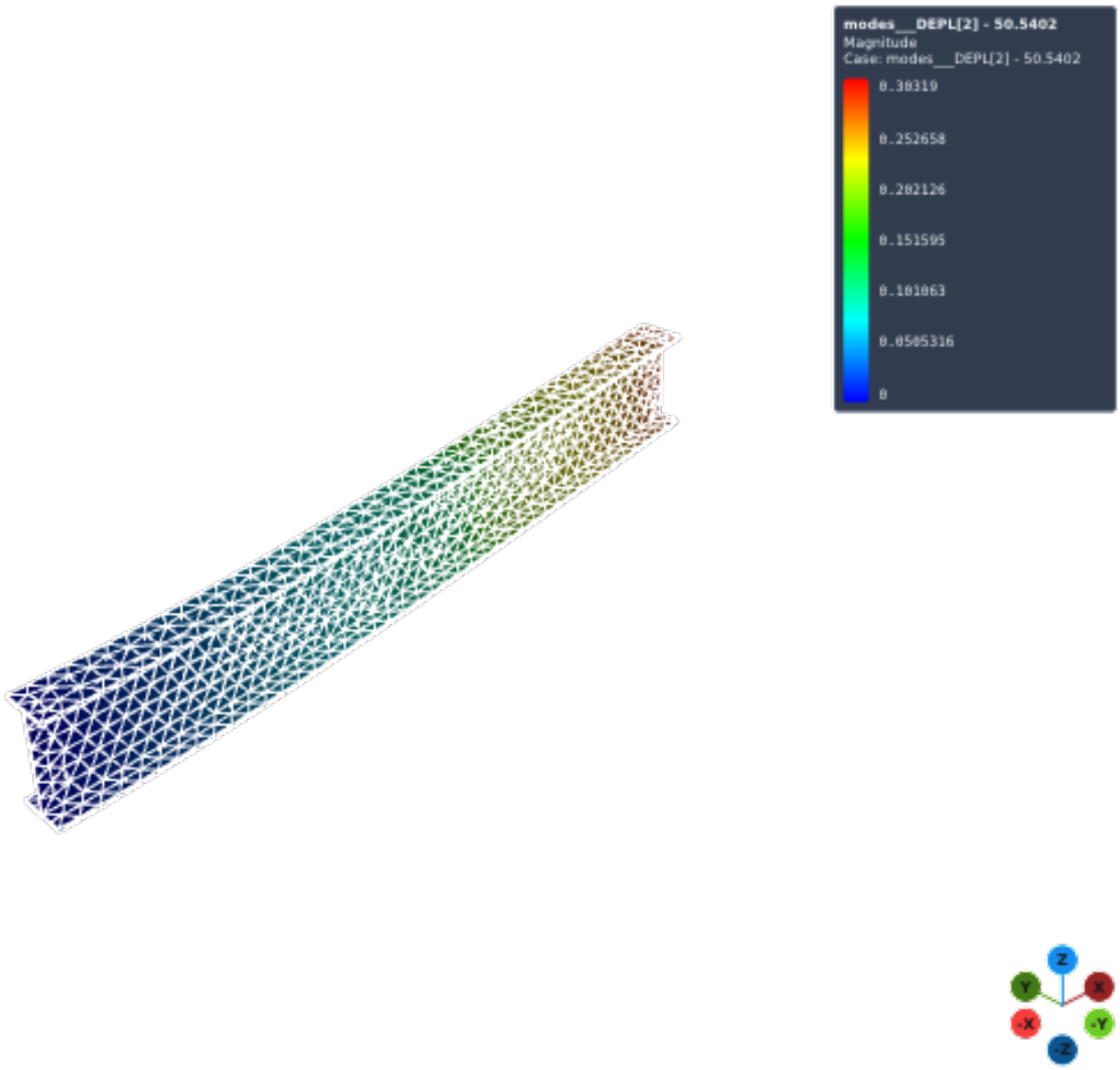

Figure 434: Code_Aster — mode 3, 50.540 Hz

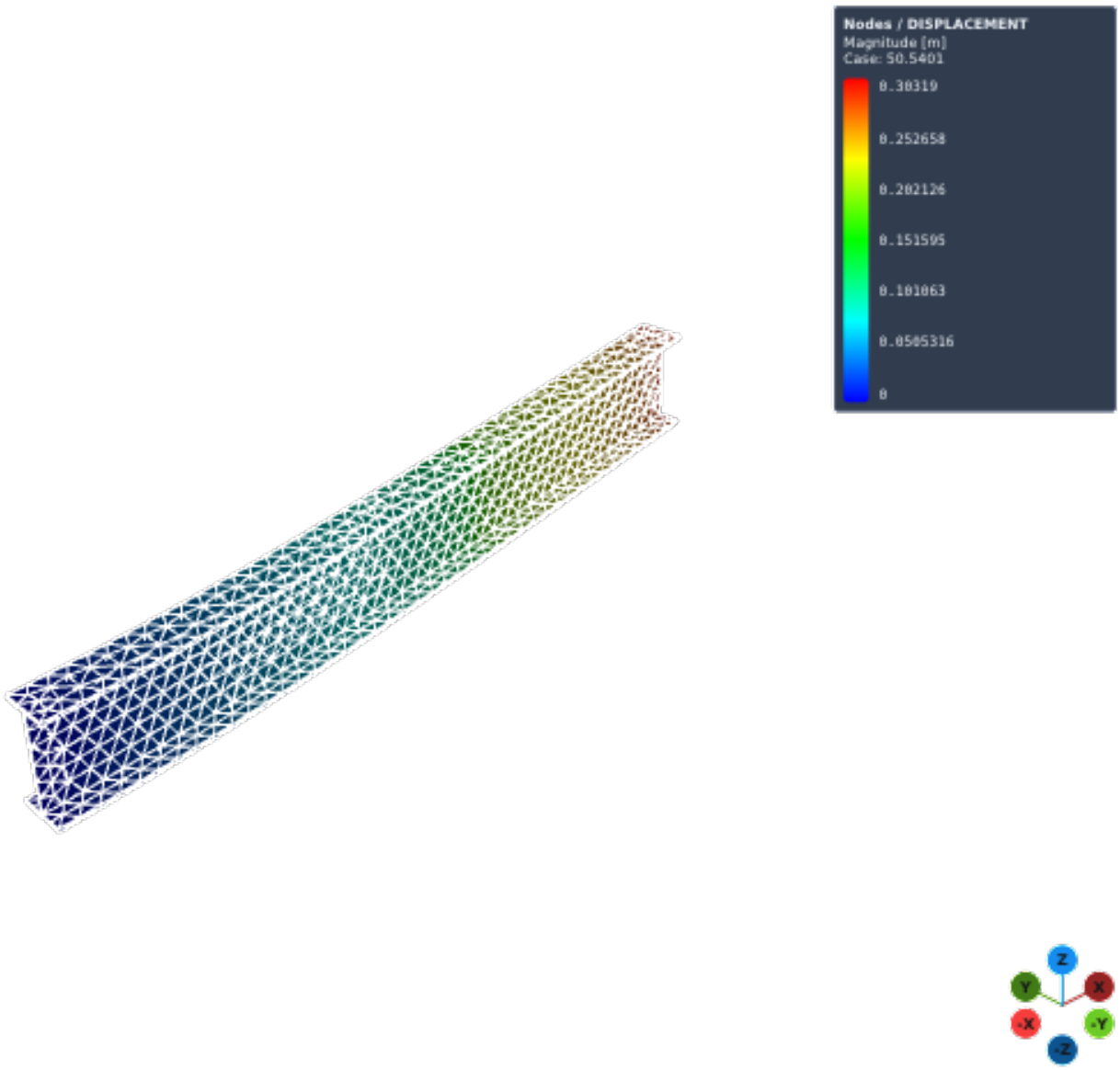

Figure 435: Sesam — mode 3, 50.540 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.540	-1.75e-04	100840	3.09e-10	1.03e-08	118.35	1.76e-05	-1.01e-04	10.879
Calculix	50.540	1.22e-04	100840	4.32e-10	6.11e-09	118.35	-2.08e-05	-7.82e-05	10.879
Code_Aster	50.540	1.29e-04	100840	4.42e-10	6.08e-09	118.35	-2.10e-05	-7.80e-05	10.879
Sesam	50.540	-7.59e-05	100840	3.10e-10	1.03e-08	118.35	1.76e-05	-1.01e-04	10.879

Mode 4

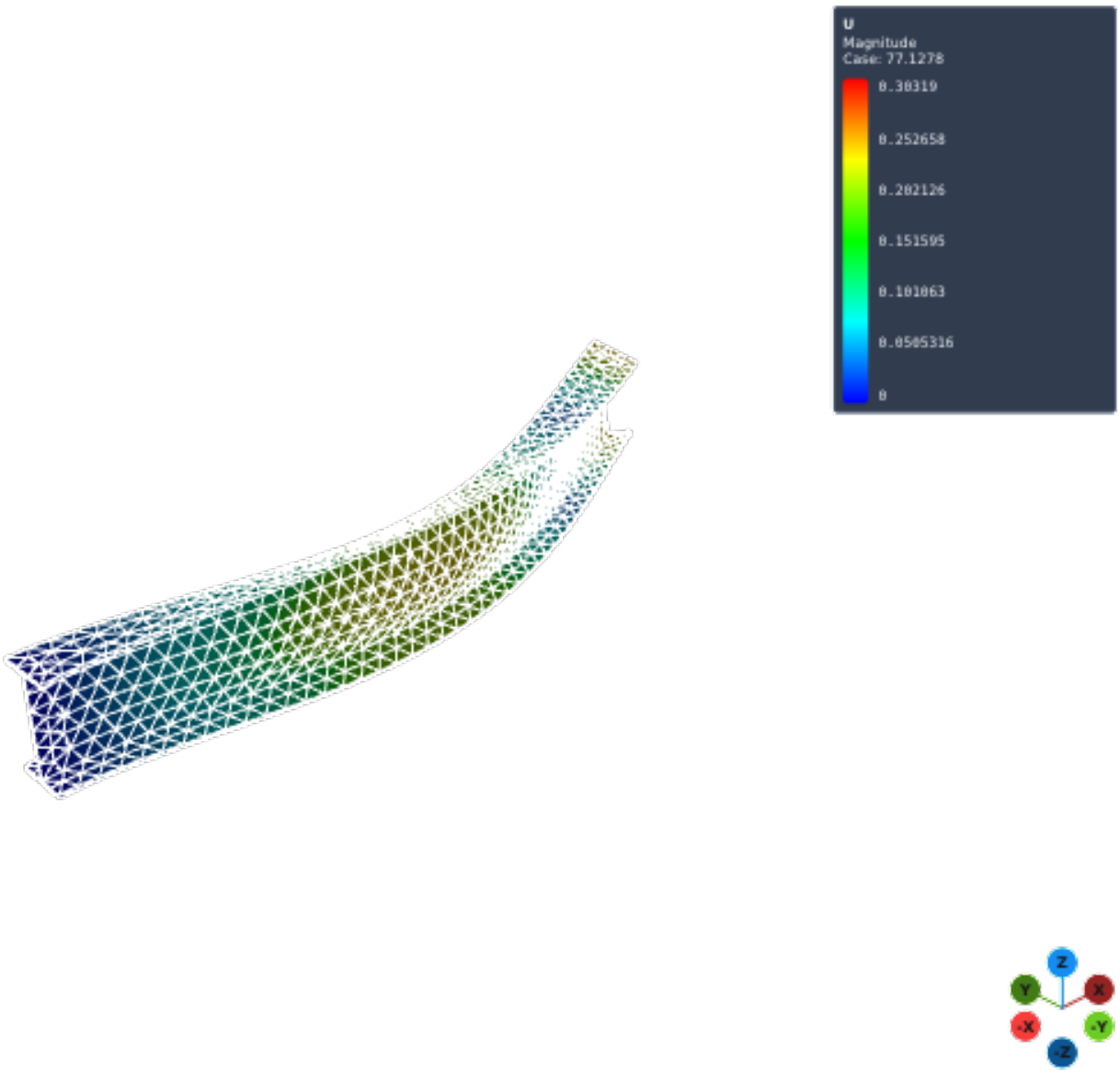

Figure 436: Abaqus — mode 4, 77.128 Hz

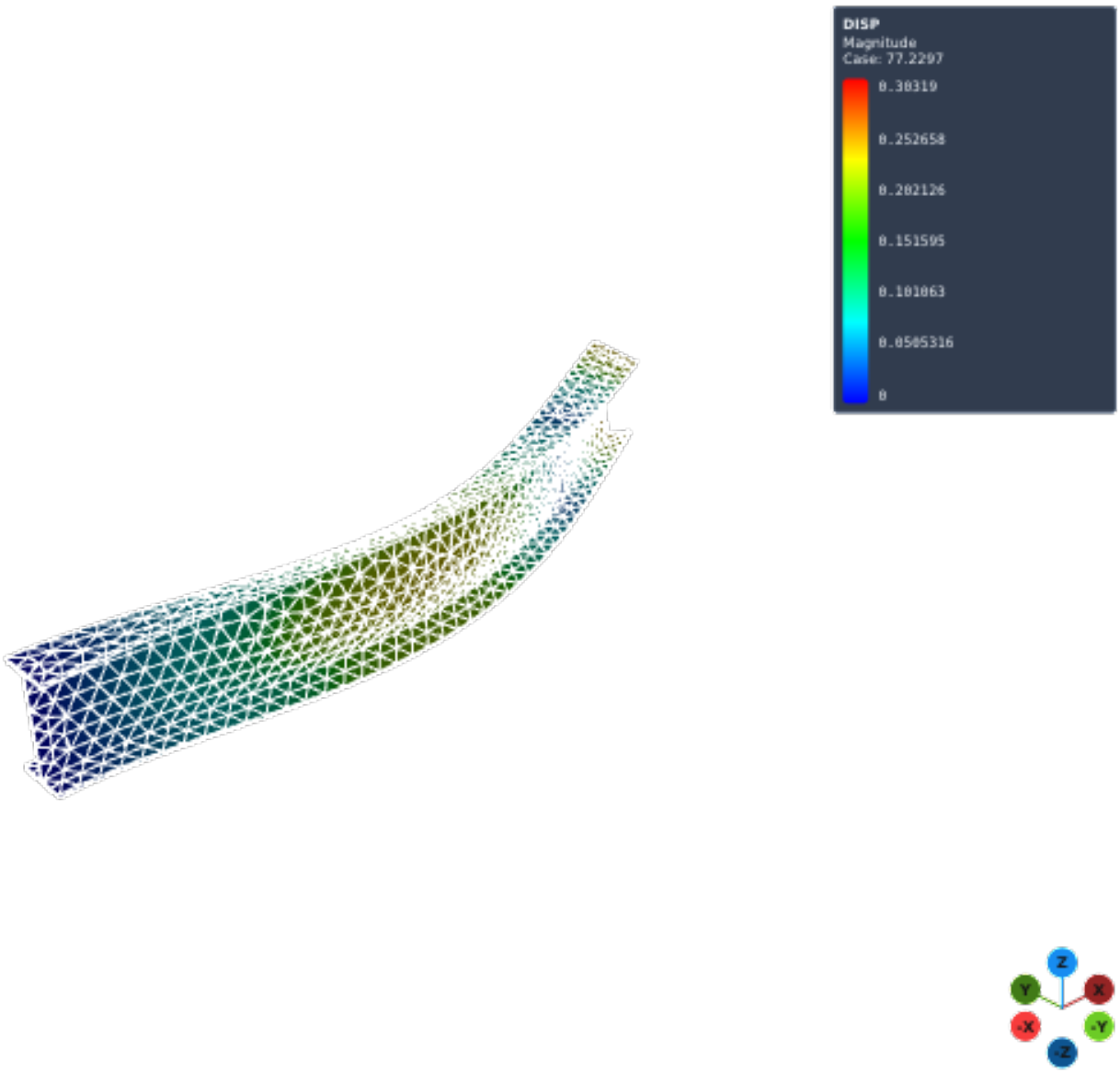

Figure 437: Calculix — mode 4, 77.230 Hz

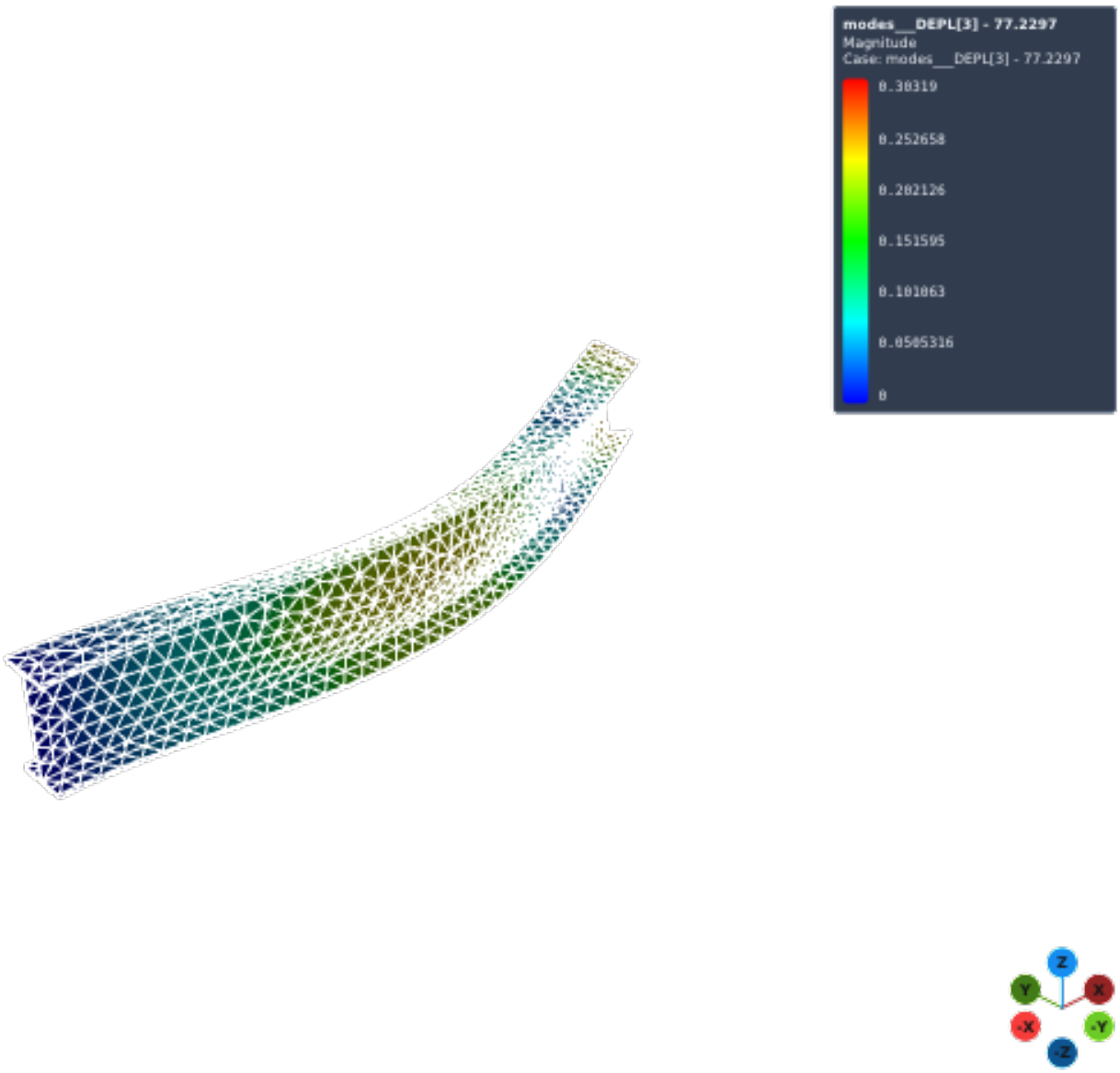

Figure 438: Code_Aster — mode 4, 77.230 Hz

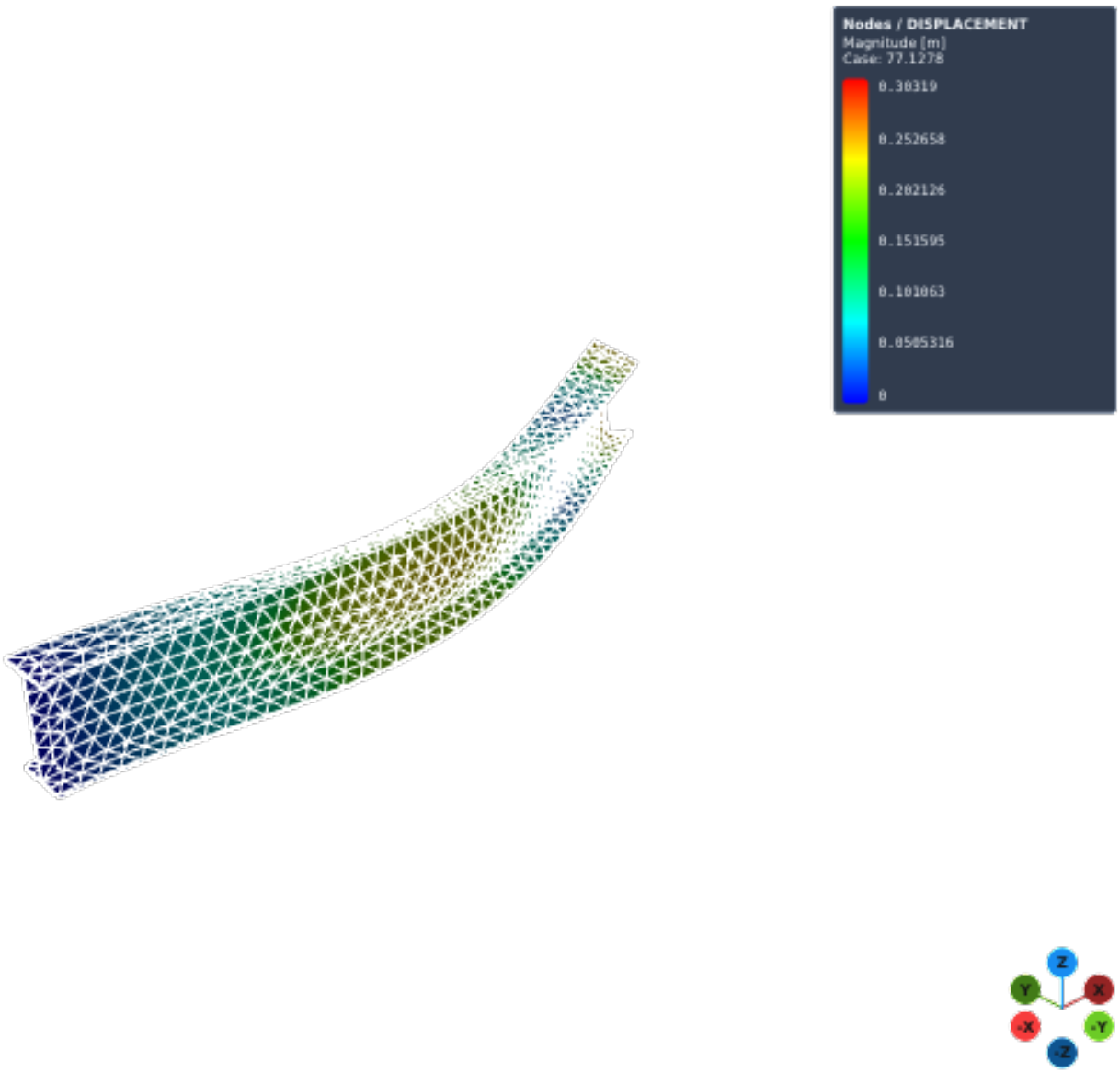

Figure 439: Sesam — mode 4, 77.128 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	77.128	-0.07	234845	6.40e-12	35.97	1.94e-09	2.53e-06	-5.997	-4.41e-05
Calculix	77.230	+0.07	235466	4.79e-11	36.04	7.89e-10	-6.92e-06	-6.004	2.81e-05
Code_Aster	77.230	+0.07	235466	4.90e-11	36.04	7.59e-10	-7.00e-06	-6.004	2.75e-05
Sesam	77.128	-0.07	234845	1.04e-11	35.97	1.89e-09	3.22e-06	-5.997	-4.35e-05

Mode 5

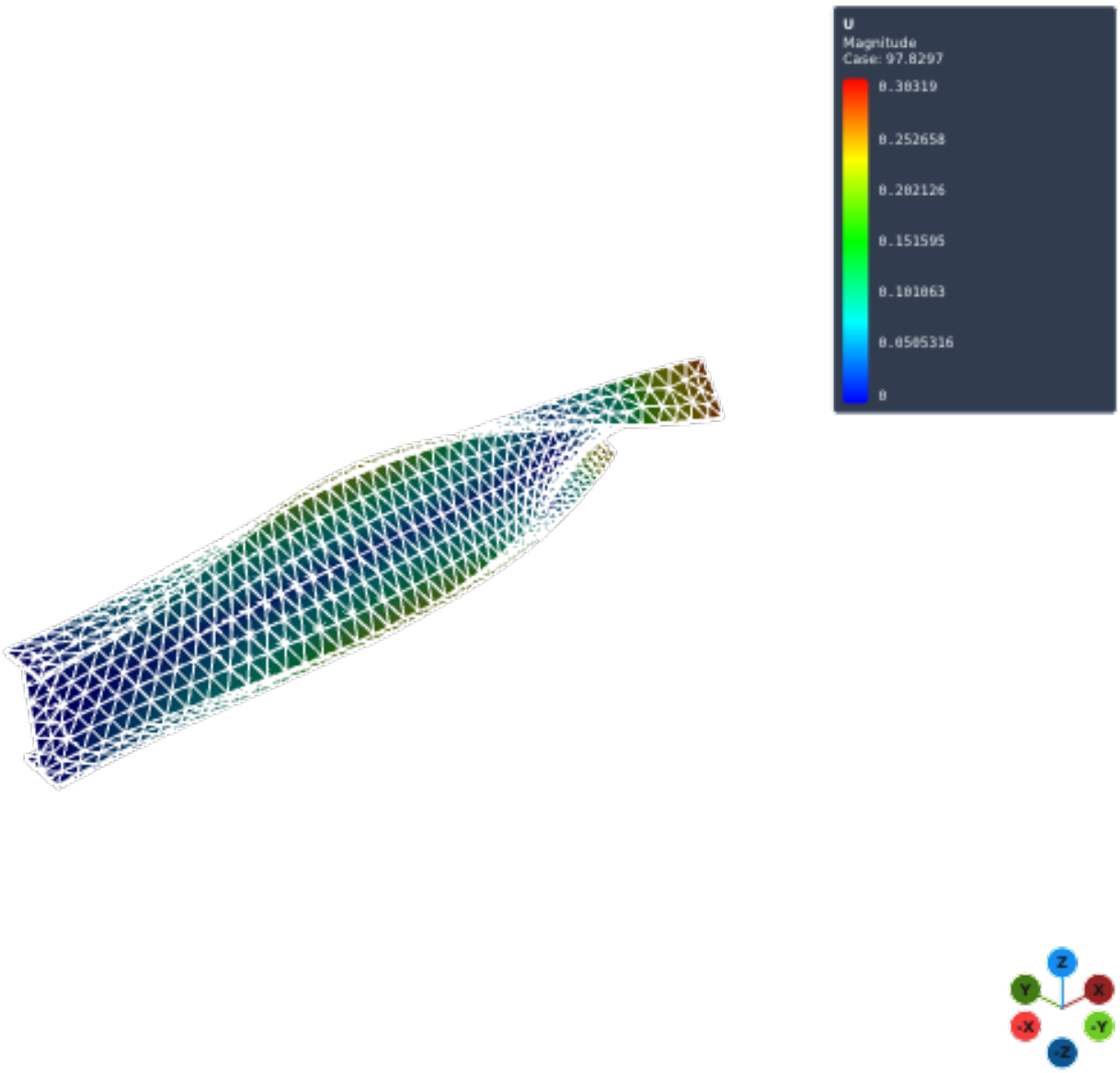

Figure 440: Abaqus — mode 5, 97.830 Hz

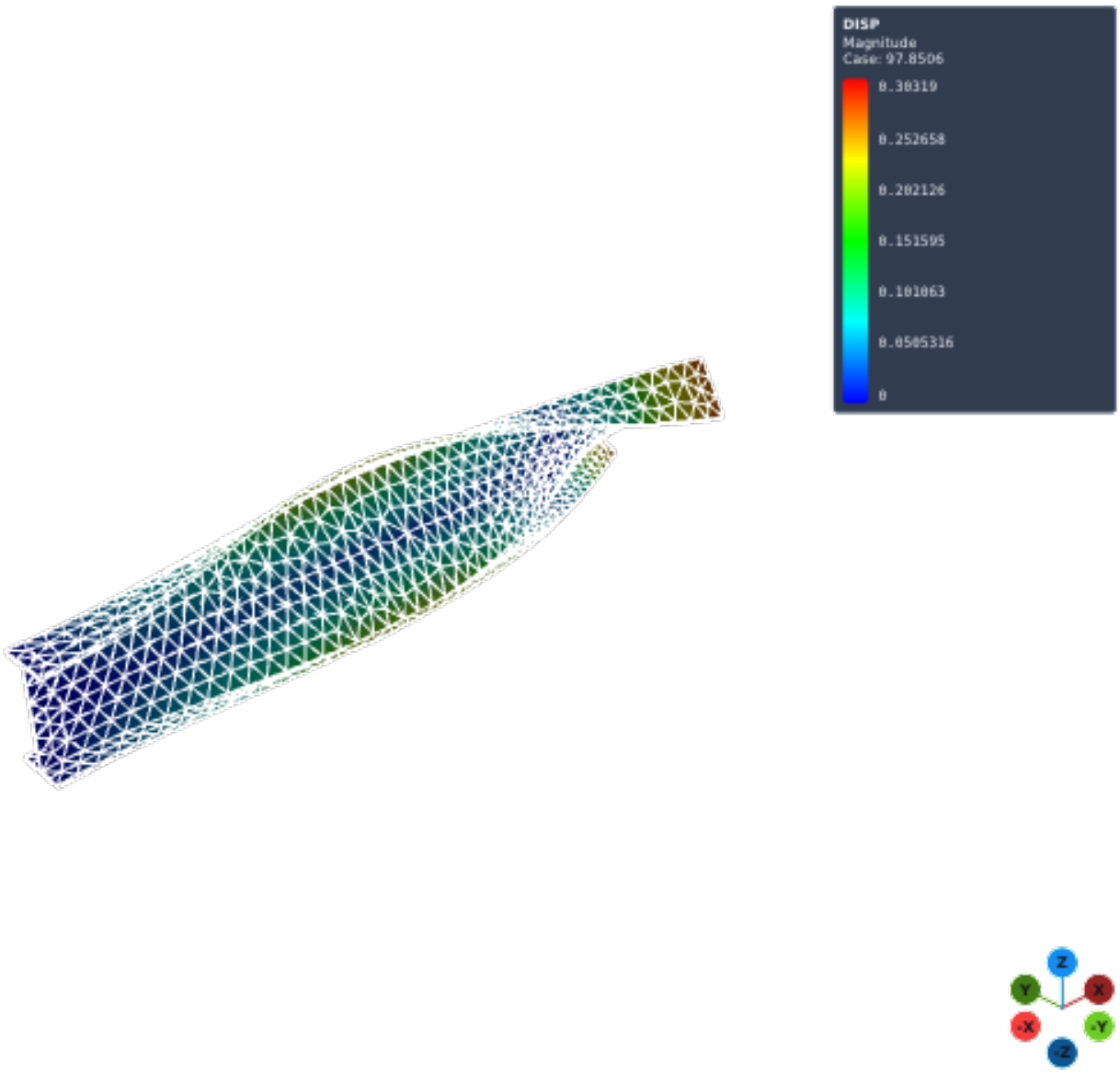

Figure 441: Calculix — mode 5, 97.851 Hz

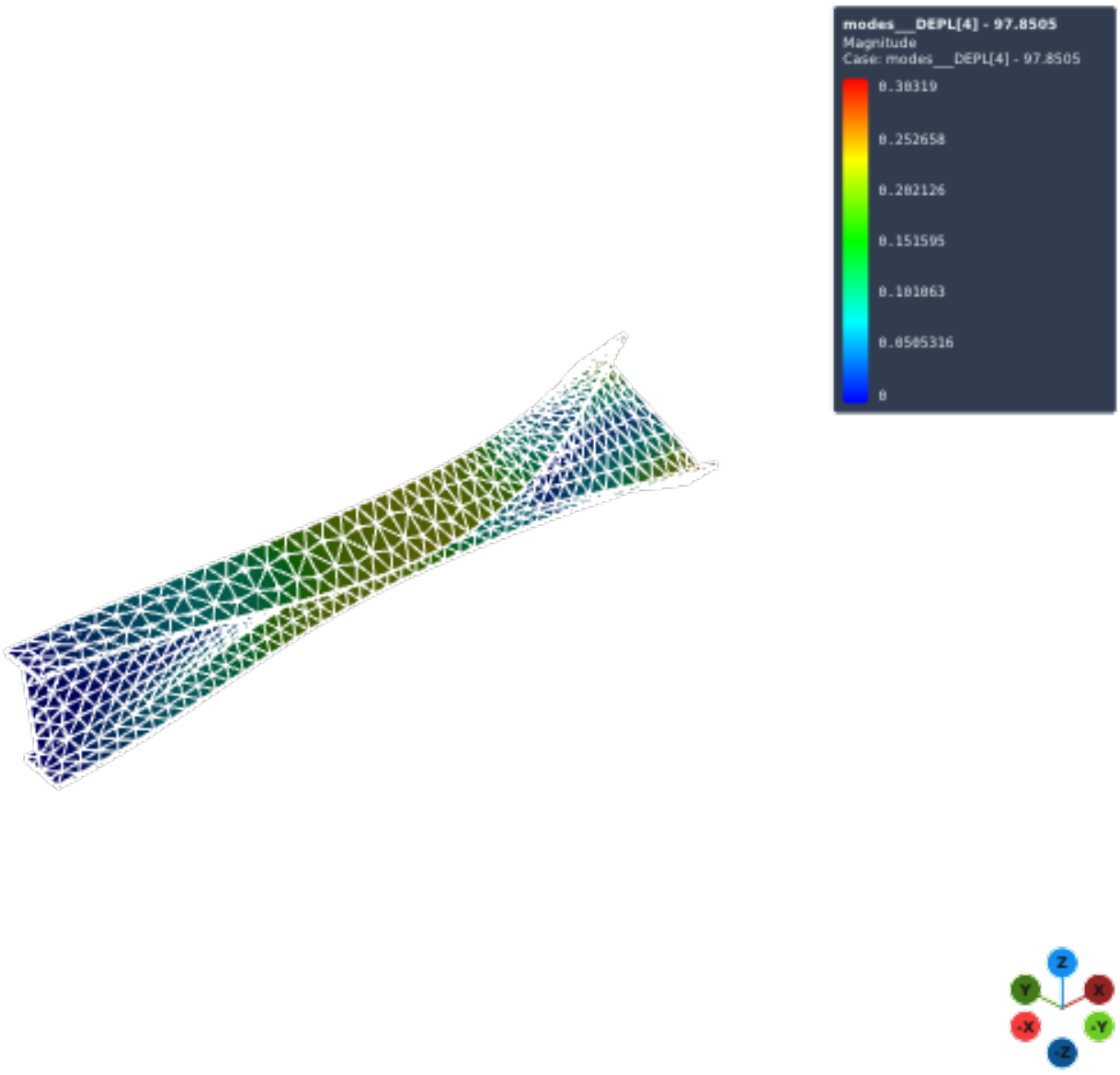

Figure 442: Code_Aster — mode 5, 97.851 Hz

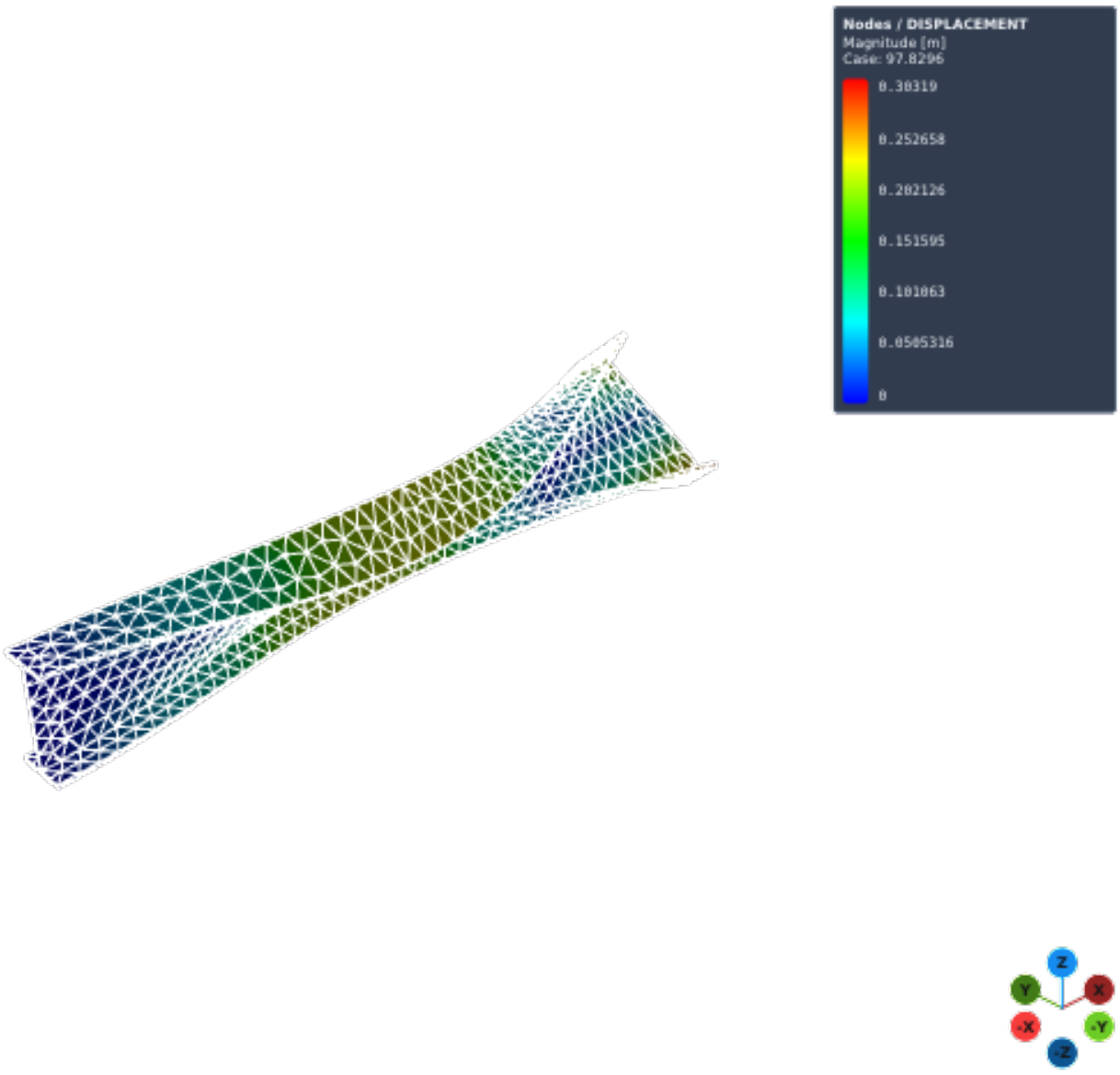

Figure 443: Sesam — mode 5, 97.830 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	97.830	-0.01	377834	1.93e-10	3.14e-07	6.49e-09	-1.39e-05	-0.001	8.06e-05
Calculix	97.851	+0.01	377996	7.91e-10	6.66e-07	5.29e-10	-2.81e-05	0.001	-2.30e-05
Code_Aster	97.851	+0.01	377995	8.39e-10	6.65e-07	5.24e-10	2.90e-05	-0.001	2.29e-05
Sesam	97.830	-0.01	377833	1.99e-10	3.14e-07	6.59e-09	1.41e-05	0.001	-8.12e-05

Mode 6

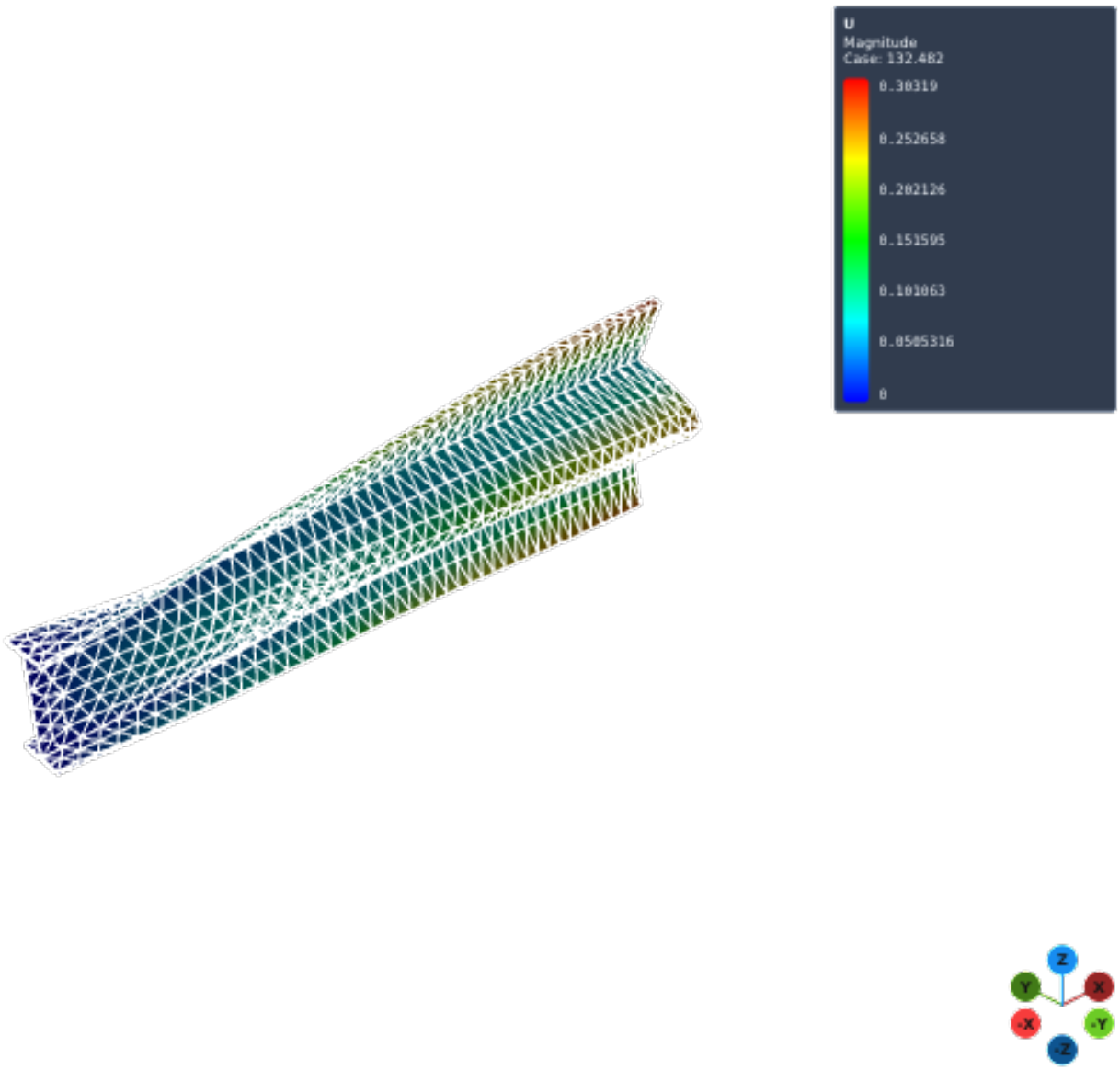

Figure 444: Abaqus — mode 6, 132.482 Hz

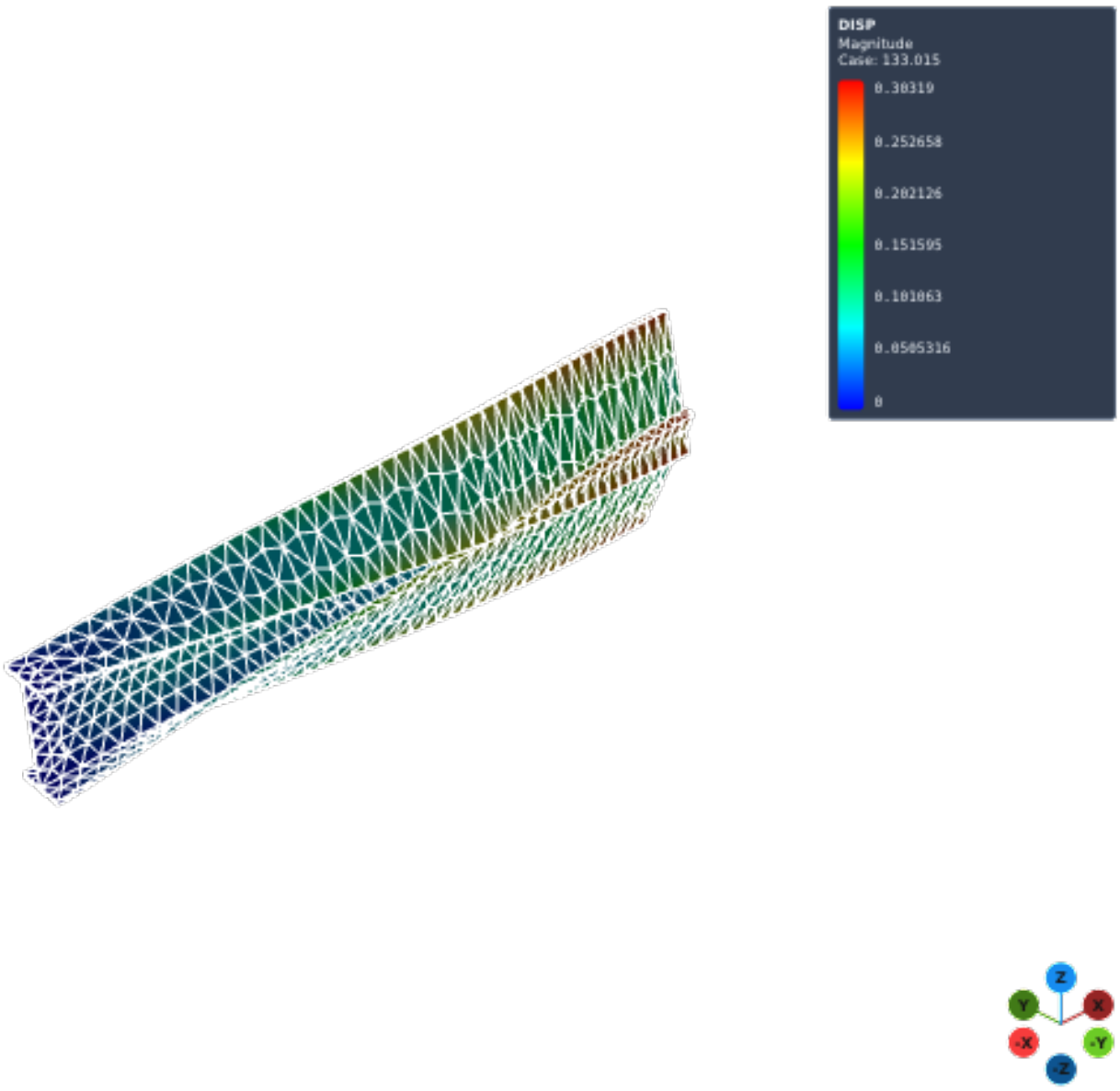

Figure 445: Calculix — mode 6, 133.015 Hz

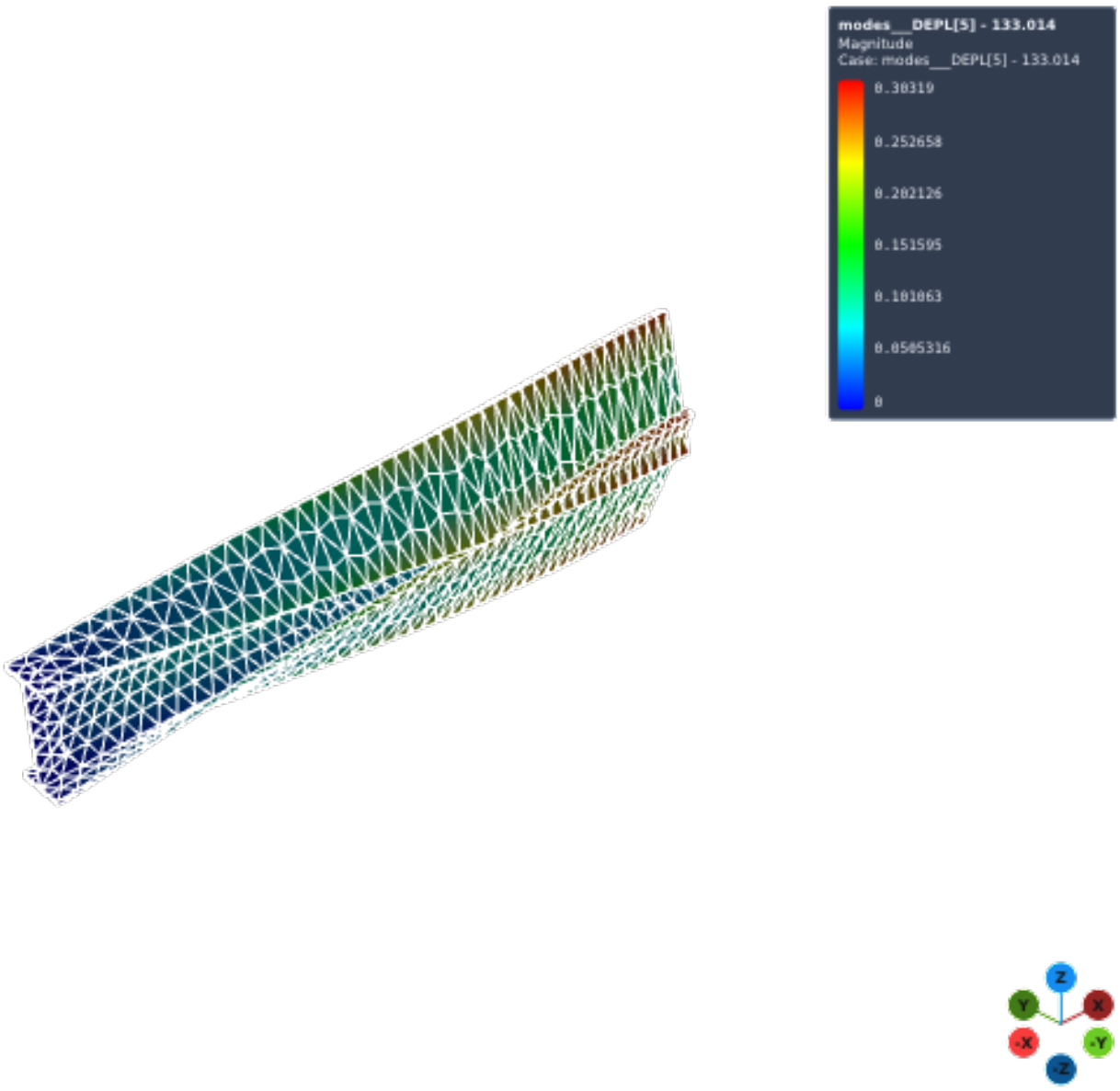

Figure 446: Code_Aster — mode 6, 133.014 Hz

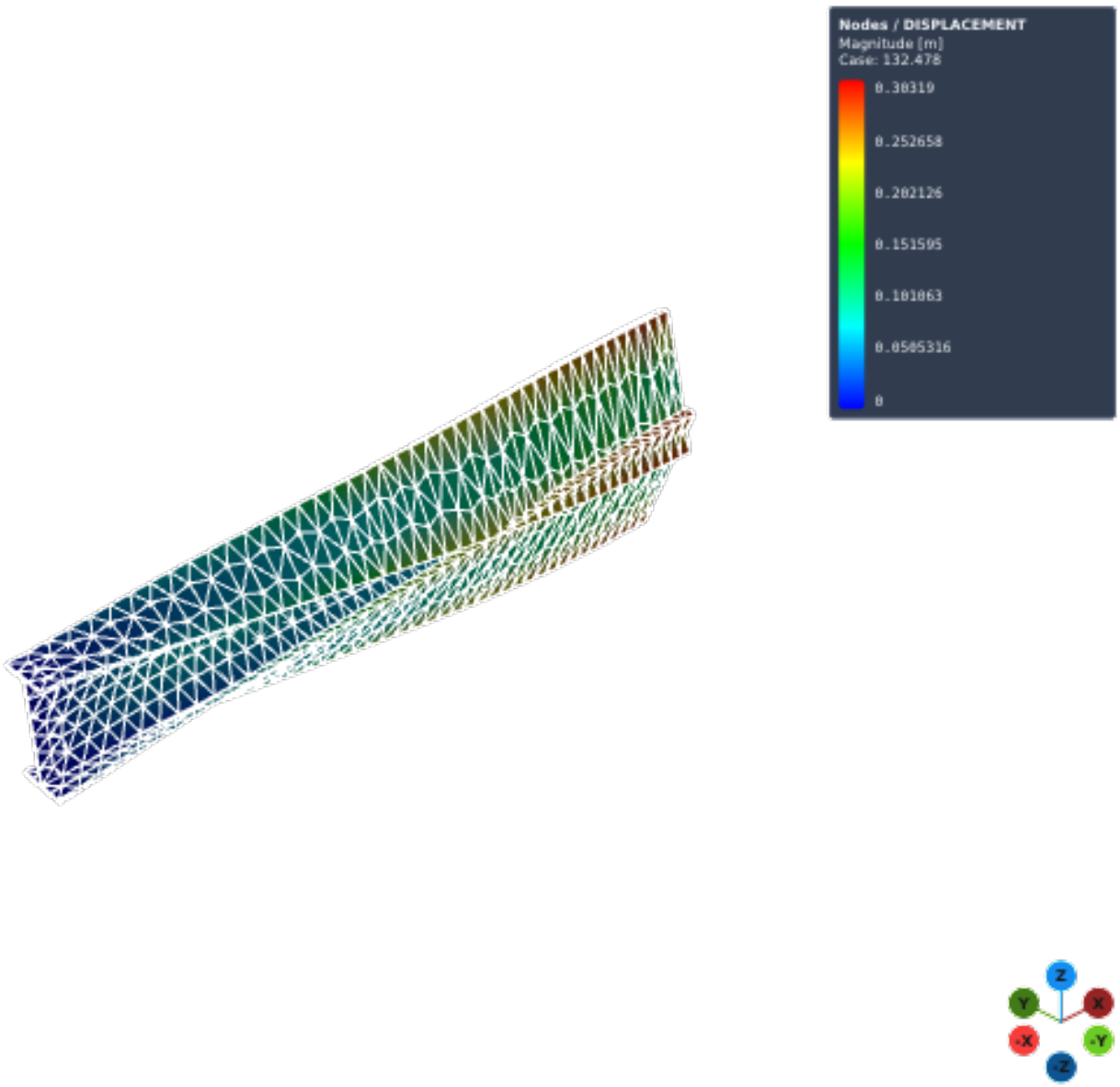

Figure 447: Sesam — mode 6, 132.478 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	132.480	-0.20	692901	1.05e-09	0.01	2.04e-09	3.23e-05	0.119	-4.51e-05
Calculix	133.015	+0.20	698496	3.94e-10	0.01	2.21e-09	1.98e-05	-0.093	4.70e-05
Code_Aster	133.014	+0.20	698479	4.00e-10	0.01	2.27e-09	2.00e-05	-0.093	4.76e-05
Sesam	132.478	-0.20	692867	1.14e-09	0.01	2.04e-09	-3.38e-05	-0.119	4.52e-05

Mode 7

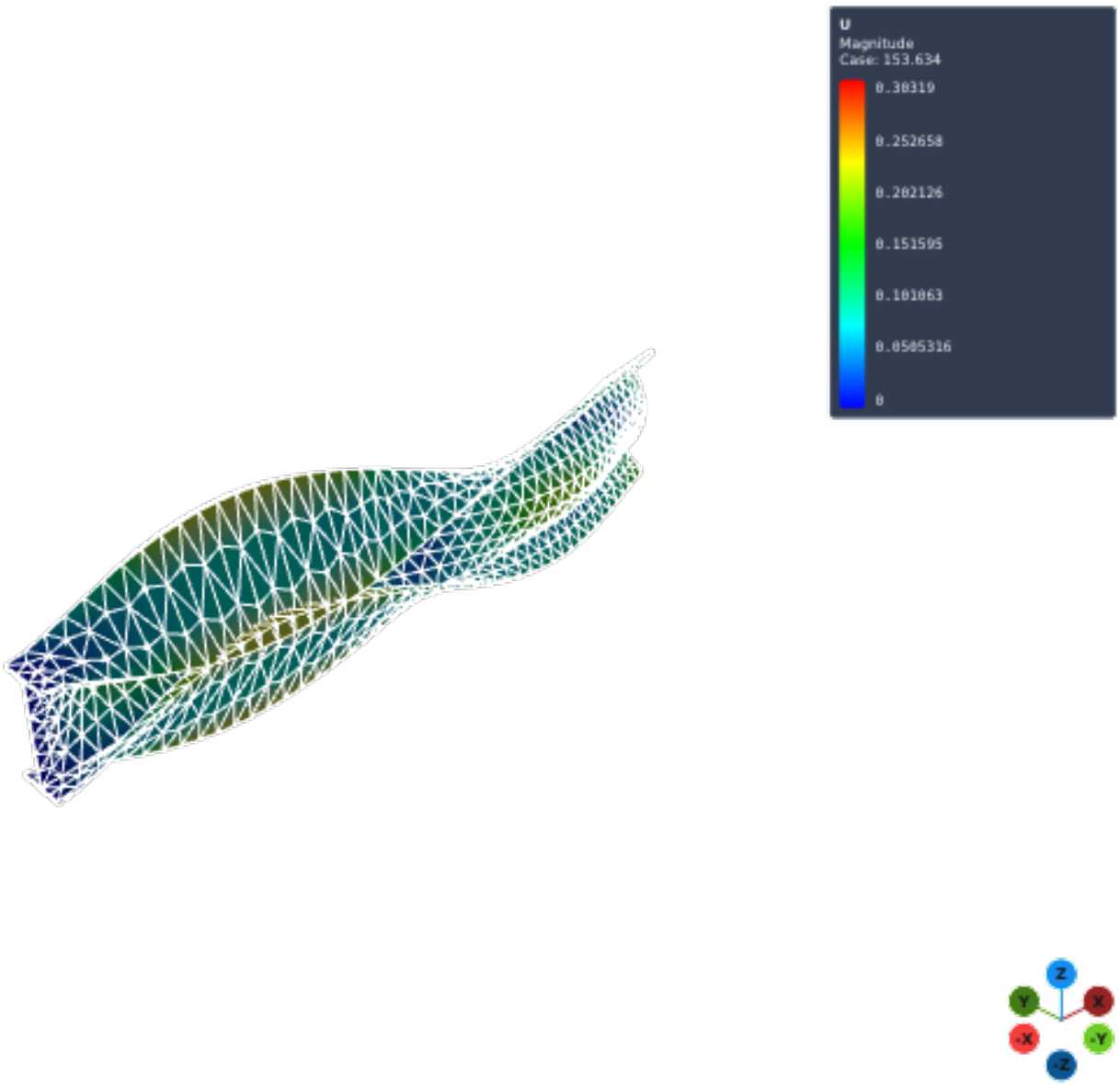

Figure 448: Abaqus — mode 7, 153.634 Hz

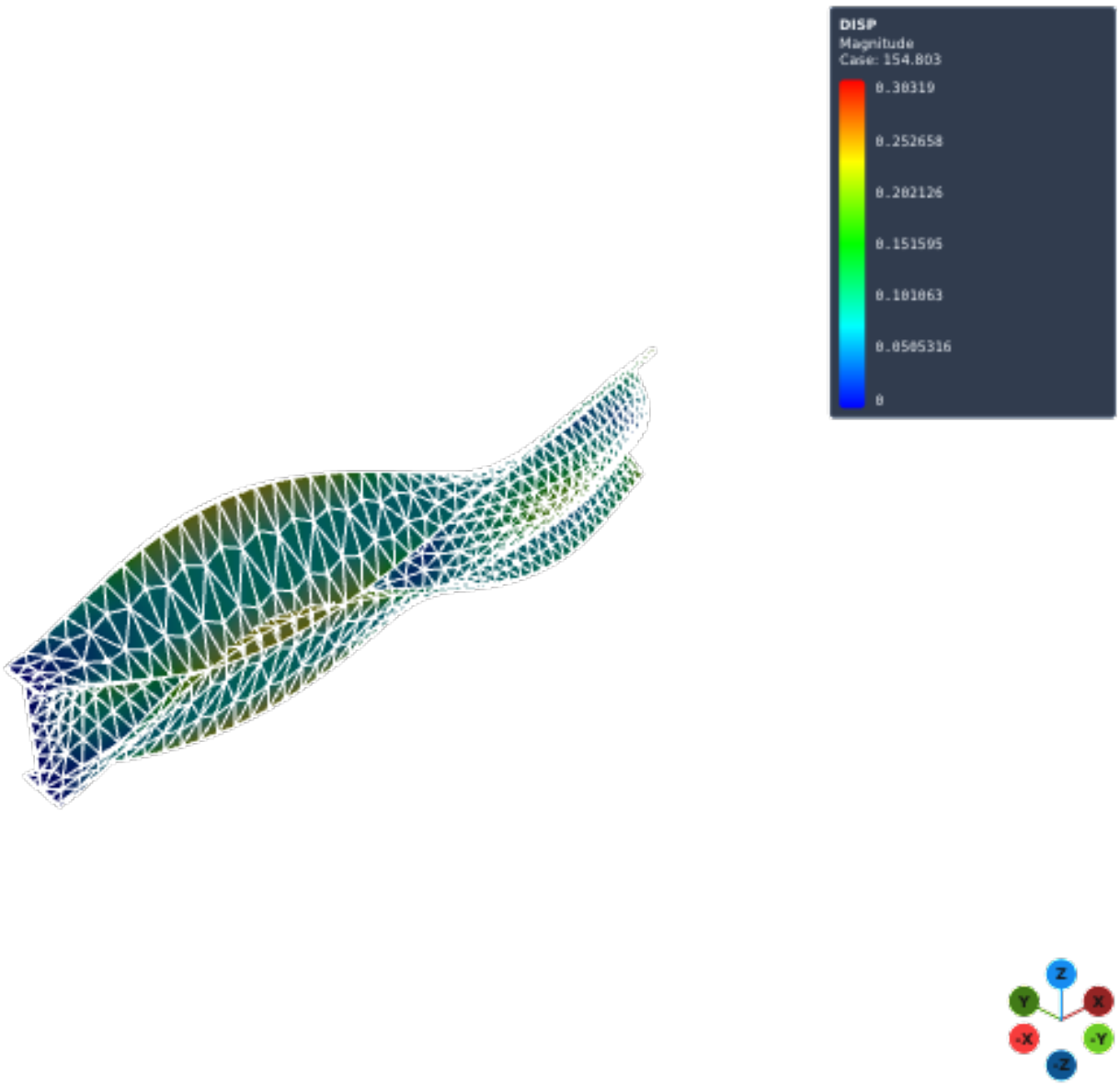

Figure 449: Calculix — mode 7, 154.803 Hz

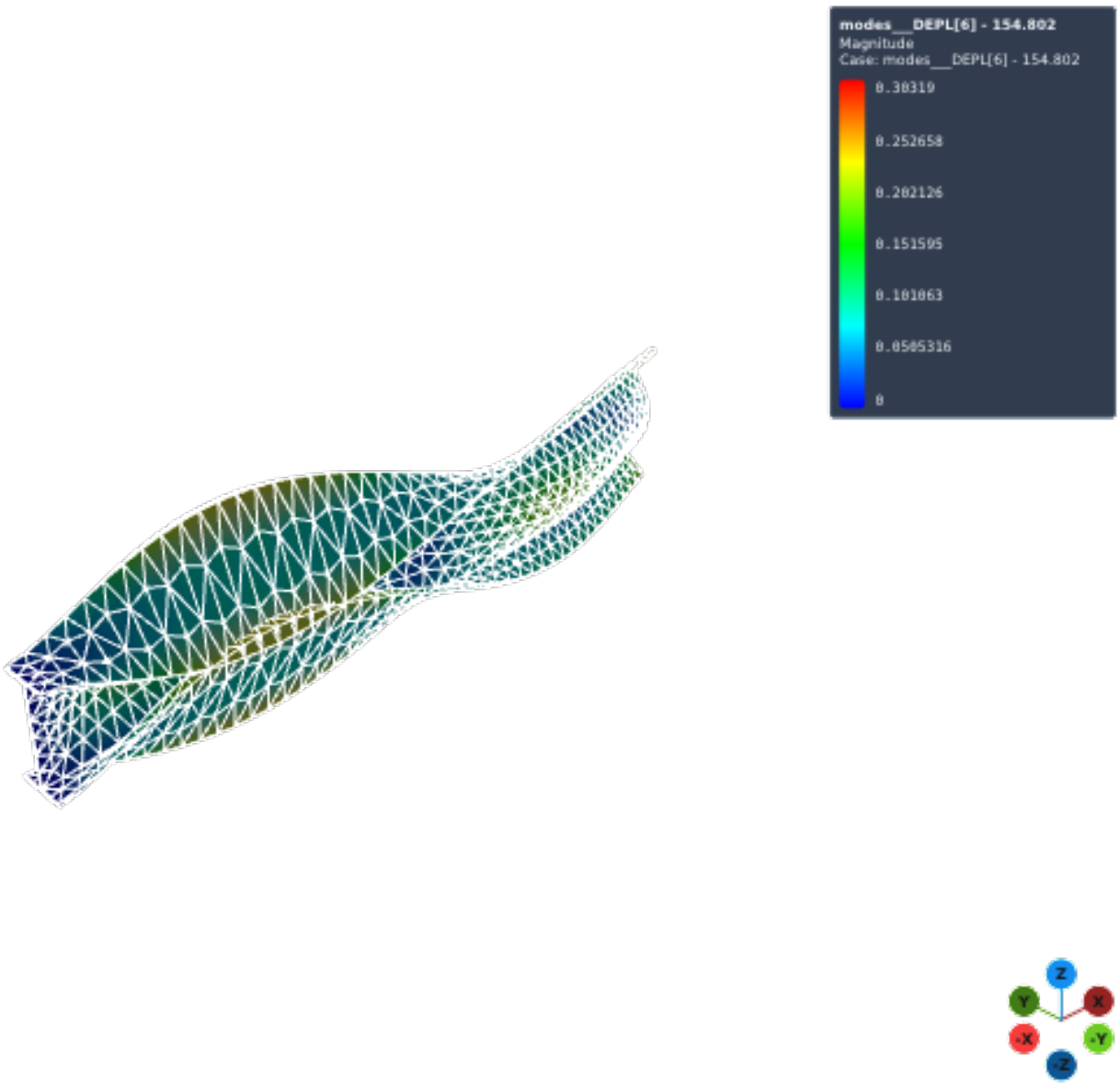

Figure 450: Code_Aster — mode 7, 154.802 Hz

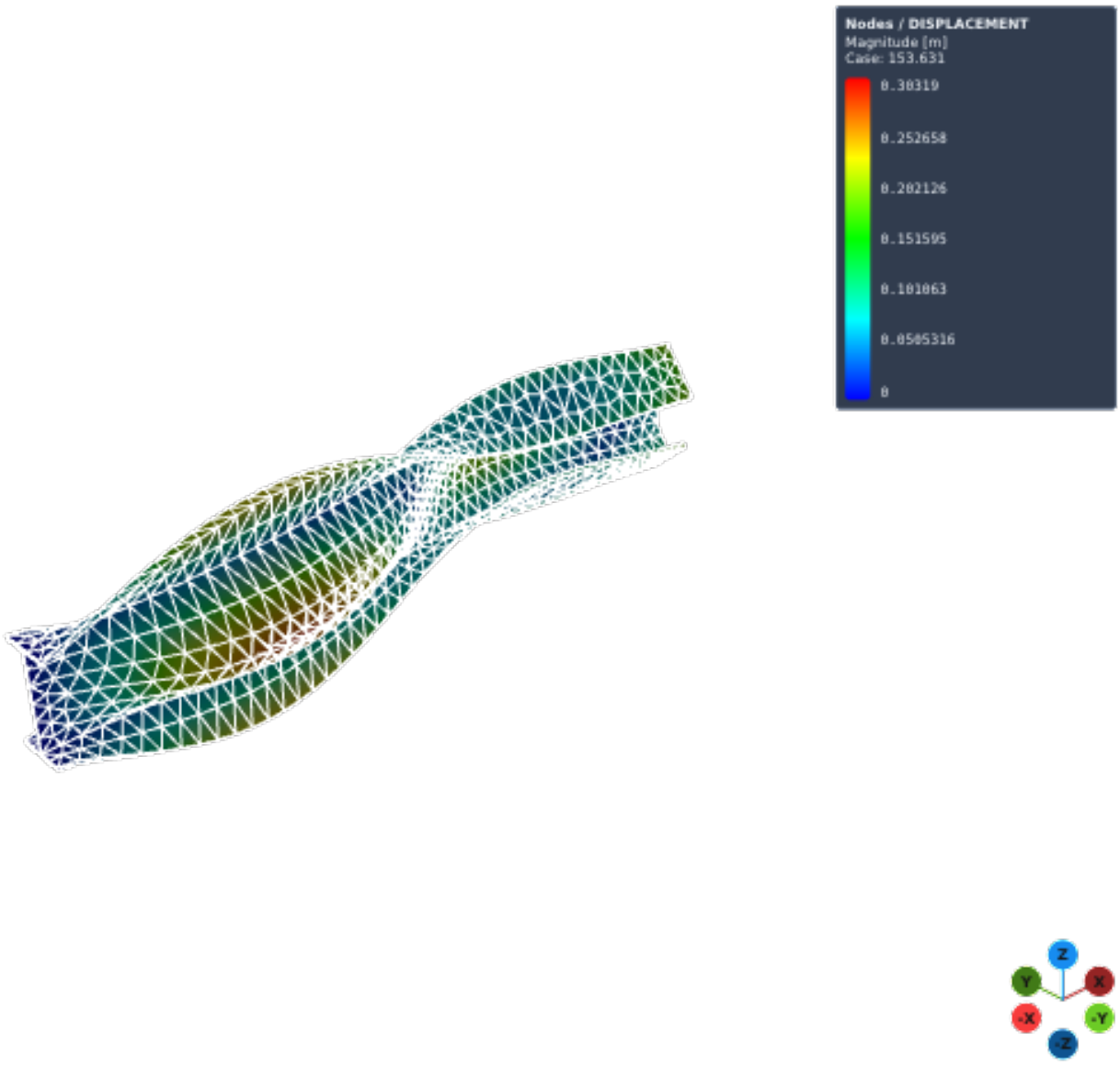

Figure 451: Sesam — mode 7, 153.631 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	153.630	-0.38	931828	1.43e-08	3.12	2.11e-09	-1.19e-04	1.768	-4.59e-05
Calculix	154.804	+0.38	946065	6.91e-09	3.24	7.22e-08	8.31e-05	1.799	2.69e-04
Code_Aster	154.802	+0.38	946046	7.01e-09	3.23	7.26e-08	8.37e-05	1.799	2.69e-04
Sesam	153.631	-0.38	931794	1.56e-08	3.12	2.15e-09	1.25e-04	-1.768	4.63e-05

Mode 8

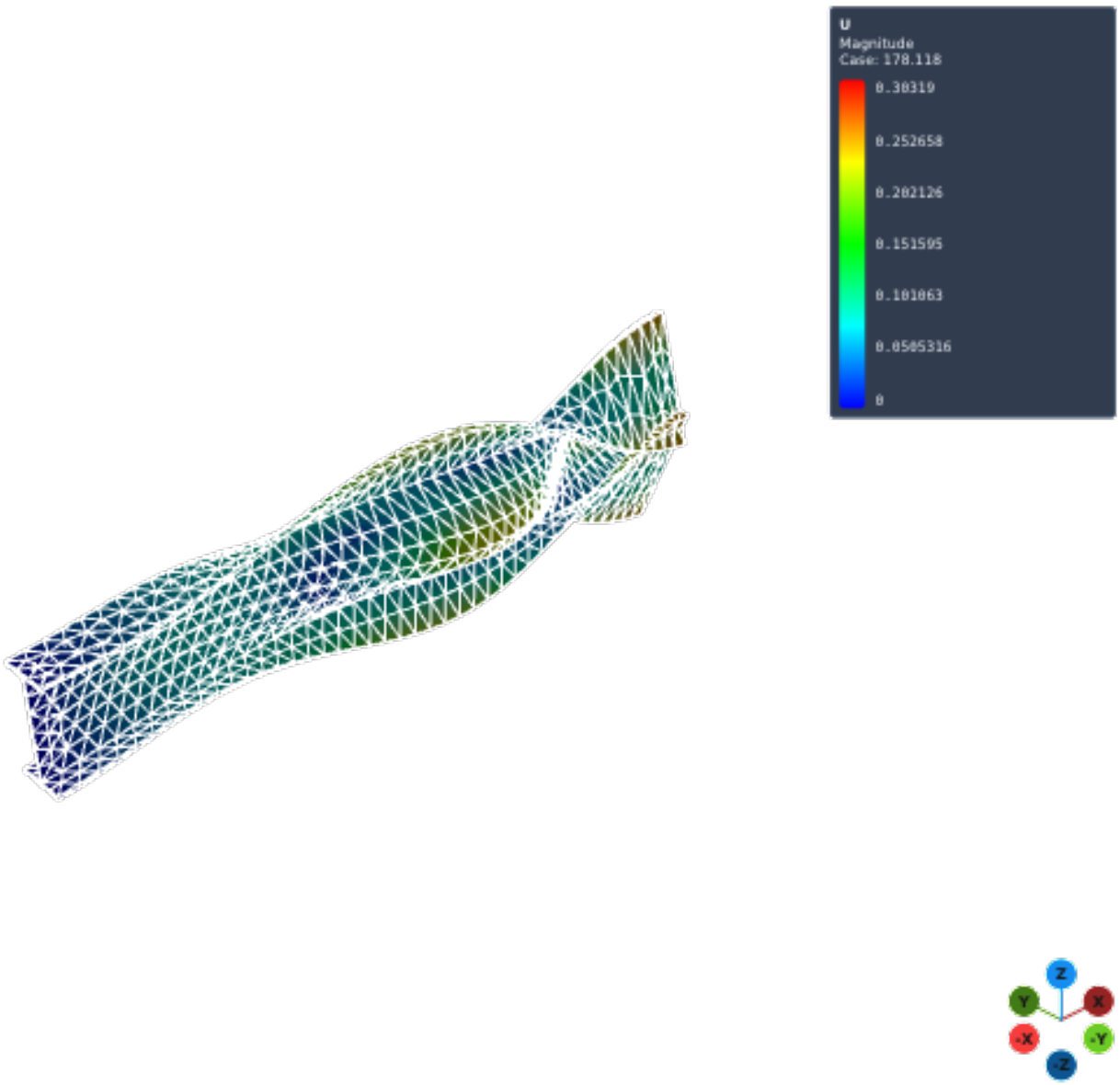

Figure 452: Abaqus — mode 8, 178.118 Hz

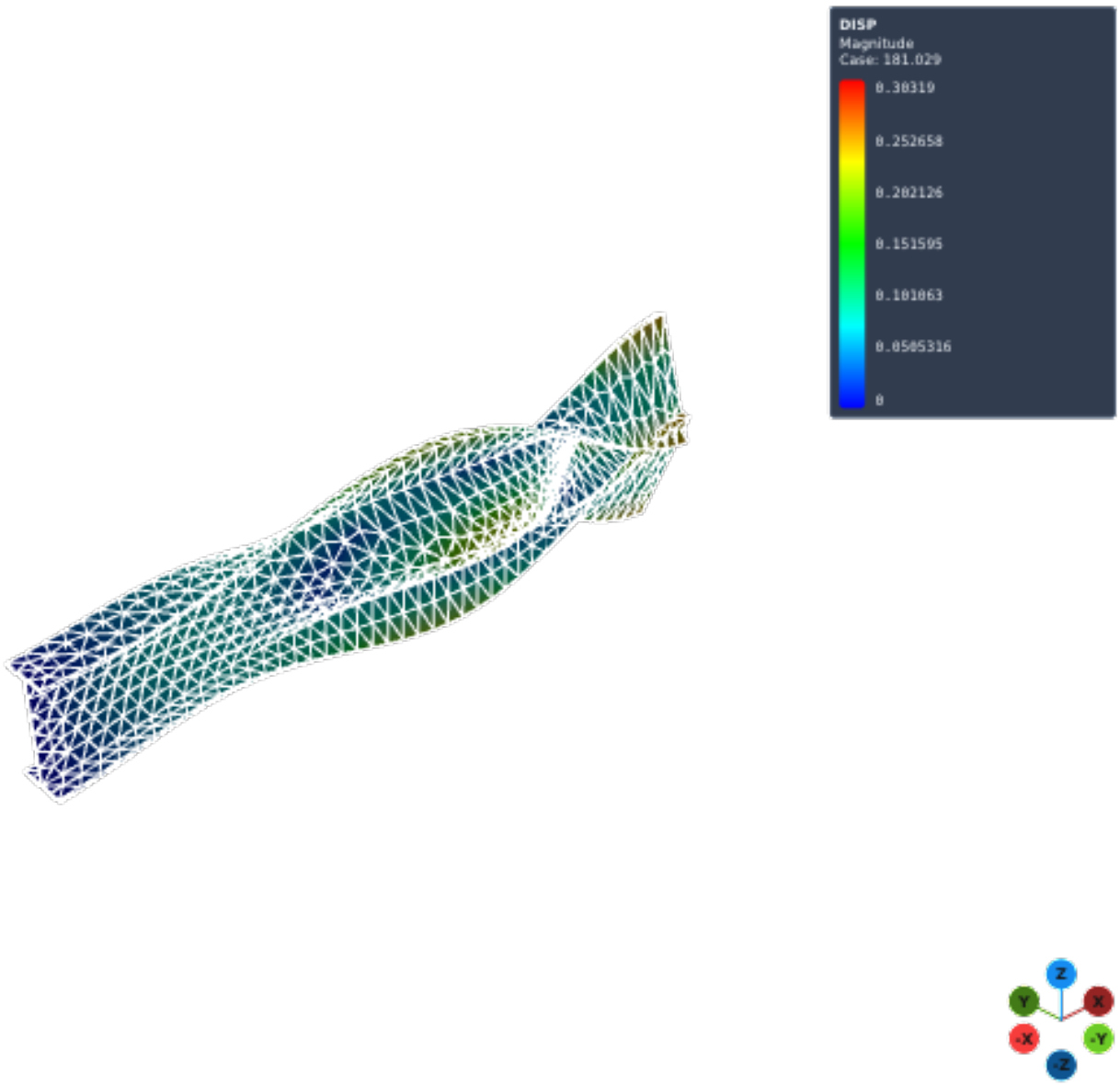

Figure 453: Calculix — mode 8, 181.029 Hz

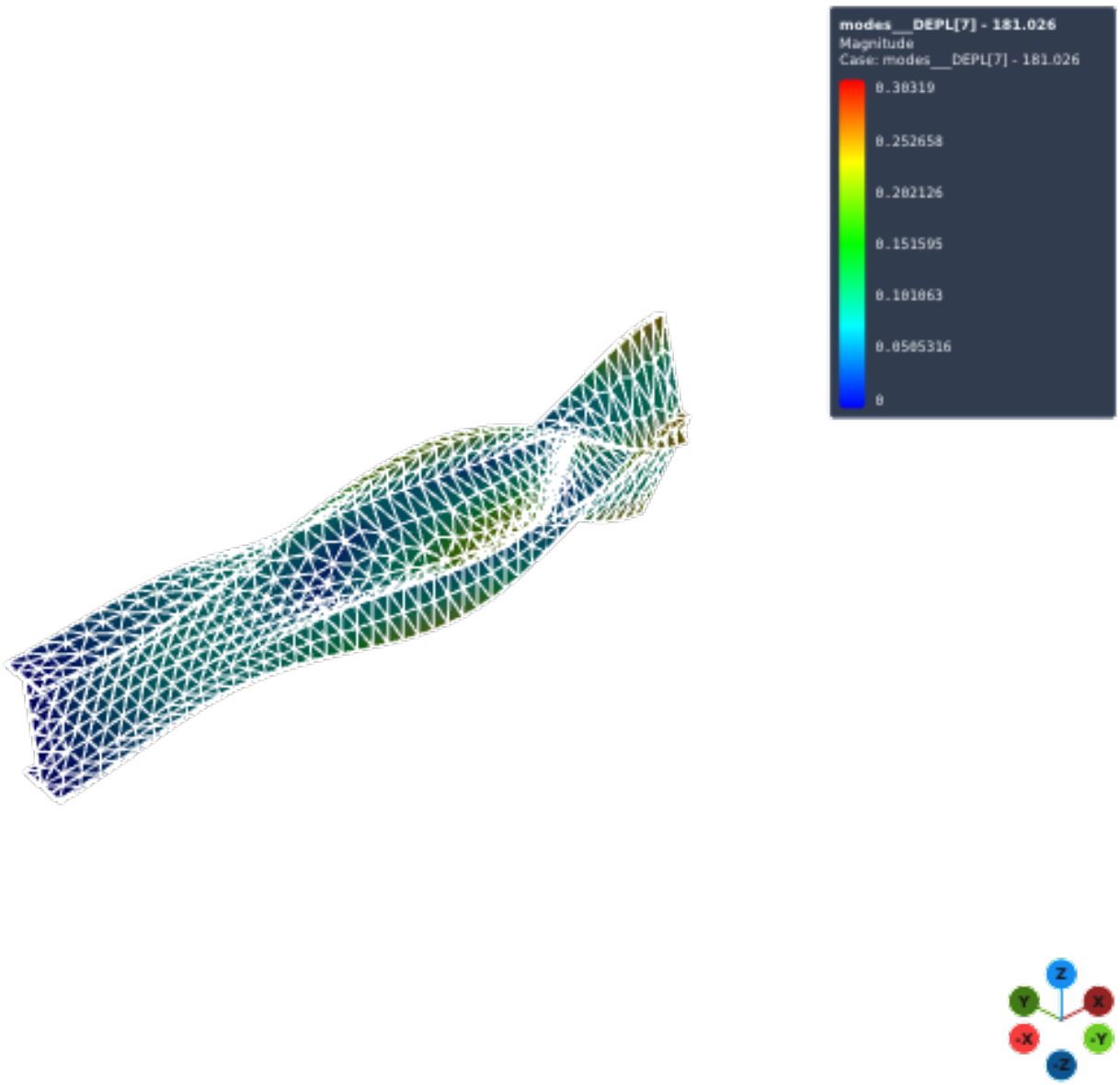

Figure 454: Code_Aster — mode 8, 181.026 Hz

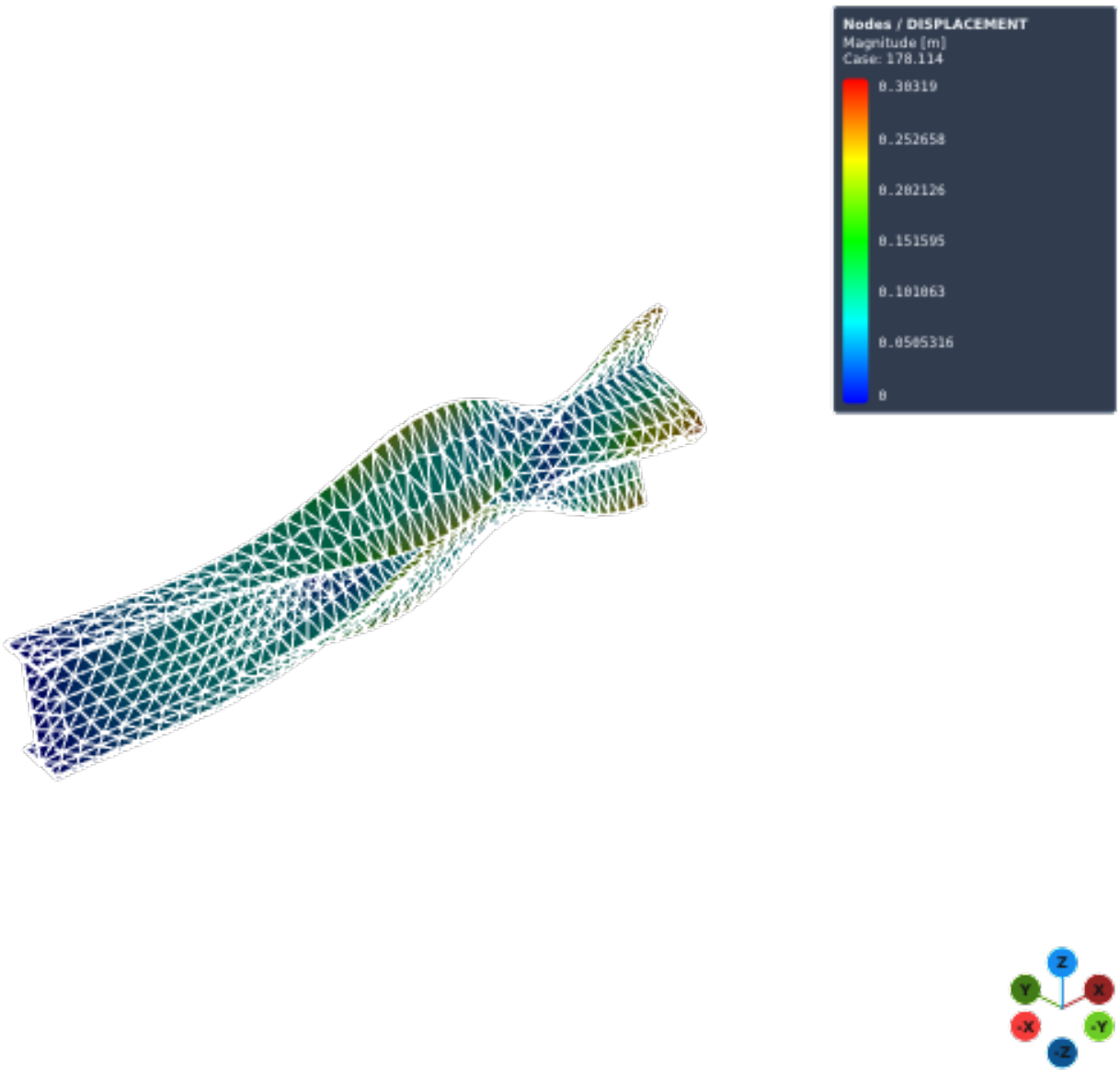

Figure 455: Sesam — mode 8, 178.114 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	178.120	-0.81	1.2525e+06	2.65e-10	5.79	5.99e-10	-1.63e-05	2.405	-2.45e-05
Calculix	181.029	+0.81	1.29376e+06	6.86e-10	5.93	6.55e-09	2.62e-05	2.436	-8.09e-05
Code_Aster	181.026	+0.81	1.29373e+06	6.94e-10	5.93	6.39e-09	2.63e-05	2.435	-7.99e-05
Sesam	178.114	-0.81	1.25243e+06	3.03e-10	5.79	6.40e-10	1.74e-05	-2.406	2.53e-05

Mode 9

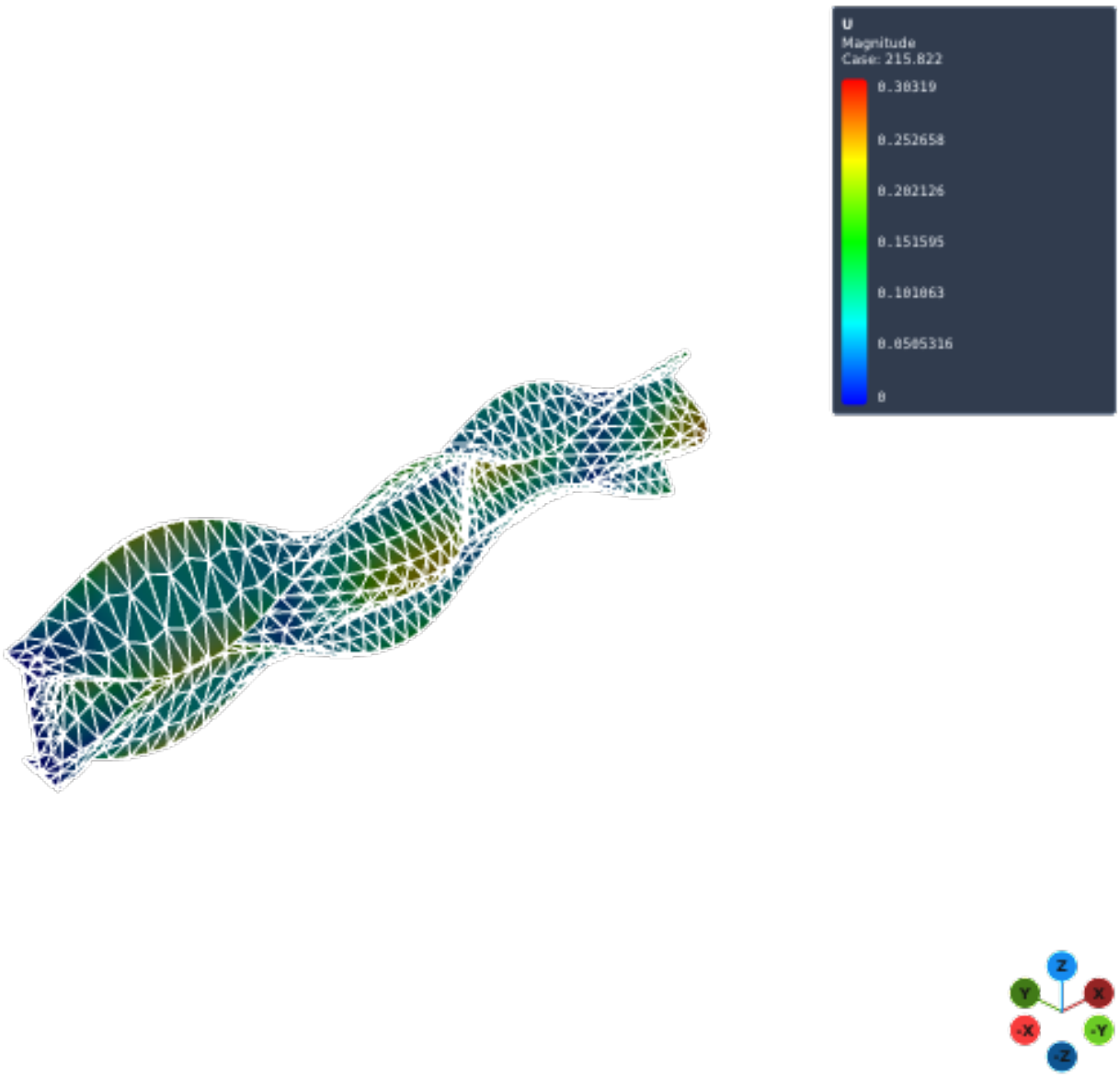

Figure 456: Abaqus — mode 9, 215.822 Hz

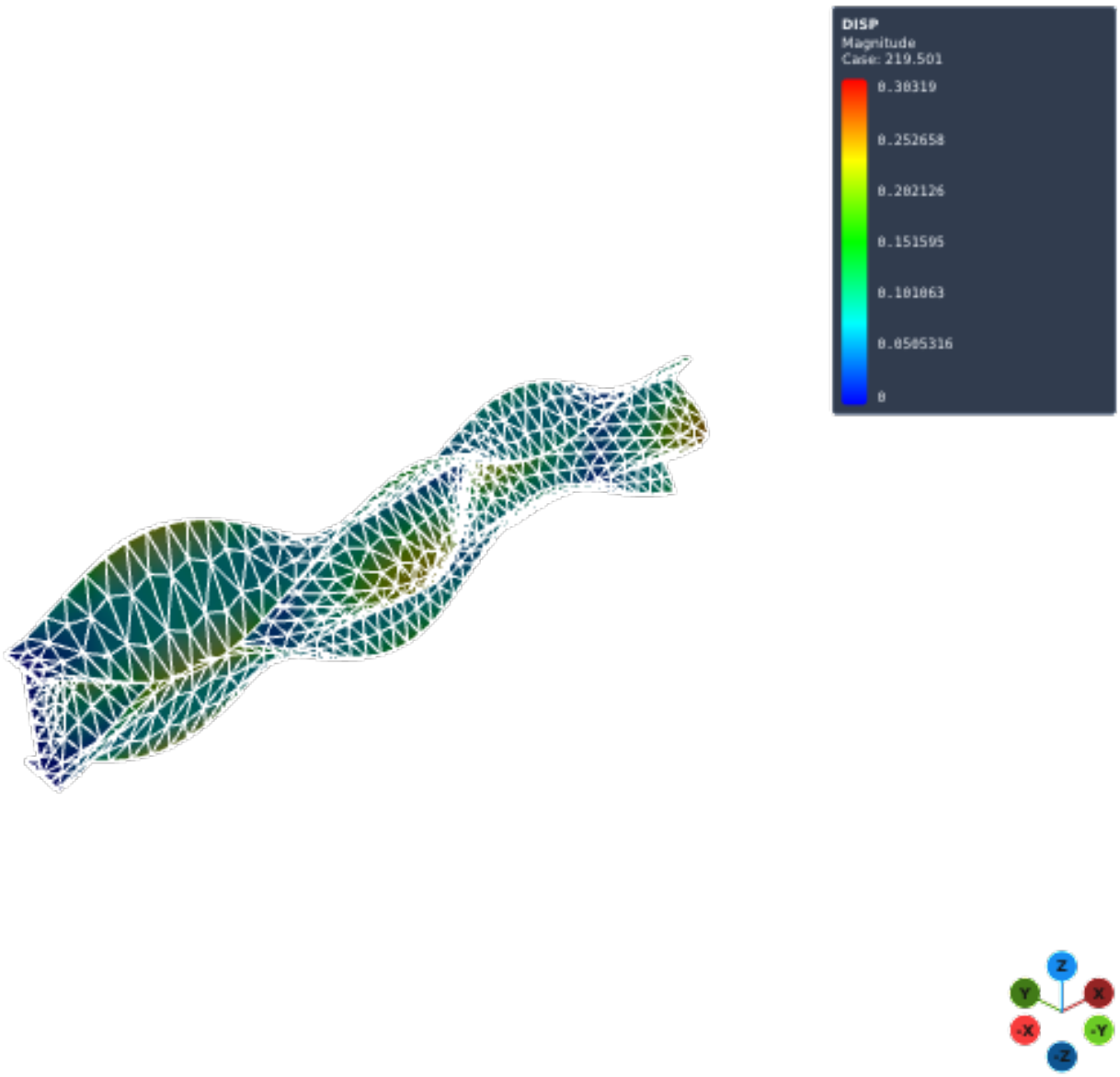

Figure 457: Calculix — mode 9, 219.501 Hz

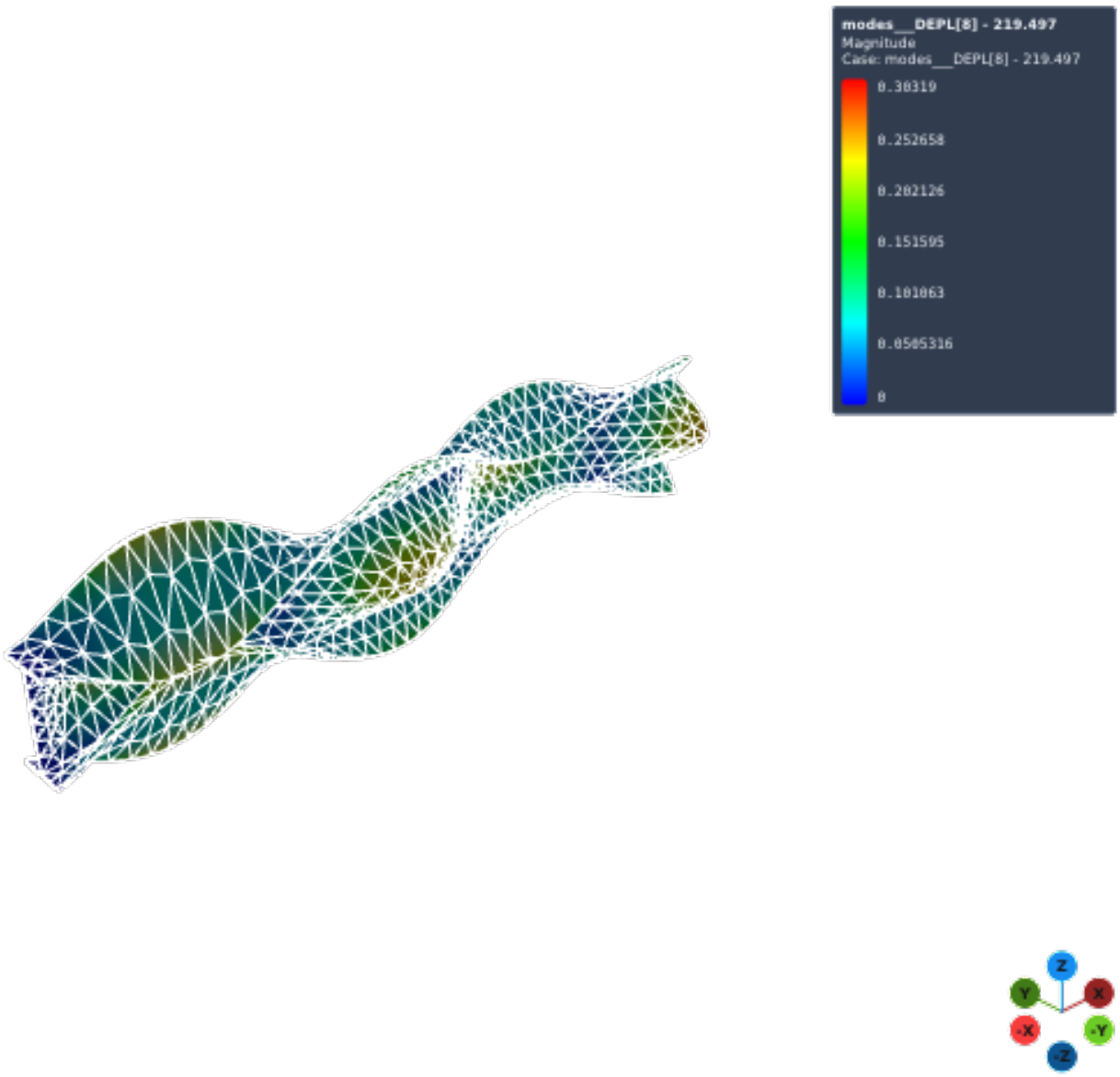

Figure 458: Code_Aster — mode 9, 219.497 Hz

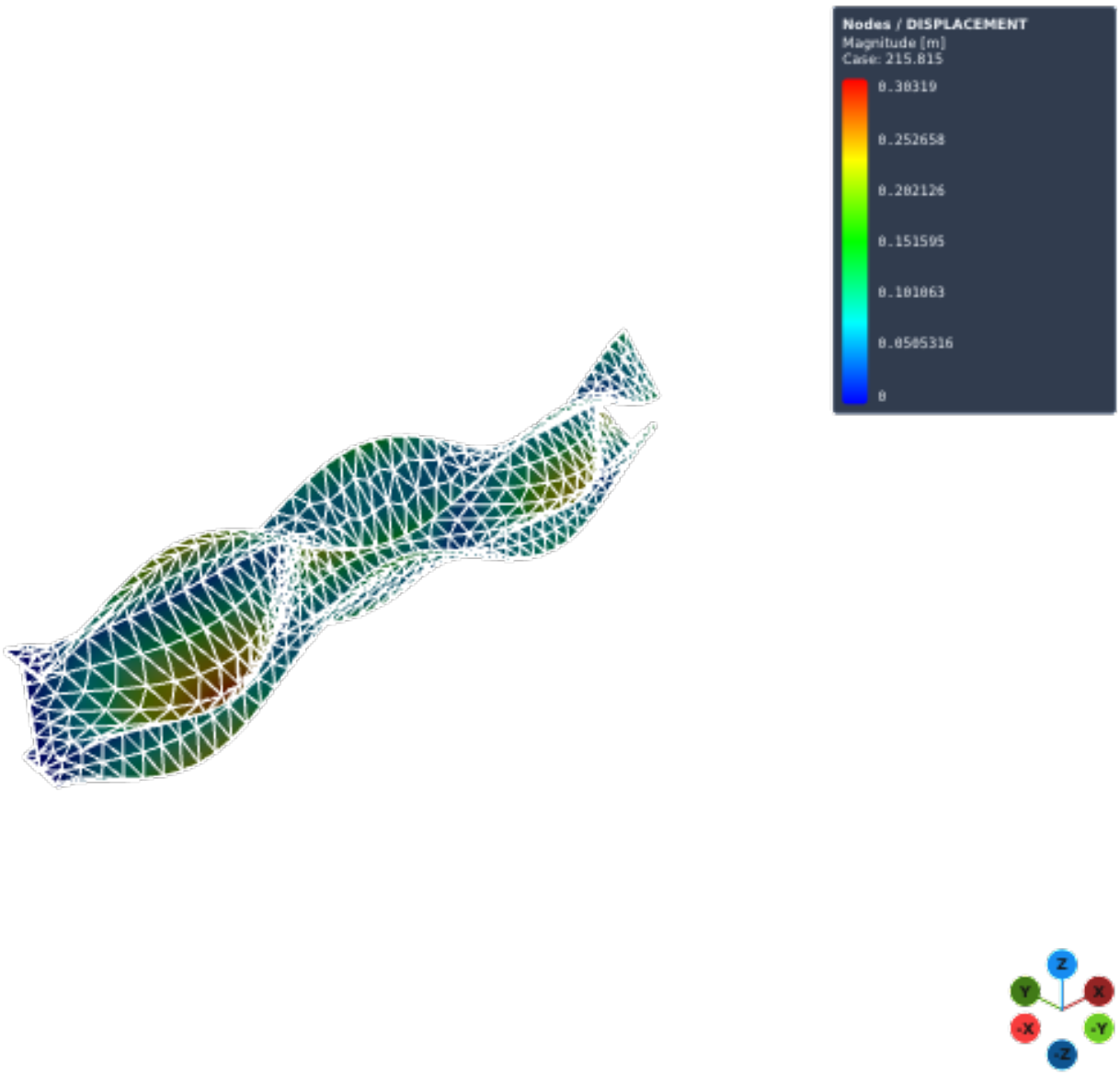

Figure 459: Sesam — mode 9, 215.815 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	215.820	-0.84	1.83887e+06	8.48e-08	0.60	8.50e-08	-2.91e-04	0.774	2.91e-04
Calculix	219.501	+0.85	1.9021e+06	5.19e-08	0.48	8.50e-08	2.28e-04	0.690	-2.92e-04
Code_Aster	219.497	+0.84	1.90203e+06	5.25e-08	0.48	8.41e-08	2.29e-04	0.690	-2.90e-04
Sesam	215.815	-0.85	1.83875e+06	9.03e-08	0.60	8.47e-08	3.00e-04	-0.774	-2.91e-04

Mode 10

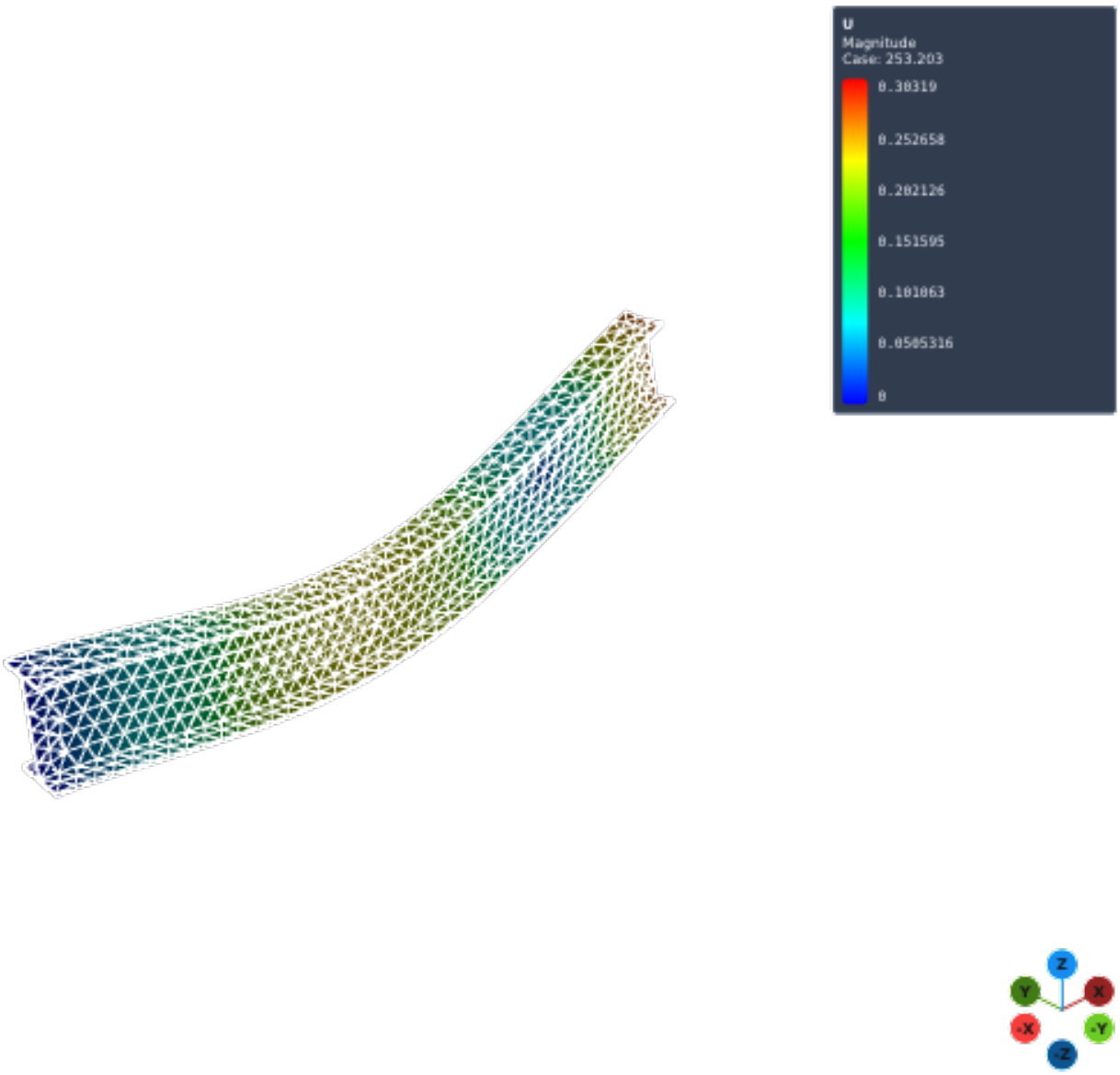

Figure 460: Abaqus — mode 10, 253.203 Hz

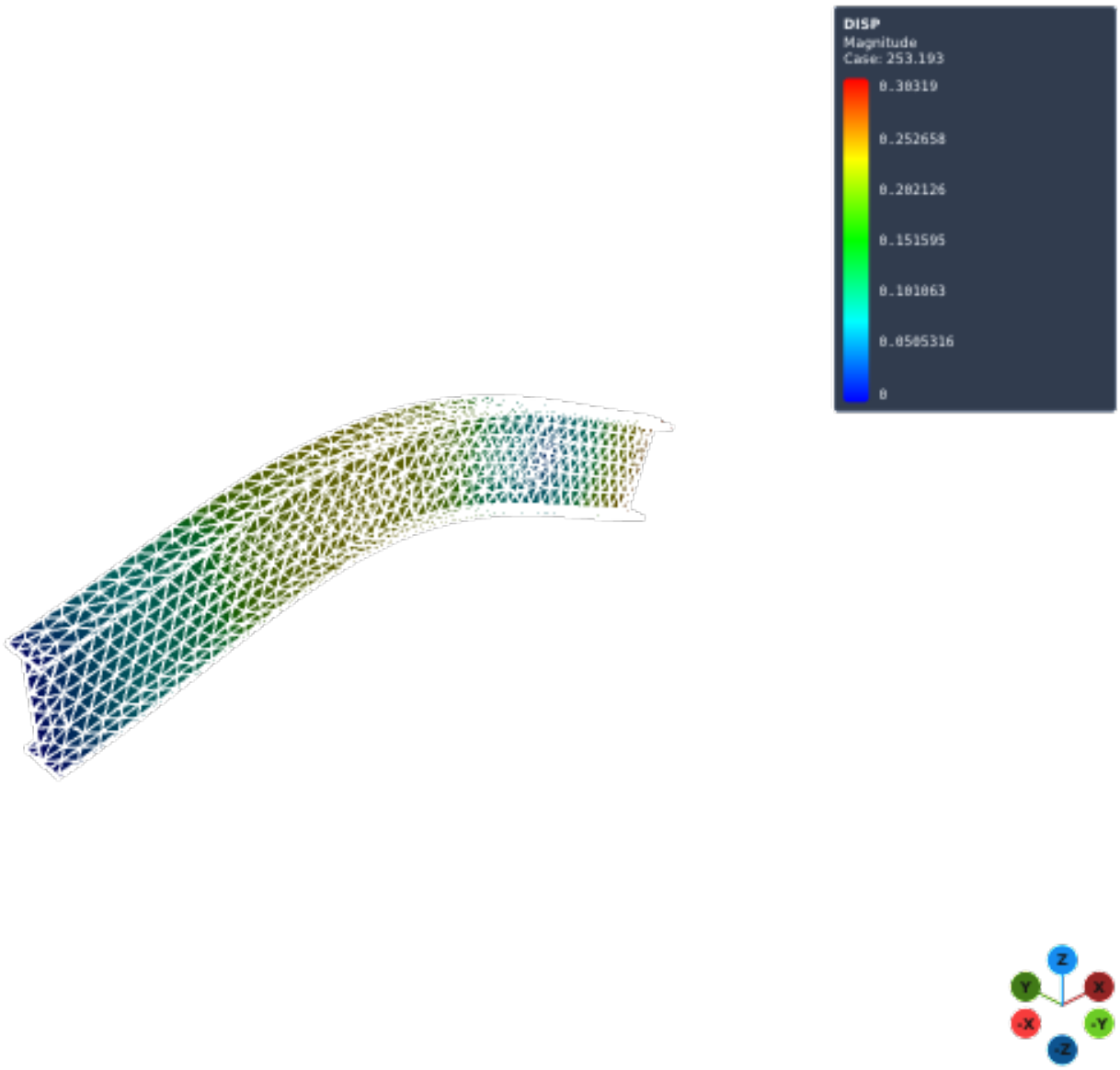

Figure 461: Calculix — mode 10, 253.193 Hz

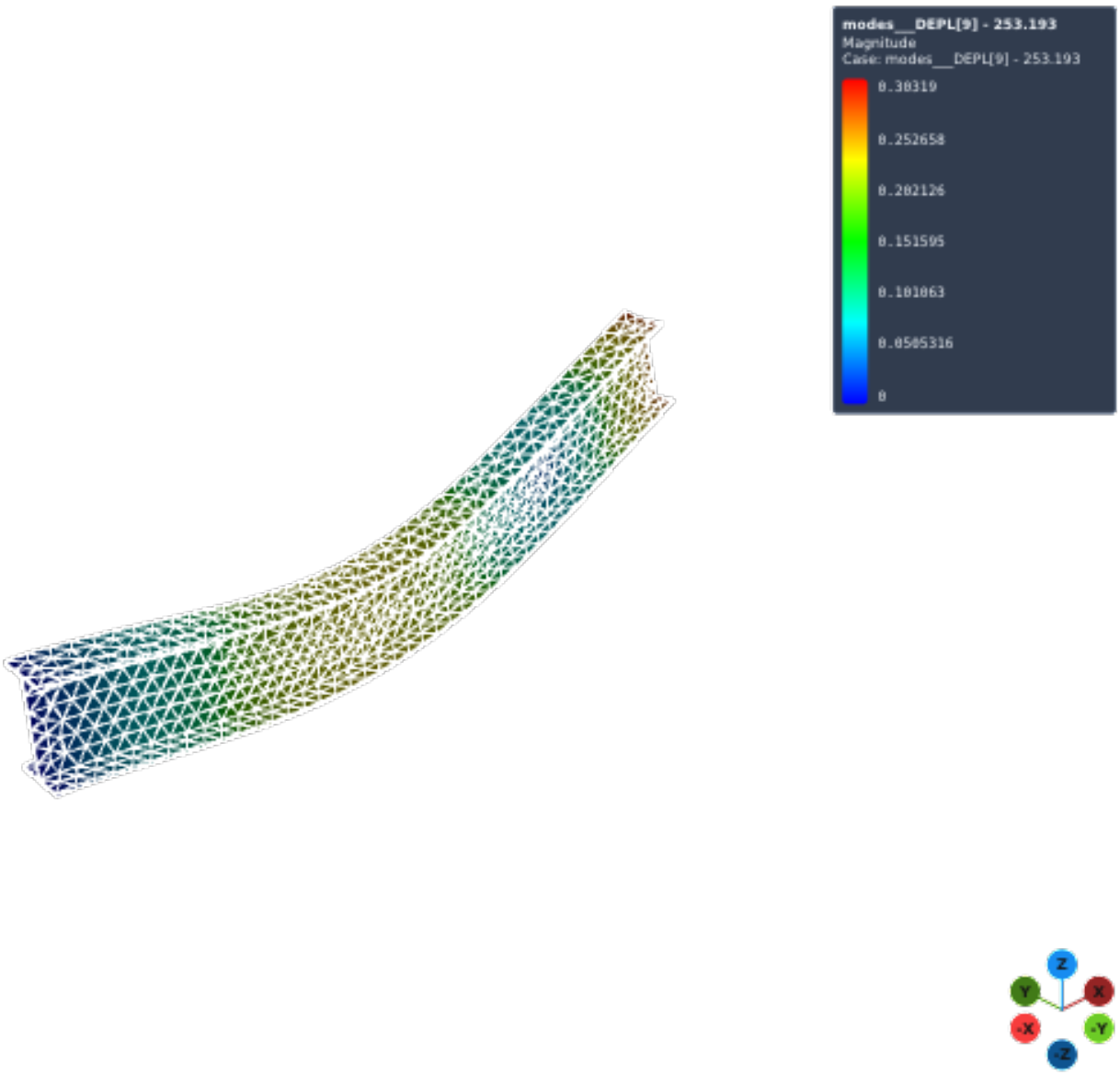

Figure 462: Code_Aster — mode 10, 253.193 Hz

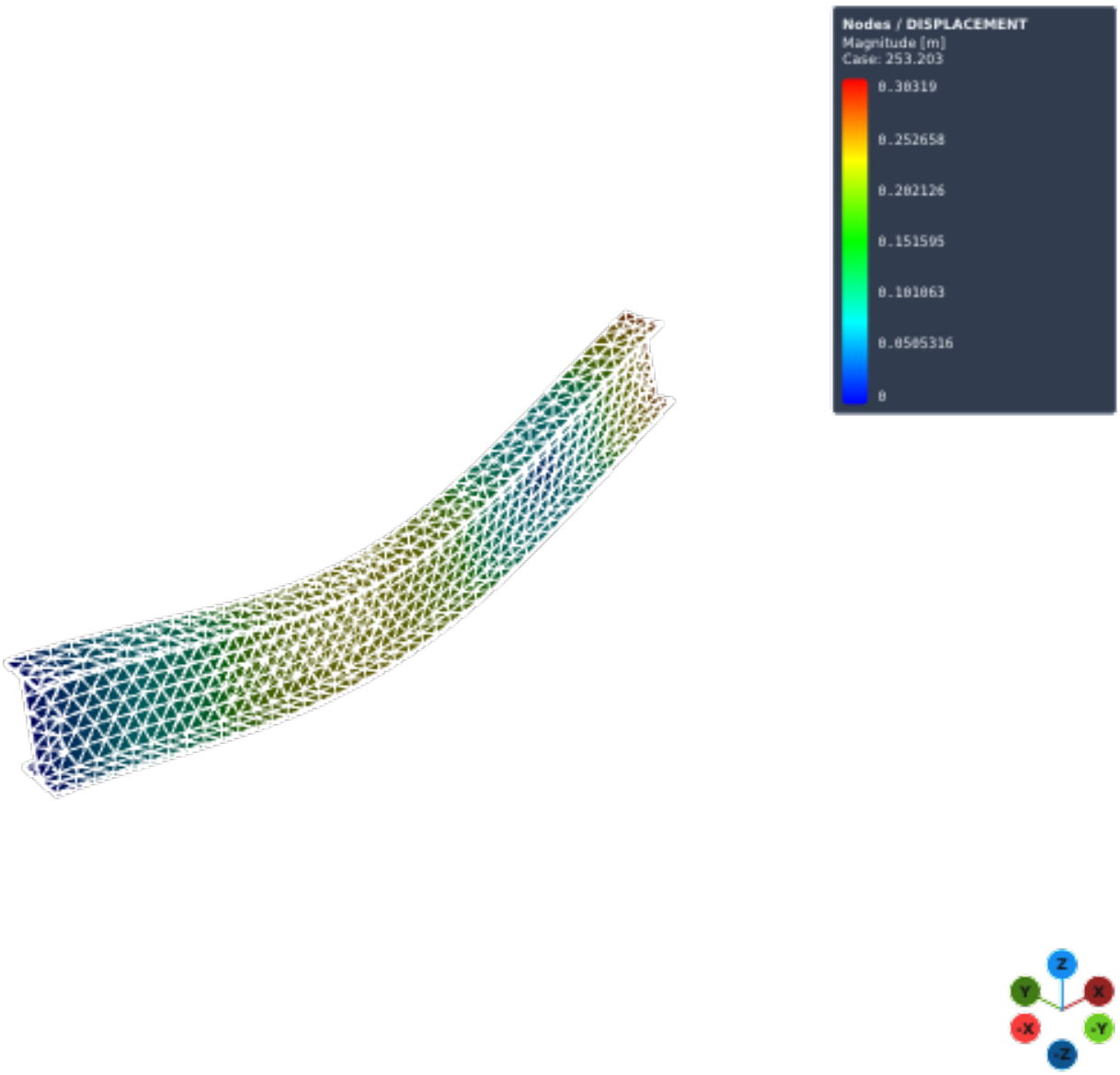

Figure 463: Sesam — mode 10, 253.203 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	253.200	1.16e-03	2.53103e+06	8.86e-09	1.97e-08	40.25	-9.41e-05	1.40e-04	-6.344
Calculix	253.193	-1.72e-03	2.53082e+06	1.34e-08	1.74e-07	40.25	-1.16e-04	-4.17e-04	6.344
Code_Aster	253.193	-1.74e-03	2.53082e+06	1.36e-08	1.74e-07	40.25	1.17e-04	4.17e-04	-6.344
Sesam	253.203	2.31e-03	2.53103e+06	9.13e-09	1.99e-08	40.25	-9.56e-05	1.41e-04	-6.344

Mode 11

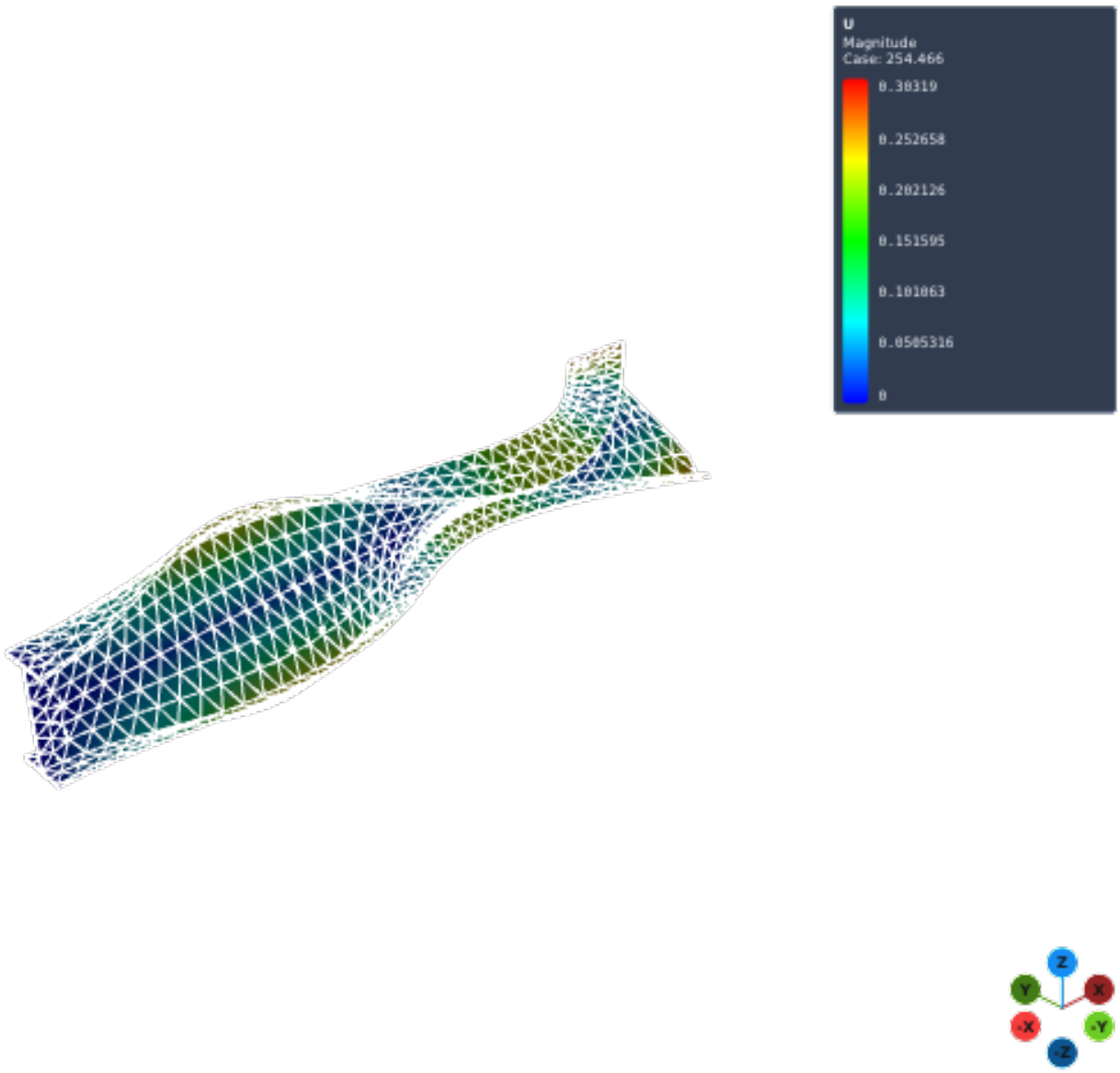

Figure 464: Abaqus — mode 11, 254.466 Hz

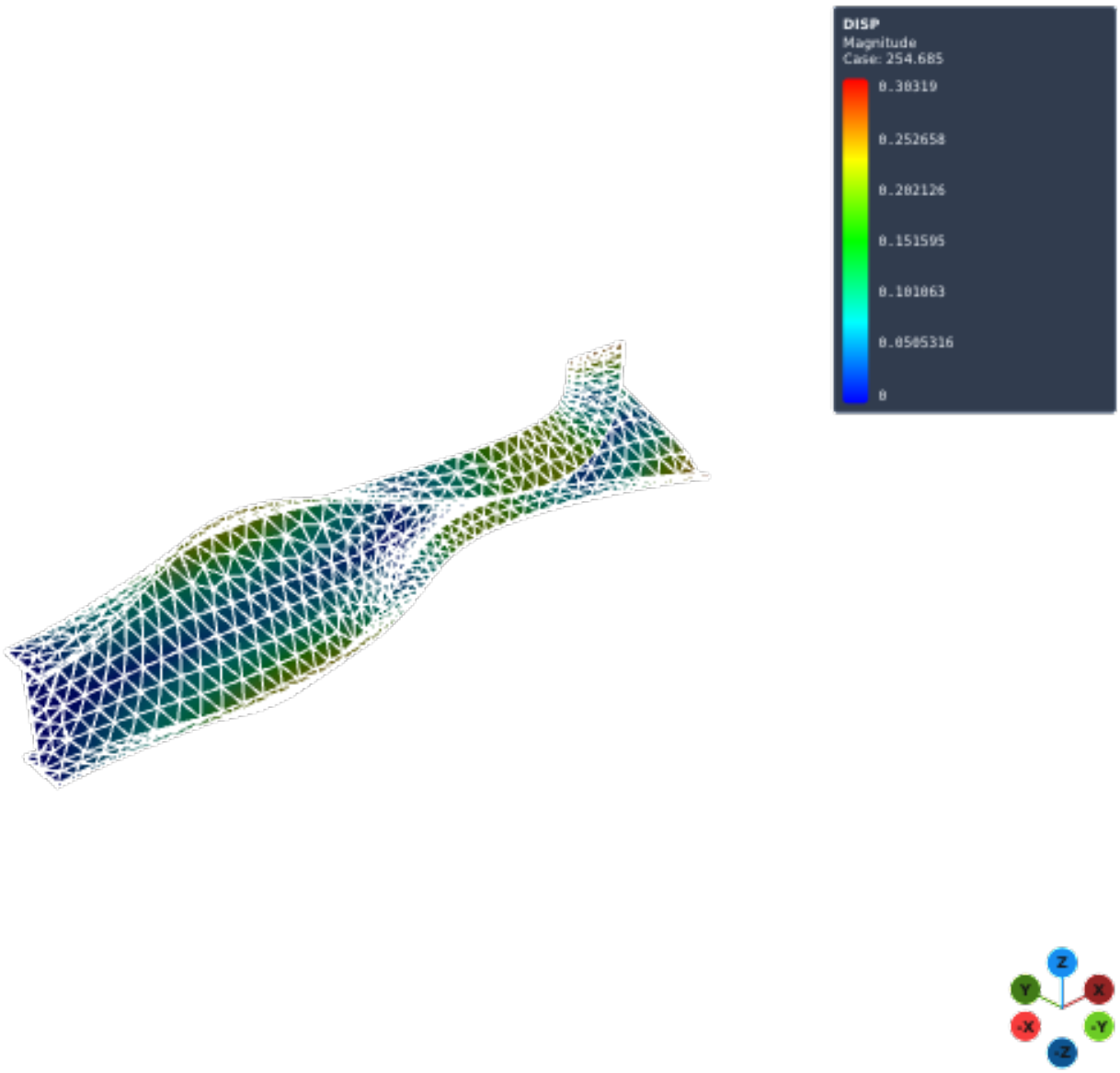

Figure 465: Calculix — mode 11, 254.685 Hz

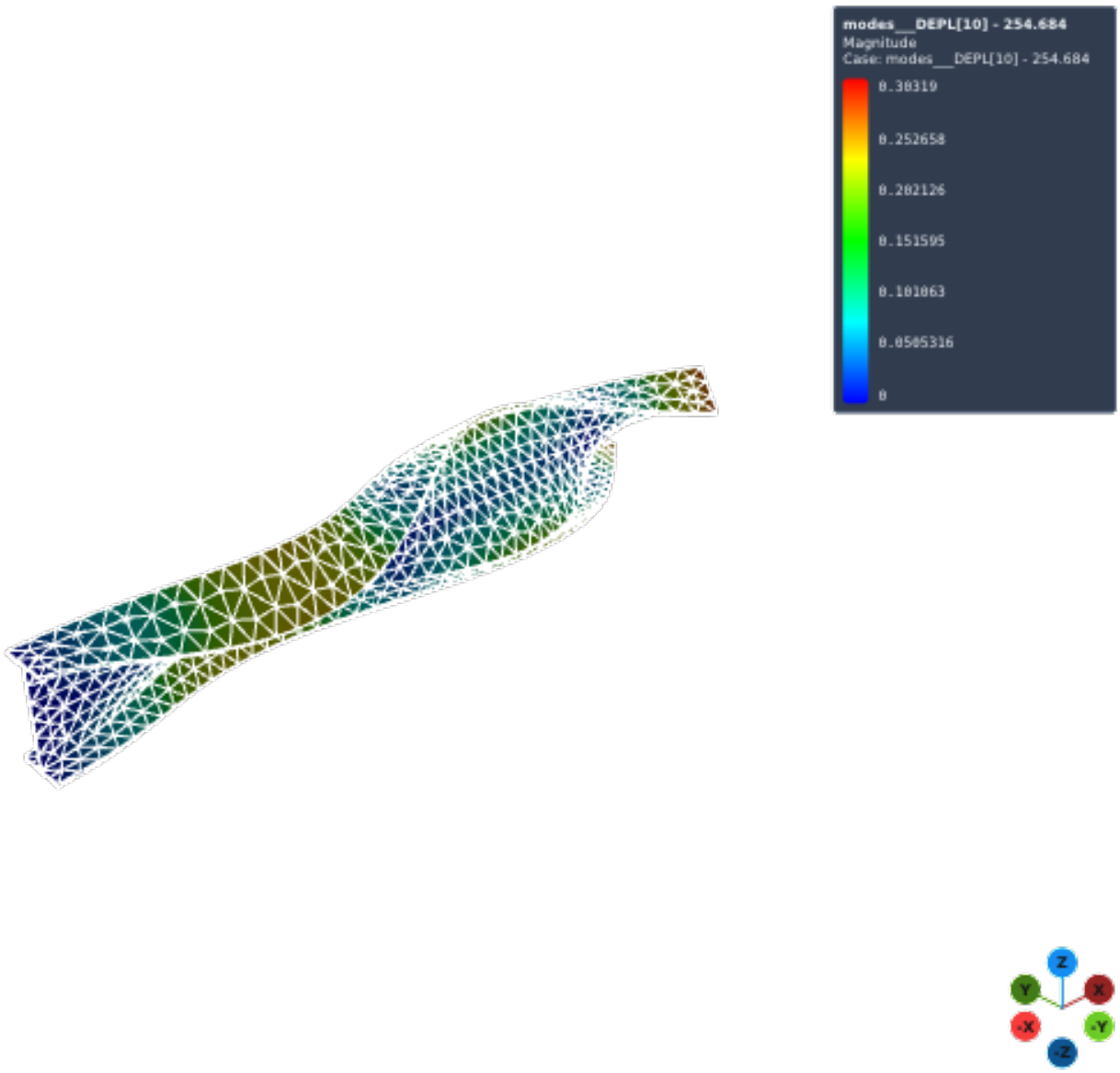

Figure 466: Code_Aster — mode 11, 254.684 Hz

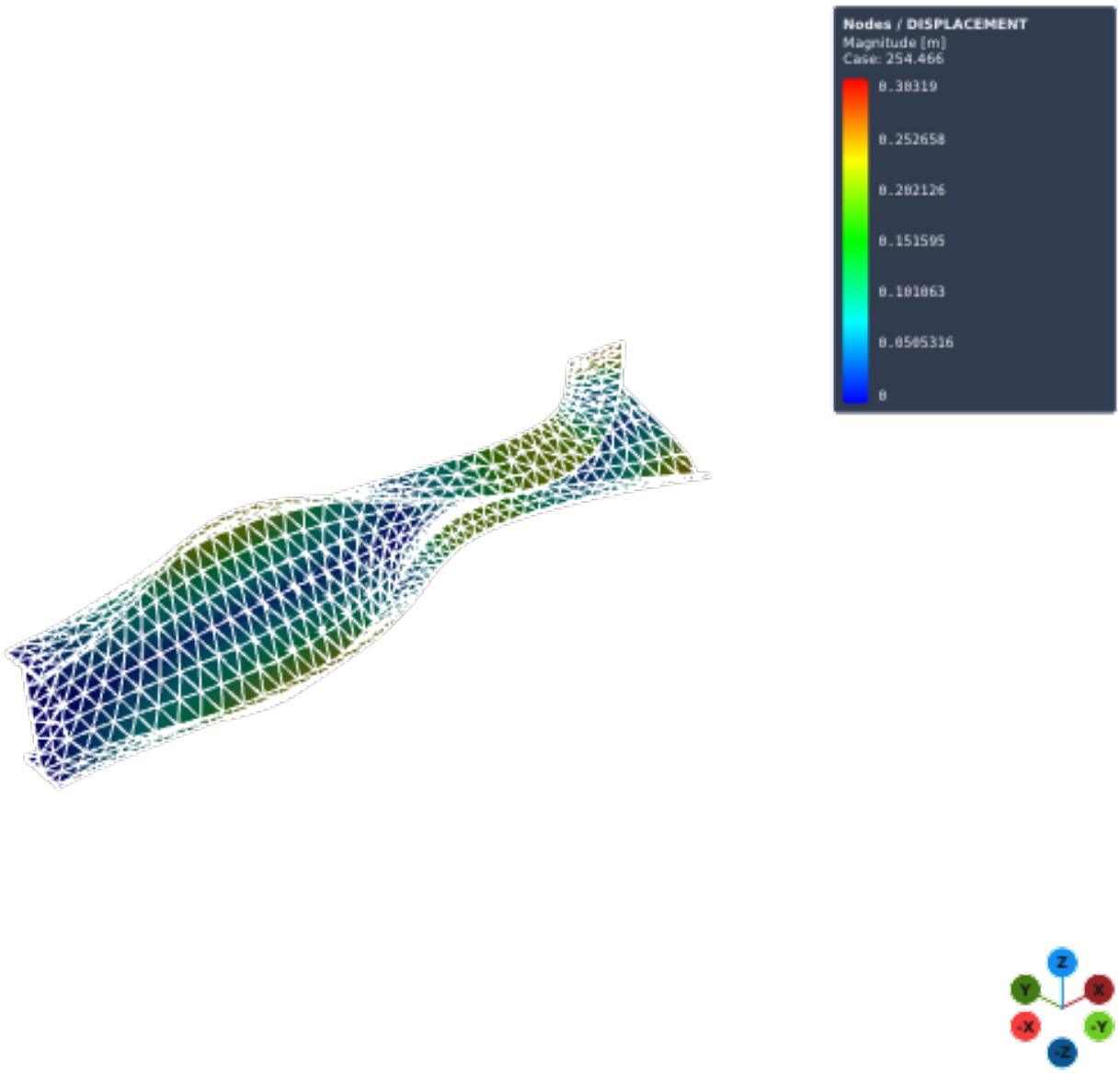

Figure 467: Sesam — mode 11, 254.466 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	254.470	-0.04	2.55635e+06	1.63e-09	6.96e-07	4.16e-05	-4.04e-05	-0.001	-0.006
Calculix	254.685	+0.04	2.56074e+06	9.09e-09	6.08e-05	1.61e-05	-9.53e-05	-0.008	0.004
Code_Aster	254.684	+0.04	2.56073e+06	9.51e-09	6.08e-05	1.60e-05	9.75e-05	0.008	-0.004
Sesam	254.466	-0.04	2.55634e+06	1.66e-09	6.95e-07	4.17e-05	-4.07e-05	-0.001	-0.006

Solid, 2nd order, HEX

Mode 1

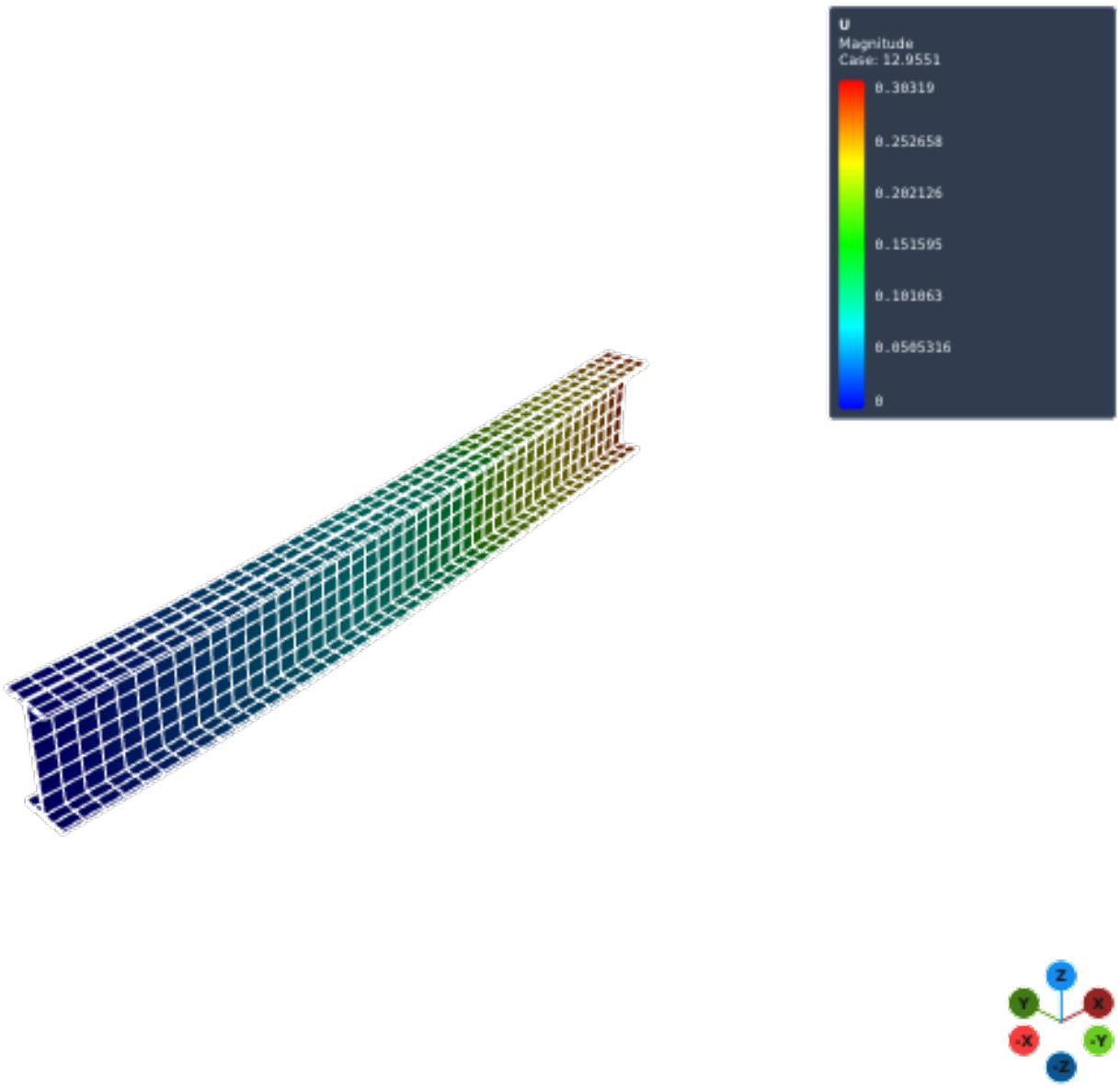

Figure 468: Abaqus — mode 1, 12.955 Hz

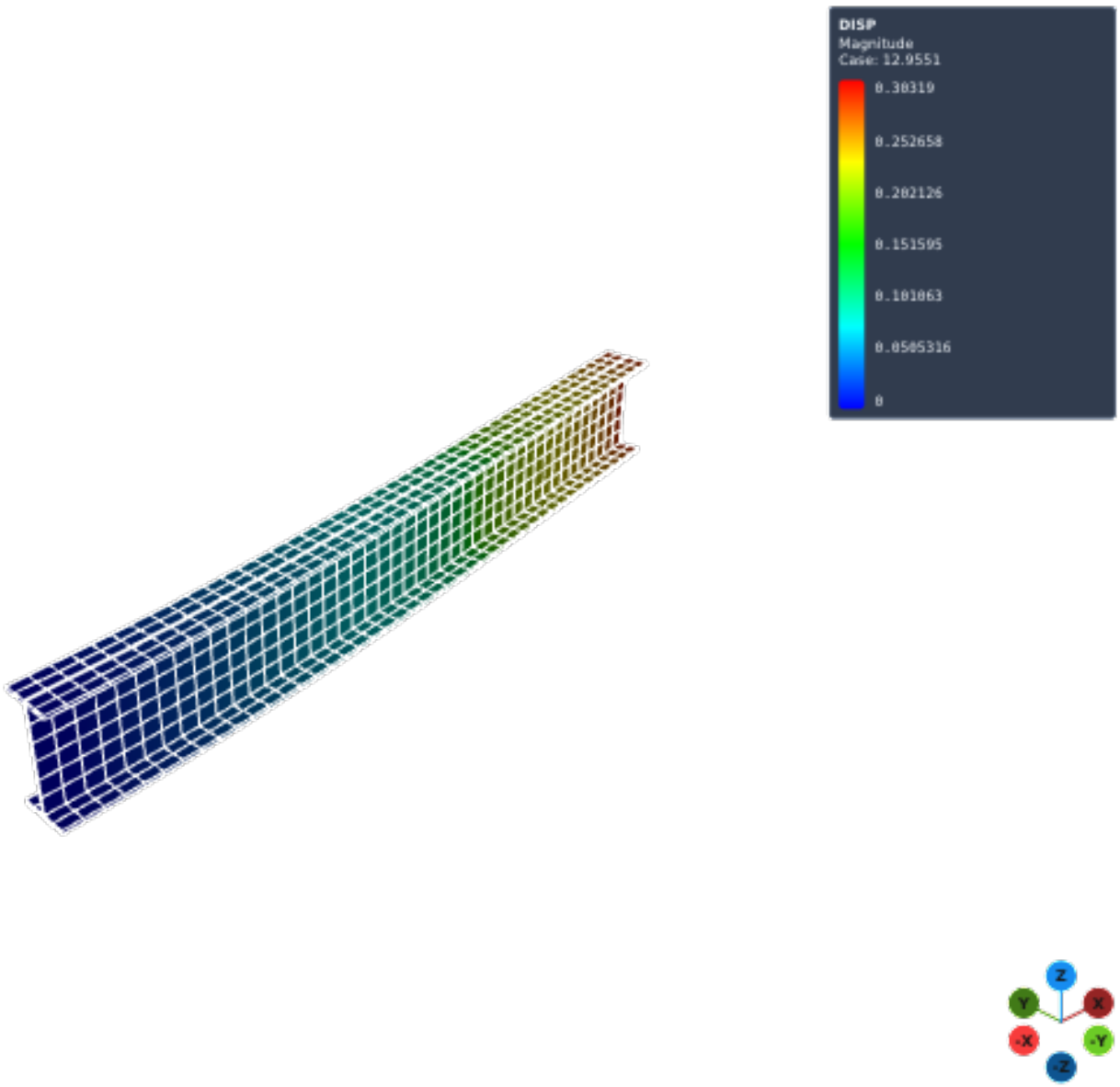

Figure 469: Calculix — mode 1, 12.955 Hz

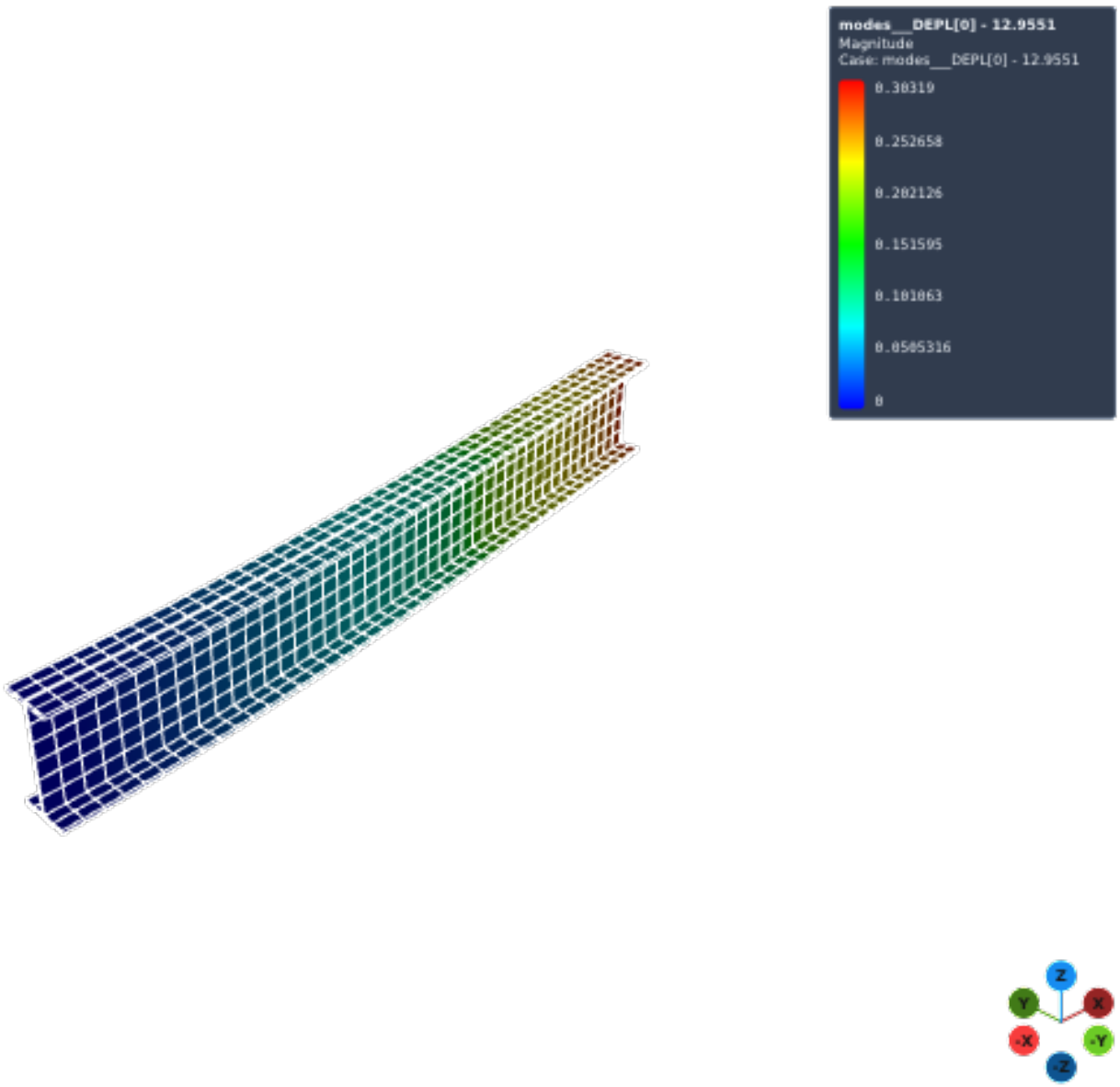

Figure 470: Code_Aster — mode 1, 12.955 Hz

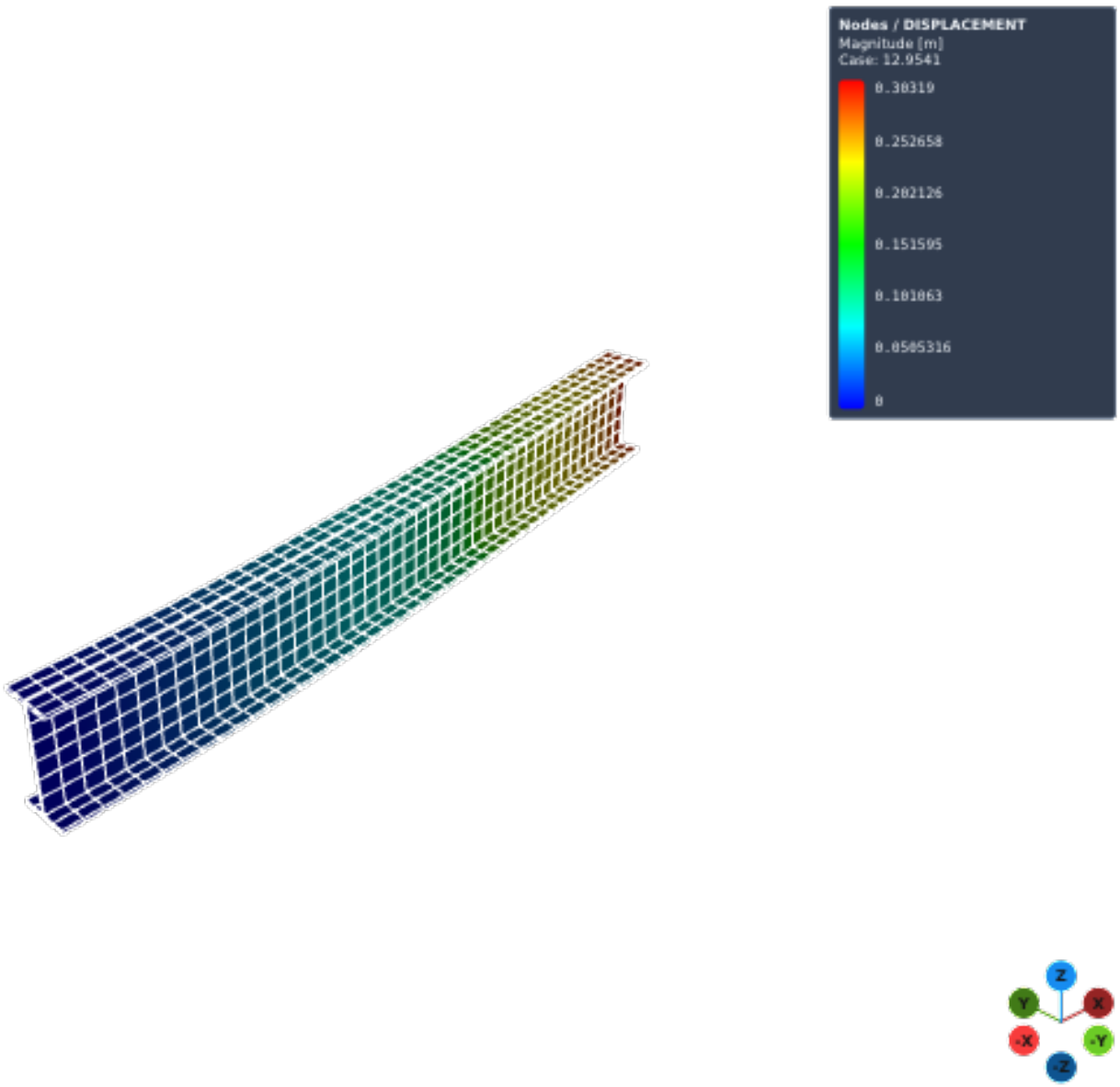

Figure 471: Sesam — mode 1, 12.954 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.955	1.39e-03	6625.8	3.08e-24	116.50	5.64e-20	1.76e-12	10.794	2.37e-10
Calculix	12.955	1.93e-03	6625.81	2.50e-24	116.50	6.61e-20	-1.58e-12	10.794	-2.57e-10
Code_Aster	12.955	1.93e-03	6625.81	1.58e-24	116.50	2.27e-21	-1.26e-12	10.794	4.77e-11
Sesam	12.954	-0.01	6624.86	4.71e-24	116.51	4.62e-21	2.17e-12	10.794	-6.80e-11

Mode 2

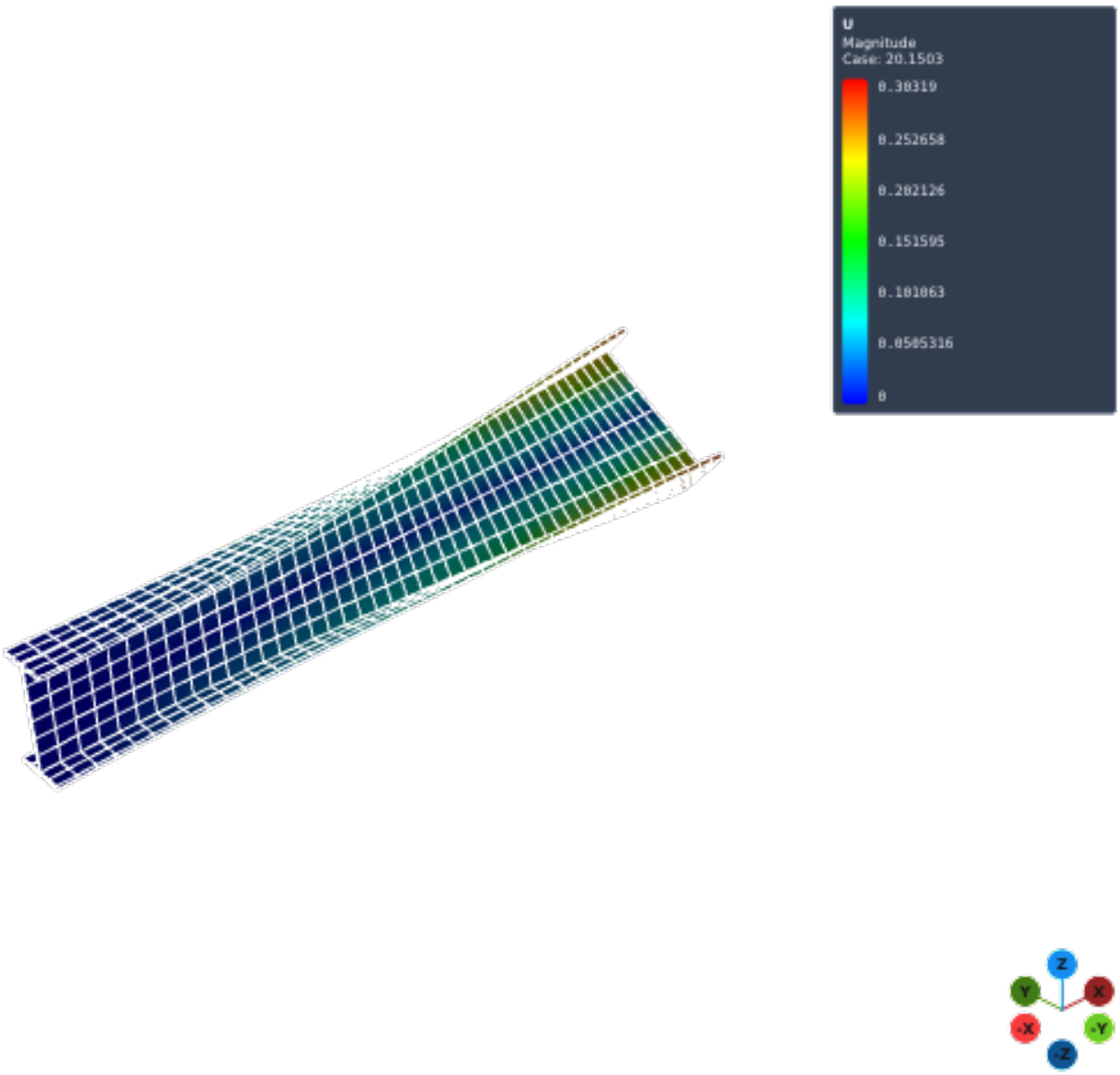

Figure 472: Abaqus — mode 2, 20.150 Hz

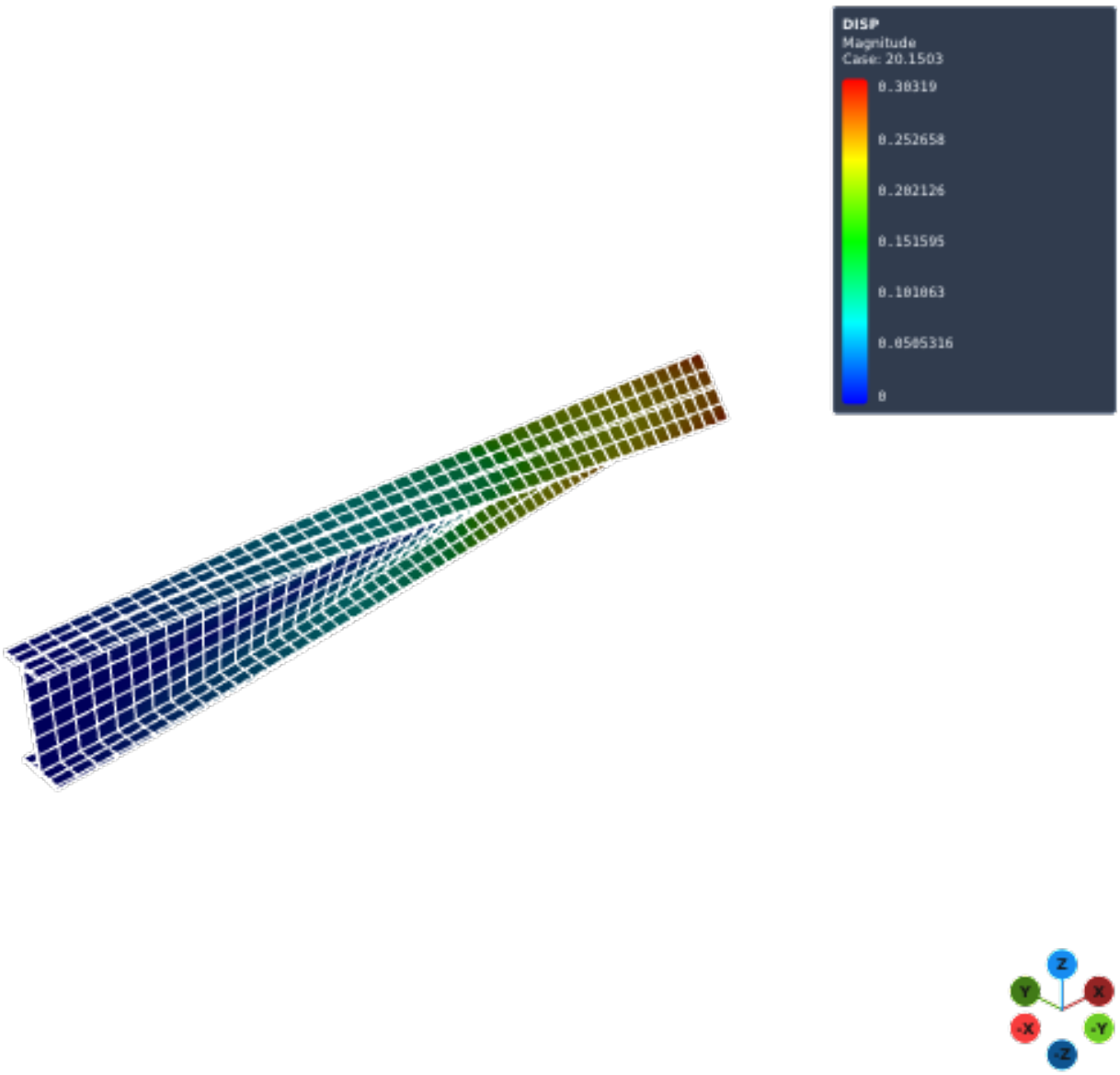

Figure 473: Calculix — mode 2, 20.150 Hz

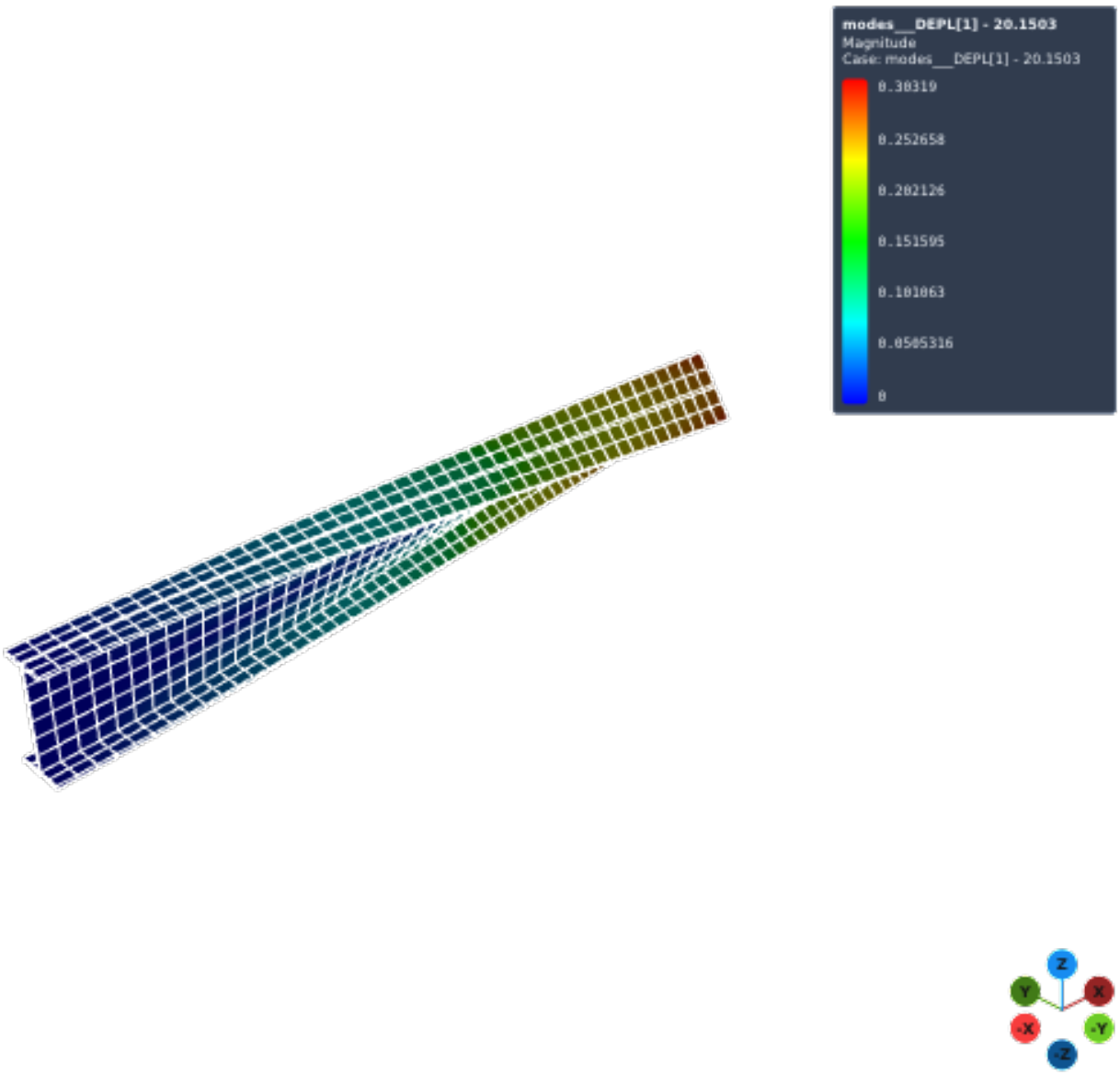

Figure 474: Code_Aster — mode 2, 20.150 Hz

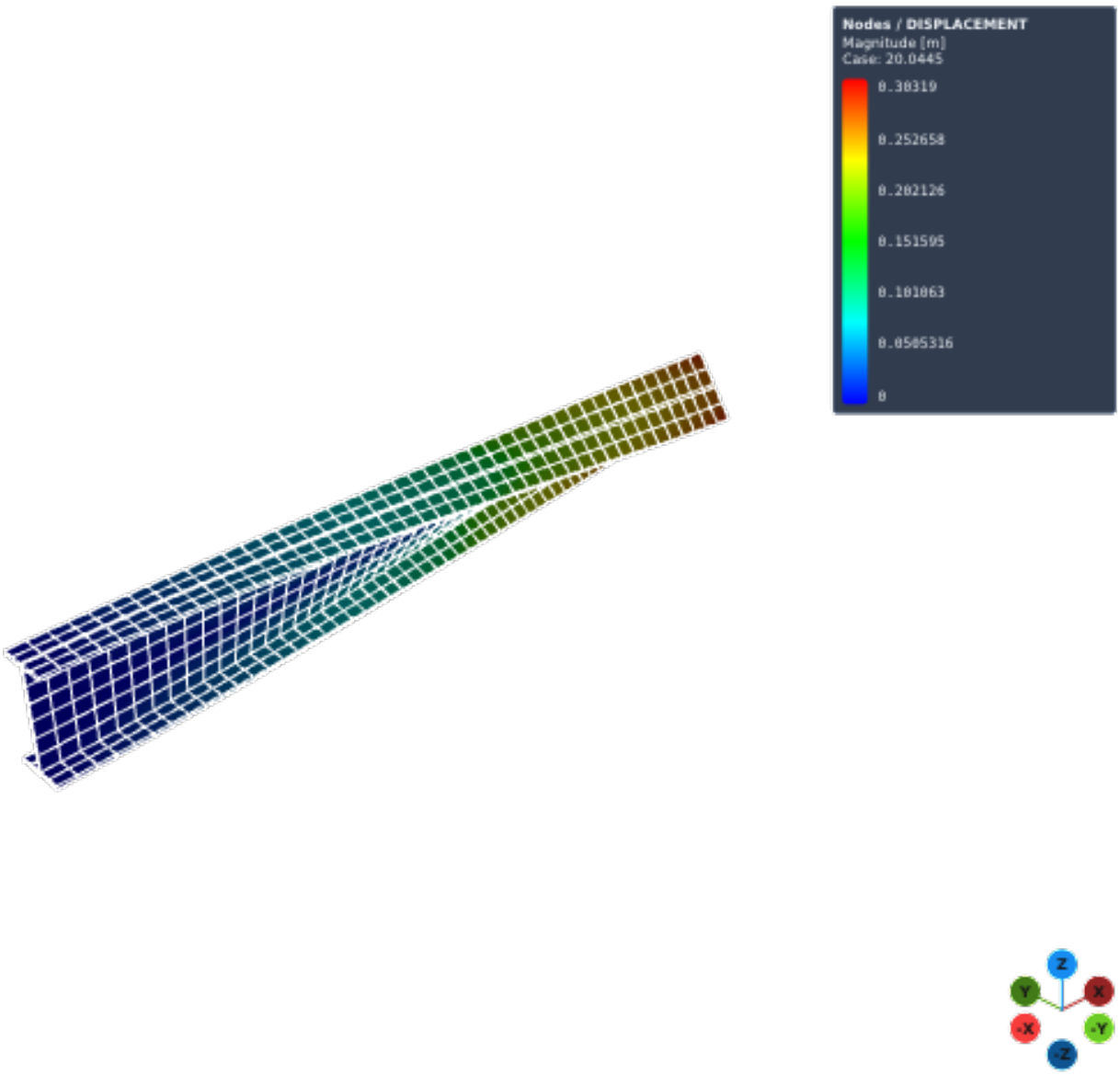

Figure 475: Sesam — mode 2, 20.044 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	20.150	+0.13	16030	2.26e-25	4.71e-16	2.32e-18	-4.76e-13	2.17e-08	1.52e-09
Calculix	20.150	+0.13	16029.6	7.08e-26	6.44e-16	1.72e-18	-2.66e-13	2.54e-08	1.31e-09
Code_Aster	20.150	+0.13	16029.6	1.72e-24	2.30e-17	2.99e-21	-1.31e-12	4.79e-09	-5.46e-11
Sesam	20.044	-0.39	15861.7	2.58e-24	5.48e-15	8.69e-19	1.61e-12	-7.40e-08	9.32e-10

Mode 3

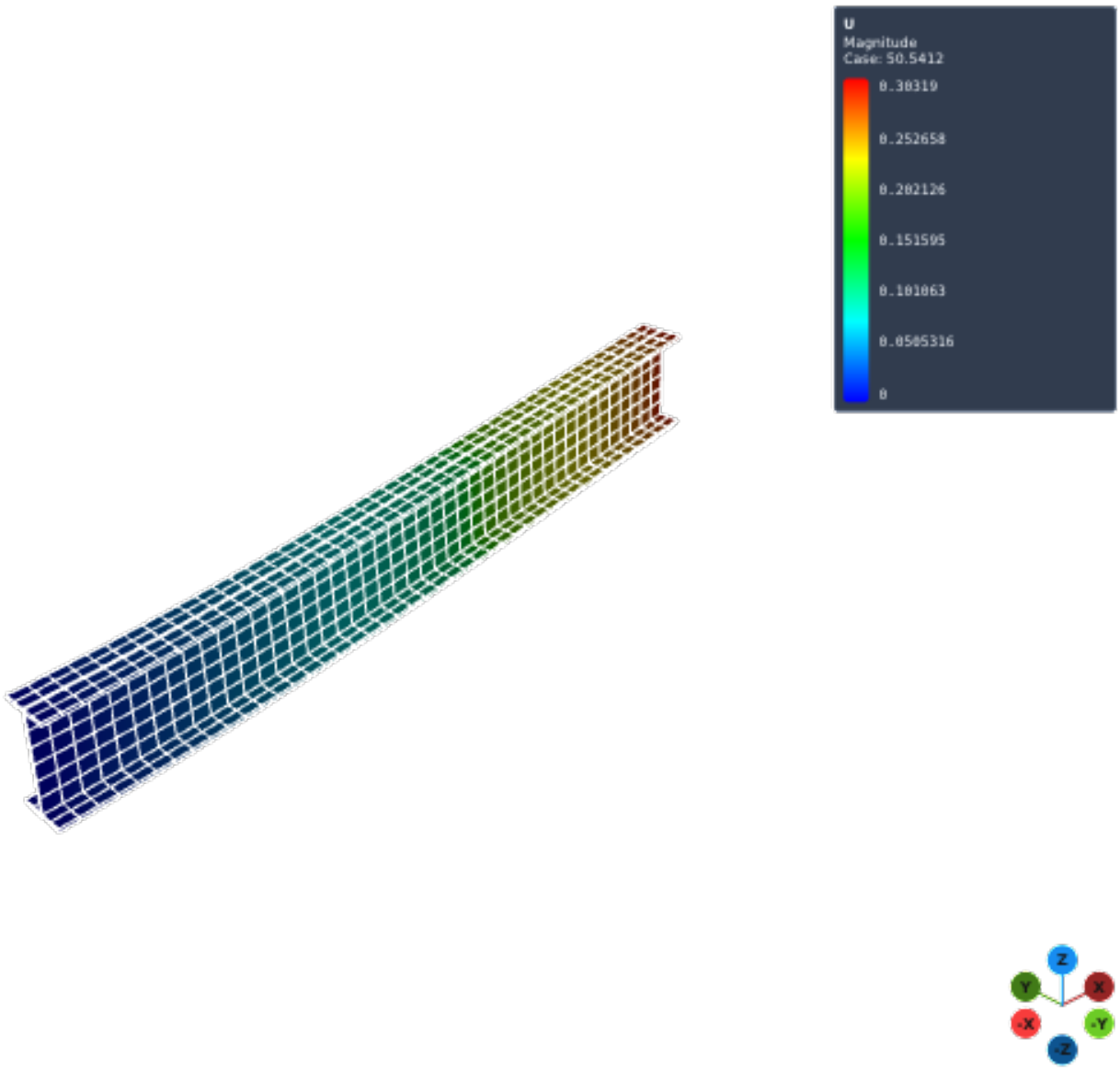

Figure 476: Abaqus — mode 3, 50.541 Hz

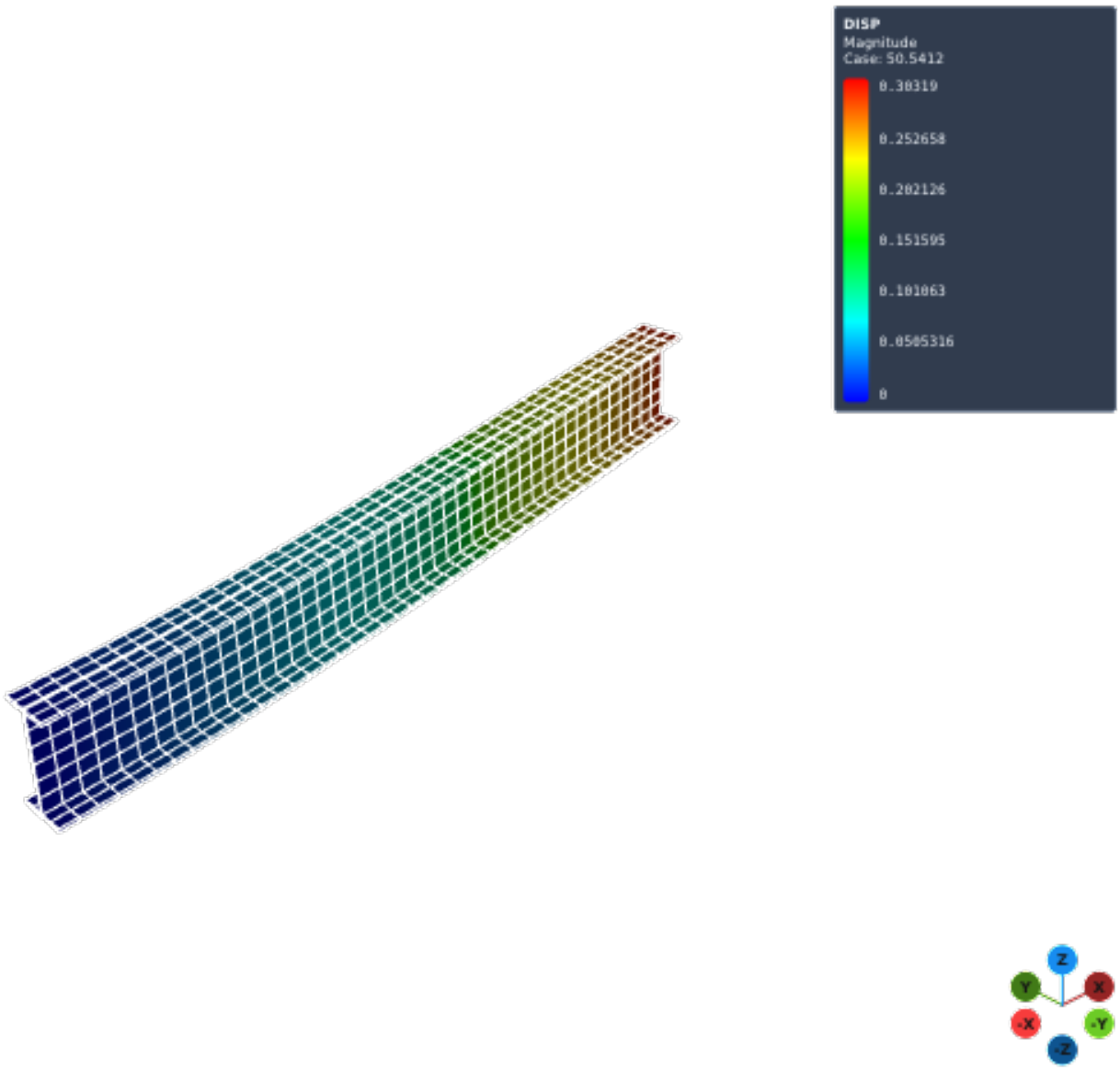

Figure 477: Calculix — mode 3, 50.541 Hz

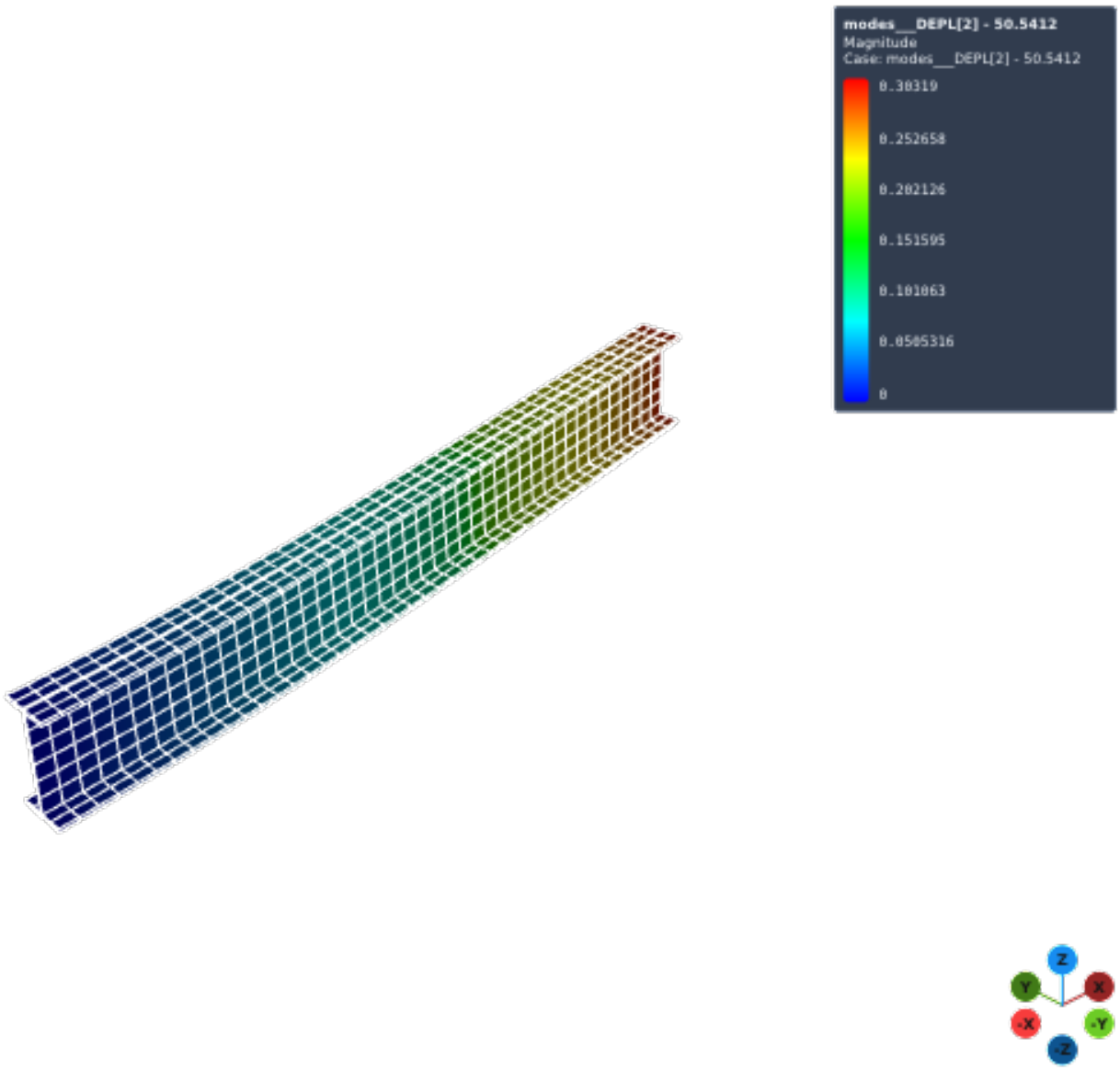

Figure 478: Code_Aster — mode 3, 50.541 Hz

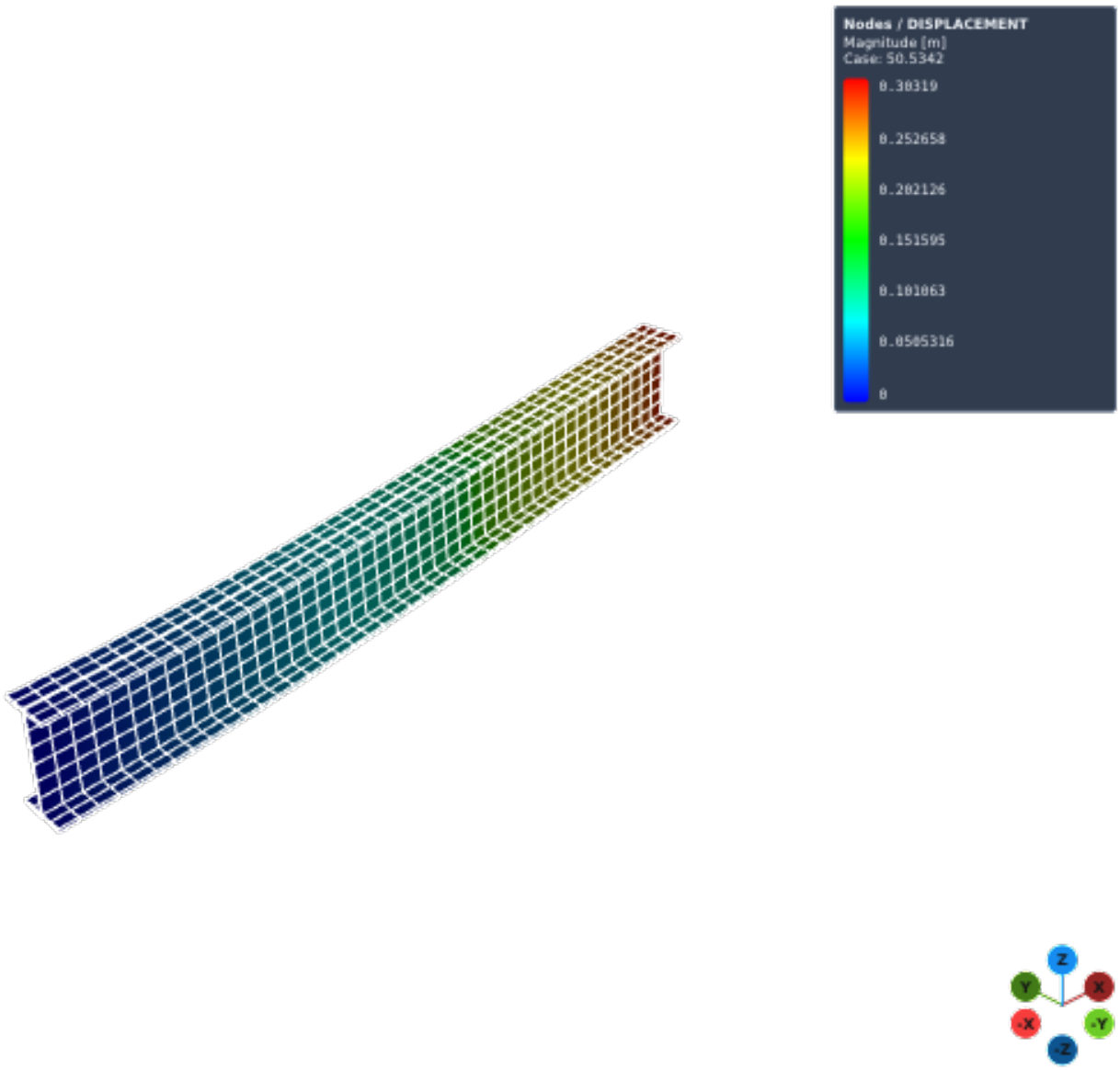

Figure 479: Sesam — mode 3, 50.534 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.541	3.14e-03	100844	1.71e-23	8.62e-20	118.36	-4.13e-12	-2.94e-10	10.879
Calculix	50.541	3.61e-03	100844	1.81e-24	1.19e-19	118.36	-1.35e-12	3.44e-10	10.879
Code_Aster	50.541	3.62e-03	100844	3.72e-24	1.03e-20	118.36	-1.93e-12	-1.01e-10	10.879
Sesam	50.534	-0.01	100816	6.44e-24	9.81e-21	118.37	2.54e-12	9.90e-11	10.880

Mode 4

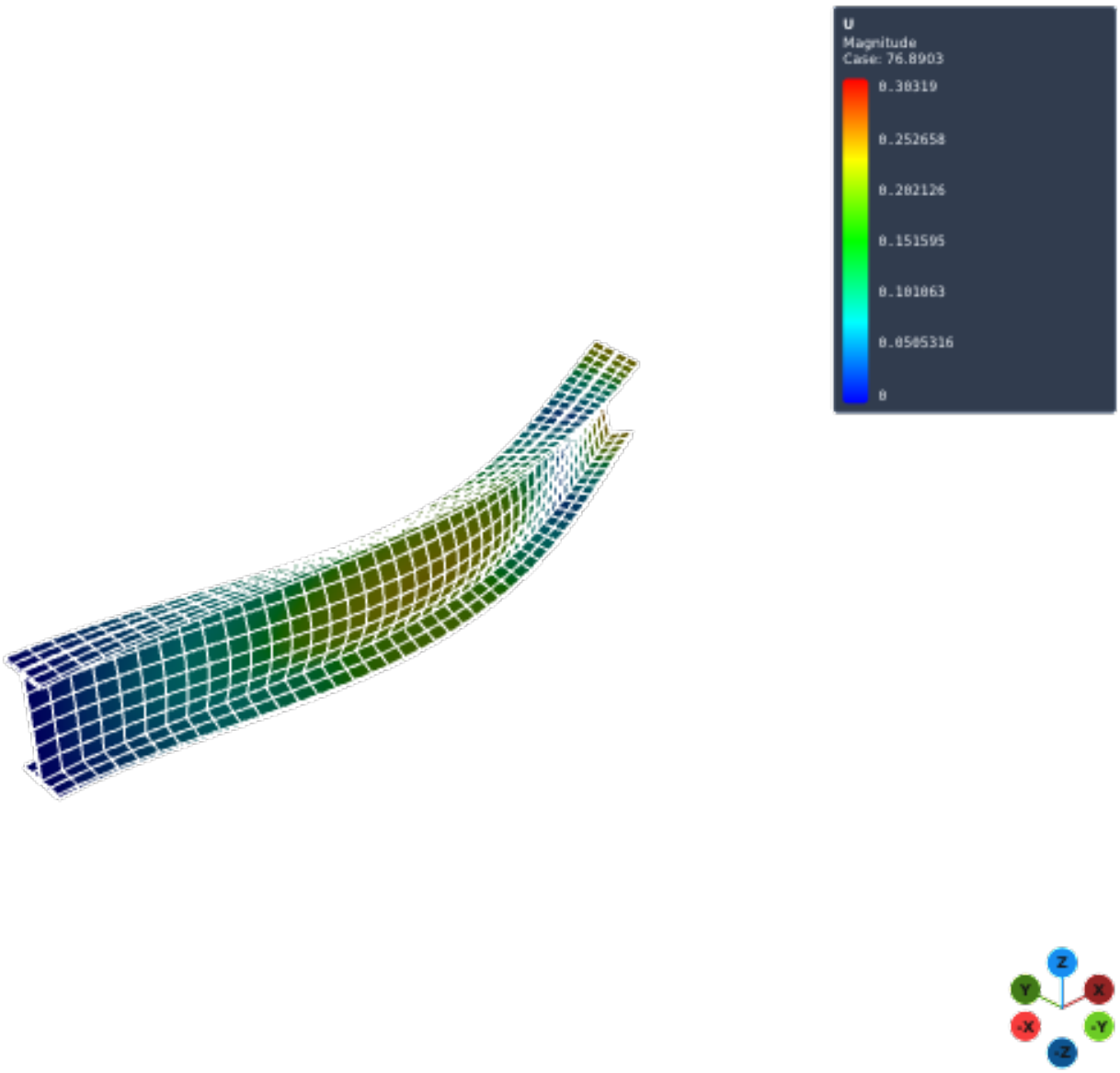

Figure 480: Abaqus — mode 4, 76.890 Hz

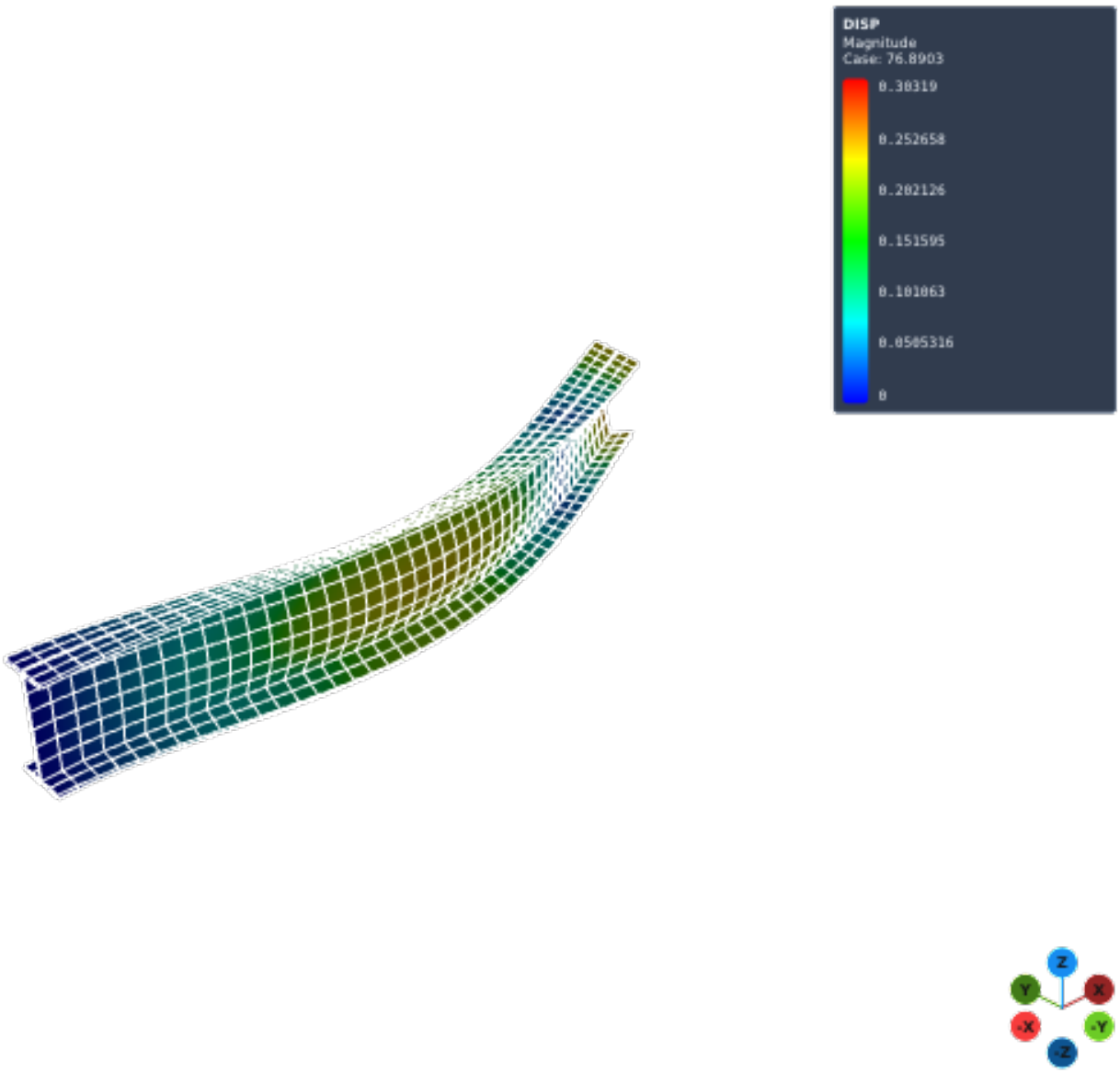

Figure 481: Calculix — mode 4, 76.890 Hz

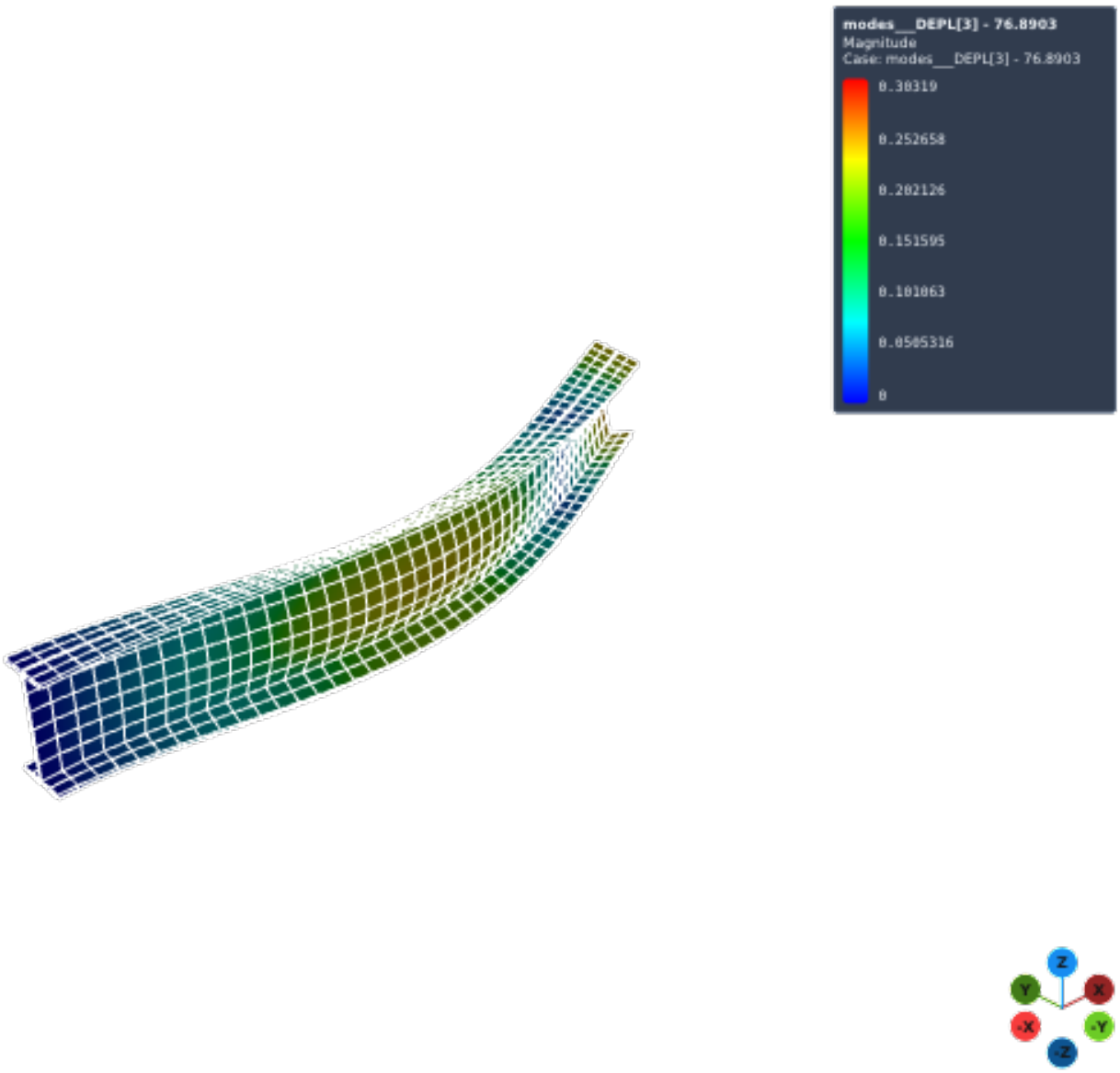

Figure 482: Code_Aster — mode 4, 76.890 Hz

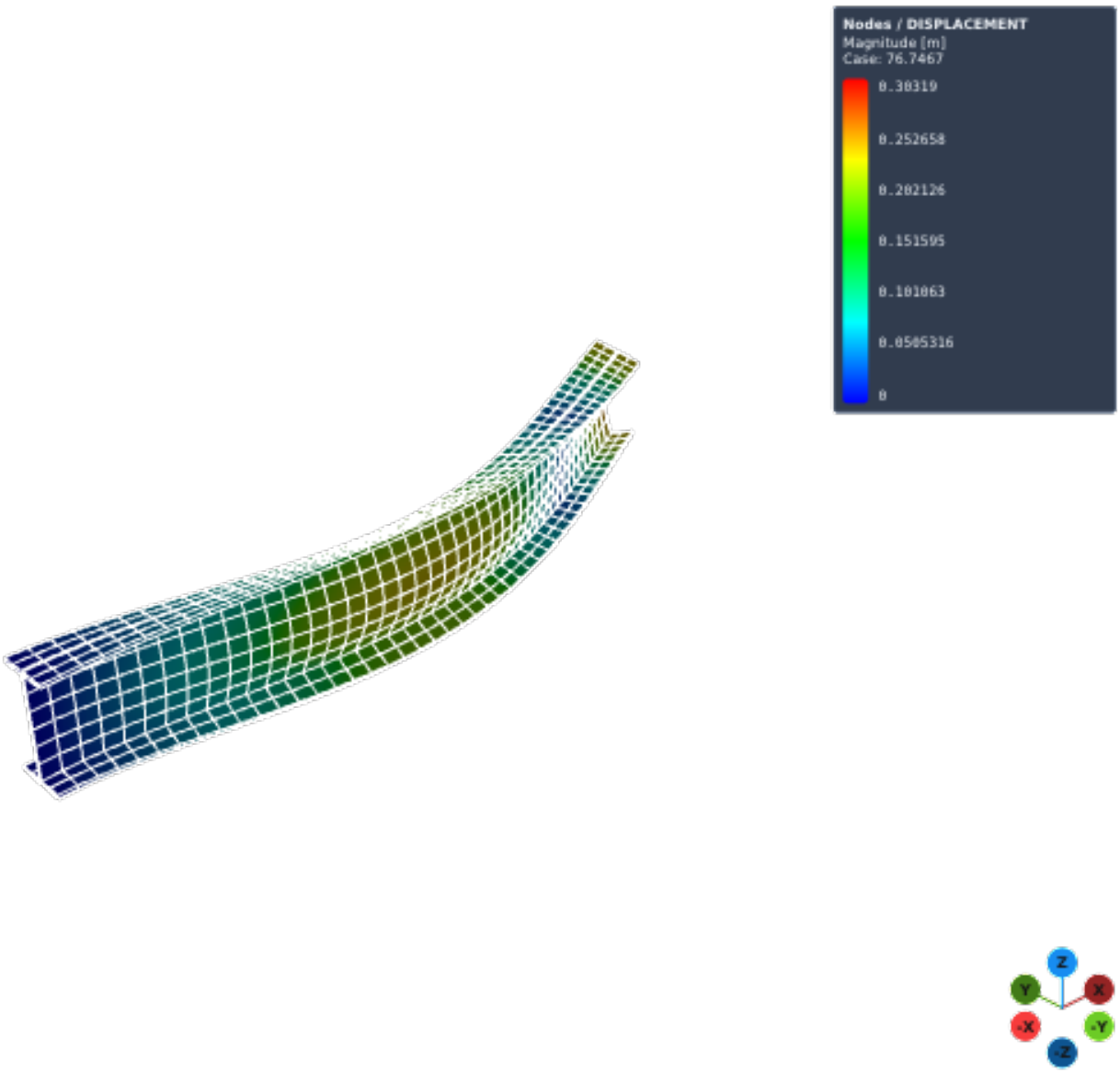

Figure 483: Sesam — mode 4, 76.747 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	76.890	+0.05	233401	9.37e-28	35.91	1.14e-20	-3.06e-14	-5.993	-1.07e-10
Calculix	76.890	+0.05	233401	4.29e-24	35.91	2.88e-20	2.07e-12	-5.993	1.70e-10
Code_Aster	76.890	+0.05	233401	1.25e-25	35.91	7.59e-21	3.54e-13	-5.993	-8.71e-11
Sesam	76.747	-0.14	232530	2.14e-25	35.89	4.07e-21	-4.63e-13	-5.991	6.38e-11

Mode 5

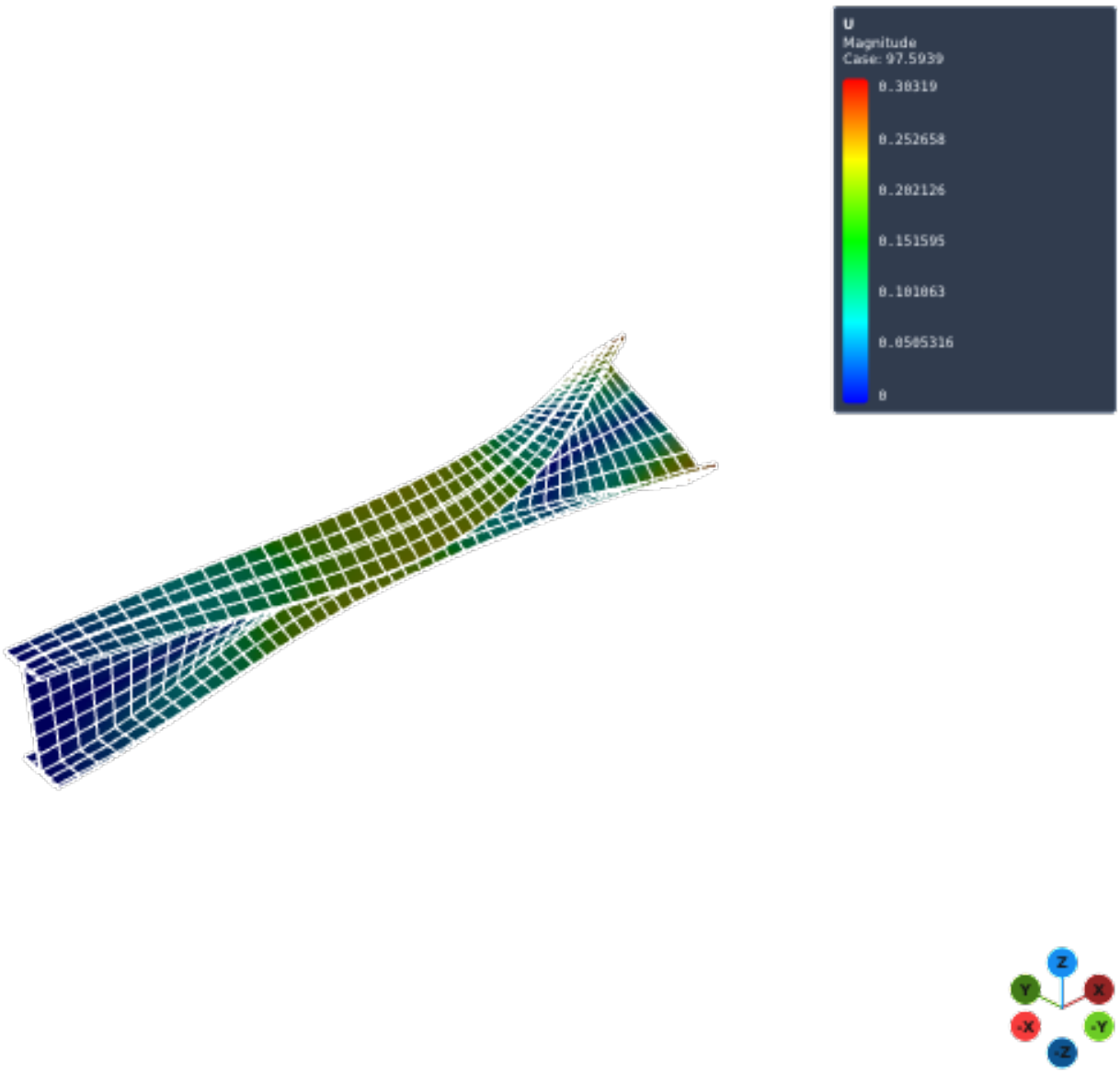

Figure 484: Abaqus — mode 5, 97.594 Hz

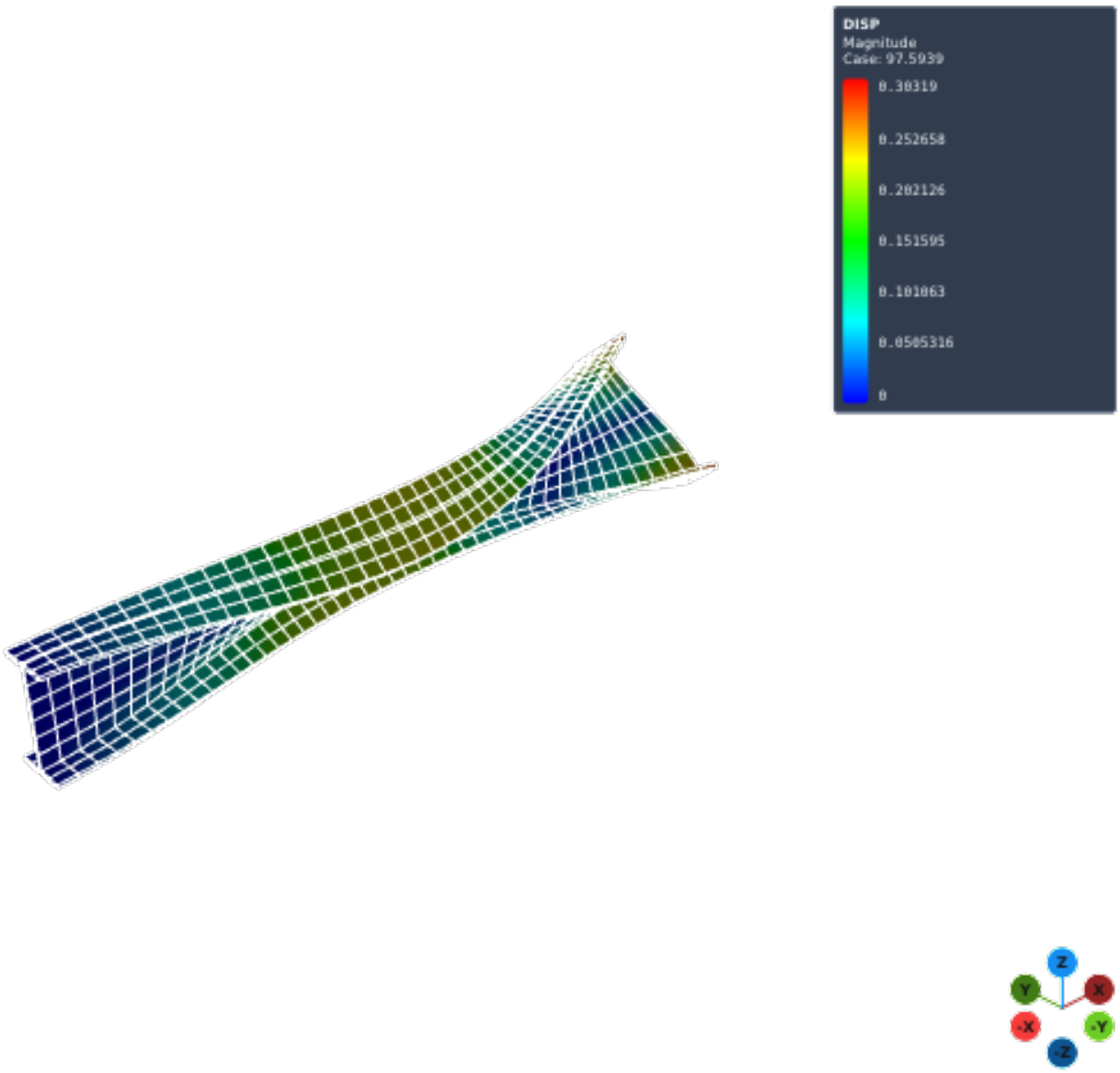

Figure 485: Calculix — mode 5, 97.594 Hz

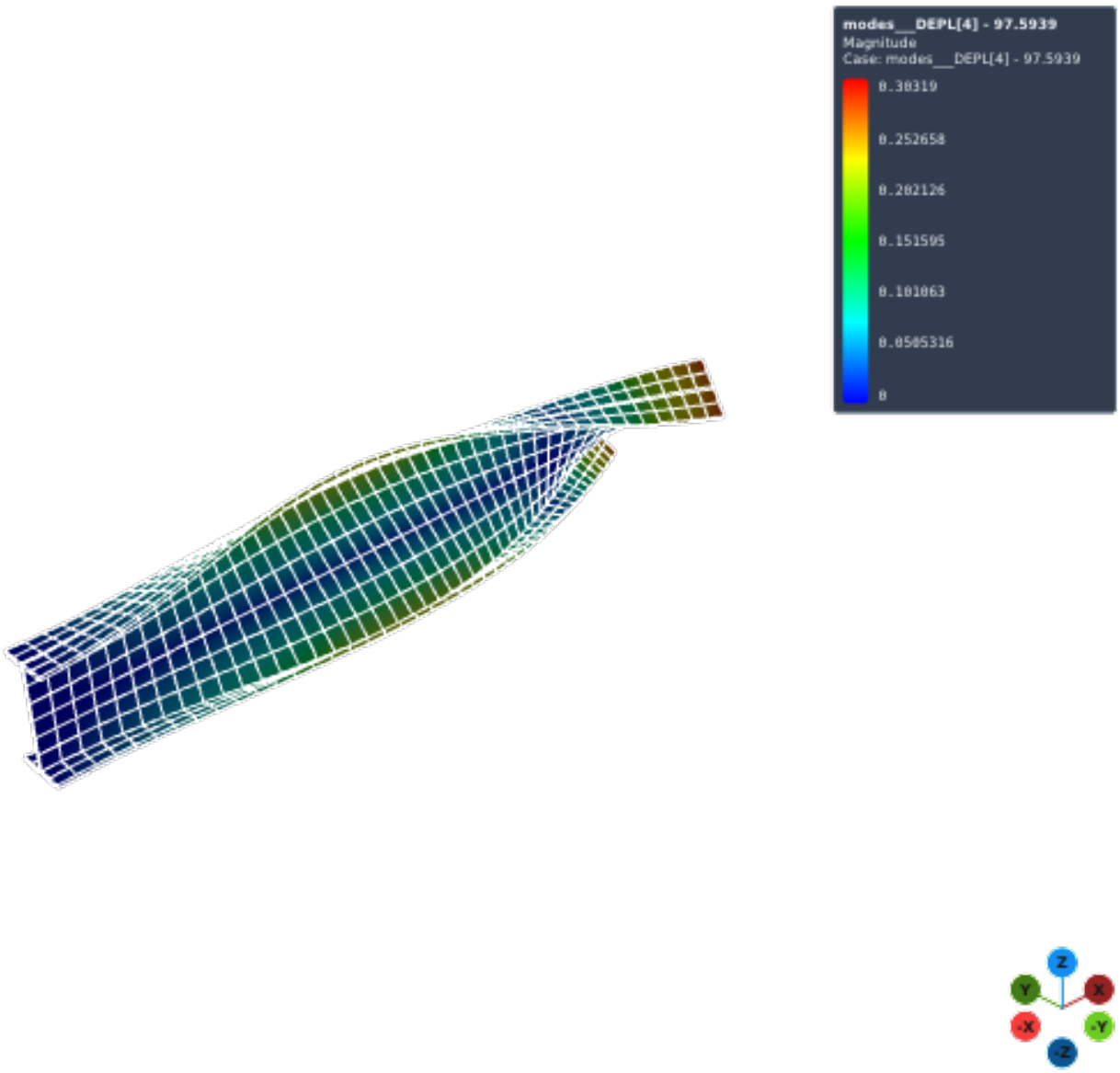

Figure 486: Code_Aster — mode 5, 97.594 Hz

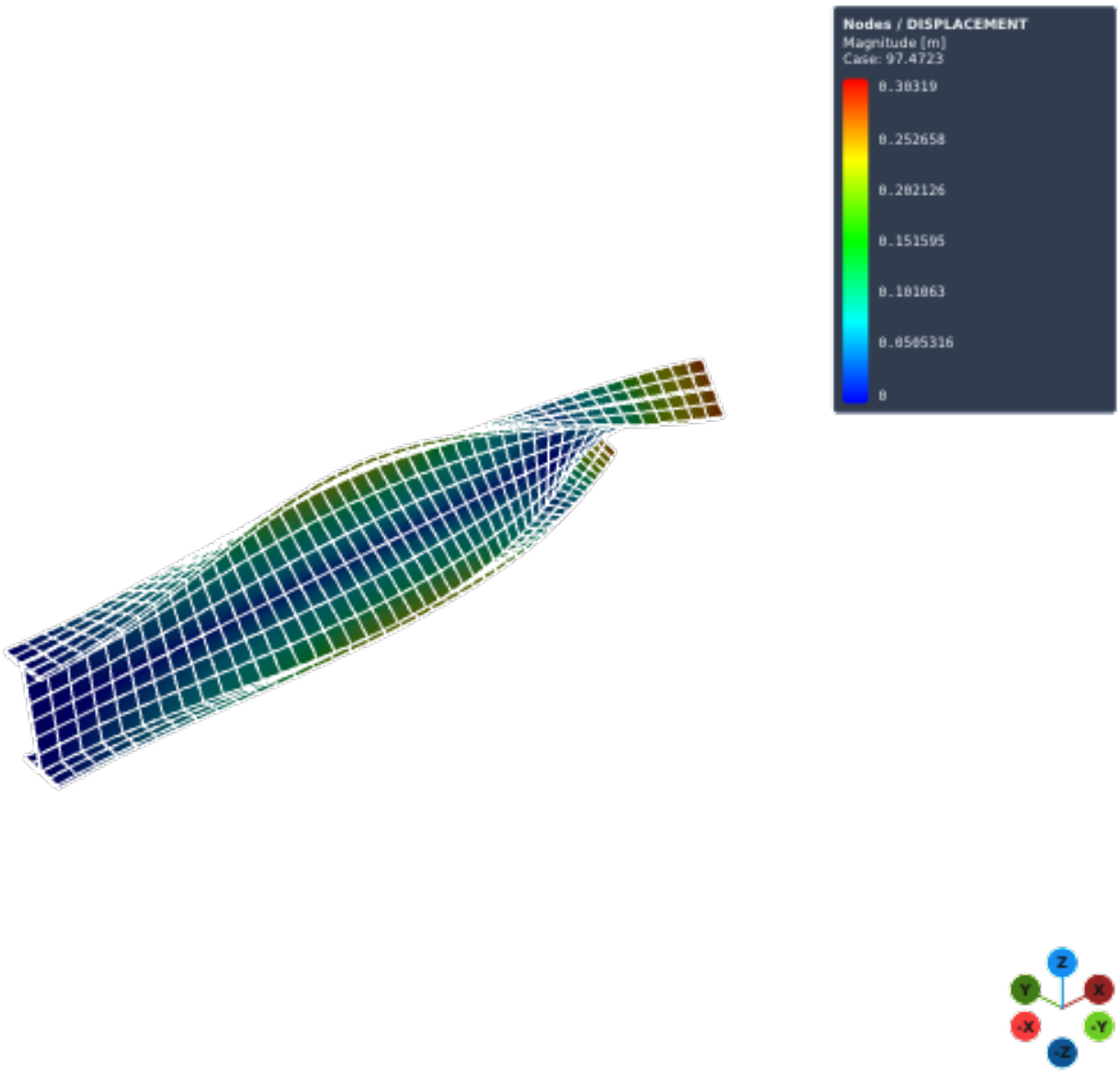

Figure 487: Sesam — mode 5, 97.472 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	97.594	+0.03	376015	1.21e-24	3.12e-19	1.62e-21	1.10e-12	-5.59e-10	-4.02e-11
Calculix	97.594	+0.03	376015	2.79e-24	6.84e-19	2.66e-21	1.67e-12	8.27e-10	-5.15e-11
Code_Aster	97.594	+0.03	376015	2.71e-27	7.32e-20	4.63e-21	-5.21e-14	-2.71e-10	6.80e-11
Sesam	97.472	-0.09	375078	2.82e-24	3.90e-18	4.03e-21	-1.68e-12	1.97e-09	-6.35e-11

Mode 6

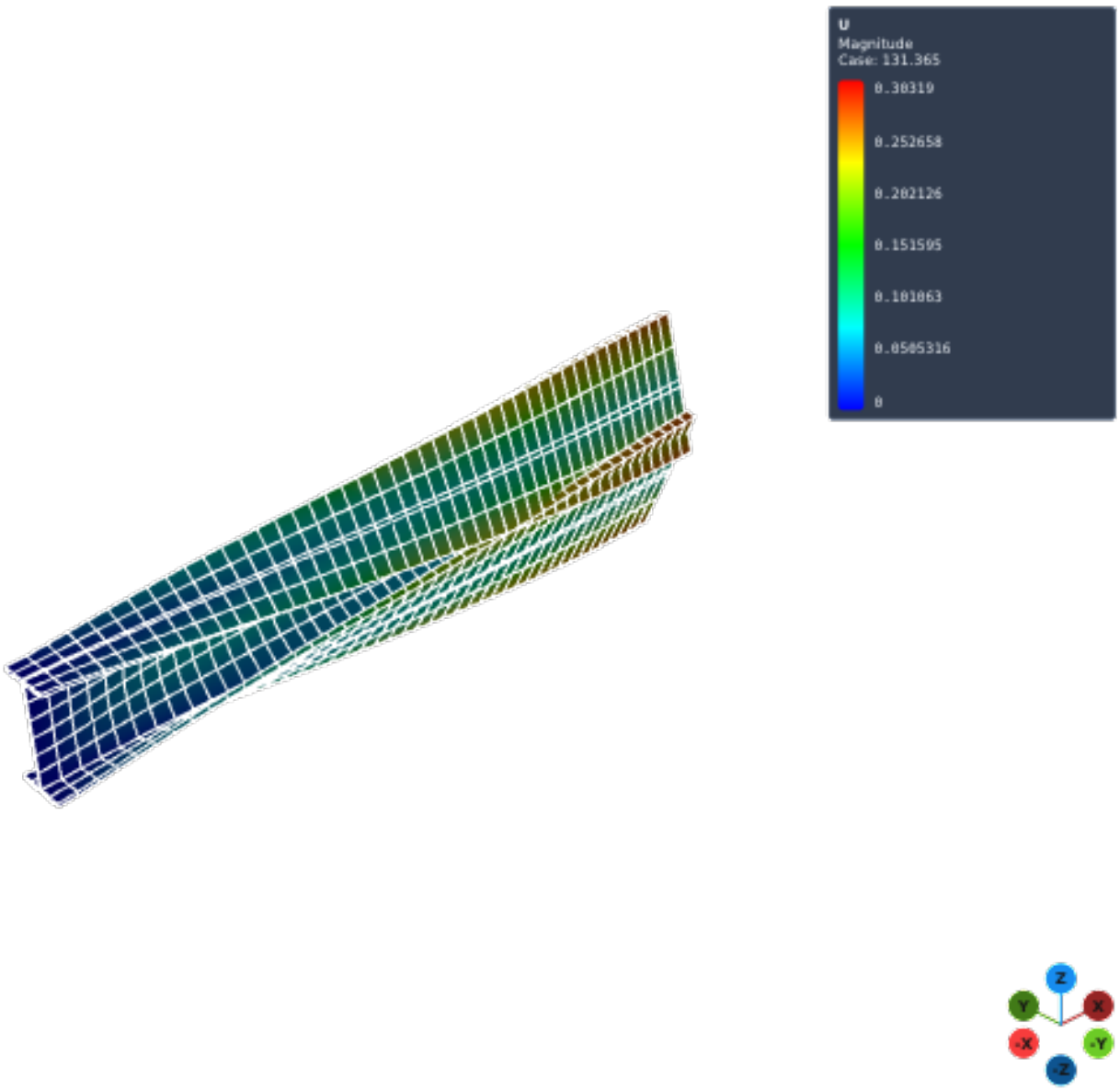

Figure 488: Abaqus — mode 6, 131.365 Hz

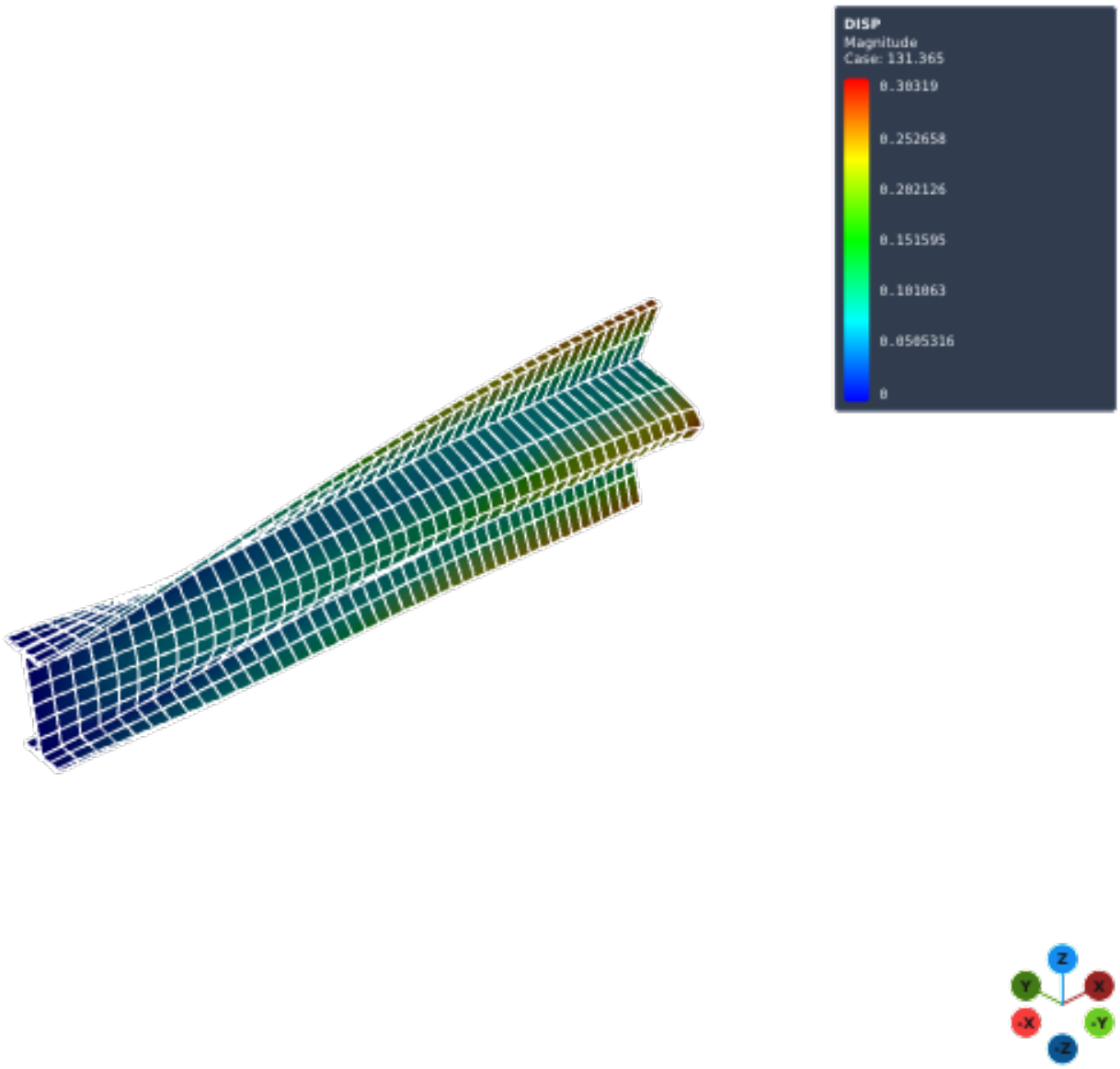

Figure 489: Calculix — mode 6, 131.365 Hz

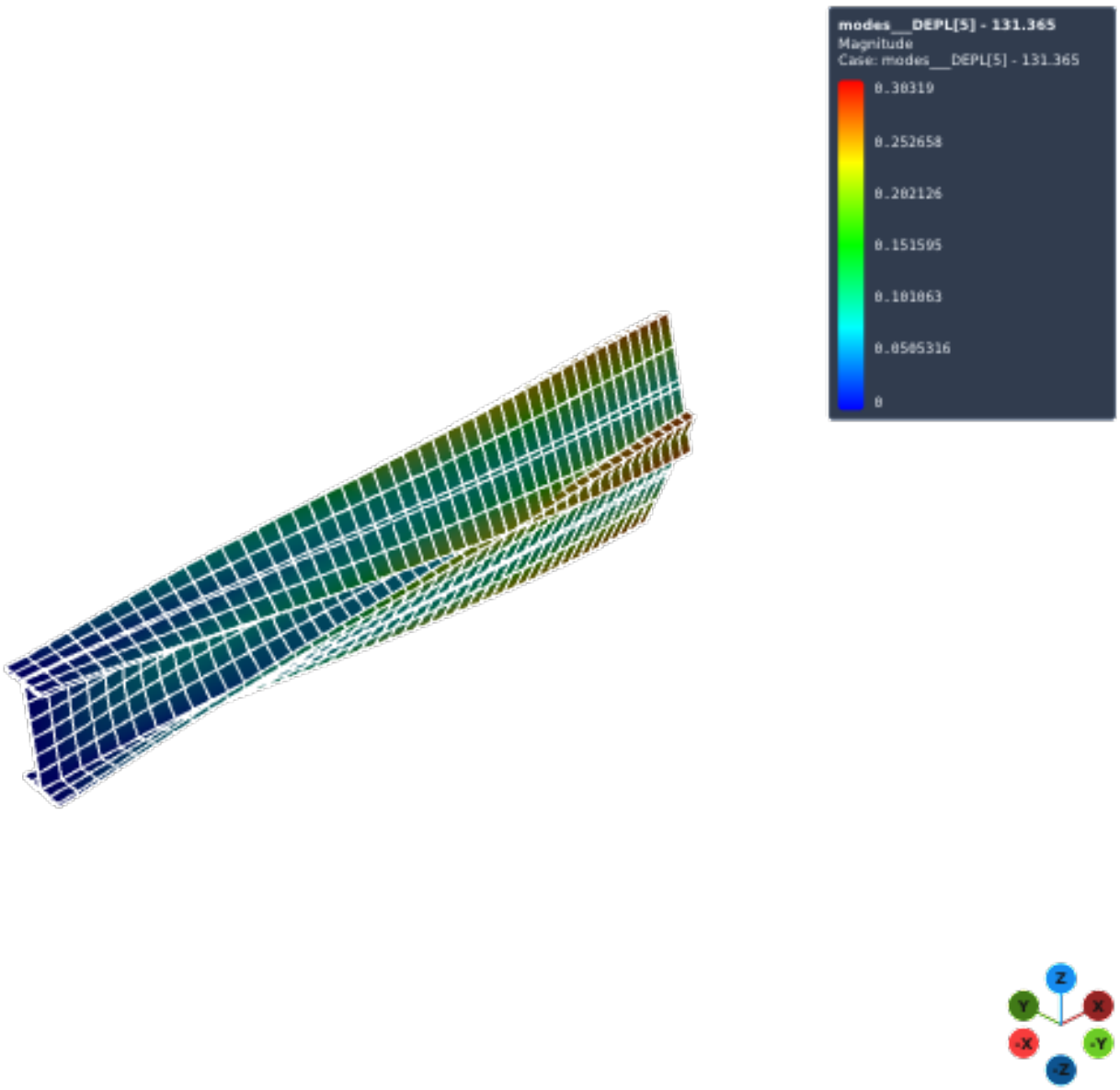

Figure 490: Code_Aster — mode 6, 131.365 Hz

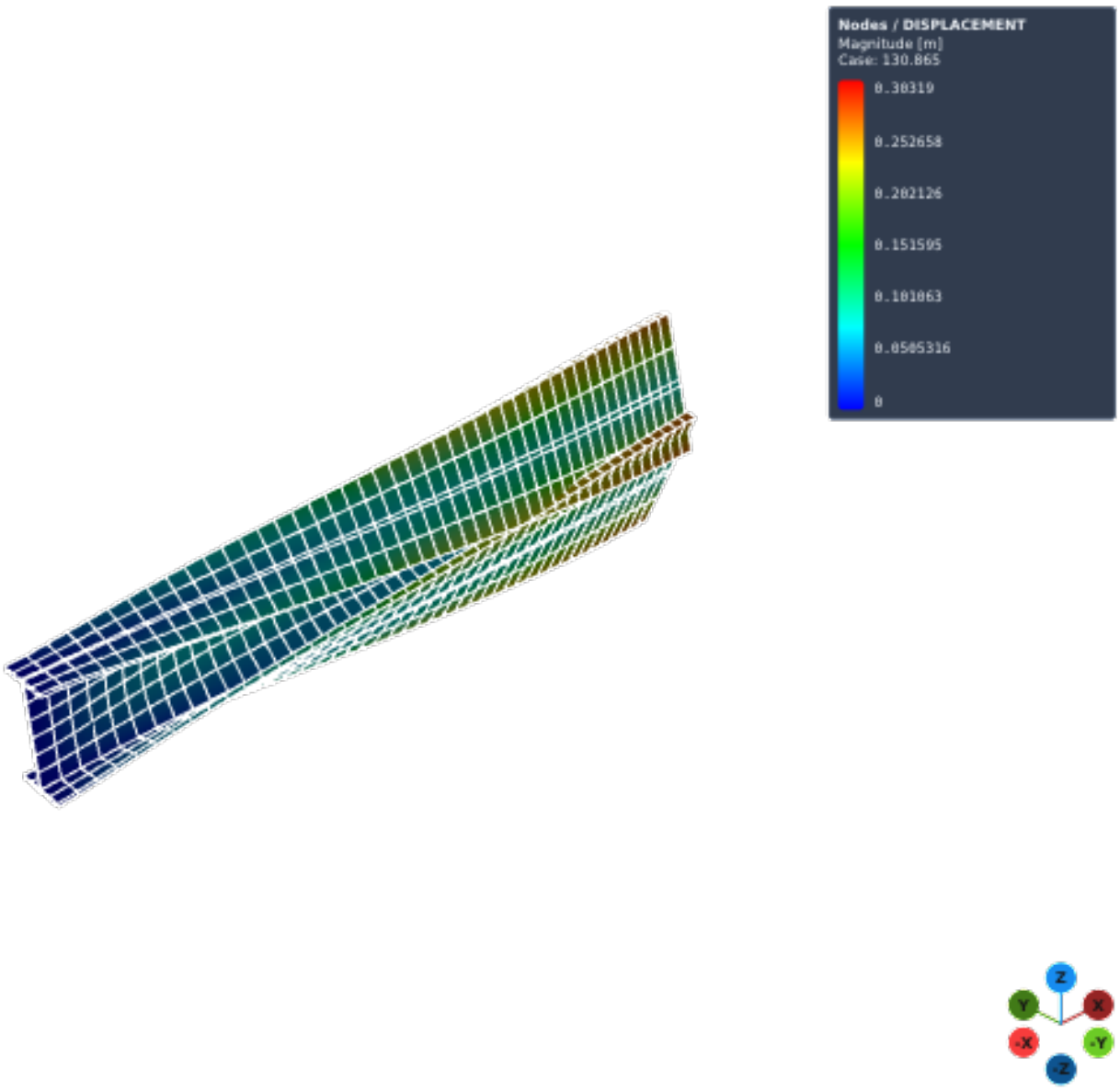

Figure 491: Sesam — mode 6, 130.865 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	131.370	+0.10	681273	2.05e-24	0.01	1.33e-19	1.43e-12	-0.103	-3.65e-10
Calculix	131.365	+0.09	681274	8.04e-24	0.01	6.41e-19	2.84e-12	0.103	8.01e-10
Code_Aster	131.365	+0.09	681273	1.58e-23	0.01	9.32e-21	-3.98e-12	-0.103	-9.65e-11
Sesam	130.865	-0.29	676090	4.18e-25	0.01	8.04e-22	6.46e-13	-0.096	2.84e-11

Mode 7

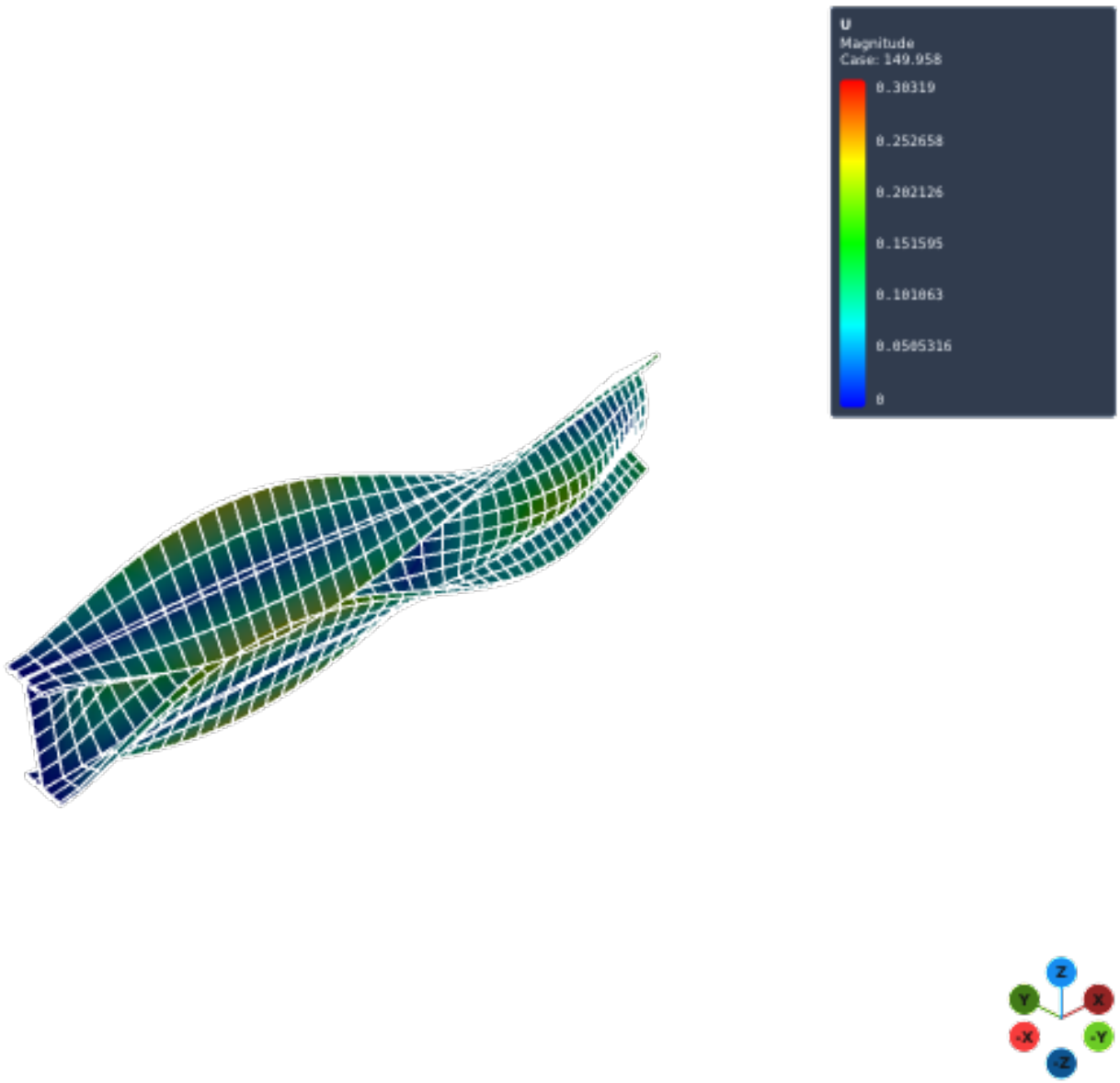

Figure 492: Abaqus — mode 7, 149.958 Hz

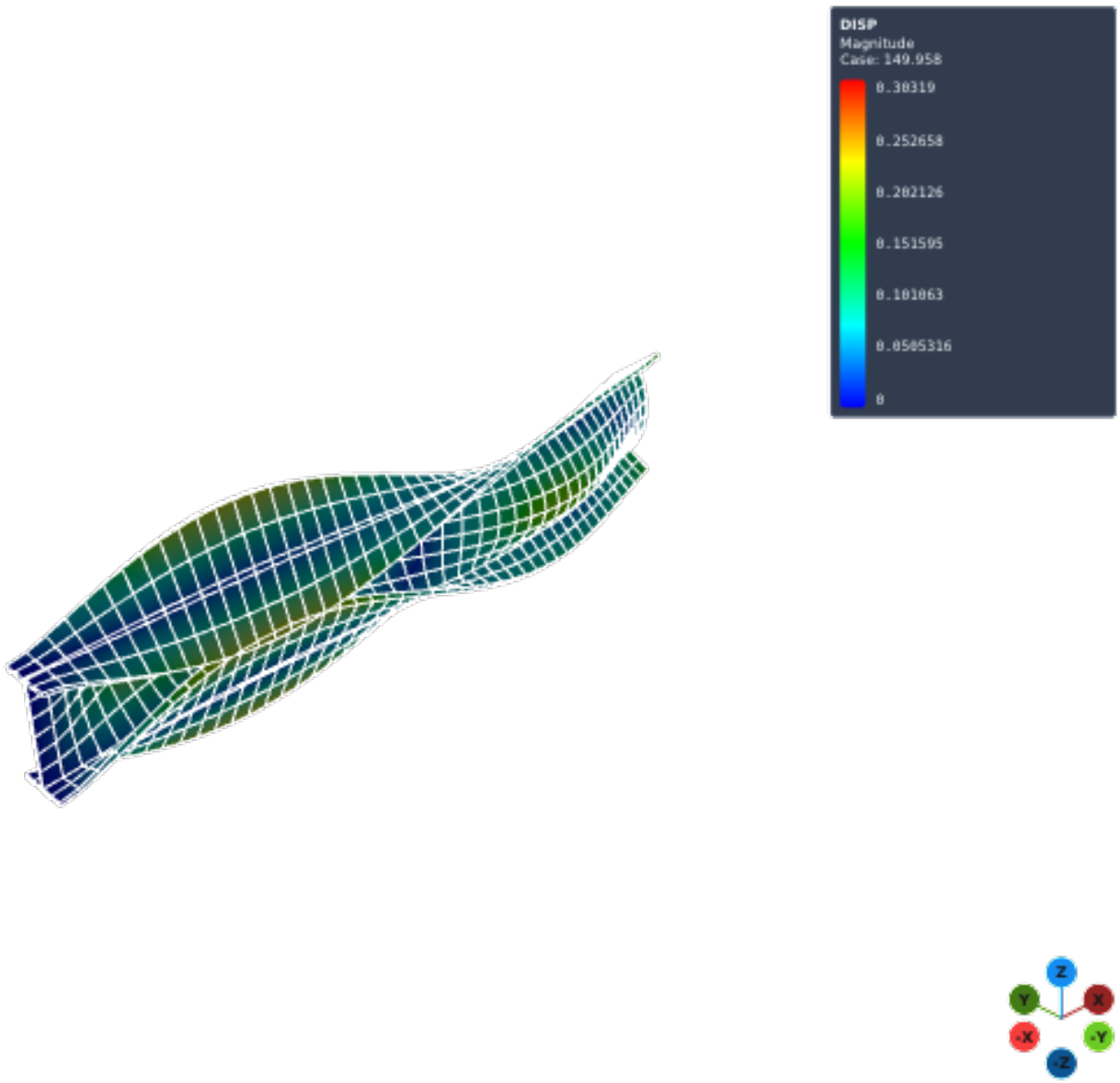

Figure 493: Calculix — mode 7, 149.958 Hz

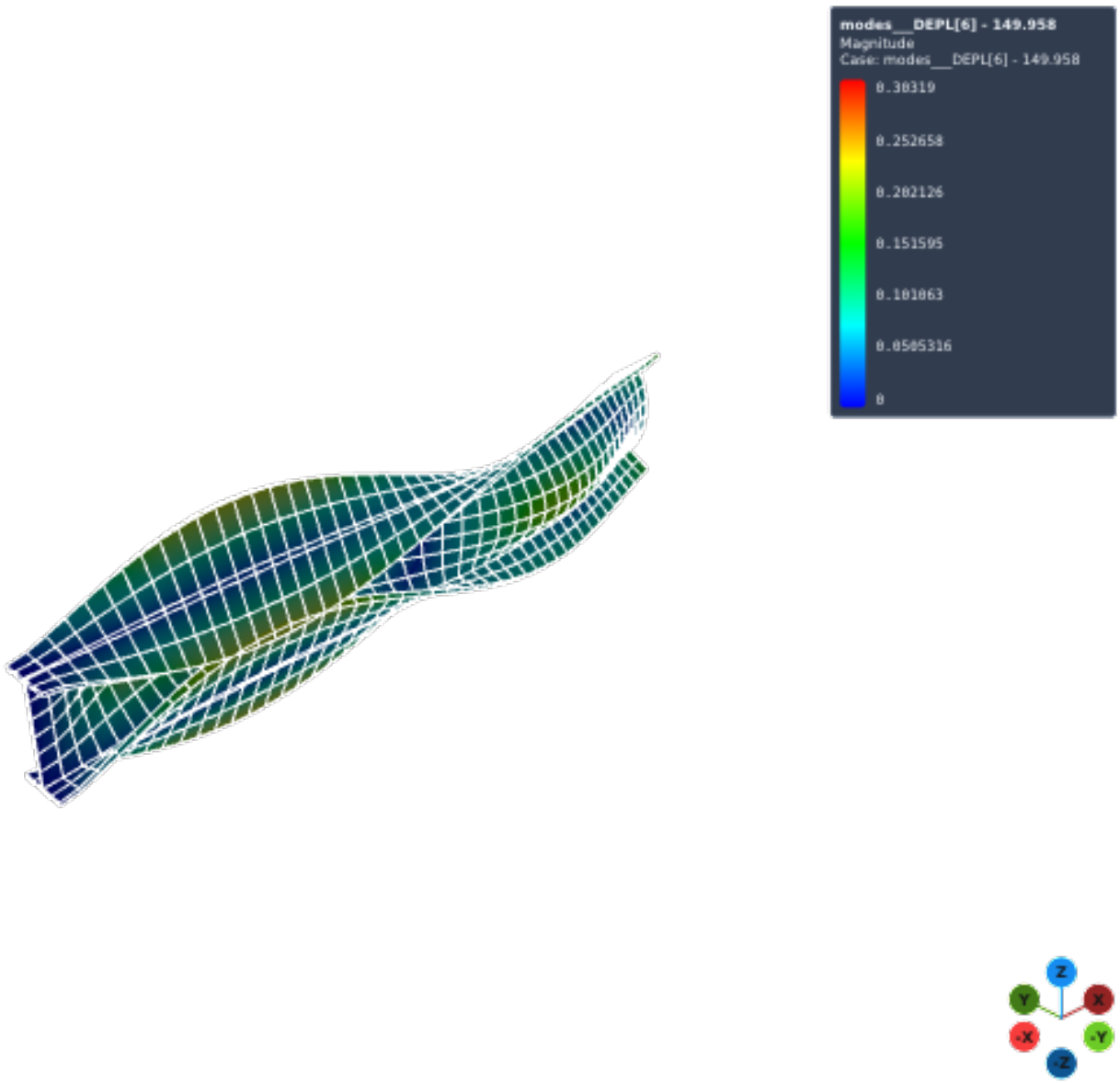

Figure 494: Code_Aster — mode 7, 149.958 Hz

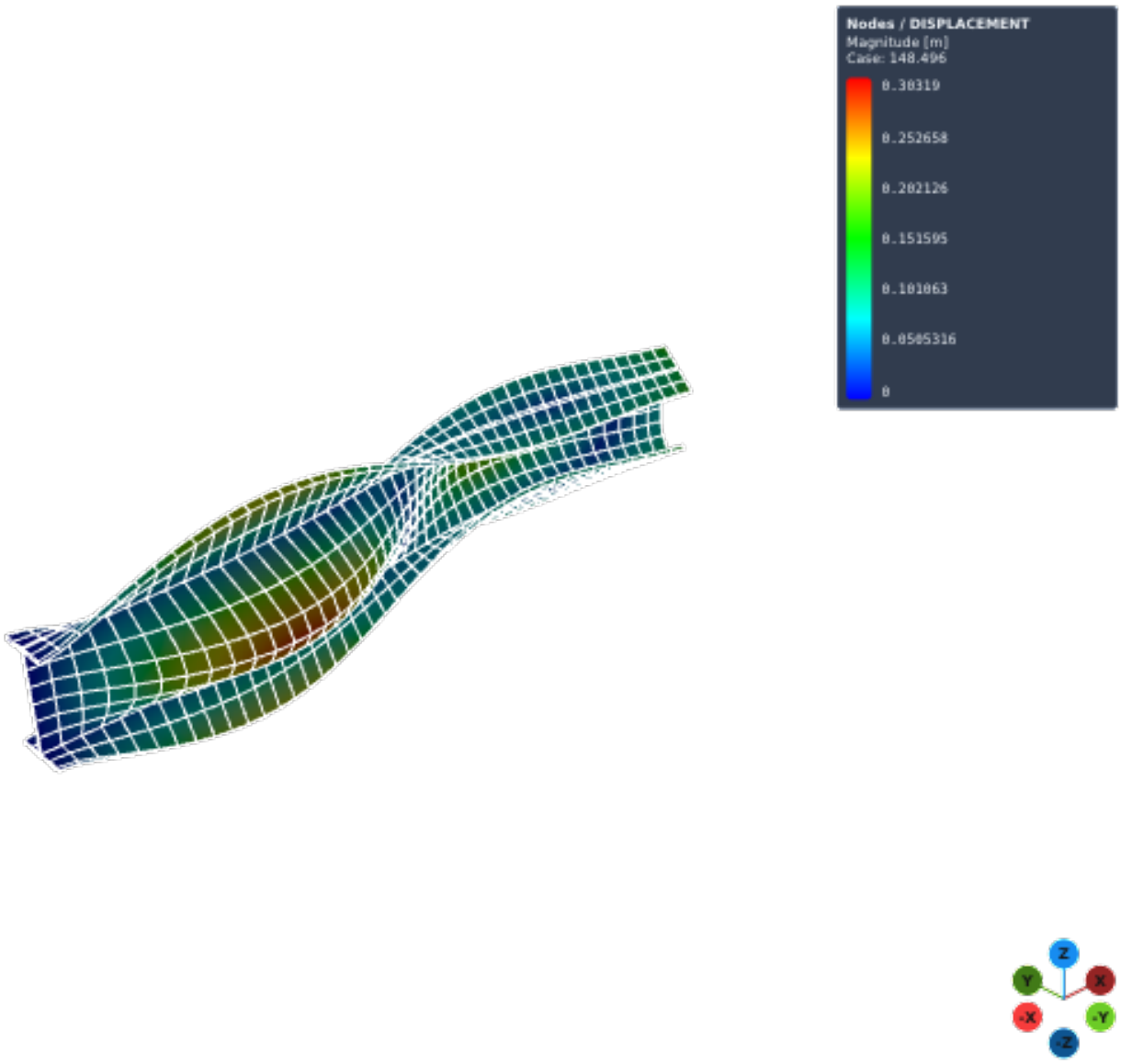

Figure 495: Sesam — mode 7, 148.496 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	149.960	+0.25	887766	2.76e-24	3.31	8.83e-21	1.66e-12	1.818	9.40e-11
Calculix	149.958	+0.24	887766	1.74e-27	3.31	4.35e-20	-4.17e-14	1.818	2.08e-10
Code_Aster	149.958	+0.24	887766	1.23e-24	3.31	2.03e-22	1.11e-12	1.818	1.43e-11
Sesam	148.496	-0.73	870539	8.20e-26	3.38	3.30e-22	2.86e-13	-1.839	1.82e-11

Mode 8

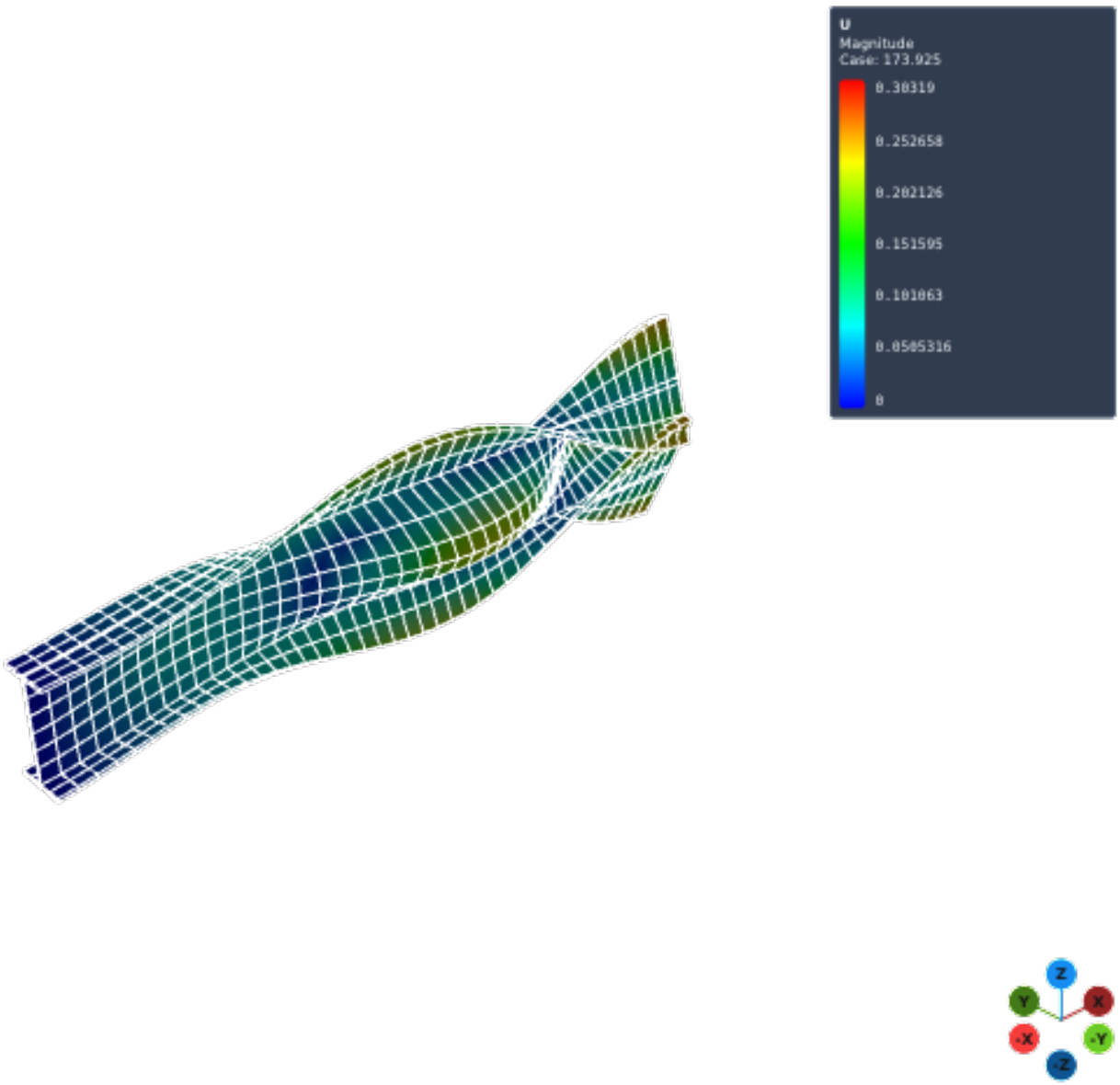

Figure 496: Abaqus — mode 8, 173.925 Hz

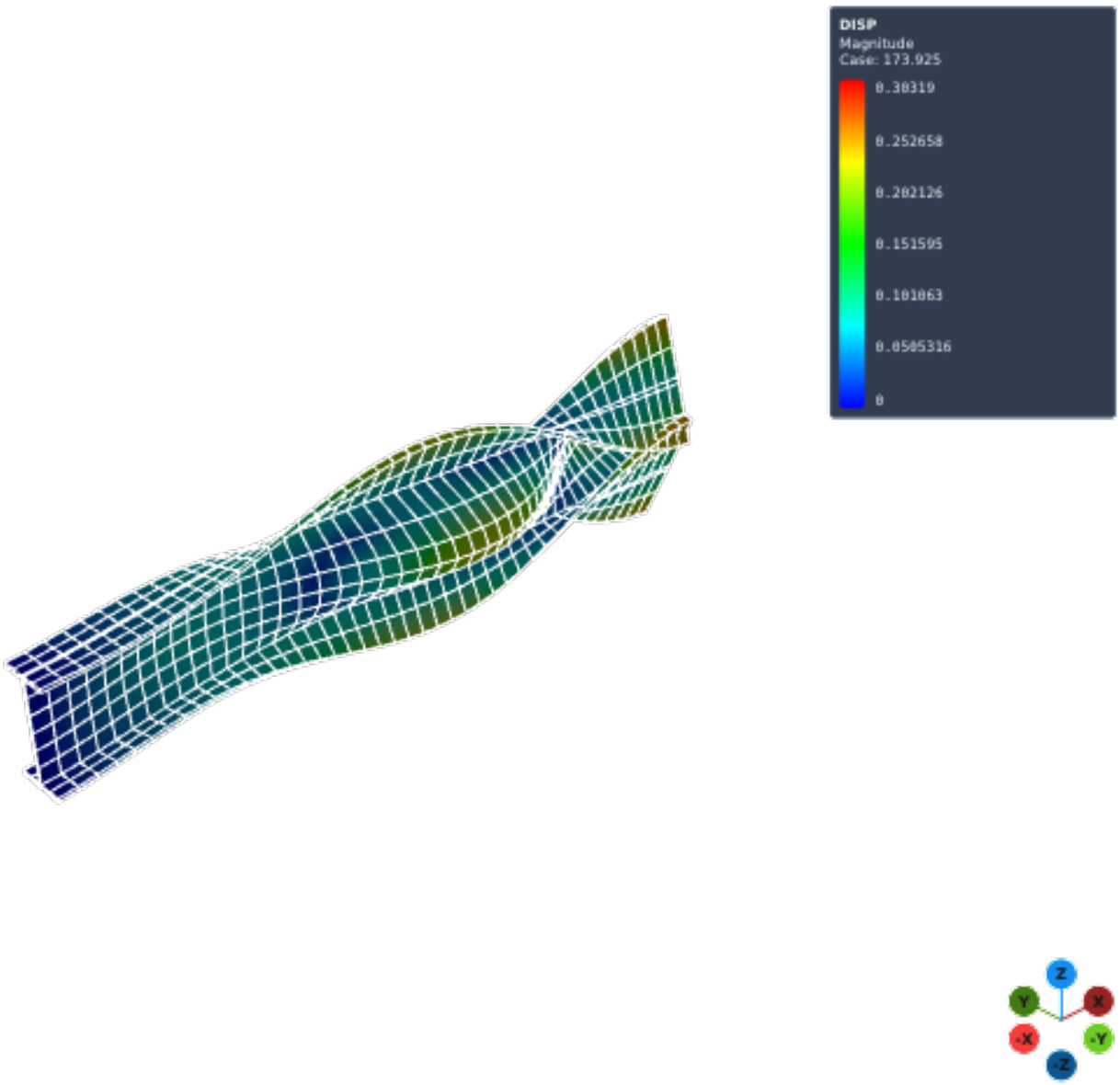

Figure 497: Calculix — mode 8, 173.925 Hz

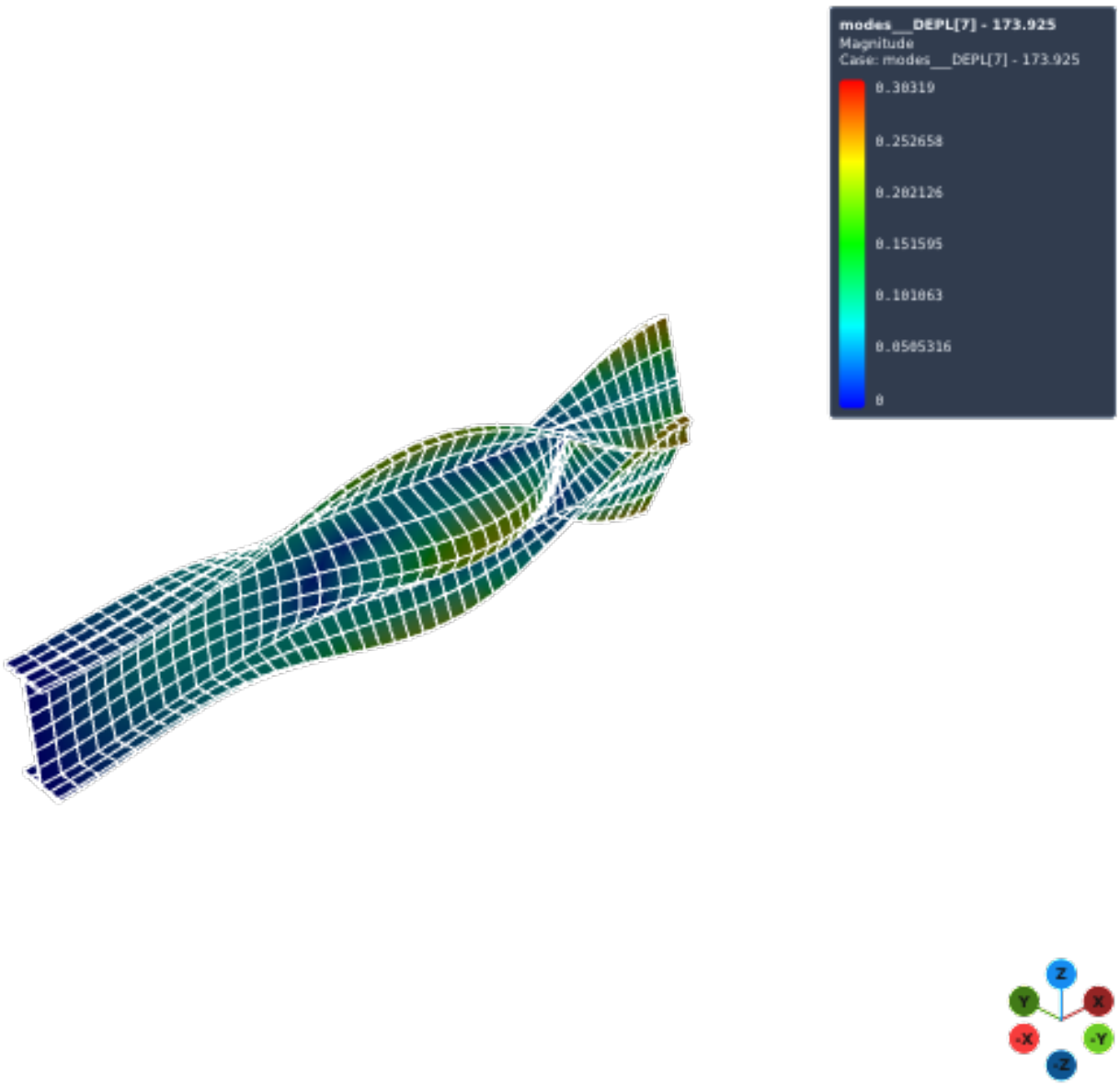

Figure 498: Code_Aster — mode 8, 173.925 Hz

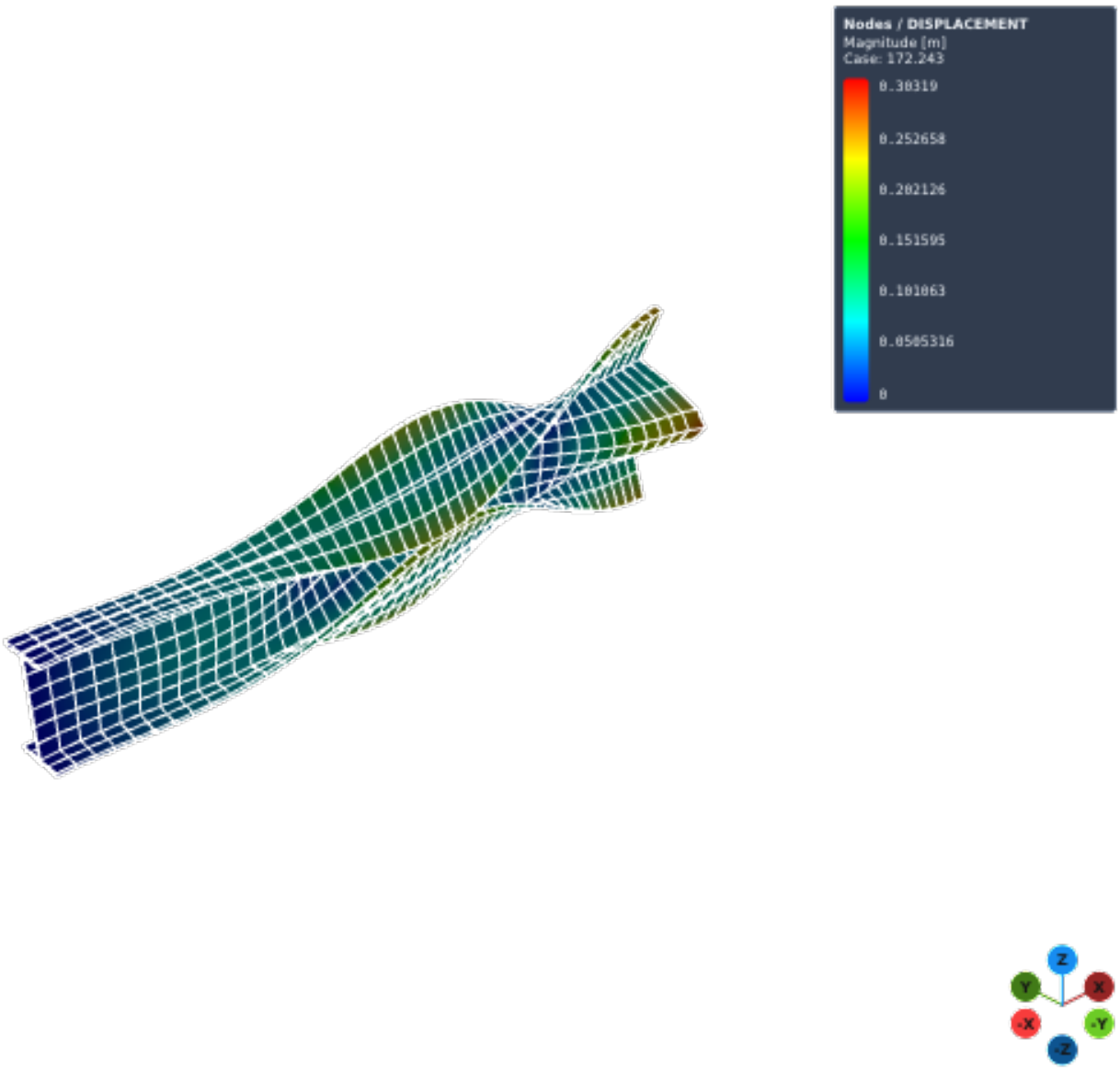

Figure 499: Sesam — mode 8, 172.243 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	173.920	+0.24	1.19422e+06	4.36e-26	5.42	1.76e-20	2.09e-13	2.328	-1.33e-10
Calculix	173.925	+0.24	1.19422e+06	4.25e-25	5.42	4.87e-20	-6.52e-13	2.328	-2.21e-10
Code_Aster	173.925	+0.24	1.19422e+06	2.37e-25	5.42	2.61e-23	-4.87e-13	2.328	-5.11e-12
Sesam	172.243	-0.73	1.17124e+06	1.58e-26	5.31	8.78e-22	-1.26e-13	-2.305	-2.96e-11

Mode 9

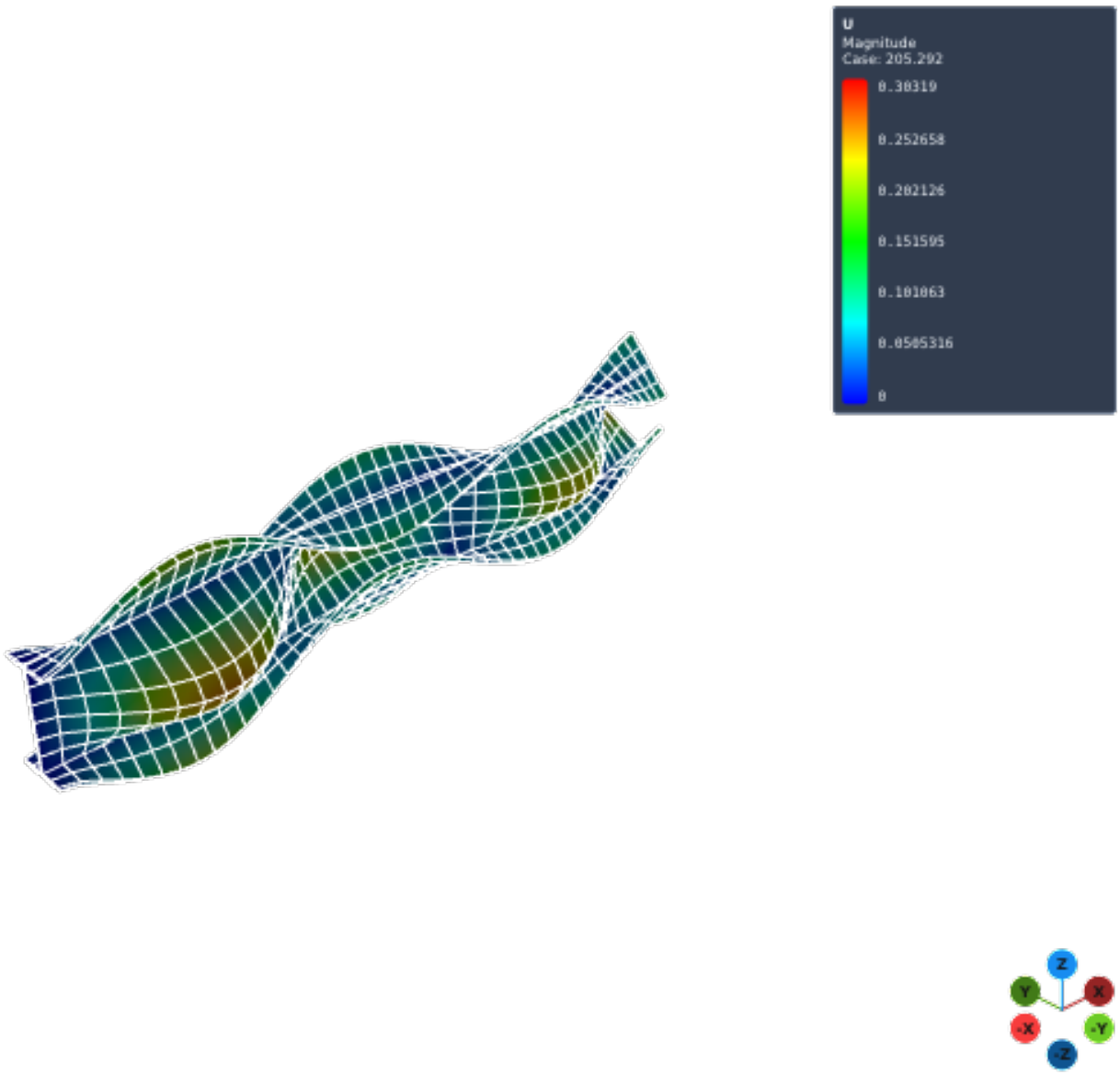

Figure 500: Abaqus — mode 9, 205.292 Hz

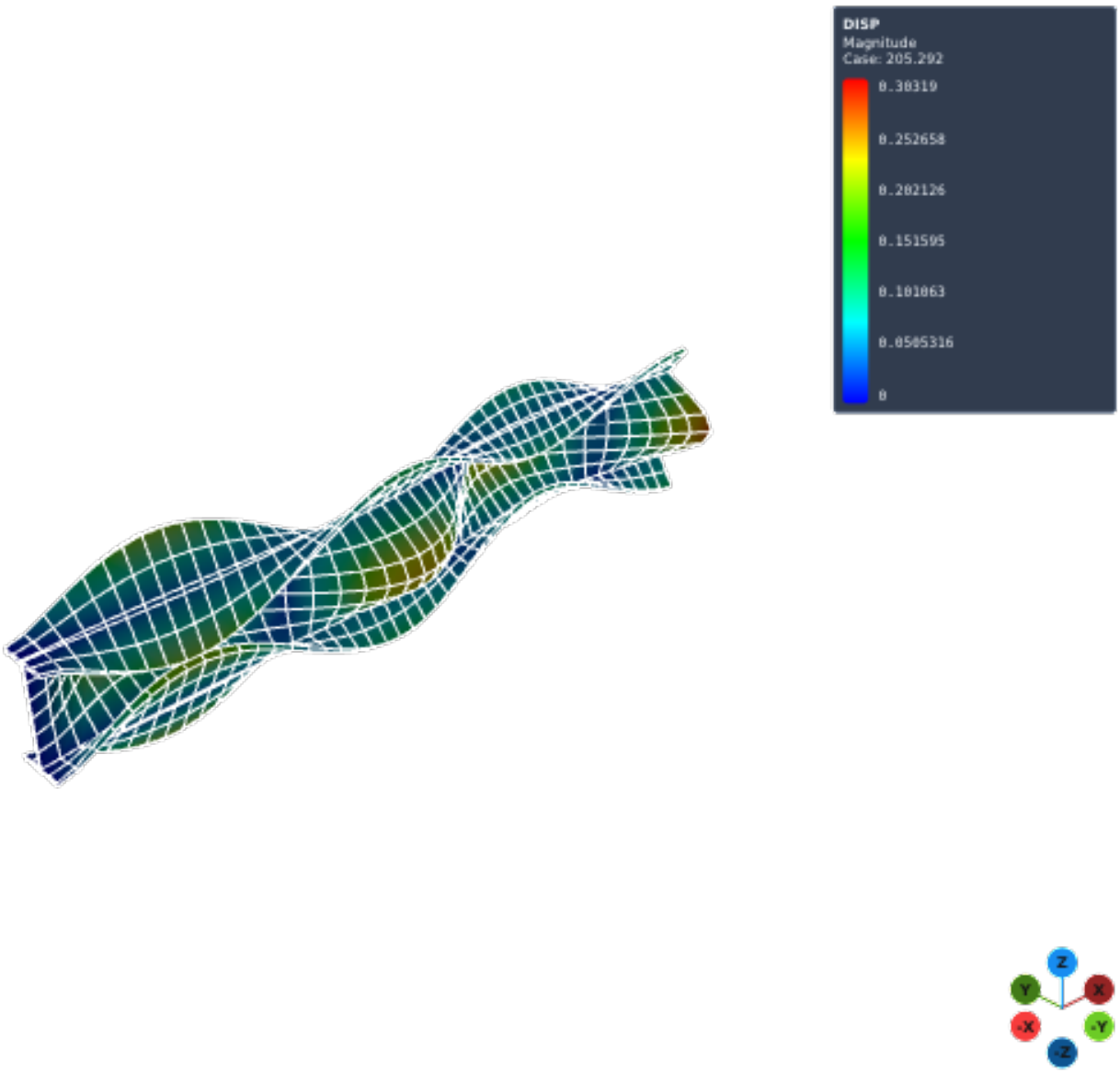

Figure 501: Calculix — mode 9, 205.292 Hz

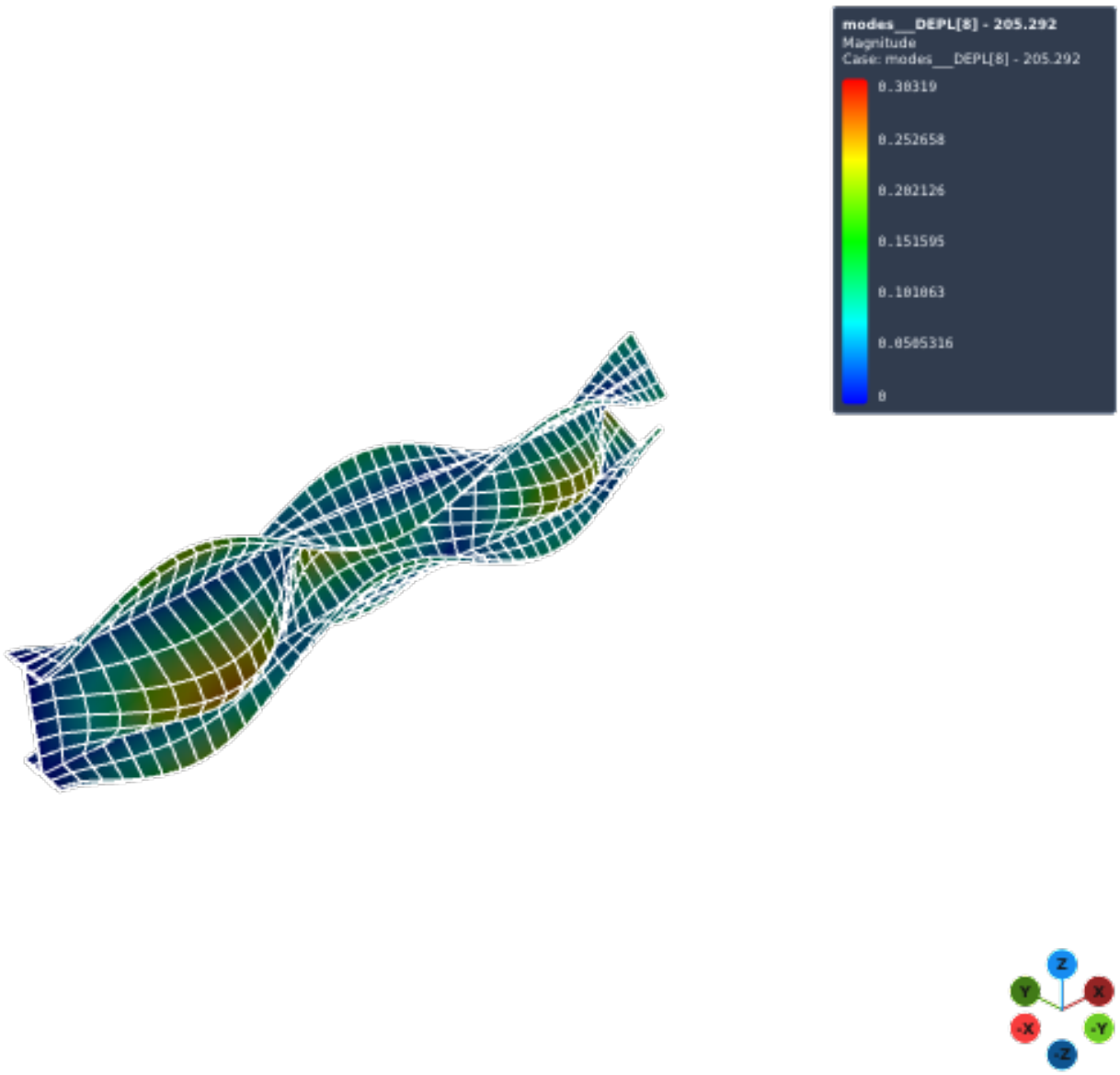

Figure 502: Code_Aster — mode 9, 205.292 Hz

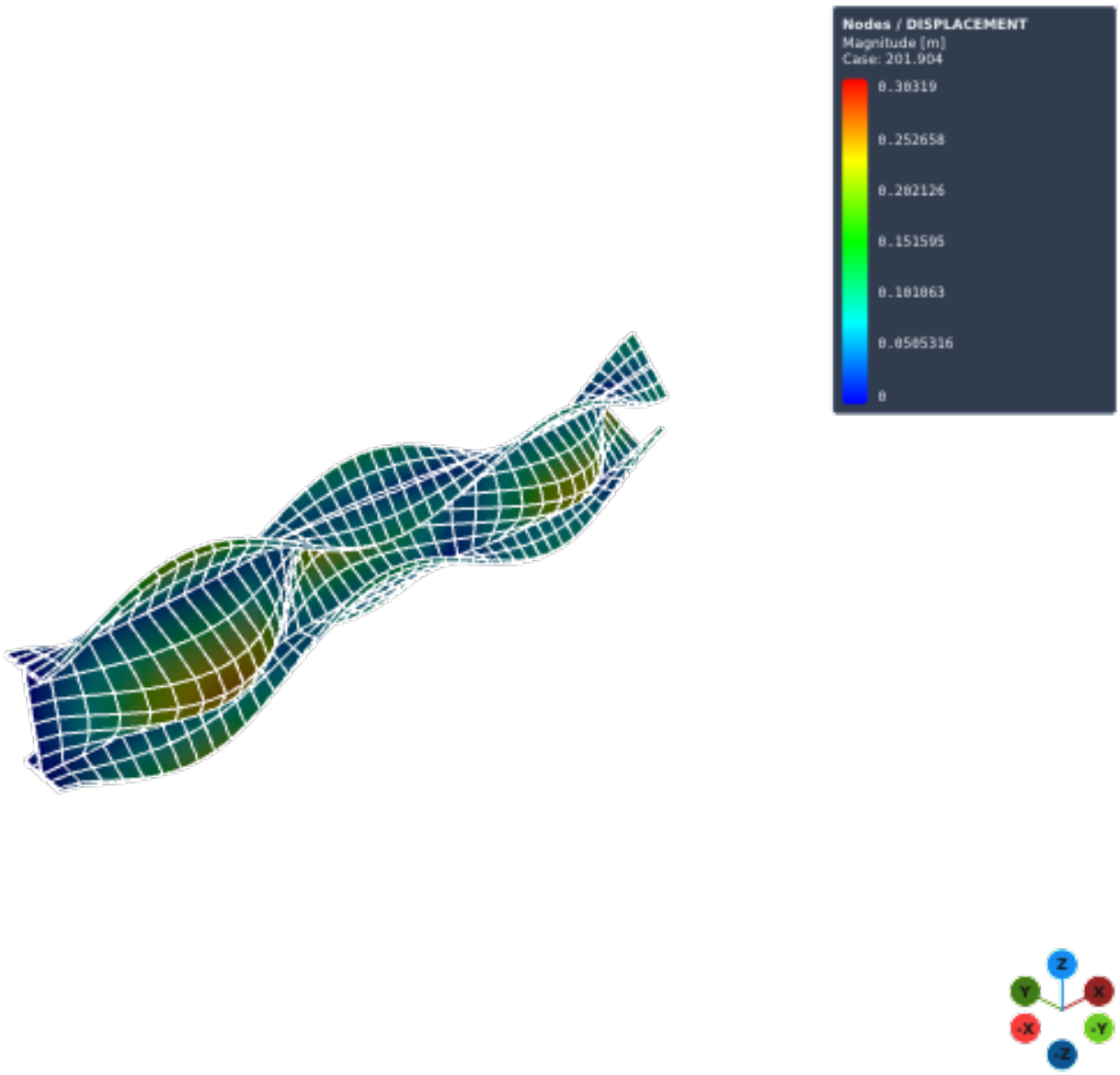

Figure 503: Sesam — mode 9, 201.904 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	205.290	+0.41	1.66381e+06	4.78e-27	0.52	2.91e-22	6.91e-14	-0.719	1.71e-11
Calculix	205.292	+0.41	1.6638e+06	5.77e-25	0.52	1.64e-21	-7.59e-13	0.720	4.04e-11
Code_Aster	205.292	+0.41	1.66381e+06	7.17e-24	0.52	1.15e-22	2.68e-12	-0.719	-1.07e-11
Sesam	201.904	-1.24	1.60935e+06	5.31e-25	0.56	3.60e-22	-7.28e-13	-0.746	1.90e-11

Mode 10

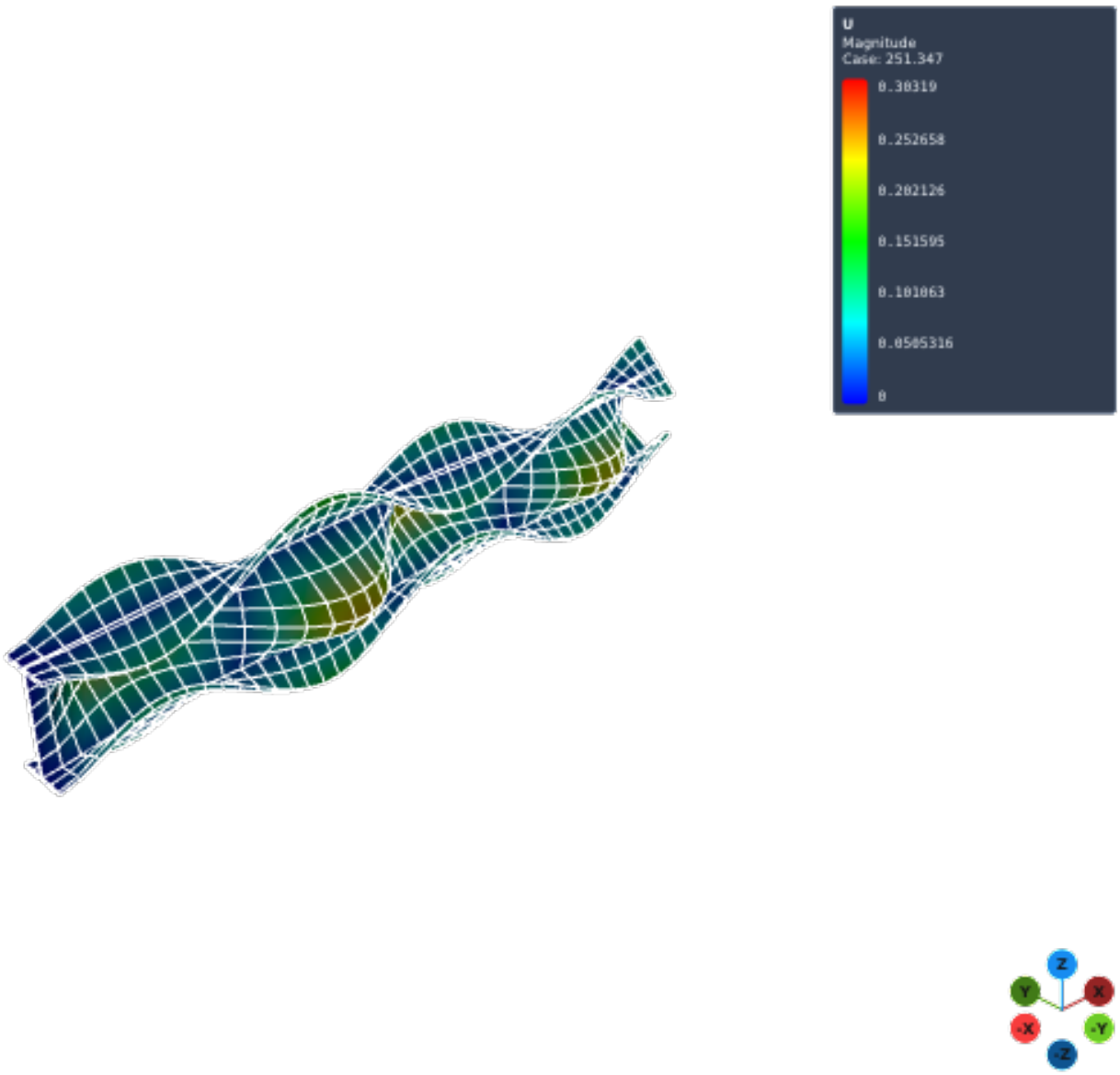

Figure 504: Abaqus — mode 10, 251.347 Hz

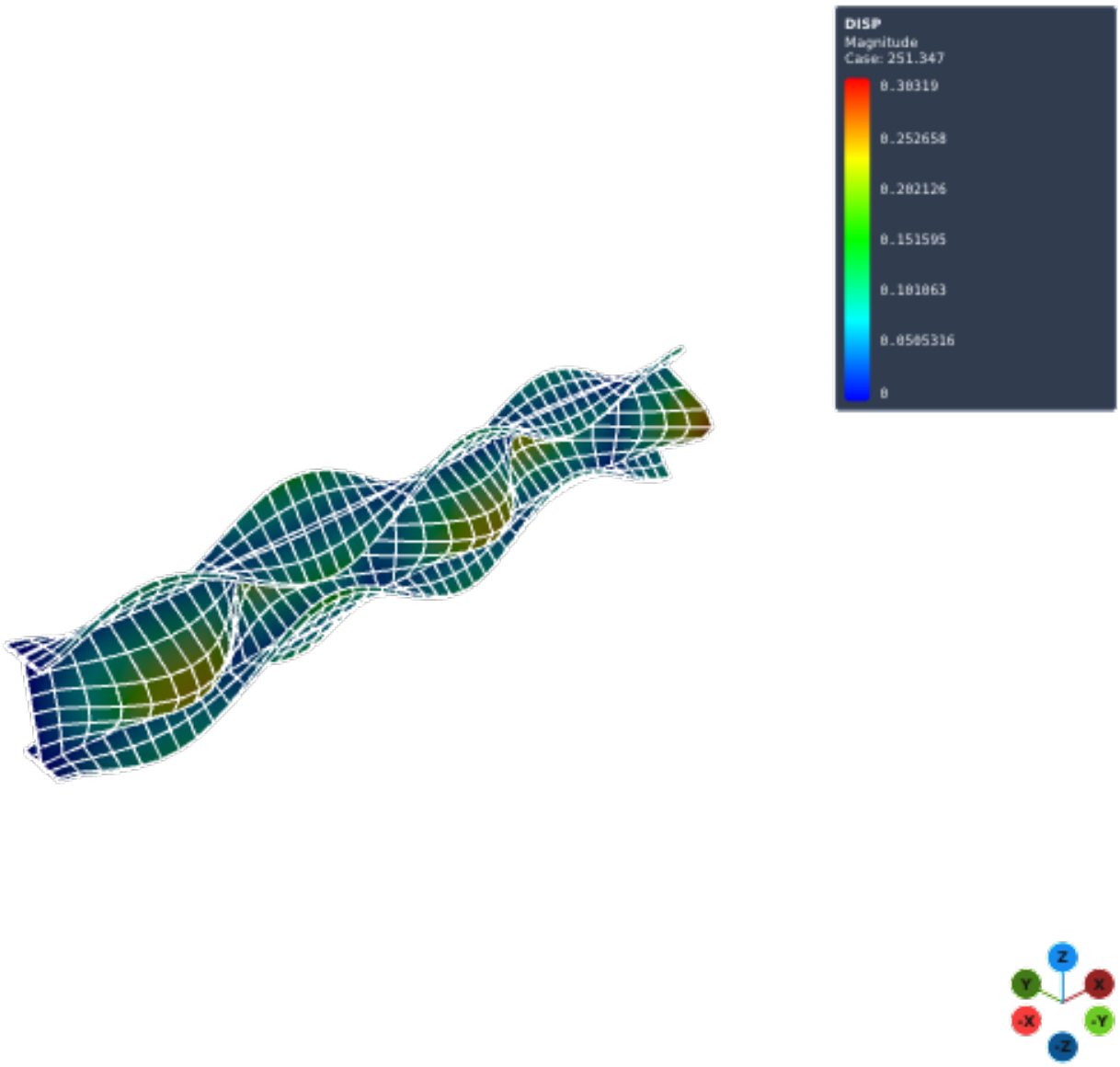

Figure 505: Calculix — mode 10, 251.347 Hz

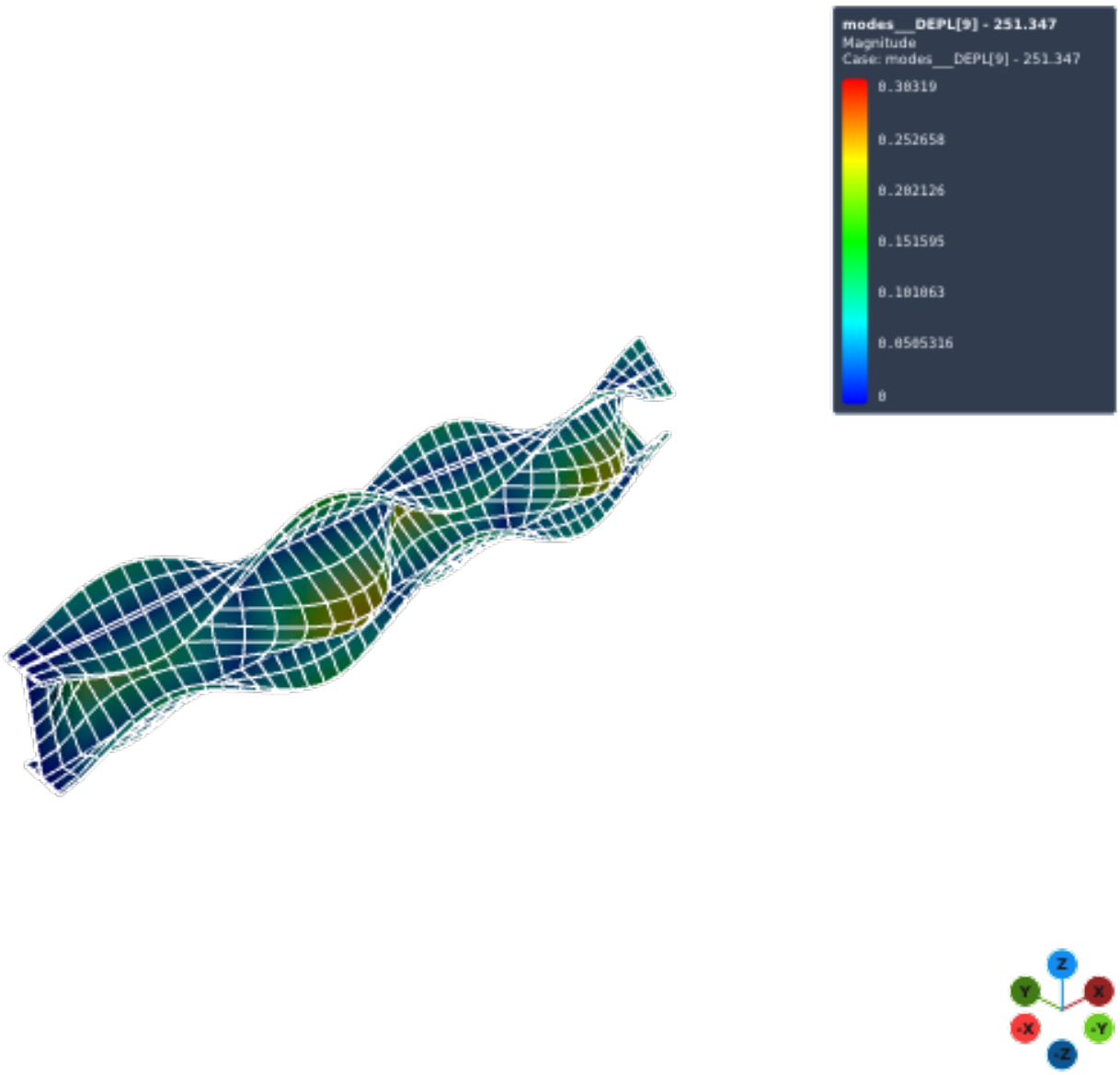

Figure 506: Code_Aster — mode 10, 251.347 Hz

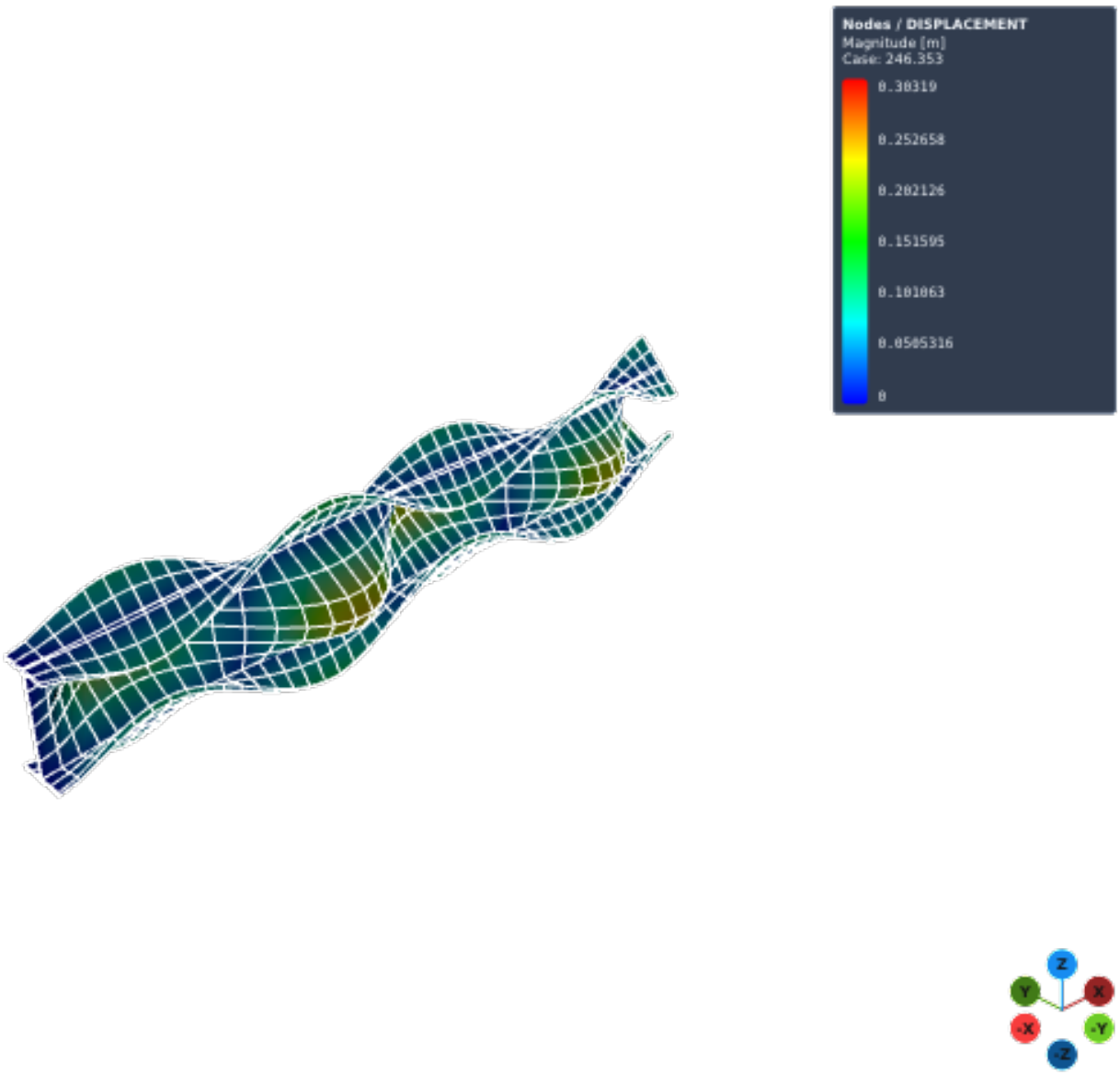

Figure 507: Sesam — mode 10, 246.353 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	251.350	+0.50	2.49407e+06	1.08e-21	1.60	3.13e-19	3.29e-11	1.267	-5.59e-10
Calculix	251.347	+0.50	2.49406e+06	1.19e-25	1.61	7.95e-19	-3.45e-13	-1.267	8.92e-10
Code_Aster	251.347	+0.50	2.49407e+06	7.51e-25	1.60	1.30e-18	-8.67e-13	1.267	-1.14e-09
Sesam	246.353	-1.50	2.39594e+06	1.32e-18	1.47	7.64e-19	-1.15e-09	1.211	8.74e-10

Mode 11

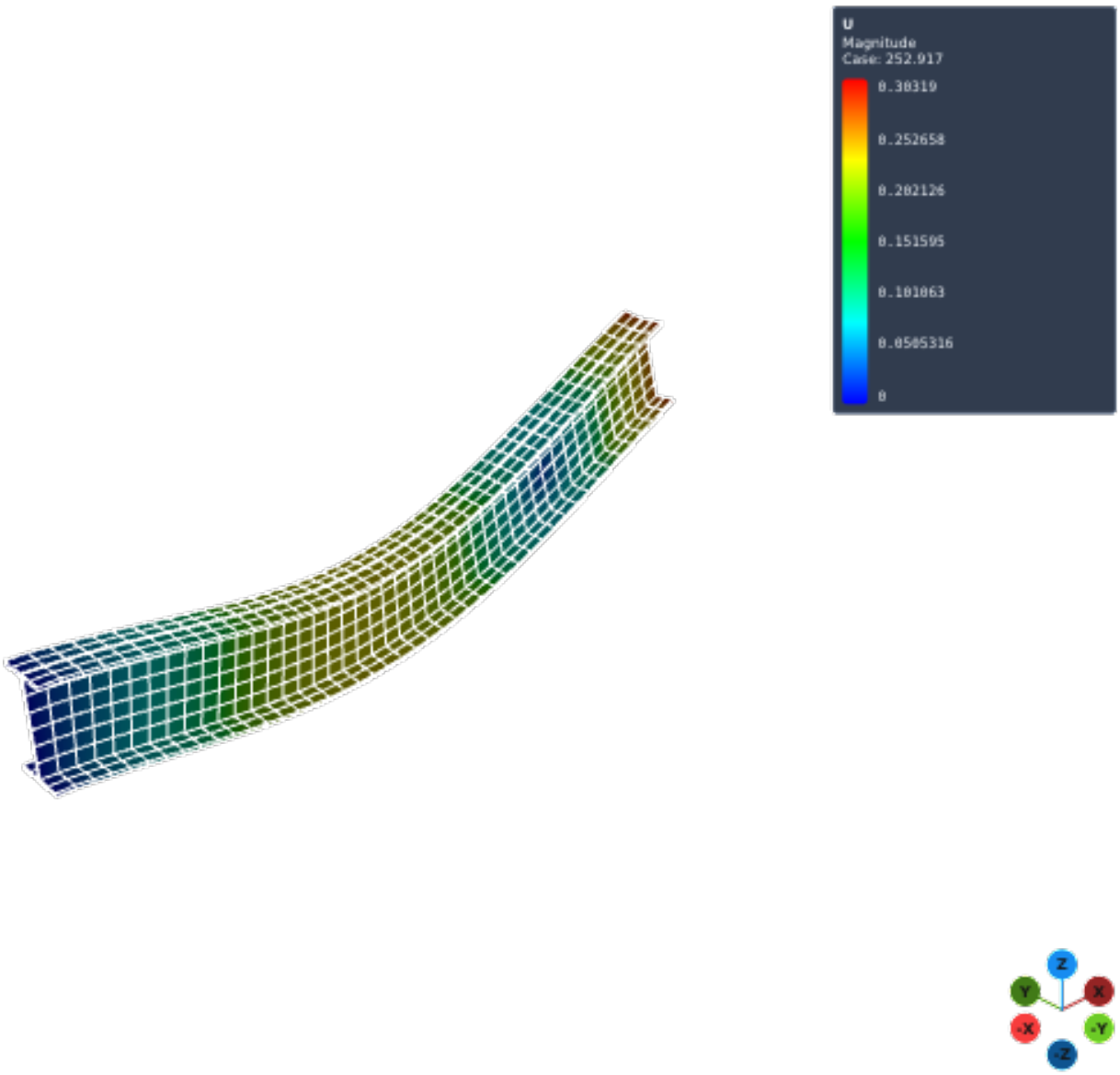

Figure 508: Abaqus — mode 11, 252.917 Hz

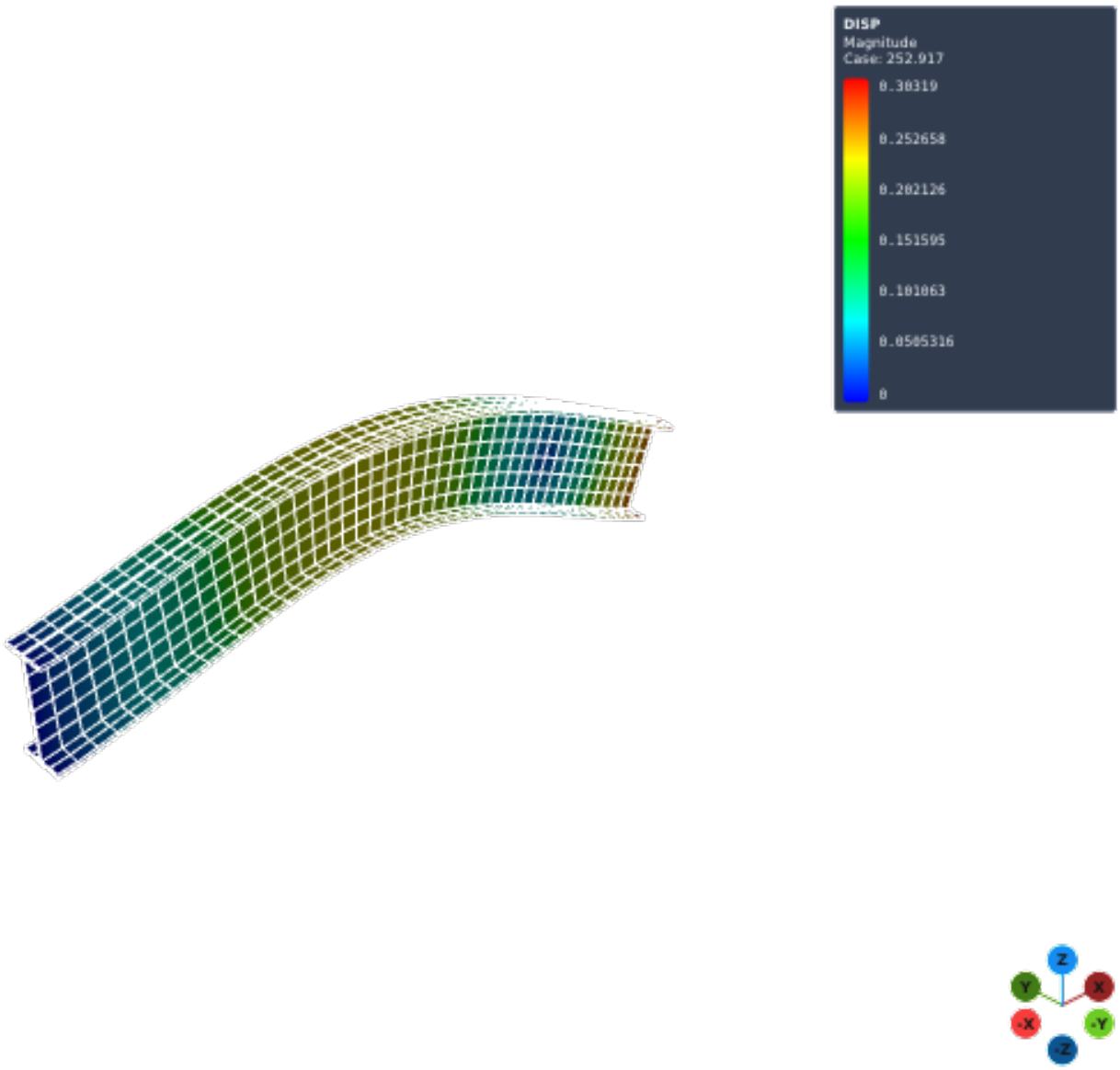

Figure 509: Calculix — mode 11, 252.917 Hz

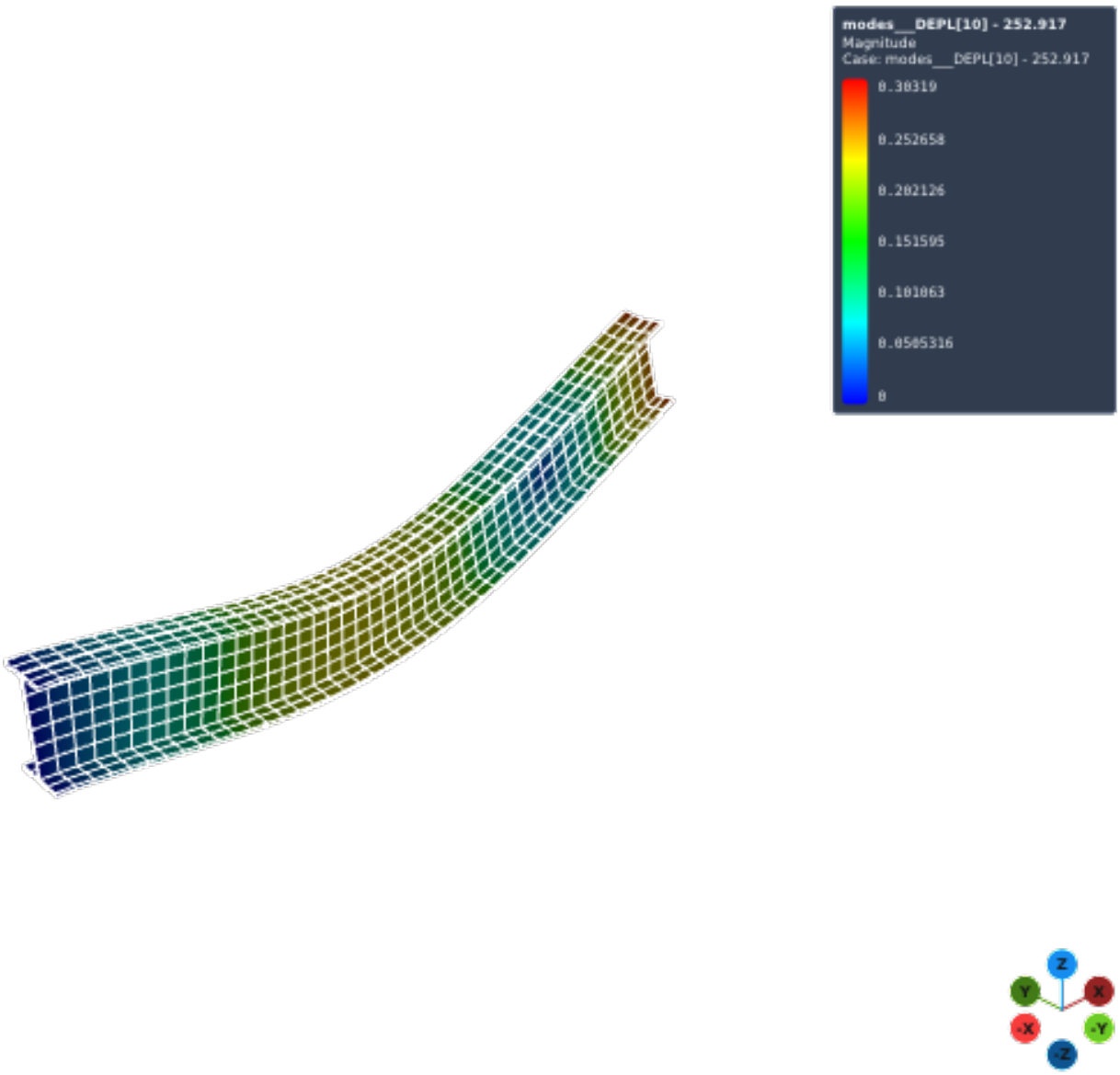

Figure 510: Code_Aster — mode 11, 252.917 Hz

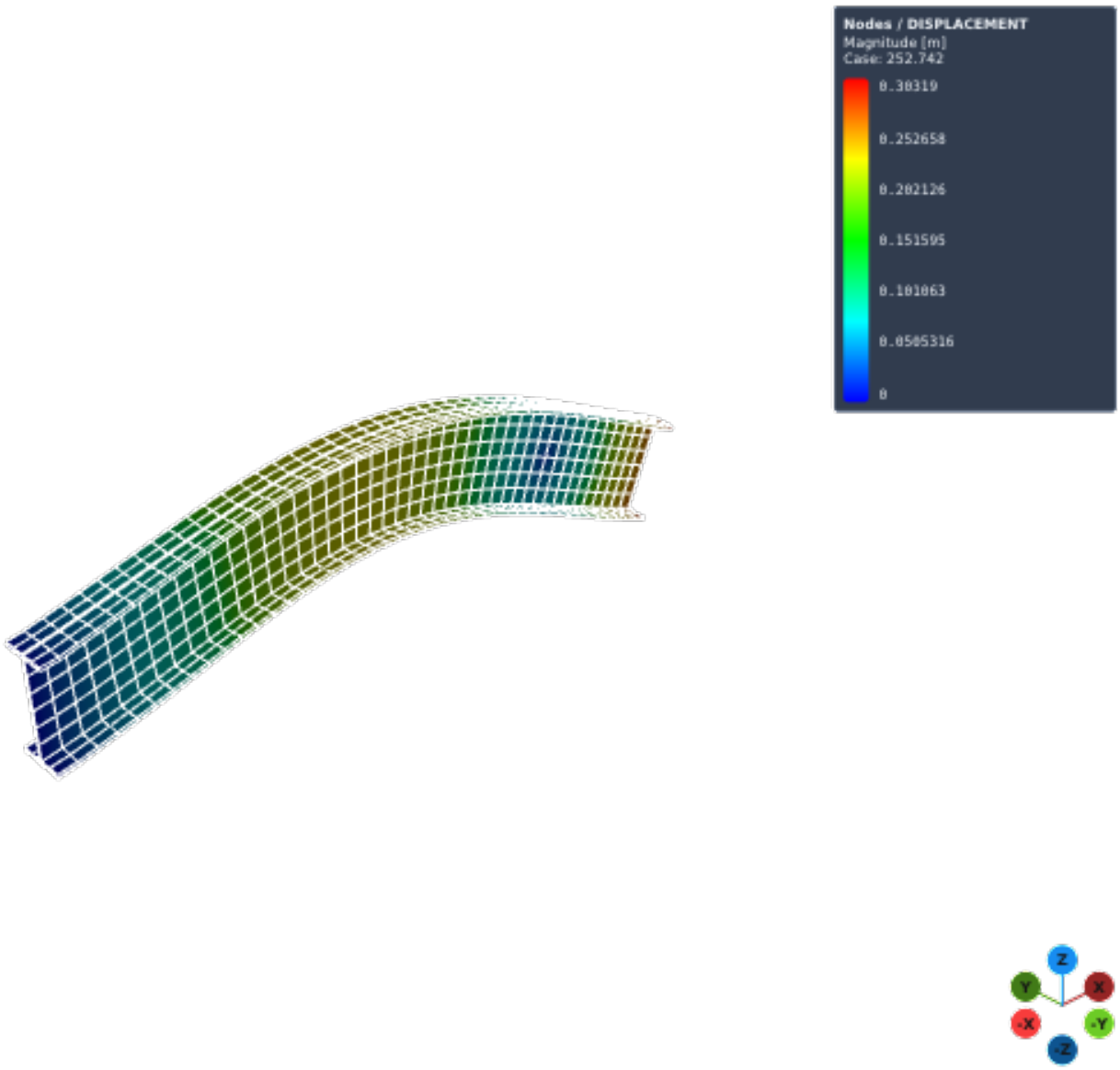

Figure 511: Sesam — mode 11, 252.742 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	252.920	+0.02	2.52532e+06	4.22e-22	4.70e-20	40.25	2.05e-11	2.17e-10	-6.344
Calculix	252.917	+0.02	2.52532e+06	2.59e-23	3.32e-20	40.25	-5.09e-12	1.82e-10	6.345
Code_Aster	252.917	+0.02	2.52532e+06	9.19e-24	5.59e-20	40.25	-3.03e-12	-2.37e-10	-6.344
Sesam	252.742	-0.05	2.52182e+06	2.26e-15	2.81e-16	40.28	-4.75e-08	1.68e-08	6.346

Solid, 2nd order, HEX, reduced integration

Mode 1

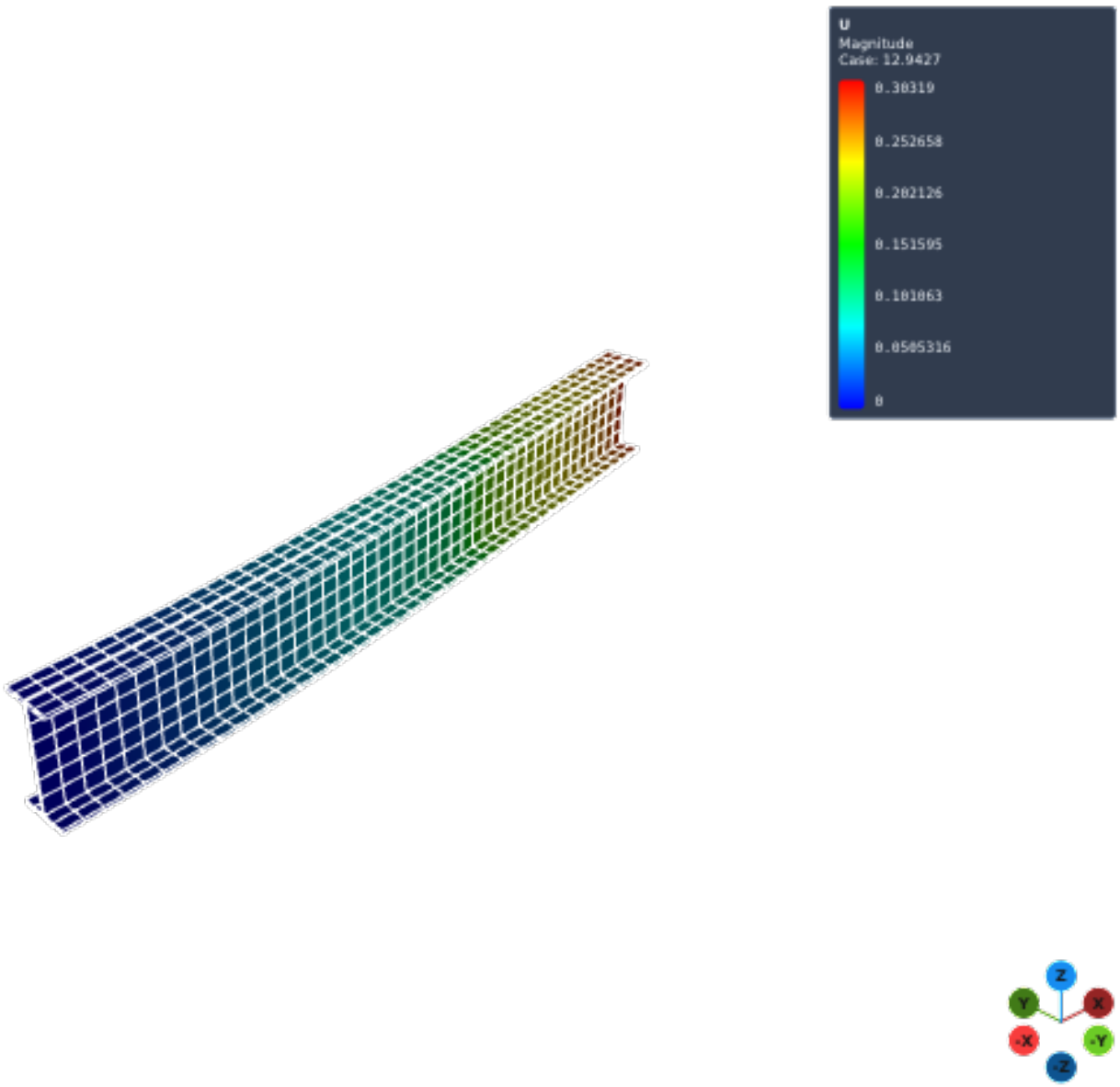

Figure 512: Abaqus — mode 1, 12.943 Hz

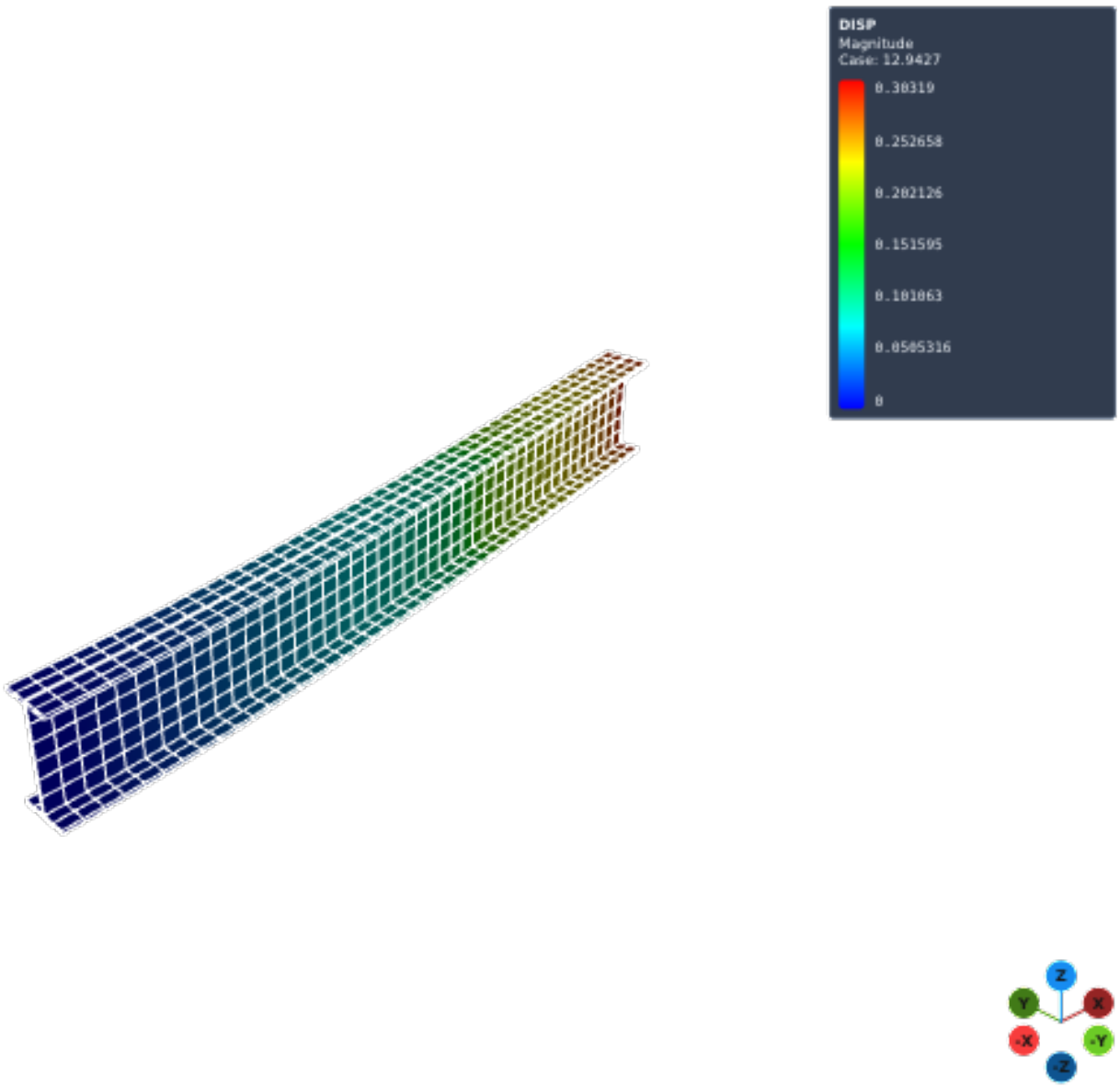

Figure 513: Calculix — mode 1, 12.943 Hz

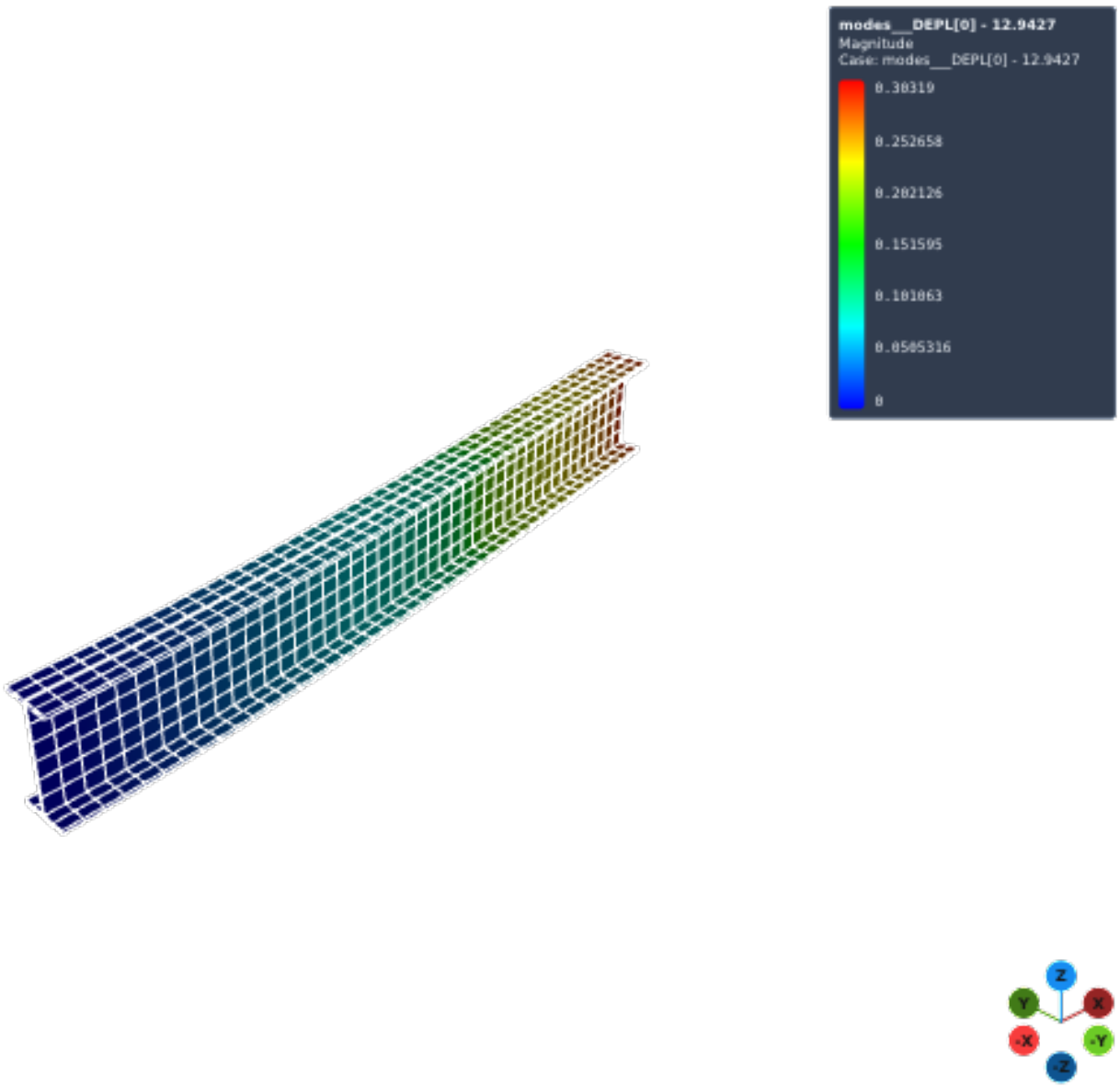

Figure 514: Code_Aster — mode 1, 12.943 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	12.943	1.29e-03	6613.2	5.11e-26	116.56	2.93e-20	-2.26e-13	10.796	1.71e-10
Calculix	12.943	-6.37e-04	6613.22	1.66e-24	116.56	6.19e-20	1.29e-12	10.796	2.49e-10
Code_Aster	12.943	-6.57e-04	6613.22	1.19e-25	116.56	5.87e-21	-3.45e-13	10.796	-7.66e-11

Mode 2

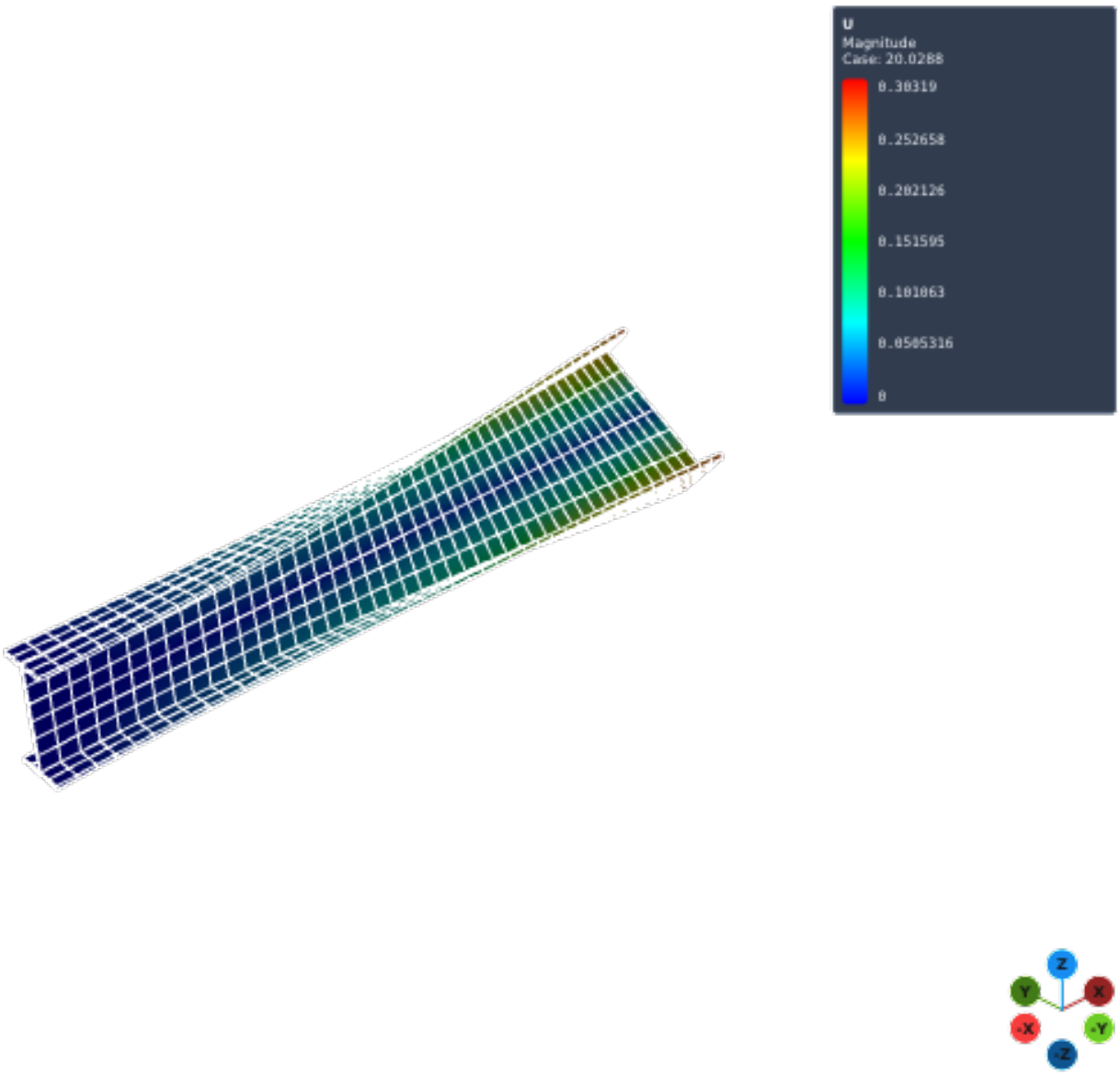

Figure 515: Abaqus — mode 2, 20.029 Hz

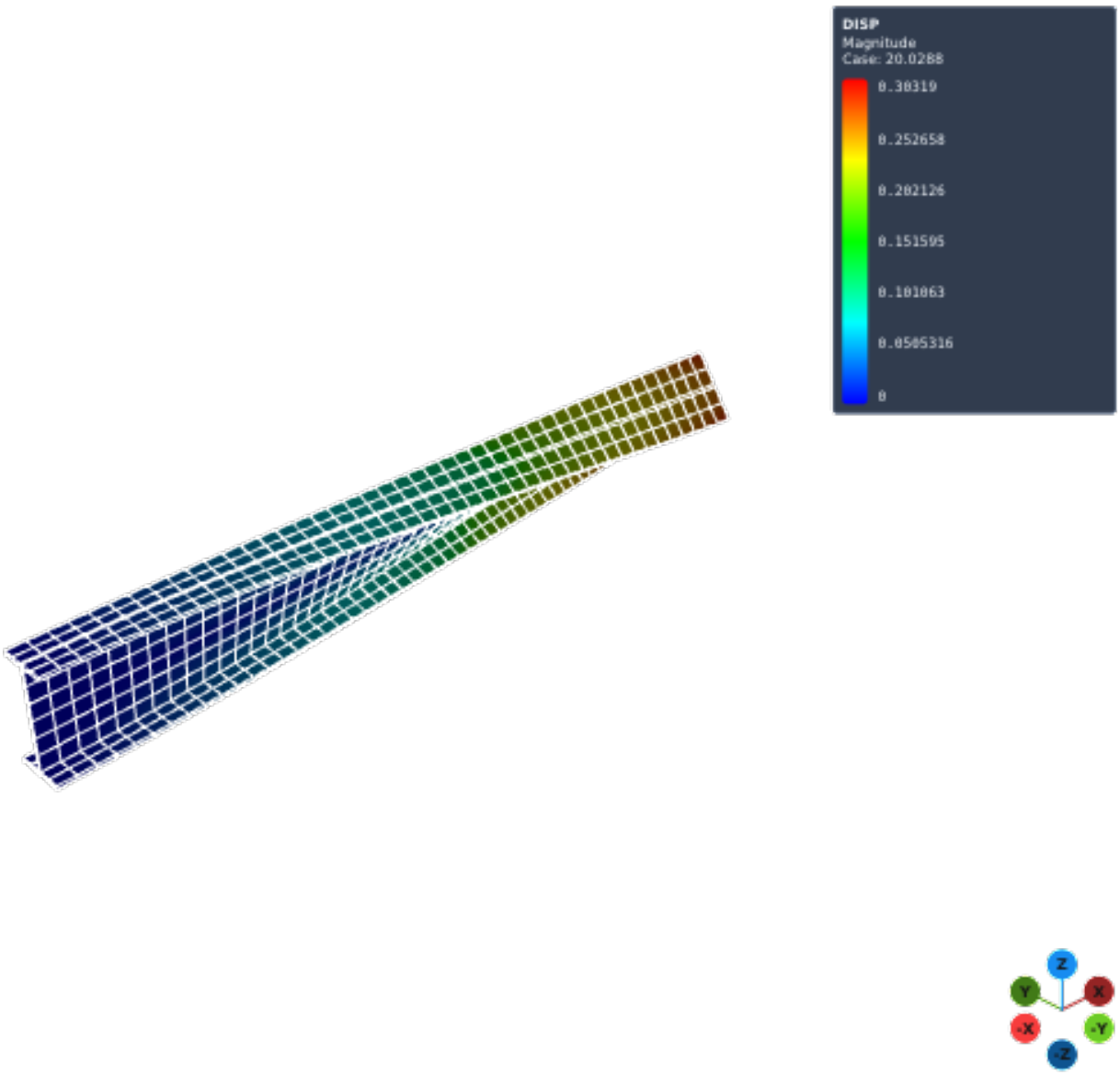

Figure 516: Calculix — mode 2, 20.029 Hz

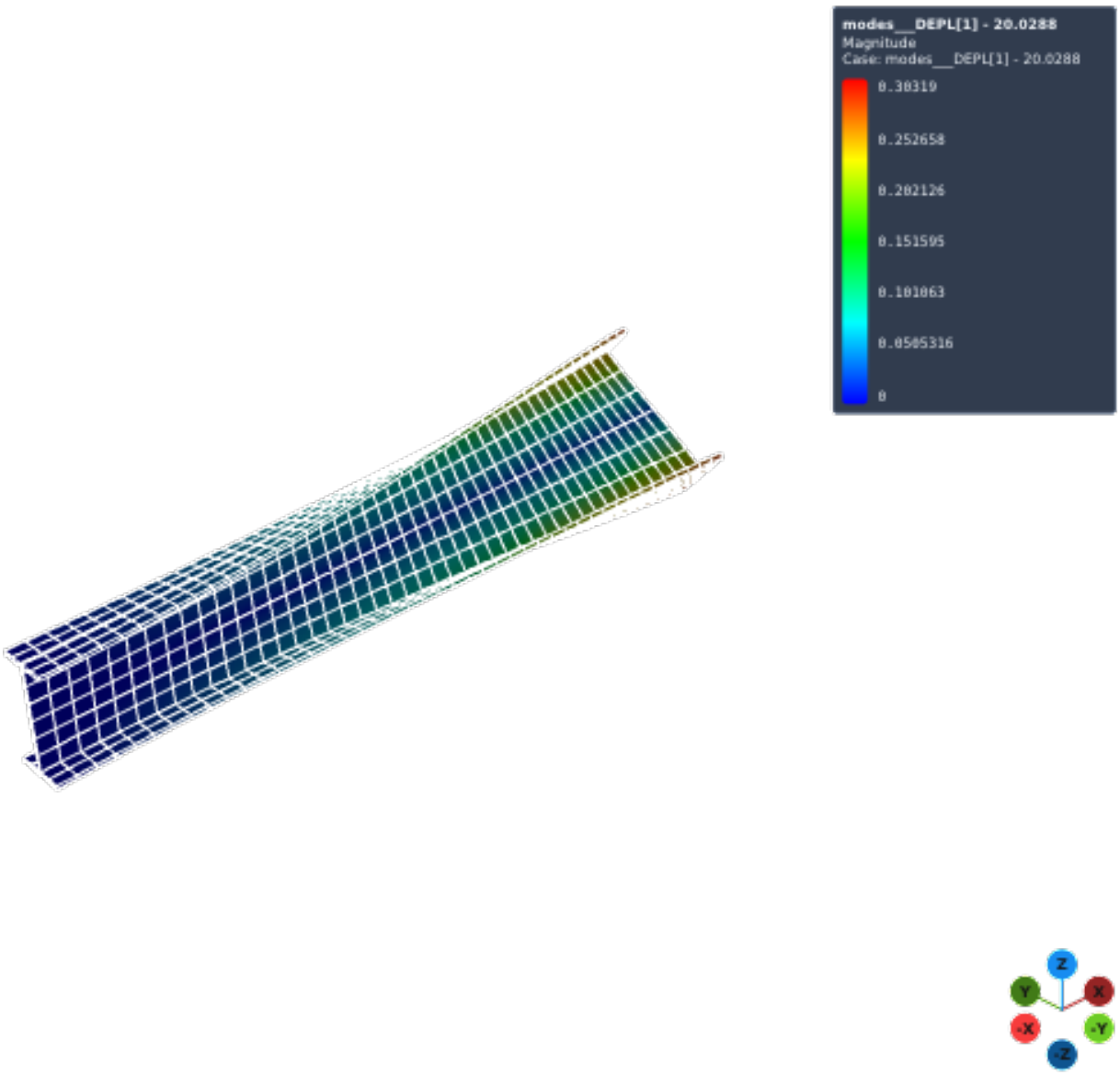

Figure 517: Code_Aster — mode 2, 20.029 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	20.029	8.03e-04	15837	2.00e-26	8.47e-16	7.79e-20	-1.41e-13	2.91e-08	-2.79e-10
Calculix	20.029	-3.95e-04	15836.8	1.31e-25	7.93e-16	7.72e-19	-3.62e-13	-2.82e-08	8.79e-10
Code_Aster	20.029	-4.08e-04	15836.8	5.43e-25	1.48e-17	1.39e-19	-7.37e-13	3.85e-09	3.73e-10

Mode 3

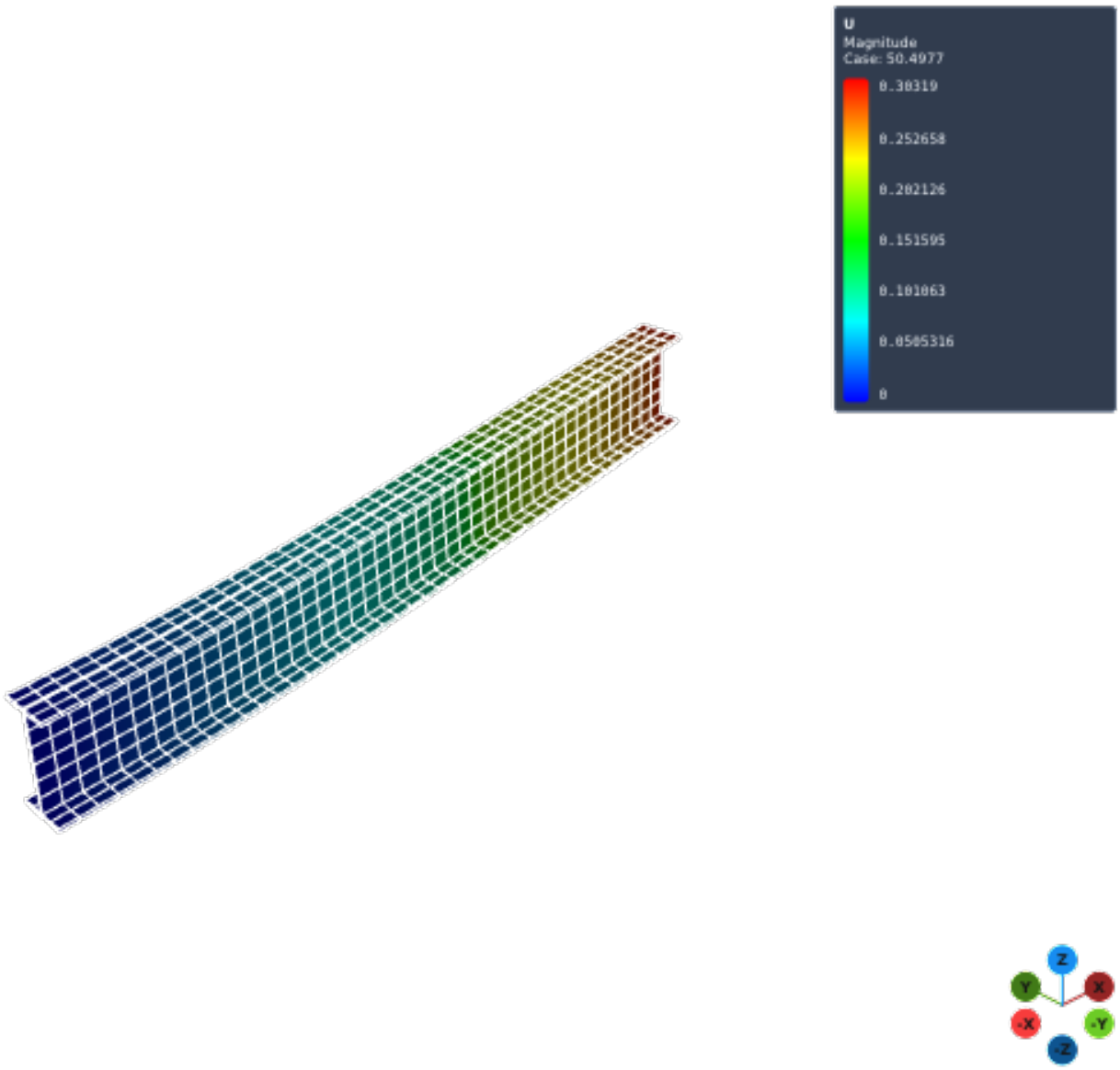

Figure 518: Abaqus — mode 3, 50.498 Hz

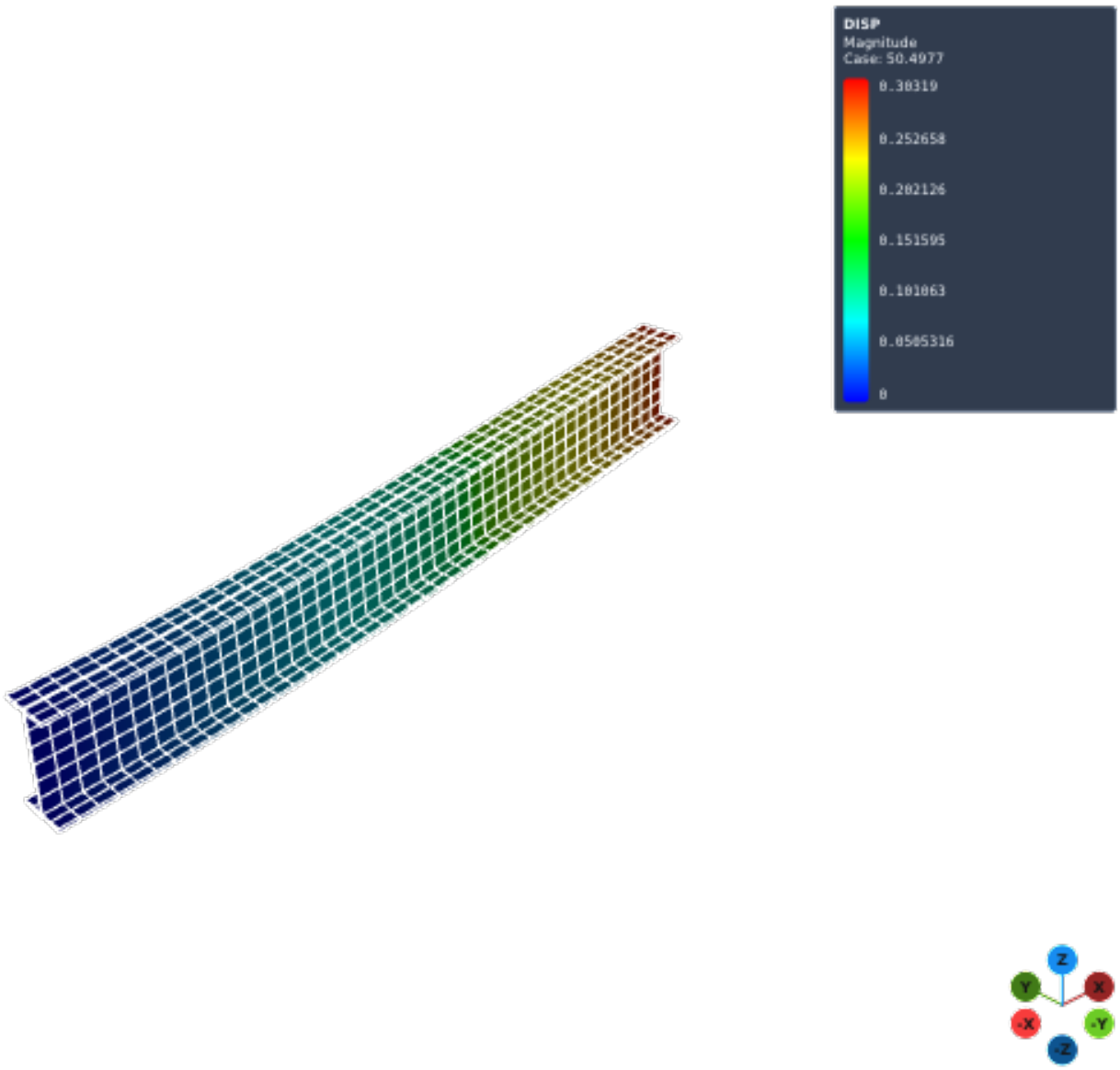

Figure 519: Calculix — mode 3, 50.498 Hz

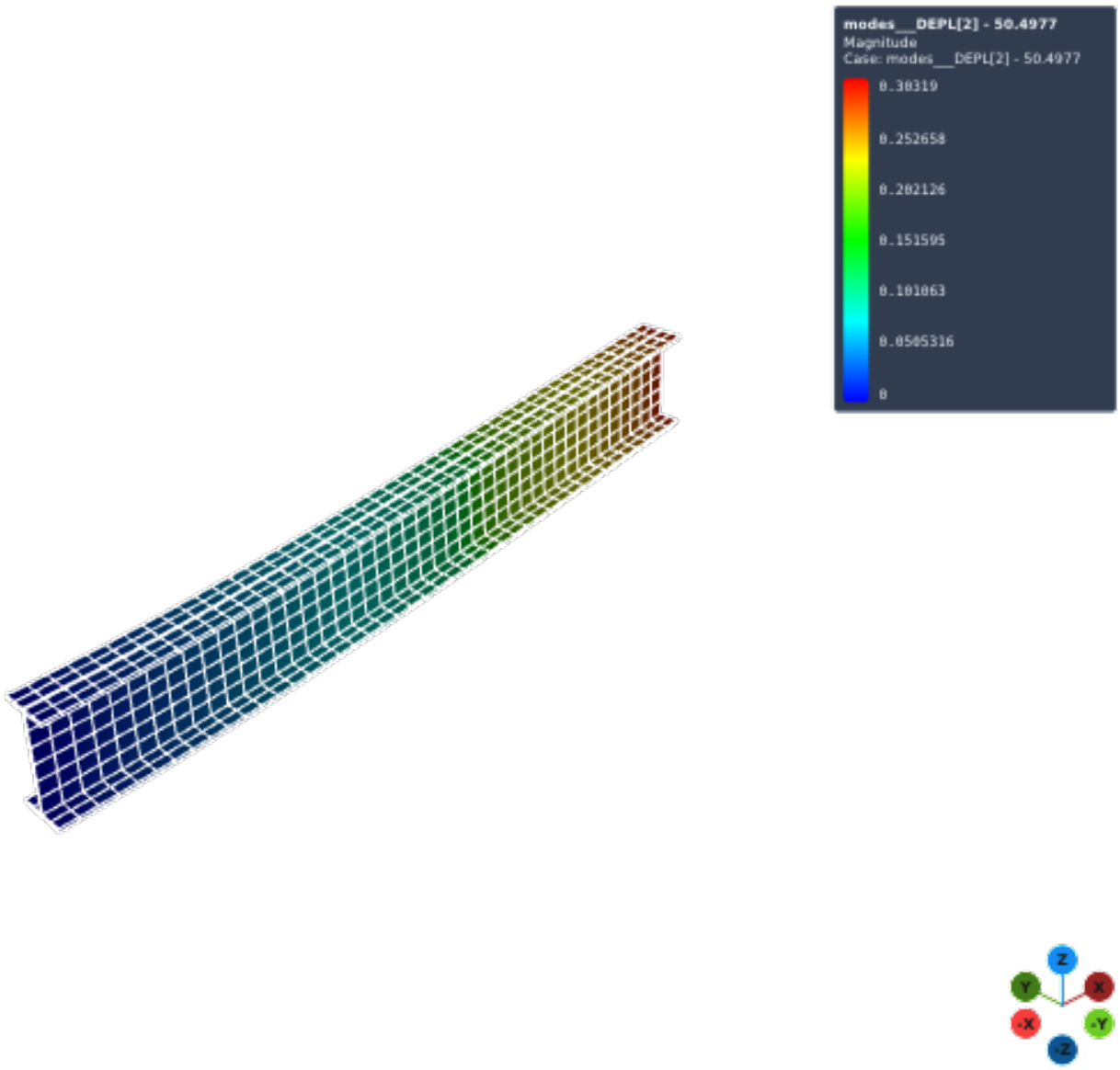

Figure 520: Code_Aster — mode 3, 50.498 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	50.498	3.96e-04	100671	1.64e-24	3.78e-20	118.41	-1.28e-12	-1.94e-10	10.882
Calculix	50.498	-1.98e-04	100671	4.96e-24	8.00e-20	118.41	-2.23e-12	-2.83e-10	10.882
Code_Aster	50.498	-1.98e-04	100671	2.17e-24	1.42e-21	118.41	-1.47e-12	3.77e-11	10.882

Mode 4

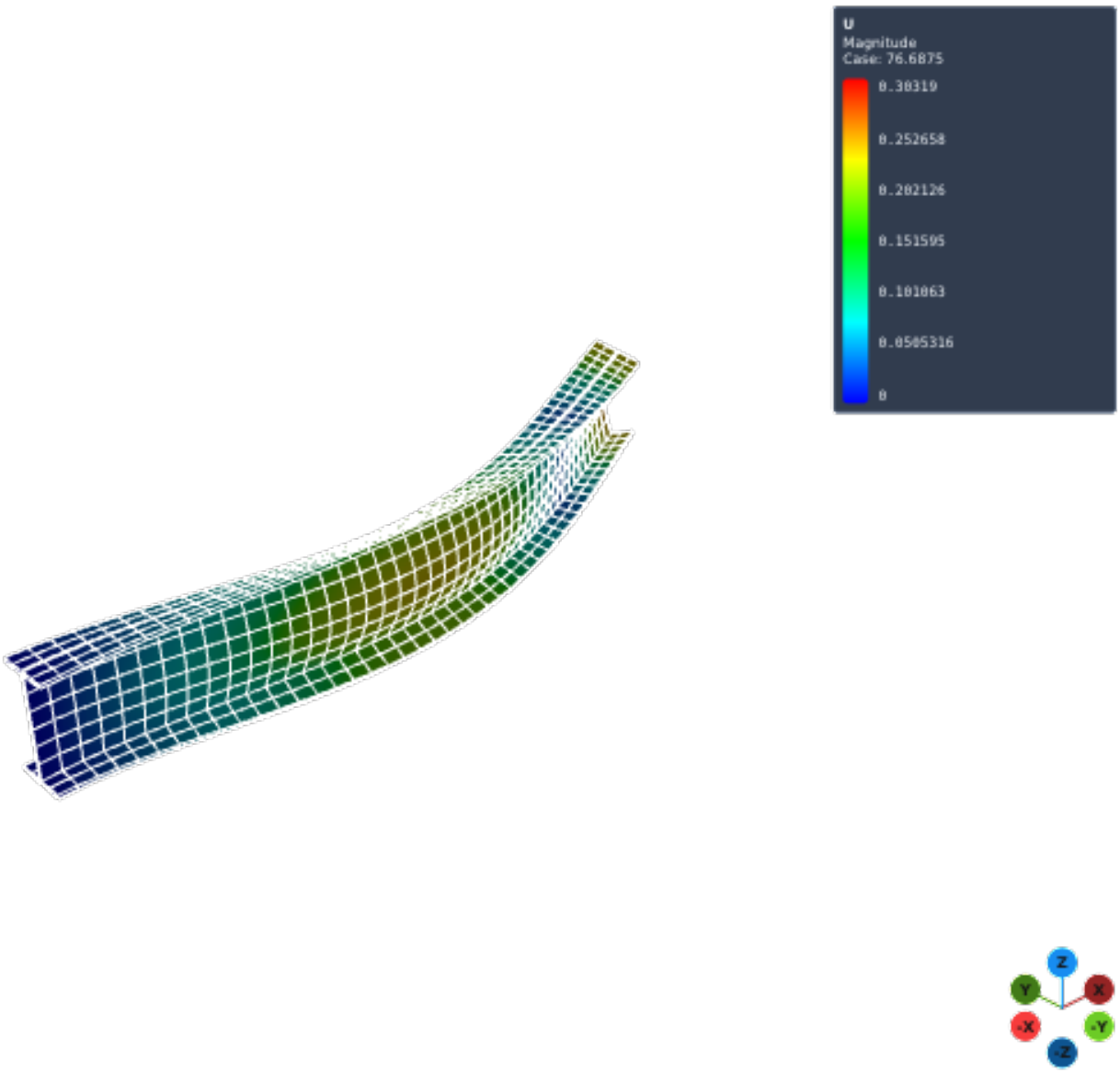

Figure 521: Abaqus — mode 4, 76.687 Hz

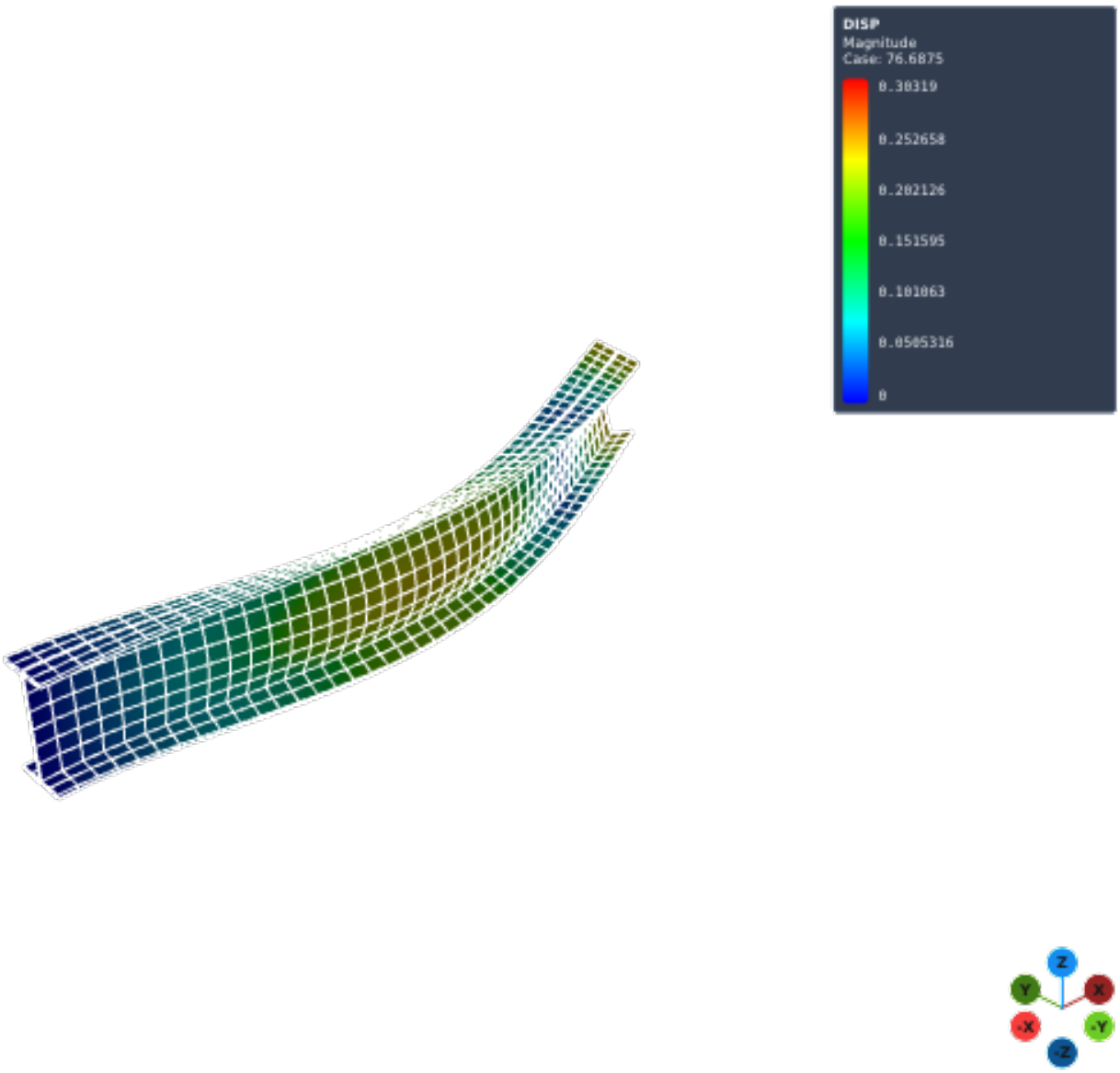

Figure 522: Calculix — mode 4, 76.688 Hz

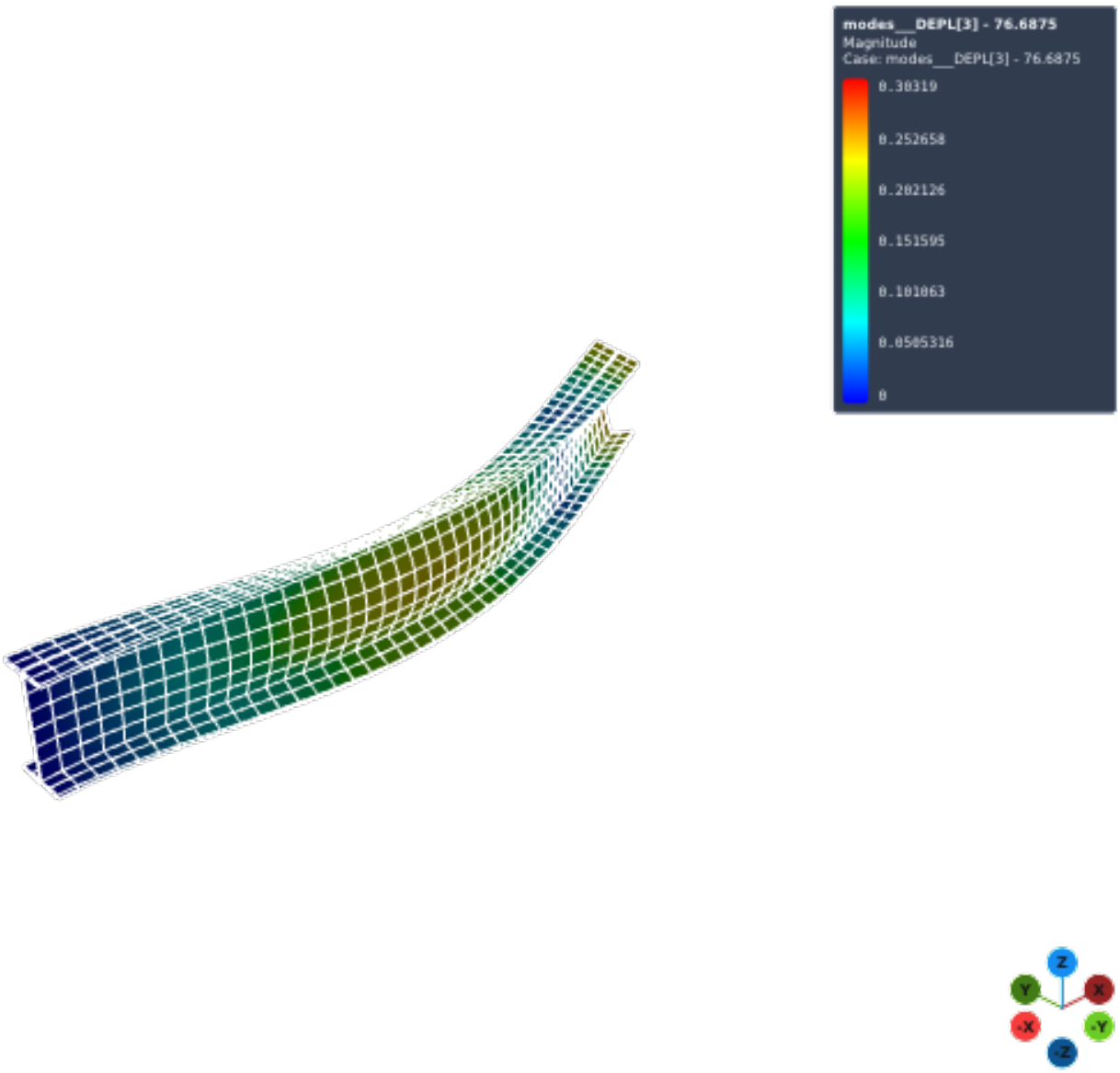

Figure 523: Code_Aster — mode 4, 76.687 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	76.687	-4.40e-04	232171	2.11e-25	35.90	8.07e-22	-4.60e-13	-5.992	-2.84e-11
Calculix	76.688	2.77e-04	232172	1.40e-24	35.90	3.37e-21	1.18e-12	-5.992	-5.81e-11
Code_Aster	76.687	1.64e-04	232171	6.43e-25	35.90	1.04e-20	-8.02e-13	-5.992	-1.02e-10

Mode 5

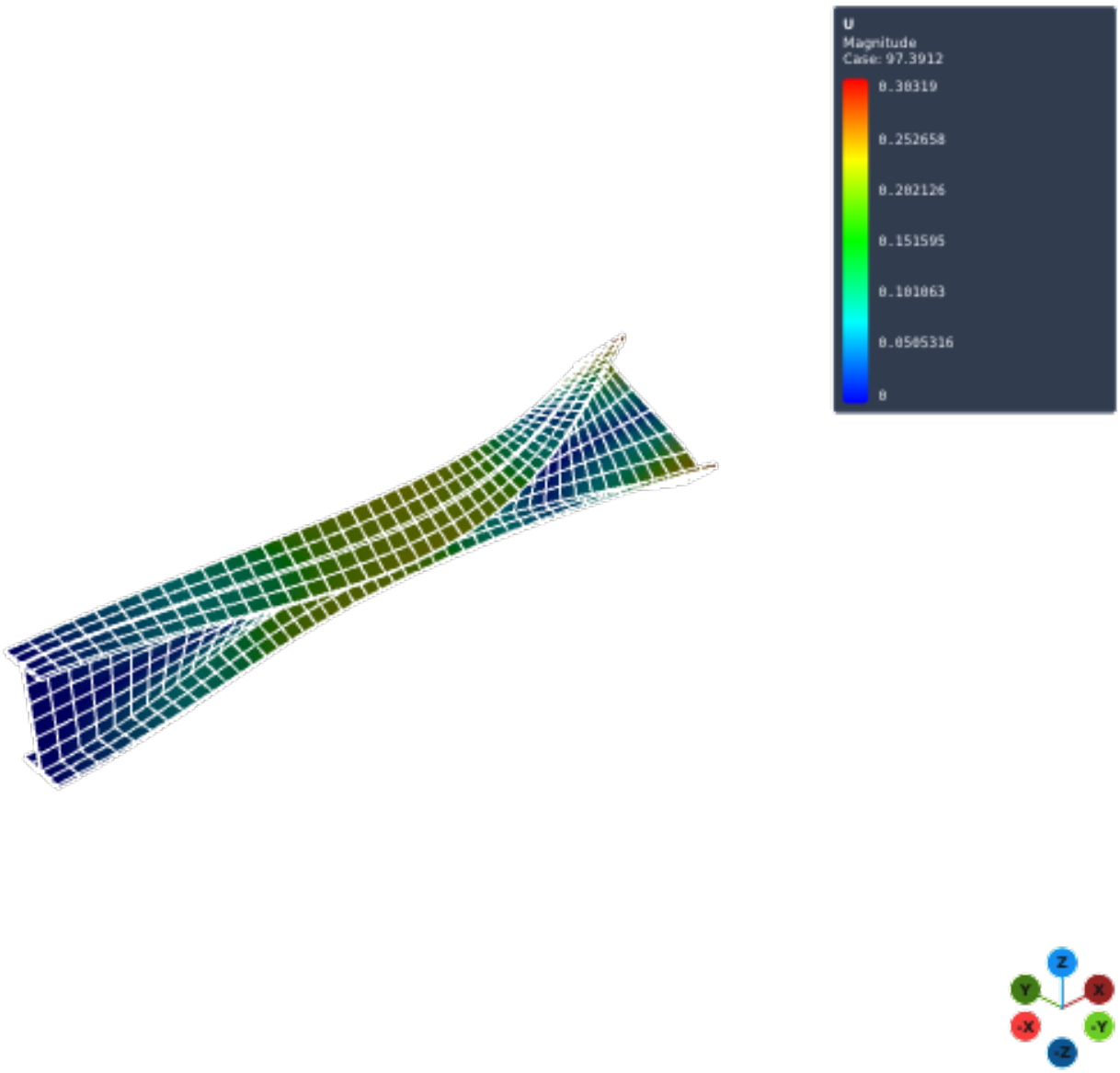

Figure 524: Abaqus — mode 5, 97.391 Hz

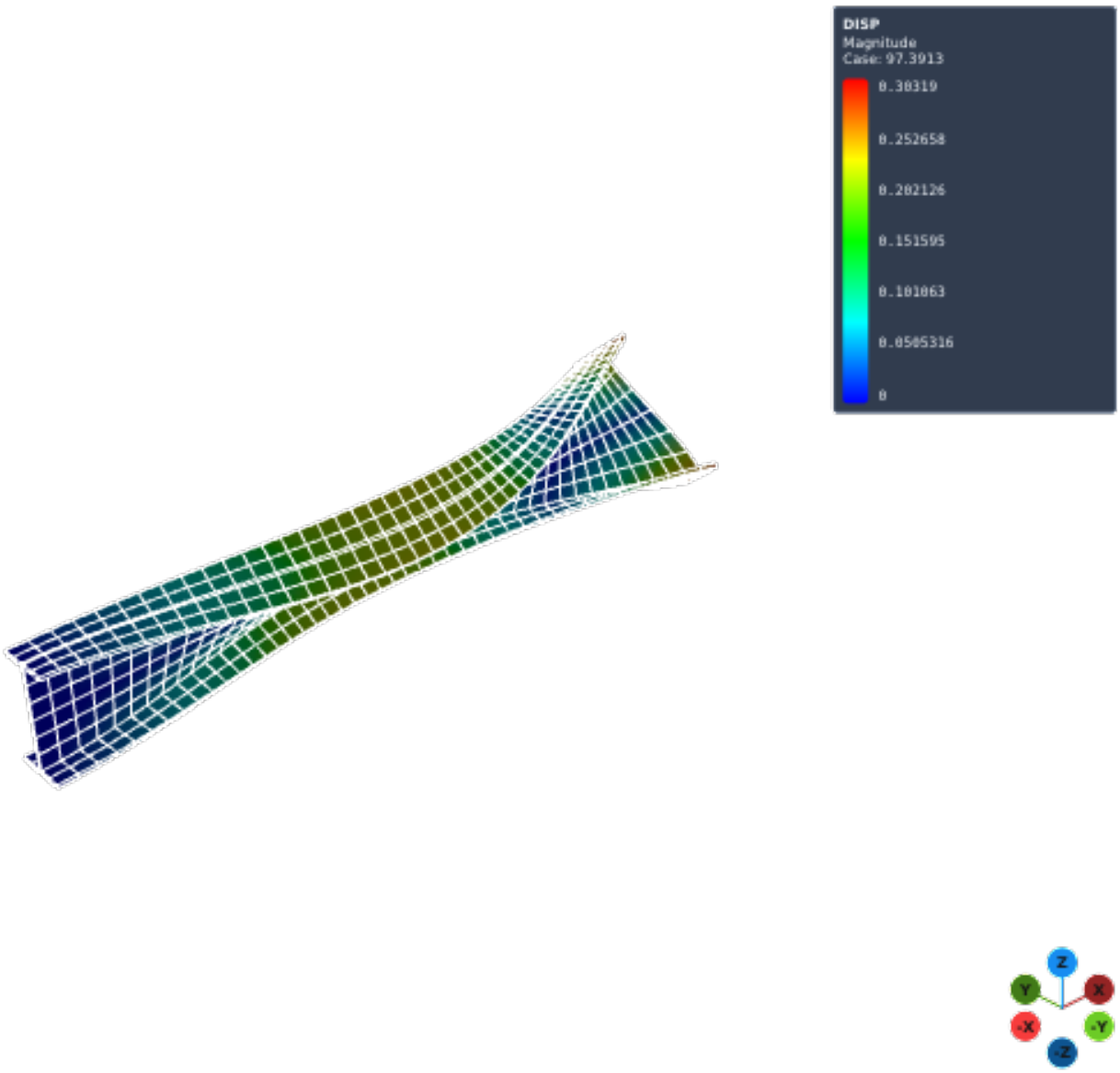

Figure 525: Calculix — mode 5, 97.391 Hz

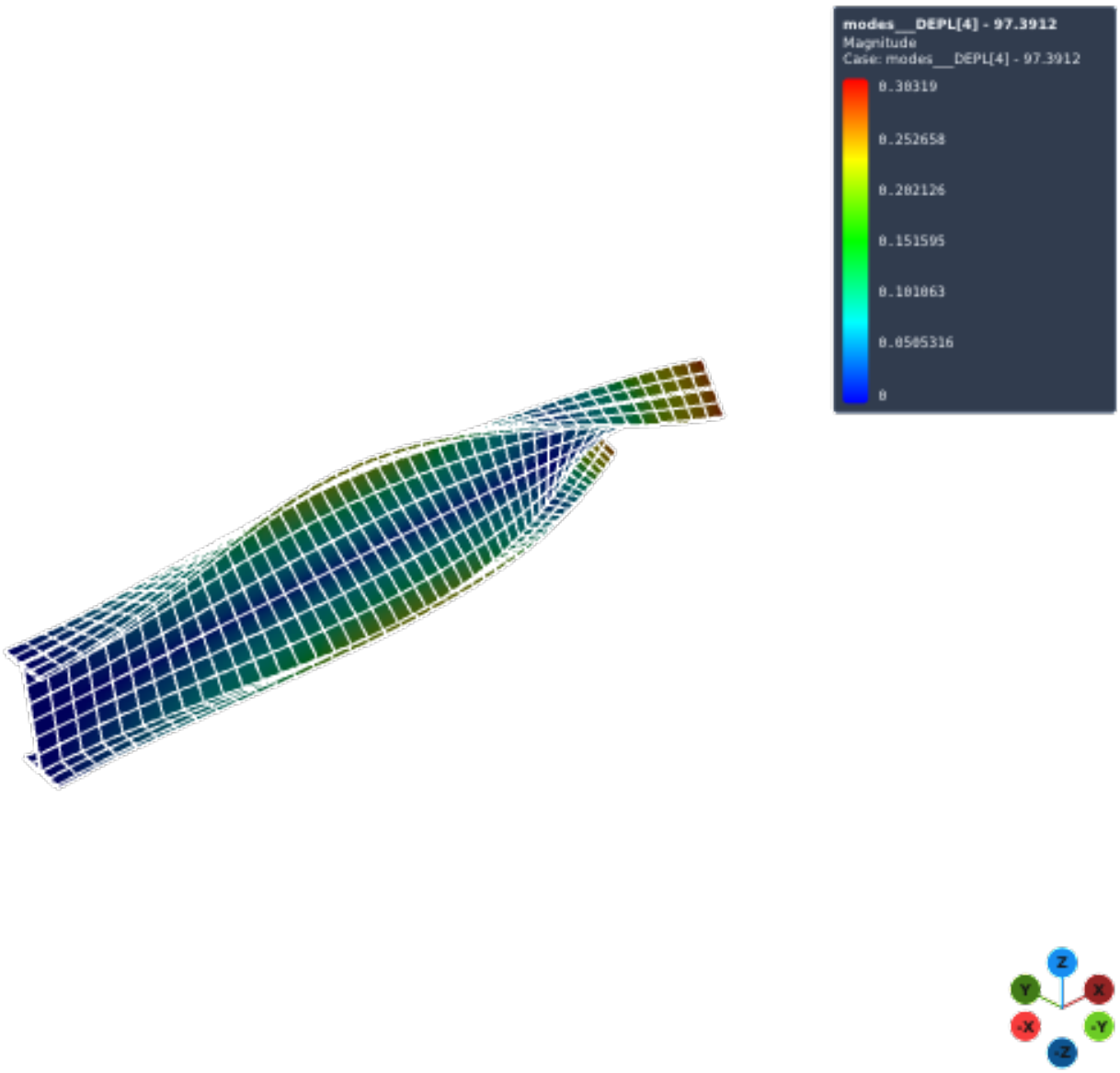

Figure 526: Code_Aster — mode 5, 97.391 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	97.391	-1.71e-04	374455	1.68e-25	8.25e-19	2.52e-24	4.10e-13	-9.08e-10	-1.59e-12
Calculix	97.391	1.06e-04	374455	5.59e-25	2.56e-19	9.90e-21	-7.47e-13	-5.06e-10	9.95e-11
Code_Aster	97.391	6.57e-05	374455	1.30e-24	1.42e-18	4.98e-21	-1.14e-12	-1.19e-09	7.06e-11

Mode 6

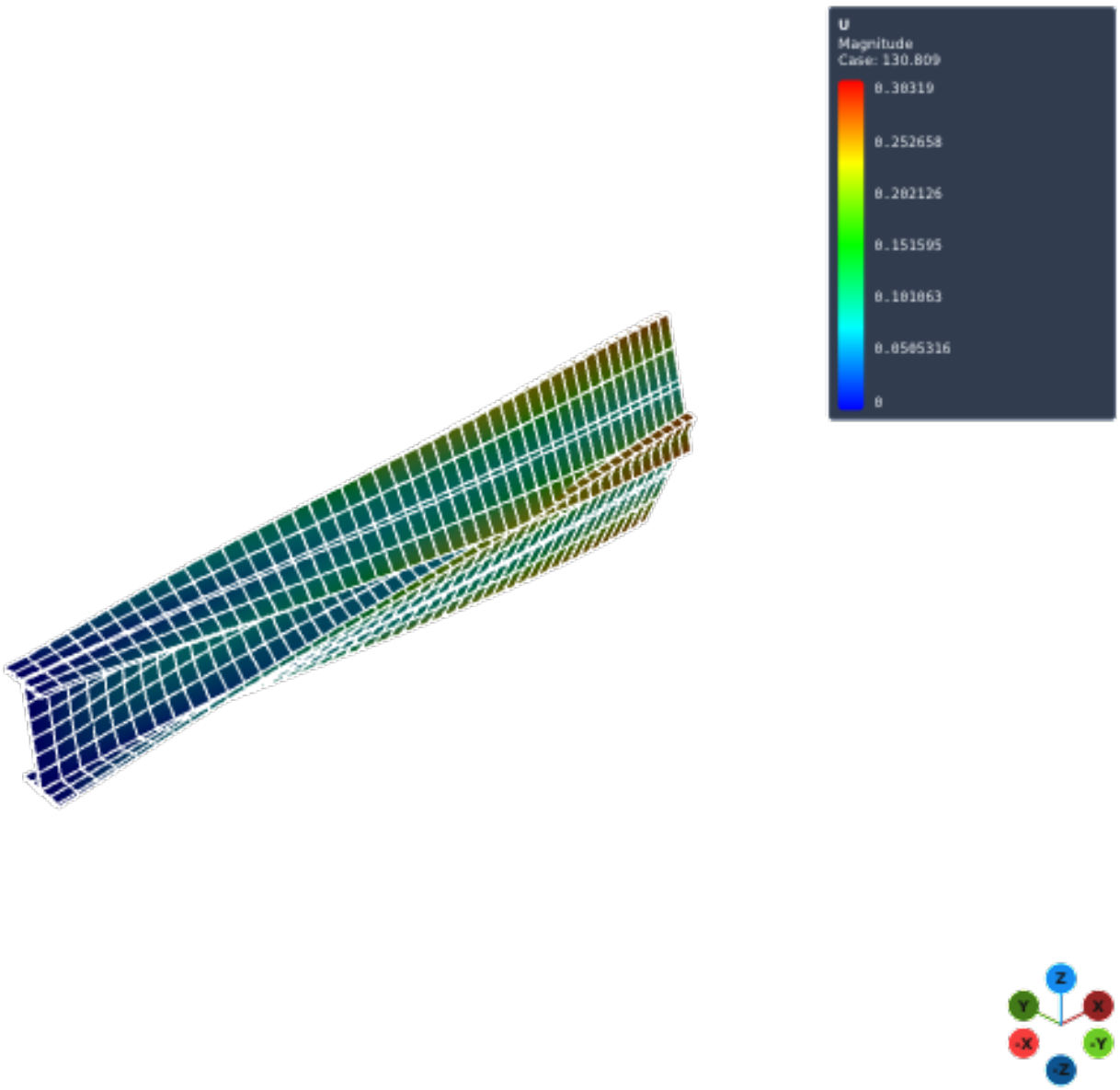

Figure 527: Abaqus — mode 6, 130.809 Hz

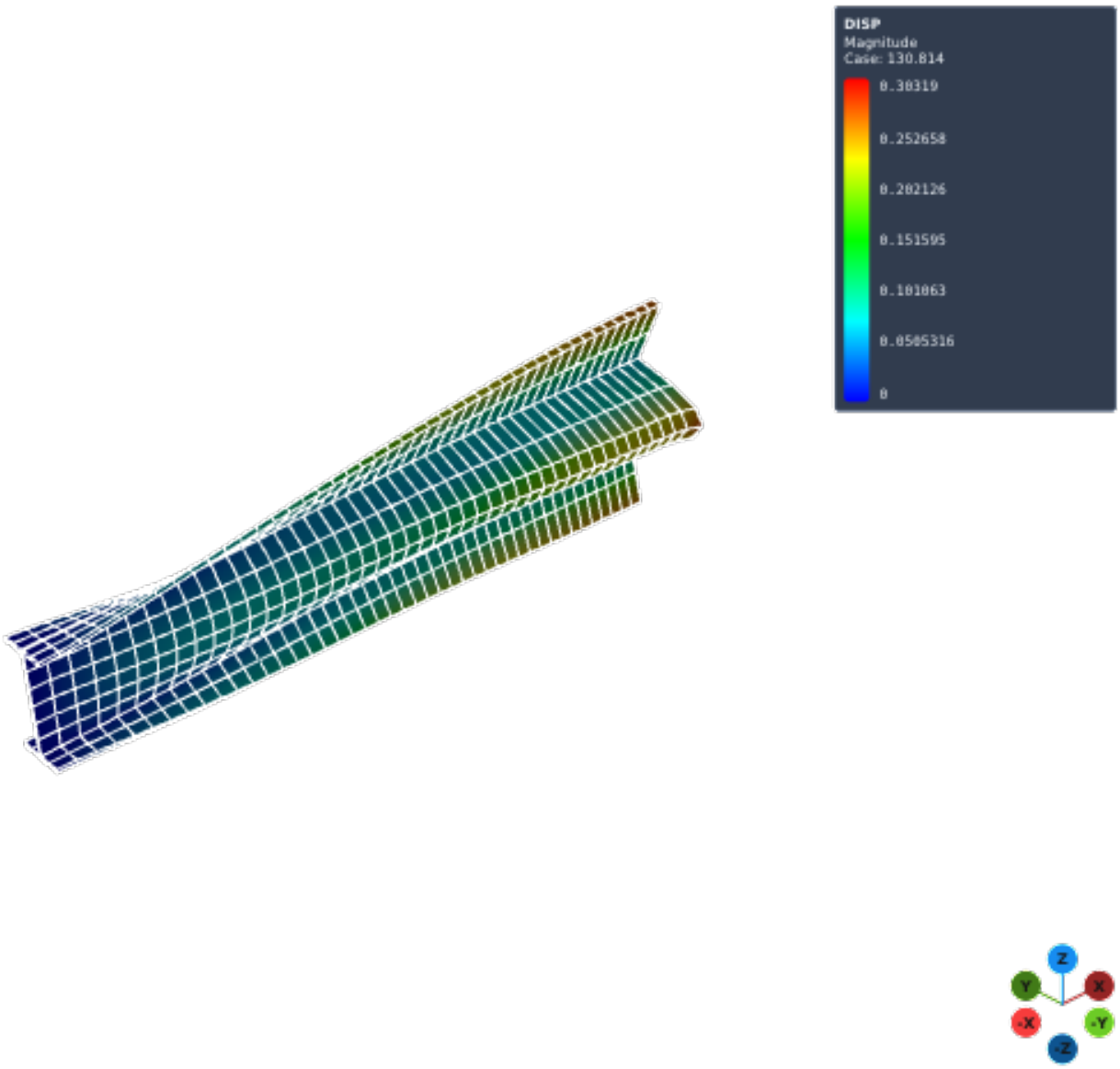

Figure 528: Calculix — mode 6, 130.814 Hz

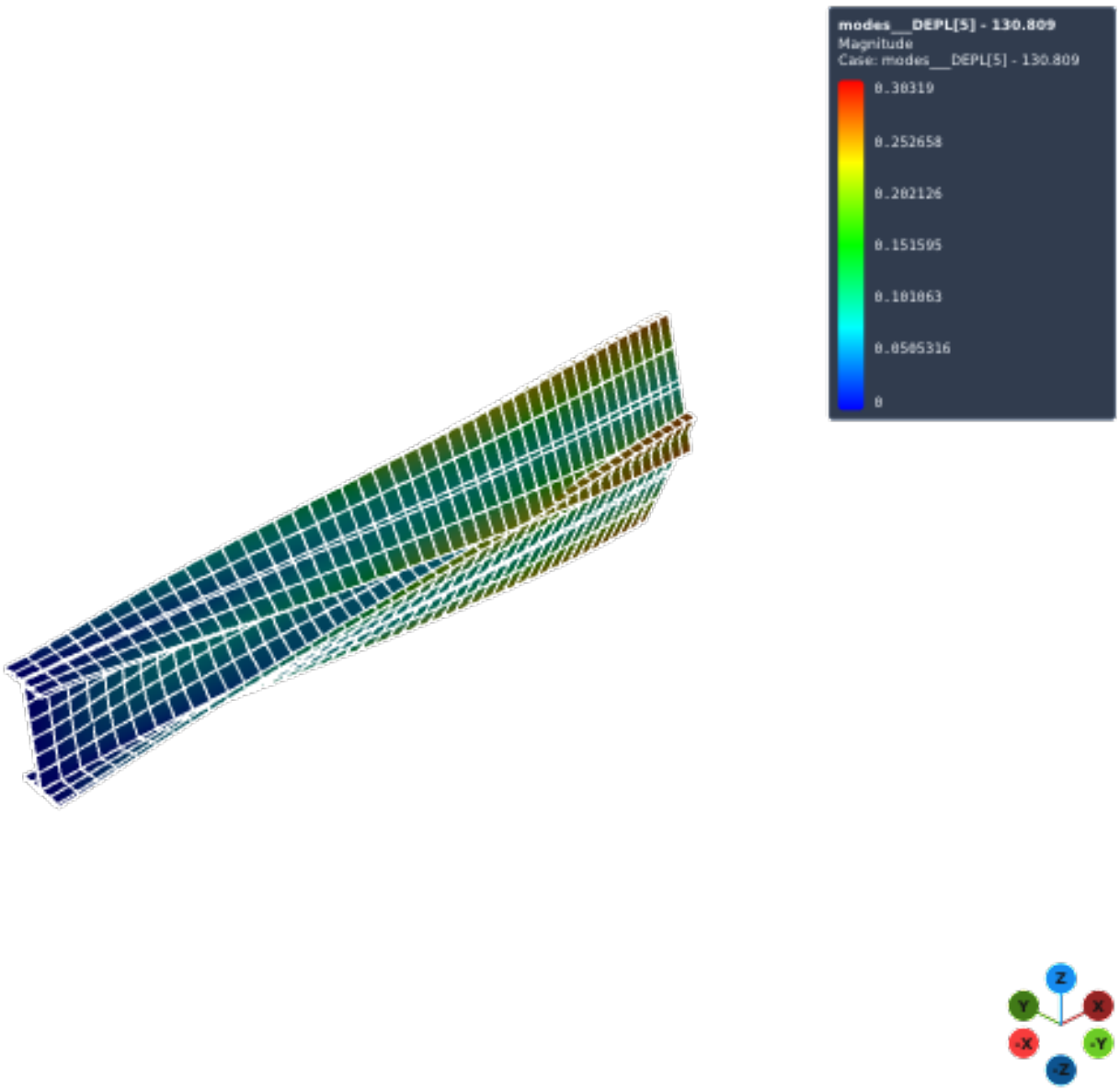

Figure 529: Code_Aster — mode 6, 130.809 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	130.810	-7.26e-04	675515	5.64e-26	0.01	6.07e-21	-2.37e-13	-0.097	7.79e-11
Calculix	130.814	2.18e-03	675564	4.82e-25	0.01	4.46e-21	-6.94e-13	0.096	6.68e-11
Code_Aster	130.809	-1.45e-03	675515	3.85e-25	0.01	6.15e-21	6.20e-13	-0.097	-7.84e-11

Mode 7

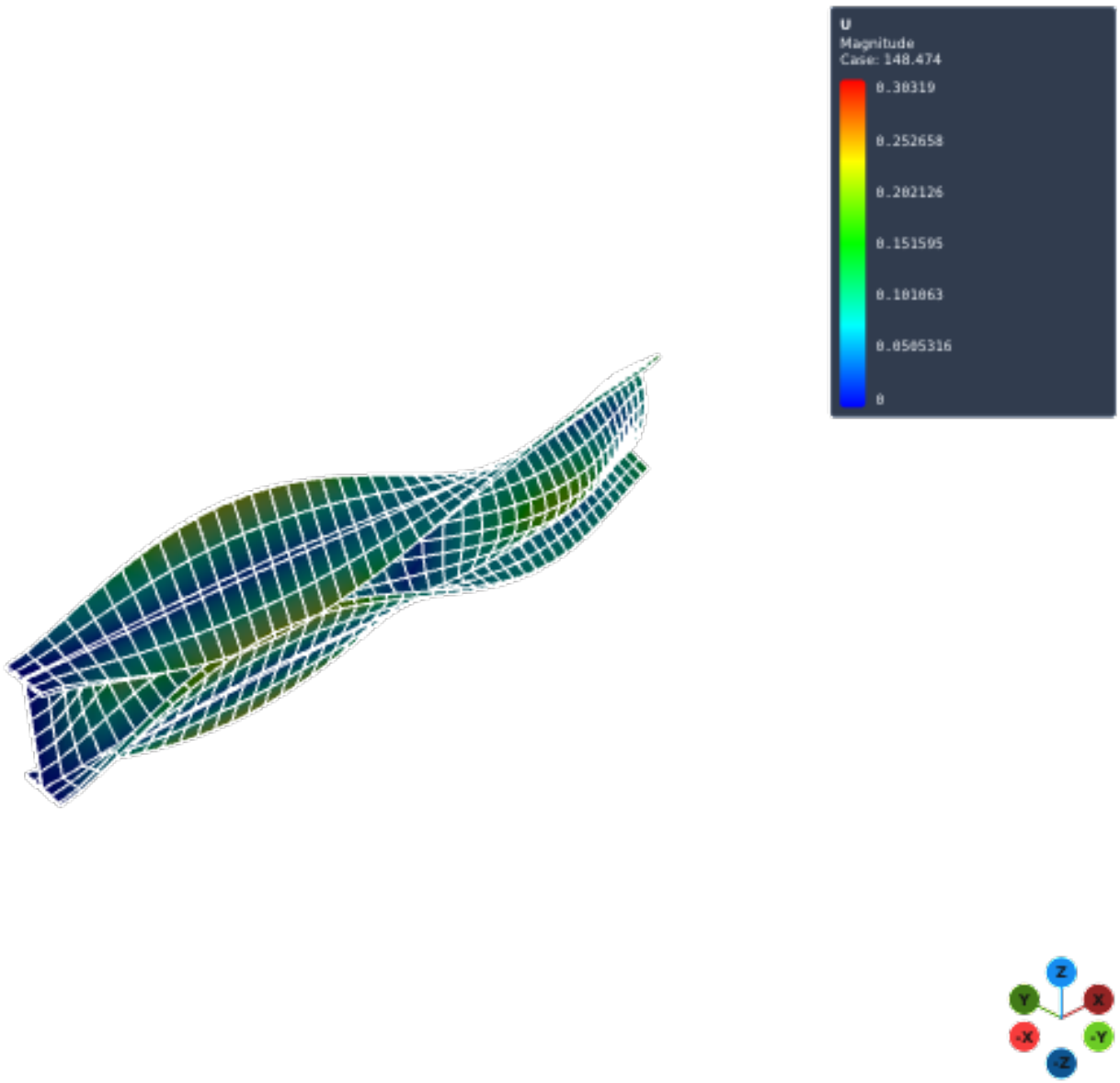

Figure 530: Abaqus — mode 7, 148.474 Hz

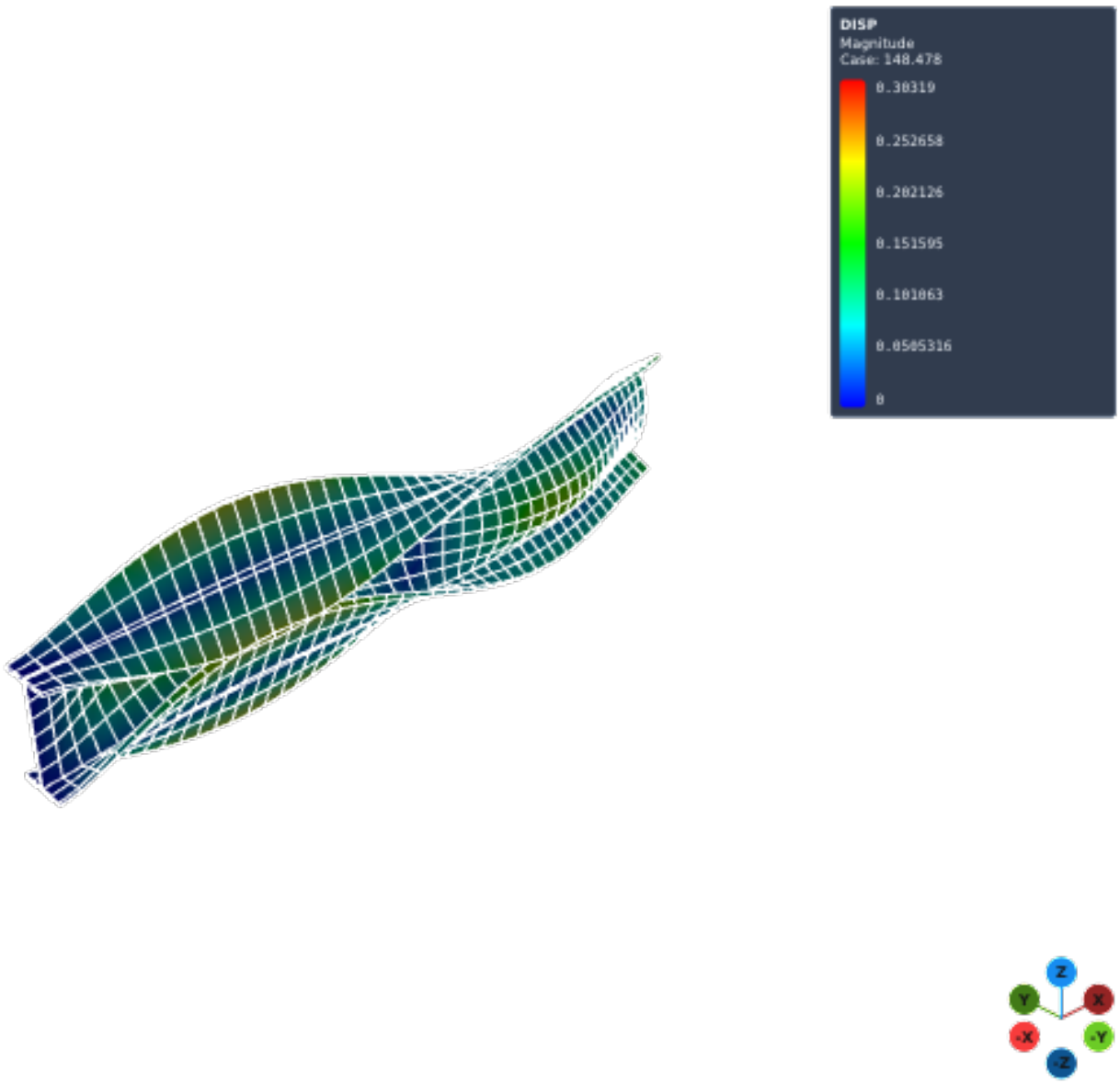

Figure 531: Calculix — mode 7, 148.478 Hz

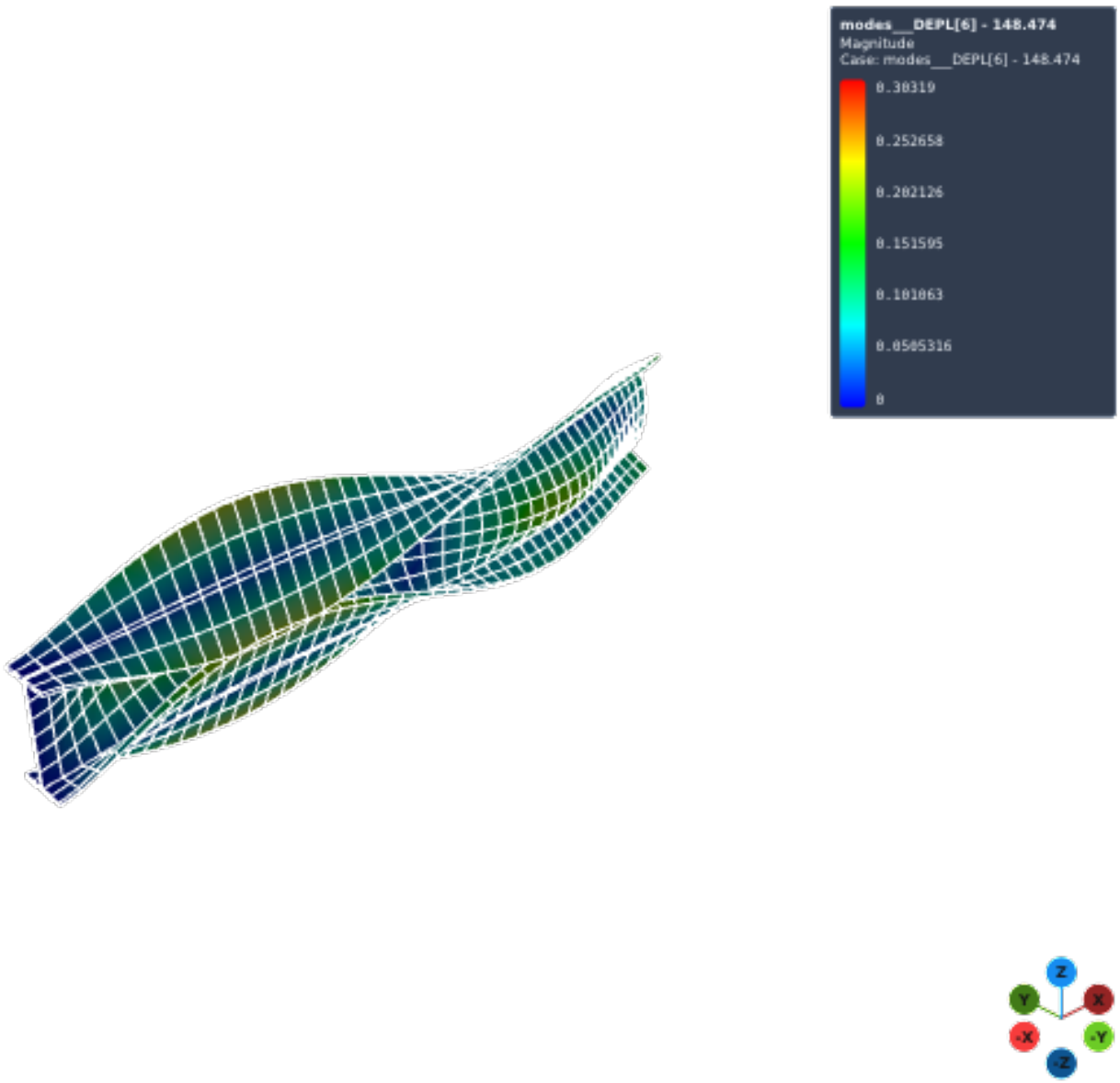

Figure 532: Code_Aster — mode 7, 148.474 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	148.470	-2.76e-03	870287	1.93e-24	3.39	2.12e-22	-1.39e-12	1.840	-1.46e-11
Calculix	148.478	2.63e-03	870330	4.10e-25	3.39	8.55e-22	6.41e-13	1.840	2.92e-11
Code_Aster	148.474	1.37e-04	870287	1.52e-25	3.39	1.11e-22	-3.90e-13	1.840	1.05e-11

Mode 8

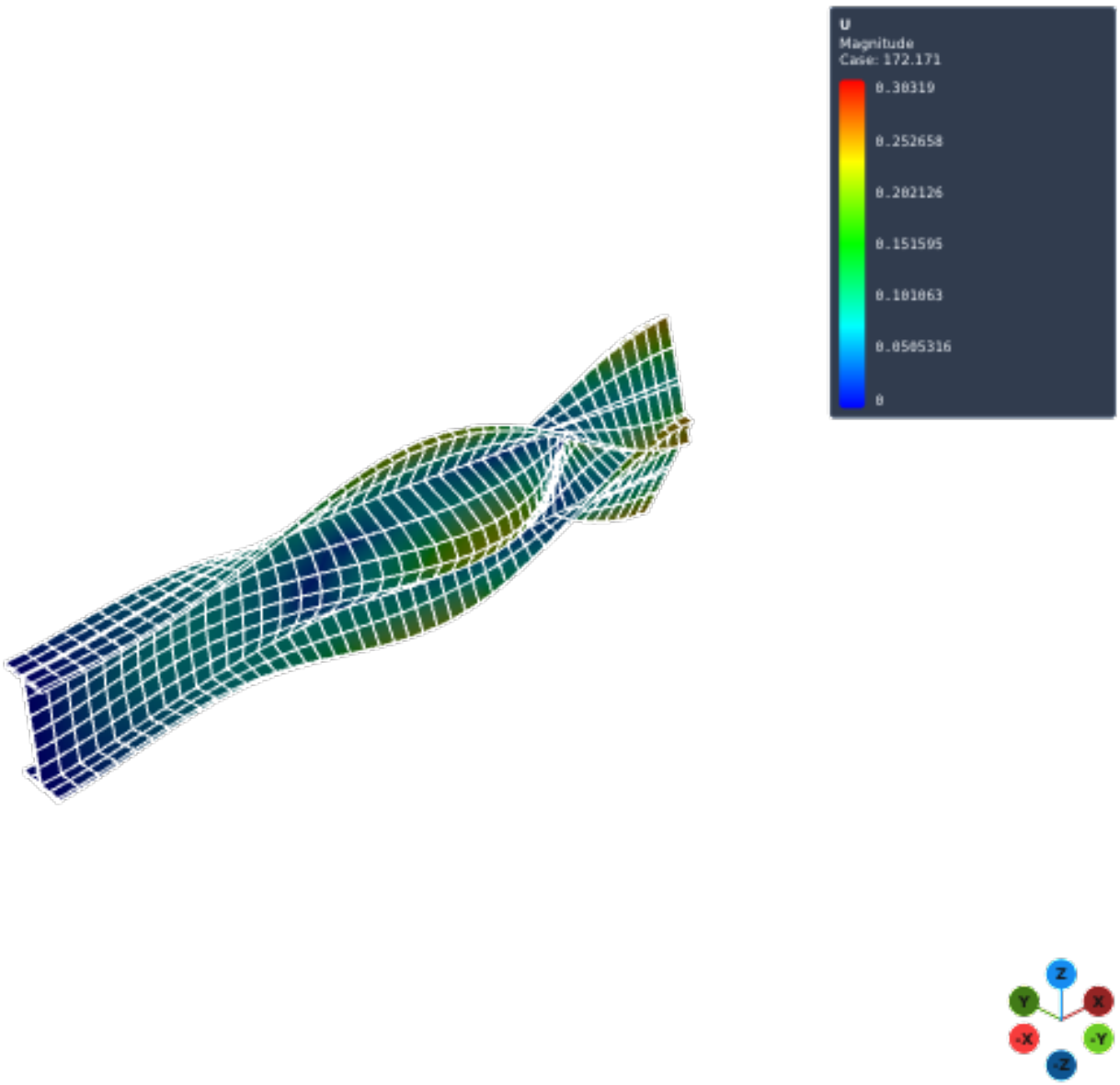

Figure 533: Abaqus — mode 8, 172.171 Hz

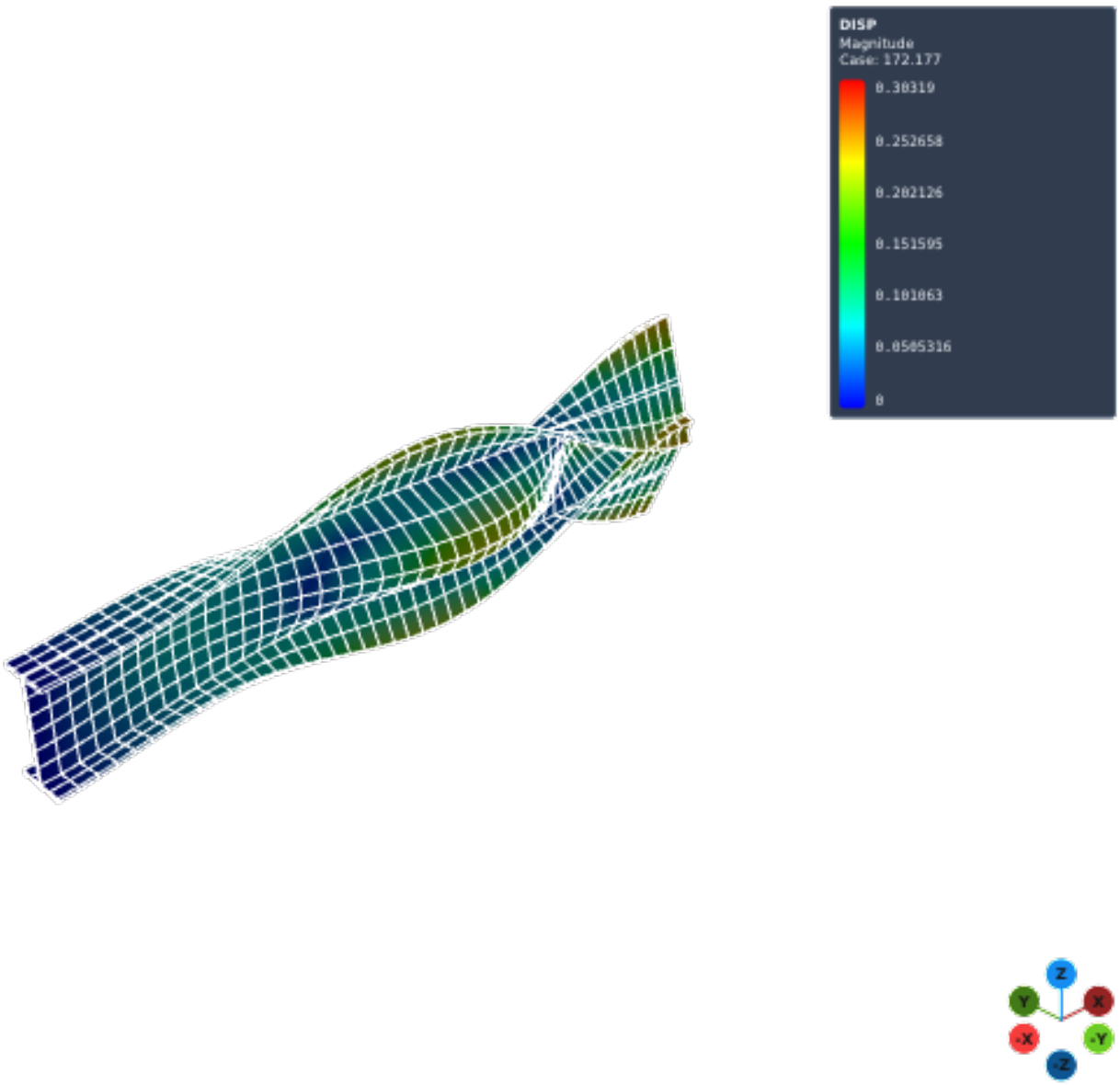

Figure 534: Calculix — mode 8, 172.177 Hz

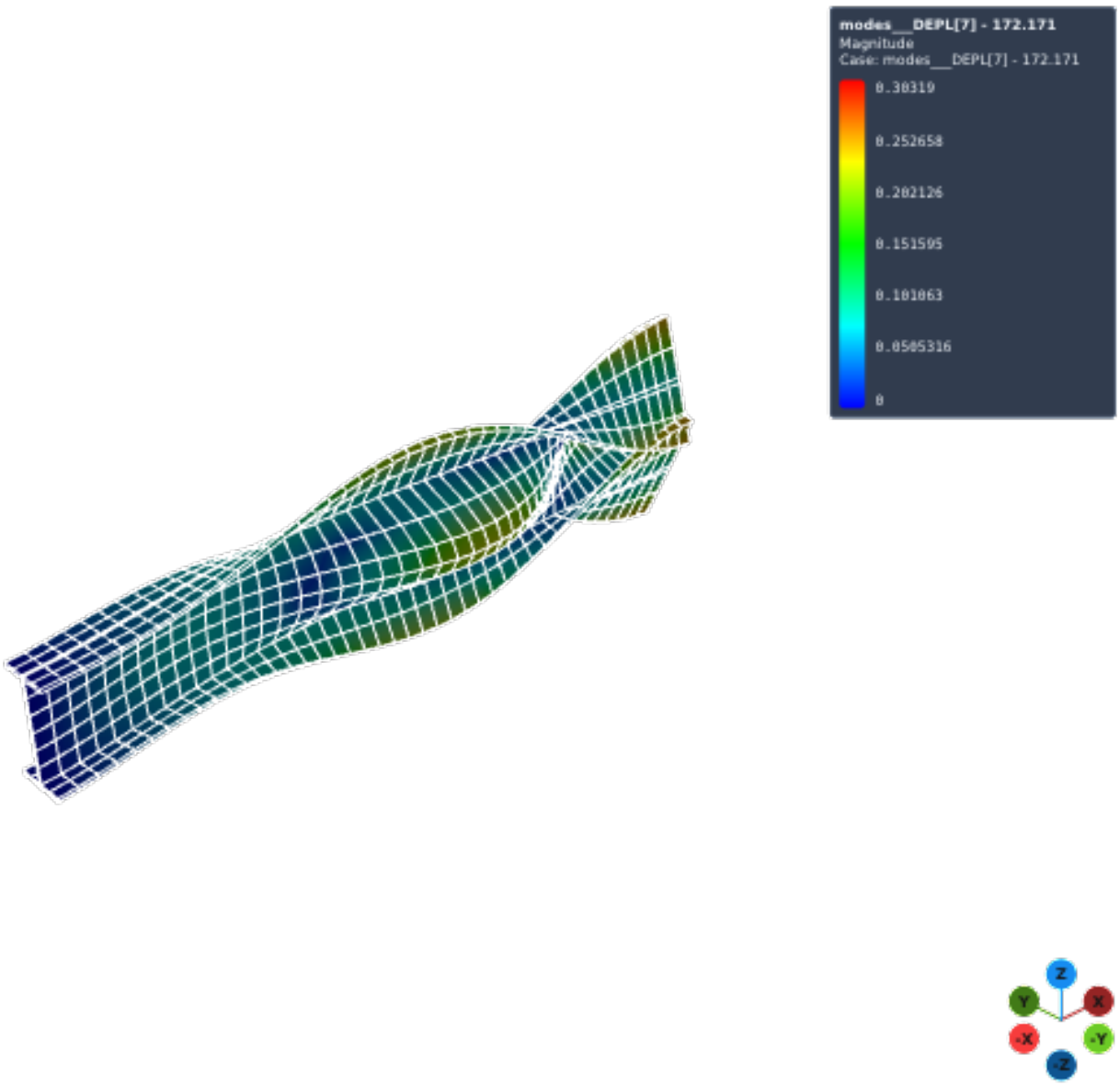

Figure 535: Code_Aster — mode 8, 172.171 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	172.170	-1.58e-03	1.17026e+06	1.27e-25	5.31	9.73e-21	3.57e-13	2.304	9.86e-11
Calculix	172.177	2.37e-03	1.17033e+06	1.37e-24	5.31	1.97e-22	1.17e-12	2.304	1.40e-11
Code_Aster	172.171	-7.89e-04	1.17026e+06	2.56e-24	5.31	1.73e-20	-1.60e-12	2.304	-1.32e-10

Mode 9

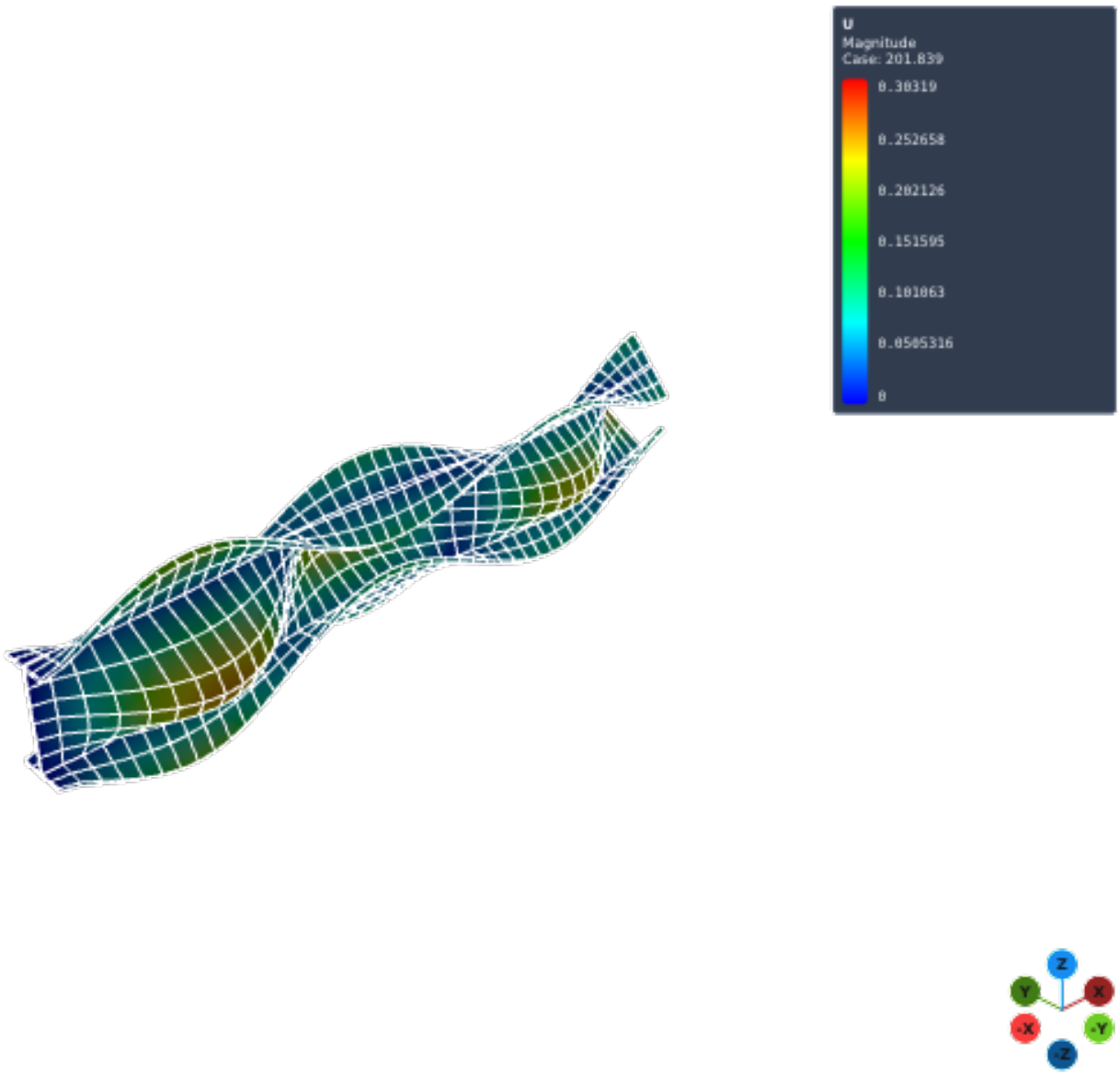

Figure 536: Abaqus — mode 9, 201.839 Hz

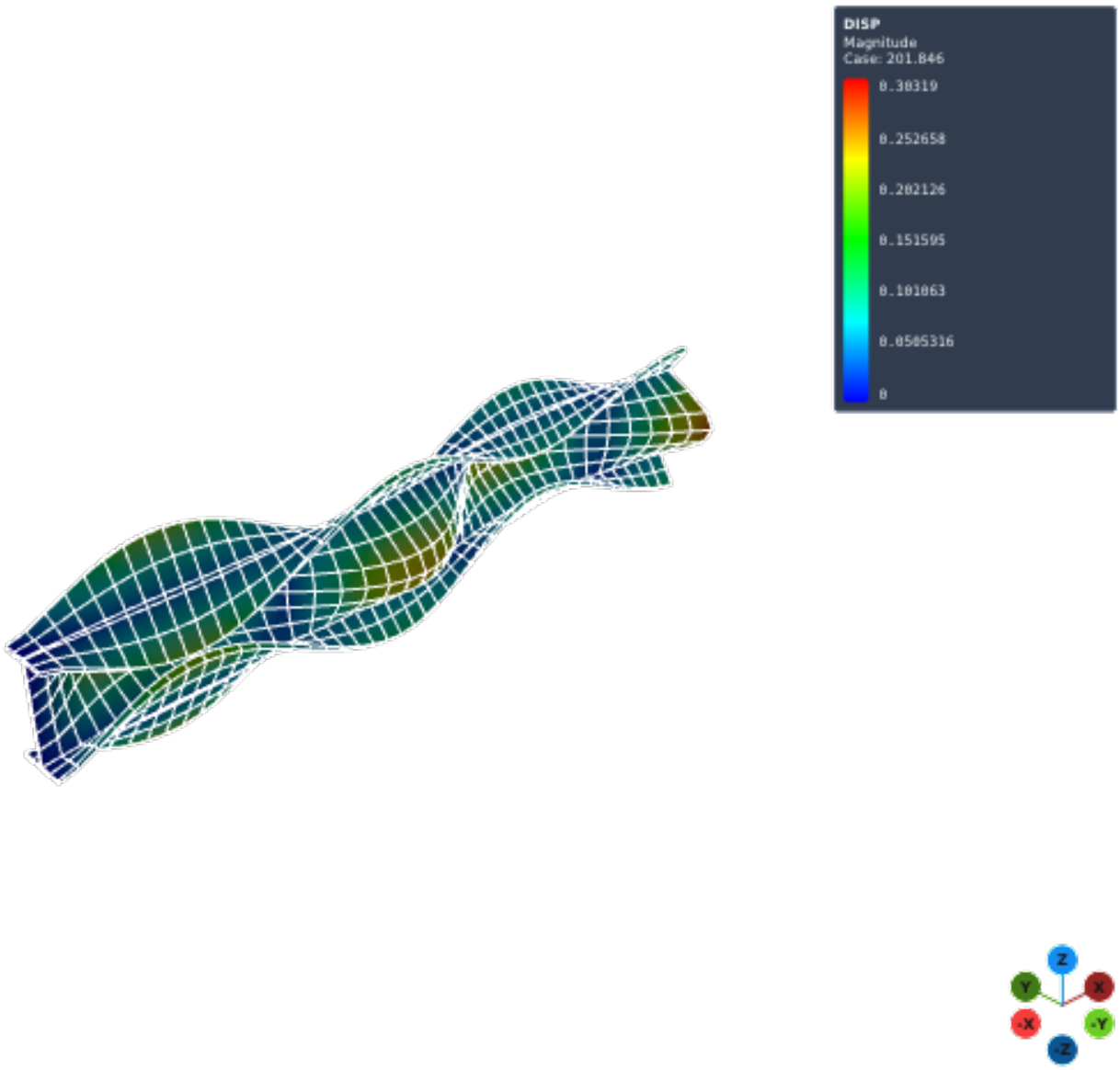

Figure 537: Calculix — mode 9, 201.846 Hz

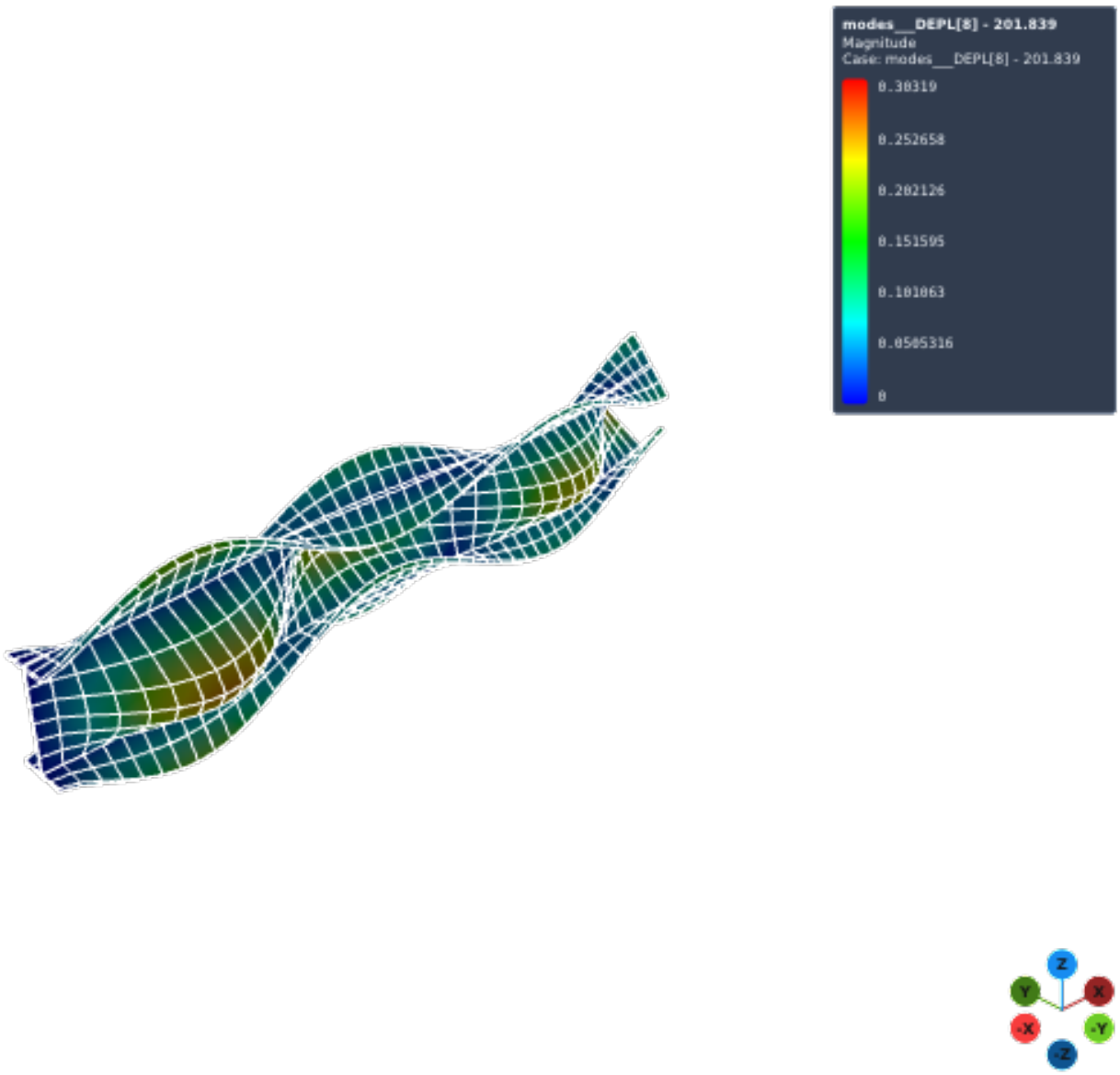

Figure 538: Code_Aster — mode 9, 201.839 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	201.840	-8.14e-04	1.60831e+06	9.86e-25	0.56	1.17e-21	-9.93e-13	-0.747	3.42e-11
Calculix	201.846	2.01e-03	1.60842e+06	3.90e-25	0.56	7.43e-22	-6.24e-13	0.747	2.73e-11
Code_Aster	201.839	-1.20e-03	1.60831e+06	3.60e-24	0.56	4.60e-21	-1.90e-12	-0.747	-6.78e-11

Mode 10

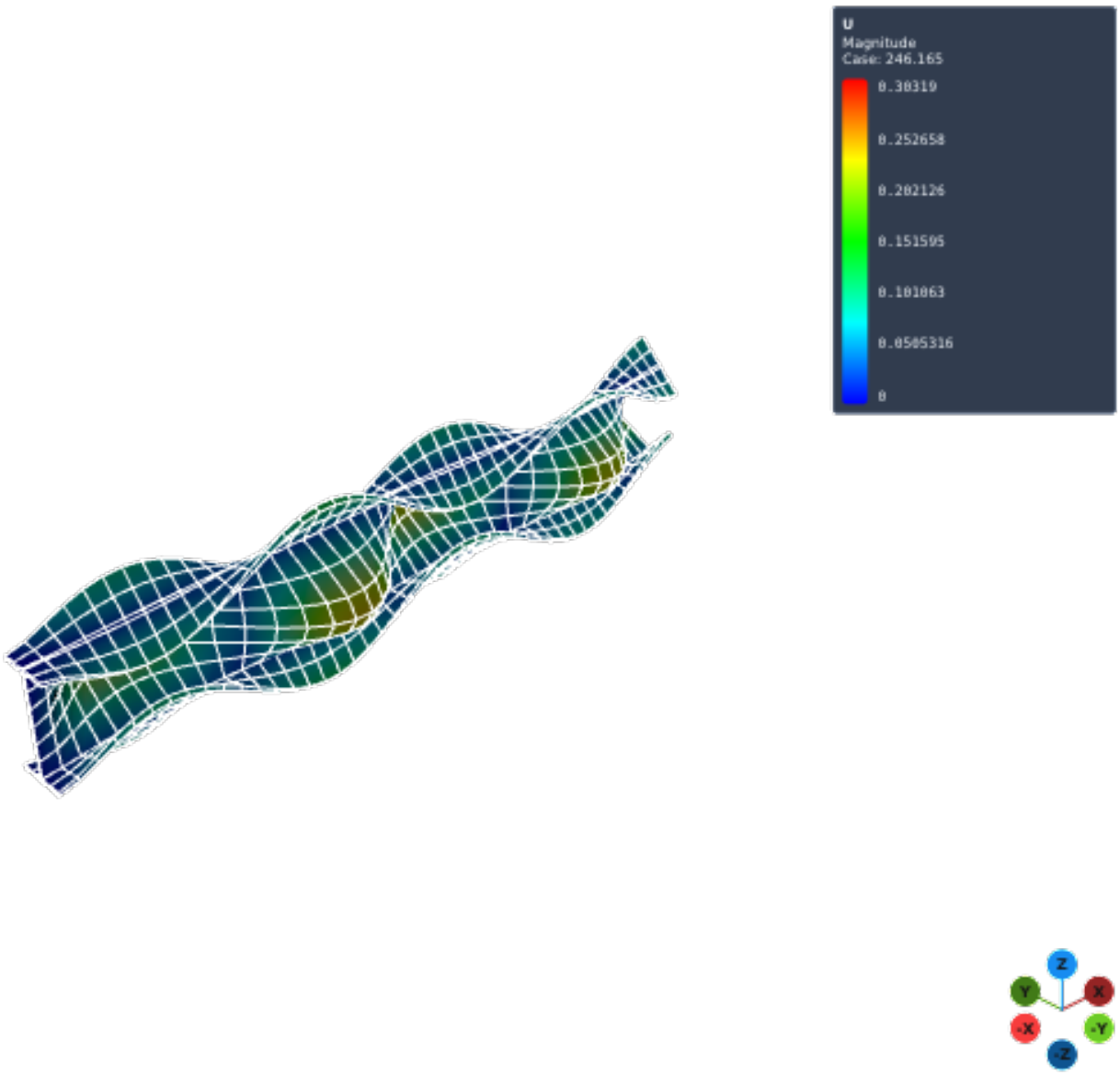

Figure 539: Abaqus — mode 10, 246.165 Hz

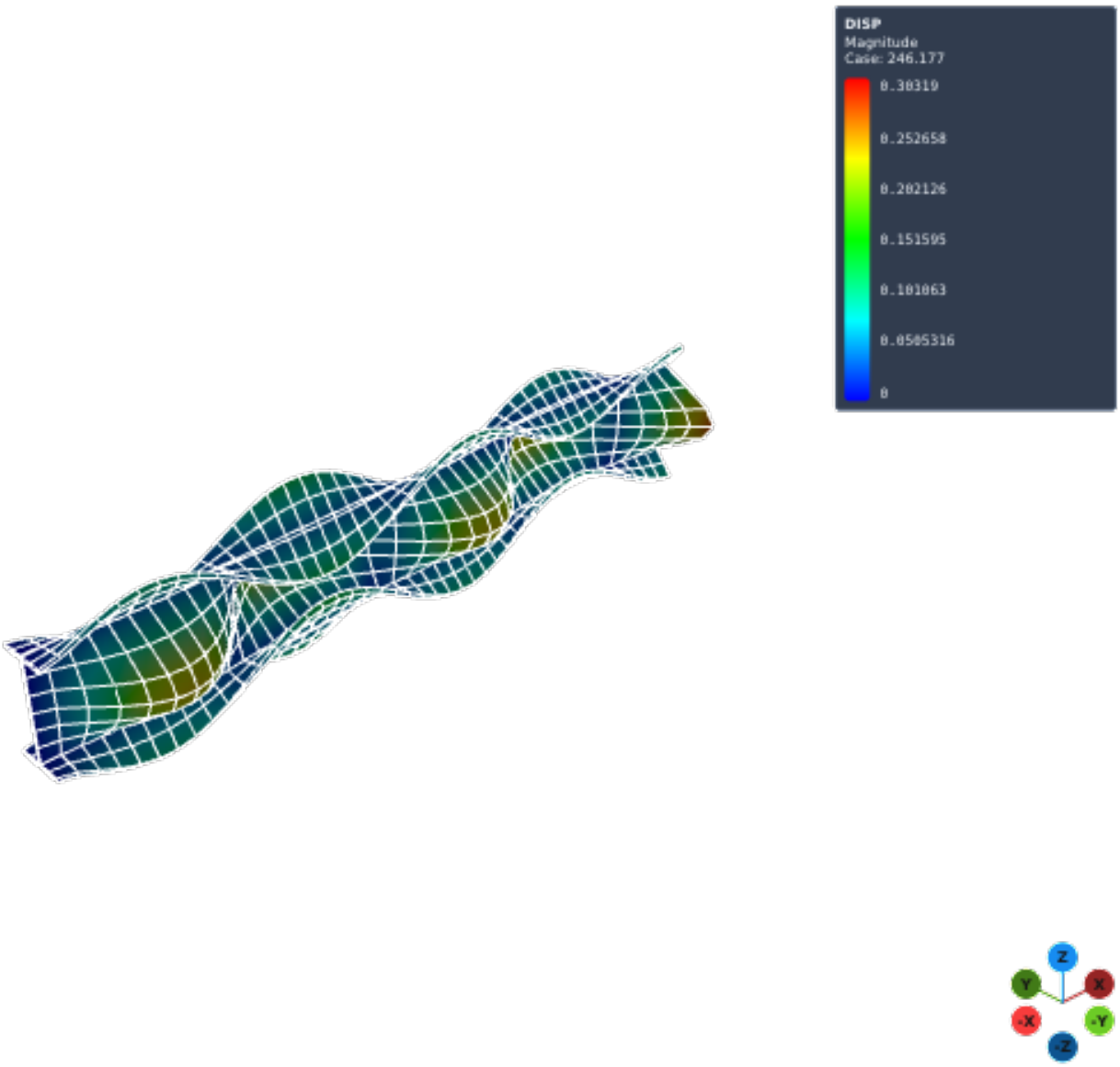

Figure 540: Calculix — mode 10, 246.177 Hz

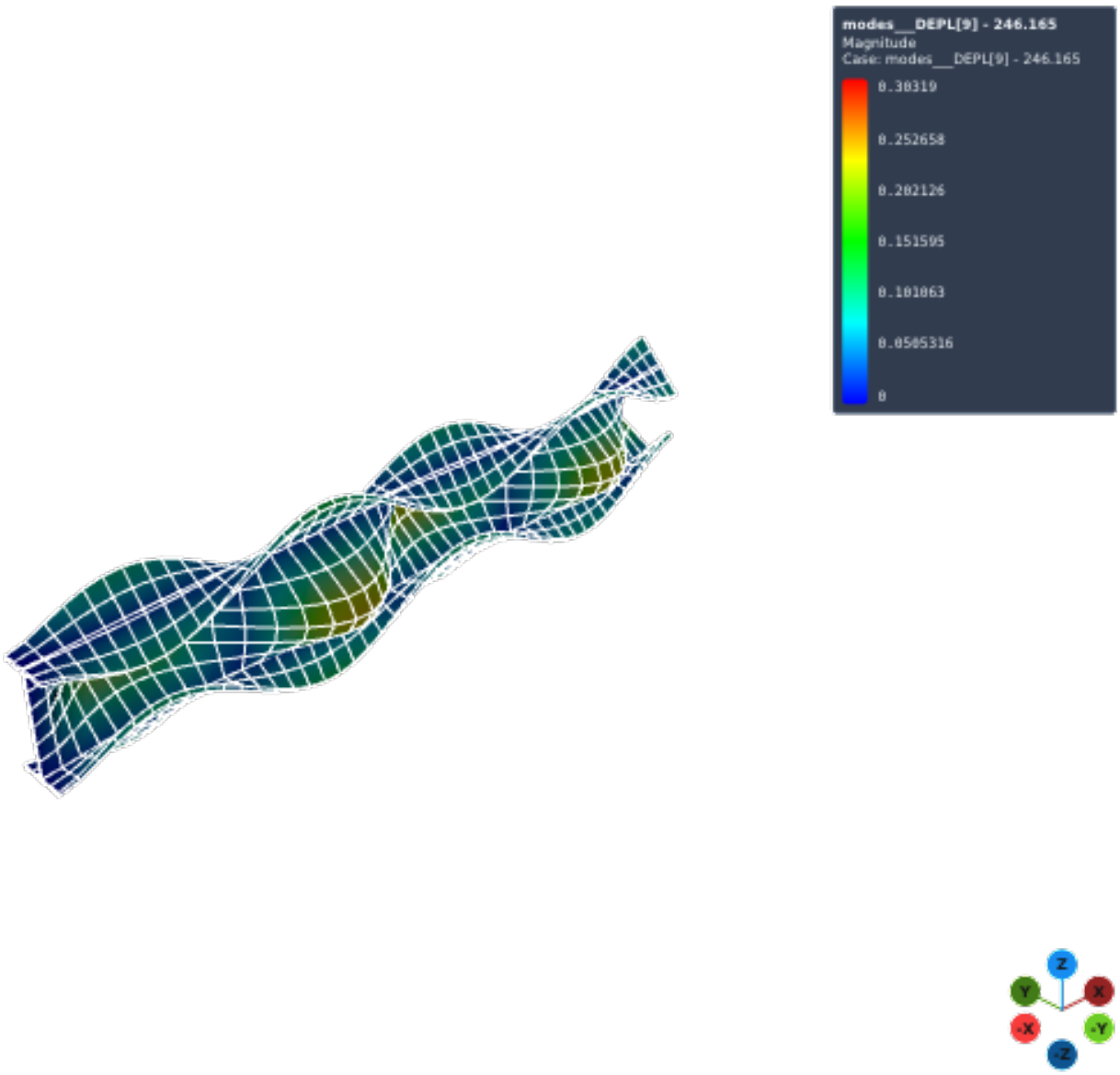

Figure 541: Code_Aster — mode 10, 246.165 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	246.170	-2.59e-04	2.39228e+06	3.32e-21	1.47	1.28e-24	5.76e-11	1.212	-1.13e-12
Calculix	246.177	2.54e-03	2.39251e+06	4.40e-29	1.47	7.89e-21	-6.63e-15	-1.213	8.88e-11
Code_Aster	246.165	-2.28e-03	2.39228e+06	4.39e-25	1.47	1.15e-19	6.62e-13	1.212	-3.38e-10

Mode 11

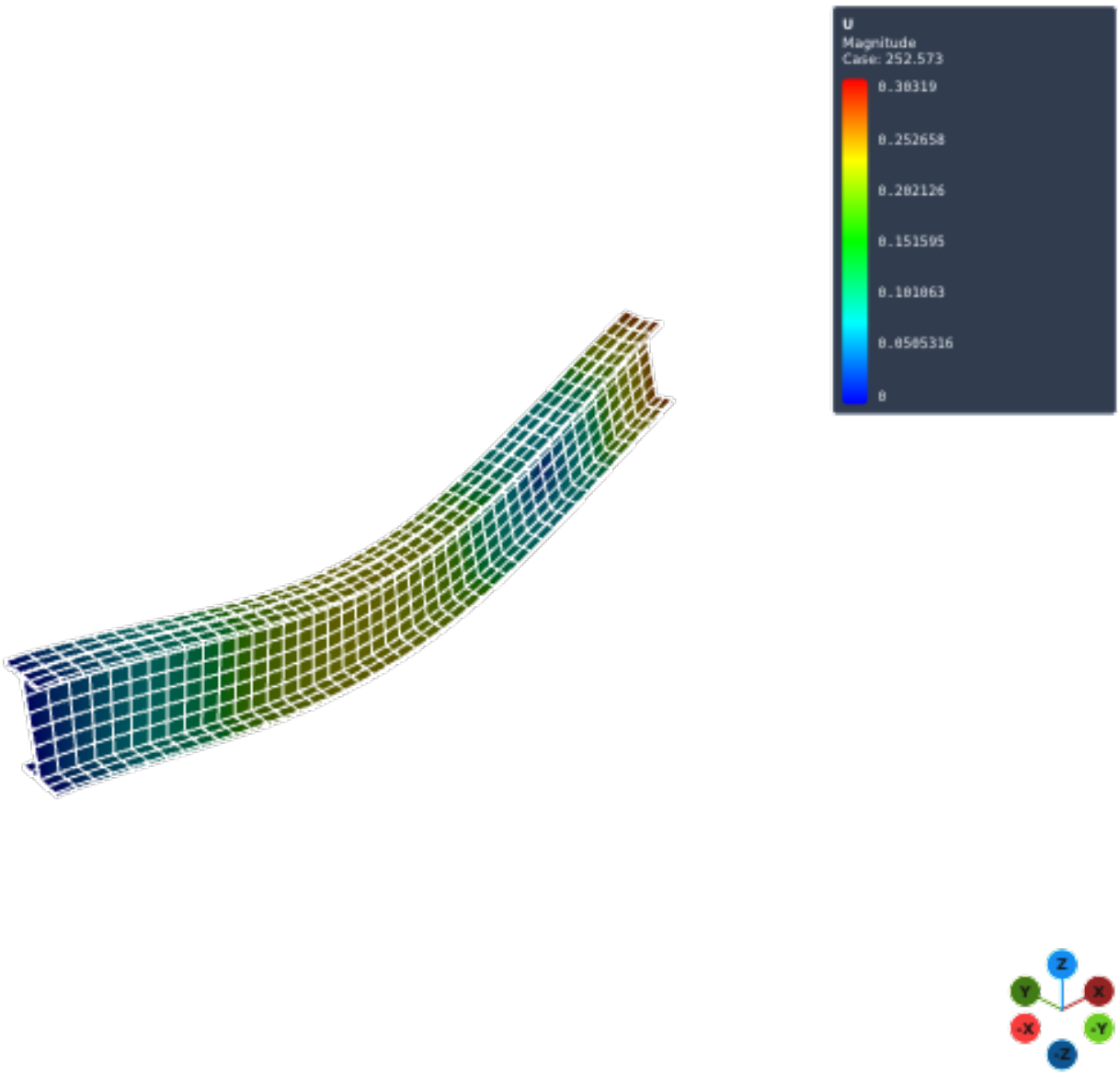

Figure 542: Abaqus — mode 11, 252.573 Hz

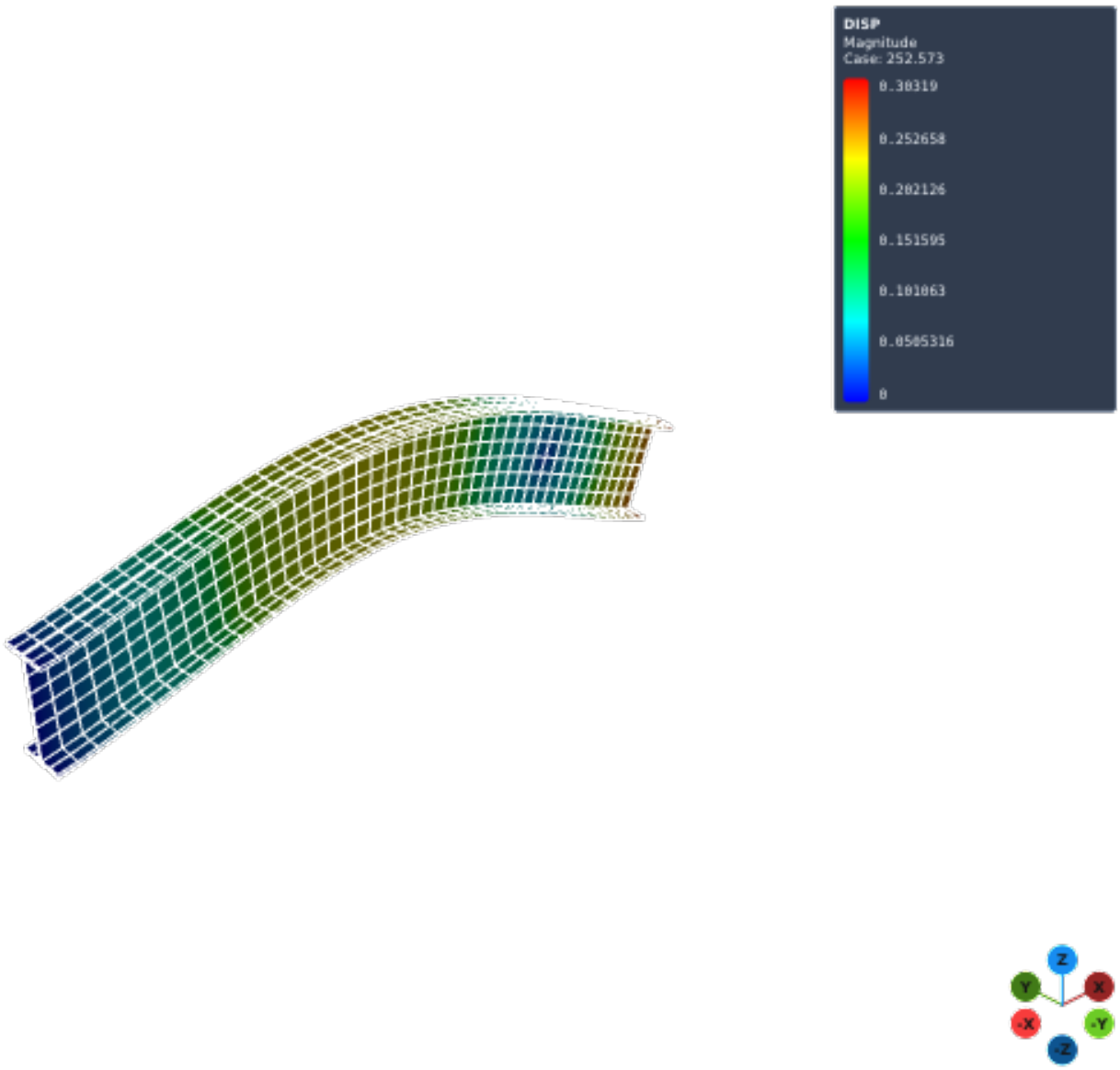

Figure 543: Calculix — mode 11, 252.573 Hz

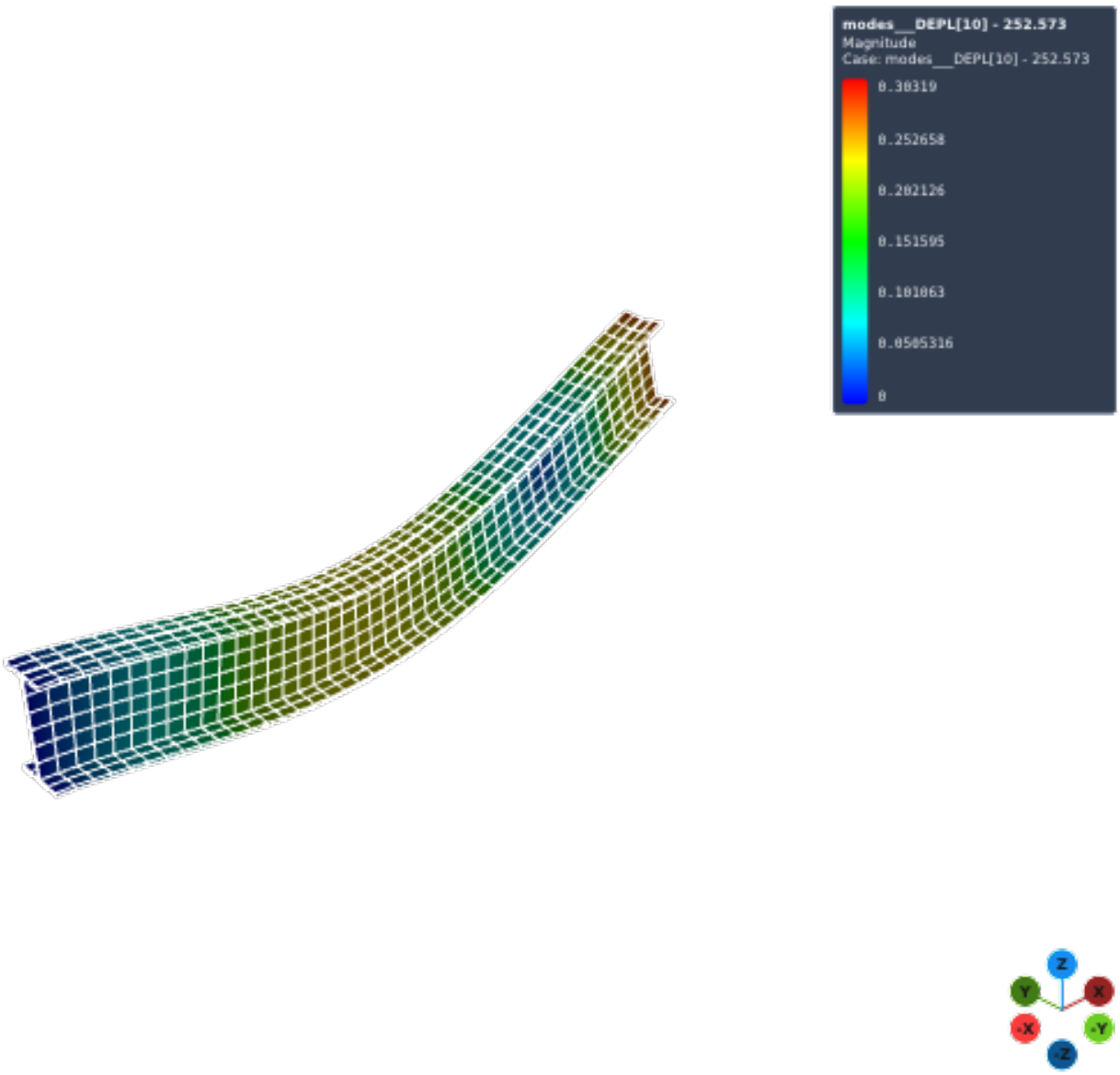

Figure 544: Code_Aster — mode 11, 252.573 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	252.570	-8.59e-04	2.51846e+06	4.57e-21	2.25e-19	40.26	6.76e-11	4.75e-10	-6.345
Calculix	252.573	4.48e-04	2.51846e+06	1.29e-24	5.04e-23	40.26	1.14e-12	-7.10e-12	6.345
Code_Aster	252.573	4.11e-04	2.51846e+06	1.85e-24	3.66e-21	40.26	-1.36e-12	-6.05e-11	-6.345

Plate eigenvalue analysis

The plain and the stiffened strip, at the eigen case’s 0.0625 m seed.

Shell, 1st order, plain

Mode 1

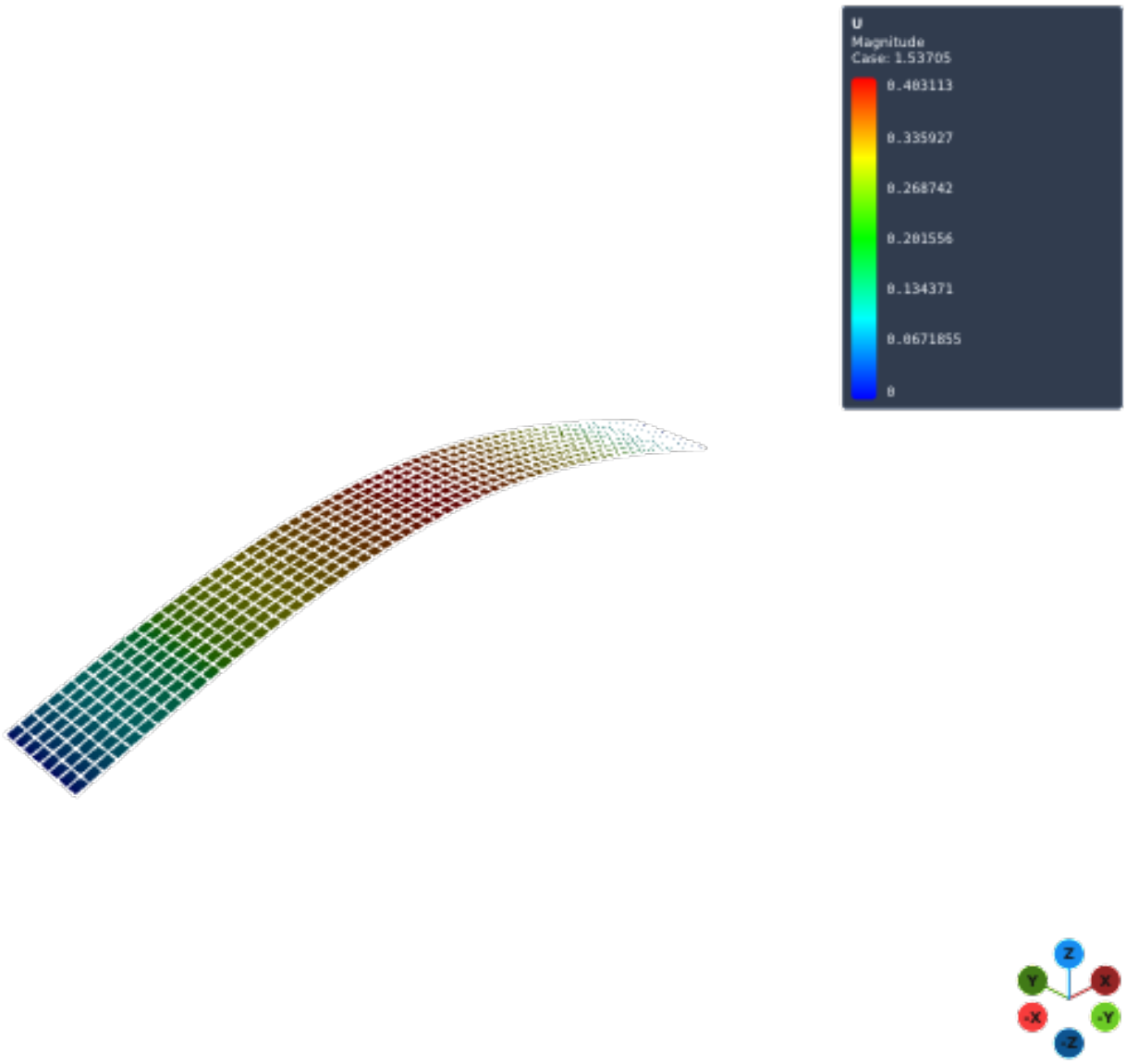

Figure 545: Abaqus — mode 1, 1.537 Hz

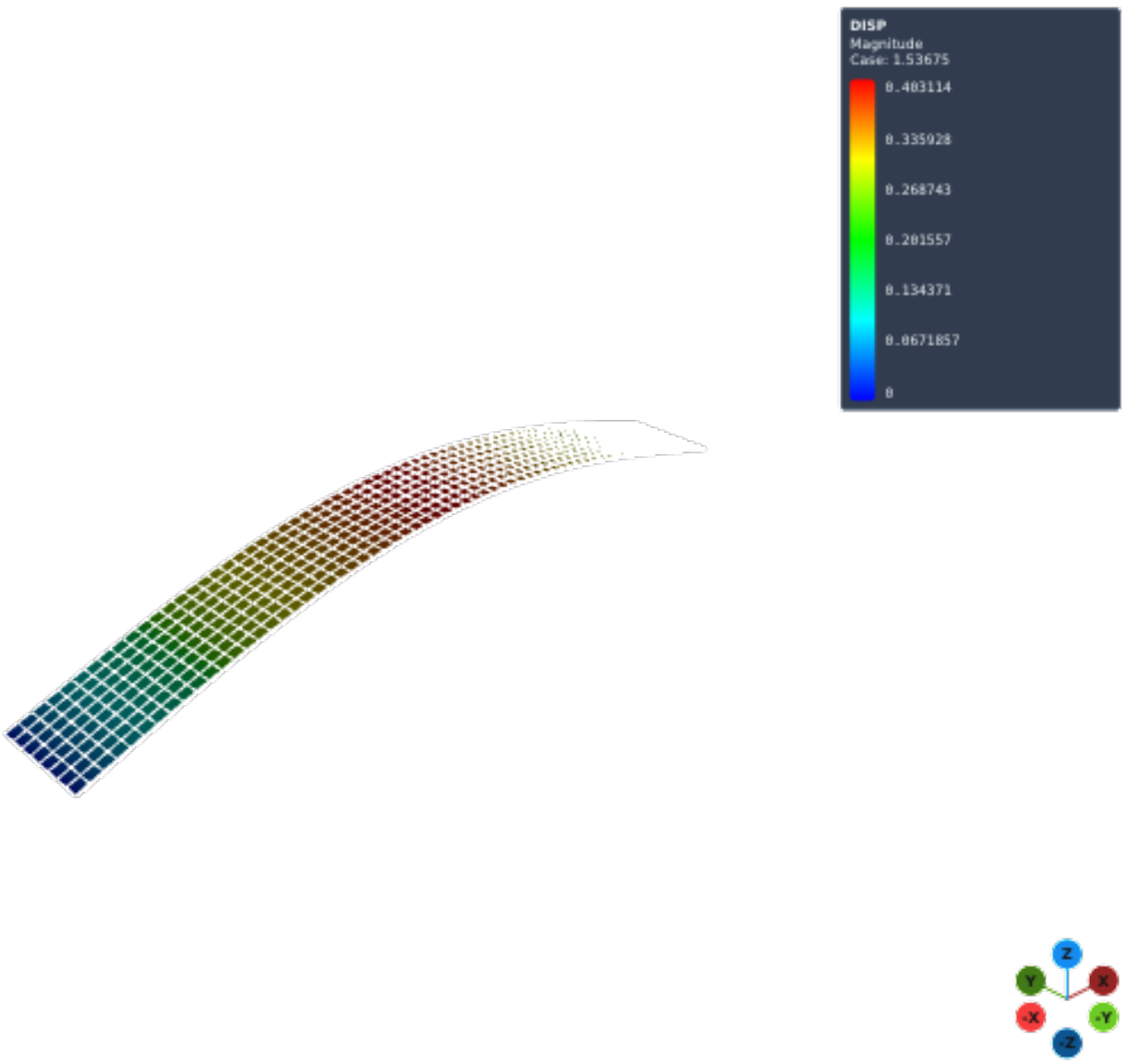

Figure 546: Calculix — mode 1, 1.537 Hz

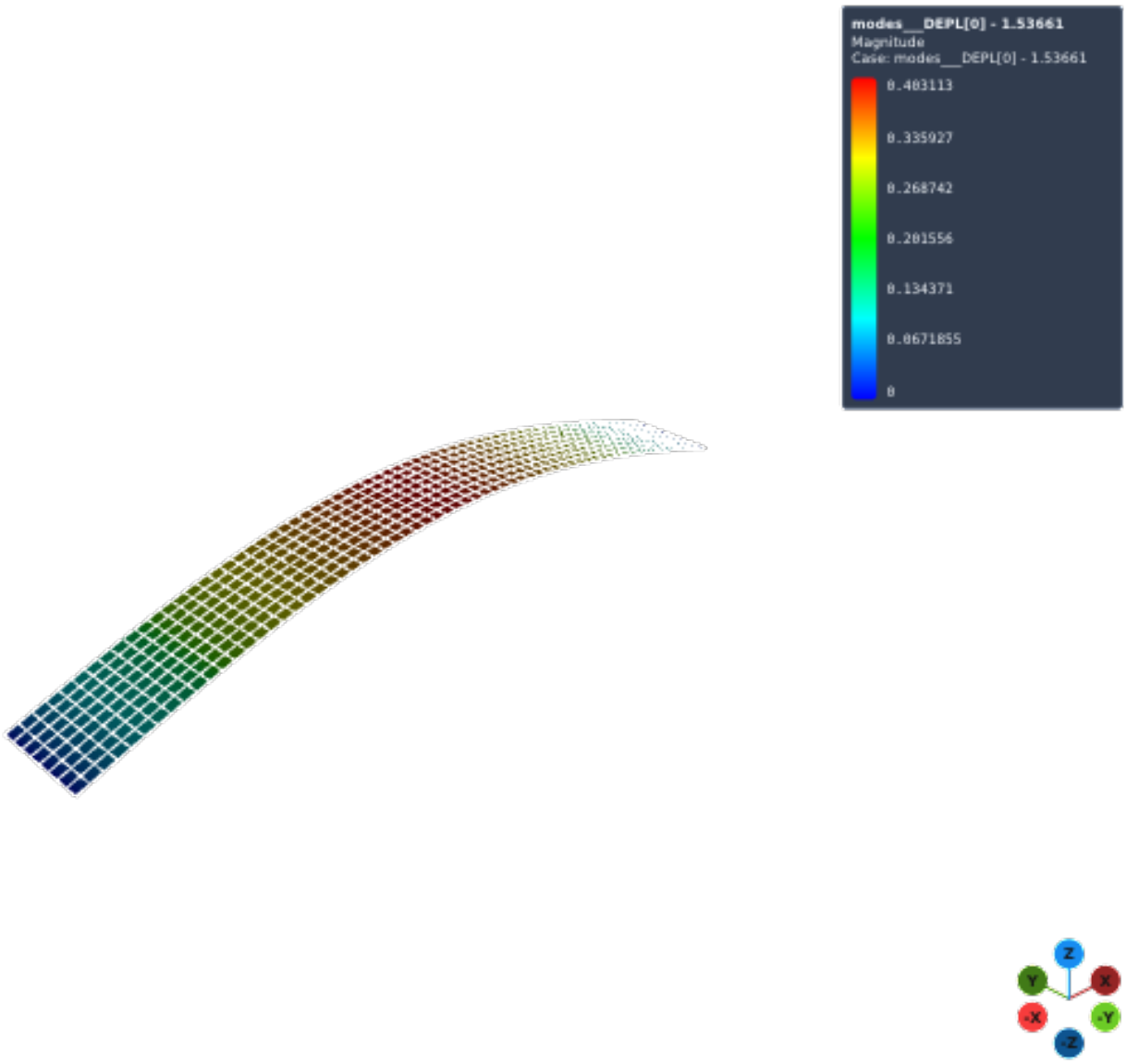

Figure 547: Code_Aster — mode 1, 1.537 Hz

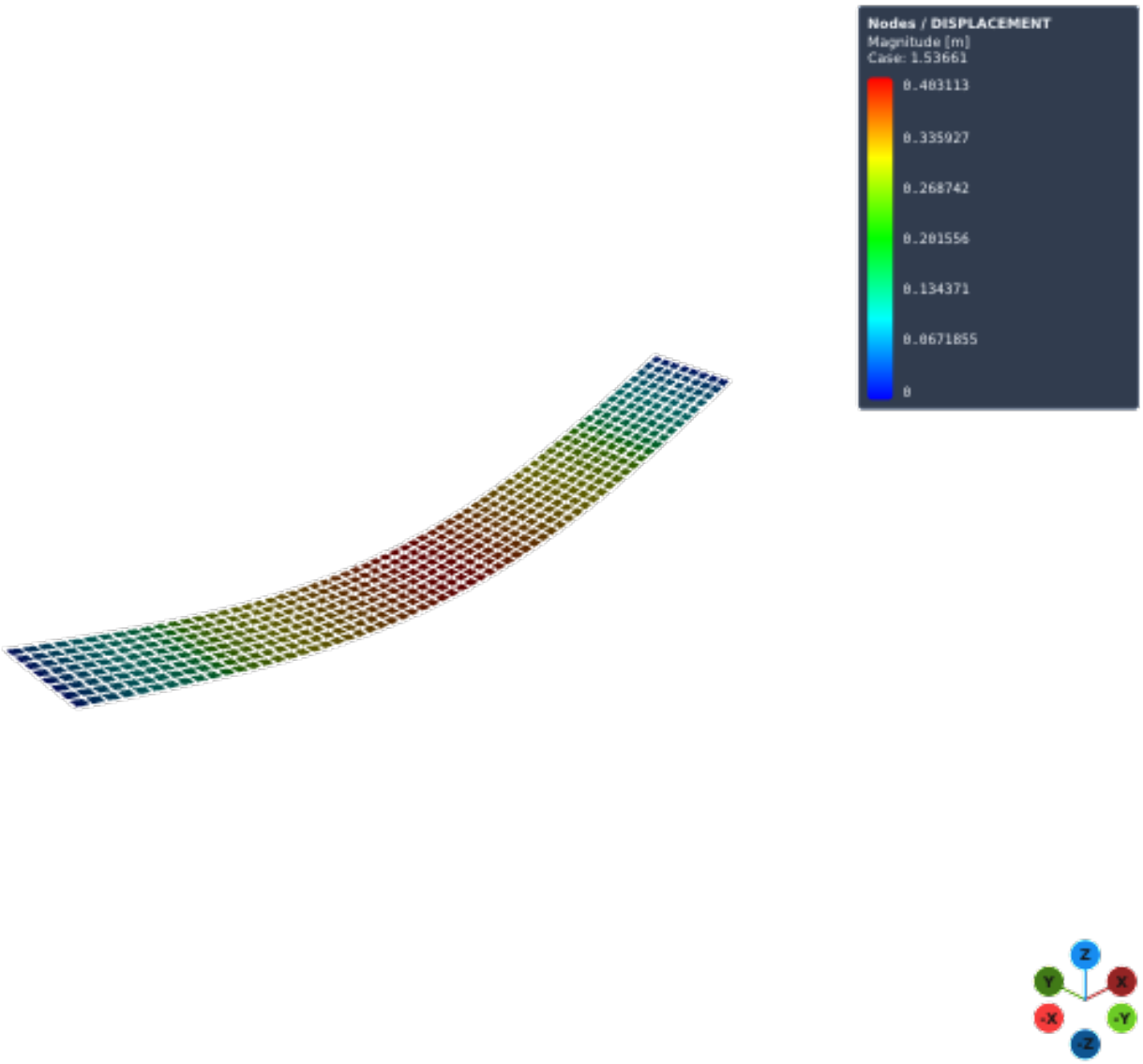

Figure 548: Sesam — mode 1, 1.537 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	1.537	+0.03	93.269	6.40e-31	3.03e-35	127.26	-8.00e-16	5.50e-18	11.281
Calculix	1.537	+0.01	93.2317	3.65e-09	1.07e-29	127.22	6.04e-05	-3.26e-15	11.279
Code_Aster	1.537	2.03e-07	93.2152	2.24e-49	1.19e-57	127.26	-4.74e-25	-3.46e-29	11.281
Sesam	1.537	-4.98e-06	93.2152	1.43e-40	1.48e-45	127.17	-1.19e-20	3.85e-23	-11.277
Closed form	1.537								

Mode 2

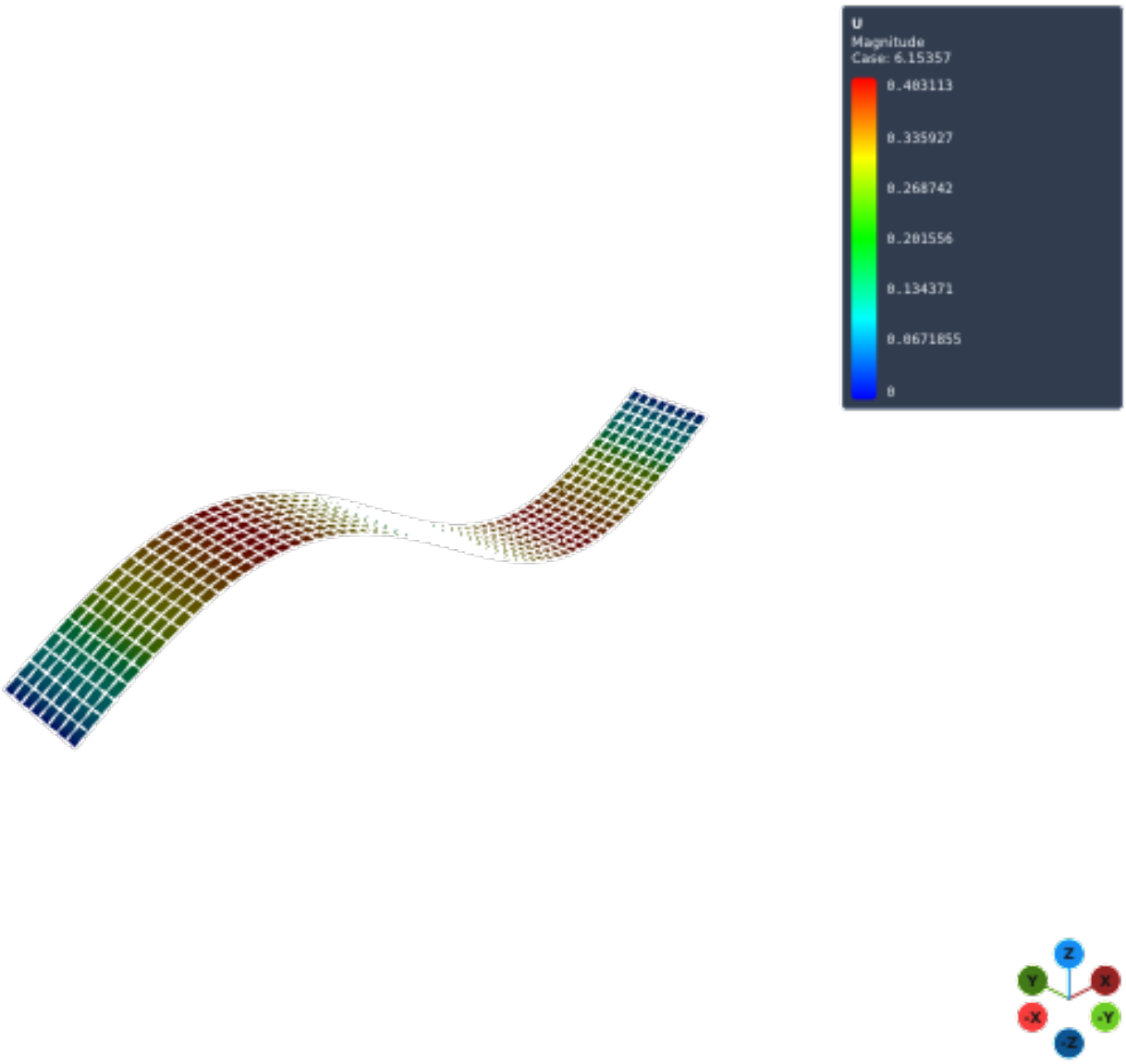

Figure 549: Abaqus — mode 2, 6.154 Hz

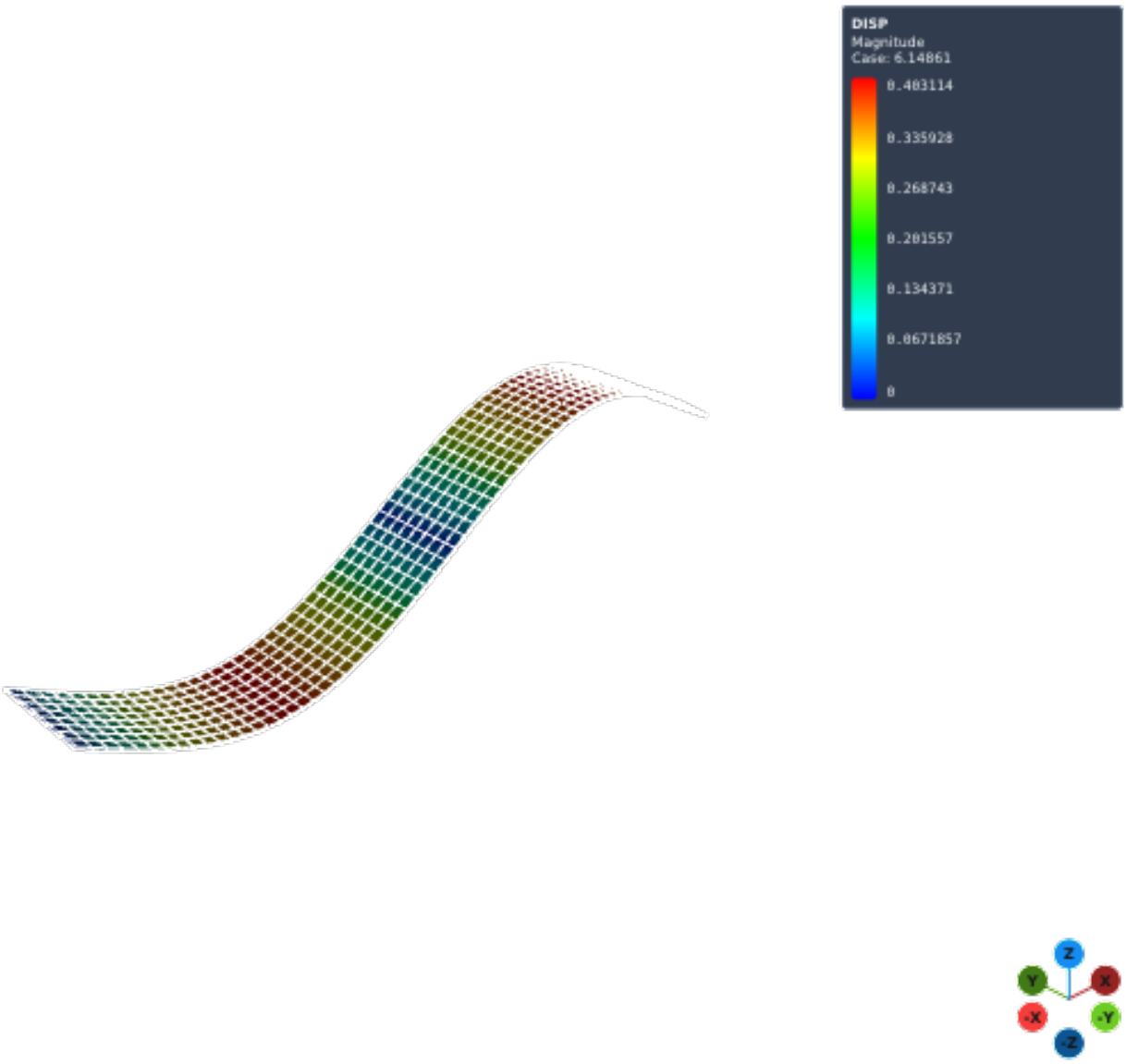

Figure 550: Calculix — mode 2, 6.149 Hz

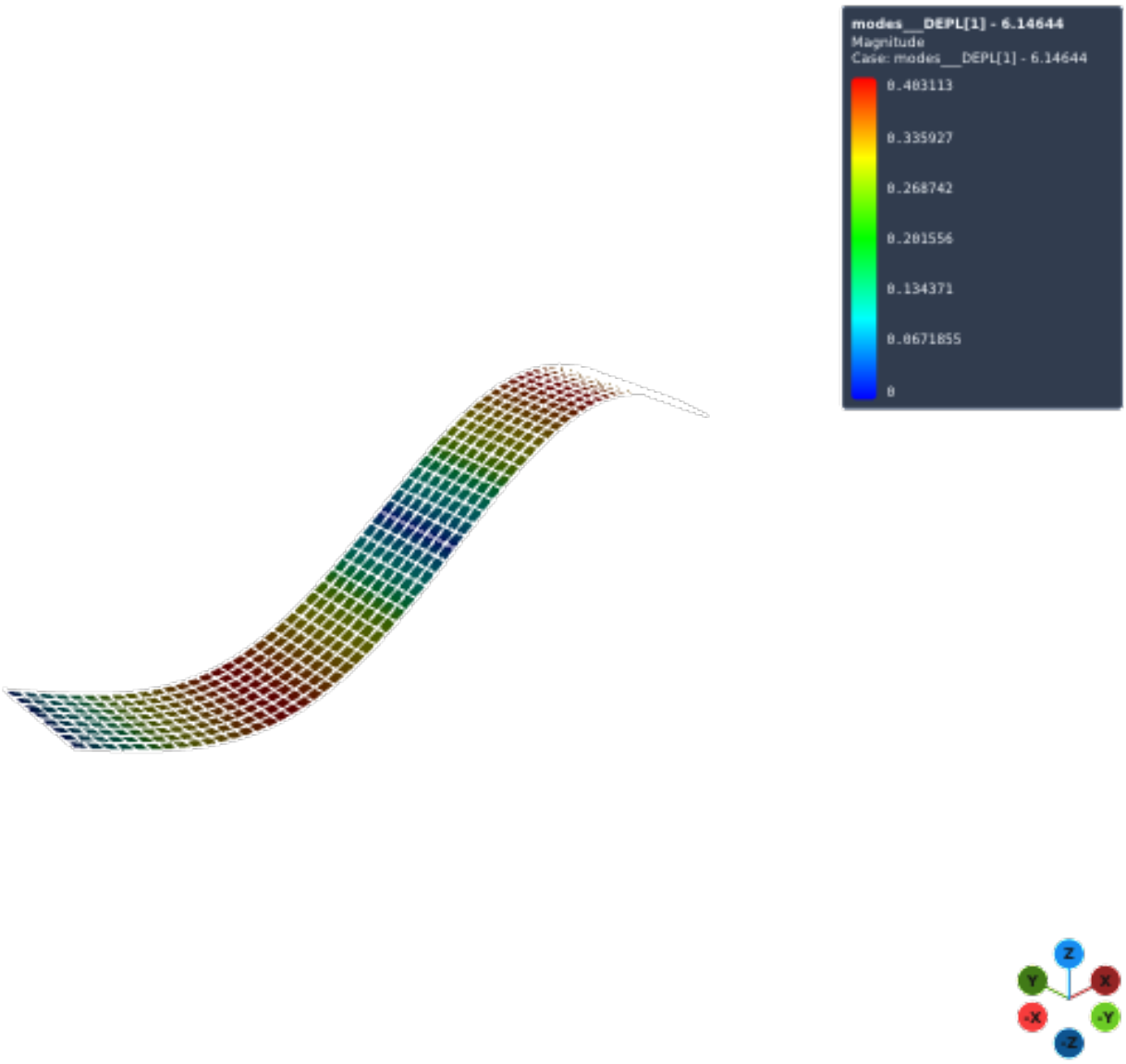

Figure 551: Code_Aster — mode 2, 6.146 Hz

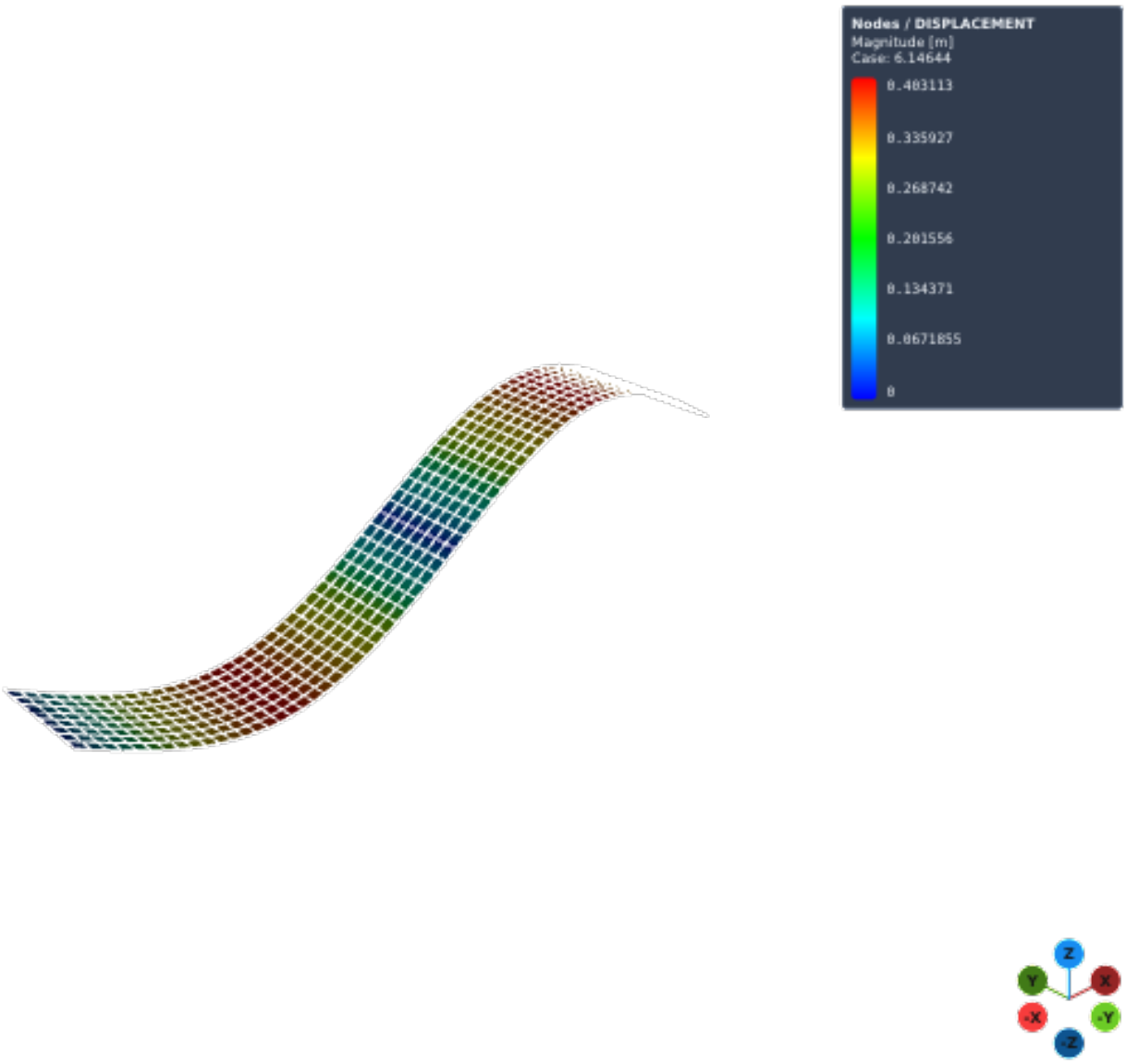

Figure 552: Sesam — mode 2, 6.146 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	6.154	+0.12	1494.9	5.42e-31	1.34e-33	5.32e-21	7.36e-16	-3.66e-17	7.29e-11
Calculix	6.149	+0.04	1492.5	1.46e-08	9.33e-31	4.33e-15	-1.21e-04	9.66e-16	-6.58e-08
Code_Aster	6.146	4.29e-06	1491.44	1.65e-44	1.01e-52	1.29e-22	-1.29e-22	-1.00e-26	1.13e-11
Sesam	6.146	1.13e-05	1491.44	1.40e-34	3.63e-39	9.71e-23	1.18e-17	-6.03e-20	-9.85e-12
Closed form	6.146								

Mode 3

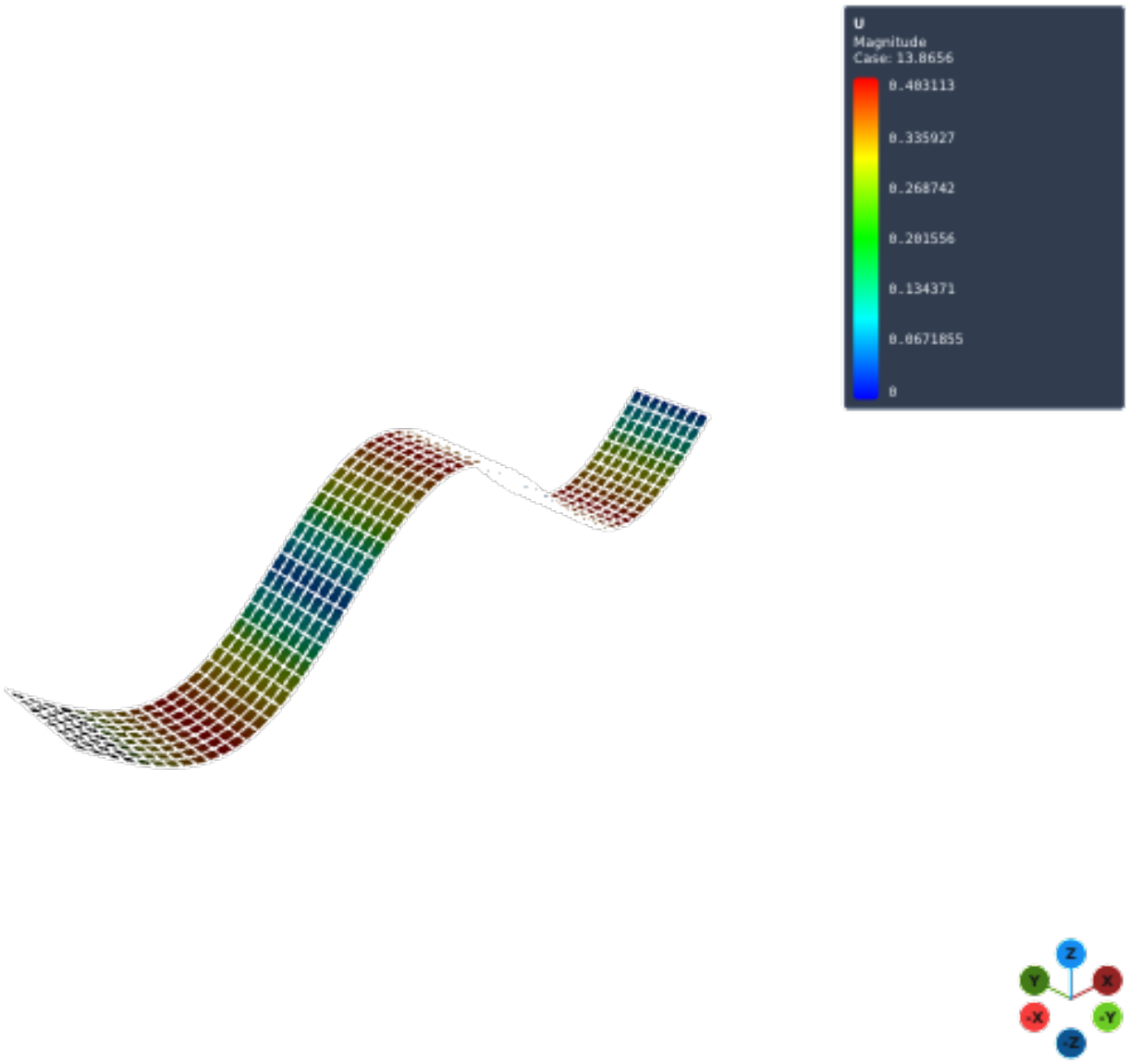

Figure 553: Abaqus — mode 3, 13.866 Hz

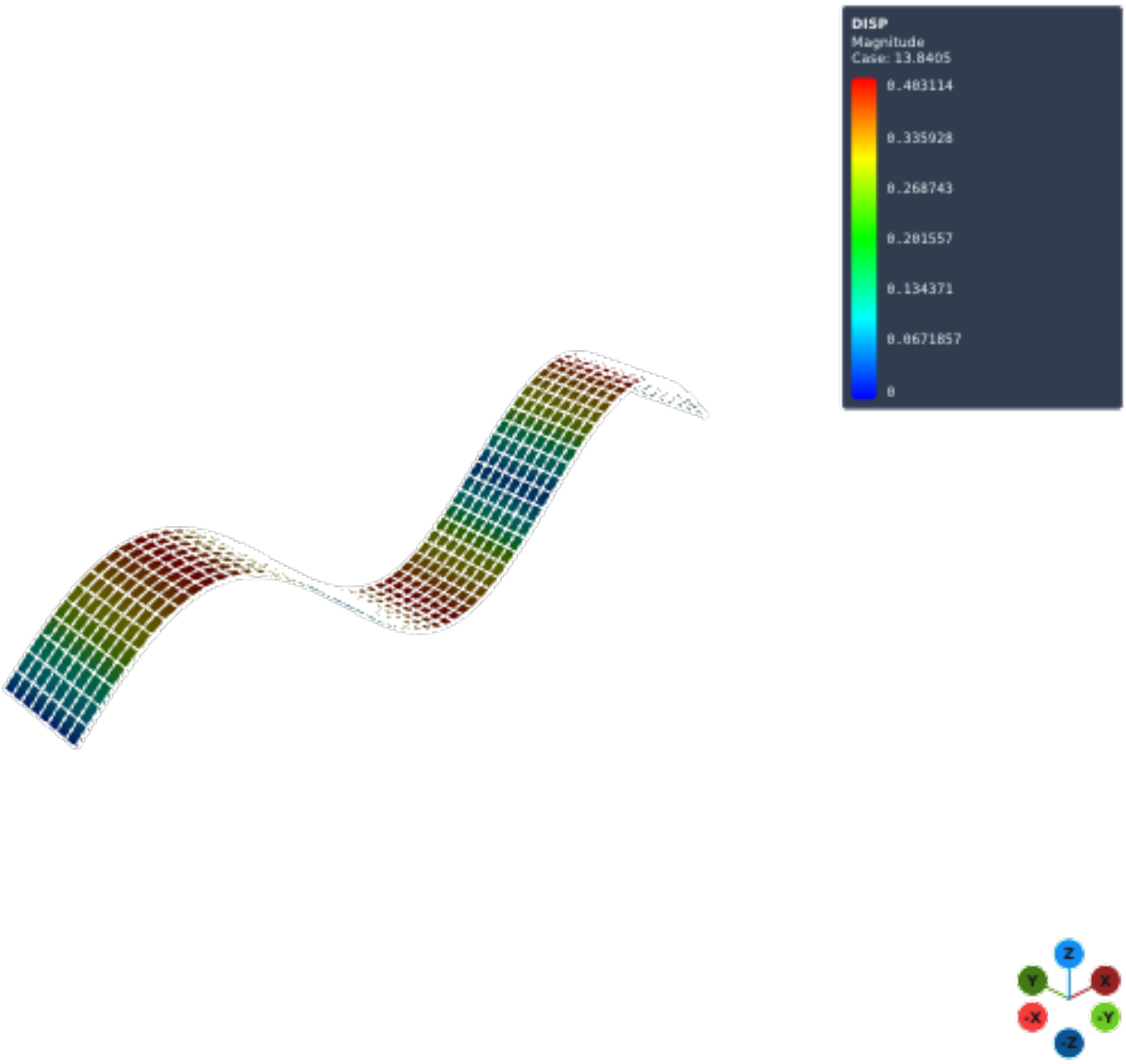

Figure 554: Calculix — mode 3, 13.840 Hz

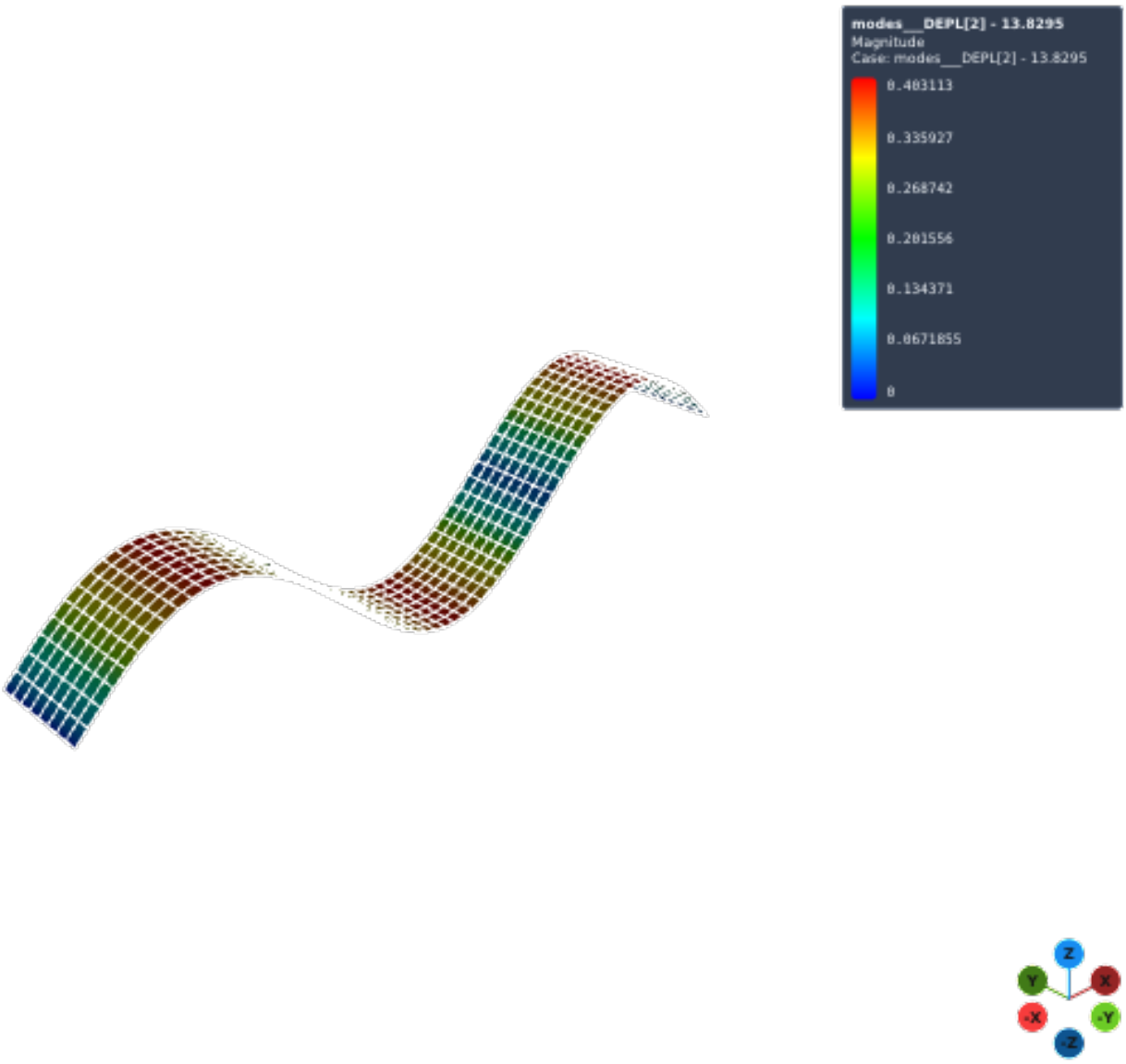

Figure 555: Code_Aster — mode 3, 13.829 Hz

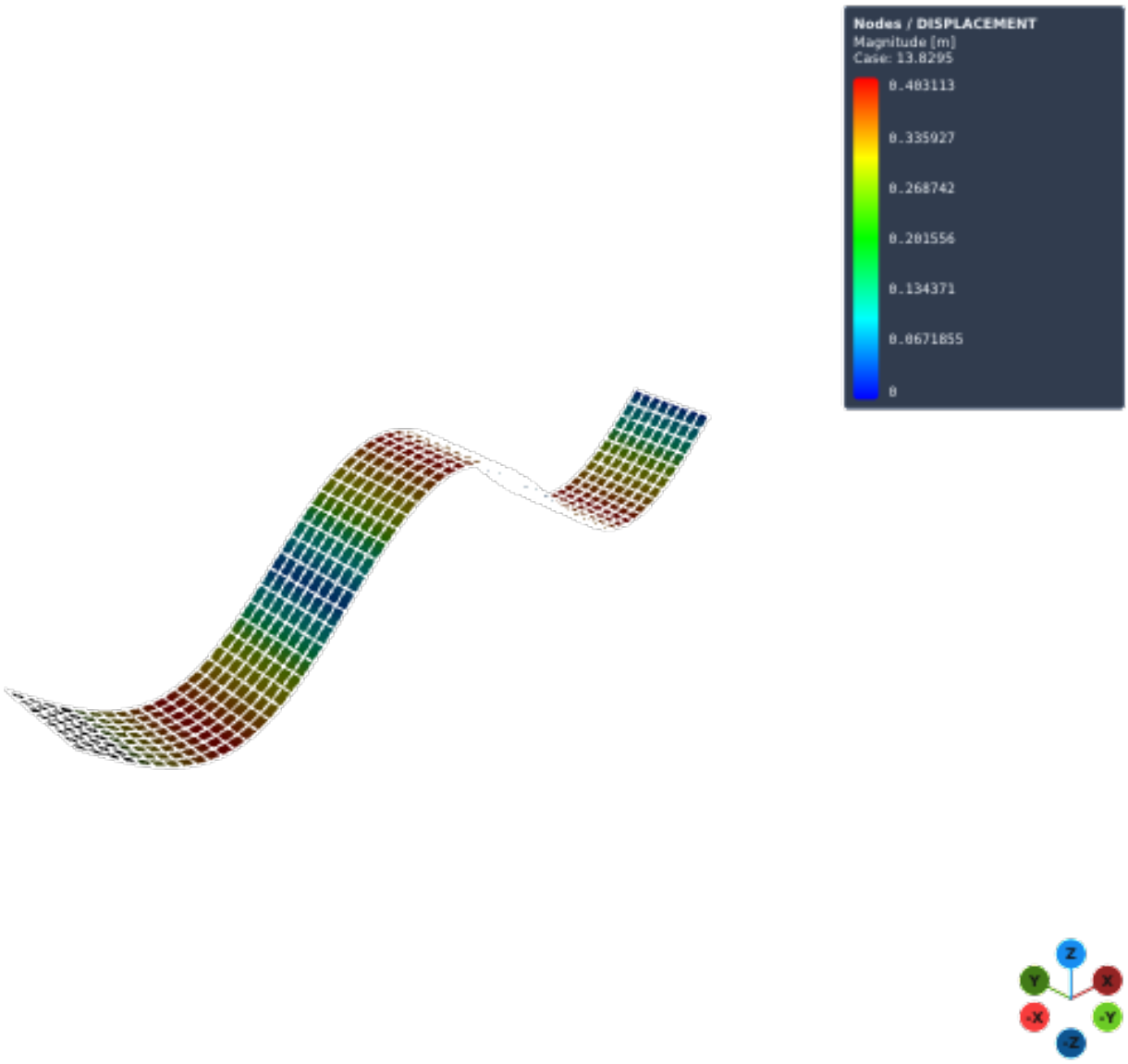

Figure 556: Sesam — mode 3, 13.829 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	13.866	+0.26	7589.9	4.43e-30	5.80e-32	14.14	2.10e-15	-2.41e-16	-3.760
Calculix	13.840	+0.08	7562.44	3.27e-08	4.89e-31	14.11	1.81e-04	6.99e-16	3.756
Code_Aster	13.829	2.17e-05	7550.43	2.99e-41	9.78e-50	14.14	-5.47e-21	-3.13e-25	3.760
Sesam	13.829	6.01e-05	7550.43	1.99e-35	2.24e-40	14.05	-4.46e-18	1.50e-20	-3.748
Closed form	13.829								

Mode 4

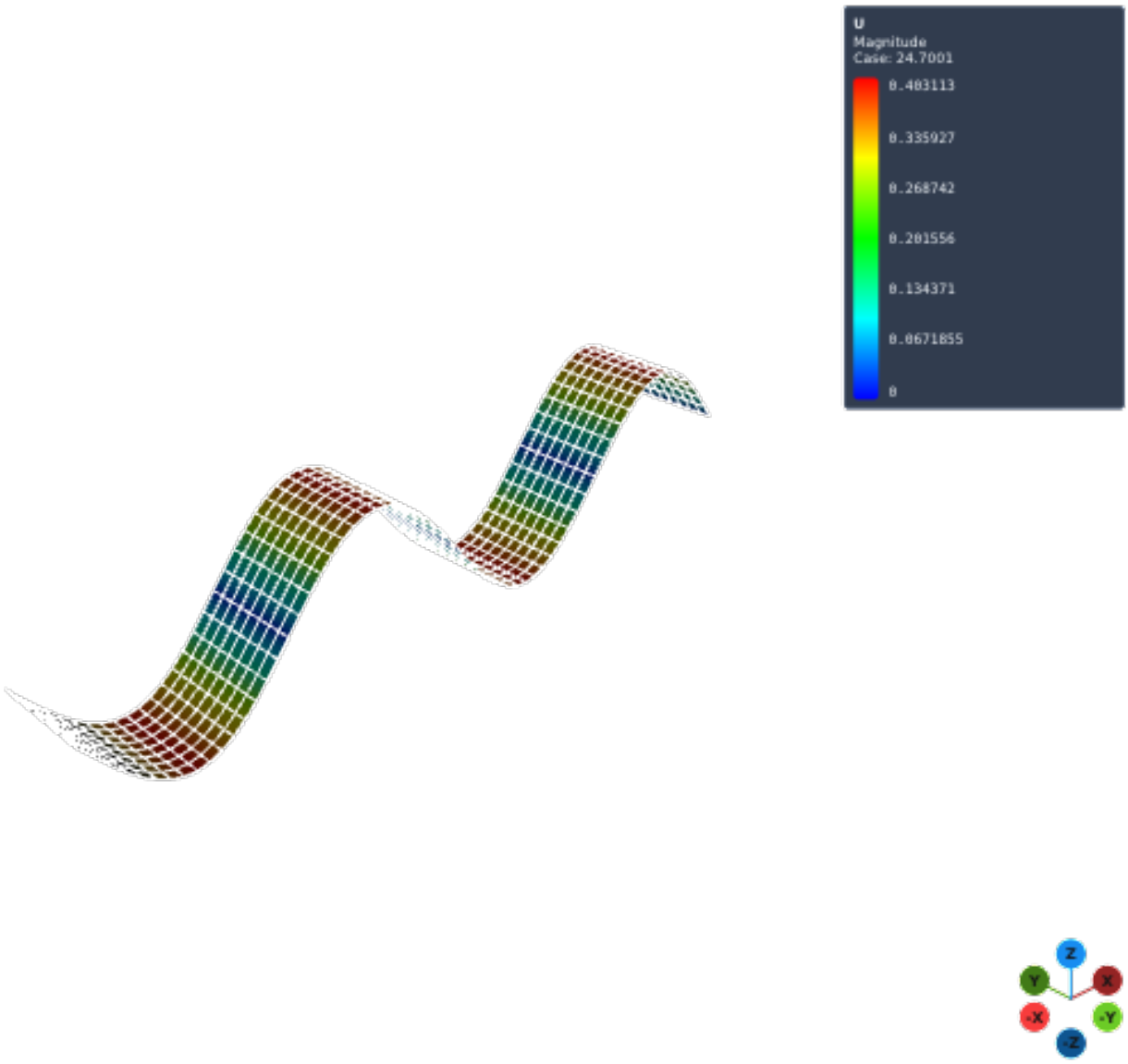

Figure 557: Abaqus — mode 4, 24.700 Hz

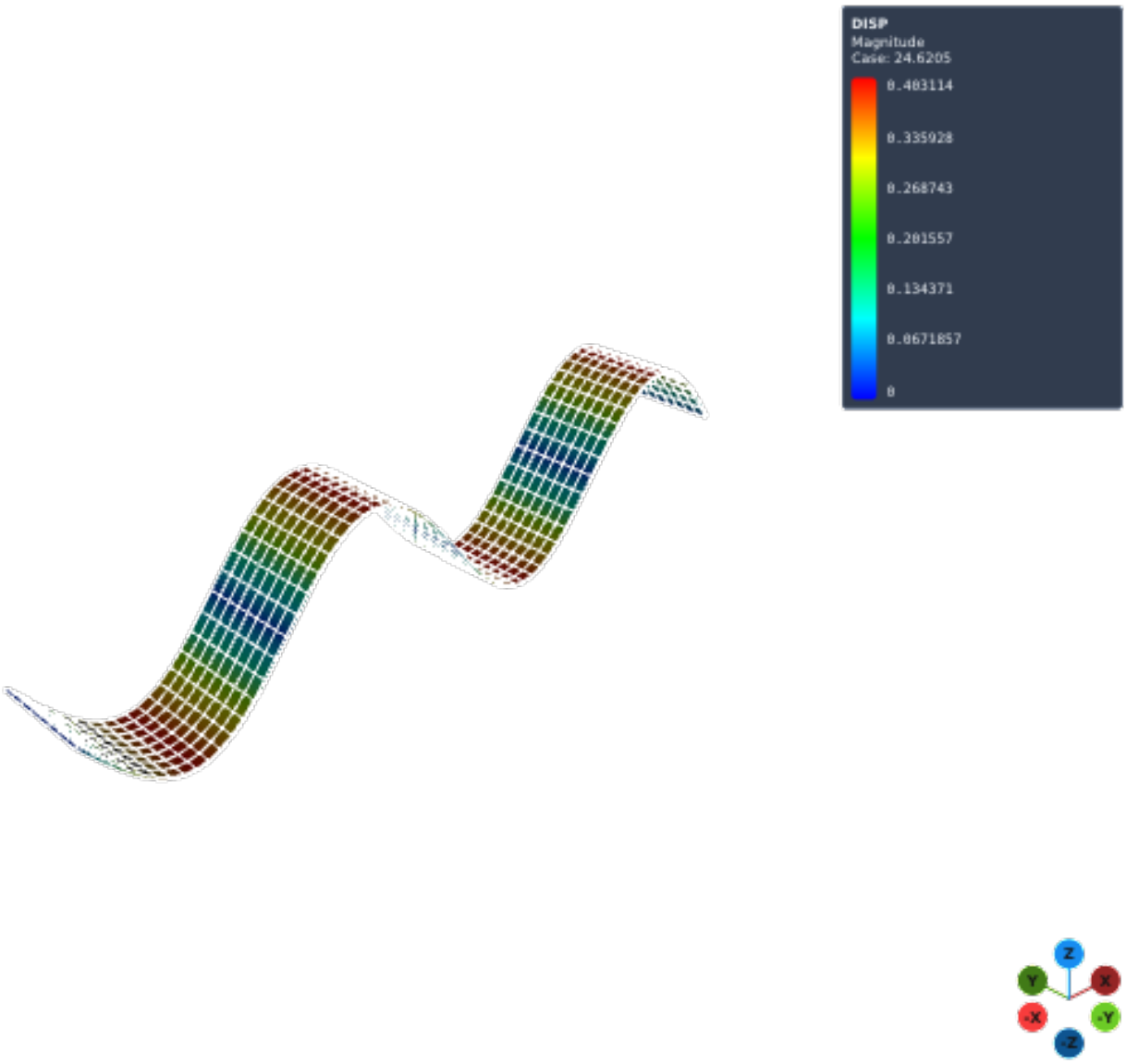

Figure 558: Calculix — mode 4, 24.621 Hz

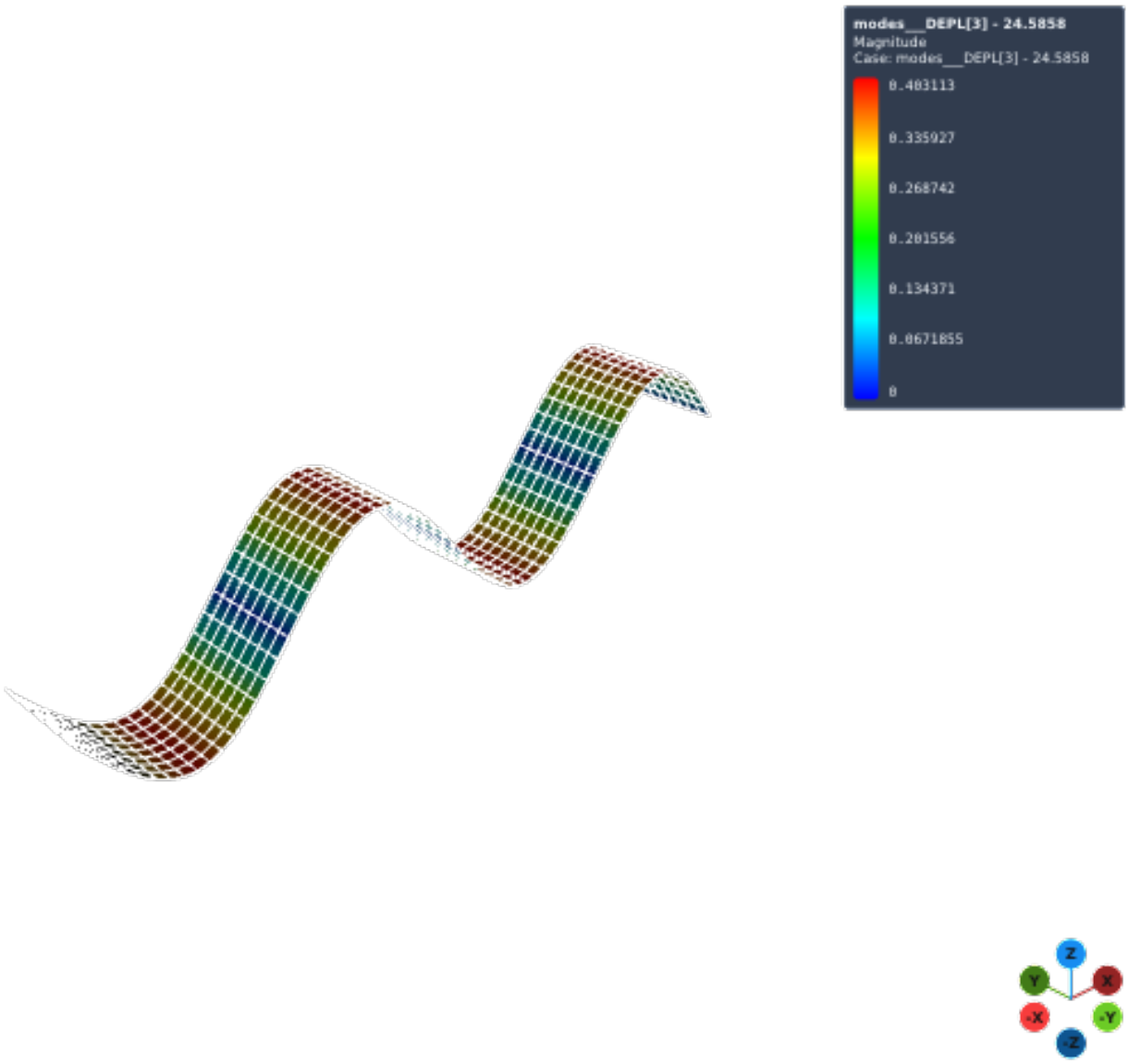

Figure 559: Code_Aster — mode 4, 24.586 Hz

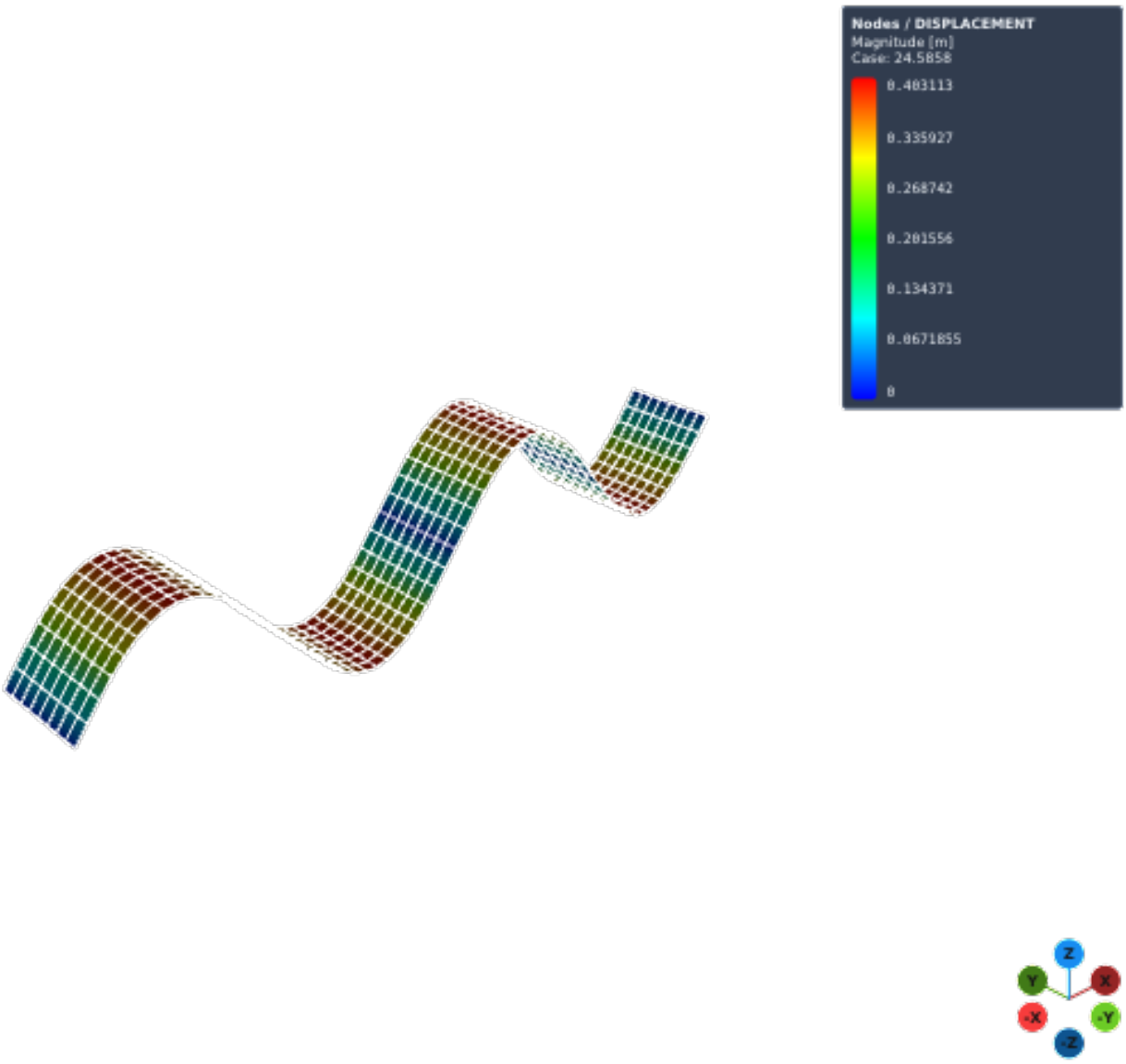

Figure 560: Sesam — mode 4, 24.586 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	24.700	+0.46	24086	1.04e-28	7.36e-32	1.83e-23	-1.02e-14	2.71e-16	4.27e-12
Calculix	24.621	+0.14	23930.6	5.80e-08	9.05e-31	5.81e-21	-2.41e-04	-9.51e-16	-7.62e-11
Code_Aster	24.586	6.86e-05	23863.1	5.65e-40	4.68e-49	3.83e-24	-2.38e-20	6.84e-25	1.96e-12
Sesam	24.586	1.01e-04	23863.1	9.91e-32	6.38e-36	1.14e-23	-3.15e-16	2.53e-18	3.37e-12
Closed form	24.586								

Mode 5

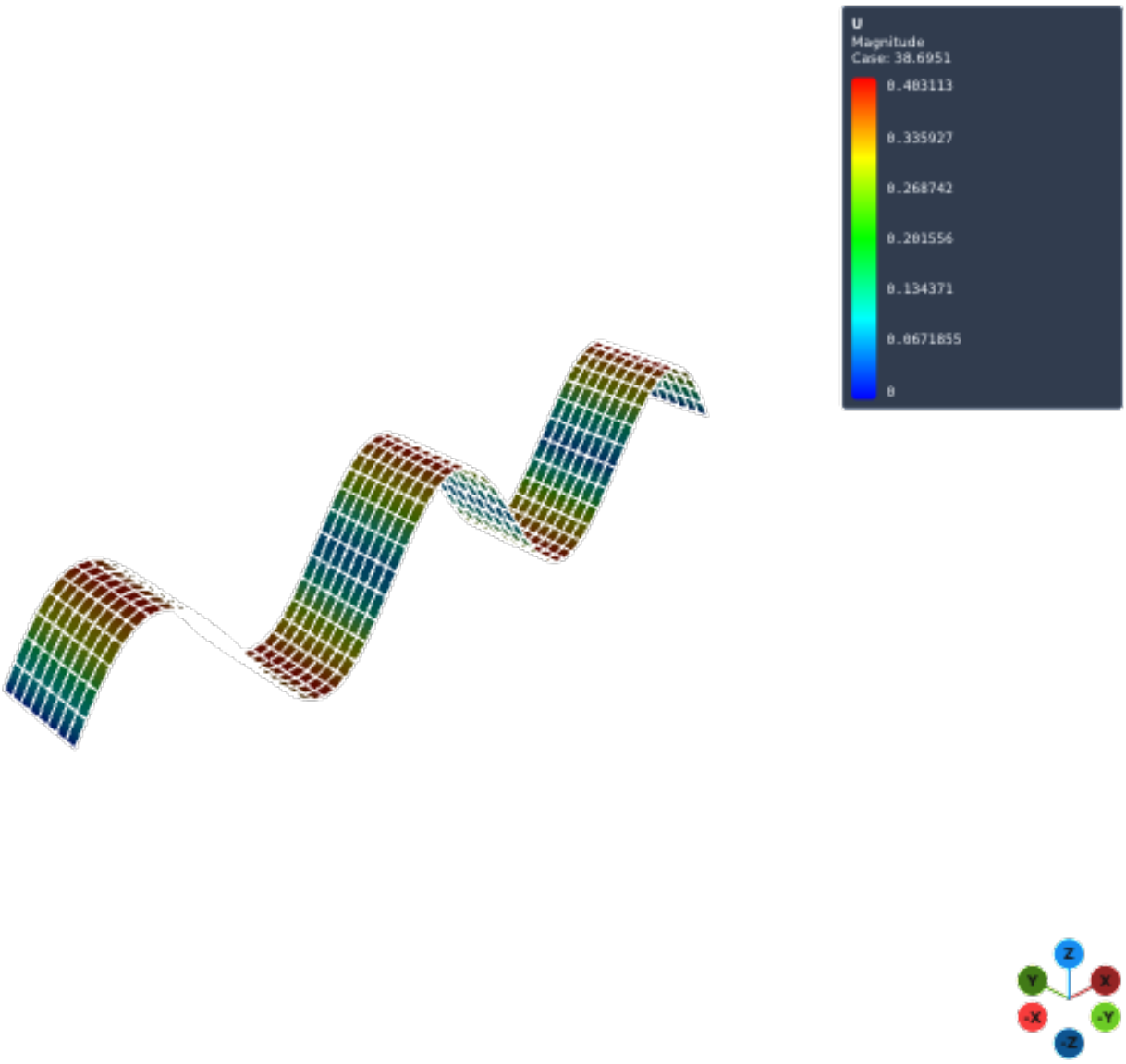

Figure 561: Abaqus — mode 5, 38.695 Hz

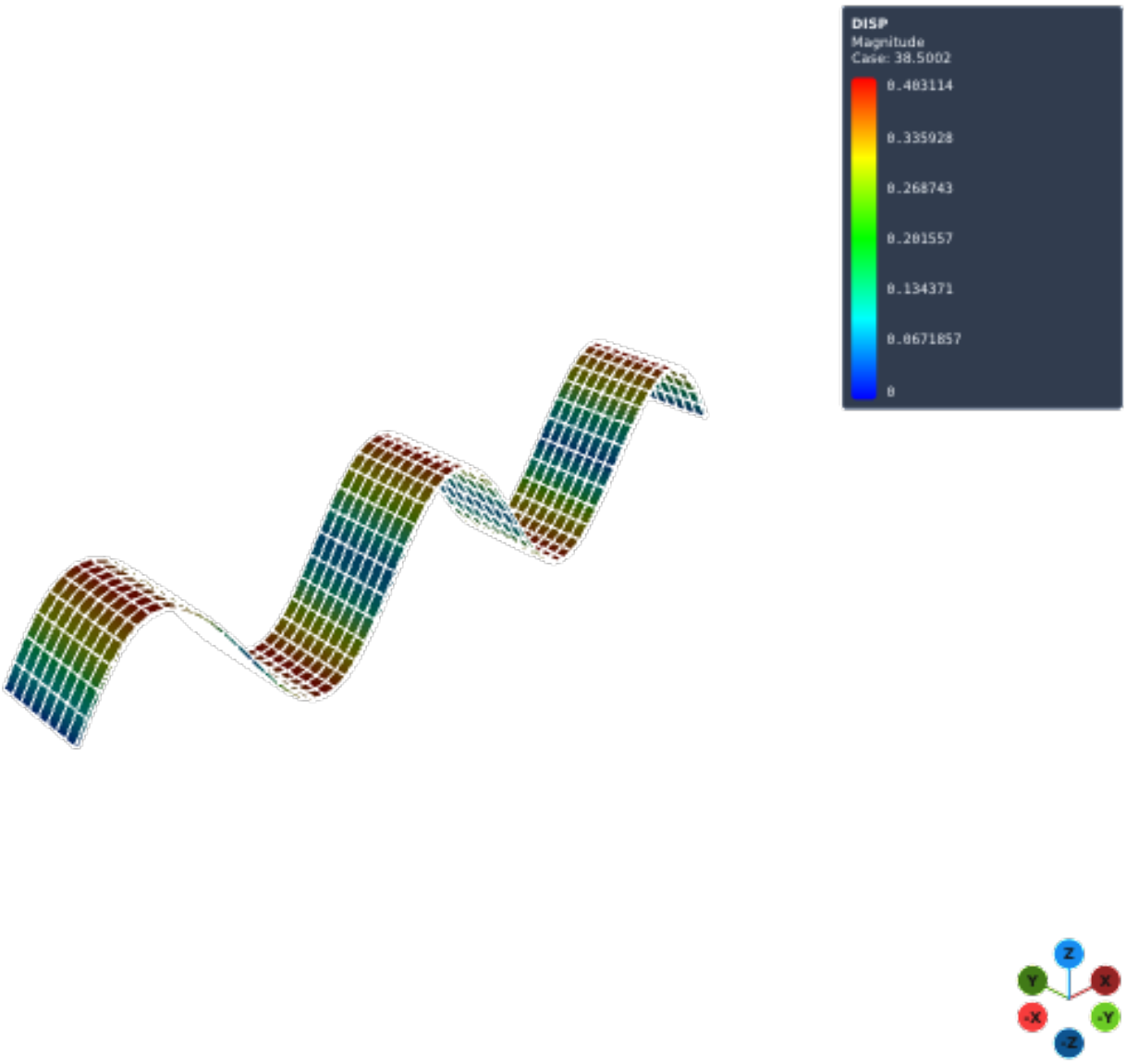

Figure 562: Calculix — mode 5, 38.500 Hz

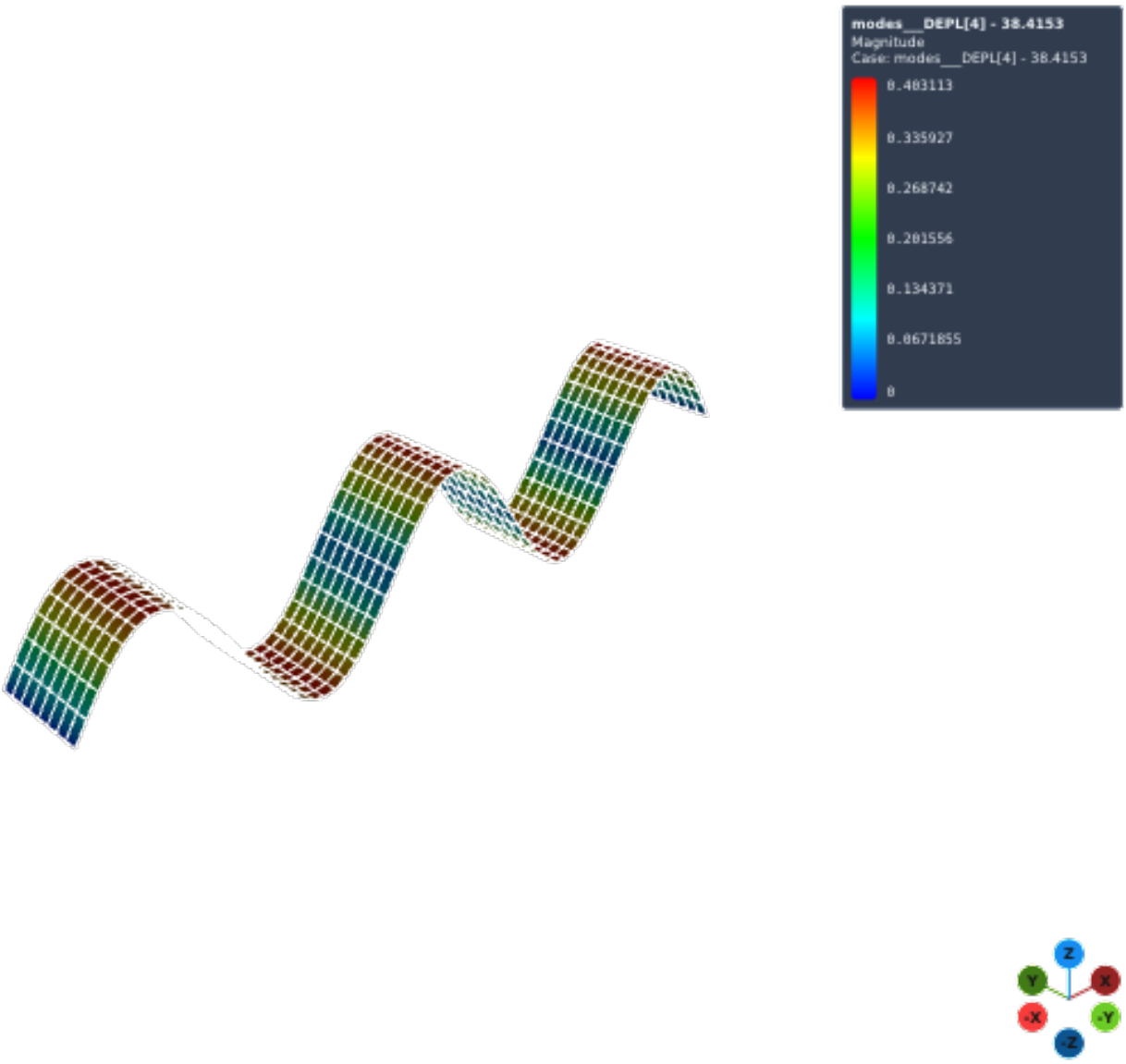

Figure 563: Code_Aster — mode 5, 38.415 Hz

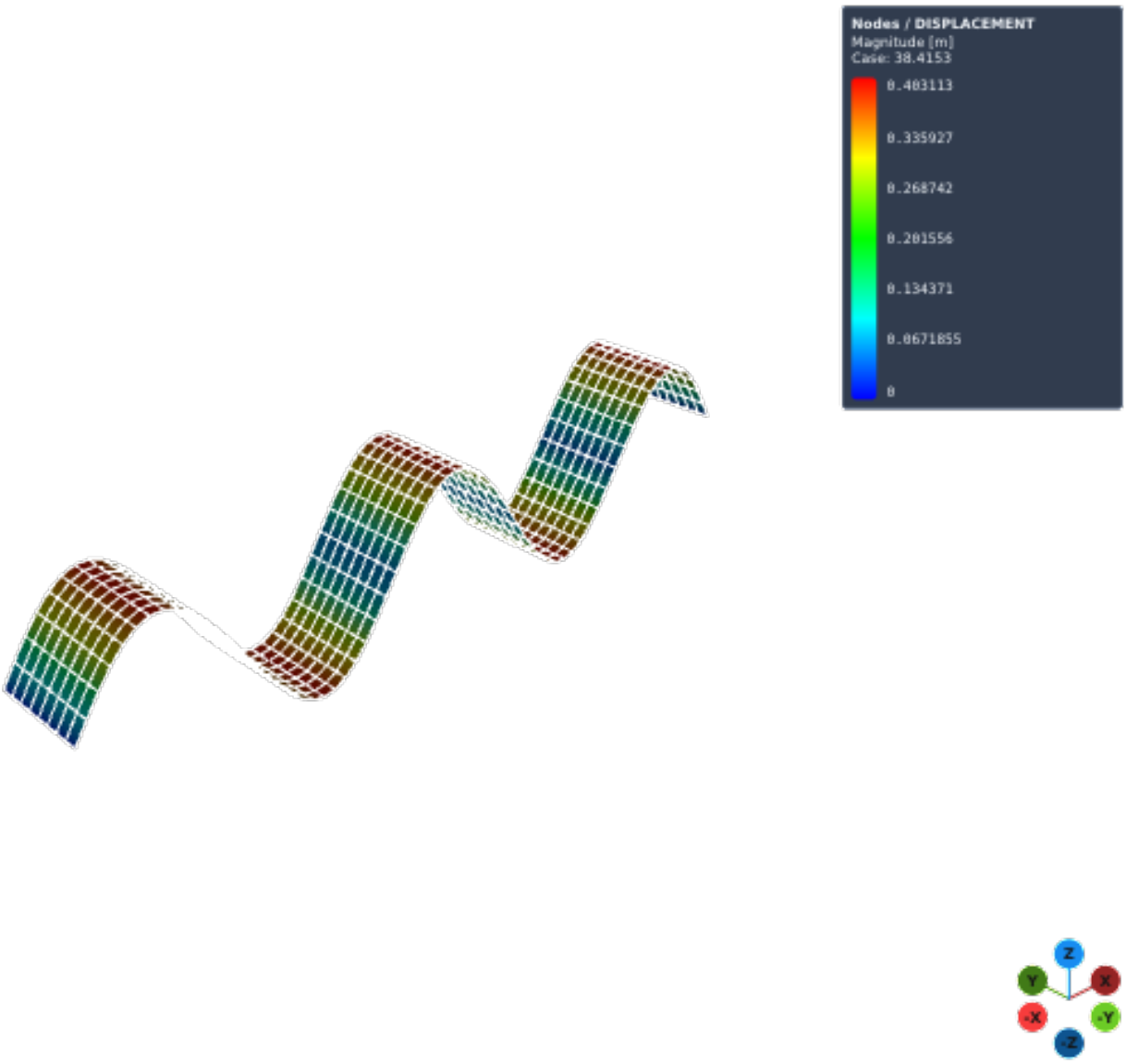

Figure 564: Sesam — mode 5, 38.415 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	38.695	+0.73	59111	1.02e-28	1.84e-30	5.09	1.01e-14	-1.36e-15	2.256
Calculix	38.500	+0.22	58517.4	9.02e-08	1.18e-30	5.06	3.00e-04	1.09e-15	2.249
Code_Aster	38.415	1.67e-04	58259.7	6.94e-39	7.45e-45	5.09	-8.33e-20	8.63e-23	2.256
Sesam	38.415	2.42e-04	58259.8	1.79e-26	3.30e-27	5.00	1.34e-13	5.75e-14	2.236
Closed form	38.415								

Mode 6

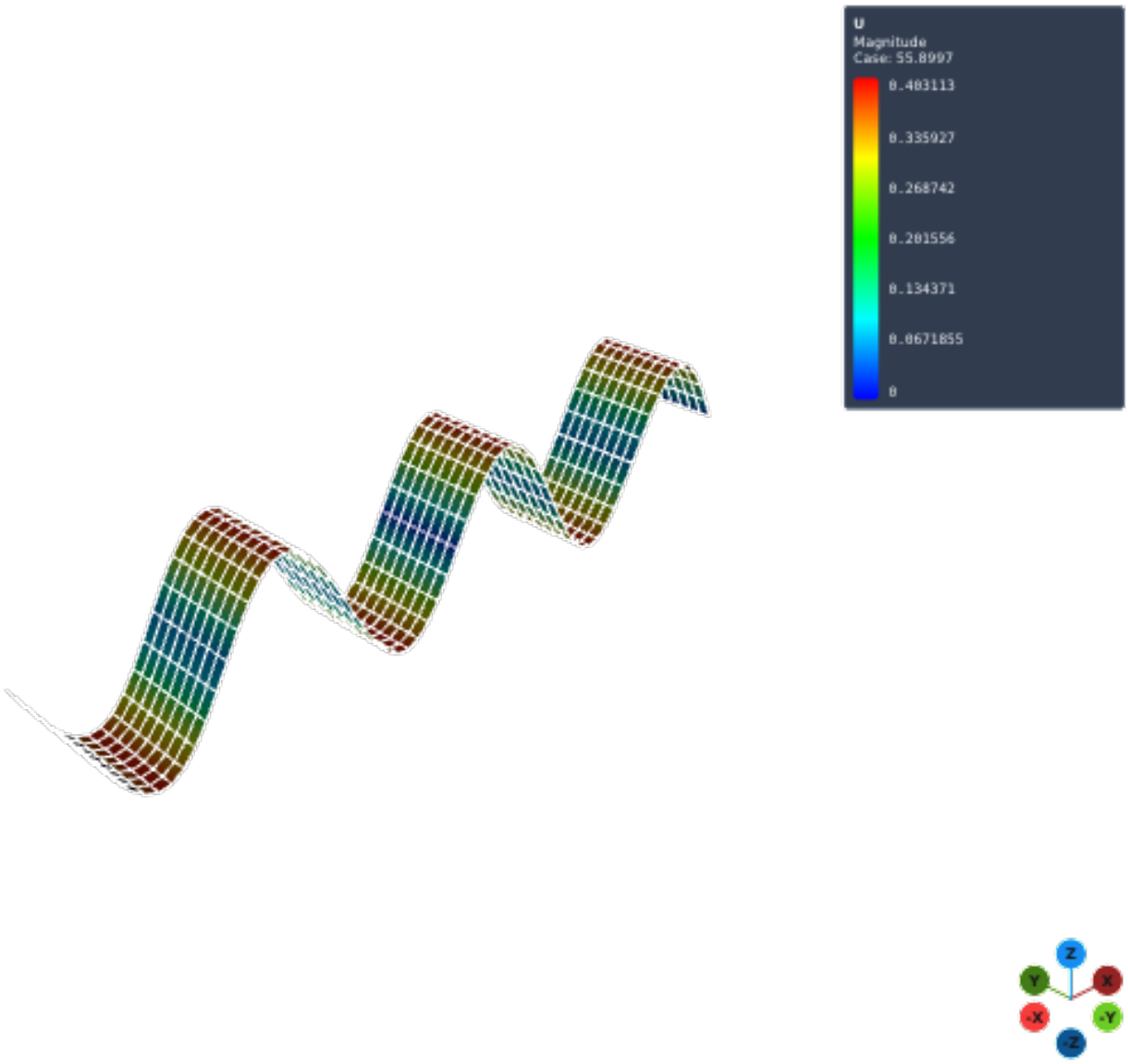

Figure 565: Abaqus — mode 6, 55.900 Hz

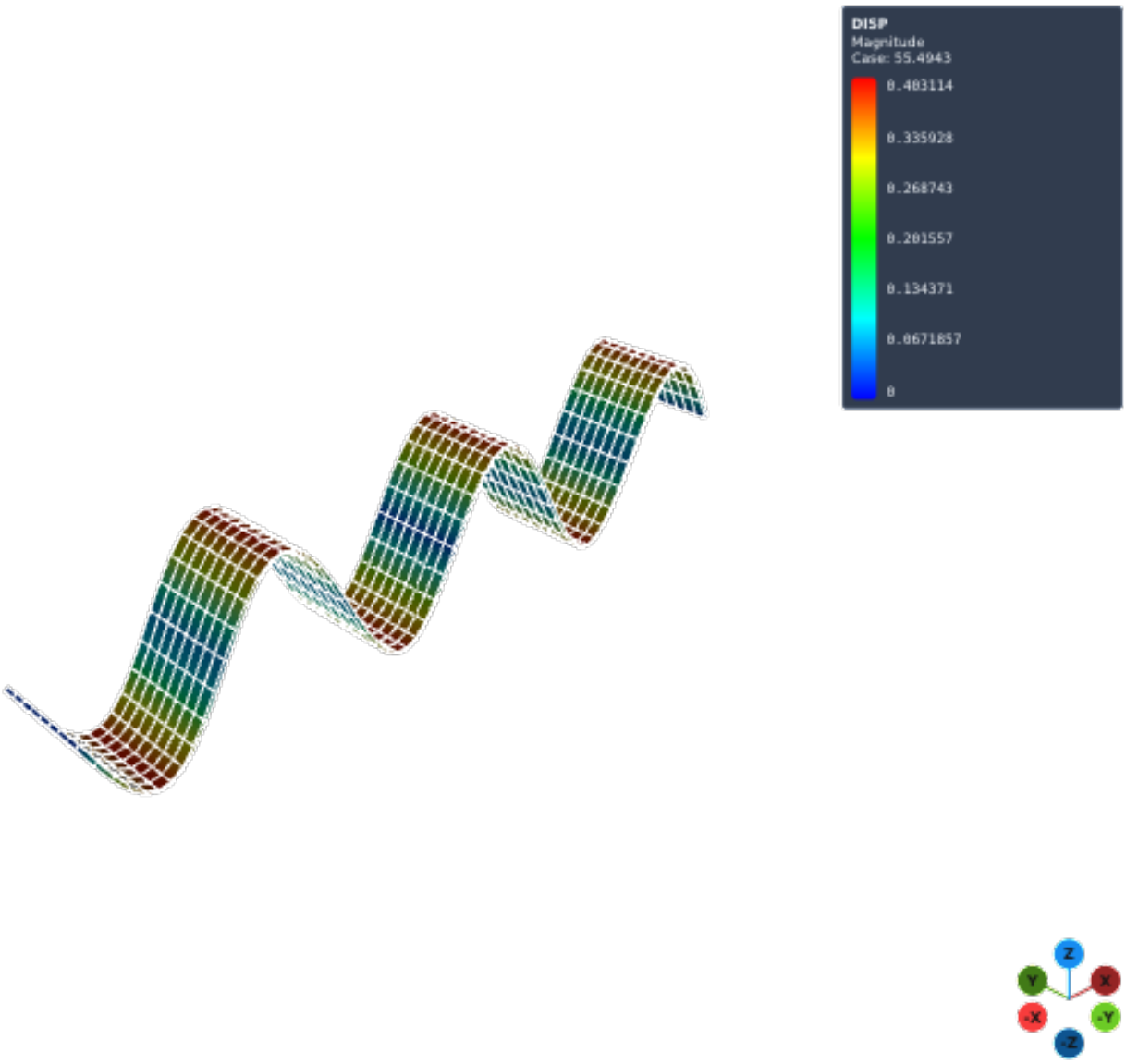

Figure 566: Calculix — mode 6, 55.494 Hz

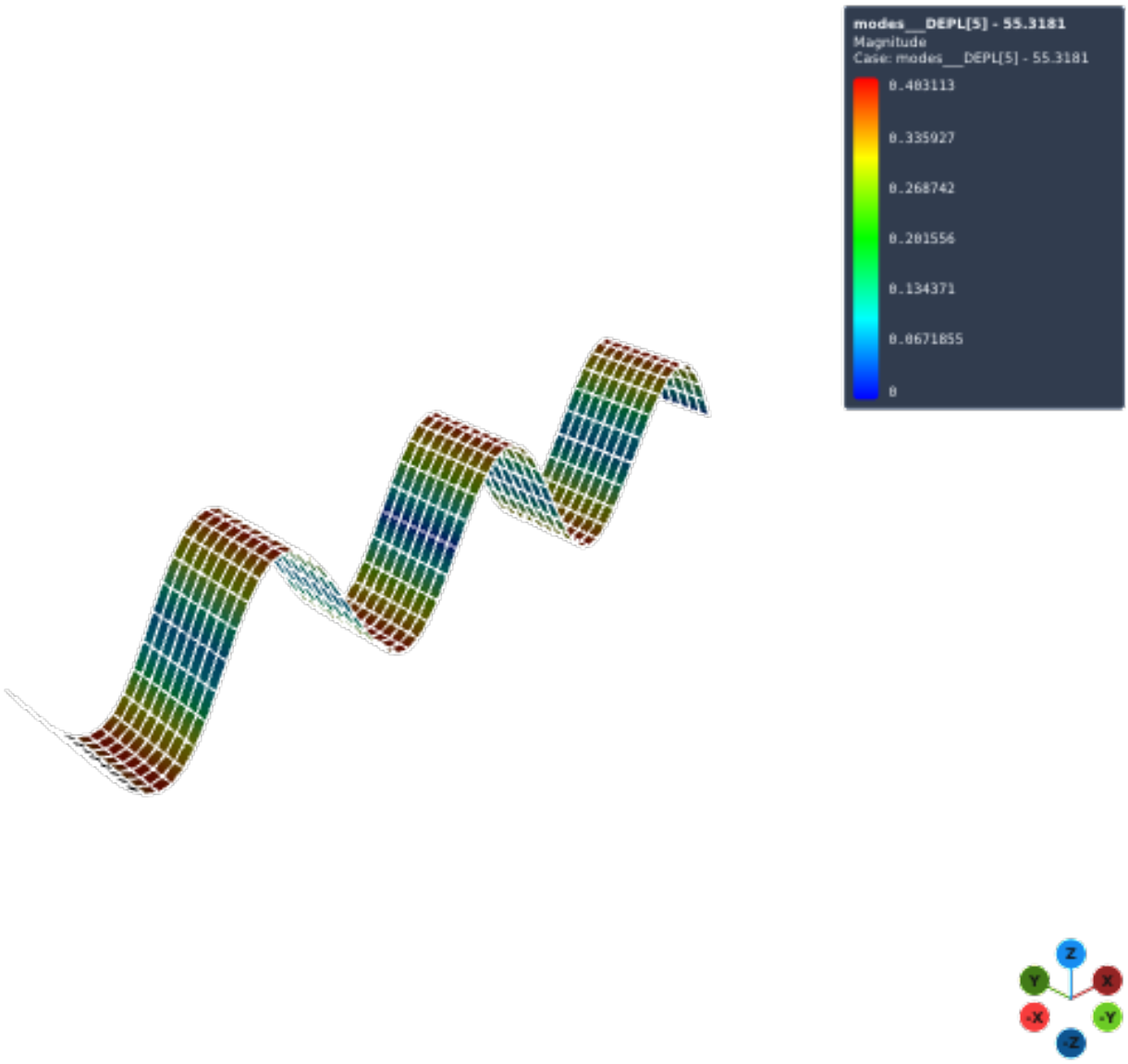

Figure 567: Code_Aster — mode 6, 55.318 Hz

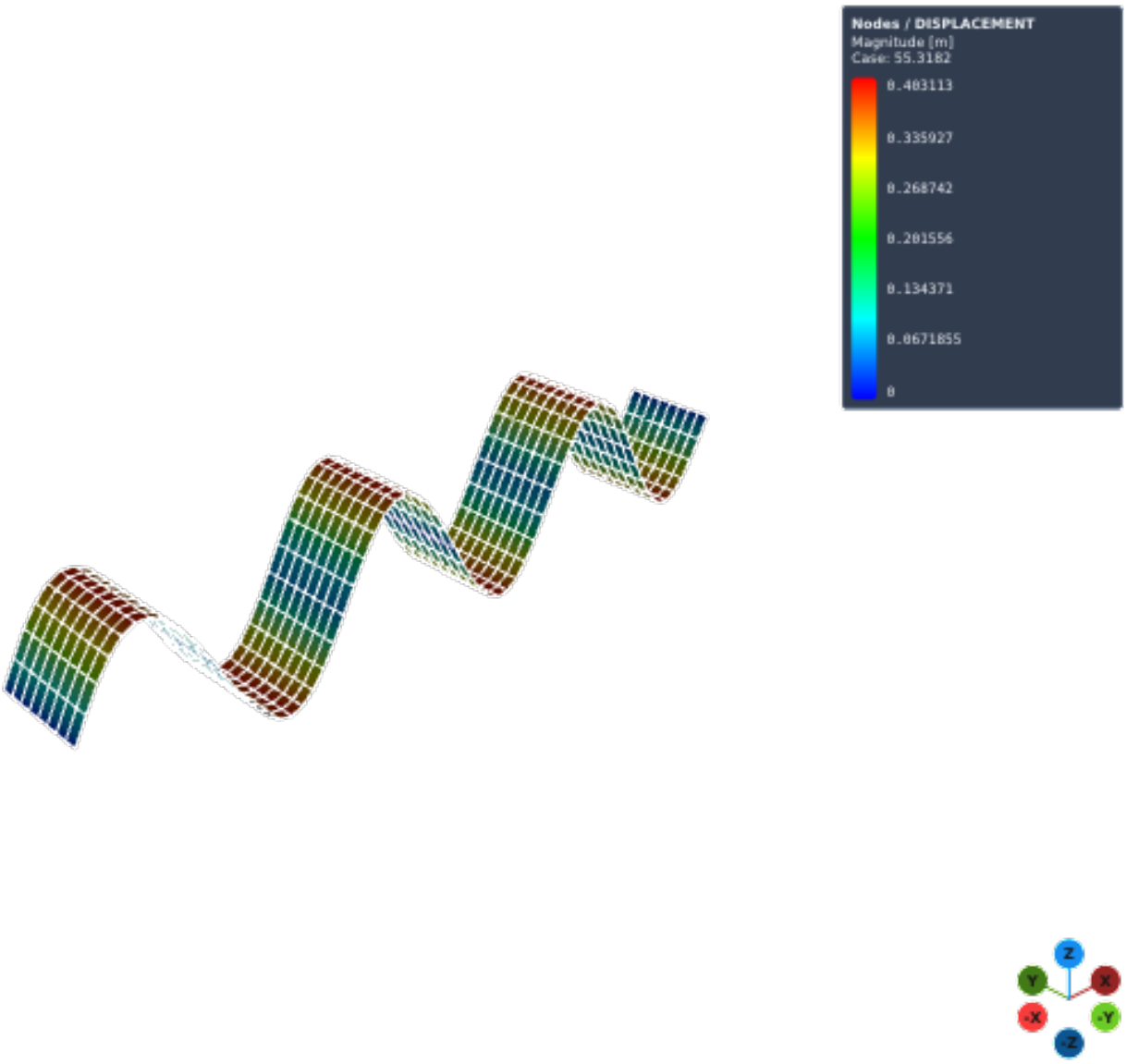

Figure 568: Sesam — mode 6, 55.318 Hz

Solver	f [Hz]	Δf vs closed form [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	55.900	+1.05	123361	4.94e-20	7.80e-26	3.09e-20	-2.22e-10	-2.79e-13	-1.76e-10
Calculix	55.494	+0.32	121578	1.29e-07	1.46e-30	2.42e-19	-3.60e-04	1.21e-15	4.92e-10
Code_Aster	55.318	3.46e-04	120808	1.66e-37	2.85e-42	5.43e-26	-4.07e-19	-1.69e-21	2.33e-13
Sesam	55.318	5.30e-04	120808	1.43e-17	2.77e-18	6.63e-16	3.78e-09	1.66e-09	-2.58e-08
Closed form	55.318								

Shell, 1st order, stiffened

Mode 1

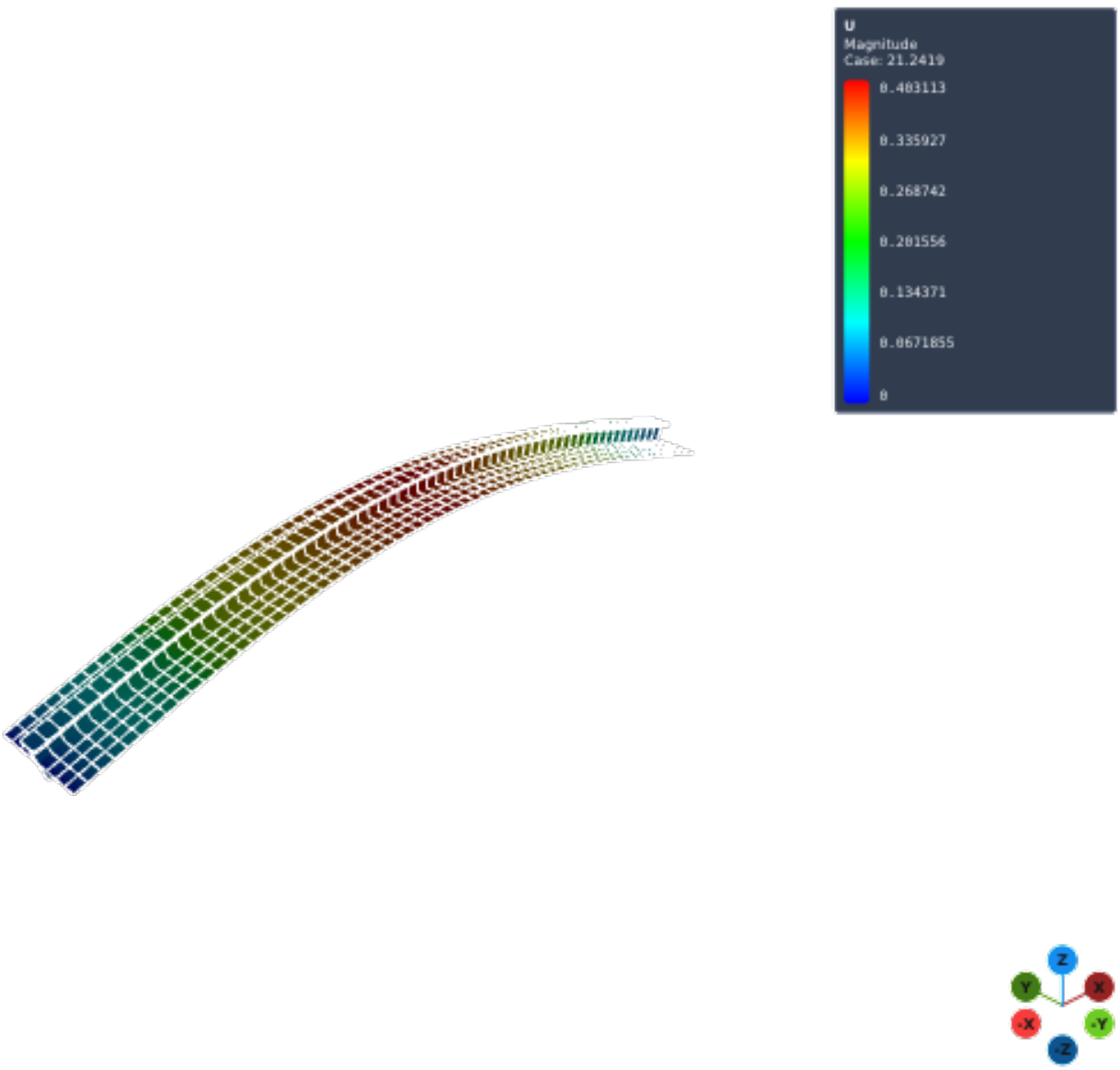

Figure 569: Abaqus — mode 1, 21.242 Hz

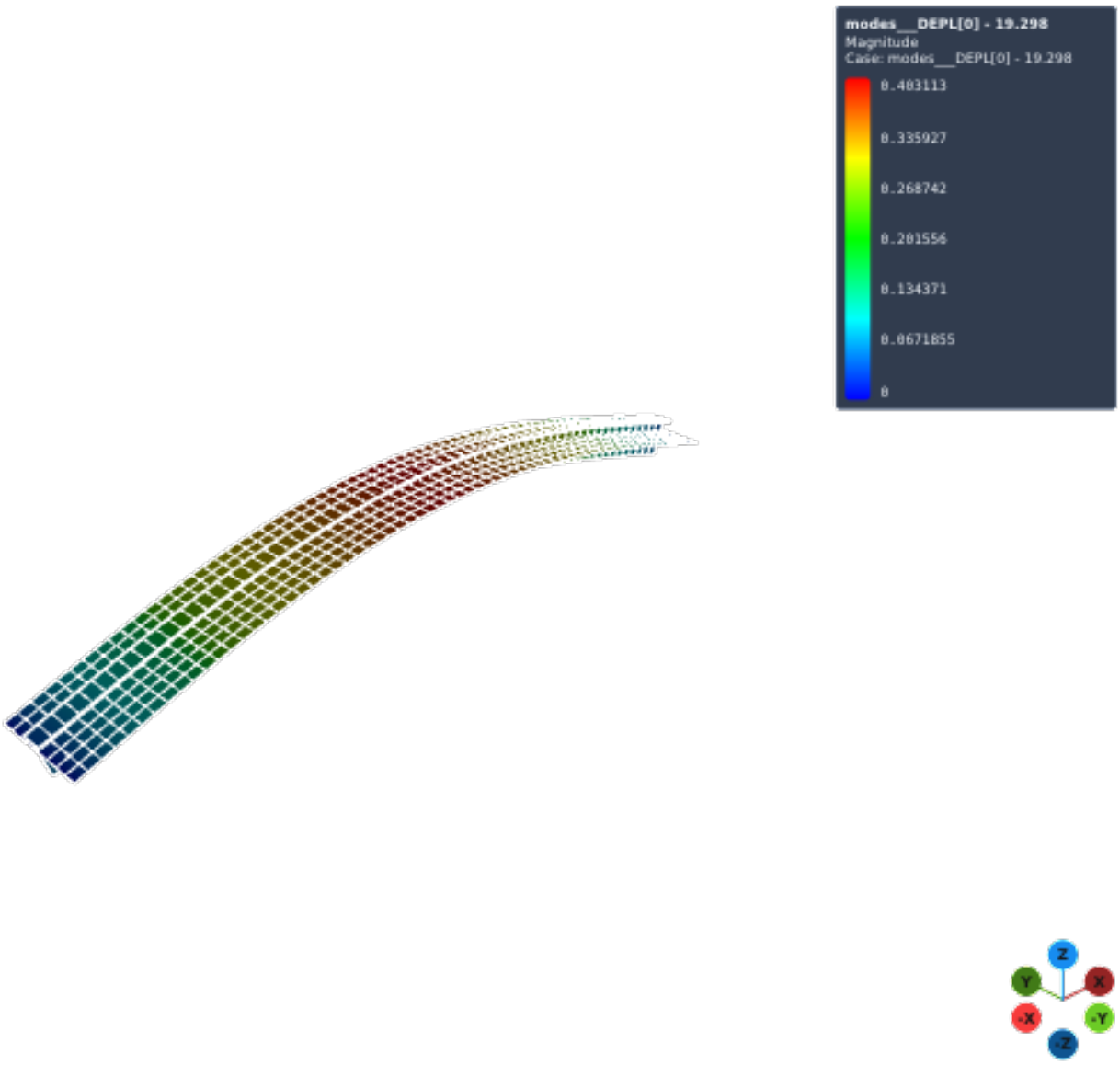

Figure 570: Code_Aster — mode 1, 19.298 Hz

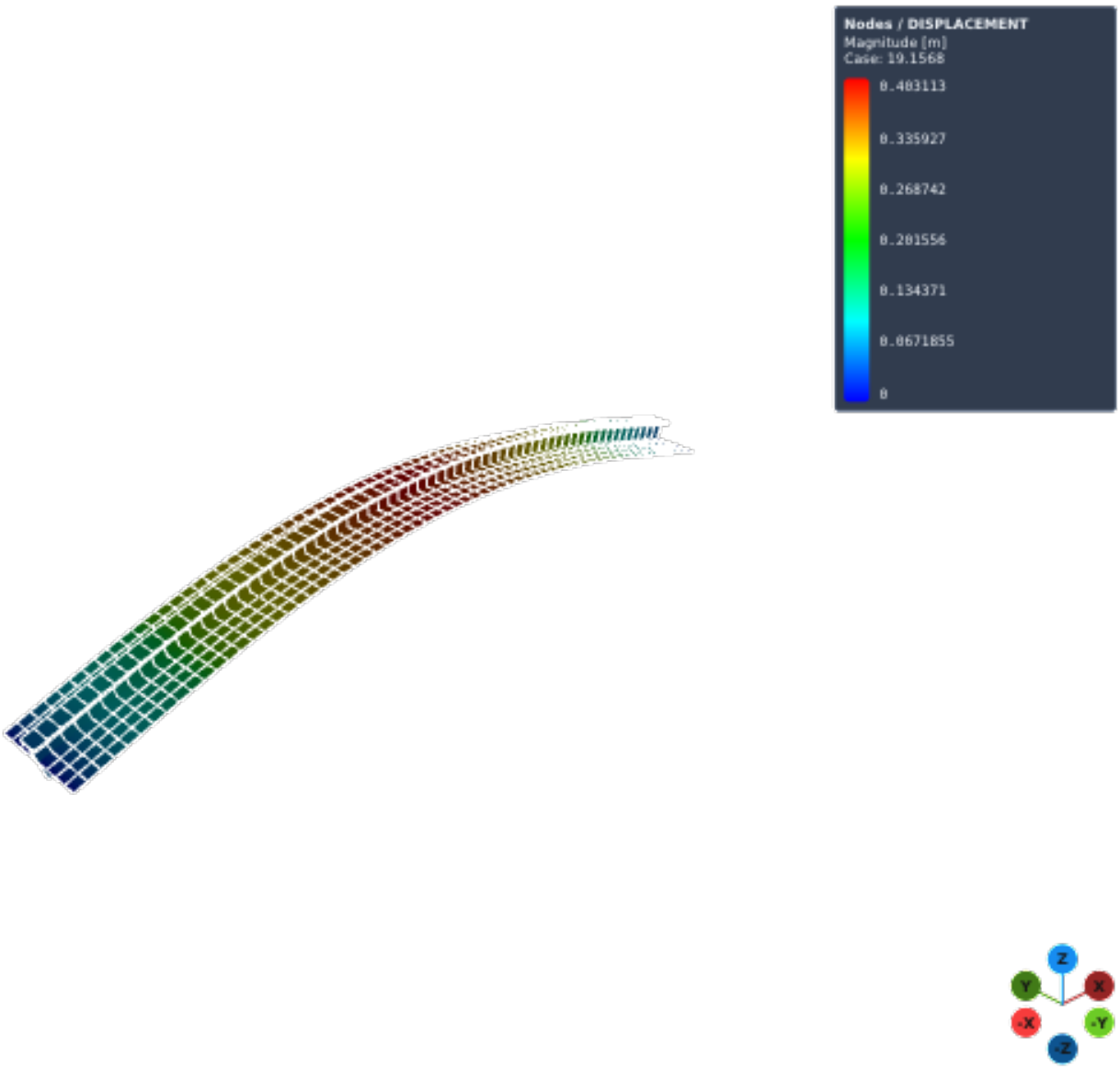

Figure 571: Sesam — mode 1, 19.157 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	21.242	+6.75	17813	0.05	9.45e-26	195.80	-0.214	-3.07e-13	13.993
Code_Aster	19.298	-3.02	14702.3	1.78e-39	5.54e-47	196.08	4.22e-20	7.44e-24	14.003
Sesam	19.157	-3.73	14488	2.36e-35	2.32e-35	195.94	4.86e-18	4.82e-18	13.998

Mode 2

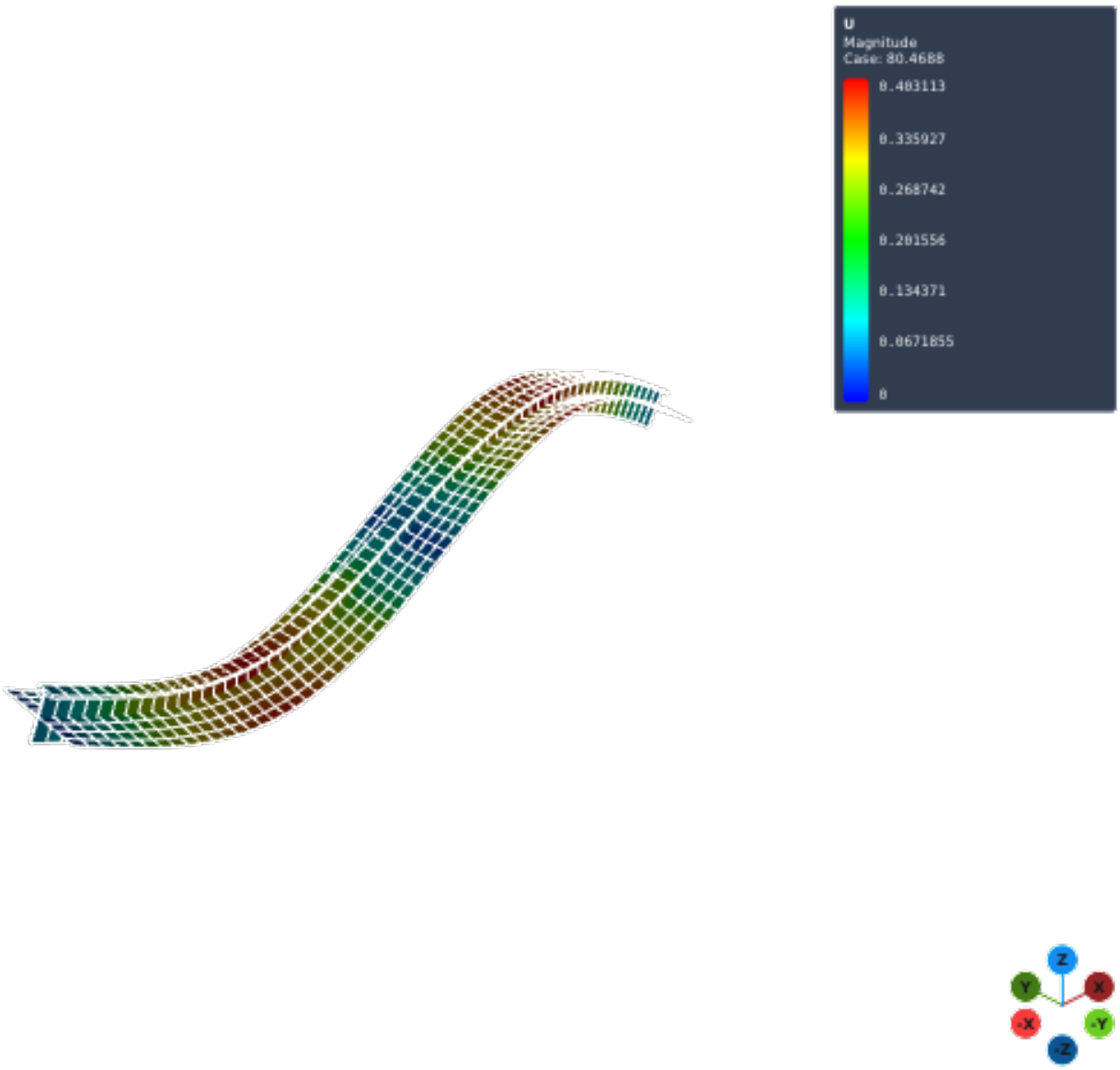

Figure 572: Abaqus — mode 2, 80.469 Hz

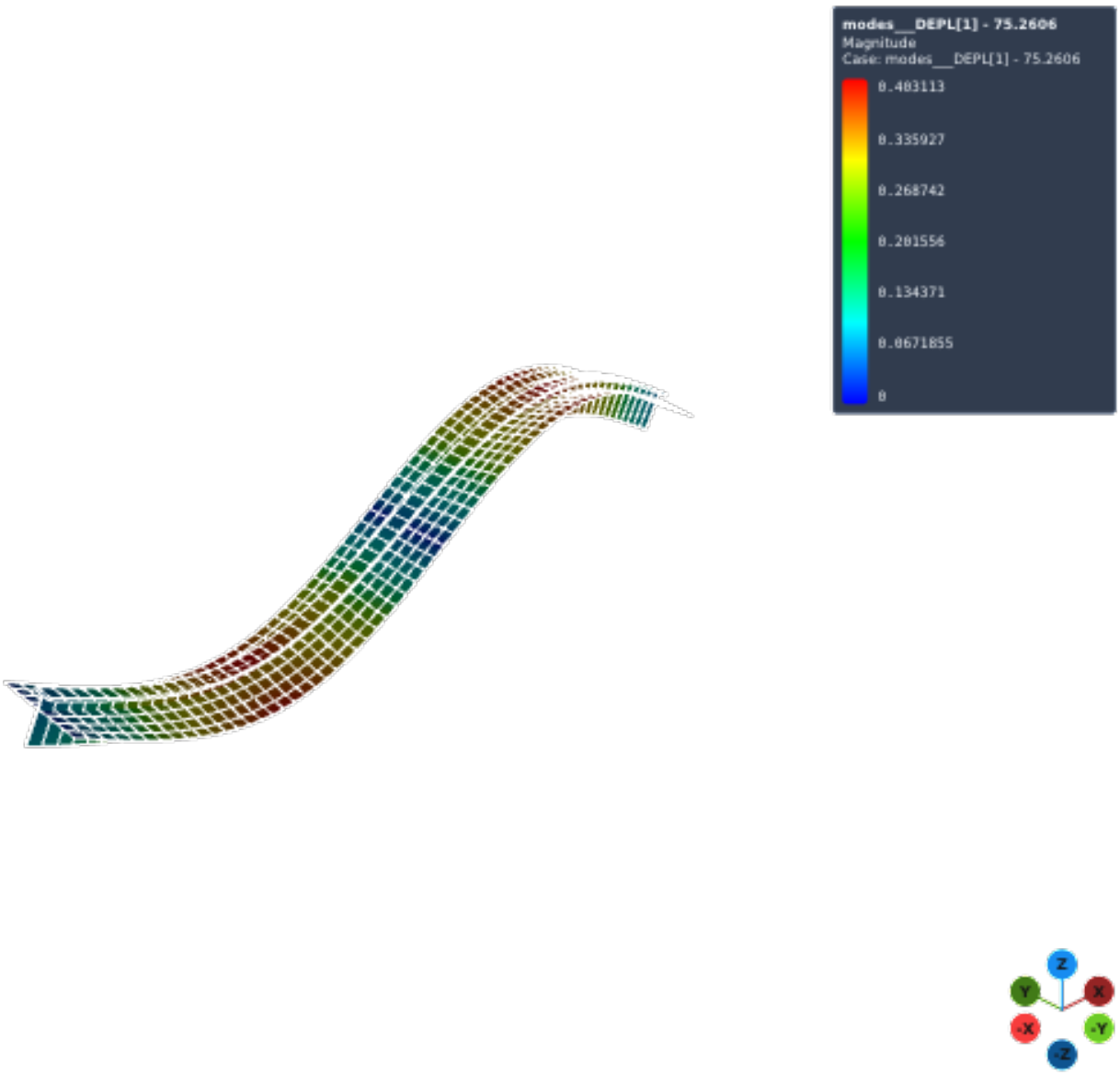

Figure 573: Code_Aster — mode 2, 75.261 Hz

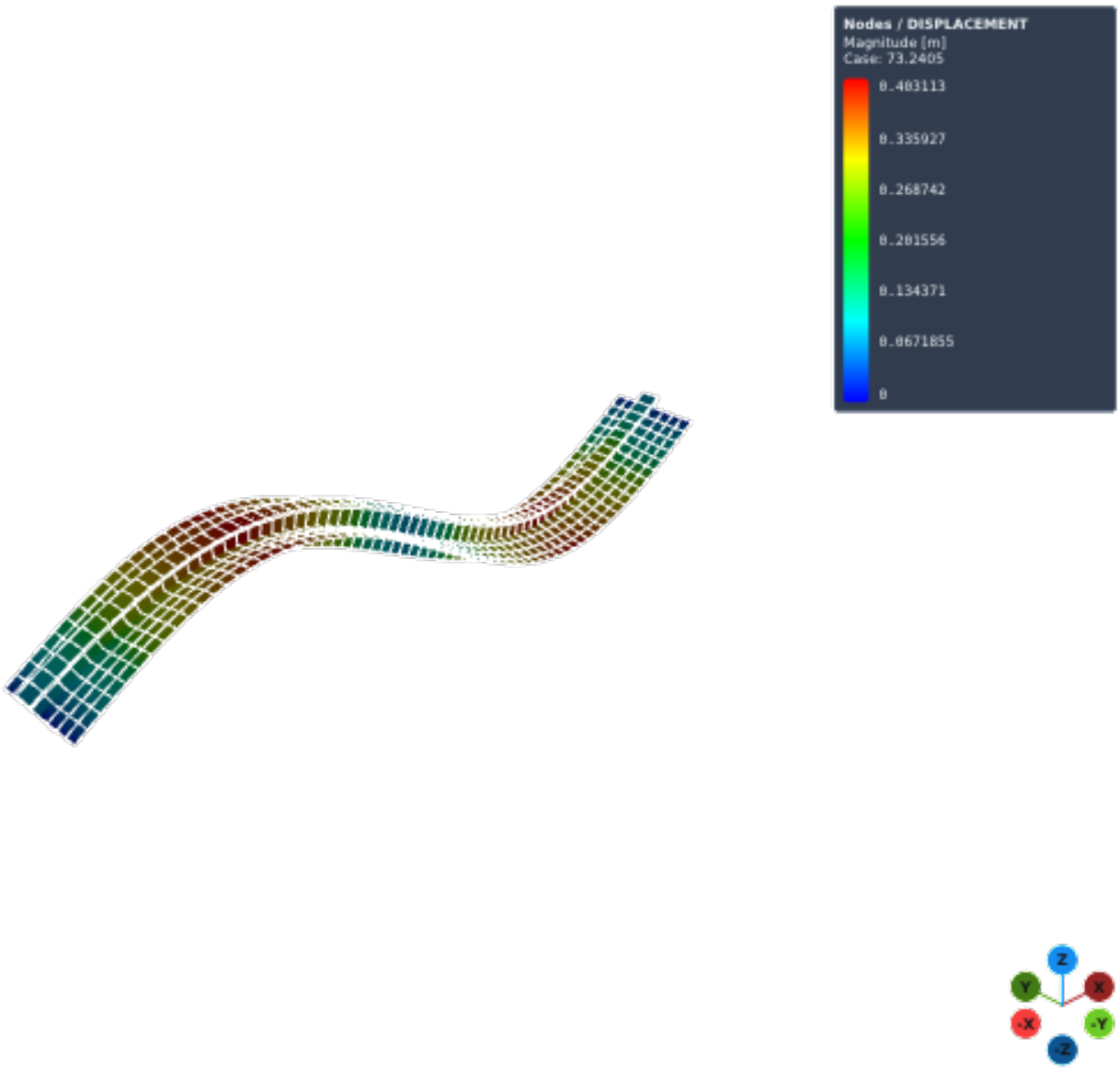

Figure 574: Sesam — mode 2, 73.241 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	80.469	+5.43	255632	0.16	1.91e-25	3.42e-05	0.403	4.37e-13	0.006
Code_Aster	75.261	-1.39	223612	6.65e-36	1.89e-42	2.29e-23	-2.58e-18	-1.38e-21	4.78e-12
Sesam	73.241	-4.04	211769	1.97e-32	2.23e-33	1.18e-23	1.40e-16	4.72e-17	-3.44e-12

Mode 3

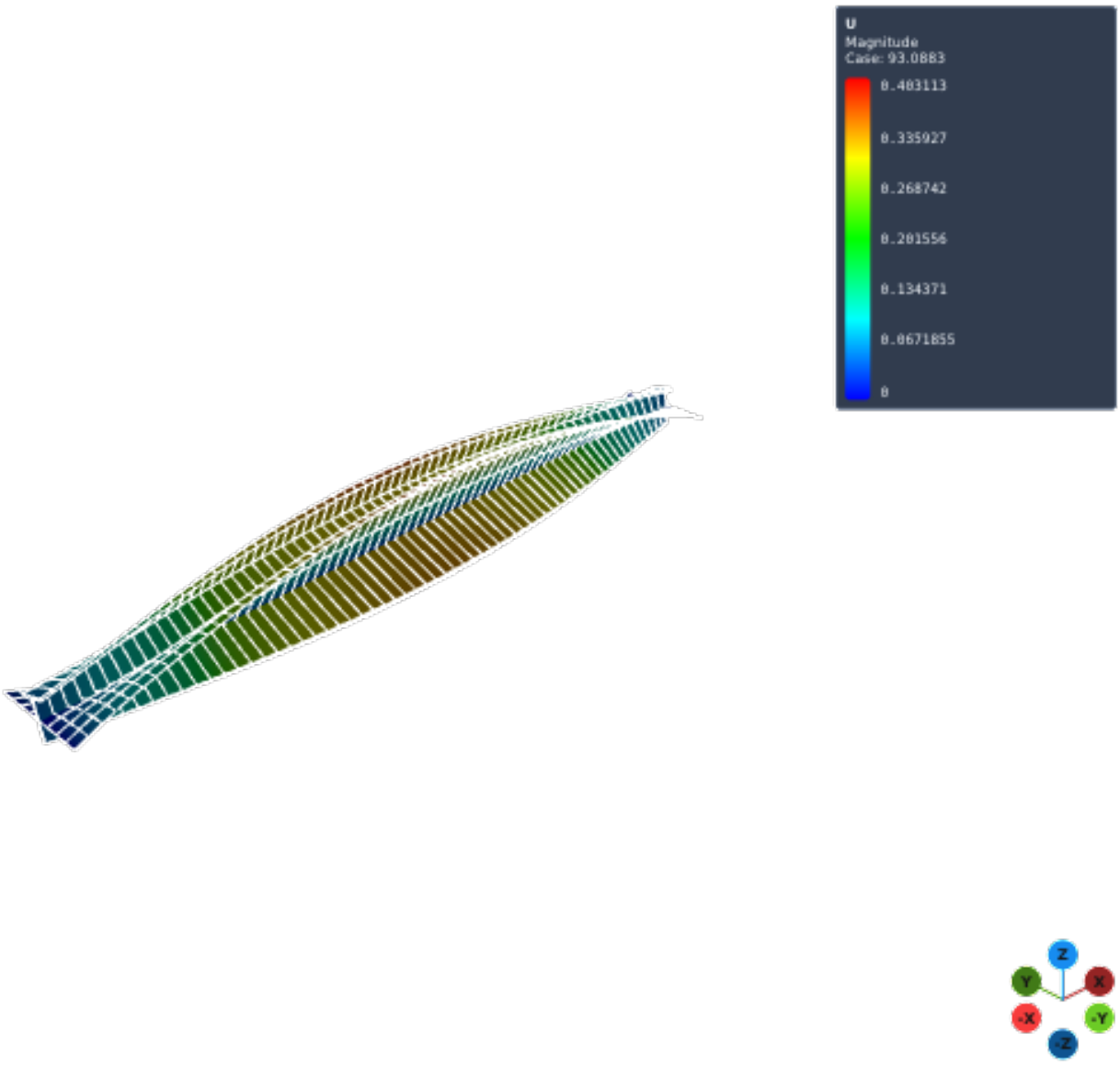

Figure 575: Abaqus — mode 3, 93.088 Hz

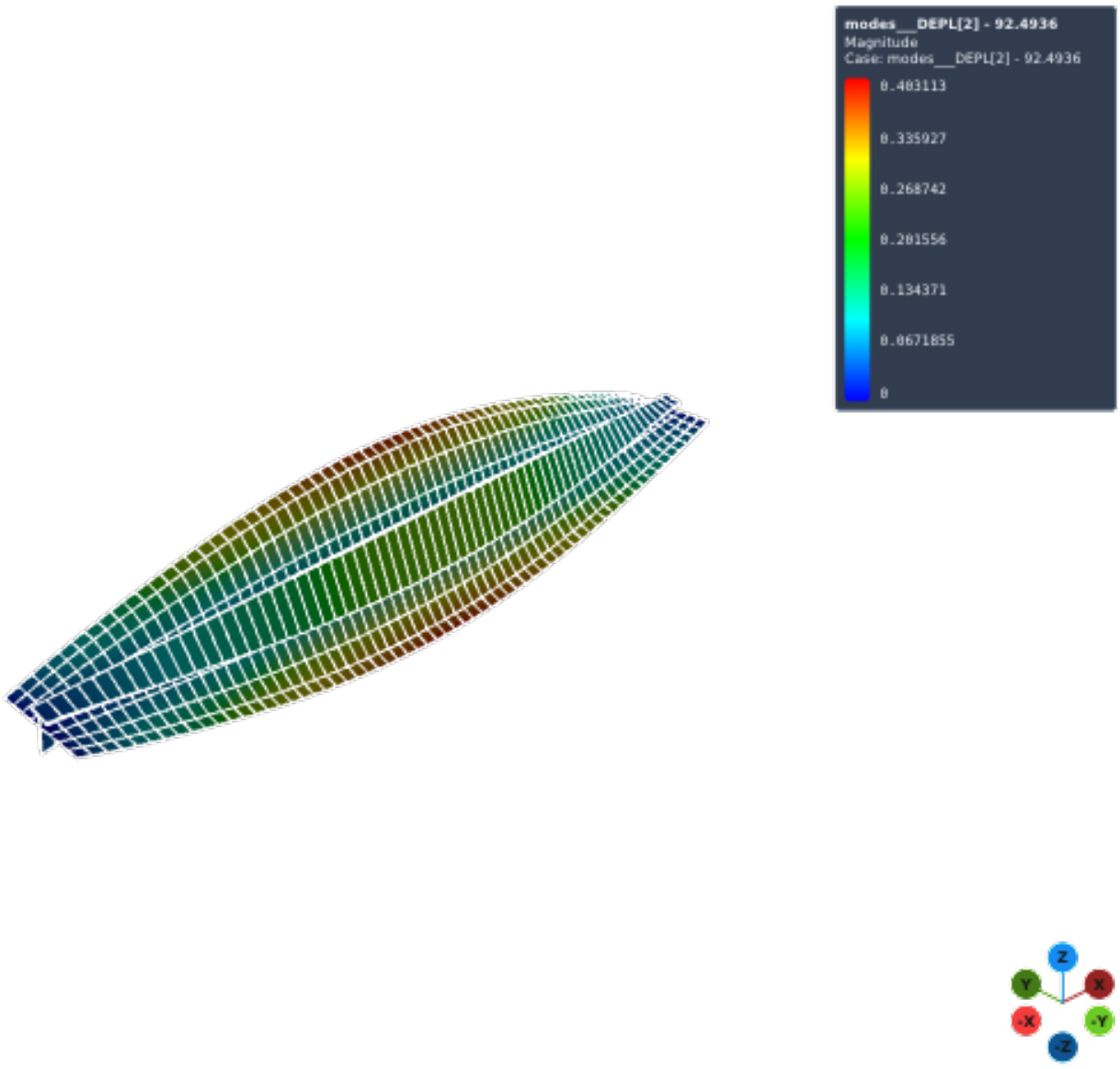

Figure 576: Code_Aster — mode 3, 92.494 Hz

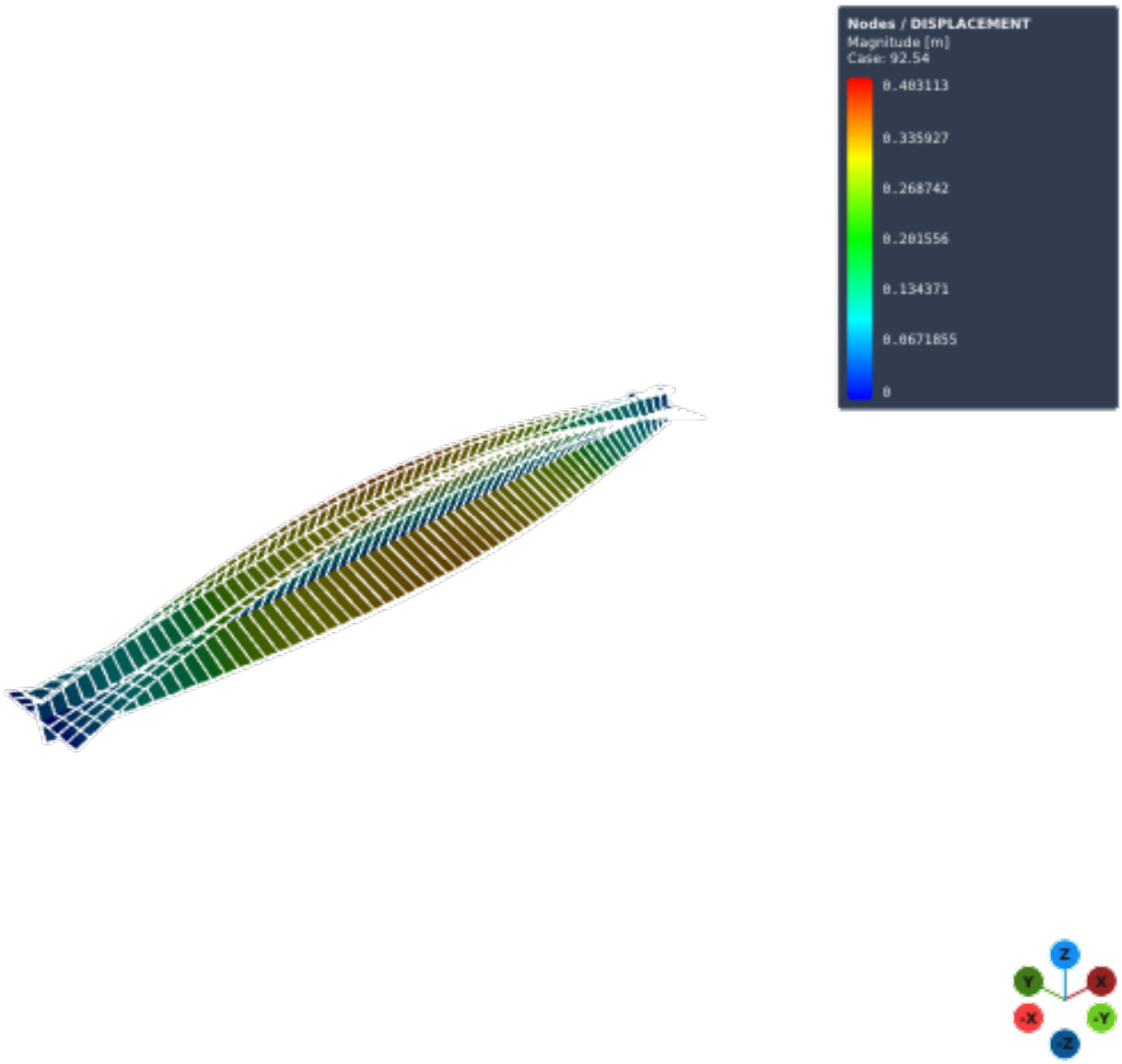

Figure 577: Sesam — mode 3, 92.540 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	93.088	+0.41	342097	2.75e-24	3.95	1.03e-23	1.66e-12	1.988	3.22e-12
Code_Aster	92.494	-0.23	337741	4.55e-35	3.26e-41	2.73e-26	6.75e-18	-5.71e-21	-1.65e-13
Sesam	92.540	-0.18	338080	2.00e-31	1.99e-07	1.09e-25	4.48e-16	4.46e-04	-3.29e-13

Mode 4

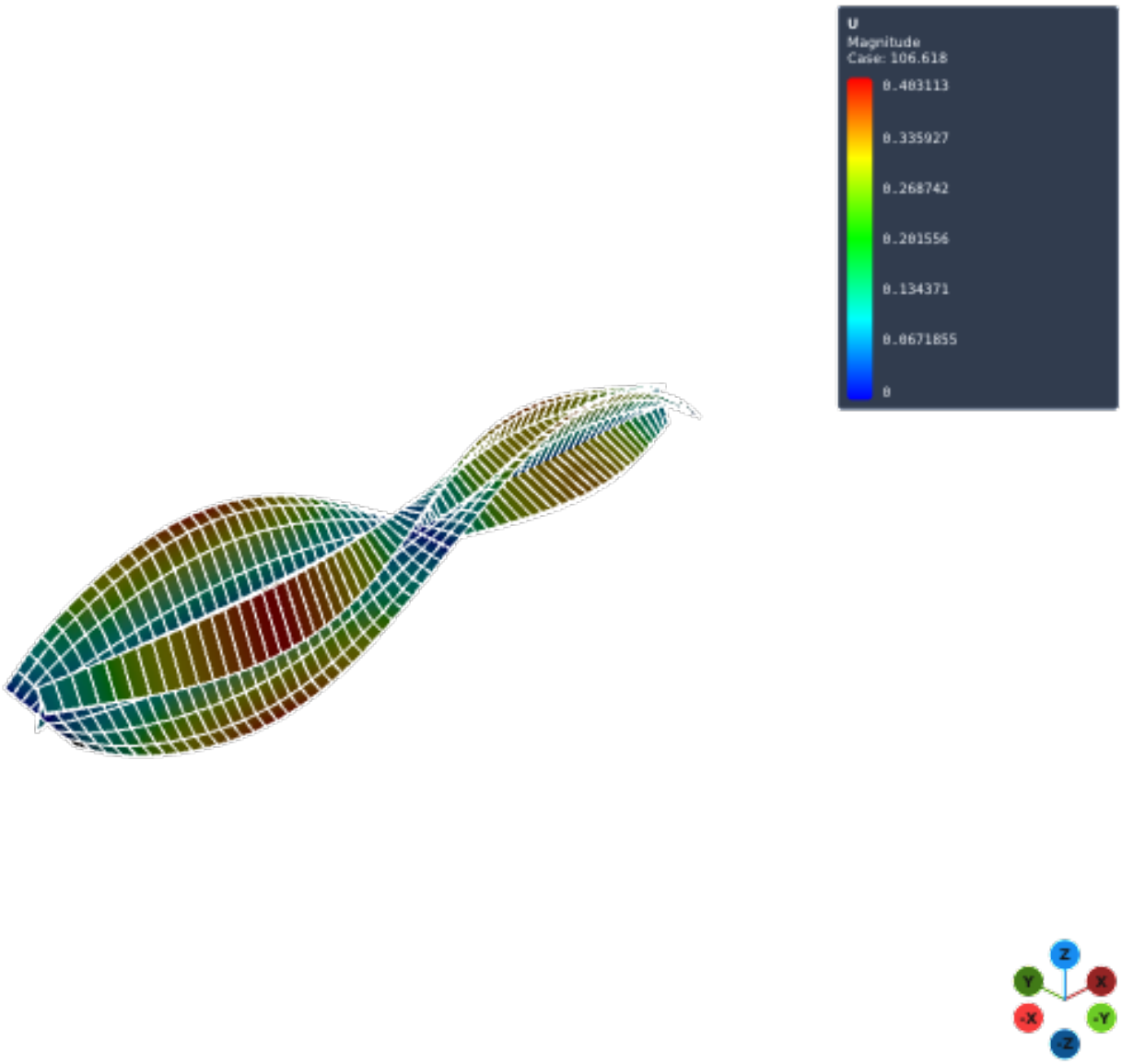

Figure 578: Abaqus — mode 4, 106.618 Hz

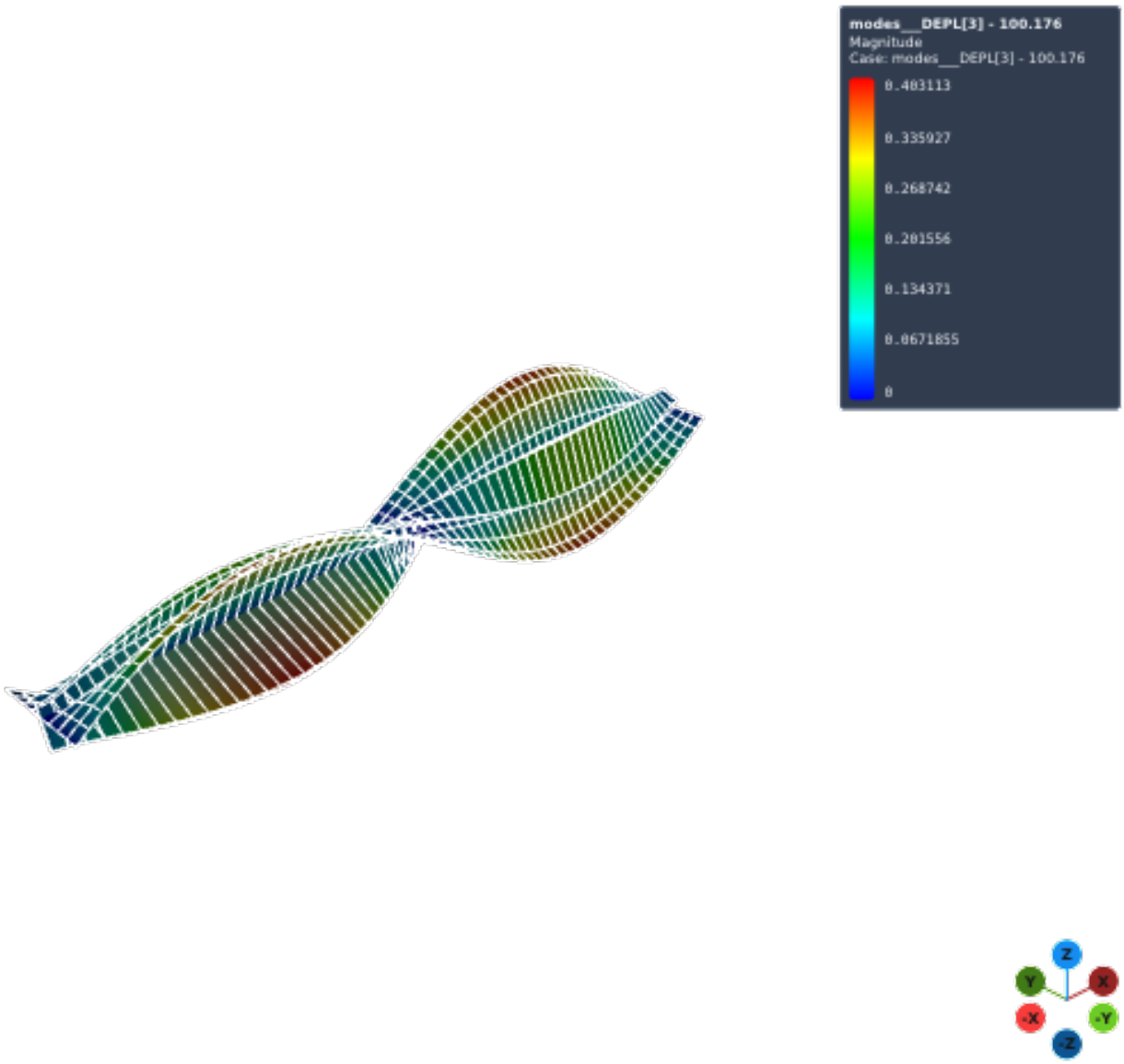

Figure 579: Code_Aster — mode 4, 100.176 Hz

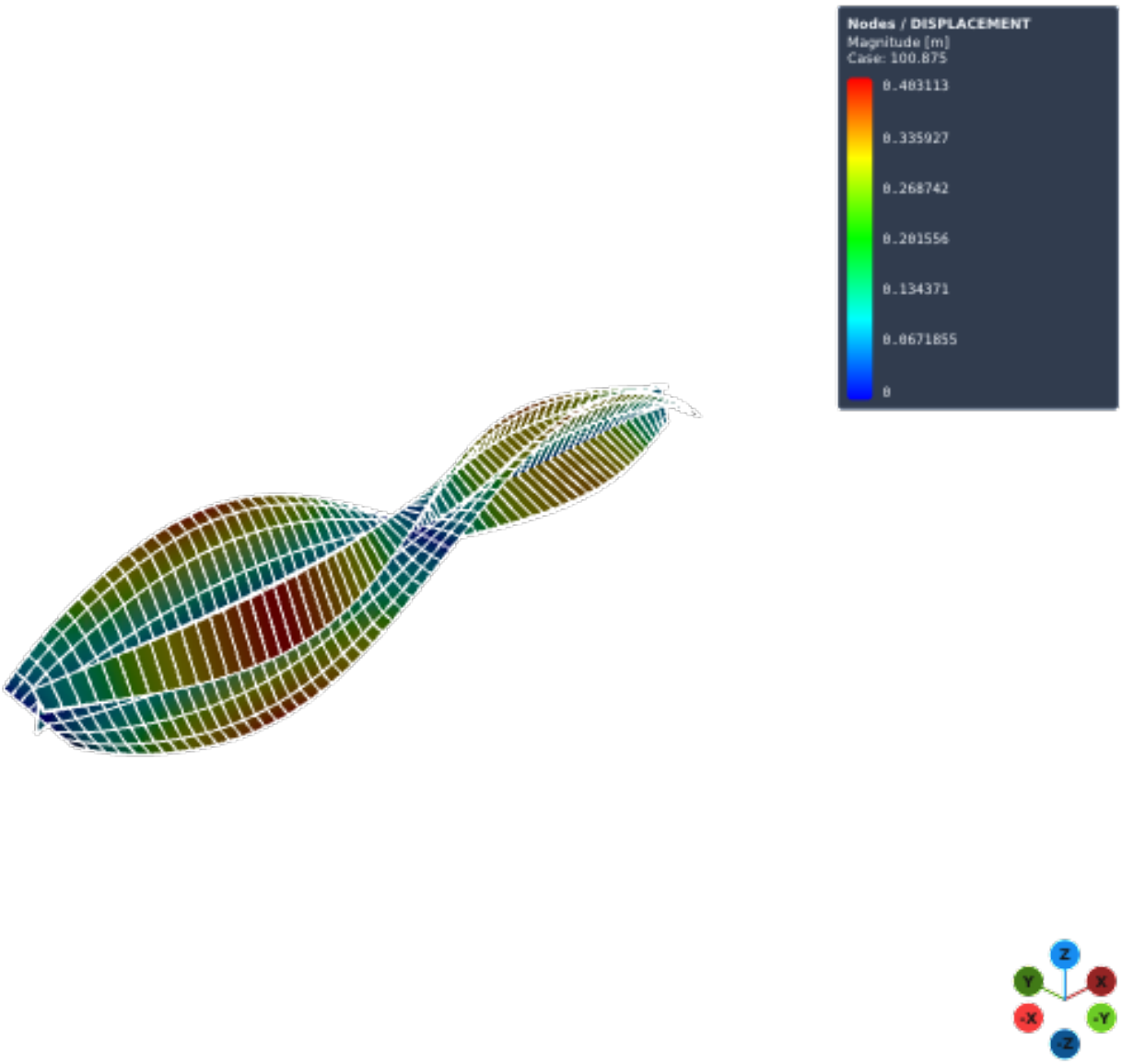

Figure 580: Sesam — mode 4, 100.875 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	106.620	+3.96	448765	1.14e-22	2.35e-08	5.91e-22	1.07e-11	1.53e-04	-2.43e-11
Code_Aster	100.176	-2.32	396174	2.37e-35	2.54e-40	5.09e-26	-4.87e-18	1.59e-20	-2.26e-13
Sesam	100.875	-1.64	401723	1.35e-30	9.30e-10	9.03e-26	-1.16e-15	3.05e-05	3.01e-13

Mode 5

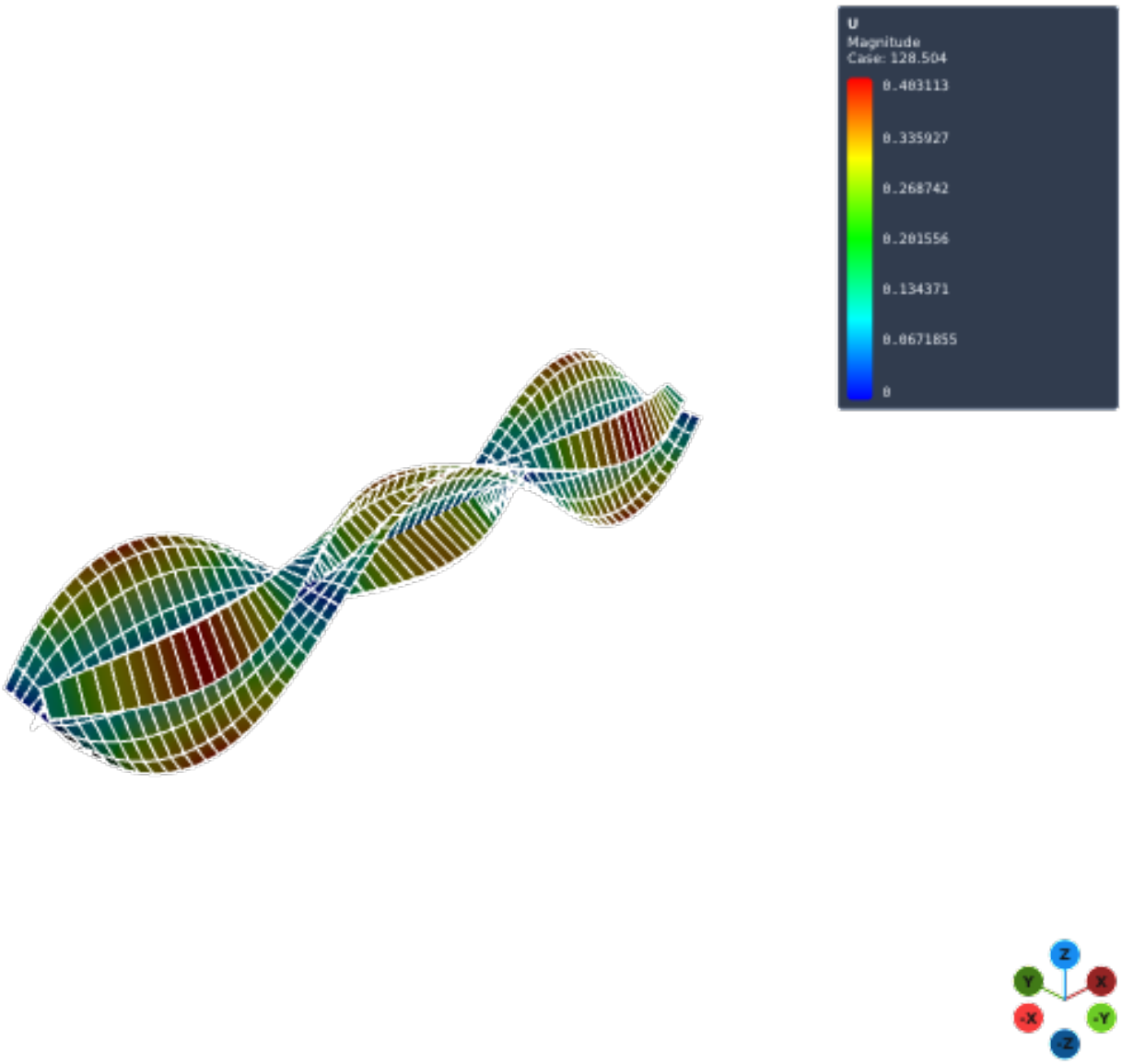

Figure 581: Abaqus — mode 5, 128.504 Hz

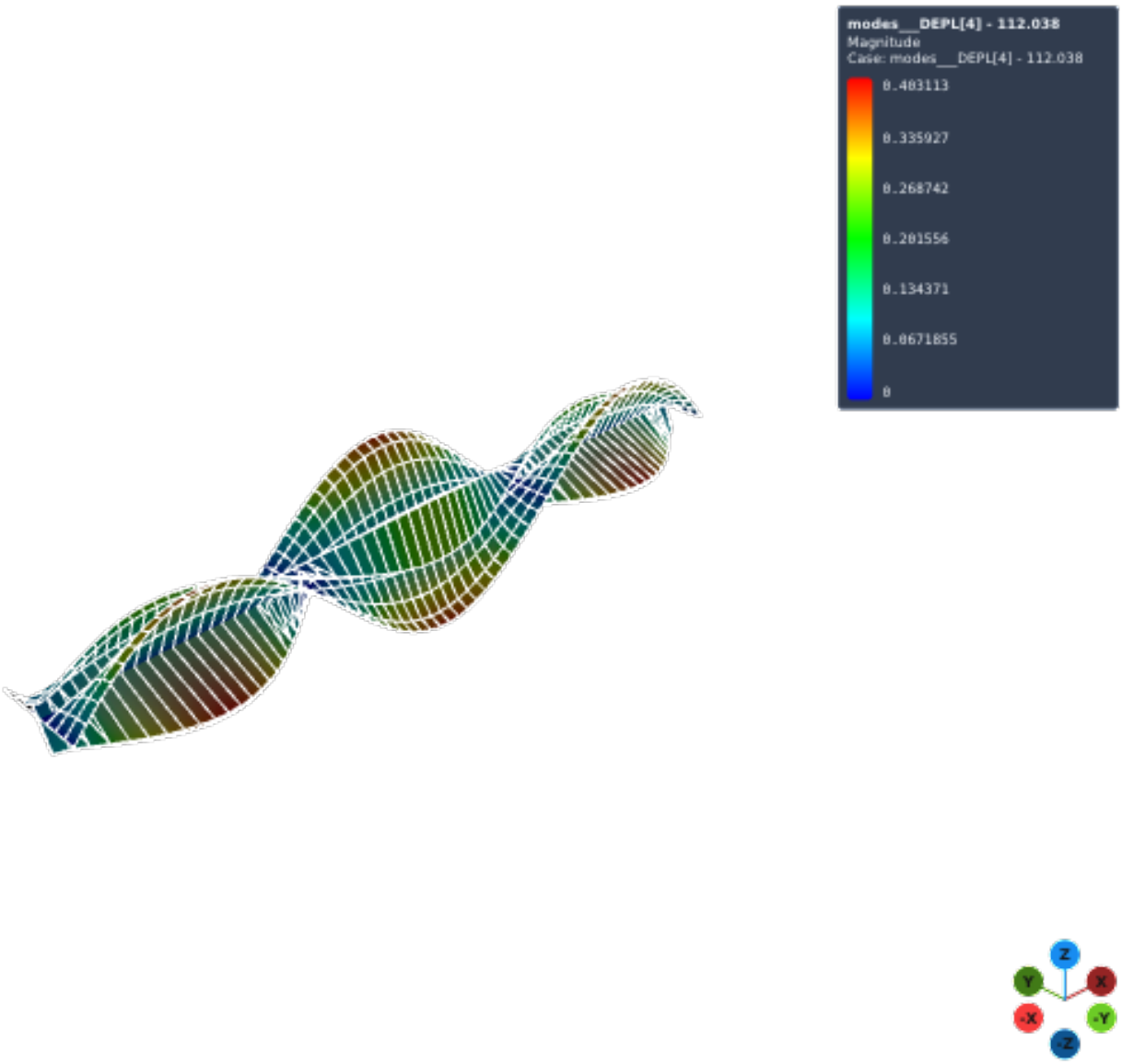

Figure 582: Code_Aster — mode 5, 112.038 Hz

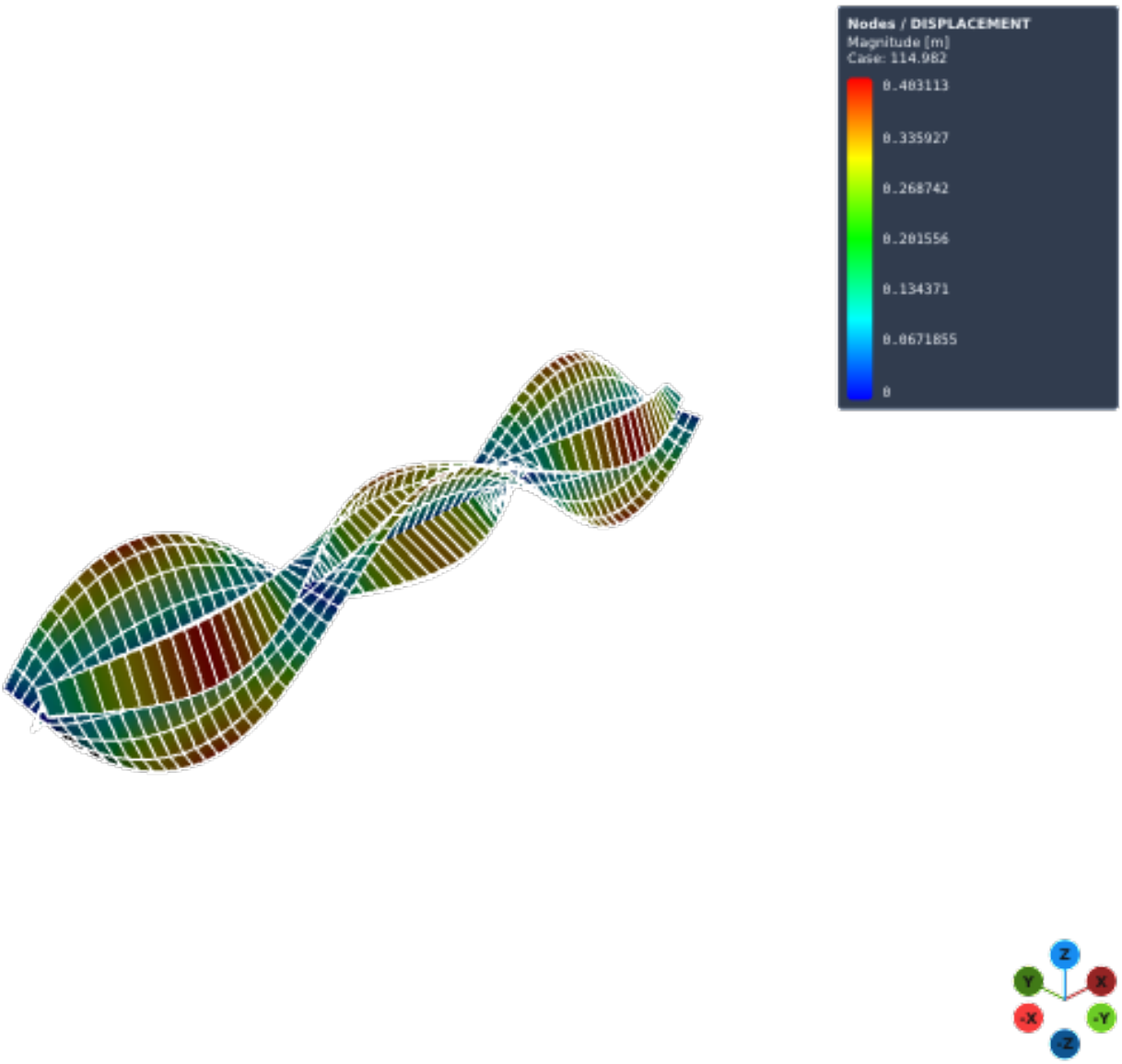

Figure 583: Sesam — mode 5, 114.982 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	128.500	+8.43	651919	5.66e-20	0.39	1.44e-19	2.38e-10	-0.624	3.79e-10
Code_Aster	112.038	-5.46	495550	7.84e-34	1.81e-41	8.00e-27	2.80e-17	4.26e-21	-8.94e-14
Sesam	114.982	-2.97	521937	2.16e-24	4.46e-07	8.33e-24	-1.47e-12	-0.001	2.89e-12

Mode 6

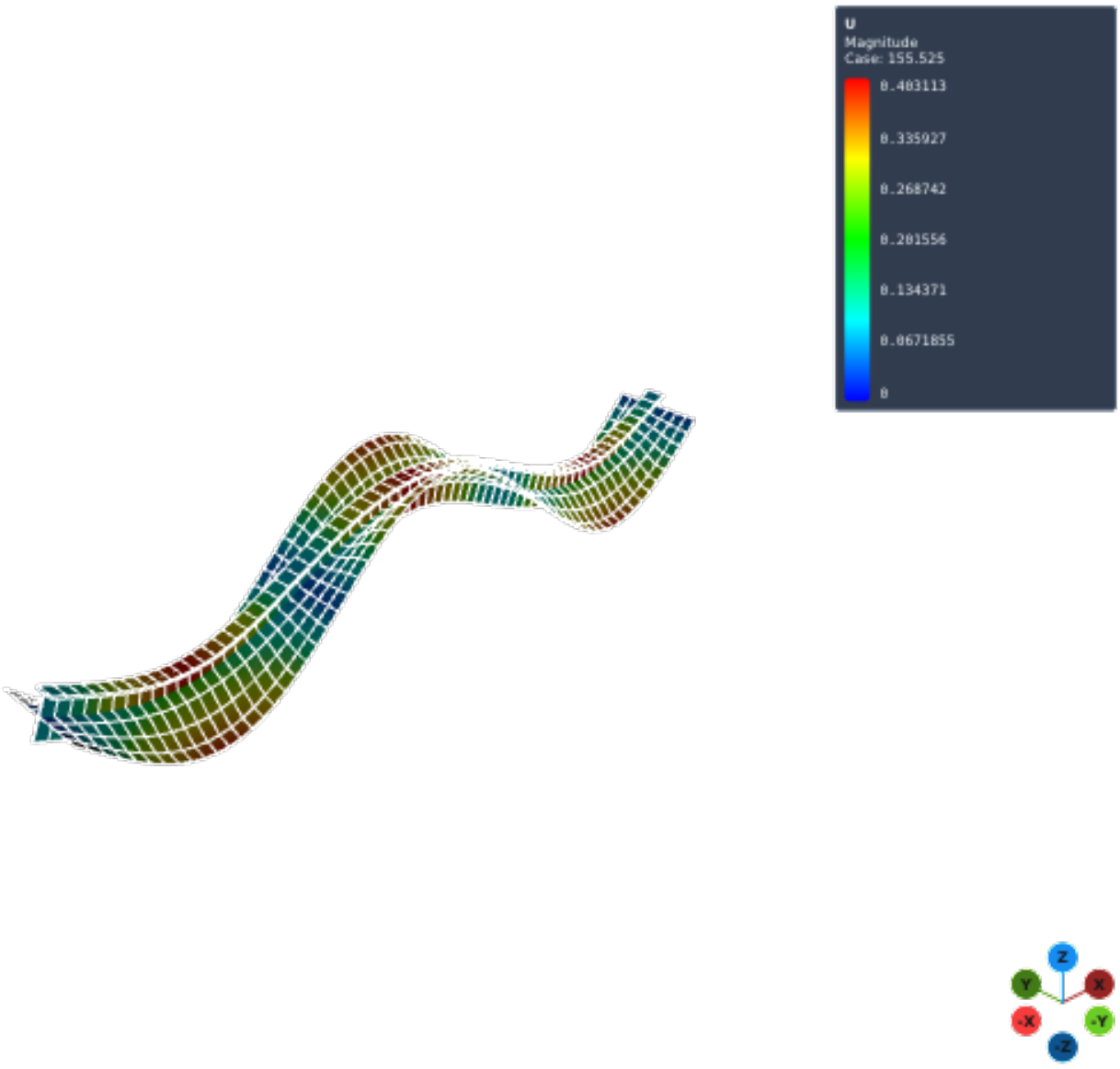

Figure 584: Abaqus — mode 6, 155.525 Hz

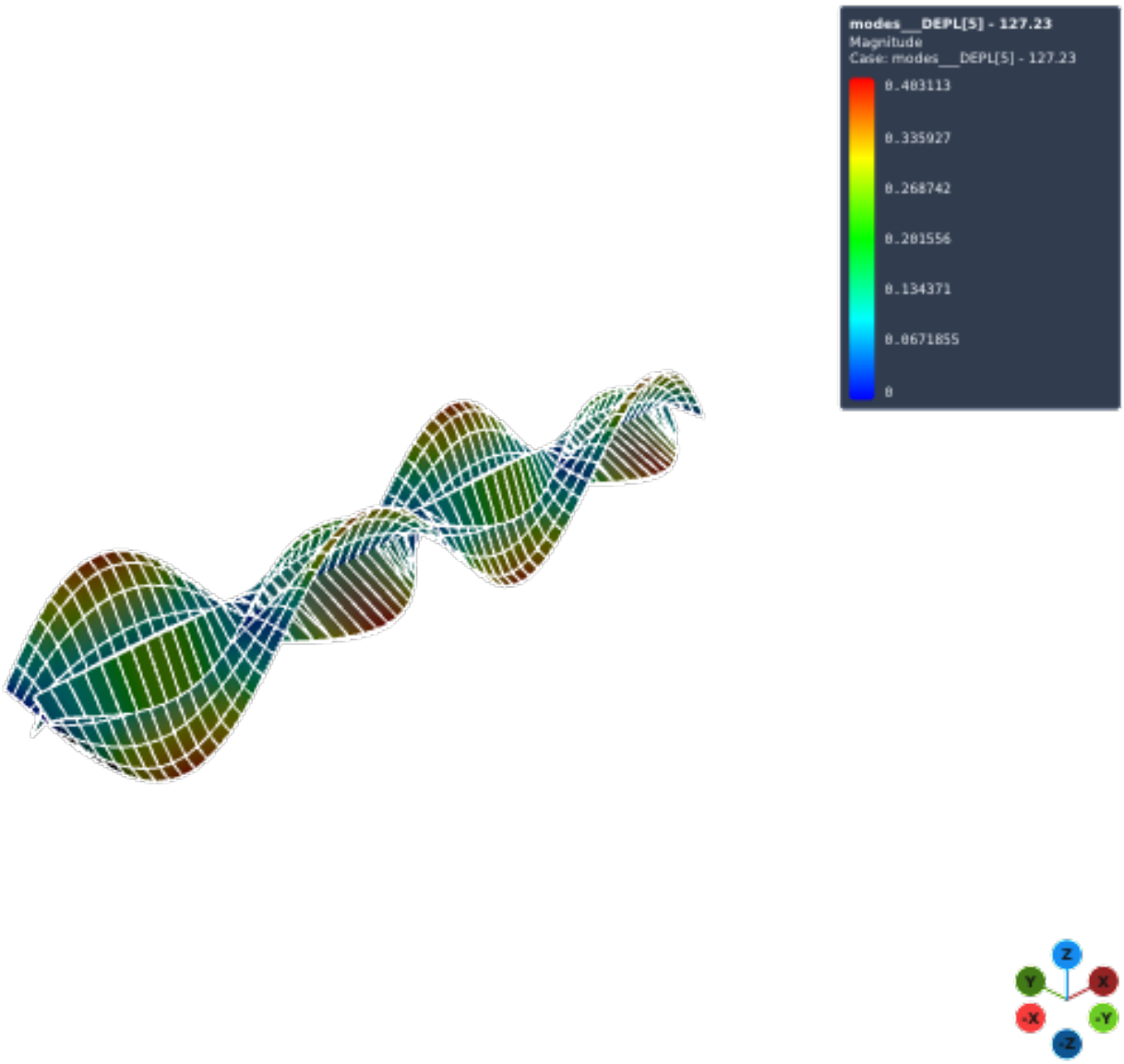

Figure 585: Code_Aster — mode 6, 127.230 Hz

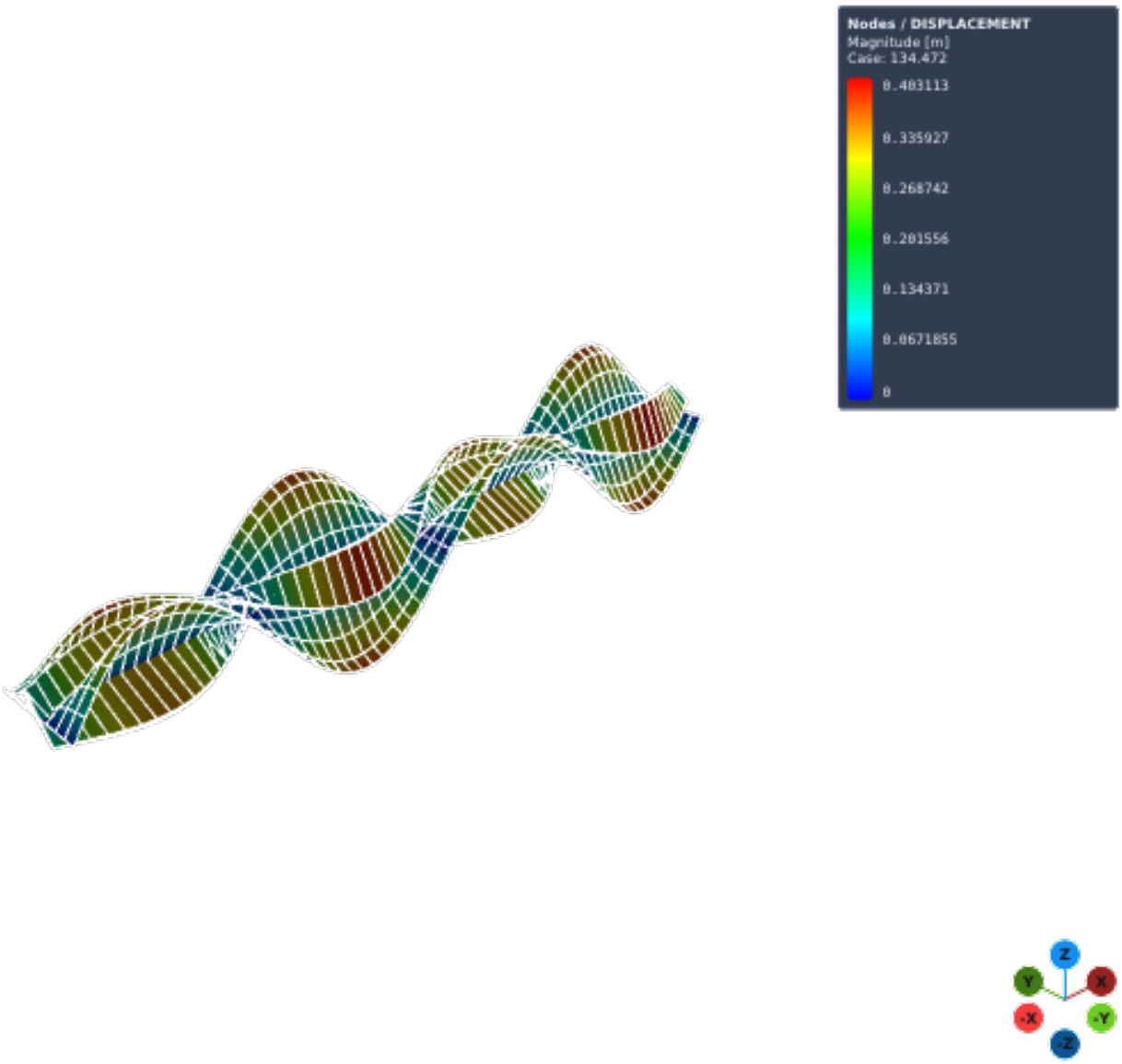

Figure 586: Sesam — mode 6, 134.472 Hz

Solver	f [Hz]	Δf vs mean [%]	λ [rad²/s²]	Meff X [kg]	Meff Y [kg]	Meff Z [kg]	Γ X	Γ Y	Γ Z
Abaqus	155.520	+11.83	954899	0.26	3.11e-16	20.18	0.507	1.76e-08	-4.492
Code_Aster	127.230	-8.52	639054	7.78e-34	1.87e-39	8.70e-26	-2.79e-17	4.33e-20	-2.95e-13
Sesam	134.472	-3.31	713882	4.52e-18	4.40e-11	2.57e-17	-2.13e-09	6.63e-06	5.07e-09

Plate static analysis

The deflected strip under 1 kPa, plain and stiffened, at the same 0.0625 m seed. The other seeds deform into the same shape; their convergence is in the summary table.

Shell, 1st order, plain, seed 0.0625 m

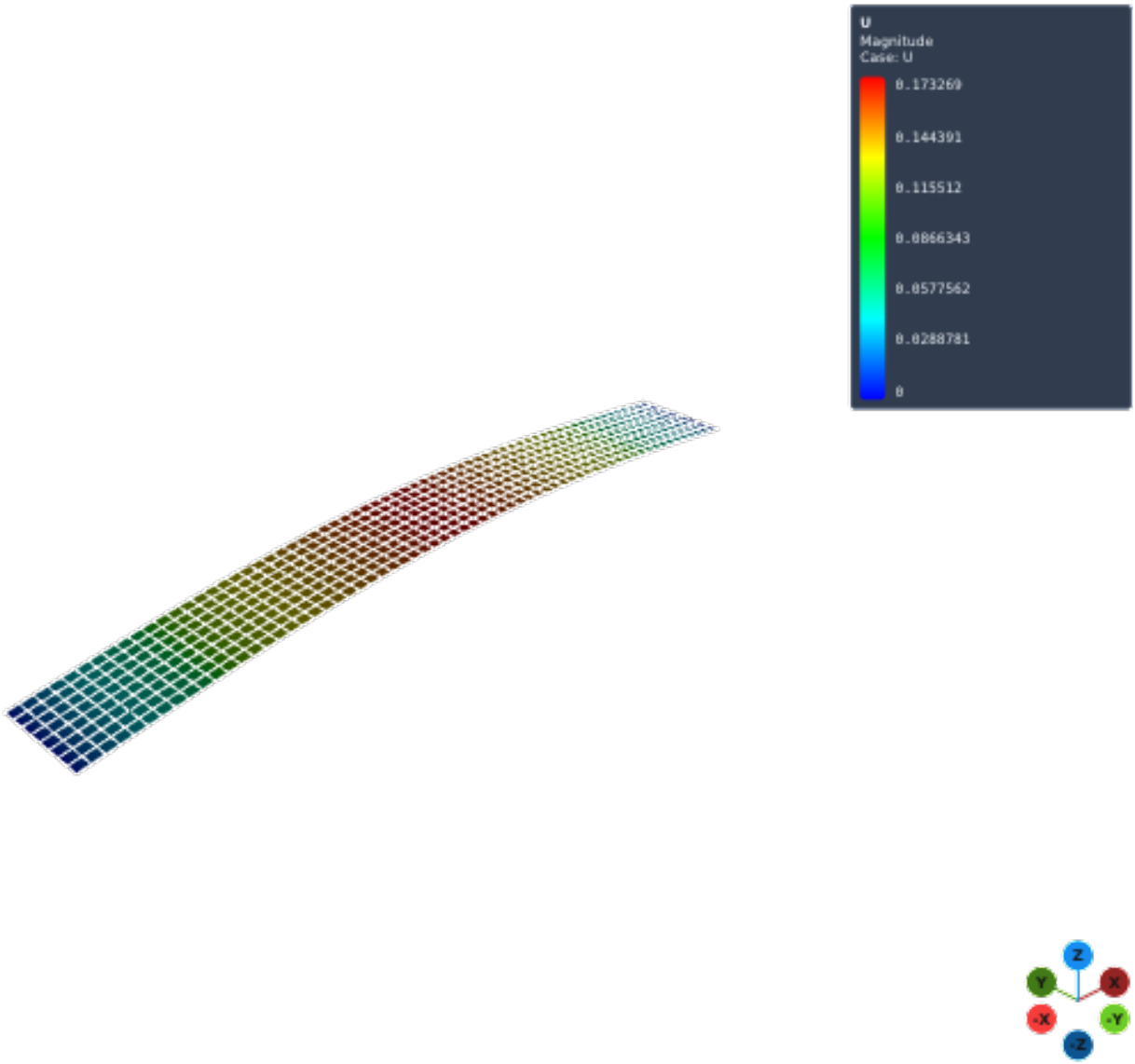

Figure 587: Abaqus — displacement

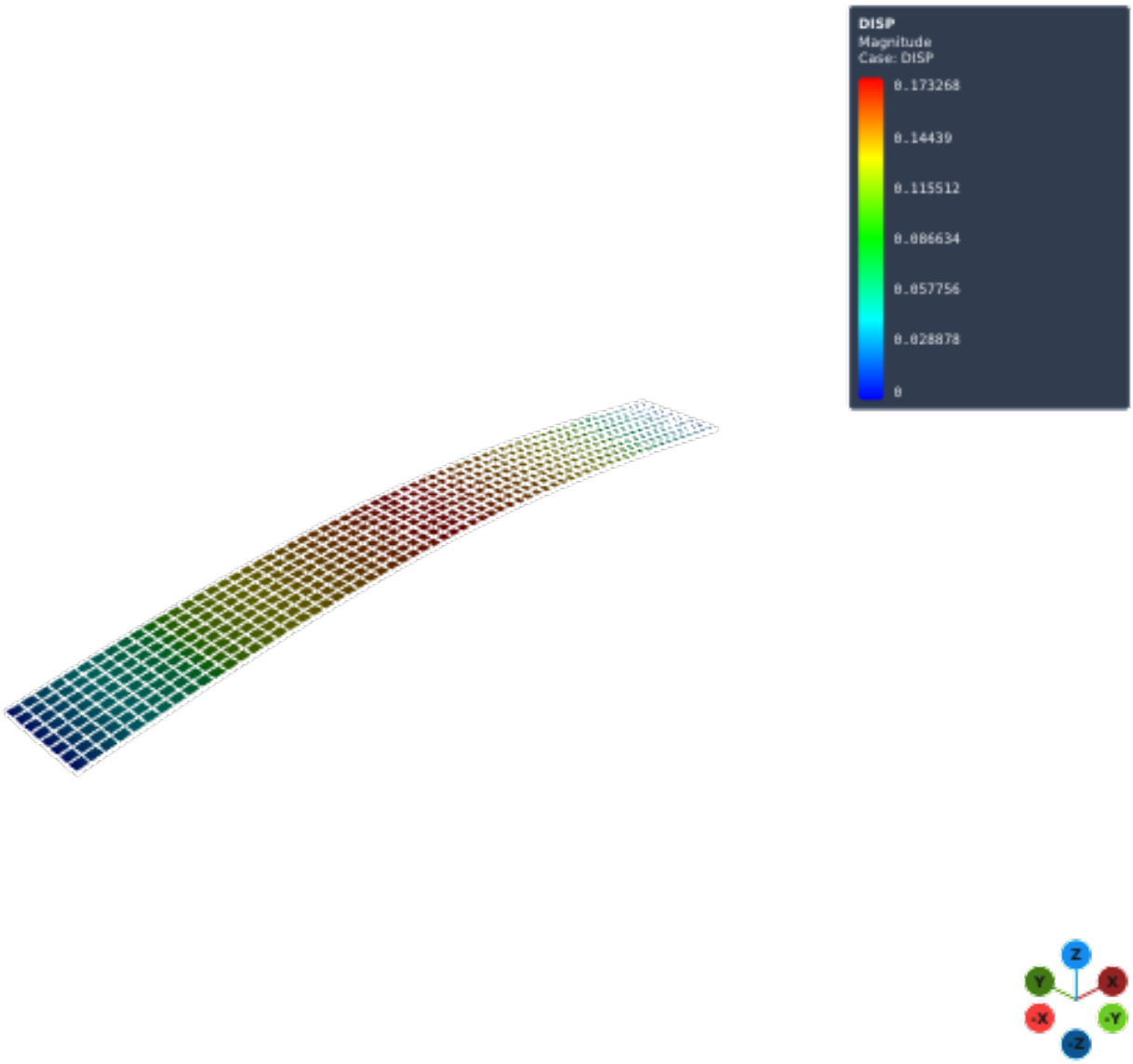

Figure 588: Calculix — displacement

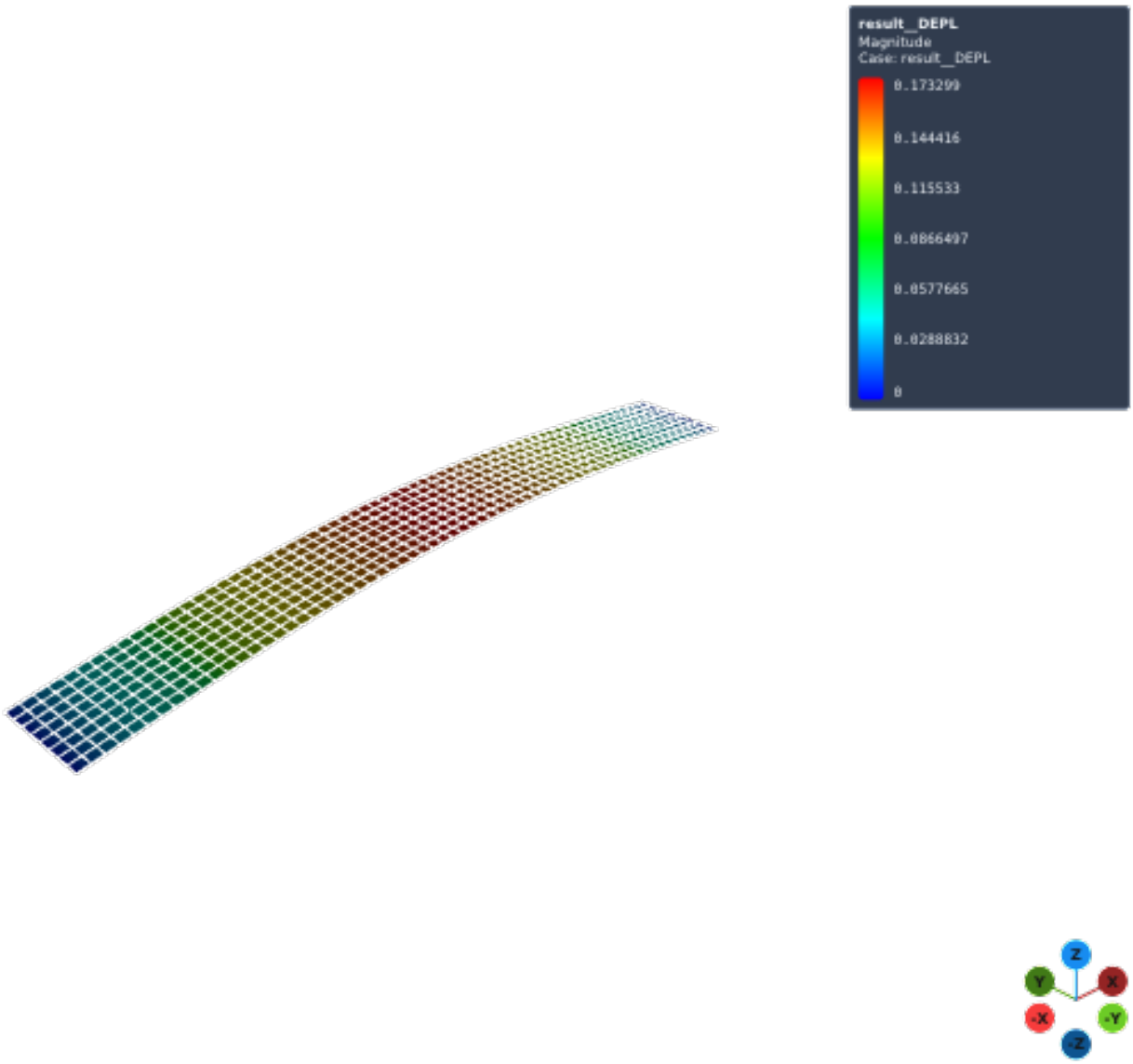

Figure 589: Code_Aster — displacement

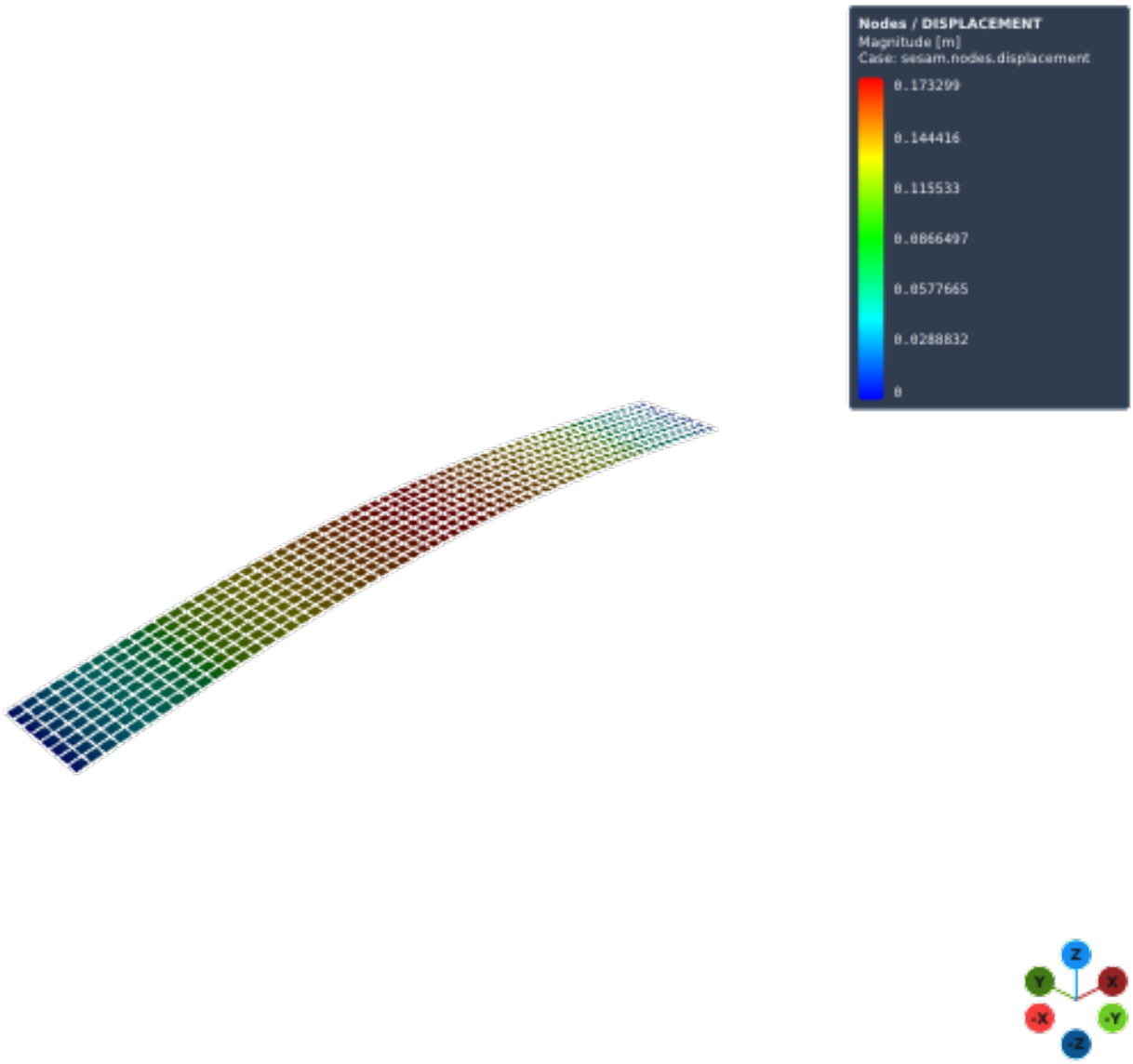

Figure 590: Sesam — displacement

Solver	Mid-span u_z [mm]	Δ vs closed form [%]	Peak u_z [mm]	Peak u_x [mm]	Peak u_y [mm]
Abaqus	173.2686	-0.04	173.2686	5.59e-17	4.12e-29
Calculix	173.2680	-0.04	173.2680	0.6932	2.94e-10
Code_Aster	173.2995	-0.02	173.2995	0.0000	0.0000
Sesam	173.2995	-0.02	173.2995	0.0000	0.0000
Closed form	173.3333				

Shell, 1st order, stiffened, seed 0.0625 m

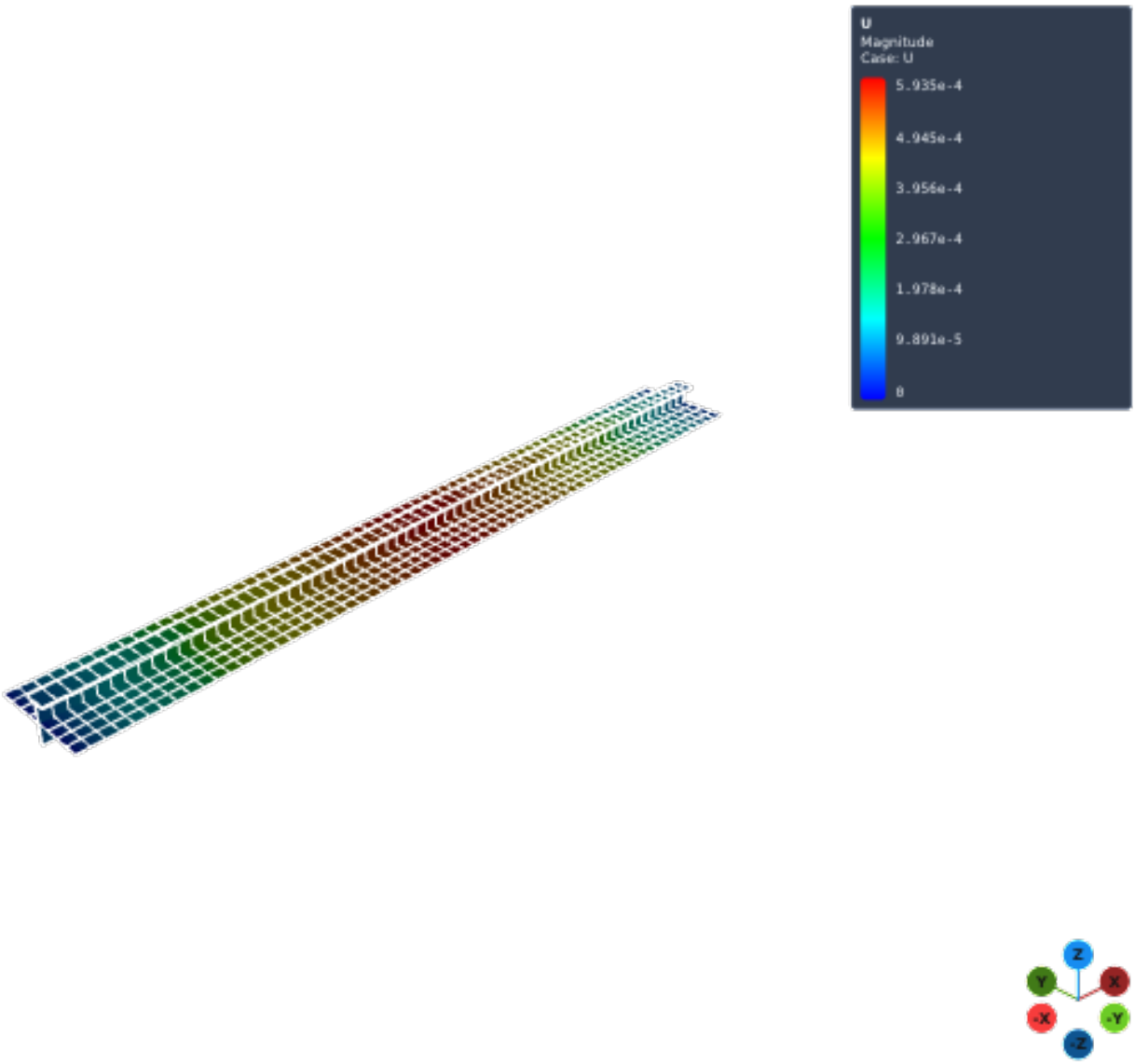

Figure 591: Abaqus — displacement

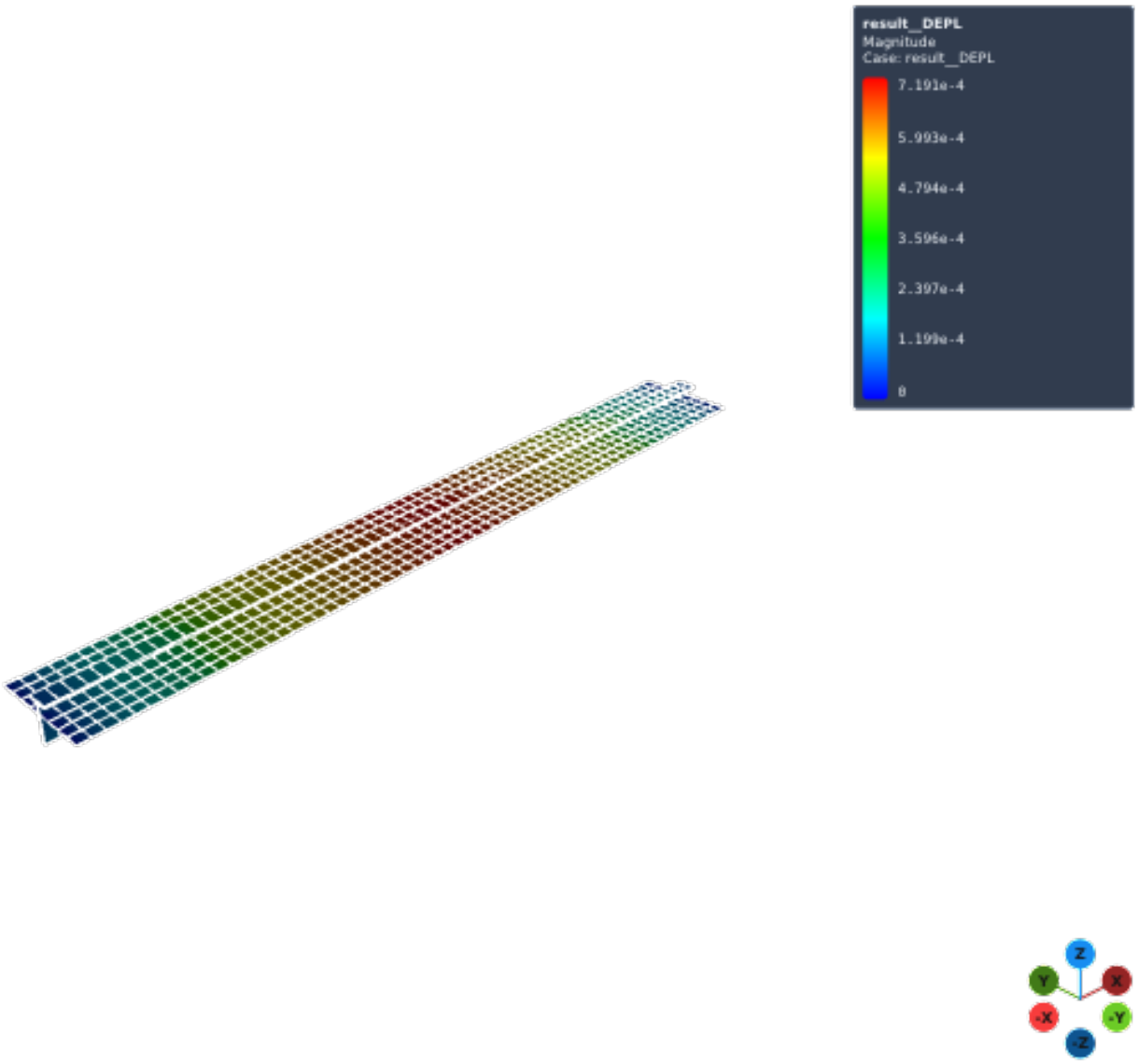

Figure 592: Code_Aster — displacement

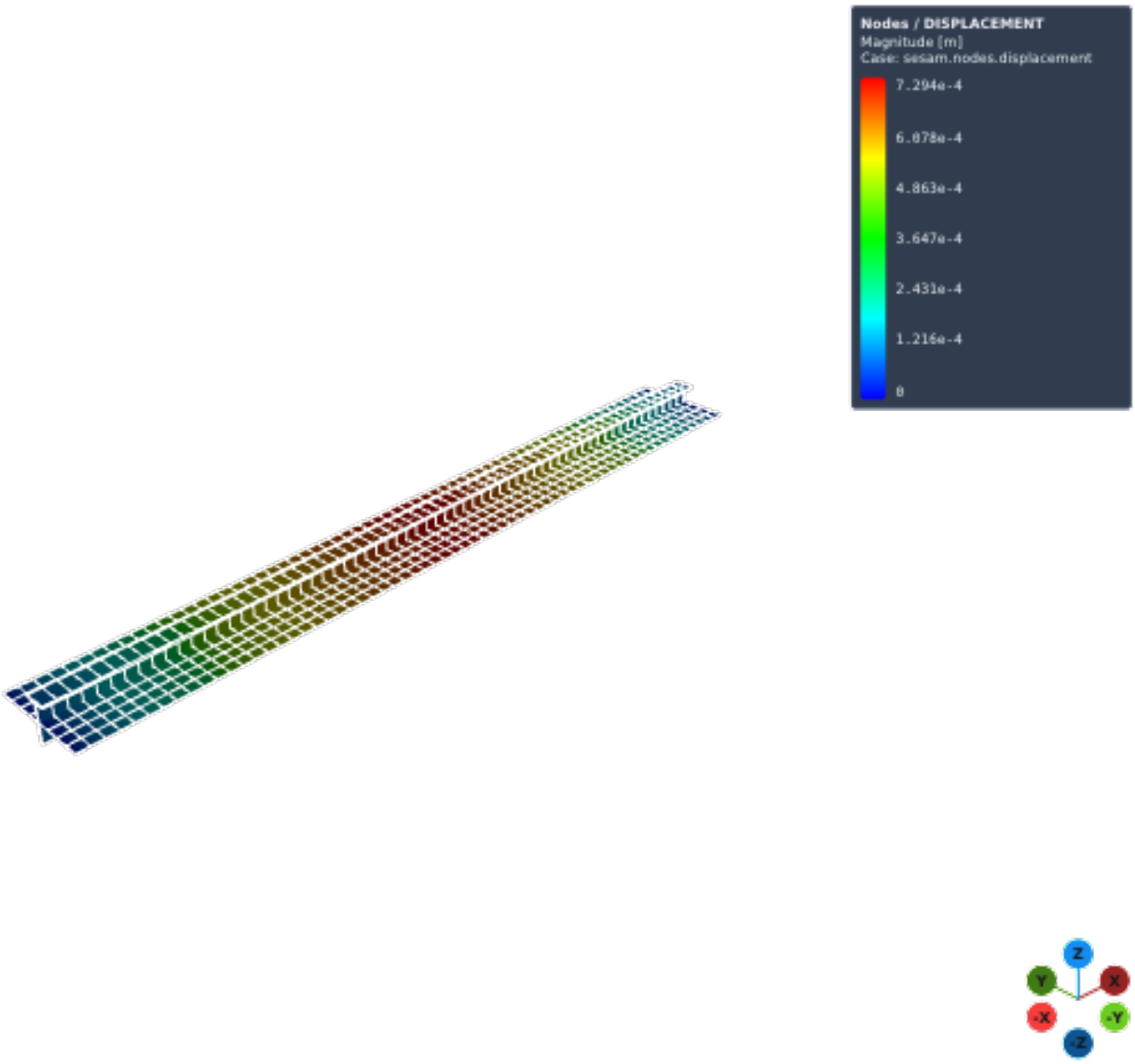

Figure 593: Sesam — displacement

Solver	Mid-span u_z [mm]	Δ vs mean [%]	Peak u_z [mm]	Peak u_x [mm]	Peak u_y [mm]
Abaqus	0.5855	-12.93	0.5934	0.0118	2.12e-05
Code_Aster	0.7107	+5.70	0.7191	0.0000	0.0000
Sesam	0.7210	+7.23	0.7294	1.18e-17	4.08e-17